Theory of Mathematics

Volume VI: Algebraic Geometry and Sheaf Theory

Contents

Part I

Classical Affine Geometry

Chapter 1

Algebraic Sets

1.1 Geometry from polynomial equations

Algebraic geometry begins with a simple operation: write down polynomial equations and study their common solutions. The decisive change of viewpoint is that we do not regard a single equation in isolation. We allow arbitrary families of polynomials, remember the field over which solutions are sought, and organize the resulting solution sets so that algebraic operations on polynomials translate into geometric operations on sets.

Let k be a field. Throughout this chapter,

$$k[x_1, \dots, x_n]$$

denotes the polynomial ring in n variables over k, and

$$\mathbb{A}^n_k := k^n$$

denotes affine n-space over k.

The set $\mathbb{A}^n_k$ is initially just a set of n-tuples. Its geometry will be created by declaring polynomial zero sets to be the distinguished subsets.

Learning goals

After this chapter, the reader should be able to:

- form the common zero locus of a family of polynomials;
- pass freely between a family of equations and the ideal it generates;
- prove the fundamental identities satisfied by algebraic sets;
- construct the vanishing ideal of an arbitrary subset of affine space;
- explain the order-reversing relation between equations and solution sets;
- recognize how the geometry changes when the ground field changes;
- distinguish a set of equations from the geometric set that it defines.

The chapter at a glance

The basic machine of classical affine algebraic geometry has two directions. Equations produce sets of points, while sets of points produce all equations that vanish on them.

$$
\begin{tikzcd}
\boxed{\begin{array}{c}\text{equations}\\ S \subseteq k[x_1,\ldots,x_n]\end{array}} \arrow[r, "\text{generate}"] & \boxed{\begin{array}{c}\text{generated ideal}\\ (S)\end{array}} \arrow[r, "\text{solve}"] & \boxed{\begin{array}{c}\text{zero locus}\\ V(S) = V((S))\end{array}} \\
\boxed{\begin{array}{c}\text{algebraic envelope}\\ V(I(X))\end{array}} & \boxed{\begin{array}{c}\text{vanishing ideal}\\ I(X)\end{array}} \arrow[l, "\text{solve}"'] & \boxed{\begin{array}{c}\text{subset of points}\\ X \subseteq \mathbb{A}^n_k\end{array}} \arrow[l, "\text{vanish on}"']
\end{tikzcd}
$$

Two warnings should be visible from the beginning:

1. the arrows reverse inclusions;
2. the round trip $I \mapsto V(I) \mapsto I(V(I))$ can enlarge an ideal, because an ordinary zero set forgets multiplicity and nilpotent information.

1.2 Affine space and polynomial functions

Definition 1.1 (Affine n-space)**.** For a field k, the *affine n-space over k* is

$$\mathbb{A}^n_k = k^n.$$

A point $a \in \mathbb{A}^n_k$ is written

$$a = (a_1, \ldots, a_n).$$

Every polynomial

$$f \in k[x_1, \ldots, x_n]$$

determines a function

$$f : \mathbb{A}^n_k \longrightarrow k, \qquad a \longmapsto f(a).$$

The distinction between a polynomial as a formal algebraic expression and the function it induces is harmless over infinite fields for many elementary purposes, but it becomes important over finite fields; see section 1.10.

Example 1.2. In $\mathbb{A}^2_{\mathbb{R}}$, the polynomial

$$f(x, y) = x^2 + y^2 - 1$$

vanishes exactly on the unit circle. The same formula over $\mathbb{C}$ defines a different geometric object: its points now have complex coordinates.

1.3 Common zero loci

Definition 1.3 (Vanishing set)**.** Let $S \subseteq k[x_1, \ldots, x_n]$. The *vanishing set* or *zero locus* of S is

$$V(S) := \{a \in \mathbb{A}^n_k : f(a) = 0 \text{ for every } f \in S\}.$$

If $S = \{f_1, \ldots, f_r\}$, we also write

$$V(f_1, \ldots, f_r).$$

Thus $V(S)$ is the simultaneous solution set of all equations

$$f = 0, \qquad f \in S.$$

Proposition 1.4 (Only the generated ideal matters)**.** *For every subset $S \subseteq k[x_1, \ldots, x_n]$,*

$$V(S) = V((S)),$$

where (S) is the ideal generated by S.

Proof. Since $S \subseteq (S)$, every point that annihilates all polynomials in (S) certainly annihilates all elements of S. Hence

$$V((S)) \subseteq V(S).$$

Conversely, let $a \in V(S)$. Every $g \in (S)$ has the form

$$g = h_1 f_1 + \cdots + h_r f_r$$

for finitely many $f_i \in S$ and $h_i \in k[x_1, \ldots, x_n]$. Evaluating at a,

$$g(a) = h_1(a) f_1(a) + \cdots + h_r(a) f_r(a) = 0.$$

Therefore $a \in V((S))$. □

This proposition is the first indication that ideals, rather than isolated equations, are the natural algebraic objects of affine algebraic geometry.

Example 1.5 (A reducible plane algebraic set)**.** Over a field k, the equation $xy = 0$ cuts out the union of the coordinate axes in $\mathbb{A}^2_k$. Thus $V(xy) = V(x) \cup V(y)$. The factorization of the polynomial already predicts the reducible geometry.

1.4 Algebraic sets

Definition 1.6 (Algebraic set)**.** A subset $X \subseteq \mathbb{A}^n_k$ is called an *algebraic set* if there exists a subset $S \subseteq k[x_1, \ldots, x_n]$ such that

$$X = V(S).$$

Equivalently, by theorem 1.4, there is an ideal $I \subseteq k[x_1, \ldots, x_n]$ with

$$X = V(I).$$

The term *affine algebraic set* is also common. Later chapters will distinguish irreducible algebraic sets, affine varieties, affine schemes, and more general schemes. At this stage, no irreducibility or smoothness assumption is imposed.

Example 1.7 (A line)**.** For $a, b, c \in k$, not both a, b zero,

$$V(ax + by - c) \subseteq \mathbb{A}^2_k$$

is an affine line.

Example 1.8 (A parabola)**.** In $\mathbb{A}^2_k$,

$$V(y - x^2)$$

is the parabola $y = x^2$.

Example 1.9 (A reducible-looking equation)**.** The equation

$$xy = 0$$

defines

$$V(xy) = V(x) \cup V(y),$$

the union of the two coordinate axes. The algebraic reason for this union formula is proved in theorem 1.13.

1.5 Set-theoretic identities

The operation V reverses containment: more equations mean fewer solutions.

Proposition 1.10 (Order reversal). *If $I \subseteq J$ are ideals of $k[x_1, \dots, x_n]$, then*

$$V(J) \subseteq V(I).$$

Proof. A point satisfying every equation in J satisfies, in particular, every equation in I. □

Proposition 1.11 (Extreme cases). *In $\mathbb{A}^n_k$,*

$$V((0)) = \mathbb{A}^n_k, \qquad V((1)) = \varnothing.$$

Proof. The zero polynomial vanishes everywhere, while the constant polynomial 1 vanishes nowhere. □

Proposition 1.12 (Arbitrary intersections). *For any family of ideals $\{I_\lambda\}_{\lambda\in\Lambda}$,*

$$\bigcap_{\lambda\in\Lambda} V(I_\lambda) = V\left(\sum_{\lambda\in\Lambda} I_\lambda\right).$$

Proof. A point lies in every $V(I_\lambda)$ precisely when every polynomial belonging to every I_λ vanishes there. This is equivalent to the vanishing of every finite sum of such polynomials, hence to vanishing of the ideal $\sum_\lambda I_\lambda$. □

Proposition 1.13 (Finite unions). *For ideals $I, J \subseteq k[x_1, \dots, x_n]$,*

$$V(I) \cup V(J) = V(IJ) = V(I \cap J).$$

Proof. Because $IJ \subseteq I \cap J$, order reversal gives

$$V(I \cap J) \subseteq V(IJ).$$

Also $I \cap J \subseteq I$ and $I \cap J \subseteq J$, so

$$V(I) \cup V(J) \subseteq V(I \cap J).$$

It remains to prove $V(IJ) \subseteq V(I) \cup V(J)$.

Take $a \notin V(I) \cup V(J)$. Then there exist $f \in I$ and $g \in J$ with $f(a) \neq 0$ and $g(a) \neq 0$. Since $fg \in IJ$,

$$(fg)(a) = f(a)g(a) \neq 0.$$

Thus $a \notin V(IJ)$. Taking contrapositives yields

$$V(IJ) \subseteq V(I) \cup V(J).$$

□

These identities will become the axioms of the Zariski topology in the next chapter: arbitrary intersections of algebraic sets are algebraic, and finite unions of algebraic sets are algebraic.

1.6 The ideal of a subset

We can reverse the construction. Instead of starting with equations and asking for their solutions, start with points and ask for all polynomial equations that vanish on them.

Definition 1.14 (Vanishing ideal)**.** For a subset $X \subseteq \mathbb{A}_k^n$, define

$$I(X) := \{f \in k[x_1, \dots, x_n] : f(a) = 0 \text{ for every } a \in X\}.$$

Proposition 1.15. *For every $X \subseteq \mathbb{A}_k^n$, the set $I(X)$ is an ideal of $k[x_1, \dots, x_n]$.*

Proof. The zero polynomial vanishes on X. If $f, g \in I(X)$, then

$$(f + g)(a) = f(a) + g(a) = 0$$

for every $a \in X$, so $f + g \in I(X)$. If $h \in k[x_1, \dots, x_n]$ and $f \in I(X)$, then

$$(hf)(a) = h(a)f(a) = 0$$

for every $a \in X$, hence $hf \in I(X)$. □

Proposition 1.16 (Order reversal for vanishing ideals)**.** *If $X \subseteq Y \subseteq \mathbb{A}_k^n$, then*

$$I(Y) \subseteq I(X).$$

Proof. A polynomial vanishing on all of Y certainly vanishes on the smaller set X. □

Example 1.17 (A finite algebraic set on the line)**.** In $\mathbb{A}_k^1$, if $a \neq b$ then $V((x - a)(x - b)) = \{a, b\}$. Over an algebraically closed field the zeros are exactly the linear factors, so factorization converts directly into geometry.

1.7 The V-I correspondence

The two constructions

$$I \longmapsto V(I), \qquad X \longmapsto I(X)$$

are both order reversing.

Proposition 1.18 (Closure inequalities)**.** *For every ideal $J \subseteq k[x_1, \dots, x_n]$ and every subset $X \subseteq \mathbb{A}_k^n$,*

$$J \subseteq I(V(J))$$

and

$$X \subseteq V(I(X)).$$

Proof. Every $f \in J$ vanishes by definition at every point of $V(J)$, so $f \in I(V(J))$. This proves the first inclusion.

For the second, let $a \in X$. Every polynomial in $I(X)$ vanishes at every point of X, hence in particular at a. Therefore $a \in V(I(X))$. □

Proposition 1.19 (Smallest algebraic set containing a subset)**.** *For every $X \subseteq \mathbb{A}_k^n$, the set*

$$V(I(X))$$

is the smallest algebraic set containing X.

Proof. By theorem 1.18, it contains X. Suppose $X \subseteq V(J)$ for some ideal J. Since every element of J vanishes on X,

$$J \subseteq I(X).$$

Order reversal for V then gives

$$V(I(X)) \subseteq V(J).$$

□

This construction anticipates Zariski closure. In Chapter VI/02, once algebraic sets are declared closed, the closure of X will be exactly $V(I(X))$.

1.8 Radicals appear naturally

Different ideals can define exactly the same algebraic set.

Example 1.20. In $\mathbb{A}^1_k$,

$$V(x) = V(x^2) = V(x^m)$$

for every integer $m \geq 1$.

The reason is that taking powers does not change whether a value is zero.

Definition 1.21 (Radical of an ideal)**.** For an ideal I in a commutative ring,

$$\sqrt{I} := \{f : f^m \in I \text{ for some } m \geq 1\}.$$

Proposition 1.22. *For every ideal $I \subseteq k[x_1, \ldots, x_n]$,*

$$V(I) = V(\sqrt{I}).$$

Proof. Since $I \subseteq \sqrt{I}$, we have

$$V(\sqrt{I}) \subseteq V(I).$$

Conversely, take $a \in V(I)$ and $f \in \sqrt{I}$. Then $f^m \in I$ for some $m \geq 1$, hence

$$0 = (f^m)(a) = f(a)^m.$$

Because k is a field, $f(a) = 0$. Thus every $f \in \sqrt{I}$ vanishes at a, so $a \in V(\sqrt{I})$. □

Corollary 1.23. *For every ideal $I \subseteq k[x_1, \ldots, x_n]$,*

$$\sqrt{I} \subseteq I(V(I)).$$

Proof. If $f \in \sqrt{I}$, then f vanishes on $V(I) = V(\sqrt{I})$. □

Over an algebraically closed field, Hilbert's Nullstellensatz gives the deep converse

$$I(V(I)) = \sqrt{I}.$$

We do not use that theorem here. Its role is to explain precisely which ideals encode the same affine algebraic set.

1.9 Finitely many equations

The definition of $V(S)$ allows an arbitrary family S of polynomial equations. In finite-dimensional affine space over a field, this apparent infinitude is unnecessary.

Proposition 1.24 (Finite description, using Hilbert's basis theorem)**.** *Every algebraic subset of $\mathbb{A}^n_k$ is the common zero locus of finitely many polynomials.*

Proof. Let $X = V(S)$. By theorem 1.4,

$$X = V((S)).$$

Hilbert's basis theorem says that $k[x_1, \ldots, x_n]$ is Noetherian, so the ideal (S) is finitely generated:

$$(S) = (f_1, \ldots, f_r).$$

Therefore

$$X = V(f_1, \ldots, f_r).$$

□

The Noetherian theorem belongs naturally to the commutative-algebra volume; here we record its geometric consequence.

Example 1.25 (Empty and full zero sets)**.** For any affine space, $V(1) = \varnothing$ because the constant polynomial 1 never vanishes, whereas $V(0) = \mathbb{A}^n_k$. These are the extremal cases of the order-reversing correspondence between ideals and zero sets.

1.10 Dependence on the ground field

The notation $V(f)$ is incomplete unless the field is understood.

Example 1.26. Consider

$$f(x) = x^2 + 1.$$

Then

$$V_{\mathbb{R}}(f) = \varnothing,$$

while

$$V_{\mathbb{C}}(f) = \{i, -i\}.$$

Example 1.27. In two variables,

$$V_{\mathbb{R}}(x^2 + y^2) = \{(0,0)\},$$

but over $\mathbb{C}$,

$$x^2 + y^2 = (x + iy)(x - iy),$$

so the zero locus is the union of two complex lines.

Finite fields produce another phenomenon. Over $\mathbb{F}_p$, the nonzero polynomial

$$x^p - x$$

vanishes at every point of $\mathbb{A}^1_{\mathbb{F}_p}$. Thus distinct polynomials may determine the same polynomial function on a finite field.

These examples are not pathologies. They show that the chosen base field is part of the geometric data.

Field-dependence chart

The polynomial is only half of the geometric datum. For the single equation

$$x^2 + 1 = 0,$$

the point set changes with the field:

Field	**Zero locus**	**Reason**
$\mathbb{R}$	$\varnothing$	no real square equals -1
$\mathbb{C}$	$\{i, -i\}$	$x^2 + 1 = (x - i)(x + i)$
$\mathbb{F}_2$	$\{1\}$	$-1 = 1$
$\mathbb{F}_5$	$\{2, 3\}$	$2^2 = 3^2 = 4 = -1$

This is why we write $\mathbb{A}^n_k$ and, when ambiguity is possible, $V_k(I)$.

1.11 A reusable dictionary

The basic algebra–geometry dictionary developed so far is:

Algebra	Geometry
$S \subseteq k[x_1, \dots, x_n]$	$V(S) \subseteq \mathbb{A}^n_k$
ideal generated by S	same zero locus as S
$I \subseteq J$	$V(J) \subseteq V(I)$
$I + J$	$V(I) \cap V(J)$
IJ or $I \cap J$	$V(I) \cup V(J)$
$\sqrt{I}$	same zero locus as I
$X \subseteq Y$	$I(Y) \subseteq I(X)$
$I(X)$	all equations true on X
$V(I(X))$	smallest algebraic set containing X

The reversal of arrows is fundamental: adding algebraic constraints removes geometric points, while adding geometric points removes polynomial equations.

1.12 Common pitfalls

Warning 1.28 (One equation is not one component). The zero locus of one polynomial can be a union of several pieces. The example $V(xy) = V(x) \cup V(y)$ is the simplest case.

Warning 1.29 (Different ideals may define the same set). The equality $V(x) = V(x^2)$ shows that geometric zero sets forget multiplicity and nilpotent information. Schemes will later restore precisely this information.

Warning 1.30 (The field matters). A polynomial may have no real zeros and many complex zeros. Always keep the base field visible when changing fields.

Warning 1.31 (Do not confuse $V(I)$ with Spec). This chapter studies solutions in k^n. Prime spectra introduce a different space whose points are prime ideals. That transition is postponed to Chapters VI/06–VI/08.

1.13 Worked examples

Example 1.32 (From equations to a single point)**.** Let $a, b \in k$. The equations

$$x - a = 0, \qquad y - b = 0$$

cut out exactly one point:

$$V(x - a, y - b) = \{(a, b)\}.$$

This is the simplest illustration of simultaneous equations producing an algebraic set.

Example 1.33 (One equation can encode several pieces)**.** In $\mathbb{A}^2_k$,

$$V\big(x(y-1)\big) = V(x) \cup V(y-1).$$

Geometrically this is the union of the vertical line $x = 0$ and the horizontal line $y = 1$. The factorization of the polynomial is already predicting the union identity $V(IJ) = V(I) \cup V(J)$.

Example 1.34 (Intersection by adding equations)**.** Let

$$L = V(y - x), \qquad P = V(y - x^2).$$

Then

$$L \cap P = V(y - x, y - x^2).$$

Subtracting the equations gives

$$x^2 - x = x(x-1) = 0.$$

Hence, in every field with $0 \neq 1$,

$$L \cap P = \{(0,0), (1,1)\}.$$

The geometric intersection is obtained by imposing both systems at once.

Example 1.35 (The coordinate axes in three-space)**.** Consider

$$X = V(xy, xz, yz) \subseteq \mathbb{A}^3_k.$$

At a point of X, no two coordinates can be simultaneously nonzero. Therefore

$$X = V(y, z) \cup V(x, z) \cup V(x, y),$$

the union of the three coordinate axes. This example will return as a harder problem when we ask for its full vanishing ideal.

Example 1.36 (Different ideals, identical geometry)**.** The ideals

$$I = (x, y), \qquad J = (x^2, xy, y^2)$$

are not equal, but

$$V(I) = V(J) = \{(0,0)\}.$$

Indeed, $J \subsetneq I$ while $\sqrt{J} = I$. Ordinary algebraic sets remember where polynomials vanish, but not how strongly they vanish.

Example 1.37 (The algebraic envelope of infinitely many points)**.** Assume k is infinite and put

$$S = \{(t, 0) : t \in k\} \subseteq \mathbb{A}^2_k.$$

Every polynomial vanishing on S is divisible by y. To see the key point, write

$$f(x, y) = a_0(x) + a_1(x)y + \cdots + a_m(x)y^m.$$

If $f(t, 0) = 0$ for every $t \in k$, then $a_0(t) = 0$ for infinitely many t, so $a_0 = 0$. Hence

$$I(S) = (y), \qquad V(I(S)) = V(y).$$

Thus an infinite set of points can force an entire algebraic line.

Example 1.38 (A parametrized cusp)**.** Let

$$C = V(y^2 - x^3) \subseteq \mathbb{A}^2_k.$$

The map

$$t \longmapsto (t^2, t^3)$$

lands in C because

$$(t^3)^2 - (t^2)^3 = 0.$$

Conversely, if $(x, y) \in C$ and $x \neq 0$, then

$$t = \frac{y}{x}$$

satisfies $x = t^2$ and $y = t^3$. This is a first glimpse of how equations and parametrizations can describe the same geometric object in different ways.

Example 1.39 (The ground field changes the point set)**.** For the equation

$$x^2 + 1 = 0,$$

we obtain

$$V_{\mathbb{R}}(x^2 + 1) = \varnothing, \qquad V_{\mathbb{C}}(x^2 + 1) = \{i, -i\}.$$

Over $\mathbb{F}_5$, the solutions are 2 and 3, because

$$2^2 = 3^2 = 4 = -1.$$

The polynomial expression has not changed; the geometry has.

Example 1.40 (A finite-field polynomial that vanishes everywhere)**.** Over $\mathbb{F}_p$, Fermat's identity gives

$$a^p - a = 0$$

for every $a \in \mathbb{F}_p$. Hence the nonzero polynomial

$$x^p - x$$

induces the zero function on $\mathbb{A}^1_{\mathbb{F}_p}$. This is why formal polynomials and polynomial functions must be distinguished over finite fields.

Example 1.41 (A closure calculation without topology)**.** Let k be infinite and

$$S = \{(t, t^2) : t \in k\} \subseteq \mathbb{A}^2_k.$$

Certainly

$$y - x^2 \in I(S),$$

so

$$V(I(S)) \subseteq V(y - x^2).$$

The reverse inclusion follows once one proves

$$I(S) = (y - x^2),$$

which is developed in Problem VI.01.P2. Thus the smallest algebraic set containing the parametrized point set is exactly the parabola.

1.14 Exercises

The exercises are deliberately pedagogical rather than archival: they progress from direct manipulation of zero loci to the V-I closure mechanism and field dependence.

Exercise VI.01.1 — Zero-locus warm-up

Describe the following subsets of affine space:

$$V(0), \qquad V(1), \qquad V(x) \subseteq \mathbb{A}^2_k, \qquad V(x, y) \subseteq \mathbb{A}^2_k, \qquad V\big(x(y-1)\big) \subseteq \mathbb{A}^2_k.$$

Exercise VI.01.2 — Changing generators without changing solutions

Let $f, g, h \in k[x_1, \ldots, x_n]$. Prove

$$V(f, g) = V(f, g + hf).$$

Explain why elementary row operations on a finite list of equations do not change its common zero locus when they preserve the generated ideal.

Exercise VI.01.3 — Intersections are sums of ideals

Let

$$X = V(x), \qquad Y = V(y - 1)$$

in $\mathbb{A}^2_k$. Compute $X \cap Y$ directly and verify

$$X \cap Y = V\big((x) + (y - 1)\big).$$

Then prove the corresponding formula for arbitrary ideals I, J.

Exercise VI.01.4 — A union in three variables

Describe

$$V(xy, xz) \subseteq \mathbb{A}^3_k$$

as a union of simpler algebraic sets, and verify your answer using the identities for finite unions.

Exercise VI.01.5 — Vanishing ideal of a coordinate subspace

Let
$$L = \{(t, 0, 0) : t \in k\} \subseteq \mathbb{A}^3_k.$$
Assume k is infinite. Prove
$$I(L) = (y, z).$$

Exercise VI.01.6 — Order reversal in both directions

Prove
$$I \subseteq J \Longrightarrow V(J) \subseteq V(I),$$
and
$$X \subseteq Y \Longrightarrow I(Y) \subseteq I(X).$$
Give a geometric sentence explaining why both arrows reverse.

Exercise VI.01.7 — The algebraic-envelope operator

Define
$$\mathrm{Alg}(X) := V(I(X)).$$
Prove that this operator is extensive, monotone, and idempotent:
$$X \subseteq \mathrm{Alg}(X),$$
$$X \subseteq Y \Longrightarrow \mathrm{Alg}(X) \subseteq \mathrm{Alg}(Y),$$
and
$$\mathrm{Alg}(\mathrm{Alg}(X)) = \mathrm{Alg}(X).$$

Exercise VI.01.8 — Radicals do not change zero loci

Prove
$$V(I) = V(\sqrt{I})$$
for every ideal $I \subseteq k[x_1, \ldots, x_n]$. Identify exactly where the fact that a field has no nonzero nilpotent elements is used.

Exercise VI.01.9 — Same set, different ideals

Show that
$$(x, y) \neq (x^2, xy, y^2)$$
but that the two ideals define the same subset of $\mathbb{A}^2_k$. Compute the radical of the second ideal.

Exercise VI.01.10 — A field-dependence table

Determine $V(x^2 + 1) \subseteq \mathbb{A}^1_k$ for
$$k = \mathbb{R}, \qquad \mathbb{C}, \qquad \mathbb{F}_2, \qquad \mathbb{F}_3, \qquad \mathbb{F}_5.$$

Exercise VI.01.11 — Polynomial functions over a finite field

Over $\mathbb{F}_p$, prove that $x^p - x$ vanishes at every point. Deduce that the map from formal polynomials to polynomial functions on $\mathbb{F}_p$ is not injective.

Exercise VI.01.12 — Every finite subset of the affine line is algebraic

Let $a_1, \ldots, a_r \in k$ be distinct. Show

$$\{a_1, \ldots, a_r\} = V\left(\prod_{j=1}^{r}(x - a_j)\right) \subseteq \mathbb{A}^1_k.$$

No algebraic-closedness hypothesis is needed.

Exercise VI.01.13 — Line meets parabola

Determine

$$V(y - x) \cap V(y - x^2)$$

in $\mathbb{A}^2_k$. Explain why the answer is valid in every field and what changes, if anything, in characteristic 2.

Exercise VI.01.14 — Parametrizing the cusp away from the origin

Let

$$C = V(y^2 - x^3).$$

Verify that $(t^2, t^3) \in C$ for every $t \in k$. If $(x, y) \in C$ and $x \neq 0$, prove that $t = y/x$ recovers the point.

Exercise VI.01.15 — Infinite points force a line

Assume k is infinite and let

$$S = \{(t, 0) : t \in k\} \subseteq \mathbb{A}^2_k.$$

Prove

$$I(S) = (y)$$

and hence

$$V(I(S)) = V(y).$$

Exercise VI.01.16 — Infinitely many equations reduce to finitely many

Assume Hilbert's basis theorem. If $S \subseteq k[x_1, \ldots, x_n]$ is arbitrary, prove that there exists a finite subset $S_0 \subseteq S$ such that

$$V(S) = V(S_0).$$

Exercise VI.01.17 — Extending the ground field

Let K/k be a field extension and let $I \subseteq k[x_1, \ldots, x_n]$. Show that

$$V_k(I) = V_K(I) \cap k^n,$$

where the same polynomials are evaluated in K. Give an example where $V_K(I)$ is strictly larger.

Exercise VI.01.18 — Reconstruct the dictionary

Without looking back at the summary table, fill in the geometric counterpart of each algebraic operation:

$$I + J, \qquad IJ, \qquad I \cap J, \qquad \sqrt{I},$$

and the algebraic counterpart of

$$X \subseteq Y, \qquad X \cup Y.$$

Prove every identity you use.

1.15 Problems

The problems are longer synthesis tasks. In the student edition only the statement is shown. In the complete edition each problem expands in place to include orientation, prerequisites, guided hints, a full solution, commentary, extensions, and provenance.

Problem VI.01.P1 — The three coordinate axes and their vanishing ideal

Problem. Assume k is infinite and let

$$X = V(xy, xz, yz) \subseteq \mathbb{A}^3_k.$$

Prove first that X is exactly the union of the three coordinate axes. Then prove the stronger statement

$$I(X) = (xy, xz, yz).$$

Problem VI.01.P2 — Recovering the equation of a parametrized parabola

Problem. Assume k is infinite and set

$$S = \{(t, t^2) : t \in k\} \subseteq \mathbb{A}^2_k.$$

Prove

$$I(S) = (y - x^2).$$

Deduce that the smallest algebraic set containing S is the parabola

$$V(y - x^2).$$

Problem VI.01.P3 — The two closure machines

Problem. For subsets $X \subseteq \mathbb{A}^n_k$ and ideals $J \subseteq k[x_1, \ldots, x_n]$, define

$$C(X) := V(I(X)), \qquad R(J) := I(V(J)).$$

Prove that both C and R are extensive, monotone, and idempotent. Prove also that

$$X = C(X)$$

if and only if X is algebraic, while

$$J = R(J)$$

if and only if J is the vanishing ideal of some subset of affine space.

Problem VI.01.P4 — What fails over a non-algebraically-closed field?

Problem. Work in $\mathbb{R}[x]$ with

$$I = (x^2 + 1).$$

Show that I is radical but

$$I(V_{\mathbb{R}}(I)) = \mathbb{R}[x].$$

Then repeat the calculation over $\mathbb{C}$ and explain why the outcome changes.

Problem VI.01.P5 — Finite point sets in affine space

Problem. Let

$$F = \{p^{(1)}, \dots, p^{(r)}\} \subseteq \mathbb{A}_k^n$$

be a finite set of distinct points. For $p = (a_1, \dots, a_n)$, put

$$\mathfrak{m}_p = (x_1 - a_1, \dots, x_n - a_n).$$

Prove

$$\{p\} = V(\mathfrak{m}_p), \qquad F = V\left(\prod_{p \in F} \mathfrak{m}_p\right),$$

and

$$I(F) = \bigcap_{p \in F} \mathfrak{m}_p.$$

Problem VI.01.P6 — All polynomial functions on a finite field

Problem. Let $k = \mathbb{F}_q$. Consider the evaluation map

$$\mathrm{ev} : k[x] \longrightarrow \mathrm{Fun}(k, k), \qquad f \longmapsto (a \mapsto f(a)).$$

Prove that ev is surjective and that

$$\ker(\mathrm{ev}) = (x^q - x).$$

Deduce

$$k[x]/(x^q - x) \cong \mathrm{Fun}(k, k).$$

Problem VI.01.P7 — An infinite system has a finite subsystem

Problem. Assume Hilbert's basis theorem. Let

$$S \subseteq k[x_1, \dots, x_n]$$

be an arbitrary, possibly infinite, family of equations. Prove that there is a finite subset $S_0 \subseteq S$ such that

$$(S_0) = (S)$$

and therefore

$$V(S_0) = V(S).$$

Explain why finite generation of (S) alone does not immediately say the chosen generators belong to S, and how to repair that point.

Problem VI.01.P8 — Vanishing ideals of unions and intersections

Problem. For arbitrary subsets $X, Y \subseteq \mathbb{A}_k^n$, prove

$$I(X \cup Y) = I(X) \cap I(Y).$$

Also prove

$$I(X) + I(Y) \subseteq I(X \cap Y).$$

Show that the second inclusion can be strict: assume k is infinite and take

$$X = V(x), \qquad Y = V(x - y^2)$$

in $\mathbb{A}_k^2$. Compute all three relevant ideals.

1.16 Challenges

The challenges push the chapter's ideas beyond the minimum prerequisites and preview later algebraic geometry and commutative algebra.

Challenge VI.01.C1 — The vanishing ideal of finite affine space

Challenge. Let $k = \mathbb{F}_q$. Prove

$$I(k^n) = (x_1^q - x_1, \ldots, x_n^q - x_n) \subseteq k[x_1, \ldots, x_n].$$

Challenge VI.01.C2 — Multivariate interpolation on an arbitrary finite set

Challenge. Let $F \subseteq k^n$ be finite. Prove that every function

$$\varphi : F \to k$$

is the restriction of a polynomial in $k[x_1, \ldots, x_n]$. Construct the polynomial explicitly.

Challenge VI.01.C3 — Why the coefficient field matters to the union formula

Challenge. Suppose one tries to define zero loci in R^n for a commutative ring R with identity, using the same evaluation definition. Prove that

$$V(IJ) = V(I) \cup V(J)$$

for all polynomial ideals $I, J \subseteq R[x_1, \ldots, x_n]$ if R is an integral domain. Show that the statement can fail when R has zero divisors by taking

$$R = \mathbb{Z}/6\mathbb{Z}, \qquad I = (2), \qquad J = (3).$$

In fact, show that validity of the formula for all constant principal ideals forces R to be a domain.

1.17 Chapter summary

An affine algebraic set is a common zero locus

$$V(I) \subseteq \mathbb{A}_k^n$$

of polynomial equations. The ideal generated by a family of equations does not change its zero locus. Arbitrary intersections and finite unions of algebraic sets remain algebraic:

$$\bigcap_\lambda V(I_\lambda) = V\left(\sum_\lambda I_\lambda\right), \qquad V(I) \cup V(J) = V(IJ).$$

Every subset X has a vanishing ideal $I(X)$, and

$$V(I(X))$$

is the smallest algebraic set containing X. The operations V and I both reverse inclusions, and radicals do not alter zero loci:

$$V(I) = V(\sqrt{I}).$$

These facts supply exactly the closure properties needed for the next chapter, where algebraic sets become the closed sets of the Zariski topology.

1.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 1.42 (Union of axes)**.** Compute $V(xy)$ in $\mathbb{A}_k^2$.

Hint. Use zero sets, vanishing ideals, and polynomial equations. ☐

Solution. It is $V(x) \cup V(y)$, the two coordinate axes. ☐

Exercise 1.43 (Parabola)**.** Describe $V(y - x^2)$ in $\mathbb{A}_k^2$.

Hint. Use zero sets, vanishing ideals, and polynomial equations. ☐

Solution. It is the affine parabola $\{(a, a^2) : a \in k\}$. ☐

Exercise 1.44 (Two points)**.** Find $V((x-1)(x+1))$ in $\mathbb{A}_k^1$ when $1 \neq -1$.

Hint. Use zero sets, vanishing ideals, and polynomial equations. ☐

Solution. It is $\{1, -1\}$. ☐

Exercise 1.45 (Unit ideal)**.** Compute $V(1)$.

Hint. Use zero sets, vanishing ideals, and polynomial equations. ☐

Solution. It is empty. ☐

Exercise 1.46 (Zero ideal)**.** Compute $V(0)$.

Hint. Use zero sets, vanishing ideals, and polynomial equations. ☐

Solution. It is all affine space. ☐

Proofs

Exercise 1.47 (Union formula)**.** Prove: Prove $V(IJ) = V(I) \cup V(J)$ for ideals in a polynomial ring over a field.

Hint. Work directly from zero sets, vanishing ideals, and polynomial equations. ☐

Solution. A point annihilates every product fg with $f \in I$, $g \in J$ iff it annihilates all of I or all of J; otherwise choose one nonvanishing element from each ideal. ☐

Exercise 1.48 (Sum formula)**.** Prove: Prove $V(I + J) = V(I) \cap V(J)$.

Hint. Work directly from zero sets, vanishing ideals, and polynomial equations. ☐

Solution. Vanishing on the sum is equivalent to vanishing on each summand ideal. ☐

Exercise 1.49 (Order reversal)**.** Prove: Prove $I \subseteq J$ implies $V(J) \subseteq V(I)$.

Hint. Work directly from zero sets, vanishing ideals, and polynomial equations. ☐

Solution. Every polynomial required to vanish by I also lies in J, so a common zero of J is a common zero of I. ☐

Exercise 1.50 (Radical invariance)**.** Prove: Prove $V(I) = V(\sqrt{I})$.

Hint. Work directly from zero sets, vanishing ideals, and polynomial equations. ☐

Solution. A polynomial and every positive power have the same pointwise zero set; use the definition of radical membership. ☐

Counterexamples and hypothesis tests

Exercise 1.51 (Infinite union)**.** Test the claim: an arbitrary union of algebraic sets is algebraic.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general; Zariski closed sets are closed under finite, not arbitrary, unions. ☐

Exercise 1.52 (Equation dependence)**.** Test the claim: different ideals always define different algebraic sets.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; I and $\sqrt{I}$ define the same zero set. ☐

Exercise 1.53 (Field dependence)**.** Test the claim: $V(x^2 + 1)$ is the same over $\mathbb{R}$ and $\mathbb{C}$.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False: it is empty over $\mathbb{R}$ and has two points over $\mathbb{C}$. ☐

Applications and investigations

Exercise 1.54 (Constraint modeling)**.** Why are algebraic sets natural solution spaces for polynomial constraints?

Hint. Translate the question into zero sets, vanishing ideals, and polynomial equations. ☐

Solution. They collect exactly the simultaneous zeros of the defining equations. ☐

Exercise 1.55 (Factorization geometry)**.** What does factorization of a defining polynomial often reveal?

Hint. Translate the question into zero sets, vanishing ideals, and polynomial equations. ☐

Solution. It often decomposes a hypersurface into a union of algebraic components. ☐

Challenge problems

Exercise 1.56 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Algebraic Sets in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Algebraic Sets, translates it through zero sets, vanishing ideals, and polynomial equations, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 1.57 (Hypothesis challenge)**.** Choose one structural statement from Algebraic Sets, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 2

The Zariski Topology

Purpose and learning goals

Chapter VI/01 built algebraic subsets of affine space from polynomial equations. The next step is conceptual: instead of treating those subsets merely as solution sets, we declare them to be the *closed sets* of a topology. This converts algebra into a language of closure, density, neighborhoods, continuity, and separation.

The topology is much coarser than Euclidean topology. That is not a defect. Its open sets are designed to record where polynomial conditions fail, and its closure operator records exactly which polynomial equations a subset forces.

Throughout, k is a field and

$$\mathbb{A}^n_k = k^n.$$

When an infinite-field hypothesis is needed, it is stated explicitly.

After this chapter the reader should be able to:

- derive the topology axioms from identities for zero loci;
- compute Zariski closures from vanishing ideals;
- work with the induced topology on affine algebraic sets;
- use principal opens $D_X(f)$ as a basis;
- distinguish finite-field, infinite-field, Euclidean, and product-topology behavior;
- prove density by showing that no nonzero polynomial vanishes identically;
- recognize determinant and minor nonvanishing as open rank conditions;
- explain why classical affine space is T_1 but usually far from Hausdorff.

2.1 From equations to topology

For ideals $I, J \subseteq k[x_1, \ldots, x_n]$, Chapter VI/01 established

$$V((0)) = \mathbb{A}^n_k, \qquad V((1)) = \varnothing,$$

$$\bigcap_\lambda V(I_\lambda) = V\left(\sum_\lambda I_\lambda\right), \qquad V(I) \cup V(J) = V(IJ).$$

These are exactly the axioms required of closed subsets.

Theorem 2.1 (Zariski topology on affine space)**.** *The algebraic subsets*

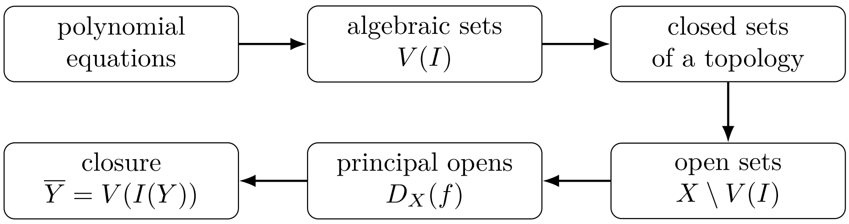
$$\{V(I) \subseteq \mathbb{A}_k^n \mid I \subseteq k[x_1, \ldots, x_n] \text{ an ideal}\}$$

are the closed sets of a topology on $\mathbb{A}_k^n$*.*

Proof. The whole space and the empty set occur as $V((0))$ and $V((1))$. Arbitrary intersections are algebraic because they correspond to sums of ideals, and finite unions are algebraic because they correspond to products of ideals. These are precisely the closed-set axioms. ☐

Definition 2.2 (Zariski topology)**.** The topology of theorem 2.1 is the *Zariski topology* on $\mathbb{A}_k^n$.

The main conceptual pipeline is worth keeping visible.



closure is determined by all equations true on Y

2.2 The induced topology on an algebraic set

Let $X \subseteq \mathbb{A}_k^n$ be algebraic. Its Zariski topology is the subspace topology. Thus $Y \subseteq X$ is closed precisely when

$$Y = X \cap Z$$

for some algebraic $Z \subseteq \mathbb{A}_k^n$.

Definition 2.3 (Relative zero locus)**.** For $S \subseteq k[x_1, \ldots, x_n]$, define

$$V_X(S) = \{x \in X \mid f(x) = 0 \text{ for all } f \in S\}.$$

Proposition 2.4. *For every* S*,*

$$V_X(S) = X \cap V(S).$$

Consequently the closed subsets of X *are exactly the relative zero loci* $V_X(S)$*.*

Proof. The equality is immediate from the definitions. It identifies relative zero loci with closed subsets in the subspace topology in both directions. ☐

Example 2.5 (Closure of an infinite subset of the line)**.** Over an infinite field, any nonzero polynomial in one variable has only finitely many roots. Hence an infinite subset of $\mathbb{A}_k^1$ is Zariski dense: the only algebraic set containing it is the whole line.

2.3 Closure is the algebraic envelope

For an arbitrary subset $Y \subseteq \mathbb{A}^n_k$, let

$$I(Y) = \{f \in k[x_1, \dots, x_n] \mid f(y) = 0 \text{ for every } y \in Y\}.$$

Theorem 2.6 (Closure formula)**.** *For every $Y \subseteq \mathbb{A}^n_k$,*

$$\overline{Y}^{\mathrm{Zar}} = V(I(Y)).$$

Proof. Chapter VI/01 showed that $V(I(Y))$ is the smallest algebraic subset containing Y. The algebraic subsets are now exactly the closed subsets, so this smallest algebraic set is precisely the topological closure. □

Corollary 2.7 (Density criterion)**.** *A subset $Y \subseteq \mathbb{A}^n_k$ is Zariski dense if and only if*

$$I(Y) = I(\mathbb{A}^n_k).$$

If k is infinite, then $I(\mathbb{A}^n_k) = (0)$, so density is equivalent to:

no nonzero polynomial vanishes at every point of Y.

Proof. Density means $V(I(Y)) = \mathbb{A}^n_k$. Over an infinite field, a polynomial that vanishes on all of k^n is zero; prove this by induction on n, using the one-variable root bound. □

Proposition 2.8 (A nonzero principal open is dense)**.** *Assume k is infinite. If $0 \neq f \in k[x_1, \dots, x_n]$, then*

$$D(f) = \{a \in \mathbb{A}^n_k \mid f(a) \neq 0\}$$

is Zariski dense in $\mathbb{A}^n_k$.

Proof. Suppose g vanishes on $D(f)$. Then fg vanishes at every point of $\mathbb{A}^n_k$: on $D(f)$ because $g = 0$, and on $V(f)$ because $f = 0$. Since k is infinite, fg is the zero polynomial. The polynomial ring over a field is a domain and $f \neq 0$, so $g = 0$. Hence $I(D(f)) = (0)$, and theorem 2.7 gives density. □

2.4 Principal opens form a basis

Definition 2.9 (Principal open)**.** For $f \in k[x_1, \dots, x_n]$ and algebraic $X \subseteq \mathbb{A}^n_k$, set

$$D_X(f) = X \setminus V_X(f) = \{x \in X \mid f(x) \neq 0\}.$$

The notation here concerns classical affine algebraic sets. The scheme-theoretic $D(f) \subseteq \operatorname{Spec} A$ is treated later in VI/08.

Proposition 2.10 (Principal-open calculus)**.** *For $f, g \in k[x_1, \dots, x_n]$,*

$$D_X(1) = X, \qquad D_X(0) = \varnothing,$$

$$D_X(f) \cap D_X(g) = D_X(fg), \qquad D_X(f) = D_X(f^m) \quad (m \geq 1).$$

Proof. All statements follow by evaluating at a point and using that a field has no zero divisors and no nonzero nilpotents. □

Theorem 2.11 (Principal opens are a basis)**.** *The family $\{D_X(f)\}$ is a basis for the Zariski topology on an affine algebraic set X.*

Proof. Let $U \subseteq X$ be open and $x \in U$. Write

$$X \setminus U = V_X(S).$$

Since $x \notin V_X(S)$, some $f \in S$ satisfies $f(x) \neq 0$. Then

$$x \in D_X(f) \subseteq U.$$

Thus every point of every open set lies in a principal open contained in that open set. □

2.5 Closed points and separation

Proposition 2.12 (Classical affine points are closed)**.** *For $a = (a_1, \dots, a_n) \in \mathbb{A}_k^n$,*

$$\{a\} = V(x_1 - a_1, \dots, x_n - a_n).$$

Hence every finite subset is closed and classical affine space is T_1.

Remark 2.13. This is a statement about k-rational points. Later, Spec A will contain nonclosed prime ideals and generic points. Do not import that scheme-theoretic behavior into this classical chapter.

Example 2.14 (A basic open set)**.** In $\mathbb{A}_k^2$, the complement of $V(x)$ is $D(x) = \{(a, b) : a \neq 0\}$. It is Zariski open and geometrically removes the y-axis.

2.6 The affine line: infinite versus finite fields

Theorem 2.15 (Infinite field: cofinite topology)**.** *If k is infinite, the proper Zariski closed subsets of $\mathbb{A}_k^1$ are exactly the finite subsets. Equivalently, every nonempty open set has finite complement.*

Proof. If $V(I) \neq \mathbb{A}_k^1$, choose $0 \neq f \in I$. Then $V(I) \subseteq V(f)$, and a nonzero one-variable polynomial has finitely many roots. Conversely, every finite set $\{a_1, \dots, a_r\}$ is

$$V\left(\prod_{j=1}^{r} (x - a_j)\right).$$

□

Corollary 2.16. *If k is infinite, $\mathbb{A}_k^1$ is T_1 but not Hausdorff, and every infinite subset is dense.*

Proof. Two nonempty cofinite open subsets must intersect, so distinct points cannot have disjoint open neighborhoods. An infinite subset cannot lie in a proper closed subset because all proper closed subsets are finite. □

Proposition 2.17 (Finite field: discrete topology)**.** *If k is finite, the Zariski topology on the finite set $\mathbb{A}_k^n = k^n$ is discrete.*

Proof. Every singleton is closed, hence every subset, being a finite union of singletons, is closed. Therefore every subset is also open. □

Space	**Proper closed sets**	**Hausdorff?**	**Typical dense set**
$\mathbb{A}_k^1$, k infinite	finite subsets	no	every infinite subset
$\mathbb{A}_k^n$, k finite	arbitrary subsets	yes (discrete)	only the whole space
$\mathbb{R}$, Euclidean	many infinite closed sets	yes	$\mathbb{Q}$, for example

2.7 Euclidean topology versus Zariski topology

For $k = \mathbb{R}$ or $\mathbb{C}$, polynomial functions are Euclidean continuous, so zero loci are Euclidean closed.

Proposition 2.18. *Every Zariski closed subset of k^n, for $k = \mathbb{R}$ or $\mathbb{C}$, is Euclidean closed. Thus the Zariski topology is coarser than the Euclidean topology.*

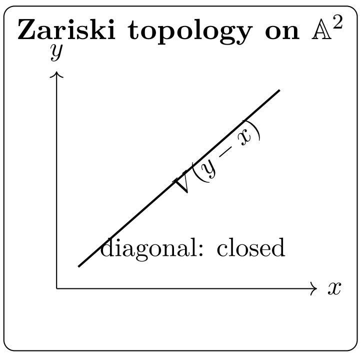
The inclusion is strict. For instance $\mathbb{Z} \subseteq \mathbb{C}$ is Euclidean closed but Zariski dense in $\mathbb{A}^1_{\mathbb{C}}$.

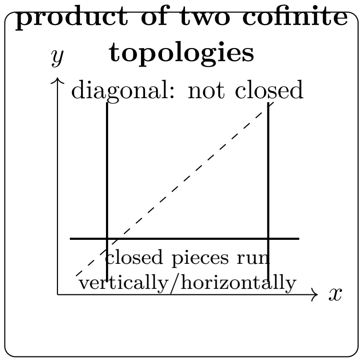
The lesson is structural: Euclidean closure measures metric limiting behavior, whereas Zariski closure measures polynomial identities.

2.8 The product topology is not the affine-plane Zariski topology

This distinction is subtle enough to deserve its own picture. Assume here that k is infinite. Give each copy of $\mathbb{A}^1_k$ its cofinite Zariski topology and form the ordinary product topology on the set $\mathbb{A}^1_k \times \mathbb{A}^1_k$.

A proper closed subset in the product topology is contained in a finite union of vertical and horizontal lines; after removing any whole such lines, only finitely many points remain. By contrast, the Zariski topology on $\mathbb{A}^2_k$ contains arbitrary plane algebraic curves as closed sets.





Proposition 2.19 (Strict comparison)**.** *Assume k is infinite. The product topology on $\mathbb{A}^1_k \times \mathbb{A}^1_k$ is strictly coarser than the Zariski topology on $\mathbb{A}^2_k$.*

Proof. A basic product open has the form $U \times V$, where U and V are cofinite. Write

$$U = D(f(x)), \qquad V = D(g(y))$$

for suitable one-variable polynomials. Then

$$U \times V = D(f(x)g(y)),$$

which is Zariski open in $\mathbb{A}^2_k$. Thus the product topology is no finer. It is strictly coarser because the diagonal

$$\Delta = V(y - x)$$

is Zariski closed, while its complement is not product open: any product neighborhood of a point off the diagonal contains a diagonal point, since two cofinite subsets of an infinite set intersect. $\square$

Example 2.20 (Dense open complement)**.** The set $D(xy)$ equals $D(x) \cap D(y)$ and consists of points with both coordinates nonzero. Its complement is the union of the two axes, a proper closed subset, so $D(xy)$ is dense in the irreducible plane.

2.9 Rank conditions are topological conditions

Identify the affine space $\mathbb{A}_k^{mn}$ with the set of $m \times n$ matrices. Determinants and minors are polynomial functions in the matrix entries.

Proposition 2.21 (Rank loci)**.** *Fix r. The set*

$$\{M \mid \operatorname{rank} M \leq r\}$$

is Zariski closed, because it is the common zero locus of all $(r+1) \times (r+1)$ minors. Therefore

$$\{M \mid \operatorname{rank} M \geq r+1\}$$

is Zariski open.

For square matrices this gives

$$GL_n(k) = D(\det).$$

Over an infinite field, det is a nonzero polynomial, so theorem 2.8 shows that invertible matrices are dense in the space of all matrices.

$$\begin{tikzcd}
\boxed{\begin{array}{c}\text{matrix space}\\ \mathbb{A}^{mn}\end{array}} \arrow[r] & \boxed{\begin{array}{c}\text{all } (r+1)\text{-minors vanish}\\ \operatorname{rank} M \leq r\text{: closed}\end{array}} \arrow[d, "\text{complement}"] \\
\text{minors are polynomial functions} & \boxed{\begin{array}{c}\text{some } (r+1)\text{-minor nonzero}\\ \operatorname{rank} M \geq r+1\text{: open}\end{array}}
\end{tikzcd}$$

2.10 A reusable topology dictionary

Algebraic statement	**Topological meaning**
$V(I)$	closed set
$X \setminus V_X(I)$	open set
$D_X(f)$	basic open neighborhood where $f \neq 0$
$V(I(Y))$	Zariski closure of Y
$I(Y) = (0)$, k infinite	Y is dense in affine space
all $(r+1)$-minors vanish	rank-at-most-r closed condition
a minor is nonzero	rank lower bound on an open set

2.11 Common pitfalls

Warning 2.22 (Do not visualize Zariski opens as small balls). A basic Zariski open is generally huge. In $\mathbb{A}_k^1$ over an infinite field it is all but finitely many points.

Warning 2.23 (Finite fields behave differently). Statements such as "every infinite subset of $\mathbb{A}^1$ is dense" presuppose an infinite field. Over a finite field the classical rational-point topology is discrete.

Warning 2.24 (Product topology is a different construction). Although the underlying set $\mathbb{A}^2$ equals $\mathbb{A}^1 \times \mathbb{A}^1$, the Zariski topology on the affine plane is not, over an infinite field, the product of the Zariski topologies on the two affine lines.

Warning 2.25 (Classical $D_X(f)$ versus scheme $D(f)$). The notation looks identical because the ideas are related, but scheme distinguished opens carry ring-theoretic information and belong to VI/08.

2.12 Worked examples

Example 2.26 (The coordinate axes as a closed set)**.** Let

$$X = V(xy) \subseteq \mathbb{A}^2_k.$$

Since $xy = 0$ in a field exactly when $x = 0$ or $y = 0$,

$$X = V(x) \cup V(y).$$

Inside X, each axis is relatively closed, while deleting one axis gives a relative open subset of the other component.

Example 2.27 (A principal open in the affine plane)**.** The set

$$D(x) = \left\{(a, b) \in k^2 \mid a \neq 0\right\}$$

is the complement of the y-axis. Similarly,

$$D(xy) = D(x) \cap D(y)$$

is the set where both coordinates are nonzero.

Example 2.28 (Finite and infinite closures in the affine line)**.** Assume k is infinite. A finite subset

$$F = \{a_1, \dots, a_r\}$$

is closed because

$$F = V\left(\prod_{j=1}^{r}(x - a_j)\right).$$

By contrast, every infinite $Y \subseteq k$ is dense: no nonzero one-variable polynomial has infinitely many roots.

Example 2.29 (The integers inside the complex line)**.** The subset $\mathbb{Z} \subseteq \mathbb{C}$ is Euclidean closed and discrete. In the Zariski topology on $\mathbb{A}^1_{\mathbb{C}}$, however, it is dense, because a complex polynomial vanishing at every integer has infinitely many roots and is therefore zero.

Example 2.30 (A relative open subset of a parabola)**.** Let

$$X = V(y - x^2) \subseteq \mathbb{A}^2_k.$$

Then

$$D_X(x) = X \setminus V_X(x)$$

is the parabola with the origin removed, because $x = 0$ on X forces $y = 0$.

Example 2.31 (A finite field gives a discrete rational-point topology)**.** For $k = \mathbb{F}_3$,

$$\mathbb{A}^1_k = \{0, 1, 2\}.$$

Every singleton is closed, so every subset is closed and open. Thus the Zariski topology on these three rational points is discrete.

Example 2.32 (The diagonal separates Zariski from product topology)**.** Assume k is infinite. The diagonal

$$\Delta = \{(t, t) \mid t \in k\} = V(y - x)$$

is closed in the Zariski topology on $\mathbb{A}^2_k$. It is not closed in the product of the two cofinite topologies on $\mathbb{A}^1_k$: every product neighborhood of a point off the diagonal meets Δ.

Example 2.33 (A dense Cartesian grid)**.** Let $A, B \subseteq k$ be infinite and k infinite. If

$$f(x, y) \in k[x, y]$$

vanishes on $A \times B$, fix $a \in A$. The polynomial $f(a, y)$ has infinitely many roots B, so it is zero. Therefore every coefficient of f, viewed as a polynomial in y, vanishes on all $a \in A$, hence is zero. Thus $A \times B$ is dense in $\mathbb{A}^2_k$.

Example 2.34 (Invertible matrices are open and dense)**.** In $\mathbb{A}^{n^2}_k$, identified with $n \times n$ matrices,

$$GL_n(k) = D(\det).$$

The determinant polynomial is nonzero. If k is infinite, this principal open is dense.

Example 2.35 (Rank at most one for 2×3 matrices)**.** Write

$$M = \begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}.$$

Then $\operatorname{rank} M \leq 1$ exactly when all 2×2 minors vanish:

$$ae - bd = 0, \qquad af - cd = 0, \qquad bf - ce = 0.$$

Hence the rank-at-most-one locus is Zariski closed, while the full-rank locus is open.

2.13 Exercises

The exercises are new or substantially redesigned to train the topology directly. Legacy problems are retained separately as Problems, with provenance and full dossiers.

Exercise VI.02.1 — Verify the topology axioms

Starting only from the identities for zero loci proved in VI/01, verify the three closed-set axioms for the Zariski topology on $\mathbb{A}^n_k$.

Exercise VI.02.2 — Relative closed sets

Let $X = V(I) \subseteq \mathbb{A}^n_k$. Prove that every closed subset of X is $V_X(J)$ for some ideal $J \subseteq k[x_1, \ldots, x_n]$.

Exercise VI.02.3 — Closure from equations

For $Y \subseteq \mathbb{A}^n_k$, prove directly that $V(I(Y))$ is the smallest Zariski closed set containing Y.

Exercise VI.02.4 — Principal-open intersection calculus

Prove

$$D_X(f) \cap D_X(g) = D_X(fg),$$

and generalize to a finite family $f_1, \ldots, f_r$.

Exercise VI.02.5 — Powers do not change principal opens

Prove $D_X(f) = D_X(f^m)$ for every $m \geq 1$. Compare with $V_X(f) = V_X(f^m)$.

Exercise VI.02.6 — Closed points

For $a = (a_1, \dots, a_n)$, prove

$$\{a\} = V(x_1 - a_1, \dots, x_n - a_n),$$

and deduce that every finite subset of affine space is closed.

Exercise VI.02.7 — The affine line is cofinite

Assume k is infinite. Prove that every proper closed subset of $\mathbb{A}^1_k$ is finite and every finite subset is closed.

Exercise VI.02.8 — Why the affine line is not Hausdorff

Assume k is infinite. Show that any two nonempty open subsets of $\mathbb{A}^1_k$ intersect. Deduce that distinct points cannot be separated by disjoint neighborhoods.

Exercise VI.02.9 — Infinite subsets are dense

Assume k is infinite. Prove that every infinite subset of $\mathbb{A}^1_k$ is Zariski dense.

Exercise VI.02.10 — Finite fields are discrete

Let k be finite. Prove that every subset of $\mathbb{A}^n_k$ is both open and closed.

Exercise VI.02.11 — Relative topology on a parabola

Let $X = V(y - x^2) \subseteq \mathbb{A}^2_k$. Describe

$$V_X(x), \quad D_X(x), \quad V_X(x-1), \quad D_X(x-1).$$

Exercise VI.02.12 — Euclidean closed versus Zariski closed

For $k = \mathbb{R}$ or $\mathbb{C}$, prove every Zariski closed subset of k^n is Euclidean closed. Give an infinite Euclidean closed subset of $\mathbb{A}^1_k$ that is not Zariski closed.

Exercise VI.02.13 — Density criterion

Assume k is infinite. Prove that $Y \subseteq \mathbb{A}^n_k$ is dense if and only if no nonzero polynomial vanishes at every point of Y.

Exercise VI.02.14 — A dense grid

Let $A_1, \dots, A_n \subseteq k$ be infinite subsets of an infinite field. Prove

$$A_1 \times \cdots \times A_n$$

is Zariski dense in $\mathbb{A}^n_k$.

Exercise VI.02.15 — Union of two principal opens

Show

$$D_X(f) \cup D_X(g) = X \setminus V_X(f, g).$$

Explain why this identity alone does not say that the union is principal.

Exercise VI.02.16 — Product-open rectangles are Zariski open

Assume k is infinite. Let $U, V \subseteq \mathbb{A}^1_k$ be nonempty Zariski open sets. Prove that $U \times V$ is Zariski open in $\mathbb{A}^2_k$.

Exercise VI.02.17 — A rank locus is closed

For $m \times n$ matrices, prove that the locus $\operatorname{rank} M \leq r$ is Zariski closed by writing explicit equations in minors.

Exercise VI.02.18 — Dense torus

Assume k is infinite. Prove that

$$(k^\times)^n = D(x_1 x_2 \cdots x_n)$$

is a dense open subset of $\mathbb{A}^n_k$.

2.14 Problems

The first three Problems recover the substantive VI/02 legacy problems. The remaining Problems extend the chapter while staying within the topology developed here.

Problem VI.02.P1 — Zariski topology versus product topology

Problem. Assume k is infinite. Describe the proper closed subsets of $\mathbb{A}^1_k$ and use this to describe the proper closed subsets of $\mathbb{A}^1_k \times \mathbb{A}^1_k$ equipped with the ordinary product topology. Compare that topology with the Zariski topology on $\mathbb{A}^2_k$. In particular, determine whether the diagonal

$$\Delta = \{(t, t) : t \in k\}$$

is closed in each topology.

Problem VI.02.P2 — Principal opens form a basis

Problem. Let $X \subseteq \mathbb{A}^n_k$ be an affine algebraic set. Prove that the collection

$$\{D_X(f) : f \in k[x_1, \ldots, x_n]\}$$

is a basis for the Zariski topology on X. Also prove

$$D_X(f) \cap D_X(g) = D_X(fg).$$

Problem VI.02.P3 — Density of invertible and full-rank matrices

Problem. Identify $\mathbb{A}^{mn}_{\mathbb{C}}$ with the space of complex $m \times n$ matrices. Prove that $GL_n(\mathbb{C})$ is Zariski dense in $\mathbb{A}^{n^2}_{\mathbb{C}}$. More generally, prove that the set of full-rank $m \times n$ matrices is Zariski open and dense.

Problem VI.02.P4 — Dense integer lattices

Problem. Prove that $\mathbb{Z}^n$ is Zariski dense in $\mathbb{A}^n_{\mathbb{C}}$. More generally, if $S_1, \ldots, S_n \subseteq k$ are infinite subsets of an infinite field k, prove that $S_1 \times \cdots \times S_n$ is dense in $\mathbb{A}^n_k$.

Problem VI.02.P5 — Finite fields and polynomial point separators

Problem. Let $k = \mathbb{F}_q$. Prove the rational-point Zariski topology on k^n is discrete. Then construct, for each $a = (a_1, \ldots, a_n) \in k^n$, a polynomial δ_a satisfying

$$\delta_a(a) = 1, \qquad \delta_a(b) = 0 \quad (b \neq a).$$

Problem VI.02.P6 — Quasi-compactness from finite generation

Problem. Prove that every open cover of $\mathbb{A}_k^n$ has a finite subcover. More generally, prove the same for every affine algebraic set $X \subseteq \mathbb{A}_k^n$. You may use Hilbert's basis theorem.

Problem VI.02.P7 — Closure of a punctured hyperbola

Problem. Assume k is infinite and let

$$Y = \left\{ (t, t^{-1}) \mid t \in k^\times \right\} \subseteq \mathbb{A}_k^2.$$

Prove that

$$\overline{Y}^{\mathrm{Zar}} = V(xy - 1).$$

Problem VI.02.P8 — Hyperplane arrangements and a dense complement

Problem. Let $\ell_1, \ldots, \ell_r \in k[x_1, \ldots, x_n]$ be nonzero affine-linear polynomials over an infinite field k, and let

$$H_i = V(\ell_i).$$

Prove

$$\mathbb{A}_k^n \setminus \bigcup_{i=1}^r H_i = D(\ell_1 \cdots \ell_r),$$

and deduce that this complement is open and dense.

2.15 Challenges

Challenge VI.02.C1 — The product-topology failure in every dimension

Challenge. Assume k is infinite and $n \geq 2$. Prove that the Zariski topology on $\mathbb{A}_k^n$ is strictly finer than the n-fold product of the cofinite Zariski topologies on $\mathbb{A}_k^1$.

Challenge VI.02.C2 — Noetherianity of affine Zariski space

Challenge. Using Hilbert's basis theorem, prove that every descending chain of Zariski closed subsets

$$X_1 \supseteq X_2 \supseteq X_3 \supseteq \cdots$$

of $\mathbb{A}_k^n$ eventually stabilizes.

Challenge VI.02.C3 — Closure of the cusp parametrization

Challenge. Assume k is infinite and let

$$Y = \left\{ (t^2, t^3) : t \in k \right\} \subseteq \mathbb{A}_k^2.$$

Prove directly that

$$\overline{Y}^{\mathrm{Zar}} = V(y^2 - x^3).$$

2.16 Chapter summary

The Zariski topology turns polynomial equations into topology:

$$\text{closed} = V(I), \qquad \text{basic open} = D_X(f), \qquad \overline{Y} = V(I(Y)).$$

Over an infinite field, density can be tested by the absence of nonzero polynomial relations. This makes infinite subsets of the affine line dense, makes nonzero principal opens dense in affine space, and makes invertibility or full-rank conditions naturally open and often dense.

The topology depends strongly on context. The affine line over an infinite field is cofinite and non-Hausdorff; over a finite field the rational-point topology is discrete; and the Zariski topology on $\mathbb{A}^2$ is strictly finer than the product of the two cofinite affine-line topologies. These contrasts are not anomalies: they are the first sign that algebraic geometry organizes spaces by polynomial relations rather than metric proximity.

2.17 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 2.36 (Basic open)**.** Describe $D(x)$ in $\mathbb{A}^2_k$.

Hint. Use Zariski closed sets, basic opens, and closure. □

Solution. It is the set of points whose first coordinate is nonzero. □

Exercise 2.37 (Intersection)**.** Compute $D(f) \cap D(g)$.

Hint. Use Zariski closed sets, basic opens, and closure. □

Solution. It is $D(fg)$. □

Exercise 2.38 (Closed union)**.** Compute $V(f) \cup V(g)$ as one zero set.

Hint. Use Zariski closed sets, basic opens, and closure. □

Solution. It is $V(fg)$. □

Exercise 2.39 (Closure on line)**.** What is the Zariski closure of an infinite subset of $\mathbb{A}^1_k$ over an infinite field?

Hint. Use Zariski closed sets, basic opens, and closure. □

Solution. The whole affine line. □

Exercise 2.40 (Point closure)**.** Over an algebraically closed field, what is the closure of a point in affine space?

Hint. Use Zariski closed sets, basic opens, and closure. □

Solution. The point itself. □

Proofs

Exercise 2.41 (Topology axioms)**.** Prove: Prove algebraic sets define a topology by taking them as closed sets.

Hint. Work directly from Zariski closed sets, basic opens, and closure. ☐

Solution. Use $V(0)$ and $V(1)$ for extremal sets, $V(\sum I_\alpha) = \bigcap V(I_\alpha)$ for arbitrary intersections, and products for finite unions. ☐

Exercise 2.42 (Basic-open intersection)**.** Prove: Prove $D(f) \cap D(g) = D(fg)$.

Hint. Work directly from Zariski closed sets, basic opens, and closure. ☐

Solution. A point avoids both zero sets exactly when neither value is zero, equivalently when the product is nonzero. ☐

Exercise 2.43 (Closure-ideal formula)**.** Prove: Prove the closure of a subset Y is $V(I(Y))$.

Hint. Work directly from Zariski closed sets, basic opens, and closure. ☐

Solution. $V(I(Y))$ is closed and contains Y; every closed set containing Y is defined by polynomials that lie in $I(Y)$. ☐

Exercise 2.44 (Density criterion)**.** Prove: Prove Y is dense iff $I(Y) = I(\mathbb{A}_k^n)$.

Hint. Work directly from Zariski closed sets, basic opens, and closure. ☐

Solution. Apply the closure formula and compare with the full affine-space zero set. ☐

Counterexamples and hypothesis tests

Exercise 2.45 (Hausdorff)**.** Test the claim: the Zariski topology on $\mathbb{A}_k^1$ is Hausdorff for infinite k.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; nonempty opens are large and cannot separate distinct points in the usual way. ☐

Exercise 2.46 (Finite sets)**.** Test the claim: finite subsets of affine space over an algebraically closed field need not be closed.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; finite unions of closed points are closed. ☐

Exercise 2.47 (Open balls)**.** Test the claim: Euclidean open balls are typically Zariski open.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; the Zariski topology is much coarser. ☐

Applications and investigations

Exercise 2.48 (Generic behavior)**.** Why are nonempty Zariski opens interpreted as generic conditions on irreducible spaces?

Hint. Translate the question into Zariski closed sets, basic opens, and closure. ☐

Solution. Their complements are proper algebraic exceptional loci. ☐

Exercise 2.49 (Polynomial nonvanishing)**.** What does $D(f)$ encode?

Hint. Translate the question into Zariski closed sets, basic opens, and closure. ☐

Solution. It is precisely the locus where the polynomial condition $f \neq 0$ holds. ☐

Challenge problems

Exercise 2.50 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Zariski Topology in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Zariski Topology, translates it through Zariski closed sets, basic opens, and closure, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 2.51 (Hypothesis challenge)**.** Choose one structural statement from Zariski Topology, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 3

Coordinate Rings

Purpose and learning goals

The first two chapters moved from polynomial equations to algebraic sets and then to the Zariski topology. We now reverse direction. Given an affine algebraic set X, we ask: *which polynomial functions actually live on X, and what algebraic relations do they satisfy?*

The answer is the coordinate ring

$$k[X] = k[x_1, \dots, x_n]/I(X).$$

This quotient is not merely a convenient presentation. It is the ring of polynomial functions on X, and it is the first object that makes the algebra–geometry dictionary behave systematically.

Throughout, k is a field and $X \subseteq \mathbb{A}^n_k$ is an affine algebraic set. Whenever a computation identifies $I(V(J))$ with a displayed ideal J, the necessary field hypothesis is stated explicitly.

After this chapter the reader should be able to:

- construct $k[X]$ from restriction of ambient polynomials;
- recognize $I(X)$ as the kernel of restriction and as the complete set of relations;
- compute coordinate rings of lines, parabolas, hyperbolas, finite sets, and unions;
- distinguish a ring from a chosen presentation by generators and relations;
- use evaluation maps to encode points by maximal ideals;
- turn a closed inclusion $Y \subseteq X$ into a quotient $k[X] \twoheadrightarrow k[Y]$;
- explain why classical coordinate rings are reduced and why nilpotents require a richer geometry;
- understand how finite fields alter the ring of polynomial functions;
- solve a nontrivial legacy example: the monomial curve (t^3, t^4, t^5).

3.1 From ambient polynomials to polynomial functions

Every ambient polynomial restricts to a function on X. The central point is that many different ambient polynomials can restrict to the same function.

Definition 3.1 (Polynomial function)**.** A function $\varphi : X \to k$ is a *polynomial function* if there exists $f \in k[x_1, \dots, x_n]$ such that $\varphi(a) = f(a)$ for every $a \in X$.

Consider the restriction homomorphism

$$\rho_X : k[x_1, \ldots, x_n] \longrightarrow \mathrm{Fun}(X, k), \qquad f \longmapsto f|_X.$$

Its image is exactly the ring of polynomial functions on X.

Proposition 3.2 (Kernel of restriction)**.**

$$\ker \rho_X = I(X).$$

Hence f and g induce the same polynomial function on X if and only if $f - g \in I(X)$.

Proof. A polynomial belongs to the kernel precisely when it vanishes at every point of X, which is the definition of $I(X)$. The second statement follows by applying this to $f - g$. $\square$

The entire construction can be summarized by the first isomorphism theorem.

$$\begin{tikzcd}
\boxed{\begin{array}{c}\text{ambient polynomials}\\ k[x_1,\ldots,x_n]\end{array}} \arrow[r, "\text{restrict}"] \arrow[d, "\text{kernel}"'] \arrow[dr, "\text{quotient}"] & \boxed{\begin{array}{c}\text{polynomial functions}\\ \text{on } X\end{array}} \\
\boxed{\begin{array}{c}\text{functions vanishing on } X\\ I(X) = \ker \rho_X\end{array}} & \boxed{\begin{array}{c}\text{coordinate ring}\\ k[X] = k[x]/I(X)\end{array}} \arrow[u, "\text{canonical isomorphism}"']
\end{tikzcd}$$

3.2 Definition of the coordinate ring

Definition 3.3 (Coordinate ring)**.** The *coordinate ring* of $X \subseteq \mathbb{A}^n_k$ is

$$k[X] := k[x_1, \ldots, x_n]/I(X).$$

By theorem 3.2, $k[X]$ is canonically the ring of polynomial functions on X. An element can therefore be read in three equivalent ways:

$$\boxed{[f] \quad \longleftrightarrow \quad f|_X \quad \longleftrightarrow \quad \text{a polynomial function } X \to k.}$$

Write $\bar{x}_i$ for the class of x_i in $k[X]$. Then

$$k[X] = k[\bar{x}_1, \ldots, \bar{x}_n].$$

The ideal $I(X)$ is exactly the set of polynomial relations among these coordinate functions.

Proposition 3.4 (Relations among coordinates)**.** *For $F \in k[t_1, \ldots, t_n]$,*

$$F(\bar{x}_1, \ldots, \bar{x}_n) = 0 \text{ in } k[X] \quad \Longleftrightarrow \quad F(x_1, \ldots, x_n) \in I(X).$$

Proof. The quotient map $k[x_1, \ldots, x_n] \to k[X]$ has kernel $I(X)$. $\square$

3.3 First computations

When k is infinite,

$$k[\mathbb{A}^n_k] = k[x_1, \ldots, x_n].$$

For a rational point $a = (a_1, \ldots, a_n)$,

$$k[\{a\}] = \frac{k[x_1, \ldots, x_n]}{(x_1 - a_1, \ldots, x_n - a_n)} \cong k.$$

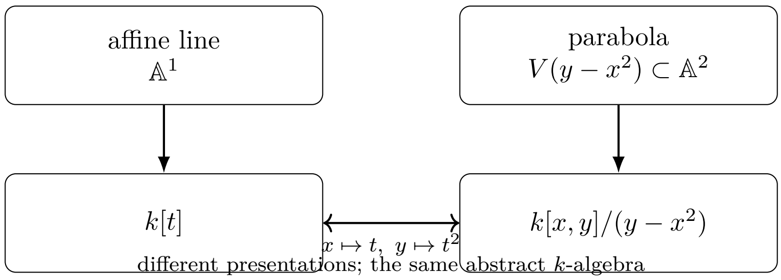
A line and a parabola already show why the presentation of a coordinate ring is not intrinsic. Assume k is infinite. For

$$L = V(y) \subseteq \mathbb{A}^2_k, \qquad P = V(y - x^2) \subseteq \mathbb{A}^2_k,$$

one has

$$k[L] \cong k[t], \qquad k[P] = k[x, y]/(y - x^2) \cong k[t].$$

The two plane embeddings look different, but their polynomial-function rings are isomorphic.



Example 3.5 (Coordinate ring of a parabola)**.** For $X = V(y - x^2) \subset \mathbb{A}^2_k$, the coordinate ring is $k[x, y]/(y - x^2)$. Substituting $y = x^2$ gives an isomorphism with $k[x]$, showing algebraically that the parabola is an affine line.

3.4 Finitely generated algebras and presentations

Definition 3.6 (Finitely generated k-algebra)**.** A k-algebra A is finitely generated if

$$A = k[a_1, \ldots, a_m]$$

for finitely many elements $a_i \in A$.

Proposition 3.7. *Every classical affine coordinate ring is a finitely generated k-algebra.*

Proof. The classes $\bar{x}_1, \ldots, \bar{x}_n$ generate the quotient $k[x_1, \ldots, x_n]/I(X)$. □

Conversely, every finitely generated commutative k-algebra admits a presentation

$$A \cong k[x_1, \ldots, x_n]/J.$$

This does *not* mean every such quotient is the coordinate ring of the classical point set $V(J)$: the classical point set forgets nilpotent and, over non-algebraically-closed fields, sometimes further information. Schemes will repair this loss later.

3.5 Reducedness and what classical geometry forgets

Since $I(X)$ is radical, classical coordinate rings contain no nonzero nilpotents.

Proposition 3.8 (Classical coordinate rings are reduced)**.** *If $u^m = 0$ in $k[X]$ for some $m \geq 1$, then $u = 0$.*

Proof. Write $u = [f]$. Then $f^m \in I(X)$. Radicality of $I(X)$ gives $f \in I(X)$, so $[f] = 0$. □

The basic warning is visible in the one-point example:

$$\frac{k[x]}{(x)} \qquad \text{versus} \qquad \frac{k[x]}{(x^2)}.$$

Both ideals have the same classical zero set $\{0\}$, but the class of x in the second ring is nonzero and nilpotent.

	Classical point	**Thickened point (later)**
ideal	(x)	(x^2)
set of rational zeros	$\{0\}$	$\{0\}$
quotient ring	k	$k[x]/(x^2)$
nonzero nilpotent	no	yes: $[x]^2 = 0$

3.6 Points as evaluation homomorphisms

For $a \in X$, evaluation descends to a k-algebra homomorphism

$$\mathrm{ev}_a : k[X] \to k, \qquad [f] \mapsto f(a).$$

It is surjective because constants lie in $k[X]$. Its kernel

$$\mathfrak{m}_a = \{\varphi \in k[X] \mid \varphi(a) = 0\}$$

is maximal, since $k[X]/\mathfrak{m}_a \cong k$.

This is the first signal that points should be encoded by ideals. The full spectrum of a ring is postponed to VI/07.

Example 3.9 (Coordinate ring of two axes)**.** For $X = V(xy)$, one has $k[X] = k[x, y]/(xy)$. The classes of x and y are nonzero zero divisors, reflecting the two components of the reducible variety.

3.7 Closed subsets become quotient rings

If $Y \subseteq X \subseteq \mathbb{A}^n_k$ are algebraic, then $I(X) \subseteq I(Y)$. Hence there is a natural surjection

$$k[X] \twoheadrightarrow k[Y].$$

Proposition 3.10 (Closed subsets give quotient rings)**.** *Define*

$$I_X(Y) := I(Y)/I(X) \subseteq k[X].$$

Then

$$k[Y] \cong k[X]/I_X(Y).$$

Proof. Apply the first isomorphism theorem to

$$\frac{k[x_1, \ldots, x_n]}{I(X)} \longrightarrow \frac{k[x_1, \ldots, x_n]}{I(Y)}.$$

Its kernel is $I(Y)/I(X)$. ☐

The direction reversal is fundamental:

$$\boxed{\begin{matrix}\text{geometry}\\ Y \subseteq X\end{matrix}} \xrightarrow{\text{reverse the arrow}} \boxed{\begin{matrix}\text{algebra}\\ k[X] \twoheadrightarrow k[Y]\end{matrix}}$$

This contravariance becomes even more important in VI/04, where maps of algebraic sets induce homomorphisms of coordinate rings in the opposite direction.

3.8 Finite sets and the Chinese remainder picture

Let $X = \{a_1, \ldots, a_r\} \subseteq \mathbb{A}_k^n$ be distinct rational points. Their maximal ideals $\mathfrak{m}_{a_i}$ are pairwise comaximal and

$$I(X) = \bigcap_{i=1}^{r} \mathfrak{m}_{a_i}.$$

Therefore the Chinese remainder theorem gives

$$k[X] \cong \prod_{i=1}^{r} k.$$

Every function on a finite set of rational points is consequently polynomial.

$$\begin{tikzcd} \begin{array}{c}\text{r distinct}\\ \text{rational points}\end{array} \arrow[r] & \begin{array}{c}\text{pairwise comaximal}\\ \mathfrak{m}_1, \ldots, \mathfrak{m}_r\end{array} \arrow[r] & \begin{array}{c}\text{CRT}\\ k[X] \cong k^r\end{array} \arrow[d] \\ & & \begin{array}{c}\text{all functions}\\ X \to k\end{array} \end{tikzcd}$$

3.9 Zero divisors as geometric signals

Assume k is algebraically closed, and let

$$X = V(xy) = V(x) \cup V(y) \subseteq \mathbb{A}_k^2.$$

Then

$$k[X] = k[x, y]/(xy).$$

The nonzero classes $\bar{x}$ and $\bar{y}$ satisfy

$$\bar{x}\bar{y} = 0.$$

The zero divisors record the fact that functions supported on different components can annihilate each other. The precise relation between irreducibility and integral domains is proved in VI/05.

Example 3.11 (A point)**.** For the point (a, b), the ideal $(x-a, y-b)$ gives $k[x, y]/(x-a, y-b) \cong k$. Evaluation at the point is the quotient map.

3.10 Finite fields and polynomial function rings

Over an infinite field, no nonzero polynomial vanishes on all of k^n. Over a finite field this fails dramatically. If $k = \mathbb{F}_q$, then

$$x^q - x$$

vanishes at every element of $\mathbb{F}_q$, and

$$I(\mathbb{A}^1_{\mathbb{F}_q}) = (x^q - x).$$

Hence

$$\mathbb{F}_q[\mathbb{A}^1_{\mathbb{F}_q}] \cong \mathbb{F}_q[x]/(x^q - x) \cong \prod_{a \in \mathbb{F}_q} \mathbb{F}_q.$$

Thus the coordinate ring of the *finite rational-point set* is the full algebra of functions on that finite set.

For several variables,

$$I(\mathbb{F}_q^n) = (x_1^q - x_1, \dots, x_n^q - x_n),$$

a result developed further in the Challenges.

3.11 A reusable coordinate-ring dictionary

Geometry	**Algebra**
$X \subseteq \mathbb{A}_k^n$	$k[X] = k[x_1, \dots, x_n]/I(X)$
polynomial function on X	residue class $[f] \in k[X]$
relation true on every point	element of $I(X)$
coordinate functions	generators $\bar{x}_1, \dots, \bar{x}_n$
point $a \in X$	evaluation $k[X] \to k$, kernel $\mathfrak{m}_a$
closed subset $Y \subseteq X$	quotient $k[X] \twoheadrightarrow k[Y]$
finite set of r rational points	product ring k^r
reducible union (preview)	zero divisors can appear
classical algebraic set	reduced finitely generated k-algebra

3.12 Common pitfalls

Warning 3.12 (Use $I(X)$, not an arbitrary ideal cutting out X)*.* If $X = V(J)$, the classical coordinate ring is

$$k[X] = k[x_1, \dots, x_n]/I(X),$$

not automatically $k[x_1, \dots, x_n]/J$. Different ideals can define the same point set.

Warning 3.13 (A presentation is not the ring itself)*.* The rings $k[x, y]/(y - x^2)$ and $k[t]$ are isomorphic. Generators and relations are a presentation, not an intrinsic coordinate system.

Warning 3.14 (Classical coordinate rings are reduced)*.* A quotient with nonzero nilpotents carries more information than the classical zero set. That extra information belongs to scheme theory.

Warning 3.15 (Finite fields change polynomial functions)*.* Over $\mathbb{F}_q$, distinct polynomials can define the same function even on all of affine space. The relations $x_i^q - x_i$ must be taken into account.

3.13 Worked examples

Example 3.16 (Affine space)**.** If k is infinite, then $I(\mathbb{A}_k^n) = (0)$, so

$$k[\mathbb{A}_k^n] = k[x_1, \dots, x_n].$$

The coordinate functions are algebraically independent.

Example 3.17 (A rational point)**.** For $a = (a_1, \dots, a_n)$,

$$k[\{a\}] \cong \frac{k[x_1, \dots, x_n]}{(x_1 - a_1, \dots, x_n - a_n)} \cong k.$$

Every polynomial function on a one-point space is constant.

Example 3.18 (A non-coordinate line)**.** Assume $b \neq 0$ and

$$L = V(ax + by - c) \subseteq \mathbb{A}^2_k.$$

Solving for y gives

$$y = \frac{c - ax}{b},$$

so $k[L] \cong k[x] \cong k[t]$. A linear change of ambient coordinates does not alter the abstract coordinate ring of a line.

Example 3.19 (Parabola)**.** For infinite k,

$$P = V(y - x^2), \qquad k[P] = k[x, y]/(y - x^2).$$

The map $[x] \mapsto t$, $[y] \mapsto t^2$ gives an isomorphism $k[P] \cong k[t]$. Thus the plane parabola and affine line have isomorphic coordinate rings.

Example 3.20 (Hyperbola and a Laurent polynomial ring)**.** Assume k is infinite and let $H = V(xy - 1)$. On H, x and y are mutually inverse. Hence

$$k[H] \cong k[t, t^{-1}],$$

with $t \leftrightarrow \bar{x}$ and $t^{-1} \leftrightarrow \bar{y}$.

Example 3.21 (Two coordinate axes)**.** Over an algebraically closed field,

$$X = V(xy), \qquad k[X] = k[x, y]/(xy).$$

The classes $\bar{x}$ and $\bar{y}$ are nonzero, but their product is zero. This is the algebraic shadow of the union of two components.

Example 3.22 (Two rational points)**.** For distinct $a, b \in k$,

$$X = \{a, b\} \subseteq \mathbb{A}^1_k, \qquad I(X) = ((x - a)(x - b)).$$

The factors are comaximal, so

$$k[X] \cong k[x]/(x - a) \times k[x]/(x - b) \cong k \times k.$$

The two idempotents produced by interpolation select the two points independently.

Example 3.23 (A cusp as a monomial subalgebra)**.** Assume k is infinite and let $C = V(y^2 - x^3)$. Substituting $x = t^2$, $y = t^3$ gives

$$k[C] \cong k[t^2, t^3] \subseteq k[t].$$

The ring remembers that the available polynomial functions are generated by t^2 and t^3, not by the parameter t itself.

Example 3.24 (A closed subset as a quotient)**.** Let $X = \mathbb{A}^2_k$ and $Y = V(y - x^2)$. Then

$$k[X] = k[x, y] \twoheadrightarrow k[Y] = k[x, y]/(y - x^2).$$

Geometric inclusion becomes a surjection of function rings.

Example 3.25 (The affine line over a finite field)**.** For $k = \mathbb{F}_q$,

$$I(\mathbb{A}^1_k) = (x^q - x), \qquad k[\mathbb{A}^1_k] = \mathbb{F}_q[x]/(x^q - x).$$

Since $x^q - x = \prod_{a \in \mathbb{F}_q}(x - a)$ has distinct linear factors,

$$k[\mathbb{A}^1_k] \cong \prod_{a \in \mathbb{F}_q} \mathbb{F}_q.$$

Thus every set-theoretic function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a polynomial.

3.14 Exercises

The Exercises are primarily new and are ordered from direct quotient computations to structural uses of evaluation, Chinese remainders, zero divisors, and finite fields.

Exercise 3.26. Prove directly that the kernel of restriction

$$k[x_1, \dots, x_n] \to \mathrm{Fun}(X, k)$$

is $I(X)$.

Exercise 3.27. Show that $f, g \in k[x_1, \dots, x_n]$ define the same polynomial function on X if and only if $f - g \in I(X)$.

Exercise 3.28. Assume k is infinite. Prove $k[\mathbb{A}^n_k] = k[x_1, \dots, x_n]$.

Exercise 3.29. For $a = (a_1, \dots, a_n)$, compute $I(\{a\})$ and $k[\{a\}]$.

Exercise 3.30. Assume k is infinite. Let $L = V(ax + by - c) \subseteq \mathbb{A}^2_k$ with $b \neq 0$. Construct an explicit isomorphism $k[L] \cong k[t]$.

Exercise 3.31. Assume k is infinite. Construct mutually inverse homomorphisms

$$k[x, y]/(y - x^2) \cong k[t].$$

Exercise 3.32. Assume k is infinite. Show

$$k[V(xy - 1)] \cong k[t, t^{-1}].$$

Exercise 3.33. Prove $k[X] = k[\bar{x}_1, \dots, \bar{x}_n]$, and show that the polynomial relations among the $\bar{x}_i$ are exactly the elements of $I(X)$.

Exercise 3.34. Prove that every classical affine coordinate ring is reduced.

Exercise 3.35. For $a \in X$, prove that evaluation $\mathrm{ev}_a : k[X] \to k$ is a surjective k-algebra homomorphism and that its kernel is maximal.

Exercise 3.36. Let $Y \subseteq X \subseteq \mathbb{A}^n_k$ be algebraic. Prove

$$k[Y] \cong k[X]/(I(Y)/I(X)).$$

Exercise 3.37. For distinct $a, b \in k$, prove

$$k[\{a, b\}] \cong k \times k.$$

Find the two idempotents corresponding to the characteristic functions of a and b.

Exercise 3.38. Let $X = \{a_1, \dots, a_r\} \subseteq \mathbb{A}^1_k$ be distinct. Prove

$$k[X] \cong k^r.$$

Exercise 3.39. Assume k is algebraically closed and $X = V(xy)$. Show that $k[X]$ contains nonzero zero divisors and identify a natural pair.

Exercise 3.40. Compare $A = k[x]/(x)$ and $B = k[x]/(x^2)$. Show that B has a nonzero nilpotent and explain why B cannot be the classical coordinate ring of the one-point set $V(x)$.

Exercise 3.41. Prove
$$I(\mathbb{A}^1_{\mathbb{F}_q}) = (x^q - x)$$
and deduce $\mathbb{F}_q[\mathbb{A}^1] \cong \mathbb{F}_q[x]/(x^q - x)$.

Exercise 3.42. Let X and Y be finite rational-point sets with $|X| = r$, $|Y| = s$. Using only the fact that every function on a finite rational-point set is polynomial, prove
$$k[X \times Y] \cong k^{rs}.$$

Exercise 3.43. Let $k = \mathbb{C}$ and $X = V(x^2 + y^2 - 1)$. In $k[X]$, put $u = \bar{x} + i\bar{y}$. Show that u is a unit and compute u^{-1}.

3.15 Problems

Problem VI.03.P1 recovers the substantive legacy coordinate-ring problem. The remaining Problems are new synthesis problems designed around the same level of structural reasoning.

Problem VI.03.P1 — The monomial curve (t^3, t^4, t^5)

Problem. Assume k is infinite and let
$$X = \left\{(t^3, t^4, t^5) : t \in k\right\} \subseteq \mathbb{A}^3_k.$$
Prove that
$$I(X) = (y^2 - xz,\ x^3 - yz,\ z^2 - x^2y).$$
Show moreover that this ideal cannot be generated by only two polynomials.

Problem VI.03.P2 — Polynomial interpolation on a finite affine set

Problem. Let $X = \{a_1, \dots, a_r\} \subseteq \mathbb{A}^n_k$ be distinct rational points. Prove that the evaluation map
$$k[x_1, \dots, x_n] \to k^r, \qquad f \mapsto (f(a_1), \dots, f(a_r))$$
is surjective and deduce $k[X] \cong k^r$.

Problem VI.03.P3 — Different embeddings, the same coordinate ring

Problem. Assume k is infinite. Let $L = \mathbb{A}^1_k$ and $P = V(y - x^2) \subseteq \mathbb{A}^2_k$. Construct an explicit isomorphism $k[P] \cong k[L]$. Identify which coordinate functions on P correspond to t and t^2 on L.

Problem VI.03.P4 — The affine line and hyperbola are distinguished by units

Problem. Assume k is infinite. Prove
$$k[t]^\times = k^\times, \qquad k[t, t^{-1}]^\times = \left\{ct^m : c \in k^\times,\ m \in \mathbb{Z}\right\}.$$
Deduce that $k[t] \not\cong k[t, t^{-1}]$ as k-algebras, and hence that the affine line and hyperbola $V(xy - 1)$ cannot have isomorphic coordinate rings.

Problem VI.03.P5 — The union of two axes as glued polynomial pairs

Problem. Assume k is algebraically closed and set $X = V(xy) \subseteq \mathbb{A}^2_k$. Prove

$$k[X] \cong \{(f, g) \in k[x] \times k[y] : f(0) = g(0)\}.$$

Problem VI.03.P6 — A tower of closed subsets and quotient maps

Problem. Let $Z \subseteq Y \subseteq X \subseteq \mathbb{A}^n_k$ be algebraic. Describe the natural surjections

$$k[X] \twoheadrightarrow k[Y] \twoheadrightarrow k[Z]$$

and prove that their composition is the natural quotient map $k[X] \twoheadrightarrow k[Z]$. Identify all three kernels in terms of vanishing ideals.

Problem VI.03.P7 — Every function on $\mathbb{F}_q$ is polynomial

Problem. Prove that every function $\mathbb{F}_q \to \mathbb{F}_q$ is represented by a unique polynomial of degree $< q$. Deduce

$$\mathbb{F}_q[x]/(x^q - x) \cong \prod_{a \in \mathbb{F}_q} \mathbb{F}_q.$$

Problem VI.03.P8 — The cusp as $k[t^2, t^3]$

Problem. Assume k is infinite and let $C = V(y^2 - x^3) \subseteq \mathbb{A}^2_k$. Prove that

$$k[C] \cong k[t^2, t^3] \subseteq k[t]$$

under $x \mapsto t^2$, $y \mapsto t^3$. Show that every element of this subalgebra is a linear combination of 1 and monomials t^m with $m \geq 2$.

3.16 Challenges

Challenge VI.03.C1 — The Boolean cube as a product algebra

Challenge. Let $X = \{0, 1\}^n \subseteq \mathbb{A}^n_k$. Prove

$$I(X) = (x_1^2 - x_1, \ldots, x_n^2 - x_n)$$

and

$$k[X] \cong k^{2^n}.$$

Construct explicit idempotents selecting individual vertices.

Challenge VI.03.C2 — Polynomial functions on $\mathbb{F}_q^n$

Challenge. Prove

$$I(\mathbb{F}_q^n) = (x_1^q - x_1, \ldots, x_n^q - x_n)$$

and deduce that every function $\mathbb{F}_q^n \to \mathbb{F}_q$ is represented uniquely by a polynomial whose degree in each variable is $< q$.

Challenge VI.03.C3 — The union of all coordinate axes

Challenge. Let $X \subseteq \mathbb{A}^n_k$ be the union of the n coordinate axes. Assume k is algebraically closed. Prove

$$I(X) = (x_i x_j : 1 \leq i < j \leq n)$$

and show

$$k[X] \cong \{(f_1, \ldots, f_n) \in k[t]^n : f_1(0) = \cdots = f_n(0)\}.$$

3.17 Chapter summary

The coordinate ring

$$k[X] = k[x_1, \ldots, x_n]/I(X)$$

is canonically the ring of polynomial functions on X. The ambient coordinate classes generate the ring, while $I(X)$ records every relation among them. Points produce evaluation maps to k; closed subsets produce quotient rings in the reverse direction; finite rational-point sets produce product rings; and reducible geometry can manifest as zero divisors.

Classical coordinate rings are finitely generated and reduced, so they deliberately forget nilpotent thickening information. Over finite fields the ring of polynomial functions is smaller than the polynomial ring because $x_i^q - x_i$ vanishes identically on rational points. These facts prepare the next chapter, where maps of affine algebraic sets will become homomorphisms of coordinate rings in the opposite direction.

3.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 3.44 (Parabola quotient)**.** Simplify $k[x, y]/(y - x^2)$.

Hint. Use coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. It is isomorphic to $k[x]$. ☐

Exercise 3.45 (Point quotient)**.** Compute $k[x, y]/(x - a, y - b)$.

Hint. Use coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. It is k. ☐

Exercise 3.46 (Axes quotient)**.** Identify zero divisors in $k[x, y]/(xy)$.

Hint. Use coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. The nonzero classes of x and y multiply to zero. ☐

Exercise 3.47 (Line)**.** Compute the coordinate ring of $V(y)$.

Hint. Use coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. It is $k[x]$. ☐

Exercise 3.48 (Empty set)**.** What quotient corresponds to the empty algebraic set?

Hint. Use coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. The unit ideal gives the zero ring. ☐

Proofs

Exercise 3.49 (Evaluation kernel)**.** Prove: Prove evaluation at $(a_1, \ldots, a_n)$ has kernel $(x_1 - a_1, \ldots, x_n - a_n)$.

Hint. Work directly from coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. Repeatedly divide by the linear factors or translate coordinates and use the zero constant term criterion. ☐

Exercise 3.50 (Coordinate ring invariance)**.** Prove: Prove replacing an ideal by its radical does not change the reduced coordinate ring of its zero set over an algebraically closed field.

Hint. Work directly from coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. Use the Nullstellensatz relation $I(V(I)) = \sqrt{I}$. ☐

Exercise 3.51 (Parabola isomorphism)**.** Prove: Prove $k[x, y]/(y - x^2) \cong k[x]$.

Hint. Work directly from coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. Map the class of x to x and the class of y to x^2; compute kernel and show surjectivity. ☐

Exercise 3.52 (Product detection)**.** Prove: Prove $k[x, y]/(xy)$ is not a domain.

Hint. Work directly from coordinate rings, vanishing ideals, and quotient calculations. ☐

Solution. The classes of x and y are both nonzero and have zero product. ☐

Counterexamples and hypothesis tests

Exercise 3.53 (Coordinate domain)**.** Test the claim: every coordinate ring is a domain.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; reducible algebraic sets give zero divisors. ☐

Exercise 3.54 (Presentation uniqueness)**.** Test the claim: a coordinate ring has a unique polynomial presentation.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; isomorphic quotient presentations can describe the same ring. ☐

Exercise 3.55 (Point ring)**.** Test the claim: every closed point has coordinate ring equal to the ground field over arbitrary nonclosed fields.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; residue fields can be nontrivial extensions. ☐

Applications and investigations

Exercise 3.56 (Geometry from ring)**.** What geometric feature do zero divisors in a coordinate ring often signal?

Hint. Translate the question into coordinate rings, vanishing ideals, and quotient calculations. □

Solution. Reducibility or multiple components. □

Exercise 3.57 (Parametrization)**.** How can a quotient-ring isomorphism reveal a parametrization?

Hint. Translate the question into coordinate rings, vanishing ideals, and quotient calculations. □

Solution. Expressing eliminated coordinates as polynomials in free coordinates gives an explicit map. □

Challenge problems

Exercise 3.58 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Coordinate Rings in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Coordinate Rings, translates it through coordinate rings, vanishing ideals, and quotient calculations, and checks the correspondence in both algebraic and geometric language. □

Exercise 3.59 (Hypothesis challenge)**.** Choose one structural statement from Coordinate Rings, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 4

Morphisms of Affine Algebraic Sets

Purpose and learning goals

The previous chapter associated a ring $k[X]$ to every affine algebraic set X. We now extend this correspondence from objects to maps.

If

$$\varphi : X \longrightarrow Y$$

is given by polynomial coordinate functions, then every polynomial function on Y can be pulled back along φ. This produces a k-algebra homomorphism

$$\varphi^* : k[Y] \longrightarrow k[X].$$

The direction reverses.

This reversal is one of the central structural ideas of algebraic geometry. Geometric maps go one way; functions go the other way.

Throughout, k is a field,

$$X \subseteq \mathbb{A}^n_k, \qquad Y \subseteq \mathbb{A}^m_k$$

are affine algebraic sets, and their coordinate rings are understood in the classical sense developed in VI/03.

After this chapter, the reader should be able to:

- define polynomial maps between affine algebraic sets;
- verify that such maps are Zariski continuous;
- construct the pullback homomorphism on coordinate rings;
- prove contravariance under composition;
- recover a polynomial map from a k-algebra homomorphism of coordinate rings;
- prove the affine correspondence

$$\mathrm{Mor}(X, Y) \cong \mathrm{Hom}_{k\text{-alg}}(k[Y], k[X]);$$

- characterize affine isomorphisms algebraically;
- understand closed embeddings as quotient maps on coordinate rings.

4.1 Polynomial maps

Definition 4.1 (Polynomial map)**.** A map

$$\varphi : X \longrightarrow \mathbb{A}_k^m$$

is a *polynomial map* if there exist polynomial functions

$$f_1, \ldots, f_m \in k[X]$$

such that

$$\varphi(x) = \big(f_1(x), \ldots, f_m(x)\big)$$

for every $x \in X$.

Each $f_i \in k[X]$ may be represented by an ambient polynomial in $k[x_1, \ldots, x_n]$, but the map depends only on its class in the coordinate ring.

Definition 4.2 (Morphisms of affine algebraic sets)**.** Let $Y \subseteq \mathbb{A}_k^m$ be algebraic. A *morphism*

$$\varphi : X \longrightarrow Y$$

is a polynomial map $X \to \mathbb{A}_k^m$ whose image lies in Y.

In this classical affine setting, we use *morphism* and *polynomial morphism* interchangeably. Later scheme theory will provide the intrinsic locally ringed-space definition.

Example 4.3 (A polynomial self-map of the affine line)**.** The map

$$\varphi : \mathbb{A}_k^1 \longrightarrow \mathbb{A}_k^1, \qquad t \longmapsto t^2 + t$$

is a morphism.

Example 4.4 (Parametrizing the parabola)**.** Let

$$P = V(y - x^2) \subseteq \mathbb{A}_k^2.$$

Then

$$\alpha : \mathbb{A}_k^1 \longrightarrow P, \qquad t \longmapsto (t, t^2)$$

is a morphism.

4.2 Testing whether a polynomial map lands in a target

Suppose

$$Y = V(J) \subseteq \mathbb{A}_k^m,$$

where

$$J \subseteq k[y_1, \ldots, y_m].$$

Given polynomial functions

$$f_1, \ldots, f_m \in k[X],$$

we would like to know whether

$$x \longmapsto (f_1(x), \ldots, f_m(x))$$

has image in Y.

Proposition 4.5 (Equation test for the target)**.** *Let*

$$\varphi = (f_1, \dots, f_m) : X \to \mathbb{A}^m_k.$$

Then $\varphi(X) \subseteq Y$ *if and only if*

$$g(f_1, \dots, f_m) = 0 \quad \text{in } k[X]$$

for every

$$g \in I(Y).$$

Proof. The image lies in Y exactly when, for every $x \in X$ and every $g \in I(Y)$,

$$g(f_1(x), \dots, f_m(x)) = 0.$$

That is exactly the statement that the polynomial function

$$g(f_1, \dots, f_m)$$

is the zero element of $k[X]$. □

Example 4.6. For the parabola

$$P = V(y - x^2),$$

the map

$$t \longmapsto (t, t^2)$$

lands in P because

$$t^2 - t^2 = 0.$$

Algebraically, the defining relation $y - x^2$ becomes zero after substituting

$$x \mapsto t, \qquad y \mapsto t^2.$$

4.3 Pullback of polynomial functions

Let

$$\varphi : X \to Y$$

be a morphism.

Every polynomial function

$$h : Y \to k$$

can be composed with φ:

$$h \circ \varphi : X \to k.$$

Because φ is polynomial, $h \circ \varphi$ is again a polynomial function on X.

Definition 4.7 (Pullback)**.** The *pullback* of functions along φ is

$$\varphi^* : k[Y] \longrightarrow k[X], \qquad h \longmapsto h \circ \varphi.$$

Proposition 4.8 (Pullback is a k-algebra homomorphism)**.** *For every morphism*

$$\varphi : X \to Y,$$

the map

$$\varphi^* : k[Y] \to k[X]$$

is a k-algebra homomorphism.

Proof. For $f, g \in k[Y]$, $c \in k$, and $x \in X$,

$$\varphi^*(f+g)(x) = (f+g)(\varphi(x)) = f(\varphi(x)) + g(\varphi(x)),$$

so

$$\varphi^*(f+g) = \varphi^*(f) + \varphi^*(g).$$

Similarly,

$$\varphi^*(fg) = \varphi^*(f)\varphi^*(g),$$

and constant functions are fixed:

$$\varphi^*(c) = c.$$

□

4.4 Coordinate description of the pullback

Write the coordinate functions on

$$Y \subseteq \mathbb{A}_k^m$$

as

$$\bar{y}_1, \ldots, \bar{y}_m \in k[Y].$$

If

$$\varphi = (f_1, \ldots, f_m),$$

then

$$\varphi^*(\bar{y}_i) = f_i.$$

Thus the entire pullback homomorphism is determined by the images of the coordinate functions.

Proposition 4.9 (Pullback by substitution)**.** *If*

$$[h] \in k[Y]$$

is represented by

$$h(y_1, \ldots, y_m),$$

then

$$\varphi^*([h]) = h(f_1, \ldots, f_m) \quad \textit{in } k[X].$$

Proof. At $x \in X$,

$$\varphi^*([h])(x) = h(\varphi(x)) = h\big(f_1(x), \ldots, f_m(x)\big).$$

□

This is simply substitution, but phrased intrinsically in coordinate rings.

Example 4.10 (A polynomial parametrization)**.** The map $\phi : \mathbb{A}_k^1 \to V(y - x^2)$ given by $t \mapsto (t, t^2)$ is a morphism. Its pullback sends $x \mapsto t$ and $y \mapsto t^2$, and the inverse morphism is projection onto the first coordinate.

4.5 Why the pullback is well-defined in quotient coordinates

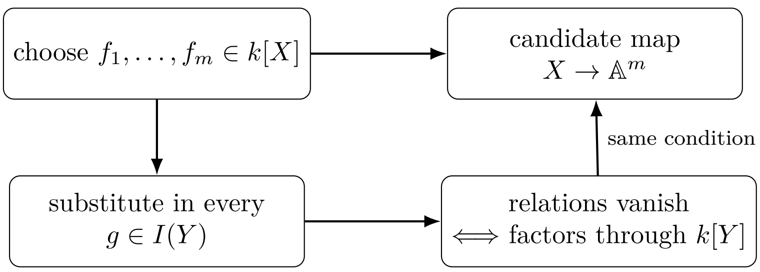
Suppose
$$k[Y] = k[y_1, \dots, y_m]/I(Y).$$
To define a homomorphism out of this quotient by choosing images $f_i \in k[X]$, the relations in $I(Y)$ must vanish after substitution.

That is exactly the condition from theorem 4.5.

Proposition 4.11 (Universal quotient criterion)**.** *Choose*
$$f_1, \dots, f_m \in k[X].$$
The assignment
$$\bar{y}_i \longmapsto f_i$$
extends to a k-algebra homomorphism
$$k[Y] \longrightarrow k[X]$$
if and only if
$$g(f_1, \dots, f_m) = 0$$
for every
$$g \in I(Y).$$

Proof. A homomorphism
$$k[y_1, \dots, y_m] \to k[X]$$
is determined by the images of the variables. It factors through
$$k[y_1, \dots, y_m]/I(Y)$$
exactly when every element of $I(Y)$ lies in its kernel. □

Compare this with theorem 4.5: the algebraic factorization condition is the same as the geometric condition that the map lands in Y.



The upper route is geometric and the lower route algebraic; the target equations make the two tests equivalent.

4.6 Contravariance

Suppose
$$X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z$$
are morphisms.

Then a function on Z pulls back first to Y, then to X.

Theorem 4.12 (Contravariance of coordinate rings)**.** *For composable morphisms*

$$X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z,$$

one has

$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^*.$$

Also,

$$(\mathrm{id}_X)^* = \mathrm{id}_{k[X]}\,.$$

Proof. For $h \in k[Z]$,

$$(\psi \circ \varphi)^*(h) = h \circ \psi \circ \varphi = \varphi^*(\psi^*(h)).$$

The identity statement follows from

$$h \circ \mathrm{id}_X = h.$$

□

Thus

$$X \longmapsto k[X]$$

reverses arrows.

$$X \xrightarrow{\quad\varphi\quad} Y$$

$$k[X] \xleftarrow{\quad\varphi^*\quad} k[Y]$$

The diagram is deliberately asymmetric: a point travels from X to Y, while a function on Y travels backward to a function on X.

4.7 Recovering a morphism from a ring homomorphism

The remarkable fact is that no information is lost.

Let

$$\Phi : k[Y] \longrightarrow k[X]$$

be a k-algebra homomorphism.

Let

$$\bar{y}_1, \ldots, \bar{y}_m$$

be the coordinate functions on Y. Define

$$f_i := \Phi(\bar{y}_i) \in k[X].$$

These polynomial functions determine a candidate map

$$\varphi_\Phi : X \to \mathbb{A}^m_k, \qquad x \longmapsto (f_1(x), \ldots, f_m(x)).$$

Proposition 4.13 (The reconstructed map lands in Y)**.** *The map φ_Φ has image contained in Y.*

Proof. Let $g \in I(Y)$. In $k[Y]$,

$$g(\bar{y}_1, \ldots, \bar{y}_m) = 0.$$

Applying Φ,

$$g(f_1, \ldots, f_m) = \Phi\big(g(\bar{y}_1, \ldots, \bar{y}_m)\big) = 0$$

in $k[X]$. By theorem 4.5, the image lies in Y. ☐

Therefore every k-algebra homomorphism

$$k[Y] \to k[X]$$

comes from a morphism

$$X \to Y.$$

4.8 The affine algebra–geometry correspondence

Theorem 4.14 (Affine morphism correspondence)**.** *For affine algebraic sets X and Y, pullback gives a natural bijection*

$$\mathrm{Mor}_k(X, Y) \;\cong\; \mathrm{Hom}_{k\text{-}alg}(k[Y], k[X]).$$

The bijection is contravariant in X and Y.

Proof. A morphism $\varphi : X \to Y$ produces

$$\varphi^* : k[Y] \to k[X].$$

Conversely, a k-algebra homomorphism

$$\Phi : k[Y] \to k[X]$$

produces the morphism

$$\varphi_\Phi = \big(\Phi(\bar{y}_1), \ldots, \Phi(\bar{y}_m)\big).$$

Starting with $\varphi = (f_1, \ldots, f_m)$, the reconstructed map has coordinates

$$\varphi^*(\bar{y}_i) = f_i,$$

so it is exactly φ.

Starting with Φ, the pullback of the reconstructed map sends

$$\bar{y}_i \longmapsto \Phi(\bar{y}_i).$$

Since the $\bar{y}_i$ generate $k[Y]$ as a k-algebra, the two homomorphisms are equal. ☐

This theorem is the central result of the chapter.

4.9 Morphisms are Zariski continuous

The algebraic definition automatically implies topological continuity.

Theorem 4.15 (Continuity of morphisms)**.** *Every morphism of affine algebraic sets*

$$\varphi : X \to Y$$

is continuous for the Zariski topologies.

Proof. It is enough to compute the inverse image of a closed set.

Let
$$W = V_Y(S) \subseteq Y$$
for some set
$$S \subseteq k[Y].$$
Then
$$x \in \varphi^{-1}(W)$$
exactly when
$$s(\varphi(x)) = 0$$
for every $s \in S$. Equivalently,
$$\varphi^*(s)(x) = 0$$
for every $s \in S$. Hence
$$\varphi^{-1}(W) = V_X(\varphi^*(S)),$$
which is closed in X. □

For a principal open,
$$D_Y(f) = \{y \in Y \mid f(y) \neq 0\},$$
one obtains the useful formula
$$\varphi^{-1}(D_Y(f)) = D_X(\varphi^* f).$$

Example 4.16 (Squaring map)**.** The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$, $t \mapsto t^2$, corresponds contravariantly to the ring map $k[u] \to k[t]$ with $u \mapsto t^2$. Fibers are read by solving $t^2 = a$.

4.10 The kernel of pullback and the closure of the image

The pullback records exactly which functions on the target vanish on the image.

Proposition 4.17 (Kernel–image relation)**.** *For a morphism $\varphi : X \to Y$,*
$$\ker(\varphi^*) = I_Y(\varphi(X)),$$
the ideal of polynomial functions on Y vanishing on the set-theoretic image. Hence
$$V_Y(\ker \varphi^*) = \overline{\varphi(X)}^{\mathrm{Zar}}.$$
In particular, $\varphi(X)$ is Zariski dense in Y if and only if φ^* is injective.

Proof. A function $h \in k[Y]$ belongs to the kernel exactly when
$$0 = \varphi^*(h) = h \circ \varphi,$$
which says precisely that h vanishes at every point of $\varphi(X)$. The closure formula from VI/02 then gives the second assertion. Density means that the vanishing ideal inside $k[Y]$ is zero. □

Geometry of $\varphi : X \to Y$	**Algebra of** $\varphi^* : k[Y] \to k[X]$
isomorphism	isomorphism
closed embedding	surjective
dense image	injective
functions vanishing on image	$\ker \varphi^*$
closure of image	$V_Y(\ker \varphi^*)$

4.11 Isomorphisms

Definition 4.18 (Isomorphism)**.** A morphism

$$\varphi : X \to Y$$

is an *isomorphism* if there exists a morphism

$$\psi : Y \to X$$

such that

$$\psi \circ \varphi = \mathrm{id}_X, \qquad \varphi \circ \psi = \mathrm{id}_Y .$$

Theorem 4.19 (Isomorphism criterion)**.** *A morphism*

$$\varphi : X \to Y$$

is an isomorphism if and only if

$$\varphi^* : k[Y] \to k[X]$$

is an isomorphism of k-algebras.

Proof. If φ has inverse ψ, contravariance gives

$$\varphi^* \circ \psi^* = (\psi \circ \varphi)^* = \mathrm{id}_{k[X]},$$

and

$$\psi^* \circ \varphi^* = (\varphi \circ \psi)^* = \mathrm{id}_{k[Y]} .$$

Thus φ^* is an isomorphism.

Conversely, if φ^* is an isomorphism, its inverse

$$(\varphi^*)^{-1} : k[X] \to k[Y]$$

corresponds by theorem 4.14 to a morphism

$$\psi : Y \to X.$$

The identities for the two ring homomorphisms translate, again by the correspondence, into the two geometric inverse identities. $\square$

Example 4.20 (The parabola is isomorphic to the affine line)**.** Let

$$P = V(y - x^2).$$

The maps

$$\alpha : \mathbb{A}^1_k \to P, \qquad t \mapsto (t, t^2),$$

and

$$\beta : P \to \mathbb{A}^1_k, \qquad (x, y) \mapsto x$$

are inverse morphisms.

On coordinate rings,

$$\alpha^* : k[x, y]/(y - x^2) \to k[t]$$

sends

$$\bar{x} \mapsto t, \qquad \bar{y} \mapsto t^2,$$

while

$$\beta^* : k[t] \to k[P]$$

sends

$$t \mapsto \bar{x}.$$

4.12 Closed embeddings

A closed inclusion gives a quotient of coordinate rings, as already observed in VI/03.

Definition 4.21 (Closed embedding in the affine classical setting)**.** A morphism

$$i : Y \to X$$

is a *closed embedding* if it identifies Y isomorphically with a Zariski closed subset of X.

Theorem 4.22 (Closed embeddings and surjections)**.** *A morphism*

$$i : Y \to X$$

is a closed embedding if and only if the pullback

$$i^* : k[X] \to k[Y]$$

is surjective.

Proof. If Y is identified with a closed subset of X, then

$$k[Y] \cong k[X]/I_X(Y),$$

so the restriction map

$$i^* : k[X] \twoheadrightarrow k[Y]$$

is surjective.

Conversely, suppose i^* is surjective. Let

$$J = \ker(i^*).$$

Then

$$k[Y] \cong k[X]/J.$$

The quotient corresponds to the closed algebraic subset of X defined by the functions in J. Under the affine correspondence, the quotient map is the pullback of the canonical closed inclusion, and because pullback determines the morphism uniquely, i identifies Y with that closed subset. □

Remark 4.23. Over a classical point-set presentation one must remember that closed subsets are governed by vanishing ideals, hence radical ideals. Scheme theory will later remove this reduction and allow arbitrary quotient ideals to define closed subschemes.

4.13 Projections and coordinate inclusions

Example 4.24 (Projection)**.** The projection

$$\pi : \mathbb{A}^2_k \to \mathbb{A}^1_k, \qquad (x, y) \mapsto x$$

is a morphism.

Its pullback is

$$\pi^* : k[t] \to k[x, y], \qquad t \mapsto x.$$

Example 4.25 (A coordinate axis)**.** Let

$$i : \mathbb{A}^1_k \to \mathbb{A}^2_k, \qquad t \mapsto (t, 0).$$

Then

$$i^* : k[x, y] \to k[t]$$

is

$$x \mapsto t, \qquad y \mapsto 0.$$

It is surjective, and its kernel is

$$(y).$$

Thus the map is a closed embedding with image $V(y)$.

4.14 Maps to affine space

Affine space is particularly simple because its coordinate ring is freely generated.

Assume k is infinite so that

$$k[\mathbb{A}^m_k] = k[y_1, \ldots, y_m].$$

Proposition 4.26. *Giving a morphism*

$$\varphi : X \to \mathbb{A}^m_k$$

is equivalent to choosing m elements

$$f_1, \ldots, f_m \in k[X].$$

Proof. A morphism is exactly an m-tuple of polynomial functions. Algebraically, a homomorphism

$$k[y_1, \ldots, y_m] \to k[X]$$

is freely determined by the images of the variables y_i. □

In particular,

$$\mathrm{Mor}(X, \mathbb{A}^1_k) \cong k[X].$$

A regular function is therefore the same thing as a morphism to the affine line.

Example 4.27 (Restriction to a closed subset)**.** A polynomial map on affine space restricts to an algebraic subset when the target equations vanish after substitution. Algebraically, this is exactly the condition that the pullback sends the target vanishing ideal into the source vanishing ideal.

4.15 Products and graphs

Products provide a first example of a geometric construction that becomes an algebraic construction in the opposite direction. If k is algebraically closed and $X \subseteq \mathbb{A}^m$, $Y \subseteq \mathbb{A}^n$ are affine algebraic sets, then

$$X \times Y \subseteq \mathbb{A}^{m+n}$$

is algebraic and

$$k[X \times Y] \cong k[X] \otimes_k k[Y].$$

The two projections correspond to the two canonical maps into the tensor product.

For morphisms $f : Z \to X$ and $g : Z \to Y$, the paired map

$$(f, g) : Z \to X \times Y$$

is a morphism. It is uniquely characterized by the two projection identities.

$$\begin{tikzcd} & Z \arrow[dl, "f"'] \arrow[d, dashed, "{(f,g)}"] \arrow[dr, "g"] & \\ X & X \times Y \arrow[l, "\pi_X"] \arrow[r, "\pi_Y"'] & Y \end{tikzcd}$$

Every morphism $f : X \to Y$ also has a graph

$$\Gamma_f = \{(x, f(x)) : x \in X\} \subseteq X \times Y.$$

Because equality of polynomial coordinates is a closed condition, Γ_f is closed, and

$$x \longmapsto (x, f(x))$$

is a closed embedding. This observation will reappear repeatedly in later geometric constructions.

4.16 Endomorphisms and automorphisms

A morphism

$$X \to X$$

is an *endomorphism*; an isomorphism $X \to X$ is an *automorphism*.

Contravariance gives

$$\mathrm{Aut}(X) \cong \mathrm{Aut}_{k\text{-alg}}(k[X])^{\mathrm{op}},$$

where the opposite notation records reversal of composition.

As a set, however, an automorphism is completely determined by the corresponding k-algebra automorphism of the coordinate ring.

Example 4.28 (Affine transformations of the line)**.** For

$$a \in k^{\times}, \qquad b \in k,$$

the map

$$\varphi(t) = at + b$$

is an automorphism of $\mathbb{A}^1_k$, with inverse

$$t \mapsto a^{-1}(t - b).$$

Its pullback is the automorphism

$$k[t] \to k[t], \qquad t \mapsto at + b.$$

4.17 Composition as substitution

Let

$$\varphi : X \to Y, \qquad \psi : Y \to Z$$

be given in coordinates by

$$\varphi = (f_1, \dots, f_m), \qquad \psi = (g_1, \dots, g_r).$$

Then

$$\psi \circ \varphi = (g_1(f_1, \dots, f_m), \dots, g_r(f_1, \dots, f_m)).$$

The coordinate-ring statement is

$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^*.$$

Thus ordinary polynomial substitution and categorical contravariance are two descriptions of the same operation.

4.18 A first categorical viewpoint

The affine correspondence may be summarized as

$$\boxed{\text{affine algebraic sets over } k \quad \longleftrightarrow \quad \text{their coordinate } k\text{-algebras, with arrows reversed.}}$$

At the classical level, coordinate rings are finitely generated reduced k-algebras of the appropriate form. Scheme theory will eventually replace an algebraic set X by

$$\operatorname{Spec} A$$

and thereby allow arbitrary commutative rings A, including rings with nilpotents.

The present chapter is therefore not a side calculation. It is the prototype for the scheme-theoretic equivalence

$$\mathrm{Hom}_{\mathsf{Sch}}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \mathrm{Hom}_{\mathsf{Ring}}(A, B),$$

which will appear later in Volume VI.

4.19 Common pitfalls

Warning 4.29 (Continuity alone is not enough). A Zariski-continuous map need not be a morphism. A morphism must arise from polynomial coordinate functions, equivalently from a homomorphism of coordinate rings.

Warning 4.30 (The pullback reverses direction). For

$$\varphi : X \to Y,$$

the induced map is

$$\varphi^* : k[Y] \to k[X],$$

not $k[X] \to k[Y]$.

Warning 4.31 (Relations in the target must be respected). Choosing arbitrary $f_1, \dots, f_m \in k[X]$ defines a map to $\mathbb{A}^m_k$, but it defines a map to

$$Y = V(I(Y))$$

only when every relation in $I(Y)$ vanishes after substitution.

Warning 4.32 (Bijective is not automatically isomorphic). A bijective morphism need not have a polynomial inverse. Isomorphism means that both directions are morphisms, or equivalently that the pullback on coordinate rings is an isomorphism.

Warning 4.33 (Classical closed subsets are reduced). A surjective coordinate-ring map in this classical chapter should be interpreted through the reduced closed subset determined by its vanishing ideal. Arbitrary nonradical quotient ideals belong naturally to closed subschemes later.

	Parabola parametrization	**Cusp parametrization**
map	$t \mapsto (t, t^2)$	$t \mapsto (t^2, t^3)$
set-theoretic behavior	bijective	bijective on k-points
pullback image	$k[t]$	$k[t^2, t^3] \subsetneq k[t]$
conclusion	isomorphism	not an isomorphism

4.20 Worked examples

The examples emphasize how the same map is read geometrically and through the reversed coordinate-ring homomorphism.

Example 4.34 (Parabola isomorphism)**.** Let $P = V(y - x^2)$. The maps

$$\alpha : \mathbb{A}^1 \to P, \quad t \mapsto (t, t^2), \qquad \beta : P \to \mathbb{A}^1, \quad (x, y) \mapsto x$$

are inverse morphisms. On rings,

$$\alpha^*(\bar{x}) = t, \quad \alpha^*(\bar{y}) = t^2, \qquad \beta^*(t) = \bar{x}.$$

Example 4.35 (Projection from a curve)**.** For $C = V(y^2 - x^3)$, projection

$$\pi : C \to \mathbb{A}^1, \qquad (x, y) \mapsto x$$

has pullback $\pi^* : k[t] \to k[C]$, $t \mapsto \bar{x}$. Its image is the subalgebra $k[\bar{x}] \subseteq k[C]$.

Example 4.36 (A closed embedding)**.** The map $i : \mathbb{A}^1 \to \mathbb{A}^2$, $t \mapsto (t, 0)$, has pullback

$$i^* : k[x, y] \to k[t], \qquad x \mapsto t, \ y \mapsto 0.$$

It is surjective with kernel (y), so the image is the closed axis $V(y)$.

Example 4.37 (Morphisms into the hyperbola)**.** For $H = V(xy - 1)$, a pair $(f, g) \in k[X]^2$ defines $X \to H$ exactly when $fg = 1$. Thus morphisms into H are controlled by the unit group $k[X]^\times$.

Example 4.38 (Squaring on the affine line)**.** For $s(t) = t^2$,

$$s^* : k[u] \to k[t], \qquad u \mapsto t^2.$$

When k is infinite this is injective but not surjective, so s is not an isomorphism even though its image may be Zariski dense.

Example 4.39 (Translation)**.** For $a \in k$, $\tau_a(t) = t + a$ is an automorphism and

$$\tau_a^*(t) = t + a, \qquad (\tau_a^{-1})^*(t) = t - a.$$

Example 4.40 (Composition is substitution)**.** Let $\varphi(t) = (t, t^2)$ and $\psi(x, y) = x + y$. Then

$$(\psi \circ \varphi)(t) = t + t^2$$

and

$$(\varphi^* \circ \psi^*)(u) = \varphi^*(x + y) = t + t^2.$$

Example 4.41 (A map constrained by a target relation)**.** For $Y = V(v - u^2)$, choosing $f \in k[X]$ and setting

$$u \mapsto f, \qquad v \mapsto f^2$$

automatically defines a morphism $X \to Y$, since the target relation becomes zero.

Example 4.42 (The cusp: bijective but not isomorphic)**.** Let $C = V(y^2 - x^3)$ and

$$\nu : \mathbb{A}^1 \to C, \qquad t \mapsto (t^2, t^3).$$

The map is bijective on k-rational points: if $x \neq 0$, then $t = y/x$; the origin comes from $t = 0$. But

$$\nu^* : k[C] \hookrightarrow k[t]$$

has image $k[t^2, t^3]$, which does not contain t. Thus ν is not an isomorphism.

Example 4.43 (The graph of a morphism)**.** For $f : \mathbb{A}^1 \to \mathbb{A}^1$, $f(t) = t^3 - t$, the graph is

$$\Gamma_f = V(y - x^3 + x) \subseteq \mathbb{A}^2.$$

The graph embedding $t \mapsto (t, t^3 - t)$ is closed, and projection to the first coordinate is its inverse onto Γ_f.

4.21 Exercises

The Exercises are primarily new, short-to-medium training problems. The three substantive legacy affine-morphism problems have been promoted to Problem Dossiers below rather than left among routine exercises.

Exercise 4.44. Let $Y = V(I) \subseteq \mathbb{A}^m_k$ and $\varphi = (f_1, \ldots, f_m) : X \to \mathbb{A}^m_k$. Prove that $\varphi(X) \subseteq Y$ iff $g(f_1, \ldots, f_m) = 0$ in $k[X]$ for every $g \in I$.

Exercise 4.45. Prove directly that $\varphi^*(h) = h \circ \varphi$ defines a k-algebra homomorphism $k[Y] \to k[X]$.

Exercise 4.46. If $\varphi = (f_1, \ldots, f_m) : X \to Y$, show that $\varphi^*(\bar{y}_i) = f_i$. Deduce that φ^* is uniquely determined by the coordinate functions of φ.

Exercise 4.47. For $X \xrightarrow{\varphi} Y \xrightarrow{\psi} Z$, prove $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$.

Exercise 4.48. Given a k-algebra homomorphism $\Phi : k[Y] \to k[X]$, reconstruct the corresponding morphism $X \to Y$ from the images of the coordinate classes on Y.

Exercise 4.49. Prove $\varphi^{-1}(V_Y(S)) = V_X(\varphi^* S)$, and deduce that every morphism is Zariski continuous.

Exercise 4.50. For $f \in k[Y]$, prove $\varphi^{-1}(D_Y(f)) = D_X(\varphi^* f)$.

Exercise 4.51. Prove that $t \mapsto (t, t^2)$ is an isomorphism $\mathbb{A}^1 \cong V(y - x^2)$. Write both inverse morphisms and both pullbacks.

Exercise 4.52. For $i(t) = (t, 0) : \mathbb{A}^1 \to \mathbb{A}^2$, compute i^*, its kernel, and its image; verify algebraically that i is a closed embedding.

Exercise 4.53. For $\varphi(t) = at + b$ with $a \neq 0$, compute the inverse morphism and the inverse k-algebra automorphism.

Exercise 4.54. Assume k is infinite. For $s(t) = t^2$, compute $s^* : k[u] \to k[t]$ and show that it is injective but not surjective.

Exercise 4.55. Assume k is infinite. Prove $\mathrm{Mor}(X, \mathbb{A}^1) \cong k[X]$.

Exercise 4.56. If $\varphi : X \to Y$ has φ^* a k-algebra isomorphism, construct the inverse morphism to φ.

Exercise 4.57. If $Y \subseteq X$ is closed, show that inclusion induces $k[X] \twoheadrightarrow k[Y]$ and identify its kernel.

Exercise 4.58. Show that a morphism $Z \to X \times Y$ is equivalent to a pair of morphisms $Z \to X$ and $Z \to Y$. Express the construction in coordinates.

Exercise 4.59. Let $X \neq \varnothing$ and $y \in Y$. Show that the constant map $c_y : X \to Y$ is a morphism with pullback $h \mapsto h(y) \cdot 1$. Conversely, if the image of φ^* is contained in the constant subfield $k \subseteq k[X]$, show that φ is constant.

Exercise 4.60. For $f = (f_1, \ldots, f_m) : X \to \mathbb{A}^m$, show that its graph in $X \times \mathbb{A}^m$ is cut out by $y_i - f_i = 0$. Deduce that the graph map is a closed embedding.

Exercise 4.61. Show directly that $\ker \varphi^*$ consists exactly of the polynomial functions on Y that vanish on $\varphi(X)$. Deduce $V_Y(\ker \varphi^*) = \overline{\varphi(X)}^{\mathrm{Zar}}$.

4.22 Problems

Problems VI.04.P1–P3 preserve the substantive legacy affine-morphism problems. Problems VI.04.P4–P8 are new synthesis problems selected to deepen the algebra–geometry dictionary.

Problem VI.04.P1 — The hyperbola is not the affine line

Problem. Let $H = V(xy - 1) \subseteq \mathbb{A}^2_k$. Show that $H \not\cong \mathbb{A}^1_k$, and determine every morphism $\mathbb{A}^1_k \to H$.

Problem VI.04.P2 — Products of affine algebraic sets

Problem. Assume k is algebraically closed. Let $X \subseteq \mathbb{A}^m$ and $Y \subseteq \mathbb{A}^n$ be affine algebraic sets. Prove that $X \times Y$ is affine algebraic and that

$$k[X \times Y] \cong k[X] \otimes_k k[Y].$$

If X and Y are irreducible, prove that $X \times Y$ is irreducible.

Problem VI.04.P3 — No nonconstant polynomial map to a smooth cubic

Problem. Assume k is algebraically closed and $\mathrm{char}(k) \neq 2$. Let

$$E = V(y^2 - x^3 + x) \subseteq \mathbb{A}^2_k.$$

Show that every morphism $\mathbb{A}^1_k \to E$ is constant.

Problem VI.04.P4 — A bijective morphism that is not an isomorphism

Problem. Let $C = V(y^2 - x^3)$ and $\nu(t) = (t^2, t^3)$. Prove that $\nu : \mathbb{A}^1 \to C$ is bijective on k-rational points but is not an isomorphism of affine algebraic sets.

Problem VI.04.P5 — Dense image and injective pullback

Problem. For a morphism $\varphi : X \to Y$, prove

$$\ker \varphi^* = I_Y(\varphi(X)), \qquad \overline{\varphi(X)} = V_Y(\ker \varphi^*).$$

Deduce that φ has dense image iff φ^* is injective. Apply this to the projection $\mathbb{A}^2 \to \mathbb{A}^1$ and, over an infinite field, to squaring $t \mapsto t^2$.

Problem VI.04.P6 — Graphs are closed embeddings

Problem. Let $f : X \to Y \subseteq \mathbb{A}^m$ be a morphism. Prove that its graph $\Gamma_f \subseteq X \times Y$ is closed and that

$$\gamma_f : X \to X \times Y, \qquad x \mapsto (x, f(x))$$

is a closed embedding. Describe the induced map on coordinate rings.

Problem VI.04.P7 — All automorphisms of the affine line

Problem. Prove that every automorphism of $\mathbb{A}^1_k$ has the form

$$t \longmapsto at + b, \qquad a \in k^\times,\ b \in k.$$

Problem VI.04.P8 — Equalizers and fixed-point loci

Problem. Let $f, g : X \to Y \subseteq \mathbb{A}^m$ be morphisms. Prove that

$$E = \{x \in X : f(x) = g(x)\}$$

is Zariski closed in X. If $f_i, g_i \in k[X]$ are their coordinate functions, show

$$E = V_X(f_1 - g_1, \dots, f_m - g_m).$$

Deduce that the fixed-point set of every endomorphism $X \to X$ is closed.

4.23 Challenges

Challenge VI.04.C1 — Endomorphisms and automorphisms of the hyperbola

Challenge. Write $H = V(xy - 1) \cong \mathbb{G}_m$. Classify all morphisms $H \to H$ and all automorphisms of H.

Challenge VI.04.C2 — Power maps have no polynomial section

Challenge. Let $d > 1$ and $p_d : \mathbb{A}^1 \to \mathbb{A}^1$, $t \mapsto t^d$. Show that there is no morphism $s : \mathbb{A}^1 \to \mathbb{A}^1$ with $p_d \circ s = \mathrm{id}$, and no morphism r with $r \circ p_d = \mathrm{id}$.

Challenge VI.04.C3 — Idempotent endomorphisms of the affine line

Challenge. Classify all morphisms $e : \mathbb{A}^1 \to \mathbb{A}^1$ satisfying

$$e \circ e = e.$$

4.24 Chapter summary

A morphism of affine algebraic sets

$$\varphi : X \to Y$$

is a polynomial map whose image lies in Y. Pulling back polynomial functions gives a k-algebra homomorphism in the reverse direction,

$$\varphi^* : k[Y] \to k[X].$$

Composition becomes substitution, and every homomorphism of coordinate rings arises from a unique affine morphism:

$$\mathrm{Mor}_k(X, Y) \cong \mathrm{Hom}_{k\text{-alg}}(k[Y], k[X]).$$

Thus isomorphisms correspond to ring isomorphisms, closed embeddings to surjective pullbacks, and dense images to injective pullbacks. Products become tensor products, while graphs become closed embeddings. The next chapter studies irreducibility, components, and connectedness, where these map-theoretic ideas interact with zero divisors, prime ideals, and idempotents.

4.25 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 4.62 (Pullback)**.** Find the pullback of u under $t \mapsto t^3$.

Hint. Use polynomial maps and pullback homomorphisms. ☐

Solution. It is t^3. ☐

Exercise 4.63 (Parabola map)**.** Write the coordinate-ring map for $t \mapsto (t, t^2)$.

Hint. Use polynomial maps and pullback homomorphisms. ☐

Solution. $x \mapsto t$, $y \mapsto t^2$. ☐

Exercise 4.64 (Composition)**.** What happens to pullbacks under composition of morphisms?

Hint. Use polynomial maps and pullback homomorphisms. □

Solution. They compose in the reverse order. □

Exercise 4.65 (Constant map)**.** What ring map corresponds to the constant point a on $\mathbb{A}^1_k$?

Hint. Use polynomial maps and pullback homomorphisms. □

Solution. Evaluation $k[x] \to k$, $x \mapsto a$. □

Exercise 4.66 (Identity)**.** What pullback corresponds to the identity morphism?

Hint. Use polynomial maps and pullback homomorphisms. □

Solution. The identity ring homomorphism. □

Proofs

Exercise 4.67 (Composition)**.** Prove: Prove composition of affine morphisms is a morphism.

Hint. Work directly from polynomial maps and pullback homomorphisms. □

Solution. Coordinate polynomials remain polynomial after substitution, equivalently pullback ring maps compose. □

Exercise 4.68 (Contravariance)**.** Prove: Prove a morphism $X \to Y$ induces a k-algebra map $k[Y] \to k[X]$.

Hint. Work directly from polynomial maps and pullback homomorphisms. □

Solution. Pull back regular coordinate functions and check that target relations vanish on X. □

Exercise 4.69 (Isomorphism criterion)**.** Prove: Prove inverse affine morphisms induce inverse coordinate-ring maps.

Hint. Work directly from polynomial maps and pullback homomorphisms. □

Solution. Apply pullback to both compositions and use contravariance. □

Exercise 4.70 (Parabola isomorphism)**.** Prove: Prove the standard parametrization identifies $\mathbb{A}^1$ with $V(y - x^2)$.

Hint. Work directly from polynomial maps and pullback homomorphisms. □

Solution. Exhibit projection as inverse and check both compositions. □

Counterexamples and hypothesis tests

Exercise 4.71 (Set map)**.** Test the claim: every set map between algebraic sets is a morphism.

Hint. Test the claim on a concrete affine or local model before generalizing. □

Solution. False; morphisms must be polynomial or regular. □

Exercise 4.72 (Same direction)**.** Test the claim: morphisms and coordinate-ring homomorphisms point in the same direction.

Hint. Test the claim on a concrete affine or local model before generalizing. □

Solution. False; the correspondence is contravariant. ☐

Exercise 4.73 (Bijective)**.** Test the claim: every bijective morphism of affine algebraic sets is automatically an isomorphism.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general; the inverse need not be regular. ☐

Applications and investigations

Exercise 4.74 (Parametric curves)**.** How are polynomial parametrizations expressed algebraically?

Hint. Translate the question into polynomial maps and pullback homomorphisms. ☐

Solution. By pullback homomorphisms from the curve coordinate ring to a polynomial ring. ☐

Exercise 4.75 (Equation verification)**.** How do you verify a polynomial map lands in a target algebraic set?

Hint. Translate the question into polynomial maps and pullback homomorphisms. ☐

Solution. Substitute the map into the target defining equations and check they vanish. ☐

Challenge problems

Exercise 4.76 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Morphisms of Affine Algebraic Sets in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Morphisms of Affine Algebraic Sets, translates it through polynomial maps and pullback homomorphisms, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 4.77 (Hypothesis challenge)**.** Choose one structural statement from Morphisms of Affine Algebraic Sets, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 5

Irreducibility, Components and Connectedness

Purpose and learning goals

Algebraic sets can split into simpler pieces. Some decompositions are genuine and intrinsic, while others merely reflect a poor choice of equations. This chapter develops two complementary notions that measure whether a space splits:

- *irreducibility*, which forbids decomposition into two proper closed pieces;
- *connectedness*, which forbids decomposition into two disjoint nonempty open pieces.

For affine algebraic sets, irreducibility is controlled by prime ideals and integral domains, while connectedness is controlled by idempotents and product decompositions of coordinate rings. The two notions are related but not equivalent.

Throughout, k is a field and
$$X \subseteq \mathbb{A}^n_k$$
is an affine algebraic set. When the connectedness–idempotent correspondence is invoked in its strongest classical point-set form, we explicitly assume that k is algebraically closed.

After this chapter, the reader should be able to:

- characterize irreducible spaces using nonempty open subsets;
- prove that nonempty open subsets of irreducible spaces are dense and irreducible;
- translate irreducibility of X into primality of $I(X)$ and domain structure of $k[X]$;
- prove existence and uniqueness of irreducible components in affine algebraic sets;
- distinguish irreducible components from connected components;
- characterize disconnected affine algebraic sets using nontrivial idempotents;
- pass between minimal primes and irreducible components;
- use component-intersection graphs to test connectedness;
- analyze explicit reducible algebraic sets by factoring equations and computing minimal components.

5.1 Irreducible topological spaces

Definition 5.1 (Irreducible space)**.** A nonempty topological space X is *irreducible* if whenever

$$X = F \cup G$$

with $F, G \subseteq X$ closed, then

$$F = X \qquad \text{or} \qquad G = X.$$

Equivalently, X cannot be written as a union of two proper closed subsets.

Proposition 5.2 (Open-set criterion)**.** *For a nonempty topological space X, the following are equivalent:*

1. *X is irreducible;*
2. *every two nonempty open subsets of X intersect;*
3. *every nonempty open subset of X is dense.*

Proof. Suppose X is irreducible and let $U, V \subseteq X$ be nonempty open subsets. If $U \cap V = \varnothing$, then

$$X = (X \setminus U) \cup (X \setminus V)$$

is a union of two proper closed subsets, contradiction. Hence (1) implies (2).

Assume (2), and let $U \subseteq X$ be nonempty open. If

$$\overline{U} \neq X,$$

then

$$X \setminus \overline{U}$$

is a nonempty open set disjoint from U, contradicting (2). Thus U is dense, so (2) implies (3).

Finally, assume every nonempty open subset is dense. If

$$X = F \cup G$$

with F, G proper closed subsets, then

$$U = X \setminus F$$

is nonempty open. Since $U \subseteq G$ and G is closed,

$$\overline{U} \subseteq G \neq X,$$

contradicting density. Thus (3) implies (1). □

Corollary 5.3. *Every nonempty open subset of an irreducible space is irreducible and dense in the ambient space.*

Proof. Density is part of theorem 5.2. Let $U \subseteq X$ be nonempty open and let $U_1, U_2 \subseteq U$ be nonempty open subsets in the subspace topology. Write

$$U_i = U \cap V_i$$

with V_i open in X. Then $U \cap V_1$ and $U \cap V_2$ are nonempty open subsets of X, so their intersection is nonempty. Hence $U_1 \cap U_2 \neq \varnothing$, and U is irreducible. □

5.2 Closure preserves irreducibility

Proposition 5.4. *If Y is an irreducible subset of a topological space X, then its closure*

$$\overline{Y}$$

is irreducible.

Proof. Suppose

$$\overline{Y} = F \cup G$$

with F, G closed in $\overline{Y}$. Then

$$Y = (Y \cap F) \cup (Y \cap G).$$

The subsets $Y \cap F$ and $Y \cap G$ are closed in Y. Since Y is irreducible, one of them is all of Y. If, for example, $Y \subseteq F$, then by closedness of F in $\overline{Y}$,

$$\overline{Y} \subseteq F.$$

Hence $F = \overline{Y}$. □

This fact is especially useful for parametrized subsets: if the parameter space is irreducible and the image is dense in its closure, the closure is irreducible.

Example 5.5 (Two axes are reducible)**.** The set $V(xy) \subset \mathbb{A}^2_k$ decomposes as $V(x) \cup V(y)$, so it is reducible. The two axes are its irreducible components, corresponding to the minimal primes (x) and (y).

5.3 Irreducibility and coordinate rings

The topological definition becomes algebraic for affine algebraic sets.

Theorem 5.6 (Prime-ideal criterion)**.** *For an affine algebraic set $X \subseteq \mathbb{A}^n_k$, the following are equivalent:*

1. *X is irreducible;*
2. *$I(X) \subseteq k[x_1, \ldots, x_n]$ is prime;*
3. *$k[X]$ is an integral domain.*

Proof. The equivalence of (2) and (3) is the standard quotient criterion:

$$k[X] = k[x_1, \ldots, x_n]/I(X)$$

is a domain exactly when $I(X)$ is prime.

Assume X is irreducible. Suppose

$$fg \in I(X).$$

Then fg vanishes on X, so

$$X \subseteq V(fg) = V(f) \cup V(g).$$

Inside X,

$$X = V_X(f) \cup V_X(g).$$

Irreducibility implies

$$X = V_X(f) \qquad \text{or} \qquad X = V_X(g).$$

Thus

$$f \in I(X) \qquad \text{or} \qquad g \in I(X),$$

so $I(X)$ is prime.

Conversely, assume $I(X)$ is prime and suppose

$$X = Y \cup Z$$

with Y, Z closed in X. We may write

$$Y = V_X(J), \qquad Z = V_X(K).$$

If both are proper, choose

$$f \in I(Y) \setminus I(X), \qquad g \in I(Z) \setminus I(X).$$

Since every point of X lies in Y or Z,

$$fg$$

vanishes on all of X. Hence

$$fg \in I(X).$$

Primality forces $f \in I(X)$ or $g \in I(X)$, contradiction. Thus one of Y, Z equals X. □

Corollary 5.7. *If k is infinite, then*

$$\mathbb{A}_k^n$$

is irreducible.

Proof. For infinite k,

$$I(\mathbb{A}_k^n) = (0),$$

and the polynomial ring $k[x_1, \ldots, x_n]$ is a domain. □

Remark 5.8 (Classical terminology). Many texts reserve the word *affine variety* for an irreducible affine algebraic set. Under that convention,

$$X \text{ is an affine variety} \quad \Longleftrightarrow \quad k[X] \text{ is a domain.}$$

Other texts allow "variety" to be reducible. We will always state irreducibility explicitly when it matters.

5.4 Reducibility and zero divisors

The contrapositive of theorem 5.6 is often computationally useful.

Corollary 5.9. *An affine algebraic set X is reducible if and only if its coordinate ring $k[X]$ has nonzero zero divisors.*

Example 5.10. For

$$X = V(xy) \subseteq \mathbb{A}_k^2,$$

the classes $\bar{x}, \bar{y} \in k[X]$ are nonzero and satisfy

$$\bar{x}\bar{y} = 0.$$

Geometrically,

$$X = V(x) \cup V(y),$$

the union of the two coordinate axes.

Zero divisors detect reducibility, but they do *not* by themselves detect disconnectedness. The crossing axes are reducible yet connected.

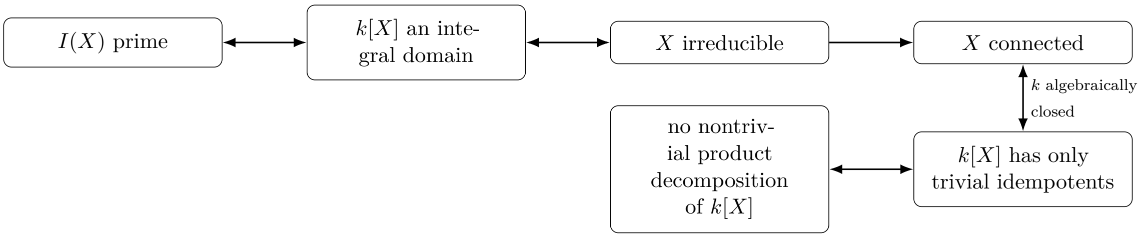


Figure 5.1: The two "one-piece" dictionaries. Irreducibility is detected by primality and the absence of zero divisors; connectedness is detected by idempotents and product splittings.

5.5 Noetherianity of affine algebraic sets

To obtain a finite decomposition into irreducible components, we use the Noetherian property.

Definition 5.11 (Noetherian topological space)**.** A topological space X is *Noetherian* if every descending chain of closed subsets

$$F_1 \supseteq F_2 \supseteq F_3 \supseteq \cdots$$

stabilizes.

Theorem 5.12. *Every affine algebraic set is a Noetherian topological space in the Zariski topology.*

Proof. Let

$$X \subseteq \mathbb{A}^n_k.$$

A descending chain of closed subsets of X,

$$F_1 \supseteq F_2 \supseteq \cdots ,$$

gives an ascending chain of vanishing ideals

$$I_X(F_1) \subseteq I_X(F_2) \subseteq \cdots$$

inside $k[X]$. The ring $k[X]$ is a quotient of $k[x_1, \ldots, x_n]$, hence is Noetherian by Hilbert's basis theorem. Therefore the ideal chain stabilizes. Applying V_X shows that the closed-set chain stabilizes. □

Corollary 5.13 (Quasi-compactness)**.** *Every affine algebraic set is quasi-compact: every open cover has a finite subcover.*

Proof. This is true for every Noetherian topological space. If an open cover had no finite subcover, the complements would yield a family of closed sets with the finite intersection property but empty total intersection. Equivalently, one can choose a maximal closed subset that cannot be finitely covered and obtain a contradiction by covering one of its points. □

Example 5.14 (A parabola is irreducible)**.** The coordinate ring $k[x, y]/(y - x^2) \cong k[x]$ is a domain. Therefore the parabola $V(y - x^2)$ is irreducible.

5.6 Irreducible components

Definition 5.15 (Irreducible component)**.** An *irreducible component* of a topological space X is a maximal irreducible closed subset of X.

Theorem 5.16 (Existence of a finite irreducible decomposition)**.** *Every nonempty affine algebraic set X can be written as a finite union*

$$X = X_1 \cup \cdots \cup X_r$$

where each X_i is irreducible and no X_i is contained in another.

Proof. Because X is Noetherian, consider the collection of closed subsets that cannot be written as a finite union of irreducible closed subsets. If this collection were nonempty, choose a minimal member F. The set F is not irreducible, so

$$F = F_1 \cup F_2$$

with $F_1, F_2 \subsetneq F$ closed. By minimality, each F_i is a finite union of irreducible closed subsets, hence so is F, contradiction. Thus every closed subset, and in particular X, has a finite irreducible decomposition. Removing redundant pieces gives the noncontainment condition. ☐

Theorem 5.17 (Uniqueness of irreducible components)**.** *Suppose*

$$X = X_1 \cup \cdots \cup X_r = Y_1 \cup \cdots \cup Y_s$$

are two decompositions into irreducible closed subsets with no redundant containments. Then

$$r = s,$$

and after reindexing,

$$X_i = Y_i$$

for all i.

Proof. Fix i. Since

$$X_i \subseteq Y_1 \cup \cdots \cup Y_s$$

and X_i is irreducible, repeated use of the defining property shows

$$X_i \subseteq Y_j$$

for some j. Similarly, $Y_j \subseteq X_\ell$ for some ℓ. Therefore

$$X_i \subseteq X_\ell.$$

The noncontainment assumption forces $\ell = i$, hence

$$X_i = Y_j.$$

Thus each X_i appears among the Y_j. By symmetry, each Y_j appears among the X_i. The two collections therefore coincide up to reordering. ☐

5.7 Minimal primes and components

Assume now that k is algebraically closed.

For a radical ideal
$$I = I(X) \subseteq k[x_1, \dots, x_n],$$
one can write
$$I = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$$
where the $\mathfrak{p}_i$ are the minimal prime ideals containing I.

Theorem 5.18 (Components from minimal primes)**.** *The irreducible components of $X = V(I)$ are exactly*
$$V(\mathfrak{p}_1), \dots, V(\mathfrak{p}_r),$$
where $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ are the minimal primes over I.

Proof. Each $V(\mathfrak{p}_i)$ is irreducible by theorem 5.6. Since
$$I = \bigcap_i \mathfrak{p}_i,$$
we have
$$V(I) = \bigcup_i V(\mathfrak{p}_i).$$
Minimality of the primes corresponds, under order reversal, to maximality of the associated irreducible closed subsets. $\square$

This gives a practical algebraic method: find the minimal primes of $I(X)$, then take their zero sets.

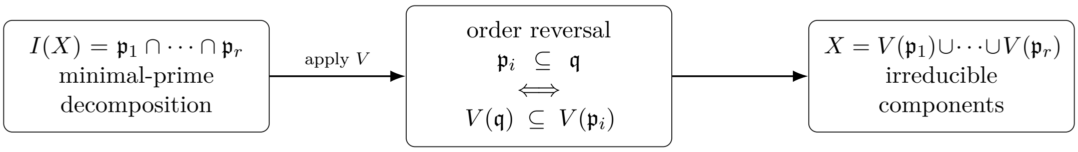


Figure 5.2: Minimal primes and irreducible components are the same decomposition viewed through the order-reversing zero-set correspondence.

5.8 Connectedness

Definition 5.19 (Connected space)**.** A topological space X is *connected* if it cannot be written as a disjoint union
$$X = U \sqcup V$$
of two nonempty open subsets.

Equivalently, the only subsets of X that are both open and closed are
$$\varnothing \qquad \text{and} \qquad X.$$

Proposition 5.20. *Every irreducible space is connected.*

Proof. If
$$X = U \sqcup V$$
with U, V nonempty open, then U and V are also closed, because each is the complement of the other. Thus
$$X = U \cup V$$
is a union of two proper closed subsets, contradicting irreducibility. □

The converse fails.

Example 5.21 (Reducible but connected)**.** The algebraic set
$$X = V(xy) \subseteq \mathbb{A}^2_k$$
is the union
$$V(x) \cup V(y)$$
of two irreducible components. They meet at the origin, so X is connected.

Example 5.22 (A finite set)**.** A finite algebraic set with at least two closed points is reducible: split it into two nonempty proper finite subsets, each closed. Thus the irreducible finite closed subsets are single points.

5.9 Idempotents and disconnected affine algebraic sets

We now assume that k is algebraically closed.

Definition 5.23 (Idempotent)**.** An element e of a ring A is *idempotent* if
$$e^2 = e.$$
It is *nontrivial* if
$$e \neq 0, 1.$$

Theorem 5.24 (Connectedness–idempotent criterion)**.** *Let X be an affine algebraic set over an algebraically closed field k. Then the following are equivalent:*

1. *X is disconnected;*
2. *$k[X]$ contains a nontrivial idempotent;*
3. *there exist nonzero k-algebras A_1, A_2 such that*
$$k[X] \cong A_1 \times A_2.$$

Proof. Assume $e \in k[X]$ is a nontrivial idempotent. At every point $x \in X$,
$$e(x)^2 = e(x),$$
so
$$e(x) \in \{0, 1\}.$$
Define
$$X_0 = V_X(e), \qquad X_1 = V_X(e-1).$$
Then
$$X = X_0 \sqcup X_1.$$

Both subsets are closed. Each is also open because it is the complement of the other. Nontriviality of e ensures both are nonempty. Hence X is disconnected.

Conversely, suppose

$$X = Y \sqcup Z$$

with Y, Z nonempty and clopen. Since they are closed algebraic subsets,

$$V(I(Y) + I(Z)) = Y \cap Z = \varnothing.$$

By the Nullstellensatz,

$$I(Y) + I(Z) = k[x_1, \dots, x_n].$$

Choose

$$f \in I(Y), \qquad g \in I(Z)$$

with

$$f + g = 1.$$

On Y, $f = 0$ and hence $g = 1$; on Z, $g = 0$. Therefore the class

$$e = [g] \in k[X]$$

takes only the values 0 and 1, so

$$e^2 = e.$$

It is nontrivial because both Y and Z are nonempty.

Finally, if e is a nontrivial idempotent in a ring A, then

$$A \longrightarrow eA \times (1-e)A, \qquad a \longmapsto (ea, (1-e)a)$$

is an isomorphism. Conversely, in a nontrivial product

$$A_1 \times A_2,$$

the element

$$(1, 0)$$

is a nontrivial idempotent. ☐

Corollary 5.25. *If $k[X]$ is a domain, then X is connected.*

This also follows from irreducibility, but the idempotent criterion gives a different algebraic perspective.

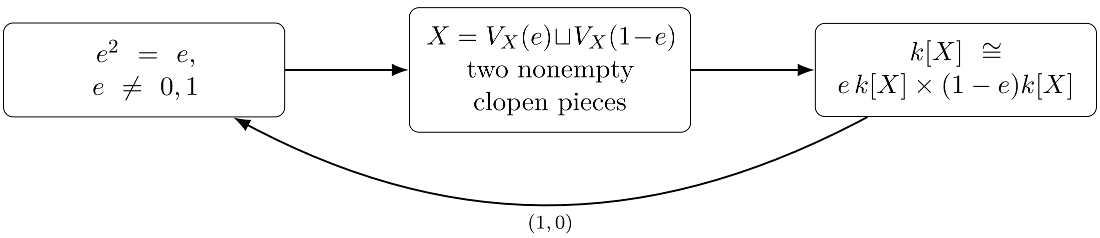


Figure 5.3: An idempotent is simultaneously a 0/1-valued function, a clopen separation, and a product decomposition of the coordinate ring.

Geometry	**Ring signal**	**Prototype**
Irreducible	domain; no nonzero zero divisors	$\mathbb{A}^1$, parabola
Reducible but connected	zero divisors; no nontrivial idempotent	$V(xy)$
Disconnected	nontrivial idempotent; product ring	$V(x(x-1))$

5.10 Connected components versus irreducible components

Irreducible components need not be disjoint. Connected components, by contrast, form a partition into maximal connected subsets.

For affine algebraic sets, there are finitely many irreducible components. Their incidence controls connectedness.

Definition 5.26 (Component-intersection graph)**.** Let

$$X = X_1 \cup \cdots \cup X_r$$

be the irreducible-component decomposition. The *component-intersection graph* Γ_X has one vertex for each X_i, with an edge between X_i and X_j when

$$X_i \cap X_j \neq \varnothing.$$

Theorem 5.27 (Intersection-graph criterion)**.** *An affine algebraic set X is connected if and only if its component-intersection graph Γ_X is connected.*

Proof. Suppose Γ_X is disconnected. Partition its vertices into two nonempty sets A, B with no edges between them. Let

$$Y = \bigcup_{i \in A} X_i, \qquad Z = \bigcup_{j \in B} X_j.$$

These are finite unions of closed sets, hence closed. The absence of edges gives

$$Y \cap Z = \varnothing,$$

and

$$X = Y \sqcup Z.$$

Thus X is disconnected.

Conversely, suppose

$$X = U \sqcup V$$

is a separation into nonempty clopen subsets. Each irreducible component X_i is connected, hence must lie entirely in U or entirely in V. If

$$X_i \cap X_j \neq \varnothing,$$

then X_i and X_j must lie on the same side. Therefore no graph edge crosses from the collection of components in U to those in V. Since both collections are nonempty, Γ_X is disconnected. □

This criterion makes the difference between irreducible and connected geometry very concrete: connectedness allows several irreducible pieces, provided the pieces are linked through intersections.

5.11 Common pitfalls

Warning 5.28 (Reducible does not mean disconnected)*.* The crossing axes

$$V(xy)$$

are reducible but connected.

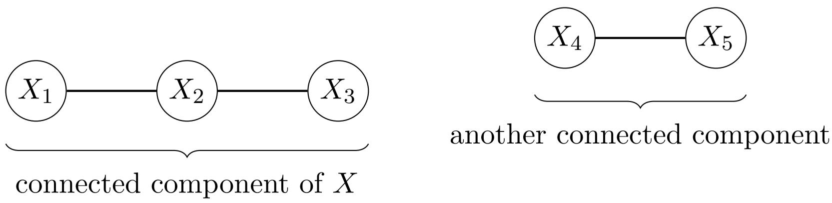


Figure 5.4: Once the irreducible components are known, connectedness becomes finite graph theory: graph-components correspond to connected components of the algebraic set.

Warning 5.29 (Connected does not mean irreducible). Connectedness only forbids a separation into disjoint clopen pieces. Components may still overlap.

Warning 5.30 (Zero divisors and idempotents detect different phenomena). Zero divisors detect reducibility of $k[X]$; nontrivial idempotents detect disconnectedness. A ring may have zero divisors but no nontrivial idempotents.

Warning 5.31 (Irreducible components are maximal, not merely any irreducible subsets). A line contained in an irreducible surface is irreducible, but it is not an irreducible component of the surface.

Warning 5.32 (Classical point sets versus schemes). Later, $\operatorname{Spec} A$ may contain generic points and nilpotent structure not visible in the classical set $V(I) \subseteq k^n$. The scheme-theoretic notions of irreducibility and connectedness will be revisited in that broader setting.

5.12 Worked examples

Example 1: Affine space is irreducible

Assume k is infinite. Since
$$I(\mathbb{A}^n_k) = (0)$$
and $k[x_1, \dots, x_n]$ is an integral domain, theorem 5.6 gives $\mathbb{A}^n_k$ irreducible. Consequently every finite intersection of nonempty Zariski open subsets of affine space is nonempty.

Example 2: A parabola is one component

Let $P = V(y - x^2) \subseteq \mathbb{A}^2_k$. Its coordinate ring is
$$k[P] \cong k[t],$$
a domain. Hence P is irreducible. The same conclusion follows from the parametrization $t \mapsto (t, t^2)$: the image of an irreducible space is irreducible.

Example 3: Crossing axes — reducible but connected

Let $X = V(xy) \subseteq \mathbb{A}^2_k$. Then
$$X = V(x) \cup V(y),$$
so X is reducible. The two components meet at the origin, hence the component-intersection graph has two vertices joined by an edge and X is connected. Algebraically, $k[X] = k[x, y]/(xy)$ has zero divisors but no nontrivial idempotent.

Example 4: Two parallel components — reducible and disconnected

Let $X = V(x(x-1)) \subseteq \mathbb{A}^2_k$. Then

$$X = V(x) \sqcup V(x-1).$$

The ideals (x) and $(x-1)$ are comaximal, so

$$k[X] \cong k[y] \times k[y].$$

The class of x is idempotent because $x^2 - x = 0$ in $k[X]$. Thus the two geometric pieces are not merely irreducible components; they are connected components.

Example 5: A plane plus a line

Consider

$$X = V(xy, xz) \subseteq \mathbb{A}^3_k.$$

Since both equations have the common factor x,

$$X = V(x) \cup V(y, z).$$

The components are the plane $x = 0$ and the x-axis. Their intersection is the origin, so X is connected but reducible. Algebraically,

$$(xy, xz) = (x) \cap (y, z).$$

Example 6: Three lines through the origin

Assume $\operatorname{char} k \neq 2$ and set

$$X = V\big(xy(x-y)\big) \subseteq \mathbb{A}^2_k.$$

Then

$$X = V(x) \cup V(y) \cup V(x-y).$$

Each component meets the other two at the origin, so the component graph is a triangle. Thus X is connected although it has three irreducible components.

Example 7: A chain of components

Suppose $X = X_1 \cup X_2 \cup X_3$ with

$$X_1 \cap X_2 \neq \varnothing, \qquad X_2 \cap X_3 \neq \varnothing, \qquad X_1 \cap X_3 = \varnothing.$$

The component graph is the path $X_1 --- X_2 --- X_3$. A chain of overlapping connected sets is connected, so X is connected even though not every pair of components meets.

Example 8: A disconnected finite algebraic set

Let $X = \{a, b\} \subseteq \mathbb{A}^1_k$ with $a \neq b$. Then

$$k[X] \cong k \times k.$$

The element $(1, 0)$ is a nontrivial idempotent. The two points are both open and closed, so $X = \{a\} \sqcup \{b\}$ is disconnected.

Example 9: The three coordinate axes in affine three-space

Let
$$X = V(xy, xz, yz) \subseteq \mathbb{A}^3_k.$$
A point in X has at most one nonzero coordinate, so
$$X = V(y, z) \cup V(x, z) \cup V(x, y).$$
These are the three coordinate axes. Their prime ideals are the minimal primes of (xy, xz, yz). All three axes meet at the origin, so the component graph is complete and X is connected.

Example 10: Two disjoint parabolas

Let
$$X = V\big((y - x^2)(y - x^2 - 1)\big) \subseteq \mathbb{A}^2_k.$$
Its two irreducible components are
$$P_0 = V(y - x^2), \qquad P_1 = V(y - x^2 - 1).$$
They are disjoint because a common point would force $0 = 1$. Moreover
$$(y - x^2) + (-(y - x^2 - 1)) = 1,$$
so the defining component ideals are comaximal and
$$k[X] \cong k[P_0] \times k[P_1] \cong k[t] \times k[t].$$
Thus disconnectedness is not tied to linear components; it is a ring-theoretic splitting.

5.13 Exercises

Exercise VI.05.1 — Three quick irreducibility tests

Assume k is algebraically closed. Decide which of
$$V(y - x^2), \qquad V(xy), \qquad V(xy - 1) \subseteq \mathbb{A}^2_k$$
are irreducible. Justify each answer using the coordinate ring.

Exercise VI.05.2 — Dense principal opens

Let X be irreducible and let $f \in k[X]$. If $D_X(f) \neq \varnothing$, prove that $D_X(f)$ is dense in X.

Exercise VI.05.3 — Polynomial identities from dense subsets

Let X be irreducible and $U \subseteq X$ a nonempty open subset. If $f, g \in k[X]$ agree on U, prove that $f = g$ in $k[X]$.

Exercise VI.05.4 — A parametrized cusp

Assume k is infinite. Let
$$C = V(y^2 - x^3) \subseteq \mathbb{A}^2_k.$$
Use $t \mapsto (t^2, t^3)$ to show that C is irreducible.

Exercise VI.05.5 — Reading components from a factorization

Assume $\operatorname{char} k \neq 2$. Determine the irreducible components of

$$X = V\big(xy(x-y)(x+y)\big) \subseteq \mathbb{A}^2_k.$$

Is X connected?

Exercise VI.05.6 — A reducible but connected surface

For $X = V(xy) \subseteq \mathbb{A}^3_k$, describe the irreducible components, their intersection, and the component graph. Deduce that X is connected but reducible.

Exercise VI.05.7 — Zero divisors versus idempotents

In $A = k[x,y]/(xy)$, exhibit nonzero zero divisors. Assuming k algebraically closed, prove that A has no nontrivial idempotents. Explain the two different geometric phenomena detected by these facts.

Exercise VI.05.8 — Two disjoint lines

Let $X = V(x(x-1)) \subseteq \mathbb{A}^2_k$. Use the Chinese remainder theorem to prove

$$k[X] \cong k[y] \times k[y].$$

Construct an explicit nontrivial idempotent and identify the corresponding clopen pieces.

Exercise VI.05.9 — A chain of components

Let $X = X_1 \cup X_2 \cup X_3 \cup X_4$, where each X_i is irreducible and

$$X_1 \cap X_2 \neq \varnothing, \quad X_2 \cap X_3 \neq \varnothing, \quad X_3 \cap X_4 \neq \varnothing.$$

Prove that X is connected.

Exercise VI.05.10 — Continuous images of irreducible spaces

Prove that the continuous image of an irreducible topological space is irreducible. Deduce that if $\phi : X \to Y$ is a morphism and X is irreducible, then $\overline{\phi(X)}$ is irreducible.

Exercise VI.05.11 — Removing a proper closed set

Let X be irreducible and $Z \subsetneq X$ closed. Prove that $X \setminus Z$ is nonempty, irreducible, and dense in X.

Exercise VI.05.12 — Compare four one-piece behaviors

Assume k is algebraically closed and choose three distinct elements $a, b, c \in k$. Determine irreducibility and connectedness for

$$\mathbb{A}^1_k, \qquad V(xy), \qquad V(x(x-1)), \qquad \{a, b, c\}.$$

For each, state the corresponding feature of its coordinate ring.

Exercise VI.05.13 — Minimal primes of a plane plus a line

Show

$$(xy, xz) = (x) \cap (y, z) \subseteq k[x, y, z].$$

Deduce the irreducible components of $V(xy, xz)$, and decide whether the set is connected.

Exercise VI.05.14 — Three disjoint vertical lines

Let $a \in k \setminus \{0, 1\}$ and

$$X = V(x(x-1)(x-a)) \subseteq \mathbb{A}^2_k.$$

Describe its irreducible and connected components. Show that $k[X] \cong k[y]^3$.

Exercise VI.05.15 — Continuous images of connected spaces

Prove that a continuous image of a connected space is connected. Deduce that the image of a connected affine algebraic set under a morphism is connected in its image topology.

Exercise VI.05.16 — Order reversal and components

Assume k algebraically closed and

$$I = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_r$$

is an irredundant intersection of distinct prime ideals, none containing another. Prove that the irreducible components of $V(I)$ are exactly $V(\mathfrak{p}_i)$.

Exercise VI.05.17 — The graph of the legacy three-component example

For

$$Y = V(x^2 - yz,\ xz - x) \subseteq \mathbb{A}^3_k,$$

use the three components identified in Problem VI.05.P3 to compute all pairwise intersections. Draw the component graph and decide whether Y is connected.

Exercise VI.05.18 — Finite intersections of dense opens

Let X be irreducible and let $U_1, \ldots, U_r \subseteq X$ be nonempty open subsets. Prove that

$$U_1 \cap \cdots \cap U_r$$

is nonempty and dense in X. Apply this to principal opens $D_X(f_1), \ldots, D_X(f_r)$.

5.14 Problems

Problem VI.05.P1 — Dense open subsets of irreducible sets

Problem. Let X be an irreducible algebraic set and let

$$\varnothing \neq U \subseteq X$$

be open. Prove that U is both dense in X and irreducible.

Problem VI.05.P2 — Uniqueness of irreducible decomposition

Problem. Suppose
$$X = X_1 \cup \cdots \cup X_r$$
and
$$X = X'_1 \cup \cdots \cup X'_s$$
are two decompositions of an affine algebraic set into irreducible closed subsets. Assume there are no redundant containments:
$$X_i \not\subseteq X_j, \qquad X'_i \not\subseteq X'_j \qquad (i \neq j).$$
Prove that
$$r = s,$$
and after reindexing,
$$X_i = X'_i$$
for all i.

Problem VI.05.P3 — Three irreducible components

Problem. Let
$$Y = V(x^2 - yz,\ xz - x) \subseteq \mathbb{A}^3_k.$$
Show that Y has exactly three irreducible components. Describe the components and their prime ideals.

Problem VI.05.P4 — Plane curves without a common component

Problem. Assume k is algebraically closed. Let
$$f, g \in k[x, y]$$
have no common factor.

1. Show that there exist
$$u, v \in k[x, y]$$
such that
$$uf + vg$$
is a nonzero polynomial in $k[x]$.
2. Assume f is irreducible. Show that any proper algebraic subset of
$$V(f)$$
is finite.
3. Deduce that two affine plane curves with no common irreducible component meet in only finitely many points.
4. Prove that $\mathbb{A}^2_k$ is quasi-compact in the Zariski topology.

You may use the resultant in part (1), and Hilbert's basis theorem in part (4).

Problem VI.05.P5 — Connectedness, idempotents and product rings

Problem. Assume k is algebraically closed and X is an affine algebraic set. Prove that the following are equivalent:

1. X is disconnected;
2. $k[X]$ contains a nontrivial idempotent;
3. $$k[X] \cong A_1 \times A_2$$
   for nonzero k-algebras A_1, A_2.

Then apply the criterion to:

$$V(xy) \subseteq \mathbb{A}^2_k$$

and

$$V(x(x-1)) \subseteq \mathbb{A}^2_k.$$

Problem VI.05.P6 — Three coordinate axes: components, graph and ring

Problem. Let

$$X = V(xy, xz, yz) \subseteq \mathbb{A}^3_k.$$

Determine all irreducible components and their prime ideals. Compute the component graph, prove that X is connected, and explain algebraically why $k[X]$ has zero divisors but no nontrivial idempotents when k is algebraically closed.

Problem VI.05.P7 — Disjoint algebraic pieces and the Chinese remainder theorem

Problem. Assume k is algebraically closed. Let $Y, Z \subseteq \mathbb{A}^n_k$ be nonempty algebraic sets. Prove that

$$Y \cap Z = \varnothing \quad \Longleftrightarrow \quad I(Y) + I(Z) = (1).$$

If these conditions hold, prove

$$k[Y \cup Z] \cong k[Y] \times k[Z],$$

and construct the idempotent that is 1 on Y and 0 on Z.

Problem VI.05.P8 — Squarefree plane curves and their component graph

Problem. Assume k is algebraically closed and let

$$f = f_1 \cdots f_r \in k[x, y]$$

where the f_i are pairwise nonassociate irreducible polynomials. Prove that the irreducible components of $V(f)$ are exactly $V(f_i)$. Show that $V(f)$ is connected if and only if the graph joining i and j whenever $V(f_i, f_j) \neq \varnothing$ is connected. Apply this to

$$f = xy(x-1)(y-1).$$

Challenges

Challenge VI.05.C1 — The component-intersection graph

Problem. Let
$$X = X_1 \cup \cdots \cup X_r$$
be the decomposition of an affine algebraic set into irreducible components. Form the graph whose vertices are the X_i, joining X_i and X_j when
$$X_i \cap X_j \neq \varnothing.$$
Prove that the connected components of this graph correspond exactly to the connected components of X.

Challenge VI.05.C2 — Primitive idempotents and connected components

Problem. Assume k is algebraically closed and let X be an affine algebraic set with connected components $C_1, \dots, C_s$. Prove that there exist pairwise orthogonal idempotents $e_1, \dots, e_s \in k[X]$ satisfying
$$e_1 + \cdots + e_s = 1,$$
with $e_i = 1$ on C_i and $e_i = 0$ on every other component. Deduce
$$k[X] \cong \prod_{i=1}^{s} k[C_i].$$
Show that the e_i are primitive idempotents and are unique up to reordering.

Challenge VI.05.C3 — The rank-one determinantal variety is irreducible

Problem. Assume k is algebraically closed and let $m, n \geq 2$. Identify $\mathbb{A}_k^{mn}$ with the space of $m \times n$ matrices. Let $R_{m,n}$ be the subset of matrices of rank at most 1.

1. Show that $R_{m,n}$ is algebraic, cut out by the 2×2 minors.
2. Consider the morphism
$$\Phi : \mathbb{A}_k^m \times \mathbb{A}_k^n \longrightarrow \mathbb{A}_k^{mn}, \qquad (u, v) \longmapsto uv^{\mathsf{T}}.$$
Show that Φ is surjective onto $R_{m,n}$.
3. Deduce that $R_{m,n}$ is irreducible.

5.15 Chapter summary

Irreducibility is the strongest basic notion of "one piece" in algebraic geometry. For an affine algebraic set,
$$X \text{ irreducible} \iff I(X) \text{ prime} \iff k[X] \text{ a domain.}$$

Every affine algebraic set is Noetherian and therefore decomposes into finitely many irreducible components,
$$X = X_1 \cup \cdots \cup X_r,$$
uniquely up to order.

Connectedness is weaker:

$$X \text{ irreducible} \Longrightarrow X \text{ connected},$$

but not conversely. Over an algebraically closed field,

$$X \text{ disconnected} \quad \Longleftrightarrow \quad k[X] \text{ has a nontrivial idempotent} \quad \Longleftrightarrow \quad k[X] \cong A_1 \times A_2.$$

Finally, if $X_1, \ldots, X_r$ are the irreducible components, then X is connected exactly when the graph recording which components intersect is connected.

The next chapters change viewpoint from k-rational zero sets to prime ideals as geometric points, beginning the passage from classical affine algebraic geometry to Spec and affine schemes.

5.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 5.33 (Axes)**.** Find the irreducible components of $V(xy)$.

Hint. Use irreducibility, prime ideals, and component decompositions. □

Solution. $V(x)$ and $V(y)$. □

Exercise 5.34 (Domain test)**.** Is $V(y - x^2)$ irreducible?

Hint. Use irreducibility, prime ideals, and component decompositions. □

Solution. Yes, its coordinate ring is a domain. □

Exercise 5.35 (Product polynomial)**.** What components are visible in $V(x(x-1))$?

Hint. Use irreducibility, prime ideals, and component decompositions. □

Solution. The two points or hyperplanes $V(x)$ and $V(x-1)$ depending on ambient dimension. □

Exercise 5.36 (Prime ideal)**.** What ring property corresponds to irreducibility of $V(P)$?

Hint. Use irreducibility, prime ideals, and component decompositions. □

Solution. Primality of P in the reduced affine setting. □

Exercise 5.37 (Point)**.** Is a closed point irreducible?

Hint. Use irreducibility, prime ideals, and component decompositions. □

Solution. Yes. □

Proofs

Exercise 5.38 (Domain criterion)**.** Prove: Prove an affine algebraic set is irreducible iff its coordinate ring is a domain.

Hint. Work directly from irreducibility, prime ideals, and component decompositions. ☐

Solution. Translate a decomposition into a product of vanishing functions and use the prime-ideal criterion. ☐

Exercise 5.39 (Closure irreducibility)**.** Prove: Prove the closure of an irreducible subset is irreducible.

Hint. Work directly from irreducibility, prime ideals, and component decompositions. ☐

Solution. If the closure lies in a union of two closed sets, intersect with the dense irreducible subset and use irreducibility there. ☐

Exercise 5.40 (Image irreducibility)**.** Prove: Prove the continuous image of an irreducible space is irreducible.

Hint. Work directly from irreducibility, prime ideals, and component decompositions. ☐

Solution. Pull a proposed closed decomposition back to the source. ☐

Exercise 5.41 (Components maximality)**.** Prove: Prove irreducible components are maximal irreducible closed subsets.

Hint. Work directly from irreducibility, prime ideals, and component decompositions. ☐

Solution. Use the definition and maximality under inclusion; in Noetherian spaces existence follows by finite decomposition. ☐

Counterexamples and hypothesis tests

Exercise 5.42 (Union)**.** Test the claim: a union of two distinct irreducible closed sets is always irreducible.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False if neither contains the other. ☐

Exercise 5.43 (Domain reducible)**.** Test the claim: a domain can correspond to a reducible affine variety.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False for the standard affine coordinate-ring criterion. ☐

Exercise 5.44 (Unique decomposition)**.** Test the claim: irreducible-component decompositions can have arbitrary redundant components.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. Minimal irreducible components are uniquely determined in Noetherian spaces. ☐

Applications and investigations

Exercise 5.45 (Component detection)**.** How can minimal primes help find irreducible components?

Hint. Translate the question into irreducibility, prime ideals, and component decompositions. □

Solution. Each minimal prime over the defining ideal determines an irreducible component. □

Exercise 5.46 (Equation factorization)**.** When does factorization help identify components?

Hint. Translate the question into irreducibility, prime ideals, and component decompositions. □

Solution. Factors often define component hypersurfaces after passing to radicals. □

Challenge problems

Exercise 5.47 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Irreducibility and Components in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Irreducibility and Components, translates it through irreducibility, prime ideals, and component decompositions, and checks the correspondence in both algebraic and geometric language. □

Exercise 5.48 (Hypothesis challenge)**.** Choose one structural statement from Irreducibility and Components, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Part II

Prime Spectra

Chapter 6

Prime Ideals as Geometric Points

Purpose and learning goals

The first five chapters used ordinary points

$$a = (a_1, \dots, a_n) \in k^n$$

and algebraic subsets of affine space. That viewpoint is concrete, but it has an important limitation: an irreducible algebraic locus contains more information than any one ordinary coordinate point. Algebra suggests a natural way to package that information. A prime ideal behaves like the algebraic fingerprint of one irreducible geometric piece.

This chapter makes that idea precise enough to motivate the next step. We will not yet construct the full topological space $\operatorname{Spec} A$; that is the subject of VI/07. Instead we ask a simpler question:

> Why are prime ideals the right objects to promote to geometric points?

The answer comes from three correspondences:

$$\text{ordinary point} \longleftrightarrow \text{maximal ideal},$$

$$\text{irreducible closed locus} \longleftrightarrow \text{prime ideal},$$

and

$$\text{containment of loci} \longleftrightarrow \text{reverse containment of ideals}.$$

Unless stated otherwise, rings are commutative with identity. For the cleanest classical geometry dictionary we assume that k is algebraically closed. Several examples then show why prime ideals remain meaningful even when k is not algebraically closed.

After this chapter, the reader should be able to:

- test whether an ideal is prime using the quotient-domain criterion;
- distinguish prime ideals from maximal ideals;
- identify ordinary affine points with maximal ideals over an algebraically closed field;
- identify irreducible affine algebraic subsets with prime vanishing ideals;
- interpret a chain of prime ideals as a reverse chain of irreducible loci;
- use the prime correspondence for quotient rings;
- explain why minimal primes encode irreducible components;

- explain why the zero ideal is the natural "generic" point of an integral affine locus;
- recognize geometric information that is invisible if one keeps only k-rational points;
- state clearly what is postponed to $\operatorname{Spec} A$, generic points, residue fields, and base change.

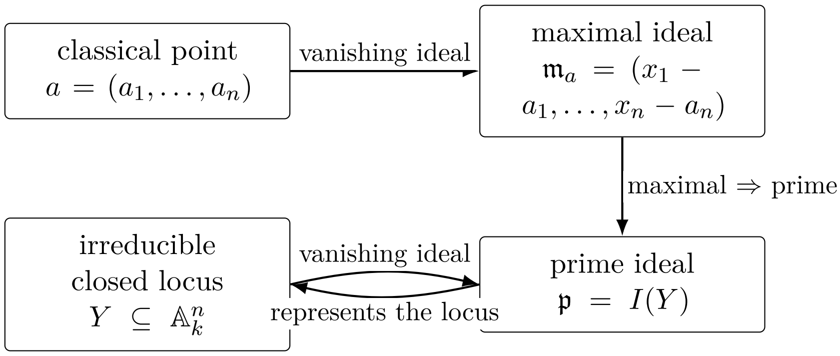


Figure 6.1: The bridge from classical affine geometry to prime ideals as geometric points. Over an algebraically closed field, closed points correspond to maximal ideals and irreducible closed loci correspond to prime ideals.

6.1 Prime ideals: the algebraic definition

Definition 6.1 (Prime ideal)**.** Let A be a ring. A proper ideal $\mathfrak{p} \subsetneq A$ is *prime* if

$$ab \in \mathfrak{p} \quad \Longrightarrow \quad a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}$$

for every $a, b \in A$.

The definition says that modulo a prime ideal, a product cannot vanish unless one factor already vanishes. This is exactly the algebraic behavior expected of an irreducible geometric locus: if an irreducible locus lies inside the union of two hypersurfaces, then it must lie inside one of them.

Theorem 6.2 (Quotient-domain criterion)**.** *For a proper ideal $\mathfrak{p} \subsetneq A$, the following are equivalent:*

1. *$\mathfrak{p}$ is prime;*
2. *$A/\mathfrak{p}$ is an integral domain.*

Proof. Suppose $\mathfrak{p}$ is prime. If

$$(a + \mathfrak{p})(b + \mathfrak{p}) = 0 + \mathfrak{p},$$

then $ab \in \mathfrak{p}$, so primality gives $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. Hence one factor is zero in the quotient.

Conversely, suppose $A/\mathfrak{p}$ is a domain and $ab \in \mathfrak{p}$. Then

$$(a + \mathfrak{p})(b + \mathfrak{p}) = 0$$

in the quotient. Since the quotient has no zero divisors, one factor is zero, so $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. $\square$

Corollary 6.3 (The zero ideal)**.** *The zero ideal $(0) \subseteq A$ is prime if and only if A is an integral domain.*

Corollary 6.4 (Maximal ideals are prime)**.** *Every maximal ideal is prime.*

Proof. If $\mathfrak{m}$ is maximal, then $A/\mathfrak{m}$ is a field, hence an integral domain. Apply theorem 6.2. ☐

The converse is false. For example,

$$(0) \subset k[t]$$

is prime because $k[t]$ is a domain, but it is not maximal because $k[t]$ is not a field.

6.2 Ordinary affine points give maximal ideals

Let

$$a = (a_1, \ldots, a_n) \in \mathbb{A}^n_k.$$

Evaluation at a defines a surjective ring homomorphism

$$\mathrm{ev}_a : k[x_1, \ldots, x_n] \longrightarrow k, \qquad f \longmapsto f(a).$$

Its kernel is

$$\mathfrak{m}_a = (x_1 - a_1, \ldots, x_n - a_n).$$

Therefore

$$k[x_1, \ldots, x_n]/\mathfrak{m}_a \cong k,$$

so $\mathfrak{m}_a$ is maximal.

Proposition 6.5 (Point ideals)**.** *Every ordinary point $a \in \mathbb{A}^n_k$ determines a maximal ideal*

$$\mathfrak{m}_a = (x_1 - a_1, \ldots, x_n - a_n).$$

Distinct points determine distinct maximal ideals.

Theorem 6.6 (Weak Nullstellensatz, geometric form)**.** *Assume k is algebraically closed. Every maximal ideal of $k[x_1, \ldots, x_n]$ is uniquely of the form*

$$(x_1 - a_1, \ldots, x_n - a_n)$$

for a point $a \in k^n$.

Thus, over an algebraically closed field,

$$\boxed{\text{ordinary affine points} \longleftrightarrow \text{maximal ideals}}.$$

Remark 6.7 (Why algebraic closedness matters)*.* Over $\mathbb{R}$, the ideal

$$(t^2 + 1) \subset \mathbb{R}[t]$$

is maximal, but it is not $(t - a)$ for any real number a. The maximal-ideal picture therefore contains closed geometric information that the set of real coordinate points alone does not see.

Example 6.8 (Closed versus generic point)**.** In $\operatorname{Spec} k[x]$, the prime (0) is the generic point: its closure is the whole spectrum. Maximal ideals such as $(x - a)$ are closed points.

6.3 Irreducible loci give prime ideals

Chapter VI/05 established the central criterion

$$Y \text{ irreducible} \iff I(Y) \text{ prime} \iff k[Y] \text{ is a domain.}$$

We now reinterpret this criterion as a statement about geometric points.

Theorem 6.9 (Prime ideals classify irreducible affine loci)**.** *Assume k is algebraically closed. The assignments*

$$Y \longmapsto I(Y) \qquad \textit{and} \qquad \mathfrak{p} \longmapsto V(\mathfrak{p})$$

give inverse correspondences between:

- *irreducible algebraic subsets $Y \subseteq \mathbb{A}^n_k$;*
- *prime ideals $\mathfrak{p} \subseteq k[x_1, \ldots, x_n]$.*

Proof. If Y is irreducible, VI/05 shows that $I(Y)$ is prime. Conversely, if $\mathfrak{p}$ is prime, then it is radical. Hilbert's Nullstellensatz gives

$$I(V(\mathfrak{p})) = \sqrt{\mathfrak{p}} = \mathfrak{p}.$$

The prime-ideal criterion then implies that $V(\mathfrak{p})$ is irreducible. The usual I-V identities show that the assignments are inverse. $\square$

This is the decisive conceptual step. A maximal ideal remembers one closed point. A nonmaximal prime ideal remembers an entire irreducible locus in a compressed algebraic form. Promoting the prime ideal itself to a point will allow that whole locus to acquire a single "generic" representative.

6.4 Containment reverses

If $I \subseteq J$, then every common zero of J is automatically a common zero of I. Hence

$$I \subseteq J \implies V(J) \subseteq V(I).$$

For prime ideals this gives a geometric hierarchy.

Example 6.10 (Plane, line, point)**.** In $k[x, y]$,

$$(0) \subset (x) \subset (x, y).$$

Assuming k algebraically closed, all three ideals are prime and

$$\mathbb{A}^2_k = V(0) \supset V(x) \supset V(x, y) = \{(0, 0)\}.$$

Thus moving upward in the ideal lattice means moving downward to a smaller geometric locus.

Proposition 6.11 (Incidence by ideal containment)**.** *Assume k algebraically closed. Let $I \subseteq k[x_1, \ldots, x_n]$ be an ideal and let $\mathfrak{p}$ be prime. Then*

$$I \subseteq \mathfrak{p} \iff V(\mathfrak{p}) \subseteq V(I).$$

ideal containment **geometric containment reverses**

$$\begin{tikzcd}
\boxed{(0)} \arrow[r, dashed] \arrow[d, "\subset"'] & \boxed{\mathbb{A}^2_k = V(0)} \arrow[d, "\supset"] \\
\boxed{(x)} \arrow[r, dashed] \arrow[d, "\subset"'] & \boxed{V(x)\text{: the } y\text{-axis}} \arrow[d, "\supset"] \\
\boxed{(x,y)} \arrow[r, dashed] & \boxed{V(x,y) = \{(0,0)\}}
\end{tikzcd}$$

Figure 6.2: A chain of prime ideals encodes a chain of irreducible loci in the opposite direction.

Proof. The forward implication is the order-reversing property of V. Conversely, if $V(\mathfrak{p}) \subseteq V(I)$, then

$$I \subseteq I(V(I)) \subseteq I(V(\mathfrak{p})) = \mathfrak{p},$$

where the last equality uses that $\mathfrak{p}$ is radical and the Nullstellensatz. □

So the innocent-looking relation

$$I \subseteq \mathfrak{p}$$

has a geometric meaning: the irreducible locus represented by $\mathfrak{p}$ lies on the closed condition defined by I.

6.5 Prime ideals in quotient rings

Closed subspaces are described algebraically by quotients. The relevant prime ideals pass through the quotient exactly as expected.

Theorem 6.12 (Prime correspondence for quotients)**.** *Let $I \subseteq A$ be an ideal and let*

$$\pi : A \twoheadrightarrow A/I$$

be the quotient map. The assignments

$$\mathfrak{p} \longmapsto \mathfrak{p}/I \qquad \textit{and} \qquad \mathfrak{q} \longmapsto \pi^{-1}(\mathfrak{q})$$

give inverse inclusion-preserving bijections between:

- *prime ideals $\mathfrak{p} \subseteq A$ containing I;*
- *prime ideals $\mathfrak{q} \subseteq A/I$.*

Proof. If $I \subseteq \mathfrak{p}$, then

$$(A/I)/(\mathfrak{p}/I) \cong A/\mathfrak{p}.$$

Thus $\mathfrak{p}/I$ is prime exactly when $\mathfrak{p}$ is prime. Conversely, the inverse image of a prime ideal under a ring homomorphism is prime. The two operations are inverse by the correspondence theorem for ideals. □

Corollary 6.13 (Prime ideals on an affine algebraic set)**.** *Let $X = V(I) \subseteq \mathbb{A}^n_k$. Prime ideals of its coordinate ring*

$$k[X] = k[x_1, \ldots, x_n]/I$$

correspond to prime ideals of $k[x_1, \ldots, x_n]$ that contain I.

This statement is the algebraic prototype of a future topological fact: the prime spectrum of a quotient will identify with the closed subset cut out by the quotient ideal.

Example 6.14 (Prime chain on the plane)**.** In $k[x, y]$, the chain $(0) \subset (x) \subset (x, y)$ corresponds to a generic point of the plane, the generic point of the y-axis, and the closed origin.

6.6 Minimal primes remember irreducible components

Suppose $X = V(I) \subseteq \mathbb{A}^n_k$ over an algebraically closed field. If

$$X = X_1 \cup \cdots \cup X_r$$

is the decomposition into irreducible components, then

$$\mathfrak{p}_i = I(X_i)$$

are prime ideals. They are exactly the prime ideals minimal among those containing $I(X)$.

Theorem 6.15 (Components and minimal primes)**.** *Let $X \subseteq \mathbb{A}^n_k$ be algebraic over an algebraically closed field. The irreducible components of X correspond bijectively to the minimal prime ideals over $I(X)$.*

Proof. By VI/05, every irreducible component X_i has prime vanishing ideal $\mathfrak{p}_i = I(X_i)$, and

$$I(X) \subseteq \mathfrak{p}_i.$$

If $I(X) \subseteq \mathfrak{q} \subsetneq \mathfrak{p}_i$ with $\mathfrak{q}$ prime, then

$$X_i = V(\mathfrak{p}_i) \subsetneq V(\mathfrak{q}) \subseteq X,$$

contradicting maximality of X_i as an irreducible subset. Thus $\mathfrak{p}_i$ is minimal over $I(X)$.

Conversely, if $\mathfrak{p}$ is minimal over $I(X)$, then $V(\mathfrak{p})$ is an irreducible subset of X. It lies in some irreducible component X_i, so

$$I(X) \subseteq I(X_i) \subseteq \mathfrak{p}.$$

Minimality forces $I(X_i) = \mathfrak{p}$. □

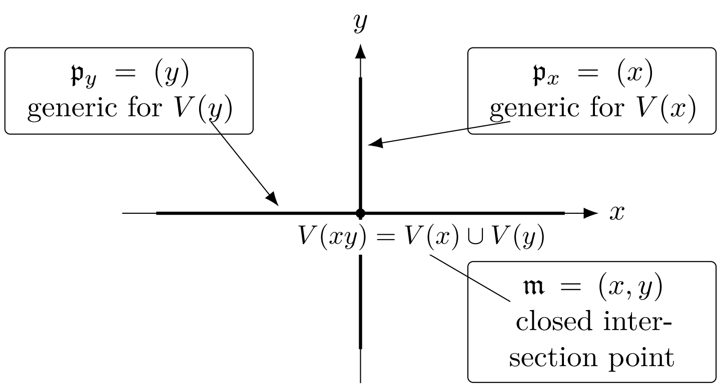


Figure 6.3: In the crossing axes, the two minimal prime ideals encode the two irreducible components, while the maximal ideal (x, y) encodes their ordinary intersection point.

6.7 The generic-point idea

A prime ideal does not merely describe equations. It can be regarded as a placeholder for the whole irreducible locus it defines.

Consider a domain A. The zero ideal is prime. In the future spectrum $\operatorname{Spec} A$, the point represented by (0) will have closure equal to the whole space. For $A = k[t]$, this point is not any coordinate value $t = a$; it represents the affine line "at a generic location," where no nonzero polynomial equation has been imposed.

More generally, a prime $\mathfrak{p}$ will become a point whose closure is the irreducible closed set represented by $\mathfrak{p}$. This language is deliberately motivational here. VI/07 defines the spectrum and its topology, while VI/09 develops generic and closed points systematically.

Remark 6.16 (Specialization preview). If

$$\mathfrak{p} \subseteq \mathfrak{q},$$

then the future point $\mathfrak{q}$ lies in the closure of the future point $\mathfrak{p}$. Thus ideal containment becomes a specialization relation between geometric points. The chain

$$(0) \subset (x) \subset (x, y)$$

should be read as

$$\text{generic plane point} \rightsquigarrow \text{generic point of a line} \rightsquigarrow \text{closed point.}$$

Example 6.17 (Nonclosed point)**.** The prime $(x) \subset k[x, y]$ is not closed in the spectrum. Its closure is $V(x)$, containing every prime ideal that contains (x).

6.8 Why rational points are not enough

The need for prime ideals is especially visible over a field that is not algebraically closed.

In $\mathbb{R}[t]$, the ideal

$$(t^2 + 1)$$

is maximal and hence prime because

$$\mathbb{R}[t]/(t^2 + 1) \cong \mathbb{C}$$

is a field. Yet the equation

$$t^2 + 1 = 0$$

has no real solution. If geometry were defined only as a subset of $\mathbb{R}$, this prime ideal would disappear. The prime-ideal viewpoint keeps it.

After extending scalars from $\mathbb{R}$ to $\mathbb{C}$,

$$t^2 + 1 = (t - i)(t + i),$$

so the single real prime splits into two complex maximal ideals. This phenomenon is a preview of geometric fibers and base change; the systematic treatment comes later.

6.9 A comparison chart for ideals and geometry

The following table separates several notions that are easy to conflate.

$\mathbb{R}[t]$
prime/maximal
ideal $(t^2 + 1)$
no real coordinate root

$\xrightarrow{\text{extend scalars } \mathbb{R} \to \mathbb{C}}$

$\mathbb{C}[t]$
$(t^2 + 1) = (t - i)(t + i)$
two maximal ideals $(t - i), (t + i)$

Prime ideals retain a geometric point that the real coordinate-tuple picture does not see.

After base change that point may split. The systematic theory belongs to VI/24.

Figure 6.4: Why prime ideals are more robust than a description using only k-rational coordinate tuples.

Ideal type	**Quotient condition**	**Classical geometric signal**	**Future spectrum signal**
maximal $\mathfrak{m}$	$A/\mathfrak{m}$ field	ordinary closed point over algebraically closed k	closed point
prime $\mathfrak{p}$	$A/\mathfrak{p}$ domain	irreducible locus	one point representing that locus
radical I	A/I reduced	reduced algebraic set, possibly reducible	closed set with no nilpotent thickening
nonradical I	A/I has nilpotents	same ordinary zero set may occur for many ideals	extra infinitesimal structure

The hierarchy is

$$\text{maximal} \Longrightarrow \text{prime} \Longrightarrow \text{radical},$$

and neither converse holds in general.

6.10 Common pitfalls

Warning 6.18 (Prime does not mean maximal)*.* The zero ideal in $k[t]$ is prime but not maximal. Prime ideals are designed to include generic irreducible information, not only closed points.

Warning 6.19 (A prime ideal is not a coordinate tuple)*.* Over a non-algebraically closed field, a maximal ideal may have no point in k^n. The ideal $(t^2 + 1) \subset \mathbb{R}[t]$ is the basic example.

Warning 6.20 (Ideal containment points the opposite way geometrically)*.* If $\mathfrak{p} \subseteq \mathfrak{q}$, then

$$V(\mathfrak{q}) \subseteq V(\mathfrak{p}).$$

Larger ideals impose more equations and therefore define smaller loci.

Warning 6.21 (Radical is weaker than prime)*.* The ideal $(xy) \subset k[x, y]$ is radical but not prime: it defines a reduced union of two axes, not one irreducible component.

Warning 6.22 (Do not import later structure too early)*.* This chapter motivates prime ideals as points. The topology of $\operatorname{Spec} A$, distinguished opens, generic/closed points, nilpotent scheme structure, local rings and residue fields are developed separately in VI/07–VI/11.

6.11 Worked examples

Example 6.23 (The zero ideal in a polynomial ring)**.** Let $A = k[t]$. Since A is a domain, (0) is prime. It is not maximal because

$$A/(0) \cong k[t]$$

is not a field. Geometrically, (0) represents the whole irreducible affine line rather than one ordinary point.

Example 6.24 (A closed point on the affine line)**.** For $a \in k$,

$$\mathfrak{m}_a = (t - a)$$

is maximal because

$$k[t]/(t - a) \cong k.$$

Thus $(t - a)$ is the algebraic avatar of the coordinate point a.

Example 6.25 (Prime ideals in a univariate polynomial ring)**.** Because $k[t]$ is a PID, every nonzero prime ideal is generated by an irreducible polynomial. Hence the prime ideals are

$$(0) \quad \text{and} \quad (f(t)) \text{ with } f \text{ irreducible.}$$

If k is algebraically closed, the irreducibles are linear, so the nonzero primes are exactly $(t - a)$.

Example 6.26 (A non-rational maximal ideal over $\mathbb{R}$)**.** The polynomial $t^2 + 1$ is irreducible in $\mathbb{R}[t]$, so

$$(t^2 + 1)$$

is maximal and prime. It is not the ideal of any real coordinate value. This is the simplest example showing that prime/maximal ideals contain geometric information beyond $\mathbb{R}$-rational points.

Example 6.27 (A chain of geometric points in the affine plane)**.** In $k[x, y]$,

$$(0) \subset (x) \subset (x, y).$$

The quotient rings are respectively

$$k[x, y], \qquad k[y], \qquad k,$$

so each ideal is prime and the last is maximal. The corresponding loci are the plane, the y-axis, and the origin.

Example 6.28 (A reducible equation is not prime)**.** The ideal $(xy) \subset k[x, y]$ is not prime because

$$xy \in (xy), \qquad x \notin (xy), \qquad y \notin (xy).$$

Equivalently,

$$k[x, y]/(xy)$$

has zero divisors. Geometrically,

$$V(xy) = V(x) \cup V(y)$$

is reducible.

Example 6.29 (The cusp is irreducible)**.** Assume char $k \neq 2, 3$. The ideal

$$(y^2 - x^3) \subset k[x, y]$$

is prime because the polynomial $y^2 - x^3$ is irreducible. Hence the cusp

$$V(y^2 - x^3)$$

is irreducible, even though it is singular at the origin. Irreducibility and smoothness are different properties.

Example 6.30 (Prime ideals on the crossing axes)**.** Let

$$A = k[x, y]/(xy).$$

The ideals

$$(\bar{x}), \qquad (\bar{y})$$

are prime because

$$A/(\bar{x}) \cong k[y], \qquad A/(\bar{y}) \cong k[x].$$

They encode the two irreducible components. The ideal

$$(\bar{x}, \bar{y})$$

is maximal and encodes the intersection point.

Example 6.31 (Primes above a quotient ideal)**.** Take $I = (xy) \subset k[x, y]$. Any prime ideal containing I must contain x or y. Thus primes of

$$k[x, y]/(xy)$$

come from primes in the plane lying on one of the two axes. This is the quotient-prime correspondence in geometric form.

Example 6.32 (Arithmetic incidence)**.** In $\mathbb{Z}$,

$$(6) \subset (2) \qquad \text{and} \qquad (6) \subset (3).$$

The prime ideals containing (6) are therefore (2) and (3). This arithmetic example anticipates the rule that the closed condition defined by an ideal consists of the prime ideals that contain it.

6.12 Exercises

Exercise 6.33. Prove directly that an ideal $\mathfrak{p} \subsetneq A$ is prime if and only if $A/\mathfrak{p}$ is an integral domain.

Exercise 6.34. Prove that every maximal ideal is prime. Give an example of a prime ideal that is not maximal.

Exercise 6.35. Show that (0) is prime in A if and only if A is an integral domain. Apply this to $\mathbb{Z}$, $k[t]$, and $k[x, y]/(xy)$.

Exercise 6.36. For $a = (a_1, \dots, a_n) \in k^n$, compute the kernel of evaluation at a and prove that it is maximal.

Exercise 6.37. Classify the prime ideals of $k[t]$. Specialize your answer when k is algebraically closed.

Exercise 6.38. List the prime ideals of $\mathbb{R}[t]$ by type. Explain why $(t^2 + 1)$ is prime and maximal although it is not the ideal of a real coordinate point.

Exercise 6.39. Verify that

$$(0) \subset (x) \subset (x, y)$$

is a chain of prime ideals in $k[x, y]$. Compute the corresponding algebraic loci and check that their containments reverse.

Exercise 6.40. Assume k is algebraically closed. For prime ideals $\mathfrak{p}, \mathfrak{q}$, prove

$$\mathfrak{p} \subseteq \mathfrak{q} \quad \Longleftrightarrow \quad V(\mathfrak{q}) \subseteq V(\mathfrak{p}).$$

Exercise 6.41. Let $\mathfrak{p} \subset k[x, y]$ be prime and suppose $(xy) \subseteq \mathfrak{p}$. Prove that $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Interpret this geometrically.

Exercise 6.42. Prove the prime correspondence between primes of A/I and primes of A containing I, including preservation of inclusions.

Exercise 6.43. Assume k is algebraically closed. Prove that an algebraic subset $Y \subseteq \mathbb{A}^n_k$ is irreducible if and only if $I(Y)$ is prime.

Exercise 6.44. Assume k algebraically closed and use the weak Nullstellensatz to show that the maximal ideals of $k[x, y]$ are exactly $(x - a, y - b)$.

Exercise 6.45. Determine all prime ideals of $\mathbb{Z}$ that contain (6), (30), and (0). What arithmetic information is the containment relation recording?

Exercise 6.46. Show that $(xy) \subset k[x, y]$ is radical but not prime. Compare $k[x, y]/(xy)$ with a domain quotient.

Exercise 6.47. In $A = k[x, y]/(xy)$, prove that $(\bar{x})$ and $(\bar{y})$ are the minimal prime ideals. Relate them to the irreducible components of the crossing axes.

Exercise 6.48. Let $\varphi : A \to B$ be a unital ring homomorphism and $\mathfrak{q} \subset B$ prime. Prove that $\varphi^{-1}(\mathfrak{q})$ is prime.

Exercise 6.49. Let $\mathfrak{p}$ be prime and let $I, J \subseteq A$ be ideals with $IJ \subseteq \mathfrak{p}$. Prove that $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$.

Exercise 6.50. Show that $(t^2 + 1)$ is prime in $\mathbb{R}[t]$ but that its extension to $\mathbb{C}[t]$ is not prime. Factor the polynomial and identify the two resulting maximal ideals.

6.13 Problems and Challenges

Problem VI.06.P1 — Ideal containment as geometric incidence

Problem. Let A be a ring, $I \subseteq A$ an ideal, and $\mathfrak{p} \subseteq A$ a prime ideal.

1. Prove that if $I = (f_1, \ldots, f_r)$, then

$$I \subseteq \mathfrak{p} \quad \Longleftrightarrow \quad f_1, \ldots, f_r \in \mathfrak{p}.$$

2. Compute the prime ideals containing $(6) \subset \mathbb{Z}$.

3. In $k[x, y]$, show that both (x) and (y) contain (xy), and explain why every prime containing (xy) contains one of them.

4. Over algebraically closed k, interpret $I \subseteq \mathfrak{p}$ as incidence of the irreducible locus $V(\mathfrak{p})$ with the closed condition $V(I)$.

Problem VI.06.P2 — The zero ideal as a generic geometric point

Problem. Let k be a field.

1. Prove that $(0) \subset k[x]$ is prime but not maximal.
2. Show that every nonzero prime ideal of $k[x]$ is maximal.
3. If k is algebraically closed, identify the ordinary closed point corresponding to each nonzero prime.
4. Explain why (0) should represent the whole affine line generically rather than one coordinate point.

Problem VI.06.P3 — Maximal ideals and classical points

Problem. Assume k is algebraically closed. Using the weak Nullstellensatz, prove that the map

$$a = (a_1, \dots, a_n) \longmapsto \mathfrak{m}_a = (x_1 - a_1, \dots, x_n - a_n)$$

is a bijection from $\mathbb{A}^n_k$ to the maximal ideals of $k[x_1, \dots, x_n]$. Then explain why this bijection fails if one replaces an algebraically closed field by $\mathbb{R}$.

Problem VI.06.P4 — Prime correspondence for a closed algebraic subset

Problem. Let $A = k[x, y]$ and $I = (xy)$. Set $B = A/I$.

1. Prove that primes of B correspond to primes of A containing (xy).
2. Show that every such prime contains x or y.
3. Identify the two minimal primes of B.
4. Assuming k algebraically closed, interpret the result geometrically on the crossing axes.

Problem VI.06.P5 — A hierarchy of loci in three-space

Problem. In $A = k[x, y, z]$, consider

$$(0) \subset (x) \subset (x, y) \subset (x, y, z).$$

1. Prove that each ideal is prime and that the final one is maximal.
2. Assuming k algebraically closed, identify all four zero loci.
3. Explain how this chain previews specialization and dimension.

Problem VI.06.P6 — One real prime that becomes two complex points

Problem. Let

$$A = \mathbb{R}[t], \qquad \mathfrak{p} = (t^2 + 1).$$

1. Prove that $\mathfrak{p}$ is maximal.
2. Show that $V_{\mathbb{R}}(t^2 + 1) = \varnothing$ as a classical real zero set.
3. Extend scalars to $\mathbb{C}$ and prove that the extended ideal is no longer prime.
4. Identify the two complex maximal ideals that appear.

Problem VI.06.P7 — Pulling prime points backward along a ring map

Problem. Let $\varphi : A \to B$ be a unital ring homomorphism and let $\mathfrak{q} \subset B$ be prime.

1. Prove that $\varphi^{-1}(\mathfrak{q})$ is prime in A.
2. Apply this to the map $k[x, y] \to k[t]$ given by $x \mapsto t$, $y \mapsto t^2$ and the prime $(t - a) \subset k[t]$.
3. Compute the inverse image and interpret the corresponding point on the parabola.

Problem VI.06.P8 — Radical versus prime: one locus or several?

Problem. Let k be a field and

$$I = (xy) \subset k[x, y].$$

1. Prove that $I = (x) \cap (y)$.
2. Deduce that I is radical.
3. Prove that I is not prime.
4. Explain geometrically why radicality corresponds to "no nilpotent thickening" while primality corresponds to "one irreducible component."

Challenge VI.06.C1 — The radical as an intersection of prime ideals

Problem. Let A be a commutative ring and $I \subseteq A$ an ideal. Prove the theorem

$$\sqrt{I} = \bigcap_{\substack{\mathfrak{p} \supseteq I \\ \mathfrak{p} \text{ prime}}} \mathfrak{p}.$$

You may use Zorn's lemma.

Challenge VI.06.C2 — Arbitrarily long prime chains in polynomial rings

Problem. For $r \geq 1$, prove that

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \ldots, x_r)$$

is a strict chain of prime ideals in $k[x_1, \ldots, x_r]$. Interpret the corresponding chain of irreducible affine loci and explain why this suggests a numerical notion of dimension.

Challenge VI.06.C3 — When does a prime stay prime after extending the field?

Problem. Let $k \subseteq K$ be a field extension and let $f \in k[t]$ be irreducible. The ideal $(f) \subset k[t]$ is prime. Investigate the extended ideal $(f) \subset K[t]$:

1. prove that it is prime exactly when f remains irreducible in $K[t]$;
2. describe what happens when f factors into distinct irreducibles over K;
3. analyze $f = t^2 + 1$ for $\mathbb{R} \subset \mathbb{C}$.

6.14 Chapter summary

Prime ideals are the correct bridge from classical affine geometry to spectra because they encode irreducible geometric information. Over an algebraically closed field,

$$\text{closed point} \longleftrightarrow \text{maximal ideal}$$

and

$$\text{irreducible closed locus} \longleftrightarrow \text{prime ideal.}$$

The quotient criterion

$$\mathfrak{p} \text{ prime} \quad \Longleftrightarrow \quad A/\mathfrak{p} \text{ domain}$$

explains why primality is the algebraic form of irreducibility. The quotient correspondence explains how prime ideals survive passage to closed algebraic subsets, while minimal primes record irreducible components.

Finally, a nonmaximal prime ideal should be imagined as one point representing an entire irreducible locus. VI/07 makes that imagination literal by defining

$$\operatorname{Spec} A$$

as a topological space whose points are precisely the prime ideals of A.

6.15 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 6.51 (Closure)**.** What is the closure of a prime $\mathfrak{p}$ in $\operatorname{Spec} A$?

Hint. Use prime ideals and generic geometric points. ☐

Solution. $V(\mathfrak{p})$. ☐

Exercise 6.52 (Closed point)**.** When is a point $\mathfrak{p}$ closed?

Hint. Use prime ideals and generic geometric points. ☐

Solution. When $\mathfrak{p}$ is maximal. ☐

Exercise 6.53 (Generic point)**.** What is the generic point of $\operatorname{Spec} A$ when A is a domain?

Hint. Use prime ideals and generic geometric points. ☐

Solution. The zero prime. ☐

Exercise 6.54 (Chain)**.** Give a prime chain in $k[x, y]$.

Hint. Use prime ideals and generic geometric points. ☐

Solution. $(0) \subset (x) \subset (x, y)$. ☐

Exercise 6.55 (Specialization)**.** When is $\mathfrak{q}$ a specialization of $\mathfrak{p}$?

Hint. Use prime ideals and generic geometric points. ☐

Solution. When $\mathfrak{p} \subseteq \mathfrak{q}$. ☐

Proofs

Exercise 6.56 (Closure formula)**.** Prove: Prove $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$.

Hint. Work directly from prime ideals and generic geometric points. ☐

Solution. A closed set $V(I)$ contains $\mathfrak{p}$ iff $I \subseteq \mathfrak{p}$, and $V(\mathfrak{p})$ is the smallest such closed set. ☐

Exercise 6.57 (Closed maximal)**.** Prove: Prove a point of $\operatorname{Spec} A$ is closed iff the prime is maximal.

Hint. Work directly from prime ideals and generic geometric points. ☐

Solution. $V(\mathfrak{p}) = \{\mathfrak{p}\}$ exactly when no strictly larger prime contains it. ☐

Exercise 6.58 (Generic domain)**.** Prove: Prove (0) is generic in the spectrum of a domain.

Hint. Work directly from prime ideals and generic geometric points. ☐

Solution. The closure of (0) is $V(0) = \operatorname{Spec} A$. ☐

Exercise 6.59 (Specialization order)**.** Prove: Prove specialization of points agrees with inclusion of prime ideals.

Hint. Work directly from prime ideals and generic geometric points. ☐

Solution. Use the closure formula. ☐

Counterexamples and hypothesis tests

Exercise 6.60 (All closed)**.** Test the claim: every prime-spectrum point is closed.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; nonmaximal primes are nonclosed. ☐

Exercise 6.61 (Geometric point)**.** Test the claim: only maximal ideals carry geometric meaning.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; nonmaximal primes encode generic points of irreducible closed subsets. ☐

Exercise 6.62 (Reverse order)**.** Test the claim: specialization reverses ideal inclusion.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; $\mathfrak{p}$ specializes to primes containing it. ☐

Applications and investigations

Exercise 6.63 (Generic behavior)**.** What does a generic point encode?

Hint. Translate the question into prime ideals and generic geometric points. ☐

Solution. It represents an entire irreducible closed subset through its closure. ☐

Exercise 6.64 (Nested geometry)**.** How do prime chains encode nested subvarieties?

Hint. Translate the question into prime ideals and generic geometric points. ☐

Solution. Inclusions of primes correspond to specializations through nested closed sets. ☐

Challenge problems

Exercise 6.65 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Prime Ideals as Geometric Points in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Prime Ideals as Geometric Points, translates it through prime ideals and generic geometric points, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 6.66 (Hypothesis challenge)**.** Choose one structural statement from Prime Ideals as Geometric Points, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 7

The Spectrum of a Ring

Purpose and learning goals

VI/06 explained why prime ideals behave like generalized geometric points. We now place *all* prime ideals of a ring into a single topological space:

$$\operatorname{Spec} A = \{\mathfrak{p} \subsetneq A : \mathfrak{p} \text{ prime}\}.$$

This is the decisive move from classical affine algebraic sets to scheme-theoretic geometry. The topology is built directly from ideal containment, so algebraic inclusions become geometric closure relations.

This chapter develops the topology of $\operatorname{Spec} A$ itself. VI/08 will refine the open sets through the distinguished basis $D(f)$ and localization; VI/09 will study generic and closed points systematically; VI/10 will separate topological information from nilpotent scheme structure; VI/11 will attach local rings and residue fields.

After this chapter, the reader should be able to:

- construct $\operatorname{Spec} A$ from the prime ideals of a ring;
- prove that the sets $V(I)$ define the Zariski topology;
- manipulate closed sets through sums, products, intersections, and radicals of ideals;
- compute closures of points and arbitrary subsets of a spectrum;
- read prime inclusion as specialization;
- recognize quotient spectra as closed subspaces;
- analyze standard spectra such as $\operatorname{Spec} \mathbb{Z}$, $\operatorname{Spec} k[t]$, $\operatorname{Spec}(k[x,y]/(xy))$, and $\operatorname{Spec}(A \times B)$;
- construct the continuous map on spectra induced contravariantly by a ring homomorphism;
- compute the closure of the image of a spectrum map.

7.1 The underlying set of the spectrum

Definition 7.1 (Prime spectrum)**.** For a commutative ring A with identity,

$$\operatorname{Spec} A = \{\mathfrak{p} \subsetneq A : \mathfrak{p} \text{ is a prime ideal}\}.$$

The zero ring has empty spectrum, because it has no proper ideals. If K is a field, then

$$\operatorname{Spec} K = \{(0)\}.$$

Thus even the simplest rings already produce legitimate topological spaces.

Remark 7.2. A point of $\operatorname{Spec} A$ is literally a prime ideal. The geometric meaning developed in VI/06 is not metaphorical notation: the prime itself is the point.

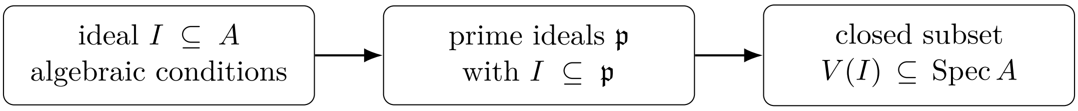
7.2 Closed sets from ideals

Definition 7.3 (Zariski closed set)**.** For an ideal $I \subseteq A$, define

$$V_A(I) = V(I) = \{\mathfrak{p} \in \operatorname{Spec} A : I \subseteq \mathfrak{p}\}.$$

The containment condition is the entire mechanism:

$$\mathfrak{p} \in V(I) \quad \Longleftrightarrow \quad I \subseteq \mathfrak{p}.$$



Theorem 7.4 (Closed-set calculus)**.** *For ideals I, J and a family $\{I_\lambda\}_{\lambda\in\Lambda}$,*

$$V((0)) = \operatorname{Spec} A, \qquad V(A) = \varnothing,$$

$$V(I) \cup V(J) = V(IJ) = V(I \cap J),$$

and

$$\bigcap_{\lambda\in\Lambda} V(I_\lambda) = V\left(\sum_{\lambda\in\Lambda} I_\lambda\right).$$

Therefore the subsets $V(I)$ are exactly the closed sets of a topology on $\operatorname{Spec} A$*.*

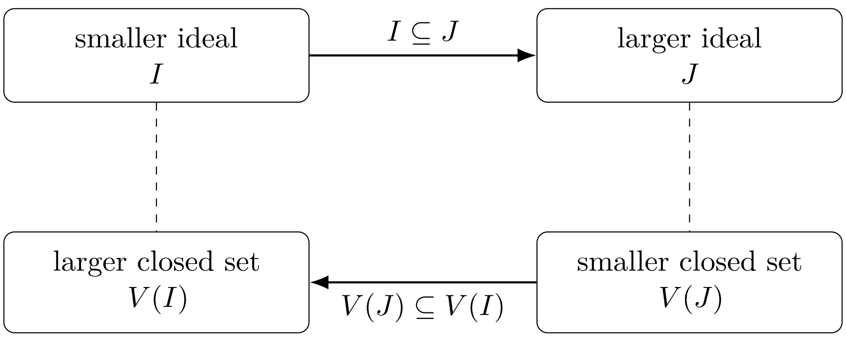
Proof. A prime contains IJ iff it contains I or J, which gives the finite-union formula. A prime contains every I_λ iff it contains their sum, which gives arbitrary intersections. The two extreme cases are immediate. □

7.3 Reversal of inclusion

If $I \subseteq J$, then every prime containing J also contains I, hence

$$I \subseteq J \quad \Longrightarrow \quad V(J) \subseteq V(I).$$

Algebra and geometry reverse order.



This is the spectrum version of a principle already visible in affine algebraic sets: adding equations makes the geometric locus smaller.

Example 7.5 (Spectrum of the integers)**.** The points of $\operatorname{Spec}\mathbb{Z}$ are (0) and the prime ideals (p). Each (p) is closed, while (0) is generic and specializes to every (p).

7.4 Radicals are the topological information

Proposition 7.6. *For every ideal I,*

$$V(I) = V(\sqrt{I}).$$

Moreover,

$$V(I) = V(J) \iff \sqrt{I} = \sqrt{J}.$$

Proof. Every prime ideal is radical, so a prime contains I iff it contains $\sqrt{I}$. For the converse statement use

$$\sqrt{I} = \bigcap_{\mathfrak{p} \supseteq I} \mathfrak{p}.$$

□

Thus the topology remembers radical ideals, not multiplicities. The deeper consequence that nilpotent thickenings can have the same underlying topological spectrum is developed in VI/10.

7.5 Closure of a point and specialization

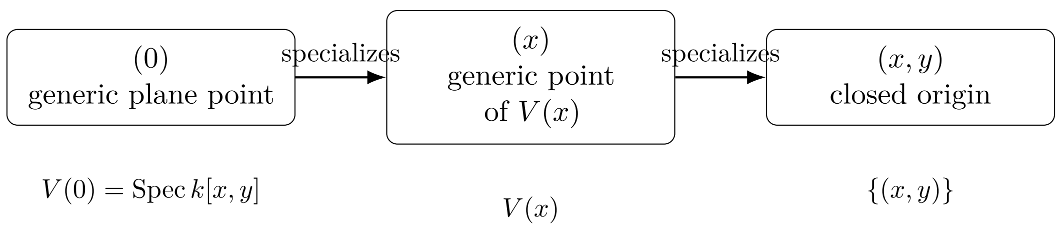
For a point $\mathfrak{p} \in \operatorname{Spec} A$,

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Theorem 7.7 (Specialization order)**.** *For $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} A$,*

$$\mathfrak{q} \in \overline{\{\mathfrak{p}\}} \iff \mathfrak{p} \subseteq \mathfrak{q}.$$

When this holds, $\mathfrak{q}$ is called a specialization *of $\mathfrak{p}$ and $\mathfrak{p}$ a* generalization *of $\mathfrak{q}$.*



A larger prime ideal represents a more specialized point. For instance,

$$(0) \subsetneq (x) \subsetneq (x, y)$$

in $k[x, y]$ becomes a specialization chain from the generic plane condition, to the $x = 0$ line condition, to the origin.

Corollary 7.8. *Every spectrum is a T_0 topological space.*

Proof. Distinct prime ideals have distinct closures because $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ and prime ideals are radical. □

A spectrum is usually not T_1: a point is closed exactly when its prime ideal is maximal. VI/09 develops that distinction in detail.

7.6 Closure of an arbitrary subset

For $S \subseteq \operatorname{Spec} A$, define
$$I(S) = \bigcap_{\mathfrak{p} \in S} \mathfrak{p}.$$

Theorem 7.9 (Closure formula)**.**
$$\overline{S} = V(I(S)).$$

Proof. A closed set $V(J)$ contains S iff $J \subseteq \mathfrak{p}$ for every $\mathfrak{p} \in S$, equivalently $J \subseteq I(S)$. By inclusion reversal, the smallest such closed set is $V(I(S))$. □

This formula is a useful bridge between topology and intersections of prime ideals.

Example 7.10 (Spectrum of a product)**.** For rings A, B, $\operatorname{Spec}(A \times B)$ is the disjoint union $\operatorname{Spec} A \sqcup \operatorname{Spec} B$. Prime ideals are exactly $P \times B$ or $A \times Q$.

7.7 Irreducible closed subsets

Proposition 7.11. *A closed set $V(I)$ is irreducible iff $\sqrt{I}$ is prime.*

Proof. Since $V(I) = V(\sqrt{I})$, reduce to a radical ideal. If $\sqrt{I} = \mathfrak{p}$ is prime, then a decomposition
$$V(\mathfrak{p}) = V(J) \cup V(K) = V(JK)$$
forces $JK \subseteq \mathfrak{p}$, hence $J \subseteq \mathfrak{p}$ or $K \subseteq \mathfrak{p}$, so one closed set equals $V(\mathfrak{p})$. The converse is the standard prime-ideal criterion for irreducibility. □

This recovers, now intrinsically inside a spectrum, the prime/irreducible dictionary of VI/06.

7.8 Quotients give closed subspaces

Let $\pi : A \twoheadrightarrow A/I$ be the quotient map. Prime correspondence gives
$$\operatorname{Spec}(A/I) \longleftrightarrow \{\mathfrak{p} \in \operatorname{Spec} A : I \subseteq \mathfrak{p}\} = V_A(I).$$

Theorem 7.12 (Closed-subspace theorem for quotients)**.** *The correspondence*
$$\mathfrak{q} \longmapsto \pi^{-1}(\mathfrak{q})$$
is a homeomorphism
$$\operatorname{Spec}(A/I) \cong V_A(I)$$
where $V_A(I)$ carries the subspace topology.

$$A \xrightarrow{\quad\pi\quad} A/I$$

$$\begin{array}{ccc} \operatorname{Spec} A & \xleftarrow{\ \pi^*\ } & \operatorname{Spec}(A/I) \\ & & \cong \\ & & V_A(I) \subseteq \operatorname{Spec} A \end{array}$$

7.9 Standard spectra

Several spectra should become mental reference models.

7.9.1 Spec $k[t]$

If k is algebraically closed,

$$\operatorname{Spec} k[t] = \{(0)\} \cup \{(t-a) : a \in k\}.$$

The zero prime specializes to every closed point. Over a nonclosed field, irreducible polynomials of higher degree contribute additional closed points.

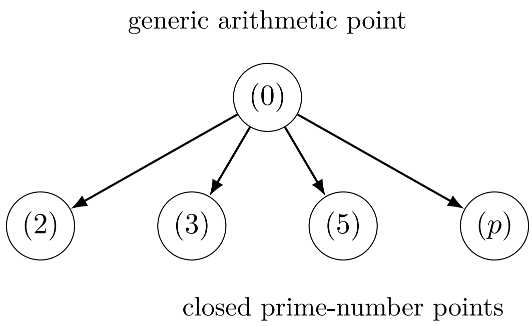
7.9.2 Spec $\mathbb{Z}$

The points are

$$(0), \quad (2), \quad (3), \quad (5), \ldots .$$

The point (0) specializes to every prime-number point. For $n \neq 0$,

$$V(n) = \{(p) : p \mid n\}.$$

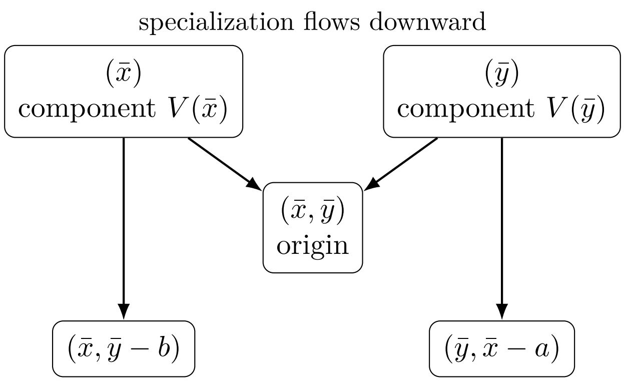


7.9.3 The crossing axes

For

$$R = k[x,y]/(xy),$$

the minimal primes $(\bar{x})$ and $(\bar{y})$ are the two component points. Closed points on the axes specialize from one of those component primes, while the origin $(\bar{x}, \bar{y})$ lies on both components.



Example 7.13 (Dual numbers)**.** For $A = k[\varepsilon]/(\varepsilon^2)$, the only prime is (ε). Thus the underlying spectrum has one point, but the ring is nonreduced and contains nilpotent structure invisible topologically.

7.10 Contravariance: ring maps reverse direction

Let $\varphi : A \to B$ be a ring homomorphism.

Proposition 7.14. *If $\mathfrak{q} \in \operatorname{Spec} B$, then $\varphi^{-1}(\mathfrak{q}) \in \operatorname{Spec} A$. Hence*

$$\varphi^* : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A, \qquad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q})$$

is well defined.

Theorem 7.15 (Continuity of the induced map)**.** *For every ideal $I \subseteq A$,*

$$(\varphi^*)^{-1}(V_A(I)) = V_B(IB),$$

where IB denotes the ideal generated by $\varphi(I)$ in B. Therefore φ^* is continuous.

Proof. A prime $\mathfrak{q}$ lies in the left side iff $I \subseteq \varphi^{-1}(\mathfrak{q})$, equivalently $\varphi(I) \subseteq \mathfrak{q}$, equivalently $IB \subseteq \mathfrak{q}$. ☐

$$A \xrightarrow{\quad\varphi\quad} B$$

$$\operatorname{Spec} A \xleftarrow{\quad\varphi^*\quad} \operatorname{Spec} B$$

$$\varphi^*(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}).$$

Identity maps and compositions behave as expected:

$$(\mathrm{id}_A)^* = \mathrm{id}_{\operatorname{Spec} A}, \qquad (\psi \circ \varphi)^* = \varphi^* \circ \psi^*.$$

Thus Spec reverses arrows.

7.11 The closure of the image of a spectrum map

Theorem 7.16 (Image-closure theorem)**.** *For $\varphi : A \to B$,*

$$\overline{\varphi^*(\operatorname{Spec} B)} = V_A(\ker \varphi).$$

Proof. The ideal of the image is

$$\bigcap_{\mathfrak{q} \in \operatorname{Spec} B} \varphi^{-1}(\mathfrak{q}) = \varphi^{-1}\Big(\sqrt{(0)_B}\Big).$$

An element $a \in A$ lies in this preimage exactly when $\varphi(a)$ is nilpotent, equivalently $a^n \in \ker \varphi$ for some n. Hence the ideal is $\sqrt{\ker \varphi}$, and the closure formula gives

$$V_A(\sqrt{\ker \varphi}) = V_A(\ker \varphi).$$

☐

Corollary 7.17. *If φ is injective, then $\varphi^*(\operatorname{Spec} B)$ is dense in* $\operatorname{Spec} A$*. If φ is surjective, φ^* is a homeomorphism onto the closed set $V(\ker \varphi)$.*

7.12 Comparison chart

Ring	**Spectrum points**	**Topological picture**
field K	(0)	one closed point
$k[t]$, k algebraically closed	(0) and $(t-a)$	one generic direction specializing to all ordinary points
$\mathbb{Z}$	(0) and (p)	arithmetic generic point plus one closed point for each prime number
$k[x,y]/(xy)$	two minimal-prime families meeting at closed points	two irreducible components with shared specializations
$A \times B$	primes from A or from B	disjoint union $\operatorname{Spec} A \sqcup \operatorname{Spec} B$
$k[\varepsilon]/(\varepsilon^2)$	(ε)	one point; topology cannot see the nilpotent thickening

7.13 Worked examples

Example 7.18 (The spectrum of a field)**.** If K is a field, its only proper ideal is (0), and that ideal is prime. Hence

$$\operatorname{Spec} K = \{(0)\}.$$

The unique point is closed.

Example 7.19 (The empty spectrum)**.** The zero ring has no proper ideals, hence

$$\operatorname{Spec}(0) = \varnothing.$$

This edge case makes $V(A) = \varnothing$ consistent with the topology.

Example 7.20 (The affine line over an algebraically closed field)**.** If k is algebraically closed,

$$\operatorname{Spec} k[t] = \{(0)\} \cup \{(t-a) : a \in k\}.$$

The closure of (0) is the whole spectrum, while each $(t-a)$ is closed.

Example 7.21 (The real affine line has extra closed points)**.** In $\mathbb{R}[t]$, the nonzero prime ideals are generated by irreducible polynomials: linear factors and irreducible quadratics. Thus (t^2+1) is a closed point of $\operatorname{Spec} \mathbb{R}[t]$ even though it is not an $\mathbb{R}$-rational point of the classical line.

Example 7.22 (Finite-field affine line)**.** The points of $\operatorname{Spec} \mathbb{F}_q[t]$ are (0) and (f) for monic irreducible polynomials f. Only degree-one irreducibles correspond to $\mathbb{F}_q$-rational points; higher-degree primes record finite field extensions.

Example 7.23 (Arithmetic line)**.** The points of $\operatorname{Spec} \mathbb{Z}$ are

$$(0), (2), (3), (5), \ldots.$$

For $n \neq 0$,

$$V(n) = \{(p) : p \mid n\}.$$

Thus arithmetic divisibility appears directly as Zariski closed geometry.

Example 7.24 (A finite arithmetic spectrum)**.** Prime ideals of $\mathbb{Z}/12\mathbb{Z}$ correspond to prime ideals of $\mathbb{Z}$ containing (12). Hence

$$\operatorname{Spec}(\mathbb{Z}/12\mathbb{Z}) = \{(2)/(12), (3)/(12)\}.$$

Both points are closed; a finite T_1 space is discrete.

Example 7.25 (Two crossing components)**.** Let $R = k[x, y]/(xy)$. The primes $(\bar{x})$ and $(\bar{y})$ are minimal. The maximal ideal $(\bar{x}, \bar{y})$ contains both, so the origin is a specialization of both component points.

Example 7.26 (A product ring)**.** Every prime of $A \times B$ is either $\mathfrak{p} \times B$ or $A \times \mathfrak{q}$. Consequently,

$$\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B.$$

Example 7.27 (A nilpotent thickening with the same topology)**.** For $R = k[\varepsilon]/(\varepsilon^2)$, the only prime ideal is $(\bar{\varepsilon})$. Thus $\operatorname{Spec} R$ is a one-point space, just like $\operatorname{Spec} k$. VI/10 explains why these schemes are nevertheless different.

Example 7.28 (A quotient as a closed subspace)**.** Let

$$A = k[x, y], \qquad I = (y - x^2).$$

Then

$$\operatorname{Spec}(A/I) \cong V_A(y - x^2).$$

The spectrum of the coordinate ring of the parabola is literally the corresponding closed subspace of $\operatorname{Spec} A$.

Example 7.29 (Three specialization levels in the affine plane)**.** For algebraically closed k,

$$(0) \subsetneq (f) \subsetneq (x - a, y - b)$$

can occur when f is irreducible and $f(a, b) = 0$. Topologically this is

$$\text{plane generic point} \rightsquigarrow \text{curve generic point} \rightsquigarrow \text{closed point}.$$

Example 7.30 (A map induced by squaring)**.** The homomorphism

$$k[t] \longrightarrow k[x], \qquad t \longmapsto x^2$$

induces

$$\operatorname{Spec} k[x] \longrightarrow \operatorname{Spec} k[t].$$

A closed point $(x - a)$ contracts to $(t - a^2)$, while (0) contracts to (0).

Example 7.31 (The generic arithmetic point from a field inclusion)**.** The inclusion

$$\mathbb{Z} \hookrightarrow \mathbb{Q}$$

induces

$$\operatorname{Spec} \mathbb{Q} \longrightarrow \operatorname{Spec} \mathbb{Z}.$$

The unique point (0) of $\operatorname{Spec} \mathbb{Q}$ maps to the generic point (0) of $\operatorname{Spec} \mathbb{Z}$.

7.14 Exercises

Exercise 7.32 (VI.07.E1 - Verify the Zariski topology)**.** Prove directly that the sets $V(I)$ satisfy the closed-set axioms:

$$V((0)) = \operatorname{Spec} A, \quad V(A) = \varnothing, \quad V(I) \cup V(J) = V(IJ), \quad \bigcap_{\lambda} V(I_\lambda) = V(\sum_{\lambda} I_\lambda).$$

Exercise 7.33 (VI.07.E2 - Radical invariance)**.** Show

$$V(I) = V(\sqrt{I})$$

for every ideal $I \subseteq A$.

Exercise 7.34 (VI.07.E3 - Equality of closed sets)**.** Prove

$$V(I) = V(J) \iff \sqrt{I} = \sqrt{J}.$$

Exercise 7.35 (VI.07.E4 - Closure of one point)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, prove

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Exercise 7.36 (VI.07.E5 - Closed points)**.** Show that $\{\mathfrak{p}\}$ is closed in $\operatorname{Spec} A$ iff $\mathfrak{p}$ is maximal.

Exercise 7.37 (VI.07.E6 - The T_0 property)**.** Prove every spectrum is T_0.

Exercise 7.38 (VI.07.E7 - Spectrum of a field and of the zero ring)**.** Compute $\operatorname{Spec} K$ for a field K and $\operatorname{Spec}(0)$ for the zero ring.

Exercise 7.39 (VI.07.E8 - Legacy CA13: points of a polynomial-ring spectrum)**.** Describe all points of $\operatorname{Spec} k[t]$. State the answer first for arbitrary k, then simplify when k is algebraically closed.

Exercise 7.40 (VI.07.E9 - The real affine line)**.** Describe the points and point closures of $\operatorname{Spec} \mathbb{R}[t]$.

Exercise 7.41 (VI.07.E10 - Legacy CA13: closed sets of $\operatorname{Spec} \mathbb{Z}$)**.** Describe all Zariski closed subsets of $\operatorname{Spec} \mathbb{Z}$.

Exercise 7.42 (VI.07.E11 - Legacy CA13: $V(xy)$)**.** In $k[x, y]$, prove

$$V(xy) = V(x) \cup V(y).$$

Interpret the two sides through prime containment.

Exercise 7.43 (VI.07.E12 - Finite arithmetic spectra)**.** For $n \geq 2$, determine the points of $\operatorname{Spec}(\mathbb{Z}/n\mathbb{Z})$ and show that the underlying topological space is finite and discrete.

Exercise 7.44 (VI.07.E13 - Crossing axes)**.** For $R = k[x, y]/(xy)$, prove every prime contains $\bar{x}$ or $\bar{y}$, and describe the closures of $(\bar{x})$, $(\bar{y})$, and $(\bar{x}, \bar{y})$.

Exercise 7.45 (VI.07.E14 - Closure of a subset)**.** For $S \subseteq \operatorname{Spec} A$, prove

$$\overline{S} = V\left(\bigcap_{\mathfrak{p} \in S} \mathfrak{p}\right).$$

Exercise 7.46 (VI.07.E15 - Quotient spectrum)**.** Show that the prime correspondence gives a homeomorphism
$$\operatorname{Spec}(A/I) \cong V_A(I).$$

Exercise 7.47 (VI.07.E16 - Continuity of contraction)**.** For $\varphi : A \to B$, prove
$$(\varphi^*)^{-1}(V_A(I)) = V_B(IB).$$

Exercise 7.48 (VI.07.E17 - Surjections)**.** If $\varphi : A \twoheadrightarrow B$, show that φ^* is a homeomorphism $\operatorname{Spec} B \cong V(\ker \varphi)$.

Exercise 7.49 (VI.07.E18 - Dense image from an injection)**.** If $\varphi : A \hookrightarrow B$ is injective, prove the image of $\operatorname{Spec} B \to \operatorname{Spec} A$ is dense.

Exercise 7.50 (VI.07.E19 - Product spectra)**.** Prove
$$\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B$$
as topological spaces.

Exercise 7.51 (VI.07.E20 - Topology ignores nilpotent powers)**.** Compare
$$\operatorname{Spec} k[x]/(x) \qquad \text{and} \qquad \operatorname{Spec} k[x]/(x^m), \quad m \geq 2,$$
as topological spaces. Do not yet claim the corresponding schemes are the same.

7.15 Problems

Problem 7.52. Describe the topological spaces $\operatorname{Spec} \mathbb{R}[x]$, $\operatorname{Spec} \mathbb{C}[x, y]$, $\operatorname{Spec} \mathbb{Z}[x]$, and $\operatorname{Spec} \mathbb{F}_p[x]$.

Problem 7.53. Prove the complete closed-set calculus for $V(I)$ and identify the radical information remembered by the topology.

Problem 7.54. Compute closure and specialization relations for arbitrary subsets of $\operatorname{Spec} A$.

Problem 7.55. Determine the topology of $\operatorname{Spec} \mathbb{Z}$ and of $\operatorname{Spec}(\mathbb{Z}/n\mathbb{Z})$.

Problem 7.56. Analyze the full specialization geometry of $\operatorname{Spec}(k[x, y]/(xy))$.

Problem 7.57. Prove that $\operatorname{Spec}(A/I)$ is homeomorphic to $V(I) \subseteq \operatorname{Spec} A$.

Problem 7.58. Prove continuity and functoriality of the map on spectra induced by a ring homomorphism.

Problem 7.59. Prove the image-closure theorem $\overline{\operatorname{im} \varphi^*} = V(\ker \varphi)$.

Problem 7.60. Describe $\operatorname{Spec}(A \times B)$ as a disjoint union.

Problem 7.61. Analyze the map $\operatorname{Spec} k[x] \to \operatorname{Spec} k[t]$ induced by $t \mapsto x^2$.

Problem 7.62. Classify the prime specialization levels of $\operatorname{Spec} k[x, y]$ for algebraically closed k.

Problem 7.63. Reconstruct the radical ideal lattice from the closed-set lattice of $\operatorname{Spec} A$.

7.16 Challenges

Problem 7.64. Show every irreducible closed subset of $\operatorname{Spec} A$ has a unique generic point.

Problem 7.65. Prove $\operatorname{Spec}(A_{\mathrm{red}}) \to \operatorname{Spec} A$ is a homeomorphism.

Problem 7.66. Analyze the contraction map $\operatorname{Spec} \mathbb{C}[x] \to \operatorname{Spec} \mathbb{R}[x]$ and the behavior of conjugate closed points.

7.17 Boundary with the next chapters

The topology of $\operatorname{Spec} A$ is now in place. The next four chapters refine different aspects without changing the underlying definition:

VI/08: $D(f)$ and localization, VI/09: generic and closed points,

VI/10: reduced versus nonreduced geometry, VI/11: local rings and residue fields.

This division is important: VI/07 constructs the topological space; the later chapters add local and scheme-theoretic structure.

7.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 7.67 (Integer points)**.** List the points of $\operatorname{Spec} \mathbb{Z}$.

Hint. Use prime spectra and the Zariski topology. □

Solution. (0) and (p) for prime integers p. □

Exercise 7.68 (Field spectrum)**.** Compute $\operatorname{Spec} k$ for a field k.

Hint. Use prime spectra and the Zariski topology. □

Solution. It is the single point (0). □

Exercise 7.69 (Product spectrum)**.** Describe $\operatorname{Spec}(A \times B)$.

Hint. Use prime spectra and the Zariski topology. □

Solution. It is the disjoint union of the two spectra. □

Exercise 7.70 (Dual numbers)**.** How many points does $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ have?

Hint. Use prime spectra and the Zariski topology. □

Solution. One. □

Exercise 7.71 (Closed set)**.** What is $V(I)$ in a spectrum?

Hint. Use prime spectra and the Zariski topology. □

Solution. The primes containing I. □

Proofs

Exercise 7.72 (Product primes)**.** Prove: Prove every prime of $A \times B$ comes from one factor.

Hint. Work directly from prime spectra and the Zariski topology. ☐

Solution. Since $(1,0)(0,1) = 0$, a prime must contain one idempotent; that forces it to contain the entire opposite factor. ☐

Exercise 7.73 (Spectrum quotient)**.** Prove: Prove $\operatorname{Spec}(A/I)$ identifies with $V(I) \subseteq \operatorname{Spec} A$.

Hint. Work directly from prime spectra and the Zariski topology. ☐

Solution. Use prime correspondence under the quotient map. ☐

Exercise 7.74 (Nilradical topology)**.** Prove: Prove A and $A/\sqrt{0}$ have homeomorphic spectra.

Hint. Work directly from prime spectra and the Zariski topology. ☐

Solution. Every prime contains the nilradical, so prime ideals correspond through the quotient. ☐

Exercise 7.75 (Zariski closed)**.** Prove: Prove the subsets $V(I)$ satisfy the closed-set axioms.

Hint. Work directly from prime spectra and the Zariski topology. ☐

Solution. Use sums for intersections and products for finite unions. ☐

Counterexamples and hypothesis tests

Exercise 7.76 (Topology detects nilpotents)**.** Test the claim: the topological space $\operatorname{Spec} A$ determines whether A is reduced.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; dual numbers and a field can have the same one-point topology. ☐

Exercise 7.77 (Field many points)**.** Test the claim: a field spectrum has many maximal points.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it has one prime ideal. ☐

Exercise 7.78 (Product connected)**.** Test the claim: $\operatorname{Spec}(A \times B)$ is connected when both factors are nonzero.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is disconnected. ☐

Applications and investigations

Exercise 7.79 (Arithmetic geometry)**.** What does $\operatorname{Spec} \mathbb{Z}$ suggest geometrically?

Hint. Translate the question into prime spectra and the Zariski topology. ☐

Solution. Prime numbers behave like closed points with a generic point above them. ☐

Exercise 7.80 (Decomposition)**.** How does a ring product appear geometrically?

Hint. Translate the question into prime spectra and the Zariski topology. ☐

Solution. As a disjoint union of spectra. ☐

Challenge problems

Exercise 7.81 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Spectrum of a Ring in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Spectrum of a Ring, translates it through prime spectra and the Zariski topology, and checks the correspondence in both algebraic and geometric language. □

Exercise 7.82 (Hypothesis challenge)**.** Choose one structural statement from Spectrum of a Ring, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 8

Basic Open Sets $D(f)$

Purpose and learning goals

VI/07 constructed the Zariski topology on $\operatorname{Spec} A$ from the closed sets $V(I)$. We now turn to the most useful open subsets:

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

These are simultaneously topological neighborhoods, complements of hypersurface-type closed sets, and the geometric manifestation of localization. Their algebra is unusually clean: multiplication becomes intersection, powers do not change the open set, and localizing at f produces exactly the spectrum carried by $D(f)$.

This chapter develops the principal-open basis and its localization calculus. VI/09 will separate generic from closed points; VI/10 will analyze nilpotent and reduced structure in its own right; VI/11 will localize at a prime and construct local rings and residue fields. Here the localization is only at a single element or a finitely generated principal-open cover.

After this chapter, the reader should be able to:

- manipulate the identities satisfied by $D(f)$;
- prove that principal opens form a basis of $\operatorname{Spec} A$;
- decide when $D(f) \subseteq D(g)$ or $D(f) = D(g)$ using radicals;
- translate open covers by $D(f_i)$ into ideal-generation statements;
- prove that every $D(f)$ is quasi-compact and hence every affine spectrum is quasi-compact;
- construct A_f and prove $\operatorname{Spec} A_f \cong D_A(f)$;
- compute nested localizations and the interaction of localization with quotients;
- compute principal opens in arithmetic, reducible, product, and nonreduced examples;
- track principal opens under maps of spectra.

8.1 Definition and first identities

Definition 8.1 (Basic or distinguished open)**.** Let A be a commutative ring and $f \in A$. The *basic open* or *distinguished open* determined by f is

$$D_A(f) = D(f) = \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

Since
$$V((f)) = \{\mathfrak{p} : f \in \mathfrak{p}\},$$
we have
$$D(f) = \operatorname{Spec} A \setminus V(f).$$

$$\begin{tikzcd} \boxed{f \in A} \arrow[r, "\text{test membership } f \in \mathfrak{p}"'] & \boxed{\begin{array}{c} V(f) = \{\mathfrak{p} : f \in \mathfrak{p}\} \\ \text{closed locus} \end{array}} \arrow[r, "\text{take the complement}"'] & \boxed{\begin{array}{c} D(f) = \operatorname{Spec} A \setminus V(f) \\ \text{principal open} \end{array}} \end{tikzcd}$$

Proposition 8.2 (Principal-open calculus)**.** *For $f, g \in A$ and $n \geq 1$,*
$$D(0) = \varnothing, \qquad D(1) = \operatorname{Spec} A,$$
$$D(fg) = D(f) \cap D(g), \qquad D(f^n) = D(f).$$

Proof. Every prime contains 0 and no proper ideal contains 1. Further, $fg \notin \mathfrak{p}$ is equivalent, by primality, to $f \notin \mathfrak{p}$ and $g \notin \mathfrak{p}$. Finally a prime contains f^n iff it contains f. ☐

$$\begin{tikzcd} \boxed{D(f)} \arrow[dr] & & \boxed{D(g)} \arrow[dl] \\ & \boxed{D(fg) = D(f) \cap D(g)} & \end{tikzcd}$$

Multiplication in the ring
becomes intersection in the topology.

Remark 8.3 (A common false rule)*.* In general,
$$D(f + g) \neq D(f) \cup D(g).$$
Addition has no comparably simple topological rule. The clean operation is multiplication, which corresponds to intersection.

8.2 Principal opens form a basis

Theorem 8.4 (Basis theorem)**.** *The sets $D(f)$, as f ranges over A, form a basis for the Zariski topology on* Spec *A. More precisely, if*
$$U = \operatorname{Spec} A \setminus V(I),$$
then
$$U = \bigcup_{f \in I} D(f).$$

Proof. A prime $\mathfrak{p}$ lies in U iff $I \not\subseteq \mathfrak{p}$, which is equivalent to there being some $f \in I$ with $f \notin \mathfrak{p}$. That is exactly $\mathfrak{p} \in D(f)$ for some $f \in I$. ☐

If $I = (f_1, \ldots, f_r)$ is finitely generated, then the same argument gives
$$\operatorname{Spec} A \setminus V(I) = D(f_1) \cup \cdots \cup D(f_r).$$
Thus finitely generated ideals produce finite principal-open covers of their complements.

8.3 Inclusion and equality criteria

The topology of principal opens is controlled by radicals.

Theorem 8.5 (Inclusion criterion)**.** *For $f, g \in A$,*

$$D(f) \subseteq D(g) \quad \Longleftrightarrow \quad f \in \sqrt{(g)}.$$

Equivalently,

$$D(f) \subseteq D(g) \quad \Longleftrightarrow \quad f^N \in (g) \textit{ for some } N \geq 1.$$

Proof. The inclusion $D(f) \subseteq D(g)$ is equivalent to $V(g) \subseteq V(f)$. By the closed-set/radical correspondence of VI/07, this is equivalent to

$$\sqrt{(f)} \subseteq \sqrt{(g)},$$

which in turn is equivalent to $f \in \sqrt{(g)}$. ☐

Corollary 8.6 (Equality criterion)**.**

$$D(f) = D(g) \quad \Longleftrightarrow \quad \sqrt{(f)} = \sqrt{(g)}.$$

In particular $D(f) = D(f^n)$ for every $n \geq 1$.

$$\begin{tikzcd}
\boxed{D(f) \subseteq D(g)} \arrow[d, "\text{complements}"] \\
\boxed{V(g) \subseteq V(f)} \arrow[d, "\text{closed-set dictionary}"] \\
\boxed{\sqrt{(f)} \subseteq \sqrt{(g)}} \arrow[d, "\text{principal radical}"] \\
\boxed{f^N \in (g) \text{ for some } N}
\end{tikzcd}$$

Example 8.7 (Removing the origin from one equation)**.** In $\operatorname{Spec} k[x]$, $D(x)$ consists of primes not containing x. It is naturally identified with $\operatorname{Spec} k[x, x^{-1}]$, the multiplicative group line.

8.4 Extremal principal opens: units and nilpotents

Proposition 8.8. *For $f \in A$,*

$$D(f) = \operatorname{Spec} A \quad \Longleftrightarrow \quad f \textit{ is a unit},$$

and

$$D(f) = \varnothing \quad \Longleftrightarrow \quad f \textit{ is nilpotent}.$$

Proof. If f is a unit, no proper ideal contains it. Conversely, if no prime contains f, then no maximal ideal contains f, so $(f) = A$ and f is a unit.

For the second statement, $D(f) = \varnothing$ means f lies in every prime ideal. The intersection of all prime ideals is the nilradical, so this is equivalent to f being nilpotent. □

The second equivalence is topologically useful here; VI/10 will study the nilpotent structure itself rather than merely its invisibility to principal opens.

8.5 Covers by principal opens

Theorem 8.9 (Cover criterion)**.** *For a family $\{f_\lambda\}_{\lambda\in\Lambda} \subseteq A$ and $f \in A$,*

$$D(f) \subseteq \bigcup_{\lambda\in\Lambda} D(f_\lambda)$$

if and only if

$$f \in \sqrt{(f_\lambda : \lambda \in \Lambda)}.$$

Equivalently, some power f^N belongs to the ideal generated by the f_λ.

Proof. The displayed inclusion is equivalent, after taking complements inside $\operatorname{Spec} A$, to

$$V(f_\lambda : \lambda \in \Lambda) \subseteq V(f).$$

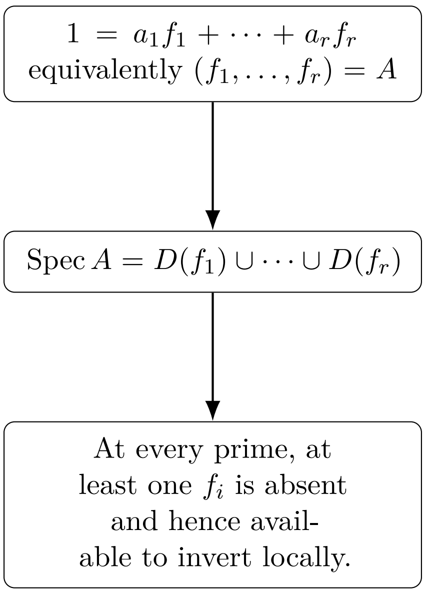
The radical/closed-set correspondence gives the stated ideal condition. □

For $f = 1$ this becomes the important special case

$$\operatorname{Spec} A = \bigcup_{\lambda} D(f_\lambda) \iff (f_\lambda : \lambda \in \Lambda) = A.$$

For a finite cover $D(f_1), \ldots, D(f_r)$, this says precisely that there are $a_1, \ldots, a_r \in A$ with

$$a_1 f_1 + \cdots + a_r f_r = 1.$$



8.6 Quasi-compactness

Theorem 8.10 (Principal opens are quasi-compact)**.** *Every* $D(f) \subseteq \operatorname{Spec} A$ *is quasi-compact. In particular,* $\operatorname{Spec} A = D(1)$ *is quasi-compact.*

Proof. Suppose

$$D(f) \subseteq \bigcup_{\lambda \in \Lambda} D(g_\lambda).$$

By the cover criterion, f^N lies in the ideal generated by the g_λ for some N. An element of an ideal is a finite linear combination, so there are finitely many indices $\lambda_1, \dots, \lambda_m$ such that

$$f^N \in (g_{\lambda_1}, \dots, g_{\lambda_m}).$$

Applying the cover criterion again gives

$$D(f) \subseteq D(g_{\lambda_1}) \cup \cdots \cup D(g_{\lambda_m}).$$

□

This finite-subcover theorem is one reason principal opens are the natural working basis for affine geometry.

Example 8.11 (Intersection rule)**.** For any $f, g \in A$, $D(f) \cap D(g) = D(fg)$. A prime avoids fg exactly when it avoids both f and g.

8.7 Localization at one element

Let

$$S_f = \{1, f, f^2, f^3, \dots\}.$$

The localization of A at f is

$$A_f = S_f^{-1} A = A\left[\frac{1}{f}\right].$$

Its elements are represented by fractions

$$\frac{a}{f^n}, \qquad a \in A,\ n \geq 0.$$

Proposition 8.12 (Universal property)**.** *A ring map* $\varphi : A \to B$ *factors uniquely through* A_f *iff* $\varphi(f)$ *is a unit in* B*.*

Thus localization is the universal algebraic operation that forces f to become invertible.

8.8 The spectrum of a localization

Theorem 8.13 (Principal-open localization theorem)**.** *Extension and contraction give mutually inverse bijections*

$$\operatorname{Spec} A_f \longleftrightarrow D_A(f),$$

$$\mathfrak{q} \longmapsto \mathfrak{q} \cap A, \qquad \mathfrak{p} \longmapsto \mathfrak{p} A_f.$$

These maps identify the Zariski topologies, hence

$$\boxed{\operatorname{Spec} A_f \cong D_A(f)}.$$

Proof. Prime ideals of A_f correspond to prime ideals of A disjoint from S_f. A prime $\mathfrak{p}$ is disjoint from S_f exactly when $f \notin \mathfrak{p}$, so the underlying bijection is exactly the one displayed. For topology, a basic closed subset of $\operatorname{Spec} A_f$ has the form $V_{A_f}(J)$; contraction identifies it with the corresponding subspace-closed set inside $D_A(f)$. Equivalently, basic opens correspond by the nested localization identity proved below. □

$$\begin{tikzcd}
\boxed{\begin{array}{c} A \longrightarrow A_f \\ \text{force } f \text{ to be a unit} \end{array}} \arrow[r, "\text{reverse arrows}"] & \boxed{\operatorname{Spec} A_f \;\cong\; D_A(f)}
\end{tikzcd}$$

$$\begin{tikzcd}
\boxed{\begin{array}{c} \mathfrak{p} \;\in\; D_A(f) \\ f \;\notin\; \mathfrak{p} \end{array}} \arrow[r, "\text{extension}"] & \boxed{\begin{array}{c} \mathfrak{p}A_f \;\in\; \operatorname{Spec} A_f \\ \text{contraction returns } \mathfrak{p} \end{array}} \arrow[l, bend left, "\text{contraction}"]
\end{tikzcd}$$

8.9 Nested principal opens

Let $g \in A$. Inside $\operatorname{Spec} A_f$, the image $g/1 \in A_f$ defines the principal open

$$D_{A_f}(g/1).$$

Under $\operatorname{Spec} A_f \cong D_A(f)$ this is precisely

$$D_A(fg).$$

Algebraically,

$$(A_f)_{g/1} \cong A_{fg}.$$

Thus shrinking from $D(f)$ to the smaller principal open cut out by g is exactly the same as inverting both f and g.

Example 8.14 (Nilpotent basic open)**.** If f is nilpotent, every prime contains f, so $D(f) = \varnothing$. Correspondingly the localization A_f is the zero ring.

8.10 Quotients and principal opens

Let $I \subseteq A$ and let $\bar{f}$ be the image of f in A/I. Then

$$(A/I)_{\bar{f}} \cong A_f/IA_f.$$

Taking spectra gives

$$D_{A/I}(\bar{f}) \cong V_A(I) \cap D_A(f).$$

$$\begin{tikzcd}
\operatorname{Spec}((A/I)_{\bar{f}}) \arrow[r, "\cong"] \arrow[d] & D_{A/I}(\bar{f}) \arrow[d, hook] \\
\operatorname{Spec}(A_f/IA_f) \arrow[r, "\cong"'] & V_A(I) \cap D_A(f)
\end{tikzcd}$$

The square records two compatible operations: impose the equations I, then invert f; or first restrict to $D(f)$, then impose the localized equations IA_f.

This formula is the basic compatibility between imposing equations and then restricting to an open region.

8.11 Behavior under maps of spectra

Let $\varphi : A \to B$ and let

$$\varphi^* : \operatorname{Spec} B \to \operatorname{Spec} A, \qquad \mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q}).$$

Then

$$(\varphi^*)^{-1}(D_A(f)) = D_B(\varphi(f)).$$

Indeed,

$$f \notin \varphi^{-1}(\mathfrak{q}) \quad \Longleftrightarrow \quad \varphi(f) \notin \mathfrak{q}.$$

This is the principal-open form of continuity and is often the fastest way to verify a map of affine spectra is continuous.

8.12 Arithmetic and geometric reference pictures

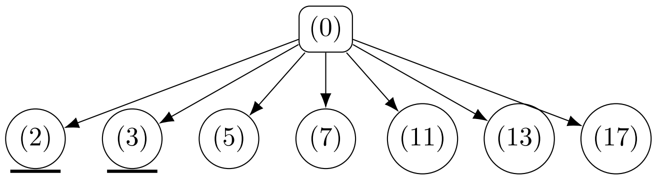
8.12.1 $D(n)$ inside $\operatorname{Spec}\mathbb{Z}$

For nonzero $n \in \mathbb{Z}$,

$$D(n) = \{(0)\} \cup \{(p) : p \nmid n\}.$$

The closed prime-number points dividing n are removed, while the generic point remains. Moreover

$$D(n) \cong \operatorname{Spec}\mathbb{Z}[1/n].$$



$V(6) = \{(2),(3)\}$ is removed. Hence $D(6) \cong \operatorname{Spec}\mathbb{Z}[1/6]$ keeps (0) and the closed points (p) with $p \nmid 6$.

8.12.2 The crossing axes

Let

$$R = k[x,y]/(xy).$$

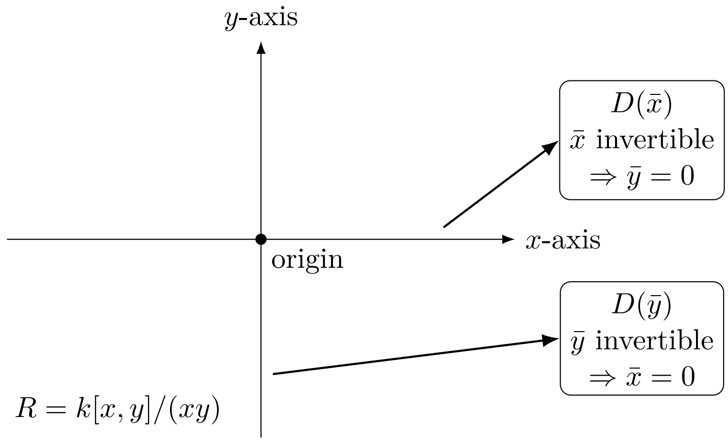
On $D(\bar{x})$ the element $\bar{x}$ becomes invertible, so the equation $\bar{x}\bar{y} = 0$ forces $\bar{y} = 0$. Hence

$$R_{\bar{x}} \cong k[x, x^{-1}].$$

Similarly,

$$R_{\bar{y}} \cong k[y, y^{-1}].$$

Localization literally removes the component on which the inverted function vanishes.



8.13 Comparison chart

Algebraic datum	**Open-set statement**	**Geometric meaning**
f unit	$D(f) = \operatorname{Spec} A$	f vanishes nowhere
f nilpotent	$D(f) = \varnothing$	every prime contains f
fg	$D(fg) = D(f) \cap D(g)$	invert both functions
f^n	$D(f^n) = D(f)$	multiplicity is invisible topologically
$f \in \sqrt{(g)}$	$D(f) \subseteq D(g)$	radical containment controls open inclusion
$(f_1, \ldots, f_r) = A$	$\bigcup_i D(f_i) = \operatorname{Spec} A$	the principal opens cover everywhere
localize at f	$\operatorname{Spec} A_f \cong D(f)$	force f to become invertible
quotient by I	$D_{A/I}(\bar{f}) \cong V(I) \cap D_A(f)$	impose equations and then restrict

8.14 Worked examples

Example 8.15 (A field)**.** Let K be a field. Since $\operatorname{Spec} K = \{(0)\}$, for $a \in K$,

$$D(a) = \begin{cases} \varnothing, & a = 0, \\ \operatorname{Spec} K, & a \neq 0. \end{cases}$$

This is the unit/nilpotent dichotomy in its simplest possible form.

Example 8.16 (The punctured affine line)**.** In $A = k[t]$,

$$D(t) = \operatorname{Spec} k[t] \setminus V(t).$$

If k is algebraically closed, this removes the closed point (t) and keeps the generic point together with all $(t - a)$ for $a \neq 0$. Algebraically,

$$A_t = k[t, t^{-1}], \qquad D(t) \cong \operatorname{Spec} k[t, t^{-1}].$$

Example 8.17 (Removing two points from an affine line)**.** Assume $a \neq b$ in an algebraically closed field k. Then

$$D((t-a)(t-b)) = D(t-a) \cap D(t-b)$$

is the affine line with the two closed points $(t-a)$ and $(t-b)$ removed. The coordinate ring of this principal open is

$$k[t]_{(t-a)(t-b)}.$$

Example 8.18 (Arithmetic localization)**.** In $\operatorname{Spec} \mathbb{Z}$,

$$D(6) = \{(0)\} \cup \{(p) : p \neq 2, 3\}.$$

The points (2) and (3) are removed because they contain 6. Correspondingly,

$$D(6) \cong \operatorname{Spec} \mathbb{Z}[1/6].$$

The localization makes both 2 and 3 units.

Example 8.19 (Coordinate axes removed)**.** In $k[x, y]$,

$$D(xy) = D(x) \cap D(y).$$

Thus $D(xy)$ is the region where both coordinates are invertible, and

$$k[x, y]_{xy} \cong k[x, x^{-1}, y, y^{-1}].$$

Geometrically it is the affine plane with both coordinate axes removed.

Example 8.20 (A principal open need not behave additively)**.** In $k[x, y]$,

$$D(x + y)$$

is the complement of the diagonal line $x + y = 0$. It is not equal to $D(x) \cup D(y)$: the prime (x, y) lies in neither side, while the prime $(x + y)$ lies in $D(x) \cup D(y)$ but not in $D(x + y)$. This illustrates why multiplication, not addition, is the natural operation for principal opens.

Example 8.21 (A two-element principal-open cover)**.** In $k[t]$,

$$(t) + (1 - t) = k[t],$$

so

$$D(t) \cup D(1 - t) = \operatorname{Spec} k[t].$$

Indeed no prime ideal can contain both t and $1 - t$, because then it would contain 1. The identity

$$t + (1 - t) = 1$$

is the algebraic certificate of the cover.

Example 8.22 (An open set that is not principal)**.** In $A = k[x, y]$,

$$U = D(x) \cup D(y) = \operatorname{Spec} A \setminus V(x, y).$$

This is the complement of the origin. It is a union of two principal opens, but in general it is not itself principal. If $U = D(f)$, then $V(f) = V(x, y)$, hence $\sqrt{(f)} = (x, y)$; over a polynomial ring in two variables this would make a height-two maximal ideal the radical of a principal ideal, contradicting the principal ideal theorem. Thus a basis element is not closed under arbitrary unions.

Example 8.23 (Crossing axes: localization removes a component)**.** Let

$$R = k[x, y]/(xy).$$

Then on $D(\bar{x})$, the element $\bar{x}$ is invertible and $\bar{x}\bar{y} = 0$ forces $\bar{y} = 0$. Therefore

$$R_{\bar{x}} \cong k[x, x^{-1}].$$

Similarly $R_{\bar{y}} \cong k[y, y^{-1}]$. Localization discards the component on which the inverted function vanishes identically.

Example 8.24 (Crossing axes minus the origin)**.** Again let $R = k[x, y]/(xy)$ and localize at $\bar{x} + \bar{y}$. In $R_{\bar{x}+\bar{y}}$ set

$$e = \frac{\bar{x}}{\bar{x} + \bar{y}}, \qquad 1 - e = \frac{\bar{y}}{\bar{x} + \bar{y}}.$$

Since $\bar{x}\bar{y} = 0$, one checks $e^2 = e$. The two idempotent pieces separate the two punctured axes, giving

$$R_{\bar{x}+\bar{y}} \cong k[x, x^{-1}] \times k[y, y^{-1}].$$

Thus removing the intersection point disconnects the reducible curve.

Example 8.25 (A quotient and a principal open)**.** Let $A = k[x, y]$, $I = (xy)$, and take $f = x + y$. Then

$$D_{A/I}(\overline{x + y}) \cong V_A(xy) \cap D_A(x + y).$$

The left side is computed by

$$(A/I)_{\overline{x+y}} \cong A_{x+y}/(xy)A_{x+y}.$$

This is the union of the two axes with their intersection point removed.

Example 8.26 (Preimage of a principal open)**.** Consider $\varphi : k[t] \to k[x]$ with $t \mapsto x^2$. The induced map

$$\varphi^* : \operatorname{Spec} k[x] \to \operatorname{Spec} k[t]$$

satisfies

$$(\varphi^*)^{-1}(D(t)) = D(x^2) = D(x).$$

Thus the inverse image of the punctured t-line is the punctured x-line.

Example 8.27 (Product rings)**.** Let $R = A \times B$ and $(a, b) \in R$. Under

$$\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B,$$

one has

$$D_{A\times B}(a, b) \cong D_A(a) \sqcup D_B(b).$$

For the idempotent $(1, 0)$, this gives

$$D(1, 0) \cong \operatorname{Spec} A,$$

while the entire $\operatorname{Spec} B$ component is removed.

Example 8.28 (Dual numbers)**.** Let

$$A = k[\varepsilon]/(\varepsilon^2).$$

The element ε is nilpotent, so

$$D(\varepsilon) = \varnothing.$$

But $1 + \varepsilon$ is a unit with inverse $1 - \varepsilon$, so

$$D(1 + \varepsilon) = \operatorname{Spec} A.$$

The one-point topology sees the two extreme principal opens but does not itself record the nilpotent thickness.

8.15 Exercises

Exercise 8.29 (VI.08.E1 - Elementary identities)**.** Prove

$$D(0) = \varnothing, \qquad D(1) = \operatorname{Spec} A, \qquad D(fg) = D(f) \cap D(g).$$

Exercise 8.30 (VI.08.E2 - Powers)**.** Show $D(f^n) = D(f)$ for every $n \geq 1$.

Exercise 8.31 (VI.08.E3 - Principal-open basis)**.** If $U = \operatorname{Spec} A \setminus V(I)$, prove

$$U = \bigcup_{f \in I} D(f).$$

Exercise 8.32 (VI.08.E4 - Finite generators)**.** If $I = (f_1, \ldots, f_r)$, show

$$\operatorname{Spec} A \setminus V(I) = D(f_1) \cup \cdots \cup D(f_r).$$

Exercise 8.33 (VI.08.E5 - Inclusion of basic opens)**.** Prove

$$D(f) \subseteq D(g) \iff f \in \sqrt{(g)}.$$

Exercise 8.34 (VI.08.E6 - Equality of basic opens)**.** Prove

$$D(f) = D(g) \iff \sqrt{(f)} = \sqrt{(g)}.$$

Exercise 8.35 (VI.08.E7 - Units)**.** Show

$$D(f) = \operatorname{Spec} A \iff f \in A^\times.$$

Exercise 8.36 (VI.08.E8 - Nilpotents)**.** Show

$$D(f) = \varnothing \iff f \text{ is nilpotent.}$$

Exercise 8.37 (VI.08.E9 - Cover of the entire spectrum)**.** For $f_1, \ldots, f_r \in A$, prove

$$\operatorname{Spec} A = \bigcup_{i=1}^{r} D(f_i) \iff (f_1, \ldots, f_r) = A.$$

Exercise 8.38 (VI.08.E10 - Relative cover criterion)**.** Let $g_1, \ldots, g_r, f \in A$. Prove

$$D(f) \subseteq \bigcup_{i=1}^{r} D(g_i) \iff f \in \sqrt{(g_1, \ldots, g_r)}.$$

Exercise 8.39 (VI.08.E11 - Quasi-compactness)**.** Prove directly from E10 that every $D(f)$ is quasi-compact.

Exercise 8.40 (VI.08.E12 - Arithmetic basic opens)**.** For nonzero $n \in \mathbb{Z}$, describe $D(n) \subseteq \operatorname{Spec} \mathbb{Z}$ and identify the points removed.

Exercise 8.41 (VI.08.E13 - Punctured line)**.** For algebraically closed k, describe $D(t) \subseteq \operatorname{Spec} k[t]$ and compute $k[t]_t$.

Exercise 8.42 (VI.08.E14 - Nested localization)**.** Prove

$$(A_f)_{g/1} \cong A_{fg}.$$

Exercise 8.43 (VI.08.E15 - Prime correspondence for A_f)**.** Show that prime ideals of A_f correspond bijectively to primes $\mathfrak{p} \subseteq A$ with $f \notin \mathfrak{p}$.

Exercise 8.44 (VI.08.E16 - Topology of the localization correspondence)**.** Upgrade E15 to a homeomorphism

$$\operatorname{Spec} A_f \cong D_A(f).$$

Exercise 8.45 (VI.08.E17 - Preimage of a basic open)**.** For $\varphi : A \to B$, prove

$$(\varphi^*)^{-1}(D_A(f)) = D_B(\varphi(f)).$$

Exercise 8.46 (VI.08.E18 - Quotient then localize)**.** For $I \subseteq A$, prove

$$(A/I)_{\bar{f}} \cong A_f/IA_f$$

and identify its spectrum with $V(I) \cap D(f)$.

Exercise 8.47 (VI.08.E19 - Crossing axes)**.** For $R = k[x, y]/(xy)$, compute $R_{\bar{x}}$ and $R_{\bar{y}}$ and interpret the corresponding basic opens geometrically.

Exercise 8.48 (VI.08.E20 - Dual numbers)**.** For $A = k[\varepsilon]/(\varepsilon^2)$, compute $D(\varepsilon)$, $D(1 + \varepsilon)$, and A_ε.

8.16 Problems

Problem 8.49. Prove that the $D(f)$ form a basis and that $D(f) = \varnothing$ iff f is nilpotent.

Problem 8.50. Prove the radical criterion for inclusion and equality of principal opens.

Problem 8.51. Prove the general cover criterion $D(f) \subseteq \bigcup D(g_\lambda)$.

Problem 8.52. Prove every principal open is quasi-compact.

Problem 8.53. Prove the homeomorphism $\operatorname{Spec} A_f \cong D_A(f)$.

Problem 8.54. Analyze $D(n) \subseteq \operatorname{Spec} \mathbb{Z}$ and identify it with $\operatorname{Spec} \mathbb{Z}[1/n]$.

Problem 8.55. Prove $(A_f)_{g/1} \cong A_{fg}$ and identify the corresponding nested open.

Problem 8.56. Prove $(A/I)_{\bar{f}} \cong A_f/IA_f$ and identify the corresponding open subset.

Problem 8.57. Prove preimages of principal opens are principal under maps of spectra.

Problem 8.58. Analyze principal-open covers of $\operatorname{Spec} A$ in terms of a unit ideal.

Problem 8.59. Analyze the principal opens of the crossing-axes ring $k[x, y]/(xy)$.

Problem 8.60. Describe principal opens in a product ring $A \times B$.

Problem 8.61. Analyze principal opens in $k[\varepsilon]/(\varepsilon^2)$.

8.17 Challenges

Problem 8.62. Show a quasi-compact open subset of $\operatorname{Spec} A$ is a finite union of principal opens.

Problem 8.63. Characterize clopen principal opens through idempotents.

Problem 8.64. Prove the equalizer/gluing statement for a finite principal-open cover.

8.18 Boundary with the next chapters

Principal opens and one-element localization are now complete. VI/09 will use the topology to distinguish generic, closed, and intermediate points. VI/10 will explain what the underlying topology misses about nilpotents. VI/11 will replace A_f by localization at a prime $A_{\mathfrak{p}}$ and construct the genuinely local algebra of a point. Later, VI/17 and VI/18 will interpret A_f as the ring of sections on $D(f)$ and use principal opens as the standard affine basis for the structure sheaf.

8.19 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 8.65 (Intersection)**.** Compute $D(f) \cap D(g)$.

Hint. Use distinguished opens and localization. ☐

Solution. $D(fg)$. ☐

Exercise 8.66 (Unit)**.** Compute $D(1)$.

Hint. Use distinguished opens and localization. ☐

Solution. The whole spectrum. ☐

Exercise 8.67 (Zero)**.** Compute $D(0)$.

Hint. Use distinguished opens and localization. ☐

Solution. The empty set. ☐

Exercise 8.68 (Nilpotent)**.** What is $D(f)$ for nilpotent f?

Hint. Use distinguished opens and localization. ☐

Solution. Empty. ☐

Exercise 8.69 (Localization)**.** Which spectrum models $D(f)$?

Hint. Use distinguished opens and localization. ☐

Solution. $\operatorname{Spec} A_f$. ☐

Proofs

Exercise 8.70 (Basis)**.** Prove: Prove the distinguished opens form a basis for the Zariski topology.

Hint. Work directly from distinguished opens and localization. ☐

Solution. Every open is a union of complements of $V(I)$, and a point outside $V(I)$ avoids some $f \in I$. ☐

Exercise 8.71 (Intersection)**.** Prove: Prove $D(f) \cap D(g) = D(fg)$.

Hint. Work directly from distinguished opens and localization. ☐

Solution. Use primality: $fg \notin \mathfrak{p}$ iff both factors avoid $\mathfrak{p}$. ☐

Exercise 8.72 (Localization primes)**.** Prove: Prove primes of A_f correspond to primes of A in $D(f)$.

Hint. Work directly from distinguished opens and localization. ☐

Solution. Use the prime correspondence for localization. ☐

Exercise 8.73 (Cover criterion)**.** Prove: Prove $D(f_1), \ldots, D(f_n)$ cover $\operatorname{Spec} A$ iff $(f_1, \ldots, f_n) = A$.

Hint. Work directly from distinguished opens and localization. ☐

Solution. The complement is $V((f_1, \ldots, f_n))$, empty exactly when the ideal is the unit ideal. ☐

Counterexamples and hypothesis tests

Exercise 8.74 (Closed)**.** Test the claim: every $D(f)$ is closed.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general; it is open. ☐

Exercise 8.75 (Empty only zero)**.** Test the claim: $D(f) = \varnothing$ implies $f = 0$.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it implies f is nilpotent. ☐

Exercise 8.76 (Union principal)**.** Test the claim: every finite union $D(f) \cup D(g)$ equals $D(fg)$.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; $D(fg)$ is the intersection. ☐

Applications and investigations

Exercise 8.77 (Localizing geometry)**.** What geometric operation corresponds to inverting f?

Hint. Translate the question into distinguished opens and localization. ☐

Solution. Restricting to the open locus where f does not vanish. ☐

Exercise 8.78 (Open cover)**.** How can algebra verify a finite distinguished-open cover?

Hint. Translate the question into distinguished opens and localization. ☐

Solution. Check that the defining elements generate the unit ideal. ☐

Challenge problems

Exercise 8.79 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Basic Open Sets in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Basic Open Sets, translates it through distinguished opens and localization, and checks the correspondence in both algebraic and geometric language. □

Exercise 8.80 (Hypothesis challenge)**.** Choose one structural statement from Basic Open Sets, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 9

Generic and Closed Points

Purpose and learning goals

The preceding chapters built the prime spectrum and its principal-open basis. We now ask what the individual points of $\operatorname{Spec} A$ mean topologically.

A prime ideal may represent a closed point, the generic point of an irreducible closed subset, or an intermediate point between these extremes. The organizing principle is the closure formula

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Because containment of prime ideals controls these closures, the specialization order is nothing more than inclusion of prime ideals.

This chapter develops that pointwise geometry carefully. It also explains why closed points need not be rational over the base field, why generic points are genuine topological points rather than informal "general positions," and why irreducible components carry distinguished generic points. We use the language of schemes only for the legacy theorem that every irreducible closed subset of a scheme has a unique generic point; local rings and residue fields remain reserved for VI/11, and function fields for VI/27.

After this chapter, the reader should be able to:

- compute the closure of a point in $\operatorname{Spec} A$;
- translate specialization and generalization into inclusion of prime ideals;
- prove that $\operatorname{Spec} A$ is a T_0 space and characterize when it is T_1;
- identify closed points with maximal ideals;
- identify generic points of irreducible closed subsets with prime ideals;
- recognize generic points of irreducible components as minimal primes;
- decide when the whole spectrum has a generic point;
- distinguish closed points from k-rational points;
- analyze generic and closed points in arithmetic, reducible, product, and nonreduced examples;
- use generic points to test density of open subsets and irreducible components.

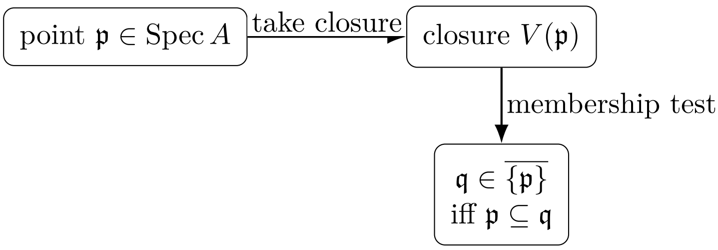
9.1 The closure of one point

Let $\mathfrak{p} \in \operatorname{Spec} A$. A closed set $V(I)$ contains $\mathfrak{p}$ exactly when $I \subseteq \mathfrak{p}$. Therefore the smallest closed set containing $\mathfrak{p}$ is $V(\mathfrak{p})$.

Proposition 9.1 (Closure formula)**.** *For every* $\mathfrak{p} \in \operatorname{Spec} A$*,*

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{p} \subseteq \mathfrak{q}\}.$$

Proof. Every closed set containing $\mathfrak{p}$ has the form $V(I)$ with $I \subseteq \mathfrak{p}$. Since $I \subseteq \mathfrak{p}$ implies $V(\mathfrak{p}) \subseteq V(I)$, the set $V(\mathfrak{p})$ is contained in every closed set containing $\mathfrak{p}$. It is itself closed and contains $\mathfrak{p}$. ☐



9.2 Specialization and generalization

Definition 9.2 (Specialization and generalization)**.** Let x, y be points of a topological space. We say that y is a *specialization* of x if

$$y \in \overline{\{x\}}.$$

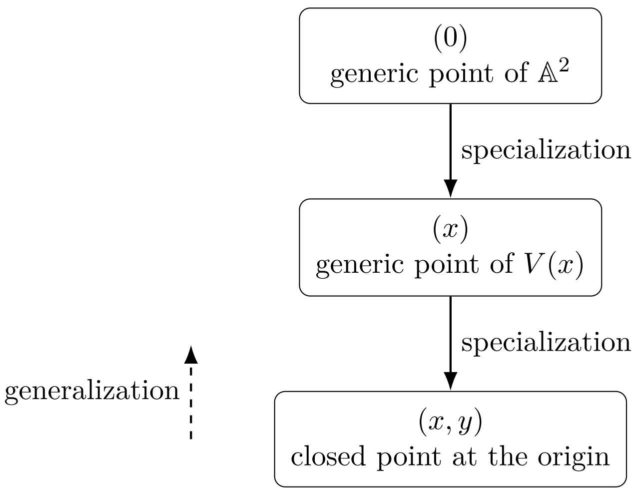
Equivalently, x is a *generalization* of y.

For spectra the order is completely algebraic.

Theorem 9.3 (Specialization order on a spectrum)**.** *For* $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} A$*,*

$$\mathfrak{q} \textit{ is a specialization of } \mathfrak{p} \iff \mathfrak{p} \subseteq \mathfrak{q}.$$

Thus smaller prime ideals are more generic and larger prime ideals are more special.



The familiar chain

$$(0) \subset (x) \subset (x, y) \qquad \text{in } k[x, y]$$

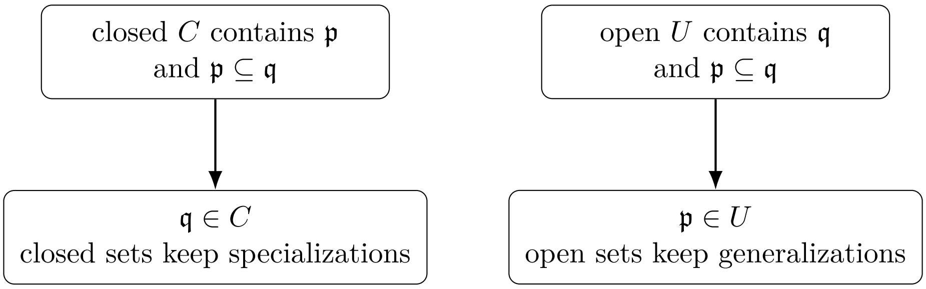
means

$$\text{generic plane point} \rightsquigarrow \text{generic point of the line } x = 0 \rightsquigarrow \text{closed origin.}$$

Proposition 9.4 (Open and closed sets respect the specialization order)**.** *In* Spec A*:*

1. *closed subsets are stable under specialization;*
2. *open subsets are stable under generalization.*

Proof. If $\mathfrak{p} \in V(I)$ and $\mathfrak{p} \subseteq \mathfrak{q}$, then $I \subseteq \mathfrak{q}$, so $\mathfrak{q} \in V(I)$. Taking complements gives the open-set statement. □



9.3 T_0 and T_1 behavior

Proposition 9.5. *Every affine spectrum* Spec A *is a* T_0 *space.*

Proof. Take distinct primes $\mathfrak{p} \neq \mathfrak{q}$. After interchanging them if necessary, choose

$$f \in \mathfrak{q} \setminus \mathfrak{p}.$$

Then $\mathfrak{p} \in D(f)$ but $\mathfrak{q} \notin D(f)$. Hence distinct points are topologically distinguishable. □

Corollary 9.6. *The closure map*

$$\mathfrak{p} \longmapsto \overline{\{\mathfrak{p}\}}$$

is injective on Spec A*.*

Proposition 9.7. *The space* Spec A *is* T_1 *if and only if every prime ideal of* A *is maximal.*

Proof. A space is T_1 exactly when every singleton is closed. By the closed-point criterion proved below, this means precisely that every prime is maximal. □

Thus affine spectra are always T_0, but are usually far from T_1.

9.4 Generic points of irreducible closed subsets

Definition 9.8 (Generic point)**.** Let Z be a closed subset of a topological space. A point $\eta \in Z$ is a *generic point* of Z if

$$\overline{\{\eta\}} = Z.$$

In an affine spectrum, prime ideals and irreducible closed subsets are two descriptions of the same data.

Theorem 9.9 (Prime ideal–generic point correspondence)**.** *For $\mathfrak{p} \in \operatorname{Spec} A$, the closed set $V(\mathfrak{p})$ is irreducible and $\mathfrak{p}$ is its unique generic point.*

Conversely, if $Z \subseteq \operatorname{Spec} A$ is irreducible and closed, then

$$Z = V(\mathfrak{p})$$

for a unique prime ideal $\mathfrak{p}$, namely the generic point of Z.

Proof. The quotient

$$A/\mathfrak{p}$$

is a domain, so $\operatorname{Spec}(A/\mathfrak{p}) \cong V(\mathfrak{p})$ is irreducible. By the closure formula, $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$, so $\mathfrak{p}$ is generic.

Conversely, write $Z = V(I)$. The closed set $V(I)$ is irreducible if and only if $\sqrt{I}$ is prime. Put $\mathfrak{p} = \sqrt{I}$. Then

$$Z = V(I) = V(\mathfrak{p}).$$

Uniqueness follows from the T_0 property, or directly from equality of radicals. $\square$

$$\begin{tikzcd} \boxed{\text{prime ideal } \mathfrak{p}} \arrow[r, leftrightarrow] & \boxed{\begin{array}{c}\text{irreducible closed set}\\ V(\mathfrak{p})\end{array}} \arrow[r, leftrightarrow] & \boxed{\begin{array}{c}\text{unique generic point}\\ \eta = \mathfrak{p}\end{array}} \end{tikzcd}$$

$A/\mathfrak{p}$ is a domain
and $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$

Remark 9.10 (Sobriety). A topological space is called *sober* if every nonempty irreducible closed subset has a unique generic point. Every affine spectrum is sober. In fact every scheme is sober; the full scheme-theoretic statement is preserved as Legacy Problem VI.09.P1 below.

Example 9.11 (Generic point of an irreducible spectrum)**.** For an integral domain A, the point (0) is generic in $\operatorname{Spec} A$: its closure is the whole spectrum. Every nonempty open contains it.

9.5 When the whole spectrum has a generic point

The entire spectrum has a generic point exactly when it is irreducible.

Theorem 9.12. *The following are equivalent:*

1. $\operatorname{Spec} A$ *is irreducible;*
2. *the nilradical $\sqrt{(0)}$ is a prime ideal;*
3. $\operatorname{Spec} A$ *has a generic point.*

When these conditions hold, the generic point is

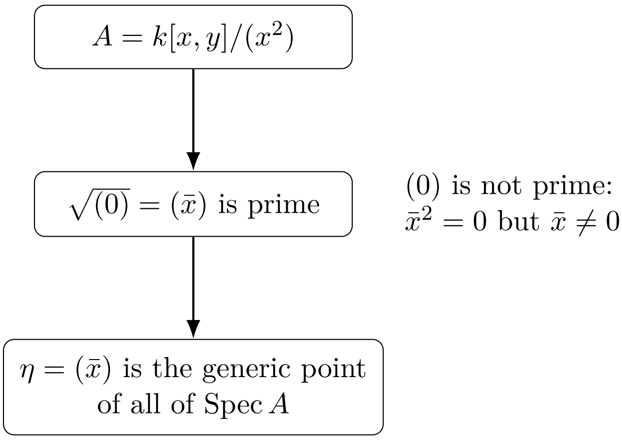
$$\eta = \sqrt{(0)}.$$

Proof. By VI/05, $\operatorname{Spec} A$ is irreducible exactly when its defining radical $\sqrt{(0)}$ is prime. If it is prime, then

$$\overline{\{\sqrt{(0)}\}} = V(\sqrt{(0)}) = V(0) = \operatorname{Spec} A.$$

Conversely, if η is generic, then $\operatorname{Spec} A = V(\eta)$ is irreducible. $\square$

If A is a domain, then $\sqrt{(0)} = (0)$, so (0) is the generic point. But in a nonreduced irreducible ring the generic point need not be the zero ideal.



9.6 Closed points

Generic points are as large as possible from the viewpoint of closure. Closed points are the opposite extreme.

Theorem 9.13 (Closed points are maximal ideals)**.** *For $\mathfrak{p} \in \operatorname{Spec} A$, the following are equivalent:*

1. *$\mathfrak{p}$ is a closed point of $\operatorname{Spec} A$;*
2. *$\overline{\{\mathfrak{p}\}} = \{\mathfrak{p}\}$;*
3. *$V(\mathfrak{p}) = \{\mathfrak{p}\}$;*
4. *$\mathfrak{p}$ is maximal.*

Proof. By the closure formula,

$$V(\mathfrak{p}) = \{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{p} \subseteq \mathfrak{q}\}.$$

This set is the singleton $\{\mathfrak{p}\}$ exactly when no proper prime ideal strictly contains $\mathfrak{p}$. A prime ideal with this property is maximal. $\square$

For a maximal ideal $\mathfrak{m}$, the quotient $A/\mathfrak{m}$ is a field. VI/11 will package this field as the residue field of the point and study the associated local ring.

9.7 Closed points versus rational points

Let A be a k-algebra. A k-rational point of $\operatorname{Spec} A$ corresponds to a k-algebra map

$$A \longrightarrow k,$$

hence to a maximal ideal $\mathfrak{m}$ for which

$$A/\mathfrak{m} \cong k.$$

A closed point need not be k-rational: it only requires $\mathfrak{m}$ to be maximal.

Theorem 9.14 (Finite-type algebraically closed case)**.** *Let k be algebraically closed and let A be a finitely generated k-algebra. By the weak Nullstellensatz, every maximal ideal has quotient k. Thus the closed points of* $\operatorname{Spec} A$ *are exactly the classical k-rational points.*

Over a non-algebraically closed or finite field, additional closed points appear.

Ring	**Closed points**	**Base-field rational?**
$\mathbb{C}[t]$	$(t-a)$, $a \in \mathbb{C}$	all closed points are $\mathbb{C}$-rational
$\mathbb{R}[t]$	$(t-a)$ and irreducible quadratics (q)	quadratic points are closed but not $\mathbb{R}$-rational
$\mathbb{F}_p[t]$	(f) for irreducible f of every degree	only degree-one points are $\mathbb{F}_p$-rational

Example 9.15 (Closed points of the affine line)**.** Over an algebraically closed field, the closed points of $\operatorname{Spec} k[x]$ are the maximal ideals $(x-a)$. The generic point (0) is the one additional nonclosed point.

9.8 Generic points of irreducible components

Let $\operatorname{Min}(A)$ denote the set of minimal prime ideals of A.

Theorem 9.16 (Minimal primes are component-generic points)**.** *The irreducible components of* $\operatorname{Spec} A$ *are exactly the closed sets*

$$V(\mathfrak{p}), \qquad \mathfrak{p} \in \operatorname{Min}(A).$$

The point $\mathfrak{p}$ is the generic point of the corresponding component.

Proof. Every irreducible component is an irreducible closed subset, hence has the form $V(\mathfrak{p})$ for a prime $\mathfrak{p}$. Maximality of the component under inclusion is equivalent, by reversal of inclusion, to minimality of $\mathfrak{p}$ among prime ideals. The generic-point statement follows from the previous theorem. $\square$

$$\begin{tikzcd}
& \boxed{\begin{array}{c}\text{minimal primes}\\ \mathfrak{p}_1,\dots,\mathfrak{p}_r\end{array}} \arrow[dl] \arrow[d] \arrow[dr] & \\
\boxed{\begin{array}{c}V(\mathfrak{p}_1)\\ \text{irreducible component}\end{array}} & \boxed{\begin{array}{c}V(\mathfrak{p}_2)\\ \text{irreducible component}\end{array}} & \boxed{\begin{array}{c}V(\mathfrak{p}_r)\\ \text{irreducible component}\end{array}}
\end{tikzcd}$$

each $\mathfrak{p}_i$ is the unique generic point of its component

This gives a particularly efficient way to read reducible geometry: the minimal primes are the generic points of the largest irreducible pieces.

9.9 The crossing axes as a specialization picture

Let

$$R = k[x,y]/(xy).$$

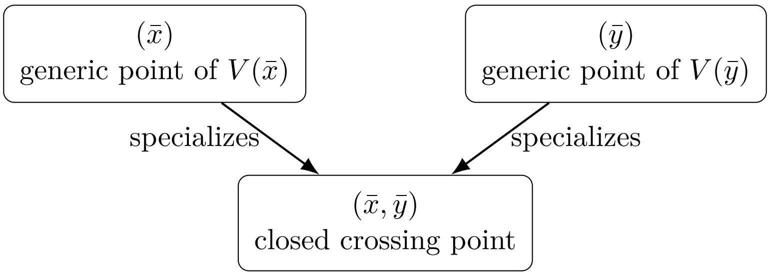
The minimal primes are

$$(\bar{x}) \qquad \text{and} \qquad (\bar{y}).$$

They are the generic points of the two axes. The maximal ideal

$$(\bar{x}, \bar{y})$$

is their common closed specialization at the crossing.



The point $(\bar{x}, \bar{y})$ lies in the closure of each component-generic point because

$$(\bar{x}) \subset (\bar{x}, \bar{y}), \qquad (\bar{y}) \subset (\bar{x}, \bar{y}).$$

9.10 Generic points and nonempty open subsets

Proposition 9.17. *Let X be an irreducible topological space with generic point η. Then every nonempty open subset $U \subseteq X$ contains η. Moreover η is the generic point of U with its subspace topology.*

Proof. If $\eta \notin U$, then the closed set $X \setminus U$ contains η, hence contains $\overline{\{\eta\}} = X$, contradicting $U \neq \varnothing$. Therefore $\eta \in U$. Furthermore

$$\overline{\{\eta\}}^{U} = U \cap \overline{\{\eta\}}^{X} = U.$$

$\square$

More generally, if C is an irreducible component with generic point η_C and an open set U meets C, then $\eta_C \in U$.

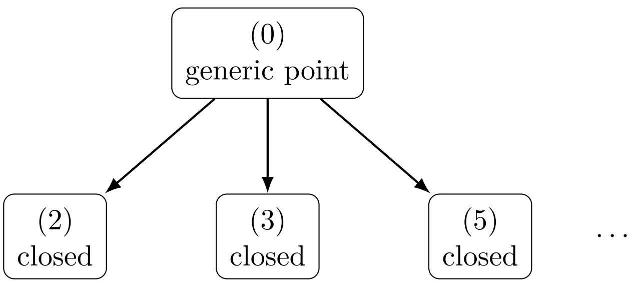
Example 9.18 (Generic point of a component)**.** In $\operatorname{Spec} k[x, y]/(xy)$, the two minimal primes (x) and (y) are generic points of the two irreducible components.

9.11 The arithmetic spectrum

The spectrum of the integers gives the cleanest example in which one generic point specializes to infinitely many closed points:

$$\operatorname{Spec} \mathbb{Z} = \{(0)\} \cup \{(p) : p \text{ prime}\}.$$

The point (0) is generic because $\mathbb{Z}$ is a domain. Every (p) is maximal and hence closed.



For nonzero n,

$$D(n) = \{(0)\} \cup \{(p) : p \nmid n\},$$

so every nonempty principal open contains the generic point and all but finitely many closed prime-number points.

9.12 Comparison chart

Point type	**Algebraic condition**	**Topological meaning**
arbitrary point	prime $\mathfrak{p}$	closure is $V(\mathfrak{p})$
specialization of $\mathfrak{p}$	$\mathfrak{p} \subseteq \mathfrak{q}$	$\mathfrak{q} \in \overline{\{\mathfrak{p}\}}$
generalization of $\mathfrak{q}$	$\mathfrak{p} \subseteq \mathfrak{q}$	$\mathfrak{p}$ is more generic
closed point	$\mathfrak{m}$ maximal	$\overline{\{\mathfrak{m}\}} = \{\mathfrak{m}\}$
generic point of $V(\mathfrak{p})$	$\mathfrak{p}$ prime	closure equals the whole irreducible closed set
component-generic point	$\mathfrak{p}$ minimal prime	generic point of an irreducible component
generic point of $\operatorname{Spec} A$	$\sqrt{(0)}$ prime	$\operatorname{Spec} A$ is irreducible
k-rational closed point	$A/\mathfrak{m} \cong k$	closed point defined over the base field

9.13 Common pitfalls

- A generic point is an actual point of the spectrum, not a vague "general point."
- A closed point need not be k-rational unless additional hypotheses such as the Nullstellensatz apply.
- The zero ideal is generic only when it is prime. In a nonreduced irreducible ring, the generic point can be a nonzero nilradical.
- Inclusion of prime ideals and inclusion of their closures run in the same direction for specializations but in the opposite direction for the defining closed sets: $\mathfrak{p} \subseteq \mathfrak{q}$ implies $V(\mathfrak{q}) \subseteq V(\mathfrak{p})$.
- Being closed in one chosen open neighborhood does not by itself mean being closed in the entire ambient scheme. In this chapter, the maximal-ideal criterion is stated for the affine spectrum under discussion.
- Generic-point existence is special: arbitrary topological spaces need not have generic points for irreducible closed subsets. Spectra and schemes do.

9.14 Worked examples

Example 9.19 (A field: one point that is both generic and closed)**.** Let K be a field. Then

$$\operatorname{Spec} K = \{(0)\}.$$

The zero ideal is maximal and prime, so the unique point is closed. Its closure is the whole one-point space, so it is also generic. Thus “generic” and “closed” are not mutually exclusive labels; they describe opposite closure behaviors only when the space has more than one point.

Example 9.20 (The affine line over an algebraically closed field)**.** Let k be algebraically closed. The points of $\operatorname{Spec} k[t]$ are

$$(0) \qquad \text{and} \qquad (t-a),\ a \in k.$$

The point (0) is generic because $k[t]$ is a domain. Each $(t-a)$ is maximal and hence closed. Moreover

$$(0) \subset (t-a),$$

so every classical point is a specialization of the generic point.

Example 9.21 (The real affine line has non-rational closed points)**.** In $\operatorname{Spec} \mathbb{R}[t]$, the nonzero prime ideals are generated by irreducible polynomials. Thus

$$(t-a), \qquad a \in \mathbb{R},$$

are closed real-rational points, but so are ideals such as

$$(t^2+1).$$

The latter is maximal because

$$\mathbb{R}[t]/(t^2+1) \cong \mathbb{C}$$

is a field. It is a closed point but not an $\mathbb{R}$-rational point.

Example 9.22 (The affine line over a finite field)**.** In $\operatorname{Spec} \mathbb{F}_p[t]$, every irreducible polynomial $f(t)$ determines a closed point (f). If $\deg f = d$, then

$$\mathbb{F}_p[t]/(f) \cong \mathbb{F}_{p^d}.$$

Only the degree-one irreducibles correspond to $\mathbb{F}_p$-rational points. The spectrum has far more closed points than the p elements of the finite field.

Example 9.23 (The arithmetic line $\operatorname{Spec} \mathbb{Z}$)**.** The points are

$$(0),\ (2),\ (3),\ (5), \ldots.$$

The point (0) is generic, while every (p) is closed. Since

$$(0) \subset (p),$$

the generic point specializes to every prime-number point.

Example 9.24 (Plane, line, point)**.** In $\operatorname{Spec} k[x,y]$,

$$(0) \subset (x) \subset (x,y).$$

Their closures are

$$\operatorname{Spec} k[x,y], \qquad V(x), \qquad \{(x,y)\},$$

respectively. One chain of prime ideals therefore records three geometric scales: the whole plane, a line, and a point.

Example 9.25 (A parabola and its generic point)**.** Assume k is algebraically closed and set

$$\mathfrak{p} = (y - x^2) \subset k[x, y].$$

Since

$$k[x, y]/(y - x^2) \cong k[x]$$

is a domain, $\mathfrak{p}$ is prime. Hence $V(y - x^2)$ is irreducible and $\mathfrak{p}$ is its generic point. Every closed point

$$(x - a, y - a^2)$$

on the parabola contains $\mathfrak{p}$, so it is a specialization of the parabola's generic point.

Example 9.26 (Two crossing axes)**.** Let

$$R = k[x, y]/(xy).$$

The minimal primes are $(\bar{x})$ and $(\bar{y})$, so the two irreducible components have distinct generic points. The crossing point

$$(\bar{x}, \bar{y})$$

contains both minimal primes and is therefore a closed specialization of both generic points.

Example 9.27 (A product of two fields)**.** For

$$A = k \times k,$$

the spectrum consists of two points,

$$k \times 0, \qquad 0 \times k.$$

Both are maximal, hence closed. Each singleton is also an irreducible component, so each point is generic for its own component. The space is discrete and has no generic point for the whole two-point spectrum.

Example 9.28 (Dual numbers)**.** Let

$$A = k[\varepsilon]/(\varepsilon^2).$$

There is one prime ideal,

$$(\bar{\varepsilon}).$$

It is maximal, so the unique point is closed; it is also generic for the whole one-point spectrum. The nilpotent element $\bar{\varepsilon}$ changes the ring but not this point-set topology, a phenomenon analyzed systematically in VI/10.

Example 9.29 (A nonreduced irreducible spectrum whose generic point is nonzero)**.** Let

$$A = k[x, y]/(x^2).$$

The nilradical is

$$\sqrt{(0)} = (\bar{x}).$$

The quotient

$$A/(\bar{x}) \cong k[y]$$

is a domain, so $(\bar{x})$ is prime. Hence $\operatorname{Spec} A$ is irreducible with generic point $(\bar{x})$. The zero ideal is not prime because $\bar{x}^2 = 0$ with $\bar{x} \neq 0$.

Example 9.30 (A generic point survives every nonempty open)**.** Let $A = k[x, y]$ and consider the generic point (0). Every nonempty principal open $D(f)$ with $f \neq 0$ contains (0) because $f \notin (0)$. Since principal opens form a basis, every nonempty open subset of $\operatorname{Spec} k[x, y]$ contains (0).

Example 9.31 (Quotient spectra preserve the generic-point dictionary)**.** Let $A = k[x, y]$ and $I = (xy)$. Under

$$\operatorname{Spec}(A/I) \cong V_A(I),$$

the minimal primes

$$(\bar{x}), \qquad (\bar{y})$$

correspond to the primes (x) and (y) of A containing I. Thus the generic points of the two components are transported exactly by the quotient-spectrum correspondence.

Example 9.32 (Contraction preserves specialization)**.** Consider

$$\varphi : k[t] \longrightarrow k[x], \qquad t \longmapsto x^2.$$

In $\operatorname{Spec} k[x]$ one has

$$(0) \subset (x - a).$$

Contraction gives

$$(0) \subseteq \varphi^{-1}(x - a).$$

If $a \neq 0$, the contraction is $(t - a^2)$; if $a = 0$, it is (t). Thus the induced map on spectra sends specialization relations to specialization relations.

9.15 Exercises

Exercise 9.33 (VI.09.E1 - Closure of a point)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, prove directly that

$$\overline{\{\mathfrak{p}\}} = V(\mathfrak{p}).$$

Exercise 9.34 (VI.09.E2 - Specialization order)**.** Show that $\mathfrak{q}$ is a specialization of $\mathfrak{p}$ if and only if

$$\mathfrak{p} \subseteq \mathfrak{q}.$$

Exercise 9.35 (VI.09.E3 - Closed sets and specialization)**.** Prove that every Zariski closed subset of $\operatorname{Spec} A$ is stable under specialization.

Exercise 9.36 (VI.09.E4 - Open sets and generalization)**.** Prove that every Zariski open subset of $\operatorname{Spec} A$ is stable under generalization.

Exercise 9.37 (VI.09.E5 - The T_0 property)**.** Prove that $\operatorname{Spec} A$ is T_0.

Exercise 9.38 (VI.09.E6 - The T_1 criterion)**.** Show that $\operatorname{Spec} A$ is T_1 if and only if every prime ideal is maximal.

Exercise 9.39 (VI.09.E7 - Closed points)**.** Show directly from the closure formula that

$$\mathfrak{p} \text{ is closed} \iff \mathfrak{p} \text{ is maximal}.$$

Exercise 9.40 (VI.09.E8 - Generic point of $V(\mathfrak{p})$)**.** If $\mathfrak{p}$ is prime, prove that $\mathfrak{p}$ is the unique generic point of $V(\mathfrak{p})$.

Exercise 9.41 (VI.09.E9 - Irreducible $V(I)$)**.** Prove that $V(I)$ is irreducible if and only if $\sqrt{I}$ is prime, and identify its generic point.

Exercise 9.42 (VI.09.E10 - Generic point of the whole spectrum)**.** Show that $\operatorname{Spec} A$ has a generic point if and only if $\sqrt{(0)}$ is prime. Identify the generic point.

Exercise 9.43 (VI.09.E11 - Minimal primes and components)**.** Show that minimal prime ideals correspond to irreducible components of $\operatorname{Spec} A$ and are their generic points.

Exercise 9.44 (VI.09.E12 - Arithmetic specializations)**.** Describe all generalizations and specializations of (0) and (p) in $\operatorname{Spec} \mathbb{Z}$.

Exercise 9.45 (VI.09.E13 - Closed points of the affine line)**.** Let k be any field. Show that the closed points of $\operatorname{Spec} k[t]$ are precisely the ideals (f) with f irreducible.

Exercise 9.46 (VI.09.E14 - Real closed points)**.** List the closed points of $\operatorname{Spec} \mathbb{R}[t]$ and distinguish the $\mathbb{R}$-rational ones.

Exercise 9.47 (VI.09.E15 - Finite-field closed points)**.** For an irreducible $f \in \mathbb{F}_p[t]$ of degree d, show (f) is a closed point and

$$\mathbb{F}_p[t]/(f) \cong \mathbb{F}_{p^d}.$$

Which such points are $\mathbb{F}_p$-rational?

Exercise 9.48 (VI.09.E16 - Crossing axes)**.** For $R = k[x, y]/(xy)$, show that $(\bar{x})$ and $(\bar{y})$ are the component-generic points and that $(\bar{x}, \bar{y})$ is a closed specialization of both.

Exercise 9.49 (VI.09.E17 - A nonreduced generic point)**.** For $A = k[x, y]/(x^2)$, prove that $\sqrt{(0)} = (\bar{x})$ is prime and hence is the generic point of $\operatorname{Spec} A$.

Exercise 9.50 (VI.09.E18 - Nonempty opens in an irreducible space)**.** Let X be irreducible with generic point η. Prove that every nonempty open $U \subseteq X$ contains η and that η is generic in U.

Exercise 9.51 (VI.09.E19 - Quotient and generic points)**.** Let $I \subseteq A$. Under $\operatorname{Spec}(A/I) \cong V_A(I)$, show that a prime $\mathfrak{p}/I$ is generic for an irreducible closed subset of $\operatorname{Spec}(A/I)$ exactly when $\mathfrak{p}$ is generic for the corresponding irreducible closed subset of $V_A(I)$.

Exercise 9.52 (VI.09.E20 - Maps preserve specialization)**.** For a ring map $\varphi : A \to B$, prove that the induced map

$$\varphi^* : \operatorname{Spec} B \to \operatorname{Spec} A$$

preserves specialization.

9.16 Problems

Problem 9.53. Prove that every irreducible closed subset of a scheme has a unique generic point.

Problem 9.54. Analyze the specialization chain $(0) \subset (x) \subset (x, y)$ in $\operatorname{Spec} k[x, y]$.

Problem 9.55. Characterize when $\operatorname{Spec} A$ has a generic point and identify it.

Problem 9.56. Prove that minimal primes are the generic points of irreducible components.

Problem 9.57. Prove that closed points of $\operatorname{Spec} A$ are exactly maximal ideals.

Problem 9.58. Classify generic and closed points of the affine line over an arbitrary field.

Problem 9.59. Analyze generic and closed points of $\operatorname{Spec}\mathbb{Z}$.

Problem 9.60. Analyze the specialization geometry of $k[x,y]/(xy)$.

Problem 9.61. Prove the T_0 and T_1 assertions for affine spectra.

Problem 9.62. Analyze the generic point of $k[x,y]/(x^2)$.

Problem 9.63. Track generic points through quotient spectra.

Problem 9.64. Show maps of spectra preserve specialization and generic images.

Problem 9.65. Prove every nonempty open of an irreducible space contains its generic point.

Problem 9.66. Compare closed and rational points over $\mathbb{C}$, $\mathbb{R}$, and finite fields.

Problem 9.67. Analyze generic and closed points in finite product rings.

9.17 Challenges

Problem 9.68. Show closed points are dense in spectra of finitely generated algebras over a field.

Problem 9.69. Characterize dense opens using generic points of irreducible components.

Problem 9.70. Show the specialization order alone does not make the Zariski topology Alexandroff.

9.18 Boundary with the next chapters

The topology of individual points is now complete. VI/10 will explain why the underlying topological space cannot detect nilpotent thickening. VI/11 will attach a local ring and a residue field to each prime and make the distinction between rational and non-rational closed points algebraically precise. Later VI/27 will return to integral schemes and complete the legacy generic-point problem by constructing the function field at the generic point.

9.19 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 9.71 (Generic domain)**.** Identify the generic point of $\operatorname{Spec} A$ for a domain.

Hint. Use generic points, closed points, and specialization. ☐

Solution. (0). ☐

Exercise 9.72 (Closed line)**.** Identify closed points of $\operatorname{Spec} k[x]$ over algebraically closed k.

Hint. Use generic points, closed points, and specialization. ☐

Solution. $(x - a)$ for $a \in k$. ☐

Exercise 9.73 (Component generics)**.** Find generic points of $\operatorname{Spec} k[x, y]/(xy)$.

Hint. Use generic points, closed points, and specialization. ☐

Solution. The minimal primes (x) and (y). ☐

Exercise 9.74 (Closure)**.** Compute the closure of $\mathfrak{p}$.

Hint. Use generic points, closed points, and specialization. ☐

Solution. $V(\mathfrak{p})$. ☐

Exercise 9.75 (Specialization)**.** When does $\mathfrak{p}$ specialize to $\mathfrak{q}$?

Hint. Use generic points, closed points, and specialization. ☐

Solution. When $\mathfrak{p} \subseteq \mathfrak{q}$. ☐

Proofs

Exercise 9.76 (Generic minimal prime)**.** Prove: Prove generic points of irreducible components are minimal primes in an affine spectrum.

Hint. Work directly from generic points, closed points, and specialization. ☐

Solution. Components are $V(\mathfrak{p})$ for minimal primes; the closure of $\mathfrak{p}$ is exactly that set. ☐

Exercise 9.77 (Nonempty open)**.** Prove: Prove every nonempty open of an irreducible space contains its generic point when one exists.

Hint. Work directly from generic points, closed points, and specialization. ☐

Solution. If the generic point were omitted, the closed complement would contain its closure, hence the whole space. ☐

Exercise 9.78 (Closed maximal)**.** Prove: Prove closed spectral points are maximal ideals.

Hint. Work directly from generic points, closed points, and specialization. ☐

Solution. Use the closure formula and absence of larger primes. ☐

Exercise 9.79 (Uniqueness generic)**.** Prove: Prove an irreducible closed subset has at most one generic point in a T_0 space such as a spectrum.

Hint. Work directly from generic points, closed points, and specialization. ☐

Solution. Two generic points would have identical closures; the T_0 property forces equality. ☐

Counterexamples and hypothesis tests

Exercise 9.80 (Every irreducible space)**.** Test the claim: every irreducible topological space has a generic point.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general, though spectra of rings are sober and irreducible closed subsets do. ☐

Exercise 9.81 (Generic closed)**.** Test the claim: a generic point is always closed.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False except in a one-point irreducible component. ☐

Exercise 9.82 (Closed generic)**.** Test the claim: every closed point is generic for the whole spectrum.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is generic only for its singleton closure. ☐

Applications and investigations

Exercise 9.83 (Component bookkeeping)**.** Why are generic points efficient labels for components?

Hint. Translate the question into generic points, closed points, and specialization. ☐

Solution. A single prime point has closure equal to the entire irreducible component. ☐

Exercise 9.84 (Degeneration)**.** How does specialization encode degeneration?

Hint. Translate the question into generic points, closed points, and specialization. ☐

Solution. Moving from a generic prime to a containing prime adds equations and produces a more special point. ☐

Challenge problems

Exercise 9.85 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Generic and Closed Points in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Generic and Closed Points, translates it through generic points, closed points, and specialization, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 9.86 (Hypothesis challenge)**.** Choose one structural statement from Generic and Closed Points, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 10

Reduced and Nonreduced Geometry

Purpose and learning goals

The preceding chapters developed the topology of prime spectra. That topology is powerful, but it deliberately forgets one kind of algebraic information: nilpotent elements. The simplest warning is already striking:

$$\operatorname{Spec} k[t]/(t) \qquad \text{and} \qquad \operatorname{Spec} k[t]/(t^2)$$

have the same underlying one-point topological space, even though the two rings are not isomorphic. The second ring contains a nonzero element whose square is zero. Scheme theory remembers this infinitesimal information even when ordinary point-set topology does not.

This chapter makes that phenomenon systematic. We study reduced rings, the nilradical, the reduced quotient, radical ideals, nilpotent thickenings, and the way reduction interacts with localization. We also separate three notions that are often confused: *reduced*, *irreducible*, and *integral*. Finally, we preview a genuinely scheme-theoretic phenomenon: reducedness may be destroyed by a field extension in positive characteristic.

The chapter remains centered on affine spectra and quotient rings. The full local-ring criterion belongs to VI/11, the global structure sheaf to VI/17, affine schemes as locally ringed spaces to VI/18, and integral schemes/function fields to VI/27.

After this chapter, the reader should be able to:

- recognize nilpotent elements and compute the nilradical;
- decide whether a ring or affine quotient is reduced;
- construct and use the reduction A_{red};
- prove the universal property of reduction;
- explain precisely why $\operatorname{Spec} A$ and $\operatorname{Spec} A_{\mathrm{red}}$ have the same topology;
- translate radical ideals into reduced closed affine structures;
- distinguish reduced, irreducible, and integral affine geometry;
- analyze doubled points, double lines, and higher nilpotent thickenings;
- prove that reduction commutes with localization;
- test reducedness locally on a principal-open cover;
- understand how products, polynomial extensions, and quotients affect reducedness;
- recognize why base change can create nilpotents in positive characteristic;

- identify mathematical information that is invisible to the underlying spectrum.

10.1 Nilpotents and the nilradical

Definition 10.1 (Nilpotent element)**.** An element $a \in A$ is *nilpotent* if

$$a^n = 0$$

for some integer $n \geq 1$.

Definition 10.2 (Nilradical)**.** The *nilradical* of A is

$$\operatorname{Nil}(A) = \sqrt{(0)} = \{a \in A : a^n = 0 \text{ for some } n \geq 1\}.$$

The nilpotent elements form an ideal. Indeed, if $a^m = 0$ and $b^n = 0$, then every term in $(a+b)^{m+n}$ contains either at least m factors of a or at least n factors of b, so $(a+b)^{m+n} = 0$. Also $(ra)^m = r^m a^m = 0$ for every $r \in A$.

From VI/08 we retain the fundamental identity

$$\operatorname{Nil}(A) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p}.$$

Thus nilpotents lie in every prime ideal. Equivalently, no point of the spectrum can separate a nilpotent element from zero.

Definition 10.3 (Reduced ring)**.** A commutative ring A is *reduced* if it has no nonzero nilpotent elements, or equivalently if

$$\operatorname{Nil}(A) = 0.$$

Proposition 10.4. *For a ring A, the following are equivalent:*

1. *A is reduced;*
2. *(0) is a radical ideal;*
3. *$a^n = 0$ implies $a = 0$ for every $a \in A$.*

10.2 The reduced quotient

Every ring has a canonical reduced quotient.

Definition 10.5 (Reduction of a ring)**.** Define

$$A_{\mathrm{red}} := A/\operatorname{Nil}(A) = A/\sqrt{(0)}.$$

Proposition 10.6. *The ring A_{red} is reduced.*

Proof. Suppose $\bar{a} \in A/\operatorname{Nil}(A)$ is nilpotent. Then $\bar{a}^n = 0$ for some n, so $a^n \in \operatorname{Nil}(A)$. Hence $(a^n)^m = 0$ for some m, and therefore $a^{nm} = 0$. Thus $a \in \operatorname{Nil}(A)$ and $\bar{a} = 0$. ☐

The reduction is canonical in a stronger sense.

Theorem 10.7 (Universal property of reduction)**.** *Let B be reduced and let $\varphi : A \to B$ be a ring homomorphism. Then φ factors uniquely through the quotient map $\pi : A \to A_{\mathrm{red}}$:*

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d, two heads, "\pi"'] & B \\ A_{\mathrm{red}} \arrow[ur, dashed, "\bar{\varphi}"'] & \end{tikzcd}$$

Proof. If $a \in \mathrm{Nil}(A)$, then $a^n = 0$ for some n, so $\varphi(a)^n = 0$. Because B is reduced, $\varphi(a) = 0$. Therefore $\mathrm{Nil}(A) \subseteq \ker \varphi$, and the universal property of quotients gives a unique map $\bar{\varphi} : A/\mathrm{Nil}(A) \to B$. □

$$\begin{tikzcd} \boxed{\begin{array}{c} A \\ \text{possibly nonreduced} \end{array}} \arrow[r, "\text{quotient}"] \arrow[d, dash, dashed] & \boxed{\begin{array}{c} A_{\mathrm{red}} = A/\mathrm{Nil}(A) \\ \text{nilpotents killed} \end{array}} \arrow[d, dash, dashed] \\ \boxed{\operatorname{Spec} A} & \boxed{\operatorname{Spec} A_{\mathrm{red}}} \arrow[l, "\text{homeomorphism}"] \end{tikzcd}$$

10.3 Why topology cannot see nilpotents

The quotient map $A \to A_{\mathrm{red}}$ does not change the prime spectrum.

Theorem 10.8 (Topological invariance under reduction)**.** *The quotient map induces a natural homeomorphism*

$$\operatorname{Spec} A_{\mathrm{red}} \xrightarrow{\sim} \operatorname{Spec} A.$$

Under this homeomorphism, a prime $\mathfrak{p}/\mathrm{Nil}(A)$ corresponds to $\mathfrak{p}$.

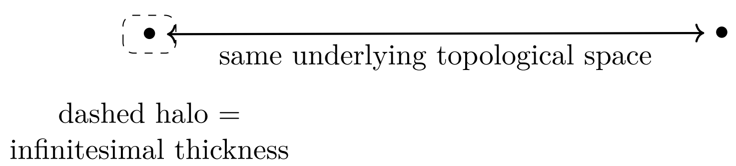
Proof. Prime ideals of $A/\mathrm{Nil}(A)$ correspond to prime ideals of A containing $\mathrm{Nil}(A)$. Every prime ideal of A contains the nilradical, so this gives a bijection on points. For any ideal $J \supseteq \mathrm{Nil}(A)$,

$$V_{A/\mathrm{Nil}(A)}(J/\mathrm{Nil}(A)) \longleftrightarrow V_A(J),$$

so the bijection preserves closed sets and is a homeomorphism. □

Thus the underlying topological space of an affine spectrum remembers all prime ideals and all specialization relations, but forgets nilpotent thickness.

$$\begin{tikzcd} \boxed{\begin{array}{c} k[t]/(t^2) \\ \text{one point} + \text{nilpotent } \bar{t} \end{array}} \arrow[r, "\text{reduction}"] & \boxed{\begin{array}{c} k[t]/(t) \cong k \\ \text{one reduced point} \end{array}} \end{tikzcd}$$



Remark 10.9. The word "same" must be used carefully. Reduction preserves the underlying spectrum as a topological space, but it changes the ring of functions whenever A has nonzero nilpotents. The full scheme therefore changes even though its points and closed subsets do not.

Example 10.10 (Dual numbers)**.** The ring $k[\varepsilon]/(\varepsilon^2)$ has the same one-point spectrum as k, but ε is a nonzero nilpotent. This is the simplest infinitesimal thickening of a point.

10.4 Radical ideals and reduced quotients

Nilpotent structure appears naturally in quotients.

Theorem 10.11. *Let $I \subseteq A$ be an ideal. Then*

$$A/I \text{ is reduced} \iff I \text{ is radical.}$$

Moreover,

$$(A/I)_{\mathrm{red}} \cong A/\sqrt{I}.$$

Proof. The nilradical of A/I is

$$\mathrm{Nil}(A/I) = \sqrt{(0)}_{A/I} = \sqrt{I}/I.$$

Thus A/I is reduced exactly when $\sqrt{I} = I$. Quotienting A/I by its nilradical gives

$$(A/I)/(\sqrt{I}/I) \cong A/\sqrt{I}.$$

□

The topological equality

$$V(I) = V(\sqrt{I})$$

therefore has a scheme-theoretic interpretation: A/I and $A/\sqrt{I}$ define the same underlying closed subset, but the former may carry nilpotent structure.

$$\begin{tikzcd}
I \arrow[r, "\text{remove nilpotent thickness}"] \arrow[d] & \sqrt{I} \arrow[d] \\
\begin{array}{c} A/I \\ \text{possibly nonreduced} \end{array} \arrow[r, "\text{reduction}"] & \begin{array}{c} A/\sqrt{I} \\ \text{reduced} \end{array}
\end{tikzcd}$$

$$V(I) = V(\sqrt{I})$$
same support, different algebra possible

10.5 Reduced, irreducible, and integral are different notions

A reduced ring need not be a domain. The basic example is

$$k[x,y]/(xy),$$

which is reduced but has zero divisors. Its spectrum is the union of two irreducible components.

Conversely, an irreducible spectrum need not come from a reduced ring. For example,

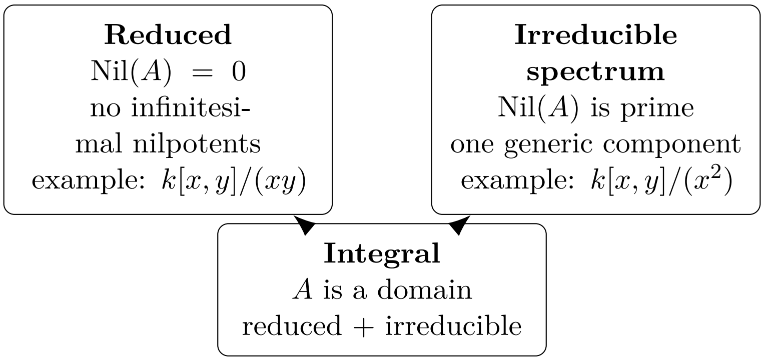
$$A = k[x,y]/(x^2)$$

has reduction $k[y]$, so $\operatorname{Spec} A$ is homeomorphic to the affine line and is irreducible, but $\bar{x} \neq 0$ and $\bar{x}^2 = 0$.

Theorem 10.12 (Affine integral criterion)**.** *For a ring A, the following are equivalent:*

1. *A is an integral domain;*
2. *A is reduced and* $\operatorname{Spec} A$ *is irreducible.*

Proof. If A is a domain, then it is reduced and (0) is prime, so $\operatorname{Spec} A$ is irreducible. Conversely, if A is reduced, then $\mathrm{Nil}(A) = 0$. If $\operatorname{Spec} A$ is irreducible, VI/05 shows that $\mathrm{Nil}(A)$ is prime. Hence (0) is prime, so A is a domain. □



Neither implication between the two top boxes holds in general.

The corresponding theorem for arbitrary schemes is a substantive legacy problem and is reserved for VI/27, where the definition of an integral scheme and its function field are developed globally.

10.6 Nilpotent thickenings

The quotient A/I depends on more than the support $V(I)$. If I is not radical, then A/I contains nilpotents and should be thought of as a *thickened* version of the reduced closed set defined by $\sqrt{I}$.

The simplest family is

$$A_n = k[t]/(t^n), \qquad n \geq 1.$$

Every A_n has a one-point spectrum, but for $n > 1$ the class of t is a nonzero nilpotent. The integer n records a depth of infinitesimal structure invisible to the underlying point.

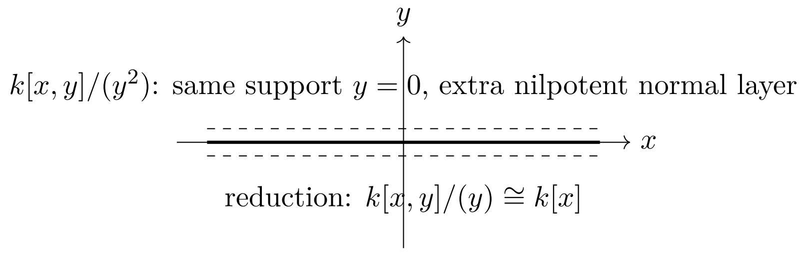
A geometric two-variable example is the double line

$$k[x,y]/(y^2).$$

Its reduction is

$$k[x,y]/(y) \cong k[x].$$

Both have the same topological support $y = 0$, but only the first remembers a first-order normal thickening of that line.



Definition 10.13 (Nilpotent thickening, affine form)**.** Let $I \subseteq A$ be a nilpotent ideal, meaning $I^N = 0$ for some N. The quotient map

$$A \twoheadrightarrow A/I$$

induces a homeomorphism

$$\operatorname{Spec}(A/I) \xrightarrow{\sim} \operatorname{Spec} A.$$

We call this an affine nilpotent thickening relationship.

The nilpotence hypothesis is stronger than necessary for the homeomorphism: it is enough that every element of I be nilpotent, i.e. $I \subseteq \operatorname{Nil}(A)$. In Noetherian rings the nilradical is nilpotent, but in arbitrary rings it need not be; Challenge VI.10.C1 explores this distinction.

Example 10.14 (Double line)**.** The quotient $k[x,y]/(y^2)$ defines a double structure supported on the line $y = 0$. Its reduction is $k[x,y]/(y) \cong k[x]$.

10.7 Reduction and localization

Reducedness behaves well under localization.

Proposition 10.15. *If A is reduced and $S \subseteq A$ is multiplicatively closed, then $S^{-1}A$ is reduced.*

Proof. Suppose $(a/s)^n = 0$ in $S^{-1}A$. Then there exists $u \in S$ such that $ua^n = 0$. Hence $(ua)^n = u^n a^n = 0$. Since A is reduced, $ua = 0$, so $a/s = 0$ in $S^{-1}A$. □

Theorem 10.16 (Reduction commutes with localization)**.** *For $f \in A$, there is a natural isomorphism*

$$(A_{\mathrm{red}})_{\bar{f}} \cong (A_f)_{\mathrm{red}}.$$

More generally, for every multiplicative set S,

$$S^{-1}(A_{\mathrm{red}}) \cong (S^{-1}A)_{\mathrm{red}}.$$

Proof. Localization commutes with quotients:

$$S^{-1}(A/\operatorname{Nil}(A)) \cong S^{-1}A/S^{-1}\operatorname{Nil}(A).$$

The nilradical of $S^{-1}A$ is exactly $S^{-1}\operatorname{Nil}(A)$. Therefore the quotient on the right is the reduction of $S^{-1}A$. □

$$\begin{tikzcd}
A \arrow[r, two heads] \arrow[d] & A_{\mathrm{red}} \arrow[d] \\
A_f \arrow[r, two heads] & (A_{\mathrm{red}})_{\bar{f}} \cong (A_f)_{\mathrm{red}}
\end{tikzcd}$$

Reducedness is also local in the Zariski topology.

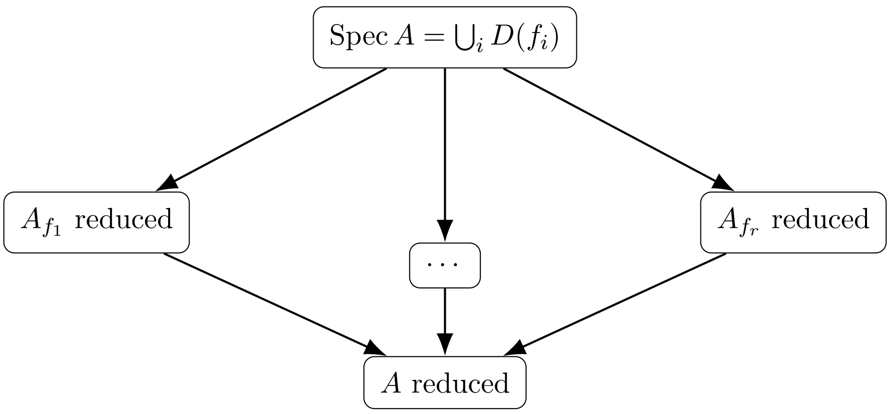
Theorem 10.17 (Local criterion for reducedness)**.** *For a ring A, the following are equivalent:*

1. *A is reduced;*
2. *$A_{\mathfrak{p}}$ is reduced for every prime $\mathfrak{p}$;*
3. *there exists a principal-open cover $\operatorname{Spec} A = \bigcup_{i=1}^{r} D(f_i)$ such that each A_{f_i} is reduced.*

Proof. If A is reduced, then every localization of A is reduced, so (1) implies (2).

Assume (2), and let $a \in A$ satisfy $a^n = 0$. Then $a/1 = 0$ in $A_{\mathfrak{p}}$ for every prime $\mathfrak{p}$. If $a \neq 0$, its annihilator $\operatorname{Ann}(a)$ is a proper ideal, so it is contained in some prime $\mathfrak{p}$. But $a/1 = 0$ in $A_{\mathfrak{p}}$ means that some $s \notin \mathfrak{p}$ satisfies $sa = 0$, hence $s \in \operatorname{Ann}(a) \subseteq \mathfrak{p}$, a contradiction. Thus $a = 0$, and (2) implies (1).

Clearly (1) implies (3), for instance by the one-set principal-open cover $D(1) = \operatorname{Spec} A$. Finally assume (3). If $a^n = 0$ in A, then $a/1 = 0$ in every A_{f_i}. Hence for each i there exists N_i with $f_i^{N_i} a = 0$. Since the $D(f_i)$ cover $\operatorname{Spec} A$, the ideal $(f_1, \ldots, f_r)$ has radical (1). Therefore some power of $(f_1, \ldots, f_r)$ is contained in the ideal $(f_1^{N_1}, \ldots, f_r^{N_r})$ after enlarging the exponents if necessary; because the radical is (1), this latter ideal is the unit ideal. Thus a linear combination of elements annihilating a equals 1, and consequently $a = 0$. Hence (3) implies (1). ☐



10.8 Basic constructions and reducedness

Proposition 10.18 (Products)**.** *A finite product*

$$A_1 \times \cdots \times A_r$$

is reduced if and only if every A_i is reduced.

Proof. An element $(a_1, \ldots, a_r)$ is nilpotent exactly when each coordinate a_i is nilpotent. ☐

Proposition 10.19 (Polynomial extensions)**.** *The polynomial ring $A[t]$ is reduced if and only if A is reduced.*

Proof. If $A[t]$ is reduced then its subring of constant polynomials is reduced. Conversely, assume A is reduced and $f(t)^n = 0$. Every prime $\mathfrak{p}$ of A gives a map

$$A[t] \to (A/\mathfrak{p})[t],$$

and the target is a domain. Hence the image of f is zero for every prime $\mathfrak{p}$, so every coefficient of f lies in every prime of A, hence in $\mathrm{Nil}(A) = 0$. Thus $f = 0$. □

Warning 10.20 (Quotients)*.* A quotient of a reduced ring need not be reduced. For example, $k[t]$ is reduced but

$$k[t]/(t^2)$$

is not. The correct criterion is that the quotient ideal be radical.

10.9 Finite reduced algebras and discrete geometry

Reducedness becomes especially concrete in finite-dimensional algebras.

Proposition 10.21. *Let k be algebraically closed and let A be a finite-dimensional commutative k-algebra. If A is reduced, then*

$$A \cong k^r$$

for some $r \geq 0$.

Proof. A finite-dimensional k-algebra is Artinian, so it has finitely many maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$, and its nilradical equals the intersection of these maximal ideals. Since A is reduced,

$$\bigcap_i \mathfrak{m}_i = 0.$$

Distinct maximal ideals are comaximal, so the Chinese remainder theorem gives

$$A \cong \prod_i A/\mathfrak{m}_i.$$

Each quotient is a finite field extension of the algebraically closed field k, hence is k. □

Thus a finite reduced affine k-scheme over an algebraically closed field is simply a finite collection of reduced points. Nonreduced finite algebras are exactly where infinitesimal multiplicity enters.

Example 10.22 (Reduction preserves points)**.** Passing from A to $A_{\mathrm{red}} = A/\sqrt{0}$ preserves the prime spectrum as a topological space because every prime contains the nilradical.

10.10 Infinitesimal information carried by dual numbers

The dual numbers

$$k[\varepsilon]/(\varepsilon^2)$$

have only one point, but they can record first-order variation. A k-algebra map

$$k[t] \longrightarrow k[\varepsilon]/(\varepsilon^2)$$

is determined by

$$t \longmapsto a + b\varepsilon.$$

All values of b induce the same map on underlying topological points, namely the point $t = a$, but different values of b are different ring maps. The coefficient b is an infinitesimal direction invisible to point-set topology.

$$\begin{tikzcd} k[t] \arrow[r, "t \mapsto a + b\varepsilon"] & k[\varepsilon]/(\varepsilon^2) \end{tikzcd}$$

$$\begin{tikzcd} \operatorname{Spec} k[t] & \bullet \arrow[l, "{(t-a),\ \text{independent of } b}"'] \end{tikzcd}$$

b survives algebraically
but is invisible topologically

This observation is only a preview. VI/32 will turn dual-number maps into tangent vectors and cotangent-space calculations.

10.11 Reducedness and base change: a warning

Localization and polynomial extension preserve reducedness, but arbitrary field extension need not.

Let

$$k = \mathbb{F}_p(s), \qquad A = k[x]/(x^p - s).$$

The polynomial $x^p - s$ is irreducible over k, so A is a field and hence reduced. Now extend scalars to

$$K = \mathbb{F}_p(s^{1/p}).$$

Writing $\alpha = s^{1/p} \in K$, characteristic p gives

$$x^p - s = (x - \alpha)^p.$$

Therefore

$$A \otimes_k K \cong K[x]/((x - \alpha)^p),$$

which is nonreduced.

$$\begin{tikzcd} \boxed{\begin{array}{c} k = \mathbb{F}_p(s) \\ A = k[x]/(x^p - s) \\ \text{field, hence reduced} \end{array}} \arrow[r, "\text{inseparable extension}"] & \boxed{\begin{array}{c} K = \mathbb{F}_p(s^{1/p}) \\ A \otimes_k K \cong K[x]/((x-\alpha)^p) \\ \text{nonreduced} \end{array}} \end{tikzcd}$$

This motivates a later refinement: a k-algebra is *geometrically reduced* if it remains reduced after suitable field extensions. The systematic theory belongs with base change in VI/24.

10.12 Comparison chart and common pitfalls

Ring	Reduced?	Irreducible?	Key feature
$k[t]$	yes	yes	domain
$k[t]/(t^2)$	no	yes	doubled point
$k[x, y]/(xy)$	yes	no	two reduced components
$k[x, y]/(x^2)$	no	yes	thickened line
$k \times k$	yes	no	two disconnected points
$\mathbb{Z}/12\mathbb{Z}$	no	no	nilpotents and several primes

Keep the following distinctions explicit:

- "Reduced" means no nonzero nilpotents. It says nothing by itself about the number of irreducible components.
- "Irreducible spectrum" is a topological condition. It does not force the ring to be reduced.
- The spectrum cannot detect whether an ideal is I or $\sqrt{I}$; quotient rings can.
- A one-point spectrum need not be a field and need not be reduced.
- A quotient of a reduced ring need not be reduced.
- Reducedness survives localization but can fail after inseparable base change.
- In a nonreduced irreducible ring, the generic point is $\mathrm{Nil}(A)$, not necessarily (0).

10.13 Worked examples

Example 10.23 (A field and a polynomial ring)**.** Every field K is reduced. More generally, every integral domain is reduced, so

$$k[t], \qquad k[x,y], \qquad \mathbb{Z}$$

are reduced. Reducedness is weaker than being a domain: later examples exhibit reduced rings with zero divisors.

Example 10.24 (Squarefree versus nonsquarefree integers)**.** The ring $\mathbb{Z}/6\mathbb{Z}$ is reduced because

$$(6) = (2) \cap (3)$$

is a radical ideal of $\mathbb{Z}$. By contrast $\mathbb{Z}/12\mathbb{Z}$ is not reduced: the nonzero class $\bar{6}$ satisfies

$$\bar{6}^2 = \overline{36} = 0.$$

More generally, $\mathbb{Z}/n\mathbb{Z}$ is reduced exactly when n is squarefree.

Example 10.25 (Dual numbers)**.** Let

$$A = k[\varepsilon]/(\varepsilon^2).$$

Then $\bar{\varepsilon} \neq 0$ but $\bar{\varepsilon}^2 = 0$, so A is nonreduced. Its nilradical is $(\bar{\varepsilon})$ and

$$A_{\mathrm{red}} \cong k.$$

Both spectra consist of one point.

Example 10.26 (Higher doubled points)**.** For

$$A_n = k[t]/(t^n), \qquad n \geq 2,$$

the class $\bar{t}$ is nilpotent of order n. The only prime ideal is $(\bar{t})$, so all A_n have the same one-point spectrum and the same reduction k. Their rings are nonisomorphic for different n because the maximum nilpotency order changes.

Example 10.27 (A double line)**.** Set

$$A = k[x, y]/(y^2).$$

The nilradical is $(\bar{y})$ and

$$A_{\text{red}} \cong k[x, y]/(y) \cong k[x].$$

Thus the double line and the ordinary line $y = 0$ have identical prime spectra but different coordinate rings.

Example 10.28 (Crossing axes: reduced but reducible)**.** The ring

$$A = k[x, y]/(xy)$$

is reduced because (xy) is radical: in the UFD $k[x, y]$, the element xy is squarefree. But A is not a domain because

$$\bar{x}\bar{y} = 0$$

with $\bar{x}, \bar{y} \neq 0$. Correspondingly,

$$\operatorname{Spec} A = V(\bar{x}) \cup V(\bar{y})$$

is reducible.

Example 10.29 (Irreducible but nonreduced)**.** Let

$$A = k[x, y]/(x^2).$$

Then $\operatorname{Nil}(A) = (\bar{x})$ and

$$A_{\text{red}} \cong k[y].$$

Hence $\operatorname{Spec} A$ is homeomorphic to the irreducible affine line. Nevertheless A is nonreduced. Its generic point is $(\bar{x})$, not (0).

Example 10.30 (Computing a reduction)**.** Let

$$A = k[x, y]/(x^2, xy).$$

The ideal (x^2, xy) has radical (x), because every common zero of x^2 and xy lies on $x = 0$, and algebraically $x^2 \in I$ implies $x \in \sqrt{I}$. Therefore

$$A_{\text{red}} \cong k[x, y]/(x) \cong k[y].$$

The nilpotent class $\bar{x}$ is killed by reduction.

Example 10.31 (A reduced product)**.** The ring

$$k \times k$$

is reduced because each factor is reduced. It has zero divisors and a disconnected spectrum consisting of two points. Thus reducedness does not imply connectedness or irreducibility.

Example 10.32 (A quotient of a reduced ring can be nonreduced)**.** The polynomial ring $k[t]$ is reduced, but

$$k[t]/(t^2)$$

is not. The issue is not the ambient ring; it is that the ideal (t^2) is not radical. Replacing it by its radical (t) produces the reduced quotient.

Example 10.33 (Localization can remove nilpotent structure)**.** Let

$$A = k[x, y]/(x^2, xy).$$

The ring is nonreduced because $\bar{x}^2 = 0$. On the principal open $D(\bar{y})$, however, $\bar{y}$ becomes invertible and the relation $\bar{x}\bar{y} = 0$ forces $\bar{x} = 0$. Thus

$$A_{\bar{y}} \cong k[y, y^{-1}],$$

which is reduced. Nilpotent structure may therefore be supported on a proper closed subset.

Example 10.34 (Reduction commutes with localization)**.** For the same ring

$$A = k[x, y]/(x^2, xy),$$

we have $A_{\mathrm{red}} \cong k[y]$. Localizing at y gives

$$(A_{\mathrm{red}})_y \cong k[y, y^{-1}].$$

On the other hand $A_y \cong k[y, y^{-1}]$ is already reduced. Hence in this example

$$(A_y)_{\mathrm{red}} \cong (A_{\mathrm{red}})_y.$$

Example 10.35 (A finite reduced algebra)**.** Assume k is algebraically closed. The quotient

$$A = k[t]/((t - a)(t - b)(t - c))$$

with a, b, c distinct is reduced. The Chinese remainder theorem gives

$$A \cong k \times k \times k.$$

Geometrically this is a finite set of three reduced points.

Example 10.36 (Reducedness can fail after field extension)**.** Let $k = \mathbb{F}_p(s)$ and

$$A = k[x]/(x^p - s).$$

The polynomial $x^p - s$ is irreducible over k, so A is a field. After adjoining $\alpha = s^{1/p}$,

$$A \otimes_k k(\alpha) \cong k(\alpha)[x]/((x - \alpha)^p),$$

which is nonreduced. This is the basic warning behind geometric reducedness.

10.14 Exercises

Exercise 10.37 (Nilpotents form an ideal)**.** Prove directly that the nilpotent elements of a commutative ring form an ideal.

Exercise 10.38 (Squarefree integers)**.** Prove that $\mathbb{Z}/n\mathbb{Z}$ is reduced if and only if n is squarefree.

Exercise 10.39 (Reduction is reduced)**.** Prove that $A/\operatorname{Nil}(A)$ is reduced without using the proposition in the text.

Exercise 10.40 (Universal property)**.** Let B be reduced. Show that every ring map $A \to B$ kills $\operatorname{Nil}(A)$ and therefore factors uniquely through A_{red}.

Exercise 10.41 (The same spectrum)**.** Show directly that prime ideals of A and A_{red} correspond bijectively and that the correspondence preserves inclusions.

Exercise 10.42 (Reduced quotient criterion)**.** Prove that A/I is reduced if and only if I is radical.

Exercise 10.43 (Compute three reductions)**.** Compute the reductions of

$$k[t]/(t^5), \qquad k[x,y]/(x^3,xy), \qquad \mathbb{Z}/72\mathbb{Z}.$$

Exercise 10.44 (Products)**.** Prove that $A \times B$ is reduced if and only if both A and B are reduced.

Exercise 10.45 (Polynomial extensions)**.** Prove that $A[t]$ is reduced if and only if A is reduced.

Exercise 10.46 (Localization preserves reducedness)**.** Let S be multiplicatively closed. Prove that A reduced implies $S^{-1}A$ reduced.

Exercise 10.47 (Reduction and localization)**.** Construct the natural isomorphism

$$S^{-1}(A_{\mathrm{red}}) \cong (S^{-1}A)_{\mathrm{red}}.$$

Exercise 10.48 (A principal-open local test)**.** Assume $D(f_1), \dots, D(f_r)$ cover $\operatorname{Spec} A$ and every A_{f_i} is reduced. Prove that A is reduced.

Exercise 10.49 (Squarefree one-variable quotients)**.** Let k be a field and $0 \neq f \in k[t]$. Show that $k[t]/(f)$ is reduced if and only if f has no repeated irreducible factor.

Exercise 10.50 (Crossing axes)**.** Prove directly that (xy) is radical in $k[x,y]$ and conclude that $k[x,y]/(xy)$ is reduced.

Exercise 10.51 (An irreducible nonreduced spectrum)**.** Show that $\operatorname{Spec} k[x,y]/(x^2)$ is irreducible and identify its generic point.

Exercise 10.52 (All thickened points have the same spectrum)**.** For $n \geq 1$, prove that

$$\operatorname{Spec} k[t]/(t^n)$$

has one point and that all these spectra are homeomorphic.

Exercise 10.53 (Maps to reduced rings)**.** Let $N \subseteq A$ be any ideal consisting entirely of nilpotent elements. Show that every map from A to a reduced ring kills N.

Exercise 10.54 (Nil ideals do not change the spectrum)**.** If $I \subseteq \operatorname{Nil}(A)$, prove that

$$\operatorname{Spec}(A/I) \to \operatorname{Spec} A$$

is a homeomorphism.

Exercise 10.55 (Prime-local criterion)**.** Prove that A is reduced if and only if $A_{\mathfrak{p}}$ is reduced for every $\mathfrak{p} \in \operatorname{Spec} A$.

Exercise 10.56 (Dual-number maps)**.** Describe all k-algebra homomorphisms

$$k[t] \longrightarrow k[\varepsilon]/(\varepsilon^2).$$

Which of them induce the same map on underlying topological points?

10.15 Problems

Problem 10.57. Give and analyze a scheme that is not reduced, preserving the reducedness subproblem of a legacy counterexample collection.

Problem 10.58. Compare $\operatorname{Spec} k[t]/(t)$ and $\operatorname{Spec} k[t]/(t^n)$ topologically and algebraically.

Problem 10.59. Prove the universal property and idempotence of reduction.

Problem 10.60. Characterize radical ideals by reduced quotients and reduced closed structures.

Problem 10.61. Analyze the double line $k[x, y]/(y^2)$ and its reduction.

Problem 10.62. Prove the affine criterion: domain if and only if reduced with irreducible spectrum.

Problem 10.63. Prove reducedness is local on prime localizations and on finite principal-open covers.

Problem 10.64. Prove reduction commutes with localization and apply it to a nonreduced example.

Problem 10.65. Characterize squarefree principal hypersurfaces as reduced.

Problem 10.66. Classify finite-dimensional reduced algebras over an algebraically closed field.

Problem 10.67. Show dual-number maps record first-order data invisible on underlying points.

Problem 10.68. Characterize quotient maps by nil ideals that leave the entire spectrum unchanged.

Problem 10.69. Show reduction preserves idempotents and connected-component decompositions.

Problem 10.70. Construct a reduced algebra that becomes nonreduced after a purely inseparable field extension.

10.16 Challenges

Problem 10.71. Construct a ring whose nilradical is not nilpotent even though its spectrum is a single point.

Problem 10.72. Characterize ideals I for which $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$ is a homeomorphism onto all of $\operatorname{Spec} A$.

Problem 10.73. Analyze the entire tower $k[t]/(t^n)$ and identify invariants that distinguish its infinitesimal layers.

10.17 Boundary with the next chapters

VI/10 has shown exactly what prime-spectrum topology forgets. VI/11 now restores local algebra at each point by attaching the local ring $A_{\mathfrak{p}}$ and residue field $\kappa(\mathfrak{p})$. VI/17 will organize regular functions into the structure sheaf, and VI/18 will package the topological space together with that sheaf as an affine scheme. Later VI/24 will return to the base-change warning and distinguish reduced from geometrically reduced objects; VI/27 will treat integral schemes and function fields globally.

10.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 10.74 (Dual reduction)**.** Reduce $k[\varepsilon]/(\varepsilon^2)$.

Hint. Use nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. The reduction is k. ☐

Exercise 10.75 (Double line)**.** Reduce $k[x, y]/(y^2)$.

Hint. Use nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. It becomes $k[x, y]/(y)$. ☐

Exercise 10.76 (Nilradical)**.** What ideal is killed in forming A_{red}?

Hint. Use nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. $\sqrt{(0)}$. ☐

Exercise 10.77 (Spectrum)**.** How does the underlying spectrum change under reduction?

Hint. Use nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. It does not change topologically. ☐

Exercise 10.78 (Reduced test)**.** When is A reduced?

Hint. Use nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. When it has no nonzero nilpotents. ☐

Proofs

Exercise 10.79 (Prime containment)**.** Prove: Prove every prime ideal contains every nilpotent element.

Hint. Work directly from nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. If $a^n = 0 \in \mathfrak{p}$, repeated primality forces $a \in \mathfrak{p}$. ☐

Exercise 10.80 (Spectrum reduction)**.** Prove: Prove $\operatorname{Spec} A_{\text{red}} \to \operatorname{Spec} A$ is a homeomorphism.

Hint. Work directly from nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. Use quotient-prime correspondence because every prime contains the nilradical. ☐

Exercise 10.81 (Double-line support)**.** Prove: Prove $V(y^2) = V(y)$ as subsets of affine space.

Hint. Work directly from nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. A scalar square vanishes exactly when the scalar itself vanishes. ☐

Exercise 10.82 (Reduced quotient)**.** Prove: Prove $A/\sqrt{0}$ is reduced.

Hint. Work directly from nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. If a coset has a power zero, its representative lies in the radical and the coset is zero. ☐

Counterexamples and hypothesis tests

Exercise 10.83 (Topology enough)**.** Test the claim: the underlying topological space detects nilpotents.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 10.84 (Nonreduced points)**.** Test the claim: a nonreduced scheme must have more topological points than its reduction.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; the point set is unchanged. ☐

Exercise 10.85 (Domain equivalence)**.** Test the claim: reduced is equivalent to integral domain.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; products of fields are reduced with zero divisors. ☐

Applications and investigations

Exercise 10.86 (Infinitesimals)**.** What do nilpotents encode geometrically?

Hint. Translate the question into nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. They encode infinitesimal thickening information invisible in the underlying point set. ☐

Exercise 10.87 (Simplification)**.** When is passing to the reduction useful?

Hint. Translate the question into nilpotents, reduced schemes, and infinitesimal thickening. ☐

Solution. When only the underlying reduced geometry or topological support matters. ☐

Challenge problems

Exercise 10.88 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Reduced and Nonreduced Geometry in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Reduced and Nonreduced Geometry, translates it through nilpotents, reduced schemes, and infinitesimal thickening, and checks the correspondence in both algebraic and geometric language. □

Exercise 10.89 (Hypothesis challenge)**.** Choose one structural statement from Reduced and Nonreduced Geometry, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 11

Local Rings and Residue Fields

Purpose and learning goals

The spectrum of a ring tells us where the prime ideals are and how they specialize, but it still treats each point largely as a topological position. Local algebra adds the next layer. Given a point $\mathfrak{p} \in \operatorname{Spec} A$, we invert every element that does not vanish at that point and obtain the local ring

$$A_{\mathfrak{p}}.$$

This ring records algebraic information visible arbitrarily near $\mathfrak{p}$. Its unique maximal ideal is $\mathfrak{p}A_{\mathfrak{p}}$, and quotienting by that maximal ideal gives the residue field

$$\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

The residue field is the correct field of values at a scheme-theoretic point.

This chapter is intentionally example-rich. In addition to the main worked-example section, there is a separate *Advanced examples* section containing arithmetic, reducible, nonreduced, and singular local rings. The examples are drawn from and expand the legacy commutative-algebra corpus, especially its detailed calculations in $k[t]$, $\mathbb{R}[t]$, $\mathbb{Z}$, finite fields, crossing components, and mixed rings such as $\mathbb{Z}[x]$.

We remain primarily in affine algebra. The formal construction of stalks from sheaves comes in VI/13, the structure sheaf in VI/17, affine schemes as locally ringed spaces in VI/18, fibers in VI/25, integral schemes and function fields in VI/27, and tangent spaces in VI/32. Where these ideas naturally appear here, we mark them as previews rather than importing the later chapters wholesale.

After this chapter, the reader should be able to:

- recognize and characterize local rings abstractly;
- compute units and the unique maximal ideal in a local ring;
- construct $A_{\mathfrak{p}}$ and determine its units and maximal ideal;
- identify $\operatorname{Spec} A_{\mathfrak{p}}$ with the generalizations of $\mathfrak{p}$;
- compute residue fields by both localization and quotient methods;
- prove $\kappa(\mathfrak{p}) \cong \operatorname{Frac}(A/\mathfrak{p})$;
- distinguish rational, non-rational closed, and generic points via residue fields;
- compute local rings at smooth points, crossings, generic points, and nonreduced points;
- compare local rings before and after restricting to a principal open;

- combine quotients and localization correctly;
- recognize local homomorphisms and the induced maps on residue fields;
- use local rings to see when components disappear after localization;
- work comfortably with arithmetic local rings such as $\mathbb{Z}_{(p)}$;
- interpret the short exact sequence $0 \to \mathfrak{m} \to A \to A/\mathfrak{m} \to 0$ in local geometry;
- avoid common confusions between A_f, $A_\mathfrak{p}$, $A/\mathfrak{p}$, and $\kappa(\mathfrak{p})$.

11.1 Local rings as algebraic objects

Definition 11.1 (Local ring)**.** A commutative ring R is *local* if it has a unique maximal ideal. We usually write

$$(R, \mathfrak{m})$$

for a local ring together with its maximal ideal.

The most useful test for locality is a statement about units.

Theorem 11.2 (Units and nonunits in a local ring)**.** *For a commutative ring R, the following are equivalent:*

1. *R is local;*
2. *the set of nonunits of R is an ideal;*
3. *there exists a proper ideal $\mathfrak{m}$ such that every element of $R \setminus \mathfrak{m}$ is a unit;*
4. *for every $a \in R$, at least one of a and $1 - a$ is a unit.*

When these conditions hold, the nonunits are exactly the unique maximal ideal $\mathfrak{m}$.

Proof. If R has unique maximal ideal $\mathfrak{m}$, every nonunit lies in some maximal ideal and therefore lies in $\mathfrak{m}$; conversely no element of $\mathfrak{m}$ is a unit. Hence the nonunits are exactly $\mathfrak{m}$ and form an ideal. This proves (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (3) is immediate.

If (3) holds and $\mathfrak{n}$ is any maximal ideal, then $\mathfrak{n}$ contains no unit, so $\mathfrak{n} \subseteq \mathfrak{m}$. Maximality forces $\mathfrak{n} = \mathfrak{m}$, proving (3) $\Rightarrow$ (1).

Finally, if R is local and both a and $1 - a$ were nonunits, then both would lie in $\mathfrak{m}$, forcing $1 \in \mathfrak{m}$. Conversely, assume condition (4). If two maximal ideals $\mathfrak{m} \neq \mathfrak{n}$ existed, choose $a \in \mathfrak{m}$ and $b \in \mathfrak{n}$ with $a + b = 1$ after using $\mathfrak{m} + \mathfrak{n} = R$. Then a and $1 - a = b$ would both be nonunits, contradiction. ☐

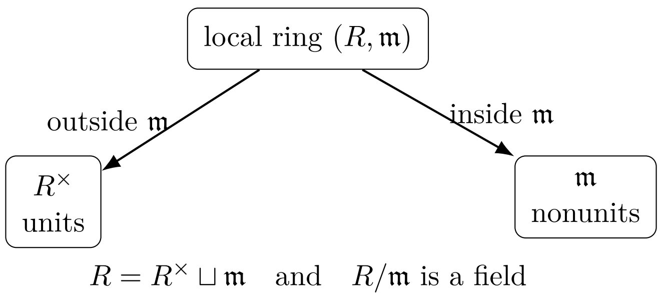


Figure 11.1: The unit/nonunit dichotomy in a local ring.

Corollary 11.3. *If* $(R, \mathfrak{m})$ *is local, then*

$$R^\times = R \setminus \mathfrak{m}.$$

In particular, every element is either a unit or lies in the maximal ideal.

11.2 Localization at a prime

Let A be a ring and $\mathfrak{p} \in \operatorname{Spec} A$. We localize at the multiplicative set

$$S = A \setminus \mathfrak{p}.$$

The resulting ring is

$$A_\mathfrak{p} = S^{-1}A.$$

An element is represented by a fraction a/s with $s \notin \mathfrak{p}$.

Theorem 11.4 (Prime localization is local)**.** *The ring* $A_\mathfrak{p}$ *is local. Its unique maximal ideal is*

$$\mathfrak{p}A_\mathfrak{p} = \left\{ \frac{a}{s} : a \in \mathfrak{p},\ s \notin \mathfrak{p} \right\}.$$

Moreover,

$$\frac{a}{s} \in A_\mathfrak{p} \quad \textit{is a unit} \quad \iff \quad a \notin \mathfrak{p}.$$

Proof. If $a \notin \mathfrak{p}$, then a belongs to the multiplicative set, so $(a/s)^{-1} = s/a$. If $a \in \mathfrak{p}$ and a/s were a unit, there would be b/t with $ab/(st) = 1$. After clearing denominators, some element outside $\mathfrak{p}$ would lie in $(a) \subseteq \mathfrak{p}$, impossible. Thus the nonunits are exactly $\mathfrak{p}A_\mathfrak{p}$, and the preceding theorem shows that this is the unique maximal ideal. $\square$

$$\begin{array}{ccccc}
A & \longrightarrow & A_\mathfrak{p} & \twoheadrightarrow & \kappa(\mathfrak{p}) \\
\Downarrow & & \Downarrow & & \Downarrow \\
\text{global functions} & & \text{functions regular near } \mathfrak{p} & & \text{values at } \mathfrak{p}
\end{array}$$

Figure 11.2: Global ring → local ring → residue field.

11.3 The local spectrum and generalizations

Localization at a prime has a clean geometric meaning.

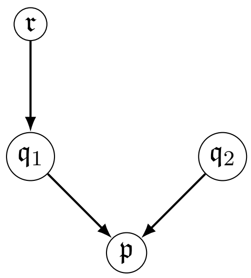
Theorem 11.5 (Prime ideals of $A_\mathfrak{p}$)**.** *Contraction and extension give an inclusion-preserving bijection*

$$\operatorname{Spec} A_\mathfrak{p} \longleftrightarrow \{\mathfrak{q} \in \operatorname{Spec} A : \mathfrak{q} \subseteq \mathfrak{p}\}.$$

Under the canonical map $\operatorname{Spec} A_\mathfrak{p} \to \operatorname{Spec} A$*, the image consists exactly of the generalizations of* $\mathfrak{p}$*.*

Proof. Prime ideals of a localization $S^{-1}A$ correspond to prime ideals of A disjoint from S. Here $S = A \setminus \mathfrak{p}$, so disjointness means every element of $\mathfrak{q}$ lies in $\mathfrak{p}$, i.e. $\mathfrak{q} \subseteq \mathfrak{p}$. $\square$

Thus the unique closed point of $\operatorname{Spec} A_\mathfrak{p}$ is $\mathfrak{p}A_\mathfrak{p}$, while smaller primes appear as more generic points.



Spec $A_{\mathfrak{p}}$ consists of
all primes $\mathfrak{q} \subseteq \mathfrak{p}$;
$\mathfrak{p}A_{\mathfrak{p}}$ is the unique closed point.

Figure 11.3: The local spectrum retains the generalizations of the chosen point.

Remark 11.6 (A local spectrum is not usually an open neighborhood). The image $\{\mathfrak{q} : \mathfrak{q} \subseteq \mathfrak{p}\}$ need not be open in Spec A. Localization at a single element f produces the open subset $D(f)$; localization at a prime is a different operation that inverts *all* elements outside $\mathfrak{p}$.

Example 11.7 (Local ring of the affine line)**.** At the point $(x - a)$ of Spec $k[x]$, the local ring is $k[x]_{(x-a)}$. Its maximal ideal is generated by $x - a$, and the residue field is k.

11.4 Residue fields

Definition 11.8 (Residue field)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, the *residue field at* $\mathfrak{p}$ is

$$\kappa(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

The canonical map

$$A \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p})$$

has kernel exactly $\mathfrak{p}$. Thus elements of A can be "evaluated" at $\mathfrak{p}$, but the values need not lie in the original ground field.

Theorem 11.9 (Computational formula for residue fields)**.** *For every prime ideal* $\mathfrak{p} \subseteq A$*,*

$$\boxed{\kappa(\mathfrak{p}) \cong \operatorname{Frac}(A/\mathfrak{p}).}$$

Proof. Localize the quotient map $A \to A/\mathfrak{p}$ at $S = A \setminus \mathfrak{p}$. This gives

$$A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong S^{-1}(A/\mathfrak{p}).$$

Since $A/\mathfrak{p}$ is a domain and the image of S is its set of nonzero elements, this localization is exactly its field of fractions. $\square$

$$\begin{tikzcd}
A \arrow[r] \arrow[d, two heads] & A_{\mathfrak{p}} \arrow[d, two heads] \\
A/\mathfrak{p} \arrow[r] & A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong \operatorname{Frac}(A/\mathfrak{p})
\end{tikzcd}$$

Figure 11.4: Residue fields are obtained by quotienting at the point and then passing to the fraction field.

Corollary 11.10 (Maximal ideals)**.** *If* $\mathfrak{m}$ *is maximal, then*

$$\kappa(\mathfrak{m}) \cong A/\mathfrak{m},$$

because $A/\mathfrak{m}$ *is already a field.*

11.5 Rational, non-rational, and generic points

Suppose A is a k-algebra. A closed point $\mathfrak{m}$ is k-rational when the structural map $k \to \kappa(\mathfrak{m})$ is an isomorphism. If k is not algebraically closed, closed points can have larger residue fields.

At the opposite extreme, if A is a domain and $\eta = (0)$ is the generic point, then

$$\kappa(\eta) = \mathrm{Frac}(A).$$

More generally, the generic point $\mathfrak{p}$ of the irreducible closed subset $V(\mathfrak{p})$ carries the field

$$\kappa(\mathfrak{p}) = \mathrm{Frac}(A/\mathfrak{p}),$$

which is the function field of that irreducible piece.

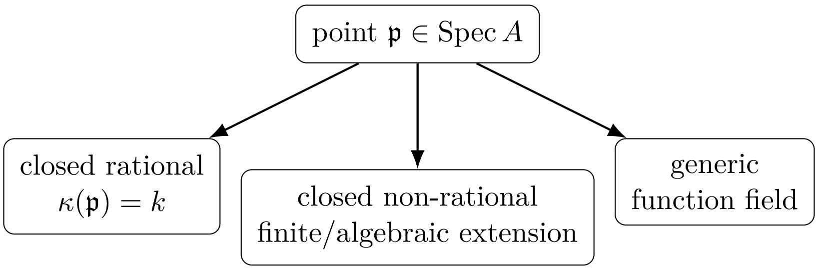


Figure 11.5: Residue fields distinguish several kinds of scheme-theoretic points.

11.6 Local rings inside principal opens

Suppose $f \notin \mathfrak{p}$. Then $\mathfrak{p} \in D(f)$ and the corresponding prime of A_f is $\mathfrak{p}A_f$.

Proposition 11.11 (Local ring does not change after shrinking around the point)**.** *If $f \notin \mathfrak{p}$, then there are canonical isomorphisms*

$$(A_f)_{\mathfrak{p}A_f} \cong A_{\mathfrak{p}}, \qquad \kappa_{A_f}(\mathfrak{p}A_f) \cong \kappa_A(\mathfrak{p}).$$

Proof. In $A_{\mathfrak{p}}$ the element f is already invertible, so the universal property of localization gives $A_f \to A_{\mathfrak{p}}$. Localizing A_f at $\mathfrak{p}A_f$ then inverts exactly the remaining elements outside $\mathfrak{p}$, giving $A_{\mathfrak{p}}$. The residue-field isomorphism follows by quotienting by the maximal ideals. ☐

$$\begin{tikzcd} A \arrow[r] \arrow[dr] & A_f \arrow[d] & & f \notin \mathfrak{p}. \\ & A_{\mathfrak{p}} \cong (A_f)_{\mathfrak{p}A_f} & & \end{tikzcd}$$

Figure 11.6: Shrinking to a principal neighborhood does not change the local ring at the point.

This proposition is the algebraic reason a local ring depends only on an arbitrarily small affine neighborhood of the point.

Example 11.12 (Generic local ring)**.** At the generic point (0) of $\operatorname{Spec} k[x]$, the local ring is the fraction field $k(x)$. Its maximal ideal is zero and its residue field is again $k(x)$.

11.7 Quotients and local rings

Let $I \subseteq \mathfrak{p}$. The point $\mathfrak{p}$ of $\operatorname{Spec} A$ corresponds to the point $\mathfrak{p}/I$ of $\operatorname{Spec}(A/I)$.

Proposition 11.13 (Localization after quotient)**.** *There is a canonical isomorphism*

$$(A/I)_{\mathfrak{p}/I} \cong A_{\mathfrak{p}}/IA_{\mathfrak{p}}.$$

The two points have the same residue field:

$$\kappa_{A/I}(\mathfrak{p}/I) \cong \kappa_A(\mathfrak{p}).$$

$$\begin{tikzcd} A \arrow[r] \arrow[d, two heads] & A_{\mathfrak{p}} \arrow[d, two heads] \\ A/I \arrow[r] & (A/I)_{\mathfrak{p}/I} \cong A_{\mathfrak{p}}/IA_{\mathfrak{p}} \end{tikzcd}$$

Figure 11.7: Quotienting and localizing at a compatible point commute.

The quotient changes the local ring by imposing the equations I, but once we reach the residue field at a point containing I, those equations already vanish.

11.8 When a fraction or germ is zero

A subtle but important localization test is the following.

Proposition 11.14 (Zero in a localization)**.** *For $a \in A$,*

$$\frac{a}{1} = 0 \text{ in } A_{\mathfrak{p}}$$

if and only if there exists $s \notin \mathfrak{p}$ such that

$$sa = 0 \text{ in } A.$$

Thus an element can vanish in the local ring without being globally zero: it is enough that some element nonvanishing at $\mathfrak{p}$ kills it. Geometrically, the element vanishes on some principal neighborhood of the point.

11.9 Maps of local rings

Let $\varphi : A \to B$ be a ring homomorphism and let $\mathfrak{q} \in \operatorname{Spec} B$. Set

$$\mathfrak{p} = \varphi^{-1}(\mathfrak{q}).$$

Every element of $A \setminus \mathfrak{p}$ maps outside $\mathfrak{q}$, so there is a canonical localized homomorphism

$$\varphi_{\mathfrak{q}} : A_{\mathfrak{p}} \longrightarrow B_{\mathfrak{q}}.$$

Definition 11.15 (Local homomorphism)**.** A homomorphism of local rings

$$(R, \mathfrak{m}) \longrightarrow (S, \mathfrak{n})$$

is *local* if the inverse image of $\mathfrak{n}$ is $\mathfrak{m}$.

Proposition 11.16. *The map $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ above is local and induces an injective field homomorphism*

$$\kappa(\mathfrak{p}) \longrightarrow \kappa(\mathfrak{q}).$$

Proof. An element a/s maps into $\mathfrak{q}B_{\mathfrak{q}}$ exactly when $\varphi(a) \in \mathfrak{q}$, equivalently $a \in \mathfrak{p}$. Hence the inverse image of the maximal ideal is $\mathfrak{p}A_{\mathfrak{p}}$. Quotienting produces a unital field homomorphism between residue fields, and every unital field homomorphism is injective. ☐

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d] & B \arrow[d] \\ A_{\mathfrak{p}} \arrow[r, "\varphi_{\mathfrak{q}}"] \arrow[d, two heads] & B_{\mathfrak{q}} \arrow[d, two heads] \\ \kappa(\mathfrak{p}) \arrow[r, hook] & \kappa(\mathfrak{q}) \end{tikzcd} \qquad \mathfrak{p} = \varphi^{-1}(\mathfrak{q}).$$

Figure 11.8: A ring map and a chosen point upstairs induce a local map and a residue-field extension.

Example 11.17 (Arithmetic residue field)**.** At the closed point (p) of $\operatorname{Spec}\mathbb{Z}$, the local ring is $\mathbb{Z}_{(p)}$ and the residue field is $\mathbb{F}_p$.

11.10 The local exact sequence

Every local ring $(R, \mathfrak{m})$ sits in the canonical short exact sequence

$$0 \longrightarrow \mathfrak{m} \longrightarrow R \longrightarrow R/\mathfrak{m} \longrightarrow 0.$$

For $R = A_{\mathfrak{p}}$ this becomes

$$0 \longrightarrow \mathfrak{p}A_{\mathfrak{p}} \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p}) \longrightarrow 0.$$

The maximal ideal consists of local functions vanishing at the point, while the quotient is their value field.

11.11 Local rings can reveal geometry invisible globally

Two nearby points on the same reducible scheme can have very different local rings. On

$$X = \operatorname{Spec} k[x, y]/(xy),$$

the origin sees both components, so its local ring has zero divisors. At a point on the x-axis with $x \neq 0$, the element x becomes invertible; the relation $xy = 0$ then forces $y = 0$, and locally the other component disappears. This is the simplest model of how local algebra isolates the geometry actually passing through a point.

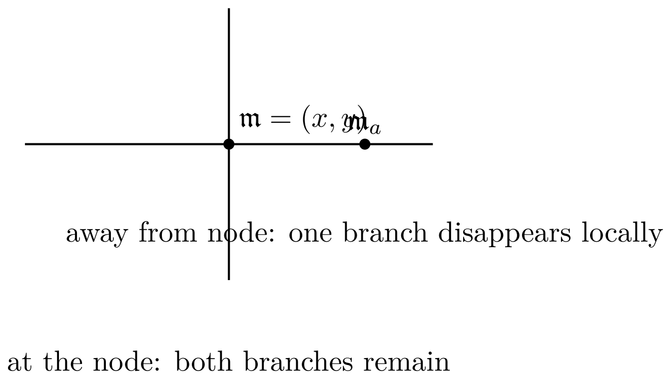


Figure 11.9: Localizing the crossing $xy = 0$ at different points changes which components survive.

11.12 A comparison chart

Point	Local ring	Maximal ideal	Residue field
$(t-a) \subset k[t]$	$k[t]_{(t-a)}$	$(t-a)$ locally	k
$(0) \subset k[t]$	$k(t)$	(0)	$k(t)$
$(p) \subset \mathbb{Z}$	$\mathbb{Z}_{(p)}$	$p\mathbb{Z}_{(p)}$	$\mathbb{F}_p$
$(t^2+1) \subset \mathbb{R}[t]$	$\mathbb{R}[t]_{(t^2+1)}$	(t^2+1) locally	$\mathbb{C}$
$(x) \subset k[x,y]/(xy)$	$k(y)$	(0)	$k(y)$
$(x,y) \subset k[x,y]/(xy)$	$A_{(x,y)}$	$(x,y)A_{(x,y)}$	k
$(\varepsilon) \subset k[\varepsilon]/(\varepsilon^2)$	same ring	(ε)	k

11.13 Common pitfalls

Keep the following distinctions explicit:

- A_f is the ring on a principal open $D(f)$; $A_{\mathfrak{p}}$ is the local ring obtained by inverting every element outside $\mathfrak{p}$.
- $A/\mathfrak{p}$ is a domain, but it need not be a field unless $\mathfrak{p}$ is maximal. The residue field is $\mathrm{Frac}(A/\mathfrak{p})$.
- A closed point need not be rational over the chosen base field.
- A local ring need not be a domain, reduced, Noetherian, or zero-dimensional.
- A one-point spectrum need not be a field: $k[\varepsilon]/(\varepsilon^2)$ is the standard counterexample.
- Localization can make a component disappear because elements outside the chosen prime become units.
- If $\mathfrak{p} \subsetneq \mathfrak{q}$, the natural map of local rings goes $A_{\mathfrak{q}} \to A_{\mathfrak{p}}$, not the other way around.
- There is generally no canonical residue-field map associated merely to a specialization $\mathfrak{p} \subsetneq \mathfrak{q}$; residue-field maps arise naturally from morphisms of schemes/ring maps together with chosen points above one another.

11.14 Worked examples

Example 11.18 (A field is already local)**.** A field K has the single maximal ideal (0), so $(K,(0))$ is local. Every nonzero element is a unit and the residue field is K itself.

Example 11.19 (The local ring of the affine line at a rational point)**.** Let $A = k[t]$ and $\mathfrak{m} = (t-a)$. Then

$$A_{\mathfrak{m}} = \left\{ \frac{f(t)}{g(t)} : g(a) \neq 0 \right\}.$$

Its units are precisely the fractions with $f(a) \neq 0$, its maximal ideal is generated by $t-a$, and

$$\kappa(\mathfrak{m}) \cong k.$$

Thus regular functions near a, modulo those vanishing at a, leave only the value at a.

Example 11.20 (The generic point of the affine line)**.** At $\mathfrak{p} = (0) \subset k[t]$ every nonzero polynomial is inverted, hence

$$k[t]_{(0)} = k(t).$$

The maximal ideal is zero, so the local ring is already its residue field:

$$\kappa((0)) = k(t).$$

Example 11.21 (A non-rational real closed point)**.** In $\mathbb{R}[t]$, the ideal $\mathfrak{m} = (t^2+1)$ is maximal. Hence

$$\kappa(\mathfrak{m}) \cong \mathbb{R}[t]/(t^2+1) \cong \mathbb{C}.$$

The point is closed but not $\mathbb{R}$-rational.

Example 11.22 (Closed and generic points of $\operatorname{Spec}\mathbb{Z}$)**.** At a prime number p,

$$\mathbb{Z}_{(p)} = \left\{ \frac{a}{b} : b \not\equiv 0 \pmod{p} \right\}$$

is local with maximal ideal $p\mathbb{Z}_{(p)}$ and residue field $\mathbb{F}_p$. At the generic point (0) the local ring and residue field are both $\mathbb{Q}$.

Example 11.23 (A rational point of the affine plane)**.** For $A = k[x,y]$ and $\mathfrak{m} = (x-a, y-b)$,

$$A/\mathfrak{m} \cong k, \qquad \kappa(\mathfrak{m}) = k.$$

The local ring $A_{\mathfrak{m}}$ consists of fractions whose denominators do not vanish at (a,b).

Example 11.24 (Generic point of a parabola)**.** Take $A = k[s,t]$ and $\mathfrak{p} = (t-s^2)$. Then

$$A/\mathfrak{p} \cong k[s], \qquad \kappa(\mathfrak{p}) = k(s).$$

The point $\mathfrak{p}$ is the generic point of the parabola $t = s^2$.

Example 11.25 (Generic point of the affine plane)**.** For $A = k[x,y]$ and $\mathfrak{p} = (0)$,

$$A_{(0)} = k(x,y), \qquad \kappa((0)) = k(x,y).$$

Every nonzero polynomial becomes a unit.

Example 11.26 (A finite-field closed point of degree d)**.** Let $f(t) \in \mathbb{F}_q[t]$ be irreducible of degree d and let $\mathfrak{m} = (f)$. Then

$$\kappa(\mathfrak{m}) = \mathbb{F}_q[t]/(f) \cong \mathbb{F}_{q^d}.$$

Thus higher-degree irreducible polynomials are genuine closed points, with residue degree d.

Example 11.27 (Crossing axes at the origin)**.** Let

$$A = k[x,y]/(xy), \qquad \mathfrak{m} = (\bar{x}, \bar{y}).$$

Then $A_\mathfrak{m}$ is local with residue field k, but it has zero divisors because

$$\bar{x}\bar{y} = 0$$

and neither factor vanishes in the local ring. Both components pass through the point.

Example 11.28 (Crossing axes away from the intersection)**.** In the same ring take $\mathfrak{m}_a = (\bar{x} - a, \bar{y})$ with $a \neq 0$. Since $\bar{x}$ is a unit in $A_{\mathfrak{m}_a}$, the relation $\bar{x}\bar{y} = 0$ forces $\bar{y} = 0$. Hence

$$A_{\mathfrak{m}_a} \cong k[x]_{(x-a)}.$$

Locally, the y-axis disappears.

Example 11.29 (Generic point of a component of the crossing)**.** At $\mathfrak{p} = (\bar{x})$, the element $\bar{y}$ becomes invertible, so $\bar{x} = 0$ and

$$A_\mathfrak{p} \cong k(y).$$

This is a field. It is both the local ring and the residue field at the generic point of the y-axis.

Example 11.30 (The dual numbers as a local ring)**.** The ring

$$R = k[\varepsilon]/(\varepsilon^2)$$

is local with unique maximal ideal $(\bar{\varepsilon})$. Its residue field is k, but the local ring contains nonzero nilpotent information invisible in the residue field.

Example 11.31 ($\mathbb{Z}/4\mathbb{Z}$ versus $\mathbb{Z}/6\mathbb{Z}$)**.** The ring $\mathbb{Z}/4\mathbb{Z}$ is local: its unique maximal ideal is $(\bar{2})$ and its residue field is $\mathbb{F}_2$. By contrast $\mathbb{Z}/6\mathbb{Z}$ has two maximal ideals, corresponding to 2 and 3, so it is not local. This is a convenient finite-ring test of the nonunit criterion.

Example 11.32 (Shrinking does not change the local ring)**.** Let $A = k[x,y]$, $\mathfrak{m} = (x-1, y)$, and $f = x$. Since $x \notin \mathfrak{m}$, $\mathfrak{m} \in D(x)$. Then

$$(k[x,y,x^{-1}])_{\mathfrak{m}A_x} \cong k[x,y]_\mathfrak{m}.$$

Restricting to the open set $x \neq 0$ has not changed the local algebra at $(1,0)$.

Example 11.33 (Quotient then localize)**.** Let $A = k[x,y]$, $I = (y - x^2)$, and $\mathfrak{m} = (x,y)$. Then

$$(A/I)_{\mathfrak{m}/I} \cong A_\mathfrak{m}/IA_\mathfrak{m} \cong k[x]_{(x)}.$$

The local ring of the parabola at the origin is the ordinary one-variable local ring.

Example 11.34 (The local zero criterion)**.** Let

$$A = k[x,y]/(xy), \qquad \mathfrak{p} = (\bar{y}).$$

The element $\bar{y}$ is globally nonzero, but in $A_\mathfrak{p}$ the element $\bar{x}$ is invertible and $\bar{x}\bar{y} = 0$, so $\bar{y}/1 = 0$. This is exactly the localization criterion: $\bar{x} \notin \mathfrak{p}$ kills $\bar{y}$.

Example 11.35 (A local map from evaluation)**.** The evaluation homomorphism

$$k[t] \longrightarrow k, \qquad f(t) \longmapsto f(a)$$

has inverse image $(t-a)$ of the zero prime of k. It induces a local homomorphism

$$k[t]_{(t-a)} \longrightarrow k$$

and the induced residue-field map is the identity $k \to k$.

11.15 Advanced examples

Example 11.36 (Advanced: a discrete valuation model)**.** The local ring

$$R = k[t]_{(t)}$$

has maximal ideal (t), and every nonzero element can be written uniquely as

$$t^n u$$

with $n \geq 0$ and u a unit. Consequently every nonzero ideal of R is (t^n) for some n. This is the basic example of a discrete valuation ring, a notion used later in divisor theory.

Example 11.37 (Advanced: the Gaussian prime $(1+i)$)**.** Let $A = \mathbb{Z}[i]$ and $\mathfrak{p} = (1+i)$. Since

$$\mathbb{Z}[i]/(1+i) \cong \mathbb{F}_2,$$

the localization $A_\mathfrak{p}$ is local with residue field $\mathbb{F}_2$. This is an arithmetic analogue of localizing a polynomial ring at a closed point.

Example 11.38 (Advanced: a mixed arithmetic-geometric point)**.** Let

$$A = \mathbb{Z}[x], \qquad \mathfrak{m} = (2, x-1).$$

Then $A/\mathfrak{m} \cong \mathbb{F}_2$, so

$$\kappa(\mathfrak{m}) = \mathbb{F}_2.$$

The local ring $\mathbb{Z}[x]_{(2,x-1)}$ simultaneously remembers arithmetic localization at 2 and geometric localization near $x = 1$.

Example 11.39 (Advanced: a product ring localizes to one factor)**.** Let $A = k \times k$. Its two prime ideals are $\mathfrak{p}_1 = 0 \times k$ and $\mathfrak{p}_2 = k \times 0$. Localizing at $\mathfrak{p}_1$ makes $(1, 0)$ a unit and annihilates the other factor, giving

$$A_{\mathfrak{p}_1} \cong k.$$

Likewise $A_{\mathfrak{p}_2} \cong k$. A disconnected global spectrum has field-valued local rings at both points.

Example 11.40 (Advanced: the cusp at its singular point)**.** Let

$$A = k[t^2, t^3] \cong k[x, y]/(y^2 - x^3)$$

and let $\mathfrak{m} = (t^2, t^3)$. The local ring

$$A_\mathfrak{m} = k[t^2, t^3]_{(t^2,t^3)}$$

has residue field k. It embeds in the larger local ring $k[t]_{(t)}$, but t itself is not in $A_\mathfrak{m}$. This is a first glimpse of normalization: the parametrizing line has a simpler local ring than the cusp.

Example 11.41 (Advanced: a specialization chain and local maps)**.** In $A = k[x, y]$ consider

$$(0) \subset (x) \subset (x, y).$$

The natural maps of local rings run in the opposite direction:

$$A_{(x,y)} \longrightarrow A_{(x)} \longrightarrow A_{(0)} = k(x, y).$$

Moving toward a more generic point means inverting more elements.

Example 11.42 (Advanced: a nonreduced local Artin ring)**.** For

$$R_n = k[\varepsilon]/(\varepsilon^n), \qquad n \geq 2,$$

there is one maximal ideal $(\bar{\varepsilon})$ and residue field k. The filtration

$$R_n \supset (\bar{\varepsilon}) \supset \cdots \supset (\bar{\varepsilon}^{n-1}) \supset 0$$

records $n - 1$ layers of infinitesimal thickness, all invisible to the one-point spectrum and to the residue field alone.

Example 11.43 (Advanced: localization at a minimal prime in a reduced ring)**.** Let $A = k[x, y]/(xy)$ and $\mathfrak{p} = (\bar{x})$. Since A is reduced and $\mathfrak{p}$ is minimal, localization kills the other component and gives the field $k(y)$. This illustrates a general phenomenon: for a reduced ring, localization at a minimal prime is a field.

Example 11.44 (Advanced: p-adic integers as a local ring)**.** The ring $\mathbb{Z}_p$ of p-adic integers is local with maximal ideal $p\mathbb{Z}_p$ and residue field

$$\mathbb{Z}_p/p\mathbb{Z}_p \cong \mathbb{F}_p.$$

Unlike $\mathbb{Z}_{(p)}$, it is complete in the p-adic topology. Completion itself is developed in Volume V; here the example is included to compare two important local rings with the same residue field.

Example 11.45 (Advanced: local exact sequence at a point)**.** For any prime $\mathfrak{p}$ there is an exact sequence

$$0 \to \mathfrak{p}A_{\mathfrak{p}} \to A_{\mathfrak{p}} \to \kappa(\mathfrak{p}) \to 0.$$

For $A = k[t]$ and $\mathfrak{p} = (t - a)$ this becomes

$$0 \to (t - a)k[t]_{(t-a)} \to k[t]_{(t-a)} \to k \to 0.$$

It formalizes the statement that local functions modulo functions vanishing at the point are just their values.

11.16 Exercises

Exercise 11.46 (VI.11.E01 — Local-ring criterion)**.** Prove that if every element a of a ring R satisfies: either a or $1 - a$ is a unit, then R is local.

Exercise 11.47 (VI.11.E02 — Units in the affine-line local ring)**.** Describe explicitly the units of $k[t]_{(t-a)}$ and identify its maximal ideal.

Exercise 11.48 (VI.11.E03 — Prime localization)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, prove directly that $a/s \in A_{\mathfrak{p}}$ is a unit iff $a \notin \mathfrak{p}$.

Exercise 11.49 (VI.11.E04 — Local spectrum)**.** Show that primes of $A_{\mathfrak{p}}$ correspond exactly to primes $\mathfrak{q} \subseteq \mathfrak{p}$ of A.

Exercise 11.50 (VI.11.E05 — Residue field formula)**.** Compute $\kappa(\mathfrak{p})$ from $A/\mathfrak{p}$ and prove $\kappa(\mathfrak{p}) \cong \mathrm{Frac}(A/\mathfrak{p})$.

Exercise 11.51 (VI.11.E06 — Affine line)**.** Compute the local rings and residue fields at (0) and $(t-a)$ in $\operatorname{Spec} k[t]$.

Exercise 11.52 (VI.11.E07 — Real quadratic point)**.** For $\mathfrak{m} = (t^2+1) \subset \mathbb{R}[t]$, compute $\mathbb{R}[t]_{\mathfrak{m}}$ and its residue field.

Exercise 11.53 (VI.11.E08 — Arithmetic line)**.** Compute $\mathbb{Z}_{(p)}$, its units, maximal ideal, and residue field.

Exercise 11.54 (VI.11.E09 — Finite-field closed point)**.** If $f \in \mathbb{F}_q[t]$ is irreducible of degree d, compute the residue field at (f).

Exercise 11.55 (VI.11.E10 — Parabola generic point)**.** Compute the local ring and residue field of $k[x,y]$ at the prime $(y-x^2)$.

Exercise 11.56 (VI.11.E11 — Crossing at origin)**.** Let $A = k[x,y]/(xy)$. Show that $A_{(x,y)}$ has zero divisors and residue field k.

Exercise 11.57 (VI.11.E12 — Crossing away from origin)**.** In the same ring, compute the local ring at $(x-a, y)$ for $a \neq 0$.

Exercise 11.58 (VI.11.E13 — Minimal prime)**.** For $A = k[x,y]/(xy)$ compute $A_{(x)}$ and $A_{(y)}$.

Exercise 11.59 (VI.11.E14 — Dual numbers)**.** Show $k[\varepsilon]/(\varepsilon^2)$ is local; compute its maximal ideal, units, and residue field.

Exercise 11.60 (VI.11.E15 — Finite local/nonlocal comparison)**.** Determine which of $\mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z}$, and $\mathbb{Z}/8\mathbb{Z}$ are local and compute residue fields of their maximal ideals.

Exercise 11.61 (VI.11.E16 — Shrinking invariance)**.** If $f \notin \mathfrak{p}$, construct explicitly the isomorphism $(A_f)_{\mathfrak{p}A_f} \cong A_{\mathfrak{p}}$.

Exercise 11.62 (VI.11.E17 — Quotient-localization)**.** If $I \subseteq \mathfrak{p}$, prove $(A/I)_{\mathfrak{p}/I} \cong A_{\mathfrak{p}}/IA_{\mathfrak{p}}$.

Exercise 11.63 (VI.11.E18 — Residue field after quotient)**.** Under the hypotheses of the preceding exercise, prove the residue fields at $\mathfrak{p}$ and $\mathfrak{p}/I$ are naturally isomorphic.

Exercise 11.64 (VI.11.E19 — Zero germ criterion)**.** Prove $a/1 = 0$ in $A_{\mathfrak{p}}$ iff $sa = 0$ for some $s \notin \mathfrak{p}$.

Exercise 11.65 (VI.11.E20 — Local map)**.** Given $\varphi : A \to B$ and $\mathfrak{q} \in \operatorname{Spec} B$, with $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$, construct $A_{\mathfrak{p}} \to B_{\mathfrak{q}}$ and prove it is local.

Exercise 11.66 (VI.11.E21 — Residue-field map)**.** Continue the preceding exercise and construct the induced injective field map $\kappa(\mathfrak{p}) \to \kappa(\mathfrak{q})$.

Exercise 11.67 (VI.11.E22 — Product ring)**.** Compute the localizations of $k \times k$ at its two prime ideals.

Exercise 11.68 (VI.11.E23 — Mixed arithmetic point)**.** For $A = \mathbb{Z}[x]$ and $\mathfrak{m} = (2, x-1)$, compute $A/\mathfrak{m}$ and $\kappa(\mathfrak{m})$, and describe which elements are units in $A_{\mathfrak{m}}$.

Exercise 11.69 (VI.11.E24 — Specialization chain)**.** For $(0) \subset (x) \subset (x,y)$ in $k[x,y]$, construct the maps $A_{(x,y)} \to A_{(x)} \to A_{(0)}$ and explain why the direction reverses the prime inclusions.

11.17 Problems

Problem 11.70. Characterize local rings by units and nonunits.

Problem 11.71. Prove that $A_{\mathfrak{p}}$ is local and determine its units.

Problem 11.72. Describe $\operatorname{Spec} A_{\mathfrak{p}}$ as the generalizations of $\mathfrak{p}$.

Problem 11.73. Derive and apply $\kappa(\mathfrak{p}) \cong \operatorname{Frac}(A/\mathfrak{p})$.

Problem 11.74. Compare local rings on the crossing $xy = 0$ at and away from the node.

Problem 11.75. Analyze the arithmetic local ring $\mathbb{Z}_{(p)}$.

Problem 11.76. Classify residue fields of closed points of $\operatorname{Spec} \mathbb{F}_q[t]$.

Problem 11.77. Prove quotient-localization compatibility at a point.

Problem 11.78. Prove local rings are unchanged after shrinking to a principal neighborhood.

Problem 11.79. Construct local homomorphisms from ring maps and determine residue-field maps.

Problem 11.80. Analyze the dual numbers as a nonreduced local ring.

Problem 11.81. Prove the local zero criterion and interpret it geometrically.

Problem 11.82. Analyze localizations of product rings at their prime ideals.

Problem 11.83. Compare the local ring of a cusp with the local ring of its parametrizing line.

Problem 11.84. Study a specialization chain through its induced maps of local rings.

Problem 11.85. Compute a mixed arithmetic-geometric local ring in $\mathbb{Z}[x]$.

11.18 Challenges

Problem 11.86. Construct and classify a family of one-point local Artin rings with the same residue field.

Problem 11.87. Show that localizing a reduced ring at all minimal primes produces the total ring of fractions in a finite-minimal-prime case.

Problem 11.88. Analyze the local ring and residue extension at a closed point of an arithmetic curve.

Problem 11.89. Relate chains in $\operatorname{Spec} A_{\mathfrak{p}}$ to chains of generalizations of $\mathfrak{p}$.

11.19 Boundary with the next chapters

VI/11 completes Part II by attaching concrete local algebra to points of spectra. VI/12 begins Part III with presheaves, VI/13 turns compatible local data into stalks, and VI/17 constructs the structure sheaf whose stalk at $\mathfrak{p}$ recovers $A_{\mathfrak{p}}$. VI/18 then packages these data into affine schemes. Later VI/25 uses residue fields to define fibers, VI/27 develops function fields globally, and VI/32 uses $\mathfrak{m}/\mathfrak{m}^2$ to build tangent spaces.

11.20 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 11.90 (Affine line stalk)**.** Compute the local ring at $(x-a)$ in $k[x]$.

Hint. Use localization at primes, local rings, and residue fields. ☐

Solution. $k[x]_{(x-a)}$. ☐

Exercise 11.91 (Residue line)**.** Compute its residue field.

Hint. Use localization at primes, local rings, and residue fields. ☐

Solution. k. ☐

Exercise 11.92 (Generic stalk)**.** Compute the local ring at (0) in $k[x]$.

Hint. Use localization at primes, local rings, and residue fields. ☐

Solution. $k(x)$. ☐

Exercise 11.93 (Integer local ring)**.** Compute the local ring at (p) in $\mathbb{Z}$.

Hint. Use localization at primes, local rings, and residue fields. ☐

Solution. $\mathbb{Z}_{(p)}$. ☐

Exercise 11.94 (Integer residue)**.** Compute the residue field at (p).

Hint. Use localization at primes, local rings, and residue fields. ☐

Solution. $\mathbb{F}_p$. ☐

Proofs

Exercise 11.95 (Localization local)**.** Prove: Prove $A_\mathfrak{p}$ is local with maximal ideal $\mathfrak{p}A_\mathfrak{p}$.

Hint. Work directly from localization at primes, local rings, and residue fields. ☐

Solution. A fraction is a unit exactly when its numerator is outside $\mathfrak{p}$. ☐

Exercise 11.96 (Residue fraction)**.** Prove: Prove $\kappa(\mathfrak{p}) \cong \mathrm{Frac}(A/\mathfrak{p})$.

Hint. Work directly from localization at primes, local rings, and residue fields. ☐

Solution. First quotient $A_\mathfrak{p}$ by its maximal ideal, then identify surviving denominators with nonzero elements of the domain $A/\mathfrak{p}$. ☐

Exercise 11.97 (Units)**.** Prove: Prove an element of a local ring is a unit iff it lies outside the maximal ideal.

Hint. Work directly from localization at primes, local rings, and residue fields. ☐

Solution. Every nonunit lies in some maximal ideal, and there is only one. ☐

Exercise 11.98 (Prime chain)**.** Prove: Prove primes of $A_{\mathfrak{p}}$ correspond to primes of A contained in $\mathfrak{p}$.

Hint. Work directly from localization at primes, local rings, and residue fields. ☐

Solution. Use localization extension-contraction and disjointness from $A \setminus \mathfrak{p}$. ☐

Counterexamples and hypothesis tests

Exercise 11.99 (All residue fields)**.** Test the claim: all residue fields of a scheme equal the base field.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general. ☐

Exercise 11.100 (Generic maximal)**.** Test the claim: the generic prime of a domain is maximal.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False unless the ring is a field. ☐

Exercise 11.101 (Local one prime)**.** Test the claim: a local ring has exactly one prime ideal.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it has exactly one maximal ideal but may have many primes. ☐

Applications and investigations

Exercise 11.102 (Local measurement)**.** What does the local ring measure?

Hint. Translate the question into localization at primes, local rings, and residue fields. ☐

Solution. It records functions defined near one prime with denominators allowed away from that prime. ☐

Exercise 11.103 (Point scalars)**.** What does the residue field represent?

Hint. Translate the question into localization at primes, local rings, and residue fields. ☐

Solution. It is the field of scalar values intrinsic to that point. ☐

Challenge problems

Exercise 11.104 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Local Rings and Residue Fields in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Local Rings and Residue Fields, translates it through localization at primes, local rings, and residue fields, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 11.105 (Hypothesis challenge)**.** Choose one structural statement from Local Rings and Residue Fields, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Part III

Sheaves

Chapter 12

Presheaves

Purpose and learning goals

Part II attached local rings and residue fields to individual points of spectra. Part III now changes viewpoint: instead of studying one point at a time, we organize algebraic data on *every open set* and compare those data when the open set shrinks. The resulting object is a presheaf.

A presheaf is deliberately weaker than a sheaf. It knows how to restrict sections, but it does not yet promise that locally compatible sections determine a unique global section. That extra locality-and-gluing principle belongs to VI/13. Keeping this distinction clear is important: many natural assignments are presheaves, some are sheaves, and some tempting assignments fail even to be presheaves because restriction is not well defined.

This chapter is intentionally example-rich. The main worked-example section contains elementary and geometric presheaves, while a separate *Advanced examples* section develops finite-space diagrams, arithmetic and affine-basis localizations, direct images, cochains, bounded-function counterexamples, and representable presheaves. These examples prepare the reader for stalks, sheafification, the structure sheaf, and later cohomology.

After this chapter, the reader should be able to:

- define a presheaf of sets, groups, rings, or modules on a topological space;
- interpret a presheaf as a contravariant functor from $\mathrm{Open}(X)$;
- write and check restriction maps, identity, and composition axioms;
- distinguish sections from their restrictions and from later notions of germs;
- construct morphisms of presheaves and verify the naturality square;
- form subpresheaves, products, kernels, images, and quotient presheaves objectwise;
- restrict a presheaf to an open subspace;
- construct direct images of presheaves along continuous maps;
- encode presheaves on finite spaces as diagrams of algebraic objects;
- recognize the naive constant presheaf and explain why it can fail the sheaf gluing axiom;
- recognize locally constant, continuous, smooth, holomorphic, harmonic, and cochain examples;
- explain why compactly supported sections do not form a presheaf under arbitrary restriction;

- construct the localization presheaf on principal opens $D(f)$;
- construct the module-localization analogue M_f on principal opens;
- distinguish "presheaf data" from the extra locality and gluing axioms of a sheaf;
- audit a proposed assignment systematically before calling it a presheaf.

12.1 The category of open sets

Let X be a topological space. Write $\mathrm{Open}(X)$ for the category whose objects are open subsets of X and with a unique morphism

$$V \longrightarrow U$$

exactly when $V \subseteq U$.

Because there is at most one morphism between any two objects, $\mathrm{Open}(X)$ is simply the poset of open subsets viewed as a category. A presheaf reverses these arrows.

$$\begin{tikzcd}
\boxed{U} \arrow[d, dash, dashed] & \boxed{V \subseteq U} \arrow[l, "\text{inclusion}"'] \arrow[d, dash, dashed] \\
\boxed{\mathcal{F}(U)} \arrow[r, "\text{restriction}"'] & \boxed{\mathcal{F}(V)}
\end{tikzcd}$$

Definition 12.1 (Presheaf)**.** Let $\mathcal{C}$ be a category. A *$\mathcal{C}$-valued presheaf* on X is a contravariant functor

$$\mathcal{F} : \mathrm{Open}(X)^{op} \longrightarrow \mathcal{C}.$$

Equivalently, it consists of:

1. an object $\mathcal{F}(U)$ of $\mathcal{C}$ for every open $U \subseteq X$;
2. for every inclusion $V \subseteq U$, a restriction morphism

$$\rho^U_V : \mathcal{F}(U) \longrightarrow \mathcal{F}(V);$$

3. the identities

$$\rho^U_U = \mathrm{id}_{\mathcal{F}(U)}, \qquad \rho^V_W \circ \rho^U_V = \rho^U_W \quad (W \subseteq V \subseteq U).$$

When $\mathcal{C} = \mathbf{Set}, \mathbf{Ab}, \mathbf{Ring}$, or $R\mathrm{-}\mathbf{Mod}$, we speak of presheaves of sets, abelian groups, rings, or R-modules.

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "\rho^U_V"] \arrow[dr, "\rho^U_W"'] & \mathcal{F}(V) \arrow[d, "\rho^V_W"] \\
& \mathcal{F}(W)
\end{tikzcd} \qquad W \subseteq V \subseteq U.$$

12.2 Sections and restriction

An element

$$s \in \mathcal{F}(U)$$

is called a *section of $\mathcal{F}$ over U*. If $V \subseteq U$, its restriction is

$$s|_V := \rho^U_V(s) \in \mathcal{F}(V).$$

The word "section" is intentionally abstract. Depending on the presheaf, a section might be a function, differential form, cochain, algebraic element, solution of an equation, or some other object defined over U.

Remark 12.2 (Restriction is structure, not notation). Writing $s|_V$ does not mean that every presheaf section is literally a function. The notation records the structure map supplied by the presheaf.

12.3 Presheaves as contravariant functors

The functorial formulation is not cosmetic. It packages the two restriction axioms into ordinary functoriality:

$$\mathcal{F}(\mathrm{id}_U) = \mathrm{id}_{\mathcal{F}(U)}$$

and

$$\mathcal{F}(W \hookrightarrow V) \circ \mathcal{F}(V \hookrightarrow U) = \mathcal{F}(W \hookrightarrow U).$$

Thus the slogan is

$$\boxed{\text{smaller open set} \quad \Longrightarrow \quad \text{restriction of data.}}$$

This reversal of arrows is the same contravariant pattern already encountered in coordinate rings and Spec.

Example 12.3 (Continuous functions presheaf)**.** For a topological space X, assigning to each open U the ring $C(U)$ of continuous functions, with ordinary restriction maps, gives a presheaf because restricting twice equals restricting once.

12.4 Morphisms of presheaves

Definition 12.4 (Morphism of presheaves)**.** Let $\mathcal{F}, \mathcal{G}$ be presheaves with values in the same category. A morphism

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

is a natural transformation. Thus for every open $U \subseteq X$ there is a morphism

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

such that for every inclusion $V \subseteq U$ the square

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "\varphi_U"] \arrow[d, "\rho^U_V"'] & \mathcal{G}(U) \arrow[d, "\rho^U_V"] \\
\mathcal{F}(V) \arrow[r, "\varphi_V"'] & \mathcal{G}(V)
\end{tikzcd}$$

commutes.

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "\varphi_U"] \arrow[d, "\mathrm{res}"'] & \mathcal{G}(U) \arrow[d, "\mathrm{res}"] \\
\mathcal{F}(V) \arrow[r, "\varphi_V"'] & \mathcal{G}(V)
\end{tikzcd}$$

The commutative square says exactly that "apply the morphism, then restrict" equals "restrict, then apply the morphism."

12.5 Subpresheaves and objectwise constructions

For presheaves of algebraic objects, many constructions are performed open set by open set.

Definition 12.5 (Subpresheaf)**.** A presheaf $\mathcal{G}$ is a *subpresheaf* of $\mathcal{F}$ if $\mathcal{G}(U) \subseteq \mathcal{F}(U)$ for every open U, and the restriction maps of $\mathcal{G}$ are the restrictions of those of $\mathcal{F}$.

For presheaves of abelian groups:

$$(\mathcal{F} \times \mathcal{G})(U) = \mathcal{F}(U) \times \mathcal{G}(U),$$

and if $\varphi : \mathcal{F} \to \mathcal{G}$,

$$(\ker \varphi)(U) = \ker(\varphi_U), \qquad (\operatorname{im}^{pre} \varphi)(U) = \operatorname{im}(\varphi_U).$$

If $\mathcal{H} \subseteq \mathcal{F}$ is a subpresheaf, the quotient presheaf is

$$(\mathcal{F}/\mathcal{H})^{pre}(U) = \mathcal{F}(U)/\mathcal{H}(U).$$

These are genuine presheaves. A major lesson of later chapters is that an objectwise image or quotient presheaf need not already be a sheaf.

12.6 Restriction to an open subset

Let $j : U \hookrightarrow X$ be an open inclusion.

Definition 12.6 (Restricted presheaf)**.** For a presheaf $\mathcal{F}$ on X, define its restriction to U by

$$(\mathcal{F}|_U)(V) = \mathcal{F}(V)$$

for every open $V \subseteq U$, with the same restriction maps.

Nothing new is created; one simply forgets all opens not contained in U.

Example 12.7 (Constant presheaf)**.** The assignment $U \mapsto A$ with identity restrictions is a presheaf. It is generally not a sheaf on disconnected opens because independently chosen local constants cannot always glue to one global constant.

12.7 Direct image of a presheaf

Let $f : X \to Y$ be continuous.

Definition 12.8 (Direct image presheaf)**.** For a presheaf $\mathcal{F}$ on X, define $f_*\mathcal{F}$ on Y by

$$(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}(V))$$

for each open $V \subseteq Y$.

If $W \subseteq V$, then

$$f^{-1}(W) \subseteq f^{-1}(V),$$

so restriction in $\mathcal{F}$ gives the required map

$$(f_*\mathcal{F})(V) \longrightarrow (f_*\mathcal{F})(W).$$

$$
\begin{tikzcd}
V \subseteq Y \arrow[d, dash, dashed] & W \subseteq V \arrow[l] \arrow[d, dash, dashed] \\
f^{-1}(V) \subseteq X & f^{-1}(W) \arrow[l]
\end{tikzcd}
$$

$$
\begin{tikzcd}
\mathcal{F}(f^{-1}V) \arrow[r] & \mathcal{F}(f^{-1}W)
\end{tikzcd}
$$

12.8 The naive constant presheaf

Let A be an abelian group. The standard naive constant presheaf is

$$
\underline{A}^{pre}(U) = \begin{cases} A, & U \neq \varnothing, \\ 0, & U = \varnothing, \end{cases}
$$

with identity restrictions between nonempty opens and the unique map to 0 when the target is empty.

This is a presheaf. However, if an open set has two disjoint nonempty components, locally different constants cannot be glued to one global element of A. Thus the naive constant presheaf is generally *not* a sheaf.

$$
\begin{tikzcd}
 & U = U_1 \sqcup U_2 & \\
U_1 :\ a_1 \arrow[ur] & & U_2 :\ a_2 \arrow[ul] \\
 & \underline{A}^{pre}(U) = A \arrow[uu, dashed, "\text{would need one global constant}"'] & a_1 \neq a_2 \Rightarrow \text{no global constant glues them}
\end{tikzcd}
$$

The presheaf of *locally constant* A-valued functions repairs this defect and will be a standard sheaf example in VI/13.

12.9 A systematic presheaf test

Given a proposed assignment $U \mapsto \mathcal{F}(U)$, check four things:

1. **Objects:** is $\mathcal{F}(U)$ defined for every open U?
2. **Restrictions:** if $V \subseteq U$, does every section over U naturally produce a section over V?
3. **Identity:** is restriction from U to itself the identity?
4. **Composition:** for $W \subseteq V \subseteq U$, is restricting in two stages the same as restricting directly?

The second test is often the subtle one. Compact support, global boundedness, normalization conditions, and choices depending on the ambient open set can fail to survive restriction.

Example 12.9 (Sections of a module)**.** On an affine open $D(f)$, the rule $D(f) \mapsto A_f$ foreshadows the structure sheaf. Restriction from $D(f)$ to $D(fg)$ is the localization map $A_f \to A_{fg}$.

12.10 Presheaves on finite spaces

Finite topological spaces make the category-theoretic content visible without analytic distractions. Consider the Sierpinski space

$$X = \{\eta, s\}, \qquad \mathrm{Open}(X) = \{\varnothing, \{\eta\}, X\}.$$

A presheaf of abelian groups, after choosing $\mathcal{F}(\varnothing) = 0$, is exactly a homomorphism

$$\mathcal{F}(X) \longrightarrow \mathcal{F}(\{\eta\}).$$

presheaf reverses arrows

$$\begin{tikzcd}
X = \{\eta, s\} & \mathcal{F}(X) \arrow[d] \\
\{\eta\} \arrow[u] & \mathcal{F}(\{\eta\}) \arrow[d] \\
\varnothing \arrow[u] & \mathcal{F}(\varnothing)
\end{tikzcd}$$

More generally, on a finite Alexandrov space a presheaf is a contravariant diagram indexed by the poset of open sets.

12.11 Presheaves on a basis: an affine preview

Let A be a ring. The principal opens $D(f) \subseteq \operatorname{Spec} A$ form a basis. On this basis one has the fundamental assignment

$$D(f) \longmapsto A_f.$$

For the especially transparent inclusion

$$D(fg) \subseteq D(f),$$

the restriction morphism is the localization map

$$A_f \longrightarrow A_{fg}.$$

Similarly, for an A-module M,

$$D(f) \longmapsto M_f$$

is the module-valued prototype.

$$\begin{array}{ccc} D(fg) \hookrightarrow D(f) & \qquad & D(fg) \hookrightarrow D(f) \\ & & \\ A_{fg} \longleftarrow A_f & \qquad & M_{fg} \longleftarrow M_f \end{array}$$

We do not yet call this the structure sheaf: extending from the basis, verifying the sheaf axioms, and identifying stalks require the machinery of VI/13 and VI/17.

12.12 What a presheaf does not promise

A presheaf guarantees only coherent restriction. It does *not* guarantee:

- that a section is determined by all of its local restrictions;
- that compatible local sections glue to a global section;
- that objectwise images or quotients behave like local images or quotients;
- that values on a basis determine values on arbitrary opens.

The first two missing properties are precisely the sheaf axioms.

$$\boxed{\begin{array}{c}\text{presheaf}\\ \text{objects on opens + restrictions}\\ \text{identity + composition}\end{array}} \xrightarrow{\text{add local axioms}} \boxed{\begin{array}{c}\text{sheaf}\\ \text{presheaf + locality + gluing}\end{array}}$$

VI/12 VI/13

12.13 Common pitfalls

Keep the following distinctions explicit:

- A presheaf is not just a collection of objects $\mathcal{F}(U)$; the restriction maps and their functoriality are part of the data.
- Restriction arrows go opposite to inclusions of open sets.
- A morphism of presheaves is not an arbitrary family of maps; it must commute with all restrictions.
- "Constant presheaf" and "locally constant functions" are different constructions.
- A natural assignment may fail to be a presheaf if its defining property is not stable under restriction.
- The fact that an assignment is a presheaf says nothing yet about gluing.
- Objectwise quotients and images are presheaves, but later they may require sheafification to become the correct sheaf-theoretic objects.
- Stalks and germs are deliberately postponed to VI/13, except for brief previews.
- The localization assignment $D(f) \mapsto A_f$ is currently a basis-level prototype; the full structure sheaf comes later.

12.14 Worked examples

Example 12.10 (Continuous real-valued functions)**.** For a topological space X, let

$$\mathcal{C}^0(U) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}.$$

Restriction is literal restriction of functions. The identity and composition axioms are immediate.

Example 12.11 (Smooth functions)**.** If M is a smooth manifold, set $\mathcal{C}^\infty(U)$ equal to the smooth real-valued functions on U. Smoothness is preserved by restriction, so this is a presheaf of rings.

Example 12.12 (Holomorphic functions)**.** On a complex manifold or open subset $X \subseteq \mathbb{C}^n$, the assignment

$$U \longmapsto \mathcal{O}(U)$$

of holomorphic functions is a presheaf of $\mathbb{C}$-algebras.

Example 12.13 (Locally constant functions)**.** For an abelian group A, let $\mathcal{L}_A(U)$ be the group of locally constant maps $U \to A$. Restriction preserves local constancy.

Example 12.14 (Naive constant presheaf)**.** For nonempty U, put $\underline{A}^{pre}(U) = A$, with identity restrictions. Put $\underline{A}^{pre}(\varnothing) = 0$. This is a presheaf, but on a disconnected open it cannot glue two different local constants to one global element.

Example 12.15 (Power-set presheaf)**.** Define

$$\mathcal{P}(U) = \{S : S \subseteq U\}.$$

For $V \subseteq U$, restrict by $S \mapsto S \cap V$. Then $(S \cap V) \cap W = S \cap W$, so the presheaf axiom is visible as associativity of intersection.

Example 12.16 (All functions with values in a set)**.** For a set A, define $\mathcal{F}(U) = A^U$, the set of all functions $U \to A$. Restriction is precomposition with the inclusion $V \hookrightarrow U$.

Example 12.17 (Vector-valued continuous functions)**.** Fix a real vector space V. Continuous maps $U \to V$ form a presheaf of real vector spaces. This is the section presheaf of the trivial bundle $X \times V \to X$.

Example 12.18 (Differential forms)**.** On a smooth manifold M, the assignment $U \mapsto \Omega^p(U)$ is a presheaf of real vector spaces. Pulling a form back along the inclusion $V \hookrightarrow U$ is exactly restriction.

Example 12.19 (Harmonic functions)**.** For an open subset $X \subseteq \mathbb{R}^n$, let

$$\mathcal{H}(U) = \{u \in C^2(U) : \Delta u = 0\}.$$

The differential equation is local and survives restriction, so $\mathcal{H}$ is a presheaf.

Example 12.20 (Solutions of a first-order ODE)**.** On $X = \mathbb{R}$, let $\mathcal{S}(U)$ be the smooth functions satisfying $u' = u$ on U. Restriction preserves the equation. On each connected component the solutions are ce^x.

Example 12.21 (Polynomial functions restricted from an ambient ring)**.** Fix $X = \mathbb{R}$. Let $\mathcal{P}_{\text{amb}}(U)$ be the set of restrictions to U of polynomials in $\mathbb{R}[x]$. Restriction is well defined and gives a presheaf of rings.

Example 12.22 (Rational functions without poles on the open)**.** On an open $U \subseteq \mathbb{R}$, let $\mathcal{R}(U)$ be rational functions p/q whose denominator has no zero on U. Shrinking U can only remove possible poles, so restriction is defined.

Example 12.23 (Bounded continuous functions)**.** Let $\mathcal{B}(U)$ be bounded continuous functions $U \to \mathbb{R}$. Restriction preserves boundedness, so $\mathcal{B}$ is a presheaf. It will later provide a clean example of a presheaf that fails gluing on an infinite cover.

Example 12.24 (Functions vanishing at a fixed point)**.** Fix $x_0 \in X$. Define

$$\mathcal{I}_{x_0}(U) = \begin{cases} \{f \in \mathcal{C}^0(U) : f(x_0) = 0\}, & x_0 \in U, \\ \mathcal{C}^0(U), & x_0 \notin U. \end{cases}$$

Restriction gives a subpresheaf of $\mathcal{C}^0$.

Example 12.25 (Skyscraper-type presheaf)**.** Fix $x \in X$ and an abelian group A. Set

$$\mathcal{K}_x(U) = \begin{cases} A, & x \in U, \\ 0, & x \notin U. \end{cases}$$

Restrictions are identity when both opens contain x and zero otherwise.

Example 12.26 (Product presheaf)**.** If $\mathcal{F}, \mathcal{G}$ are presheaves of abelian groups, then

$$U \longmapsto \mathcal{F}(U) \times \mathcal{G}(U)$$

with componentwise restriction is again a presheaf.

Example 12.27 (Kernel presheaf)**.** For a morphism $\varphi : \mathcal{F} \to \mathcal{G}$, define

$$(\ker \varphi)(U) = \ker(\varphi_U).$$

Naturality ensures that restrictions send kernels to kernels.

12.15 Advanced examples

Example 12.28 (Sierpiński space as one homomorphism)**.** Let $X = \{\eta, s\}$ with opens $\varnothing, \{\eta\}, X$. A normalized presheaf of abelian groups is exactly a homomorphism

$$A = \mathcal{F}(X) \longrightarrow B = \mathcal{F}(\{\eta\}).$$

Thus the abstract definition reduces to a single algebraic arrow.

Example 12.29 (A three-step finite topology)**.** Suppose the opens form a chain

$$\varnothing \subset U_2 \subset U_1 \subset X.$$

A normalized presheaf is a diagram

$$\mathcal{F}(X) \longrightarrow \mathcal{F}(U_1) \longrightarrow \mathcal{F}(U_2),$$

with the direct restriction $X \to U_2$ forced to be the composite.

Example 12.30 (Bounded functions: presheaf but not sheaf)**.** Cover $\mathbb{R}$ by $U_n = (-n, n)$ and take $s_n(x) = x$. Each s_n is bounded on U_n and the family is compatible. The only possible global glue is $s(x) = x$, which is unbounded on $\mathbb{R}$. Hence bounded continuous functions form a presheaf but fail the gluing axiom.

$$\boxed{\mathbb{R} = \textstyle\bigcup_{n\geq 1}(-n,n)}$$

$$\begin{tikzcd}
\boxed{(-1,1):\ s_1(x)=x} \arrow[dr] & \boxed{(-2,2):\ s_2(x)=x} \arrow[d] & \boxed{(-n,n):\ s_n(x)=x} \arrow[dl] \\
& \boxed{s(x)=x \text{ on } \mathbb{R}} & \begin{array}{l}\text{each } s_n \text{ bounded}\\ \text{global glue unbounded}\end{array}
\end{tikzcd}$$

Example 12.31 (Compact support can fail restriction)**.** Let $U = \mathbb{R}$ and choose a continuous function supported on $[0,1]$ and nonzero on $(0,1)$. Restrict to $V = (0,1)$. Its support inside V is all of $(0,1)$, which is not compact. Therefore "compactly supported continuous functions" is not a presheaf under ordinary restriction.

Example 12.32 (Singular cochains)**.** For an abelian group A, define

$$\mathcal{C}^q(U;A) = \mathrm{Hom}(C_q(U), A).$$

An inclusion $V \subseteq U$ induces $C_q(V) \to C_q(U)$, hence by precomposition a restriction

$$\mathcal{C}^q(U;A) \longrightarrow \mathcal{C}^q(V;A).$$

This is a presheaf of abelian groups.

Example 12.33 (Direct image)**.** For $f : X \to Y$ and continuous functions $\mathcal{C}^0_X$, the presheaf

$$(f_*\mathcal{C}^0_X)(V) = \mathcal{C}^0(f^{-1}V)$$

records functions on inverse images of opens in Y.

Example 12.34 (Localization on principal opens)**.** For $X = \operatorname{Spec} A$, define on the basis of principal opens

$$\mathcal{O}^{pre}(D(f)) = A_f.$$

For $D(fg) \subseteq D(f)$, restriction is $A_f \to A_{fg}$. The functoriality identity is the transitivity of localization.

Example 12.35 (Module localization)**.** If M is an A-module, the analogous basis assignment

$$\widetilde{M}^{pre}(D(f)) = M_f$$

has restrictions $M_f \to M_{fg}$. This is the prototype for quasi-coherent sheaves.

Example 12.36 (Arithmetic localization)**.** On $X = \operatorname{Spec} \mathbb{Z}$, the principal-open prototype assigns

$$D(n) \longmapsto \mathbb{Z}[1/n].$$

If $D(mn) \subseteq D(n)$, the restriction map is

$$\mathbb{Z}[1/n] \longrightarrow \mathbb{Z}[1/mn].$$

Example 12.37 (Objectwise image versus local image)**.** Given a morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$, the assignment

$$U \mapsto \mathrm{im}(\varphi_U)$$

is always a presheaf. Later, for sheaves, this objectwise image may be too small because a section can be locally in the image without being globally in the image.

Example 12.38 (Representable presheaf on opens)**.** Fix an open $U_0 \subseteq X$. Define

$$h_{U_0}(V) = \mathrm{Hom}_{\mathrm{Open}(X)}(V, U_0).$$

Because Open(X) is a poset, this is a singleton when $V \subseteq U_0$ and empty otherwise. It is a presheaf of sets and is the simplest entry point to the Yoneda viewpoint.

Example 12.39 (Agreement on a basis need not determine a presheaf)**.** Let X have a basis $\mathcal{B}$ that does not contain X itself. One may define two presheaves that agree on every $B \in \mathcal{B}$ and on all restrictions among basis opens, but assign different objects to X with restriction maps landing in the same basis data. Without a locality axiom, the basis data do not force the global value.

12.16 Exercises

Exercise 12.40 (VI.12.E01 — Restriction axioms)**.** Write the presheaf axioms explicitly for a chain $W \subseteq V \subseteq U$ and explain why they are precisely functoriality on Open$(X)^{op}$.

Exercise 12.41 (VI.12.E02 — Continuous functions)**.** Verify directly that $U \mapsto C^0(U, \mathbb{R})$ is a presheaf of rings.

Exercise 12.42 (VI.12.E03 — Power sets)**.** Verify that $U \mapsto \mathcal{P}(U)$ with $S \mapsto S \cap V$ is a presheaf of sets.

Exercise 12.43 (VI.12.E04 — Naive constant presheaf)**.** Define the naive constant presheaf of an abelian group A, including its value on $\varnothing$, and verify the axioms.

Exercise 12.44 (VI.12.E05 — Locally constant functions)**.** Show that locally constant A-valued functions form a presheaf.

Exercise 12.45 (VI.12.E06 — Bounded functions)**.** Show bounded continuous functions form a presheaf.

Exercise 12.46 (VI.12.E07 — Compact support failure)**.** Give an explicit continuous compactly supported function on $\mathbb{R}$ whose restriction to $(0, 1)$ is not compactly supported there.

Exercise 12.47 (VI.12.E08 — Morphisms)**.** For presheaves of rings, write the naturality square for a morphism $\varphi : \mathcal{F} \to \mathcal{G}$.

Exercise 12.48 (VI.12.E09 — Kernel presheaf)**.** Prove $U \mapsto \ker(\varphi_U)$ is a subpresheaf of $\mathcal{F}$.

Exercise 12.49 (VI.12.E10 — Image presheaf)**.** Prove $U \mapsto \mathrm{im}(\varphi_U)$ is a presheaf.

Exercise 12.50 (VI.12.E11 — Quotient presheaf)**.** If $\mathcal{H} \subseteq \mathcal{F}$ are presheaves of abelian groups, construct the quotient presheaf sectionwise.

Exercise 12.51 (VI.12.E12 — Restriction to an open)**.** For $j : U \hookrightarrow X$, prove $\mathcal{F}|_U$ is a presheaf and that $(\mathcal{F}|_U)|_V = \mathcal{F}|_V$ for $V \subseteq U$.

Exercise 12.52 (VI.12.E13 — Direct image)**.** For a continuous map $f : X \to Y$, verify that $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V)$ defines a presheaf.

Exercise 12.53 (VI.12.E14 — Composition of direct images)**.** For $X \xrightarrow{f} Y \xrightarrow{g} Z$, prove $(g \circ f)_*\mathcal{F} = g_*(f_*\mathcal{F})$.

Exercise 12.54 (VI.12.E15 — Sierpinski classification)**.** Classify normalized presheaves of abelian groups on the Sierpiński space in terms of one group homomorphism.

Exercise 12.55 (VI.12.E16 — Three-open chain)**.** Describe presheaves on $\varnothing \subset U_2 \subset U_1 \subset X$ as composable homomorphisms.

Exercise 12.56 (VI.12.E17 — Skyscraper-type presheaf)**.** Verify the fixed-point assignment $\mathcal{K}_x(U) = A$ if $x \in U$ and 0 otherwise is a presheaf.

Exercise 12.57 (VI.12.E18 — ODE solutions)**.** Show solutions of $u' = u$ form a presheaf and compute sections on a disconnected open subset of $\mathbb{R}$.

Exercise 12.58 (VI.12.E19 — Singular cochains)**.** Show $U \mapsto C^q(U; A)$ is contravariant under inclusions.

Exercise 12.59 (VI.12.E20 — Principal-open localization)**.** For $D(fg) \subseteq D(f)$, construct the restriction $A_f \to A_{fg}$ and verify transitivity for $D(fgh) \subseteq D(fg) \subseteq D(f)$.

Exercise 12.60 (VI.12.E21 — Module localization)**.** Repeat the preceding exercise for an A-module M and the maps $M_f \to M_{fg}$.

Exercise 12.61 (VI.12.E22 — Arithmetic principal opens)**.** On $\operatorname{Spec}\mathbb{Z}$, compute the maps attached to $D(30) \subseteq D(6) \subseteq D(2)$.

Exercise 12.62 (VI.12.E23 — Presheaf but not sheaf)**.** Use $U_n = (-n, n)$ and $s_n(x) = x$ to prove bounded continuous functions fail the gluing axiom on $\mathbb{R}$.

Exercise 12.63 (VI.12.E24 — Basis warning)**.** Construct two presheaves on a finite topological space that agree on a proper basis but differ on the whole space.

12.17 Problems

Problem 12.64. Reformulate the presheaf axioms as contravariant functoriality.

Problem 12.65. Classify presheaves of abelian groups on the Sierpinski space.

Problem 12.66. Analyze the naive constant presheaf and its gluing failure on disconnected opens.

Problem 12.67. Show that bounded continuous functions form a presheaf but not, in general, a sheaf.

Problem 12.68. Show that compactly supported continuous functions do not form a presheaf under arbitrary restriction.

Problem 12.69. Construct and analyze morphisms of presheaves using the naturality square.

Problem 12.70. Construct kernels, images, and quotients objectwise in the presheaf category.

Problem 12.71. Prove that restriction to an open subset is functorial.

Problem 12.72. Construct the direct image presheaf under a continuous map.

Problem 12.73. Analyze a presheaf of local solutions to a differential equation.

Problem 12.74. Construct the singular cochain presheaf and its restriction maps.

Problem 12.75. Build the localization presheaf on the basis of principal opens of an affine spectrum.

Problem 12.76. Build the module-localization presheaf $D(f) \mapsto M_f$.

Problem 12.77. Compare the naive constant presheaf with locally constant functions.

Problem 12.78. Construct a presheaf on a three-step finite topology and read it as a diagram.

Problem 12.79. Give two presheaves that agree on a chosen basis but differ on a larger open set.

Problem 12.80. Analyze a skyscraper-type presheaf supported at one point.

Problem 12.81. Prove a Yoneda-style classification for representable presheaves on $\mathrm{Open}(X)$.

12.18 Challenges

Problem 12.82. Classify presheaves on a finite Alexandrov space as contravariant diagrams on its specialization poset.

Problem 12.83. Prove the open-set Yoneda lemma $\mathrm{Nat}(h_U, \mathcal{F}) \cong \mathcal{F}(U)$.

Problem 12.84. Construct nonisomorphic presheaves whose restrictions to a fixed basis are isomorphic.

Problem 12.85. Develop the localization presheaf on principal opens and prove all restriction maps are independent of chosen factorizations.

12.19 Legacy-corpus boundary

The legacy corpus contains substantial sheaf-family material, but its numbered problems are not all VI/12 problems. The present chapter absorbs the presheaf definitions, restriction formalism, and legacy examples belonging here. The associated-sheaf exercise is reserved for VI/14, image and quotient problems for VI/15, stalkwise exactness and isomorphism criteria for VI/16, and left exactness of global sections for the later cohomology arc. The audit appendix records these destinations explicitly so no substantive legacy problem disappears or is counted twice.

12.20 Boundary with the next chapters

VI/12 supplies the raw contravariant data. VI/13 adds the locality and gluing axioms and develops germs and stalks. VI/14 constructs sheafification, VI/15 studies kernels, images, and quotient sheaves, VI/16 develops exact sequences, and VI/17 specializes the formalism to the structure sheaf of an affine spectrum.

12.21 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 12.86 (Identity restriction)**.** What must the restriction $F(U) \to F(U)$ be?

Hint. Use restriction maps and presheaf compatibility. ☐

Solution. The identity. ☐

Exercise 12.87 (Composition)**.** What compatibility holds for $W \subseteq V \subseteq U$?

Hint. Use restriction maps and presheaf compatibility. ☐

Solution. Restriction from U to W equals the composite through V. ☐

Exercise 12.88 (Constant presheaf)**.** What is its restriction map?

Hint. Use restriction maps and presheaf compatibility. ☐

Solution. The identity on the fixed set or ring. ☐

Exercise 12.89 (Localization restriction)**.** What map models A_f restricted to $D(fg)$?

Hint. Use restriction maps and presheaf compatibility. ☐

Solution. $A_f \to A_{fg}$. ☐

Exercise 12.90 (Empty open)**.** Does a presheaf axiom alone force a unique section over the empty set?

Hint. Use restriction maps and presheaf compatibility. ☐

Solution. No; that is part of sheaf behavior in standard categories. ☐

Proofs

Exercise 12.91 (Function presheaf)**.** Prove: Prove continuous functions with restriction form a presheaf.

Hint. Work directly from restriction maps and presheaf compatibility. ☐

Solution. Identity and transitivity of ordinary restriction give the two axioms. ☐

Exercise 12.92 (Localization transitivity)**.** Prove: Prove the localization restriction maps compose correctly.

Hint. Work directly from restriction maps and presheaf compatibility. ☐

Solution. Universal localization maps send the same fractions to the same further localized fractions. ☐

Exercise 12.93 (Morphisms)**.** Prove: Prove a morphism of presheaves respects all restrictions.

Hint. Work directly from restriction maps and presheaf compatibility. ☐

Solution. Write the commutative square for every inclusion and check the defining naturality condition. ☐

Exercise 12.94 (Kernel presheaf)**.** Prove: Prove pointwise kernels of a presheaf morphism form a presheaf.

Hint. Work directly from restriction maps and presheaf compatibility. ☐

Solution. Restrictions preserve zero and commute with the morphism, so kernel sections restrict into kernels. ☐

Counterexamples and hypothesis tests

Exercise 12.95 (Gluing)**.** Test the claim: every presheaf automatically allows compatible local sections to glue.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; gluing is the extra sheaf axiom. ☐

Exercise 12.96 (Uniqueness)**.** Test the claim: a presheaf automatically has uniqueness of local gluing.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 12.97 (Constant sheaf)**.** Test the claim: the constant presheaf is always already a sheaf.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False on spaces with suitable disconnected opens. ☐

Applications and investigations

Exercise 12.98 (Local data)**.** Why are presheaves a natural language for local information?

Hint. Translate the question into restriction maps and presheaf compatibility. ☐

Solution. They assign data to opens and provide coherent restriction to smaller regions. ☐

Exercise 12.99 (Algebra localization)**.** Why do localizations naturally form presheaf-like data?

Hint. Translate the question into restriction maps and presheaf compatibility. ☐

Solution. Further restriction corresponds to inverting additional functions. ☐

Challenge problems

Exercise 12.100 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Presheaves in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Presheaves, translates it through restriction maps and presheaf compatibility, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 12.101 (Hypothesis challenge)**.** Choose one structural statement from Presheaves, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 13

Sheaves and Stalks

Purpose and learning goals

VI/12 introduced presheaves: algebraic data attached contravariantly to open sets. The present chapter adds the decisive local-to-global principle. A *sheaf* is a presheaf in which compatible local sections glue uniquely, and a *stalk* records everything a sheaf knows arbitrarily close to one point.

These two ideas are complementary. The sheaf axiom moves from local data to a global section; the stalk moves from sections on neighborhoods to a single local object. Together they form the basic language of modern geometry. Later chapters will sheafify presheaves, construct kernels and quotients, define the structure sheaf, and use exact sequences and cohomology. All of those constructions depend on the material developed here.

This chapter retains the example-heavy policy introduced in VI/12. The ordinary example bank builds fluency with locality, gluing, germs, and stalks; a separate advanced bank studies discrete and irreducible spaces, skyscraper sheaves, local branches, support, basis criteria, and stalkwise detection.

After this chapter, the reader should be able to:

- state the locality and gluing axioms of a sheaf;
- reformulate the sheaf axiom as an equalizer condition for sheaves of groups;
- prove that continuous, smooth, holomorphic, and locally constant functions form sheaves;
- diagnose standard presheaves that fail the sheaf axiom;
- define germs explicitly as equivalence classes of local sections;
- construct the stalk $\mathcal{F}_x$ as a filtered colimit;
- test equality of germs by shrinking neighborhoods;
- compute stalks in standard analytic, topological, and algebraic examples;
- construct the stalk map induced by a morphism of sheaves;
- prove that sheaf sections and sheaf morphisms are detected by all stalks;
- prove that a morphism of sheaves is an isomorphism when every stalk map is an isomorphism;
- define and compute skyscraper sheaves;
- describe sheaves on discrete spaces directly from their stalks;

- understand why a naive constant presheaf differs from the constant sheaf;
- use a basis of the topology to verify the sheaf condition;
- define the support of a section using its germs;
- keep sheafification, quotient sheaves, and exactness in their proper later chapters.

13.1 From presheaves to sheaves

Let X be a topological space and let $\mathcal{F}$ be a presheaf of sets, groups, rings, or modules. Presheaf axioms tell us only how to restrict data. They do not say that a section is determined by its restrictions, nor that compatible local sections arise from a global one.

Definition 13.1 (Sheaf)**.** A presheaf $\mathcal{F}$ is a *sheaf* if for every open set $U \subseteq X$, every open cover $U = \bigcup_{i\in I} U_i$, and every family $s_i \in \mathcal{F}(U_i)$, the following hold.

1. **Locality:** if $s, t \in \mathcal{F}(U)$ and $s|_{U_i} = t|_{U_i}$ for all i, then $s = t$.
2. **Gluing:** if
$$s_i|_{U_i\cap U_j} = s_j|_{U_i\cap U_j} \qquad \text{for all } i, j,$$
then there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for every i.

By locality, a glued section is automatically unique.

$$\begin{tikzcd}
s_1 \in \mathcal{F}(U_1) \arrow[dr] & & s_2 \in \mathcal{F}(U_2) \arrow[dl] \\
& s_1|_{U_1\cap U_2} = s_2|_{U_1\cap U_2} \arrow[d, "\text{glue}"] & \\
& \exists!\ s \in \mathcal{F}(U_1 \cup U_2) &
\end{tikzcd}$$

The slogan is

$$\boxed{\text{compatible local data glue to one and only one global section.}}$$

13.2 The empty set and uniqueness

The empty cover of $\varnothing$ forces a small but useful convention. For a sheaf of sets, $\mathcal{F}(\varnothing)$ is a singleton. For a sheaf of groups, rings with zero object in the chosen category, or modules, it is the terminal object appropriate to the category; in particular for abelian groups and modules,
$$\mathcal{F}(\varnothing) = 0.$$

Proposition 13.2 (Uniqueness follows from locality)**.** *Suppose a presheaf satisfies locality. If a compatible family $s_i \in \mathcal{F}(U_i)$ has two gluings $s, t \in \mathcal{F}(U)$, then $s = t$.*

Proof. Both sections restrict to s_i on every U_i, so locality gives equality. $\square$

13.3 The equalizer form of the sheaf axiom

For sheaves of abelian groups, the two axioms can be packaged into one exactness statement. Let $U = \bigcup_i U_i$. Define

$$\Delta : \mathcal{F}(U) \longrightarrow \prod_i \mathcal{F}(U_i), \qquad s \longmapsto (s|_{U_i})_i.$$

The two maps to pairwise overlaps are restriction from the first and second index.

$$0 \longrightarrow \mathcal{F}(U) \xrightarrow{\Delta} \prod_i \mathcal{F}(U_i) \overset{\alpha}{\underset{\beta}{\rightrightarrows}} \prod_{i,j} \mathcal{F}(U_i \cap U_j),$$

$$\alpha((s_i)) = (s_i|_{U_i \cap U_j})_{i,j}, \qquad \beta((s_i)) = (s_j|_{U_i \cap U_j})_{i,j}.$$

Proposition 13.3 (Equalizer criterion)**.** *A presheaf of abelian groups is a sheaf if and only if for every open cover the displayed sequence is exact at the first two terms; equivalently, $\mathcal{F}(U)$ is the equalizer of α and β.*

Proof. Injectivity of Δ is locality. Its image consists precisely of families whose two restrictions to every overlap agree, and gluing says exactly that every such family lies in the image. ☐

Example 13.4 (Gluing regular functions)**.** If functions on an open cover agree on every overlap, the sheaf axiom produces a unique function on the union. For ordinary continuous functions, the glued function is defined pointwise from any local representative.

13.4 Standard sheaves of functions

The most important elementary examples are defined by local conditions on functions.

Proposition 13.5 (Continuous functions)**.** *For a topological space X, the assignment*

$$U \longmapsto C^0(U, \mathbb{R})$$

with ordinary restriction is a sheaf of rings.

Proof. Functions agreeing on a cover agree pointwise. Compatible continuous functions define a unique function $f : U \to \mathbb{R}$ pointwise; continuity is local, so f is continuous. ☐

Exactly the same proof works for smooth functions on a smooth manifold, holomorphic functions on a complex manifold, and sections of a vector bundle: the defining condition is local and compatible sections glue as ordinary functions or local bundle sections.

Proposition 13.6 (Locally constant functions)**.** *For an abelian group A, the assignment*

$$\underline{A}(U) = \{f : U \to A : f \text{ is locally constant}\}$$

is a sheaf of abelian groups. It is called the constant sheaf *with value A.*

$$\begin{tikzcd}
\boxed{U = \coprod_{\alpha \in \pi_0(U)} U_\alpha} \arrow[d] \\
\boxed{\underline{A}(U) = \{\text{locally constant } U \to A\}} \arrow[d] \\
\boxed{\underline{A}(U) \cong \prod_{\alpha \in \pi_0(U)} A}
\end{tikzcd}$$

This is different from the naive constant presheaf of VI/12. On a disconnected open set, locally constant functions may take different values on different connected components.

13.5 Presheaves that fail to be sheaves

The sheaf axiom is strong enough to rule out natural-looking global conditions.

Example 13.7 (Bounded continuous functions)**.** Let $\mathcal{B}(U)$ be the bounded continuous real-valued functions on U. Restriction makes $\mathcal{B}$ a presheaf. On $\mathbb{R}$ cover by $U_n = (-n, n)$. The functions $s_n(x) = x$ on U_n are bounded on each U_n and compatible, but their only possible gluing is $x \mapsto x$, which is unbounded on $\mathbb{R}$. Thus $\mathcal{B}$ is not a sheaf.

Example 13.8 (The naive constant presheaf)**.** On a space with a disconnected open $U = U_1 \sqcup U_2$, choose distinct $a, b \in A$ on the two components. They are compatible because $U_1 \cap U_2 = \varnothing$, but there is no single element of the naive constant presheaf on U restricting to both. Hence the naive constant presheaf generally fails gluing.

13.6 Germs of sections

A stalk is built from germs, so it is useful to define germs concretely before invoking a colimit.

Fix $x \in X$. Consider pairs (U, s) with $x \in U$ open and $s \in \mathcal{F}(U)$. Declare

$$(U, s) \sim (V, t)$$

if there exists an open neighborhood W of x with $W \subseteq U \cap V$ and

$$s|_W = t|_W.$$

This is an equivalence relation. The class of (U, s) is the *germ of s at x*, denoted s_x.

$$\begin{tikzcd}
\boxed{\begin{array}{c}(U,s)\\ x \in U\end{array}} \arrow[dr] & & \boxed{\begin{array}{c}(V,t)\\ x \in V\end{array}} \arrow[dl] \\
& \boxed{\begin{array}{c}\exists\, W \ni x,\, W \subseteq U \cap V\\ s|_W = t|_W\end{array}} \arrow[d] & \\
& \boxed{[U,s]_x = [V,t]_x \in \mathcal{F}_x} &
\end{tikzcd}$$

Warning 13.9 (Same value does not mean same germ). For function sheaves, equality $f(x) = g(x)$ is much weaker than $f_x = g_x$. For example, $f(t) = t$ and $g(t) = 0$ have the same value at 0 but different germs there.

13.7 The stalk as a filtered colimit

Definition 13.10 (Stalk)**.** The *stalk* of a presheaf $\mathcal{F}$ at $x \in X$ is

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U),$$

where neighborhoods are ordered by reverse inclusion through restriction maps.

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[rr] \arrow[dr] & & \mathcal{F}(V) \arrow[dl] \\
& \mathcal{F}_x &
\end{tikzcd}$$

Here $x \in V \subseteq U$; every neighborhood section maps to its germ in the filtered colimit.

The equivalence-class construction above and this filtered-colimit definition give the same object. If $\mathcal{F}$ is a presheaf of groups, rings, or modules, the stalk inherits the same algebraic structure.

Proposition 13.11 (Equality in a stalk)**.** *For sections $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ with $x \in U \cap V$,*

$$s_x = t_x$$

if and only if there exists a neighborhood $W \subseteq U \cap V$ of x such that $s|_W = t|_W$.

Example 13.12 (A stalk as germs)**.** The stalk F_x identifies sections defined on possibly different neighborhoods of x when they agree on some smaller neighborhood. Thus only behavior arbitrarily close to x survives.

13.8 What a stalk remembers and forgets

A germ remembers all behavior on some sufficiently small unspecified neighborhood; it forgets what happens far from the point. In particular:

- two global sections can define the same germ even when they differ elsewhere;
- a stalk is generally much richer than point evaluation;
- global topological obstructions can disappear in a stalk;
- sheafification will later preserve stalks even when it changes global sections.

For the sheaf C^0_X of continuous real-valued functions, $(C^0_X)_x$ is the ring of germs of continuous functions at x. For C^∞_M, it is the local ring of smooth germs. For the holomorphic structure sheaf on a complex manifold, it is the local ring of holomorphic germs.

13.9 Morphisms and maps on stalks

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. Define

$$\varphi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x, \qquad \varphi_x(s_x) := (\varphi_U(s))_x.$$

Naturality with restriction makes this independent of the representative.

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "\varphi_U"] \arrow[d] & \mathcal{G}(U) \arrow[d] \\
\mathcal{F}_x \arrow[r, "\varphi_x"'] & \mathcal{G}_x
\end{tikzcd}$$

Proposition 13.13 (Functoriality on stalks)**.** *For morphisms $\mathcal{F} \xrightarrow{\varphi} \mathcal{G} \xrightarrow{\psi} \mathcal{H}$,*

$$(\psi \circ \varphi)_x = \psi_x \circ \varphi_x, \qquad (\mathrm{id}_{\mathcal{F}})_x = \mathrm{id}_{\mathcal{F}_x}.$$

13.10 Stalks detect sections and morphisms

The next statements are special to sheaves; they can fail for arbitrary presheaves.

Theorem 13.14 (Sections are determined by their germs)**.** *Let $\mathcal{F}$ be a sheaf and $s, t \in \mathcal{F}(U)$. Then*

$$s = t \iff s_x = t_x \text{ for every } x \in U.$$

Proof. Only the reverse implication needs proof. For each x, equality of germs gives a neighborhood $V_x \subseteq U$ on which s and t agree. The V_x cover U, and locality gives $s = t$. □

Corollary 13.15 (Morphisms are determined stalkwise)**.** *If $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ are morphisms of sheaves and $\varphi_x = \psi_x$ for every $x \in X$, then $\varphi = \psi$.*

Theorem 13.16 (Stalkwise isomorphism criterion)**.** *A morphism of sheaves of sets, groups, rings, or modules*

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

is an isomorphism if and only if every stalk map

$$\varphi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x$$

is an isomorphism.

Proof. Necessity is immediate. Conversely, let $t \in \mathcal{G}(U)$. Stalkwise surjectivity gives, for each $x \in U$, a germ $a_x \in \mathcal{F}_x$ mapping to t_x. Represent a_x by a section s_x on a neighborhood V_x and shrink so that $\varphi(s_x) = t|_{V_x}$. On overlaps, stalkwise injectivity implies the germs of the two local lifts agree; after shrinking around every point they agree as sections, and locality shows they agree on the whole overlap. The sheaf axiom glues the s_x uniquely to a section $s \in \mathcal{F}(U)$ with $\varphi(s) = t$. The same stalkwise injectivity argument shows uniqueness. Thus every φ_U is bijective and the inverse maps commute with restrictions. □

Remark 13.17 (A boundary for VI/16)*.* The full theory of injective/surjective morphisms and exact sequences of sheaves is deferred to VI/16. In particular, sectionwise surjectivity is stronger than the natural local notion of surjectivity for sheaves.

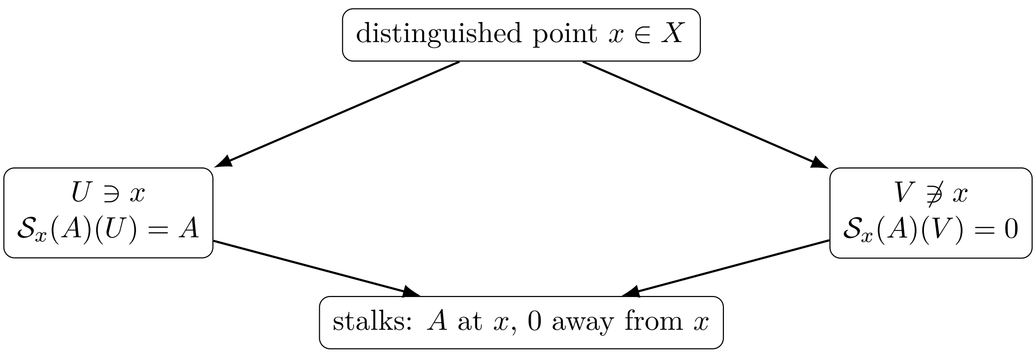
13.11 Skyscraper sheaves

Fix $x \in X$ and an abelian group A.

Definition 13.18 (Skyscraper sheaf)**.** Define

$$\mathcal{S}_x(A)(U) = \begin{cases} A, & x \in U, \\ 0, & x \notin U, \end{cases}$$

with the evident restriction maps. This is the *skyscraper sheaf at x with value A.*



Its stalk is A at x and 0 at every $y \neq x$ provided points can be separated by a neighborhood of y avoiding x; in particular this holds in T_1 spaces. On arbitrary spaces, the stalk at y is A exactly when every neighborhood of y contains x.

Example 13.19 (Structure-sheaf stalk preview)**.** For $X = \operatorname{Spec} A$, the stalk of the structure sheaf at $\mathfrak{p}$ is $A_{\mathfrak{p}}$. Fractions become equivalent precisely when they agree on a sufficiently small basic neighborhood.

13.12 Sheaves on discrete spaces

If X is discrete, every singleton is open and every open U is the disjoint union of its points. The sheaf axiom therefore gives a complete classification.

Theorem 13.20 (Discrete-space classification)**.** *For a sheaf of sets or abelian groups on a discrete space X,*

$$\mathcal{F}(U) \cong \prod_{x\in U} \mathcal{F}_x.$$

Thus a sheaf on X is determined by the family of its stalks.

$$\boxed{X \text{ discrete}, \quad \mathcal{F}(U) \;\cong\; \prod_{x\in U} \mathcal{F}_x}$$

$$\begin{tikzcd}
\boxed{\mathcal{F}(U)} \arrow[r, leftrightarrow] & \boxed{\prod_{x\in U} \mathcal{F}(\{x\})} \arrow[r, leftrightarrow] & \boxed{\prod_{x\in U} \mathcal{F}_x}
\end{tikzcd}$$

13.13 Sheaf condition on a basis

In geometry one rarely wants to verify gluing on every open set separately. A basis is enough.

Theorem 13.21 (Basis criterion)**.** *Let $\mathcal{B}$ be a basis for the topology of X, closed under taking sufficiently small basis neighborhoods inside intersections. Suppose algebraic data $\mathcal{F}(B)$ and restriction maps are given for $B \in \mathcal{B}$, and suppose locality and gluing hold for covers of basis opens by basis opens. Then the data extend uniquely, up to unique isomorphism, to a sheaf on all open subsets of X.*

$$\begin{tikzcd}
\boxed{\text{basis } \mathcal{B} \text{ of opens}} \arrow[d] \\
\boxed{\text{sections + restrictions on } B \in \mathcal{B}} \arrow[d] \\
\boxed{\text{locality + gluing for basis covers}} \arrow[d] \\
\boxed{\text{unique sheaf on all opens}}
\end{tikzcd}$$

This criterion is the bridge to the structure sheaf: on $\operatorname{Spec} A$, principal opens $D(f)$ form a basis and carry localization rings A_f.

13.14 Support of a section

Let $\mathcal{F}$ be a sheaf of abelian groups and $s \in \mathcal{F}(U)$. Define

$$\mathrm{NZ}(s) = \{x \in U : s_x \neq 0\}.$$

Because vanishing of a germ means vanishing on some neighborhood, the zero-germ locus

$$\{x \in U : s_x = 0\}$$

is open. Therefore $\mathrm{NZ}(s)$ is already closed in U.

Definition 13.22 (Support)**.** The *support* of s is the closed set

$$\mathrm{supp}(s) = \{x \in U : s_x \neq 0\}.$$

For ordinary function sheaves this equals the closure of the pointwise nonvanishing set $\{x : s(x) \neq 0\}$.

$$\mathrm{supp}(s) = \mathrm{NZ}(s) := \{x \in U : s_x \neq 0\}.$$

$$\begin{tikzcd}
\boxed{s \in \mathcal{F}(U)} \arrow[r] & \boxed{\text{germs } s_x} \arrow[r] & \boxed{\{x : s_x \neq 0\}} \arrow[d] \\
 & & \boxed{\text{closed because the zero-germ locus is open}}
\end{tikzcd}$$

For ordinary continuous or smooth functions this agrees with the usual support.

13.15 Common pitfalls

1. A presheaf is not automatically a sheaf; restriction axioms do not imply gluing.
2. Equality at a point is not equality of germs.
3. A germ is not a section on a distinguished smallest neighborhood; most spaces have no such neighborhood.
4. Stalks forget global topology. A global obstruction can vanish locally at every point.
5. A family of arbitrary germs does not necessarily come from a section; the family must vary locally in a representable way.
6. The naive constant presheaf and the constant sheaf are different on disconnected opens.
7. Stalkwise surjectivity of a morphism does not mean every section has one global lift.
8. The structure sheaf is not yet being defined; VI/17 will specialize the present theory to regular functions on spectra.

13.16 Worked examples

Example 13.23 (Continuous real-valued functions)**.** For each open $U \subseteq X$, let $\mathcal{C}^0(U) = C^0(U, \mathbb{R})$. Restrictions are ordinary function restrictions. If continuous functions agree on overlaps, the pointwise glued function is continuous because continuity is local. Hence $\mathcal{C}^0$ is a sheaf of rings.

Example 13.24 (Smooth functions)**.** On a smooth manifold M, $U \mapsto C^\infty(U)$ is a sheaf. Compatible smooth functions glue as functions, and smoothness can be checked in a neighborhood of each point.

Example 13.25 (Holomorphic functions)**.** On an open subset $X \subseteq \mathbb{C}$, let $\mathcal{O}(U)$ be the holomorphic functions on U. Compatible holomorphic functions glue uniquely, and holomorphicity is local. The stalk $\mathcal{O}_x$ is the ring of holomorphic germs at x.

Example 13.26 (Locally constant functions)**.** Let A be an abelian group. The constant sheaf $\underline{A}$ assigns to U the locally constant maps $U \to A$. If U has connected components U_α, then $\underline{A}(U) \cong \prod_\alpha A$.

Example 13.27 (The naive constant presheaf fails gluing)**.** Let $U = U_1 \sqcup U_2$ with both pieces nonempty. In the naive constant presheaf choose a on U_1 and $b \neq a$ on U_2. The overlap is empty, so compatibility is automatic, but no single element over U restricts to both. This is the standard reason the constant sheaf must use locally constant functions.

Example 13.28 (Bounded functions fail gluing)**.** On $\mathbb{R}$, define

$$\mathcal{B}(U) = \{f : U \to \mathbb{R} : f \text{ is continuous and bounded}\}.$$

This is a presheaf. On $U_n = (-n, n)$, the sections $x \mapsto x$ are compatible and individually bounded. Their only possible gluing on $\mathbb{R}$ is unbounded, so $\mathcal{B}$ is not a sheaf.

Example 13.29 (Two sections with the same value but different germs)**.** In $C^0_{\mathbb{R}}$, let $f(t) = t$ and $g(t) = 0$. Then $f(0) = g(0)$, but $f_0 \neq g_0$ because they do not agree on any neighborhood of 0.

Example 13.30 (Two different global sections with the same germ)**.** Let $f(t) = 0$ for all t and let $g(t) = 0$ on $(-1, 1)$ but $g(t) = t^2 - 1$ outside a slightly larger neighborhood, smoothed continuously across the transition. Then $f_0 = g_0$ although $f \neq g$ globally. The stalk ignores behavior far from 0.

Example 13.31 (Germs of continuous functions)**.** An element of $(C^0_X)_x$ is represented by a continuous function on some neighborhood of x. Two representatives are equal exactly when they agree after shrinking. Addition and multiplication are performed on a common smaller neighborhood.

Example 13.32 (Germs of smooth functions)**.** For $M = \mathbb{R}$ and $x = 0$, the germs of e^t, $1 + t + t^2/2 + \cdots$, and any actual smooth function agreeing with e^t near 0 define the same element of $C^\infty_{M,0}$. A Taylor series alone does not determine a smooth germ, however: distinct smooth germs can have the same Taylor series.

Example 13.33 (Holomorphic germs are rigid)**.** For holomorphic functions, the identity theorem gives extra rigidity. If two holomorphic germs agree on a set with an accumulation point in a connected representative neighborhood, the germs agree. This contrasts with smooth germs.

Example 13.34 (A germ represented on many neighborhoods)**.** The germ at 0 of $1/(1-t)$ can be represented on any interval $(-r, r)$ with $0 < r < 1$. There is no preferred representative; the stalk identifies all compatible restrictions.

Example 13.35 (Stalk of the constant sheaf)**.** For every $x \in X$, $(\underline{A})_x \cong A$. A locally constant function is constant on some neighborhood of x, so its germ is determined by one element of A, and every element occurs.

Example 13.36 (Skyscraper at a point of a Hausdorff space)**.** Let $\mathcal{S}_x(A)$ be the skyscraper sheaf. At x its stalk is A. If $y \neq x$ and X is Hausdorff (indeed T_1 suffices), choose a neighborhood of y avoiding x; on that neighborhood the sheaf is zero, so the stalk at y is zero.

Example 13.37 (Sections on a two-point discrete space)**.** Let $X = \{p, q\}$ be discrete and let a sheaf have stalks A at p and B at q. Then

$$\mathcal{F}(X) \cong A \times B, \qquad \mathcal{F}(\{p\}) \cong A, \qquad \mathcal{F}(\{q\}) \cong B.$$

The sheaf condition reconstructs global sections from the two independent point sections.

Example 13.38 (A sheaf of maps to a space)**.** Fix a topological space Y. The assignment $U \mapsto C^0(U, Y)$ is a sheaf of sets. Local continuous maps agreeing on overlaps glue to a unique continuous map.

Example 13.39 (Sections of a covering map)**.** For a covering map $p : E \to X$, assign to U the continuous sections $s : U \to E$ with $p \circ s = \mathrm{id}_U$. Compatible local sections glue as maps, so this is a sheaf of sets.

Example 13.40 (Differential forms)**.** On a smooth manifold, $U \mapsto \Omega^k(U)$ is a sheaf of vector spaces. The exterior derivative commutes with restriction and therefore gives a morphism of sheaves $d : \Omega^k \to \Omega^{k+1}$.

Example 13.41 (A sheaf morphism and its stalk map)**.** The derivative $D : C^\infty_{\mathbb{R}} \to C^\infty_{\mathbb{R}}$ is compatible with restriction. At each x, it induces D_x sending the germ of f to the germ of f'. The definition does not depend on which neighborhood represents the germ.

Example 13.42 (Support of a smooth section)**.** Let $f \in C^\infty(\mathbb{R})$ be a bump function positive on $(-1, 1)$ and zero outside $[-1, 1]$. The germ f_x is zero for $|x| > 1$ and nonzero for $|x| < 1$. Hence $\mathrm{supp}(f) = [-1, 1]$.

13.17 Advanced worked examples

Example 13.43 (Advanced: every sheaf on a discrete space is a product of stalks)**.** Let X be discrete. Cover $U \subseteq X$ by singleton opens. There are no nontrivial overlap conditions, so the sheaf axiom gives

$$\mathcal{F}(U) \xrightarrow{\sim} \prod_{x \in U} \mathcal{F}(\{x\}) \cong \prod_{x \in U} \mathcal{F}_x.$$

Thus the category of sheaves of abelian groups on a discrete X is equivalent to the product category $\prod_{x \in X} \mathbf{Ab}$.

Example 13.44 (Advanced: the naive constant presheaf on an irreducible space)**.** Suppose every two nonempty opens of X meet. Then the naive constant presheaf A (with value 0 on $\varnothing$ in the abelian-group case) is actually a sheaf: in any open cover, compatibility forces all local constants to be equal because every pair of nonempty cover members meets. Thus failure of the naive constant presheaf is controlled by the connectedness pattern of covers, not by constancy alone.

Example 13.45 (Advanced: a Sierpinski space)**.** Let $X = \{\eta, x\}$ with opens $\varnothing, \{\eta\}, X$. A presheaf of abelian groups is a single restriction map $A = \mathcal{F}(X) \to B = \mathcal{F}(\{\eta\})$, together with $\mathcal{F}(\varnothing) = 0$. Every cover of X contains X itself, so no additional gluing condition appears: every such presheaf is a sheaf. The stalks are $\mathcal{F}_x \cong A$ and $\mathcal{F}_\eta \cong B$.

Example 13.46 (Advanced: skyscraper sheaf on a non-T_1 space)**.** For the same Sierpinski space, place a skyscraper at η. Every neighborhood of x contains η, so the stalk at x is also A. This shows why the common slogan "skyscraper stalks vanish away from the support point" silently uses a separation hypothesis.

Example 13.47 (Advanced: local holomorphic logarithms on $\mathbb{C}^*$)**.** Let $\mathcal{L}(U)$ be the set of holomorphic functions g on $U \subseteq \mathbb{C}^*$ satisfying $e^g = z$. Restriction gives a sheaf of sets. Every point has a neighborhood admitting such a branch, but $\mathcal{L}(\mathbb{C}^*) = \varnothing$. Thus nonempty stalks do not imply a global section; gluing can be obstructed by incompatible choices around a loop.

Example 13.48 (Advanced: local square roots)**.** On $X = \mathbb{C}^*$, let $\mathcal{R}(U)$ be holomorphic g with $g^2 = z$. Again the stalk at every point is nonempty, while there is no global holomorphic square root on all of $\mathbb{C}^*$. On a simply connected slit domain there are two global branches. This is a concrete preview of monodromy without requiring its formal theory.

Example 13.49 (Advanced: sheaf of local solutions)**.** Let L be a linear differential operator on a manifold. Define

$$\mathcal{S}(U) = \{u \in C^\infty(U) : Lu = 0\}.$$

Because L is local, compatible local solutions glue and remain solutions. Thus $\mathcal{S}$ is a sheaf. For $L = d/dx$ on an interval, $\mathcal{S}$ is the constant sheaf $\underline{\mathbb{R}}$.

Example 13.50 (Advanced: germs of rational functions regular at a point)**.** Let k be a field and consider the affine line informally through its classical points. At a point a, rational functions f/g with $g(a) \neq 0$ form a local ring under germ equivalence. This is the geometric pattern formalized in VI/11 by $k[t]_{(t-a)}$ and later in VI/17 by the structure sheaf.

Example 13.51 (Advanced: stalk of a direct image)**.** For a continuous map $f : X \to Y$ and a sheaf $\mathcal{F}$ on X,

$$(f_*\mathcal{F})_y = \varinjlim_{y \in V} \mathcal{F}(f^{-1}V).$$

This need not equal a single stalk of $\mathcal{F}$ because $f^{-1}(V)$ can contain many points. For the map from a two-point discrete space to a point, $(f_*\mathcal{F})_* = \mathcal{F}(X) \cong \mathcal{F}_p \times \mathcal{F}_q$.

Example 13.52 (Advanced: support uses germs, not point values)**.** For the continuous function $f(x) = x$ on $\mathbb{R}$, the point value at 0 vanishes but the germ f_0 is nonzero, because f is not identically zero on any neighborhood of 0. Thus

$$\{x : f_x \neq 0\} = \mathbb{R} = \operatorname{supp}(f),$$

which is exactly the closure of the pointwise nonvanishing set $\mathbb{R} \setminus \{0\}$. This illustrates why support is naturally expressed by germs.

Example 13.53 (Advanced: sheaf condition on principal opens)**.** Let A be a ring and imagine the basis $D(f)$ of $\operatorname{Spec} A$. The rule $D(f) \mapsto A_f$ comes with restriction maps induced by further localization. To construct a sheaf on the whole spectrum, it is enough to verify locality and gluing on covers by principal opens. VI/17 will carry out this program for the structure sheaf.

Example 13.54 (Advanced: a stalk sees only eventual behavior)**.** Let $X = \mathbb{R}$ and define continuous functions f_n that agree with 0 on $(-1/n, 1/n)$ but behave arbitrarily outside. Every f_n represents the zero germ at 0. Thus an infinite amount of global information can collapse to one stalk element.

Example 13.55 (Advanced: sheaf morphisms can be locally surjective but not sectionwise surjective)**.** The exponential map of sheaves of smooth functions

$$\exp : C^\infty_{\mathbb{R}} \longrightarrow (C^\infty_{\mathbb{R}})_{>0}$$

is sectionwise surjective on intervals because positive smooth functions admit smooth logarithms. On a more general sheaf problem, stalkwise lifts can exist without a global lift; the logarithm sheaf on $\mathbb{C}^*$ illustrates the obstruction. This is why VI/16 uses a local, stalkwise notion of surjectivity.

Example 13.56 (Advanced: basis data do not determine a presheaf, but can determine a sheaf)**.** VI/12 showed that arbitrary presheaves may agree on a basis and differ on larger opens. For sheaves the gluing axiom removes this freedom: if two sheaves have naturally isomorphic restrictions to a basis and the isomorphisms commute with basis restrictions, the basis criterion extends them uniquely to an isomorphism on all opens.

13.18 Exercises

Exercise 13.57 (Locality gives uniqueness)**.** Prove that in the definition of a sheaf, once locality is assumed, a gluing (if it exists) is unique.

Exercise 13.58 (The empty set)**.** For a sheaf of abelian groups, prove that $\mathcal{F}(\varnothing) = 0$. For a sheaf of sets, prove that $\mathcal{F}(\varnothing)$ is a singleton.

Exercise 13.59 (Continuous functions)**.** Prove directly that $U \mapsto C^0(U, \mathbb{R})$ is a sheaf of rings.

Exercise 13.60 (Smooth functions)**.** Prove directly that smooth real-valued functions on a smooth manifold form a sheaf of rings.

Exercise 13.61 (Locally constant functions)**.** Let A be an abelian group. Prove that locally constant A-valued functions form a sheaf.

Exercise 13.62 (Naive constant presheaf)**.** Give an explicit disconnected open set on which the naive constant presheaf with value A fails gluing when A has at least two elements.

Exercise 13.63 (Bounded functions)**.** Verify carefully that bounded continuous functions form a presheaf on $\mathbb{R}$ but not a sheaf.

Exercise 13.64 (Germ equivalence)**.** Prove that the relation defining equality of germs at a point is an equivalence relation.

Exercise 13.65 (Same value, different germ)**.** Find two smooth functions on $\mathbb{R}$ with the same value at 0 but different germs at 0.

Exercise 13.66 (Same germ, different global sections)**.** Find two smooth functions on $\mathbb{R}$ with the same germ at 0 but which are not equal globally.

Exercise 13.67 (Algebra on stalks)**.** For a presheaf of rings, define addition and multiplication of germs and prove that they are well defined.

Exercise 13.68 (Constant-sheaf stalk)**.** Prove that $(\underline{A})_x \cong A$ for every point x.

Exercise 13.69 (Skyscraper stalks)**.** On a T_1 space, compute every stalk of the skyscraper sheaf $\mathcal{S}_x(A)$.

Exercise 13.70 (Morphism on stalks)**.** Given a morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$, prove that $\varphi_x(s_x) = (\varphi(s))_x$ is well defined.

Exercise 13.71 (Functoriality)**.** Prove $(\psi \circ \varphi)_x = \psi_x \circ \varphi_x$ and $(\mathrm{id}_{\mathcal{F}})_x = \mathrm{id}_{\mathcal{F}_x}$.

Exercise 13.72 (Equality from germs)**.** Let $\mathcal{F}$ be a sheaf. Prove that two sections over U are equal iff all their germs agree.

Exercise 13.73 (Zero section from germs)**.** For a sheaf of abelian groups, prove $s = 0$ iff $s_x = 0$ for every x.

Exercise 13.74 (Morphisms determined by stalks)**.** Prove that two morphisms of sheaves are equal if all induced stalk maps are equal.

Exercise 13.75 (Discrete space)**.** Let X be discrete. Prove $\mathcal{F}(U) \cong \prod_{x\in U} \mathcal{F}_x$ for every sheaf of abelian groups.

Exercise 13.76 (Two-point discrete classification)**.** Classify sheaves of abelian groups on a two-point discrete space in terms of two abelian groups.

Exercise 13.77 (Sierpinski stalks)**.** For $X = \{\eta, x\}$ with opens $\varnothing, \{\eta\}, X$, write a sheaf as a map $A \to B$ and compute its two stalks.

Exercise 13.78 (Support)**.** Show that for a sheaf of abelian groups, the set $\{x : s_x = 0\}$ is open. Deduce the usual formula for support.

Exercise 13.79 (Restriction and stalks)**.** Let $j : U \hookrightarrow X$ be open and $x \in U$. Prove $(\mathcal{F}|_U)_x \cong \mathcal{F}_x$.

Exercise 13.80 (Product sheaves)**.** For sheaves $\mathcal{F}, \mathcal{G}$ of abelian groups, show that the objectwise product is a sheaf and $(\mathcal{F} \times \mathcal{G})_x \cong \mathcal{F}_x \times \mathcal{G}_x$.

Exercise 13.81 (Basis locality)**.** Let $\mathcal{B}$ be a basis. Show that to prove two sheaf sections over U are equal, it suffices to prove equality on a basis cover of U.

Exercise 13.82 (Local solutions)**.** Let L be a local linear differential operator. Prove that $U \mapsto \ker(L : C^\infty(U) \to C^\infty(U))$ is a sheaf.

13.19 Problems

Problem 13.83 (The sheaf axiom as an equalizer)**.** Let $\mathcal{F}$ be a presheaf of abelian groups and $U = \bigcup_i U_i$ an open cover. Define

$$\Delta : \mathcal{F}(U) \to \prod_i \mathcal{F}(U_i),$$

and the two restriction maps α, β to $\prod_{i,j} \mathcal{F}(U_i \cap U_j)$. Prove that $\mathcal{F}$ is a sheaf exactly when, for every cover, Δ identifies $\mathcal{F}(U)$ with the equalizer of α and β.

Why this problem matters. This turns the verbal locality-and-gluing axioms into an algebraic universal property used throughout sheaf theory.

Useful ideas before starting. Separate uniqueness from existence: injectivity of Δ is locality, while its image condition is gluing.

Guided hints. Write out $\alpha((s_i))$ and $\beta((s_i))$ coordinatewise on every pairwise overlap.

Provenance. New canonical synthesis from the preferred sheaf notes and AG1.

Problem 13.84 (Continuous functions really form a sheaf)**.** Let X be any topological space and Y any topological space. Prove that

$$\mathcal{C}_Y(U) = C^0(U, Y)$$

is a sheaf of sets. Then specialize to $Y = \mathbb{R}$ and prove that the ring operations make $\mathcal{C}_{\mathbb{R}}$ a sheaf of rings.

Why this problem matters. This is the archetype: a property that is local on the domain gives a sheaf.

Useful ideas before starting. Compatible maps glue pointwise. Continuity can be checked on an open cover.

Guided hints. For continuity of the glued map f, compute $f^{-1}(V)$ for an open $V \subseteq Y$ as a union of its intersections with the cover members.

Provenance. Legacy-derived from the sheaf examples in AG2 and the preferred sheaf notes; expanded as a canonical dossier.

Problem 13.85 (The constant presheaf versus the constant sheaf)**.** Let A be a nonzero abelian group. Compare the naive constant presheaf A^{pre} with the sheaf $\underline{A}$ of locally constant A-valued functions. Prove that A^{pre} fails gluing on a disconnected open set, compute $\underline{A}(U)$ from the connected components of U, and determine when the canonical map $A^{pre}(U) \to \underline{A}(U)$ is an isomorphism.

Why this problem matters. This is the first important example where sheafification changes global sections but not the intended local values.

Useful ideas before starting. A locally constant function is constant on each connected component. A single element of A forces the same value everywhere.

Guided hints. If $U = \coprod U_\alpha$, identify a locally constant map with the tuple of its componentwise values.

Provenance. Legacy-derived from the preferred sheaf notes and AG1.

Problem 13.86 (Bounded continuous functions: a precise gluing failure)**.** On $X = \mathbb{R}$, let $\mathcal{B}(U)$ be the bounded continuous real-valued functions. Prove that $\mathcal{B}$ is a presheaf and satisfies locality but not gluing. Determine exactly which part of the sheaf axiom fails in the standard cover $U_n = (-n, n)$.

Why this problem matters. It separates restriction, locality, and gluing, showing that a genuinely global boundedness condition is not sheaf-local.

Useful ideas before starting. Restriction preserves boundedness. Equality is pointwise. Use the compatible functions $x \mapsto x$.

Guided hints. The only candidate global gluing is forced pointwise, so there is no alternative bounded section that could work.

Provenance. Legacy-derived from the preferred sheaf notes, where bounded continuous functions are used as a counterexample.

Problem 13.87 (Germs as an explicit equivalence relation)**.** Fix $x \in X$ and a presheaf $\mathcal{F}$. On pairs (U, s) with $x \in U$ and $s \in \mathcal{F}(U)$, define $(U, s) \sim (V, t)$ when s and t agree on some neighborhood of x inside $U \cap V$. Prove that this is an equivalence relation and that its set of classes has the universal property of the filtered colimit $\varinjlim_{x \in U} \mathcal{F}(U)$.

Why this problem matters. This is the concrete construction hidden behind the notation $\mathcal{F}_x$.

Useful ideas before starting. Transitivity uses intersection of witnessing neighborhoods. For the colimit property, compatible maps out of neighborhood sections must identify equivalent pairs.

Guided hints. Given compatible maps $q_U : \mathcal{F}(U) \to Q$, define $q([U, s]) = q_U(s)$ and prove well-definedness by shrinking.

Provenance. Directly develops the legacy stalk/germ definitions in AG1, AG2, and the preferred sheaf notes.

Problem 13.88 (Stalks of the constant sheaf and the skyscraper sheaf)**.** Let $\underline{A}$ be the constant sheaf on X, and let $\mathcal{S}_x(B)$ be the skyscraper sheaf at x. Compute all stalks of both sheaves. For the skyscraper, give the correct answer on an arbitrary topological space and then simplify it under the T_1 hypothesis.

Why this problem matters. These are the two most useful model computations for understanding what filtered colimits over neighborhoods actually do.

Useful ideas before starting. For $\underline{A}$, every germ has a locally constant representative, hence a single local value. For the skyscraper, ask whether neighborhoods of y can avoid x.

Guided hints. At y, the skyscraper stalk is B precisely when every neighborhood of y contains x.

Provenance. New canonical synthesis from the standard skyscraper and stalk examples; no numbered legacy problem is displaced.

Problem 13.89 (Sections of a sheaf are determined by their germs)**.** Let $\mathcal{F}$ be a sheaf and $s, t \in \mathcal{F}(U)$. Prove

$$s = t \iff s_x = t_x \text{ for all } x \in U.$$

Show by counterexample that this statement can fail for presheaves that are not sheaves.

Why this problem matters. This is one of the central local-to-global uses of stalks and the key reason stalks detect morphisms.

Useful ideas before starting. Equality of germs gives a neighborhood on which the sections agree. For the counterexample, use a presheaf with invisible global data.

Guided hints. For a counterexample, define a presheaf on a two-point discrete space with a nonzero group on the whole space and zero on each proper open.

Provenance. Legacy-derived from the stalk-detection proposition in the preferred sheaf notes.

Problem 13.90 (A morphism of sheaves is determined by its stalk maps). Let $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ be morphisms of sheaves. Prove that $\varphi_x = \psi_x$ for every x implies $\varphi = \psi$. Also prove that $\varphi = 0$ iff every $\varphi_x = 0$.

Why this problem matters. This is the morphism-level form of local detection and is used constantly in later exactness arguments.

Useful ideas before starting. Apply equality of germs to the two target sections $\varphi_U(s)$ and $\psi_U(s)$.

Guided hints. For each $s \in \mathcal{F}(U)$, compute the germ of $\varphi_U(s)$ at x.

Provenance. Legacy-derived from the preferred sheaf notes; retained here as theory-level corpus content.

Problem 13.91 (Stalkwise isomorphisms are sheaf isomorphisms). Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. Assume every φ_x is an isomorphism. Construct the inverse sheaf morphism explicitly by local lifting and gluing.

Why this problem matters. This is the strongest basic theorem showing that stalks completely detect local equivalence of sheaves.

Useful ideas before starting. Lift each germ of a target section, represent the lift on a neighborhood, shrink until it maps to the target section, then glue the local lifts.

Guided hints. On overlaps, use stalkwise injectivity first pointwise and then sheaf locality to prove the local lifts agree.

Provenance. New dossier around a legacy theorem; AG2 Problem 7 remains reserved for VI/16 and is not duplicated as a numbered legacy problem here.

Problem 13.92 (Classification of sheaves on a discrete space). Let X be discrete. Prove that the functor

$$\mathcal{F} \longmapsto (\mathcal{F}_x)_{x \in X}$$

defines an equivalence between sheaves of abelian groups on X and families of abelian groups indexed by X. Describe the inverse functor and the correspondence on morphisms.

Why this problem matters. This is a complete classification theorem in a setting where gluing becomes transparent.

Useful ideas before starting. Use the singleton cover of every open set.

Guided hints. Given groups A_x, define $\mathcal{F}(U) = \prod_{x\in U} A_x$ and use coordinate projection for restriction.

Provenance. New canonical synthesis; motivated by the legacy discrete constant-sheaf examples.

Problem 13.93 (Sheaves on the Sierpinski space)**.** Let $X = \{\eta, x\}$ with opens $\varnothing, \{\eta\}, X$. Classify sheaves of abelian groups on X. Compute their stalks, determine morphisms between them, and explain how the restriction map records specialization from η toward x.

Why this problem matters. Finite spaces convert sheaf theory into explicit algebraic diagrams and expose topology encoded in stalk relations.

Useful ideas before starting. A presheaf is one map $A = \mathcal{F}(X) \to B = \mathcal{F}(\{\eta\})$; covers impose no extra condition.

Guided hints. Every open cover of X must contain X because x belongs to no smaller nonempty open.

Provenance. New canonical synthesis from finite-space presheaf examples; no numbered legacy VI/13 problem exists.

Problem 13.94 (A sheaf of local logarithms with no global section)**.** On $X = \mathbb{C}^*$, let $\mathcal{L}(U)$ be the set of holomorphic functions g on U satisfying $e^{g(z)} = z$. Prove that $\mathcal{L}$ is a sheaf of sets, that every stalk is nonempty, and that $\mathcal{L}(X)$ is empty.

Why this problem matters. This sharply separates local existence from global existence and gives geometric meaning to stalks without yet invoking cohomology.

Useful ideas before starting. Local logarithms exist on small disks avoiding 0. A global logarithm would have derivative $1/z$ and lead to a contradiction around the unit circle.

Guided hints. If g were global, differentiate $e^g = z$ to obtain $g' = 1/z$ and integrate around $|z| = 1$.

Provenance. New advanced synthesis consistent with the local-to-global emphasis of the legacy sheaf notes.

Problem 13.95 (Sheaf of solutions of a differential equation)**.** Let M be a smooth manifold and $L : C^\infty_M \to \mathcal{E}$ a local linear differential operator into a sheaf $\mathcal{E}$. Define $\mathcal{S}(U) = \{u \in C^\infty(U) : Lu = 0\}$. Prove that $\mathcal{S}$ is a sheaf. Compute it for $L = d/dx$ on an interval and for $L = d^2/dx^2$ on an interval.

Why this problem matters. Many important sheaves arise as local solution spaces rather than as all functions.

Useful ideas before starting. The kernel condition is local. For the ODE examples, solve the differential equations on each connected interval.

Guided hints. For $u = 0$, local solutions are affine functions $ax + b$.

Provenance. New canonical synthesis from the standard function-sheaf corpus.

Problem 13.96 (Support via germs)**.** For a sheaf of abelian groups $\mathcal{F}$ and $s \in \mathcal{F}(U)$, prove that

$$Z_g(s) := \{x \in U : s_x = 0\}$$

is open. Define $\operatorname{supp}(s) = U \setminus Z_g(s) = \{x : s_x \neq 0\}$ and prove that this set is closed. For ordinary function sheaves, relate it to the closure of the pointwise nonvanishing set. Compute supports for standard continuous bump functions and for a skyscraper section.

Why this problem matters. Support is fundamentally a stalkwise notion and is crucial later for extension, cohomology, and distributions.

Useful ideas before starting. A zero germ is represented by an actual zero section on some neighborhood.

Guided hints. For a skyscraper section, use the arbitrary-space stalk computation before imposing T_1.

Provenance. New canonical synthesis; support terminology anticipates later sheaf and cohomology chapters.

Problem 13.97 (Restriction of sheaves and invariance of stalks)**.** Let $j : U \hookrightarrow X$ be an open inclusion and $\mathcal{F}$ a sheaf on X. Prove that $\mathcal{F}|_U$ is a sheaf on U and that for every $x \in U$ there is a canonical isomorphism

$$(\mathcal{F}|_U)_x \cong \mathcal{F}_x.$$

Prove naturality in $\mathcal{F}$.

Why this problem matters. Local objects should not change when the ambient space is replaced by an open neighborhood.

Useful ideas before starting. Open covers inside U are also open covers inside X. Neighborhoods contained in U are cofinal among neighborhoods of x after intersecting with U.

Guided hints. Map a germ represented by $s \in \mathcal{F}(V)$ to the same section restricted to $V \cap U$.

Provenance. New canonical synthesis from the restriction and stalk machinery.

Problem 13.98 (Basis criterion for sheaves)**.** Let $\mathcal{B}$ be a basis for X. Suppose groups $F(B)$ and restriction maps are given for basis opens, with locality and gluing for basis covers of basis opens. Construct a sheaf $\widetilde{F}$ on all opens and prove uniqueness up to unique isomorphism.

Why this problem matters. This is the practical engine behind constructing the structure sheaf from principal opens $D(f)$.

Useful ideas before starting. For an arbitrary open U, define sections as compatible families (s_B) over basis opens $B \subseteq U$, or equivalently as the inverse limit of the basis data.

Guided hints. Set

$$\widetilde{F}(U) = \varprojlim_{B \in \mathcal{B},\, B \subseteq U} F(B).$$

Then verify gluing componentwise.

Provenance. New canonical synthesis from standard sheaf-basis theory; explicitly prepares the legacy structure-sheaf material.

Problem 13.99 (A presheaf with all zero stalks but nonzero global sections)**.** Construct a presheaf $\mathcal{P}$ of abelian groups on a topological space such that $\mathcal{P}(X) \neq 0$ but $\mathcal{P}_x = 0$ for every $x \in X$. Prove that $\mathcal{P}$ cannot be a sheaf and identify exactly which sheaf axiom fails.

Why this problem matters. This shows why stalkwise detection is a theorem about sheaves, not arbitrary presheaves.

Useful ideas before starting. Use a discrete space with a nonzero group only on the whole space and zero on all smaller opens.

Guided hints. On $X = \{p, q\}$ discrete, set $\mathcal{P}(X) = A \neq 0$ and $\mathcal{P}(V) = 0$ for $V \neq X$.

Provenance. New canonical counterexample motivated by the stalk-detection theorem in the legacy sheaf notes.

Problem 13.100 (Local branches and failure of global lifting)**.** Consider the covering map $p : \mathbb{C}^* \to \mathbb{C}^*$, $p(w) = w^n$ for $n \geq 2$. Let $\mathcal{S}(U)$ be the set of continuous sections of p over U. Prove $\mathcal{S}$ is a sheaf, compute its stalk cardinalities, and show $\mathcal{S}(\mathbb{C}^*) = \varnothing$.

Why this problem matters. This packages roots as a sheaf of local choices and displays monodromy before formal covering-space language is used in this volume.

Useful ideas before starting. A small simply connected neighborhood has exactly n continuous root branches. A global section would give a continuous nth root of the identity map on the circle.

Guided hints. Restrict a hypothetical global section to S^1 and compare degrees: $\deg(z \mapsto s(z)^n) = n \deg(s)$.

Provenance. New advanced synthesis; no numbered legacy VI/13 problem is replaced.

Problem 13.101 (Sheaf data on an irreducible space)**.** Let X be irreducible and A an abelian group. Prove that the naive constant presheaf A^{pre} with value A on nonempty opens and 0 on the empty open is a sheaf. Explain why this does not contradict the need for the constant sheaf on general spaces.

Why this problem matters. The example connects sheaf gluing to irreducibility developed earlier in Volume VI.

Useful ideas before starting. Any two nonempty opens in an irreducible space intersect. Thus compatible constants on a cover must all coincide.

Guided hints. First prove that every nonempty open subset of an irreducible space is itself irreducible, so the intersection graph of a nonempty open cover is complete.

Provenance. New cross-chapter synthesis using VI/05 irreducibility and the legacy constant-presheaf example.

13.20 Challenges

Problem 13.102 (Challenge: Sheaves on finite Alexandrov spaces)**.** Let X be a finite T_0 Alexandrov space with specialization poset. Show that every point has a smallest open neighborhood U_x. Prove that a sheaf of abelian groups determines a diagram $x \mapsto \mathcal{F}(U_x)$ with maps along the specialization order, and formulate a converse construction. Identify precisely where the sheaf condition enters.

Problem 13.103 (Challenge: Local systems from covering spaces)**.** Let $p : E \to X$ be a covering space with discrete fiber. Show that the sheaf of local sections of p has stalk naturally identified with the fiber $p^{-1}(x)$ at each x. Analyze how continuation along paths can permute these stalks, and work out the nth-root covering of S^1.

Problem 13.104 (Challenge: Sheafification foreshadowed by germs)**.** Let $\mathcal{P}$ be a presheaf. Define $\mathcal{P}^{loc}(U)$ to be families $(a_x)_{x\in U}$ with $a_x \in \mathcal{P}_x$ that are locally represented by one presheaf section. Prove directly that $\mathcal{P}^{loc}$ is a sheaf and that its stalks are canonically $\mathcal{P}_x$.

Problem 13.105 (Challenge: Stalkwise equivalence versus global sections)**.** Construct two sheaves and a morphism between them that is a stalkwise isomorphism and use the theorem to prove it is a sheaf isomorphism, while the analogous map of underlying presheaf-like global presentations looks non-bijective before the correct local objects are chosen. Explain what sheafification repairs.

Problem 13.106 (Challenge: Support and specialization)**.** Let $X = \operatorname{Spec} A$ and let $\mathcal{F}$ be any sheaf of modules once the structure sheaf is available. Anticipating later chapters, investigate what conditions on a sheaf make the set $\{x : \mathcal{F}_x \neq 0\}$ stable under specialization. Test the claim on skyscraper sheaves and on sheaves associated to finitely generated modules in basic affine examples.

Summary

A sheaf is a presheaf with a precise local-to-global principle: local equality detects global equality, and compatible local sections glue uniquely. A stalk

$$\mathcal{F}_x = \varinjlim_{x\in U} \mathcal{F}(U)$$

packages all sections defined near a point into germs. Sheaf sections and sheaf morphisms are detected by stalks, and a morphism is an isomorphism exactly when every stalk map is. The examples of locally constant functions, discrete spaces, skyscraper sheaves, and local function germs show how much geometric information can be organized by these two ideas.

VI/14 now addresses the next unavoidable question: given a presheaf that fails locality or gluing, how can one construct the closest sheaf without changing its local germs?

13.21 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 13.107 (Germ)**.** What is an element of a stalk?

Hint. Use sheaf gluing, locality, and stalks. ☐

Solution. An equivalence class of a section defined on some neighborhood of the point. ☐

Exercise 13.108 (Equality germs)**.** When do two sections define the same germ?

Hint. Use sheaf gluing, locality, and stalks. ☐

Solution. When they agree on a smaller neighborhood. ☐

Exercise 13.109 (Stalk preview)**.** What is the structure-sheaf stalk at $\mathfrak{p}$?

Hint. Use sheaf gluing, locality, and stalks. ☐

Solution. $A_{\mathfrak{p}}$. ☐

Exercise 13.110 (Locality)**.** If two sheaf sections agree on a cover, what follows?

Hint. Use sheaf gluing, locality, and stalks. ☐

Solution. They are equal globally. ☐

Exercise 13.111 (Gluing)**.** What is required before local sections can glue?

Hint. Use sheaf gluing, locality, and stalks. ☐

Solution. They must agree on pairwise overlaps. ☐

Proofs

Exercise 13.112 (Germ equivalence)**.** Prove: Prove the stalk relation on local sections is an equivalence relation.

Hint. Work directly from sheaf gluing, locality, and stalks. ☐

Solution. Reflexivity and symmetry are immediate; for transitivity intersect the two neighborhoods where agreements hold. ☐

Exercise 13.113 (Stalk functor)**.** Prove: Prove a sheaf morphism induces a map on each stalk.

Hint. Work directly from sheaf gluing, locality, and stalks. ☐

Solution. Send a germ to the germ of its image and use restriction compatibility for well-definedness. ☐

Exercise 13.114 (Uniqueness)**.** Prove: Prove the uniqueness half of the sheaf axiom from locality.

Hint. Work directly from sheaf gluing, locality, and stalks. ☐

Solution. If two global candidates have identical restrictions on a cover, locality says they coincide. ☐

Exercise 13.115 (Continuous gluing)**.** Prove: Prove compatible continuous functions glue to a continuous function.

Hint. Work directly from sheaf gluing, locality, and stalks. ☐

Solution. Define pointwise using local representatives; compatibility gives well-definedness and continuity is local. ☐

Counterexamples and hypothesis tests

Exercise 13.116 (One neighborhood)**.** Test the claim: a germ remembers the entire original domain of a section.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; only sufficiently small local behavior remains. ☐

Exercise 13.117 (Stalk global)**.** Test the claim: knowing one stalk determines all global sections.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 13.118 (Presheaf enough)**.** Test the claim: every presheaf has the same gluing property as a sheaf.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Applications and investigations

Exercise 13.119 (Local equations)**.** Why are stalks useful for local geometric questions?

Hint. Translate the question into sheaf gluing, locality, and stalks. ☐

Solution. They retain exactly the functions and relations visible near a chosen point. ☐

Exercise 13.120 (Local isomorphism)**.** How can stalk maps test a sheaf morphism?

Hint. Translate the question into sheaf gluing, locality, and stalks. ☐

Solution. Many sheaf properties, including isomorphism, can be checked on all stalks. ☐

Challenge problems

Exercise 13.121 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Sheaves and Stalks in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Sheaves and Stalks, translates it through sheaf gluing, locality, and stalks, and checks the correspondence in both algebraic and geometric language. □

Exercise 13.122 (Hypothesis challenge)**.** Choose one structural statement from Sheaves and Stalks, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 14

Sheafification

Purpose and learning goals

VI/12 introduced presheaves and VI/13 imposed the local-to-global sheaf axiom. The next question is unavoidable: what should one do with a useful presheaf that fails locality or gluing? *Sheafification* is the canonical repair. It replaces a presheaf by a sheaf without changing its germs at any point, and it is characterized by a universal mapping property rather than by an arbitrary choice of construction.

This chapter treats sheafification from three complementary viewpoints:

1. the universal property, which says what the associated sheaf *is*;
2. the espace étalé construction, which proves existence concretely from stalks;
3. the plus construction, which makes the "separate, then glue" mechanism explicit.

The ordinary and advanced example banks are intentionally large. They show both ways in which sheafification changes a presheaf: it may *add* locally compatible sections, and it may *identify or kill* sections that have the same germs.

A corpus-wide legacy-problem ledger is now maintained for Volume VI. The central legacy item of this chapter is AG1 Exercise 4, a long associated-sheaf problem requiring existence, the canonical morphism, the universal property, preservation of stalks, functoriality, and the adjunction. It is preserved as VI.14.P1 at essentially its full mathematical depth and merged with the preferred reorganized sheaf notes rather than duplicated.

After this chapter, the reader should be able to:

- state the universal property of the associated sheaf;
- distinguish a construction of sheafification from its universal characterization;
- build the espace étalé $E_{\mathcal{F}} = \coprod_x \mathcal{F}_x$;
- prove that the germ images $\widetilde{s}(U)$ form a basis for a topology on $E_{\mathcal{F}}$;
- construct the associated sheaf as locally representable continuous sections of $E_{\mathcal{F}} \to X$;
- construct the canonical morphism $\eta : \mathcal{F} \to \mathcal{F}^{\#}$;
- prove $(\mathcal{F}^{\#})_x \cong \mathcal{F}_x$ for every x;
- prove and use the universal property;
- prove uniqueness of sheafification up to a unique isomorphism;
- sheafify morphisms and establish functoriality;

- express sheafification as a left adjoint $a \dashv i$;
- prove that sheafification of an already existing sheaf is canonically isomorphic to it;
- understand why the unit need be neither injective nor surjective on sections;
- compute sheafifications in constant, bounded-function, finite-space, and basis examples;
- use the plus construction and its separatedness step;
- keep quotient sheaves, image sheaves, and exactness for VI/15–VI/16, where the relevant legacy problems are reserved.

14.1 The universal problem

Let X be a topological space and let $\mathcal{F}$ be a presheaf of sets, abelian groups, rings, or modules. A sheafification should be a sheaf receiving a map from $\mathcal{F}$ through which every map from $\mathcal{F}$ to a sheaf factors uniquely.

Definition 14.1 (Sheafification)**.** A *sheafification* or *associated sheaf* of $\mathcal{F}$ is a pair $(\mathcal{F}^{\#}, \eta_{\mathcal{F}})$ consisting of a sheaf $\mathcal{F}^{\#}$ and a morphism of presheaves

$$\eta_{\mathcal{F}} : \mathcal{F} \longrightarrow \mathcal{F}^{\#}$$

such that for every sheaf $\mathcal{G}$ and every presheaf morphism $u : \mathcal{F} \to \mathcal{G}$, there exists a unique sheaf morphism $\bar{u} : \mathcal{F}^{\#} \to \mathcal{G}$ satisfying

$$u = \bar{u} \circ \eta_{\mathcal{F}}.$$

$$\begin{tikzcd} \mathcal{F} \arrow[r, "u"] \arrow[d, "\eta_{\mathcal{F}}"'] & \mathcal{G} \\ \mathcal{F}^{\#} \arrow[ur, dashed, "\exists!\,\bar{u}"'] & \end{tikzcd}$$

The phrase "closest sheaf" means precisely this universal property. It does *not* mean that $\mathcal{F}^{\#}(U)$ is numerically or set-theoretically close to $\mathcal{F}(U)$. Indeed, on some opens sheafification can add sections, while on others it can identify sections.

14.2 Uniqueness before existence

Universal properties give uniqueness automatically.

Theorem 14.2 (Uniqueness of sheafification)**.** *If $(\mathcal{A}, \eta)$ and $(\mathcal{B}, \theta)$ are two sheafifications of the same presheaf $\mathcal{F}$, there is a unique isomorphism*

$$\Phi : \mathcal{A} \xrightarrow{\sim} \mathcal{B}$$

satisfying $\Phi \circ \eta = \theta$.

Proof. Apply the universal property of $\mathcal{A}$ to $\theta : \mathcal{F} \to \mathcal{B}$ to obtain $\Phi : \mathcal{A} \to \mathcal{B}$. Similarly obtain $\Psi : \mathcal{B} \to \mathcal{A}$ with $\Psi\theta = \eta$. Then $(\Psi\Phi)\eta = \eta$. By uniqueness in the universal property of $\mathcal{A}$, $\Psi\Phi = \mathrm{id}_{\mathcal{A}}$. Likewise $\Phi\Psi = \mathrm{id}_{\mathcal{B}}$. Uniqueness of Φ is built into the same factorization property. $\square$

Thus one may use whichever construction is most convenient and still speak of *the* sheafification up to canonical isomorphism.

14.3 The espace étalé: all germs over all points

We now prove existence. For each $x \in X$, let $\mathcal{F}_x$ be the stalk of the presheaf from VI/13. Form the disjoint union

$$E_{\mathcal{F}} := \coprod_{x \in X} \mathcal{F}_x$$

and the projection

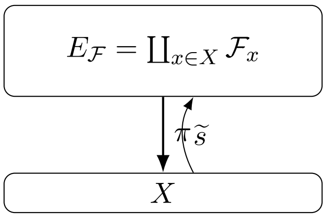
$$\pi : E_{\mathcal{F}} \longrightarrow X, \qquad a \in \mathcal{F}_x \longmapsto x.$$

For an open $U \subseteq X$ and $s \in \mathcal{F}(U)$ define

$$\widetilde{s} : U \longrightarrow E_{\mathcal{F}}, \qquad x \longmapsto s_x.$$

Its image is

$$\widetilde{s}(U) = \{s_x : x \in U\}.$$



$s \in \mathcal{F}(U)$
$\widetilde{s}(x) = s_x$

Proposition 14.3 (The germ images form a basis)**.** *The sets $\widetilde{s}(U)$, as U and $s \in \mathcal{F}(U)$ vary, form a basis for a topology on $E_{\mathcal{F}}$.*

Proof. Every point $a \in \mathcal{F}_x$ is represented by some $s \in \mathcal{F}(U)$ with $x \in U$, so $a = s_x \in \widetilde{s}(U)$. Suppose $a \in \widetilde{s}(U) \cap \widetilde{t}(V)$ and $a \in \mathcal{F}_x$. Then $s_x = t_x$. Equality of germs gives a neighborhood $W \subseteq U \cap V$ of x on which $s|_W = t|_W$. Hence

$$a \in \widetilde{s|_W}(W) \subseteq \widetilde{s}(U) \cap \widetilde{t}(V).$$

This is the basis-intersection criterion. □

With this topology,
pi is a local homeomorphism: on each basic open $\widetilde{s}(U)$, the inverse of π is $x \mapsto s_x$.

14.4 Locally representable sections

Definition 14.4 (Associated sheaf from the espace étalé)**.** For an open set $U \subseteq X$, define

$$\mathcal{F}^{\#}(U) = \{\sigma : U \to E_{\mathcal{F}} : \pi \circ \sigma = \mathrm{id}_U,\ \sigma \text{ is continuous}\}.$$

Restrictions are ordinary restrictions of maps.

Continuity has a useful algebraic meaning.

Proposition 14.5 (Local representability)**.** *A section $\sigma : U \to E_{\mathcal{F}}$ belongs to $\mathcal{F}^{\#}(U)$ if and only if every $x \in U$ has a neighborhood $V \subseteq U$ and some $s \in \mathcal{F}(V)$ such that*

$$\sigma(y) = s_y \qquad (y \in V).$$

Proof. If σ is continuous and $x \in U$, choose a basic neighborhood $\widetilde{s}(V)$ of $\sigma(x)$. Shrinking V if necessary inside U, the local homeomorphism property forces $\sigma|_V = \widetilde{s}$. Conversely, a map which is locally of the form $\widetilde{s}$ is continuous because each $\widetilde{s}$ is a homeomorphism onto a basic open. □

$$\begin{tikzcd} & \sigma \in \mathcal{F}^{\#}(U) \arrow[dl, "\text{shrink}"'] & \\ x \in V \subseteq U \arrow[rr] & & s \in \mathcal{F}(V) \arrow[ul, "\sigma|_V = \widetilde{s}"'] \end{tikzcd}$$

Theorem 14.6 (The associated presheaf is a sheaf)**.** *$U \mapsto \mathcal{F}^{\#}(U)$ is a sheaf.*

Proof. Compatible continuous sections of π glue uniquely as ordinary maps. Continuity is local on the domain, so the glued map is continuous. Because every local section satisfies $\pi\sigma = \mathrm{id}$, the gluing does also. Locality is equality of maps on an open cover. □

Example 14.7 (Constant presheaf sheafifies to locally constant functions)**.** The constant presheaf with value A does not permit independent constants on disconnected pieces. Its sheafification assigns locally constant A-valued functions, which can vary between connected components.

14.5 The canonical morphism

Define

$$\eta_U : \mathcal{F}(U) \longrightarrow \mathcal{F}^{\#}(U), \qquad s \longmapsto \widetilde{s}.$$

The compatibility

$$\widetilde{s|_V} = \widetilde{s}|_V$$

shows that these maps form a morphism of presheaves

$$\eta_{\mathcal{F}} : \mathcal{F} \longrightarrow \mathcal{F}^{\#}.$$

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\eta_U"] \arrow[d, "\rho^U_V"'] & \mathcal{F}^{\#}(U) \arrow[d, "\rho^U_V"] \\ \mathcal{F}(V) \arrow[r, "\eta_V"'] & \mathcal{F}^{\#}(V) \end{tikzcd}$$

The unit may fail to be injective on sections: two sections of a presheaf can have the same germ at every point without being equal if locality fails. It may also fail to be surjective: a locally representable family of germs can fail to arise from one original global section if gluing fails.

14.6 Sheafification preserves every stalk

This theorem is the central local invariant of sheafification.

Theorem 14.8 (Stalk preservation)**.** *For every $x \in X$, the map induced by the unit is an isomorphism*

$$(\eta_{\mathcal{F}})_x : \mathcal{F}_x \xrightarrow{\sim} (\mathcal{F}^{\#})_x.$$

Proof. Define

$$\Phi_x : (\mathcal{F}^{\#})_x \longrightarrow \mathcal{F}_x, \qquad [U, \sigma]_x \longmapsto \sigma(x).$$

This is well defined: if two sheafified sections have the same germ, they agree on some neighborhood of x and hence have the same value in the fiber $\mathcal{F}_x$.

For surjectivity, let $a \in \mathcal{F}_x$. Choose $s \in \mathcal{F}(U)$ representing a. Then $\widetilde{s} \in \mathcal{F}^{\#}(U)$ and $\Phi_x((\widetilde{s})_x) = a$.

For injectivity, suppose $\Phi_x(\sigma_x) = \Phi_x(\tau_x)$. Then $\sigma(x) = \tau(x)$ in the fiber. By local representability, after shrinking around x, write $\sigma = \widetilde{s}$ and $\tau = \widetilde{t}$. Equality at x means $s_x = t_x$ in $\mathcal{F}_x$, so after shrinking once more s and t agree. Hence $\sigma_x = \tau_x$. The inverse is induced by $a = s_x \mapsto (\widetilde{s})_x$, exactly the stalk map of $\eta_{\mathcal{F}}$. □

$$\begin{tikzcd} \mathcal{F}_x \arrow[r, "(\eta_{\mathcal{F}})_x", "\sim"'] & (\mathcal{F}^{\#})_x \\ {[s,U]} \arrow[r, mapsto] & {[\widetilde{s},U]} \end{tikzcd}$$

Thus sheafification changes global bookkeeping while leaving all local germs unchanged.

14.7 The universal property: existence of the factorization

Let $\mathcal{G}$ be a sheaf and $u : \mathcal{F} \to \mathcal{G}$ a presheaf morphism. Every x has an induced stalk map $u_x : \mathcal{F}_x \to \mathcal{G}_x$. If $\sigma \in \mathcal{F}^{\#}(U)$, define a family of germs

$$x \longmapsto u_x(\sigma(x)) \in \mathcal{G}_x.$$

Because σ is locally represented by sections of $\mathcal{F}$, this family is locally represented by sections of $\mathcal{G}$. Since $\mathcal{G}$ is a sheaf, these local representatives glue uniquely to a section of $\mathcal{G}(U)$.

Theorem 14.9 (Universal property)**.** *There exists a unique morphism of sheaves*

$$\bar{u} : \mathcal{F}^{\#} \longrightarrow \mathcal{G}$$

with $u = \bar{u}\eta_{\mathcal{F}}$.

Proof. For $\sigma \in \mathcal{F}^{\#}(U)$ choose an open cover $U = \bigcup_i U_i$ and $s_i \in \mathcal{F}(U_i)$ with $\sigma|_{U_i} = \widetilde{s}_i$. Set $g_i = u(s_i) \in \mathcal{G}(U_i)$. On $U_i \cap U_j$, equality of the sheafified sections means the germs of s_i and s_j agree at every point. Applying u gives equal germs of g_i and g_j; because $\mathcal{G}$ is a sheaf, equality of all germs implies equality of sections. Hence the g_i glue uniquely to $g \in \mathcal{G}(U)$. Define $\bar{u}_U(\sigma) = g$. This is independent of the chosen local presentation and commutes with restrictions. For $s \in \mathcal{F}(U)$, the construction applied to $\widetilde{s}$ gives $u(s)$, so $\bar{u}\eta = u$.

If $v : \mathcal{F}^{\#} \to \mathcal{G}$ is another factorization, every sheafified section is locally of the form $\widetilde{s}$, and on such a neighborhood $v(\widetilde{s}) = u(s) = \bar{u}(\widetilde{s})$. Locality in $\mathcal{G}$ gives $v = \bar{u}$. □

14.8 Functoriality

Given $f : \mathcal{F} \to \mathcal{G}$ in $\mathbf{PreSh}(X)$, apply the universal property to

$$\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{\eta_{\mathcal{G}}} \mathcal{G}^{\#}.$$

There is a unique sheaf morphism

$$f^{\#} : \mathcal{F}^{\#} \longrightarrow \mathcal{G}^{\#}$$

with

$$f^{\#}\eta_{\mathcal{F}} = \eta_{\mathcal{G}} f.$$

$$\begin{tikzcd} \mathcal{F} \arrow[r, "f"] \arrow[d, "\eta_{\mathcal{F}}"'] & \mathcal{G} \arrow[d, "\eta_{\mathcal{G}}"] \\ \mathcal{F}^{\#} \arrow[r, "f^{\#}"'] & \mathcal{G}^{\#} \end{tikzcd}$$

Proposition 14.10 (Sheafification is a functor)**.** *The assignments $\mathcal{F} \mapsto \mathcal{F}^{\#}$ and $f \mapsto f^{\#}$ satisfy*

$$(\mathrm{id}_{\mathcal{F}})^{\#} = \mathrm{id}_{\mathcal{F}^{\#}}, \qquad (gf)^{\#} = g^{\#} f^{\#}.$$

Moreover, under the canonical stalk identifications,

$$(f^{\#})_x = f_x.$$

Proof. The identity and composition formulas follow from uniqueness in the universal property: both candidate maps have the same composite with the unit. The stalk formula follows from naturality of the unit and the fact that both vertical unit maps are stalk isomorphisms. □

Example 14.11 (Germs construction)**.** Sheafification can be built from stalk germs: over an open set, take functions assigning to each point a germ that is locally represented by one presheaf section. Local representability supplies the sheaf axioms.

14.9 Adjunction

Let

$$i : \mathbf{Sh}(X) \hookrightarrow \mathbf{PreSh}(X)$$

be the inclusion functor. Sheafification is left adjoint to i.

Theorem 14.12 (Adjunction)**.** *For every presheaf $\mathcal{F}$ and sheaf $\mathcal{G}$ there is a natural bijection*

$$\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{PreSh}(X)}(\mathcal{F}, i\mathcal{G}),$$

given by precomposition with $\eta_{\mathcal{F}}$. Symbolically,

$$a : \mathbf{PreSh}(X) \rightleftarrows \mathbf{Sh}(X) : i, \qquad a \dashv i.$$

$$\begin{tikzcd} \boxed{\mathbf{PreSh}_X} \arrow[rr, bend left, "a = (-)^{\#}"] & & \boxed{\mathbf{Sh}_X} \arrow[ll, bend left, "i"] \end{tikzcd}$$

$$a \dashv i$$

This compact statement contains existence, uniqueness, and functoriality at once.

14.10 Sheafifying a sheaf does nothing

Corollary 14.13 (Idempotence)**.** *If $\mathcal{F}$ is already a sheaf, the unit*

$$\eta_{\mathcal{F}} : \mathcal{F} \longrightarrow \mathcal{F}^{\#}$$

is an isomorphism. Consequently

$$(\mathcal{F}^{\#})^{\#} \cong \mathcal{F}^{\#}.$$

Proof. Apply the universal property to the identity map $\mathcal{F} \to \mathcal{F}$. It produces a retraction $r : \mathcal{F}^{\#} \to \mathcal{F}$ with $r\eta = \mathrm{id}$. The composites ηr and $\mathrm{id}_{\mathcal{F}^{\#}}$ agree after precomposition with η, so uniqueness in the universal property gives $\eta r = \mathrm{id}$. □

14.11 Separated presheaves and the plus construction

There is another construction that makes the repair mechanism transparent. A presheaf is *separated* if locality holds, even if gluing may fail.

For an open U, consider pairs consisting of an open cover $U = \bigcup_i U_i$ and a compatible family $s_i \in \mathcal{F}(U_i)$. Two such families are equivalent if they agree after passing to a common refinement. Denote the resulting presheaf by $\mathcal{F}^+$. There is a natural map $\mathcal{F} \to \mathcal{F}^+$ sending a section to its restrictions.

Theorem 14.14 (Plus construction)**.** *For presheaves of sets or abelian groups:*

1. *$\mathcal{F}^+$ is separated;*
2. *if $\mathcal{F}$ is separated, then $\mathcal{F}^+$ is a sheaf;*
3. *therefore $\mathcal{F}^{++}$ is always a sheaf and is canonically isomorphic to $\mathcal{F}^\#$.*

$$\begin{tikzcd} \boxed{\mathcal{F}} \arrow[r, "\text{separate}"] & \boxed{\mathcal{F}^+} \arrow[r, "\text{glue}"] & \boxed{\mathcal{F}^{++}} \end{tikzcd}$$

$\mathcal{F}^+$ separated $\quad$ $\mathcal{F}^{++}$ a sheaf

The first "plus" removes failure of uniqueness; the second removes failure of existence. This notation is different from the legacy AG1 Exercise 4, where $\mathcal{F}^+$ denotes the final associated sheaf built from stalks. VI.14.P1 preserves the legacy notation inside the problem while the chapter uses $\mathcal{F}^\#$ for the final sheafification.

Example 14.15 (Stalk preservation)**.** The canonical map from a presheaf to its sheafification induces isomorphisms on stalks. Sheafification repairs gluing without changing the local germs already present.

14.12 Restriction to an open subset

If $j : U \hookrightarrow X$ is open, then sheafification commutes with restriction:

$$(\mathcal{F}|_U)^\# \cong \mathcal{F}^\#|_U.$$

This can be proved either by comparing stalks and universal properties or by observing that the espace étalé over U is simply the restriction of $E_{\mathcal{F}}$ to U.

$$\begin{tikzcd} (\mathcal{F}|_U)^\# \arrow[r, "\sim"] & (\mathcal{F}^\#)|_U \\ \mathcal{F}_x \arrow[u, "\sim"] \arrow[r, equal] & \mathcal{F}_x \arrow[u, "\sim"'] \end{tikzcd}$$

14.13 Sheafification on a basis

Let $\mathcal{B}$ be a basis for X. Suppose data are given on basis opens with compatible restrictions, but locality or gluing may fail. One can form germs using only basis neighborhoods because they are cofinal among all neighborhoods. The same espace étalé construction then produces the sheafification on all opens. This is especially important in algebraic geometry: principal opens $D(f)$ form a basis of $\operatorname{Spec} A$, and the localization data $D(f) \mapsto A_f$ will feed directly into VI/17.

14.14 What the unit can do to sections

The map

$$\eta_U : \mathcal{F}(U) \to \mathcal{F}^{\#}(U)$$

can behave in three qualitatively different ways.

1. It can be an isomorphism, as when $\mathcal{F}$ is already a sheaf.
2. It can fail to be surjective: the constant presheaf on a disconnected open has too few sections, while its sheafification contains independently chosen locally constant values.
3. It can fail to be injective: a presheaf can contain distinct sections whose restrictions coincide near every point; sheafification identifies them because their germs are equal.

$$\begin{tikzcd}
& \boxed{\text{espace-etale } a_{\mathrm{et}}(\mathcal{F})} \arrow[dd, leftrightarrow, "\text{unique } \sim"] \\
\boxed{\mathcal{F}} \arrow[ur] \arrow[dr] & \\
& \boxed{\text{two-plus } \mathcal{F}^{++}}
\end{tikzcd}$$

This explains the right mental model:

$$\boxed{\text{sheafification preserves local germs, not the original global section sets.}}$$

14.15 Common pitfalls

1. Sheafification is not defined by simply imposing an equivalence relation on each $\mathcal{F}(U)$ independently; new sections may also have to be added.
2. Equality of all stalks does not mean two presheaves are isomorphic. It does mean their sheafifications may be locally equivalent, and the universal property decides the actual comparison.
3. The canonical map need be neither injective nor surjective on a fixed open set.
4. The phrase “associated sheaf” refers to a universal property; the espace étalé and plus constructions are models of that property.
5. In the standard plus construction, one application produces a separated presheaf and two applications produce a sheaf. Do not confuse that notation with sources that call the final sheafification $\mathcal{F}^+$ directly.
6. The image presheaf and quotient presheaf of sheaves need not themselves be sheaves. Those are VI/15 problems, although their repair uses the present chapter.
7. Stalkwise exactness and sheaf surjectivity belong to VI/16; sheafification is the tool, not the final exactness theory.

14.16 Worked examples

Example 14.16 (A sheaf is fixed by sheafification)**.** Let $\mathcal{F} = C^0_X$, the sheaf of continuous real-valued functions. Every $\sigma \in \mathcal{F}^\#(U)$ is locally represented by ordinary continuous functions. These local representatives agree on overlaps because they determine the same germs, hence glue inside C^0_X. Thus every sheafified section comes from one section of $C^0_X(U)$ and

$$C^0_X \xrightarrow{\sim} (C^0_X)^\#.$$

This is the concrete version of idempotence.

Example 14.17 (Constant presheaf on a disconnected open)**.** Let A be an abelian group and let $\mathcal{P}(U) = A$ for every nonempty open U, with identity restriction maps. Suppose $U = U_1 \sqcup U_2$ with both pieces nonempty. A section of $\mathcal{P}(U)$ is one element $a \in A$, so its restrictions have the same value on both components. In the sheafification, however,

$$\mathcal{P}^\#(U) = \{\text{locally constant maps } U \to A\} \cong A \times A,$$

and (a, b) with $a \neq b$ is a new section added by sheafification.

Example 14.18 (Constant presheaf on a connected interval)**.** On an interval $I \subseteq \mathbb{R}$, every locally constant A-valued function is constant. Hence

$$\mathcal{P}^\#(I) \cong A \cong \mathcal{P}(I).$$

The constant presheaf can therefore look correct on each connected open while still failing on disconnected opens. Sheafification repairs the global bookkeeping without changing any stalk, all of which are A.

Example 14.19 (Bounded continuous functions)**.** Let $\mathcal{B}(U)$ be the bounded continuous functions $U \to \mathbb{R}$. VI/13 showed that this is not a sheaf. Every continuous function is locally bounded: if $f(x) = c$, continuity gives an open neighborhood on which f takes values in $(c-1, c+1)$. Therefore every section of C^0_X is locally represented by a section of $\mathcal{B}$, and conversely bounded continuous functions are continuous. Hence

$$\mathcal{B}^\# \cong C^0_X.$$

Sheafification removes the nonlocal condition “bounded on all of U.”

Example 14.20 (Locally polynomial functions on the real line)**.** Define $\mathcal{P}(U)$ to be restrictions to U of a single polynomial in $\mathbb{R}[t]$. This is a presheaf. On a disconnected open $U = U_1 \sqcup U_2$, one may choose different polynomials on the two components, so $\mathcal{P}$ is generally not a sheaf. Its sheafification consists of functions which are locally polynomial. On each connected open interval such a function is represented by one polynomial, since two polynomials agreeing on a nonempty open interval are equal.

Example 14.21 (Two sections become equal after sheafification)**.** Let X be a space in which every point has a proper open neighborhood. Define a presheaf of abelian groups by

$$\mathcal{F}(X) = A, \qquad \mathcal{F}(U) = 0 \quad (U \subsetneq X),$$

with the only possible restriction maps. Every germ of every global section is zero, because near each point one can restrict to a proper open. Consequently every stalk is zero and

$$\mathcal{F}^\# = 0.$$

Distinct elements of $A = \mathcal{F}(X)$ are all identified by sheafification.

Example 14.22 (A presheaf with the same stalks as the zero sheaf)**.** In the preceding example, $\mathcal{F}_x = 0$ for every x, even though $\mathcal{F}(X) = A$ can be nonzero. The canonical map

$$\mathcal{F} \to 0$$

is a stalkwise isomorphism and induces an isomorphism after sheafification. This shows that stalks are local invariants, not a complete invariant of arbitrary presheaves.

Example 14.23 (One-point space)**.** Let $X = \{x\}$. The only opens are $\varnothing$ and X. A presheaf of abelian groups with $\mathcal{F}(\varnothing) = 0$ is already a sheaf: every nonempty open cover of X contains X. Therefore

$$\mathcal{F}^{\#} \cong \mathcal{F}.$$

The espace étalé is simply the discrete fiber $\mathcal{F}_x = \mathcal{F}(X)$ over the single point.

Example 14.24 (Two-point discrete space)**.** Let $X = \{p, q\}$ be discrete and start from the naive constant presheaf A. Its stalks are A at both points, so the sheafification is the sheaf determined by these two stalks:

$$\mathcal{F}^{\#}(X) = A \times A, \qquad \mathcal{F}^{\#}(\{p\}) = A, \qquad \mathcal{F}^{\#}(\{q\}) = A.$$

The unit on X is the diagonal map $A \to A \times A$.

Example 14.25 (A Sierpiński space)**.** Let $X = \{\eta, x\}$ with opens $\varnothing, \{\eta\}, X$. A presheaf of abelian groups is a map $A = \mathcal{F}(X) \to B = \mathcal{F}(\{\eta\})$. Every cover of X contains X itself, so every such presheaf is already a sheaf. Its sheafification is therefore unchanged. This finite example emphasizes that failure of the sheaf axiom depends on the available covers.

Example 14.26 (The canonical map on stalks)**.** Let $s \in \mathcal{F}(U)$ and $x \in U$. Under

$$\mathcal{F}_x \xrightarrow{\sim} (\mathcal{F}^{\#})_x,$$

the germ s_x is sent to the germ $(\widetilde{s})_x$. The inverse evaluates a sheafified germ at x. Thus the stalk isomorphism is not merely existential; it is the obvious map suggested by the espace étalé construction.

Example 14.27 (Local factorization into a sheaf)**.** Let $u : \mathcal{F} \to C^0_X$ be a presheaf morphism and $\sigma \in \mathcal{F}^{\#}(U)$. Choose a cover $U = \bigcup U_i$ with $\sigma|_{U_i} = \widetilde{s}_i$. The sections $u(s_i)$ of $C^0_X(U_i)$ have equal germs on overlaps, hence are equal there and glue to one continuous function on U. This is exactly $\bar{u}_U(\sigma)$ in the universal property.

Example 14.28 (Naturality of the unit)**.** For a morphism $f : \mathcal{F} \to \mathcal{G}$ and a section $s \in \mathcal{F}(U)$, the two routes around the naturality square give

$$f^{\#}(\widetilde{s}) = \widetilde{f(s)}.$$

Indeed, at x both sides have value $f_x(s_x)$. Thus the commutative square for the unit is visible point by point in the fibers of the two espace étalés.

Example 14.29 (Sheafification of the naive constant presheaf on a circle)**.** Let $A = \mathbb{Z}$ and $X = S^1$. The sheafification is the locally constant integer-valued sheaf. Since S^1 is connected,

$$\underline{\mathbb{Z}}(S^1) = \mathbb{Z}.$$

But on a disconnected open consisting of two short arcs,

$$\underline{\mathbb{Z}}(U) \cong \mathbb{Z} \times \mathbb{Z}.$$

Thus global sections on the whole circle happen to agree with the naive presheaf, while the presheaf still fails to be a sheaf on smaller disconnected opens.

Example 14.30 (Sheafification commutes with open restriction)**.** Let $j : U \hookrightarrow X$. Both $(\mathcal{F}|_U)^\#$ and $\mathcal{F}^\#|_U$ have the same espace étalé over U:

$$\coprod_{x \in U} \mathcal{F}_x.$$

Their locally representable sections are therefore the same. Hence

$$(\mathcal{F}|_U)^\# \cong \mathcal{F}^\#|_U.$$

Example 14.31 (A basis computation)**.** Let $X = \mathbb{R}$ and let $\mathcal{B}$ be the basis of open intervals. Suppose a presheaf is given only on intervals and satisfies compatible restriction maps. The germ at x can be computed using only intervals containing x, because they are cofinal among neighborhoods. The espace étalé construction then produces the same sheaf as if one first extended the presheaf to all opens and then sheafified.

Example 14.32 (The first plus construction removes ambiguity)**.** Suppose $s, t \in \mathcal{F}(U)$ restrict equally on an open cover $U = \bigcup U_i$ although $s \neq t$. In the plus construction, both sections determine the same compatible family $\{s|_{U_i}\} = \{t|_{U_i}\}$, so their images in $\mathcal{F}^+(U)$ coincide. Thus $\mathcal{F}^+$ satisfies locality even if $\mathcal{F}$ does not.

Example 14.33 (The second plus construction supplies gluings)**.** Suppose $\mathcal{F}$ is already separated but compatible sections on a cover fail to glue. Their compatible family itself defines an element of $\mathcal{F}^+(U)$. Because separatedness removes ambiguity on refinements, these new local objects glue correctly; standard plus construction theory shows $\mathcal{F}^+$ is now a sheaf. For arbitrary $\mathcal{F}$, two applications give $\mathcal{F}^{++} \cong \mathcal{F}^\#$.

Example 14.34 (Idempotence through the universal property)**.** Let $\mathcal{G}$ be a sheaf. The identity map $\mathcal{G} \to \mathcal{G}$ factors uniquely through $\eta_\mathcal{G} : \mathcal{G} \to \mathcal{G}^\#$. The resulting map $r : \mathcal{G}^\# \to \mathcal{G}$ satisfies $r\eta = \mathrm{id}$. Applying uniqueness again gives $\eta r = \mathrm{id}$. This constructs the canonical inverse to the unit without inspecting any sections explicitly.

Example 14.35 (Same sheafification, different presheaves)**.** On a disconnected space, let $\mathcal{P}$ be the naive constant presheaf A and let $\underline{A}$ be the constant sheaf. They are not isomorphic as presheaves, because on a disconnected open U the first has A while the second can have a product of several copies of A. Nevertheless

$$\mathcal{P}^\# \cong \underline{A} \cong (\underline{A})^\#.$$

Sheafification forgets the defective global presentation and retains the same local germs.

14.17 Advanced worked examples

Example 14.36 (Advanced: sheafification can kill a large global group)**.** Let X be a manifold with at least two points and let A be a large abelian group. Set

$$\mathcal{F}(X) = A, \qquad \mathcal{F}(U) = 0 \quad (U \subsetneq X).$$

Every point has a proper coordinate neighborhood, so every stalk is zero and $\mathcal{F}^\# = 0$. Thus the map on global sections

$$A = \mathcal{F}(X) \longrightarrow \mathcal{F}^\#(X) = 0$$

can have arbitrarily large kernel. The unit is therefore not generally a monomorphism of presheaves.

Example 14.37 (Advanced: sheafification can add arbitrarily many global sections)**.** Let X be a discrete set with n points and let $\mathcal{P}$ be the naive constant presheaf A. Then

$$\mathcal{P}(X) = A, \qquad \mathcal{P}^{\#}(X) \cong A^n.$$

For an infinite discrete set X,

$$\mathcal{P}^{\#}(X) \cong \prod_{x \in X} A.$$

Thus the failure of surjectivity of the unit can be as large as the topology allows.

Example 14.38 (Advanced: local polynomial data and analytic rigidity)**.** On $\mathbb{C}$, let $\mathcal{P}(U)$ be restrictions of one global polynomial. Its sheafification is the sheaf of locally polynomial functions. On every connected open set a locally polynomial function is represented by one polynomial: two polynomial representatives that overlap on a nonempty open set agree identically. Hence the only new sections arise from choosing different polynomials on distinct connected components. This contrasts with bounded continuous functions, where sheafification can genuinely enlarge sections even on a connected noncompact open.

Example 14.39 (Advanced: local boundedness is the sheafification of global boundedness)**.** For continuous real-valued functions, "bounded on U" is global while "locally bounded" is automatic. The sheafification of bounded continuous functions therefore forgets the global bound. If one instead starts with locally bounded continuous functions, the presheaf is already the full sheaf C^0_X. Sheafification detects exactly the local content of the property.

Example 14.40 (Advanced: the unit is a local isomorphism)**.** For every presheaf $\mathcal{F}$, the canonical map $\eta : \mathcal{F} \to \mathcal{F}^{\#}$ is an isomorphism on every stalk. In the language used later for sheaves, it is a universal "local isomorphism." This explains why any construction depending only on germs cannot distinguish a presheaf from its sheafification. However, a construction depending on global section sets can distinguish them sharply.

Example 14.41 (Advanced: maps to every sheaf only see the sheafification)**.** Suppose $\mathcal{F}$ and $\mathcal{H}$ have isomorphic sheafifications. Then for every sheaf $\mathcal{G}$ there are natural bijections

$$\mathrm{Hom}_{\mathbf{PreSh}}(\mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{Sh}}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{Sh}}(\mathcal{H}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{PreSh}}(\mathcal{H}, \mathcal{G}).$$

Thus sheaves cannot distinguish the defective global presentation of $\mathcal{F}$ from that of $\mathcal{H}$ once both are repaired.

Example 14.42 (Advanced: a local homeomorphism from the espace étalé)**.** Take a germ $a = s_x \in \mathcal{F}_x$. The basic open $\widetilde{s}(U)$ around a maps by π homeomorphically onto U, with inverse $y \mapsto s_y$. Hence every point of $E_{\mathcal{F}}$ has a neighborhood on which π is a homeomorphism. The associated sheaf can therefore be viewed as the sheaf of continuous local sections of a local homeomorphism.

Example 14.43 (Advanced: the espace étalé of a constant sheaf)**.** For a locally connected space and the constant sheaf $\underline{A}$, the espace étalé is locally a disjoint union of copies of the base. Over a sufficiently small connected open U, every section is constant, so

$$\pi^{-1}(U) \cong U \times A$$

with A discrete. The global topology of the espace étalé records how local choices of constants can vary from component to component.

Example 14.44 (Advanced: plus construction on the constant presheaf)**.** Let $X = U_1 \sqcup U_2$ and start with the naive constant presheaf A. A compatible family on the disjoint cover is simply a pair $(a_1, a_2) \in A^2$, since there are no overlap conditions. Thus the first plus construction already produces A^2 on X. Because the constant presheaf is separated on this cover, no second correction is needed there; globally the standard theorem guarantees $\mathcal{P}^{++} \cong \underline{A}$.

Example 14.45 (Advanced: a presheaf failing locality)**.** Let $X = U \cup V$ and define a presheaf whose global group contains two elements $s \neq t$ but whose restrictions to both U and V are equal. Then locality fails. The first plus construction identifies s and t, because both determine the same compatible family on $\{U, V\}$. This is the "separation" stage of sheafification before any new gluing data are added.

Example 14.46 (Advanced: a separated presheaf failing gluing)**.** Let $\mathcal{F}$ be separated and suppose compatible local sections $s_i \in \mathcal{F}(U_i)$ have no global gluing. Their equivalence class as compatible cover data defines a new section in $\mathcal{F}^+(U)$. Since separatedness guarantees uniqueness on refinements, $\mathcal{F}^+$ satisfies the full sheaf axiom. This explains the theorem that one plus is enough once locality is already known.

Example 14.47 (Advanced: sheafification and principal-open data)**.** Let A be a ring and consider the basis $\{D(f)\}$ of $\operatorname{Spec} A$. The assignments

$$D(f) \longmapsto A_f$$

with localization maps are already compatible on the basis. Even before VI/17 proves the basis sheaf condition directly, one can always regard this as presheaf data on the basis and apply sheafification. The resulting sheaf has stalk $A_{\mathfrak{p}}$ at $\mathfrak{p}$, anticipating the structure sheaf without yet identifying all its sections.

Example 14.48 (Advanced: sheafification is local on the base)**.** Let $X = U \cup V$. To compute $\mathcal{F}^{\#}$ it is enough to compute $(\mathcal{F}|_U)^{\#}$ and $(\mathcal{F}|_V)^{\#}$ and identify them on $U \cap V$ via the canonical restriction isomorphisms. The result glues uniquely. This is a practical consequence of $(\mathcal{F}|_U)^{\#} \cong \mathcal{F}^{\#}|_U$ and is especially useful on spaces covered by simple coordinate charts.

Example 14.49 (Advanced: the adjunction explains preservation of colimits)**.** Because sheafification is a left adjoint, it preserves all colimits that exist in the presheaf category. In particular, constructions such as cokernels can be formed objectwise as presheaves and then sheafified to obtain the corresponding sheaf colimit. VI/15 will apply exactly this principle to quotient and cokernel sheaves. We record the categorical reason here but defer the concrete quotient theory to its dedicated chapter.

14.18 Exercises

Exercise 14.50 (Universal triangle)**.** State the universal property of a sheafification $(\mathcal{F}^{\#}, \eta)$ and explain why the factorization morphism is unique.

Exercise 14.51 (Uniqueness of associated sheaf)**.** Prove that any two sheafifications of the same presheaf are uniquely isomorphic compatibly with their unit maps.

Exercise 14.52 (Basis for the espace etale)**.** Prove that the sets $\widetilde{s}(U) = \{s_x : x \in U\}$ form a basis for a topology on $E_{\mathcal{F}} = \coprod_x \mathcal{F}_x$.

Exercise 14.53 (Local homeomorphism)**.** Show that $\pi : E_{\mathcal{F}} \to X$ restricts to a homeomorphism $\widetilde{s}(U) \to U$ on every basic open.

Exercise 14.54 (Local representability)**.** Prove that a section $\sigma : U \to E_{\mathcal{F}}$ is continuous iff it is locally of the form $x \mapsto s_x$ for sections s of the original presheaf.

Exercise 14.55 (Associated sections form a sheaf)**.** Using ordinary gluing of maps, prove directly that $U \mapsto \mathcal{F}^{\#}(U)$ is a sheaf.

Exercise 14.56 (The canonical morphism)**.** Prove that $s \mapsto \widetilde{s}$ is compatible with restriction and hence defines $\eta : \mathcal{F} \to \mathcal{F}^{\#}$.

Exercise 14.57 (Surjectivity on stalks)**.** Prove that $(\eta_{\mathcal{F}})_x : \mathcal{F}_x \to (\mathcal{F}^{\#})_x$ is surjective.

Exercise 14.58 (Injectivity on stalks)**.** Prove that $(\eta_{\mathcal{F}})_x$ is injective.

Exercise 14.59 (Factorization existence)**.** Let $u : \mathcal{F} \to \mathcal{G}$ with $\mathcal{G}$ a sheaf. Construct locally and then glue the factor $\bar{u} : \mathcal{F}^{\#} \to \mathcal{G}$.

Exercise 14.60 (Factorization uniqueness)**.** Prove the factor in the universal property is unique because every sheafified section is locally represented by original sections.

Exercise 14.61 (Functoriality)**.** For $f : \mathcal{F} \to \mathcal{G}$, construct $f^{\#}$ and prove $(gf)^{\#} = g^{\#} f^{\#}$.

Exercise 14.62 (Naturality of the unit)**.** Prove $f^{\#}\eta_{\mathcal{F}} = \eta_{\mathcal{G}} f$.

Exercise 14.63 (Stalk map of a sheafified morphism)**.** Under the canonical stalk identifications, prove $(f^{\#})_x = f_x$.

Exercise 14.64 (Sheafifying a sheaf)**.** If $\mathcal{G}$ is a sheaf, prove $\eta_{\mathcal{G}}$ is an isomorphism using only the universal property.

Exercise 14.65 (Idempotence)**.** Deduce $(\mathcal{F}^{\#})^{\#} \cong \mathcal{F}^{\#}$ canonically.

Exercise 14.66 (Constant presheaf on two points)**.** On a two-point discrete space, compute the sheafification of the naive constant presheaf A and identify the unit on global sections.

Exercise 14.67 (Bounded continuous functions)**.** Show that the sheafification of the presheaf of bounded continuous real functions is the full sheaf of continuous real functions.

Exercise 14.68 (A unit that is not injective)**.** For a space where every point has a proper neighborhood, analyze the presheaf $\mathcal{F}(X) = A$, $\mathcal{F}(U) = 0$ for $U \subsetneq X$, and prove its sheafification is zero.

Exercise 14.69 (Restriction to an open)**.** Let $j : U \hookrightarrow X$. Prove $(\mathcal{F}|_U)^{\#} \cong \mathcal{F}^{\#}|_U$.

Exercise 14.70 (Cofinal basis neighborhoods)**.** Let $\mathcal{B}$ be a basis. Prove that the stalk $\mathcal{F}_x$ can be computed using only basis opens $B \ni x$.

Exercise 14.71 (Separated presheaf)**.** Define separatedness for a presheaf and show every sheaf is separated.

Exercise 14.72 (First plus is separated)**.** For the standard plus construction, prove that $\mathcal{F}^{+}$ is separated.

Exercise 14.73 (Plus of a separated presheaf)**.** Assuming $\mathcal{F}$ is separated, prove that $\mathcal{F}^{+}$ satisfies gluing and is a sheaf.

Exercise 14.74 (Two pluses)**.** Deduce that $\mathcal{F}^{++}$ is always a sheaf and has the universal property of sheafification.

Exercise 14.75 (Adjunction)**.** Show that precomposition with the unit gives a natural bijection $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{PreSh}(X)}(\mathcal{F}, \mathcal{G})$ for every sheaf $\mathcal{G}$.

14.19 Problems

Problem 14.76 (Legacy AG1 Exercise 4: construct the associated sheaf)**.** Let $\mathcal{F}$ be a presheaf of abelian groups on a topological space X. In the legacy notation write $\mathcal{F}^+$ for the associated sheaf obtained from the disjoint union of stalks. Complete the construction and prove all of the following.

1. Form
$$E_{\mathcal{F}} = \coprod_{p \in X} \mathcal{F}_p, \qquad \pi : E_{\mathcal{F}} \to X,$$
endow $E_{\mathcal{F}}$ with the topology generated by the sets $\widetilde{s}(U) = \{s_p : p \in U\}$, and show that the continuous sections of π form a sheaf $\mathcal{F}^+$.

2. Construct the natural morphism
$$\theta : \mathcal{F} \longrightarrow \mathcal{F}^+, \qquad s \longmapsto \widetilde{s},$$
and prove that $(\mathcal{F}^+, \theta)$ has the universal property of the associated sheaf.

3. Prove that for every $p \in X$ the induced map on stalks is an isomorphism
$$(\mathcal{F}^+)_p \cong \mathcal{F}_p.$$

4. Given a morphism of presheaves $f : \mathcal{F} \to \mathcal{G}$, construct the induced morphism
$$f^+ : \mathcal{F}^+ \longrightarrow \mathcal{G}^+$$
and prove that, under the canonical stalk identifications,
$$(f^+)_p = f_p.$$

5. Let $i : \mathbf{Sh}_X \to \mathbf{PreSh}_X$ be the inclusion. Prove the natural adjunction
$$\mathrm{Hom}_{\mathbf{Sh}_X}(\mathcal{F}^+, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{PreSh}_X}(\mathcal{F}, i\mathcal{G})$$
for every sheaf $\mathcal{G}$; equivalently, sheafification is left adjoint to inclusion.

Why this problem matters. This is the main legacy sheafification problem in the Volume VI corpus. It does not ask for one isolated fact: it asks for the complete mechanism that turns a presheaf into a sheaf and then proves that the construction is canonical. The existence theorem, stalk preservation, functoriality, and adjunction that appear throughout modern sheaf theory are all contained in this one problem.

Useful ideas before starting. There are three recurring moves. First, equality in a stalk means equality after shrinking. Second, a section of the associated sheaf is locally represented by an original presheaf section. Third, once the target is a sheaf, equal germs force equality of local sections and compatible local sections glue uniquely.

Guided hints.

1. To prove that $\widetilde{s}(U)$ form a basis, take a germ lying in the intersection of two such sets and shrink to a neighborhood on which the two representatives agree.

2. To prove that $\mathcal{F}^+$ is a sheaf, glue the continuous maps $U_i \to E_{\mathcal{F}}$ pointwise; continuity is local on the domain.

3. For stalk preservation, define

$$\Phi_p : (\mathcal{F}^+)_p \to \mathcal{F}_p, \qquad \sigma_p \mapsto \sigma(p).$$

Use local representability for injectivity and a representative s of a given germ for surjectivity.

4. For the universal property, cover a sheafified section by opens on which it equals $\widetilde{s_i}$, apply the given morphism to s_i, and glue in the target sheaf.

5. Functoriality and adjunction are not new constructions after the universal property; they are consequences of uniqueness of factorization.

Provenance. Legacy AG1 Exercise 4, merged with the preferred sheafification notes.

Problem 14.77 (A noninjective sheafification unit)**.** Let $A \neq 0$ be an abelian group and $X = \mathbb{R}$. Define a presheaf $\mathcal{F}$ by

$$\mathcal{F}(X) = A, \qquad \mathcal{F}(U) = 0 \quad \text{for every proper open } U \subsetneq X,$$

with the only possible restriction maps. Prove that every stalk $\mathcal{F}_x$ is zero, identify $\mathcal{F}^\#$, and show that the unit $\eta_X : \mathcal{F}(X) \to \mathcal{F}^\#(X)$ is not injective.

Why this problem matters. This example isolates the part of sheafification that discards globally recorded data which is invisible in every sufficiently small neighborhood.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Every point of
mathbbR has a proper open neighborhood. Compute the filtered colimit defining the stalk before invoking stalk preservation.

Provenance. New synthesis problem, built from the unit-pathology examples in the legacy sheaf notes.

Problem 14.78 (A nonsurjective sheafification unit)**.** Let $X = \{p, q\}$ with the discrete topology and let $A \neq 0$. Let $\mathcal{F}$ be the naive constant presheaf: $\mathcal{F}(U) = A$ for nonempty U and $\mathcal{F}(\varnothing) = 0$, with identity restrictions between nonempty opens. Compute $\mathcal{F}^\#$ and the unit on X.

Why this problem matters. Sheafification may add sections even while preserving all stalks.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. A sheaf on a discrete space is the product of its point stalks. Track what a single constant section does at p and q.

Provenance. New synthesis problem from the legacy constant-presheaf example.

Problem 14.79 (Bounded continuous functions sheafify to all continuous functions)**.** On $X = \mathbb{R}$, let $\mathcal{B}(U)$ be the abelian group of bounded continuous functions $U \to \mathbb{R}$. Prove that $\mathcal{B}$ is a presheaf, generally not a sheaf, and that

$$\mathcal{B}^{\#} \cong \mathcal{C}^0_{\mathbb{R}}.$$

Why this problem matters. This is a clean model of sheafification replacing a global constraint by its local content.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Restrictions preserve boundedness. Use an unbounded continuous function for failure of gluing. Every continuous function is bounded on a sufficiently small neighborhood of each point.

Provenance. Legacy-derived from the bounded-functions counterexample in the preferred sheaf notes.

Problem 14.80 (The sheaf of locally polynomial functions)**.** Let k be a field and $X = k$ with a topology in which polynomial functions make sense (for instance $k = \mathbb{R}$ with the Euclidean topology). Define $\mathcal{P}(U)$ to be restrictions to U of a single polynomial in $k[t]$. Describe $\mathcal{P}^{\#}$ intrinsically.

Why this problem matters. The example separates global presentation by one formula from local presentation by possibly different formulas.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Use the locally representable-sections description of sheafification.

Provenance. New synthesis problem based on the local-representability principle.

Problem 14.81 (Uniqueness of sheafification from the universal property)**.** Suppose $(\mathcal{S}, \eta)$ and $(\mathcal{T}, \theta)$ are two sheafifications of the same presheaf $\mathcal{F}$. Prove there is a unique isomorphism $\alpha : \mathcal{S} \to \mathcal{T}$ satisfying $\alpha\eta = \theta$.

Why this problem matters. A universal construction is useful only if its output is canonical up to unique compatible isomorphism.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Factor θ through η and η through θ, then compare composites with identities.

Provenance. New synthesis problem distilled from the uniqueness clause in the legacy construction.

Problem 14.82 (Functoriality is forced by universality)**.** Given presheaf morphisms $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$, construct $f^\#$ and prove

$$(\mathrm{id}_{\mathcal{F}})^\# = \mathrm{id}_{\mathcal{F}^\#}, \qquad (g \circ f)^\# = g^\# \circ f^\#.$$

Why this problem matters. The associated-sheaf construction must be a functor, not a collection of unrelated choices.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Factor $\eta_{\mathcal{G}} f$ through $\eta_{\mathcal{F}}$ and use uniqueness twice.

Provenance. Legacy-derived from AG1 Exercise 4, extracted as a separate synthesis dossier; the original demand remains fully preserved in P1.

Problem 14.83 (Sheafification does not change the map on germs)**.** For a morphism of presheaves $f : \mathcal{F} \to \mathcal{G}$, prove that under the canonical identifications $(\mathcal{F}^\#)_x \cong \mathcal{F}_x$ and $(\mathcal{G}^\#)_x \cong \mathcal{G}_x$ one has

$$(f^\#)_x = f_x.$$

Why this problem matters. This is the precise sense in which sheafification changes global organization without changing local germ data.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Take stalks of the naturality square for the units.

Provenance. Legacy-derived from the final functoriality clause of AG1 Exercise 4; retained inside P1 and developed independently here.

Problem 14.84 (Why the plus construction is applied twice)**.** For a presheaf $\mathcal{F}$ of abelian groups, define $\mathcal{F}^+(U)$ as compatible families of sections on an open cover of U, modulo common refinement. Prove: (a) $\mathcal{F}^+$ is separated; (b) if $\mathcal{F}$ is separated then $\mathcal{F}^+$ is a sheaf; (c) consequently $\mathcal{F}^{++}$ is a sheaf.

Why this problem matters. This is the standard algebraic alternative to the espace-etale construction and explains the traditional “two pluses theorem.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. For separatedness, refine two representatives until they agree locally. For gluing, refine all local representatives to one common cover and use separatedness to remove choice ambiguities.

Provenance. New canonical problem from the standard plus-construction material in the preferred sheafification source.

Problem 14.85 (Equivalence of plus and espace-etale sheafification)**.** Let $a_{\mathrm{et}}(\mathcal{F})$ be the sheaf built from the espace etale of germs and let $a_+(\mathcal{F}) = \mathcal{F}^{++}$ be the two-plus construction. Assuming each satisfies the sheafification universal property, construct the canonical isomorphism between them and prove it is natural in $\mathcal{F}$.

Why this problem matters. Two very different-looking constructions should produce one canonical object.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Use P6 objectwise. For naturality, compare two maps after precomposing with the unit.

Provenance. New synthesis problem comparing the two canonical constructions developed in the legacy notes.

Problem 14.86 (Sheafification commutes with restriction to an open subset)**.** Let $j : U \hookrightarrow X$ be open and $\mathcal{F}$ a presheaf on X. Prove a natural isomorphism

$$(\mathcal{F}|_U)^\# \cong (\mathcal{F}^\#)|_U.$$

Why this problem matters. Sheafification is itself local; passing to a smaller open region should not change the repair.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Compare stalks at $x \in U$, or verify the universal property on U.

Provenance. New synthesis problem; the restriction compatibility is part of the preferred sheafification theory.

Problem 14.87 (From a sheaf on a basis to a sheaf on the whole space)**.** Let $\mathcal{B}$ be a basis of X closed under finite intersections (or replace intersections by basis refinements). Suppose F assigns groups and restrictions to members of $\mathcal{B}$ and satisfies the sheaf axiom for covers by basis elements. Show that there is a sheaf $\widetilde{F}$ on X, unique up to unique isomorphism, whose restriction to the basis is F.

Why this problem matters. Algebraic geometry usually specifies regular functions first on a convenient basis, not on every open set.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. For arbitrary U, define sections as compatible families on all basis opens $B \subseteq U$, or first form the right Kan-style presheaf and sheafify.

Provenance. New synthesis problem preparing the structure-sheaf construction without consuming VI/17.

Problem 14.88 (Sheafifying the constant presheaf)**.** For an abelian group A, let A_{pre} be the naive constant presheaf on X. Prove that its sheafification is the sheaf $\underline{A}$ of locally constant A-valued functions. Describe the unit explicitly.

Why this problem matters. This is the standard first nontrivial computation of an associated sheaf and appears repeatedly in the legacy sheaf notes.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Both have stalk A. Map a constant section to the corresponding constant function and use local constancy for local surjectivity.

Provenance. Legacy-derived from the constant-presheaf/sheafification example in the preferred notes.

Problem 14.89 (When is the naive constant presheaf already a sheaf?)**.** Let A have at least two elements. On a topological space X, define the naive constant presheaf by $A(U) = A$ for nonempty U and $A(\varnothing) = 0$. Prove it is a sheaf iff every nonempty open subset of X is connected.

Why this problem matters. This problem pinpoints exactly which topology makes "one value on every nonempty open compatible with sheaf gluing.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. A disconnected open gives an immediate counterexample. Conversely, show that the overlap graph of any open cover of a connected open is connected.

Provenance. New synthesis problem extending the legacy constant-presheaf example.

Problem 14.90 (The adjunction in full natural form)**.** Let $a : \mathbf{PreSh}_X \to \mathbf{Sh}_X$ be sheafification and $i : \mathbf{Sh}_X \to \mathbf{PreSh}_X$ inclusion. Construct explicitly the bijection

$$\Phi_{\mathcal{F},\mathcal{G}} : \mathrm{Hom}_{\mathbf{Sh}_X}(a\mathcal{F}, \mathcal{G}) \to \mathrm{Hom}_{\mathbf{PreSh}_X}(\mathcal{F}, i\mathcal{G})$$

and prove it is natural in both variables.

Why this problem matters. Adjunction packages the universal property into the categorical statement used throughout later homological sheaf theory.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. The bijection is precomposition with the unit. Naturality follows by associativity plus the defining naturality of the unit.

Provenance. Legacy INCLUDE-derived: the adjunction demand is an explicit final clause of AG1 Exercise 4 and remains fully solved in P1.

Problem 14.91 (Stalkwise equivalence of presheaves becomes an isomorphism)**.** Let $f : \mathcal{F} \to \mathcal{G}$ be a presheaf morphism such that $f_x : \mathcal{F}_x \to \mathcal{G}_x$ is an isomorphism for every x. Prove $f^{\#} : \mathcal{F}^{\#} \to \mathcal{G}^{\#}$ is an isomorphism.

Why this problem matters. Sheafification identifies presheaves that carry the same local germ information.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Use P8 and the stalkwise isomorphism criterion for sheaves.

Provenance. New synthesis problem from stalk preservation plus the VI/13 stalkwise criterion.

Problem 14.92 (Sheafification as an idempotent reflector)**.** Prove that $a \circ i \cong \mathrm{id}_{\mathbf{Sh}_X}$ and $a \circ a \cong a$. Show that the second isomorphism is compatible with the unit.

Why this problem matters. Sheafification repairs a presheaf once; repeating the repair changes nothing.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Apply the universal property to a presheaf that is already a sheaf.

Provenance. New synthesis problem emphasizing the categorical structure implicit in the preferred notes.

Problem 14.93 (Explicit sheafification on a finite discrete space)**.** Let X be a finite discrete space and $\mathcal{F}$ any presheaf of abelian groups. Prove

$$\mathcal{F}^{\#}(U) \cong \prod_{x \in U} \mathcal{F}_x$$

for every $U \subseteq X$, and describe the unit.

Why this problem matters. Finite discrete spaces turn the abstract germ construction into an exact finite formula.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. The singleton neighborhoods are final in the stalk systems, and a sheaf on a discrete space is a product over singletons.

Provenance. New advanced synthesis problem based on finite-space examples in VI/12-VI/13.

Problem 14.94 (Principal-open localization and its stalks)**.** Let A be a ring. On the basis of principal opens of $\operatorname{Spec} A$, assign

$$D(f) \longmapsto A_f,$$

with the usual localization maps on refinements. Show that the germ at $\mathfrak{p}$ is canonically $A_{\mathfrak{p}}$. Explain why any sheafification/extension of this basis presheaf must preserve these local rings.

Why this problem matters. This problem is the bridge from abstract sheafification to the structure sheaf constructed in VI/17.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Principal opens containing $\mathfrak{p}$ are indexed by $f \notin \mathfrak{p}$. Take the filtered colimit of A_f.

Provenance. New synthesis problem; deliberately only a preview of VI/17, with no structure-sheaf legacy problem consumed.

Problem 14.95 (Sheafification preserves coproducts)**.** Use the adjunction to prove that sheafification preserves arbitrary coproducts. In particular, for presheaves $\{\mathcal{F}_i\}_{i\in I}$ identify

$$a\left(\coprod_{i\in I}^{\mathbf{PreSh}} \mathcal{F}_i\right) \cong \coprod_{i\in I}^{\mathbf{Sh}} a(\mathcal{F}_i).$$

Why this problem matters. This is the first structural payoff of recognizing sheafification as a left adjoint.

Useful ideas before starting. Use the universal property, stalk preservation, and the fact that a morphism of sheaves is an isomorphism when it is so on every stalk. Keep the distinction between a presheaf statement on sections and a local statement on germs explicit.

Guided hints. Test both sides by maps into an arbitrary sheaf $\mathcal{G}$ and use the universal properties of coproduct and adjunction.

Provenance. New advanced synthesis problem from the adjunction; it does not migrate AG1 Exercise 5 or AG2 Problem 8.

14.20 Challenges

Problem 14.96 (Challenge: Sheaves of sets and etale spaces)**.** Work with sheaves of sets on a topological space X. Starting from a sheaf $\mathcal{F}$, form $E = \coprod_x \mathcal{F}_x$ with the germ topology and prove that $\pi : E \to X$ is a local homeomorphism. Conversely, for a local homeomorphism $\pi : E \to X$, prove that $U \mapsto \Gamma(U, E)$ is a sheaf of sets. Construct the natural equivalence between these two procedures.

Problem 14.97 (Challenge: Sheafification as localization at local equivalences)**.** Call a presheaf morphism a *local equivalence* if it induces an isomorphism on every stalk. Prove that the unit $\eta_{\mathcal{F}} : \mathcal{F} \to \mathcal{F}^{\#}$ is a local equivalence and that every map from $\mathcal{F}$ to a sheaf factors uniquely through this local equivalence. Show also that sheafification sends every local equivalence to an isomorphism.

Problem 14.98 (Challenge: A non-locally-connected constant-sheaf calculation)**.** Let $X = \{0\} \cup \{1/n : n \geq 1\} \subset \mathbb{R}$ with the subspace topology and let A be a nonzero abelian group. Describe the sheafification of the naive constant presheaf on X and analyze its global sections. In particular, determine which A-valued functions on X are locally constant.

Problem 14.99 (Challenge: Finite products and sheafification)**.** For presheaves of abelian groups $\mathcal{F}$ and $\mathcal{G}$, prove that the canonical map

$$(\mathcal{F} \times \mathcal{G})^{\#} \longrightarrow \mathcal{F}^{\#} \times \mathcal{G}^{\#}$$

is an isomorphism. Explain why the proof is different in spirit from preservation of coproducts by a left adjoint.

Problem 14.100 (Challenge: Algebraic structures)**.** Show that the germ/etale-space construction sheafifies presheaves of rings and presheaves of modules, not merely presheaves of abelian groups. Explain how addition, multiplication, scalar multiplication, and units are defined on locally represented germ sections and why the unit map preserves the relevant algebraic structure.

Summary

Sheafification is the universal local-to-global repair of a presheaf. The associated sheaf $\mathcal{F}^{\#}$ is characterized by

$$\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathcal{F}^{\#}, \mathcal{G}) \cong \mathrm{Hom}_{\mathbf{PreSh}(X)}(\mathcal{F}, i\mathcal{G})$$

for every sheaf $\mathcal{G}$, and it preserves all stalks:

$$(\mathcal{F}^{\#})_x \cong \mathcal{F}_x.$$

The espace étalé construction proves existence by turning germs into a local homeomorphism over X; the plus construction explains the repair as separation followed by gluing. The unit may add sections or identify sections, but it never changes the local germ seen at a point.

VI/15 uses this machinery to construct kernels, images, and quotient sheaves. The Volume VI legacy-problem ledger records AG1 Exercise 5 and AG2 Problem 8 there, so the sheafification chapter does not consume or duplicate them.

14.21 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 14.101 (Constant case)**.** What does the constant presheaf sheafify to?

Hint. Use turning compatible local data into a sheaf. ☐

Solution. The sheaf of locally constant functions. ☐

Exercise 14.102 (Stalks)**.** How do stalks change under sheafification?

Hint. Use turning compatible local data into a sheaf. ☐

Solution. They are canonically unchanged. ☐

Exercise 14.103 (Canonical map)**.** What map accompanies sheafification?

Hint. Use turning compatible local data into a sheaf. ☐

Solution. A natural presheaf morphism $F \to F^{\#}$. ☐

Exercise 14.104 (Already sheaf)**.** What happens when the presheaf is already a sheaf?

Hint. Use turning compatible local data into a sheaf. ☐

Solution. The canonical map to its sheafification is an isomorphism. ☐

Exercise 14.105 (Local representation)**.** What condition defines a section in the germ construction?

Hint. Use turning compatible local data into a sheaf. ☐

Solution. It must be locally represented by original presheaf sections. ☐

Proofs

Exercise 14.106 (Stalk preservation)**.** Prove: Prove $F_x \to (F^{\#})_x$ is an isomorphism.

Hint. Work directly from turning compatible local data into a sheaf. ☐

Solution. Every sheafified germ is locally represented by an original section, giving surjectivity; equality after sheafification occurs locally, giving injectivity. ☐

Exercise 14.107 (Universal property)**.** Prove: Prove every map from F to a sheaf factors uniquely through $F^{\#}$.

Hint. Work directly from turning compatible local data into a sheaf. ☐

Solution. Define the factor on locally represented germs and use sheaf locality for uniqueness. ☐

Exercise 14.108 (Idempotence)**.** Prove: Prove sheafifying a sheaf gives an isomorphic sheaf.

Hint. Work directly from turning compatible local data into a sheaf. ☐

Solution. Apply the universal property to the identity map and use the canonical map. ☐

Exercise 14.109 (Local constants)**.** Prove: Prove locally constant functions form the sheafification of the constant presheaf.

Hint. Work directly from turning compatible local data into a sheaf. ☐

Solution. Locally every such function is represented by one constant value, and compatible local constants glue uniquely. ☐

Counterexamples and hypothesis tests

Exercise 14.110 (New stalks)**.** Test the claim: sheafification changes stalks essentially.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; stalks are preserved. ☐

Exercise 14.111 (Destroy data)**.** Test the claim: sheafification simply discards all nongluable local data.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it reorganizes germs to enforce local gluing. ☐

Exercise 14.112 (Unique construction)**.** Test the claim: different sheafification constructions can yield nonisomorphic results.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; the universal property determines the result uniquely up to unique isomorphism. ☐

Applications and investigations

Exercise 14.113 (Repairing local data)**.** When is sheafification needed?

Hint. Translate the question into turning compatible local data into a sheaf. ☐

Solution. When a natural presheaf has correct local germs but fails the sheaf gluing axiom. ☐

Exercise 14.114 (Local-to-global)**.** What does the universal property buy?

Hint. Translate the question into turning compatible local data into a sheaf. ☐

Solution. It lets any map into a genuine sheaf be constructed through the canonical sheafification. ☐

Challenge problems

Exercise 14.115 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Sheafification in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Sheafification, translates it through turning compatible local data into a sheaf, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 14.116 (Hypothesis challenge)**.** Choose one structural statement from Sheafification, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 15

Kernels, Images and Quotients of Sheaves

Purpose and learning goals

Sheafification repaired a presheaf without changing its germs. We now need that repair in ordinary algebraic constructions. If

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

is a morphism of sheaves of abelian groups, the sectionwise kernel is already a sheaf, but the sectionwise image, cokernel, and quotient need not be. Their correct sheaf-theoretic versions are obtained by imposing a local condition or, equivalently, by sheafifying the corresponding presheaves.

This distinction is small in notation and fundamental in practice. A section may be locally in the image of a morphism on every neighborhood and still have no single global preimage. Thus the image sheaf can be strictly larger on an open set than the image of the map on that open set. Quotients exhibit the same phenomenon.

The Volume VI legacy-problem ledger is the starting point for this reconstruction. The two principal numbered corpus items are AG1 Exercise 5, on identifying the image sheaf with the subsheaf of locally liftable sections, and AG2 Problem 8, on quotient sheaves and the resulting short exact sequence. Both are preserved as full detailed dossiers, VI.15.P1 and VI.15.P2. The preferred reorganized sheaf notes are merged with them and with the chapter theory rather than counted as duplicate problems.

After this chapter, the reader should be able to:

- compute the kernel sheaf sectionwise and prove its universal property;
- distinguish the presheaf image from the image sheaf;
- characterize image-sheaf sections by local liftability;
- define cokernel and quotient sheaves by sheafification;
- prove the stalk identities

$$(\ker \varphi)_x = \ker(\varphi_x), \quad (\operatorname{im} \varphi)_x = \operatorname{im}(\varphi_x), \quad (\operatorname{coker} \varphi)_x = \operatorname{coker}(\varphi_x);$$

- prove

$$(\mathcal{F}/\mathcal{F}')_x \cong \mathcal{F}_x/\mathcal{F}'_x$$

  for every subsheaf $\mathcal{F}' \subseteq \mathcal{F}$;

- use the universal properties of kernels, quotients, and cokernels;
- explain why a sheaf epimorphism need not be surjective on sections over every open set;
- identify the canonical coimage-to-image map and understand why it is an isomorphism for sheaves of abelian groups;
- keep general exactness and the stalkwise exactness criterion for VI/16 while using the legacy quotient problem here as a motivating application.

15.1 Kernels are computed sectionwise

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups on X. For every open $U \subseteq X$ define

$$(\ker \varphi)(U) := \ker\big(\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)\big).$$

Restrictions preserve kernels because φ commutes with restriction maps.

Theorem 15.1 (Kernel sheaf)**.** *The presheaf $U \mapsto \ker(\varphi_U)$ is a sheaf. With its natural inclusion $\ker \varphi \hookrightarrow \mathcal{F}$, it is the categorical kernel of φ in the category of sheaves of abelian groups.*

Proof. Suppose $U = \bigcup_i U_i$ and $s_i \in \ker(\varphi)(U_i)$ are compatible. Since $\mathcal{F}$ is a sheaf, they glue uniquely to $s \in \mathcal{F}(U)$. On every U_i,

$$\varphi_U(s)|_{U_i} = \varphi_{U_i}(s_i) = 0.$$

By locality in $\mathcal{G}$, $\varphi_U(s) = 0$, so $s \in \ker(\varphi)(U)$. Uniqueness is inherited from $\mathcal{F}$. For the universal property, any $h : \mathcal{H} \to \mathcal{F}$ with $\varphi h = 0$ has every h_U landing in $\ker(\varphi_U)$, giving a unique factorization through the kernel sheaf. □

$$\begin{tikzcd}
\mathcal{H} \arrow[r, "\exists!"] \arrow[rr, bend right, "h"'] & \ker \varphi \arrow[r] & \mathcal{F} \arrow[r, "\varphi"] & \mathcal{G}
\end{tikzcd}$$

15.2 The image presheaf and the local image sheaf

The most tempting definition is

$$\mathrm{Im}^{\mathrm{pre}}(\varphi)(U) := \mathrm{im}\big(\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)\big).$$

This is a subpresheaf of $\mathcal{G}$, but it need not be a sheaf. The obstruction is global: compatible target sections may have local preimages without those preimages being capable of forming one global source section.

Definition 15.2 (Image sheaf)**.** The *image sheaf* of φ is

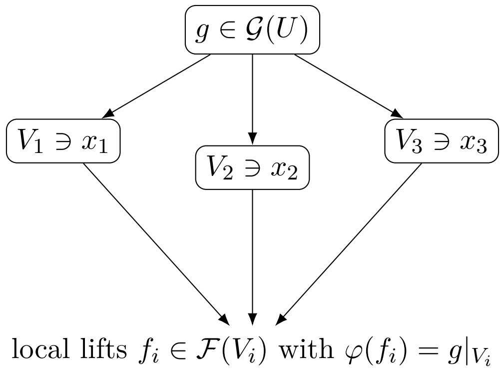
$$\mathrm{im}\,\varphi := \big(\mathrm{Im}^{\mathrm{pre}}(\varphi)\big)^{\#}.$$

Equivalently, it is the following subsheaf of $\mathcal{G}$:

$$(\mathrm{im}\,\varphi)(U) = \left\{ g \in \mathcal{G}(U) : \begin{array}{l} \text{for every } x \in U \text{ there is an open } V \subseteq U \text{ with } x \in V \\ \text{and } f \in \mathcal{F}(V) \text{ such that } g|_V = \varphi_V(f) \end{array} \right\}.$$

Theorem 15.3 (Locally in the image)**.** *The locally liftable sections above form a subsheaf of $\mathcal{G}$, and this subsheaf is canonically isomorphic to the sheafification of the image presheaf.*

Proof. Restriction preserves local liftability. Compatible locally liftable sections glue in $\mathcal{G}$; local liftability of the glued section is checked point by point inside one member of the cover. Thus they form a subsheaf $\mathcal{H} \subseteq \mathcal{G}$. The presheaf image maps into $\mathcal{H}$. Since $\mathcal{H}$ is a sheaf, the map factors through the sheafification. Conversely, every section of $\mathcal{H}$ is locally represented by a section of the presheaf image, which is exactly the local description of its sheafification. The two maps are inverse locally and hence globally. ☐



Example 15.4 (Kernel sheaf)**.** For a sheaf morphism $\phi : F \to G$, the assignment $U \mapsto \ker(F(U) \to G(U))$ is already a sheaf. Vanishing is local, so compatible kernel sections glue inside the kernel.

15.3 Images on stalks

Theorem 15.5 (Image stalk)**.** *For every $x \in X$ there is a canonical equality of subgroups of $\mathcal{G}_x$,*

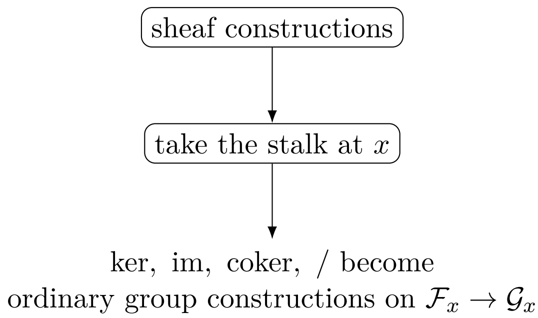
$$(\operatorname{im}\varphi)_x = \operatorname{im}(\varphi_x : \mathcal{F}_x \to \mathcal{G}_x).$$

Proof. A germ in the image-sheaf stalk is represented near x by a section that is locally in the image, so after shrinking it is literally $\varphi(f)$ and its germ lies in $\operatorname{im}\varphi_x$. Conversely, if $g_x = \varphi_x(f_x)$, choose representatives and shrink until the equality of germs becomes equality of sections. Then the representative of g_x is locally in the image, so $g_x \in (\operatorname{im}\varphi)_x$. ☐

Together with the sectionwise definition of the kernel,

$$(\ker\varphi)_x = \ker(\varphi_x).$$

This is the first indication that sheaf algebra is ordinary algebra when viewed through one stalk at a time.



15.4 Quotient presheaves and quotient sheaves

Let $\mathcal{F}' \subseteq \mathcal{F}$ be a subsheaf. Sectionwise algebra gives a quotient presheaf

$$\mathcal{Q}^{\mathrm{pre}}(U) = \mathcal{F}(U)/\mathcal{F}'(U).$$

It can fail to be a sheaf, so the correct quotient is its sheafification.

Definition 15.6 (Quotient sheaf)**.**

$$\mathcal{F}/\mathcal{F}' := (U \longmapsto \mathcal{F}(U)/\mathcal{F}'(U))^{\#}.$$

The composite

$$\mathcal{F} \longrightarrow \mathcal{Q}^{\mathrm{pre}} \longrightarrow \mathcal{F}/\mathcal{F}'$$

is the canonical quotient morphism q.

Theorem 15.7 (Stalks of a quotient)**.** *For every $x \in X$ there is a natural isomorphism*

$$(\mathcal{F}/\mathcal{F}')_x \cong \mathcal{F}_x/\mathcal{F}'_x.$$

Proof. Sheafification preserves stalks, so the left side is the stalk of the quotient presheaf. Filtered colimits of abelian groups are exact, hence localization to the stalk commutes with the cokernel of $\mathcal{F}'(U) \to \mathcal{F}(U)$. Concretely, a germ of a quotient section is represented by $[s] \in \mathcal{F}(U)/\mathcal{F}'(U)$ and is sent to the class of s_x in $\mathcal{F}_x/\mathcal{F}'_x$. Equality of classes is verified after shrinking, which proves injectivity as well as surjectivity. $\square$

$$\begin{tikzcd} \boxed{\mathcal{F}} \arrow[r, "\text{objectwise quotient}"] & \boxed{U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)} \arrow[r, "\text{sheafify}"] & \boxed{\mathcal{F}/\mathcal{F}'} \end{tikzcd}$$

15.5 Cokernels

For a general morphism $\varphi : \mathcal{F} \to \mathcal{G}$, define the cokernel presheaf by

$$U \longmapsto \operatorname{coker}(\varphi_U) = \mathcal{G}(U)/\operatorname{im}(\varphi_U).$$

Its sheafification is the cokernel sheaf:

$$\operatorname{coker}\varphi := (U \longmapsto \operatorname{coker}(\varphi_U))^{\#}.$$

Equivalently,

$$\operatorname{coker}\varphi \cong \mathcal{G}/\operatorname{im}\varphi.$$

Theorem 15.8 (Cokernel universal property)**.** *If $h : \mathcal{G} \to \mathcal{H}$ satisfies $h\varphi = 0$, then there is a unique morphism $\bar{h} : \operatorname{coker}\varphi \to \mathcal{H}$ with $h = \bar{h} \circ q$. Moreover,*

$$(\operatorname{coker}\varphi)_x \cong \operatorname{coker}(\varphi_x).$$

Proof. Sectionwise, h_U kills $\operatorname{im}\varphi_U$ and therefore factors through the cokernel presheaf. The target is already a sheaf, so the universal property of sheafification gives a unique factorization through its sheafification. Stalk preservation and the filtered-colimit calculation above give the stalk formula. $\square$

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] & \mathcal{G} \arrow[r, "q"] \arrow[dr, "h"'] & \operatorname{coker}\varphi \arrow[d, dashed, "\exists!\,\bar{h}"] \\ & & \mathcal{H} \end{tikzcd}$$

Example 15.9 (Image needs sheafification)**.** The pointwise presheaf image $U \mapsto \operatorname{im}(F(U) \to G(U))$ need not be a sheaf. The sheaf image is obtained by sheafifying it; stalkwise it has image $\operatorname{im}(F_x \to G_x)$.

15.6 The quotient by a subsheaf

The quotient map $q : \mathcal{F} \to \mathcal{F}/\mathcal{F}'$ has stalk map

$$q_x : \mathcal{F}_x \longrightarrow \mathcal{F}_x/\mathcal{F}'_x.$$

Hence it is locally onto in the precise sheaf-theoretic sense, and its kernel sheaf is $\mathcal{F}'$. This yields the short sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{F}' \longrightarrow 0.$$

The full theory of exact sequences and the criterion for exactness on all stalks are the subject of VI/16. Here the sequence is retained because it is the mathematical center of legacy AG2 Problem 8.

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{F}' \arrow[r, hook] & \mathcal{F} \arrow[r, "q"] & \mathcal{F}/\mathcal{F}' \arrow[r] & 0 \end{tikzcd}$$

15.7 Sheaf surjectivity is local, not sectionwise

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ can have image sheaf equal to $\mathcal{G}$ while

$$\mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

is not surjective for some open U. Equivalently, every target germ can lift while some global target section has no global lift.

The basic model is the exponential map of function sheaves. On a circle, every circle-valued continuous function has a local real-valued argument, but a degree-one map has no global continuous argument. Thus local liftability and global liftability are genuinely different.

$$\begin{tikzcd} & \boxed{\text{global target section } g} \arrow[d] & \\ & \boxed{\text{open cover } U = \bigcup U_i} \arrow[d] & \\ & \boxed{\text{local lifts } f_i \text{ exist}} \arrow[dl] \arrow[dr] & \\ \boxed{g \in (\operatorname{im}\varphi)(U)} & & \boxed{\text{one global lift may not exist}} \end{tikzcd}$$

Example 15.10 (Quotient sheaf)**.** Similarly, the naive quotient presheaf $U \mapsto G(U)/F(U)$ can fail gluing. The cokernel in sheaves is its sheafification, while stalks compute the expected module quotient.

15.8 Coimage and image

Define the coimage by

$$\operatorname{Coim}(\varphi) := \mathcal{F}/\ker\varphi.$$

There is a canonical morphism

$$\operatorname{Coim}(\varphi) \longrightarrow \operatorname{im}\varphi.$$

Theorem 15.11 (Coimage equals image for sheaves of abelian groups)**.** *The canonical morphism*

$$\mathcal{F}/\ker\varphi \xrightarrow{\sim} \operatorname{im}\varphi$$

is an isomorphism.

Proof. At x it becomes the ordinary first-isomorphism-theorem map

$$\mathcal{F}_x/\ker(\varphi_x) \xrightarrow{\sim} \operatorname{im}(\varphi_x).$$

A morphism of sheaves that is an isomorphism on every stalk is an isomorphism. The latter criterion was established in VI/13; VI/16 will place it into the general language of exactness. □

$$\begin{tikzcd}
\mathcal{F} \arrow[r, two heads] \arrow[dr, "\varphi"'] & \mathcal{F}/\ker\varphi \arrow[d, "\sim"] & \\
& \operatorname{im}\varphi \arrow[r, hook] & \mathcal{G}
\end{tikzcd}$$

15.9 Restriction and locality of the constructions

If $j : U \hookrightarrow X$ is open, restriction commutes with all four constructions:

$$(\ker\varphi)|_U \cong \ker(\varphi|_U), \qquad (\operatorname{im}\varphi)|_U \cong \operatorname{im}(\varphi|_U),$$

$$(\operatorname{coker}\varphi)|_U \cong \operatorname{coker}(\varphi|_U), \qquad (\mathcal{F}/\mathcal{F}')|_U \cong (\mathcal{F}|_U)/(\mathcal{F}'|_U).$$

For kernels this is sectionwise. For the others it follows either from the local description or from sheafification commuting with restriction.

15.10 Common pitfalls

1. The sectionwise image $U \mapsto \operatorname{im}\varphi_U$ is a presheaf, not necessarily a sheaf.
2. The sectionwise quotient $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$ is a presheaf, not necessarily a sheaf.
3. "Surjective morphism of sheaves" does not mean surjective on sections over every open set; the correct notion is local liftability, equivalently surjectivity on stalks.
4. Kernels are different: they are already computed sectionwise because the sheaf axiom passes to a subgroup defined by an equation.
5. Do not replace $\mathcal{F}/\mathcal{F}'$ by a sectionwise quotient in later exact-sequence arguments unless an independent reason shows that quotient presheaf is already a sheaf.
6. The identity $\operatorname{coker}\varphi \cong \mathcal{G}/\operatorname{im}\varphi$ uses the *image sheaf*, not merely the image presheaf.
7. General stalkwise exactness belongs to VI/16. The stalk computations here are the algebraic ingredients for that theorem.

15.11 Worked examples

Example 15.12 (The zero morphism)**.** For $0 : \mathcal{F} \to \mathcal{G}$, $\ker 0 = \mathcal{F}$ and $\operatorname{im} 0 = 0$. The image presheaf is already the zero sheaf, so no repair is needed.

Example 15.13 (The identity morphism)**.** For $\mathrm{id}_{\mathcal{F}}$, the kernel is 0, the image is $\mathcal{F}$, and the cokernel is 0. This is the sheaf version of the most elementary group calculation.

Example 15.14 (Multiplication by an integer)**.** On the constant sheaf $\underline{\mathbb{Z}}$, multiplication by n has kernel 0 for $n \neq 0$ and cokernel $\underline{\mathbb{Z}/n\mathbb{Z}}$. Stalkwise this is the ordinary map $\mathbb{Z} \xrightarrow{n} \mathbb{Z}$.

Example 15.15 (Restriction to an open subset)**.** If $j : U \hookrightarrow X$ and $\varphi : \mathcal{F} \to \mathcal{G}$, then $(\ker \varphi)|_U = \ker(\varphi|_U)$. The same holds for image and cokernel because these are characterized locally.

Example 15.16 (A skyscraper kernel)**.** Let $i_x(A)$ be the skyscraper sheaf at a closed point x and let $A \to B$ be a group map. The induced skyscraper morphism has kernel $i_x(\ker(A \to B))$, image $i_x(\mathrm{im}(A \to B))$, and cokernel $i_x(\mathrm{coker}(A \to B))$.

Example 15.17 (Quotient of skyscraper sheaves)**.** If $B \subseteq A$, then $i_x(A)/i_x(B) \cong i_x(A/B)$. Every nonzero stalk occurs only at x, so the quotient calculation is completely pointwise.

Example 15.18 (Discrete spaces)**.** If X is discrete, a sheaf is a product of its stalks: $\mathcal{F}(U) \cong \prod_{x \in U} \mathcal{F}_x$. Kernels, images, and quotients are then computed componentwise and the naive sectionwise constructions are already sheaves.

Example 15.19 (A kernel as vanishing locus of a restriction)**.** Let $V \subseteq U$ be open. For the restriction morphism of sheaves $\mathcal{C}^0_U \to j_*\mathcal{C}^0_V$, the kernel consists of continuous functions whose restriction to V is zero. This condition is local and therefore automatically defines a sheaf.

Example 15.20 (A quotient with no correction needed)**.** On a discrete two-point space, let $\mathcal{F}$ have stalks $\mathbb{Z}$ and let $2\mathcal{F} \subseteq \mathcal{F}$. Then $(\mathcal{F}/2\mathcal{F})(U)$ is exactly the product of copies of $\mathbb{Z}/2\mathbb{Z}$ over points of U.

Example 15.21 (Locally constant functions modulo n)**.** For the locally constant sheaf $\underline{\mathbb{Z}}$, the quotient by $n\underline{\mathbb{Z}}$ is $\underline{\mathbb{Z}/n\mathbb{Z}}$. This can be checked on stalks or by locally constant representatives.

Example 15.22 (Image of an inclusion)**.** If $\mathcal{F}' \hookrightarrow \mathcal{F}$ is literally a subsheaf inclusion, then the image sheaf is $\mathcal{F}'$. Its sectionwise image is already a sheaf because the map identifies $\mathcal{F}'(U)$ with a subgroup of $\mathcal{F}(U)$ compatible with gluing.

Example 15.23 (Projection from a direct sum)**.** For $\mathcal{F} \oplus \mathcal{G} \to \mathcal{G}$, the kernel is $\mathcal{F}$, the image is $\mathcal{G}$, and the quotient by the kernel is canonically $\mathcal{G}$.

Example 15.24 (Diagonal and difference)**.** The diagonal map $\Delta : \mathcal{F} \to \mathcal{F} \oplus \mathcal{F}$ has cokernel canonically isomorphic to $\mathcal{F}$ via $(a, b) \mapsto b - a$. This can be verified sectionwise because the displayed difference gives a global splitting.

Example 15.25 (Functions vanishing at a point)**.** Let X be a space and $x \in X$. The morphism $\mathcal{C}^0_X \to i_x(\mathbb{R})$ taking a function to its value germ at x has kernel the sheaf of continuous functions vanishing at x in the stalk sense. The quotient has stalk $\mathbb{R}$ at x and 0 elsewhere.

Example 15.26 (A locally liftable section)**.** Suppose $g \in \mathcal{G}(U)$ and for each $x \in U$ there is V_x and $f_x \in \mathcal{F}(V_x)$ with $\varphi(f_x) = g|_{V_x}$. Then g belongs to the image sheaf even if the f_x do not agree on overlaps and no global f exists.

Example 15.27 (Quotient classes are local)**.** A section of $\mathcal{F}/\mathcal{F}'$ over U may be represented by quotient classes $[s_i]$ on an open cover whose differences $s_i - s_j$ lie in $\mathcal{F}'(U_i \cap U_j)$. The representatives need not glue in $\mathcal{F}$; their quotient classes do.

Example 15.28 (Cokernel of an inclusion)**.** For $i : \mathcal{F}' \hookrightarrow \mathcal{F}$, $\operatorname{coker} i$ is exactly $\mathcal{F}/\mathcal{F}'$. The sectionwise quotient may fail to be a sheaf, which is why the definition uses sheafification.

Example 15.29 (Kernel and stalks)**.** If a section $s \in \mathcal{F}(U)$ has germ $s_x \in \ker \varphi_x$ for every $x \in U$, then $\varphi(s)$ has zero germ everywhere. Since $\mathcal{G}$ is a sheaf, $\varphi(s) = 0$, so $s \in (\ker \varphi)(U)$.

Example 15.30 (Image and stalks)**.** If every germ g_x of $g \in \mathcal{G}(U)$ lies in $\operatorname{im} \varphi_x$, then for each x choose a local representative lift. Hence g is locally in the image and therefore lies in $(\operatorname{im} \varphi)(U)$.

Example 15.31 (First isomorphism theorem)**.** For any $\varphi : \mathcal{F} \to \mathcal{G}$, the canonical map $\mathcal{F}/\ker \varphi \to \operatorname{im} \varphi$ is an isomorphism. On stalks it is the ordinary first isomorphism theorem for abelian groups.

15.12 Advanced worked examples

Example 15.32 (Advanced: Exponential map on the circle)**.** Let $\mathcal{C}^0_{\mathbb{R}}$ be continuous real-valued functions and $\mathcal{C}^0_{S^1}$ continuous circle-valued functions. The morphism $f \mapsto e^{2\pi i f}$ is locally surjective because every circle-valued map has local arguments. Hence its image sheaf is all of $\mathcal{C}^0_{S^1}$. Over S^1 itself, the identity map $z \mapsto z$ has no continuous real-valued lift, so the image presheaf is strictly smaller on global sections.

Example 15.33 (Advanced: The exponential quotient)**.** For the same map, the kernel is the locally constant integer-valued sheaf $\underline{\mathbb{Z}}$. Therefore $\mathcal{C}^0_{\mathbb{R}}/\underline{\mathbb{Z}} \cong \mathcal{C}^0_{S^1}$ as sheaves even though the sectionwise quotient over S^1 misses maps of nonzero winding number.

Example 15.34 (Advanced: Derivative on a circle)**.** On a smooth one-manifold, $d : \mathcal{C}^\infty \to \Omega^1$ is locally surjective by the existence of local primitives. On S^1, a one-form with nonzero integral has no global primitive. Thus the image sheaf is all of Ω^1 while the global sectionwise image is the subspace of exact one-forms.

Example 15.35 (Advanced: Holomorphic derivative on $\mathbb{C}^imes$)**.** The morphism $d : \mathcal{O} \to \Omega^1_{\mathrm{hol}}$ is locally surjective. The form dz/z has local holomorphic primitives but no global primitive on $\mathbb{C}^\times$. Again the sheaf image is larger than the sectionwise global image.

Example 15.36 (Advanced: Holomorphic exponential)**.** The map $\exp : \mathcal{O} \to \mathcal{O}^\times$ is locally surjective: a nowhere-zero holomorphic function has a local logarithm. On $\mathbb{C}^\times$, the function z has no global holomorphic logarithm. The image sheaf is $\mathcal{O}^\times$ although the map on global sections is not onto.

Example 15.37 (Advanced: A quotient described by local lifts)**.** Let $\mathcal{F}' \subseteq \mathcal{F}$. A section of $\mathcal{F}/\mathcal{F}'$ can be described by local lifts $s_i \in \mathcal{F}(U_i)$ whose differences lie in $\mathcal{F}'(U_i \cap U_j)$. Replacing s_i by $s_i + t_i$ with $t_i \in \mathcal{F}'(U_i)$ does not change the quotient section.

Example 15.38 (Advanced: Restriction commutes with quotient)**.** For $j : U \hookrightarrow X$, the natural map $(\mathcal{F}/\mathcal{F}')|_U \to (\mathcal{F}|_U)/(\mathcal{F}'|_U)$ is an isomorphism. At each $x \in U$ both stalks are $\mathcal{F}_x/\mathcal{F}'_x$.

Example 15.39 (Advanced: Direct image preserves kernels)**.** For a continuous map $f : X \to Y$, $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V)$. Therefore f_* computes kernels sectionwise and $f_*(\ker \varphi) = \ker(f_*\varphi)$. This is a concrete instance of the fact that a right adjoint preserves limits.

Example 15.40 (Advanced: Direct image need not preserve cokernels)**.** Cokernels require sheafification after an objectwise quotient, while direct image does not generally commute with that repair. This contrasts sharply with kernels and warns against treating all finite algebraic constructions as sectionwise.

Example 15.41 (Advanced: Quotient of an ideal sheaf)**.** If $\mathcal{I} \subseteq \mathcal{O}_X$ is a sheaf of ideals, the quotient $\mathcal{O}_X/\mathcal{I}$ is a sheaf of rings. Its stalk at x is $\mathcal{O}_{X,x}/\mathcal{I}_x$. This is the local algebra underlying closed subschemes, developed later in VI/22.

Example 15.42 (Advanced: Principal ideal quotient on an affine scheme - preview)**.** For $X = \operatorname{Spec} A$ and an element $f \in A$, the future structure sheaf satisfies $(\mathcal{O}_X/f\mathcal{O}_X)_{\mathfrak{p}} \cong A_{\mathfrak{p}}/fA_{\mathfrak{p}}$. The point of the present chapter is that this local quotient determines the correct quotient sheaf even when a naive global quotient is insufficient.

Example 15.43 (Advanced: Coimage-image comparison)**.** For $\varphi : \mathcal{F} \to \mathcal{G}$, form $\mathcal{F}/\ker\varphi$. Its stalk at x is $\mathcal{F}_x/\ker\varphi_x$, canonically isomorphic to $\operatorname{im}\varphi_x$. Thus the coimage-to-image map is a stalkwise isomorphism and hence a sheaf isomorphism.

Example 15.44 (Advanced: Epimorphism without global surjectivity)**.** The exponential morphism above is an epimorphism of sheaves because its image sheaf is the target. Yet it fails to be surjective on global sections of S^1. This is the canonical warning that categorical/sheaf surjectivity is a local condition.

Example 15.45 (Advanced: Failure disappears on a good cover)**.** Cover S^1 by two arcs whose overlap has two components. A degree-one circle map admits a real logarithm on each arc. The two logarithms differ by locally constant integers on the overlap. Those differences are precisely the quotient data killed by $\underline{\mathbb{Z}}$, so the local classes glue in $\mathcal{C}^0_{\mathbb{R}}/\underline{\mathbb{Z}}$ even though the lifts do not glue in $\mathcal{C}^0_{\mathbb{R}}$.

$$\begin{tikzcd} \boxed{\begin{array}{c}\text{discrete space}\\ \text{quotients are pointwise products}\end{array}} \arrow[r, leftrightarrow, "\text{contrast}"] & \boxed{\begin{array}{c}\text{general space}\\ \text{local lifts may not globalize}\end{array}} \end{tikzcd}$$

$$\begin{tikzcd} \mathcal{C}^\infty \arrow[rr, "d"] \arrow[d, Rightarrow] & & \Omega^1 \arrow[d, Rightarrow] \\ \text{local primitive exists} & & \text{global primitive may fail} \end{tikzcd}$$

15.13 Exercises

Exercise 15.46 (Kernel restrictions)**.** Prove directly that $U \mapsto \ker\varphi_U$ is a presheaf and that every restriction map is the restriction inherited from $\mathcal{F}$.

Exercise 15.47 (Kernel sheaf)**.** Prove the sheaf axiom for $\ker\varphi$ without using the theorem in the text.

Exercise 15.48 (Kernel stalk)**.** Construct the canonical isomorphism $(\ker\varphi)_x \cong \ker\varphi_x$.

Exercise 15.49 (Image presheaf)**.** Show that $U \mapsto \operatorname{im}\varphi_U$ is a subpresheaf of $\mathcal{G}$.

Exercise 15.50 (Local-image restrictions)**.** Show that a section locally in the image remains locally in the image after restriction.

Exercise 15.51 (Local-image gluing)**.** Prove that locally-in-the-image sections satisfy the sheaf gluing axiom.

Exercise 15.52 (Image stalk)**.** Prove $(\operatorname{im}\varphi)_x = \operatorname{im}\varphi_x$ directly from germs.

Exercise 15.53 (Quotient presheaf)**.** Verify that $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$ is a presheaf.

Exercise 15.54 (Quotient stalk)**.** Write down explicitly the map from the stalk of the quotient presheaf to $\mathcal{F}_x/\mathcal{F}'_x$ and prove it is an isomorphism.

Exercise 15.55 (Cokernel stalk)**.** Prove $(\operatorname{coker}\varphi)_x \cong \operatorname{coker}\varphi_x$.

Exercise 15.56 (Quotient kernel)**.** For $q : \mathcal{F} \to \mathcal{F}/\mathcal{F}'$, prove $\ker q = \mathcal{F}'$.

Exercise 15.57 (Cokernel as quotient)**.** Prove $\operatorname{coker}\varphi \cong \mathcal{G}/\operatorname{im}\varphi$.

Exercise 15.58 (Zero and identity)**.** Compute kernel, image, coimage, and cokernel for the zero morphism and the identity morphism.

Exercise 15.59 (Multiplication by n)**.** Compute these four constructions for $n : \underline{\mathbb{Z}} \to \underline{\mathbb{Z}}$.

Exercise 15.60 (Skyscraper map)**.** For a homomorphism $A \to B$, compute the kernel, image, and cokernel of the induced skyscraper-sheaf morphism at a closed point.

Exercise 15.61 (Direct sums)**.** For the projection $\mathcal{F} \oplus \mathcal{G} \to \mathcal{G}$, identify kernel and cokernel.

Exercise 15.62 (Diagonal)**.** Show that the cokernel of $\Delta : \mathcal{F} \to \mathcal{F} \oplus \mathcal{F}$ is isomorphic to $\mathcal{F}$.

Exercise 15.63 (Restriction to an open)**.** Prove that image sheaves and quotient sheaves commute with restriction to an open subset.

Exercise 15.64 (Local surjectivity)**.** Show that $\operatorname{im}\varphi = \mathcal{G}$ iff every $g \in \mathcal{G}(U)$ is locally liftable through φ.

Exercise 15.65 (Stalkwise surjectivity)**.** Show that $\operatorname{im}\varphi = \mathcal{G}$ iff every stalk map φ_x is surjective.

Exercise 15.66 (Coimage)**.** Construct the canonical map $\mathcal{F}/\ker\varphi \to \operatorname{im}\varphi$.

Exercise 15.67 (First isomorphism theorem)**.** Prove the map in the previous exercise is an isomorphism by taking stalks.

Exercise 15.68 (Continuous exponential)**.** Show that $f \mapsto e^{2\pi i f}$ from real-valued to circle-valued continuous-function sheaves is locally surjective.

Exercise 15.69 (No global argument)**.** Show that the identity $S^1 \to S^1$ has no continuous lift $S^1 \to \mathbb{R}$ through $t \mapsto e^{2\pi i t}$.

Exercise 15.70 (Derivative and local primitives)**.** Show that $d : \mathcal{C}^\infty \to \Omega^1$ on a smooth one-manifold has image sheaf all of Ω^1.

Exercise 15.71 (Ideal-sheaf quotient)**.** If $\mathcal{I} \subseteq \mathcal{O}$ is a sheaf of ideals, prove that $\mathcal{O}/\mathcal{I}$ is naturally a sheaf of rings and identify its stalks.

15.14 Problems

Problem 15.72 (Legacy AG1 Exercise 5: identify the image sheaf)**.** Let

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of sheaves of abelian groups on a topological space X. Begin with the presheaf image

$$\operatorname{Im}^{\mathrm{pre}}(f)(U) = \operatorname{im}\big(f_U : \mathcal{F}(U) \to \mathcal{G}(U)\big)$$

and let $\operatorname{im} f$ denote its sheafification. Prove that $\operatorname{im} f$ can be naturally identified with the subsheaf $\mathcal{H} \subseteq \mathcal{G}$ defined by

$$\mathcal{H}(U) = \left\{ s \in \mathcal{G}(U) : \begin{array}{l} \text{for every } p \in U \text{ there is a neighborhood } V \subseteq U \text{ of } p \\ \text{and } t \in \mathcal{F}(V) \text{ such that } s|_V = f_V(t) \end{array} \right\}.$$

Deduce that

$$(\operatorname{im} f)_p = \operatorname{im}(f_p)$$

for every $p \in X$.

Provenance. Legacy AG1 Exercise 5, merged with the preferred reorganized sheaf notes.

Problem 15.73 (Legacy AG2 Problem 8: quotient sheaves and short exact sequences)**.** Let $\mathcal{F}' \subseteq \mathcal{F}$ be a subsheaf of a sheaf of abelian groups.

1. Define $\mathcal{F}/\mathcal{F}'$ as the sheafification of the quotient presheaf $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$. Show that the natural map
$$q : \mathcal{F} \longrightarrow \mathcal{F}/\mathcal{F}'$$
has kernel $\mathcal{F}'$ and is surjective in the sheaf-theoretic sense, giving
$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \xrightarrow{q} \mathcal{F}/\mathcal{F}' \longrightarrow 0.$$

2. Conversely, suppose
$$0 \longrightarrow \mathcal{F}' \xrightarrow{i} \mathcal{F} \xrightarrow{p} \mathcal{F}'' \longrightarrow 0$$
is a short exact sequence of sheaves. Show that $\mathcal{F}'$ is isomorphic to a subsheaf of $\mathcal{F}$ and that $\mathcal{F}''$ is naturally isomorphic to the quotient of $\mathcal{F}$ by that subsheaf.

Provenance. Legacy AG2 Problem 8, preserving both directions.

Problem 15.74 (Kernel universal property)**.** Prove that the sectionwise kernel together with its inclusion is the categorical kernel in $\mathbf{Sh}(X)$.

Provenance. New synthesis; based on the preferred kernel-sheaf theory.

Problem 15.75 (Smallest subsheaf containing the presheaf image)**.** Show that $\operatorname{im}\varphi$ is the smallest subsheaf of $\mathcal{G}$ containing $\operatorname{im}(\varphi_U)$ for every open U.

Provenance. New synthesis; derived from the preferred image-sheaf source.

Problem 15.76 (A strict image-presheaf example)**.** For the exponential morphism $\mathcal{C}^0_{\mathbb{R}} \to \mathcal{C}^0_{S^1}$ on S^1, prove that the image sheaf is the whole target but the identity circle map is not in the image of global sections.

Provenance. New synthesis from the corpus quotient/image examples.

Problem 15.77 (A strict quotient-presheaf example)**.** Using the same exponential morphism, show that $\mathcal{C}^0_{\mathbb{R}}/\mathbb{Z} \cong \mathcal{C}^0_{S^1}$ as sheaves, while the sectionwise quotient over S^1 does not contain the degree-one map.

Provenance. New synthesis from AG2 quotient examples.

Problem 15.78 (Cokernel universal property)**.** Prove the universal property of $\operatorname{coker}\varphi$ directly from the objectwise cokernel presheaf and sheafification.

Provenance. New synthesis.

Problem 15.79 (Quotient universal property)**.** Let $\mathcal{F}' \subseteq \mathcal{F}$. Prove that morphisms $\mathcal{F}/\mathcal{F}' \to \mathcal{H}$ are naturally equivalent to morphisms $\mathcal{F} \to \mathcal{H}$ vanishing on $\mathcal{F}'$.

Provenance. New synthesis; companion to legacy AG2 Problem 8.

Problem 15.80 (Coimage equals image)**.** Prove $\mathcal{F}/\ker\varphi \xrightarrow{\sim} \operatorname{im}\varphi$ and identify the map on every stalk.

Provenance. New synthesis.

Problem 15.81 (Epimorphisms and local lifts)**.** Prove that φ is an epimorphism of sheaves of abelian groups iff $\operatorname{im}\varphi = \mathcal{G}$, iff every φ_x is surjective.

Provenance. New synthesis; VI/16 will integrate this into exactness.

Problem 15.82 (Monomorphisms and kernels)**.** Prove that φ is a monomorphism iff $\ker\varphi = 0$, iff every stalk map φ_x is injective.

Provenance. New synthesis.

Problem 15.83 (Restriction commutes with quotient)**.** Prove $(\mathcal{F}/\mathcal{F}')|_U \cong (\mathcal{F}|_U)/(\mathcal{F}'|_U)$ for every open $U \subseteq X$.

Provenance. New synthesis.

Problem 15.84 (Restriction commutes with image)**.** Prove $(\operatorname{im}\varphi)|_U \cong \operatorname{im}(\varphi|_U)$.

Provenance. New synthesis.

Problem 15.85 (Direct image and kernels)**.** For a continuous map $f : X \to Y$, prove $f_*(\ker\varphi) = \ker(f_*\varphi)$.

Provenance. New synthesis.

Problem 15.86 (Skyscraper quotients)**.** For $B \subseteq A$, prove $i_x(A)/i_x(B) \cong i_x(A/B)$ and compute all stalks.

Provenance. New synthesis.

Problem 15.87 (Ideal-sheaf quotient)**.** If $\mathcal{I} \subseteq \mathcal{O}$ is a sheaf of ideals, prove $\mathcal{O}/\mathcal{I}$ is a sheaf of rings and $(\mathcal{O}/\mathcal{I})_x \cong \mathcal{O}_x/\mathcal{I}_x$.

Provenance. New synthesis; previews VI/22.

Problem 15.88 (Derivative image on S^1)**.** Show that $d : \mathcal{C}^\infty \to \Omega^1$ has image sheaf all of Ω^1, but on global sections its image consists of one-forms with integral zero.

Provenance. New synthesis.

Problem 15.89 (Holomorphic logarithms)**.** For $\exp : \mathcal{O} \to \mathcal{O}^\times$ on $\mathbb{C}^\times$, prove local surjectivity and show the global section z has no holomorphic logarithm.

Provenance. New synthesis.

Problem 15.90 (A quotient descent datum)**.** Given an open cover $U = \bigcup U_i$, sections $s_i \in \mathcal{F}(U_i)$, and differences $s_i - s_j \in \mathcal{F}'(U_i \cap U_j)$, prove the classes $[s_i]$ glue to a section of $\mathcal{F}/\mathcal{F}'$.

Provenance. New synthesis.

Problem 15.91 (Compare two subsheaves by quotients)**.** If $\mathcal{F}' \subseteq \mathcal{F}'' \subseteq \mathcal{F}$, construct and prove the natural isomorphism $(\mathcal{F}/\mathcal{F}')/(\mathcal{F}''/\mathcal{F}') \cong \mathcal{F}/\mathcal{F}''$.

Provenance. New synthesis; useful preparation for filtrations and cohomology.

15.15 Challenges

Problem 15.92 (Challenge: Epimorphism but not sectionwise surjectivity)**.** Construct a fully explicit morphism of sheaves on S^1 that is an epimorphism but whose map on global sections is not surjective. Identify a global target section with no global lift and prove local lifts exist.

Problem 15.93 (Challenge: Coimage-image without invoking abelian categories)**.** Prove $\mathcal{F}/\ker\varphi \cong \operatorname{im}\varphi$ using only the definitions of quotient sheaf, image sheaf, germs, and the stalkwise isomorphism criterion.

Problem 15.94 (Challenge: Successive quotients)**.** For $\mathcal{A} \subseteq \mathcal{B} \subseteq \mathcal{F}$, establish the sheaf-theoretic third isomorphism theorem and formulate the corresponding universal property.

Problem 15.95 (Challenge: When is the quotient presheaf already a sheaf?)**.** Find useful sufficient conditions under which $U \mapsto \mathcal{F}(U)/\mathcal{F}'(U)$ is already a sheaf. Test your criteria on discrete spaces, split inclusions, and the exponential example.

Problem 15.96 (Challenge: Local algebra of a closed-subscheme prototype)**.** Let $X = \operatorname{Spec} A$ and let $I \subseteq A$. Assuming the structure-sheaf localization formula to be proved in VI/17, show that the stalks of $\mathcal{O}_X/I\mathcal{O}_X$ are $A_{\mathfrak{p}}/IA_{\mathfrak{p}}$ and predict the support of the quotient sheaf.

Summary

For a morphism of sheaves $\varphi : \mathcal{F} \to \mathcal{G}$, kernels are already local enough to be computed sectionwise, while images, quotients, and cokernels generally require sheafification:

$$(\ker\varphi)(U) = \ker\varphi_U, \qquad \operatorname{im}\varphi = (U \mapsto \operatorname{im}\varphi_U)^{\#},$$

$$\mathcal{F}/\mathcal{F}' = (U \mapsto \mathcal{F}(U)/\mathcal{F}'(U))^{\#}, \qquad \operatorname{coker}\varphi = (U \mapsto \operatorname{coker}\varphi_U)^{\#}.$$

At every point the constructions become ordinary algebra:

$$(\ker\varphi)_x = \ker\varphi_x, \quad (\operatorname{im}\varphi)_x = \operatorname{im}\varphi_x, \quad (\operatorname{coker}\varphi)_x = \operatorname{coker}\varphi_x, \quad (\mathcal{F}/\mathcal{F}')_x = \mathcal{F}_x/\mathcal{F}'_x.$$

AG1 Exercise 5 and AG2 Problem 8 are preserved here as VI.15.P1 and VI.15.P2. The updated Volume VI ledger records that migration explicitly so neither is lost when VI/16 develops the general theory of exact sequences.

15.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 15.97 (Kernel)**.** How is the kernel sheaf computed on opens?

Hint. Use sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Pointwise kernels. ☐

Exercise 15.98 (Image)**.** How is the image sheaf obtained?

Hint. Use sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Sheafify the presheaf image. ☐

Exercise 15.99 (Cokernel)**.** How is the cokernel sheaf obtained?

Hint. Use sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Sheafify the presheaf cokernel. ☐

Exercise 15.100 (Stalk kernel)**.** What is $(\ker\phi)_x$?

Hint. Use sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. $\ker(\phi_x)$. ☐

Exercise 15.101 (Stalk cokernel)**.** What is $(\operatorname{coker}\phi)_x$?

Hint. Use sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. $\operatorname{coker}(\phi_x)$. ☐

Proofs

Exercise 15.102 (Kernel sheaf)**.** Prove: Prove the presheaf kernel is a sheaf.

Hint. Work directly from sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Compatible local kernel sections glue in F, and their images glue to zero in G, so the global glue stays in the kernel. ☐

Exercise 15.103 (Stalk exactness)**.** Prove: Prove taking stalks commutes with kernels and cokernels for sheaves of modules.

Hint. Work directly from sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Stalks are filtered colimits, which are exact in modules. ☐

Exercise 15.104 (Image stalk)**.** Prove: Prove the stalk of the sheaf image equals the image of the stalk map.

Hint. Work directly from sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Sheafification preserves stalks, and the presheaf image germ condition is local. ☐

Exercise 15.105 (Quotient stalk)**.** Prove: Prove $(G/F)_x \cong G_x/F_x$ for a subsheaf of modules.

Hint. Work directly from sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. Use exactness of stalks on $0 \to F \to G \to G/F \to 0$. ☐

Counterexamples and hypothesis tests

Exercise 15.106 (Image presheaf)**.** Test the claim: the pointwise image presheaf is always a sheaf.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general. ☐

Exercise 15.107 (Cokernel pointwise)**.** Test the claim: sheaf cokernels are always computed by raw sectionwise quotients.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; sheafification can be required. ☐

Exercise 15.108 (Kernel sheafification)**.** Test the claim: kernels always require sheafification.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; pointwise kernels already satisfy the sheaf axioms. ☐

Applications and investigations

Exercise 15.109 (Local exactness)**.** Why are stalks efficient for kernel/image questions?

Hint. Translate the question into sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. They turn sheaf operations into ordinary module operations at each point. ☐

Exercise 15.110 (Quotient geometry)**.** What does a quotient sheaf encode?

Hint. Translate the question into sheaf kernels, images, cokernels, and stalkwise algebra. ☐

Solution. It measures local sections modulo a subsheaf, with gluing repaired sheaf-theoretically. ☐

Challenge problems

Exercise 15.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Kernels, Images, and Quotients of Sheaves in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Kernels, Images, and Quotients of Sheaves, translates it through sheaf kernels, images, cokernels, and stalkwise algebra, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 15.112 (Hypothesis challenge)**.** Choose one structural statement from Kernels, Images, and Quotients of Sheaves, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 16

Exact Sequences of Sheaves

Purpose and learning goals

The preceding chapters built the local algebra of sheaves: germs and stalks, sheafification, and the correct kernel, image, quotient, and cokernel constructions. Exact sequences are the language that assembles those constructions into a reusable calculus.

For sheaves of abelian groups, exactness is fundamentally local. A sequence

$$\cdots \longrightarrow \mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1} \longrightarrow \cdots$$

is exact precisely when every induced sequence of stalks is exact. This theorem is the principal legacy problem AG1 Exercise 6 and is preserved in full as VI.16.P1. The second principal corpus problem, AG2 Problem 7, proves that a morphism of sheaves is an isomorphism exactly when it is injective and surjective in the sheaf-theoretic sense; it is preserved as VI.16.P2. The preferred reorganized sheaf notes contain parallel treatments and are merged into the theory and these dossiers rather than counted as duplicate problems.

After this chapter, the reader should be able to:

- define exactness at a term using the image and kernel sheaves;
- prove and use the stalkwise exactness criterion;
- recognize monomorphisms and epimorphisms by their stalk maps;
- distinguish a sheaf epimorphism from surjectivity on sections over every open set;
- work with short exact sequences and quotient presentations;
- prove that restriction to an open subset preserves exactness;
- understand the left exactness of the section functor without prematurely developing cohomology;
- package failure of exactness into homology sheaves;
- use local lifting arguments in continuous, smooth, and holomorphic examples;
- read exactness diagrams stalkwise and perform basic diagram chases.

16.1 Exactness at a term

Let

$$\mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1}$$

be morphisms of sheaves of abelian groups.

Definition 16.1 (Exactness at a term)**.** The sequence is *exact at* $\mathcal{F}_i$ if

$$\operatorname{im}(d_{i-1}) = \ker(d_i)$$

as subsheaves of $\mathcal{F}_i$.

This equality automatically implies

$$d_i \circ d_{i-1} = 0.$$

The converse is weaker: $d_i d_{i-1} = 0$ gives only

$$\operatorname{im}(d_{i-1}) \subseteq \ker(d_i).$$

Exactness says that every local solution of $d_i(s) = 0$ is locally produced by the preceding map.

$$\boxed{\mathcal{F}_{i-1}} \xrightarrow{d_{i-1}} \boxed{\mathcal{F}_i} \xrightarrow{d_i} \boxed{\mathcal{F}_{i+1}}$$

$$\operatorname{im}(d_{i-1}) = \ker(d_i)$$

Figure 16.1: Exactness at the middle term is equality of two subsheaves.

Remark 16.2 (Local meaning of the image). Because $\operatorname{im}(d_{i-1})$ is the sheaf image, a section $s \in \mathcal{F}_i(U)$ belongs to it exactly when every $x \in U$ has a neighborhood $V \subseteq U$ and a section $t \in \mathcal{F}_{i-1}(V)$ with

$$d_{i-1}(t) = s|_V.$$

Thus exactness of sheaves means *local exactness of sections*, not necessarily exactness of every sequence of section groups.

16.2 The stalkwise exactness theorem

Theorem 16.3 (Stalkwise criterion for exactness)**.** *A sequence of sheaves of abelian groups is exact if and only if, for every $x \in X$, the induced sequence of stalks is exact.*

Proof. Suppose first that the sheaf sequence is exact at $\mathcal{F}_i$. Then

$$\operatorname{im}(d_{i-1}) = \ker(d_i).$$

Taking the stalk at x and using the identities established in VI/15,

$$(\operatorname{im} d_{i-1})_x = \operatorname{im}((d_{i-1})_x), \qquad (\ker d_i)_x = \ker((d_i)_x),$$

gives exactness at $(\mathcal{F}_i)_x$.

Conversely, suppose every stalk sequence is exact. Since $(d_i)_x(d_{i-1})_x = 0$ for every x, the morphism $d_i d_{i-1}$ has zero stalk map everywhere and hence is zero. Therefore $\operatorname{im}(d_{i-1}) \subseteq \ker(d_i)$.

Let $s \in \ker(d_i)(U)$ and $x \in U$. Exactness of the stalk sequence gives a germ $t_x \in (\mathcal{F}_{i-1})_x$ satisfying

$$(d_{i-1})_x(t_x) = s_x.$$

Choose a representative $t \in \mathcal{F}_{i-1}(V)$ on a neighborhood V of x. Equality of germs allows us to shrink V so that

$$d_{i-1}(t) = s|_V.$$

Hence s is locally in the image of d_{i-1} at every point, so $s \in (\operatorname{im} d_{i-1})(U)$. Thus the reverse inclusion holds and the sequence is exact. $\square$

$$\begin{tikzcd}
\mathcal{F}_{i-1} \arrow[r, "d_{i-1}"] \arrow[d, "(-)_x"'] & \mathcal{F}_i \arrow[r, "d_i"] \arrow[d, "(-)_x"'] & \mathcal{F}_{i+1} \arrow[d, "(-)_x"] \\
(\mathcal{F}_{i-1})_x \arrow[r, "(d_{i-1})_x"'] & (\mathcal{F}_i)_x \arrow[r, "(d_i)_x"'] & (\mathcal{F}_{i+1})_x
\end{tikzcd}$$

Figure 16.2: Exactness is detected after passing to every stalk.

Example 16.4 (Exactness checked on stalks)**.** A sequence $F' \to F \to F''$ of sheaves of modules is exact iff every stalk sequence $F'_x \to F_x \to F''_x$ is exact. Thus global sheaf exactness is a local algebra condition.

16.3 Monomorphisms, epimorphisms, and isomorphisms

For a morphism $f : \mathcal{F} \to \mathcal{G}$, the following conditions are equivalent:

$$\ker f = 0 \quad\iff\quad f_x : \mathcal{F}_x \to \mathcal{G}_x \text{ is injective for all } x$$

and these are also equivalent to injectivity of $f_U : \mathcal{F}(U) \to \mathcal{G}(U)$ for every open U.

Surjectivity is subtler.

Theorem 16.5 (Local lifting criterion for epimorphisms)**.** *For $f : \mathcal{F} \to \mathcal{G}$, the following are equivalent:*

1. $\operatorname{im} f = \mathcal{G}$*;*
2. $\operatorname{coker} f = 0$*;*
3. *every stalk map f_x is surjective;*
4. *every section $g \in \mathcal{G}(U)$ is locally liftable: each $x \in U$ has a neighborhood $V \subseteq U$ and $s \in \mathcal{F}(V)$ such that $f(s) = g|_V$.*

A morphism satisfying these conditions is called surjective as a morphism of sheaves, or an epimorphism in this abelian category. It need not be surjective on $\mathcal{F}(U) \to \mathcal{G}(U)$ for a large open set U.

Corollary 16.6 (Stalkwise isomorphism criterion)**.** *A morphism $f : \mathcal{F} \to \mathcal{G}$ is an isomorphism if and only if every f_x is an isomorphism. Equivalently, f is an isomorphism if and only if it is both injective and surjective as a morphism of sheaves.*

16.4 Short exact sequences

Definition 16.7 (Short exact sequence)**.** A sequence

$$0 \longrightarrow \mathcal{F}' \xrightarrow{i} \mathcal{F} \xrightarrow{q} \mathcal{F}'' \longrightarrow 0$$

is short exact if it is exact at each of the three nonzero terms.

$$\begin{tikzcd}
 & \boxed{g \in \mathcal{G}(U)} \arrow[dl] \arrow[dr] & \\
\boxed{\begin{matrix} s_1 \in \mathcal{F}(V_1) \\ f(s_1) = g|_{V_1} \end{matrix}} & & \boxed{\begin{matrix} s_2 \in \mathcal{F}(V_2) \\ f(s_2) = g|_{V_2} \end{matrix}}
\end{tikzcd}$$

$U = \bigcup_x V_x$: local lifts suffice for a sheaf epimorphism

Figure 16.3: Surjectivity of sheaves means local liftability.

Equivalently, i identifies $\mathcal{F}'$ with $\ker q$, the map q is locally surjective, and

$$\mathcal{F}'' \cong \mathcal{F}/\mathcal{F}'.$$

By the stalkwise criterion, short exactness is equivalent to exactness of

$$0 \longrightarrow \mathcal{F}'_x \longrightarrow \mathcal{F}_x \longrightarrow \mathcal{F}''_x \longrightarrow 0$$

for every x.

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}' \arrow[r, hook, "i"] & \mathcal{F} \arrow[r, two heads, "q"] & \mathcal{F}'' \arrow[r] & 0
\end{tikzcd}$$

$$\ker q = \operatorname{im} i \cong \mathcal{F}', \qquad \mathcal{F}'' \cong \mathcal{F}/\mathcal{F}'.$$

Figure 16.4: The structure of a short exact sequence.

Example 16.8 (Restriction sequence)**.** For a subsheaf $F' \subseteq F$, the sequence $0 \to F' \to F \to F/F' \to 0$ is exact as sheaves even though the map on sections over a fixed open need not be surjective.

16.5 Restriction preserves exactness

Let $j : U \hookrightarrow X$ be the inclusion of an open subset. Restriction of a sheaf to U does not change the stalks at points of U:

$$(\mathcal{F}|_U)_x \cong \mathcal{F}_x \qquad (x \in U).$$

Therefore restricting an exact sequence of sheaves on X yields an exact sequence on U. This is one of the most useful manifestations of locality: exactness can be checked after passing to a sufficiently small neighborhood of every point.

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}' \arrow[r] \arrow[d, "|_U"'] & \mathcal{F} \arrow[r] \arrow[d, "|_U"'] & \mathcal{F}'' \arrow[r] \arrow[d, "|_U"] & 0 \\
0 \arrow[r] & \mathcal{F}'|_U \arrow[r] & \mathcal{F}|_U \arrow[r] & \mathcal{F}''|_U \arrow[r] & 0
\end{tikzcd}$$

Figure 16.5: Restriction to an open subset preserves exactness.

16.6 Sections are left exact, but not generally exact

Let

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

be short exact. For every open $U \subseteq X$, the induced sequence

$$0 \longrightarrow \mathcal{F}'(U) \longrightarrow \mathcal{F}(U) \longrightarrow \mathcal{F}''(U)$$

is exact at the first two terms. The final map need not be surjective.

Theorem 16.9 (Left exactness of sections)**.** *For every open U, the functor $\Gamma(U, -)$ preserves kernels and therefore sends a short exact sequence of sheaves of abelian groups to a left exact sequence of abelian groups.*

The failure of the last map to be surjective measures a genuinely global obstruction. The systematic study of that failure belongs to VI/48; here we use it only as a warning and as a source of examples.

short exact sheaves
$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$

$\downarrow$

sections on U
$0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U)$

exact at the first two terms;
the final arrow need not be onto

Figure 16.6: Global sections are left exact, not generally exact.

Example 16.10 (Local surjectivity)**.** A sheaf morphism can be surjective while some section of the target over a large open has no global preimage: surjectivity only requires preimages locally around each point.

16.7 Homology sheaves

Suppose $d_i d_{i-1} = 0$. Define the ith homology sheaf of the complex by

$$\mathcal{H}_i := \ker(d_i)/\operatorname{im}(d_{i-1}).$$

Because quotients are sheaf quotients, this is again a sheaf.

Theorem 16.11 (Homology is computed on stalks)**.** *For every $x \in X$,*

$$(\mathcal{H}_i)_x \cong \frac{\ker((d_i)_x)}{\operatorname{im}((d_{i-1})_x)}.$$

Consequently, the complex is exact at $\mathcal{F}_i$ if and only if $\mathcal{H}_i = 0$.

$$\boxed{\ker d_i} \longrightarrow \boxed{\mathcal{H}_i = \ker d_i/\operatorname{im} d_{i-1}}$$

$$\mathcal{H}_i = 0 \iff \text{exact at } \mathcal{F}_i$$

Figure 16.7: The homology sheaf records failure of exactness.

16.8 Common pitfalls

1. Exactness of sheaves is equality of the *sheaf image* and the kernel sheaf.
2. A sheaf epimorphism need not be surjective on sections over every open set.
3. Exactness on stalks is equivalent to exactness of sheaves; exactness of global-section sequences is stronger at the right end.
4. The relation $d_i d_{i-1} = 0$ is necessary for exactness but is not sufficient.
5. If a section lies in the kernel, exactness gives local preimages; those preimages need not glue globally.
6. Restriction to an open subset preserves exactness, while direct image need not preserve surjectivity.
7. Do not consume AG2 Problem 9 here: the failure of right exactness of $\Gamma(U, -)$ is only previewed; the corpus problem is reserved for VI/48.

16.9 Worked examples

Example 16.12 (The zero sequence)**.** The sequence $0 \to \mathcal{F} \xrightarrow{\mathrm{id}} \mathcal{F} \to 0$ is exact. Every stalk is the ordinary exact sequence $0 \to \mathcal{F}_x \xrightarrow{\mathrm{id}} \mathcal{F}_x \to 0$.

Example 16.13 (A split short exact sequence)**.** For any $\mathcal{F}$ and $\mathcal{G}$,

$$0 \to \mathcal{F} \xrightarrow{s \mapsto (s,0)} \mathcal{F} \oplus \mathcal{G} \xrightarrow{(s,t) \mapsto t} \mathcal{G} \to 0$$

is short exact and even splits globally.

Example 16.14 (Multiplication by an integer)**.** For $n \neq 0$,

$$0 \to \underline{\mathbb{Z}} \xrightarrow{n} \underline{\mathbb{Z}} \to \underline{\mathbb{Z}/n\mathbb{Z}} \to 0$$

is exact. This follows immediately from the stalk sequence $0 \to \mathbb{Z} \xrightarrow{n} \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z} \to 0$.

Example 16.15 (A skyscraper short exact sequence)**.** If $0 \to A' \to A \to A'' \to 0$ is exact and i_x denotes a skyscraper sheaf at a closed point, then $0 \to i_x(A') \to i_x(A) \to i_x(A'') \to 0$ is exact because the only nonzero stalk is at x.

Example 16.16 (Exactness at the middle term)**.** If $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ is exact at $\mathcal{G}$, then a section $s \in \mathcal{G}(U)$ with $g(s) = 0$ need only be *locally* of the form $f(t)$. Exactness does not promise one global $t \in \mathcal{F}(U)$.

Example 16.17 (Kernel detection)**.** A morphism $f : \mathcal{F} \to \mathcal{G}$ is injective if and only if f_x is injective for every x. If $f(s) = 0$, all germs s_x vanish, hence $s = 0$ by the sheaf axiom.

Example 16.18 (Local surjectivity)**.** A morphism $f : \mathcal{F} \to \mathcal{G}$ is surjective as a sheaf morphism exactly when every section of $\mathcal{G}$ admits preimages on a neighborhood of every point. This is equivalent to surjectivity of all stalk maps.

Example 16.19 (Restriction of a short exact sequence)**.** If $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is short exact on X and $U \subseteq X$ is open, then $0 \to \mathcal{F}'|_U \to \mathcal{F}|_U \to \mathcal{F}''|_U \to 0$ is short exact on U.

Example 16.20 (Finite discrete space)**.** On a finite discrete space, a sheaf is simply a family of stalk groups. A sequence of sheaves is exact precisely when the corresponding sequence is exact at each point, and here sectionwise exactness over every open also follows componentwise.

Example 16.21 (Continuous exponential)**.** The sequence

$$0 \to \underline{\mathbb{Z}} \to \mathcal{C}^0_{\mathbb{R}} \xrightarrow{f \mapsto e^{2\pi i f}} \mathcal{C}^0_{S^1} \to 0$$

is exact as a sequence of sheaves: every circle-valued function has local continuous arguments.

Example 16.22 (Derivative on an interval)**.** On an open interval $I \subseteq \mathbb{R}$, the map $d : \mathcal{C}^\infty_I \to \Omega^1_I$ is surjective on sections because every smooth one-form has a global primitive on an interval. Its kernel is the locally constant sheaf $\underline{\mathbb{R}}$.

Example 16.23 (A complex with zero homology sheaf)**.** If $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ is exact at $\mathcal{G}$, then $\ker g / \operatorname{im} f = 0$. Conversely, the vanishing of this homology sheaf is exactly the statement of exactness at $\mathcal{G}$.

Example 16.24 (Detecting nonexactness at one point)**.** If for some x the sequence $\mathcal{F}_x \to \mathcal{G}_x \to \mathcal{H}_x$ has a kernel element not in the preceding image, then the sheaf sequence is not exact. One bad stalk is enough.

Example 16.25 (Exactness on a neighborhood)**.** Suppose every $x \in X$ has an open neighborhood U_x on which a given sequence restricts to an exact sequence. Then the original sequence is exact, because each stalk can be computed inside one such neighborhood.

Example 16.26 (Inclusion and quotient)**.** For a subsheaf $\mathcal{F}' \subseteq \mathcal{F}$, the quotient sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}/\mathcal{F}' \to 0$ is exact. This is the basic source of short exact sequences.

Example 16.27 (Composition vanishes)**.** In any exact pair $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$, one has $gf = 0$ because $\operatorname{im} f \subseteq \ker g$.

Example 16.28 (Exactness and supports)**.** For a complex with homology sheaf $\mathcal{H}_i$, exactness holds on an open set U exactly when $\mathcal{H}_i|_U = 0$. Thus $\operatorname{Supp} \mathcal{H}_i$ records where exactness fails.

Example 16.29 (Locally constant sheaves)**.** An exact sequence $0 \to A' \to A \to A'' \to 0$ of abelian groups induces a short exact sequence of locally constant sheaves on a locally connected space. Every stalk is the original group sequence.

Example 16.30 (Left exactness of sections)**.** From $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, the sequence $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U)$ is exact. A kernel section in $\mathcal{F}(U)$ is globally a section of the kernel subsheaf $\mathcal{F}'$.

Example 16.31 (A sheaf epimorphism with no global lift)**.** For the continuous exponential on S^1, the identity map $z \mapsto z$ is a global section of $\mathcal{C}^0_{S^1}(S^1)$ with no global real-valued argument. The map is nevertheless a sheaf epimorphism because local arguments exist.

16.10 Advanced worked examples

Example 16.32 (Advanced: Holomorphic exponential sequence)**.** On a complex manifold,

$$0 \to 2\pi i\,\underline{\mathbb{Z}} \to \mathcal{O} \xrightarrow{\exp} \mathcal{O}^{\times} \to 1$$

is exact: a nowhere-zero holomorphic function has a local logarithm. On $\mathbb{C}^{\times}$, the function z has no global holomorphic logarithm, so global surjectivity fails.

Example 16.33 (Advanced: The de Rham complex)**.** On a smooth manifold, the Poincare lemma says that

$$0 \to \underline{\mathbb{R}} \to \mathcal{C}^{\infty} \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \xrightarrow{d} \cdots$$

is locally exact. Hence it is an exact complex of sheaves in positive degrees, although global closed forms need not be globally exact.

Example 16.34 (Advanced: A closed one-form on the circle)**.** The form $d\theta$ on S^1 is locally df but has no global smooth primitive. Thus $d : \mathcal{C}^{\infty} \to \Omega^1$ is a sheaf epimorphism on a one-manifold even when the map on global sections is not onto.

Example 16.35 (Advanced: Direct image is only left exact)**.** For $f : X \to Y$, $(f_*\mathcal{F})(V) = \mathcal{F}(f^{-1}V)$. Because sections are left exact, f_* preserves kernels and injections. It need not preserve a sheaf epimorphism: taking Y to be a point turns f_* into global sections, where the exponential example gives a counterexample.

Example 16.36 (Advanced: Inverse image preserves exactness)**.** For sheaves of abelian groups and a continuous map $f : X \to Y$, the stalk identity

$$(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$$

shows immediately that f^{-1} carries exact sequences on Y to exact sequences on X.

Example 16.37 (Advanced: Sheafification is exact for abelian groups)**.** An objectwise exact sequence of presheaves of abelian groups remains exact after sheafification. Indeed, stalks are filtered colimits, filtered colimits of abelian groups are exact, and sheafification does not change stalks.

Example 16.38 (Advanced: Direct sums preserve exactness)**.** If for every λ the sequence $0 \to \mathcal{F}'_{\lambda} \to \mathcal{F}_{\lambda} \to \mathcal{F}''_{\lambda} \to 0$ is exact, then the direct-sum sequence is exact. At each stalk this is the direct sum of exact sequences of abelian groups.

Example 16.39 (Advanced: Homology sheaf and the failure locus)**.** For a complex $\mathcal{F}^{\bullet}$, the support of $\mathcal{H}^i(\mathcal{F}^{\bullet})$ is exactly the set of points at which the stalk complex fails to be exact in degree i. Exactness is therefore encoded geometrically by the vanishing locus of a sheaf.

Example 16.40 (Advanced: Exactness on the Sierpinski space)**.** On the Sierpinski space $\{\eta, x\}$ with opens $\varnothing, \{x\}, \{\eta, x\}$, a sheaf can be encoded by a restriction map $A \to B$. A sequence of such sheaves is exact exactly when the induced sequences at the two stalks are exact, making the stalkwise theorem completely explicit.

Example 16.41 (Advanced: A locally split quotient)**.** If a short exact sequence admits local splittings $s_i : \mathcal{F}''|_{U_i} \to \mathcal{F}|_{U_i}$, then locally $\mathcal{F}|_{U_i} \cong \mathcal{F}'|_{U_i} \oplus \mathcal{F}''|_{U_i}$. The splittings need not agree on overlaps, so local splitting is weaker than global splitting.

Example 16.42 (Advanced: An ideal-sheaf exact sequence - preview)**.** For a sheaf of ideals $\mathcal{I} \subseteq \mathcal{O}_X$ one has

$$0 \to \mathcal{I} \to \mathcal{O}_X \to \mathcal{O}_X/\mathcal{I} \to 0.$$

At a point x this becomes the ordinary exact quotient sequence $0 \to \mathcal{I}_x \to \mathcal{O}_{X,x} \to \mathcal{O}_{X,x}/\mathcal{I}_x \to 0$.

Example 16.43 (Advanced: Exactness can be tested after shrinking)**.** To prove exactness at $\mathcal{G}$, it is enough to show: whenever $g \in \mathcal{G}(U)$ maps to zero and $x \in U$, there is a neighborhood V of x and a preceding section mapping to $g|_V$. No global lifting argument is required.

Example 16.44 (Advanced: Five-lemma arguments become stalkwise)**.** Given a commutative diagram with exact rows, any statement such as the short five lemma can be proved on each stalk using ordinary group theory and then promoted back to a sheaf isomorphism by the stalkwise isomorphism criterion.

Example 16.45 (Advanced: Global lifting obstruction as a cocycle preview)**.** Given local lifts s_i of a quotient section, the differences $s_i - s_j$ lie in the kernel sheaf on overlaps. Their failure to be removable by changing the lifts is the prototype of a cohomological obstruction. VI/48 develops this systematically.

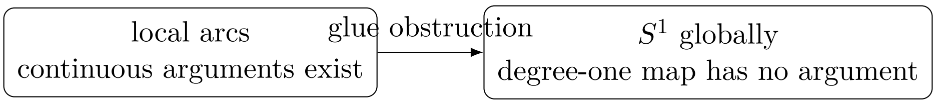


Figure 16.8: The exponential map is locally surjective but need not be globally surjective.

$$\underline{\mathbb{R}} \longrightarrow \mathcal{C}^\infty \xrightarrow{d} \Omega^1 \xrightarrow{d} \Omega^2 \longrightarrow \cdots$$

Poincare lemma: exact after shrinking to a coordinate ball.

Figure 16.9: The de Rham complex is a model of local exactness.

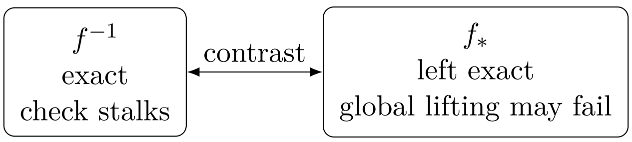


Figure 16.10: Inverse and direct image have different exactness behavior.

16.11 Exercises

Exercise 16.46 (Exactness at a term)**.** Let $\mathcal{F} \xrightarrow{f} \mathcal{G} \xrightarrow{g} \mathcal{H}$ be a sequence. Rewrite exactness at $\mathcal{G}$ as a local lifting statement for sections in $\ker g$.

Exercise 16.47 (Composition zero)**.** Prove that exactness at $\mathcal{G}$ implies $g \circ f = 0$.

Exercise 16.48 (Stalkwise forward direction)**.** Assuming $\operatorname{im} f = \ker g$, prove $\operatorname{im} f_x = \ker g_x$ for every x.

Exercise 16.49 (Stalkwise converse)**.** Assume $\operatorname{im} f_x = \ker g_x$ for every x. Prove $\operatorname{im} f = \ker g$.

Exercise 16.50 (Injectivity on stalks)**.** Show that $f : \mathcal{F} \to \mathcal{G}$ is injective iff every stalk map f_x is injective.

Exercise 16.51 (Injectivity on sections)**.** Show that a sheaf morphism is injective iff f_U is injective for every open U.

Exercise 16.52 (Epimorphism criterion)**.** Prove that f is surjective as a sheaf morphism iff every f_x is surjective.

Exercise 16.53 (Local lifting criterion)**.** Prove that stalkwise surjectivity is equivalent to local liftability of every target section.

Exercise 16.54 (Isomorphism criterion)**.** Prove that f is an isomorphism iff every f_x is an isomorphism.

Exercise 16.55 (Short exact stalks)**.** Show that a sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ is short exact iff its stalk sequence is short exact for every point.

Exercise 16.56 (Quotient recognition)**.** In a short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, prove $\mathcal{F}'' \cong \mathcal{F}/\mathcal{F}'$.

Exercise 16.57 (Restriction)**.** Prove directly from stalks that restriction to an open subset preserves exactness.

Exercise 16.58 (Local exactness)**.** Show that if an open cover $\{U_i\}$ has exact restricted sequences on every U_i, then the original sequence is exact.

Exercise 16.59 (Left exact sections)**.** From a short exact sequence prove $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U)$ is exact.

Exercise 16.60 (Failure at the right end)**.** For the continuous exponential sequence on S^1, exhibit a section of the target with no global lift.

Exercise 16.61 (Continuous exponential kernel)**.** Show that the kernel of $f \mapsto e^{2\pi i f}$ is the locally constant integer-valued sheaf.

Exercise 16.62 (Holomorphic logarithms)**.** Show locally that $\exp : \mathcal{O} \to \mathcal{O}^\times$ is surjective on stalks.

Exercise 16.63 (Derivative sheaf)**.** On a smooth one-manifold prove $0 \to \underline{\mathbb{R}} \to \mathcal{C}^\infty \xrightarrow{d} \Omega^1 \to 0$ is exact.

Exercise 16.64 (Homology stalk)**.** For a complex, prove

$$(\ker d_i / \operatorname{im} d_{i-1})_x \cong \ker((d_i)_x)/\operatorname{im}((d_{i-1})_x).$$

Exercise 16.65 (Vanishing homology)**.** Prove that a complex is exact at $\mathcal{F}_i$ iff its ith homology sheaf is zero.

Exercise 16.66 (Direct sums)**.** Show that a direct sum of short exact sequences of sheaves of abelian groups is short exact.

Exercise 16.67 (Skyscraper exactness)**.** Given an exact sequence of abelian groups, prove the induced skyscraper sequence at a closed point is exact.

Exercise 16.68 (Locally constant exactness)**.** Given a short exact sequence of abelian groups, show the associated locally constant sheaves form a short exact sequence.

Exercise 16.69 (Direct image left exactness)**.** For $f : X \to Y$, prove f_* preserves kernels and therefore is left exact.

Exercise 16.70 (Inverse image exactness)**.** Using $(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$, prove that f^{-1} preserves exact sequences of sheaves of abelian groups.

Exercise 16.71 (Sheafification exactness)**.** Show that sheafification carries an objectwise exact sequence of presheaves of abelian groups to an exact sequence of sheaves.

16.12 Problems

Problem 16.72 (Legacy AG1 Exercise 6: exactness is stalkwise)**.** Show that a sequence of sheaves

$$\cdots \to \mathcal{F}_{i-1} \xrightarrow{d_{i-1}} \mathcal{F}_i \xrightarrow{d_i} \mathcal{F}_{i+1} \to \cdots$$

is exact if and only if, for every $p \in X$, the corresponding stalk sequence

$$\cdots \to (\mathcal{F}_{i-1})_p \xrightarrow{(d_{i-1})_p} (\mathcal{F}_i)_p \xrightarrow{(d_i)_p} (\mathcal{F}_{i+1})_p \to \cdots$$

is exact.

Provenance. **Legacy INCLUDE/MERGE.** AG1 Exercise 6 is retained at full theorem-and-proof depth. The preferred reorganized sheaf notes contain the same stalkwise criterion and are merged into this dossier and the chapter theory, not duplicated as a second canonical problem.

Problem 16.73 (Legacy AG2 Problem 7: injective and surjective implies isomorphism)**.** Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves of abelian groups. Prove that φ is an isomorphism if and only if it is injective and surjective in the sheaf-theoretic sense. In the nontrivial direction, construct the inverse on sections by lifting target germs locally, prove the local lifts agree on overlaps using injectivity, glue them, and verify compatibility with restrictions.

Provenance. **Legacy INCLUDE/MERGE.** AG2 Problem 7 is preserved with its explicit germ-lifting and gluing proof. The later theorem-form repetition in AG2 is treated as a duplicate theorem statement and merged here rather than counted separately.

Problem 16.74 (Local lifting criterion for a sheaf epimorphism)**.** Let $f : \mathcal{F} \to \mathcal{G}$. Prove that the following are equivalent: f is a sheaf epimorphism; every f_x is surjective; and every target section is locally liftable.

Provenance. New synthesis; built from the preferred exactness notes and VI/15 local-image theory.

Problem 16.75 (The continuous exponential short exact sequence)**.** On a topological space X, analyze the morphism $\exp(2\pi i-) : \mathcal{C}^0_{\mathbb{R}} \to \mathcal{C}^0_{S^1}$. Prove the resulting sequence with kernel $\underline{\mathbb{Z}}$ is exact as sheaves and explain the failure of global surjectivity on S^1.

Provenance. New synthesis from preferred corpus examples; no separate numbered legacy problem.

Problem 16.76 (The holomorphic exponential sequence)**.** For a complex manifold X, prove the local exactness of $0 \to 2\pi i\underline{\mathbb{Z}} \to \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \to 1$, and show why $z \in \mathcal{O}^\times(\mathbb{C}^\times)$ has no global holomorphic logarithm.

Provenance. New synthesis from preferred corpus examples.

Problem 16.77 (Global sections are left exact but not right exact)**.** Given a short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$, prove $0 \to \mathcal{F}'(U) \to \mathcal{F}(U) \to \mathcal{F}''(U)$ is exact. Then give a concrete example where the last map is not surjective.

Provenance. New synthesis/bridge; does not consume AG2 Problem 9.

Problem 16.78 (Direct image is left exact)**.** For a continuous map $f : X \to Y$, prove that f_* preserves kernels and is left exact. Give a counterexample to preservation of epimorphisms.

Provenance. New advanced synthesis.

Problem 16.79 (Inverse image preserves exactness)**.** Let $f : X \to Y$. Using the stalk identity $(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$, prove that inverse image preserves exact sequences of sheaves of abelian groups.

Provenance. New advanced synthesis.

Problem 16.80 (Exactness is preserved by restriction)**.** Prove that if a sequence is exact on X, then its restriction to every open U is exact. Prove conversely that exactness can be checked on an open cover.

Provenance. New synthesis.

Problem 16.81 (Homology sheaves detect exactness)**.** For a complex of sheaves define $\mathcal{H}_i = \ker d_i / \operatorname{im} d_{i-1}$. Prove $(\mathcal{H}_i)_x$ is the ordinary homology of the stalk complex and deduce that exactness at degree i is equivalent to $\mathcal{H}_i = 0$.

Provenance. New synthesis.

Problem 16.82 (Support of the failure of exactness)**.** Let $\mathcal{H}_i$ be a homology sheaf. Prove that the complex is exact on $X \setminus \operatorname{Supp} \mathcal{H}_i$, and that every point of $\operatorname{Supp} \mathcal{H}_i$ is a point where the stalk complex fails to be exact.

Provenance. New synthesis.

Problem 16.83 (Exactness of sheafification)**.** Let $0 \to \mathcal{P}' \to \mathcal{P} \to \mathcal{P}'' \to 0$ be objectwise exact presheaves of abelian groups. Prove their sheafifications form a short exact sequence.

Provenance. New advanced synthesis.

Problem 16.84 (Split short exact sequences)**.** Prove that if $0 \to \mathcal{F}' \xrightarrow{i} \mathcal{F} \xrightarrow{q} \mathcal{F}'' \to 0$ admits a morphism $s : \mathcal{F}'' \to \mathcal{F}$ with $qs = \mathrm{id}$, then $\mathcal{F} \cong \mathcal{F}' \oplus \mathcal{F}''$.

Provenance. New synthesis.

Problem 16.85 (Direct sums of exact sequences)**.** Prove that arbitrary direct sums of short exact sequences of sheaves of abelian groups are short exact.

Provenance. New synthesis.

Problem 16.86 (Skyscraper exact sequences)**.** Let i_x be the skyscraper functor at a closed point x. Show that an exact sequence of abelian groups gives an exact sequence of skyscraper sheaves and conversely the exactness at x recovers the group sequence.

Provenance. New synthesis.

Problem 16.87 (Locally constant exact sequences)**.** Starting from $0 \to A' \to A \to A'' \to 0$, prove the associated locally constant sheaves form a short exact sequence on a locally connected space.

Provenance. New synthesis.

Problem 16.88 (A point-evaluation short exact sequence)**.** For a closed point x of a reasonable space, let $\mathcal{I}_x \subseteq \mathcal{C}^0_X$ be the sheaf of continuous functions whose germ vanishes at x. Prove $0 \to \mathcal{I}_x \to \mathcal{C}^0_X \to i_x(\mathbb{R}) \to 0$ is exact.

Provenance. New synthesis.

Problem 16.89 (Exactness on a finite Sierpinski space)**.** Describe sheaves of abelian groups on the Sierpinski space by a restriction map $A \to B$. Translate exactness of a sequence of such sheaves into exactness of the two associated stalk sequences.

Provenance. New synthesis.

Problem 16.90 (Local splitting versus global splitting)**.** Suppose a short exact sequence admits splittings after restricting to an open cover. Prove that it is locally a direct sum. Show that a global splitting exists exactly when the local splittings can be modified to agree on overlaps.

Provenance. New synthesis.

Problem 16.91 (Local lifts and the global obstruction)**.** Let $0 \to \mathcal{F}' \to \mathcal{F} \xrightarrow{q} \mathcal{F}'' \to 0$ be short exact and $t \in \mathcal{F}''(U)$. Choose local lifts $s_i \in \mathcal{F}(U_i)$. Prove $s_i - s_j$ lies in $\mathcal{F}'(U_i \cap U_j)$ and derive the precise condition for the local lifts to be adjustable to one global lift.

Provenance. New synthesis/bridge; deliberately does not consume the later cohomology problem.

16.13 Challenges

Problem 16.92 (Challenge: Short five lemma for sheaves)**.** Consider a commutative diagram of short exact sequences

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}' \arrow[r] \arrow[d, "a"] & \mathcal{F} \arrow[r] \arrow[d, "b"] & \mathcal{F}'' \arrow[r] \arrow[d, "c"] & 0 \\
0 \arrow[r] & \mathcal{G}' \arrow[r] & \mathcal{G} \arrow[r] & \mathcal{G}'' \arrow[r] & 0.
\end{tikzcd}$$

Prove that if a and c are isomorphisms, then b is an isomorphism.

Hint. Take stalks and apply the ordinary short five lemma, then use stalkwise detection of isomorphisms. Test exactness on stalks; for a connecting or lifting question, identify the local preimages first and then measure the obstruction to gluing them. ☐

Problem 16.93 (Challenge: Snake lemma for sheaves)**.** Formulate and prove the snake lemma for a commutative diagram of sheaves with exact rows, constructing the connecting morphism from the kernel on the right to the cokernel on the left.

Hint. Apply the ordinary snake lemma to every stalk. Then explain why the stalkwise connecting maps are induced by a unique sheaf morphism; alternatively construct the connecting map locally by lifting and chasing. Test exactness on stalks; for a connecting or lifting question, identify the local preimages first and then measure the obstruction to gluing them. ☐

Problem 16.94 (Challenge: Exactness of the de Rham complex)**.** On a smooth manifold, prove the sheaf de Rham complex is exact in positive degrees using the Poincare lemma, and identify its degree-zero kernel.

Hint. Use local contractible coordinate balls and the stalkwise criterion. The degree-zero kernel consists of locally constant functions. Test exactness on stalks; for a connecting or lifting question, identify the local preimages first and then measure the obstruction to gluing them. ☐

Problem 16.95 (Challenge: Inverse and direct image contrast)**.** Prove that inverse image is exact for sheaves of abelian groups while direct image is only left exact in general. Produce an explicit direct-image failure of surjectivity.

Hint. Use stalks for inverse image. For direct image use the description on opens and the exponential sequence with the target space a point. Test exactness on stalks; for a connecting or lifting question, identify the local preimages first and then measure the obstruction to gluing them. ☐

Problem 16.96 (Challenge: From local lifts to the connecting obstruction)**.** For a short exact sequence, organize the overlap differences of local lifts into a Cech-style cocycle and prove that changing local lifts changes it by a coboundary-style term. Explain why vanishing of the resulting class would be equivalent to existence of a global lift once Cech cohomology is available.

Hint. This is a preview only: verify the cocycle identity on triple overlaps and the transformation law under changing lifts. Do not assume the later cohomology theory. Test exactness on stalks; for a connecting or lifting question, identify the local preimages first and then measure the obstruction to gluing them. ☐

Summary

The central theorem of this chapter is

$$\boxed{\text{a sequence of sheaves is exact} \iff \text{every sequence of stalks is exact.}}$$

It converts sheaf-theoretic exactness into ordinary exactness of abelian groups, while the local-image construction explains why a surjective sheaf morphism can fail to be surjective on global sections. Short exact sequences are therefore locally ordinary short exact sequences, restrictions preserve exactness, and homology sheaves detect the precise locus of failure of exactness.

The Volume VI ledger records AG1 Exercise 6 as VI.16.P1 and AG2 Problem 7 as VI.16.P2. Their parallel variants in the preferred notes and repeated theorem statements are merged rather than duplicated. AG2 Problem 9 remains reserved for VI/48.

16.14 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 16.97 (Stalk test)**.** How do you test exactness of a sheaf sequence?

Hint. Use exactness of sheaves and stalkwise tests. ☐

Solution. Check exactness at every stalk. ☐

Exercise 16.98 (Monomorphism)**.** When is a sheaf map injective?

Hint. Use exactness of sheaves and stalkwise tests. ☐

Solution. When every stalk map is injective. ☐

Exercise 16.99 (Epimorphism)**.** When is a sheaf map surjective?

Hint. Use exactness of sheaves and stalkwise tests. ☐

Solution. When every stalk map is surjective, equivalently target sections lift locally. ☐

Exercise 16.100 (Quotient)**.** What sequence accompanies a subsheaf?

Hint. Use exactness of sheaves and stalkwise tests. ☐

Solution. $0 \to F' \to F \to F/F' \to 0$. ☐

Exercise 16.101 (Sections caveat)**.** Does sheaf surjectivity imply surjectivity on sections over every open?

Hint. Use exactness of sheaves and stalkwise tests. ☐

Solution. No. ☐

Proofs

Exercise 16.102 (Stalk exactness)**.** Prove: Prove exactness of sheaves of modules is equivalent to exactness on all stalks.

Hint. Work directly from exactness of sheaves and stalkwise tests. ☐

Solution. Kernel and image sheaves have stalks equal to kernel and image of the stalk maps. ☐

Exercise 16.103 (Local surjectivity)**.** Prove: Prove a surjective sheaf morphism is locally surjective on sections.

Hint. Work directly from exactness of sheaves and stalkwise tests. ☐

Solution. A germ preimage exists at each point; choose a representative on a neighborhood. ☐

Exercise 16.104 (Injectivity sections)**.** Prove: Prove a sheaf monomorphism is injective on sections over every open.

Hint. Work directly from exactness of sheaves and stalkwise tests. ☐

Solution. If two sections have the same image, their difference has zero germ everywhere, hence is zero by sheaf locality. ☐

Exercise 16.105 (Quotient exact)**.** Prove: Prove $0 \to F' \to F \to F/F' \to 0$ is exact.

Hint. Work directly from exactness of sheaves and stalkwise tests. ☐

Solution. Check on stalks, where it becomes the ordinary exact quotient sequence of modules. ☐

Counterexamples and hypothesis tests

Exercise 16.106 (Sectionwise exact)**.** Test the claim: applying global sections to a short exact sequence of sheaves is always exact on the right.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; global sections is left exact but need not preserve surjectivity. ☐

Exercise 16.107 (Surjective sections)**.** Test the claim: a surjective sheaf map is surjective on every open's sections.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 16.108 (Local irrelevant)**.** Test the claim: sheaf exactness is fundamentally global and cannot be checked locally.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is stalkwise. ☐

Applications and investigations

Exercise 16.109 (Cohomology preview)**.** Why does failure of global-section surjectivity matter?

Hint. Translate the question into exactness of sheaves and stalkwise tests. ☐

Solution. It is one source of higher cohomological obstruction. ☐

Exercise 16.110 (Local algebra)**.** Why are exact sheaf sequences manageable?

Hint. Translate the question into exactness of sheaves and stalkwise tests. ☐

Solution. They reduce pointwise to exact sequences of local modules. ☐

Challenge problems

Exercise 16.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Exact Sequences of Sheaves in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Exact Sequences of Sheaves, translates it through exactness of sheaves and stalkwise tests, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 16.112 (Hypothesis challenge)**.** Choose one structural statement from Exact Sequences of Sheaves, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 17

The Structure Sheaf

Purpose and learning goals

A spectrum is not only a topological space of prime ideals. Its geometry is carried by a sheaf of rings that records which algebraic functions are available on each open region. For an affine spectrum $X = \operatorname{Spec} A$, that sheaf is the *structure sheaf* $\mathcal{O}_X$. On a principal open $D(f)$ it has the fundamental form

$$\mathcal{O}_X(D(f)) \cong A_f,$$

and at a point $\mathfrak{p}$ its stalk is

$$\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}.$$

These two formulas are the bridge from the sheaf theory of VI/12–VI/16 to affine schemes in VI/18.

The corpus audit for this chapter is deliberately conservative. The explicit VI/17 source stream in `theory-of-algebraic-geometry-3.tex` contains structure-sheaf theory and regular-function examples, but no numbered Problem whose unique mathematical center is the structure sheaf alone. Four numbered Problems in the same legacy file are tracked rather than absorbed by the fallback rule: Problem 11 is reserved for VI/18, Problem 12 for VI/19, Problem 13 for VI/21, and Problem 14 for VI/20. The structure-sheaf examples in the commutative-algebra corpus are merged into this chapter as examples, not promoted to false legacy Problems.

After this chapter, the reader should be able to:

- define $\mathcal{O}_{\operatorname{Spec} A}$ by locally representable fractions;
- prove that the definition gives a sheaf of rings;
- compute sections on $D(f)$ as A_f;
- identify the stalk at $\mathfrak{p}$ with $A_{\mathfrak{p}}$ and its residue field with $\operatorname{Frac}(A/\mathfrak{p})$;
- interpret restriction maps as further localization;
- evaluate regular functions in the varying residue fields $\kappa(\mathfrak{p})$;
- distinguish topology from scheme-theoretic function data, especially in nonreduced examples;
- compute structure sheaves on $\operatorname{Spec} \mathbb{Z}$, polynomial spectra, product rings, and reducible rings;
- use the sheaf equalizer condition to compute sections on unions of principal opens;

- recognize exactly what remains for the definition of an affine scheme in VI/18.

$$\text{Presheaves VI/12} \longrightarrow \text{Sheaves and stalks VI/13–VI/16} \longrightarrow \text{Structure sheaf } \mathcal{O}_{\operatorname{Spec} A} \longrightarrow \text{Affine schemes VI/18}$$

17.1 From a spectrum to functions

Let A be a commutative ring and let

$$X = \operatorname{Spec} A.$$

A point of X is a prime ideal $\mathfrak{p}$. The ring A supplies global algebraic expressions, but an open subset may admit expressions with denominators that are invertible there. This is why localization is the correct local algebra.

For $f \in A$ the principal open is

$$D(f) = \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}\}.$$

On $D(f)$ the element f is nowhere contained in a prime, so it is natural to allow powers of f in denominators. The expected ring of functions is therefore A_f.

$$A \xrightarrow{\text{spectrum}} X = \operatorname{Spec} A \xrightarrow{\text{open region}} D(f) \xrightarrow{\text{functions}} A_f$$

17.2 Definition by locally representable fractions

For each $\mathfrak{p} \in X$, let $A_{\mathfrak{p}}$ be the localization of A at $A \setminus \mathfrak{p}$. For an open subset $U \subseteq X$, define $\mathcal{O}_X(U)$ to be the set of families

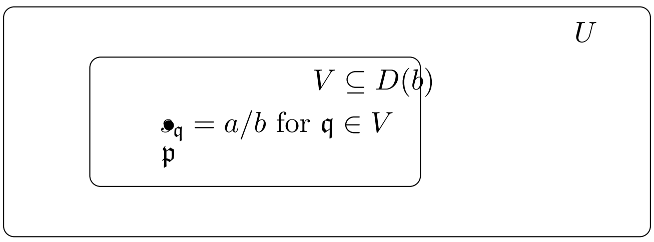
$$s = (s_{\mathfrak{p}})_{\mathfrak{p} \in U}, \qquad s_{\mathfrak{p}} \in A_{\mathfrak{p}},$$

that are *locally represented by one fraction from A*: for every $\mathfrak{p} \in U$ there exist an open neighborhood $V \subseteq U$ of $\mathfrak{p}$ and elements $a, b \in A$ such that

$$V \subseteq D(b) \quad \text{and} \quad s_{\mathfrak{q}} = \frac{a}{b} \in A_{\mathfrak{q}} \qquad (\mathfrak{q} \in V).$$

Addition and multiplication are defined pointwise in the localizations $A_{\mathfrak{p}}$.

Definition 17.1 (Structure sheaf of an affine spectrum)**.** The assignment $U \mapsto \mathcal{O}_X(U)$ with restriction given by deleting components of the family outside the smaller open set is the *structure sheaf* of $X = \operatorname{Spec} A$.



Theorem 17.2 (Sheaf property)**.** *The assignment $U \mapsto \mathcal{O}_X(U)$ is a sheaf of commutative rings.*

Proof. Locality is immediate: two families equal on an open cover have the same component at every point. For gluing, let $U = \bigcup_i U_i$ and let $s_i \in \mathcal{O}_X(U_i)$ agree on overlaps. Define $s_\mathfrak{p}$ using any i with $\mathfrak{p} \in U_i$. Compatibility makes this well defined. Each s_i is locally represented by a fraction a/b, so the glued family is also locally represented by such fractions. Hence it is a section of $\mathcal{O}_X(U)$, and uniqueness follows pointwise. □

Example 17.3 (Sections on a distinguished open)**.** For $X = \operatorname{Spec} A$ and $f \in A$, the structure sheaf satisfies $\mathcal{O}_X(D(f)) \cong A_f$. A regular function on this basic open is represented by a fraction whose denominator is a power of f.

17.3 Sections on principal opens

The central computation is the following.

Theorem 17.4 (Principal-open formula)**.** *For every $f \in A$ there is a natural isomorphism*

$$\mathcal{O}_X(D(f)) \cong A_f.$$

Under this isomorphism, a/f^n is the section whose germ at $\mathfrak{p} \in D(f)$ is the same fraction in $A_\mathfrak{p}$.

The theorem says that a principal open carries exactly the algebra obtained by making its defining function invertible. In particular,

$$\mathcal{O}_X(X) = \mathcal{O}_X(D(1)) \cong A.$$

Thus the global ring can be recovered from its affine structure sheaf.

$$\begin{tikzcd}
D(f) \arrow[r, mapsto, "\text{regular sections}"] & A_f & \\
& & \boxed{\mathcal{O}_{\operatorname{Spec} A}(D(f)) \cong A_f} \\
\mathfrak{p} \arrow[r, mapsto] & \dfrac{a}{f^n} \in A_\mathfrak{p} &
\end{tikzcd}$$

Remark 17.5 (What is not being consumed here)*.* The stronger legacy statement that the restricted locally ringed space $(D(f), \mathcal{O}_X|_{D(f)})$ is canonically isomorphic to $\operatorname{Spec} A_f$ is AG3 Problem 11. It is reserved for VI/18, where principal opens are treated as affine schemes. This chapter uses only the structure-sheaf computations needed to prepare that result.

17.4 Restriction maps are further localization

If $g \in A$, then

$$D(fg) = D(f) \cap D(g) \subseteq D(f).$$

Under the identifications from Theorem 17.4, the restriction map is

$$A_f \longrightarrow A_{fg},$$

the localization map that also makes g invertible. Composition of restrictions is therefore associativity of localization.

$$\begin{tikzcd}
D(fg) \arrow[r, hook] & D(f) \\
\mathcal{O} & \\
A_{fg} & A_f \arrow[l, "\text{loc}"']
\end{tikzcd}$$

17.5 Stalks are local rings

Fix $\mathfrak{p} \in \operatorname{Spec} A$. Principal opens $D(f)$ with $f \notin \mathfrak{p}$ form a neighborhood basis at $\mathfrak{p}$, hence

$$\mathcal{O}_{X,\mathfrak{p}} = \varinjlim_{\mathfrak{p} \in D(f)} \mathcal{O}_X(D(f)) \cong \varinjlim_{f \notin \mathfrak{p}} A_f.$$

The colimit is exactly $A_{\mathfrak{p}}$.

Theorem 17.6 (Stalk formula)**.** *For every prime* $\mathfrak{p}$,

$$\boxed{\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}.}$$

The unique maximal ideal is $\mathfrak{p}A_{\mathfrak{p}}$.

$$\begin{tikzcd}
\cdots \arrow[r] & A_f \arrow[r] & A_{fg} \arrow[r] & \cdots \arrow[r] & A_{\mathfrak{p}} \\
& \mathfrak{p} \in D(f) & \mathfrak{p} \in D(fg) & & \mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}
\end{tikzcd}$$

Example 17.7 (Stalk at a prime)**.** At $\mathfrak{p} \in \operatorname{Spec} A$, the stalk $\mathcal{O}_{X,\mathfrak{p}}$ is $A_{\mathfrak{p}}$. A function germ may use any denominator outside $\mathfrak{p}$.

17.6 Residue fields and values of regular functions

The residue field at $\mathfrak{p}$ is

$$\kappa(\mathfrak{p}) := \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{p}\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

Since $A/\mathfrak{p}$ is a domain,

$$\boxed{\kappa(\mathfrak{p}) \cong \operatorname{Frac}(A/\mathfrak{p}).}$$

A section $s \in \mathcal{O}_X(U)$ therefore has a value at each $\mathfrak{p} \in U$, namely the residue class of its germ

$$s(\mathfrak{p}) \in \kappa(\mathfrak{p}).$$

The target field varies with the point. This is one of the basic differences between a scheme and an ordinary set carrying scalar-valued functions.

17.7 Generic and closed points revisited

If A is a domain, the generic point (0) has stalk

$$\mathcal{O}_{X,(0)} \cong \operatorname{Frac}(A),$$

so the function field is already present as the local ring at the generic point. If $\mathfrak{m}$ is maximal, then the closed point has residue field $A/\mathfrak{m}$.

For $A = k[t]$ with k algebraically closed,

$$\mathcal{O}_{X,(0)} = k(t), \qquad \mathcal{O}_{X,(t-a)} = k[t]_{(t-a)}, \qquad \kappa(t-a) = k.$$

$$\begin{tikzcd}
\boxed{\begin{array}{c} \mathfrak{p} = (0) \\ \mathcal{O}_{X,\mathfrak{p}} = k(t) \\ \text{generic} \end{array}} \arrow[r, "\text{specialization}"] & \boxed{\begin{array}{c} \mathfrak{m} = (t-a) \\ \mathcal{O}_{X,\mathfrak{m}} = k[t]_{(t-a)} \\ \kappa(\mathfrak{m}) = k \end{array}}
\end{tikzcd}$$

17.8 The arithmetic line Spec $\mathbb{Z}$

For $X = \operatorname{Spec} \mathbb{Z}$,

$$\mathcal{O}_X(D(n)) = \mathbb{Z}[1/n].$$

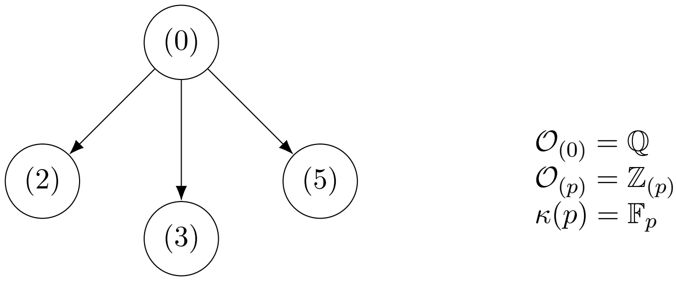
At a closed point (p),

$$\mathcal{O}_{X,(p)} = \mathbb{Z}_{(p)}, \qquad \kappa(p) = \mathbb{F}_p,$$

while the generic point has

$$\mathcal{O}_{X,(0)} = \mathbb{Q}.$$

Thus one sheaf simultaneously remembers rational generic information and modular special fibers.



17.9 Nilpotents are invisible topologically but visible sheaf-theoretically

Let

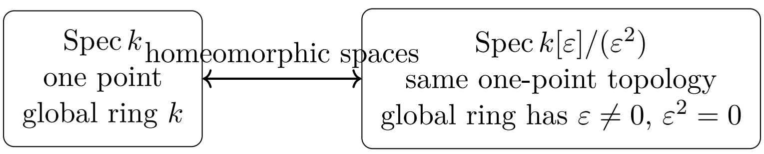
$$A = k[\varepsilon]/(\varepsilon^2).$$

Its spectrum has a single point, just like Spec k, but

$$\mathcal{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) = A$$

contains the nonzero nilpotent section ε. Hence the same underlying topological space can support different structure sheaves.

More generally, A and $A_{\mathrm{red}} = A/\operatorname{Nil}(A)$ have homeomorphic spectra, but their structure sheaves differ by nilpotent functions.



Example 17.8 (Dual-number structure)**.** On $X = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$, the underlying space has one point, but its structure sheaf has global ring $k[\varepsilon]/(\varepsilon^2)$, retaining the nilpotent infinitesimal direction.

17.10 Gluing regular functions

If $U = D(f) \cup D(g)$, a section over U is determined by sections

$$s_f \in A_f, \qquad s_g \in A_g$$

that agree after restriction to

$$D(fg) = D(f) \cap D(g).$$

Thus

$$\mathcal{O}_X(U) \cong \left\{(s_f, s_g) \in A_f \times A_g : s_f|_{A_{fg}} = s_g|_{A_{fg}}\right\}.$$

This is the equalizer form of the sheaf axiom.

$$\mathcal{O}_X(D(f) \cup D(g)) \longrightarrow A_f \times A_g \underset{\rho_g}{\overset{\rho_f}{\rightrightarrows}} A_{fg}$$

$$\mathcal{O}_X(D(f) \cup D(g)) = \mathrm{Eq}(A_f \times A_g \rightrightarrows A_{fg}).$$

17.11 Units and local vanishing

For $a \in A$, the germ $a/1 \in A_{\mathfrak{p}}$ is a unit if and only if $a \notin \mathfrak{p}$. Hence the unit locus of the global section a is precisely

$$D(a).$$

By contrast,

$$\frac{a}{1} = 0 \text{ in } A_{\mathfrak{p}}$$

if and only if there exists $s \notin \mathfrak{p}$ with $sa = 0$ in A. In a ring with zero divisors, local vanishing is therefore subtler than membership $a \in \mathfrak{p}$.

17.12 Common pitfalls

1. A regular section over $D(f)$ need not come from an element of A; it may use powers of f in the denominator.
2. The value of a regular function at $\mathfrak{p}$ lies in $\kappa(\mathfrak{p})$, not in one fixed base field in general.
3. The topology of $\operatorname{Spec} A$ does not determine A: nilpotents can change the structure sheaf without changing the underlying topological space.
4. The stalk at $\mathfrak{p}$ is $A_{\mathfrak{p}}$, not the residue field $\kappa(\mathfrak{p})$; the latter is a quotient of the stalk.
5. A fraction a/b represents a section only where b is invertible, i.e. on an open contained in $D(b)$.
6. The full statement $D(f) \cong \operatorname{Spec} A_f$ as locally ringed spaces is reserved for VI/18; the universal property of affine schemes is reserved for VI/19.
7. The fallback migration rule for the legacy AG3 file must not send Problems 11–14 to VI/17; the ledger records their later destinations explicitly.

17.13 Worked examples

Example 17.9 (A one-point affine spectrum)**.** For $A = k$ a field, $X = \operatorname{Spec} k$ has one point (0) and only two opens. Since $X = D(1)$,

$$\mathcal{O}_X(X) = k.$$

The unique stalk is also k, so the structure sheaf adds no extra localization in this case.

Example 17.10 (The punctured affine line)**.** Let $A = k[x]$. On

$$D(x) = \mathbb{A}^1_k \setminus V(x)$$

we obtain

$$\mathcal{O}_X(D(x)) = k[x]_x = k[x, x^{-1}].$$

Thus $1/x$ is regular on $D(x)$ although it is not a global polynomial on X.

Example 17.11 (Deleting one closed point)**.** For $A = k[x]$ and $a \in k$,

$$\mathcal{O}_X(D(x-a)) = k[x, (x-a)^{-1}].$$

The denominator $x - a$ is permitted precisely because it does not vanish on the chosen open.

Example 17.12 (Inverting a product)**.** For $A = k[x, y]$,

$$\mathcal{O}_X(D(xy)) = A_{xy}.$$

Since $x^{-1} = y/(xy)$ and $y^{-1} = x/(xy)$ in A_{xy},

$$A_{xy} = k[x, y, x^{-1}, y^{-1}].$$

Example 17.13 (An arithmetic principal open)**.** For $X = \operatorname{Spec} \mathbb{Z}$,

$$\mathcal{O}_X(D(6)) = \mathbb{Z}[1/6].$$

The primes (2) and (3) have been removed, exactly the primes at which 6 fails to be a unit.

Example 17.14 (The local ring at a prime integer)**.** At $(p) \in \operatorname{Spec} \mathbb{Z}$,

$$\mathcal{O}_{X,(p)} = \mathbb{Z}_{(p)} = \left\{ \frac{a}{b} : b \not\equiv 0 \pmod{p} \right\}.$$

Its maximal ideal is $p\mathbb{Z}_{(p)}$ and its residue field is $\mathbb{F}_p$.

Example 17.15 (The generic arithmetic stalk)**.** At $(0) \in \operatorname{Spec} \mathbb{Z}$, every nonzero integer is inverted, hence

$$\mathcal{O}_{X,(0)} = \mathbb{Q}.$$

The generic point therefore sees rational functions rather than modular functions.

Example 17.16 (A closed point of the affine line)**.** For $X = \operatorname{Spec} k[x]$ and $\mathfrak{m} = (x - a)$,

$$\mathcal{O}_{X,\mathfrak{m}} = k[x]_{(x-a)}, \qquad \kappa(\mathfrak{m}) = k[x]_{(x-a)}/(x-a) \cong k.$$

Fractions are allowed exactly when their denominators are nonzero at a.

Example 17.17 (The generic point of $\mathbb{A}^1$)**.** If $A = k[x]$, then

$$\mathcal{O}_{X,(0)} = k(x).$$

Thus the same sheaf contains both local polynomial rings at closed points and the rational-function field at the generic point.

Example 17.18 (A closed point with a larger residue field)**.** Let $A = \mathbb{R}[x]$ and $\mathfrak{m} = (x^2 + 1)$. Then

$$\mathcal{O}_{X,\mathfrak{m}} = \mathbb{R}[x]_{(x^2+1)}, \qquad \kappa(\mathfrak{m}) \cong \mathbb{R}[x]/(x^2+1) \cong \mathbb{C}.$$

So values of regular functions need not lie in the base field $\mathbb{R}$.

Example 17.19 (A nonreduced point)**.** For $A = k[\varepsilon]/(\varepsilon^2)$, the spectrum has one point, but

$$\mathcal{O}_X(X) = A$$

contains the nonzero nilpotent section ε. The structure sheaf sees infinitesimal information that the topology misses.

Example 17.20 (A product ring)**.** For $A = k \times k$, $\operatorname{Spec} A$ is the disjoint union of two points. The idempotents $e_1 = (1, 0)$ and $e_2 = (0, 1)$ cut out the two clopen pieces, and the restrictions of the structure sheaf to them have section ring k.

Example 17.21 (The crossing $xy = 0$)**.** Let $A = k[x, y]/(xy)$ and let $\mathfrak{m} = (x, y)$. The stalk

$$\mathcal{O}_{X,\mathfrak{m}} = A_{(x,y)}$$

remembers both branches and the relation $xy = 0$. Its residue field is k.

Example 17.22 (Restriction means one more localization)**.** For $A = k[x, y]$,

$$D(xy) \subseteq D(x).$$

The restriction

$$\mathcal{O}_X(D(x)) = A_x \longrightarrow \mathcal{O}_X(D(xy)) = A_{xy}$$

is obtained by additionally inverting y.

Example 17.23 (A rational expression regular on its natural domain)**.** The fraction

$$\frac{x + y}{y}$$

defines a section on $D(y) \subseteq \operatorname{Spec} k[x, y]$ because y is invertible there. Its value at a prime $\mathfrak{p} \in D(y)$ is the residue of $(x + y)/y$ in $\kappa(\mathfrak{p})$.

Example 17.24 (Global sections are the original ring)**.** Because $X = D(1)$,

$$\mathcal{O}_X(X) = A_1 \cong A.$$

Thus the global regular section represented by $a \in A$ is the family $a/1 \in A_{\mathfrak{p}}$ for all primes $\mathfrak{p}$.

Example 17.25 (A principal open can be empty)**.** If f is nilpotent, then $D(f) = \varnothing$. Localizing at f forces a nilpotent element to be a unit, hence forces $1 = 0$; therefore

$$A_f = 0 = \mathcal{O}_X(\varnothing).$$

Example 17.26 (The unit locus of a section)**.** For $a \in A$, the germ $a/1 \in A_{\mathfrak{p}}$ is invertible iff $a \notin \mathfrak{p}$. Consequently the set of points where the global section a is a unit is exactly $D(a)$.

Example 17.27 (Local vanishing in a ring with zero divisors)**.** Let $A = k[x, y]/(xy)$ and consider $a = x$. At the prime (x) the element $y \notin (x)$ and $yx = 0$, so

$$x/1 = 0 \quad \text{in } A_{(x)}.$$

Thus local vanishing can occur because an annihilator becomes invertible.

Example 17.28 (A section determined by compatible principal pieces)**.** Let $U = D(f) \cup D(g)$. If $s_f \in A_f$ and $s_g \in A_g$ have the same image in A_{fg}, the sheaf axiom produces a unique

$$s \in \mathcal{O}_X(U)$$

restricting to them. This is the equalizer description of sections on U.

17.14 Advanced worked examples

Example 17.29 (Local fraction representatives need not be global)**.** A section over a general open U need not admit one fraction a/b valid everywhere. The definition requires only that every point have a neighborhood on which one fraction works. Compatible local fractions then glue by the sheaf axiom.

Example 17.30 (The punctured affine plane)**.** Let $A = k[x, y]$ and

$$U = D(x) \cup D(y) = \mathbb{A}^2_k \setminus \{(x, y)\}.$$

Then

$$\mathcal{O}_X(U) = A_x \times_{A_{xy}} A_y.$$

Inside $k(x, y)$ one has $A_x \cap A_y = A$, because a denominator that is simultaneously a power of x and a power of y must cancel in the UFD $k[x, y]$. Hence

$$\mathcal{O}_X(U) \cong k[x, y].$$

Example 17.31 (Removing the crossing point separates branches)**.** Let $A = k[x, y]/(xy)$ and $U = D(x) \cup D(y)$. Since

$$D(x) \cap D(y) = D(xy) = \varnothing,$$

a section on U is an arbitrary pair

$$(s_x, s_y) \in A_x \times A_y.$$

The two branches can therefore carry independent functions once the intersection point has been removed.

Example 17.32 (Generic stalks of the two components)**.** For $A = k[x, y]/(xy)$, the minimal primes are (x) and (y). Localizing at (x) makes y invertible and forces $x = 0$, hence

$$A_{(x)} \cong k(y).$$

Similarly $A_{(y)} \cong k(x)$. Each irreducible component has its own function field encoded in a generic stalk.

Example 17.33 (The reduced quotient changes the sheaf)**.** For $A = k[\varepsilon]/(\varepsilon^2)$,

$$A_{\text{red}} = k.$$

The underlying spectra are homeomorphic, but on the unique open nonempty set the section rings are A and k. The quotient map kills precisely the nilpotent infinitesimal direction.

Example 17.34 (Clopen pieces and idempotents)**.** If $e^2 = e$, then

$$D(e) \cap D(1 - e) = \varnothing, \qquad D(e) \cup D(1 - e) = X.$$

The sheaf therefore decomposes into independent section theories on two clopen pieces. Algebraically this reflects the product decomposition

$$A \cong A_e \times A_{1-e}.$$

Example 17.35 (A germ represented on a principal neighborhood)**.** Given $\alpha \in A_{\mathfrak{p}}$, choose a representation a/s with $s \notin \mathfrak{p}$. Then $\mathfrak{p} \in D(s)$ and

$$\frac{a}{s} \in A_s = \mathcal{O}_X(D(s))$$

represents the germ α. Thus every stalk element is represented by a regular section on a principal neighborhood.

Example 17.36 (Equality of germs)**.** Two fractions a/s and b/t define the same germ in $A_{\mathfrak{p}}$ iff there exists $u \notin \mathfrak{p}$ with

$$u(at - bs) = 0.$$

Equivalently, after restricting to $D(ust)$ near $\mathfrak{p}$, the corresponding regular sections agree.

Example 17.37 (The value field varies under specialization)**.** In $\operatorname{Spec}\mathbb{Z}$, the generic point has value field $\mathbb{Q}$, while the closed point (p) has value field $\mathbb{F}_p$. A global integer n therefore has generic value $n \in \mathbb{Q}$ and special value $n \bmod p \in \mathbb{F}_p$.

Example 17.38 (A section can vanish locally without being zero globally)**.** In $A = k[x,y]/(xy)$, the global section x has zero germ at the generic point (x) of the y-axis, but it is nonzero at the generic point (y) of the x-axis. The structure sheaf records componentwise behavior invisible to a single scalar evaluation.

Example 17.39 (Principal covers and the sheaf equalizer)**.** If $D(f_i)$ cover X, then regular sections satisfy

$$A \cong \operatorname{Eq}\left(\prod_i A_{f_i} \rightrightarrows \prod_{i,j} A_{f_i f_j}\right).$$

For a finite cover this is the algebraic form of gluing local fractions into a global element of A.

Example 17.40 (Local rings detect the closed point)**.** In $A_{\mathfrak{p}}$ the nonunits are exactly the fractions whose numerator lies in $\mathfrak{p}$. They form the unique maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. Hence the stalk remembers which prime of A is the center of the localization.

Example 17.41 (Residue field as a fraction field)**.** For a prime $\mathfrak{p}$, localization and quotient give

$$A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong (A/\mathfrak{p})_{(0)} \cong \operatorname{Frac}(A/\mathfrak{p}).$$

This gives a uniform formula valid for generic and closed points alike.

Example 17.42 (Structure sheaf versus the next chapter)**.** The data

$$(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

are now fully constructed. VI/18 gives this locally ringed space a name—the affine scheme associated to A—and studies its open affine pieces. Keeping this boundary explicit prevents the legacy basic-open-subsheme problem from being duplicated here.

17.15 Exercises

Exercise 17.43 (Global sections)**.** Prove directly from $D(1) = \operatorname{Spec} A$ that $\mathcal{O}_X(X) \cong A$.

Exercise 17.44 (A basic-open section)**.** For $A = k[x]$, compute $\mathcal{O}_X(D(x^3))$ and compare it with $\mathcal{O}_X(D(x))$.

Exercise 17.45 (Restriction)**.** For $A = k[x, y]$, write explicitly the restriction map $A_x \to A_{xy}$.

Exercise 17.46 (Stalk from principal neighborhoods)**.** Show that $\varinjlim_{f \notin \mathfrak{p}} A_f \cong A_{\mathfrak{p}}$.

Exercise 17.47 (Unique maximal ideal)**.** Prove that $A_{\mathfrak{p}}$ is local with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$.

Exercise 17.48 (Residue field)**.** Prove $\kappa(\mathfrak{p}) \cong \mathrm{Frac}(A/\mathfrak{p})$.

Exercise 17.49 (Generic point)**.** If A is a domain, compute $\mathcal{O}_{X,(0)}$.

Exercise 17.50 (Closed point on the affine line)**.** Compute the stalk and residue field at $(x - a) \in \operatorname{Spec} k[x]$.

Exercise 17.51 (Arithmetic stalk)**.** Compute $\mathcal{O}_{\operatorname{Spec}\mathbb{Z},(p)}$ and its residue field.

Exercise 17.52 (Arithmetic principal open)**.** Compute $\mathcal{O}_{\operatorname{Spec}\mathbb{Z}}(D(30))$.

Exercise 17.53 (Nilpotent principal open)**.** Show that if f is nilpotent then $D(f) = \varnothing$ and $A_f = 0$.

Exercise 17.54 (Unit locus)**.** Show that the unit locus of $a \in A = \mathcal{O}_X(X)$ is $D(a)$.

Exercise 17.55 (Local zero criterion)**.** Show $a/1 = 0$ in $A_{\mathfrak{p}}$ iff some $s \notin \mathfrak{p}$ satisfies $sa = 0$.

Exercise 17.56 (One-point nonreduced scheme)**.** Compare the structure sheaves of $\operatorname{Spec} k$ and $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$.

Exercise 17.57 (Product ring)**.** Describe $\mathcal{O}_{\operatorname{Spec}(k\times k)}$ on all open subsets.

Exercise 17.58 (Equalizer on two opens)**.** Prove $\mathcal{O}_X(D(f) \cup D(g)) \cong A_f \times_{A_{fg}} A_g$.

Exercise 17.59 (Empty intersection)**.** For $A = k[x, y]/(xy)$, compute sections on $D(x) \cup D(y)$.

Exercise 17.60 (Local representative)**.** Given $\alpha \in A_{\mathfrak{p}}$, find a principal neighborhood on which α is represented by a section.

Exercise 17.61 (Equality of germs)**.** Give an algebraic criterion for a/s and b/t to have the same germ at $\mathfrak{p}$.

Exercise 17.62 (Values at points)**.** For $A = \mathbb{Z}$ and global section n, compute its value at (0) and at (p).

Exercise 17.63 (Irreducible component stalk)**.** For $A = k[x, y]/(xy)$ compute the stalks at the minimal primes (x) and (y).

Exercise 17.64 (Reduced quotient)**.** Describe the induced map on stalks from $A \to A_{\mathrm{red}}$.

Exercise 17.65 (Principal-cover gluing)**.** Suppose $D(f) \cup D(g) = X$. Show compatible elements of A_f and A_g glue to a unique element of A.

Exercise 17.66 (Locality of regularity)**.** Show that if a family of germs is locally represented by fractions, then its restriction to any open subset has the same property.

Exercise 17.67 (Sheaf locality)**.** Prove two sections of $\mathcal{O}_X(U)$ are equal iff all of their germs are equal.

Exercise 17.68 (Bridge to affine schemes)**.** State precisely which extra assertion is needed to pass from the structure-sheaf formula on $D(f)$ to the legacy statement $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$.

17.16 Problems

Problem 17.69 (Construct the structure sheaf from local fractions)**.** Let $X = \operatorname{Spec} A$. Define $\mathcal{O}_X(U)$ to be families $s_{\mathfrak{p}} \in A_{\mathfrak{p}}$ locally represented by a fraction a/b on neighborhoods contained in $D(b)$. Prove that this assignment is a sheaf of rings.

Provenance. New synthesis based on the VI/17 structure-sheaf source stream; no numbered legacy Problem is claimed.

Problem 17.70 (Compute sections on a principal open)**.** Prove that for $X = \operatorname{Spec} A$ and $f \in A$, the canonical map $A_f \to \mathcal{O}_X(D(f))$ is an isomorphism.

Provenance. New canonical proof of the core VI/17 theorem, derived from the regular-function material in AG3 and CA13.

Problem 17.71 (Compute the stalk of the structure sheaf)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, prove $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$.

Provenance. New synthesis of the stalk formula explicitly present in the legacy structure-sheaf discussion.

Problem 17.72 (Recover the ring from global sections)**.** Show that the canonical map $A \to \Gamma(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ is an isomorphism.

Provenance. New synthesis; the corresponding universal morphism statement is deliberately reserved for AG3 Problem 12 in VI/19.

Problem 17.73 (Residue fields from local rings)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, prove

$$\kappa(\mathfrak{p}) = \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{p}\mathcal{O}_{X,\mathfrak{p}} \cong \operatorname{Frac}(A/\mathfrak{p}).$$

Provenance. New synthesis of corpus residue-field and structure-sheaf material.

Problem 17.74 (Regular functions on the punctured affine line)**.** Let $X = \operatorname{Spec} k[x]$. Compute $\mathcal{O}_X(D(x))$ and prove that $1/x$ is regular on $D(x)$ but does not extend to a global regular section of X.

Provenance. Legacy-derived synthesis from the explicit regular-function example in AG3/CA13; not a numbered legacy Problem.

Problem 17.75 (The structure sheaf of $\operatorname{Spec} \mathbb{Z}$)**.** Compute $\mathcal{O}(D(n))$, the stalk at (p), the stalk at (0), and the corresponding residue fields for $X = \operatorname{Spec} \mathbb{Z}$.

Provenance. New synthesis using recurring corpus examples for $\operatorname{Spec} \mathbb{Z}$**.**

Problem 17.76 (A nonreduced point is detected by its structure sheaf)**.** Compare $(\operatorname{Spec} k, \mathcal{O})$ with $(\operatorname{Spec} k[\varepsilon]/(\varepsilon^2), \mathcal{O})$. Prove that the underlying spaces are homeomorphic but the ringed spaces are not isomorphic.

Provenance. New synthesis of legacy reduced/nonreduced and structure-sheaf themes.

Problem 17.77 (Product rings and clopen decomposition)**.** For $A = A_1 \times A_2$, describe $\operatorname{Spec} A$ and its structure sheaf in terms of the two factors.

Provenance. New synthesis, consistent with earlier component/idempotent corpus material.

Problem 17.78 (Gluing on two principal opens)**.** Let $U = D(f) \cup D(g)$. Prove

$$\mathcal{O}_X(U) \cong A_f \times_{A_{fg}} A_g.$$

Provenance. New synthesis of VI/12–VI/17 sheaf machinery.

Problem 17.79 (The generic stalk is the function field)**.** Let A be a domain and $X = \operatorname{Spec} A$. Prove $\mathcal{O}_{X,(0)} \cong \operatorname{Frac}(A)$ and explain why every nonempty principal open contains (0).

Provenance. New synthesis. The function-field part of legacy AG6 Problem 7 remains reserved for VI/27 and is not claimed here.

Problem 17.80 (Global functions on the punctured affine plane)**.** Let $A = k[x, y]$ and $U = D(x) \cup D(y)$. Prove $\mathcal{O}_X(U) \cong A$.

Provenance. New advanced synthesis; not a corpus Problem.

Problem 17.81 (Restriction maps and transitivity)**.** For $f, g, h \in A$, identify the restriction maps along $D(fgh) \subseteq D(fg) \subseteq D(f)$ and prove their composition equals the direct restriction $A_f \to A_{fgh}$.

Provenance. New synthesis from the principal-open structure-sheaf calculus.

Problem 17.82 (Units in a stalk)**.** Prove that $a/s \in A_\mathfrak{p}$ is a unit iff $a \notin \mathfrak{p}$, and deduce that the maximal ideal is $\mathfrak{p}A_\mathfrak{p}$.

Provenance. New synthesis of local-ring and structure-sheaf material.

Problem 17.83 (A nonrational closed point)**.** For $A = \mathbb{R}[x]$ and $\mathfrak{m} = (x^2 + 1)$, compute $\mathcal{O}_{X,\mathfrak{m}}$ and $\kappa(\mathfrak{m})$, and interpret the value of a real polynomial at this scheme point.

Provenance. New synthesis of a recurring corpus residue-field example.

Problem 17.84 (The nilradical sheaf)**.** Let $N = \operatorname{Nil}(A)$ and let $X = \operatorname{Spec} A \cong \operatorname{Spec}(A/N)$ topologically. Show that at each $\mathfrak{p}$ the map on structure-sheaf stalks is

$$A_\mathfrak{p} \to A_\mathfrak{p}/N_\mathfrak{p},$$

and explain how nilpotent sections are removed.

Provenance. New synthesis of VI/10 and VI/17; no later closed-subscheme corpus Problem is consumed.

Problem 17.85 (Detecting zero germs algebraically)**.** For $a \in A$ and $\mathfrak{p} \in \operatorname{Spec} A$, prove

$$a/1 = 0 \text{ in } \mathcal{O}_{X,\mathfrak{p}} \iff \exists s \notin \mathfrak{p} \text{ with } sa = 0.$$

Give a nontrivial example in $A = k[x, y]/(xy)$.

Provenance. New synthesis of localization and structure-sheaf behavior.

Problem 17.86 (Reconstruct the sheaf from principal-open data)**.** Suppose a sheaf of rings $\mathcal{F}$ on $\operatorname{Spec} A$ is equipped with compatible isomorphisms $\mathcal{F}(D(f)) \cong A_f$ for every $f \in A$. Prove $\mathcal{F} \cong \mathcal{O}_{\operatorname{Spec} A}$.

Provenance. New synthesis. It prepares but does not consume AG3 Problem 11.

Problem 17.87 (Local fractions and the basis-sheaf construction agree)**.** Compare two constructions of $\mathcal{O}_{\operatorname{Spec} A}$: (i) locally representable families in the stalks $A_\mathfrak{p}$, and (ii) the sheaf determined on the basis $D(f)$ by A_f. Prove they are canonically isomorphic.

Provenance. New synthesis of the presheaf/basis material from VI/12 with the current chapter.

Problem 17.88 (A principal cover of the whole spectrum)**.** Assume $D(f) \cup D(g) = \operatorname{Spec} A$. Prove that compatible elements $u \in A_f$ and $v \in A_g$ arise from a unique element $a \in A$.

Provenance. New synthesis, with the cohomological continuation reserved for VI/47–VI/49.

Problem 17.89 (Part III synthesis)**.** Explain why the pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ has local rings at every point and why this is exactly the data needed before defining an affine scheme. Do not use the affine-scheme universal property.

Provenance. New transition problem. AG3 Problems 11–14 remain reserved for later chapters and are not duplicated here.

17.17 Challenges

Problem 17.90 (Challenge: Finite distinguished-cover equalizer)**.** Let $D(f_1), \dots, D(f_n)$ cover $\operatorname{Spec} A$. Prove

$$A \longrightarrow \prod_i A_{f_i} \rightrightarrows \prod_{i,j} A_{f_i f_j}$$

is an equalizer diagram. Give a direct algebraic proof as well as the sheaf-theoretic proof.

Problem 17.91 (Challenge: Algebraic Hartogs precursor)**.** For $A = k[x_1, \dots, x_n]$ with $n \geq 2$, let $U = \mathbb{A}^n_k \setminus \{0\}$. Prove $\Gamma(U, \mathcal{O}) = A$ by a principal-open cover and intersection of localizations.

Problem 17.92 (Challenge: Nilpotent thickening)**.** Let A be a ring and $I \subset A$ a nilpotent ideal. Compare the structure sheaves on $\operatorname{Spec} A$ and $\operatorname{Spec}(A/I)$ stalk by stalk, and describe precisely the sheaf kernel.

Problem 17.93 (Challenge: Idempotents from clopen decompositions)**.** Prove that a decomposition $\operatorname{Spec} A = U \sqcup V$ into two nonempty clopen subsets determines a nontrivial idempotent $e \in A$ with $U = D(e)$ and $V = D(1-e)$.

Problem 17.94 (Challenge: Affine rigidity from global sections)**.** Suppose two affine locally ringed spaces $(\operatorname{Spec} A, \mathcal{O}_A)$ and $(\operatorname{Spec} B, \mathcal{O}_B)$ are isomorphic. Prove $A \cong B$ by passing to global sections. Explain why this statement does not yet prove the full anti-equivalence between rings and affine schemes.

Summary

For $X = \operatorname{Spec} A$, the structure sheaf is the local algebraic function theory determined by localization:

$$\boxed{\mathcal{O}_X(D(f)) = A_f, \qquad \mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}, \qquad \kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}).}$$

The sheaf glues locally representable fractions, recovers A from global sections, records nilpotents that topology cannot see, and makes every point carry a local ring. This completes Part III: the sheaf machinery is now attached to spectra. VI/18 packages the pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ as an affine scheme.

The Volume VI ledger records that no numbered legacy Problem is consumed by VI/17 itself. AG3 Problems 11–14 are reserved exactly once for VI/18, VI/19, VI/21, and VI/20, respectively, while the structure-sheaf theory/examples from AG3 and CA13 are merged here.

17.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 17.95 (Basic sections)**.** Compute $\mathcal{O}_X(D(f))$ for affine $X = \operatorname{Spec} A$.

Hint. Use the structure sheaf, local fractions, and stalks. □

Solution. A_f. □

Exercise 17.96 (Global sections)**.** Compute $\Gamma(X, \mathcal{O}_X)$ for affine X.

Hint. Use the structure sheaf, local fractions, and stalks. □

Solution. A. □

Exercise 17.97 (Stalk)**.** Compute $\mathcal{O}_{X,\mathfrak{p}}$.

Hint. Use the structure sheaf, local fractions, and stalks. □

Solution. $A_{\mathfrak{p}}$. □

Exercise 17.98 (Residue)**.** What field comes from the stalk at $\mathfrak{p}$?

Hint. Use the structure sheaf, local fractions, and stalks. □

Solution. $\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$. □

Exercise 17.99 (Unit open)**.** What is $\mathcal{O}_X(D(1))$?

Hint. Use the structure sheaf, local fractions, and stalks. □

Solution. A. □

Proofs

Exercise 17.100 (Basic-open formula)**.** Prove: Prove $\mathcal{O}_X(D(f)) \cong A_f$ for the affine structure sheaf.

Hint. Work directly from the structure sheaf, local fractions, and stalks. ☐

Solution. Use the construction by locally represented fractions and the distinguished-open basis. ☐

Exercise 17.101 (Stalk formula)**.** Prove: Prove $\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}}$.

Hint. Work directly from the structure sheaf, local fractions, and stalks. ☐

Solution. Pass to the direct limit over distinguished neighborhoods $D(f)$ with $f \notin \mathfrak{p}$. ☐

Exercise 17.102 (Local ring)**.** Prove: Prove each structure-sheaf stalk is local.

Hint. Work directly from the structure sheaf, local fractions, and stalks. ☐

Solution. Under the localization identification the unique maximal ideal is $\mathfrak{p}A_{\mathfrak{p}}$. ☐

Exercise 17.103 (Restriction localization)**.** Prove: Prove the restriction $A_f \to A_{fg}$ is the usual localization map.

Hint. Work directly from the structure sheaf, local fractions, and stalks. ☐

Solution. Both sheaf restriction and algebraic localization send a fraction to the same further-localized fraction. ☐

Counterexamples and hypothesis tests

Exercise 17.104 (Topology enough)**.** Test the claim: the structure sheaf is determined by the underlying point set alone.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; reduced and nonreduced schemes can share the same topology. ☐

Exercise 17.105 (Polynomial only)**.** Test the claim: regular functions on $D(f)$ must come from elements of A without denominators.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; they are elements of A_f. ☐

Exercise 17.106 (All stalks field)**.** Test the claim: every structure-sheaf stalk is a field.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is a local ring, a field only at special generic situations. ☐

Applications and investigations

Exercise 17.107 (Local functions)**.** What does the structure sheaf add to the spectrum?

Hint. Translate the question into the structure sheaf, local fractions, and stalks. ☐

Solution. It records which algebraic functions are available on each open and at each germ. ☐

Exercise 17.108 (Scheme distinction)**.** Why can the structure sheaf distinguish identical topological spectra?

Hint. Translate the question into the structure sheaf, local fractions, and stalks. ☐

Solution. It retains nilpotents and local algebra invisible to topology. ☐

Challenge problems

Exercise 17.109 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Structure Sheaf in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Structure Sheaf, translates it through the structure sheaf, local fractions, and stalks, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 17.110 (Hypothesis challenge)**.** Choose one structural statement from Structure Sheaf, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Part IV

Affine and General Schemes

Chapter 18

Affine Schemes

Purpose and learning goals

The previous chapter attached a structure sheaf to every spectrum. We now package the pair

$$(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

as one geometric object: an *affine scheme*. This is the basic local model for all scheme theory. Its points are prime ideals, its open sets are Zariski opens, its regular functions are local fractions, and its local ring at $\mathfrak{p}$ is $A_{\mathfrak{p}}$.

The chapter begins from the Volume VI legacy-problem ledger rather than from a desired problem count. Two substantive corpus items belong here and are preserved in full: AG3 Problem 11, proving that a distinguished open $D(f)$ is the affine scheme $\operatorname{Spec} A_f$ as a locally ringed space, and the affineness subproblem AG6 Problem 6.4, which exhibits $\mathbb{P}^1_k$ as a non-affine scheme. The neighboring AG3 Problems 12–14 remain reserved for VI/19, VI/21, and VI/20 respectively.

After this chapter, the reader should be able to:

- define affine schemes as spectra equipped with their structure sheaves;
- pass freely among the ring A, its spectrum, and the corresponding locally ringed space $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$;
- compute global sections, stalks, residue fields, and distinguished opens;
- prove that $D(f)$ is canonically $\operatorname{Spec} A_f$ as a locally ringed space;
- recognize that distinguished opens form an affine basis;
- prove that every affine scheme is quasi-compact;
- recover the coordinate ring of an affine scheme from global sections;
- work with affine schemes arising from fields, polynomial rings, product rings, quotients, localizations, arithmetic rings, and nonreduced rings;
- distinguish "affine" from "locally affine" and explain why open subsets need not be affine;
- prove the legacy example that $\mathbb{P}^1_k$ is not affine;
- keep scheme-morphism and gluing results properly reserved for VI/19 and VI/20.

$$\begin{tikzcd}
\boxed{\begin{matrix}\text{Structure sheaf}\\ \text{VI/17}\end{matrix}} \arrow[r] & \boxed{\begin{matrix}\text{Affine schemes}\\ \text{VI/18}\end{matrix}} \arrow[r] & \boxed{\begin{matrix}\text{Affine morphisms}\\ \text{VI/19}\end{matrix}} \arrow[d] \\
 & & \boxed{\begin{matrix}\text{Gluing and general schemes}\\ \text{VI/20–VI/21}\end{matrix}}
\end{tikzcd}$$

18.1 Locally ringed spaces and the affine model

A *locally ringed space* is a topological space X equipped with a sheaf of rings $\mathcal{O}_X$ such that every stalk $\mathcal{O}_{X,x}$ is a local ring. VI/17 proved that for $X = \operatorname{Spec} A$,

$$\mathcal{O}_{X,\mathfrak{p}} \cong A_{\mathfrak{p}},$$

and $A_{\mathfrak{p}}$ is local with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$. Thus every spectrum with its structure sheaf is naturally a locally ringed space.

Definition 18.1 (Affine scheme)**.** An *affine scheme* is a locally ringed space isomorphic to

$$\operatorname{Spec} A := (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$$

for some commutative ring A with identity.

The adjective “affine” refers to the fact that one ring controls the whole space. Later, a general scheme will be a locally ringed space that is only *locally* affine.

$$\begin{tikzcd}
\boxed{\text{ring } A} \arrow[r] & \boxed{\text{space } \operatorname{Spec} A} \arrow[r] & \boxed{\text{sheaf } \mathcal{O}_{\operatorname{Spec} A}} \arrow[r] & \boxed{\begin{matrix}\text{affine scheme}\\ (\operatorname{Spec} A, \mathcal{O})\end{matrix}}
\end{tikzcd}$$

18.2 The three layers of an affine scheme

For $X = \operatorname{Spec} A$, one should keep three layers visible simultaneously:

$$\boxed{\begin{gathered}\text{topology: prime ideals and Zariski opens,}\\ \text{functions: } \mathcal{O}_X(D(f)) = A_f,\\ \text{local algebra: } \mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}.\end{gathered}}$$

The residue field at a point is

$$\kappa(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong \operatorname{Frac}(A/\mathfrak{p}).$$

Thus an affine scheme is much richer than its underlying topological space.

18.3 Global sections recover the affine coordinate ring

Since $X = D(1)$, VI/17 gives

$$\Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X) \cong A.$$

Consequently, if X is affine then its coordinate ring can be recovered intrinsically:

$$\boxed{A \cong \Gamma(X, \mathcal{O}_X).}$$

In particular, if $X \cong \operatorname{Spec} A$, then

$$X \cong \operatorname{Spec} \Gamma(X, \mathcal{O}_X).$$

This elementary recovery principle is one of the strongest tests for non-affineness used later in the chapter.

$$A \xrightarrow{\sim} \Gamma(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}) \xmapsto{\operatorname{Spec}} \operatorname{Spec} \Gamma(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}) \cong \operatorname{Spec} A$$

Example 18.2 (Distinguished open as an affine scheme)**.** For $X = \operatorname{Spec} A$, the open $D(f)$ with its restricted structure sheaf is canonically $\operatorname{Spec} A_f$. For $A = k[x]$ and $f = x$, this gives $D(x) \cong \operatorname{Spec} k[x, x^{-1}]$.

18.4 Distinguished opens are affine

Let $X = \operatorname{Spec} A$ and $f \in A$. Topologically, primes of A_f correspond exactly to primes of A not containing f. Hence

$$\operatorname{Spec} A_f \longleftrightarrow D(f).$$

The structure sheaves agree under this correspondence because, for $g \in A$,

$$\mathcal{O}_{\operatorname{Spec} A}|_{D(f)}(D(fg)) = A_{fg}$$

and

$$\mathcal{O}_{\operatorname{Spec} A_f}(D(g/1)) = (A_f)_{g/1} \cong A_{fg}.$$

Therefore one has the fundamental geometric localization theorem

$$\boxed{(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f.}$$

This is the full content of legacy AG3 Problem 11 and is proved in detail in Problem VI.18.P1.

$$\begin{array}{ccc} \operatorname{Spec} A_f & \xrightarrow{\sim} & D(f) \subseteq \operatorname{Spec} A \\ & & \qquad\qquad (A_f)_{g/1} \cong A_{fg}. \\ \mathfrak{q} & \longmapsto & \mathfrak{q} \cap A \end{array}$$

Corollary 18.3 (Affine basis)**.** *Every affine scheme has a basis of open subsets that are themselves affine schemes.*

Indeed, the distinguished opens $D(f)$ form a basis of $\operatorname{Spec} A$, and every one of them is an affine scheme $\operatorname{Spec} A_f$.

18.5 Finite principal covers and quasi-compactness

Suppose

$$\operatorname{Spec} A = \bigcup_{i \in I} D(f_i).$$

The complement is

$$V((f_i)_{i\in I}),$$

so the cover condition is equivalent to the ideal generated by the f_i being all of A. Thus there exist finitely many indices $i_1, \dots, i_r$ and elements $a_j \in A$ such that

$$1 = a_1 f_{i_1} + \cdots + a_r f_{i_r}.$$

Hence

$$\operatorname{Spec} A = D(f_{i_1}) \cup \cdots \cup D(f_{i_r}).$$

Theorem 18.4 (Affine schemes are quasi-compact)**.** *Every open cover of an affine scheme has a finite subcover.*

Proof. Refine the cover by distinguished opens. The refined distinguished-open cover admits a finite subcover by the preceding ideal argument; the corresponding members of the original cover form a finite subcover as well. ☐

$$\begin{tikzcd}
\boxed{\operatorname{Spec} A = \bigcup_i D(f_i)} \arrow[r] & \boxed{(f_i)_i = A} \arrow[d] \\
\boxed{\text{finite subcover}} & \boxed{1 = \sum_{j=1}^{r} a_j f_{i_j}} \arrow[l]
\end{tikzcd}$$

18.6 Basic affine schemes

The basic examples include

$$\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \dots, x_n], \qquad \operatorname{Spec} \mathbb{Z}, \qquad \operatorname{Spec} k[\varepsilon]/(\varepsilon^2),$$

and reducible spectra such as

$$\operatorname{Spec} k[x, y]/(xy).$$

The multiplicative group is the distinguished open

$$\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}] \cong D(t) \subseteq \mathbb{A}^1_k.$$

Product rings give disconnected affine schemes:

$$\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B$$

as locally ringed spaces.

$$\begin{tikzcd}
\boxed{\begin{array}{c}\mathbb{A}^n_k \\ \operatorname{Spec} k[x_1, \dots, x_n]\end{array}} & \boxed{\begin{array}{c}\operatorname{Spec} \mathbb{Z} \\ \text{arithmetic line}\end{array}} \\
\boxed{\begin{array}{c}\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \\ \text{nonreduced point}\end{array}} & \boxed{\begin{array}{c}\operatorname{Spec} k[x, y]/(xy) \\ \text{reducible cross}\end{array}}
\end{tikzcd}$$

Example 18.5 (A nonreduced affine scheme)**.** The affine scheme $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ has one topological point but a nonreduced global ring. Its structure sheaf remembers the nilpotent section ε.

18.7 The empty affine scheme

The zero ring has no prime ideals, so

$$\operatorname{Spec} 0 = \varnothing.$$

With its unique sheaf of rings, this is the empty affine scheme. It is important to allow this object: it appears naturally as $D(f)$ whenever f is nilpotent, since then $A_f = 0$.

18.8 Affine schemes can have the same underlying space

The spectra

$$\operatorname{Spec} k \qquad \text{and} \qquad \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$$

both have one point, but their global rings are different. Since an affine scheme recovers its global ring, they are not isomorphic as affine schemes. Affine geometry therefore remembers nilpotent directions that the underlying topology forgets.

18.9 Local affine schemes

If $(A, \mathfrak{m})$ is a local ring, then $\operatorname{Spec} A$ has a unique closed point, namely $\mathfrak{m}$. Conversely, if an affine scheme $\operatorname{Spec} A$ has a unique closed point, then A has a unique maximal ideal and is local. Thus

$$\boxed{A \text{ local} \iff \operatorname{Spec} A \text{ has exactly one closed point.}}$$

This gives a useful geometric interpretation of local rings without confusing a local scheme with a one-point space: a local ring can have many nonclosed prime ideals.

Example 18.6 (Disconnected affine scheme)**.** For $A = k \times k$, $\operatorname{Spec} A$ is two disjoint points. The orthogonal idempotents $(1, 0)$ and $(0, 1)$ split the ring and the affine scheme into two open-and-closed components.

18.10 Affine algebraic sets versus affine schemes

If k is algebraically closed and

$$V = V(I) \subseteq k^n, \qquad A = k[x_1, \ldots, x_n]/I,$$

then the classical points of V correspond to maximal ideals of A of the form $(x_1 - a_1, \ldots, x_n - a_n)/I$. The affine scheme $\operatorname{Spec} A$ also contains nonclosed prime ideals, generic points, residue-field information, and nilpotent structure when I is not radical. Thus affine schemes extend rather than merely rename classical affine algebraic sets.

18.11 Open subsets of affine schemes need not be affine

Distinguished opens are affine, but arbitrary open subsets of affine schemes need not be. The standard example is the punctured affine plane

$$U = \mathbb{A}^2_k \setminus \{(0, 0)\} = D(x) \cup D(y) \subseteq \operatorname{Spec} k[x, y].$$

The two pieces are affine and their intersection is $D(xy)$. The sheaf condition gives

$$\Gamma(U, \mathcal{O}_U) \cong k[x, y]_x \cap k[x, y]_y \subseteq k(x, y).$$

Because x and y are relatively prime in the UFD $k[x,y]$, this intersection is exactly

$$\boxed{\Gamma(U,\mathcal{O}_U)=k[x,y].}$$

If U were affine, affine reconstruction would identify it with

$$\operatorname{Spec}\Gamma(U,\mathcal{O}_U)=\operatorname{Spec}k[x,y]=\mathbb{A}^2_k.$$

Under the displayed identification of global sections, the canonical morphism is the original open immersion $U\hookrightarrow\mathbb{A}^2_k$, which omits the origin and therefore cannot be an isomorphism. Hence U is not affine.

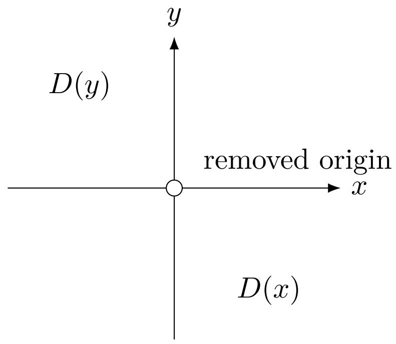


Figure 18.1: The punctured affine plane $\mathbb{A}^2_k\setminus\{0\}=D(x)\cup D(y)$ is covered by two affine distinguished opens but is not affine.

18.12 An affine cover does not imply affineness

A second, logically different phenomenon is that a scheme may admit an affine open cover without itself being affine. The projective line is the standard example:

$$\mathbb{P}^1_k=U_0\cup U_1,\qquad U_0\cong U_1\cong\mathbb{A}^1_k.$$

In the advanced examples and Problem VI.18.P2 we compute

$$\Gamma(\mathbb{P}^1_k,\mathcal{O})=k$$

and deduce that affineness would force

$$\mathbb{P}^1_k\cong\operatorname{Spec}k,$$

which is impossible. Thus

$$\boxed{\text{covered by affine opens}\not\Rightarrow\text{affine.}}$$

The formal gluing construction of general schemes is reserved for VI/20–VI/21.

$$\begin{tikzcd}
\boxed{\begin{array}{c}U_0\cong\mathbb{A}^1\\ \text{coordinate } t\end{array}} \arrow[r, leftrightarrow] & \boxed{\begin{array}{c}U_0\cap U_1\cong\mathbb{G}_m\\ u=t^{-1}\end{array}} \arrow[r, leftrightarrow] & \boxed{\begin{array}{c}U_1\cong\mathbb{A}^1\\ \text{coordinate } u\end{array}}
\end{tikzcd}$$

$$\Gamma(\mathbb{P}^1,\mathcal{O})=k[t]\cap k[t^{-1}]=k$$

18.13 Common pitfalls

1. An affine scheme is not only the topological space $\operatorname{Spec} A$; the structure sheaf is part of the object.
2. "Affine" means globally isomorphic to one spectrum. Being covered by affine opens is weaker.
3. Not every open subset of an affine scheme is affine, although every distinguished open $D(f)$ is affine.
4. The coordinate ring of an affine scheme is recovered as $\Gamma(X, \mathcal{O}_X)$; this is a powerful invariant.
5. $\operatorname{Spec}(A/I)$ is certainly affine, but interpreting it as a *closed subscheme* of $\operatorname{Spec} A$ is reserved for VI/22.
6. A ring homomorphism induces a contravariant map of spectra, but the full definition and universal property of affine-scheme morphisms are reserved for VI/19 and AG3 Problem 12.
7. The scheme-gluing theorem AG3 Problem 14 remains reserved for VI/20.
8. A local affine scheme need not be a one-point space; it has one closed point but may have many generic or intermediate points.

18.14 Worked examples

Example 18.7 (The spectrum of a field)**.** For a field k, $\operatorname{Spec} k$ has one point and

$$\Gamma(\operatorname{Spec} k, \mathcal{O}) = k.$$

Its unique local ring and residue field are both k.

Example 18.8 (The affine line)**.** The affine line over k is

$$\mathbb{A}^1_k = \operatorname{Spec} k[t].$$

It has generic point (0), closed points from maximal ideals, and structure sheaf $\mathcal{O}(D(f)) = k[t]_f$.

Example 18.9 (Affine n-space)**.**

$$\mathbb{A}^n_k = \operatorname{Spec} k[x_1, \dots, x_n].$$

When k is algebraically closed, its closed points are the maximal ideals $(x_1 - a_1, \dots, x_n - a_n)$, while nonmaximal primes encode irreducible subvarieties.

Example 18.10 (The arithmetic affine line)**.** $X = \operatorname{Spec} \mathbb{Z}$ is affine. Its generic point has local ring $\mathbb{Q}$, and the closed point (p) has local ring $\mathbb{Z}_{(p)}$ and residue field $\mathbb{F}_p$.

Example 18.11 (A nonreduced affine point)**.** For $A = k[\varepsilon]/(\varepsilon^2)$, $\operatorname{Spec} A$ has one point, but $\Gamma(X, \mathcal{O}_X) = A$ contains the nonzero nilpotent ε.

Example 18.12 (Two crossing affine lines)**.**

$$X = \operatorname{Spec} k[x, y]/(xy)$$

is affine and reducible. Its minimal primes (x) and (y) are the generic points of the two components.

Example 18.13 (A disconnected affine scheme)**.** For $A = k \times k$,

$$\operatorname{Spec} A \cong \operatorname{Spec} k \sqcup \operatorname{Spec} k.$$

The two points are both open and closed, and the global ring is $k \times k$.

Example 18.14 (The empty affine scheme)**.** For the zero ring,

$$\operatorname{Spec} 0 = \varnothing.$$

This is the empty affine scheme.

Example 18.15 (A punctured affine line is affine)**.** Inside $\mathbb{A}^1_k$,

$$D(t) \cong \operatorname{Spec} k[t, t^{-1}].$$

Thus removing the origin from the affine line gives the affine multiplicative group $\mathbb{G}_m$.

Example 18.16 (Removing two coordinate axes)**.** In $\mathbb{A}^2_k = \operatorname{Spec} k[x, y]$,

$$D(xy) \cong \operatorname{Spec} k[x, y, x^{-1}, y^{-1}].$$

Both x and y become units.

Example 18.17 (An arithmetic distinguished open)**.**

$$D(6) \subseteq \operatorname{Spec} \mathbb{Z}$$

is affine with coordinate ring $\mathbb{Z}[1/6]$. The points (2) and (3) are removed.

Example 18.18 (A unit gives the whole affine scheme)**.** If $u \in A^\times$, then $D(u) = \operatorname{Spec} A$ and $A_u \cong A$. The localization theorem gives no change of affine scheme.

Example 18.19 (A nilpotent gives the empty affine scheme)**.** If f is nilpotent, then $D(f) = \varnothing$ and $A_f = 0$, so the localization theorem reads

$$\varnothing \cong \operatorname{Spec} 0.$$

Example 18.20 (Iterated localization)**.** For $f, g \in A$,

$$D(g/1) \subseteq \operatorname{Spec} A_f$$

corresponds to $D(fg) \subseteq \operatorname{Spec} A$, and

$$(A_f)_{g/1} \cong A_{fg}.$$

Example 18.21 (An affine local scheme)**.** For the local ring $A = k[x]_{(x)}$, the affine scheme $\operatorname{Spec} A$ has unique closed point $(x)A$, but it also has the generic point (0).

Example 18.22 (Recovering a ring from its affine scheme)**.** If $X = \operatorname{Spec} k[x, y]/(x^2 - y^3)$, then

$$\Gamma(X, \mathcal{O}_X) = k[x, y]/(x^2 - y^3).$$

Thus the coordinate ring is an invariant of the affine scheme.

Example 18.23 (A quotient ring still gives an affine scheme)**.** For any ideal $I \subseteq A$, the ring A/I defines the affine scheme $\operatorname{Spec}(A/I)$. The interpretation as a closed subscheme of $\operatorname{Spec} A$ is postponed to VI/22.

Example 18.24 (Affine scheme attached to a classical conic)**.** The classical curve $x^2 + y^2 = 1$ has coordinate ring

$$A = k[x, y]/(x^2 + y^2 - 1),$$

and its affine-scheme enhancement is $\operatorname{Spec} A$.

Example 18.25 (Closed points need not have residue field k)**.** For $X = \operatorname{Spec} \mathbb{R}[t]$, the maximal ideal $(t^2 + 1)$ defines a closed point with

$$\kappa(t^2 + 1) \cong \mathbb{C}.$$

The affine scheme naturally remembers this extension field.

Example 18.26 (A principal affine neighborhood)**.** Let $\mathfrak{p} \in \operatorname{Spec} A$ and choose $f \notin \mathfrak{p}$. Then

$$\mathfrak{p} \in D(f) \cong \operatorname{Spec} A_f.$$

Thus every point of an affine scheme has arbitrarily small affine neighborhoods from the distinguished-open basis.

18.15 Advanced worked examples

Example 18.27 (Advanced: Product decompositions)**.** For a finite product $A = A_1 \times \cdots \times A_r$,

$$\operatorname{Spec} A \cong \coprod_{i=1}^{r} \operatorname{Spec} A_i.$$

Each component is open and closed, and the structure sheaf restricts to the corresponding factor. Affine disconnectedness is therefore encoded by idempotents.

Example 18.28 (Advanced: One-point spectra with different infinitesimal thickness)**.** For every $n \geq 2$, the ring $k[\varepsilon]/(\varepsilon^n)$ has a one-point spectrum, but its affine scheme is distinguished by its global ring and nilpotent filtration.

Example 18.29 (Advanced: Distinguished opens are stable under intersection)**.** Using $D(f) \cap D(g) = D(fg)$,

$$\operatorname{Spec} A_f \cap \operatorname{Spec} A_g \quad \text{inside } \operatorname{Spec} A$$

is represented by $\operatorname{Spec} A_{fg}$. This makes principal affine charts exceptionally easy to compare.

Example 18.30 (Advanced: A finite principal cover)**.** If $f_1, \ldots, f_r$ generate the unit ideal, then

$$\operatorname{Spec} A = D(f_1) \cup \cdots \cup D(f_r).$$

For $A = k[x]$ one has $D(x) \cup D(1 - x) = \mathbb{A}^1_k$ because $x + (1 - x) = 1$.

Example 18.31 (Advanced: Why affine schemes are quasi-compact)**.** Even if an affine scheme has many points, an arbitrary open cover can be refined to principal opens and then reduced to finitely many by the unit-ideal criterion. This is an algebraic compactness phenomenon, not Hausdorff compactness.

Example 18.32 (Advanced: A local ring with many points)**.** Let $A = k[x, y]_{(x,y)}$. The closed point is $(x, y)A$, but the primes (0), (x), (y), and many height-one primes also remain. Local means one closed point, not one point total.

Example 18.33 (Advanced: Generic point of an irreducible affine scheme)**.** If A is a domain, (0) is a point of $\operatorname{Spec} A$ whose closure is all of $\operatorname{Spec} A$. Its local ring is $\operatorname{Frac}(A)$. The broader integral-scheme theory is reserved for VI/27.

Example 18.34 (Advanced: Affineness survives localization)**.** Any iterated localization $S^{-1}A$ defines an affine scheme. If S is generated by one element this is a distinguished open; for general S it is the inverse limit of the corresponding principal-open localizations at the ring level, while its spectrum consists of primes disjoint from S.

Example 18.35 (Advanced: Topology does not determine affineness data)**.** $\operatorname{Spec} k$ and $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ are homeomorphic, but the global section rings differ. Any test of affine-scheme isomorphism that ignores the sheaf therefore fails immediately.

Example 18.36 (Advanced: The canonical affine reconstruction)**.** If X is known to be affine, choose an isomorphism $X \cong \operatorname{Spec} A$. Then $\Gamma(X, \mathcal{O}_X) \cong A$, so composing with the induced spectrum isomorphism gives

$$X \cong \operatorname{Spec} \Gamma(X, \mathcal{O}_X).$$

No arbitrary choice of coordinate ring survives in the final expression.

Example 18.37 (Advanced: The projective line has only constant global functions)**.** Cover $\mathbb{P}^1_k$ by $U_0 \cong \operatorname{Spec} k[t]$ and $U_1 \cong \operatorname{Spec} k[u]$, with $u = t^{-1}$ on the overlap. A global regular function is a pair

$$f(t) \in k[t], \qquad g(u) \in k[u]$$

that agree in $k[t, t^{-1}]$. Hence f belongs to both $k[t]$ and $k[t^{-1}]$, so $f \in k$. Thus $\Gamma(\mathbb{P}^1_k, \mathcal{O}) = k$.

Example 18.38 (Advanced: Non-affineness of $\mathbb{P}^1$)**.** If $\mathbb{P}^1_k$ were affine, affine reconstruction would give

$$\mathbb{P}^1_k \cong \operatorname{Spec} \Gamma(\mathbb{P}^1_k, \mathcal{O}) = \operatorname{Spec} k,$$

a one-point scheme. Since $\mathbb{P}^1_k$ has many points, this is impossible.

Example 18.39 (Advanced: The punctured affine plane)**.** Let $U = \mathbb{A}^2_k \setminus \{(x, y)\} = D(x) \cup D(y)$. The sheaf equalizer gives

$$\Gamma(U, \mathcal{O}) = k[x, y]_x \cap k[x, y]_y = k[x, y]$$

inside $k(x, y)$. This is the starting calculation for the classical fact that U is not affine; a complete proof is posed as a Challenge because it uses the canonical map to $\operatorname{Spec} \Gamma(U, \mathcal{O})$.

Example 18.40 (Advanced: Affine charts as a basis for later scheme theory)**.** The theorem $D(f) \cong \operatorname{Spec} A_f$ shows that an affine scheme already carries many overlapping affine models. VI/20 will reverse this perspective: compatible affine models can be glued to construct spaces that need not themselves be affine.

18.16 Exercises

Exercise 18.41 (Definition check)**.** State the data of the affine scheme $\operatorname{Spec} A$ and identify its stalk at $\mathfrak{p}$.

Exercise 18.42 (Global sections)**.** Prove $\Gamma(\operatorname{Spec} A, \mathcal{O}) = A$.

Exercise 18.43 (Affine reconstruction)**.** If X is affine, show $X \cong \operatorname{Spec} \Gamma(X, \mathcal{O}_X)$.

Exercise 18.44 (Field spectrum)**.** Describe $\operatorname{Spec} k$ as an affine scheme for a field k.

Exercise 18.45 (Empty affine scheme)**.** Show $\operatorname{Spec} 0 = \varnothing$.

Exercise 18.46 (Affine line)**.** List the generic and closed points of $\operatorname{Spec} k[t]$ when k is algebraically closed.

Exercise 18.47 (Arithmetic line)**.** Describe the points and residue fields of $\operatorname{Spec} \mathbb{Z}$.

Exercise 18.48 (Principal open)**.** For $A = k[x, y]$, identify $D(x)$ with an affine spectrum.

Exercise 18.49 (Intersection)**.** Show $D(f) \cap D(g) \cong \operatorname{Spec} A_{fg}$.

Exercise 18.50 (Unit principal open)**.** Show $D(u) = \operatorname{Spec} A$ for $u \in A^\times$.

Exercise 18.51 (Nilpotent principal open)**.** Show $D(f) = \varnothing$ for nilpotent f and identify A_f.

Exercise 18.52 (Finite cover criterion)**.** Prove that the distinguished opens $D(f_1), \ldots, D(f_r)$ cover $\operatorname{Spec} A$ exactly when the ideal $(f_1, \ldots, f_r)$ is A.

Exercise 18.53 (Quasi-compactness)**.** Deduce that every affine scheme is quasi-compact.

Exercise 18.54 (Product ring)**.** Describe $\operatorname{Spec}(A \times B)$ and its two open-and-closed pieces.

Exercise 18.55 (Local ring geometry)**.** Prove A is local iff $\operatorname{Spec} A$ has a unique closed point.

Exercise 18.56 (Dual numbers)**.** Compare $\operatorname{Spec} k$ and $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ as spaces and as affine schemes.

Exercise 18.57 (Laurent line)**.** Show $D(t) \subseteq \operatorname{Spec} k[t]$ is $\operatorname{Spec} k[t, t^{-1}]$.

Exercise 18.58 (Arithmetic localization)**.** Identify $D(30) \subseteq \operatorname{Spec} \mathbb{Z}$ with a spectrum and list the closed points removed.

Exercise 18.59 (Iterated localization)**.** Prove $(A_f)_{g/1} \cong A_{fg}$.

Exercise 18.60 (Residue field)**.** Compute the residue field at $(t^2 + 1) \in \operatorname{Spec} \mathbb{R}[t]$.

Exercise 18.61 (Classical affine set)**.** For $A = k[x, y]/(y - x^2)$, describe the closed points and generic point of $\operatorname{Spec} A$.

Exercise 18.62 (One-point criterion)**.** Give a necessary and sufficient condition on the prime ideals of A for $\operatorname{Spec} A$ to have one point.

Exercise 18.63 (Principal neighborhoods)**.** Show every point of an affine scheme has a basis of affine neighborhoods.

Exercise 18.64 (Recover an isomorphism invariant)**.** Explain why nonisomorphic global rings imply nonisomorphic affine schemes.

Exercise 18.65 (Projective-line global functions)**.** Using the two standard affine charts, prove $\Gamma(\mathbb{P}^1_k, \mathcal{O}) = k$.

Exercise 18.66 (Boundary to VI/19)**.** State what extra data beyond a continuous map of spectra are required for a morphism of affine schemes, without proving the VI/19 universal property.

18.17 Problems

Problem 18.67 (Legacy AG3 Problem 11: The basic open subscheme $D(f)$)**.** Let A be a ring, $f \in A$, and $X = \operatorname{Spec} A$. Prove, as locally ringed spaces,

$$\boxed{(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f.}$$

Your proof must identify the underlying topological spaces, compare the structure sheaves on a basis, verify compatibility of restriction maps, and identify the local rings on stalks.

Provenance. Legacy recovery: theory-of-algebraic-geometry-3.tex, Problem 11. Preserved in full and expanded without changing its mathematical center.

Problem 18.68 (Legacy AG6 Problem 6.4: A scheme that is not affine)**.** Show that the projective line $\mathbb{P}^1_k$ is not affine. Give a self-contained proof using its two standard affine charts and the affine reconstruction principle $X \cong \operatorname{Spec}\Gamma(X, \mathcal{O}_X)$ valid for affine X.

Provenance. Legacy recovery: theory-of-algebraic-geometry-6.tex, Problem 6, subproblem 4. The multi-property corpus problem is split; only the affineness subproblem is consumed here.

Problem 18.69 (Recover the coordinate ring)**.** Let X be affine. Prove that the isomorphism class of $\Gamma(X, \mathcal{O}_X)$ is determined by X, and that any presentation $X \cong \operatorname{Spec} A$ yields $A \cong \Gamma(X, \mathcal{O}_X)$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.70 (Principal opens give an affine basis)**.** Using the principal-open basis theorem from VI/08 together with VI.18.P1, prove that every affine scheme has a basis of affine open subschemes of the form

$$D(f) \cong \operatorname{Spec} A_f.$$

Provenance. New canonical synthesis combining VI.08.P1 (legacy AG1 Problem 1) with VI.18.P1. The earlier basis theorem is cited rather than re-migrated; no additional legacy Problem is consumed.

Problem 18.71 (Affine schemes are quasi-compact)**.** Prove directly that every affine scheme is quasi-compact by converting a principal-open cover into an equation $1 = \sum a_i f_i$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.72 (Finite principal-cover criterion)**.** For $f_1, \ldots, f_r \in A$, prove $\operatorname{Spec} A = \bigcup_i D(f_i)$ iff $(f_1, \ldots, f_r) = A$, and formulate the infinite-family version.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.73 (Product rings and disjoint unions)**.** Prove $\operatorname{Spec}(A \times B) \cong \operatorname{Spec} A \sqcup \operatorname{Spec} B$ as affine schemes, including the structure sheaf.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.74 (The empty affine scheme)**.** Show that $\operatorname{Spec} A$ is empty iff A is the zero ring.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.75 (Local rings and unique closed points)**.** Prove that A is local iff $\operatorname{Spec} A$ has exactly one closed point.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.76 (One-point affine schemes)**.** Characterize rings whose spectra have exactly one point and give three nonisomorphic examples with the same one-point topology.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.77 (Nilpotent thickenings)**.** Compare $\operatorname{Spec} A$ and $\operatorname{Spec} A_{\mathrm{red}}$ topologically and explain, using global sections, why they need not be isomorphic as affine schemes.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.78 (The affine cross)**.** Analyze $X = \operatorname{Spec} k[x, y]/(xy)$: find its two generic component points, their closures, the origin, and principal affine opens $D(x)$ and $D(y)$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.79 (Arithmetic affine geometry)**.** For $X = \operatorname{Spec} \mathbb{Z}$, identify $D(n)$, its coordinate ring, its closed points, and the local rings at the remaining closed points.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.80 (Laurent tori as distinguished opens)**.** Show that the distinguished open

$$D(x_1 \cdots x_n) \subseteq \mathbb{A}^n_k$$

is the affine scheme

$$\operatorname{Spec} k[x_1^{\pm 1}, \ldots, x_n^{\pm 1}].$$

Provenance. New canonical derivative of the VI.18.P1 distinguished-open/localization theorem; no additional legacy Problem is consumed.

Problem 18.81 (Affine neighborhood calculations)**.** Given $\mathfrak{p} \in \operatorname{Spec} A$ and $a/s \in A_{\mathfrak{p}}$, construct a distinguished affine neighborhood on which the germ is represented by a regular section.

Provenance. New canonical derivative of the VI.18.P1 distinguished-open/localization theorem; no additional legacy Problem is consumed.

Problem 18.82 (Residue fields across an affine scheme)**.** For $A = \mathbb{Z}[t]$, compute residue fields at (0), (p), and $(p, t - a)$, explaining how the fields vary with specialization.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.83 (Classical affine variety versus spectrum)**.** For an algebraically closed field k and radical ideal $I \subseteq k[x_1, \ldots, x_n]$, compare $V(I)$ with $\operatorname{Spec}(k[x_1, \ldots, x_n]/I)$ and identify exactly which points correspond to classical points.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.84 (Affine schemes with identical topology)**.** Construct two affine schemes with homeomorphic underlying spaces but nonisomorphic structure sheaves and prove they are not isomorphic.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 18.85 (Distinguished-open transitivity)**.** Starting with $D(f) \cong \operatorname{Spec} A_f$, prove that a distinguished open $D(g/1)$ inside it corresponds to $D(fg)$ inside $\operatorname{Spec} A$, including the sheaf identification.

Provenance. New canonical transitivity variant of VI.18.P1; this is not a second migration of AG3 Problem 11.

Problem 18.86 (Affine cover does not imply affine)**.** Explain precisely why the standard two-affine cover of $\mathbb{P}^1$ does not contradict its non-affineness, and identify which global invariant fails to match a single spectrum.

Provenance. New canonical reinforcement of VI.18.P2; this is not a second migration of AG6 Problem 6.4.

18.18 Challenges

Problem 18.87 (Challenge: The punctured affine plane)**.** Let $U = \mathbb{A}^2_k \setminus \{(0,0)\} = D(x) \cup D(y)$. Prove

$$\Gamma(U, \mathcal{O}_U) = k[x,y]$$

by an explicit intersection calculation. Then, using the canonical map to $\operatorname{Spec}\Gamma(U, \mathcal{O}_U)$, prove that U is not affine.

Problem 18.88 (Challenge: Projective space is not affine)**.** Generalize the $\mathbb{P}^1_k$ argument to show that $\mathbb{P}^n_k$ is not affine for $n \geq 1$, after proving that every global regular function is constant.

Problem 18.89 (Challenge: Finite affine schemes)**.** Classify the topology of $\operatorname{Spec} A$ when A is an Artinian ring. Show that it is a finite discrete set of closed points and relate the decomposition to the product decomposition of A into Artinian local rings.

Problem 18.90 (Challenge: Reconstruct the sheaf from the principal basis)**.** Starting only from the assignments $D(f) \mapsto A_f$ and the localization restriction maps, reconstruct the entire structure sheaf on $\operatorname{Spec} A$ and prove uniqueness.

Problem 18.91 (Challenge: When is a principal open all or nothing?)**.** Prove

$$D(f) = \operatorname{Spec} A \iff f \in A^\times, \qquad D(f) = \varnothing \iff f \in \sqrt{(0)}.$$

Interpret both statements as extreme affine localizations.

Summary

An affine scheme is one globally controlled by a ring:

$$\boxed{X = \operatorname{Spec} A = (\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A}).}$$

Its fundamental computations are

$$\Gamma(X, \mathcal{O}_X) = A, \qquad \mathcal{O}_{X,\mathfrak{p}} = A_{\mathfrak{p}}, \qquad \kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}),$$

and its basic open charts satisfy

$$\boxed{D(f) \cong \operatorname{Spec} A_f.}$$

Distinguished opens form an affine basis and affine schemes are quasi-compact. The ring is recoverable from global sections, while topology alone is insufficient to recover the scheme. The legacy ledger now records AG3 Problem 11 as VI.18.P1 and the non-affineness subproblem AG6 Problem 6.4 as VI.18.P2, while AG3 Problems 12–14 remain reserved for their later canonical destinations.

18.19 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 18.92 (Global ring)**.** Compute global sections of $\operatorname{Spec} A$.

Hint. Use affine schemes, global sections, and distinguished opens. □

Solution. A. □

Exercise 18.93 (Basic open)**.** Identify $D(f)$ as an affine scheme.

Hint. Use affine schemes, global sections, and distinguished opens. □

Solution. $\operatorname{Spec} A_f$. □

Exercise 18.94 (Field)**.** Describe $\operatorname{Spec} k$.

Hint. Use affine schemes, global sections, and distinguished opens. □

Solution. A one-point affine scheme. □

Exercise 18.95 (Product)**.** Describe $\operatorname{Spec}(A \times B)$.

Hint. Use affine schemes, global sections, and distinguished opens. □

Solution. The disjoint union of the two affine spectra. □

Exercise 18.96 (Zero ring)**.** Describe $\operatorname{Spec} 0$.

Hint. Use affine schemes, global sections, and distinguished opens. □

Solution. The empty affine scheme. □

Proofs

Exercise 18.97 (Distinguished affine)**.** Prove: Prove $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$.

Hint. Work directly from affine schemes, global sections, and distinguished opens. □

Solution. Use prime correspondence for localization and compare the structure sheaves on the distinguished-open bases. □

Exercise 18.98 (Global recovery)**.** Prove: Prove $\Gamma(\operatorname{Spec} A, \mathcal{O}) = A$.

Hint. Work directly from affine schemes, global sections, and distinguished opens. □

Solution. Use $D(1) = \operatorname{Spec} A$ and the basic-open section formula. □

Exercise 18.99 (Quasi-compact)**.** Prove: Prove every affine scheme is quasi-compact.

Hint. Work directly from affine schemes, global sections, and distinguished opens. □

Solution. Refine an open cover by distinguished opens and use the unit-ideal finite-generation argument. □

Exercise 18.100 (Product disjoint)**.** Prove: Prove Spec($A \times B$) is the disjoint union of the two spectra as locally ringed spaces.

Hint. Work directly from affine schemes, global sections, and distinguished opens. ☐

Solution. Classify primes using idempotents and compare localizations on each component. ☐

Counterexamples and hypothesis tests

Exercise 18.101 (Locally affine)**.** Test the claim: every locally affine open subset of an affine scheme is itself affine.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general. ☐

Exercise 18.102 (Topology determines ring)**.** Test the claim: an affine scheme is determined by its underlying topological space.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; the structure sheaf is essential. ☐

Exercise 18.103 (All opens affine)**.** Test the claim: every open subset of an affine scheme is affine.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Applications and investigations

Exercise 18.104 (Local models)**.** Why are affine schemes the local building blocks of schemes?

Hint. Translate the question into affine schemes, global sections, and distinguished opens. ☐

Solution. General schemes are covered by opens isomorphic to affine spectra. ☐

Exercise 18.105 (Computational bridge)**.** Why are affine schemes computationally convenient?

Hint. Translate the question into affine schemes, global sections, and distinguished opens. ☐

Solution. Global geometry is encoded by one ring and localization describes basic opens. ☐

Challenge problems

Exercise 18.106 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Affine Schemes in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Affine Schemes, translates it through affine schemes, global sections, and distinguished opens, and checks the correspondence in both algebraic and geometric language. □

Exercise 18.107 (Hypothesis challenge)**.** Choose one structural statement from Affine Schemes, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 19

Morphisms of Affine Schemes

Purpose and learning goals

VI/18 constructed affine schemes from rings. The next question is how maps between these geometric objects are encoded algebraically. The answer is one of the central reversals of algebraic geometry:

$$\boxed{\varphi : A \longrightarrow B \qquad \Longleftrightarrow \qquad \operatorname{Spec} B \longrightarrow \operatorname{Spec} A.}$$

The direction reverses because a point of $\operatorname{Spec} B$ is a prime ideal of B, and a ring map $A \to B$ sends it back to its contraction in A.

The chapter is organized around the one numbered legacy Problem assigned to the direct VI/19 corpus scope: AG3 Problem 12, the universal property

$$\boxed{\operatorname{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} A) \cong \operatorname{Hom}_{\mathrm{Ring}}(A, \Gamma(X, \mathcal{O}_X)).}$$

For affine $X = \operatorname{Spec} B$, this becomes the familiar affine anti-equivalence

$$\operatorname{Hom}_{\mathrm{AffSch}}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \operatorname{Hom}_{\mathrm{Ring}}(A, B).$$

The proof is carried out at locally-ringed-space level, so it does not require the general scheme definition reserved for VI/20–VI/21.

After this chapter, the reader should be able to:

- define morphisms of locally ringed spaces and affine schemes;
- construct $\operatorname{Spec}(\varphi)$ from a ring homomorphism $\varphi : A \to B$;
- prove $\operatorname{Spec}(\varphi)^{-1}(D(a)) = D(\varphi(a))$;
- construct the induced morphism of structure sheaves and the local maps on stalks;
- prove functoriality and contravariance of Spec;
- recover a morphism of affine schemes uniquely from its map on global sections;
- prove the AG3 Problem 12 universal property in full;
- recognize maps to the universal affine line $\operatorname{Spec} \mathbb{Z}[t]$ as global regular functions;
- detect affine-scheme isomorphisms by ring isomorphisms;
- compute morphisms induced by polynomial, quotient, and localization maps;
- compute the induced maps of residue fields;

- distinguish underlying continuous maps from morphisms of locally ringed spaces;
- keep gluing, structure morphisms, closed subschemes, fiber products, and base change at their later canonical destinations.

$$\begin{tikzcd}
\boxed{\begin{matrix}\text{Affine schemes}\\ \text{VI/18}\end{matrix}} \arrow[r] & \boxed{\begin{matrix}\text{Affine morphisms}\\ \text{VI/19}\end{matrix}} \arrow[r] & \boxed{\begin{matrix}\text{Gluing}\\ \text{VI/20}\end{matrix}} \arrow[d] \\
& & \boxed{\begin{matrix}\text{Schemes and points}\\ \text{VI/21}\end{matrix}}
\end{tikzcd}$$

19.1 Morphisms of locally ringed spaces

A morphism of ringed spaces

$$f : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

consists of a continuous map $f : X \to Y$ together with a morphism of sheaves of rings

$$f^\sharp : \mathcal{O}_Y \longrightarrow f_*\mathcal{O}_X.$$

Equivalently, for every open $V \subseteq Y$ there is a compatible ring homomorphism

$$f^\sharp_V : \mathcal{O}_Y(V) \longrightarrow \mathcal{O}_X(f^{-1}V).$$

It is a morphism of *locally* ringed spaces when, for each $x \in X$, the induced map on stalks

$$f^\sharp_x : \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x}$$

is local: the inverse image of the maximal ideal of $\mathcal{O}_{X,x}$ is the maximal ideal of $\mathcal{O}_{Y,f(x)}$.

An affine-scheme morphism is simply a morphism of locally ringed spaces between affine schemes.

$$\begin{tikzcd}
(X, \mathcal{O}_X) \arrow[r, "f"] & (Y, \mathcal{O}_Y) \\
\mathcal{O}_Y \arrow[r, "f^\sharp"'] & f_*\mathcal{O}_X
\end{tikzcd}
\qquad \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x} \text{ local.}$$

19.2 A ring homomorphism gives a map of spectra

Let

$$\varphi : A \longrightarrow B$$

be a ring homomorphism. For each $\mathfrak{p} \in \operatorname{Spec} B$, contraction gives a prime ideal

$$\varphi^{-1}(\mathfrak{p}) \in \operatorname{Spec} A.$$

Thus define

$$f = \operatorname{Spec}(\varphi) : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A, \qquad \mathfrak{p} \longmapsto \varphi^{-1}(\mathfrak{p}).$$

The point map already exhibits the contravariance: arrows of rings and arrows of affine schemes point in opposite directions.

$$\begin{tikzcd}
A \arrow[r, "\varphi"] & B \\
\operatorname{Spec} A & \operatorname{Spec} B \arrow[l, "\operatorname{Spec}(\varphi)"']
\end{tikzcd}
\qquad \mathfrak{p} \longmapsto \varphi^{-1}(\mathfrak{p}).$$

19.3 Continuity and principal opens

The point map is continuous for an especially transparent reason. If $a \in A$, then

$$\begin{aligned} \mathfrak{p} \in f^{-1}(D(a)) &\iff a \notin \varphi^{-1}(\mathfrak{p}) \\ &\iff \varphi(a) \notin \mathfrak{p} \\ &\iff \mathfrak{p} \in D(\varphi(a)). \end{aligned}$$

Hence

$$\boxed{f^{-1}(D(a)) = D(\varphi(a)).}$$

Since principal opens form a basis, f is continuous.

$$\begin{tikzcd} D(\varphi(a)) \subseteq \operatorname{Spec} B \arrow[r, hook] \arrow[d] & \operatorname{Spec} B \arrow[d, "\operatorname{Spec}(\varphi)"] \\ D(a) \subseteq \operatorname{Spec} A \arrow[r, hook] & \operatorname{Spec} A \end{tikzcd} \qquad f^{-1}(D(a)) = D(\varphi(a)).$$

Example 19.1 (Squaring morphism)**.** The ring map $k[u] \to k[t]$, $u \mapsto t^2$, induces the scheme morphism $\mathbb{A}^1_k \to \mathbb{A}^1_k$ sending a prime $\mathfrak{p}$ to its inverse image. On closed k-points this is the familiar map $t \mapsto t^2$.

19.4 The map on structure sheaves

On a principal open $D(a) \subseteq \operatorname{Spec} A$, the structure sheaf gives A_a. Its inverse image is $D(\varphi(a)) \subseteq \operatorname{Spec} B$, whose section ring is $B_{\varphi(a)}$. The universal property of localization therefore gives

$$\varphi_a : A_a \longrightarrow B_{\varphi(a)}, \qquad \frac{r}{a^n} \longmapsto \frac{\varphi(r)}{\varphi(a)^n}.$$

These maps commute with further localization, hence with restriction maps. They therefore define the sheaf morphism

$$f^\sharp : \mathcal{O}_{\operatorname{Spec} A} \longrightarrow f_*\mathcal{O}_{\operatorname{Spec} B}.$$

19.5 The maps on stalks are local

Let $\mathfrak{p} \in \operatorname{Spec} B$ and put $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$. Passing to stalks gives

$$A_\mathfrak{q} \longrightarrow B_\mathfrak{p}, \qquad \frac{a}{s} \longmapsto \frac{\varphi(a)}{\varphi(s)}.$$

If the image lies in the maximal ideal $\mathfrak{p}B_\mathfrak{p}$, then $\varphi(a) \in \mathfrak{p}$, hence $a \in \mathfrak{q}$. Conversely $a \in \mathfrak{q}$ clearly maps into $\mathfrak{p}B_\mathfrak{p}$. Thus the inverse image of the maximal ideal is $\mathfrak{q}A_\mathfrak{q}$, so the stalk map is local.

$$\begin{tikzcd} A_\mathfrak{q} \arrow[r] & B_\mathfrak{p} \\ \mathfrak{q}A_\mathfrak{q} \arrow[r, mapsto] & \mathfrak{p}B_\mathfrak{p} \end{tikzcd} \qquad \mathfrak{q} = \varphi^{-1}(\mathfrak{p}).$$

19.6 Functoriality of the spectrum construction

For ring maps

$$A \xrightarrow{\varphi} B \xrightarrow{\psi} C$$

one has

$$\operatorname{Spec}(\psi \circ \varphi) = \operatorname{Spec}(\varphi) \circ \operatorname{Spec}(\psi),$$

and $\operatorname{Spec}(\mathrm{id}_A) = \mathrm{id}_{\operatorname{Spec} A}$. The same identities hold for the sheaf maps because localization respects composition. Thus

$$\operatorname{Spec} : (\mathbf{CRing})^{\mathrm{op}} \longrightarrow \mathbf{AffSch}$$

is a contravariant functor.

$$A \xrightarrow{\quad\varphi\quad} B \xrightarrow{\quad\psi\quad} C$$

$$\operatorname{Spec} A \xleftarrow{\operatorname{Spec}\varphi} \operatorname{Spec} B \xleftarrow{\operatorname{Spec}\psi} \operatorname{Spec} C$$

Example 19.2 (Closed immersion)**.** A quotient map $A \to A/I$ induces $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$. Its image is $V(I)$ and the morphism is a closed immersion.

19.7 Affine schemes and rings form opposite categories

If $X = \operatorname{Spec} B$ and $Y = \operatorname{Spec} A$, every ring map $A \to B$ gives an affine-scheme morphism $X \to Y$. Conversely, a morphism $f : X \to Y$ induces on global sections a ring map

$$\Gamma(Y, \mathcal{O}_Y) = A \longrightarrow \Gamma(X, \mathcal{O}_X) = B.$$

The two constructions are inverse. Therefore

$$\boxed{\mathrm{Hom}_{\mathrm{AffSch}}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \mathrm{Hom}_{\mathrm{Ring}}(A, B).}$$

In particular, an affine-scheme morphism is completely determined by its pullback on global regular functions.

$$\boxed{\begin{array}{c}\mathbf{CRing}^{\mathrm{op}}\\ \text{ring homomorphisms}\end{array}} \xleftrightarrow{\ \operatorname{Spec},\,\Gamma \text{ quasi-inverse on affine objects}\ } \boxed{\begin{array}{c}\mathbf{AffSch}\\ \text{affine-scheme morphisms}\end{array}}$$

19.8 The universal property of $\operatorname{Spec} A$ — AG3 Problem 12

The affine statement admits a stronger formulation. Let $(X, \mathcal{O}_X)$ be any locally ringed space. A morphism

$$f : X \longrightarrow \operatorname{Spec} A$$

induces a ring map on global sections

$$A = \Gamma(\operatorname{Spec} A, \mathcal{O}) \xrightarrow{\Gamma(f^\sharp)} \Gamma(X, \mathcal{O}_X).$$

Conversely, given

$$\alpha : A \longrightarrow \Gamma(X, \mathcal{O}_X),$$

for each $x \in X$ compose with the germ map to obtain

$$A \xrightarrow{\alpha} \Gamma(X, \mathcal{O}_X) \longrightarrow \mathcal{O}_{X,x}.$$

Since $\mathcal{O}_{X,x}$ is local with maximal ideal $\mathfrak{m}_x$, define

$$f_\alpha(x) = \{a \in A : \alpha(a)_x \in \mathfrak{m}_x\}.$$

This is a prime ideal. The section $\alpha(a)$ is a unit at x exactly when $x \in f_\alpha^{-1}(D(a))$, so

$$f_\alpha^{-1}(D(a)) = \{x : \alpha(a)_x \in \mathcal{O}_{X,x}^\times\},$$

which is open. Localizing α on these opens constructs the required sheaf map, and the stalk maps are local by construction. Uniqueness follows because principal opens form a basis and a local stalk map forces the image prime at every point.

Thus

$$\boxed{\mathrm{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} A) \cong \mathrm{Hom}_{\mathrm{Ring}}(A, \Gamma(X, \mathcal{O}_X)).}$$

Problem VI.19.P1 gives the complete proof with all continuity, sheaf, stalk, uniqueness, and naturality steps. This is the canonical migration of legacy AG3 Problem 12.

$$\begin{tikzcd} X \arrow[r, "f"] & \operatorname{Spec} A \\ \Gamma(X, \mathcal{O}_X) & A \arrow[l, "f^\sharp_X"'] \end{tikzcd} \qquad f \longleftrightarrow (A \to \Gamma(X, \mathcal{O}_X)).$$

19.9 Maps to affine space and global functions

Taking $A = \mathbb{Z}[t]$ in the universal property gives, for any locally ringed space X,

$$\mathrm{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} \mathbb{Z}[t]) \cong \Gamma(X, \mathcal{O}_X).$$

If one works over a fixed field k and restricts to ring maps fixing k, the same idea says that a map to affine n-space is encoded by n global functions. In the affine case $X = \operatorname{Spec} B$, a tuple $(b_1, \ldots, b_n) \in B^n$ corresponds to

$$k[x_1, \ldots, x_n] \longrightarrow B, \qquad x_i \longmapsto b_i.$$

$$\begin{tikzcd} k[x_1, \ldots, x_n] \arrow[r, "x_i \mapsto b_i"] & B \\ \mathbb{A}^n_k & \operatorname{Spec} B \arrow[l] \end{tikzcd}$$

19.10 Isomorphisms

Because the Hom correspondence is functorial, a ring isomorphism $A \cong B$ induces an affine-scheme isomorphism $\operatorname{Spec} B \cong \operatorname{Spec} A$, and conversely every affine-scheme isomorphism induces an isomorphism on global sections. Hence

$$\boxed{\operatorname{Spec} A \cong \operatorname{Spec} B \text{ as affine schemes} \iff A \cong B \text{ as rings.}}$$

This strengthens VI/18's reconstruction principle from objects to morphisms.

Example 19.3 (Open localization map)**.** The localization map $A \to A_f$ induces $\operatorname{Spec} A_f \to \operatorname{Spec} A$, identifying the source with the open subset $D(f)$.

19.11 Localization and quotient maps

Two important ring maps already have familiar topological meanings.

For localization $A \to A_f$,

$$\operatorname{Spec} A_f \longrightarrow \operatorname{Spec} A$$

identifies the source with the distinguished open $D(f)$, recovering VI.18.P1 now as a morphism statement.

For a quotient $A \to A/I$,

$$\operatorname{Spec}(A/I) \longrightarrow \operatorname{Spec} A$$

is injective on points with image $V(I)$. This chapter uses this only as an affine morphism and a statement about the underlying topology. The language and universal property of *closed subschemes* are deliberately reserved for VI/22.

$A \to A_f$ $\operatorname{Spec} A_f \cong D(f)$	$A \to A/I$ image $V(I)$
open-chart morphism	closed-set morphism (subscheme theory: VI/22)

19.12 Residue fields under a morphism

For $f = \operatorname{Spec}(\varphi) : \operatorname{Spec} B \to \operatorname{Spec} A$ and $\mathfrak{p} \in \operatorname{Spec} B$, with $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$, the local map

$$A_{\mathfrak{q}} \longrightarrow B_{\mathfrak{p}}$$

induces a field homomorphism

$$\boxed{\kappa(\mathfrak{q}) \longrightarrow \kappa(\mathfrak{p}).}$$

Thus scheme morphisms do not merely send points to points; they also compare the fields in which functions take values at those points.

19.13 The closure of the image

If $f = \operatorname{Spec}(\varphi) : \operatorname{Spec} B \to \operatorname{Spec} A$, then every point of the image contains $\ker \varphi$. More precisely,

$$\boxed{\overline{f(\operatorname{Spec} B)} = V(\ker \varphi).}$$

Indeed, the ideal of functions vanishing on the image is $\varphi^{-1}(\sqrt{(0)_B}) = \sqrt{\ker \varphi}$, which defines the same closed set as $\ker \varphi$. In particular, the image is dense exactly when $\ker \varphi$ is nilpotent in the radical sense $\sqrt{\ker \varphi} = \sqrt{(0)}$; equivalently $\ker \varphi \subseteq \sqrt{(0)_A}$.

19.14 Common pitfalls

1. Ring maps reverse direction under Spec: $A \to B$ gives $\operatorname{Spec} B \to \operatorname{Spec} A$.
2. A continuous map of spectra is not by itself an affine-scheme morphism; the compatible sheaf map and local stalk condition are essential.
3. The point map is contraction of primes, not extension of primes.

4. On stalks the relevant map is $A_{\varphi^{-1}(\mathfrak{p})} \to B_{\mathfrak{p}}$.
5. A morphism of affine schemes is determined by the global ring map, but this fact uses both the affine reconstruction theorem and the universal property.
6. Quotient maps are discussed here only as affine morphisms with image $V(I)$; the closed-subscheme formalism belongs to VI/22.
7. The canonical structure morphism and the statement that $\operatorname{Spec} \mathbb{Z}$ is final are AG3 Problem 13 and remain reserved for VI/21.
8. Gluing local affine morphisms into general scheme morphisms belongs to VI/20–VI/21; AG3 Problem 14 remains reserved for VI/20.
9. Fiber products and base change are not hidden inside the affine Hom correspondence; they receive their own treatment in VI/23–VI/24.

19.15 Worked examples

Example 19.4 (Identity map)**.** The identity ring map $A \to A$ induces $\mathrm{id}_{\operatorname{Spec} A}$. Both the point map and all localized section maps are identities.

Example 19.5 (A polynomial projection)**.** The inclusion $k[t] \hookrightarrow k[x, y]$, $t \mapsto x$, induces $\mathbb{A}^2_k \to \mathbb{A}^1_k$. On closed k-points it is $(a, b) \mapsto a$.

Example 19.6 (The squaring map on the affine line)**.** The homomorphism $k[t] \to k[x]$, $t \mapsto x^2$, induces $\mathbb{A}^1_k \to \mathbb{A}^1_k$. Over an algebraically closed field it sends the closed point $(x - a)$ to $(t - a^2)$.

Example 19.7 (A translation)**.** The automorphism $k[t] \to k[t]$, $t \mapsto t + c$, induces an automorphism of $\mathbb{A}^1_k$. On a closed point $(t - a)$, contraction gives $(t - (a + c))$, so the corresponding geometric point moves by $a \mapsto a + c$.

Example 19.8 (A quotient map)**.** For $A \to A/I$, the induced map $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$ sends $\mathfrak{p}/I$ to $\mathfrak{p}$ and has image $V(I)$.

Example 19.9 (A localization map)**.** The localization $A \to A_f$ induces $\operatorname{Spec} A_f \to \operatorname{Spec} A$ with image exactly $D(f)$; VI/18 identifies it with the inclusion of that open affine chart.

Example 19.10 (An arithmetic map)**.** The inclusion $\mathbb{Z} \to \mathbb{Z}[i]$ induces $\operatorname{Spec} \mathbb{Z}[i] \to \operatorname{Spec} \mathbb{Z}$. A prime of $\mathbb{Z}[i]$ is sent to its contraction in $\mathbb{Z}$. The general structure-morphism viewpoint is reserved for VI/21.

Example 19.11 (Dual numbers to a reduced point)**.** The quotient $k[\varepsilon]/(\varepsilon^2) \to k$ induces $\operatorname{Spec} k \to \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. Both spaces have one point, but the morphism is not an isomorphism because the ring map is not.

Example 19.12 (Reduction map)**.** The quotient $A \to A_{\mathrm{red}} = A/\sqrt{(0)}$ induces a morphism $\operatorname{Spec} A_{\mathrm{red}} \to \operatorname{Spec} A$ that is a homeomorphism on underlying spaces, but need not be an isomorphism of affine schemes.

Example 19.13 (Projection from a polynomial extension)**.** The inclusion $A \to A[t]$ induces $\operatorname{Spec} A[t] \to \operatorname{Spec} A$. Fibers of this map are postponed to VI/25, but contractions of primes can already be computed.

Example 19.14 (Evaluation at a scalar)**.** For $a \in k$, evaluation $k[t] \to k$, $t \mapsto a$, induces the map $\operatorname{Spec} k \to \mathbb{A}^1_k$ whose unique point lands at the closed point $(t - a)$.

Example 19.15 (A map to affine two-space)**.** A pair $b_1, b_2 \in B$ determines $k[x,y] \to B$ by $x \mapsto b_1$, $y \mapsto b_2$, hence a morphism $\operatorname{Spec} B \to \mathbb{A}^2_k$.

Example 19.16 (Pulling back a principal open)**.** For $\varphi : A \to B$ and $a \in A$, the inverse image of $D(a)$ is $D(\varphi(a))$. If $\varphi(a)$ is a unit, the entire source maps into $D(a)$.

Example 19.17 (Pulling back a global function)**.** A morphism $f : \operatorname{Spec} B \to \operatorname{Spec} A$ corresponding to $\varphi : A \to B$ sends the global section $a \in A$ to $\varphi(a) \in B$.

Example 19.18 (Residue field at an evaluation point)**.** For $k[t] \to k$, $t \mapsto a$, the induced point has residue-field map $k \to k$, the identity when the homomorphism fixes k.

Example 19.19 (A noninjective ring map with dense spectrum map)**.** If $A = k[\varepsilon]/(\varepsilon^2) \to k$ kills ε, then the induced map has dense (indeed full) one-point image because the kernel is nilpotent.

Example 19.20 (A surjective map has closed image)**.** If $A \twoheadrightarrow B$ has kernel I, then $\operatorname{Spec} B \to \operatorname{Spec} A$ has image $V(I)$, a closed subset.

Example 19.21 (An injective map need not be surjective on spectra)**.** The inclusion $\mathbb{Z} \hookrightarrow \mathbb{Q}$ induces $\operatorname{Spec} \mathbb{Q} \to \operatorname{Spec} \mathbb{Z}$ whose image is only the generic point (0).

Example 19.22 (Composition)**.** If $A \to B \to C$, then the morphism $\operatorname{Spec} C \to \operatorname{Spec} A$ equals the composite $\operatorname{Spec} C \to \operatorname{Spec} B \to \operatorname{Spec} A$.

Example 19.23 (Two morphisms with the same underlying point map)**.** On $A = k[\varepsilon]/(\varepsilon^2)$, the identity and the endomorphism $\varepsilon \mapsto 0$ induce the same map of the one-point topological space $\operatorname{Spec} A$, but different morphisms of affine schemes.

19.16 Advanced worked examples

Example 19.24 (Advanced: Affine anti-equivalence as a reconstruction theorem)**.** The assignments $A \mapsto \operatorname{Spec} A$ and $X \mapsto \Gamma(X, \mathcal{O}_X)$ are inverse up to canonical isomorphism on affine objects. Hence all affine-scheme morphism questions may be translated into ring-homomorphism questions and back.

Example 19.25 (Advanced: Universal property beyond affine sources)**.** For a locally ringed space X, a ring map $A \to \Gamma(X, \mathcal{O}_X)$ already determines a unique locally-ringed-space morphism $X \to \operatorname{Spec} A$. No affine hypothesis on X is required.

Example 19.26 (Advanced: Maps to affine line detect global sections)**.** Taking $A = \mathbb{Z}[t]$ gives $\operatorname{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} \mathbb{Z}[t]) \cong \Gamma(X, \mathcal{O}_X)$. Thus a global function may be viewed geometrically as a map to the universal affine line.

Example 19.27 (Advanced: Factorization through a quotient)**.** For $I \subseteq A$ and $f : X \to \operatorname{Spec} A$ corresponding to $\alpha : A \to \Gamma(X, \mathcal{O}_X)$, the map factors through $\operatorname{Spec}(A/I)$ exactly when $I \subseteq \ker \alpha$. Closed-subscheme language is postponed to VI/22.

Example 19.28 (Advanced: Factorization through a principal open)**.** The same f factors through $D(a) = \operatorname{Spec} A_a$ exactly when $\alpha(a)$ is a unit in $\Gamma(X, \mathcal{O}_X)$. This is the localization universal property translated geometrically.

Example 19.29 (Advanced: Closure of image)**.** For $\varphi : A \to B$, the closure of the image of $\operatorname{Spec} B \to \operatorname{Spec} A$ is $V(\ker \varphi)$. Hence a spectrum map may be dense even when the ring map is not injective, provided its kernel is nilpotent.

Example 19.30 (Advanced: Same topology, different sheaf maps)**.** The endomorphisms $\varepsilon \mapsto \varepsilon$ and $\varepsilon \mapsto 0$ of $k[\varepsilon]/(\varepsilon^2)$ give identical continuous maps on the one-point spectrum but different pullbacks on global sections. The sheaf component is essential.

Example 19.31 (Advanced: Automorphisms of the affine line)**.** A k-algebra automorphism of $k[t]$ has $t \mapsto at + b$ with $a \in k^\times$. Therefore affine-line automorphisms over k are exactly the corresponding affine coordinate changes.

Example 19.32 (Advanced: Morphisms into affine space as tuples)**.** For a k-algebra B, morphisms $\operatorname{Spec} B \to \mathbb{A}^n_k$ correspond to n-tuples of elements of B. Composition becomes substitution of polynomials.

Example 19.33 (Advanced: Contraction and specialization)**.** If $\mathfrak{p} \subseteq \mathfrak{p}'$ in B, then $\varphi^{-1}(\mathfrak{p}) \subseteq \varphi^{-1}(\mathfrak{p}')$ in A. Thus a spectrum morphism preserves specialization order.

Example 19.34 (Advanced: Local maps force the correct point)**.** Given a locally-ringed-space morphism $f : X \to \operatorname{Spec} A$, the point $f(x)$ can be recovered from the global pullback: it consists exactly of those $a \in A$ whose germ at x lands in the maximal ideal of $\mathcal{O}_{X,x}$.

Example 19.35 (Advanced: Residue-field extensions at non-rational points)**.** A morphism may send a point with residue field K to one with residue field k while inducing a nontrivial embedding $k \hookrightarrow K$. This is why scheme-valued points retain arithmetic information beyond the topological image.

Example 19.36 (Advanced: Localization transitivity)**.** For $A \to A_f \to (A_f)_{g/1} \cong A_{fg}$, the corresponding morphisms compose as $D(fg) \hookrightarrow D(f) \hookrightarrow \operatorname{Spec} A$. This is functoriality plus VI/18 localization.

Example 19.37 (Advanced: Boundary to gluing)**.** VI/19 explains affine morphisms entirely algebraically. VI/20 will ask when morphisms defined on overlapping affine charts agree and glue. The universal property here is the local algebraic engine for that later construction.

19.17 Exercises

Exercise 19.38 (Direction reversal)**.** For a ring map $\varphi : A \to B$, state the induced map of spectra and explain why its direction is reversed.

Exercise 19.39 (Contraction is prime)**.** Prove that $\varphi^{-1}(\mathfrak{p})$ is prime whenever $\mathfrak{p} \in \operatorname{Spec} B$.

Exercise 19.40 (Principal-open inverse image)**.** Show $\operatorname{Spec}(\varphi)^{-1}(D(a)) = D(\varphi(a))$.

Exercise 19.41 (Continuity)**.** Deduce that $\operatorname{Spec}(\varphi)$ is continuous.

Exercise 19.42 (Localized section map)**.** Write explicitly the map $A_a \to B_{\varphi(a)}$ induced by φ.

Exercise 19.43 (Stalk map)**.** For $\mathfrak{p} \in \operatorname{Spec} B$ and $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$, write the induced map $A_\mathfrak{q} \to B_\mathfrak{p}$.

Exercise 19.44 (Locality of the stalk map)**.** Prove the stalk map in the previous exercise is local.

Exercise 19.45 (Identity functoriality)**.** Show $\operatorname{Spec}(\mathrm{id}_A) = \mathrm{id}_{\operatorname{Spec} A}$ including the sheaf map.

Exercise 19.46 (Composition)**.** For $A \xrightarrow{\varphi} B \xrightarrow{\psi} C$, prove $\operatorname{Spec}(\psi\varphi) = \operatorname{Spec}(\varphi) \circ \operatorname{Spec}(\psi)$.

Exercise 19.47 (Polynomial projection)**.** Compute the map $\mathbb{A}^2_k \to \mathbb{A}^1_k$ induced by $k[t] \to k[x, y]$, $t \mapsto x$, on closed k-points.

Exercise 19.48 (Squaring map)**.** For $k[t] \to k[x]$, $t \mapsto x^2$, compute the image of the closed point $(x - a)$.

Exercise 19.49 (Localization image)**.** Show that $A \to A_f$ induces a spectrum map with image $D(f)$.

Exercise 19.50 (Quotient image)**.** Show that $A \to A/I$ induces an injective map on spectra with image $V(I)$.

Exercise 19.51 (Reduction map)**.** Explain why $\operatorname{Spec} A_{\mathrm{red}} \to \operatorname{Spec} A$ is a homeomorphism but may fail to be an affine-scheme isomorphism.

Exercise 19.52 (Maps to affine line)**.** Show that ring maps $\mathbb{Z}[t] \to B$ are in bijection with elements of B.

Exercise 19.53 (Maps to affine n-space)**.** For a k-algebra B, show

$$\operatorname{Hom}_{k\text{-alg}}(k[x_1, \dots, x_n], B) \cong B^n.$$

Interpret the bijection as evaluation on the coordinate generators.

Exercise 19.54 (Isomorphism criterion)**.** Prove that a morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ is an isomorphism iff its corresponding ring map $A \to B$ is an isomorphism.

Exercise 19.55 (Residue fields)**.** Construct the induced field map $\kappa(\varphi^{-1}(\mathfrak{p})) \to \kappa(\mathfrak{p})$.

Exercise 19.56 (Dense image criterion)**.** Show that the closure of the image of $\operatorname{Spec} B \to \operatorname{Spec} A$ is $V(\ker \varphi)$.

Exercise 19.57 (Dense but noninjective)**.** Give a noninjective ring map whose induced map on spectra has dense image.

Exercise 19.58 (Factor through a quotient)**.** For $\alpha : A \to B$, prove that $\operatorname{Spec} B \to \operatorname{Spec} A$ factors through $\operatorname{Spec}(A/I)$ iff $I \subseteq \ker \alpha$.

Exercise 19.59 (Factor through a principal open)**.** For $\alpha : A \to B$, prove $\operatorname{Spec} B \to \operatorname{Spec} A$ factors through $D(f) = \operatorname{Spec} A_f$ iff $\alpha(f)$ is a unit in B.

Exercise 19.60 (Same continuous map, different scheme maps)**.** Find two different endomorphisms of $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ with the same underlying continuous map.

Exercise 19.61 (Recover a morphism from global sections)**.** Let $f, g : \operatorname{Spec} B \to \operatorname{Spec} A$ induce the same map $A \to B$ on global sections. Prove $f = g$.

Exercise 19.62 (Universal property for a locally ringed source)**.** Let X be a locally ringed space and $s \in \Gamma(X, \mathcal{O}_X)$. Describe explicitly the morphism $X \to \operatorname{Spec} \mathbb{Z}[t]$ corresponding to $t \mapsto s$.

Exercise 19.63 (Boundary check)**.** State which neighboring legacy problems remain reserved after VI/19 and where they go.

19.18 Problems

Problem 19.64 (Legacy AG3 Problem 12: Universal property of affine schemes)**.** Let $(X, \mathcal{O}_X)$ be a locally ringed space and let A be a ring. Prove the natural bijection

$$\boxed{\mathrm{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} A) \cong \mathrm{Hom}_{\mathrm{Ring}}(A, \Gamma(X, \mathcal{O}_X)).}$$

Your proof must construct the point map from a ring homomorphism $A \to \Gamma(X, \mathcal{O}_X)$, prove continuity using principal opens, construct the sheaf morphism by localization, verify locality on stalks, prove uniqueness, and check that the two constructions are inverse.

Provenance. Legacy recovery: theory-of-algebraic-geometry-3.tex, Problem 12. Preserved at full universal-property depth and expanded without changing its mathematical center.

Problem 19.65 (Construct the affine morphism from a ring map)**.** Let $\varphi : A \to B$. Construct $\operatorname{Spec} B \to \operatorname{Spec} A$ as a morphism of locally ringed spaces, including the point map, maps on principal-open sections, and maps on stalks.

Provenance. Canonical derivative of AG3 Problem 12 / VI.19.P1; not an additional legacy migration.

Problem 19.66 (Functoriality of Spec)**.** For $A \xrightarrow{\varphi} B \xrightarrow{\psi} C$, prove $\operatorname{Spec}(\psi \circ \varphi) = \operatorname{Spec}(\varphi) \circ \operatorname{Spec}(\psi)$ as morphisms of locally ringed spaces, and prove the identity law.

Provenance. Canonical derivative of AG3 Problem 12 / VI.19.P1; not an additional legacy migration.

Problem 19.67 (Affine anti-equivalence)**.** Prove the natural bijection

$$\mathrm{Hom}_{\mathrm{AffSch}}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \mathrm{Hom}_{\mathrm{Ring}}(A, B),$$

and show that composition corresponds to reversed composition of ring maps.

Provenance. Canonical derivative of AG3 Problem 12 / VI.19.P1; not an additional legacy migration.

Problem 19.68 (Projection $\mathbb{A}^2 \to \mathbb{A}^1$)**.** Let $k[t] \to k[x, y]$ send t to x. Describe the induced map of spectra, its action on closed k-points, the inverse image of $D(t)$, and the map on the stalk at $(x - a, y - b)$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.69 (Polynomial maps and substitution)**.** Given polynomials $p_1, \ldots, p_m \in k[x_1, \ldots, x_n]$, construct the affine morphism $\mathbb{A}^n_k \to \mathbb{A}^m_k$ and prove that composition corresponds to substitution of polynomials.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.70 (Localization as an open affine morphism)**.** For $f \in A$, analyze the morphism induced by $A \to A_f$ and prove it identifies $\operatorname{Spec} A_f$ with the open affine subscheme $D(f)$ from VI/18.

Provenance. Canonical application of VI.18.P1 and derivative of VI.19.P1; no additional legacy migration.

Problem 19.71 (Quotients and their underlying closed images)**.** Let $I \subseteq A$. Prove that $A \to A/I$ induces an injective spectrum map with image $V(I)$ and describe the map on local rings at a prime containing I.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.72 (Residue fields under affine morphisms)**.** Let $f = \operatorname{Spec}(\varphi) : \operatorname{Spec} B \to \operatorname{Spec} A$. For $\mathfrak{p} \in \operatorname{Spec} B$, prove the local stalk map induces a field homomorphism $\kappa(f(\mathfrak{p})) \to \kappa(\mathfrak{p})$ and compute it for an evaluation map $k[t] \to K$, $t \mapsto a$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.73 (Maps to the affine line are global functions)**.** For a locally ringed space X, prove $\operatorname{Hom}_{\mathrm{LRS}}(X, \operatorname{Spec} \mathbb{Z}[t]) \cong \Gamma(X, \mathcal{O}_X)$ and identify explicitly the function corresponding to a morphism.

Provenance. Canonical specialization of AG3 Problem 12 / VI.19.P1; not an additional legacy migration.

Problem 19.74 (Affine-scheme isomorphisms are ring isomorphisms)**.** Prove $\operatorname{Spec} A \cong \operatorname{Spec} B$ as affine schemes if and only if $A \cong B$ as rings, and show a given affine morphism is an isomorphism iff its global pullback is.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.75 (Automorphisms of the affine line)**.** Let k be a field. Prove every k-algebra automorphism of $k[t]$ has $t \mapsto at + b$ with $a \in k^\times$, and translate this to automorphisms of $\mathbb{A}^1_k$ whose pullback fixes k.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.76 (Closure of the image)**.** For $\varphi : A \to B$, prove $\overline{\operatorname{im}(\operatorname{Spec} B \to \operatorname{Spec} A)} = V(\ker \varphi)$ and deduce a dense-image criterion.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.77 (Dense image without injectivity)**.** Give and analyze an explicit noninjective ring map whose spectrum map is surjective on points, and explain why this does not contradict affine anti-equivalence.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.78 (Factorization through a quotient)**.** Let $\alpha : A \to B$ and $I \subseteq A$. Prove the associated morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ factors through $\operatorname{Spec}(A/I)$ if and only if $I \subseteq \ker \alpha$, and prove uniqueness of the factorization.

Provenance. Canonical application of VI.19.P1; the closed-subscheme formulation remains reserved for VI/22.

Problem 19.79 (Factorization through a distinguished open)**.** Let $\alpha : A \to B$ and $f \in A$. Prove the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ factors through $D(f) = \operatorname{Spec} A_f$ if and only if $\alpha(f)$ is a unit in B.

Provenance. Canonical derivative/application of AG3 Problem 12 together with VI.18.P1; not an additional legacy migration.

Problem 19.80 (Two different affine morphisms with the same point map)**.** Let $A = k[\varepsilon]/(\varepsilon^2)$. Show that the ring endomorphisms $\varepsilon \mapsto \varepsilon$ and $\varepsilon \mapsto 0$ induce the same continuous map $\operatorname{Spec} A \to \operatorname{Spec} A$ but different affine-scheme morphisms.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.81 (Morphisms preserve specialization)**.** Let $f = \operatorname{Spec}(\varphi) : \operatorname{Spec} B \to \operatorname{Spec} A$. Prove that if $\mathfrak{p} \subseteq \mathfrak{p}'$ in B, then $f(\mathfrak{p}) \subseteq f(\mathfrak{p}')$ in A, and interpret this as preservation of specialization order.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 19.82 (Recover the point map from global pullback)**.** Let $f : X \to \operatorname{Spec} A$ be a morphism of locally ringed spaces and let $\alpha : A \to \Gamma(X, \mathcal{O}_X)$ be its global pullback. Prove

$$f(x) = \{a \in A : \alpha(a)_x \in \mathfrak{m}_x\}.$$

Provenance. Canonical extraction from the uniqueness step of AG3 Problem 12 / VI.19.P1; not an additional legacy migration.

Problem 19.83 (Naturality of the universal property)**.** Show the bijection of VI.19.P1 is natural in both X and A: precomposition by a locally-ringed-space morphism corresponds to pulling global sections forward along it, and a ring map $A \to A'$ corresponds to postcomposition with $\operatorname{Spec} A' \to \operatorname{Spec} A$.

Provenance. Canonical completion of the naturality content of AG3 Problem 12 / VI.19.P1; not an additional legacy migration.

19.19 Challenges

Problem 19.84 (Challenge VI.19.C1 – The affine anti-equivalence as an equivalence of categories)**.** Build the two functors

$$\operatorname{Spec} : (\mathbf{CRing})^{\mathrm{op}} \to \mathbf{AffSch}, \qquad \Gamma : \mathbf{AffSch} \to (\mathbf{CRing})^{\mathrm{op}},$$

and construct the natural isomorphisms showing they are quasi-inverse. Your argument must identify the unit/counit maps explicitly on objects and morphisms.

Problem 19.85 (Challenge VI.19.C2 – Classify endomorphisms of the dual-number point)**.** Let $A = k[\varepsilon]/(\varepsilon^2)$. Classify all k-algebra endomorphisms of A, hence all affine-scheme endomorphisms of $\operatorname{Spec} A$ over k in the pullback-on-k sense. Determine which are automorphisms and explain why every one has the same continuous point map.

Problem 19.86 (Challenge VI.19.C3 – Image closure and nilpotent kernels)**.** Prove that for every ring map $\varphi : A \to B$, the induced spectrum map is dense exactly when $\ker \varphi \subseteq \sqrt{(0)_A}$. Construct three examples: injective dense, noninjective dense, and nondense.

Problem 19.87 (Challenge VI.19.C4 – Global functions as maps and equations)**.** Let X be a locally ringed space and $s_1, \ldots, s_n \in \Gamma(X, \mathcal{O}_X)$. Construct the associated map $X \to \operatorname{Spec} \mathbb{Z}[t_1, \ldots, t_n]$. For an ideal $I \subseteq \mathbb{Z}[t_1, \ldots, t_n]$, prove that the map factors through $\operatorname{Spec}(\mathbb{Z}[t_1, \ldots, t_n]/I)$ exactly when every polynomial in I vanishes after substituting the sections s_i.

Problem 19.88 (Challenge VI.19.C5 – From affine compatibility to the gluing boundary)**.** Let $X = D(f) \cup D(g) \subseteq \operatorname{Spec} A$ and let $Y = \operatorname{Spec} B$. Suppose morphisms $D(f) \to Y$ and $D(g) \to Y$ correspond to ring maps $B \to A_f$ and $B \to A_g$ whose composites to A_{fg} agree. Prove there is a unique ring map $B \to \Gamma(X, \mathcal{O}_X)$ producing both local morphisms, and explain precisely what additional theorem VI/20 will use to turn this into a gluing statement for general schemes.

Summary

A ring homomorphism $\varphi : A \to B$ produces the affine morphism

$$\operatorname{Spec}(\varphi) : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A, \qquad \mathfrak{p} \longmapsto \varphi^{-1}(\mathfrak{p}),$$

with

$$\operatorname{Spec}(\varphi)^{-1}(D(a)) = D(\varphi(a)), \qquad A_a \longrightarrow B_{\varphi(a)}, \qquad A_{\varphi^{-1}(\mathfrak{p})} \longrightarrow B_{\mathfrak{p}}.$$

The construction is functorial and yields the anti-equivalence

$$\operatorname{Hom}_{\mathsf{AffSch}}(\operatorname{Spec} B, \operatorname{Spec} A) \cong \operatorname{Hom}_{\mathsf{Ring}}(A, B).$$

The stronger AG3 Problem 12 universal property identifies maps from any locally ringed space X into $\operatorname{Spec} A$ with ring maps $A \to \Gamma(X, \mathcal{O}_X)$. The Volume VI ledger now records AG3 Problem 12 uniquely as VI.19.P1; AG3 Problems 13 and 14 remain reserved for VI/21 and VI/20.

19.20 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 19.89 (Direction)**.** What scheme morphism comes from $A \to B$?

Hint. Use contravariant ring maps and locally ringed-space morphisms. ☐

Solution. $\operatorname{Spec} B \to \operatorname{Spec} A$. ☐

Exercise 19.90 (Quotient)**.** What is the image of $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$?

Hint. Use contravariant ring maps and locally ringed-space morphisms. ☐

Solution. $V(I)$. ☐

Exercise 19.91 (Localization)**.** What is the image of $\operatorname{Spec} A_f \to \operatorname{Spec} A$?

Hint. Use contravariant ring maps and locally ringed-space morphisms. ☐

Solution. $D(f)$. ☐

Exercise 19.92 (Point map)**.** How does a prime $\mathfrak{q} \subset B$ map under $A \to B$?

Hint. Use contravariant ring maps and locally ringed-space morphisms. ☐

Solution. To its inverse image in A. ☐

Exercise 19.93 (Composition)**.** What ring map corresponds to a composite of affine scheme morphisms?

Hint. Use contravariant ring maps and locally ringed-space morphisms. ☐

Solution. The composite ring homomorphism in reverse direction. ☐

Proofs

Exercise 19.94 (Affine anti-equivalence)**.** Prove: Prove morphisms $\operatorname{Spec} B \to \operatorname{Spec} A$ correspond to ring maps $A \to B$.

Hint. Work directly from contravariant ring maps and locally ringed-space morphisms. ☐

Solution. Construct the morphism on points and local sheaves from a ring map, then recover the ring map from global sections. ☐

Exercise 19.95 (Closed immersion quotient)**.** Prove: Prove a quotient map induces a closed immersion.

Hint. Work directly from contravariant ring maps and locally ringed-space morphisms. ☐

Solution. Use the homeomorphism with $V(I)$ and identify the structure sheaf as the quotient sheaf locally. ☐

Exercise 19.96 (Open immersion localization)**.** Prove: Prove $A \to A_f$ induces the distinguished-open immersion.

Hint. Work directly from contravariant ring maps and locally ringed-space morphisms. ☐

Solution. Use the affine identification of $D(f)$ with $\operatorname{Spec} A_f$. ☐

Exercise 19.97 (Residue fields)**.** Prove: Show a scheme morphism induces a field homomorphism between residue fields in the correct direction at a point.

Hint. Work directly from contravariant ring maps and locally ringed-space morphisms. ☐

Solution. Use the local homomorphism of stalks and then pass to residue fields. ☐

Counterexamples and hypothesis tests

Exercise 19.98 (Covariant)**.** Test the claim: Spec is covariant on rings.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is contravariant. ☐

Exercise 19.99 (Every continuous map)**.** Test the claim: every continuous map of affine spectra is a scheme morphism.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; a compatible local sheaf morphism is also required. ☐

Exercise 19.100 (Quotient open)**.** Test the claim: quotient maps of rings generally induce open immersions.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; they induce closed immersions. ☐

Applications and investigations

Exercise 19.101 (Equation inclusion)**.** How does adding equations appear scheme-theoretically?

Hint. Translate the question into contravariant ring maps and locally ringed-space morphisms. □

Solution. As a quotient ring and hence a closed immersion. □

Exercise 19.102 (Nonvanishing restriction)**.** How does restricting to $f \neq 0$ appear?

Hint. Translate the question into contravariant ring maps and locally ringed-space morphisms. □

Solution. As localization and an open immersion. □

Challenge problems

Exercise 19.103 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Morphisms of Affine Schemes in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Morphisms of Affine Schemes, translates it through contravariant ring maps and locally ringed-space morphisms, and checks the correspondence in both algebraic and geometric language. □

Exercise 19.104 (Hypothesis challenge)**.** Choose one structural statement from Morphisms of Affine Schemes, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 20

Gluing Affine Schemes

Purpose and learning goals

Affine schemes are local building blocks, not the whole subject. The passage from affine geometry to general scheme geometry is made by gluing compatible local models along open subschemes. The central theorem of this chapter is the full legacy AG3 Problem 14: a family of schemes equipped with compatible open-overlap isomorphisms can be assembled into a scheme, unique up to unique isomorphism.

A second legacy problem belongs here as well. AG5 Problem 2 proves that if two affine open subschemes $U = \operatorname{Spec} A$ and $V = \operatorname{Spec} B$ lie in one scheme, then $U \cap V$ can be covered by open affines that are distinguished in both U and V. This is exactly the local algebraic refinement needed to make gluing computations explicit.

After this chapter, the reader should be able to:

- state gluing data and the inverse/cocycle compatibility conditions;
- construct the quotient topological space underlying a glued scheme;
- prove each original chart embeds homeomorphically as an open subset;
- define the glued structure sheaf by compatible families of local sections;
- verify the sheaf axiom and identify the stalks with chart stalks;
- prove the resulting locally ringed space is a scheme;
- prove uniqueness up to unique isomorphism;
- use the universal mapping property for morphisms out of a glued scheme;
- reduce two-chart affine gluing to ring isomorphisms on localized coordinate rings;
- construct the doubled-origin line and a two-chart model later recognized as $\mathbb{P}^1$;
- recover a scheme from any affine open cover;
- refine intersections of affine opens by common distinguished affine opens;
- compute global sections of a two-chart gluing as an equalizer;
- distinguish gluing from later closed-subscheme, fiber-product, and base-change constructions.

$$\begin{tikzcd} \boxed{\begin{array}{c}\text{affine/scheme charts}\\ X_i\end{array}} \arrow[r] & \boxed{\begin{array}{c}\text{open overlaps}\\ U_{ij}\end{array}} \arrow[r] & \boxed{\begin{array}{c}\text{transition maps}\\ \varphi_{ij}\end{array}} \arrow[r, "\text{cocycle}"] & \boxed{\begin{array}{c}\text{glued scheme}\\ X\end{array}} \end{tikzcd}$$

20.1 Gluing data

Let $\{X_i\}_{i\in I}$ be schemes. For each pair i,j, choose an open subscheme $U_{ij} \subseteq X_i$ and an isomorphism
$$\varphi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}.$$
It is convenient to put $U_{ii} = X_i$ and $\varphi_{ii} = \mathrm{id}_{X_i}$. The gluing conditions are
$$\varphi_{ji} = \varphi_{ij}^{-1}$$
and, on every triple overlap where both sides are defined,
$$\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}.$$
Equivalently, φ_{ij} must carry the part of U_{ij} meeting U_{ik} onto the corresponding part of U_{ji} meeting U_{jk}, and the transition maps satisfy the cocycle law.

$$\begin{tikzcd} U_{ij}\cap U_{ik} \arrow[r, "\varphi_{ij}"] \arrow[dr, "\varphi_{ik}"'] & U_{ji}\cap U_{jk} \arrow[d, "\varphi_{jk}"] \\ & U_{ki}\cap U_{kj} \end{tikzcd} \qquad \varphi_{ik} = \varphi_{jk}\circ\varphi_{ij}.$$

20.2 The quotient set and topology

Start from the disjoint union of underlying sets
$$Y = \coprod_{i\in I} X_i.$$
For $x \in X_i$ and $y \in X_j$, declare $x \sim y$ exactly when $x \in U_{ij}$ and $y = \varphi_{ij}(x)$. Identity, inverse, and cocycle conditions make this an equivalence relation. Define
$$X = Y/\sim$$
and let $\psi_i : X_i \to X$ be the quotient maps.

A subset $V \subseteq X$ is declared open precisely when every $\psi_i^{-1}(V)$ is open in X_i. This final topology is designed so that all chart maps are continuous. Moreover,
$$\psi_j^{-1}(\psi_i(X_i)) = U_{ji},$$
so $\psi_i(X_i)$ is open, and $\psi_i : X_i \to \psi_i(X_i)$ is a homeomorphism. The images cover X, and
$$\psi_i(X_i) \cap \psi_j(X_j) = \psi_i(U_{ij}) = \psi_j(U_{ji}).$$

$$\coprod_i X_i \longrightarrow X = (\coprod_i X_i)/\sim, \qquad x \sim \varphi_{ij}(x) \text{ for } x \in U_{ij}.$$

20.3 Gluing the structure sheaf

For an open set $V \subseteq X$, put $V_i = \psi_i^{-1}(V)$. Define
$$\mathcal{O}_X(V) = \left\{ (s_i)_i \in \prod_i \mathcal{O}_{X_i}(V_i) : s_i|_{V_i\cap U_{ij}} = \varphi_{ij}^{\sharp}(s_j|_{V_j\cap U_{ji}}) \text{ for all } i,j \right\}.$$

Restrictions are componentwise. Because each $\mathcal{O}_{X_i}$ is a sheaf and the transition maps satisfy the cocycle law, compatible local families glue uniquely; hence $\mathcal{O}_X$ is a sheaf. On the chart $\psi_i(X_i)$ the restriction of $\mathcal{O}_X$ identifies canonically with $\mathcal{O}_{X_i}$.

$$\mathcal{O}_X(V) \longrightarrow \prod_i \mathcal{O}_{X_i}(V_i) \rightrightarrows \prod_{i,j} \mathcal{O}_{X_i}(V_i \cap U_{ij}),$$

where the equalizer condition is agreement after transport by $\varphi_{ij}^\sharp$.

Example 20.1 (The projective line from two affines)**.** Glue $U_0 = \operatorname{Spec} k[t]$ and $U_1 = \operatorname{Spec} k[s]$ along $D(t)$ and $D(s)$ via $s = t^{-1}$. The result is $\mathbb{P}^1_k$, covered by two affine lines.

20.4 Local rings and the scheme condition

If $x \in X$, choose a chart $\psi_i(X_i)$ containing it and a point $x_i \in X_i$ with $\psi_i(x_i) = x$. The chart identification gives

$$\mathcal{O}_{X,x} \cong \mathcal{O}_{X_i,x_i}.$$

The ring on the right is local, so the glued ringed space is locally ringed. Since the opens $\psi_i(X_i)$ cover X and are schemes, X is a scheme. Thus gluing is genuinely local: all stalk and affine-neighborhood questions may be checked inside one chart.

$$\begin{tikzcd}
X_i \arrow[r, "\sim"] & \psi_i(X_i) \subseteq X \\
x_i \arrow[r, mapsto] & x
\end{tikzcd} \qquad \mathcal{O}_{X,x} \cong \mathcal{O}_{X_i,x_i}.$$

20.5 Uniqueness and the universal mapping property

Suppose X and X' are two gluings of the same data. On every chart, the identity of X_i produces an isomorphism between the corresponding open subsets of X and X'. These chartwise isomorphisms agree on overlaps, hence glue to a unique isomorphism $X \xrightarrow{\sim} X'$. Therefore the gluing is unique up to unique isomorphism compatible with all charts.

A useful equivalent form is the mapping property. If Y is any scheme, then giving a morphism $f : X \to Y$ is equivalent to giving morphisms $f_i : X_i \to Y$ such that

$$f_i|_{U_{ij}} = f_j \circ \varphi_{ij}$$

on every overlap. Thus morphisms out of a glued scheme are exactly compatible families of morphisms out of its charts.

$$\operatorname{Hom}(X, Y) \cong \left\{(f_i)_i : \ f_i : X_i \to Y,\ f_i|_{U_{ij}} = f_j \circ \varphi_{ij}\right\}.$$

20.6 Two-chart affine gluing

Let $X_1 = \operatorname{Spec} A$ and $X_2 = \operatorname{Spec} B$. Suppose principal opens $D(f) \subseteq X_1$ and $D(g) \subseteq X_2$ are identified by an affine-scheme isomorphism. By VI/18–VI/19 this is equivalently a ring isomorphism

$$B_g \xrightarrow{\sim} A_f$$

with the direction reversed geometrically. Gluing the two affine schemes along these opens produces a scheme covered by two affine charts.

This is the basic computational model: topology says which points are identified; localization says what the functions are on the overlap; the isomorphism of localized rings tells how functions transform between charts.

$$A_f \xleftarrow{\sim} B_g$$

$$D(f) \subseteq \operatorname{Spec} A \xrightarrow{\sim} D(g) \subseteq \operatorname{Spec} B$$

Example 20.2 (Doubled origin)**.** Glue two copies of $\mathbb{A}^1_k$ along $D(x)$ by the identity. The two origins remain distinct while every nonzero point is identified, producing the classical nonseparated doubled-origin scheme.

20.7 The doubled-origin line

Take two copies $X_1 = X_2 = \mathbb{A}^1_k = \operatorname{Spec} k[t]$ and glue

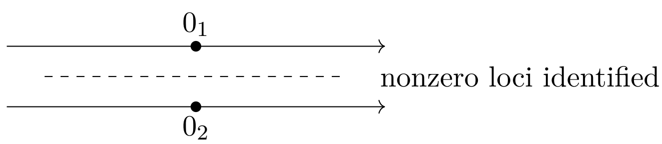
$$D(t) \subseteq X_1 \qquad \text{to} \qquad D(t) \subseteq X_2$$

by the identity. Every nonzero point is identified across the two charts, but the two origins are not. The result is the affine line with doubled origin.

Its global sections are the equalizer

$$\Gamma(X, \mathcal{O}_X) = \{(p,q) \in k[t] \times k[t] : p = q \text{ in } k[t,t^{-1}]\} \cong k[t].$$

Nevertheless the glued scheme is not $\operatorname{Spec} k[t]$: its underlying space has two distinct origins. Thus a scheme covered by affine opens need not itself be affine.



20.8 A two-chart projective-line model

Take $X_0 = \operatorname{Spec} k[t]$ and $X_1 = \operatorname{Spec} k[u]$. Identify $D(t)$ with $D(u)$ by

$$u \longmapsto t^{-1}$$

at the level of overlap rings $k[u,u^{-1}] \cong k[t,t^{-1}]$. Gluing produces a scheme covered by two affine lines. Later projective geometry will identify this model canonically with $\mathbb{P}^1_k$; here its role is simply to show that non-affine global geometry can emerge from elementary affine charts and localization.

$$\begin{tikzcd}
\operatorname{Spec} k[t,t^{-1}] \arrow[r, "u=t^{-1}", "\sim"'] & \operatorname{Spec} k[u,u^{-1}] \\
\operatorname{Spec} k[t] \arrow[u, hook] & \operatorname{Spec} k[u] \arrow[u, hook]
\end{tikzcd}$$

20.9 Intersections of affine open subschemes

Let X be a scheme and let

$$U = \operatorname{Spec} A, \qquad V = \operatorname{Spec} B$$

be affine opens. For every $x \in U \cap V$, the basis theorem for distinguished opens lets us choose a distinguished neighborhood inside U contained in $U \cap V$, then refine inside V, and finally refine once more so the same open is distinguished in both charts. Consequently,

$$\boxed{U \cap V = \bigcup_{\lambda} W_\lambda, \qquad W_\lambda = D(f_\lambda) \subseteq U = D(g_\lambda) \subseteq V.}$$

This is legacy AG5 Problem 2 and is the practical bridge from abstract overlap isomorphisms to localization computations.

$$x \in W = D(f) \subseteq U = \operatorname{Spec} A, \qquad W = D(g) \subseteq V = \operatorname{Spec} B, \qquad W \subseteq U \cap V.$$

Example 20.3 (Transition cocycle)**.** With three affine pieces, transition isomorphisms must agree on triple overlaps. The cocycle condition ensures that identifying a point through different intermediate charts gives the same result.

20.10 Reconstructing a scheme from an affine cover

Let X be any scheme with affine open cover $X = \bigcup_i U_i$. On $U_i \cap U_j$, use the identity as the transition isomorphism between the two copies of the overlap. The cocycle condition is automatic, so the gluing theorem reconstructs a scheme canonically isomorphic to X. Thus the slogan "schemes are obtained by gluing affine schemes" is literally a reconstruction theorem, not merely a heuristic.

20.11 Global sections as an equalizer

For a two-open cover $X = U \cup V$, the sheaf axiom yields

$$0 \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(U, \mathcal{O}_X) \times \Gamma(V, \mathcal{O}_X) \rightrightarrows \Gamma(U \cap V, \mathcal{O}_X),$$

where the two arrows are the two restrictions. Hence global sections are pairs of local functions that agree on the overlap. This formula is especially effective when U, V, and their overlap are affine principal opens.

20.12 Common pitfalls

1. Pairwise identifications are not enough: triple-overlap cocycle compatibility is essential.
2. The quotient topology alone does not make a scheme; the structure sheaf must also be glued.
3. The sheaf is not a direct product of chart sheaves: sections must satisfy overlap compatibility.
4. The chart maps are open embeddings because inverse images of chart images are the prescribed open overlaps.

5. A glued scheme can be non-affine even when every chart and overlap is affine.
6. An isomorphism of overlap spaces is insufficient; one needs an isomorphism of schemes, including the sheaves.
7. When affines overlap inside a scheme, the whole intersection need not be one principal open, but it has a cover by opens principal in both charts.
8. AG3 Problem 13, the finality of $\operatorname{Spec}\mathbb{Z}$ and structure morphism, remains reserved for VI/21.
9. Closed subschemes remain centered in VI/22; quotient-ring examples here are used only when needed to understand open-chart gluing.
10. Fiber products and base change are different universal constructions and remain reserved for VI/23–VI/24.

20.13 Worked examples

Example 20.4 (Disjoint union)**.** If every U_{ij} with $i \neq j$ is empty, the gluing is simply the disjoint union $\coprod_i X_i$ as a scheme.

Example 20.5 (Gluing identical charts)**.** Gluing two copies of a scheme X along all of X by the identity produces a scheme canonically isomorphic to X; the quotient identifies corresponding points and the sheaves agree everywhere.

Example 20.6 (Two affine lines along the punctured line)**.** Two copies of $\operatorname{Spec} k[t]$ glued along $D(t)$ by the identity give the doubled-origin line. The nonzero loci are identified while the origins remain distinct.

Example 20.7 (Two affine lines by inversion)**.** Glue $\operatorname{Spec} k[t]$ and $\operatorname{Spec} k[u]$ by $u = t^{-1}$ on the punctured lines. The result has two affine charts whose overlap ring is $k[t, t^{-1}]$.

Example 20.8 (A redundant chart)**.** If X_3 is an open subscheme already contained in X_1 and the transition maps are the obvious inclusions, adding X_3 to the gluing data does not change the resulting scheme up to unique isomorphism.

Example 20.9 (Global sections on a union)**.** For $X = U \cup V$, a global function is a pair (s_U, s_V) with equal restrictions to $U \cap V$. This is the sheaf equalizer formula.

Example 20.10 (Common distinguished neighborhood)**.** If $x \in U \cap V$ with $U = \operatorname{Spec} A$ and $V = \operatorname{Spec} B$, AG5-P2 gives an affine neighborhood W of x with $W = D(f)$ in U and $W = D(g)$ in V.

Example 20.11 (Empty overlap)**.** Gluing X_1 and X_2 along the empty open gives $X_1 \amalg X_2$; the cocycle conditions are vacuous.

Example 20.12 (Full overlap)**.** If $U_{12} = X_1$, $U_{21} = X_2$, and φ_{12} is an isomorphism, the gluing identifies the two whole charts and yields either one up to canonical isomorphism.

Example 20.13 (Gluing principal opens)**.** An isomorphism $A_f \cong B_g$ determines an isomorphism $D(g) \subseteq \operatorname{Spec} B \cong D(f) \subseteq \operatorname{Spec} A$ and hence a two-chart gluing.

Example 20.14 (A chart stalk)**.** At a point represented by $x_i \in X_i$, the stalk of the glued sheaf is $\mathcal{O}_{X_i,x_i}$; choosing another representative gives the same local ring through the transition isomorphism.

Example 20.15 (Compatible maps glue)**.** If $f_i : X_i \to Y$ agree on overlaps after the transition identifications, there is exactly one $f : X \to Y$ restricting to all f_i.

Example 20.16 (Cocycle on three copies)**.** For three charts, choosing $\varphi_{13} = \varphi_{23} \circ \varphi_{12}$ on the triple overlap ensures that identifying through chart 2 or directly through chart 3 gives the same equivalence class.

Example 20.17 (Failure without cocycle)**.** If three transition maps disagree on a triple overlap, the attempted chart identifications are not coherent; one cannot simultaneously recover the prescribed pairwise identifications as embeddings into a single glued scheme.

Example 20.18 (Restriction of the glued sheaf)**.** On $\psi_i(X_i)$ the compatible-family definition of $\mathcal{O}_X$ recovers $\mathcal{O}_{X_i}$ because every other component is forced from the ith component on overlaps.

Example 20.19 (Affine cover reconstruction)**.** Given an affine open cover $X = \bigcup U_i$, gluing copies of the U_i along their literal intersections using identity maps reconstructs X.

Example 20.20 (Refinement does not change the gluing)**.** Replacing an overlap by a cover of smaller opens and using the restricted transition maps produces the same glued scheme once the pieces collectively encode the same identifications.

Example 20.21 (Doubled-origin global functions)**.** For the doubled-origin line, compatible pairs of polynomials satisfy $p = q$ in $k[t, t^{-1}]$, hence $p = q$ already in $k[t]$; therefore $\Gamma(X, \mathcal{O}_X) \cong k[t]$.

Example 20.22 (Overlap ring by localization)**.** If $W = D(f) \subseteq \operatorname{Spec} A$, then $\Gamma(W, \mathcal{O}) = A_f$. A gluing isomorphism of such overlaps is therefore encoded by a ring isomorphism of localizations.

Example 20.23 (Local property survives gluing)**.** Any property defined purely by the local rings and stable under isomorphism can be checked chartwise on a glued scheme, because every stalk comes from a chart stalk.

20.14 Advanced worked examples

Example 20.24 (Advanced: Gluing as descent for open covers)**.** The compatible-family sheaf formula is the simplest instance of descent: objects and morphisms defined locally can be reconstructed globally when transition data satisfy a cocycle condition.

Example 20.25 (Advanced: The gluing universal property)**.** The glued scheme is characterized by the fact that maps from it to any scheme Y are exactly compatible chart maps $X_i \to Y$. This gives uniqueness without repeating the quotient construction.

Example 20.26 (Advanced: Associativity of finite gluing)**.** For finitely many charts, gluing them in stages produces the same result up to unique isomorphism, provided all original transition data and cocycles are preserved.

Example 20.27 (Advanced: Affine overlaps need not be globally principal)**.** Even when U and V are affine, $U \cap V$ need not be a single distinguished open of either. AG5-P2 supplies a cover by opens distinguished in both, which is enough for all local computations.

Example 20.28 (Advanced: Equalizers and global functions)**.** For a two-chart affine gluing $X = \operatorname{Spec} A \cup \operatorname{Spec} B$ with affine overlap W, global sections are the equalizer of $A \times B \rightrightarrows \Gamma(W, \mathcal{O}_W)$. This may fail to make X affine.

Example 20.29 (Advanced: Doubled origin as a topology/sheaf warning)**.** The doubled-origin line and $\mathbb{A}^1$ have the same global section ring $k[t]$ but are not isomorphic. Global functions alone reconstruct a scheme only under the affine hypothesis.

Example 20.30 (Advanced: Two-chart compactification model)**.** The inversion gluing of two affine lines adds one new point beyond either chart while retaining the punctured line as overlap. Projective geometry later identifies the result with $\mathbb{P}^1$.

Example 20.31 (Advanced: Chart-independent stalks)**.** If a point has representatives $x_i \in X_i$ and $x_j = \varphi_{ij}(x_i)$, then the transition isomorphism induces an isomorphism $\mathcal{O}_{X_i,x_i} \cong \mathcal{O}_{X_j,x_j}$, so the glued local ring is independent of chart.

Example 20.32 (Advanced: Morphisms may be checked locally on the source)**.** Two morphisms $f, g : X \to Y$ from a glued scheme are equal if their restrictions to all chart opens are equal. Existence is likewise local provided restrictions agree on overlaps.

Example 20.33 (Advanced: Refining to principal overlap algebra)**.** AG5-P2 means that every point of the intersection of two affine charts has a neighborhood whose coordinate ring is simultaneously a localization of both chart rings. This converts transition geometry into explicit ring isomorphisms.

Example 20.34 (Advanced: Reconstruction from an affine atlas)**.** An affine atlas together with the induced overlap transition maps contains enough data to reconstruct the entire scheme. The ambient scheme can therefore be forgotten and recovered from the atlas.

Example 20.35 (Advanced: Gluing and non-affineness)**.** Gluing is a mechanism for producing schemes beyond the affine category. The failure of the global-section functor to recover such schemes is not a defect; it is exactly why local affine geometry must be glued.

Example 20.36 (Advanced: Boundary to structure morphisms)**.** The present theorem builds spaces from local charts. VI/21 adds the canonical map of every scheme to $\operatorname{Spec} \mathbb{Z}$ and develops the point/structure-morphism viewpoint; AG3-P13 is therefore intentionally untouched here.

Example 20.37 (Advanced: Boundary to fiber constructions)**.** Open gluing identifies already-existing open pieces. Fiber products and base change solve different universal problems and are deferred to VI/23–VI/24.

20.15 Exercises

Exercise 20.38 (Identity transition)**.** Explain why setting $U_{ii} = X_i$ and $\varphi_{ii} = \mathrm{id}$ makes the gluing conditions uniform.

Exercise 20.39 (Symmetry of the relation)**.** Show $\varphi_{ji} = \varphi_{ij}^{-1}$ implies the point-identification relation is symmetric.

Exercise 20.40 (Transitivity)**.** Use the cocycle law to prove transitivity of the gluing relation.

Exercise 20.41 (Open chart image)**.** Prove $\psi_i(X_i)$ is open in the quotient topology.

Exercise 20.42 (Chart injectivity)**.** Prove $\psi_i : X_i \to X$ is injective.

Exercise 20.43 (Overlap identity)**.** Show $\psi_i(X_i) \cap \psi_j(X_j) = \psi_i(U_{ij})$.

Exercise 20.44 (Componentwise restrictions)**.** Show the compatible-family definition of $\mathcal{O}_X$ is stable under restriction.

Exercise 20.45 (Sheaf locality)**.** Prove uniqueness in the sheaf axiom for the glued presheaf.

Exercise 20.46 (Sheaf gluing)**.** Prove existence in the sheaf axiom for $\mathcal{O}_X$.

Exercise 20.47 (Stalk identification)**.** For $x = \psi_i(x_i)$, prove $\mathcal{O}_{X,x} \cong \mathcal{O}_{X_i,x_i}$.

Exercise 20.48 (Locality)**.** Why is the glued ringed space locally ringed?

Exercise 20.49 (Scheme condition)**.** Why is the glued locally ringed space a scheme?

Exercise 20.50 (Two-chart global sections)**.** For $X = U \cup V$, derive the equalizer formula for $\Gamma(X, \mathcal{O}_X)$.

Exercise 20.51 (Doubled origin sections)**.** Compute global sections of the doubled-origin line.

Exercise 20.52 (Doubled origin non-affine)**.** Use the previous exercise to show the doubled-origin line is not affine.

Exercise 20.53 (Inversion overlap)**.** Write the ring isomorphism used to glue two affine lines by inversion.

Exercise 20.54 (Common distinguished refinement)**.** Let $x \in U \cap V$ for affine opens U, V. Outline the shrinking argument producing $W = D(f) \subseteq U = D(g) \subseteq V$.

Exercise 20.55 (Common overlap ring)**.** If $W = D(f) \subseteq \operatorname{Spec} A = D(g) \subseteq \operatorname{Spec} B$, identify its coordinate ring in two ways.

Exercise 20.56 (Compatible maps)**.** Given $f_i : X_i \to Y$ agreeing on overlaps, explain how the point maps glue.

Exercise 20.57 (Uniqueness of a glued map)**.** Why is a morphism $X \to Y$ determined by its restrictions to the chart cover?

Exercise 20.58 (Reconstruct from affine cover)**.** Given $X = \bigcup U_i$, specify the transition maps that reconstruct X by gluing.

Exercise 20.59 (Disjoint union as gluing)**.** Recover the disjoint union construction from the gluing theorem.

Exercise 20.60 (Full-overlap gluing)**.** If $X_1 \cong X_2$ and the full schemes are glued by that isomorphism, identify the result.

Exercise 20.61 (Triple cocycle)**.** State precisely what can fail if $\varphi_{13} \neq \varphi_{23} \circ \varphi_{12}$ on a triple overlap.

Exercise 20.62 (Inversion gluing global sections)**.** Show that the two affine lines glued by $u = t^{-1}$ have only constant global functions.

Exercise 20.63 (Boundary audit)**.** Which numbered AG3 problem remains reserved immediately after VI/20, and where is it assigned?

20.16 Problems

Problem 20.64 (Legacy AG3 Problem 14: Gluing schemes)**.** Let $\{X_i\}_{i \in I}$ be schemes. For each $i \neq j$, let $U_{ij} \subseteq X_i$ be open and let $\varphi_{ij} : U_{ij} \xrightarrow{\sim} U_{ji}$ be an isomorphism. Assume $\varphi_{ji} = \varphi_{ij}^{-1}$ and the full triple-overlap compatibility/cocycle conditions. Construct a scheme X with open embeddings $\psi_i : X_i \to X$ whose images cover X, satisfy $\psi_i(X_i) \cap \psi_j(X_j) = \psi_i(U_{ij})$, and obey $\psi_i = \psi_j \circ \varphi_{ij}$ on U_{ij}. Prove the glued scheme is unique up to unique isomorphism compatible with the charts.

Provenance. Legacy recovery: theory-of-algebraic-geometry-3.tex, Problem 14. Full arbitrary-family gluing theorem preserved, including quotient topology, sheaf, stalks, scheme verification, and uniqueness.

Problem 20.65 (Legacy AG5 Problem 2: Intersections of affine open subschemes)**.** Let X be a scheme and $U = \operatorname{Spec} A$, $V = \operatorname{Spec} B$ affine open subschemes. Prove $U \cap V$ can be covered by affine opens W_λ for which

$$W_\lambda = D(f_\lambda) \subseteq U, \qquad W_\lambda = D(g_\lambda) \subseteq V$$

for suitable $f_\lambda \in A$, $g_\lambda \in B$.

Provenance. Legacy recovery: theory-of-algebraic-geometry-5.tex, Problem 2. Newly added to the legacy ledger during the VI/20 corpus audit.

Problem 20.66 (Two-chart gluing theorem)**.** Specialize VI.20.P1 to two schemes and prove the construction directly, including the overlap and sheaf formulas.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.67 (Universal mapping property of a gluing)**.** Prove $\operatorname{Hom}(X, Y)$ is the set of compatible families $f_i : X_i \to Y$ for a glued scheme X.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.68 (Reconstruct a scheme from an affine cover)**.** Prove every scheme is canonically recovered by gluing the members of any affine open cover along their intersections.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.69 (Gluing along principal affine opens)**.** Let $A_f \cong B_g$. Construct the scheme obtained by gluing $\operatorname{Spec} A$ and $\operatorname{Spec} B$ along $D(f) \cong D(g)$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.70 (Doubled-origin line)**.** Construct the doubled-origin line by gluing two copies of $\mathbb{A}^1_k$ along $D(t)$ by the identity; compute its global sections and prove it is not affine.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.71 (Two-chart projective-line model)**.** Glue $\operatorname{Spec} k[t]$ and $\operatorname{Spec} k[u]$ along punctured lines using $u = t^{-1}$. Describe the points and overlap functions.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.72 (Global sections of the inversion gluing)**.** Compute the global sections of the two affine lines glued by inversion.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.73 (Associativity of finite gluing)**.** Show that gluing three charts in two stages gives a scheme uniquely isomorphic to the one-step gluing, provided the original cocycle data are respected.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.74 (Refinement invariance)**.** Show refining an affine cover and using restricted transition maps reconstructs the same scheme.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.75 (Common refinement of two affine atlases)**.** Given two affine covers of one scheme, construct a common affine refinement using VI.20.P2.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.76 (Chart-independent local rings)**.** If a glued point is represented by $x_i \in X_i$ and $x_j \in X_j$, prove the transition map identifies the two chart local rings.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.77 (Gluing compatible morphisms to an affine target)**.** Let X be glued from X_i and let $Y = \operatorname{Spec} A$. Show compatible maps $X_i \to Y$ correspond to ring maps $A \to \Gamma(X_i, \mathcal{O})$ compatible under overlap restriction.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.78 (Equalizer for two affine charts)**.** Let $X = U \cup V$ with $U = \operatorname{Spec} A$, $V = \operatorname{Spec} B$, and affine overlap $W = \operatorname{Spec} C$. Prove $\Gamma(X, \mathcal{O}_X)$ is the equalizer of $A \times B \rightrightarrows C$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.79 (Gluing with an empty overlap)**.** Prove gluing two schemes along the empty opens gives their disjoint union as a scheme.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.80 (Gluing along the whole chart)**.** If $\varphi : X_1 \to X_2$ is an isomorphism, glue along $U_{12} = X_1$, $U_{21} = X_2$ and identify the result.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.81 (Why the cocycle is necessary)**.** Construct an abstract three-chart scenario showing that pairwise inverse maps without the cocycle need not define the prescribed gluing.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.82 (Gluing is local on the source)**.** Prove a morphism from a glued scheme is uniquely determined by its restrictions to any open cover refining the original charts.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

Problem 20.83 (Affine atlas as localization data)**.** Show an affine atlas may be refined so every transition is described locally by isomorphisms between localizations $A_f \cong B_g$.

Provenance. New canonical synthesis; no numbered legacy Problem is claimed.

20.17 Challenges

Problem 20.84 (Challenge 1: Four-chart cocycle audit)**.** Design four affine charts with specified principal-open overlaps. Write a minimal set of transition isomorphisms from which all remaining ones are forced by inverse and cocycle laws, and prove the resulting gluing data are coherent.

Problem 20.85 (Challenge 2: Doubled-origin morphisms)**.** Classify morphisms from the doubled-origin line to $\mathbb{A}^1_k$ in terms of compatible chart polynomials and compare with its global section ring.

Problem 20.86 (Challenge 3: Atlas comparison)**.** Given two finite affine covers of a scheme, build an explicit common distinguished refinement and write all transition maps as localization isomorphisms.

Problem 20.87 (Challenge 4: Non-affine two-chart criterion experiment)**.** Construct several two-chart gluings with affine overlap, compute their global-section equalizers, and determine which can or cannot be affine using affine reconstruction.

Problem 20.88 (Challenge 5: Gluing theorem from the mapping property)**.** Starting only from the quotient topology and the compatible-family sheaf, prove the universal mapping property, then use it to rederive uniqueness, reconstruction from affine covers, and associativity of staged gluing.

Summary

Compatible open-subscheme isomorphisms satisfying inverse and cocycle laws determine a scheme obtained from the disjoint union of the charts by quotienting points and gluing functions. Each chart survives as an open subscheme, the stalks are the original chart stalks, and the result is unique up to unique isomorphism. This is AG3 Problem 14, now recovered uniquely as VI.20.P1.

The corpus audit also recovers AG5 Problem 2 as VI.20.P2: intersections of affine opens possess a cover by affine opens distinguished in both charts. This turns abstract overlap geometry into explicit localization algebra and closes VI/20 without consuming AG3 Problem 13, which remains reserved for VI/21.

20.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 20.89 (Projective overlap)**.** What is the overlap of the two standard affine charts of $\mathbb{P}^1$?

Hint. Use scheme gluing and transition isomorphisms. ☐

Solution. $\operatorname{Spec} k[t, t^{-1}]$. ☐

Exercise 20.90 (Transition)**.** What is the coordinate change between t and s?

Hint. Use scheme gluing and transition isomorphisms. ☐

Solution. $s = t^{-1}$. ☐

Exercise 20.91 (Doubled origin)**.** How many origins survive in the doubled-origin scheme?

Hint. Use scheme gluing and transition isomorphisms. ☐

Solution. Two. ☐

Exercise 20.92 (Cocycle)**.** What condition is required on triple overlaps?

Hint. Use scheme gluing and transition isomorphisms. ☐

Solution. Transition maps must compose consistently. ☐

Exercise 20.93 (Local structure)**.** Why is a glued space still locally affine?

Hint. Use scheme gluing and transition isomorphisms. ☐

Solution. Each original affine piece maps to an open chart. ☐

Proofs

Exercise 20.94 (Gluing equivalence)**.** Prove: Prove compatible open-subscheme identifications define an equivalence relation on the disjoint union.

Hint. Work directly from scheme gluing and transition isomorphisms. ☐

Solution. Identity, inverse, and cocycle conditions give reflexivity, symmetry, and transitivity. ☐

Exercise 20.95 (Sheaf gluing)**.** Prove: Explain why the structure sheaves glue along the transition isomorphisms.

Hint. Work directly from scheme gluing and transition isomorphisms. ☐

Solution. Use the sheaf gluing theorem with the compatible isomorphisms on overlaps. ☐

Exercise 20.96 (Projective line)**.** Prove: Prove the two-chart construction yields a scheme covered by two affine lines with transition $t \mapsto t^{-1}$.

Hint. Work directly from scheme gluing and transition isomorphisms. ☐

Solution. Apply the gluing theorem and verify the overlap localizations are isomorphic. ☐

Exercise 20.97 (Doubled origin nonseparated preview)**.** Prove: Show the two origins in the doubled-origin construction cannot be separated by disjoint opens.

Hint. Work directly from scheme gluing and transition isomorphisms. ☐

Solution. Every neighborhood of either origin contains nonzero points, and the nonzero loci have been identified. ☐

Counterexamples and hypothesis tests

Exercise 20.98 (Arbitrary gluing)**.** Test the claim: arbitrary identifications of open subsets always produce a scheme.

Hint. Test the claim on a concrete affine or local model before generalizing. □

Solution. False; compatibility and cocycle conditions are needed. □

Exercise 20.99 (Affine result)**.** Test the claim: gluing affine schemes always produces an affine scheme.

Hint. Test the claim on a concrete affine or local model before generalizing. □

Solution. False; $\mathbb{P}^1$ is the standard counterexample. □

Exercise 20.100 (Hausdorff intuition)**.** Test the claim: all schemes obtained by gluing are Hausdorff in the ordinary topology.

Hint. Test the claim on a concrete affine or local model before generalizing. □

Solution. False; scheme topology and separatedness differ. □

Applications and investigations

Exercise 20.101 (Global construction)**.** Why is gluing indispensable in algebraic geometry?

Hint. Translate the question into scheme gluing and transition isomorphisms. □

Solution. Many global spaces, including projective varieties, are not affine but are assembled from affine charts. □

Exercise 20.102 (Coordinate transitions)**.** What role do transition maps play?

Hint. Translate the question into scheme gluing and transition isomorphisms. □

Solution. They identify local coordinate rings consistently on overlaps. □

Challenge problems

Exercise 20.103 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Gluing Affine Schemes in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Gluing Affine Schemes, translates it through scheme gluing and transition isomorphisms, and checks the correspondence in both algebraic and geometric language. □

Exercise 20.104 (Hypothesis challenge)**.** Choose one structural statement from Gluing Affine Schemes, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 21

Schemes and Their Points

A scheme is simultaneously a topological space, a sheaf of rings, and a system of local rings. Its points therefore carry more information than locations alone. Each point has a local ring, a residue field, a specialization position, and functorial behavior under scheme morphisms. This chapter makes that pointwise language explicit and places every scheme over the absolute arithmetic base $\operatorname{Spec}\mathbb{Z}$.

21.1 The absolute structure morphism

The universal property proved in VI/19 gives, for every scheme X,

$$\mathrm{Hom}_{\mathbf{Sch}}(X, \operatorname{Spec}\mathbb{Z}) \cong \mathrm{Hom}_{\mathbf{Rings}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)).$$

There is exactly one unital map $\mathbb{Z} \to R$ for every ring R. Hence:

Theorem 21.1 (Absolute structure morphism)**.** *For every scheme X there is a unique morphism*

$$\pi_X : X \longrightarrow \operatorname{Spec}\mathbb{Z}.$$

Thus $\operatorname{Spec}\mathbb{Z}$ *is final in* **Sch**.

Remark 21.2. The theorem is the scheme-theoretic reflection of the fact that $\mathbb{Z}$ is initial among unital rings. It is the content of legacy AG3 Problem 13, reconstructed in full as VI.21.P1.

$$\begin{tikzcd} X \arrow[rr, "\pi_X"] & & \operatorname{Spec}\mathbb{Z} \end{tikzcd}$$

Figure 21.1: Absolute structure morphism.

21.2 What a point of a scheme carries

Let $x \in X$. The stalk $\mathcal{O}_{X,x}$ is a local ring with maximal ideal $\mathfrak{m}_x$, and the residue field is

$$\kappa(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x.$$

The quotient map $\mathcal{O}_{X,x} \to \kappa(x)$ packages evaluation at x in the local language.

Proposition 21.3 (Canonical residue-field point)**.** *Every point $x \in X$ determines a canonical morphism*

$$i_x : \operatorname{Spec} \kappa(x) \longrightarrow X$$

whose unique source point maps to x.

Proof. Choose an affine neighbourhood $U = \operatorname{Spec} A$ of x, corresponding to a prime $\mathfrak{p} \subset A$. The field map $A \to A_{\mathfrak{p}} \to \kappa(\mathfrak{p})$ induces $\operatorname{Spec} \kappa(\mathfrak{p}) \to \operatorname{Spec} A = U \to X$. Independence of U follows because on a smaller affine neighbourhood the maps are induced by the same local homomorphism $\mathcal{O}_{X,x} \to \kappa(x)$. □

$$\begin{tikzcd} \operatorname{Spec} \kappa(x) \arrow[r, "i_x"] & X \end{tikzcd}$$

Figure 21.2: Canonical point morphism.

Example 21.4 (A rational point as a morphism)**.** A k-rational point of a k-scheme X is a morphism $\operatorname{Spec} k \to X$. For $X = \mathbb{A}^1_k$, such morphisms correspond to choices $a \in k$ through $k[x] \to k$, $x \mapsto a$.

21.3 Field-valued points

The canonical point morphism has a useful converse.

Theorem 21.5 (Field-valued points)**.** *Let K be a field. To give a morphism*

$$u : \operatorname{Spec} K \longrightarrow X$$

is equivalent to giving a pair (x, ι) consisting of a point $x \in X$ and a field homomorphism

$$\iota : \kappa(x) \longrightarrow K.$$

Proof. A morphism u has a unique image point x. The local stalk map $\mathcal{O}_{X,x} \to K$ kills $\mathfrak{m}_x$, hence factors uniquely through $\kappa(x)$. Conversely, compose the canonical map $\operatorname{Spec} K \to \operatorname{Spec} \kappa(x)$ induced by ι with i_x. The constructions are inverse. □

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] & \operatorname{Spec} \kappa(x) \arrow[r] & X \end{tikzcd}$$

Figure 21.3: Field-valued point factorization.

21.4 Morphisms on points, stalks, and residue fields

A scheme morphism $f : X \to Y$ gives three compatible layers of data:

$$x \longmapsto f(x), \qquad f_x^{\#} : \mathcal{O}_{Y,f(x)} \longrightarrow \mathcal{O}_{X,x}, \qquad \kappa(f(x)) \longrightarrow \kappa(x).$$

The stalk homomorphism is local, so it descends to the indicated residue-field map.

Proposition 21.6 (Residue-field functoriality)**.** *For morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ and $x \in X$, the residue-field map for $g \circ f$ is the composite*

$$\kappa(g(f(x))) \longrightarrow \kappa(f(x)) \longrightarrow \kappa(x).$$

$$\begin{tikzcd}
\mathcal{O}_{Y,f(x)} \arrow[r] \arrow[d] & \mathcal{O}_{X,x} \arrow[d] \\
\kappa(f(x)) \arrow[r] & \kappa(x)
\end{tikzcd}$$

Figure 21.4: Stalk and residue-field maps.

21.5 Specialization and generalization

A point x is a *specialization* of y if $x \in \overline{\{y\}}$; equivalently y is a generalization of x. In an affine scheme $\operatorname{Spec} A$ this is exactly reverse inclusion of prime ideals:

$$\mathfrak{p} \rightsquigarrow \mathfrak{q} \quad \Longleftrightarrow \quad \mathfrak{p} \subseteq \mathfrak{q},$$

where $\mathfrak{q}$ is a specialization of $\mathfrak{p}$.

Proposition 21.7 (Morphisms preserve specialization)**.** *If x is a specialization of y in X and $f : X \to Y$ is a scheme morphism, then $f(x)$ is a specialization of $f(y)$.*

Proof. Continuity gives $f(x) \in f(\overline{\{y\}}) \subseteq \overline{\{f(y)\}}$. □

$$y \xrightarrow{\text{specializes to}} x$$

Figure 21.5: Specialization.

Example 21.8 (A non-rational closed point)**.** In $\operatorname{Spec} \mathbb{R}[x]$, the maximal ideal $(x^2 + 1)$ is a closed point with residue field $\mathbb{C}$. It is a genuine scheme point but not an $\mathbb{R}$-rational point.

21.6 The arithmetic image of a point

The structure morphism detects the characteristic of residue fields.

Proposition 21.9 (Residue characteristic)**.** *Let $x \in X$. Under $\pi_X : X \to \operatorname{Spec} \mathbb{Z}$,*

$$\pi_X(x) = \ker(\mathbb{Z} \to \kappa(x)).$$

Hence $\pi_X(x) = (0)$ when $\operatorname{char} \kappa(x) = 0$, and $\pi_X(x) = (p)$ when $\operatorname{char} \kappa(x) = p > 0$.

$$x \in X \xrightarrow{\ker(\mathbb{Z} \to \kappa(x))} (0) \text{ or } (p) \in \operatorname{Spec} \mathbb{Z}$$

Figure 21.6: Residue characteristic.

21.7 Affine formulas

For $X = \operatorname{Spec} A$ and $x = \mathfrak{p}$, one has

$$\mathcal{O}_{X,x} = A_{\mathfrak{p}}, \qquad \kappa(x) = \operatorname{Frac}(A/\mathfrak{p})$$

when $A/\mathfrak{p}$ is a domain, with the second expression understood as its fraction field. If A is a k-algebra, then k-rational points of X are the morphisms $\operatorname{Spec} k \to X$ over $\operatorname{Spec} k$, equivalently the k-algebra maps $A \to k$.

$$\mathfrak{p} \in \operatorname{Spec} A \longrightarrow A_{\mathfrak{p}} \longrightarrow \kappa(\mathfrak{p})$$

Figure 21.7: Affine point dictionary.

21.8 Schemes over a base

The absolute structure morphism is automatic, but much of algebraic geometry is relative to a chosen base.

Definition 21.10 (S-scheme)**.** Let S be a scheme. An *S-scheme* is a scheme X equipped with a specified morphism

$$p_X : X \longrightarrow S.$$

A morphism of S-schemes $f : X \to Y$ is a scheme morphism satisfying

$$p_Y \circ f = p_X.$$

Equivalently, the category of S-schemes is the slice category $\mathbf{Sch}/S$.

When $S = \operatorname{Spec} k$ for a field k, one speaks of a *k-scheme.* If K/k is a field extension, the inclusion $k \hookrightarrow K$ gives $\operatorname{Spec} K \to \operatorname{Spec} k$, and one defines

$$\boxed{X(K) = \operatorname{Hom}_{\operatorname{Spec} k}(\operatorname{Spec} K, X).}$$

In particular,

$$X(k) = \operatorname{Hom}_{\operatorname{Spec} k}(\operatorname{Spec} k, X)$$

is the set of k-rational points.

This relative notation is stricter than the absolute set $\operatorname{Hom}(\operatorname{Spec} K, X)$: the triangle over $\operatorname{Spec} k$ must commute. For an affine k-scheme $X = \operatorname{Spec} A$,

$$X(K) \cong \operatorname{Hom}_{k\text{-alg}}(A, K).$$

The relative viewpoint is the natural language for fiber products, base change, fibers, and geometric points in the following chapters.

Example 21.11 (Generic point as a field-valued point)**.** For an integral scheme with function field K, the generic point has residue field K. The map $\operatorname{Spec} K \to X$ lands at that generic point and records generic rational functions.

21.9 Local schemes around a point

The local ring itself has a geometric spectrum. There is a canonical morphism

$$\lambda_x : \operatorname{Spec} \mathcal{O}_{X,x} \longrightarrow X$$

whose closed point maps to x. Its points encode generalizations of x visible in sufficiently small affine neighbourhoods. This construction is local and should not be confused with the closed/open subscheme formalism deferred to VI/22.

21.10 Why the local condition matters

A morphism of schemes is a morphism of *locally* ringed spaces: for each x, the map $f_x^\#$ must carry the maximal ideal of $\mathcal{O}_{Y,f(x)}$ into the maximal ideal of $\mathcal{O}_{X,x}$, or equivalently pull the latter maximal ideal back to the former. This is precisely what makes residue-field maps functorial and makes the image point algebraically correct.

$$\operatorname{Spec}\mathcal{O}_{X,x} \xrightarrow{\lambda_x} X$$

Figure 21.8: Local scheme.

$$\mathfrak{m}_{f(x)} \xrightarrow{f_x^\#} \mathfrak{m}_x$$

Figure 21.9: Locality condition.

21.11 Examples

Example 21.12 (The unique map from an affine scheme)**.** For $X = \operatorname{Spec} A$, the structure morphism is induced by the unique unital map $\mathbb{Z} \to A$. On points it sends $\mathfrak{p}$ to $\mathfrak{p} \cap \mathbb{Z}$.

Example 21.13 (A characteristic-zero point)**.** The generic point $(0) \in \operatorname{Spec}\mathbb{Z}$ has residue field $\mathbb{Q}$, so its image under the structure morphism is (0).

Example 21.14 (A characteristic-p point)**.** The closed point $(p) \in \operatorname{Spec}\mathbb{Z}$ has residue field $\mathbb{F}_p$ and maps to (p).

Example 21.15 (Points of the affine line)**.** A point $\mathfrak{p} \in \operatorname{Spec} k[t]$ carries the local ring $k[t]_{\mathfrak{p}}$ and residue field $\kappa(\mathfrak{p})$.

Example 21.16 (A rational point)**.** For $a \in k$, evaluation $k[t] \to k$, $t \mapsto a$, gives a k-point $\operatorname{Spec} k \to \mathbb{A}^1_k$ with image $(t - a)$.

Example 21.17 (A non-rational closed point)**.** Over $k = \mathbb{R}$, the prime $(t^2 + 1)$ has residue field $\mathbb{C}$; it is closed but is not an $\mathbb{R}$-rational point.

Example 21.18 (The generic point of an integral affine scheme)**.** If A is a domain, the point (0) has residue field $\operatorname{Frac}(A)$ and specializes to every point of $\operatorname{Spec} A$.

Example 21.19 (Specialization in $\operatorname{Spec}\mathbb{Z}$)**.** The generic point (0) specializes to every closed point (p).

Example 21.20 (Residue field under a localization morphism)**.** For $A \to A_f$ and $\mathfrak{q} \in \operatorname{Spec} A_f$, the residue field of $\mathfrak{q}$ is canonically isomorphic to the residue field of its contraction in $\operatorname{Spec} A$.

Example 21.21 (Residue field under a quotient morphism)**.** For $A \to A/I$ and $\mathfrak{q} \supset I$, the corresponding point of $\operatorname{Spec}(A/I)$ has the same residue field as $\mathfrak{q} \in \operatorname{Spec} A$.

Example 21.22 (A point morphism)**.** For $x \in X$, the map $\operatorname{Spec}\kappa(x) \to X$ is canonical even when x is not closed.

Example 21.23 (Extension of residue fields)**.** If $K/\kappa(x)$ is a field extension, then $\operatorname{Spec} K \to X$ has image x.

Example 21.24 (Two field-valued points with the same image)**.** Different embeddings $\kappa(x) \to K$ can define distinct morphisms $\operatorname{Spec} K \to X$ with the same underlying image point.

Example 21.25 (A local map on stalks)**.** If $f : X \to Y$ and $x \mapsto y$, then $f_x^\# : \mathcal{O}_{Y,y} \to \mathcal{O}_{X,x}$ sends nonunits to nonunits in the precise local-homomorphism sense.

Example 21.26 (Characteristic map on an arithmetic affine line)**.** A prime $\mathfrak{p} \subset \mathbb{Z}[t]$ maps to $\mathfrak{p} \cap \mathbb{Z}$, either (0) or (p).

Example 21.27 (A scheme entirely in characteristic p)**.** For an $\mathbb{F}_p$-scheme, every residue field has characteristic p, so the absolute structure morphism factors through $\operatorname{Spec}\mathbb{F}_p \to \operatorname{Spec}\mathbb{Z}$.

Example 21.28 (One-point nonreduced scheme)**.** $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ and $\operatorname{Spec} k$ have the same one-point topology but different local rings. Point sets alone do not determine schemes.

Example 21.29 (Continuous maps are not enough)**.** A continuous map of underlying spaces does not determine the sheaf homomorphism, and a sheaf homomorphism without local stalk maps need not be a scheme morphism.

Example 21.30 (Closed versus rational)**.** Closedness is topological; rationality is residue-field-theoretic. A closed point may have residue field strictly larger than the base field.

Example 21.31 (Naturality of the absolute base)**.** For every scheme morphism $f : X \to Y$, uniqueness gives $\pi_Y \circ f = \pi_X$.

Example 21.32 (Field-valued point classification)**.** For any field K, $X(K) = \operatorname{Hom}(\operatorname{Spec} K, X)$ decomposes as pairs $(x, \kappa(x) \to K)$. This separates the topological image from the residue-field embedding.

Example 21.33 (Geometric points)**.** When K is algebraically closed, a morphism $\operatorname{Spec} K \to X$ is often called a geometric point. Its image need not be closed without additional finiteness hypotheses.

Example 21.34 (The local scheme at a point)**.** $\operatorname{Spec}\mathcal{O}_{X,x} \to X$ packages all sufficiently local affine information around x; its closed point maps to x and its other points represent generalizations.

Example 21.35 (Specialization order and prime inclusion)**.** On $\operatorname{Spec} A$, the specialization preorder is exactly inclusion of primes. This order is functorial under every scheme morphism.

Example 21.36 (Residue-characteristic stratification of points)**.** The point set of any scheme is partitioned by the primes of $\mathbb{Z}$ according to residue characteristic, with characteristic zero over (0) and characteristic p over (p).

Example 21.37 (Factorization through $\operatorname{Spec}\mathbb{F}_p$)**.** The absolute structure morphism factors through $\operatorname{Spec}\mathbb{F}_p$ exactly when the global section $p \cdot 1$ is zero; equivalently it is zero in every stalk.

Example 21.38 (Affine points after field extension)**.** For a k-algebra A and field extension K/k, affine points are described by

$$\operatorname{Hom}_k(\operatorname{Spec} K, \operatorname{Spec} A) \cong \operatorname{Hom}_{k\text{-alg}}(A, K).$$

Example 21.39 (Residue fields and composition)**.** For $X \to Y \to Z$, the induced residue-field embeddings compose contravariantly along the image chain.

Example 21.40 (Underlying topology forgets nilpotents)**.** Reduction $X_{\mathrm{red}} \to X$ is a homeomorphism on point sets while changing local rings. This is a canonical reminder that scheme points are enriched by sheaf data.

Example 21.41 (The generic point as a field-valued point)**.** For an integral scheme with generic point η, the canonical $\operatorname{Spec}\kappa(\eta) \to X$ is the generic-point morphism and $\kappa(\eta)$ is the function field when that terminology is introduced later.

Example 21.42 (Arithmetic movement under specialization)**.** If x specializes to y, then $\pi_X(x)$ specializes to $\pi_X(y)$ in $\operatorname{Spec}\mathbb{Z}$. Along a specialization $x \rightsquigarrow y$, residue characteristic may change $0 \to p$, but never $p \to 0$. In particular, a positive-characteristic point cannot specialize to a characteristic-zero point.

Example 21.43 (Point detection by affine neighbourhoods)**.** The image of $\operatorname{Spec} K \to X$ can be studied in any affine neighbourhood of its image, reducing point questions to kernels of ring maps into fields.

Example 21.44 (Locality of image primes)**.** Given $A \to K$ to a field, the image point in $\operatorname{Spec} A$ is the kernel. Localizing at that kernel produces the unique local map $A_{\mathfrak{p}} \to K$.

Example 21.45 (Absolute versus relative viewpoints)**.** Every scheme is automatically over $\operatorname{Spec}\mathbb{Z}$. Choosing another base S adds structure; it does not replace the absolute structure morphism, which is obtained by composition $X \to S \to \operatorname{Spec}\mathbb{Z}$.

21.12 Problem dossiers

VI.21.P1: AG3 Problem 13: $\operatorname{Spec}\mathbb{Z}$ is final

Problem 21.46. Let X be a scheme. Prove that there exists a unique morphism

$$X \longrightarrow \operatorname{Spec}\mathbb{Z},$$

identify it as the absolute structure morphism, describe it on affine charts and on points, and prove its naturality in X.

Provenance. Legacy recovery: theory-of-algebraic-geometry-3.tex, Problem 13. Preserved as the unique canonical migration and expanded to full global, affine, pointwise, and naturality depth.

VI.21.P2: The affine formula for the structure morphism

Problem 21.47. Let $X = \operatorname{Spec} A$. Show that π_X is induced by $\mathbb{Z} \to A$ and prove

$$\pi_X(\mathfrak{p}) = \mathfrak{p} \cap \mathbb{Z}$$

for every $\mathfrak{p} \in \operatorname{Spec} A$.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P3: The canonical morphism of a point

Problem 21.48. For $x \in X$, construct canonically a morphism

$$i_x : \operatorname{Spec}\kappa(x) \longrightarrow X$$

with image x, and prove independence of the affine neighbourhood used in the construction.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P4: Classify morphisms from a field spectrum

Problem 21.49. Let K be a field. Prove that morphisms $\operatorname{Spec} K \to X$ are naturally in bijection with pairs

$$(x, \iota), \qquad x \in X, \quad \iota : \kappa(x) \to K$$

where ι is a field homomorphism. State the relative version over a base field.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P5: Residue fields under a scheme morphism

Problem 21.50. Let $f : X \to Y$ and $x \in X$. Construct canonically a field homomorphism

$$\kappa(f(x)) \longrightarrow \kappa(x)$$

and prove that it is induced by the local stalk homomorphism.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P6: Composition on residue fields

Problem 21.51. For morphisms $X \xrightarrow{f} Y \xrightarrow{g} Z$ and $x \in X$, prove that the residue-field map for $g \circ f$ is the composite of the residue-field maps for g and f.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P7: Specialization in an affine spectrum

Problem 21.52. For primes $\mathfrak{p}, \mathfrak{q} \subset A$, prove

$$\mathfrak{q} \in \overline{\{\mathfrak{p}\}} \iff \mathfrak{p} \subseteq \mathfrak{q}.$$

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P8: Morphisms preserve specialization

Problem 21.53. If $x \in \overline{\{y\}}$ and $f : X \to Y$ is a scheme morphism, prove $f(x) \in \overline{\{f(y)\}}$.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P9: Residue characteristic and the absolute base

Problem 21.54. For $x \in X$, prove

$$\pi_X(x) = \ker(\mathbb{Z} \to \kappa(x)),$$

and deduce the characteristic-zero/characteristic-p alternatives.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P10: k-rational points of an affine k-scheme

Problem 21.55. Let $X = \operatorname{Spec} A$ for a k-algebra A. Prove

$$X(k) = \operatorname{Hom}_{\operatorname{Spec} k}(\operatorname{Spec} k, X) \cong \operatorname{Hom}_{k\text{-alg}}(A, k).$$

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P11: Closed does not mean rational

Problem 21.56. Give a closed point of $\mathbb{A}^1_{\mathbb{R}}$ that is not $\mathbb{R}$-rational, and prove both assertions.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P12: The generic point morphism

Problem 21.57. Let A be a domain. Describe the canonical morphism attached to the generic point $(0) \in \operatorname{Spec} A$ and identify its residue field.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P13: The local scheme at a point

Problem 21.58. Construct the canonical morphism

$$\lambda_x : \operatorname{Spec} \mathcal{O}_{X,x} \longrightarrow X$$

and prove that the closed point of the source maps to x, independently of the affine chart.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P14: Generalizations visible in the local scheme

Problem 21.59. For $x = \mathfrak{p} \in \operatorname{Spec} A$, identify the image of

$$\operatorname{Spec} A_{\mathfrak{p}} \longrightarrow \operatorname{Spec} A$$

and interpret it in the specialization order.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P15: Factorization through characteristic p

Problem 21.60. Prove that the absolute structure morphism π_X factors through

$$\operatorname{Spec} \mathbb{F}_p \longrightarrow \operatorname{Spec} \mathbb{Z}$$

if and only if $p \cdot 1 = 0$ in $\Gamma(X, \mathcal{O}_X)$, and prove uniqueness of the factorization.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P16: Pointwise residue characteristic versus scheme-theoretic characteristic

Problem 21.61. Assume every $x \in X$ has residue field of characteristic p. Determine the point-set image of $X \to \operatorname{Spec}\mathbb{Z}$, give an example showing that this need not imply factorization through $\operatorname{Spec}\mathbb{F}_p$, and state the correct criterion.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P17: Topological points do not determine a scheme

Problem 21.62. Compare $X = \operatorname{Spec} k$ and $Y = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. Prove their underlying spaces are homeomorphic but the schemes are not isomorphic.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P18: Why local stalk maps are required

Problem 21.63. Explain precisely why the local-homomorphism condition on stalks is essential in the definition of a scheme morphism.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P19: Naturality of the structure morphism

Problem 21.64. For every scheme morphism $f : X \to Y$, prove that

$$\pi_X = \pi_Y \circ f,$$

and interpret this as a commutative square over the absolute base.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

VI.21.P20: Recovering the image point from a field map

Problem 21.65. Let $U = \operatorname{Spec} A \subseteq X$ and let $u : \operatorname{Spec} K \to X$ have image in U. Prove that its image point in U corresponds to the kernel of the induced ring map $A \to K$.

Provenance. Canonical VI/21 derivative/application; not an additional legacy migration.

21.13 Exercises

Exercise 21.66. Compute the image in $\operatorname{Spec}\mathbb{Z}$ of every point of $\operatorname{Spec}\mathbb{F}_p[t]$.

Exercise 21.67. Compute the image of $(2,t)$ and (0) in $\operatorname{Spec}\mathbb{Z}[t]$ under the absolute structure morphism.

Exercise 21.68. For $x \in X$, prove directly that $\operatorname{Spec}\kappa(x) \to X$ is unchanged after replacing an affine neighbourhood by a smaller principal neighbourhood.

Exercise 21.69. Classify morphisms $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{R}[t]$ whose image is (t^2+1).

Exercise 21.70. Describe the specialization order on $\operatorname{Spec} k[t]$.

Exercise 21.71. Show that a closed point can have a nontrivial residue-field extension of the base field.

Exercise 21.72. For $A \to B$ and $\mathfrak{q} \in \operatorname{Spec} B$, write the induced map $\kappa(\mathfrak{q} \cap A) \to \kappa(\mathfrak{q})$.

Exercise 21.73. Prove that an isomorphism of schemes induces isomorphisms on all residue fields.

Exercise 21.74. For a domain A, identify the residue field at the generic point.

Exercise 21.75. Check that $\operatorname{Spec} A_{\mathfrak{p}} \to \operatorname{Spec} A$ sends its closed point to $\mathfrak{p}$.

Exercise 21.76. Identify all points in the image of $\operatorname{Spec} A_{\mathfrak{p}} \to \operatorname{Spec} A$.

Exercise 21.77. Prove that the image of a specialization chain under a scheme morphism is again a specialization chain, possibly with repetitions.

Exercise 21.78. Show that $X(k)$ for $X = \mathbb{A}^n_k$ is naturally k^n.

Exercise 21.79. For $X = \operatorname{Spec} k[x,y]/(xy)$, describe the canonical residue-field morphisms at the two generic points of the irreducible components.

Exercise 21.80. Explain why $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ has the same point set as $\operatorname{Spec} k$ but different stalk data.

Exercise 21.81. Prove that π_X is compatible with restriction to an open subscheme.

Exercise 21.82. If every point of X has residue characteristic p, determine the image of the underlying point map $X \to \operatorname{Spec}\mathbb{Z}$.

Exercise 21.83. Give an example of two distinct K-valued points with the same image point.

Exercise 21.84. Show that a k-rational point x has a k-algebra homomorphism $\kappa(x) \to k$.

Exercise 21.85. When k is algebraically closed, prove that a closed point of an affine finite-type k-scheme has residue field k only if you may invoke the appropriate Nullstellensatz; record the dependency rather than proving it here.

Exercise 21.86. Compute the residue characteristic of the point $(3, t^2+1) \in \operatorname{Spec}\mathbb{Z}[t]$.

Exercise 21.87. For $X \to Y \to Z$, draw the stalk and residue-field diagrams at a point $x \in X$.

Exercise 21.88. Prove that the canonical map $\operatorname{Spec}\kappa(x) \to X$ factors through every open neighbourhood of x.

Exercise 21.89. For $A \to K$ with K a field, prove that the induced local map $A_\mathfrak{p} \to K$ exists for $\mathfrak{p} = \ker(A \to K)$.

Exercise 21.90. Explain why the final-object property of $\operatorname{Spec} \mathbb{Z}$ is stronger than the affine statement alone.

Exercise 21.91. Summarize in one diagram the data attached to $x \in X$: stalk, maximal ideal, residue field, and point morphism.

21.14 Challenges

Problem 21.92 (Field-valued points as a disjoint union)**.** Prove a natural set-level decomposition

$$X(K) \cong \coprod_{x \in X} \operatorname{Hom}_{\mathrm{Fields}}(\kappa(x), K),$$

where only field homomorphisms compatible with any specified base structure are retained in the relative case. Explain exactly where locality of stalk maps enters.

Problem 21.93 (Specialization posets are functorial)**.** Regard the point set of a scheme as a preorder under specialization. Prove that every scheme morphism is monotone. Determine when two distinct specialization points may collapse to one image point.

Problem 21.94 (Point sets versus scheme structure)**.** Construct several affine schemes with homeomorphic underlying point spaces but nonisomorphic structure sheaves. Use nilpotents and residue-field variation to show precisely what topological points forget.

Problem 21.95 (Absolute characteristic portrait)**.** For a scheme X, study the map of point preorders $|X| \to |\operatorname{Spec} \mathbb{Z}|$. Show that it records residue characteristic and is compatible with specialization. Give examples containing both characteristic-zero and characteristic-p points.

Problem 21.96 (Local schemes and generalizations)**.** For $x \in X$, prove locally that the image of $\operatorname{Spec} \mathcal{O}_{X,x} \to X$ is exactly the set of generalizations of x. Check compatibility on overlaps and explain why this result is independent of an affine chart.

21.15 Chapter synthesis

The pointwise data of a scheme are organized by the diagram

$$\text{point } x \rightsquigarrow \mathcal{O}_{X,x} \rightsquigarrow \kappa(x) \rightsquigarrow \operatorname{Spec} \kappa(x) \to X,$$

while every morphism transports this structure by local maps. The absolute structure morphism then places every point over a unique prime of $\mathbb{Z}$, recording its residue characteristic.

$$x \longrightarrow \mathcal{O}_{X,x} \longrightarrow \kappa(x) \longrightarrow \operatorname{Spec} \kappa(x) \to X$$

Figure 21.10: Pointwise architecture.

21.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 21.97 (Affine rational point)**.** What ring map represents $a \in \mathbb{A}^1(k)$?

Hint. Use scheme points, residue fields, and functorial points. ☐

Solution. $k[x] \to k$, $x \mapsto a$. ☐

Exercise 21.98 (Residue field)**.** What is the residue field at $(x^2+1) \subset \mathbb{R}[x]$?

Hint. Use scheme points, residue fields, and functorial points. ☐

Solution. $\mathbb{C}$. ☐

Exercise 21.99 (Generic residue)**.** What is the residue field at the generic point of an integral affine scheme?

Hint. Use scheme points, residue fields, and functorial points. ☐

Solution. Its fraction field. ☐

Exercise 21.100 (Underlying point)**.** What ring-theoretic object is a point of $\operatorname{Spec} A$?

Hint. Use scheme points, residue fields, and functorial points. ☐

Solution. A prime ideal. ☐

Exercise 21.101 (Closed affine point)**.** When is a spectral point closed?

Hint. Use scheme points, residue fields, and functorial points. ☐

Solution. When its prime ideal is maximal. ☐

Proofs

Exercise 21.102 (Rational points)**.** Prove: Prove $\mathbb{A}^n_k(k) \cong k^n$.

Hint. Work directly from scheme points, residue fields, and functorial points. ☐

Solution. Ring maps $k[x_1, \ldots, x_n] \to k$ are determined freely by the images of the coordinate variables. ☐

Exercise 21.103 (Point factorization)**.** Prove: Prove a morphism $\operatorname{Spec} K \to X$ has an underlying image point together with an embedding of its residue field into K.

Hint. Work directly from scheme points, residue fields, and functorial points. ☐

Solution. Use the local map on stalks at the image point and pass to residue fields. ☐

Exercise 21.104 (Closed point residue)**.** Prove: Prove the residue field at a maximal ideal $\mathfrak{m}$ of an affine scheme is $\operatorname{Frac}(A/\mathfrak{m}) = A/\mathfrak{m}$.

Hint. Work directly from scheme points, residue fields, and functorial points. ☐

Solution. The quotient is already a field. ☐

Exercise 21.105 (Generic function field)**.** Prove: Prove the residue field at the generic point of $\operatorname{Spec} A$ for a domain is $\operatorname{Frac} A$.

Hint. Work directly from scheme points, residue fields, and functorial points. ☐

Solution. Localizing at the zero prime inverts every nonzero element. ☐

Counterexamples and hypothesis tests

Exercise 21.106 (All rational)**.** Test the claim: every closed point of a k-scheme is k-rational.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; residue fields can be finite or algebraic extensions. ☐

Exercise 21.107 (Points maximal)**.** Test the claim: all scheme points are maximal ideals.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; generic and intermediate primes are essential. ☐

Exercise 21.108 (Same point functors)**.** Test the claim: underlying topological points capture all field-valued points.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; residue-field embeddings carry additional data. ☐

Applications and investigations

Exercise 21.109 (Field extensions)**.** Why consider K-valued points for extensions K/k?

Hint. Translate the question into scheme points, residue fields, and functorial points. ☐

Solution. They reveal geometric points and solutions not rational over the base field. ☐

Exercise 21.110 (Moduli intuition)**.** Why formulate points as morphisms from spectra of fields?

Hint. Translate the question into scheme points, residue fields, and functorial points. ☐

Solution. This description is functorial and extends naturally to families over arbitrary test schemes. ☐

Challenge problems

Exercise 21.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Schemes and Points in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Schemes and Points, translates it through scheme points, residue fields, and functorial points, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 21.112 (Hypothesis challenge)**.** Choose one structural statement from Schemes and Points, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 22

Open and Closed Subschemes

Subschemes are the scheme-theoretic way to isolate a region or impose equations. Open subschemes are obtained by restriction. Closed subschemes require more information than a closed subset: an ideal of functions records not only where the subscheme lies, but also its possible nilpotent and infinitesimal structure. This chapter develops the local affine construction, the global gluing principle, and the language of open, closed, and locally closed immersions without consuming the later theories of fiber products or quasi-coherent sheaves.

22.1 Open subschemes

Let X be a scheme and $U \subseteq X$ open. The pair

$$(U, \mathcal{O}_X|_U)$$

is again a scheme. Its inclusion

$$j : U \hookrightarrow X$$

is an *open immersion*: it is a homeomorphism onto an open subset and identifies the structure sheaf of U with the restricted structure sheaf of X.

Proposition 22.1 (Uniqueness of the open subscheme structure)**.** *For a fixed open subset $U \subseteq X$, there is a unique scheme structure on U for which the inclusion is an open immersion, up to the unique isomorphism compatible with the inclusion.*

$$U \xrightarrow{\text{open immersion}} X$$

22.2 Principal opens revisited

If $X = \operatorname{Spec} A$ and $f \in A$, then VI/18 gives

$$D(f) \cong \operatorname{Spec} A_f.$$

Thus the canonical open subscheme on a principal open is simultaneously the localization construction. For $g \in A$,

$$D(f) \cap D(g) = D(fg), \qquad (A_f)_{g/1} \cong A_{fg}.$$

This chapter uses that theorem; it does not re-migrate it.

$$A_f \longleftarrow A$$

$$\operatorname{Spec} A_f \cong D(f) \longrightarrow \operatorname{Spec} A$$

Example 22.2 (Closed subscheme from one equation)**.** Inside $\mathbb{A}^2_k = \operatorname{Spec} k[x,y]$, the quotient by $(y - x^2)$ defines the parabola as a closed subscheme. Its structure sheaf is the quotient by the ideal sheaf generated by $y - x^2$.

22.3 Closed subschemes in the affine case

Let A be a ring and $I \subseteq A$ an ideal. The quotient map

$$A \twoheadrightarrow A/I$$

induces

$$i : \operatorname{Spec}(A/I) \longrightarrow \operatorname{Spec} A.$$

Its image is the closed subset

$$V(I) = \{\mathfrak{p} \in \operatorname{Spec} A : I \subseteq \mathfrak{p}\}.$$

On points, $\mathfrak{q} \subseteq A/I$ is sent to its inverse image in A.

Theorem 22.3 (Affine closed subscheme)**.** *The morphism* $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$ *is a homeomorphism onto* $V(I)$*. For* $\mathfrak{p} \supseteq I$*, the induced local ring is*

$$\mathcal{O}_{\operatorname{Spec}(A/I),\mathfrak{p}/I} \cong A_{\mathfrak{p}}/IA_{\mathfrak{p}}.$$

$$A/I \longleftarrow A$$

$$\operatorname{Spec}(A/I) \xrightarrow[\text{closed immersion}]{} \operatorname{Spec} A$$

22.4 A closed subset is not a closed subscheme

The topological set $V(I)$ depends only on $\sqrt{I}$, but the closed subscheme $\operatorname{Spec}(A/I)$ depends on I itself. Hence

$$V(I) = V(\sqrt{I})$$

while

$$\operatorname{Spec}(A/I) \quad \text{and} \quad \operatorname{Spec}(A/\sqrt{I})$$

need not be isomorphic. For example, (x) and (x^2) define the same closed point of $\mathbb{A}^1_k$ but different closed subschemes.

Definition 22.4 (Reduced induced closed subscheme)**.** For an affine closed set $V(I) \subseteq \operatorname{Spec} A$, its reduced induced closed subscheme is $\operatorname{Spec}(A/\sqrt{I})$.

$$(x) \xrightarrow{\text{same radical}} (x^2)$$

$$\operatorname{Spec} k \xrightarrow[\text{same support, different sheaf}]{} \operatorname{Spec} k[x]/(x^2)$$

22.5 Closed subschemes on a general scheme

A closed subscheme $Z \hookrightarrow X$ is locally affine in the following precise sense. For an affine open cover $X = \bigcup U_i$ with $U_i = \operatorname{Spec} A_i$, each inverse image $Z \cap U_i$ is of the form

$$\operatorname{Spec}(A_i/I_i) \hookrightarrow \operatorname{Spec} A_i,$$

and the ideals agree after restriction to overlaps. These compatible local quotients glue by VI/20. Equivalently, one may package the compatible ideals into an ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_X$ and write

$$\mathcal{O}_Z \cong (\mathcal{O}_X/\mathcal{I})|_Z.$$

The full theory of quasi-coherent ideal sheaves is deferred to VI/33; only the local affine compatibility needed for closed subschemes is used here.

$$\begin{tikzcd} Z \cap U \arrow[r] \arrow[d, dashed, dash, "\text{agree on overlap}"'] & U \\ Z \cap V \arrow[r] & V \end{tikzcd}$$

Example 22.5 (Nonreduced closed subscheme)**.** The ideal (y^2) defines a double line supported on $V(y)$. The closed subscheme and the reduced line have the same underlying closed set but different structure sheaves.

22.6 Closed immersions

A morphism $i : Z \to X$ is a *closed immersion* if it identifies Z with a closed subscheme of X.

Theorem 22.6 (Affine-local criterion)**.** *A morphism $i : Z \to X$ is a closed immersion if and only if X has an affine open cover $U_\alpha = \operatorname{Spec} A_\alpha$ such that each $i^{-1}(U_\alpha)$ is affine, say $\operatorname{Spec} B_\alpha$, and each induced map*

$$A_\alpha \longrightarrow B_\alpha$$

is surjective.

$$\begin{tikzcd}
B = A/I & A \arrow[l] \\
Z \arrow[r] & X
\end{tikzcd}$$

22.7 Restriction of closed subschemes to opens

If $Z \hookrightarrow X$ is closed and $U \subseteq X$ is open, then

$$Z \cap U \hookrightarrow U$$

is a closed subscheme. In an affine chart $X = \operatorname{Spec} A$, $Z = \operatorname{Spec}(A/I)$ and $U = D(f)$, one obtains

$$Z \cap D(f) \cong \operatorname{Spec}((A/I)_{\bar{f}}) \cong \operatorname{Spec}(A_f/IA_f).$$

This compatibility is the local mechanism by which closed subschemes glue.

$$\begin{tikzcd}
\operatorname{Spec}(A_f/IA_f) \arrow[r] \arrow[d] & D(f) \arrow[d] \\
\operatorname{Spec}(A/I) \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

22.8 Open, closed, and locally closed immersions

An *immersion* is a morphism that factors as a closed immersion followed by an open immersion. Equivalently, it identifies the source with a closed subscheme of some open subscheme of the target. Its image is therefore locally closed as a topological subset.

Remark 22.7. This chapter treats the basic immersion formalism only. Projective closed embeddings and their homogeneous ideals are reserved for VI/38.

$$\begin{tikzcd}
Z \arrow[r, "\text{closed}"] \arrow[rr, bend right, "\text{immersion}"'] & U \arrow[r, "\text{open}"] & X
\end{tikzcd}$$

Example 22.8 (Principal open subscheme)**.** The open subscheme $D(x) \subset \mathbb{A}^2_k$ is affine with coordinate ring $k[x, y, x^{-1}]$. Open immersion corresponds to restricting the structure sheaf rather than quotienting it.

22.9 Nilpotent thickenings and the reduced subscheme

For any affine $X = \operatorname{Spec} A$, the quotient by the nilradical gives

$$X_{\text{red}} = \operatorname{Spec}(A/\sqrt{(0)}) \hookrightarrow X.$$

It has the same underlying topological space as X but no nilpotents. These affine reductions are compatible on overlaps, so they define a canonical closed subscheme for every scheme.

Proposition 22.9 (Universal property of reduction)**.** *If T is reduced, every morphism $T \to X$ factors uniquely through $X_{\mathrm{red}} \hookrightarrow X$.*

$$X_{\mathrm{red}} \xrightarrow{\text{same points; nilpotents removed}} X$$

22.10 Equations and factorization

For $X = \operatorname{Spec} A$, $Z = \operatorname{Spec}(A/I)$, and an affine scheme $T = \operatorname{Spec} B$, a morphism $T \to X$ corresponding to $A \to B$ factors through Z exactly when every element of I maps to zero in B. Thus closed subschemes represent the condition that specified functions vanish scheme-theoretically, not merely pointwise.

22.11 Examples

Example 22.10 (A principal open)**.** In $X = \operatorname{Spec} k[x, y]$, the open subscheme $D(x)$ is affine, with

$$D(x) \cong \operatorname{Spec} k[x, y, x^{-1}].$$

Example 22.11 (The complement of the origin in the affine line)**.** $D(t) \subset \operatorname{Spec} k[t]$ is the multiplicative group $\operatorname{Spec} k[t, t^{-1}]$.

Example 22.12 (A reduced closed point)**.** The ideal $(x) \subset k[x]$ defines the closed subscheme $\operatorname{Spec} k$ at the origin.

Example 22.13 (A doubled closed point)**.** The ideal (x^2) defines a nonreduced closed subscheme supported at the same origin.

Example 22.14 (A coordinate axis)**.** In $\mathbb{A}^2_k = \operatorname{Spec} k[x, y]$, the ideal (x) defines the y-axis $\operatorname{Spec} k[y]$.

Example 22.15 (The union of coordinate axes)**.** The ideal (xy) defines $V(xy) = V(x) \cup V(y)$ with coordinate ring $k[x, y]/(xy)$.

Example 22.16 (A thickened axis)**.** $(x^2) \subset k[x, y]$ defines an infinitesimal thickening of the y-axis.

Example 22.17 (Restriction of a closed point)**.** If $Z = V(x) \subset \operatorname{Spec} k[x]$ and $U = D(x)$, then $Z \cap U = \varnothing$ because $(k[x]/(x))_x = 0$.

Example 22.18 (Restriction of an axis)**.** For $Z = V(x) \subset \mathbb{A}^2$ and $U = D(y)$, $Z \cap U = \operatorname{Spec} k[y, y^{-1}]$.

Example 22.19 (A quotient-induced closed immersion)**.** A surjection $A \twoheadrightarrow B$ always induces a closed immersion $\operatorname{Spec} B \hookrightarrow \operatorname{Spec} A$.

Example 22.20 (An open immersion from localization)**.** $A \to A_f$ induces the open immersion $D(f) \hookrightarrow \operatorname{Spec} A$.

Example 22.21 (Same support, different subschemes)**.** (x) and (x^n) have the same radical for $n \geq 2$ but different quotient rings.

Example 22.22 (Reduction of a doubled point)**.** $(\operatorname{Spec} k[\varepsilon]/(\varepsilon^2))_{\mathrm{red}} = \operatorname{Spec} k$.

Example 22.23 (Reduction of a reducible affine scheme)**.** $k[x,y]/(x^2y)$ reduces to $k[x,y]/(xy)$ because $\sqrt{(x^2y)} = (xy)$.

Example 22.24 (A locally closed point)**.** A closed point in an open subscheme is a locally closed subscheme of the ambient scheme.

Example 22.25 (A punctured closed curve)**.** If $C = V(y - x^2) \subset \mathbb{A}^2$ and $U = D(x)$, then $C \cap U$ is closed in U but generally not closed in all of $\mathbb{A}^2$.

Example 22.26 (Vanishing criterion)**.** A map $\operatorname{Spec} B \to \operatorname{Spec} A$ factors through $V(I)$ exactly when the ring map kills I.

Example 22.27 (Closed immersion and residue fields)**.** For $\operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec} A$, the residue field at $\mathfrak{p}/I$ is canonically the residue field at $\mathfrak{p}$.

Example 22.28 (Open immersion and stalks)**.** For $j : U \hookrightarrow X$ and $u \in U$, the stalk map $\mathcal{O}_{X,u} \to \mathcal{O}_{U,u}$ is an isomorphism.

Example 22.29 (Closed immersion and local rings)**.** At $z \in Z \hookrightarrow X$, the local ring $\mathcal{O}_{Z,z}$ is a quotient of $\mathcal{O}_{X,z}$.

Example 22.30 (Two closed structures on one support)**.** For $I = (x)$ and $J = (x^3)$ in $k[x]$, the supports agree while the conormal nilpotent structure differs.

Example 22.31 (Scheme-theoretic intersection in an affine chart)**.** Closed subschemes defined by I and J impose both sets of equations; locally their common closed subscheme is defined by $I + J$.

Example 22.32 (Scheme-theoretic union in an affine chart)**.** The smallest closed subscheme containing both $V(I)$ and $V(J)$ is defined by $I \cap J$.

Example 22.33 (Set-theoretic versus scheme-theoretic union)**.** $V(IJ) = V(I \cap J)$ topologically, although the quotient schemes defined by IJ and $I \cap J$ can differ.

Example 22.34 (A first infinitesimal neighbourhood)**.** If $Z = V(I) \subset X$ is affine, $V(I^2)$ has the same support but remembers first-order normal information.

Example 22.35 (Closed immersion detected locally)**.** If a morphism becomes a quotient map on coordinate rings after restricting to an affine cover of the target, the pieces glue to a closed immersion.

Example 22.36 (Open immersion detected locally)**.** If a morphism identifies inverse images of an open cover with open subschemes and the local maps agree, it is an open immersion.

Example 22.37 (Immersion into an affine scheme)**.** A locally closed subscheme $Z \subset U \subset \operatorname{Spec} A$ is affine-locally a quotient of a localization A_f.

Example 22.38 (The reduced induced structure)**.** Among closed subschemes with support $V(I)$, $\operatorname{Spec}(A/\sqrt{I})$ is the unique reduced one.

Example 22.39 (Nilpotents are invisible to points)**.** The canonical closed immersion $X_{\mathrm{red}} \hookrightarrow X$ is a homeomorphism on underlying spaces.

Example 22.40 (Vanishing after localization)**.** The restriction of $V(I)$ to $D(f)$ is empty exactly when $IA_f = A_f$, equivalently some power of f lies in I.

Example 22.41 (Closed subscheme of a closed subscheme)**.** If $I \subset J \subset A$, then $\operatorname{Spec}(A/J) \hookrightarrow \operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec} A$ is a chain of closed immersions.

Example 22.42 (Composition of open immersions)**.** The composite of two open immersions is an open immersion because restrictions of restrictions are restrictions.

Example 22.43 (Composition of closed immersions)**.** Affine-locally it is induced by a composite of surjective ring maps, hence is again a closed immersion.

22.12 Problem dossiers

VI.22.P1: The canonical open subscheme

Problem 22.44. Let $U \subseteq X$ be open. Prove that $(U, \mathcal{O}_X|_U)$ is a scheme and that $U \hookrightarrow X$ is an open immersion.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P2: Uniqueness of the open subscheme structure

Problem 22.45. Suppose a scheme structure on the open set $U \subseteq X$ makes the inclusion an open immersion. Prove it is uniquely isomorphic to $(U, \mathcal{O}_X|_U)$ over X.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P3: Affine closed subschemes from ideals

Problem 22.46. Let $I \subseteq A$. Prove that $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$ is a homeomorphism onto $V(I)$ and compute the stalks.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P4: Same support, different closed subschemes

Problem 22.47. Show that (x) and (x^2) in $k[x]$ define the same closed subset but nonisomorphic closed subschemes of $\mathbb{A}^1_k$.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P5: Restriction of a closed subscheme to a principal open

Problem 22.48. For $Z = \operatorname{Spec}(A/I) \hookrightarrow X = \operatorname{Spec} A$ and $U = D(f)$, prove $Z \cap U \cong \operatorname{Spec}(A_f/IA_f)$.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P6: Local gluing of closed subschemes

Problem 22.49. Let $X = U \cup V$ with U, V affine. Suppose closed subschemes $Z_U \hookrightarrow U$ and $Z_V \hookrightarrow V$ agree on $U \cap V$. Show they glue uniquely to a closed subscheme of X.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P7: The affine-local criterion for closed immersions

Problem 22.50. Prove that a morphism that is induced by surjective ring maps on an affine cover of the target is a closed immersion.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P8: Factorization through a closed subscheme

Problem 22.51. Let $A \to B$ define $f : \operatorname{Spec} B \to \operatorname{Spec} A$ and let $Z = \operatorname{Spec}(A/I)$. Prove f factors through Z iff I maps to zero in B.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P9: The reduced subscheme of an affine scheme

Problem 22.52. For $X = \operatorname{Spec} A$, show $X_{\mathrm{red}} = \operatorname{Spec}(A/\sqrt{(0)})$ has the same underlying space as X and is reduced.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P10: Universal property of reduction

Problem 22.53. Let T be reduced. Prove every morphism $T \to X$ factors uniquely through X_{red}.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P11: Empty restriction criterion

Problem 22.54. Let $Z = V(I) \subset \operatorname{Spec} A$. Prove $Z \cap D(f) = \varnothing$ iff $IA_f = A_f$, equivalently iff $f^n \in I$ for some $n \geq 0$.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P12: Closed subschemes inside closed subschemes

Problem 22.55. If $I \subseteq J \subseteq A$, identify $\operatorname{Spec}(A/J)$ as a closed subscheme of $\operatorname{Spec}(A/I)$.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P13: Scheme-theoretic intersection

Problem 22.56. Let $Z_1 = V(I)$ and $Z_2 = V(J)$ be affine closed subschemes of $\operatorname{Spec} A$. Show the closed subscheme imposing both sets of equations is $V(I + J)$.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P14: Scheme-theoretic union

Problem 22.57. Show that the smallest affine closed subscheme containing both $V(I)$ and $V(J)$ is defined by $I \cap J$.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P15: Open immersions compose

Problem 22.58. Prove the composite of two open immersions is an open immersion.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P16: Closed immersions compose

Problem 22.59. Prove the composite of two closed immersions is a closed immersion.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P17: A locally closed immersion

Problem 22.60. Let $U \subseteq X$ be open and $Z \hookrightarrow U$ closed. Show $Z \to X$ is an immersion and its image is locally closed.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P18: Closedness is not determined by point image alone

Problem 22.61. Explain why knowing only the subset $|Z| \subseteq |X|$ cannot determine a closed subscheme structure.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P19: Closed immersion and residue fields

Problem 22.62. For $i : Z \hookrightarrow X$ closed and $z \in Z$, prove the induced map $\kappa(i(z)) \to \kappa(z)$ is an isomorphism.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

VI.22.P20: Open immersion and local rings

Problem 22.63. For $j : U \hookrightarrow X$ open and $u \in U$, prove $\mathcal{O}_{X,u} \to \mathcal{O}_{U,u}$ is an isomorphism.

Provenance. Canonical VI/22 dossier developed from the AG5 T02 theory scope; no numbered legacy problem is claimed.

22.13 Exercises

Exercise 22.64. For $U \subseteq X$ open, show every affine open of U is also an open subscheme of X.

Exercise 22.65. Compute the closed subscheme of $\operatorname{Spec} \mathbb{Z}$ defined by (n).

Exercise 22.66. Compare the closed subschemes of $\mathbb{A}^1_k$ defined by (x), (x^2), and (x^3).

Exercise 22.67. For $I = (x, y) \subset k[x, y]$, compute $\mathcal{O}_{V(I),0}$.

Exercise 22.68. Prove $V(I) = V(\sqrt{I})$ as subsets of $\operatorname{Spec} A$.

Exercise 22.69. Give two ideals with the same radical but nonisomorphic quotient rings.

Exercise 22.70. Compute $V(x) \cap D(y)$ scheme-theoretically in $\operatorname{Spec} k[x, y]$.

Exercise 22.71. For $Z = V(I)$, prove $Z \cap D(f)$ is empty iff some power of f belongs to I.

Exercise 22.72. Prove a quotient of a local ring by a proper ideal is local.

Exercise 22.73. Show a surjective ring map induces isomorphisms on residue fields at corresponding primes.

Exercise 22.74. Prove the reduction of a reduced scheme is itself.

Exercise 22.75. Compute the reduction of $\operatorname{Spec} k[x, y]/(x^2y^3)$.

Exercise 22.76. Show $(X_{mred})_{mred} = X_{mred}$.

Exercise 22.77. Prove every map from a reduced affine scheme to $\operatorname{Spec} A$ kills the nilradical of A.

Exercise 22.78. For ideals I, J, compare $V(I + J)$ with $V(I) \cap V(J)$ topologically.

Exercise 22.79. For ideals I, J, compare $V(I \cap J)$ with $V(I) \cup V(J)$ topologically.

Exercise 22.80. Compute the scheme-theoretic intersection of the axes in $\mathbb{A}^2_k$.

Exercise 22.81. Compute the scheme-theoretic union of the axes in $\mathbb{A}^2_k$.

Exercise 22.82. Give an immersion whose image is not closed in the ambient scheme.

Exercise 22.83. Prove an open immersion is injective on points.

Exercise 22.84. Prove a closed immersion is injective on points.

Exercise 22.85. Show a morphism that is both open and closed immersion identifies the source with an open-and-closed subscheme.

Exercise 22.86. For $I \subset J$, identify the ideal of $\mathrm{Spec}(A/J)$ inside $\mathrm{Spec}(A/I)$.

Exercise 22.87. Check that localization commutes with quotient: $(A/I)_f \cong A_f/IA_f$.

Exercise 22.88. Explain why the open complement of a closed subset has a canonical open subscheme structure even when the closed subset has many closed subscheme structures.

Exercise 22.89. Give an example of a nilpotent thickening with unchanged underlying topological space.

22.14 Challenges

Challenge 1: Classify closed subschemes of a one-point affine scheme

Problem 22.90. Let A be local Artinian with one prime ideal. Describe closed subschemes of $\mathrm{Spec}\, A$ in terms of ideals of A, and explain why they can all have the same underlying point.

Challenge 2: Reduction is global

Problem 22.91. Prove that the affine reductions on an arbitrary affine cover of X glue independently of the cover.

Challenge 3: Closed structures with fixed support

Problem 22.92. For $X = \mathrm{Spec}\, A$ and a closed subset $V(J)$, show closed subscheme structures with this support correspond to ideals I with $\sqrt{I} = \sqrt{J}$.

Challenge 4: Immersions are locally affine quotients of localizations

Problem 22.93. Show that an immersion into an affine scheme is locally represented by a surjection $A_f \twoheadrightarrow B$.

Challenge 5: Detecting scheme-theoretic vanishing

Problem 22.94. Let $I \subseteq A$ and $f : \mathrm{Spec}\, B \to \mathrm{Spec}\, A$. Compare the conditions that f maps all points into $V(I)$ and that f factors through $\mathrm{Spec}(A/I)$. Give an example showing the first is weaker.

22.15 Chapter synthesis

Open subschemes restrict the structure sheaf; closed subschemes quotient it by compatible local ideals. The distinction between a closed subset and a closed subscheme is the distinction between radical information and full ideal-theoretic information. Immersions combine the two operations, and nilpotent thickenings show why scheme structure is strictly richer than topology.

$$\text{open} = \text{restrict} \longrightarrow \text{closed} = \text{quotient} \longrightarrow \text{immersion} = \text{closed then open}$$

22.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 22.95 (Parabola closed)**.** What quotient ring defines the parabola?

Hint. Use open immersions, closed immersions, and ideal sheaves. ☐

Solution. $k[x, y]/(y - x^2)$. ☐

Exercise 22.96 (Double line)**.** What quotient defines the double line?

Hint. Use open immersions, closed immersions, and ideal sheaves. ☐

Solution. $k[x, y]/(y^2)$. ☐

Exercise 22.97 (Principal open)**.** What ring defines $D(x)$?

Hint. Use open immersions, closed immersions, and ideal sheaves. ☐

Solution. $k[x, y, x^{-1}]$. ☐

Exercise 22.98 (Support)**.** What is the support of the double line?

Hint. Use open immersions, closed immersions, and ideal sheaves. ☐

Solution. The ordinary line $V(y)$. ☐

Exercise 22.99 (Open sheaf)**.** What structure sheaf does an open subscheme use?

Hint. Use open immersions, closed immersions, and ideal sheaves. ☐

Solution. The restriction of the ambient structure sheaf. ☐

Proofs

Exercise 22.100 (Closed immersion affine criterion)**.** Prove: Prove $\operatorname{Spec}(A/I) \to \operatorname{Spec} A$ is a closed immersion with image $V(I)$.

Hint. Work directly from open immersions, closed immersions, and ideal sheaves. ☐

Solution. Use quotient-prime correspondence and the local quotient structure sheaf. ☐

Exercise 22.101 (Open restriction)**.** Prove: Prove an open subset with restricted structure sheaf is a scheme.

Hint. Work directly from open immersions, closed immersions, and ideal sheaves. ☐

Solution. Intersect an affine cover with the open subset and refine by distinguished affine opens. ☐

Exercise 22.102 (Nilpotent support)**.** Prove: Prove $V(I) = V(\sqrt{I})$ but the closed subschemes defined by I and $\sqrt{I}$ can differ.

Hint. Work directly from open immersions, closed immersions, and ideal sheaves. ☐

Solution. Zero sets depend only on radicals, while quotient rings can retain nilpotents. ☐

Exercise 22.103 (Ideal sheaf)**.** Prove: Show a quasi-coherent ideal sheaf on an affine scheme determines a closed subscheme locally by quotients.

Hint. Work directly from open immersions, closed immersions, and ideal sheaves. ☐

Solution. On distinguished affine opens, take the quotient ring by the corresponding localized ideal and verify compatibility. ☐

Counterexamples and hypothesis tests

Exercise 22.104 (Closed reduced)**.** Test the claim: every closed subscheme is reduced.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; (y^2) defines a nonreduced double line. ☐

Exercise 22.105 (Same support same scheme)**.** Test the claim: closed subschemes with the same underlying closed set are equal.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; nilpotent thickenings differ. ☐

Exercise 22.106 (Open quotient)**.** Test the claim: open subschemes are generally obtained by quotient rings.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; they arise by restriction/localization. ☐

Applications and investigations

Exercise 22.107 (Equations)**.** How are equations imposed scheme-theoretically?

Hint. Translate the question into open immersions, closed immersions, and ideal sheaves. ☐

Solution. By quotienting the structure sheaf by an ideal sheaf, producing a closed subscheme. ☐

Exercise 22.108 (Removing loci)**.** How is a nonvanishing condition imposed?

Hint. Translate the question into open immersions, closed immersions, and ideal sheaves. ☐

Solution. By passing to an open subscheme such as a distinguished open. ☐

Challenge problems

Exercise 22.109 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Open and Closed Subschemes in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Open and Closed Subschemes, translates it through open immersions, closed immersions, and ideal sheaves, and checks the correspondence in both algebraic and geometric language. □

Exercise 22.110 (Hypothesis challenge)**.** Choose one structural statement from Open and Closed Subschemes, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Part V

Morphisms and Families

Chapter 23

Fiber Products

Fiber products are the basic mechanism for imposing compatibility between two maps to a common base. They simultaneously generalize ordinary products, intersections defined by equations, graphs, diagonals, and the pullback squares that dominate later scheme theory. The defining principle is categorical: the fiber product is characterized not by coordinates but by a universal property. Coordinates enter afterward, and for affine schemes they turn the construction into a tensor product of rings.

23.1 The universal property

Let $f : X \to S$ and $g : Y \to S$ be morphisms of schemes. A *fiber product* of X and Y over S is a scheme P with morphisms $p_X : P \to X$ and $p_Y : P \to Y$ such that

$$f \circ p_X = g \circ p_Y,$$

and such that for every scheme T with morphisms $a : T \to X$ and $b : T \to Y$ satisfying $f \circ a = g \circ b$, there exists a unique $u : T \to P$ with

$$p_X \circ u = a, \qquad p_Y \circ u = b.$$

When it exists, we write $P = X \times_S Y$.

Theorem 23.1 (Uniqueness up to unique isomorphism)**.** *Any two fiber products of the same pair $X \to S \leftarrow Y$ are uniquely isomorphic in a way compatible with both projections.*

$$\begin{tikzcd} X \times_S Y \arrow[r, "p_Y"] \arrow[d, "p_X"'] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & S \end{tikzcd}$$

Figure 23.1: The defining Cartesian square for a fiber product.

23.2 Cartesian squares

A commutative square

$$\begin{tikzcd} P \arrow[r] \arrow[d] & Y \arrow[d, "g"] \\ X \arrow[r, "f"'] & S \end{tikzcd}$$

is *Cartesian* if P, together with the displayed maps, is a fiber product $X \times_S Y$. Equivalently, for every T one has a natural bijection

$$\mathrm{Hom}(T, P) \cong \mathrm{Hom}(T, X) \times_{\mathrm{Hom}(T,S)} \mathrm{Hom}(T, Y).$$

This Hom-set formula is the main recognition tool throughout the chapter.

$$\begin{tikzcd}
& T \arrow[dl, "a"'] \arrow[dr, "b"] & \\
X \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\
& S &
\end{tikzcd}$$

Figure 23.2: Compatible maps from a test scheme determine a unique map to the fiber product.

23.3 The affine construction

Suppose

$$S = \operatorname{Spec} A, \qquad X = \operatorname{Spec} B, \qquad Y = \operatorname{Spec} C,$$

where B and C are A-algebras. Then

$$X \times_S Y \cong \operatorname{Spec}(B \otimes_A C).$$

The two projections correspond to the canonical maps

$$B \longrightarrow B \otimes_A C, \quad b \mapsto b \otimes 1, \qquad C \longrightarrow B \otimes_A C, \quad c \mapsto 1 \otimes c.$$

Theorem 23.2 (Affine fiber product)**.** *For affine schemes over an affine base,*

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} C \cong \operatorname{Spec}(B \otimes_A C).$$

$$\begin{tikzcd}
\operatorname{Spec}(B \otimes_A C) \arrow[r] \arrow[d] & \operatorname{Spec} C \arrow[d] \\
\operatorname{Spec} B \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

Figure 23.3: Affine fiber products are spectra of tensor products.

Example 23.3 (Affine fiber product)**.** For affine schemes $\operatorname{Spec} B \to \operatorname{Spec} A \leftarrow \operatorname{Spec} C$, the fiber product is $\operatorname{Spec}(B \otimes_A C)$. Tensor product is the algebraic pushout corresponding to the geometric pullback.

23.4 Why the tensor product appears

For any ring R, a morphism

$$\operatorname{Spec} R \to \operatorname{Spec}(B \otimes_A C)$$

corresponds to a ring homomorphism $B \otimes_A C \to R$. By the tensor-product universal property, this is equivalent to a pair of ring maps

$$B \to R, \qquad C \to R$$

whose restrictions to A agree. Reversing arrows gives exactly the universal property of the scheme fiber product.

23.5 Existence for arbitrary schemes

Fiber products of schemes always exist. The construction is local on the three schemes involved. Choose affine opens $U = \operatorname{Spec} A \subseteq S$, $V = \operatorname{Spec} B \subseteq f^{-1}(U)$, and $W = \operatorname{Spec} C \subseteq g^{-1}(U)$. Form

$$V \times_U W = \operatorname{Spec}(B \otimes_A C).$$

On smaller common affine opens these constructions agree by localization and the uniqueness of fiber products. The affine pieces therefore glue, using VI/20, to a scheme $X \times_S Y$.

Theorem 23.4 (Existence of fiber products of schemes)**.** *The category of schemes admits fiber products.*

$$\boxed{\text{affine pullbacks}} \longrightarrow \boxed{\text{localize on overlaps}} \longrightarrow \boxed{\text{glue by VI/20}} \longrightarrow \boxed{X \times_S Y}$$

Figure 23.4: Existence of scheme fiber products is obtained locally and then glued.

23.6 Products as fiber products

Because $\operatorname{Spec} \mathbb{Z}$ is final by VI/21, the categorical product of two schemes is

$$X \times Y := X \times_{\operatorname{Spec} \mathbb{Z}} Y.$$

If X and Y are schemes over a field k, their product over k means

$$X \times_k Y := X \times_{\operatorname{Spec} k} Y.$$

For example,

$$\mathbb{A}^m_k \times_k \mathbb{A}^n_k \cong \mathbb{A}^{m+n}_k.$$

$$\begin{tikzcd}
\mathbb{A}^m_k \times_k \mathbb{A}^n_k \arrow[rr, "\sim"] & & \mathbb{A}^{m+n}_k \\
\operatorname{Spec}(k[x_1, \dots, x_m] \otimes_k k[y_1, \dots, y_n]) \arrow[u, "\sim"] \arrow[rr, "\sim"'] & & \operatorname{Spec} k[x_1, \dots, x_m, y_1, \dots, y_n] \arrow[u, "\sim"']
\end{tikzcd}$$

Figure 23.5: Products of affine spaces combine coordinate variables.

Example 23.5 (Intersection over affine space)**.** The fiber product of $V(x) \hookrightarrow \mathbb{A}^2$ and $V(y) \hookrightarrow \mathbb{A}^2$ has coordinate ring $k[x,y]/(x) \otimes_{k[x,y]} k[x,y]/(y) \cong k$, giving their scheme-theoretic intersection at the origin.

23.7 Diagonal morphisms

For $X \to S$, the pair $(\mathrm{id}_X, \mathrm{id}_X)$ induces the diagonal

$$\Delta_{X/S} : X \longrightarrow X \times_S X.$$

It is the universal morphism imposing equality of the two projections. In the affine case $X = \operatorname{Spec} B$, $S = \operatorname{Spec} A$, it corresponds to multiplication

$$B \otimes_A B \longrightarrow B, \qquad b_1 \otimes b_2 \longmapsto b_1 b_2.$$

The later theory of separated morphisms will place geometric conditions on this diagonal; here we only establish the construction.

$$\begin{tikzcd} X \arrow[r, "\Delta_{X/S}"] \arrow[dr, "\mathrm{id}_X"'] & X \times_S X \arrow[d, "p_2"] \\ & X \end{tikzcd}$$

Figure 23.6: The diagonal makes both product projections equal to the identity.

23.8 Graphs as pullbacks of diagonals

Let $f : X \to Y$ be a morphism over S. Its graph is

$$\Gamma_f = (\mathrm{id}_X, f) : X \to X \times_S Y.$$

The square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\Gamma_f"'] & Y \arrow[d, "\Delta_{Y/S}"] \\ X \times_S Y \arrow[r, "h"'] & Y \times_S Y \end{tikzcd}$$

is Cartesian. This is one of the central Cartesian constructions in the AG7 source stream.

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\Gamma_f"'] & Y \arrow[d, "\Delta_{Y/S}"] \\ X \times_S Y \arrow[r, "h"'] & Y \times_S Y \end{tikzcd}$$

Figure 23.7: The graph is the pullback of the diagonal.

23.9 Changing the equality condition

Suppose $X \to S \to T$ and $Y \to S \to T$. The diagonal $S \to S \times_T S$ forces two maps to S that already agree over T to become equal over S. Consequently

$$X \times_S Y \cong (X \times_T Y) \times_{S \times_T S} S.$$

Equivalently, the square

$$\begin{tikzcd} X \times_S Y \arrow[r] \arrow[d] & S \arrow[d, "\Delta_{S/T}"] \\ X \times_T Y \arrow[r] & S \times_T S \end{tikzcd}$$

is Cartesian. This is the second central Cartesian construction in the AG7 source stream.

$$\begin{tikzcd} X \times_S Y \arrow[r] \arrow[d] & S \arrow[d, "\Delta_{S/T}"] \\ X \times_T Y \arrow[r] & S \times_T S \end{tikzcd}$$

Figure 23.8: Pulling back the diagonal strengthens equality over T to equality over S.

Example 23.6 (Base field product)**.** Over a field k, $\mathbb{A}_k^m \times_k \mathbb{A}_k^n \cong \mathbb{A}_k^{m+n}$ because $k[x_1, \dots, x_m] \otimes_k k[y_1, \dots, y_n]$ is the polynomial ring in all variables.

23.10 The points of a fiber product

For schemes, $X \times_S Y$ is not generally just the set of pairs (x, y) with equal image in S. If $x \mapsto s$ and $y \mapsto s$, points of the fiber product lying over the pair (x, y) are controlled by

$$\mathrm{Spec}(\kappa(x) \otimes_{\kappa(s)} \kappa(y)),$$

where the tensor product uses the residue-field homomorphisms induced by the two structure maps. Thus one compatible pair can give zero, one, or several scheme points. The fields need not be literally presented as common subfields: the two maps out of $\kappa(s)$ are the structure data that define the tensor product. This residue-field refinement is essential whenever field extensions are present.

$$\begin{tikzcd} \boxed{\text{compatible pair } (x, y) \text{ over } s} \arrow[d] \\ \boxed{\kappa(x) \otimes_{\kappa(s)} \kappa(y)} \arrow[d] \\ \boxed{\text{its prime ideals}} \arrow[d] \\ \boxed{\text{points of } X \times_S Y \text{ above } (x, y)} \end{tikzcd}$$

Figure 23.9: Scheme points refine compatible pairs by residue-field algebra.

23.11 Elementary formal identities

The universal property gives canonical isomorphisms

$$X \times_S Y \cong Y \times_S X,$$

$$(X \times_S Y) \times_S Z \cong X \times_S (Y \times_S Z),$$

and

$$X \times_S S \cong X.$$

These isomorphisms are not arbitrary coordinate tricks: each is the unique map induced by the same compatible families of projections.

23.12 Boundary with the next chapters

This chapter develops the fiber-product construction itself. VI/24 will study *base change* as a systematic operation on morphisms and properties. VI/25 will specialize the construction to scheme-theoretic fibers

$$X_s = X \times_S \operatorname{Spec} \kappa(s)$$

and analyze their geometry. The reserved CA13 tensor-product exercises and the detailed AG7 fiber examples therefore remain untouched here.

23.13 Examples

Example 23.7 (Product of two affine lines)**.** Over a field k, $k[x] \otimes_k k[y] \cong k[x, y]$, so $\mathbb{A}^1_k \times_k \mathbb{A}^1_k \cong \mathbb{A}^2_k$.

Example 23.8 (Product with a point)**.** For a k-scheme X, $X \times_k \operatorname{Spec} k \cong X$ by the universal property.

Example 23.9 (A polynomial surface as a pullback)**.** The maps $\operatorname{Spec} k[x, z] \to \operatorname{Spec} k[t]$ with $t \mapsto x$ and $\operatorname{Spec} k[y] \to \operatorname{Spec} k[t]$ with $t \mapsto y^2$ give fiber product $\operatorname{Spec} k[x, y, z]/(x - y^2)$.

Example 23.10 (Two quotient maps)**.** For A and ideals I, J, $\operatorname{Spec}(A/I) \times_{\operatorname{Spec} A} \operatorname{Spec}(A/J) \cong \operatorname{Spec}(A/(I + J))$.

Example 23.11 (An open restriction)**.** For $X = \operatorname{Spec} B \to \operatorname{Spec} A$ and $D(f) \subseteq \operatorname{Spec} A$, the pullback is $\operatorname{Spec}(B \otimes_A A_f) \cong \operatorname{Spec} B_{\varphi(f)}$. This identity is used here only as a fiber-product computation; systematic base change is VI/24.

Example 23.12 (Symmetry)**.** The flip $b \otimes c \mapsto c \otimes b$ induces $\operatorname{Spec}(B \otimes_A C) \cong \operatorname{Spec}(C \otimes_A B)$, matching symmetry of fiber products.

Example 23.13 (Associativity)**.** The canonical associativity isomorphism $(B \otimes_A C) \otimes_A D \cong B \otimes_A (C \otimes_A D)$ realizes associativity of affine fiber products.

Example 23.14 (The diagonal of affine line)**.** For $X = \mathbb{A}^1_k$, $X \times_k X = \operatorname{Spec} k[x_1, x_2]$, and Δ corresponds to $k[x_1, x_2] \to k[x]$, $x_1, x_2 \mapsto x$.

Example 23.15 (Graph of a square map)**.** For $f : \mathbb{A}^1_k \to \mathbb{A}^1_k$, $t \mapsto t^2$, the graph is $\operatorname{Spec} k[x, y]/(y - x^2)$ inside $\mathbb{A}^2_k$.

Example 23.16 (Identity pullback)**.** The square obtained by pulling $X \to S$ against id_S has upper-left object canonically isomorphic to X.

Example 23.17 (Empty scheme)**.** Because the empty scheme is initial, $\varnothing \times_S Y \cong \varnothing$.

Example 23.18 (Product over Spec Z)**.** $\operatorname{Spec} \mathbb{Z}[x] \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Z}[y] \cong \operatorname{Spec} \mathbb{Z}[x, y]$.

Example 23.19 (Common zero locus)**.** Inside $\operatorname{Spec} A$, the scheme-theoretic intersection of $V(I)$ and $V(J)$ is the fiber product and is cut out by $I + J$.

Example 23.20 (Diagonal imposes equality)**.** A map $T \to X \times_S X$ factors through $\Delta_{X/S}$ exactly when its two component maps $T \to X$ are equal.

Example 23.21 (Graph criterion)**.** A map $T \to X \times_S Y$ factors through Γ_f exactly when its Y-component equals f composed with its X-component.

Example 23.22 (A field extension product)**.** If K/k is a field extension, $\operatorname{Spec} K \times_k \mathbb{A}^1_k \cong \operatorname{Spec} K[t]$.

Example 23.23 (Two copies over a common affine base)**.** Maps $A \to B$ and $A \to C$ may identify equations from the base; the tensor product $B \otimes_A C$ is precisely the quotient that enforces those identifications universally.

Example 23.24 (A repeated factor)**.** $X \times_S X$ carries two projections to X and a diagonal section; these are distinct unless $X \to S$ is itself an isomorphism in a very strong sense.

Example 23.25 (Fiber product of open charts)**.** If $U \subseteq X$ and $V \subseteq Y$ map into a common open $W \subseteq S$, then $U \times_W V$ is an open subscheme of $X \times_S Y$.

Example 23.26 (Uniqueness in practice)**.** To construct an isomorphism between two candidate pullbacks, use the projections to obtain maps in both directions; uniqueness in the universal property forces the composites to be identities.

Example 23.27 (Compatible pair but several points)**.** For a field k and finite extensions K, L, points over the unique pair in $\operatorname{Spec} K \times_k \operatorname{Spec} L$ are the primes of $K \otimes_k L$, which can be more than one.

Example 23.28 (Separable splitting phenomenon)**.** If L/k is finite separable and Ω contains all k-embeddings of L, then $L \otimes_k \Omega$ can split as a product of copies of Ω, yielding several points above one set-theoretic pair.

Example 23.29 (Inseparable tensor product)**.** In characteristic p, tensor products of purely inseparable extensions may be nonreduced, showing that a fiber product can acquire nilpotents even when both factors are fields.

Example 23.30 (Equalizers from diagonals)**.** For $f, g : T \rightrightarrows X$ over S, the equalizer condition $f = g$ is imposed by pulling the map $(f, g) : T \to X \times_S X$ back along $\Delta_{X/S}$.

Example 23.31 (Intersections are scheme-theoretic)**.** When two closed subschemes of X meet, their fiber product remembers the sum of defining ideals, including nilpotent structure invisible to the set-theoretic intersection.

Example 23.32 (Localization commutes with tensor product)**.** For $f \in A$, $(B \otimes_A C) \otimes_A A_f \cong B_f \otimes_{A_f} C_f$, explaining compatibility of affine pullbacks with restriction to principal opens.

Example 23.33 (Descent of the affine construction to overlaps)**.** On overlapping affine charts, the two tensor-product descriptions localize to canonically isomorphic rings. Uniqueness supplies the cocycle needed for gluing.

Example 23.34 (Graph and closedness preview)**.** The graph is always a pullback of the diagonal. Therefore any future property of the diagonal stable under pullback transfers to graphs; this observation underlies separatedness theory.

Example 23.35 (Products and final objects)**.** The existence of the final object $\operatorname{Spec}\mathbb{Z}$ converts binary products in schemes into fiber products over an absolute base.

Example 23.36 (Iterated compatibility)**.** An iterated fiber product represents triples of maps with one common image in the base, explaining canonical associativity without choosing coordinates.

Example 23.37 (Functor-of-points viewpoint)**.** For every T, $(X \times_S Y)(T)$ is the set-theoretic fiber product $X(T) \times_{S(T)} Y(T)$. This does not mean underlying topological points form a naive fiber product.

Example 23.38 (Tensor product zero)**.** It is possible for a tensor product over a base ring to be the zero ring; then the affine fiber product is empty. Compatibility over a base can therefore eliminate all geometric points.

Example 23.39 (Residue fields at a product point)**.** A product point determines maps from $\kappa(x)$ and $\kappa(y)$ into its residue field compatible over the base. This is the field-theoretic refinement of the pair (x, y).

Example 23.40 (Locality of existence)**.** The global existence theorem is a model local-to-global argument: solve the universal problem on affine charts, verify compatibility on overlaps, then glue the representing objects.

23.14 Problem dossiers

VI.23.P1: The AG7 diagonal Cartesian square

Problem 23.41. Suppose $X \to S \to T$ and $Y \to S \to T$. Prove that

$$\begin{tikzcd} X \times_S Y \arrow[r] \arrow[d] & S \arrow[d, "\Delta_{S/T}"] \\ X \times_T Y \arrow[r] & S \times_T S \end{tikzcd}$$

is Cartesian.

Provenance. Canonical dossier reconstructing the first central Cartesian construction of the AG7 T01 source stream; the audited source contains no standalone numbered Problem 10 block.

VI.23.P2: The AG7 graph Cartesian square

Problem 23.42. Let $f : X \to Y$ be a morphism over S. Prove that

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\Gamma_f"'] & Y \arrow[d, "\Delta_{Y/S}"] \\ X \times_S Y \arrow[r, "h"'] & Y \times_S Y \end{tikzcd}$$

is Cartesian.

Provenance. Canonical dossier reconstructing the second central Cartesian construction of the AG7 T01 source stream.

VI.23.P3: Affine fiber product

Problem 23.43. Let $A \to B$ and $A \to C$ be ring maps. Prove

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} C \cong \operatorname{Spec}(B \otimes_A C).$$

Provenance. Canonical affine realization of the AG7 fiber-product theory stream.

VI.23.P4: Uniqueness of fiber products

Problem 23.44. Given two fiber products P and P' of $X \to S \leftarrow Y$, construct the unique projection-compatible isomorphism $P \cong P'$.

Provenance. Canonical categorical derivative.

VI.23.P5: Product of affine spaces

Problem 23.45. Prove $\mathbb{A}^m_k \times_k \mathbb{A}^n_k \cong \mathbb{A}^{m+n}_k$.

Provenance. Canonical affine application.

VI.23.P6: Fiber product with the base

Problem 23.46. Show $X \times_S S \cong X$ when the second map is id_S.

Provenance. Canonical universal-property derivative.

VI.23.P7: Symmetry

Problem 23.47. Construct the canonical isomorphism $X \times_S Y \cong Y \times_S X$ and prove its square is the identity.

Provenance. Canonical formal derivative.

VI.23.P8: Associativity

Problem 23.48. For $X, Y, Z \to S$, prove

$$(X \times_S Y) \times_S Z \cong X \times_S (Y \times_S Z).$$

Provenance. Canonical formal derivative.

VI.23.P9: Closed intersections by fiber product

Problem 23.49. Let $I, J \subseteq A$. Compute

$$\operatorname{Spec}(A/I) \times_{\operatorname{Spec} A} \operatorname{Spec}(A/J).$$

Provenance. Canonical bridge to VI/22 closed subschemes, without re-migrating VI/22.

VI.23.P10: Pullback of a principal open

Problem 23.50. Let $A \to B$ and $f \in A$. Compute

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} A_f.$$

Provenance. Canonical fiber-product computation; systematic base-change language is reserved for VI/24.

VI.23.P11: Pullback of a closed immersion in the affine case

Problem 23.51. For $A \to B$ and $I \subseteq A$, compute

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec}(A/I).$$

Provenance. Canonical affine pullback computation; no general base-change preservation theorem is claimed here.

VI.23.P12: The affine diagonal

Problem 23.52. Let $A \to B$. Identify the ring map corresponding to $\Delta : \operatorname{Spec} B \to \operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} B$.

Provenance. Canonical diagonal computation.

VI.23.P13: The graph in affine coordinates

Problem 23.53. Let $f : \operatorname{Spec} B \to \operatorname{Spec} C$ be a morphism over $\operatorname{Spec} A$, corresponding to $C \to B$. Identify the graph map on rings.

Provenance. Canonical graph computation.

VI.23.P14: Equalizer as a pullback of the diagonal

Problem 23.54. Let $f, g : T \rightrightarrows X$ be morphisms over S. Show that the pullback of $(f, g) : T \to X \times_S X$ along $\Delta_{X/S}$ represents maps $Q \to T$ on which f and g become equal.

Provenance. Canonical equality-condition derivative.

VI.23.P15: Construction by affine gluing

Problem 23.55. Outline a proof that fiber products of arbitrary schemes exist using affine charts.

Provenance. Canonical existence theorem from the AG7 fiber-product stream plus VI/20 gluing.

VI.23.P16: Compatible affine refinements

Problem 23.56. Explain why shrinking $\operatorname{Spec} A$ to $D(f)$ and the factors to the corresponding inverse images does not change the local fiber-product construction except by localization.

Provenance. Canonical local-compatibility derivative.

VI.23.P17: Points are not merely compatible pairs

Problem 23.57. Explain why the underlying set of $X \times_S Y$ cannot in general be identified with $|X| \times_{|S|} |Y|$.

Provenance. Canonical point-theoretic refinement; detailed fibers are reserved for VI/25.

VI.23.P18: A fiber product can be empty

Problem 23.58. Give an affine criterion for $\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} C$ to be empty.

Provenance. Canonical tensor-product diagnostic.

VI.23.P19: Functor of points

Problem 23.59. Prove that for every scheme T,

$$(X \times_S Y)(T) \cong X(T) \times_{S(T)} Y(T).$$

Provenance. Canonical categorical reformulation.

VI.23.P20: Boundary preview: fibers without computing them

Problem 23.60. Let $X \to S$ and $s \in S$. Explain formally why $X \times_S \operatorname{Spec} \kappa(s)$ is the correct object to call the fiber over s, without performing any detailed fiber computation.

Provenance. Boundary dossier only; the AG7 Problem 9 fiber corpus remains reserved for VI/25.

23.15 Exercises

Exercise 23.61. State the universal property of $X \times_S Y$ without using coordinates.

Exercise 23.62. Show that a Cartesian square remains Cartesian after replacing its upper-left object by an isomorphic scheme compatibly with the arrows.

Exercise 23.63. Prove $X \times_S S \cong X$.

Exercise 23.64. Prove $X \times_S Y \cong Y \times_S X$.

Exercise 23.65. Compute $\mathbb{A}^1_k \times_k \mathbb{A}^2_k$.

Exercise 23.66. Compute $\operatorname{Spec}(A/I) \times_{\operatorname{Spec} A} \operatorname{Spec}(A/J)$.

Exercise 23.67. Compute $\operatorname{Spec} B \times_{\operatorname{Spec} A} D(f)$ for $A \to B$.

Exercise 23.68. Show $\varnothing \times_S Y \cong \varnothing$.

Exercise 23.69. Write the ring map defining the diagonal of $\operatorname{Spec} B$ over $\operatorname{Spec} A$.

Exercise 23.70. Write the ring map defining the graph of an affine morphism $\operatorname{Spec} B \to \operatorname{Spec} C$ over $\operatorname{Spec} A$.

Exercise 23.71. Show the two projections from $X \times_S Y$ have equal composites to S.

Exercise 23.72. Use the universal property to construct the flip isomorphism twice and prove the composite is the identity.

Exercise 23.73. Show iterated fiber products represent triples of maps with a common image in S.

Exercise 23.74. For k a field, compute $\operatorname{Spec} k[x] \times_k \operatorname{Spec} k[y, z]$.

Exercise 23.75. For ideals $I, J, K \subseteq A$, compute $V(I) \times_{\operatorname{Spec} A} V(J) \times_{\operatorname{Spec} A} V(K)$ in affine coordinates.

Exercise 23.76. Show a map $T \to X \times_S X$ factors through the diagonal iff its two component maps to X are equal.

Exercise 23.77. Show a map to $X \times_S Y$ factors through Γ_f iff its Y-component is f of its X-component.

Exercise 23.78. Explain why $|X \times_S Y|$ need not equal the set-theoretic fiber product of underlying point sets.

Exercise 23.79. Give an example where one compatible pair of underlying points yields more than one product point.

Exercise 23.80. Prove $B \otimes_A A \cong B$ as A-algebras.

Exercise 23.81. Prove $(B \otimes_A C)_f \cong B_f \otimes_{A_f} C_f$ in the common localization situation.

Exercise 23.82. Show $\operatorname{Spec} \mathbb{Z}[x] \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Z}[y] \cong \operatorname{Spec} \mathbb{Z}[x, y]$.

Exercise 23.83. Explain how uniqueness of fiber products supplies cocycle identities when affine pullbacks are glued.

Exercise 23.84. Describe the Hom-set criterion for a commutative square to be Cartesian.

Exercise 23.85. Explain why the construction $X_s = X \times_S \operatorname{Spec} \kappa(s)$ belongs formally to fiber products but its detailed study belongs to VI/25.

Exercise 23.86. Explain why the reserved tensor-product base-change exercises CA13-E6/E7 should not be counted as VI/23 legacy problems.

23.16 Challenges

Challenge 1: Existence from gluing

Problem 23.87. Give a complete local-to-global proof that schemes admit fiber products. Your proof should specify an affine cover of the prospective product, identify all overlap maps, use localization to verify compatibility, and then prove the global universal property.

Challenge 2: Residue-field point classification

Problem 23.88. Develop a precise description of the points above a compatible pair $x \in X$, $y \in Y$ over $s \in S$ in terms of primes of a suitable tensor product of residue fields. Explain what changes when the maps $\kappa(s) \to \kappa(x), \kappa(y)$ are not embeddings in a naive sense.

Challenge 3: Diagonal and equalizers

Problem 23.89. Use the diagonal to construct equalizers of two scheme morphisms whenever the relevant fiber products exist. Work out the affine coordinate ring and compare the resulting scheme with the set-theoretic equalizer.

Challenge 4: Tensor products and geometric surprises

Problem 23.90. Find examples in which the affine fiber product of reduced schemes is (a) reducible, (b) nonreduced, and (c) empty. For each example, isolate the algebraic mechanism inside the tensor product.

Challenge 5: Functorial calculus without coordinates

Problem 23.91. Prove symmetry, associativity, identity, graph, and the AG7 diagonal-change Cartesian square using only universal properties. Then explain why this proof is more robust than choosing affine coordinates.

23.17 Chapter synthesis

Fiber products convert compatibility conditions into geometric objects. Their categorical universal property explains uniqueness and formal identities; their affine incarnation as tensor products makes them computable; and gluing makes them available for arbitrary schemes. Diagonals and graphs show how equality conditions become pullback squares, while the residue-field description warns that the underlying points carry more information than a naive set-theoretic product.

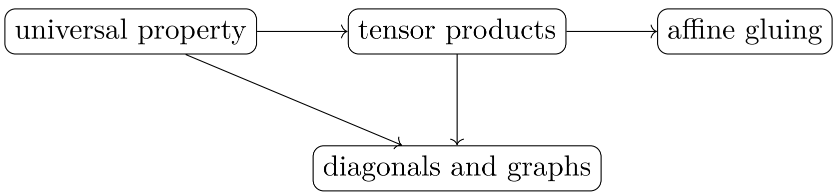


Figure 23.10: The four organizing ideas of VI/23.

23.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 23.92 (Affine product)**.** Compute $\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} C$.

Hint. Use fiber products and tensor-product coordinate rings. $\square$

Solution. $\operatorname{Spec}(B \otimes_A C)$. ☐

Exercise 23.93 (Affine spaces)**.** Compute $\mathbb{A}^2_k \times_k \mathbb{A}^3_k$.

Hint. Use fiber products and tensor-product coordinate rings. ☐

Solution. $\mathbb{A}^5_k$. ☐

Exercise 23.94 (Axes intersection)**.** Compute the fiber product of the two coordinate axes inside $\mathbb{A}^2$.

Hint. Use fiber products and tensor-product coordinate rings. ☐

Solution. The origin $\operatorname{Spec} k$. ☐

Exercise 23.95 (Identity)**.** Compute $X \times_S S$.

Hint. Use fiber products and tensor-product coordinate rings. ☐

Solution. X. ☐

Exercise 23.96 (Point fiber)**.** What is $X \times_S \operatorname{Spec} \kappa(s)$ called?

Hint. Use fiber products and tensor-product coordinate rings. ☐

Solution. The fiber over s. ☐

Proofs

Exercise 23.97 (Affine formula)**.** Prove: Prove the affine fiber-product formula using universal properties.

Hint. Work directly from fiber products and tensor-product coordinate rings. ☐

Solution. Maps into $\operatorname{Spec}(B \otimes_A C)$ correspond contravariantly to compatible ring maps from B and C, exactly the fiber-product condition. ☐

Exercise 23.98 (Symmetry)**.** Prove: Prove $X \times_S Y \cong Y \times_S X$.

Hint. Work directly from fiber products and tensor-product coordinate rings. ☐

Solution. Swap the two projections and use the universal property. ☐

Exercise 23.99 (Associativity)**.** Prove: Prove iterated fiber products are canonically associative when bases are compatible.

Hint. Work directly from fiber products and tensor-product coordinate rings. ☐

Solution. Both objects represent the same compatible triples of maps. ☐

Exercise 23.100 (Base change)**.** Prove: Prove pulling a morphism back along $S' \to S$ is expressed by a fiber product.

Hint. Work directly from fiber products and tensor-product coordinate rings. ☐

Solution. Take $X \times_S S'$ and verify the universal property. ☐

Counterexamples and hypothesis tests

Exercise 23.101 (Set product)**.** Test the claim: scheme fiber products are always just Cartesian products of underlying point sets.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; residue fields and tensor products create subtler points. ☐

Exercise 23.102 (Ring product)**.** Test the claim: affine fiber products correspond to direct products of rings.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; they correspond to tensor products. ☐

Exercise 23.103 (Empty intersection)**.** Test the claim: disjoint underlying closed subsets can never have a nontrivial scheme-theoretic fiber product.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. For genuinely disjoint closed immersions the fiber product is empty, but multiplicities can matter when supports meet. ☐

Applications and investigations

Exercise 23.104 (Intersections)**.** Why are fiber products used for scheme-theoretic intersections?

Hint. Translate the question into fiber products and tensor-product coordinate rings. ☐

Solution. Closed embeddings pulled back against each other retain multiplicity and nilpotent information. ☐

Exercise 23.105 (Families)**.** Why are fiber products central to families?

Hint. Translate the question into fiber products and tensor-product coordinate rings. ☐

Solution. They implement change of base and formation of fibers. ☐

Challenge problems

Exercise 23.106 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Fiber Products in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Fiber Products, translates it through fiber products and tensor-product coordinate rings, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 23.107 (Hypothesis challenge)**.** Choose one structural statement from Fiber Products, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 24

Base Change

Base change is the systematic operation of replacing the base of a scheme or morphism while preserving all compatibility data. If $X \to S$ is a scheme over S and $S' \to S$ is another morphism, the new scheme

$$X_{S'} := X \times_S S'$$

is the same family viewed over the new base S'. VI/23 constructed fiber products; this chapter turns that construction into a working calculus. In affine coordinates, base change is scalar extension by tensor product.

24.1 Definition and the base-change square

Let $f : X \to S$ and $g : S' \to S$. The *base change of X along g* is

$$X_{S'} = X \times_S S'.$$

The second projection gives the base-changed structure morphism

$$f_{S'} : X_{S'} \longrightarrow S'.$$

By definition the square

$$\begin{tikzcd} X_{S'} \arrow[r] \arrow[d, "f_{S'}"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S \end{tikzcd}$$

is Cartesian.

$$\begin{tikzcd} X_{S'} \arrow[r] \arrow[d, "f_{S'}"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S \end{tikzcd}$$

Figure 24.1: Base change is a Cartesian square.

24.2 Universal property after changing the base

For every $T \to S'$, a morphism $T \to X_{S'}$ over S' is equivalent to a morphism $T \to X$ over S whose structure map to S factors through the prescribed $T \to S' \to S$. Thus base change is not a new ad hoc construction: it is the universal pullback from VI/23, now regarded as an operation depending on the map $S' \to S$.

Proposition 24.1 (Identity base change)**.** *For every $X \to S$ there is a canonical isomorphism*

$$X \times_S S \cong X,$$

where the second map $S \to S$ is the identity.

Proposition 24.2 (Iterated base change)**.** *If $S'' \to S' \to S$, then there is a canonical isomorphism*

$$(X_{S'})_{S''} \cong X_{S''}.$$

$$\begin{tikzcd}
\boxed{X/S} \arrow[r, "S' \to S"] \arrow[rr, bend right, "S'' \to S"'] & \boxed{X_{S'}/S'} \arrow[r, "S'' \to S'"] & \boxed{X_{S''}/S''}
\end{tikzcd}$$

Figure 24.2: Iterated base change agrees canonically with direct base change.

24.3 Affine base change is scalar extension

Suppose

$$S = \operatorname{Spec} A, \qquad X = \operatorname{Spec} B, \qquad S' = \operatorname{Spec} A',$$

with maps induced by $A \to B$ and $A \to A'$. By the affine fiber-product formula,

$$X_{S'} \cong \operatorname{Spec}(B \otimes_A A').$$

The new coordinate ring is therefore obtained by extending scalars from A to A'.

Theorem 24.3 (Affine base-change formula)**.** *For an A-algebra B and an A-algebra A',*

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} A' \cong \operatorname{Spec}(B \otimes_A A').$$

$$\begin{tikzcd}
\operatorname{Spec}(B \otimes_A A') \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
\operatorname{Spec} A' \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

Figure 24.3: Affine base change is scalar extension by a tensor product.

Example 24.4 (Field extension)**.** Base changing $\mathbb{A}^1_{\mathbb{R}}$ to $\mathbb{C}$ gives $\mathbb{A}^1_{\mathbb{C}}$. On rings this is $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}[x] \cong \mathbb{C}[x]$.

24.4 Changing coefficients in equations

Let $k \to K$ be a field extension and

$$X = \operatorname{Spec} k[x_1, \dots, x_n]/I.$$

Then

$$X_K := X \times_{\operatorname{Spec} k} \operatorname{Spec} K \cong \operatorname{Spec}\big(K[x_1, \dots, x_n]/IK[x_1, \dots, x_n]\big).$$

Thus base change from k to K leaves the equations formally unchanged while allowing their coefficients, factorizations, and solutions to live in the larger field. This is why geometric features can become visible only after extending the base field.

$$\boxed{k[x_1, \dots, x_n]/I} \longrightarrow \boxed{\otimes_k K} \longrightarrow \boxed{K[x_1, \dots, x_n]/IK[x_1, \dots, x_n]}$$

Figure 24.4: Changing the base field extends the coefficients of the same equations.

24.5 Quotients commute with scalar extension

If $I \subseteq B$ is an ideal and $A \to A'$ is a ring map, right exactness of tensor product gives

$$(B/I) \otimes_A A' \cong (B \otimes_A A')/ \operatorname{im}(I \otimes_A A').$$

When B is an A-algebra and the ideal is extended in the usual way, this is written

$$(B/I) \otimes_A A' \cong (B \otimes_A A')/I(B \otimes_A A').$$

This identity is the algebra behind base change of affine closed subschemes.

24.6 Localization commutes with base change

If $b \in B$, then

$$B_b \otimes_A A' \cong (B \otimes_A A')_{b\otimes 1}.$$

Geometrically, pulling back the principal open $D(b) \subseteq \operatorname{Spec} B$ gives the principal open $D(b \otimes 1)$ in the base-changed affine scheme. This is the local algebra behind stability of open immersions under base change.

$$\begin{tikzcd}
D(b \otimes 1) \arrow[r, hook] \arrow[d, "\sim"'] & \operatorname{Spec}(B \otimes_A A') \\
\operatorname{Spec}(B_b \otimes_A A') \arrow[ur, hook]
\end{tikzcd}$$

Figure 24.5: Localization commutes with scalar extension.

Example 24.5 (Splitting after base change)**.** The real scheme $\operatorname{Spec} \mathbb{R}[x]/(x^2 + 1)$ becomes $\operatorname{Spec} \mathbb{C}[x]/((x - i)(x + i))$ after base change to $\mathbb{C}$, revealing two complex points.

24.7 Base change of morphisms

Suppose $u : X \to Y$ is a morphism of schemes over S. For $S' \to S$, the universal property gives an induced morphism

$$u_{S'} : X_{S'} \longrightarrow Y_{S'}.$$

Identity morphisms and compositions are preserved:

$$(\mathrm{id}_X)_{S'} = \mathrm{id}_{X_{S'}}, \qquad (v \circ u)_{S'} = v_{S'} \circ u_{S'}.$$

Hence base change defines a functor from schemes over S to schemes over S'.

24.8 Properties visibly preserved by base change

Several fundamental classes of morphisms are stable for formal or affine-local reasons.

- Isomorphisms remain isomorphisms after any base change.
- Open immersions remain open immersions, by localization on affine charts.
- Closed immersions remain closed immersions, because quotient rings remain quotient rings after scalar extension.
- Monomorphisms remain monomorphisms, directly from the universal property.

Later chapters treat deeper finiteness and geometric properties. VI/24 establishes the language and the elementary stable classes without consuming VI/26.

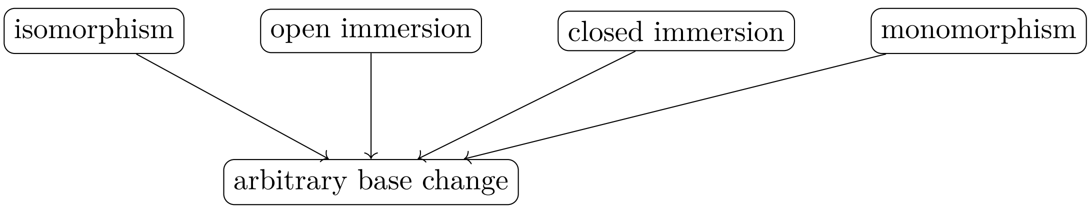


Figure 24.6: Elementary morphism classes proved stable under base change in VI/24.

24.9 Products commute with base change

If $X \to S$ and $Y \to S$, then for every $S' \to S$ there is a canonical isomorphism

$$(X \times_S Y) \times_S S' \cong X_{S'} \times_{S'} Y_{S'}.$$

Both sides represent triples of compatible maps to X, Y, and S'. In affine coordinates the identity is the associativity of tensor products.

$$(X \times_S Y)_{S'} \xrightarrow{\ \sim\ } X_{S'} \times_{S'} Y_{S'}$$

Figure 24.7: Fiber products commute with change of base.

Example 24.6 (Base-changed morphism)**.** If $X \to S$ is a morphism and $S' \to S$ is any map, the base change is $X_{S'} = X \times_S S' \to S'$. Every construction is controlled by the same pullback square.

24.10 Diagonals and graphs under base change

For $X \to S$, the diagonal of the base-changed scheme is canonically the pullback of the original diagonal:

$$\Delta_{X_{S'}/S'} \cong (\Delta_{X/S})_{S'}.$$

Likewise, if $u : X \to Y$ is over S, then the graph of $u_{S'}$ is the base change of the graph of u. These identities are formal consequences of the Cartesian constructions in VI/23.

$$\begin{tikzcd} X_{S'} \arrow[r, "\Delta"] \arrow[d] & X_{S'} \times_{S'} X_{S'} \arrow[d, "\sim"] \\ X \arrow[r, "\Delta"'] & X \times_S X \end{tikzcd}$$

Figure 24.8: The diagonal construction is compatible with base change.

24.11 Field extension can reveal hidden geometry

The underlying shape of a scheme can change after extending the field. For example,

$$\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} = \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) \cong \operatorname{Spec}(\mathbb{C} \times \mathbb{C}),$$

so one point over $\mathbb{R}$ splits into two points after base change to $\mathbb{C}$. Conversely, purely inseparable extensions can create nilpotents after base change. Base change therefore preserves the pullback structure, not every topological or reducedness property.

$$\begin{tikzcd} \operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \arrow[r, "\sim"] & \operatorname{Spec}(\mathbb{C} \times \mathbb{C}) \\ & \operatorname{Spec} \mathbb{C} \amalg \operatorname{Spec} \mathbb{C} \arrow[u, "\sim"'] \end{tikzcd}$$

Figure 24.9: Complex base change splits the real point with residue field $\mathbb{C}$.

24.12 Boundary with fibers and finiteness

A fiber is a special base change

$$X_s = X \times_S \operatorname{Spec} \kappa(s),$$

but VI/25 is devoted to the detailed geometry of these schemes and their residue-field extensions. Similarly, VI/26 studies finite-type and Noetherian conditions systematically. Here these appear only as forward references; the chapter's center is the operation of changing the base itself.

24.13 Examples

Example 24.7 (Identity base change)**.** For $X \to S$, pulling back along id_S gives $X \times_S S \cong X$.

Example 24.8 (Affine line over a larger field)**.** If $k \subseteq K$, then $\mathbb{A}^1_k \times_k K \cong \operatorname{Spec}(K[x]) = \mathbb{A}^1_K$.

Example 24.9 (Affine space)**.** More generally $\mathbb{A}^n_k \times_k K \cong \mathbb{A}^n_K$ because $k[x_1, \dots, x_n] \otimes_k K \cong K[x_1, \dots, x_n]$.

Example 24.10 (Scalar extension of a quotient)**.** For $f \in k[x]$, $(k[x]/(f)) \otimes_k K \cong K[x]/(f)$; this is the legacy CA13-E7 computation.

Example 24.11 (Complexification of a real point)**.** $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec}(\mathbb{C} \times \mathbb{C})$, so the base-changed point splits.

Example 24.12 (A principal open)**.** For $D(b) \subseteq \operatorname{Spec} B$, base change by $A \to A'$ gives $D(b \otimes 1) \subseteq \operatorname{Spec}(B \otimes_A A')$.

Example 24.13 (A closed subscheme)**.** If $Z = \operatorname{Spec}(B/I) \hookrightarrow \operatorname{Spec} B$, then $Z_{A'} \cong \operatorname{Spec}((B \otimes_A A')/I(B \otimes_A A'))$.

Example 24.14 (A reducible conic after extension)**.** $\operatorname{Spec} \mathbb{R}[x]/(x^2 + 1)$ is one closed point over $\mathbb{R}$, while after base change to $\mathbb{C}$ the polynomial factors and the scheme becomes two reduced points.

Example 24.15 (Dual numbers)**.** $k[\varepsilon]/(\varepsilon^2) \otimes_k K \cong K[\varepsilon]/(\varepsilon^2)$, so this nilpotent thickening stays nonreduced after a field extension.

Example 24.16 (Base-changing a morphism)**.** The k-morphism $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $y = x^2$ becomes the same formula over K after $k \subseteq K$.

Example 24.17 (Base change of a product)**.** $(\mathbb{A}^1_k \times_k \mathbb{A}^1_k)_K \cong \mathbb{A}^1_K \times_K \mathbb{A}^1_K \cong \mathbb{A}^2_K$.

Example 24.18 (Base change of the diagonal)**.** The equation $x_1 - x_2 = 0$ defining the diagonal of $\mathbb{A}^1_k$ extends to the same equation in $K[x_1, x_2]$.

Example 24.19 (Base change of a graph)**.** The graph $y - x^2 = 0$ in $\mathbb{A}^2_k$ becomes the graph $y - x^2 = 0$ in $\mathbb{A}^2_K$.

Example 24.20 (Changing from integers to rationals)**.** $\operatorname{Spec} \mathbb{Z}[x] \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Q} \cong \operatorname{Spec} \mathbb{Q}[x]$. This is scalar extension, while its interpretation as the generic fiber is deferred to VI/25.

Example 24.21 (Reduction modulo p as scalar extension)**.** $\operatorname{Spec} \mathbb{Z}[x]/(F) \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{F}_p \cong \operatorname{Spec} \mathbb{F}_p[x]/(\overline{F})$. Detailed fiber analysis is VI/25.

Example 24.22 (An empty scalar extension)**.** If $A' = 0$, then $B \otimes_A A' = 0$ and the base-changed affine scheme is empty.

Example 24.23 (Disjoint unions)**.** Base change commutes with disjoint unions because both are characterized componentwise by universal properties.

Example 24.24 (Two equations)**.** A scheme $V(f, g) \subseteq \mathbb{A}^n_k$ base changes to the scheme cut out by the same two polynomials regarded in $K[x_1, \dots, x_n]$.

Example 24.25 (Linear systems)**.** A matrix with entries in k defines a linear subscheme; after $k \subseteq K$ the same matrix defines its scalar extension over K.

Example 24.26 (Base-changing an isomorphism)**.** If $X \to Y$ is an isomorphism over S, then $X_{S'} \to Y_{S'}$ is an isomorphism with inverse obtained by base-changing the original inverse.

Example 24.27 (Tensor products and quotients)**.** For an A-algebra B and ideal $I \subseteq B$, scalar extension sends B/I to $(B \otimes_A A')/I(B \otimes_A A')$.

Example 24.28 (Tensor products and localization)**.** The canonical map $B_b \otimes_A A' \to (B \otimes_A A')_{b\otimes 1}$ is an isomorphism.

Example 24.29 (Iterated scalar extension)**.** For $A \to A' \to A''$, associativity gives $(B \otimes_A A') \otimes_{A'} A'' \cong B \otimes_A A''$.

Example 24.30 (Products after scalar extension)**.** Associativity and symmetry of tensor products yield $(B \otimes_A C) \otimes_A A' \cong (B \otimes_A A') \otimes_{A'} (C \otimes_A A')$.

Example 24.31 (Base change can disconnect)**.** $\operatorname{Spec} \mathbb{C}$ is connected as an $\mathbb{R}$-scheme, but after base change to $\mathbb{C}$ it becomes $\operatorname{Spec}(\mathbb{C} \times \mathbb{C})$, which is disconnected.

Example 24.32 (Base change can change irreducibility)**.** The real scheme $\operatorname{Spec} \mathbb{R}[x]/(x^2+1)$ is irreducible, but its complexification has two irreducible components.

Example 24.33 (Purely inseparable base change can create nilpotents)**.** In characteristic p, let $k = \mathbb{F}_p(s^p)$ and $L = \mathbb{F}_p(s)$. Then $L \otimes_k L \cong L[u]/((u-s)^p)$, which is nonreduced.

Example 24.34 (Separable splitting)**.** For a finite separable extension L/k and an algebraic closure $\overline{k}$, $L \otimes_k \overline{k}$ is a product of copies of $\overline{k}$, one for each k-embedding $L \hookrightarrow \overline{k}$.

Example 24.35 (Closed immersion stability)**.** A surjection $B \twoheadrightarrow C$ remains surjective after tensoring with A'; therefore affine closed immersions survive arbitrary base change.

Example 24.36 (Open immersion stability)**.** Localization commutes with scalar extension, so principal open immersions survive base change; affine localization then proves the general statement locally.

Example 24.37 (Monomorphism stability)**.** If $X \to Y$ is a monomorphism, then $X_{S'} \to Y_{S'}$ is a monomorphism because equality of maps can be tested after composing with the Cartesian projections.

Example 24.38 (The diagonal commutes with base change)**.** Using the universal property twice identifies $X_{S'} \times_{S'} X_{S'}$ with $(X \times_S X)_{S'}$ and the two diagonal maps correspond.

Example 24.39 (Graphs commute with base change)**.** Since graphs are pullbacks of diagonals, their compatibility with base change follows from associativity of pullbacks.

Example 24.40 (A descent warning)**.** A property visible after one base change need not hold before base change without a descent theorem. Geometric connectedness and geometric irreducibility are deliberately stronger notions than connectedness and irreducibility over the original field.

24.14 Problem dossiers

VI.24.P1: Legacy CA13-E6: compute $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$

Problem 24.41. Compute $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ and interpret the result geometrically as a base change.

Provenance. Legacy CA13-E6. This is a reserved tensor-product/base-change anchor assigned to VI/24.

VI.24.P2: Legacy CA13-E7: scalar extension of a polynomial quotient

Problem 24.42. Let $k \subseteq K$ be a field extension and $f \in k[x]$. Prove

$$(k[x]/(f)) \otimes_k K \cong K[x]/(f).$$

Provenance. **Legacy CA13-E7.** This reserved scalar-extension exercise belongs to VI/24, not VI/23.

VI.24.P3: Construct the base-change square

Problem 24.43. Given $X \to S$ and $S' \to S$, state and prove the universal property of $X_{S'} = X \times_S S'$.

Provenance. Canonical base-change dossier from the VI/23 fiber-product universal property.

VI.24.P4: Identity and iterated base change

Problem 24.44. Prove $X_S \cong X$ and $(X_{S'})_{S''} \cong X_{S''}$ for $S'' \to S' \to S$.

Provenance. Canonical functoriality dossier.

VI.24.P5: Affine scalar-extension formula

Problem 24.45. For $A \to B$ and $A \to A'$, derive the coordinate ring of the base change of $\operatorname{Spec} B \to \operatorname{Spec} A$ along $\operatorname{Spec} A' \to \operatorname{Spec} A$.

Provenance. Canonical affine realization of base change.

VI.24.P6: Base change of equations

Problem 24.46. Let $X = \operatorname{Spec} k[x_1, \ldots, x_n]/I$ and $k \subseteq K$. Compute X_K.

Provenance. Canonical equations-under-scalar-extension dossier.

VI.24.P7: Open immersions are stable

Problem 24.47. Prove that the base change of an open immersion is an open immersion.

Provenance. Canonical elementary stability theorem.

VI.24.P8: Closed immersions are stable

Problem 24.48. Prove that the base change of a closed immersion is a closed immersion.

Provenance. Canonical elementary stability theorem.

VI.24.P9: Isomorphisms are stable

Problem 24.49. Prove that an isomorphism remains an isomorphism after arbitrary base change.

Provenance. Canonical formal stability theorem.

VI.24.P10: Products commute with base change

Problem 24.50. Show

$$(X \times_S Y)_{S'} \cong X_{S'} \times_{S'} Y_{S'}.$$

Provenance. Canonical pullback-associativity theorem.

VI.24.P11: Base change of a diagonal

Problem 24.51. Identify the base change of $\Delta_{X/S} : X \to X \times_S X$ along $S' \to S$.

Provenance. Canonical diagonal/base-change compatibility theorem.

VI.24.P12: Base change of a graph

Problem 24.52. Let $u : X \to Y$ be over S. Show that the graph of $u_{S'}$ is the base change of the graph of u.

Provenance. Canonical graph/base-change compatibility theorem.

VI.24.P13: Scalar extension of affine space

Problem 24.53. Prove $\mathbb{A}^n_k \times_k K \cong \mathbb{A}^n_K$.

Provenance. Canonical affine-space computation.

VI.24.P14: Geometric interpretation of $\mathbb{C}/\mathbb{R}$ splitting

Problem 24.54. Explain why P1 shows that connectedness and irreducibility can change under field extension.

Provenance. Canonical geometric interpretation of legacy CA13-E6.

VI.24.P15: Nilpotents under field extension

Problem 24.55. Show that dual numbers remain nonreduced after any field extension $k \subseteq K$.

Provenance. Canonical persistence-of-nilpotents example.

VI.24.P16: Purely inseparable creation of nilpotents

Problem 24.56. In characteristic p, let $k = \mathbb{F}_p(s^p)$ and $L = \mathbb{F}_p(s)$. Show $L \otimes_k L$ is nonreduced.

Provenance. Canonical inseparability diagnostic; detailed fiber language remains for VI/25.

VI.24.P17: Base change of a closed subscheme

Problem 24.57. If $Z \hookrightarrow X = \operatorname{Spec} B$ is defined by $I \subseteq B$, describe $Z_{A'} \hookrightarrow X_{A'}$.

Provenance. Canonical bridge from VI/22 closed subschemes to base change.

VI.24.P18: Base change of a principal open

Problem 24.58. For $X = \operatorname{Spec} B$ and $D(b) \subseteq X$, identify its pullback to $X_{A'}$.

Provenance. Canonical bridge from localization to base change.

VI.24.P19: Base-change functoriality

Problem 24.59. Show that $X \mapsto X_{S'}$ and $u \mapsto u_{S'}$ define a functor $\mathrm{Sch}/S \to \mathrm{Sch}/S'$.

Provenance. Canonical categorical calculus of base change.

VI.24.P20: Fiber preview without consuming VI/25

Problem 24.60. Explain why $X_s = X \times_S \operatorname{Spec} \kappa(s)$ is a base change and state what remains for VI/25.

Provenance. Boundary dossier: the AG7 detailed fiber corpus remains assigned to VI/25.

24.15 Exercises

Exercise 24.61. Write the Cartesian square defining $X_{S'}$.

Exercise 24.62. Prove $X \times_S S \cong X$.

Exercise 24.63. Compute $k[x] \otimes_k K$.

Exercise 24.64. Compute $(k[x]/(x^2+1)) \otimes_k K$.

Exercise 24.65. Show $(B/I) \otimes_A A'$ is a quotient of $B \otimes_A A'$.

Exercise 24.66. Show localization commutes with base change.

Exercise 24.67. Base-change $D(x) \subseteq \mathbb{A}^1_k$ to K.

Exercise 24.68. Base-change $V(x^2) \subseteq \mathbb{A}^1_k$ to K.

Exercise 24.69. Prove base change of an inverse is the inverse of the base-changed map.

Exercise 24.70. Prove iterated base change algebraically in the affine case.

Exercise 24.71. Compute $\mathbb{R}[x]/(x^2+1) \otimes_{\mathbb{R}} \mathbb{C}$.

Exercise 24.72. Show $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$ has nontrivial idempotents.

Exercise 24.73. Base-change $\mathbb{Z}[x]/(x^2-2)$ to $\mathbb{Q}$.

Exercise 24.74. Base-change the same ring to $\mathbb{F}_2$.

Exercise 24.75. Show base change commutes with binary products over S.

Exercise 24.76. Identify the base change of a diagonal.

Exercise 24.77. Identify the base change of a graph.

Exercise 24.78. Give a property not preserved by field extension.

Exercise 24.79. Give a nonreduced scalar extension from an inseparable extension.

Exercise 24.80. Explain why closed immersions survive base change.

Exercise 24.81. Explain why open immersions survive base change.

Exercise 24.82. Show base change preserves monomorphisms.

Exercise 24.83. Compute the scalar extension of $k[x,y]/(xy)$ to K.

Exercise 24.84. Compute the scalar extension of $\mathbb{A}^2_k$ along $k \to K$.

Exercise 24.85. State the fiber as a special base change.

Exercise 24.86. Distinguish VI/24 from VI/25 in one sentence.

24.16 Challenges

Problem 24.87 (Challenge: Explicit splitting idempotents)**.** Using the isomorphism $\mathbb{C}\otimes_{\mathbb{R}}\mathbb{C} \cong \mathbb{C} \times \mathbb{C}$, find explicit idempotents in the tensor product corresponding to $(1, 0)$ and $(0, 1)$ and verify them directly.

Problem 24.88 (Challenge: Finite separable splitting)**.** Let L/k be finite separable. Show after base change to an algebraic closure that $L\otimes_k \overline{k}$ is a product indexed by the k-embeddings $L \hookrightarrow \overline{k}$. Interpret the components geometrically.

Problem 24.89 (Challenge: Inseparable thickening)**.** Work out in detail the example $k = \mathbb{F}_p(s^p) \subset L = \mathbb{F}_p(s)$ and describe the nilpotent structure of $\operatorname{Spec}(L \otimes_k L)$.

Problem 24.90 (Challenge: Base change and scheme-theoretic intersection)**.** For closed subschemes $Z_1, Z_2 \hookrightarrow X$ of an affine S-scheme, compare the scalar extension of $Z_1 \cap Z_2$ with the intersection of their scalar extensions. Prove the expected equality using ideals and tensor products.

Problem 24.91 (Challenge: Stability/descent audit)**.** Prepare a table separating properties proved in VI/24 to be stable under arbitrary base change from properties that require later hypotheses or descent theorems. Include isomorphism, open immersion, closed immersion, connectedness, irreducibility, and reducedness.

24.17 Chapter synthesis

Base change takes a geometric object over S and transports it to a new base S' by a Cartesian square. In affine charts this is exactly extension of scalars. Identity and composition laws make base change functorial; quotients and localizations explain the stability of closed and open immersions; and field extension examples show why the operation is geometrically powerful. The next chapter specializes this calculus to fibers over points.

$$\begin{tikzcd}
\boxed{\text{VI/23 fiber products}} \arrow[r] & \boxed{\text{VI/24 base change}} \arrow[r] \arrow[d] & \boxed{\text{VI/25 fibers}} \\
 & \boxed{\text{VI/26 finiteness}} &
\end{tikzcd}$$

Figure 24.10: VI/24 turns pullbacks into a systematic operation, then hands specializations to later chapters.

24.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 24.92 (Affine line)**.** Base change $\mathbb{A}^1_k$ along a field extension K/k.

Hint. Use base change of schemes and tensor products. ☐

Solution. It becomes $\mathbb{A}^1_K$. ☐

Exercise 24.93 (Polynomial)**.** Compute $K \otimes_k k[x]$.

Hint. Use base change of schemes and tensor products. ☐

Solution. $K[x]$. ☐

Exercise 24.94 (Quadratic)**.** What happens to $x^2 + 1$ from $\mathbb{R}$ to $\mathbb{C}$?

Hint. Use base change of schemes and tensor products. ☐

Solution. It factors as $(x - i)(x + i)$. ☐

Exercise 24.95 (Notation)**.** What is $X_{S'}$?

Hint. Use base change of schemes and tensor products. ☐

Solution. $X \times_S S'$. ☐

Exercise 24.96 (Identity)**.** What is base change along $S \to S$?

Hint. Use base change of schemes and tensor products. ☐

Solution. The original $X \to S$. ☐

Proofs

Exercise 24.97 (Affine base change)**.** Prove: Prove base change of $\operatorname{Spec} B \to \operatorname{Spec} A$ along $\operatorname{Spec} A' \to \operatorname{Spec} A$ is $\operatorname{Spec}(B \otimes_A A')$.

Hint. Work directly from base change of schemes and tensor products. ☐

Solution. Apply the affine fiber-product formula. ☐

Exercise 24.98 (Composition)**.** Prove: Prove successive base changes are canonically equivalent to base change along the composite.

Hint. Work directly from base change of schemes and tensor products. ☐

Solution. Use associativity of fiber products or tensor products. ☐

Exercise 24.99 (Property transport example)**.** Prove: Prove an open immersion remains an open immersion after base change.

Hint. Work directly from base change of schemes and tensor products. ☐

Solution. Check locally where it is localization; tensoring a localization gives the corresponding localization after base change. ☐

Exercise 24.100 (Closed immersion stability)**.** Prove: Prove a closed immersion remains closed after base change.

Hint. Work directly from base change of schemes and tensor products. ☐

Solution. Affinely, a surjection $A \to B$ remains surjective after tensoring with the new base. ☐

Counterexamples and hypothesis tests

Exercise 24.101 (Irreducibility)**.** Test the claim: irreducibility is always preserved by arbitrary field extension.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; $x^2 + 1$ over $\mathbb{R}$ splits over $\mathbb{C}$. ☐

Exercise 24.102 (Point count)**.** Test the claim: base change cannot change the number of geometric points.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 24.103 (Same ring)**.** Test the claim: base change leaves coordinate rings literally unchanged.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; coefficients are tensor-extended. ☐

Applications and investigations

Exercise 24.104 (Geometric points)**.** Why base change to an algebraic closure?

Hint. Translate the question into base change of schemes and tensor products. ☐

Solution. It reveals geometric components and points hidden over the original field. ☐

Exercise 24.105 (Family transport)**.** What does base change do to a family?

Hint. Translate the question into base change of schemes and tensor products. ☐

Solution. It pulls the family to a new base while preserving the defining Cartesian relation. ☐

Challenge problems

Exercise 24.106 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Base Change in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Base Change, translates it through base change of schemes and tensor products, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 24.107 (Hypothesis challenge)**.** Choose one structural statement from Base Change, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 25

Fibers and Geometric Fibers

A morphism $f : X \to S$ is a family of schemes parametrized by the points of S. VI/23 constructed fiber products and VI/24 developed base change. The most important specialization of that calculus is obtained by changing the base from S to the residue field of one point $s \in S$. The result is the *scheme-theoretic fiber*

$$X_s := X \times_S \operatorname{Spec} \kappa(s).$$

It remembers more than the set of points above s: it remembers nilpotents, multiplicities, residue fields, components, and the equations after specialization. Extending $\kappa(s)$ further to an algebraically closed field produces a geometric fiber and reveals geometry hidden over the original residue field.

25.1 The scheme-theoretic fiber

Let $f : X \to S$ and $s \in S$. The canonical point morphism $\operatorname{Spec} \kappa(s) \to S$ defines the Cartesian square in figure 25.1. By definition, X_s is a scheme over the field $\kappa(s)$.

$$\begin{tikzcd}
X_s \arrow[r] \arrow[d] & X \arrow[d, "f"] \\
\operatorname{Spec} \kappa(s) \arrow[r] & S
\end{tikzcd}$$

Figure 25.1: A fiber is base change to the residue field of a point.

Definition 25.1 (Fiber)**.** The *fiber of f over s* is $X_s = X \times_S \operatorname{Spec} \kappa(s)$.

The projection $X_s \to X$ lands on the set-theoretic preimage of s. The scheme structure on X_s, however, may contain nilpotents or residue-field information that the bare set cannot record.

25.2 Affine fiber formula

Suppose $S = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, and f is induced by $A \to B$. For $\mathfrak{p} \in \operatorname{Spec} A$,

$$X_{\mathfrak{p}} \cong \operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p})).$$

Since

$$\kappa(\mathfrak{p}) = \operatorname{Frac}(A/\mathfrak{p}),$$

the computation has two steps: impose the equations in $\mathfrak{p}$, then invert every nonzero element of $A/\mathfrak{p}$.

Theorem 25.2 (Affine fiber formula)**.** *For an A-algebra B and $\mathfrak{p} \in \operatorname{Spec} A$,*

$$\operatorname{Spec} B \times_{\operatorname{Spec} A} \operatorname{Spec} \kappa(\mathfrak{p}) \cong \operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p})).$$

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ \kappa(\mathfrak{p}) \arrow[r] & B \otimes_A \kappa(\mathfrak{p}) \end{tikzcd}$$

Figure 25.2: The affine fiber ring is obtained by tensoring with the residue field.

Example 25.3 (Fiber of a polynomial map)**.** For $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $t \mapsto t^2$, the fiber over $a \in k$ has coordinate ring $k[t]/(t^2 - a)$. Its splitting and multiplicity depend on a and on the field.

25.3 Specializing equations

For a family

$$X = \operatorname{Spec} k[t, x_1, \ldots, x_n]/I \longrightarrow \operatorname{Spec} k[t],$$

the fiber over the closed point $(t - a)$ is obtained by substituting $t = a$:

$$X_a \cong \operatorname{Spec} k[x_1, \ldots, x_n]/I_a.$$

This elementary operation can change factorization, incidence, reducedness, or the number of components. The legacy degeneration DG4-P12 is the cleanest example: all fibers are reduced unions of two affine lines, but the two lines intersect for $a \neq 0$ and become parallel at $a = 0$.

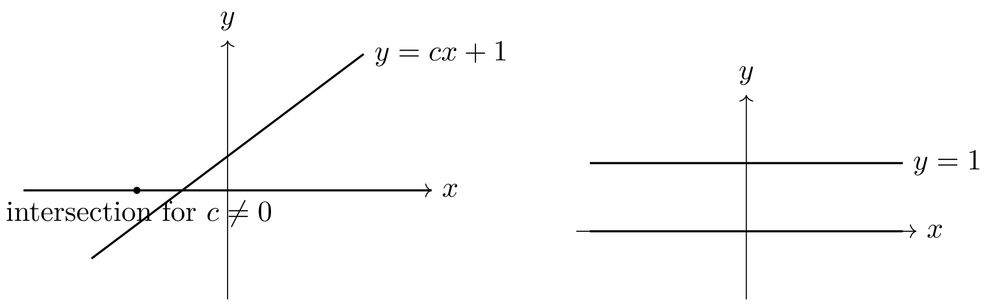


Figure 25.3: DG4-P12: intersecting lines for $c \neq 0$ and parallel lines at $c = 0$.

25.4 Generic and special fibers over $\operatorname{Spec} \mathbb{Z}$

For $X \to \operatorname{Spec} \mathbb{Z}$, the generic point (0) has residue field $\mathbb{Q}$, whereas the closed point (p) has residue field $\mathbb{F}_p$. Hence

$$X_{\mathbb{Q}} = X \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{Q}, \qquad X_p = X \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} \mathbb{F}_p.$$

A single arithmetic scheme therefore packages a characteristic-zero generic fiber together with fibers in every prime characteristic.

$$\begin{tikzcd}
X_{\mathbb{Q}} \arrow[dd] & X_p \arrow[dd] \\
 & \operatorname{Spec}\mathbb{Z} \\
\operatorname{Spec}\mathbb{Q} \arrow[ur] & \operatorname{Spec}\mathbb{F}_p \arrow[u]
\end{tikzcd}$$

Figure 25.4: Generic and special fibers over $\operatorname{Spec}\mathbb{Z}$.

25.5 Reducedness and irreducibility can vary

The ring of a fiber must be analyzed after tensoring with the residue field. Typical outcomes are

$$k, \qquad k \times k, \qquad k[\varepsilon]/(\varepsilon^2), \qquad 0.$$

These correspond respectively to one reduced point, two reduced points, one nonreduced doubled point, and the empty scheme. Thus reducedness and irreducibility are independent questions.

Remark 25.4. The empty scheme is reduced. We do not call it irreducible under the standard convention that an irreducible topological space is nonempty.

$$\boxed{k} \qquad\qquad \boxed{k \times k} \qquad\qquad \boxed{k[\varepsilon]/(\varepsilon^2)}$$

one reduced point two reduced points one doubled point

Figure 25.5: Three basic fiber rings encode different geometry.

25.6 Geometric fibers

Let $s \in S$ and choose an algebraically closed extension field Ω of $\kappa(s)$. The *geometric fiber over the chosen geometric point* $\bar{s} : \operatorname{Spec}\Omega \to S$ is

$$X_{\bar{s}} := X_s \times_{\operatorname{Spec}\kappa(s)} \operatorname{Spec}\Omega.$$

It is the ordinary fiber with its ground field enlarged enough to expose algebraic splitting. Different choices of algebraically closed overfield can be compared after passage to a common overfield; the purpose is to distinguish geometry intrinsic after field extension from accidents of the original residue field.

Definition 25.5 (Geometric fiber)**.** For a geometric point $\bar{s} : \operatorname{Spec}\Omega \to S$ with image s and Ω algebraically closed, $X_{\bar{s}} := X \times_S \operatorname{Spec}\Omega \cong X_s \times_{\kappa(s)} \Omega$.

Example 25.6 (Nonreduced special fiber)**.** For the family $\operatorname{Spec} k[t,x]/(x^2 - t) \to \operatorname{Spec} k[t]$, the fiber at $t = 0$ is $\operatorname{Spec} k[x]/(x^2)$, a nonreduced double point, while fibers at nonzero parameters can be reduced.

$$\begin{tikzcd}
X_{\bar{s}} \arrow[r] \arrow[d] & X_s \arrow[r] \arrow[d] & X \arrow[d] \\
\operatorname{Spec}\Omega \arrow[r] & \operatorname{Spec}\kappa(s) \arrow[r] & S
\end{tikzcd}$$

Figure 25.6: A geometric fiber is a further field extension of the ordinary fiber.

25.7 Why ordinary and geometric fibers differ

Over $\mathbb{R}$, $\operatorname{Spec}\mathbb{C}$ is a single point and is irreducible. After extension to $\mathbb{C}$,

$$\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C} \times \mathbb{C},$$

so its geometric fiber consists of two points. Thus an ordinary fiber may be irreducible over its residue field but become reducible after algebraic closure. Purely inseparable field extensions can also make a geometric fiber nonreduced.

$$\begin{tikzcd}
\boxed{\operatorname{Spec}\mathbb{C} \text{ over } \mathbb{R}} \arrow[r, "\text{base change to } \mathbb{C}"] & \boxed{\operatorname{Spec}(\mathbb{C} \times \mathbb{C})} \\
\text{one point} & \text{two points}
\end{tikzcd}$$

Figure 25.7: An ordinary irreducible fiber can split geometrically.

25.8 Fibers after a further base change

Suppose $S' \to S$, $s' \in S'$ maps to $s \in S$, and $X' = X \times_S S'$. There is a canonical isomorphism

$$X'_{s'} \cong X_s \times_{\operatorname{Spec}\kappa(s)} \operatorname{Spec}\kappa(s').$$

Thus taking a fiber after base change means first taking the old fiber and then extending its residue field. This formula is the bridge between VI/24 and the geometric-fiber viewpoint.

$$\begin{tikzcd}
(X \times_S S')_{s'} \arrow[r, "\sim"] \arrow[d] & X_s \times_{\kappa(s)} \kappa(s') \arrow[d] \\
\operatorname{Spec}\kappa(s') \arrow[r] & \operatorname{Spec}\kappa(s)
\end{tikzcd}$$

Figure 25.8: Fibers after base change are residue-field extensions of the old fiber.

Example 25.7 (Geometric fiber)**.** The geometric fiber at $s \in S$ is obtained by extending the ordinary fiber from $\kappa(s)$ to an algebraic closure $\overline{\kappa(s)}$. This separates field-arithmetic effects from intrinsic geometric structure.

25.9 Arithmetic fibers and factorization

For

$$X = \operatorname{Spec}\mathbb{Z}[T]/(T^2+1) \longrightarrow \operatorname{Spec}\mathbb{Z},$$

the fiber at p is

$$X_p = \operatorname{Spec} \mathbb{F}_p[T]/(T^2 + 1).$$

At $p = 2$ the polynomial is $(T + 1)^2$ and the fiber is nonreduced. If $p \equiv 1 \pmod 4$ it splits into two distinct linear factors, whereas if $p \equiv 3 \pmod 4$ it remains irreducible and gives $\operatorname{Spec} \mathbb{F}_{p^2}$. Scheme-theoretic fibers therefore record ramification, splitting, and inertia in one uniform construction.

$\boxed{p = 2}$	$\boxed{p \equiv 1\ (4)}$	$\boxed{p \equiv 3\ (4)}$
doubled point	two points	$\operatorname{Spec} \mathbb{F}_{p^2}$

Figure 25.9: Three fiber types for the Gaussian integers.

25.10 Boundary with adjacent chapters

VI/23 supplies the universal fiber-product construction and VI/24 supplies arbitrary base change. VI/25 specializes them to points and geometric points and analyzes the resulting schemes. We do not yet develop finite-type or Noetherian morphisms (VI/26), integral schemes and function fields (VI/27), or normalization (VI/28).

25.11 Examples

Example 25.8 (Affine line over a field)**.** For $\mathbb{A}^1_k \to \operatorname{Spec} k$, the only fiber is $\mathbb{A}^1_k$ itself.

Example 25.9 (Projection from the affine plane)**.** For $\operatorname{Spec} k[t, x] \to \operatorname{Spec} k[t]$, the fiber at $t = a$ is $\operatorname{Spec} k[x] \cong \mathbb{A}^1_k$.

Example 25.10 (Hyperbola projection)**.** For $X = \operatorname{Spec} k[t, u]/(tu - 1) \to \operatorname{Spec} k[t]$, the fiber at 0 is empty, while for $a \neq 0$ it is $\operatorname{Spec} k$.

Example 25.11 (A doubled special fiber)**.** For $\operatorname{Spec} k[t, u]/(u^2 - t) \to \operatorname{Spec} k[t]$, the fiber at 0 is $\operatorname{Spec} k[u]/(u^2)$, a doubled point.

Example 25.12 (Two reduced points)**.** If $\operatorname{char} k \neq 2$, the fiber of $u^2 - t^2 = 0$ at $t = a \neq 0$ is $\operatorname{Spec}(k \times k)$ after the two distinct linear factors are identified.

Example 25.13 (Legacy line degeneration)**.** For $y(y - tx - 1) = 0$, fibers at $t \neq 0$ are two intersecting lines and the fiber at 0 is two parallel lines.

Example 25.14 (Affine line over the integers)**.** $\operatorname{Spec} \mathbb{Z}[x] \to \operatorname{Spec} \mathbb{Z}$ has generic fiber $\mathbb{A}^1_{\mathbb{Q}}$ and p-fiber $\mathbb{A}^1_{\mathbb{F}_p}$.

Example 25.15 (A fiber can be empty)**.** $\operatorname{Spec} \mathbb{Q} \to \operatorname{Spec} \mathbb{Z}$ has empty fiber over every closed point (p) because $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{F}_p = 0$.

Example 25.16 (A point with larger residue field)**.** The $\mathbb{R}$-scheme $\operatorname{Spec} \mathbb{C}$ is one ordinary fiber over $\operatorname{Spec} \mathbb{R}$, but its geometric base change to $\mathbb{C}$ splits into two points.

Example 25.17 (Gaussian fiber at two)**.** $\mathbb{F}_2[T]/(T^2 + 1) = \mathbb{F}_2[T]/((T + 1)^2)$, so the fiber at 2 of $\operatorname{Spec} \mathbb{Z}[i]$ is nonreduced.

Example 25.18 (Gaussian split prime)**.** For $p \equiv 1 \pmod 4$, the p-fiber of $\operatorname{Spec} \mathbb{Z}[i]$ consists of two reduced points.

Example 25.19 (Gaussian inert prime)**.** For $p \equiv 3 \pmod 4$, the p-fiber is $\operatorname{Spec} \mathbb{F}_{p^2}$: one point with a quadratic residue field.

Example 25.20 (Closed immersion fiber)**.** For $\operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec} A$, the fiber over $\mathfrak{p}$ is empty if $I \not\subseteq \mathfrak{p}$ and is $\operatorname{Spec} \kappa(\mathfrak{p})$ if $I \subseteq \mathfrak{p}$.

Example 25.21 (Open immersion fiber)**.** For $D(f) \hookrightarrow \operatorname{Spec} A$, the fiber over $\mathfrak{p}$ is a point exactly when $f \notin \mathfrak{p}$, and is empty otherwise.

Example 25.22 (Constant family)**.** For a k-scheme Y, the projection $Y \times_k S \to S$ has fiber over s equal to $Y \times_k \kappa(s)$.

Example 25.23 (Fiber of an isomorphism)**.** An isomorphism $X \to S$ has every fiber equal to the one-point scheme $\operatorname{Spec} \kappa(s)$.

Example 25.24 (Fiber of a product family)**.** The fiber of $X \times_S Y \to S$ at s is canonically $X_s \times_{\kappa(s)} Y_s$.

Example 25.25 (Specializing a hypersurface)**.** For $F(t, x, y) = 0$, the fiber at $t = a$ is the hypersurface $F(a, x, y) = 0$ over $\kappa(a)$.

Example 25.26 (Generic point of an affine base)**.** For a domain A with fraction field K, the generic fiber of $\operatorname{Spec} B \to \operatorname{Spec} A$ is $\operatorname{Spec}(B \otimes_A K)$.

Example 25.27 (Geometric fiber notation)**.** Choosing an algebraic closure $\overline{\kappa(s)}$ gives $X_{\bar{s}} = X_s \times_{\kappa(s)} \overline{\kappa(s)}$.

Example 25.28 (Residue fields after geometric extension)**.** If $x \in X_s$, points of the geometric fiber above x are controlled by primes of $\kappa(x) \otimes_{\kappa(s)} \Omega$; one ordinary point can therefore split geometrically.

Example 25.29 (Purely inseparable nonreduced geometric fiber)**.** In characteristic p, let $k = \mathbb{F}_p(a^p)$ and $L = k(a)$. Then $L \otimes_k L$ contains nilpotents because $T^p - a^p = (T - a)^p$ after extending scalars to L.

Example 25.30 (Fiber of a composition)**.** For $X \to S' \to S$, the fiber over $s \in S$ can be studied by first base-changing S' to $\kappa(s)$ and then pulling back X; this is associativity of fiber products.

Example 25.31 (Fiber after base change)**.** If $s' \mapsto s$, then $(X \times_S S')_{s'} \cong X_s \times_{\kappa(s)} \kappa(s')$.

Example 25.32 (Generic fiber detects vertical equations)**.** If an equation is annihilated by a nonzero element of the domain A, it can disappear after tensoring with $\operatorname{Frac}(A)$; special fibers may retain it.

Example 25.33 (A family with constant dimension but changing incidence)**.** DG4-P12 has two one-dimensional components in every fiber, yet the intersection pattern jumps at the special parameter.

Example 25.34 (A family with changing number of points)**.** For $u^2 - t = 0$ over a field of characteristic not 2, a nonzero square parameter yields two rational points, a nonsquare may yield one degree-two point, and $t = 0$ yields a doubled point.

Example 25.35 (Geometric splitting removes arithmetic ambiguity)**.** After passing to an algebraic closure, a separable polynomial splits into linear factors; the geometric fiber separates the corresponding geometric points.

Example 25.36 (Nilpotents survive algebraic closure)**.** If a fiber already contains a nonzero nilpotent, every faithfully flat field extension preserves its nonzeroness, so algebraic closure does not repair nonreducedness.

Example 25.37 (Products of fibers)**.** Associativity gives $(X \times_S Y)_{\bar{s}} \cong X_{\bar{s}} \times_\Omega Y_{\bar{s}}$ for a geometric point with field Ω.

Example 25.38 (Empty geometric fibers)**.** If $X_s = \varnothing$, every geometric fiber over s is empty; scalar extension of the zero ring stays zero.

Example 25.39 (Geometric connectedness warning)**.** A connected scheme over a non-algebraically-closed field can become disconnected after scalar extension, as $\operatorname{Spec}\mathbb{C}$ over $\mathbb{R}$ demonstrates.

Example 25.40 (Arithmetic factorization as fiber geometry)**.** For monic $F \in \mathbb{Z}[T]$, the decomposition of F modulo p controls the components and residue fields of the p-fiber of $\operatorname{Spec}\mathbb{Z}[T]/(F)$.

Example 25.41 (Fiber data is local on source and base)**.** Fiber computations may be performed on affine neighborhoods of s and affine opens of X lying over them, then glued; the tensor formula is the local engine.

25.12 Problem dossiers

VI.25.P1: Legacy DG4-P12: intersecting fibers degenerating to parallel lines

Problem 25.42. Let k be a field and let $\pi : \mathbb{A}^3_k \to \mathbb{A}^1_k$ be projection to z. Preserve the legacy construction by taking

$$X = V\big(y(y - zx - 1)\big) \subset \mathbb{A}^3_k.$$

Show that for $c \neq 0$ the fiber X_c is the union of two intersecting lines, while X_0 is the union of two parallel lines.

Provenance. Legacy DG4-P12. The degeneration is preserved as the chapter's principal legacy fiber example.

VI.25.P2: Affine fiber formula

Problem 25.43. Let $A \to B$ and $\mathfrak{p} \in \operatorname{Spec} A$. Prove

$$(\operatorname{Spec} B)_{\mathfrak{p}} \cong \operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p})).$$

Provenance. Canonical affine-fiber theorem derived from VI/23 and VI/24.

VI.25.P3: Hyperbola projection: empty and one-point fibers

Problem 25.44. For

$$X = \operatorname{Spec} k[t, u]/(tu - 1) \to \operatorname{Spec} k[t],$$

compute every closed fiber.

Provenance. Canonical affine fiber computation.

VI.25.P4: A double cover with a doubled special fiber

Problem 25.45. Assume $\operatorname{char} k \neq 2$ and study

$$X = \operatorname{Spec} k[t,u]/(u^2 - t^2) \to \operatorname{Spec} k[t].$$

Provenance. Canonical degeneration dossier.

VI.25.P5: Generic and special fibers of the affine line over $\mathbb{Z}$

Problem 25.46. Compute the generic fiber and every closed fiber of

$$\operatorname{Spec} \mathbb{Z}[x] \to \operatorname{Spec} \mathbb{Z}.$$

Provenance. Canonical arithmetic-fiber computation.

VI.25.P6: Gaussian integers: ramified, split, inert

Problem 25.47. Analyze the fibers of

$$\operatorname{Spec} \mathbb{Z}[T]/(T^2 + 1) \to \operatorname{Spec} \mathbb{Z}.$$

Provenance. Canonical arithmetic fiber dossier.

VI.25.P7: The fibers of $\operatorname{Spec} \mathbb{Q} \to \operatorname{Spec} \mathbb{Z}$

Problem 25.48. Compute the fiber over (0) and over every (p).

Provenance. Canonical localization/fiber dossier.

VI.25.P8: Reduced versus nonreduced fibers

Problem 25.49. Give one family whose special fiber is nonreduced although nearby fibers are reduced.

Provenance. Canonical reducedness-jump example.

VI.25.P9: Ordinary irreducibility versus geometric irreducibility

Problem 25.50. View $X = \operatorname{Spec} \mathbb{C}$ as an $\mathbb{R}$-scheme. Compare the ordinary fiber with its base change to $\mathbb{C}$.

Provenance. Canonical geometric-fiber splitting example.

VI.25.P10: Constructing a geometric fiber

Problem 25.51. Let $f : X \to S$, $s \in S$, and $\Omega/\kappa(s)$ be algebraically closed. Show

$$X \times_S \operatorname{Spec} \Omega \cong X_s \times_{\operatorname{Spec} \kappa(s)} \operatorname{Spec} \Omega.$$

Provenance. Canonical geometric-fiber identity.

VI.25.P11: A real conic after geometric extension

Problem 25.52. Study
$$X = \operatorname{Spec} \mathbb{R}[x, y]/(x^2 + y^2)$$
before and after scalar extension to $\mathbb{C}$.

Provenance. Canonical geometric-splitting dossier.

VI.25.P12: Fibers after base change

Problem 25.53. Let $S' \to S$, let $s' \mapsto s$, and set $X' = X \times_S S'$. Prove
$$X'_{s'} \cong X_s \times_{\kappa(s)} \kappa(s').$$

Provenance. Canonical VI/24-to-VI/25 comparison.

VI.25.P13: Fiber of a closed immersion

Problem 25.54. Let
$$i : \operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec} A.$$
Compute the fiber at $\mathfrak{p} \in \operatorname{Spec} A$.

Provenance. Canonical closed-immersion fiber calculation.

VI.25.P14: Fiber of an open immersion

Problem 25.55. For
$$D(f) \hookrightarrow \operatorname{Spec} A,$$
compute the fiber over $\mathfrak{p}$.

Provenance. Canonical open-immersion fiber calculation.

VI.25.P15: Reduction modulo 2 creates a doubled fiber

Problem 25.56. Compute the fiber at 2 of
$$X = \operatorname{Spec} \mathbb{Z}[x]/(x^2 - 2) \to \operatorname{Spec} \mathbb{Z}.$$

Provenance. Canonical arithmetic ramification example.

VI.25.P16: Geometric splitting of a real point

Problem 25.57. Show explicitly that the geometric fiber of
$$\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} \mathbb{R}$$
over the unique point has two points.

Provenance. Canonical geometric splitting computation.

VI.25.P17: Purely inseparable geometric nonreducedness

Problem 25.58. In characteristic p, let

$$k = \mathbb{F}_p(a^p), \qquad L = k(a).$$

Analyze

$$\operatorname{Spec} L \times_{\operatorname{Spec} k} \operatorname{Spec} L.$$

Provenance. Canonical purely inseparable geometric-fiber example.

VI.25.P18: Fiber of a product family

Problem 25.59. For $X \to S$ and $Y \to S$, prove

$$(X \times_S Y)_s \cong X_s \times_{\kappa(s)} Y_s.$$

Provenance. Canonical product/fiber compatibility.

VI.25.P19: Factorization controls a finite arithmetic fiber

Problem 25.60. Let $F \in \mathbb{Z}[T]$ be monic. Explain how the factorization of $\overline{F}$ in $\mathbb{F}_p[T]$ controls the p-fiber of

$$\operatorname{Spec} \mathbb{Z}[T]/(F).$$

Provenance. Canonical finite arithmetic-fiber classifier.

VI.25.P20: Ordinary fiber, geometric fiber, and set-theoretic inverse image

Problem 25.61. Explain the three levels of information carried by $f^{-1}(s)$, X_s, and $X_{\bar{s}}$.

Provenance. Canonical synthesis dossier for VI/25.

25.13 Exercises

Exercise 25.62. Write the Cartesian square defining X_s.

Exercise 25.63. Compute the affine fiber ring for $A \to B$ at $\mathfrak{p}$.

Exercise 25.64. Compute the fiber of $k[t,u]/(tu-1)$ at $t=0$.

Exercise 25.65. Compute the same fiber at $t = a \neq 0$.

Exercise 25.66. Compute the 0-fiber of $k[t,u]/(u^2-t)$.

Exercise 25.67. For DG4-P12, find the intersection point when $c \neq 0$.

Exercise 25.68. For DG4-P12, write the two components when $c = 0$.

Exercise 25.69. Compute the generic fiber of $\operatorname{Spec} \mathbb{Z}[x] \to \operatorname{Spec} \mathbb{Z}$.

Exercise 25.70. Compute its fiber at (p).

Exercise 25.71. Compute the 2-fiber of $\operatorname{Spec} \mathbb{Z}[i]$.

Exercise 25.72. Describe the p-fiber for $p \equiv 3 \pmod 4$.

Exercise 25.73. Show $\operatorname{Spec}\mathbb{Q} \to \operatorname{Spec}\mathbb{Z}$ has empty closed fibers.

Exercise 25.74. Give a fiber that is irreducible but nonreduced.

Exercise 25.75. Give a fiber that is reduced but reducible.

Exercise 25.76. Define a geometric fiber over s.

Exercise 25.77. Compute $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$.

Exercise 25.78. Explain why $\operatorname{Spec}\mathbb{C}$ over $\mathbb{R}$ is not geometrically irreducible.

Exercise 25.79. Prove the fiber-after-base-change formula.

Exercise 25.80. Compute the fiber of $D(f) \hookrightarrow \operatorname{Spec} A$ at $\mathfrak{p}$.

Exercise 25.81. Compute the fiber of $\operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec} A$ at $\mathfrak{p}$.

Exercise 25.82. Compute the 2-fiber of $\operatorname{Spec}\mathbb{Z}[x]/(x^2 - 2)$.

Exercise 25.83. Give a purely inseparable scalar extension producing nilpotents.

Exercise 25.84. Show fibers commute with products over the base.

Exercise 25.85. Explain the difference between a degree-two point and two rational points.

Exercise 25.86. State one phenomenon detected by geometric rather than ordinary fibers.

Exercise 25.87. Distinguish VI/24 from VI/25 in one sentence.

25.14 Challenges

Problem 25.88 (Challenge: Complete degeneration audit)**.** For $X = V(y(y - tx - 1)) \to \mathbb{A}^1_t$, prove every fiber is reduced, determine its irreducible components, and describe exactly how the incidence relation between the components changes at $t = 0$.

Problem 25.89 (Challenge: Arithmetic fiber classifier)**.** For $X = \operatorname{Spec}\mathbb{Z}[T]/(T^3 - 2) \to \operatorname{Spec}\mathbb{Z}$, classify several primes by the factorization of $T^3 - 2$ modulo p, recording residue degrees and repeated factors scheme-theoretically.

Problem 25.90 (Challenge: Geometric splitting versus nilpotence)**.** Construct one finite field extension whose geometric scalar extension splits into several reduced points and one purely inseparable extension whose scalar extension becomes nonreduced. Explain the algebraic cause of the contrast.

Problem 25.91 (Challenge: Fiber comparison under a tower)**.** Given $S'' \to S' \to S$ and $s'' \mapsto s' \mapsto s$, derive canonical comparison isomorphisms among the fiber of $X_{S'}$, the fiber of $X_{S''}$, and successive residue-field extensions of X_s.

Problem 25.92 (Challenge: Design a family with controlled special fiber)**.** Construct an affine family over $\mathbb{A}^1$ whose generic closed fibers are two distinct reduced points but whose special fiber at 0 is one doubled point. Then modify the construction so the special fiber is empty instead.

25.15 Chapter synthesis

A fiber is not merely a set-theoretic inverse image. It is a base change to a residue field, and its coordinate ring can retain nilpotents, multiplicity, field extensions, and changing component structure. Geometric fibers then extend the residue field to expose splitting and inseparability. Together the ordinary and geometric fibers turn a morphism into a family whose members can be studied with the full algebraic structure intact.

$$\begin{tikzcd}
\boxed{\text{set-theoretic preimage}} \arrow[d, "\text{add scheme structure}"] \\
\boxed{\text{scheme fiber } X_s} \arrow[d, "\text{extend residue field}"] \\
\boxed{\text{geometric fiber } X_{\bar{s}}}
\end{tikzcd}$$

Figure 25.10: Three increasingly refined views of a member of a family.

25.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 25.93 (Square map fiber)**.** Write the coordinate ring of the fiber of $t \mapsto t^2$ over a.

Hint. Use scheme fibers, residue fields, and geometric fibers. ☐

Solution. $k[t]/(t^2 - a)$. ☐

Exercise 25.94 (Special fiber)**.** Compute the fiber of $x^2 = t$ at $t = 0$.

Hint. Use scheme fibers, residue fields, and geometric fibers. ☐

Solution. $k[x]/(x^2)$. ☐

Exercise 25.95 (Fiber definition)**.** What is X_s?

Hint. Use scheme fibers, residue fields, and geometric fibers. ☐

Solution. $X \times_S \operatorname{Spec} \kappa(s)$. ☐

Exercise 25.96 (Geometric fiber)**.** What field is used for a geometric fiber?

Hint. Use scheme fibers, residue fields, and geometric fibers. ☐

Solution. An algebraic closure of $\kappa(s)$. ☐

Exercise 25.97 (Generic fiber)**.** What point is used for the generic fiber of an integral base?

Hint. Use scheme fibers, residue fields, and geometric fibers. ☐

Solution. The generic point. ☐

Proofs

Exercise 25.98 (Affine fiber formula)**.** Prove: Prove the fiber of $\operatorname{Spec} B \to \operatorname{Spec} A$ over $\mathfrak{p}$ is $\operatorname{Spec}(B \otimes_A \kappa(\mathfrak{p}))$.

Hint. Work directly from scheme fibers, residue fields, and geometric fibers. ☐

Solution. Apply the affine fiber-product formula to $\operatorname{Spec} \kappa(\mathfrak{p}) \to \operatorname{Spec} A$. ☐

Exercise 25.99 (Geometric base change)**.** Prove: Prove the geometric fiber is a base change of the ordinary fiber.

Hint. Work directly from scheme fibers, residue fields, and geometric fibers. ☐

Solution. Associate the two fiber products along $\operatorname{Spec} \overline{\kappa(s)} \to \operatorname{Spec} \kappa(s)$. ☐

Exercise 25.100 (Special nonreduced)**.** Prove: Prove the fiber of $x^2 - t$ at $t = 0$ is nonreduced.

Hint. Work directly from scheme fibers, residue fields, and geometric fibers. ☐

Solution. The coordinate ring contains the nonzero nilpotent class of x. ☐

Exercise 25.101 (Fiber compatibility)**.** Prove: Prove fibers commute with further base change.

Hint. Work directly from scheme fibers, residue fields, and geometric fibers. ☐

Solution. Use associativity of fiber products. ☐

Counterexamples and hypothesis tests

Exercise 25.102 (All fibers same)**.** Test the claim: all fibers of a finite-type morphism are isomorphic.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; geometry can jump with the parameter. ☐

Exercise 25.103 (Reduced family)**.** Test the claim: a reduced total space forces every fiber to be reduced.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 25.104 (Rational enough)**.** Test the claim: ordinary fibers over residue fields always reveal all geometric components.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; components may split after algebraic closure. ☐

Applications and investigations

Exercise 25.105 (Degeneration)**.** How do fibers describe degeneration?

Hint. Translate the question into scheme fibers, residue fields, and geometric fibers. ☐

Solution. They compare the geometry of members of a family at different base points. ☐

Exercise 25.106 (Geometric irreducibility)**.** Why inspect geometric fibers?

Hint. Translate the question into scheme fibers, residue fields, and geometric fibers. ☐

Solution. They test properties after removing residue-field arithmetic obstructions. ☐

Challenge problems

Exercise 25.107 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Fibers and Geometric Fibers in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Fibers and Geometric Fibers, translates it through scheme fibers, residue fields, and geometric fibers, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 25.108 (Hypothesis challenge)**.** Choose one structural statement from Fibers and Geometric Fibers, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 26

Finite-Type and Noetherian Morphisms

The preceding chapters built fiber products, base change, and fibers. We now impose a finiteness condition on morphisms. Algebraically, a map of affine schemes

$$\operatorname{Spec} B \longrightarrow \operatorname{Spec} A$$

is of finite type when B is generated by finitely many elements as an A-algebra. Geometrically, this says that the source is described over the base by finitely many coordinates, locally on the target. The second finiteness theme is Noetherianity: ascending chains of algebraic conditions eventually stabilize. These two notions interact through Hilbert's basis theorem and provide the standard bridge from arbitrary schemes to manageable algebraic families.

26.1 Locally of finite type and finite type

Definition 26.1 (Locally of finite type)**.** A morphism $f : X \to Y$ is *locally of finite type* if Y has an affine open cover $V_i = \operatorname{Spec} B_i$ such that each $f^{-1}(V_i)$ has an affine open cover $U_{ij} = \operatorname{Spec} A_{ij}$ with every A_{ij} a finitely generated B_i-algebra.

Definition 26.2 (Finite type)**.** A morphism $f : X \to Y$ is *of finite type* if it is locally of finite type and quasi-compact. Equivalently, over every affine open $V = \operatorname{Spec} B \subseteq Y$, the inverse image $f^{-1}(V)$ admits a *finite* affine open cover $\operatorname{Spec} A_j$ with each A_j finitely generated over B.

$$\begin{tikzcd} \boxed{\text{locally finite type}} \arrow[dr] & & \boxed{\text{quasi-compact}} \arrow[dl] \\ & \boxed{\text{finite type}} & \end{tikzcd}$$

finite type = LFT + QC

Figure 26.1: The two ingredients of finite type.

26.2 The affine algebra behind finite type

If $A \to B$ is a ring map, then B is a finitely generated A-algebra precisely when there are finitely many elements $b_1, \ldots, b_n \in B$ and a surjection

$$A[x_1, \ldots, x_n] \twoheadrightarrow B, \qquad x_i \mapsto b_i.$$

Thus every affine finite-type morphism factors as a closed immersion into affine n-space over the base followed by the projection to the base. No finite-generation hypothesis on the kernel is required here; that stronger condition belongs to finite presentation, not finite type.

$$\begin{tikzcd}
\boxed{A[x_1, \ldots, x_n]} \arrow[r, two heads, "\text{surjection}"] & \boxed{B} \\
\boxed{\operatorname{Spec} A} \arrow[r] & \boxed{\operatorname{Spec} B}
\end{tikzcd}$$

Figure 26.2: Finitely many algebra generators give a finite-type affine morphism.

26.3 Checking locally finite type on every affine open

The legacy Problems AG6-P5 and AG5-P5 establish an important invariance: the definition may begin with one affine cover of the target, but once it holds, it holds over *every* affine open $V = \operatorname{Spec} B \subseteq Y$. The mechanism is simultaneous distinguished shrinking. Around $y \in V \cap V_i$ choose

$$W = D(b) \subseteq V, \qquad W = D(c) \subseteq V_i.$$

Above W, the coordinate rings are localizations of finitely generated algebras and hence remain finitely generated. Since $B_b \cong B[t]/(bt - 1)$ is itself of finite type over B, transitivity of finite generation finishes the argument.

$$\begin{tikzcd}
\operatorname{Spec}(A_{ij})_{\varphi(c)} \arrow[r, hook] \arrow[d] & \operatorname{Spec} A_{ij} \arrow[d] \\
W = D(b) = D(c) \arrow[r, hook] & V_i
\end{tikzcd}$$

Figure 26.3: Common distinguished shrinking in the AG6-P5 criterion.

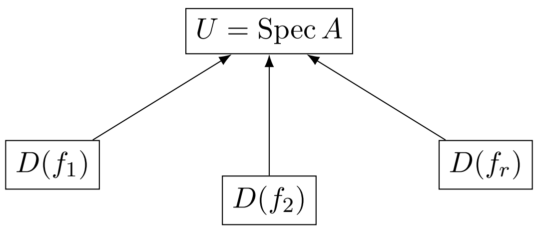
Example 26.3 (Affine finite-type morphism)**.** The projection $\mathbb{A}^n_A = \operatorname{Spec} A[x_1, \ldots, x_n] \to \operatorname{Spec} A$ is of finite type because its coordinate algebra is generated by finitely many variables.

26.4 Locally Noetherian schemes

Definition 26.4 (Locally Noetherian and Noetherian)**.** A scheme X is *locally Noetherian* if it has an affine open cover by spectra of Noetherian rings. It is *Noetherian* if it is locally Noetherian and quasi-compact.

The recovered legacy Problem AG5-P4 proves that local Noetherianity is independent of the chosen affine cover: X is locally Noetherian if and only if every affine open $U = \operatorname{Spec} A \subseteq X$ has A Noetherian. The proof uses distinguished refinements and the local-global criterion for finite generation of ideals.

Theorem 26.5 (Affine-open criterion for locally Noetherian schemes)**.** *A scheme X is locally Noetherian if and only if every affine open $\operatorname{Spec} A \subseteq X$ has A Noetherian.*



A_{f_i} Noetherian $\Rightarrow$ A Noetherian

Figure 26.4: Finite distinguished descent of Noetherianity.

26.5 Finite type equals local finite type plus quasi-compactness

The distinction between locally finite type and finite type is global rather than algebraic. Every point may have a finite-coordinate neighborhood while infinitely many such neighborhoods are needed globally. The standard example is

$$\coprod_{n\geq 1} \operatorname{Spec} k \longrightarrow \operatorname{Spec} k,$$

which is locally of finite type but not quasi-compact, hence not of finite type.

Theorem 26.6 (Finite type criterion)**.** *A morphism is of finite type if and only if it is locally of finite type and quasi-compact.*

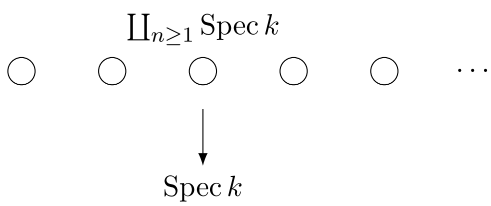


Figure 26.5: Locally finite type but not quasi-compact.

26.6 A morphism that is not locally of finite type

The opposite failure is genuinely algebraic. The morphism

$$\operatorname{Spec} k[x_1, x_2, x_3, \dots] \longrightarrow \operatorname{Spec} k$$

$$\boxed{k} \longrightarrow \boxed{k[x_1, x_2, x_3, \dots]}$$

infinitely many algebra generators

Figure 26.6: The basic non-locally-finite-type affine example.

is not locally of finite type. No neighborhood of the generic point can be described by a finitely generated k-algebra: localizing at one polynomial only involves finitely many variables and leaves infinitely many independent variables present.

Example 26.7 (Closed subscheme finite type)**.** If $B = A[x_1, \dots, x_n]/I$, then $\operatorname{Spec} B \to \operatorname{Spec} A$ is finite type. When I is finitely generated it is finitely presented.

26.7 Stability under composition

If $A \to B \to C$ are ring maps with B finitely generated over A and C finitely generated over B, then C is finitely generated over A. After affine localization this proves that locally finite type morphisms are closed under composition. Combining this with quasi-compactness gives the same result for finite-type morphisms.

Proposition 26.8 (Composition)**.** *If $X \to Y$ and $Y \to Z$ are locally of finite type, then $X \to Z$ is locally of finite type. If both are of finite type, then the composite is of finite type.*

$$X \xrightarrow{\;f\;} Y \xrightarrow{\;g\;} Z$$

Figure 26.7: Composition preserves local finite type and finite type.

26.8 Stability under base change

Let $f : X \to S$ be locally of finite type and let $S' \to S$ be arbitrary. Affine-locally the base change replaces an A-algebra B by $B \otimes_A A'$. If $B = A[b_1, \dots, b_n]$, then

$$B \otimes_A A' = A'[b_1 \otimes 1, \dots, b_n \otimes 1],$$

so finite generation survives. Quasi-compactness also survives base change.

Theorem 26.9 (Base-change stability)**.** *Locally finite type and finite type morphisms are stable under arbitrary base change.*

$$\begin{tikzcd} X \times_S S' \arrow[r] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r] & S \end{tikzcd}$$

Figure 26.8: Finite type and local finite type are stable under base change.

26.9 Noetherian consequences of finite type

Hilbert's basis theorem says that a finitely generated algebra over a Noetherian ring is Noetherian. Therefore finite-coordinate geometry over a Noetherian base remains Noetherian.

Theorem 26.10 (Finite type over a Noetherian base)**.** *If Y is locally Noetherian and $f : X \to Y$ is locally of finite type, then X is locally Noetherian. If Y is Noetherian and f is of finite type, then X is Noetherian.*

$$\boxed{A \text{ Noetherian}} \xrightarrow{\text{Hilbert basis}} \boxed{B = A[b_1, \ldots, b_n]}$$

B Noetherian

Figure 26.9: Finite type over a Noetherian affine base remains Noetherian.

Example 26.11 (Noetherian inheritance)**.** If A is Noetherian and B is a finitely generated A-algebra, Hilbert's basis theorem implies B is Noetherian. Thus affine finite-type schemes over a Noetherian base are Noetherian.

26.10 Fibers of a locally finite type morphism

By VI/25 a fiber is a base change to $\operatorname{Spec} \kappa(s)$. Base-change stability therefore gives an immediate consequence: if $f : X \to S$ is locally of finite type, every fiber X_s is locally of finite type over the field $\kappa(s)$; if f is of finite type, every fiber is a finite-type $\kappa(s)$-scheme. This is the basic reason that finite-type morphisms really do behave like algebraic families of finite-dimensional coordinate complexity.

26.11 Boundary with adjacent chapters

VI/23–VI/25 supplied fiber products, base change, and fibers. VI/26 adds finiteness of morphisms and the Noetherian consequences. Integral schemes and function fields remain for VI/27, and normalization for VI/28. We do not conflate finite type with finite presentation: finite type controls generators of an algebra, whereas finite presentation also controls generators of the relations.

26.12 Examples

Example 26.12 (Affine line over a field)**.** The structure morphism $\mathbb{A}^1_k = \operatorname{Spec} k[t] \to \operatorname{Spec} k$ is of finite type because $k[t]$ is generated by t as a k-algebra.

Example 26.13 (Affine n-space)**.** $\mathbb{A}^n_A = \operatorname{Spec} A[x_1, \ldots, x_n] \to \operatorname{Spec} A$ is of finite type; the coordinate algebra has the displayed finite generating set.

Example 26.14 (A hypersurface family)**.** For $B = A[x_1, \ldots, x_n]/(F)$, the morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ is of finite type. The number of equations is irrelevant to finite type; only the number of algebra generators matters.

Example 26.15 (An arbitrary affine quotient)**.** For any ideal $I \subseteq A[x_1, \ldots, x_n]$, even one not finitely generated, $A[x_1, \ldots, x_n]/I$ is a finite-type A-algebra.

Example 26.16 (A principal localization)**.** $A_f \cong A[t]/(ft-1)$ is finitely generated over A, so $D(f) \to \operatorname{Spec} A$ is of finite type.

Example 26.17 (The multiplicative group as an affine scheme)**.** $\operatorname{Spec} k[t,t^{-1}] \to \operatorname{Spec} k$ is of finite type because $k[t,t^{-1}] \cong k[t,u]/(tu-1)$.

Example 26.18 (Polynomial algebra in infinitely many variables)**.** $\operatorname{Spec} k[x_1,x_2,\ldots] \to \operatorname{Spec} k$ is not locally of finite type; infinitely many algebra generators remain near the generic point.

Example 26.19 (Infinite disjoint union of points)**.** $\coprod_{n\geq 1} \operatorname{Spec} k \to \operatorname{Spec} k$ is locally of finite type but not finite type because the source is not quasi-compact.

Example 26.20 (A finite disjoint union of points)**.** $\coprod_{i=1}^{r} \operatorname{Spec} k \to \operatorname{Spec} k$ is finite type: it is locally finite type and the source has a finite affine cover.

Example 26.21 (Coordinate projection)**.** $\operatorname{Spec} k[x,y] \to \operatorname{Spec} k[x]$ is of finite type because $k[x,y] = k[x][y]$.

Example 26.22 (Restriction to a basic open)**.** Restricting the preceding projection over $D(x)$ gives $\operatorname{Spec} k[x,x^{-1},y] \to \operatorname{Spec} k[x,x^{-1}]$, again of finite type.

Example 26.23 (Scalar extension)**.** If B is finite type over A and $A \to A'$ is any ring map, then $B \otimes_A A'$ is finite type over A'.

Example 26.24 (A fiber of affine space)**.** The fiber of $\mathbb{A}^n_S \to S$ at s is $\mathbb{A}^n_{\kappa(s)}$, a finite-type scheme over the residue field.

Example 26.25 (The integers are Noetherian)**.** $\operatorname{Spec} \mathbb{Z}$ is Noetherian because $\mathbb{Z}$ is a principal ideal domain, hence Noetherian.

Example 26.26 (Affine space over the integers)**.** $\mathbb{Z}[x_1,\ldots,x_n]$ is Noetherian by Hilbert's basis theorem, so $\mathbb{A}^n_{\mathbb{Z}}$ is Noetherian.

Example 26.27 (A distinguished open in a Noetherian affine scheme)**.** If A is Noetherian, then A_f is Noetherian; hence every distinguished open $D(f) \subseteq \operatorname{Spec} A$ is Noetherian.

Example 26.28 (A non-Noetherian affine scheme)**.** $\operatorname{Spec} k[x_1,x_2,\ldots]$ is not Noetherian because $(x_1) \subsetneq (x_1,x_2) \subsetneq \cdots$ is a strictly ascending chain.

Example 26.29 (Composition)**.** $\operatorname{Spec} k[x,y] \to \operatorname{Spec} k[x] \to \operatorname{Spec} k$ is a composite of finite-type morphisms, hence finite type.

Example 26.30 (Field extension)**.** For any extension K/k, base changing a finite-type k-scheme X gives the finite-type K-scheme X_K.

Example 26.31 (Finite type over an affine Noetherian base)**.** If A is Noetherian and $B = A[b_1,\ldots,b_n]$, then B is Noetherian; thus $\operatorname{Spec} B \to \operatorname{Spec} A$ preserves local Noetherianity.

Example 26.32 (Advanced: Hilbert basis mechanism)**.** Starting from a Noetherian ring A, repeated application of Hilbert's basis theorem shows $A[x_1,\ldots,x_n]$ Noetherian; quotients and localizations then remain Noetherian.

Example 26.33 (Advanced: Finite affine local-global Noetherianity)**.** If $D(f_1),\ldots,D(f_r)$ cover $\operatorname{Spec} A$ and every A_{f_i} is Noetherian, then A is Noetherian. Finite generation of an arbitrary ideal can be checked on this finite cover and descended.

Example 26.34 (Advanced: Transitivity of algebra generation)**.** If $B = A[b_1, \ldots, b_m]$ and $C = B[c_1, \ldots, c_n]$, then $C = A[b_1, \ldots, b_m, c_1, \ldots, c_n]$. This is the affine core of composition stability.

Example 26.35 (Advanced: Base change of generators)**.** If $B = A[b_1, \ldots, b_n]$, then $B \otimes_A A'$ is generated over A' by the elements $b_i \otimes 1$.

Example 26.36 (Advanced: Base change of quasi-compactness)**.** A quasi-compact morphism remains quasi-compact after base change. Combined with local finite type stability this proves finite type is stable under base change.

Example 26.37 (Advanced: Locally finite type over locally Noetherian)**.** On affine charts $A \to B$ with A Noetherian and B finitely generated, Hilbert's basis theorem makes B Noetherian. Hence the source is locally Noetherian.

Example 26.38 (Advanced: Finite type over Noetherian)**.** If the base is Noetherian and the morphism finite type, then the source is locally Noetherian and quasi-compact, hence Noetherian.

Example 26.39 (Advanced: Fiber finiteness)**.** For f locally finite type, $X_s \to \operatorname{Spec} \kappa(s)$ is locally finite type because it is a base change of f. If f is finite type, the fiber is finite type.

Example 26.40 (Advanced: Why the infinite disjoint union fails)**.** Every point component is an affine finite-type neighborhood, but no finite subcollection covers the source. The missing hypothesis is exactly quasi-compactness.

Example 26.41 (Advanced: Why infinitely many variables fail locally)**.** At the generic point of $k[x_1, x_2, \ldots]$, any basic neighborhood inverts one polynomial involving only finitely many variables; infinitely many untouched variables remain algebraically independent.

Example 26.42 (Advanced: Finite type does not mean finitely many relations)**.** The quotient $A[x_1, \ldots, x_n]/I$ is finite type for arbitrary I. Finite generation of I is a finite-presentation issue.

Example 26.43 (Advanced: Localization preserves finite generation)**.** If B is finite type over A, then localizing at an element from A or B yields a finite-type algebra over the corresponding localized base.

Example 26.44 (Advanced: Affine-open criterion)**.** The AG6-P5 refinement argument converts one target affine cover into a statement over every affine open by simultaneous distinguished shrinking.

Example 26.45 (Advanced: Quasi-compact source over an affine target)**.** If $X \to \operatorname{Spec} A$ is locally finite type and X is quasi-compact, finitely many finite-type affine source charts cover X, so the morphism is finite type.

26.13 Problem dossiers

Problem 26.46 (Legacy AG5-P4: every affine open of a locally Noetherian scheme is Noetherian)**.** Let X be a scheme. Prove the legacy theorem that X is locally Noetherian if and only if every affine open $U = \operatorname{Spec} A \subseteq X$ has A Noetherian.

Provenance. Recovered numbered legacy Problem AG5-P4; previously routed by VI-SCOPE-041 but absent from the ledger.

Problem 26.47 (Legacy AG6-P5 with AG5-P5 merged: locally finite type on every affine open)**.** Prove that $f : X \to Y$ is locally of finite type if and only if for every affine open $V = \operatorname{Spec} B \subseteq Y$, the inverse image $f^{-1}(V)$ has an affine open cover $\operatorname{Spec} A_j$ with each A_j finitely generated over B.

Provenance. Legacy AG6-P5; AG5-P5 is a duplicate/variant merged here, not a second canonical problem.

Problem 26.48 (Legacy AG6-P6e: the three finiteness counterexamples)**.** Over a field k, verify: (a) $\operatorname{Spec} k[x_1, x_2, \ldots]$ is not Noetherian; (b) its structure morphism to $\operatorname{Spec} k$ is not locally of finite type; (c) $\coprod_{n\geq 1} \operatorname{Spec} k \to \operatorname{Spec} k$ is locally of finite type but not of finite type.

Provenance. Preserves the finite-type/Noetherian subparts grouped in legacy row AG6-P6e.

Problem 26.49 (Finite type is locally finite type plus quasi-compact)**.** Prove Theorem 26.6 from the affine-cover definition of finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.50 (Affine criterion for finite type)**.** Let $A \to B$ be a ring map. Show that $\operatorname{Spec} B \to \operatorname{Spec} A$ is finite type if B is finitely generated as an A-algebra.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.51 (Restriction to open subschemes)**.** Show that restricting a locally finite type morphism to an open subscheme of the target remains locally finite type. Show the analogous statement for an open restriction of the source.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.52 (Composition)**.** Prove that composites of locally finite type morphisms are locally finite type, and composites of finite-type morphisms are finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.53 (Base change of locally finite type)**.** Prove that a base change of a locally finite type morphism is locally finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.54 (Base change of finite type)**.** Prove that a base change of a finite-type morphism is finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.55 (Affine space is finite type)**.** Show that $\mathbb{A}^n_S \to S$ is finite type for every scheme S.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.56 (Closed subschemes of affine space)**.** Let $Z \hookrightarrow \mathbb{A}^n_S$ be a closed subscheme. Show $Z \to S$ is finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.57 (Finite-type schemes over a field)**.** Let X be quasi-compact and locally of finite type over a field k. Show that $X \to \operatorname{Spec} k$ is finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.58 (Locally finite type over locally Noetherian)**.** Prove the first assertion of Theorem 26.10.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.59 (Finite type over Noetherian)**.** If Y is Noetherian and $X \to Y$ is finite type, prove X Noetherian.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.60 (Fibers inherit finite type)**.** Let $f : X \to S$ be locally finite type. Prove that every fiber X_s is locally finite type over $\kappa(s)$; if f is finite type, prove X_s finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.61 (The quasi-compactness obstruction)**.** Explain precisely why $\coprod_{n\geq 1} \operatorname{Spec} k \to \operatorname{Spec} k$ fails to be finite type even though every component is finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.62 (Localization is finite type)**.** Show $A \to A_f$ is a finite-type ring map and interpret geometrically.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.63 (Finite type is local on the target)**.** Suppose $Y = \bigcup_i V_i$ is an open cover and each restricted morphism $f^{-1}(V_i) \to V_i$ is finite type. Show f is finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.64 (Quasi-compact source over an affine target)**.** Let $X \to \operatorname{Spec} A$ be locally finite type and assume X is quasi-compact. Show the morphism is finite type.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

Problem 26.65 (Noetherian schemes have quasi-compact opens)**.** Show that every open subset of a Noetherian topological space is quasi-compact, and explain the scheme-theoretic consequence.

Provenance. Canonical VI/26 dossier developed from the audited AG5/AG6 finiteness theory streams; no additional numbered legacy problem is claimed.

26.14 Exercises

Exercise 26.66. Show that a quotient of a finitely generated A-algebra is finitely generated over A.

Exercise 26.67. Show A_f is a finite-type A-algebra.

Exercise 26.68. If B is finite type over A and C finite type over B, show C finite type over A.

Exercise 26.69. Show $A[x_1, \ldots, x_n]/I$ is finite type over A for arbitrary I.

Exercise 26.70. Show that finite type need not imply finite presentation over a non-Noetherian base.

Exercise 26.71. Prove a finite-type morphism is quasi-compact.

Exercise 26.72. Show an affine morphism locally of finite type is finite type.

Exercise 26.73. Show $\mathbb{A}^n_S \to S$ remains finite type after any base change.

Exercise 26.74. Let A be Noetherian. Show every localization $S^{-1}A$ is Noetherian.

Exercise 26.75. Let A be Noetherian. Show every quotient A/I is Noetherian.

Exercise 26.76. Use Hilbert's basis theorem to show $A[x_1, \ldots, x_n]$ Noetherian when A is.

Exercise 26.77. Show a finite-type algebra over a Noetherian ring is Noetherian.

Exercise 26.78. Show the ascending chain $(x_1) \subset (x_1, x_2) \subset \cdots$ in $k[x_1, x_2, \ldots]$ is strict.

Exercise 26.79. Explain why the infinite disjoint union of points is not quasi-compact.

Exercise 26.80. Show a finite disjoint union of finite-type k-schemes is finite type over k.

Exercise 26.81. Show a fiber of a finite-type morphism is finite type over its residue field.

Exercise 26.82. Show local finite type is preserved by composition.

Exercise 26.83. Show finite type is preserved by composition.

Exercise 26.84. Show local finite type is preserved by base change.

Exercise 26.85. Show finite type is preserved by base change.

Exercise 26.86. If X is locally Noetherian, show every open subscheme is locally Noetherian.

Exercise 26.87. If X is Noetherian, show every open subscheme is quasi-compact.

Exercise 26.88. Show that a finite-type scheme over a field is Noetherian.

Exercise 26.89. Give a locally finite type but non-finite-type morphism different in notation from the chapter example.

Exercise 26.90. Give a finite-type morphism that is not finite as a morphism.

Exercise 26.91. Explain why finite type is a statement about generators, not about the number of points in a fiber.

26.15 Challenges

Problem 26.92 (Challenge 1: Local-global Noetherian descent)**.** Give a complete proof that if $D(f_1), \ldots, D(f_r)$ cover $\operatorname{Spec} A$ and each A_{f_i} is Noetherian, then A is Noetherian. Make the annihilation step for I/J explicit.

Problem 26.93 (Challenge 2: Finite type versus finite presentation)**.** Construct an explicit non-Noetherian ring A and infinitely generated ideal $I \subset A[x]$ for which $A[x]/I$ is finite type but not finitely presented over A. Prove both assertions.

Problem 26.94 (Challenge 3: Finiteness through a Cartesian square)**.** In a Cartesian square, assume the right vertical morphism is finite type. Prove the left vertical morphism finite type directly from affine tensor products, then compare your proof with the abstract base-change theorem.

Problem 26.95 (Challenge 4: Noetherian family over arithmetic base)**.** Let $X = \operatorname{Spec} \mathbb{Z}[x, y]/(y^2 - x^3 - x)$. Prove $X \to \operatorname{Spec} \mathbb{Z}$ finite type and X Noetherian. Describe why every fiber is finite type without analyzing its factorization.

Problem 26.96 (Challenge 5: Detect the exact missing hypothesis)**.** Classify which of the implications among locally finite type, finite type, quasi-compact, locally Noetherian source, and Noetherian source require hypotheses on the base. For each false converse, give a counterexample.

26.16 Chapter synthesis

Finite type is the condition that a morphism uses finitely many algebraic coordinates locally on its target; quasi-compactness upgrades that local statement to a finite global description over each affine base chart. Noetherianity is the companion stabilization condition on ideals. Together they are robust under the operations already developed in this volume – restriction, composition, base change, and passage to fibers – and they form the finiteness framework for the scheme theory that follows.

$$\begin{tikzcd}
\text{VI/23 fiber products} \arrow[r] & \text{VI/24 base change} \arrow[r] & \text{VI/25 fibers} \arrow[dl] \\
\text{VI/27 integral/function fields} & \text{VI/26 finiteness} \arrow[l] \arrow[r] & \text{VI/28 normalization}
\end{tikzcd}$$

Figure 26.10: VI/26 in the Morphisms and Families arc.

26.17 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 26.97 (Affine space)**.** Is $\mathbb{A}^5_A \to \operatorname{Spec} A$ finite type?

Hint. Use finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Yes. ☐

Exercise 26.98 (One equation)**.** Is $A[x, y]/(xy - 1)$ finite type over A?

Hint. Use finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Yes. ☐

Exercise 26.99 (Finite presentation)**.** When is $A[x_1, \dots, x_n]/I$ finitely presented?

Hint. Use finite type, finite presentation, and Noetherian finiteness. ☐

Solution. When I is finitely generated. ☐

Exercise 26.100 (Noetherian algebra)**.** If A is Noetherian and B finite type over A, is B Noetherian?

Hint. Use finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Yes. ☐

Exercise 26.101 (Infinite polynomial)**.** Is $A[x_1, x_2, \dots]$ finite type over A?

Hint. Use finite type, finite presentation, and Noetherian finiteness. ☐

Solution. No. ☐

Proofs

Exercise 26.102 (Composition finite type)**.** Prove: Prove composition of finite-type morphisms is finite type.

Hint. Work directly from finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Affinely, a finitely generated algebra over a finitely generated algebra is finitely generated over the base. ☐

Exercise 26.103 (Base change finite type)**.** Prove: Prove finite type is preserved by base change.

Hint. Work directly from finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Tensor a finite generating set with the new base ring. ☐

Exercise 26.104 (Closed subscheme)**.** Prove: Prove a closed subscheme of a finite-type scheme over a base is finite type.

Hint. Work directly from finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Affinely, quotient a finitely generated algebra. ☐

Exercise 26.105 (Noetherian finite type)**.** Prove: Prove a finite-type algebra over a Noetherian ring is Noetherian.

Hint. Work directly from finite type, finite presentation, and Noetherian finiteness. ☐

Solution. Apply Hilbert's basis theorem and quotient stability. ☐

Counterexamples and hypothesis tests

Exercise 26.106 (Infinite variables)**.** Test the claim: a polynomial algebra in infinitely many variables is finite type.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 26.107 (Noetherian base unnecessary)**.** Test the claim: every finite-type algebra over every ring is Noetherian.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; the base itself may be non-Noetherian. ☐

Exercise 26.108 (Finite type finite module)**.** Test the claim: finite type as an algebra means finite as a module.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; polynomial rings are finite type but generally infinite as modules. ☐

Applications and investigations

Exercise 26.109 (Algebraic families)**.** Why is finite type the standard algebraicity condition for morphisms?

Hint. Translate the question into finite type, finite presentation, and Noetherian finiteness. ☐

Solution. It ensures only finitely many algebraic generators are needed locally. ☐

Exercise 26.110 (Algorithmic finiteness)**.** What does Noetherianity provide?

Hint. Translate the question into finite type, finite presentation, and Noetherian finiteness. ☐

Solution. It forces ideals and submodules to admit finite generating data. ☐

Challenge problems

Exercise 26.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Finite-Type and Noetherian Morphisms in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Finite-Type and Noetherian Morphisms, translates it through finite type, finite presentation, and Noetherian finiteness, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 26.112 (Hypothesis challenge)**.** Choose one structural statement from Finite-Type and Noetherian Morphisms, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ∎

Chapter 27

Integral Schemes and Function Fields

The finiteness conditions of VI/26 tell us when a scheme is built from finitely many algebraic coordinates. We now isolate the class on which one can speak unambiguously about a single field of rational functions. An integral scheme is simultaneously reduced and irreducible. Its unique generic point, constructed in VI/09, carries a field as its local ring; that field is the function field of the entire scheme.

27.1 Integral schemes

Definition 27.1 (Integral scheme)**.** A scheme X is *integral* if it is nonempty, reduced, and irreducible. Equivalently, every nonempty affine open $U = \operatorname{Spec} A \subseteq X$ has A an integral domain.

The nonempty qualifier is important when one phrases the condition using rings of sections: the zero ring on the empty open is not treated as an integral domain.

Theorem 27.2 (Reduced plus irreducible)**.** *For a nonempty scheme X, the following are equivalent:*

(i) X is integral;

(ii) X is reduced and irreducible;

(iii) every nonempty affine open $\operatorname{Spec} A \subseteq X$ *has A a domain;*

(iv) for every nonempty open $U \subseteq X$, the ring $\mathcal{O}_X(U)$ is a domain.

$$X \text{ integral} \Longrightarrow X \text{ reduced}$$

$$X \text{ integral} \Longrightarrow X \text{ irreducible}$$

Figure 27.1: Integrality combines reducedness with irreducibility.

27.2 The generic point revisited

An integral scheme is irreducible, so by VI/09 it has a unique generic point η with

$$\overline{\{\eta\}} = X.$$

Every nonempty open subset contains η. Consequently every nonempty open subset of an integral scheme is again integral and dense.

Proposition 27.3 (Generic point on affine charts)**.** *If X is integral and $U = \operatorname{Spec} A \subseteq X$ is nonempty affine, then A is a domain and the generic point η corresponds to the prime ideal $(0) \subset A$.*

$$\begin{tikzcd} & \eta \arrow[dl, hook] \arrow[dr, hook] & \\ U = \operatorname{Spec} A \arrow[rr, hook] & & X \end{tikzcd}$$

Figure 27.2: Every nonempty open of an integral scheme contains the generic point; on U, $\eta = (0)$.

Example 27.4 (Affine integral scheme)**.** For a domain A, $X = \operatorname{Spec} A$ is irreducible and reduced, hence integral. Its generic point is (0) and its function field is $\operatorname{Frac} A$.

27.3 The function field

Definition 27.5 (Function field)**.** Let X be integral with generic point η. Its *function field* is

$$K(X) := \mathcal{O}_{X,\eta}.$$

Because η is the zero prime on every nonempty affine chart $U = \operatorname{Spec} A$, localization gives

$$K(X) \cong A_{(0)} = \operatorname{Frac}(A).$$

Thus the field is independent of the affine chart used to compute it.

Theorem 27.6 (Affine computation of $K(X)$)**.** *For every nonempty affine open $U = \operatorname{Spec} A \subseteq X$ of an integral scheme,*

$$K(X) \cong \operatorname{Frac}(A).$$

If $V = \operatorname{Spec} B$ is another nonempty affine open, then $\operatorname{Frac}(A)$ and $\operatorname{Frac}(B)$ are canonically identified through $\mathcal{O}_{X,\eta}$.

$$\begin{tikzcd} A \arrow[r, hook] \arrow[dr] & \operatorname{Frac}(A) \arrow[d, "\sim"] \\ & \mathcal{O}_{X,\eta} = K(X) \end{tikzcd}$$

Figure 27.3: The function field is the fraction field of every nonempty affine coordinate ring.

27.4 Rational functions as germs on dense opens

A rational function may be represented by a pair (U, s) where $U \subseteq X$ is nonempty open and $s \in \mathcal{O}_X(U)$. Two representatives (U, s) and (V, t) are equivalent if they agree on some nonempty open subset of $U \cap V$. On an integral scheme this equivalence class is exactly a germ at the generic point and hence an element of $K(X)$.

Proposition 27.7 (Sections embed in the function field)**.** *For every nonempty open $U \subseteq X$ of an integral scheme, restriction to the generic stalk induces an injective ring map*

$$\mathcal{O}_X(U) \hookrightarrow K(X).$$

Hence two regular functions on U that agree on a nonempty open subset are equal on all of U.

$$\begin{tikzcd} \mathcal{O}_X(U) \arrow[r, hook] \arrow[dr] & K(X) \arrow[d, equal] \\ & \mathcal{O}_{X,\eta} \end{tikzcd}$$

Figure 27.4: Regular sections on a nonempty open embed in the generic stalk.

27.5 Regularity of a rational function at a point

After identifying $\mathcal{O}_{X,x}$ with a subring of $K(X)$, a rational function $r \in K(X)$ is *regular at x* when $r \in \mathcal{O}_{X,x}$. On an affine chart $U = \operatorname{Spec} A$ with $x \leftrightarrow \mathfrak{p}$, writing $r = a/b \in \operatorname{Frac}(A)$, this means that r admits a representation with denominator not in $\mathfrak{p}$.

Proposition 27.8 (Regular locus)**.** *For $r \in K(X)$, the set of points at which r is regular is open and contains the generic point. In particular, every rational function is represented by an honest regular function on some dense open subset.*

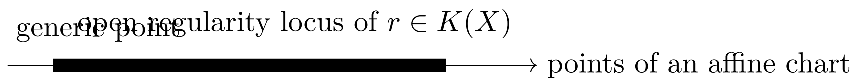


Figure 27.5: A rational function is regular on a dense open neighborhood of the generic point.

Example 27.9 (Affine line function field)**.** The function field of $\mathbb{A}^1_k$ is $k(t)$. A rational function such as $(t^2+1)/(t-1)$ is regular away from the pole $t = 1$.

27.6 Dominant morphisms and function fields

Let $f : X \to Y$ be a morphism of integral schemes. If f is dominant, then $f(\eta_X) = \eta_Y$. Passing to the local maps at generic points yields a homomorphism of fields

$$f^* : K(Y) \longrightarrow K(X).$$

Every unital map of fields is injective.

Theorem 27.10 (Contravariance under dominance)**.** *A dominant morphism $f : X \to Y$ of integral schemes induces a canonical field embedding*

$$f^* : K(Y) \hookrightarrow K(X).$$

Composition of dominant morphisms corresponds to composition of the induced embeddings in the opposite direction.

$$
\begin{tikzcd}
X \arrow[r, two heads, "f"] & Y
\end{tikzcd}
$$

$$
\begin{tikzcd}
K(X) & K(Y) \arrow[l, hook', "f^*"']
\end{tikzcd}
$$

Figure 27.6: Dominant geometry produces an embedding of function fields in the opposite direction.

27.7 Standard function fields

For affine n-space over a field k,

$$K(\mathbb{A}^n_k) = k(t_1, \ldots, t_n).$$

For an irreducible affine hypersurface $X = \operatorname{Spec} k[x_1, \ldots, x_n]/(F)$ with (F) prime, the function field is the fraction field of its coordinate ring. For the parabola $y = x^2$, this fraction field is $k(x)$ even though the chosen affine presentation uses two coordinates.

$$
\begin{tikzcd}
k[x,y]/(y-x^2) \arrow[r, hook] \arrow[d, "y\mapsto x^2"'] & K(X) \\
k[x] \arrow[ur, hook]
\end{tikzcd}
$$

Figure 27.7: The parabola has function field $k(x)$.

27.8 Finite-type integral schemes and transcendence degree

Suppose X is integral and of finite type over k. If $U = \operatorname{Spec} A \subseteq X$ is nonempty affine, then A is a finitely generated k-domain and

$$K(X) = \operatorname{Frac}(A).$$

Therefore $\operatorname{trdeg}_k K(X)$ is a chart-independent invariant. The AG8-P2c legacy slice uses exactly this invariant. The deeper theorem

$$\dim X = \operatorname{trdeg}_k K(X)$$

belongs to the dimension arc and is proved canonically in VI/30; VI/27 records the function-field side without consuming that later proof.

$$A \text{ finitely generated } k\text{-domain} \longrightarrow \operatorname{Frac}(A) = K(X) \longrightarrow \operatorname{trdeg}_k K(X)$$

Figure 27.8: The transcendence degree of the function field is independent of the chosen affine chart.

Example 27.11 (Plane curve function field)**.** For an irreducible affine curve $C = V(f) \subset \mathbb{A}^2_k$, the coordinate ring $k[x,y]/(f)$ is a domain and the function field is its fraction field.

27.9 What integrality does and does not say

Integrality is global: having all local rings domains does not by itself force irreducibility. For example, $\operatorname{Spec} k \amalg \operatorname{Spec} k$ has domain local rings at every point but is not irreducible. Conversely, an irreducible scheme may fail to be integral if it is nonreduced, as with $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$.

The affine scheme

$$\operatorname{Spec} k[x, y]/(xy)$$

is reduced but reducible, hence not integral. This is the exact counterexample reserved from AG6-P6c.

$$\operatorname{Spec} k[x, y]/(xy) \longrightarrow \{x = 0\} \cup \{y = 0\}$$

$$\text{reduced} \longrightarrow \text{reducible} \longrightarrow \text{not integral}$$

Figure 27.9: The AG6-P6c counterexample: zero divisors encode two components.

27.10 Boundary with normalization and dimension

VI/27 stops at the field of rational functions and its basic functoriality. VI/28 will enlarge an integral scheme inside its own function field by integral closure, producing the normalization. VI/29–VI/32 then study chains of prime ideals, dimension, codimension, and local geometry. Rational maps and birational geometry will use the language developed here later in the volume.

27.11 Examples

Example 27.12 (Affine line)**.** For $X = \operatorname{Spec} k[t]$, the zero ideal is prime, X is integral, and $K(X) = k(t)$.

Example 27.13 (Affine space)**.** $\mathbb{A}^n_k = \operatorname{Spec} k[t_1, \ldots, t_n]$ is integral and has $K(X) = k(t_1, \ldots, t_n)$.

Example 27.14 (Parabola)**.** $k[x, y]/(y - x^2) \cong k[x]$, so the parabola is integral and its function field is $k(x)$.

Example 27.15 (Irreducible hypersurface)**.** If $F \in k[x_1, \ldots, x_n]$ generates a prime ideal, then $\operatorname{Spec} k[x_1, \ldots, x_n]/(F)$ is integral.

Example 27.16 (Two axes)**.** $\operatorname{Spec} k[x, y]/(xy)$ is reduced when k is a field, but it is reducible and therefore not integral.

Example 27.17 (Doubled point)**.** $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ is irreducible but nonreduced, hence not integral.

Example 27.18 (Disjoint integral pieces)**.** $\operatorname{Spec} k \amalg \operatorname{Spec} k$ has field-valued local rings but is reducible; local domains alone do not imply integrality.

Example 27.19 (A dense principal open)**.** For integral $X = \operatorname{Spec} A$ and $0 \neq f \in A$, $D(f)$ is nonempty, dense, integral, and has the same function field $\operatorname{Frac}(A)$.

Example 27.20 (A rational function)**.** On $\mathbb{A}^1_k$, $1/t \in k(t)$ is regular exactly on $D(t)$, not at the closed point (t).

Example 27.21 (Agreement on a dense open)**.** If $f, g \in A$ for a domain A and their restrictions agree on some nonempty $D(h)$, then $h^N(f-g) = 0$ for some N; hence $f = g$.

Example 27.22 (Localization does not change fractions)**.** For a domain A and $0 \neq f \in A$, $\mathrm{Frac}(A_f) = \mathrm{Frac}(A)$ canonically.

Example 27.23 (Generic stalk)**.** At the generic point (0) of $\operatorname{Spec} A$ for a domain A, the local ring is $A_{(0)} = \mathrm{Frac}(A)$.

Example 27.24 (Dominant polynomial map)**.** The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$, $t \mapsto t^m$, is dominant and induces $k(u) \hookrightarrow k(t)$ by $u \mapsto t^m$.

Example 27.25 (Closed immersion is usually not dominant)**.** A proper closed subscheme of an integral scheme does not contain the generic point, so its inclusion is not dominant.

Example 27.26 (Function field after shrinking)**.** Removing finitely many closed points from an integral curve leaves the function field unchanged.

Example 27.27 (Integral finite-type curve)**.** If $A = k[x, y]/(F)$ is a finitely generated domain, then $K(\operatorname{Spec} A) = \mathrm{Frac}(A)$ and its transcendence degree over k is chart-independent.

Example 27.28 (Finite extension)**.** For $X = \operatorname{Spec} k[t]$ and $Y = \operatorname{Spec} k[t^m]$, the dominant finite map gives $K(Y) = k(t^m) \subset k(t) = K(X)$.

Example 27.29 (Generic point belongs to every nonempty open)**.** If a nonempty open U missed η, then its complement would be a proper closed set containing η, contradicting density of η.

Example 27.30 (Irreducibility of a nonempty open)**.** A nonempty open subset of an irreducible space is irreducible; together with reducedness this makes every nonempty open subscheme of an integral scheme integral.

Example 27.31 (Same rational function, different formulas)**.** On overlapping affine charts, two fractions represent the same element of $K(X)$ precisely when their germs agree at the generic point.

Example 27.32 (Advanced 1: Cusp)**.** For $A = k[t^2, t^3] \subset k[t]$, A is a domain and $K(\operatorname{Spec} A) = k(t)$. The fact that A is not integrally closed belongs to VI/28.

Example 27.33 (Advanced 2: Node as an integral curve)**.** If $y^2 - x^2(x+1)$ is irreducible over k, the coordinate ring of the nodal affine curve is a domain even though the curve is singular.

Example 27.34 (Advanced 3: Field extension as function-field extension)**.** A dominant morphism of integral curves can induce a nontrivial finite field extension $K(Y) \subset K(X)$; degree becomes a birational invariant under additional hypotheses.

Example 27.35 (Advanced 4: Purely transcendental field)**.** The function field of $\mathbb{A}^n_k$ has transcendence basis $t_1, \ldots, t_n$ over k.

Example 27.36 (Advanced 5: Generic regularity)**.** Given $a/b \in \mathrm{Frac}(A)$ with A a domain and $b \neq 0$, the function is regular on $D(b)$, a nonempty dense open.

Example 27.37 (Advanced 6: Intersection of local rings)**.** For a domain A, every $A_{\mathfrak{p}}$ embeds in $\mathrm{Frac}(A)$. Their intersections inside the fraction field encode where a rational function is simultaneously regular.

Example 27.38 (Advanced 7: Dominance from injectivity in the affine case)**.** For a map $A \to B$ of domains, $\operatorname{Spec} B \to \operatorname{Spec} A$ is dominant iff the kernel is zero; hence dominance is equivalent to an injection $A \hookrightarrow B$.

Example 27.39 (Advanced 8: Generic point under dominance)**.** If $f : X \to Y$ is dominant between integral schemes, continuity gives $\overline{\{f(\eta_X)\}} = Y$, so $f(\eta_X) = \eta_Y$.

Example 27.40 (Advanced 9: Non-dominant map and absent field map)**.** A morphism landing in a proper closed subset need not yield $K(Y) \to K(X)$ because the generic point of Y is not hit.

Example 27.41 (Advanced 10: Integral scheme over a field)**.** If X is integral of finite type over k, the structure map gives an embedding $k \hookrightarrow K(X)$, making $K(X)$ a finitely generated field extension of k.

Example 27.42 (Advanced 11: Transcendence degree survives affine shrinking)**.** Since A_f and A have the same fraction field for $f \neq 0$, their associated function field has the same transcendence degree.

Example 27.43 (Advanced 12: Reduced but not geometrically integral)**.** An integral k-scheme can become reducible after a field extension. Geometric integrality is a stronger base-change condition and is not assumed here.

Example 27.44 (Advanced 13: Integral does not mean normal)**.** $k[t^2, t^3]$ is a domain, so its spectrum is integral, but it is not normal. Integrality is therefore strictly weaker than normality.

Example 27.45 (Advanced 14: Rational functions form a field)**.** Nonzero germs at the generic point are invertible because $\mathcal{O}_{X,\eta}$ is the localization of a domain at (0).

27.12 Problem dossiers

Problem 27.46 (Legacy AG5-P3: integral iff reduced and irreducible)**.** Let X be a nonempty scheme. Prove the legacy theorem that X is integral if and only if it is reduced and irreducible. Reconcile the source formulation using domains of sections on nonempty opens with the affine-local formulation.

Provenance. Numbered legacy AG5-P3; full theorem preserved.

Problem 27.47 (Legacy AG6-P6c: a scheme that is not integral)**.** Give an explicit scheme that is not integral and explain exactly which part of integrality fails.

Provenance. AG6 Problem 6, integral-scheme subpart; other subparts remain in their assigned chapters.

Problem 27.48 (Legacy AG6-P7b: function field and generic stalk)**.** Let X be integral with generic point η. Prove that $\mathcal{O}_{X,\eta}$ is a field and that for every nonempty affine open $U = \operatorname{Spec} A \subseteq X$ one has $\mathcal{O}_{X,\eta} \cong \operatorname{Frac}(A)$.

Provenance. Function-field half of split legacy AG6-P7; the generic-point half was already merged into VI/09.

Problem 27.49 (Legacy AG8-P2c: the function-field slice of the variety dimension problem)**.** Let X be an integral finite-type k-scheme and $U = \operatorname{Spec} A \subseteq X$ nonempty affine. Show that $K(X) = \operatorname{Frac}(A)$ is a finitely generated field extension of k and that $\operatorname{trdeg}_k K(X)$ is independent of U. State how this interfaces with the later equality $\dim X = \operatorname{trdeg}_k K(X)$.

Provenance. Function-field/transcendence-degree slice of split legacy AG8-P2; dimension and codimension slices remain reserved for VI/30–VI/32.

Problem 27.50 (Affine criterion for integrality)**.** Prove that a nonempty scheme is integral iff it admits an affine open cover by spectra of domains and is irreducible.

Provenance. Canonical VI/27 dossier.

Problem 27.51 (Nonempty opens have the same function field)**.** If $U \subseteq X$ is nonempty open in an integral scheme, prove U integral and $K(U) = K(X)$.

Provenance. Canonical VI/27 dossier.

Problem 27.52 (Sections inject into $K(X)$)**.** Prove that $\mathcal{O}_X(U) \to K(X)$ is injective for every nonempty open U of an integral scheme.

Provenance. Canonical VI/27 dossier.

Problem 27.53 (Equality on a dense open)**.** Let $s, t \in \mathcal{O}_X(U)$ on a nonempty open of an integral scheme. Show that if they agree on some nonempty open $V \subseteq U$, then $s = t$.

Provenance. Canonical VI/27 dossier.

Problem 27.54 (Regular locus of a rational function)**.** For $r \in K(X)$, prove that the points where r is regular form a nonempty open subset.

Provenance. Canonical VI/27 dossier.

Problem 27.55 (Local rings inside the function field)**.** For $x \in X$ integral, construct the canonical injection $\mathcal{O}_{X,x} \hookrightarrow K(X)$.

Provenance. Canonical VI/27 dossier.

Problem 27.56 (Dominance sends generic to generic)**.** Let $f : X \to Y$ be dominant between integral schemes. Show $f(\eta_X) = \eta_Y$.

Provenance. Canonical VI/27 dossier.

Problem 27.57 (Dominant morphism gives field embedding)**.** Deduce that a dominant morphism $f : X \to Y$ of integral schemes induces $K(Y) \hookrightarrow K(X)$.

Provenance. Canonical VI/27 dossier.

Problem 27.58 (Affine dominance criterion)**.** For a homomorphism of domains $A \to B$, show $\operatorname{Spec} B \to \operatorname{Spec} A$ is dominant iff $A \to B$ is injective.

Provenance. Canonical VI/27 dossier.

Problem 27.59 (Function field of affine space)**.** Compute $K(\mathbb{A}^n_k)$.

Provenance. Canonical VI/27 dossier.

Problem 27.60 (Function field of an irreducible hypersurface)**.** Let $F \in k[x_1, \dots, x_n]$ generate a prime ideal. Compute the function field of $X = V(F)$.

Provenance. Canonical VI/27 dossier.

Problem 27.61 (A local-domain counterexample to global integrality)**.** Show that all local rings of $\operatorname{Spec} k \amalg \operatorname{Spec} k$ are domains although the scheme is not integral.

Provenance. Canonical VI/27 dossier.

Problem 27.62 (Shrinking preserves transcendence degree)**.** Let X be integral finite type over k and $U \subseteq X$ nonempty open. Show $\operatorname{trdeg}_k K(U) = \operatorname{trdeg}_k K(X)$.

Provenance. Canonical VI/27 dossier.

Problem 27.63 (The field map from a power map)**.** For $f : \mathbb{A}^1_k \to \mathbb{A}^1_k$ induced by $k[u] \to k[t]$, $u \mapsto t^m$, compute the induced map on function fields.

Provenance. Canonical VI/27 dossier.

Problem 27.64 (Rational function regularity in affine coordinates)**.** Let A be a domain, $x = \mathfrak{p} \in \operatorname{Spec} A$, and $r \in \operatorname{Frac}(A)$. Prove r is regular at x iff $r = a/b$ for some $a, b \in A$ with $b \notin \mathfrak{p}$.

Provenance. Canonical VI/27 dossier.

Problem 27.65 (Why normalization needs $K(X)$)**.** Explain why the normalization of an integral affine scheme naturally lives inside its function field.

Provenance. Canonical bridge to VI/28.

27.13 Exercises

Exercise 27.66. Show that every nonempty open subset of an integral scheme is dense.

Exercise 27.67. Prove that every nonempty open subscheme of an integral scheme is integral.

Exercise 27.68. For a domain A, identify the generic point of $\operatorname{Spec} A$.

Exercise 27.69. Compute the local ring at the generic point of $\operatorname{Spec} k[x, y]$.

Exercise 27.70. Compute $K(\mathbb{A}^2_k)$.

Exercise 27.71. Compute the function field of $V(y - x^3) \subset \mathbb{A}^2_k$.

Exercise 27.72. Show that $\operatorname{Frac}(A_f) = \operatorname{Frac}(A)$ for a domain A and $0 \neq f \in A$.

Exercise 27.73. Show that $\operatorname{Spec} k[x, y]/(xy)$ is reduced but not integral.

Exercise 27.74. Show that $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ is irreducible but not integral.

Exercise 27.75. Show that a disjoint union of two integral schemes need not be integral.

Exercise 27.76. Prove that $\mathcal{O}_X(U) \to K(X)$ is injective for nonempty open U in integral X.

Exercise 27.77. If two sections on a nonempty open agree on a nonempty subopen, prove they agree everywhere.

Exercise 27.78. For $r = a/b \in \operatorname{Frac}(A)$, identify an open subset on which r is regular.

Exercise 27.79. Prove that the regular locus of a rational function is open.

Exercise 27.80. Embed $A_{\mathfrak{p}}$ into $\operatorname{Frac}(A)$ for a domain A.

Exercise 27.81. Show that a dominant morphism of integral schemes sends generic point to generic point.

Exercise 27.82. Show that a dominant morphism induces an injection of function fields.

Exercise 27.83. For domains $A \to B$, prove the affine dominance criterion using the kernel.

Exercise 27.84. Compute the field map induced by $u \mapsto t^3$ from $\mathbb{A}^1$ to itself.

Exercise 27.85. Show that a proper closed immersion into an integral scheme is not dominant.

Exercise 27.86. Show that $k[x, y]/(y - x^2)$ has function field $k(x)$.

Exercise 27.87. Show that $k[t^2, t^3]$ and $k[t]$ have the same fraction field.

Exercise 27.88. For an integral finite-type k-scheme, show that $K(X)$ is finitely generated as a field extension of k.

Exercise 27.89. Show that transcendence degree of $K(X)$ is unchanged by replacing X by a nonempty open subscheme.

Exercise 27.90. Explain why all nonzero rational functions are invertible in $K(X)$.

Exercise 27.91. Give an integral scheme that is not normal and identify its function field.

27.14 Challenges

Problem 27.92 (Challenge 1: Reconstruct a function field from an affine atlas)**.** Let X be integral with a finite affine cover $U_i = \operatorname{Spec} A_i$. Starting only from the rings and restriction maps on overlaps, construct canonical isomorphisms $\operatorname{Frac}(A_i) \cong \operatorname{Frac}(A_j)$ satisfying the cocycle condition and recover $K(X)$.

Problem 27.93 (Challenge 2: Regular functions as an intersection)**.** For an integral affine scheme $X = \operatorname{Spec} A$, investigate the identity $A = \bigcap_{\mathfrak{m}\in \operatorname{Max} A} A_{\mathfrak{m}}$ inside $\operatorname{Frac}(A)$ and interpret it geometrically.

Problem 27.94 (Challenge 3: Dominant maps and algebraic dependence)**.** For dominant $f : X \to Y$ between integral finite-type k-schemes, relate algebraic dependence in $K(Y)$ to its image in $K(X)$ and compare transcendence degrees.

Problem 27.95 (Challenge 4: Geometric integrality warning)**.** Find an integral k-scheme that becomes reducible after a field extension, and explain why the ordinary function field does not by itself encode geometric integrality.

Problem 27.96 (Challenge 5: From rational functions to normalization)**.** For $A = k[t^2, t^3]$, work entirely inside $K(X) = k(t)$: identify elements integral over A, locate the missing element t, and predict the affine normalization before VI/28 proves the general theorem.

27.15 Chapter synthesis

An integral scheme has one generic point and one ambient field of rational functions. Every nonempty affine chart is a domain, every nonempty open set is dense, and every regular section embeds into the same field $K(X)$. This single field is what makes later normalization and birational constructions possible: local coordinate rings can all be compared inside one common algebraic object.

$$\text{integral scheme } X \longrightarrow K(X) \longrightarrow \text{normalization } \widetilde{X} \longrightarrow \text{birational geometry}$$

Figure 27.10: VI/27 supplies the common field in which normalization and later birational constructions live.

27.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 27.97 (Affine line)**.** Find the function field of $\mathbb{A}^1_k$.

Hint. Use integral schemes, generic points, and rational functions. □

Solution. $k(t)$. □

Exercise 27.98 (Affine space)**.** Find the function field of $\mathbb{A}^n_k$.

Hint. Use integral schemes, generic points, and rational functions. □

Solution. $k(x_1, \ldots, x_n)$. ☐

Exercise 27.99 (Domain generic)**.** Identify the generic point of Spec A for a domain.

Hint. Use integral schemes, generic points, and rational functions. ☐

Solution. (0). ☐

Exercise 27.100 (Curve field)**.** What is the function field of irreducible $V(f)$?

Hint. Use integral schemes, generic points, and rational functions. ☐

Solution. $\mathrm{Frac}(k[x, y]/(f))$. ☐

Exercise 27.101 (Integral criterion)**.** What two conditions define an integral scheme?

Hint. Use integral schemes, generic points, and rational functions. ☐

Solution. Reduced and irreducible. ☐

Proofs

Exercise 27.102 (Affine criterion)**.** Prove: Prove Spec A is integral iff A is a domain.

Hint. Work directly from integral schemes, generic points, and rational functions. ☐

Solution. Reduced corresponds to no nilpotents and irreducible to the nilradical being prime; together this is the zero ideal prime. ☐

Exercise 27.103 (Function field local)**.** Prove: Prove the stalk at the generic point of an integral scheme is a field.

Hint. Work directly from integral schemes, generic points, and rational functions. ☐

Solution. On an affine domain chart it is localization at the zero prime, the fraction field. ☐

Exercise 27.104 (Affine consistency)**.** Prove: Prove different nonempty affine opens of an integral scheme have canonically isomorphic fraction fields.

Hint. Work directly from integral schemes, generic points, and rational functions. ☐

Solution. Both embed into the generic stalk and localize to the same field. ☐

Exercise 27.105 (Dense open)**.** Prove: Prove every nonempty open in an integral scheme contains the generic point.

Hint. Work directly from integral schemes, generic points, and rational functions. ☐

Solution. The generic point's closure is the whole irreducible space. ☐

Counterexamples and hypothesis tests

Exercise 27.106 (Irreducible enough)**.** Test the claim: irreducible implies integral.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; nilpotent thickenings can be irreducible but nonreduced. ☐

Exercise 27.107 (Reduced enough)**.** Test the claim: reduced implies integral.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; a union of components can be reduced and reducible. ☐

Exercise 27.108 (Function regular)**.** Test the claim: every function-field element is globally regular.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; rational functions may have poles. ☐

Applications and investigations

Exercise 27.109 (Birational geometry)**.** Why is the function field central to birational geometry?

Hint. Translate the question into integral schemes, generic points, and rational functions. ☐

Solution. Birational integral schemes have isomorphic function fields. ☐

Exercise 27.110 (Generic computation)**.** What does the function field capture?

Hint. Translate the question into integral schemes, generic points, and rational functions. ☐

Solution. It records rational functions defined on some dense open neighborhood of the generic point. ☐

Challenge problems

Exercise 27.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Integral Schemes and Function Fields in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Integral Schemes and Function Fields, translates it through integral schemes, generic points, and rational functions, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 27.112 (Hypothesis challenge)**.** Choose one structural statement from Integral Schemes and Function Fields, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 28

Normalization

VI/27 placed every nonempty affine chart of an integral scheme inside one common function field. Normalization uses that field to repair failures of integral closedness without changing the rational functions. On an affine integral scheme $X = \operatorname{Spec} A$, one replaces A by its integral closure $\overline{A}$ in $K(X) = \operatorname{Frac}(A)$. The compatibility of integral closure with localization makes these affine repairs glue.

28.1 Normal domains and normal schemes

Definition 28.1 (Normal domain)**.** A domain A with fraction field K is *normal* if it is integrally closed in K: every $z \in K$ integral over A already lies in A.

Definition 28.2 (Normal integral scheme)**.** An integral scheme X is *normal* if every local ring $\mathcal{O}_{X,x}$ is an integrally closed domain. Equivalently, every affine open $U = \operatorname{Spec} A \subseteq X$ has A a normal domain.

Normality is thus strictly stronger than integrality. The cusp is integral but not normal; affine space over a field is normal.

$$\text{normal} \Longrightarrow \text{integral} \Longrightarrow \text{reduced and irreducible}$$

Figure 28.1: Normality strengthens integrality.

28.2 Integral closure inside the function field

Let A be a domain and $K = \operatorname{Frac}(A)$. Its integral closure in K is

$$\overline{A} = \{z \in K : z \text{ is integral over } A\}.$$

The algebraic theory of integral dependence and integral closure belongs to Volume V. Here we use its geometric consequence: $A \subseteq \overline{A} \subseteq K$, and $\overline{A}$ is a normal domain with the same fraction field K.

Proposition 28.3 (Minimal normal enlargement)**.** *If $A \subseteq B \subseteq K$ and B is integrally closed in K, then $\overline{A} \subseteq B$. Thus $\overline{A}$ is the smallest normal intermediate domain containing A inside K.*

Example 28.4 (Normalization of the cusp)**.** For $A = k[t^2, t^3]$, the element t is integral over A but not in A. The integral closure of A in $k(t)$ is $k[t]$, so $\mathbb{A}^1_k \to \operatorname{Spec} A$ is the normalization of the cusp.

$$A \hookrightarrow \overline{A} \hookrightarrow K = \mathrm{Frac}(A)$$

Figure 28.2: Integral closure is the smallest normal intermediate domain inside the same fraction field.

28.3 Affine normalization

Definition 28.5 (Affine normalization)**.** For an integral affine scheme $X = \operatorname{Spec} A$ with $K = \mathrm{Frac}(A)$, define

$$X^\nu := \operatorname{Spec} \overline{A},$$

where $\overline{A}$ is the integral closure of A in K. The inclusion $A \hookrightarrow \overline{A}$ induces the *normalization morphism*

$$\nu : X^\nu \longrightarrow X.$$

The morphism ν is integral and dominant; by lying over it is surjective. It preserves the function field, so it is birational in the function-field sense. In the finite-type setting of varieties, it is moreover an isomorphism over a nonempty dense open. If A is already normal, ν is an isomorphism.

$$\begin{tikzcd} X^\nu = \operatorname{Spec} \overline{A} \arrow[r, "\nu"] & X = \operatorname{Spec} A \\ K(X^\nu) \arrow[r, equal] & K(X) \end{tikzcd}$$

Figure 28.3: Affine normalization changes the model but not the function field.

28.4 Normalization commutes with localization

The decisive gluing fact is

$$\overline{S^{-1}A} = S^{-1}\overline{A}$$

for every multiplicative set $S \subseteq A$. In particular,

$$\overline{A_f} = (\overline{A})_f.$$

Hence the normalization of $D(f) \subseteq \operatorname{Spec} A$ is the inverse image of $D(f)$ inside $\operatorname{Spec} \overline{A}$.

Theorem 28.6 (Restriction to opens)**.** *If X is integral with normalization $\nu : X^\nu \to X$ and $U \subseteq X$ is nonempty open, then $\nu^{-1}(U) \to U$ is the normalization of U.*

$$\begin{tikzcd} \operatorname{Spec}(\overline{A})_f \arrow[r, hook] \arrow[d] & \operatorname{Spec} \overline{A} \arrow[d, "\nu"] \\ D(f) = \operatorname{Spec} A_f \arrow[r, hook] & \operatorname{Spec} A \end{tikzcd}$$

Figure 28.4: Normalization restricts compatibly to distinguished opens.

28.5 Global normalization by gluing

Let X be integral and choose an affine cover $X = \bigcup_i U_i$ with $U_i = \operatorname{Spec} A_i$. Normalize each chart:

$$U_i^\nu = \operatorname{Spec} \overline{A_i}.$$

On overlaps, common distinguished refinements from VI/20 and localization compatibility identify the normalized pieces canonically. These identifications satisfy the cocycle condition, so the U_i^ν glue to a normal integral scheme X^ν and the affine maps glue to

$$\nu : X^\nu \to X.$$

The construction is independent of the chosen affine cover.

$$\begin{tikzcd} U_i^\nu \arrow[r, shift left] & (U_i \cap U_j)^\nu \arrow[l, shift left] \arrow[r, shift left] & U_j^\nu \arrow[l, shift left] \\ & \text{glue} & \\ & X^\nu & \end{tikzcd}$$

Figure 28.5: Localization compatibility supplies canonical overlap identifications for global normalization.

Example 28.7 (Normal affine space)**.** Polynomial rings over a field are UFDs and hence integrally closed. Thus affine space $\mathbb{A}_k^n$ is normal.

28.6 Universal property

Normalization is characterized by a factorization property.

Theorem 28.8 (Universal property of normalization)**.** *Let X be integral, let $\nu : X^\nu \to X$ be its normalization, and let Z be a normal integral scheme. Every dominant morphism $f : Z \to X$ factors uniquely through ν:*

$$\begin{tikzcd} Z \arrow[rr, dashed, "\widetilde{f}"] \arrow[dr, "f"'] & & X^\nu \arrow[dl, "\nu"] \\ & X & \end{tikzcd}$$

On affine charts, an element of $\overline{A}$ is integral over A; after mapping into $K(Z)$ it is integral over the corresponding normal coordinate ring and therefore lies in that ring. Uniqueness follows from agreement on the common function field.

$$\begin{tikzcd} Z \arrow[rr, dashed, "\widetilde{f}"] \arrow[dr, "f"'] & & X^\nu \arrow[dl, "\nu"] \\ & X & \end{tikzcd}$$

Figure 28.6: The universal property: dominant maps from normal integral schemes factor uniquely through normalization.

28.7 The cusp: adjoining the missing parameter

Let
$$A = k[t^2, t^3] \subset k[t].$$
Then $K = \mathrm{Frac}(A) = k(t)$ and t is integral over A because it satisfies
$$T^2 - t^2 = 0.$$
Thus A is not normal. Its integral closure in $k(t)$ is $k[t]$, so the normalization is
$$\mathbb{A}^1_k = \operatorname{Spec} k[t] \longrightarrow \operatorname{Spec} k[t^2, t^3].$$
Geometrically this parametrizes the cusp $y^2 = x^3$ by $x = t^2$, $y = t^3$.

$$\begin{tikzcd} k[t^2, t^3] \arrow[r, hook] & k[t] \arrow[r, hook] & k(t) \\ \operatorname{Spec} k[t] \arrow[r, "parametrization"'] & \text{cusp} & \end{tikzcd}$$

Figure 28.7: Normalization of the cusp adjoins the missing parameter t.

28.8 Normalization can separate branches

For a nodal integral curve, the normalization need not be injective on points. For example, when the displayed curve is integral (for instance over a field of characteristic not 2), the nodal curve
$$y^2 = x^2(x + 1)$$
admits the parametrization
$$x = t^2 - 1, \qquad y = t(t^2 - 1).$$
The two values $t = 1$ and $t = -1$ map to the node $(0, 0)$. Thus normalization preserves the function field while it can separate distinct branches through a singular point.

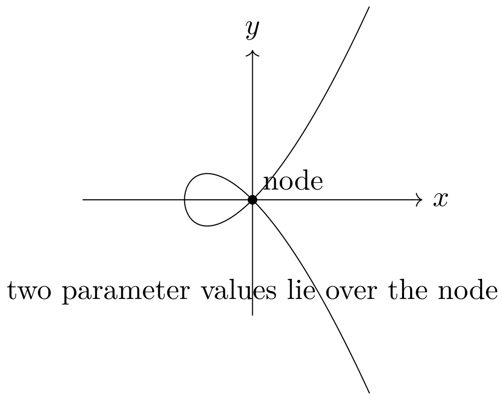


Figure 28.8: Normalization can separate branches meeting at a singular point.

Example 28.9 (Finite normalization in a basic curve example)**.** The map $k[t^2, t^3] \hookrightarrow k[t]$ is finite because $k[t]$ is generated as a module by 1 and t over the cusp ring. This makes the normalization morphism finite in this example.

28.9 Finite normalization

The normalization morphism is always integral, but it is not automatically finite for an arbitrary domain. A central finite-type case is safe and important:

Theorem 28.10 (Finite normalization over a field)**.** *If A is a finitely generated domain over a field k, then its integral closure in* $\mathrm{Frac}(A)$ *is a finite A-module. Consequently the normalization of an integral finite-type k-scheme is finite.*

This theorem uses the finiteness theory of integral closure developed in commutative algebra. It is the bridge from an abstract universal construction to a finite geometric map for algebraic varieties.

$$A \text{ finite type over } k \longrightarrow \overline{A} \text{ finite } A\text{-module} \longrightarrow X^\nu \to X \text{ finite}$$

Figure 28.9: For integral finite-type schemes over a field, normalization is a finite morphism.

28.10 Warnings and boundaries

Normalization is not the same as reduction: reduction removes nilpotents without changing the topological space, whereas normalization applies to an integral scheme and may change the point set above singularities. Normality can also fail after arbitrary field extension in positive characteristic, so normalization does not commute naively with every base change.

VI/28 does not reprove the algebraic theory of integral extensions from Volume V, and it does not yet analyze codimension-one criteria, discrete valuations, or divisor class groups. Those belong to later chapters. The next chapter, VI/29, begins the dimension arc.

28.11 Examples

Example 28.11 (A field)**.** Every field is integrally closed in itself, so $\operatorname{Spec} K$ is normal.

Example 28.12 (The integers)**.** $\mathbb{Z}$ is a UFD and hence integrally closed in $\mathbb{Q}$; therefore $\operatorname{Spec}\mathbb{Z}$ is normal.

Example 28.13 (Affine space)**.** $k[x_1, \ldots, x_n]$ is a UFD, hence normal, so $\mathbb{A}^n_k$ is normal.

Example 28.14 (Principal opens)**.** If A is normal and $f \in A$, then A_f is normal; every distinguished open of a normal affine scheme is normal.

Example 28.15 (The cusp)**.** $A = k[t^2, t^3]$ is integral but not normal because $t \in \mathrm{Frac}(A)$ is integral over A and $t \notin A$.

Example 28.16 (Normalization of the cusp)**.** The integral closure of $k[t^2, t^3]$ in $k(t)$ is $k[t]$, giving $\mathbb{A}^1_k \to V(y^2 - x^3)$.

Example 28.17 (A quadratic order)**.** The integral closure of $\mathbb{Z}[\sqrt{5}]$ in $\mathbb{Q}(\sqrt{5})$ is $\mathbb{Z}[(1+\sqrt{5})/2]$.

Example 28.18 (A DVR)**.** Every discrete valuation ring is integrally closed, hence its spectrum is normal.

Example 28.19 (Already normal)**.** If $A = \overline{A}$, then $\operatorname{Spec}\overline{A} \to \operatorname{Spec} A$ is the identity up to canonical isomorphism.

Example 28.20 (Localization of a cusp)**.** Away from the singular point, the cusp is already normal; normalization is an isomorphism over that open set.

Example 28.21 (Function field unchanged)**.** For any affine normalization $A \subseteq \overline{A} \subseteq \mathrm{Frac}(A)$, one has $\mathrm{Frac}(\overline{A}) = \mathrm{Frac}(A)$.

Example 28.22 (Surjectivity)**.** Because $\overline{A}$ is integral over A, lying over makes $\operatorname{Spec} \overline{A} \to \operatorname{Spec} A$ surjective.

Example 28.23 (Dominance)**.** The inclusion $A \hookrightarrow \overline{A}$ has zero kernel, so the normalization morphism is dominant.

Example 28.24 (Nodal curve)**.** The normalization of $y^2 = x^2(x + 1)$ separates the two branches through the node; two points of the normal curve lie over the singular point.

Example 28.25 (Normal curve chart)**.** A nonsingular affine line chart $k[t]$ is already normal, so normalization does nothing there.

Example 28.26 (Integral but not normal)**.** Integrality only asks for a domain; $k[t^2, t^3]$ shows that a domain can fail integral closedness.

Example 28.27 (Finite normalization)**.** If A is a finitely generated k-domain, then $\overline{A}$ is finite over A, so normalization is a finite morphism.

Example 28.28 (Reduction is different)**.** $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ has a reduction but is not integral, so the integral-scheme normalization construction is not the same operation.

Example 28.29 (Normality is local)**.** If an integral scheme has an affine cover by spectra of normal domains, then all of its local rings are integrally closed.

Example 28.30 (Normal open subsets)**.** Every nonempty open subscheme of a normal integral scheme is again normal and has the same function field.

Example 28.31 (Advanced 1: Semigroup ring)**.** For $A = k[t^m, t^n]$ with $\gcd(m, n) = 1$, the fraction field is $k(t)$ and the normalization is $k[t]$.

Example 28.32 (Advanced 2: Branch separation)**.** Normalization can be finite and birational without being injective on points; a node is the basic curve example.

Example 28.33 (Advanced 3: Unibranch cusp)**.** The cusp normalization is bijective on geometric points in characteristic zero but is still not an isomorphism because the local rings differ at the cusp.

Example 28.34 (Advanced 4: Integral closure after localization)**.** For a domain A and multiplicative set S, normalization inside $\mathrm{Frac}(A)$ satisfies $\overline{S^{-1}A} = S^{-1}\overline{A}$.

Example 28.35 (Advanced 5: Universal factorization)**.** A dominant map from a normal integral affine scheme $\operatorname{Spec} B$ to $\operatorname{Spec} A$ extends uniquely from A to $\overline{A}$ inside the induced embedding of fraction fields.

Example 28.36 (Advanced 6: Normalization is idempotent)**.** Since X^ν is normal, normalizing it again gives a canonical isomorphism $(X^\nu)^\nu \cong X^\nu$.

Example 28.37 (Advanced 7: Birational invariance)**.** X and X^ν have canonically isomorphic function fields, so normalization stays in the same birational class.

Example 28.38 (Advanced 8: Finite type over a field)**.** For a finitely generated k-domain, finiteness of integral closure makes normalization proper over affine charts because finite morphisms are affine and proper.

Example 28.39 (Advanced 9: Positive-characteristic warning)**.** A normal k-scheme need not remain normal after an arbitrary inseparable field extension; geometric normality is a stronger condition.

Example 28.40 (Advanced 10: Base-change warning)**.** Because normality may fail after field extension, forming X^ν and then base-changing need not equal normalizing the base change.

Example 28.41 (Advanced 11: Normal versus regular)**.** Regular schemes are normal under the standard Noetherian hypotheses, but normal schemes can still have singularities in dimension at least two.

Example 28.42 (Advanced 12: Codimension-one preview)**.** Later divisor theory detects normality using height-one local data together with a depth condition; VI/28 does not use that criterion yet.

Example 28.43 (Advanced 13: Normalization of an order)**.** The inclusion $\mathbb{Z}[\sqrt{5}] \subset \mathbb{Z}[(1+\sqrt{5})/2]$ is finite and birational on spectra, illustrating arithmetic normalization.

Example 28.44 (Advanced 14: Nonfinite warning)**.** For arbitrary non-Noetherian domains, the integral closure in the fraction field need not be finite as a module, although the normalization morphism remains integral.

28.12 Problem dossiers

Problem 28.45 (VI.28.P1: AG7 T05 reconstruction: affine normalization)**.** Let A be a domain, $K = \mathrm{Frac}(A)$, and $\overline{A}$ the integral closure of A in K. Prove that $\operatorname{Spec} \overline{A} \to \operatorname{Spec} A$ is integral, dominant, surjective, normal on the source, and induces an isomorphism of function fields.

Provenance. Canonical reconstruction of the affine normalization construction in AG7 T05; no numbered legacy Problem 8 block is claimed.

Problem 28.46 (VI.28.P2: AG7 T05 reconstruction: localization, gluing, and global normalization)**.** Let X be integral. Normalize every affine chart $U_i = \operatorname{Spec} A_i$. Using compatibility of integral closure with localization and common distinguished refinements of overlaps, prove that the normalized affine charts glue to a normal integral scheme X^ν with a morphism $\nu : X^\nu \to X$ independent of the chosen cover.

Provenance. Canonical reconstruction of AG7 T05 global normalization theory; no numbered legacy Problem 8 block is present in the audited source snapshot.

Problem 28.47 (VI.28.P3: Universal property of normalization)**.** Prove that a dominant morphism $f : Z \to X$ from a normal integral scheme factors uniquely through X^ν.

Provenance. Canonical consequence of the AG7 normalization stream.

Problem 28.48 (VI.28.P4: Normalize the cusp)**.** Compute the normalization of $A = k[x,y]/(y^2 - x^3)$ by identifying it with $k[t^2, t^3] \subset k[t]$.

Provenance. Canonical cusp computation derived from the AG7 normalization theory.

Problem 28.49 (VI.28.P5: Normality iff normalization is trivial)**.** Show that an integral scheme X is normal if and only if its normalization morphism $X^\nu \to X$ is an isomorphism.

Provenance. Canonical dossier.

Problem 28.50 (VI.28.P6: Normalization is idempotent)**.** Prove $(X^\nu)^\nu \cong X^\nu$ canonically.

Provenance. Canonical dossier.

Problem 28.51 (VI.28.P7: Restriction to a nonempty open)**.** For nonempty open $U \subseteq X$, prove that $\nu^{-1}(U) \to U$ is the normalization of U.

Provenance. Canonical dossier.

Problem 28.52 (VI.28.P8: Local rings upstairs)**.** Let $y \in X^\nu$ lie over $x \in X$. Explain how $\mathcal{O}_{X^\nu,y}$ sits inside $K(X)$ and why it is integral over the image of $\mathcal{O}_{X,x}$.

Provenance. Canonical dossier.

Problem 28.53 (VI.28.P9: Same function field)**.** Prove globally that $K(X^\nu) \cong K(X)$ canonically.

Provenance. Canonical dossier.

Problem 28.54 (VI.28.P10: Birationality of normalization)**.** Assume X is an integral finite-type scheme over a field. Using the definition that two integral schemes are birational when they contain isomorphic nonempty dense opens, show that the normalization $X^\nu \to X$ is birational.

Provenance. Canonical dossier; the later birational-geometry arc develops this language further.

Problem 28.55 (VI.28.P11: Normality is affine-local)**.** Assume an integral scheme admits an affine open cover by spectra of normal domains. Prove every local ring is an integrally closed domain.

Provenance. Canonical dossier.

Problem 28.56 (VI.28.P12: UFDs give normal schemes)**.** Show that if A is a UFD then $\operatorname{Spec} A$ is normal.

Provenance. Canonical bridge to standard algebra examples.

Problem 28.57 (VI.28.P13: A quadratic arithmetic normalization)**.** Compute the integral closure of $\mathbb{Z}[\sqrt{5}]$ in $\mathbb{Q}(\sqrt{5})$.

Provenance. Canonical arithmetic example; algebraic integral-closure theory belongs to Volume V.

Problem 28.58 (VI.28.P14: Finite normalization over a field)**.** Let X be integral and finite type over a field k. Explain why $X^\nu \to X$ is finite.

Provenance. Canonical finite-type consequence.

Problem 28.59 (VI.28.P15: Normalization need not be a homeomorphism)**.** Explain using a nodal curve why a normalization morphism can have more than one point over a singular point.

Provenance. Canonical geometric warning.

Problem 28.60 (VI.28.P16: Cusp normalization can be bijective but not an isomorphism)**.** Over an algebraically closed field of characteristic zero, explain why the cusp normalization can be bijective on points while failing to be an isomorphism.

Provenance. Canonical geometric warning.

Problem 28.61 (VI.28.P17: Normalization versus reduction)**.** Compare reduction and normalization and give an example showing they are different operations.

Provenance. Canonical boundary dossier.

Problem 28.62 (VI.28.P18: Normality after field extension: warning)**.** Explain why normality over k need not survive arbitrary field extension in positive characteristic.

Provenance. Canonical base-change warning; systematic base change was VI/24.

Problem 28.63 (VI.28.P19: Normalization and arbitrary base change)**.** Why can one not assert $(X^\nu)_{S'} \cong (X_{S'})^\nu$ for arbitrary base change without hypotheses?

Provenance. Canonical warning dossier.

Problem 28.64 (VI.28.P20: Boundary to dimension and divisors)**.** State precisely what VI/28 establishes that later dimension and divisor chapters may use.

Provenance. Canonical boundary dossier.

28.13 Exercises

Exercise 28.65 (VI.28.E1: Normal domain)**.** State the definition of a normal domain.

Exercise 28.66 (VI.28.E2: Affine normality)**.** Show $\operatorname{Spec} A$ is normal when A is integrally closed.

Exercise 28.67 (VI.28.E3: Field)**.** Show every field is normal.

Exercise 28.68 (VI.28.E4: Localization)**.** Show a localization of a normal domain is normal.

Exercise 28.69 (VI.28.E5: Fraction fields)**.** Show $\operatorname{Frac}(\overline{A}) = \operatorname{Frac}(A)$.

Exercise 28.70 (VI.28.E6: Integral morphism)**.** Why is $\operatorname{Spec} \overline{A} \to \operatorname{Spec} A$ integral?

Exercise 28.71 (VI.28.E7: Surjectivity)**.** Why is the affine normalization morphism surjective?

Exercise 28.72 (VI.28.E8: Normal fixed point)**.** Show a normal affine scheme equals its normalization.

Exercise 28.73 (VI.28.E9: Cusp element)**.** For $A = k[t^2, t^3]$, prove t is integral over A.

Exercise 28.74 (VI.28.E10: Cusp closure)**.** Show $k[t]$ is the integral closure of $k[t^2, t^3]$ in $k(t)$.

Exercise 28.75 (VI.28.E11: Principal open)**.** Compute the normalization of $D(f) \subseteq \operatorname{Spec} A$ from $\overline{A}$.

Exercise 28.76 (VI.28.E12: Open restriction)**.** Explain why normalization restricts to nonempty opens.

Exercise 28.77 (VI.28.E13: Generic point)**.** Show X and X^ν have corresponding generic points.

Exercise 28.78 (VI.28.E14: Function field)**.** Show $K(X^\nu) = K(X)$.

Exercise 28.79 (VI.28.E15: Idempotence)**.** Show normalization is idempotent.

Exercise 28.80 (VI.28.E16: UFD)**.** Explain why a UFD is normal.

Exercise 28.81 (VI.28.E17: Polynomial ring)**.** Deduce that $\mathbb{A}^n_k$ is normal.

Exercise 28.82 (VI.28.E18: Quadratic integer)**.** Show $(1 + \sqrt{5})/2$ is integral over $\mathbb{Z}[\sqrt{5}]$.

Exercise 28.83 (VI.28.E19: Node preimages)**.** Under $x = t^2 - 1$, $y = t(t^2 - 1)$, find the preimages of $(0, 0)$.

Exercise 28.84 (VI.28.E20: Reduction comparison)**.** Give a reduced nonnormal affine scheme.

Exercise 28.85 (VI.28.E21: Finite type)**.** State when normalization is finite for a finitely generated k-domain.

Exercise 28.86 (VI.28.E22: Dominant factorization)**.** State the universal property of normalization.

Exercise 28.87 (VI.28.E23: Local ring upstairs)**.** Describe $\mathcal{O}_{X^\nu,y}$ on an affine normalization.

Exercise 28.88 (VI.28.E24: Base-change warning)**.** Why need normality not survive inseparable scalar extension?

Exercise 28.89 (VI.28.E25: Birationality)**.** Why is normalization birational?

Exercise 28.90 (VI.28.E26: Boundary)**.** Name one topic deliberately deferred beyond VI/28.

28.14 Challenges

Problem 28.91 (Challenge 1: Normalize a family of monomial curves)**.** For coprime $m, n \geq 2$, prove that the normalization of $\operatorname{Spec} k[t^m, t^n]$ is $\operatorname{Spec} k[t]$ and analyze which singular point is repaired.

Problem 28.92 (Challenge 2: Normalization of a node in detail)**.** Starting from $A = k[x, y]/(y^2 - x^2(x + 1))$ in characteristic not 2, identify its fraction field with $k(t)$ via $t = y/x$, prove t is integral, and determine the integral closure.

Problem 28.93 (Challenge 3: Universal property from rings only)**.** Reprove the global universal property by constructing the affine extensions $\overline{A} \to B$, checking compatibility on common distinguished refinements, and gluing without invoking a prepackaged universal-property theorem.

Problem 28.94 (Challenge 4: Finite versus merely integral normalization)**.** Survey which hypothesis in the finite-type-over-a-field theorem guarantees module finiteness of $\overline{A}$ and produce or research a non-Noetherian warning where integral closure is not finite.

Problem 28.95 (Challenge 5: Base change and geometric normality)**.** Construct explicitly a purely inseparable field extension for which a normal k-scheme becomes nonreduced after scalar extension, and explain exactly why normalization and base change fail to commute in that example.

28.15 Chapter synthesis

Normalization keeps the function field fixed and enlarges every affine coordinate ring only by elements already forced integrally inside that field. Localization compatibility turns this local algebra into a global scheme, and the universal property makes the result canonical. In finite-type geometry the normalization map is finite, so singular integral models can be compared to a normal model without leaving the same birational class.

$$X \text{ integral} \longrightarrow K(X) \longrightarrow X^\nu \text{ normal} \longrightarrow \text{same birational class}$$

Figure 28.10: Normalization is the canonical normal model inside the function field.

28.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 28.96 (Cusp closure)**.** Normalize $k[t^2, t^3]$ in $k(t)$.

Hint. Use integral closure and normalization of schemes. ☐

Solution. $k[t]$. ☐

Exercise 28.97 (Witness)**.** What element witnesses nonnormality of the cusp ring?

Hint. Use integral closure and normalization of schemes. ☐

Solution. t. ☐

Exercise 28.98 (Polynomial)**.** Is $k[x, y]$ normal?

Hint. Use integral closure and normalization of schemes. ☐

Solution. Yes. ☐

Exercise 28.99 (Definition)**.** What ring is used on an affine normalization?

Hint. Use integral closure and normalization of schemes. ☐

Solution. The integral closure in the function field. ☐

Exercise 28.100 (Map direction)**.** What scheme map comes from $A \subset \overline{A}$?

Hint. Use integral closure and normalization of schemes. ☐

Solution. $\operatorname{Spec} \overline{A} \to \operatorname{Spec} A$. ☐

Proofs

Exercise 28.101 (Cusp integrality)**.** Prove: Prove t is integral over $k[t^2, t^3]$.

Hint. Work directly from integral closure and normalization of schemes. ☐

Solution. It satisfies the monic equation $X^2 - t^2 = 0$. ☐

Exercise 28.102 (UFD normal)**.** Prove: Prove a UFD is integrally closed.

Hint. Work directly from integral closure and normalization of schemes. ☐

Solution. Write an integral fraction in lowest terms and use divisibility of the monic relation to force the denominator to be a unit. ☐

Exercise 28.103 (Normalization birational)**.** Prove: Prove affine normalization of a domain has the same fraction field.

Hint. Work directly from integral closure and normalization of schemes. ☐

Solution. The integral closure is taken inside the original fraction field and contains the original domain. ☐

Exercise 28.104 (Normal fixed)**.** Prove: Prove normalization of a normal domain is itself.

Hint. Work directly from integral closure and normalization of schemes. ☐

Solution. Normality means the domain already equals its integral closure in the fraction field. ☐

Counterexamples and hypothesis tests

Exercise 28.105 (Every domain normal)**.** Test the claim: every integral domain is normal.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; the cusp ring is a standard counterexample. ☐

Exercise 28.106 (Normalization topology)**.** Test the claim: normalization must be a homeomorphism on underlying spaces.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general. ☐

Exercise 28.107 (Fraction field changes)**.** Test the claim: normalization changes the function field.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is birational and keeps the same fraction field. ☐

Applications and investigations

Exercise 28.108 (Repair singularities)**.** What geometric role can normalization play?

Hint. Translate the question into integral closure and normalization of schemes. ☐

Solution. It replaces a nonnormal integral scheme by a normal birational model. ☐

Exercise 28.109 (Curve parametrization)**.** How does the cusp normalization relate to parametrization?

Hint. Translate the question into integral closure and normalization of schemes. ☐

Solution. The parameter t gives the normalization map to the singular curve. ☐

Challenge problems

Exercise 28.110 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Normalization in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Normalization, translates it through integral closure and normalization of schemes, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 28.111 (Hypothesis challenge)**.** Choose one structural statement from Normalization, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Part VI

Dimension

Chapter 29

Krull Dimension

The first invariant in the dimension arc is entirely order-theoretic: how long can a strict chain of prime ideals be? For an affine scheme this same number measures the longest chain of irreducible closed subsets. VI/29 develops this ring-theoretic and affine-topological core. VI/30 will globalize dimension to arbitrary schemes and prove the finite-type dimension formulas; VI/31 will isolate codimension, and VI/32 will study local dimension and tangent data.

29.1 Prime chains and Krull dimension

Definition 29.1 (Krull dimension)**.** Let A be a nonzero ring. The *Krull dimension* of A is

$$\dim A = \sup\{n : \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n \text{ is a chain of prime ideals of } A\}.$$

The length of the displayed chain is n, the number of strict inclusions, not the number $n+1$ of primes.

A field has dimension 0. The ring $k[x]$ has dimension 1, while $k[x, y]$ has dimension 2. The infinite polynomial ring $k[x_1, x_2, \ldots]$ has infinite dimension because

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots$$

contains chains of arbitrary finite length.

$$\begin{tikzcd}
\mathfrak{p}_n \\
\cdots \arrow[u, "\subsetneq"'] \\
\mathfrak{p}_1 \arrow[u, "\subsetneq"'] \\
\mathfrak{p}_0 \arrow[u, "\subsetneq"']
\end{tikzcd}$$

Figure 29.1: Krull dimension counts strict inclusions in prime chains.

29.2 Irreducible closed subsets of an affine scheme

Every irreducible closed subset of $\operatorname{Spec} A$ is $V(\mathfrak{p})$ for a unique prime ideal $\mathfrak{p}$. Inclusion is reversed:

$$\mathfrak{p} \subseteq \mathfrak{q} \iff V(\mathfrak{q}) \subseteq V(\mathfrak{p}).$$

Therefore a prime chain

$$\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$

corresponds to the irreducible-closed chain

$$V(\mathfrak{p}_n) \subsetneq \cdots \subsetneq V(\mathfrak{p}_0).$$

Thus the topological dimension of $\operatorname{Spec} A$ equals $\dim A$.

Proposition 29.2 (Affine chain dictionary)**.** *For every nonzero ring A, the supremum of lengths of strict chains of prime ideals equals the supremum of lengths of strict chains of irreducible closed subsets of* $\operatorname{Spec} A$*.*

$$\begin{array}{cc}
\mathfrak{p}_0 & V(\mathfrak{p}_0) \\
\downarrow & \uparrow \\
\mathfrak{p}_1 & V(\mathfrak{p}_1) \\
\downarrow & \uparrow \\
\mathfrak{p}_2 & V(\mathfrak{p}_2)
\end{array}$$

Figure 29.2: Prime inclusion becomes reverse inclusion of irreducible closed subsets.

Example 29.3 (Dimension of a field)**.** A field has only the prime ideal (0), so there is no strict nontrivial chain of primes. Its Krull dimension is 0.

29.3 Height of a prime

Definition 29.4 (Height)**.** For $\mathfrak{p} \in \operatorname{Spec} A$, the *height* of $\mathfrak{p}$ is

$$\operatorname{ht}(\mathfrak{p}) = \sup\{n : \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p}\}.$$

Prime ideals below $\mathfrak{p}$ are exactly the primes of the localization $A_{\mathfrak{p}}$. Hence

Proposition 29.5 (Height as local Krull dimension)**.**

$$\operatorname{ht}(\mathfrak{p}) = \dim A_{\mathfrak{p}}.$$

This is a ring-theoretic statement. The geometric interpretation of $\dim \mathcal{O}_{X,x}$ as local dimension belongs to VI/32.

29.4 Quotients and closed subsets

Prime ideals of A/I correspond to primes $\mathfrak{p} \supseteq I$ of A. Consequently

Proposition 29.6 (Dimension of a quotient)**.**

$$\dim(A/I) \leq \dim A.$$

More precisely, $\dim(A/I)$ *is the supremum of lengths of chains of primes of A all containing I.*

Geometrically $\operatorname{Spec}(A/I)$ is the closed subspace $V(I) \subseteq \operatorname{Spec} A$ with its scheme structure.

$$
\begin{tikzcd}
A \arrow[r, "\text{localize}"] & A_{\mathfrak{p}} \arrow[d] \\
& \dim A_{\mathfrak{p}} = \mathrm{ht}(\mathfrak{p})
\end{tikzcd}
$$

Figure 29.3: Localizing at $\mathfrak{p}$ keeps exactly the primes below $\mathfrak{p}$.

$$
\begin{tikzcd}
\mathrm{Spec}(A/I) \arrow[r, hook] \arrow[d, leftrightarrow] & \mathrm{Spec}\, A \arrow[d, leftrightarrow] \\
\{\mathfrak{p} \supseteq I\} \arrow[r, hook] & \{\text{all primes}\}
\end{tikzcd}
$$

Figure 29.4: Quotients restrict prime chains to the closed subset $V(I)$.

29.5 Localization and open subsets

Prime ideals of $S^{-1}A$ correspond to primes of A disjoint from S, with inclusions preserved. Therefore

Proposition 29.7 (Localization cannot increase dimension)**.**

$$\dim S^{-1}A \leq \dim A.$$

For a principal localization, $\dim A_f$ *is computed from prime chains entirely inside* $D(f)$*.*

The inequality can be strict: localizing $k[x]$ at the multiplicative set of all nonzero polynomials gives $k(x)$, whose dimension is 0.

$$
\begin{tikzcd}
\mathrm{Spec}\, S^{-1}A \arrow[r, hook] \arrow[d, leftrightarrow] & \mathrm{Spec}\, A \arrow[d, leftrightarrow] \\
\{\mathfrak{p} : \mathfrak{p} \cap S = \varnothing\} \arrow[r, hook] & \{\text{all primes}\}
\end{tikzcd}
$$

Figure 29.5: Localization keeps only primes disjoint from the multiplicative set.

Example 29.8 (Dimension of a polynomial line)**.** In $k[x]$, the chain $(0) \subset (x - a)$ has length 1, and no longer prime chain exists. Thus $\dim k[x] = 1$.

29.6 Nilpotents do not affect dimension

The reduction map $A \to A_{\mathrm{red}} = A/\sqrt{(0)}$ induces a homeomorphism on spectra and a bijection on prime ideals preserving inclusions. Hence

Proposition 29.9 (Dimension ignores nilpotents)**.**

$$\dim A = \dim A_{\mathrm{red}}.$$

Dimension measures the specialization order of points; nilpotent thickening changes the structure sheaf, not this order.

$$\begin{tikzcd} A \arrow[r] & A_{\mathrm{red}} = A/\sqrt{(0)} \end{tikzcd}$$

$$\begin{tikzcd} \operatorname{Spec} A & \operatorname{Spec} A_{\mathrm{red}} \arrow[l, "\sim"'] \end{tikzcd}$$

Figure 29.6: Reduction changes nilpotents but not the ordered set of prime ideals.

29.7 Products and disconnected spectra

Since

$$\operatorname{Spec}(A \times B) = \operatorname{Spec} A \sqcup \operatorname{Spec} B,$$

no prime chain crosses from one component to the other. Thus

Proposition 29.10 (Dimension of a product)**.**

$$\dim(A \times B) = \max\{\dim A, \dim B\}.$$

This simple example anticipates a warning for VI/30: global dimension is a supremum over components, so local or open pieces need not all have the same dimension without irreducibility hypotheses.

$$\boxed{\operatorname{Spec} A} \qquad \boxed{\operatorname{Spec} B}$$

$$\operatorname{Spec}(A \times B) = \operatorname{Spec} A \sqcup \operatorname{Spec} B$$

Figure 29.7: Prime chains in a product ring remain in one connected component.

29.8 Integral extensions

Let $A \subseteq B$ be integral. Going up lifts prime chains from A to B, while incomparability prevents a strict chain in B from contracting two comparable primes to the same prime of A. Therefore

Theorem 29.11 (Dimension under integral extensions)**.** *If $A \subseteq B$ is integral, then*

$$\dim B = \dim A.$$

In particular, normalization does not change Krull dimension. This links VI/28 to the dimension arc without reopening the normalization construction.

$$\begin{tikzcd} A \arrow[r, hook, "\text{integral}"] & B \\ \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n \arrow[d, mapsto] & \mathfrak{q}_0 \subsetneq \cdots \subsetneq \mathfrak{q}_n \arrow[l, mapsto] \\ \text{chain in } A & \text{chain in } B \end{tikzcd}$$

Figure 29.8: Going up and incomparability make integral extensions dimension-preserving.

Example 29.12 (A two-dimensional chain)**.** In $k[x, y]$, $(0) \subset (x) \subset (x, y)$ is a chain of length 2. The polynomial ring has Krull dimension 2.

29.9 Polynomial rings and finite-type algebras

For a field k,

$$\dim k[x_1, \ldots, x_n] = n.$$

The lower bound is the chain

$$(0) \subsetneq (x_1) \subsetneq (x_1, x_2) \subsetneq \cdots \subsetneq (x_1, \ldots, x_n),$$

and the upper bound is the classical dimension theorem for polynomial rings. Quotients therefore give many finite-dimensional affine schemes. The stronger equality between dimension and transcendence degree for integral finite-type k-algebras is used systematically in VI/30.

$$(0) \longrightarrow (x_1) \longrightarrow (x_1, x_2) \longrightarrow \cdots \longrightarrow (x_1, \ldots, x_n)$$

Figure 29.9: The coordinate-prime chain realizes the lower bound $\dim k[x_1, \ldots, x_n] \geq n$.

29.10 Boundaries of VI/29

VI/29 defines and computes Krull dimension through prime chains and affine topology. It deliberately does not yet prove the global variety statements from AG8 Problem 2: equality with local dimension at closed points, additivity with codimension, invariance on nonempty opens of an irreducible variety, or the full transcendence-degree formula. Those are routed to VI/30–VI/32. Likewise AG8 Problem 3 tests failures of those global statements for general schemes; the VI/29 instance audit finds no Krull-dimension-only numbered subproblem there.

29.11 Examples

Example 29.13 (Example 1: Field)**.** For a field K, the only prime is (0), so $\dim K = 0$.

Example 29.14 (Example 2: Integers)**.** The chains $(0) \subsetneq (p)$ show $\dim \mathbb{Z} = 1$; every nonzero prime is maximal.

Example 29.15 (Example 3: Affine line)**.** For $k[x]$, $(0) \subsetneq (x - a)$ is a maximal chain, hence $\dim k[x] = 1$.

Example 29.16 (Example 4: Affine plane)**.** In $k[x, y]$, $(0) \subsetneq (x) \subsetneq (x, y)$ has length 2.

Example 29.17 (Example 5: Affine n-space)**.** A coordinate-prime chain has length n, and the polynomial dimension theorem gives $\dim k[x_1, \ldots, x_n] = n$.

Example 29.18 (Example 6: A hypersurface quotient)**.** $k[x, y]/(x)$ is isomorphic to $k[y]$, so the quotient has dimension 1.

Example 29.19 (Example 7: Artinian ring)**.** Every prime in an Artinian ring is maximal; therefore every nonzero Artinian ring has Krull dimension 0.

Example 29.20 (Example 8: Dual numbers)**.** $k[\varepsilon]/(\varepsilon^2)$ has the same spectrum as k, hence dimension 0.

Example 29.21 (Example 9: Reducible curve)**.** $k[x, y]/(xy)$ has dimension 1: its two irreducible components are affine lines.

Example 29.22 (Example 10: Localization to a field)**.** Localizing $k[x]$ at all nonzero elements gives $k(x)$ and drops dimension from 1 to 0.

Example 29.23 (Example 11: Localizing at a maximal ideal)**.** $k[x,y]_{(x,y)}$ has dimension 2 because $(0) \subsetneq (x) \subsetneq (x,y)$ survives.

Example 29.24 (Example 12: Product ring)**.** $\dim(k \times k[x]) = \max\{0,1\} = 1$.

Example 29.25 (Example 13: Reduction)**.** $k[x,\varepsilon]/(\varepsilon^2)$ and $k[x]$ have the same prime chains, hence both have dimension 1.

Example 29.26 (Example 14: Closed point quotient)**.** $k[x,y]/(x,y) \cong k$ has dimension 0.

Example 29.27 (Example 15: Principal open)**.** $k[x,y]_x$ still has dimension 2: for example $(0) \subsetneq (y) \subsetneq (y, x-1)$ survives when x is inverted.

Example 29.28 (Example 16: Infinite polynomial ring)**.** $k[x_1, x_2, \ldots]$ has infinite dimension from chains of coordinate ideals of arbitrary length.

Example 29.29 (Example 17: Height one prime)**.** In $k[x,y]$, the prime (x) has height 1 and $\dim k[x,y]_{(x)} = 1$.

Example 29.30 (Example 18: Maximal ideal height)**.** In $k[x,y]$, (x,y) has height 2.

Example 29.31 (Example 19: Integral extension)**.** $k[t^2,t^3] \subset k[t]$ is integral; both rings have dimension 1.

Example 29.32 (Example 20: Empty-dimensional warning)**.** Dimension conventions for the zero ring or empty spectrum vary; VI/29 avoids using those conventions in computations.

Example 29.33 (Advanced 1: Saturated prime chains)**.** A maximal-length chain need not be unique. Dimension records only the supremal length, not the branching pattern of $\operatorname{Spec} A$.

Example 29.34 (Advanced 2: Noetherian does not mean finite-dimensional)**.** There exist Noetherian rings of infinite Krull dimension; finite-dimensionality needs additional hypotheses such as finite type over a field.

Example 29.35 (Advanced 3: Catenary preview)**.** Even when endpoint primes are fixed, different saturated chains can have different lengths in noncatenary rings. Krull dimension alone does not detect this pathology.

Example 29.36 (Advanced 4: Equidimensionality preview)**.** A reducible ring can have irreducible components of different dimensions; the ring dimension is the maximum component dimension.

Example 29.37 (Advanced 5: Dimension after quotient)**.** The inequality $\dim A/I \le \dim A$ can be strict or an equality, depending on which chains remain above I.

Example 29.38 (Advanced 6: Dimension after localization)**.** Localization selects primes disjoint from S; it may preserve dimension or lower it drastically.

Example 29.39 (Advanced 7: Integral closure)**.** An integral domain and its normalization have equal Krull dimension even when the normalization separates branches geometrically.

Example 29.40 (Advanced 8: Minimal primes)**.** Every prime chain begins above some minimal prime, so $\dim A$ is the supremum of the dimensions of the irreducible components $V(\mathfrak{p}_{\min})$.

Example 29.41 (Advanced 9: Height versus coheight)**.** $\mathrm{ht}(\mathfrak{p})$ measures chains below $\mathfrak{p}$, whereas $\dim(A/\mathfrak{p})$ measures chains above it.

Example 29.42 (Advanced 10: Dimension formula warning)**.** The equality $\mathrm{ht}(\mathfrak{p}) + \dim(A/\mathfrak{p}) = \dim A$ is not a formal consequence of the definition for arbitrary rings.

Example 29.43 (Advanced 11: Valuation domains)**.** Prime ideals in a valuation domain are totally ordered; their order type makes valuation domains a rich source of prescribed Krull dimensions.

Example 29.44 (Advanced 12: Polynomial extension)**.** For Noetherian A, one has $\dim A[x] = \dim A+1$ under the classical polynomial dimension theorem; VI/29 uses the field case explicitly.

Example 29.45 (Advanced 13: Specialization order)**.** In $\operatorname{Spec} A$, $\mathfrak{q}$ is a specialization of $\mathfrak{p}$ exactly when $\mathfrak{p} \subseteq \mathfrak{q}$.

Example 29.46 (Advanced 14: Topological invariance)**.** Because Krull dimension of an affine spectrum depends only on the specialization order, nilpotent thickenings leave it unchanged.

29.12 Problem dossiers

Problem 29.47 (VI.29.P1: Prime-chain definition)**.** State the Krull-dimension definition carefully and explain why a chain with $n+1$ primes has length n.

Provenance. Canonical VI/29 development.

Problem 29.48 (VI.29.P2: Affine chain dictionary)**.** Prove that prime chains in A and irreducible-closed chains in $\operatorname{Spec} A$ have the same possible lengths.

Provenance. Canonical VI/29 development.

Problem 29.49 (VI.29.P3: Dimension of a field)**.** Prove that a field has Krull dimension 0.

Provenance. Canonical VI/29 development.

Problem 29.50 (VI.29.P4: Dimension of $\mathbb{Z}$)**.** Compute $\dim \mathbb{Z}$.

Provenance. Canonical VI/29 development.

Problem 29.51 (VI.29.P5: Dimension of the affine-line coordinate ring)**.** Show $\dim k[x] = 1$.

Provenance. Canonical VI/29 development.

Problem 29.52 (VI.29.P6: Coordinate chain in a polynomial ring)**.** Construct a prime chain of length n in $k[x_1, \ldots, x_n]$.

Provenance. Canonical VI/29 development.

Problem 29.53 (VI.29.P7: Height equals localized dimension)**.** Prove $\mathrm{ht}(\mathfrak{p}) = \dim A_{\mathfrak{p}}$.

Provenance. Canonical VI/29 development.

Problem 29.54 (VI.29.P8: Quotient chains)**.** Show that $\dim(A/I)$ is the supremum of chain lengths among primes of A containing I.

Provenance. Canonical VI/29 development.

Problem 29.55 (VI.29.P9: Localization chains)**.** Show that $\dim S^{-1}A \leq \dim A$.

Provenance. Canonical VI/29 development.

Problem 29.56 (VI.29.P10: Dimension and reduction)**.** Prove $\dim A = \dim A_{\mathrm{red}}$.

Provenance. Canonical VI/29 development.

Problem 29.57 (VI.29.P11: Dimension of a product)**.** Prove $\dim(A \times B) = \max\{\dim A, \dim B\}$.

Provenance. Canonical VI/29 development.

Problem 29.58 (VI.29.P12: A dimension drop by localization)**.** Give a localization of a one-dimensional domain that has dimension 0.

Provenance. Canonical VI/29 development.

Problem 29.59 (VI.29.P13: A dimension-preserving localization)**.** Show $\dim k[x, y]_x = 2$.

Provenance. Canonical VI/29 development.

Problem 29.60 (VI.29.P14: Infinite Krull dimension)**.** Show $k[x_1, x_2, \dots]$ has infinite Krull dimension.

Provenance. Canonical VI/29 development.

Problem 29.61 (VI.29.P15: Integral extensions preserve dimension)**.** Assuming going up and incomparability, prove that an integral extension $A \subseteq B$ has $\dim A = \dim B$.

Provenance. Canonical VI/29 development.

Problem 29.62 (VI.29.P16: Normalization preserves dimension)**.** Deduce that an integral domain and its normalization have the same Krull dimension.

Provenance. Canonical VI/29 development.

Problem 29.63 (VI.29.P17: Component dimensions)**.** Let $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ be the minimal primes of a ring with finitely many minimal primes. Explain why $\dim A = \max_i \dim(A/\mathfrak{p}_i)$.

Provenance. Canonical VI/29 development.

Problem 29.64 (VI.29.P18: AG6 dimension-preview reconstruction)**.** Explain how the dimension-preview material routed from AG6 is absorbed into VI/29 without creating a false numbered legacy problem.

Provenance. Canonical reconstruction of the AG6 T05 dimension-preview stream; no numbered legacy problem is claimed.

Problem 29.65 (VI.29.P19: AG8 Krull stream reconstruction)**.** Explain why the Krull-dimension definitions in AG8 are tracked as a theory stream rather than as a new numbered problem.

Provenance. Canonical reconstruction of AG8 T01 Krull-dimension theory; no numbered legacy problem is claimed.

Problem 29.66 (VI.29.P20: AG8 Problem 3 routing audit)**.** Why is no part of AG8 Problem 3 consumed by VI/29?

Provenance. Instance audit of AG8 Problem 3; the numbered problem remains reserved for VI/30–VI/32.

29.13 Exercises

Exercise 29.67 (VI.29.E1: Chain length)**.** What is the length of $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2$?

Exercise 29.68 (VI.29.E2: Field)**.** Compute $\dim k$ for a field k.

Exercise 29.69 (VI.29.E3: Integers)**.** Compute $\dim \mathbb{Z}$.

Exercise 29.70 (VI.29.E4: Affine line)**.** Compute $\dim k[x]$.

Exercise 29.71 (VI.29.E5: Affine plane lower bound)**.** Give a prime chain of length 2 in $k[x, y]$.

Exercise 29.72 (VI.29.E6: Height)**.** Define $\mathrm{ht}(\mathfrak{p})$.

Exercise 29.73 (VI.29.E7: Localization)**.** Why can localization not increase Krull dimension?

Exercise 29.74 (VI.29.E8: Quotient)**.** Why can quotienting not increase dimension?

Exercise 29.75 (VI.29.E9: Reduction)**.** Show nilpotents do not change dimension.

Exercise 29.76 (VI.29.E10: Product)**.** Compute $\dim(A \times B)$ from $\dim A$ and $\dim B$.

Exercise 29.77 (VI.29.E11: Dual numbers)**.** Compute $\dim k[\varepsilon]/(\varepsilon^2)$.

Exercise 29.78 (VI.29.E12: Reducible curve)**.** Compute $\dim k[x,y]/(xy)$.

Exercise 29.79 (VI.29.E13: Height in the plane)**.** Compute $\mathrm{ht}(x)$ in $k[x,y]$.

Exercise 29.80 (VI.29.E14: Maximal height)**.** Compute $\mathrm{ht}(x,y)$ in $k[x,y]$.

Exercise 29.81 (VI.29.E15: Infinite variables)**.** Why does $k[x_1, x_2, \ldots]$ have infinite dimension?

Exercise 29.82 (VI.29.E16: Normalization)**.** Why does normalization preserve Krull dimension?

Exercise 29.83 (VI.29.E17: Principal open)**.** Interpret $\dim A_f$ in terms of $\operatorname{Spec} A$.

Exercise 29.84 (VI.29.E18: Closed subscheme)**.** Interpret $\dim(A/I)$ in terms of $V(I)$.

Exercise 29.85 (VI.29.E19: Specialization)**.** When is $\mathfrak{q}$ a specialization of $\mathfrak{p}$ in $\operatorname{Spec} A$?

Exercise 29.86 (VI.29.E20: Irreducible chain)**.** Translate $(0) \subsetneq (x) \subsetneq (x,y)$ into closed subsets of $\operatorname{Spec} k[x,y]$.

Exercise 29.87 (VI.29.E21: Polynomial dimension)**.** State $\dim k[x_1, \ldots, x_n]$.

Exercise 29.88 (VI.29.E22: Artinian ring)**.** What is the dimension of a nonzero Artinian ring?

Exercise 29.89 (VI.29.E23: Noetherian warning)**.** Does Noetherian imply finite Krull dimension?

Exercise 29.90 (VI.29.E24: Boundary VI/30)**.** Name one global dimension theorem deliberately deferred to VI/30.

Exercise 29.91 (VI.29.E25: Boundary VI/31)**.** Which next chapter isolates codimension?

Exercise 29.92 (VI.29.E26: Boundary VI/32)**.** Which next chapter treats local dimension geometrically?

29.14 Challenges

Problem 29.93 (Challenge 1: Dimension filtration)**.** For a Noetherian affine scheme $X = \operatorname{Spec} A$, organize points by height and explain how the filtration by possible heights reflects specialization chains. Give the filtration explicitly for $\operatorname{Spec} k[x,y]$.

Hint. Separate generic points (height 0), codimension-one points (height 1), and closed points of height 2; justify each layer using prime chains.

Problem 29.94 (Challenge 2: Mixed-dimensional components)**.** Construct a finite-type k-algebra whose spectrum has one irreducible component of dimension 2 and another of dimension 1, and compute the ring dimension.

Hint. Use a product such as $k[x, y] \times k[t]$ or a connected reducible example if desired; dimension is the maximum component dimension.

Problem 29.95 (Challenge 3: Localization design)**.** Starting with $k[x, y, z]$, choose multiplicative sets producing localizations of dimensions 3, 2, 1, and 0.

Hint. Select opens retaining prime chains of the desired lengths; for dimension 0, pass to the fraction field.

Problem 29.96 (Challenge 4: Integral extension chain lifting)**.** Work through a concrete integral extension, for example $k[t^2, t^3] \subset k[t]$, and compare all prime-chain lengths explicitly.

Hint. Use contraction, lying over, going up, and the fact both rings are one-dimensional domains.

Problem 29.97 (Challenge 5: Infinite dimension)**.** Analyze the coordinate-prime chains in $k[x_1, x_2, \ldots]$ and explain precisely why every finite n occurs as a chain length even though no single displayed finite chain proves an infinite value by itself.

Hint. For each n, exhibit a separate length-n chain; then take the supremum over all n.

29.15 Chapter synthesis

Krull dimension records the depth of the specialization order. For affine schemes, prime chains and chains of irreducible closed subsets are the same data read in opposite directions. Quotients restrict to closed subsets, localizations restrict to opens, nilpotents leave the order unchanged, products take maxima, and integral extensions preserve the dimension. These mechanisms supply the algebraic engine for the global dimension theory of VI/30.

$$\begin{tikzcd}
\text{prime chains} \arrow[r, leftrightarrow] \arrow[d] & \text{irreducible closed chains} \arrow[d] \\
\text{quotients / closed} \arrow[r, leftrightarrow] & \text{localizations / open}
\end{tikzcd}$$

Figure 29.10: VI/29 organizes dimension through the specialization order and its affine operations.

29.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 29.98 (Field)**.** Compute $\dim k$.

Hint. Use chains of prime ideals and Krull dimension. ☐

Solution. 0. ☐

Exercise 29.99 (One variable)**.** Compute $\dim k[x]$.

Hint. Use chains of prime ideals and Krull dimension. ☐

Solution. 1. ☐

Exercise 29.100 (Two variables)**.** Compute $\dim k[x, y]$.

Hint. Use chains of prime ideals and Krull dimension. ☐

Solution. 2. ☐

Exercise 29.101 (Quotient line)**.** Compute $\dim k[x, y]/(x)$.

Hint. Use chains of prime ideals and Krull dimension. ☐

Solution. 1. ☐

Exercise 29.102 (Chain length)**.** What length has $(0) \subset (x) \subset (x, y)$?

Hint. Use chains of prime ideals and Krull dimension. ☐

Solution. 2. ☐

Proofs

Exercise 29.103 (Quotient chains)**.** Prove: Prove primes of A/I correspond to primes of A containing I, preserving chains.

Hint. Work directly from chains of prime ideals and Krull dimension. ☐

Solution. Use quotient extension-contraction. ☐

Exercise 29.104 (Localization chains)**.** Prove: Prove primes of $A_{\mathfrak{p}}$ correspond to primes of A contained in $\mathfrak{p}$.

Hint. Work directly from chains of prime ideals and Krull dimension. ☐

Solution. Use localization prime correspondence. ☐

Exercise 29.105 (Dimension lower bound)**.** Prove: Prove the displayed prime chain gives a lower bound for dimension.

Hint. Work directly from chains of prime ideals and Krull dimension. ☐

Solution. Krull dimension is the supremum of lengths of strict prime chains. ☐

Exercise 29.106 (Field dimension)**.** Prove: Prove a field has dimension zero.

Hint. Work directly from chains of prime ideals and Krull dimension. ☐

Solution. The zero ideal is its only prime, so no strict inclusion chain has positive length. ☐

Counterexamples and hypothesis tests

Exercise 29.107 (Number of generators)**.** Test the claim: dimension always equals the minimal number of ring generators.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 29.108 (Finite ring)**.** Test the claim: every finite ring has positive dimension.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; Artinian rings have dimension zero. ☐

Exercise 29.109 (Quotient larger)**.** Test the claim: quotienting can increase Krull dimension.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. Generally quotient prime chains are restricted to those containing the ideal, so dimension does not increase in the standard Noetherian setting. ☐

Applications and investigations

Exercise 29.110 (Geometric dimension)**.** Why do prime chains measure dimension?

Hint. Translate the question into chains of prime ideals and Krull dimension. ☐

Solution. They encode nested irreducible closed subsets. ☐

Exercise 29.111 (Local dimension)**.** How is local dimension at a prime related to localization?

Hint. Translate the question into chains of prime ideals and Krull dimension. ☐

Solution. It is measured by prime chains inside the local ring. ☐

Challenge problems

Exercise 29.112 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Krull Dimension in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Krull Dimension, translates it through chains of prime ideals and Krull dimension, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 29.113 (Hypothesis challenge)**.** Choose one structural statement from Krull Dimension, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 30

Dimension of Schemes

VI/29 built the affine engine: Krull dimension is the supremum of lengths of prime chains, equivalently chains of irreducible closed subsets of an affine scheme. VI/30 globalizes that invariant to schemes and isolates the extra hypotheses under which dimension behaves uniformly on open subsets. The central source theorem is the variety case: if X is an integral scheme of finite type over a field and $U \subseteq X$ is nonempty open, then $\dim U = \dim X$. This is false for general schemes, and even for some Noetherian integral schemes.

30.1 Global dimension

Definition 30.1 (Dimension of a scheme)**.** For a nonempty scheme X, define

$$\dim X = \sup\{n : Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \text{ are irreducible closed subsets of } X\}.$$

The empty scheme is assigned dimension $-\infty$ when a convention is needed.

For $X = \operatorname{Spec} A$, the affine chain dictionary from VI/29 gives

$$\dim X = \dim A.$$

Thus the global definition extends Krull dimension without changing the affine case.

$$Z_0 \xrightarrow{\text{strict}} Z_1 \longrightarrow \cdots \xrightarrow{\text{strict}} Z_n$$

Figure 30.1: Global dimension counts strict chains of irreducible closed subsets.

30.2 Closed subsets, components, and reduction

If $Y \subseteq X$ is closed, every irreducible closed chain in Y is one in X, hence

Proposition 30.2 (Closed subsets do not increase dimension)**.**

$$\dim Y \leq \dim X.$$

If X is Noetherian with irreducible components $X_1, \ldots, X_r$, every irreducible closed subset is contained in some X_i, so

Proposition 30.3 (Dimension from irreducible components)**.**

$$\dim X = \max_i \dim X_i.$$

The reduction $X_{\mathrm{red}} \to X$ is a homeomorphism, so

Proposition 30.4 (Nilpotents do not change global dimension)**.**

$$\dim X_{\mathrm{red}} = \dim X.$$

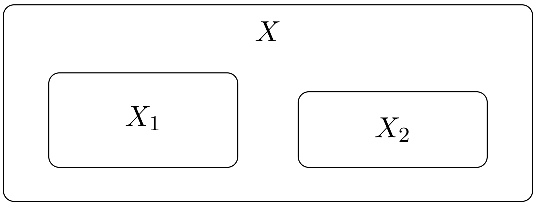


Figure 30.2: For a Noetherian scheme, global dimension is the maximum of the component dimensions.

Example 30.5 (Affine space)**.** The scheme $\mathbb{A}^n_k$ has dimension n because its coordinate ring $k[x_1, \ldots, x_n]$ has Krull dimension n.

30.3 Open subsets: the general inequality

For an open subset $U \subseteq X$, the closure in X of an irreducible closed subset of U is irreducible, and a strict chain in U gives a strict chain of closures in X. Therefore

Proposition 30.6 (Open subsets cannot increase dimension)**.**

$$\dim U \leq \dim X.$$

Equality need not hold. The exact AG8 Problem 3 counterexample is

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k, \qquad U = \operatorname{Spec} k.$$

Then U is nonempty and open, but $\dim U = 0$ whereas $\dim X = 1$.

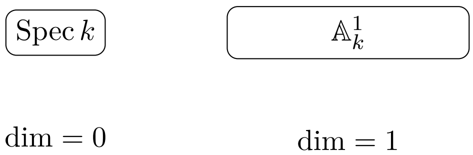


Figure 30.3: AG8 Problem 3(d): a nonempty open component can have smaller dimension than the whole scheme.

30.4 Why irreducibility alone is not enough

Even Noetherian and integral do not force every nonempty open to have full dimension. Let

$$A = k[x, y]_{(x)}.$$

Since y is inverted, $A \cong k(y)[x]_{(x)}$ and $\dim A = 1$. But

$$D(x) \subseteq \operatorname{Spec} A$$

contains only the generic point, so $\dim D(x) = 0$. The missing hypothesis in the source theorem is the finite-type variety setting, not merely irreducibility or Noetherianity.

$$\begin{tikzcd} \boxed{(0)} \arrow[r, "\text{specializes}"] & \boxed{(x)} \end{tikzcd}$$

$$D(x) \qquad \operatorname{Spec} k[x,y]_{(x)}$$

Figure 30.4: An integral Noetherian scheme can have a nonempty open subset of smaller dimension.

30.5 Varieties and nonempty open subsets

Here a variety over k means an integral scheme of finite type over the field k. If $V = \operatorname{Spec} A$ is a nonempty affine open of a variety X, then A is a finitely generated k-domain and its fraction field is $K(X)$. The finite-type dimension theorem gives

$$\dim A = \operatorname{trdeg}_k \operatorname{Frac}(A) = \operatorname{trdeg}_k K(X).$$

Consequently every nonempty affine open has the same dimension. Every nonempty open $U \subseteq X$ contains a nonempty affine open V, so

$$\dim X \geq \dim U \geq \dim V = \dim X.$$

Thus:

Theorem 30.7 (Nonempty-open invariance for varieties). *If X is a variety over k and $\varnothing \neq U \subseteq X$ is open, then*

$$\dim U = \dim X.$$

This is the VI/30 part of AG8 Problem 2.

$$\begin{tikzcd} \boxed{V = \operatorname{Spec} A \subseteq U} \arrow[r, "\dim = \operatorname{trdeg}_k K(X)"] & \boxed{U \subseteq X} \end{tikzcd}$$

Figure 30.5: A nonempty affine chart inside an open subset of a variety forces full dimension.

Example 30.8 (Hypersurface dimension). The plane curve $V(y - x^2)$ has coordinate ring $k[x]$, so its scheme dimension is 1 even though it sits inside the two-dimensional affine plane.

30.6 Dimension and the function field

VI/27 established that every nonempty affine chart of an integral scheme has the same fraction field $K(X)$. In the finite-type setting, combining that fact with the affine dimension theorem yields:

Theorem 30.9 (Dimension–transcendence-degree theorem). *For a variety X over k,*

$$\dim X = \operatorname{trdeg}_k K(X).$$

The function-field material itself remains canonically assigned to VI/27. This theorem is its dimension-side consequence and is not a second destination for that legacy fragment.

$$\begin{tikzcd} \boxed{\operatorname{Spec} A} \arrow[r, "\dim A"] & \boxed{K(X) = \operatorname{Frac}(A)} \\ & \operatorname{trdeg}_k K(X) \end{tikzcd}$$

Figure 30.6: For a variety, affine dimension is the transcendence degree of the common function field.

30.7 Basic computations

Since $\dim k[x_1, \dots, x_n] = n$,

$$\dim \mathbb{A}^n_k = n.$$

More generally, if A is a finitely generated k-domain, then

$$\dim \operatorname{Spec} A = \operatorname{trdeg}_k \operatorname{Frac}(A).$$

For a finite disjoint union,

$$\dim(X_1 \sqcup \cdots \sqcup X_r) = \max_i \dim X_i,$$

and for an arbitrary disjoint union the maximum is replaced by the supremum.

$$\begin{tikzcd} \boxed{\mathbb{A}^0} \arrow[r] & \boxed{\mathbb{A}^1} \arrow[r] & \boxed{\mathbb{A}^2} \arrow[r, dashed] & \boxed{\mathbb{A}^n} \end{tikzcd}$$

Figure 30.7: Affine n-space has dimension n.

30.8 Products of varieties

For varieties X, Y over k, every irreducible component of $X \times_k Y$ has the expected dimension; in the standard geometrically integral situation the product is itself a variety and

$$\dim(X \times_k Y) = \dim X + \dim Y.$$

For affine integral charts this is the transcendence-degree computation for the relevant tensor-product component. One must not suppress hypotheses: tensor products of domains over a non-algebraically-closed field need not remain domains.

$$\begin{tikzcd} X & & Y \\ & X \times_k Y \arrow[ul] \arrow[ur] & \end{tikzcd}$$

Figure 30.8: Under suitable geometric-integrality hypotheses, dimensions add in products of varieties.

Example 30.10 (Reducible dimension)**.** For $X = V(xy) \subset \mathbb{A}^2$, both irreducible components are lines, so X has dimension 1. Scheme dimension takes the supremum over all irreducible chains.

30.9 Dimension under integral and finite morphisms

VI/29 proved that integral extensions preserve Krull dimension. Hence an integral surjective affine morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ has source and target of the same dimension. In particular normalization preserves dimension. For finite surjective morphisms between Noetherian schemes, this affine statement globalizes componentwise:

$$\dim X = \dim Y.$$

This is a dimension consequence, not a reopening of the integral-extension or normalization chapters.

$$\begin{tikzcd} \boxed{X^\nu} \arrow[r, "\text{integral}", "\dim X^\nu = \dim X"'] & \boxed{X} \end{tikzcd}$$

Figure 30.9: Normalization preserves dimension because it is affine-locally an integral extension.

30.10 Boundaries of VI/30

Global dimension must be kept distinct from two later invariants. VI/31 studies codimension and the formula $\dim Y + \operatorname{codim}(Y, X) = \dim X$ under suitable hypotheses. VI/32 studies local dimension $\dim \mathcal{O}_{X,x}$ and tangent-space geometry. Accordingly AG8 Problem 2(a)–(c) and the corresponding Problem 3 counterexamples are not consumed here. VI/30 owns the global/open-dimension statement and the dimension-side transcendence-degree theorem.

30.11 Examples

Example 30.11 (A point)**.** $\operatorname{Spec} k$ has dimension 0: there is one prime and no strict prime inclusion.

Example 30.12 (The affine line)**.** $\mathbb{A}^1_k = \operatorname{Spec} k[t]$ has dimension 1 from $(0) \subsetneq (t - a)$, and no longer prime chain exists.

Example 30.13 (The affine plane)**.** $\mathbb{A}^2_k$ has dimension 2, witnessed by $(0) \subsetneq (x) \subsetneq (x, y)$.

Example 30.14 (Reduction)**.** $\operatorname{Spec} k[t]/(t^2)$ and $\operatorname{Spec} k[t]/(t)$ have the same one-point topology and both have dimension 0.

Example 30.15 (A closed line in the plane)**.** $V(x) \subset \mathbb{A}^2_k$ is isomorphic to $\mathbb{A}^1_k$, so its dimension drops from 2 to 1.

Example 30.16 (A principal open of the affine plane)**.** $D(x) \subset \mathbb{A}^2_k$ has coordinate ring $k[x, x^{-1}, y]$ and dimension 2; this agrees with nonempty-open invariance for varieties.

Example 30.17 (Two components of different dimensions)**.** $\operatorname{Spec} k \sqcup \mathbb{A}^1_k$ has dimension 1, the maximum of component dimensions.

Example 30.18 (Open component of smaller dimension)**.** In $\operatorname{Spec} k \sqcup \mathbb{A}^1_k$, the component $\operatorname{Spec} k$ is nonempty open of dimension 0, strictly below the ambient dimension 1.

Example 30.19 (Affine three-space)**.** $\dim \mathbb{A}^3_k = 3$ by the chain $(0) \subsetneq (x) \subsetneq (x, y) \subsetneq (x, y, z)$.

Example 30.20 (An affine curve)**.** If $A = k[x, y]/(f)$ is a domain with nonconstant irreducible f, then $\mathrm{Frac}(A)$ has transcendence degree 1, hence $\dim \operatorname{Spec} A = 1$.

Example 30.21 (An affine surface)**.** If $A = k[x, y, z]/(f)$ is a finitely generated domain with one nontrivial algebraic relation of the expected height, then $\operatorname{trdeg}_k \mathrm{Frac}(A) = 2$ and $\dim \operatorname{Spec} A = 2$.

Example 30.22 (Normalization)**.** The cusp $\operatorname{Spec} k[t^2, t^3]$ and its normalization $\operatorname{Spec} k[t]$ both have dimension 1.

Example 30.23 (Finite field extension)**.** For a finite extension L/k, $\operatorname{Spec} L \to \operatorname{Spec} k$ is finite and both schemes have dimension 0.

Example 30.24 (Disjoint union of affine spaces)**.** $\coprod_{n\geq 0} \mathbb{A}^n_k$ has infinite dimension because the component dimensions are unbounded.

Example 30.25 (Localization can lower dimension)**.** For $A = k[x, y]_{(x)}$, $\dim \operatorname{Spec} A = 1$ but $D(x)$ has only the generic point and dimension 0.

Example 30.26 (Dense open in a variety)**.** If X is a variety and $U \subseteq X$ is nonempty open, then U is dense and $\dim U = \dim X$.

Example 30.27 (Function field of affine space)**.** $K(\mathbb{A}^n_k) = k(x_1, \ldots, x_n)$ has transcendence degree n, recovering $\dim \mathbb{A}^n_k = n$.

Example 30.28 (Product with the affine line)**.** For a geometrically integral variety X, $\dim(X \times_k \mathbb{A}^1_k) = \dim X + 1$.

Example 30.29 (A zero-dimensional finite scheme)**.** Any finite k-scheme has dimension 0, even if it is nonreduced.

Example 30.30 (Closed subset inequality)**.** Every closed subscheme $Y \hookrightarrow X$ satisfies $\dim Y \leq \dim X$ because its underlying irreducible-closed chains are chains in X.

Example 30.31 (Advanced 1: Noetherian integral does not suffice)**.** The integral Noetherian scheme $X = \operatorname{Spec} k[x, y]_{(x)}$ has dimension 1, while $D(x)$ has dimension 0. Thus the AG8 open-dimension theorem needs its variety hypothesis.

Example 30.32 (Advanced 2: Non-equidimensional reducible scheme)**.** For $X = \mathbb{A}^2_k \sqcup \mathbb{A}^1_k$, global dimension is 2, yet an open-and-closed component has dimension 1.

Example 30.33 (Advanced 3: Infinite-dimensional scheme)**.** $\operatorname{Spec} k[x_1, x_2, \ldots]$ has infinite dimension through chains $(0) \subsetneq (x_1) \subsetneq \cdots \subsetneq (x_1, \ldots, x_n)$ of arbitrary length.

Example 30.34 (Advanced 4: Arbitrary disjoint union)**.** For $X = \coprod_i X_i$, every irreducible subset lies in one component, so $\dim X = \sup_i \dim X_i$.

Example 30.35 (Advanced 5: Dominant morphisms of varieties)**.** A dominant morphism $f : X \to Y$ of varieties induces $K(Y) \hookrightarrow K(X)$ and therefore $\dim X \geq \dim Y$.

Example 30.36 (Advanced 6: Generically finite dominant morphism)**.** If $K(X)$ is algebraic over $K(Y)$ for a dominant morphism of varieties, then the transcendence degrees agree and $\dim X = \dim Y$.

Example 30.37 (Advanced 7: Finite surjective morphism)**.** A finite surjective morphism of Noetherian schemes is integral and preserves dimension componentwise; in the affine case this is the integral-extension theorem.

Example 30.38 (Advanced 8: Tensor-product warning)**.** The formula $\dim(X \times_k Y) = \dim X + \dim Y$ requires care when $X \times_k Y$ is reducible; the clean function-field proof applies componentwise under suitable geometric-integrality hypotheses.

Example 30.39 (Advanced 9: A dense open can lose dimension outside finite type)**.** In $X = \operatorname{Spec} k[x,y]_{(x)}$, the principal open $D(x)$ is dense because X is integral, but its dimension is $0 < 1$. Density alone does not imply dimension equality.

Example 30.40 (Advanced 10: Nilpotent thickening)**.** If $X' \to X$ is a nilpotent thickening, the underlying spaces coincide, hence $\dim X' = \dim X$ even though tangent spaces and local rings may change.

Example 30.41 (Advanced 11: Component formula in Noetherian schemes)**.** Since a Noetherian scheme has finitely many irreducible components and every irreducible closed subset lies in one, global dimension is the maximum of component dimensions.

Example 30.42 (Advanced 12: Affine charts need not all have ambient dimension)**.** For general schemes an affine open chart can have smaller dimension than X; the equality of all nonempty affine charts is a special strength of varieties.

Example 30.43 (Advanced 13: Normalization of a curve)**.** For an integral finite-type curve, normalization is finite in the usual excellent setting and leaves dimension 1 unchanged while correcting nonnormal local rings.

Example 30.44 (Advanced 14: Transcendence basis viewpoint)**.** If X is a variety and $t_1, \ldots, t_d$ is a transcendence basis of $K(X)/k$, then $d = \dim X$. The integer d is independent of the chosen basis and of the chosen affine chart.

30.12 Problem dossiers

Problem 30.45 (VI.30.P1: AG8 Problem 2: nonempty opens of a variety)**.** Let X be a variety over k and let $\varnothing \neq U \subseteq X$ be open. Prove that $\dim U = \dim X$.

Provenance. Legacy: AG8-P2d, the global/open-dimension portion of Problem 2. The function-field setup is already canonical in VI/27; local dimension and codimension remain for VI/32 and VI/31.

Problem 30.46 (VI.30.P2: AG8 Problem 3(d): failure for a general scheme)**.** Set $X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k$ and $U = \operatorname{Spec} k \subseteq X$. Show that U is nonempty open but $\dim U \neq \dim X$.

Provenance. Legacy: AG8-P3d, the part-(d) counterexample split from AG8 Problem 3. Parts (a) and (c) remain reserved for VI/32 and VI/31.

Problem 30.47 (VI.30.P3: Affine compatibility)**.** Prove directly that the global topological dimension of $\operatorname{Spec} A$ equals the Krull dimension of A.

Provenance. Canonical VI/30 development from the VI/29 affine chain dictionary.

Problem 30.48 (VI.30.P4: Affine space)**.** Compute $\dim \mathbb{A}^n_k$.

Provenance. Canonical computation.

Problem 30.49 (VI.30.P5: Closed subset inequality)**.** If $Y \subseteq X$ is closed, prove $\dim Y \leq \dim X$.

Provenance. Canonical global-dimension lemma.

Problem 30.50 (VI.30.P6: Finite disjoint unions)**.** Prove $\dim(X \sqcup Y) = \max\{\dim X, \dim Y\}$.

Provenance. Canonical component computation.

Problem 30.51 (VI.30.P7: Noetherian irreducible components)**.** Let X be Noetherian with irreducible components $X_1, \dots, X_r$. Prove $\dim X = \max_i \dim X_i$.

Provenance. Canonical Noetherian component theorem.

Problem 30.52 (VI.30.P8: Reduction invariance)**.** Prove $\dim X_{\text{red}} = \dim X$.

Provenance. Canonical consequence of VI/10 and VI/29.

Problem 30.53 (VI.30.P9: Normalization invariance)**.** Let X be an integral scheme and $X^\nu \to X$ its normalization. Under the standard affine normalization construction, explain why dimension is unchanged.

Provenance. Canonical bridge from VI/28–VI/29.

Problem 30.54 (VI.30.P10: Open subset inequality)**.** Prove that every open $U \subseteq X$ satisfies $\dim U \leq \dim X$.

Provenance. Canonical global theorem.

Problem 30.55 (VI.30.P11: An integral Noetherian counterexample)**.** Let $A = k[x, y]_{(x)}$. Show $\dim \operatorname{Spec} A = 1$ but $\dim D(x) = 0$.

Provenance. Canonical warning: finite type matters.

Problem 30.56 (VI.30.P12: Dimension and transcendence degree)**.** Let X be a variety over k. Prove $\dim X = \operatorname{trdeg}_k K(X)$ without assigning a second legacy destination to VI/27's function-field problem.

Provenance. Canonical dimension-side completion using VI/27 data.

Problem 30.57 (VI.30.P13: Dominant morphism inequality)**.** If $f : X \to Y$ is dominant between varieties over k, prove $\dim X \geq \dim Y$.

Provenance. Canonical morphism consequence.

Problem 30.58 (VI.30.P14: Generically finite morphisms)**.** Under the hypotheses of Problem 13, assume $K(X)$ is algebraic over $K(Y)$. Prove $\dim X = \dim Y$.

Provenance. Canonical function-field application.

Problem 30.59 (VI.30.P15: Arbitrary disjoint unions)**.** Prove $\dim(\coprod_i X_i) = \sup_i \dim X_i$.

Provenance. Canonical non-Noetherian extension.

Problem 30.60 (VI.30.P16: An infinite-dimensional scheme)**.** Show $X = \coprod_{n \geq 0} \mathbb{A}^n_k$ has infinite dimension.

Provenance. Canonical example.

Problem 30.61 (VI.30.P17: Finite morphisms)**.** Let $f : X \to Y$ be finite and surjective between affine schemes. Explain why $\dim X = \dim Y$.

Provenance. Canonical bridge to finite morphisms.

Problem 30.62 (VI.30.P18: A finite union of closed subsets)**.** Suppose $X = Y_1 \cup \cdots \cup Y_r$ with each Y_i closed. Prove $\dim X = \max_i \dim Y_i$ if the Y_i contain all irreducible components of X.

Provenance. Canonical component argument.

Problem 30.63 (VI.30.P19: Product with affine space)**.** Let X be a geometrically integral variety over k. Show $\dim(X \times_k \mathbb{A}^m_k) = \dim X + m$.

Provenance. Canonical product computation under safe hypotheses.

Problem 30.64 (VI.30.P20: Hypothesis audit)**.** For each statement below, decide whether it holds for every scheme: (i) $\dim U \leq \dim X$ for open U; (ii) $\dim U = \dim X$ for every nonempty open U; (iii) $\dim X = \dim X_{\text{red}}$; (iv) $\dim X = \operatorname{trdeg}_k K(X)$.

Provenance. Canonical synthesis dossier.

30.13 Exercises

Exercise 30.65 (VI.30.E1)**.** Show $\dim \operatorname{Spec} k = 0$.

Exercise 30.66 (VI.30.E2)**.** Write a length-two prime chain in $k[x, y]$.

Exercise 30.67 (VI.30.E3)**.** Compute $\dim \mathbb{A}^4_k$.

Exercise 30.68 (VI.30.E4)**.** If $Y \subseteq X$ is closed and $\dim X = 2$, what can be said about $\dim Y$?

Exercise 30.69 (VI.30.E5)**.** Prove a nilpotent thickening does not change dimension.

Exercise 30.70 (VI.30.E6)**.** Compute the dimension of $\mathbb{A}^2_k \sqcup \mathbb{A}^5_k$.

Exercise 30.71 (VI.30.E7)**.** Give a nonempty open subset of $\operatorname{Spec} k[t]$ with dimension 1.

Exercise 30.72 (VI.30.E8)**.** Explain why $D(x)$ in $\operatorname{Spec} k[x, y]_{(x)}$ has only one point.

Exercise 30.73 (VI.30.E9)**.** Let X be a variety with $\dim X = 3$. What is the dimension of every nonempty open subset?

Exercise 30.74 (VI.30.E10)**.** If $K(X) = k(s, t)$ for a variety X, compute $\dim X$.

Exercise 30.75 (VI.30.E11)**.** If $K(X)$ is algebraic over k, compute the dimension of the variety X.

Exercise 30.76 (VI.30.E12)**.** Prove a dominant morphism from a curve to a surface cannot exist between varieties.

Exercise 30.77 (VI.30.E13)**.** If $f : X \to Y$ is dominant and generically finite between varieties and $\dim Y = 4$, compute $\dim X$.

Exercise 30.78 (VI.30.E14)**.** Compute $\dim(\coprod_{n=0}^{10} \mathbb{A}^n_k)$.

Exercise 30.79 (VI.30.E15)**.** Compute $\dim(\coprod_{n\geq 0} \operatorname{Spec} k)$.

Exercise 30.80 (VI.30.E16)**.** Can a dense open subset of an integral Noetherian scheme have smaller dimension?

Exercise 30.81 (VI.30.E17)**.** Let X be a variety of dimension d. Compute $\dim(X \times_k \mathbb{A}^1_k)$ under geometric-integrality hypotheses.

Exercise 30.82 (VI.30.E18)**.** Why does $\dim X = \dim X_{\text{red}}$ not imply the schemes are isomorphic?

Exercise 30.83 (VI.30.E19)**.** If $X^\nu \to X$ is normalization of an integral curve, compare dimensions.

Exercise 30.84 (VI.30.E20)**.** For an affine finite morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ with B integral over A, compare dimensions.

Exercise 30.85 (VI.30.E21)**.** Show that a closed point is a zero-dimensional irreducible closed subset.

Exercise 30.86 (VI.30.E22)**.** Does $\dim Y < \dim X$ hold for every proper closed subset $Y \subsetneq X$?

Exercise 30.87 (VI.30.E23)**.** Does $\dim U < \dim X$ hold for every proper open subset?

Exercise 30.88 (VI.30.E24)**.** Find a scheme whose dimension is the supremum ∞ of finite component dimensions.

Exercise 30.89 (VI.30.E25)**.** State the distinction between global dimension and local dimension at x.

Exercise 30.90 (VI.30.E26)**.** State the distinction between dimension and codimension.

30.14 Challenges

Problem 30.91 (Challenge VI.30.C1: Classify open-dimension equality)**.** Investigate hypotheses on a Noetherian scheme X ensuring $\dim U = \dim X$ for every nonempty open U. Compare irreducibility, equidimensionality, Jacobson hypotheses, and finite type over a field. Produce examples showing that each naive weakening can fail.

Problem 30.92 (Challenge VI.30.C2: Dimension through a dominant map)**.** Let $f : X \to Y$ be a dominant morphism of varieties. Relate $\dim X - \dim Y$ to $\operatorname{trdeg}_{K(Y)} K(X)$ and interpret the result as the generic relative dimension.

Problem 30.93 (Challenge VI.30.C3: Dimension in an infinite union)**.** Construct a scheme with infinitely many irreducible components whose dimensions are unbounded, prove that its global dimension is infinite, and compare this with the dimensions of quasi-compact open subsets.

Problem 30.94 (Challenge VI.30.C4: Products over nonclosed fields)**.** Study when the product of two integral k-schemes fails to be integral. Determine how the dimension-additivity statement should be reformulated componentwise when $X \times_k Y$ is reducible.

Problem 30.95 (Challenge VI.30.C5: Build a hypothesis map)**.** Create a theorem/counterexample table for the statements: open subsets preserve dimension; proper closed subsets lower dimension; normalization preserves dimension; finite surjective maps preserve dimension; dominant maps do not lower dimension. State the weakest hypotheses you can justify for each.

30.15 Chapter synthesis

Scheme dimension is a topological chain invariant that agrees with Krull dimension on affine charts, but its global behavior is sensitive to components and to the class of schemes under consideration. Closed and open subsets can only lower dimension. In the finite-type integral setting over a field, every nonempty open has full dimension and that common value is the transcendence degree of the function field. The failures outside this setting are not pathologies to hide: they explain exactly why the hypotheses in the variety theorem matter.

$$\begin{tikzcd}
\boxed{\text{Krull chains}} \arrow[r] & \boxed{\text{global dimension}} \arrow[r, "\text{VI/31}"] \arrow[d, "\text{VI/32}"] & \boxed{\text{codimension}} \\
& \boxed{\text{local dimension}} &
\end{tikzcd}$$

Figure 30.10: VI/30 globalizes VI/29 and prepares, but does not consume, codimension and local dimension.

30.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 30.96 (Affine n-space)**.** Compute $\dim \mathbb{A}^n_k$.

Hint. Use scheme dimension and affine-local prime chains. ☐

Solution. n. ☐

Exercise 30.97 (Parabola)**.** Compute the dimension of $V(y - x^2)$.

Hint. Use scheme dimension and affine-local prime chains. ☐

Solution. 1. ☐

Exercise 30.98 (Point)**.** Compute the dimension of $\operatorname{Spec} k$.

Hint. Use scheme dimension and affine-local prime chains. ☐

Solution. 0. ☐

Exercise 30.99 (Axes union)**.** Compute the dimension of $V(xy) \subset \mathbb{A}^2$.

Hint. Use scheme dimension and affine-local prime chains. ☐

Solution. 1. ☐

Exercise 30.100 (Affine chart)**.** How is dimension computed on an affine scheme?

Hint. Use scheme dimension and affine-local prime chains. ☐

Solution. As the Krull dimension of its coordinate ring. ☐

Proofs

Exercise 30.101 (Affine agreement)**.** Prove: Prove topological scheme dimension of $\operatorname{Spec} A$ agrees with Krull dimension of A.

Hint. Work directly from scheme dimension and affine-local prime chains. ☐

Solution. Irreducible closed subsets are closures of prime points, and strict inclusion corresponds to strict reverse inclusion of prime ideals. ☐

Exercise 30.102 (Open bound)**.** Prove: Prove the dimension of an open subscheme is at most the dimension of the ambient scheme.

Hint. Work directly from scheme dimension and affine-local prime chains. ☐

Solution. Any chain of irreducible closed subsets in the open yields a corresponding chain in their ambient closures. ☐

Exercise 30.103 (Component supremum)**.** Prove: Prove dimension of a Noetherian scheme is the supremum of dimensions of irreducible components.

Hint. Work directly from scheme dimension and affine-local prime chains. ☐

Solution. Every irreducible chain lies inside some component, while component chains are ambient chains. ☐

Exercise 30.104 (Parabola)**.** Prove: Prove the parabola has dimension one via its coordinate-ring isomorphism with $k[x]$.

Hint. Work directly from scheme dimension and affine-local prime chains. ☐

Solution. Scheme isomorphisms preserve prime chains and dimension. ☐

Counterexamples and hypothesis tests

Exercise 30.105 (Ambient dimension)**.** Test the claim: every closed subscheme of $\mathbb{A}^n$ has dimension n.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False. ☐

Exercise 30.106 (Point positive)**.** Test the claim: a closed point has dimension one.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it has dimension zero. ☐

Exercise 30.107 (Components add)**.** Test the claim: dimension of a finite union is the sum of component dimensions.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; it is the maximum in the Noetherian irreducible-component setting. ☐

Applications and investigations

Exercise 30.108 (Expected dimension)**.** Why compare equations with dimension?

Hint. Translate the question into scheme dimension and affine-local prime chains. ☐

Solution. Independent equations often lower dimension and reveal whether an intersection has expected size. ☐

Exercise 30.109 (Local models)**.** Why is affine-local dimension useful?

Hint. Translate the question into scheme dimension and affine-local prime chains. ☐

Solution. It reduces geometric dimension questions to prime-chain algebra in coordinate rings. ☐

Challenge problems

Exercise 30.110 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Dimension of Schemes in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution identifies the concrete model from Dimension of Schemes, translates it through scheme dimension and affine-local prime chains, and checks the correspondence in both algebraic and geometric language. ☐

Exercise 30.111 (Hypothesis challenge)**.** Choose one structural statement from Dimension of Schemes, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. ☐

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. ☐

Chapter 31

Codimension

VI/29 measured the absolute length of prime chains and VI/30 globalized dimension to schemes. Codimension measures the same chains *relative to a chosen closed locus*. For an irreducible closed subset $Y \subseteq X$, it records how many strict irreducible specializations can occur between an ambient component and Y. On an affine scheme this is height, and at the generic point of Y it is the dimension of the corresponding local ring. For varieties over a field, codimension and dimension fit the clean formula

$$\dim Y + \operatorname{codim}(Y, X) = \dim X.$$

The hypotheses matter: AG8 Problem 3 asks for a general scheme where this additivity fails, and that exact counterexample is preserved here.

Learning goals

By the end of the chapter, the reader should be able to:

- compute codimension from chains of irreducible closed subsets;
- translate codimension on an affine scheme into the height of a prime ideal;
- identify codimension with the Krull dimension of the local ring at the generic point;
- use the dimension–codimension formula correctly for varieties and recognize why it can fail for general schemes;
- distinguish codimension from the number of equations or generators used to present a closed locus;
- recognize codimension-one geometry as the numerical precursor of divisor theory.

Conceptual roadmap

The chapter moves through one dictionary in three forms:

$$\text{prime chains below } \mathfrak{p} \quad \longleftrightarrow \quad \text{irreducible closed chains above } V(\mathfrak{p}) \quad \longleftrightarrow \quad \dim A_{\mathfrak{p}}.$$

The variety case then adds a fourth term, $\dim(A/\mathfrak{p})$, producing the familiar decomposition of ambient dimension into dimension along the subvariety plus codimension transverse to it.

31.1 Relative chains

Definition 31.1 (Codimension of an irreducible closed subset)**.** Let $Y \subseteq X$ be irreducible and closed. Define

$$\operatorname{codim}(Y, X) = \sup\{n : Y = Z_n \subsetneq Z_{n-1} \subsetneq \cdots \subsetneq Z_0,\ Z_i \text{ irreducible closed in } X\}.$$

Thus codimension zero means that Y is an irreducible component of X.

For a closed subset $W \subseteq X$ with irreducible components W_α, we use

Definition 31.2 (Codimension of a closed subset)**.**

$$\operatorname{codim}(W, X) = \inf_\alpha \operatorname{codim}(W_\alpha, X).$$

This convention records the component of W lying closest to an ambient component.

$$\begin{tikzcd} X \\ Z_1 \arrow[u] \\ Z_2 \arrow[u] \\ Y \arrow[u] \end{tikzcd}$$

Figure 31.1: Codimension counts strict irreducible closed inclusions upward from Y.

31.2 Affine translation: codimension is height

Let $X = \operatorname{Spec} A$ and let $Y = V(\mathfrak{p})$ for a prime $\mathfrak{p}$. Reversing prime inclusions gives

$$\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{p} \quad \Longleftrightarrow \quad V(\mathfrak{p}) = V(\mathfrak{p}_n) \subsetneq \cdots \subsetneq V(\mathfrak{p}_0).$$

Hence

Proposition 31.3 (Affine codimension–height dictionary)**.**

$$\operatorname{codim}(V(\mathfrak{p}), \operatorname{Spec} A) = \operatorname{ht}(\mathfrak{p}).$$

This is the relative counterpart of $\dim \operatorname{Spec} A = \dim A$.

$$\begin{array}{cccc} & \mathfrak{p}_0 & \subsetneq \mathfrak{p}_1 & \subsetneq \mathfrak{p} \\ & & & \\ \subsetneq V(\mathfrak{p}_0) & \subsetneq V(\mathfrak{p}_1) & V(\mathfrak{p}) & \end{array}$$

Figure 31.2: Prime inclusion reverses to inclusion of irreducible closed subsets.

Example 31.4 (A line in the plane)**.** The closed irreducible subset $V(x) \subset \mathbb{A}^2_k$ has dimension 1 inside a 2-dimensional irreducible ambient space, so its codimension is 1.

31.3 The generic-point local ring

Every irreducible closed subset Y of a scheme has a unique generic point η_Y. Choose an affine neighborhood $U = \operatorname{Spec} A$ of η_Y and let $\mathfrak{p}$ be the corresponding prime. Any generalization of η_Y also lies in U, so every chain relevant to the codimension can be read inside this one affine neighborhood. Therefore

Theorem 31.5 (Codimension at the generic point)**.** *For every irreducible closed $Y \subseteq X$,*

$$\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,\eta_Y}.$$

Indeed $\mathcal{O}_{X,\eta_Y} \cong A_{\mathfrak{p}}$ and $\dim A_{\mathfrak{p}} = \operatorname{ht}(\mathfrak{p})$.

$$\boxed{Y = \overline{\{\eta_Y\}}} \xrightarrow{\text{generic point}} \boxed{\mathcal{O}_{X,\eta_Y}}$$

$$\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,\eta_Y}$$

Figure 31.3: The generic stalk converts relative geometry into local Krull dimension.

31.4 The infimum formula

If $y \in Y$, then y is a specialization of η_Y. In an affine neighborhood of y, the prime of η_Y is contained in the prime of y, so every chain below the former is also a chain below the latter. Thus

$$\dim \mathcal{O}_{X,y} \geq \dim \mathcal{O}_{X,\eta_Y}.$$

Since equality occurs at $y = \eta_Y$, we obtain the codimension statement singled out in AG8 Problem 2:

Theorem 31.6 (Local-ring characterization of codimension)**.** *For irreducible closed $Y \subseteq X$,*

$$\operatorname{codim}(Y, X) = \inf_{y \in Y} \dim \mathcal{O}_{X,y}.$$

For a general closed W, the same formula holds with the convention of theorem 31.2.

VI/32 will study $\dim \mathcal{O}_{X,x}$ as a local invariant in its own right; here it appears only as the exact bridge to codimension.

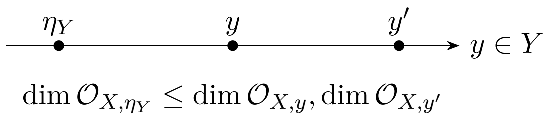


Figure 31.4: Local dimensions along Y are minimized at the generic point.

31.5 Varieties: dimension plus codimension

Let X be a variety over k and $Y \subseteq X$ an irreducible closed subvariety. Choose a nonempty affine open $U = \operatorname{Spec} A$ meeting Y, with $Y \cap U = V(\mathfrak{p})$. The finitely generated k-domain A satisfies the dimension formula

$$\operatorname{ht}(\mathfrak{p}) + \dim(A/\mathfrak{p}) = \dim A.$$

VI/30 identifies the dimensions of the nonempty affine opens with the dimensions of the varieties. Hence

Theorem 31.7 (Dimension–codimension additivity for varieties)**.** *If X is a variety over k and $Y \subseteq X$ is irreducible closed, then*

$$\boxed{\dim Y + \operatorname{codim}(Y, X) = \dim X.}$$

For a reducible closed subset W of a variety, the convention using the smallest component codimension gives the same identity $\dim W + \operatorname{codim}(W, X) = \dim X$.

$$\begin{tikzcd} \boxed{X} \arrow[r] & \dim X = \dim Y + \operatorname{codim}(Y, X) \\ \boxed{Y} \arrow[u] & \end{tikzcd}$$

Figure 31.5: For varieties, ambient dimension splits into subvariety dimension plus codimension.

Example 31.8 (A point in the plane)**.** A closed point (a, b) in $\mathbb{A}^2_k$ has dimension 0, hence codimension 2. Algebraically the corresponding maximal ideal sits at the end of a prime chain of length two.

31.6 The AG8 counterexample: additivity can fail

Set

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k, \qquad Y = \operatorname{Spec} k \subseteq X.$$

Then Y is an irreducible component, so $\operatorname{codim}(Y, X) = 0$. Also $\dim Y = 0$ while $\dim X = 1$. Thus

$$\dim Y + \operatorname{codim}(Y, X) = 0 \neq 1 = \dim X.$$

This is precisely the part-(c) counterexample requested by AG8 Problem 3. It shows that the variety theorem is not a formal identity for arbitrary schemes; non-equidimensional components already destroy it.

$$\boxed{\operatorname{Spec} k} \qquad \boxed{\mathbb{A}^1_k}$$

$$\dim X = 1, \text{ but } \dim Y = \operatorname{codim}(Y, X) = 0$$

Figure 31.6: AG8 Problem 3(c): a low-dimensional component breaks global additivity.

31.7 Codimension one

Codimension-one irreducible closed subsets are especially important because they become prime divisors in the later divisor chapters. In $\mathbb{A}^n_k$, the coordinate hyperplane $V(x_1)$ has codimension one. More generally, if A is a Noetherian domain and $0 \neq f \in A$ is a nonunit, every prime minimal over (f) has height one by the principal ideal theorem. Thus every irreducible component of a genuine principal hypersurface has codimension one. This chapter uses the fact only as a codimension computation; divisor theory remains reserved for the later divisor chapters.

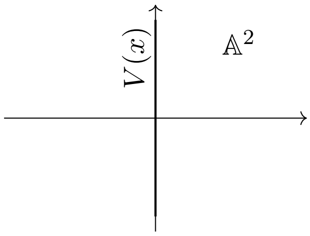


Figure 31.7: A genuine irreducible hypersurface in affine space has codimension one.

31.8 Nested subvarieties and chain additivity

For arbitrary schemes and irreducible closed subsets $Z \subseteq Y \subseteq X$, concatenating chains gives the general inequality

Proposition 31.9 (Concatenation inequality)**.**

$$\operatorname{codim}(Z, X) \geq \operatorname{codim}(Z, Y) + \operatorname{codim}(Y, X)$$

whenever the displayed codimensions are finite.

Equality is not formal. For varieties over a field, however, applying theorem 31.7 three times yields

Corollary 31.10 (Additivity along subvarieties)**.** *If $Z \subseteq Y \subseteq X$ are irreducible closed subvarieties of a variety, then*

$$\operatorname{codim}(Z, X) = \operatorname{codim}(Z, Y) + \operatorname{codim}(Y, X).$$

$$\begin{tikzcd} \boxed{Z} \arrow[r, "a"] \arrow[rr, bend right, "a+b"'] & \boxed{Y} \arrow[r, "b"] & \boxed{X} \end{tikzcd}$$

Figure 31.8: Nested codimensions add exactly for subvarieties of a variety.

Example 31.11 (A hypersurface)**.** For a nonzero nonunit irreducible polynomial $f \in k[x_1, \ldots, x_n]$, the hypersurface $V(f)$ has codimension 1 in affine n-space under the standard dimension theorem.

31.9 Standard computations

For the coordinate linear subspace

$$L = V(x_1, \ldots, x_r) \subseteq \mathbb{A}^n_k$$

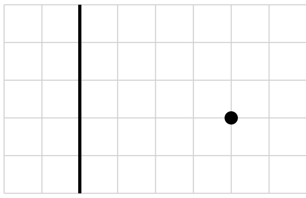
we have $\dim L = n - r$ and

$$\operatorname{codim}(L, \mathbb{A}^n_k) = r.$$

A closed point of $\mathbb{A}^n_k$ has codimension n. The diagonal $\Delta \subseteq \mathbb{A}^n_k \times_k \mathbb{A}^n_k$ has dimension n inside ambient dimension $2n$, hence codimension n. For subvarieties $Y_i \subseteq X_i$ of varieties, the same calculation gives

$$\operatorname{codim}(Y_1 \times_k Y_2, X_1 \times_k X_2) = \operatorname{codim}(Y_1, X_1) + \operatorname{codim}(Y_2, X_2)$$

whenever the indicated products are irreducible, so that the relative codimensions are being taken for irreducible closed subsets. In particular, this applies when the factors involved are geometrically integral over k.



line: codim 1; point: codim 2 in $\mathbb{A}^2$

Figure 31.9: Codimension distinguishes a divisor-like line from a closed point.

31.10 Boundaries of VI/31

VI/31 owns height as the affine model for codimension, the generic-point local-ring formula, the AG8 codimension and additivity statements, and the AG8 Problem 3(c) failure. It does not absorb the deeper local geometry of VI/32: closed-point local dimension, tangent spaces, embedding dimension, regularity, and the remaining Problem 3(a) counterexample stay there. Nor does codimension one yet mean a divisor: Weil and Cartier divisor structures belong to the later divisor arc.

31.11 Common pitfalls

- **Codimension is not the number of defining equations.** Redundant equations can leave codimension unchanged, and non-complete-intersection loci need not have codimension equal to a chosen number of generators.
- **Codimension is relative to the ambient scheme.** The same irreducible scheme can have different codimensions in different embeddings.
- **The formula** $\dim Y + \operatorname{codim}(Y, X) = \dim X$ **is not formal.** It is valid in the variety setting used here, but can fail immediately on non-equidimensional schemes.
- **Codimension one is not yet a divisor.** A codimension-one irreducible closed subset becomes a prime divisor only after the hypotheses and structures of the divisor chapters are introduced.
- **Integral extensions require care prime by prime.** Going-up and incomparability control chains under an integral map, but they do not by themselves say that every chosen prime upstairs has the same height as its contraction.

31.12 Examples

Example 31.12 (A component has codimension zero)**.** If Y is an irreducible component of X, no strictly larger irreducible closed subset lies between Y and an ambient component. Hence $\operatorname{codim}(Y, X) = 0$.

Example 31.13 (A line in the affine plane)**.** For $Y = V(x) \subset \mathbb{A}^2_k$, the prime (x) has height 1, so $\operatorname{codim}(Y, \mathbb{A}^2_k) = 1$.

Example 31.14 (The origin in the affine plane)**.** The maximal ideal (x, y) has height 2 in $k[x, y]$. Thus the origin has codimension 2 in $\mathbb{A}^2_k$.

Example 31.15 (A coordinate plane in affine four-space)**.** $V(x_1, x_2) \subset \mathbb{A}^4_k$ has dimension 2 and codimension 2.

Example 31.16 (A hypersurface)**.** If $f \in k[x_1, \dots, x_n]$ is nonconstant and irreducible, then $V(f)$ has codimension 1 in $\mathbb{A}^n_k$.

Example 31.17 (The generic point)**.** For an integral scheme X, the closed irreducible subset X itself has generic point η and codimension 0; correspondingly $\mathcal{O}_{X,\eta} = K(X)$ has dimension 0.

Example 31.18 (A closed point on a curve)**.** Every closed point of an integral curve has codimension 1.

Example 31.19 (A closed point on a surface)**.** For a surface variety, a closed point has codimension 2.

Example 31.20 (The diagonal in the affine line squared)**.** The diagonal $V(x-y) \subset \mathbb{A}^1_k \times \mathbb{A}^1_k \cong \mathbb{A}^2_k$ is a hypersurface, hence has codimension 1.

Example 31.21 (The diagonal in affine n-space)**.** $\Delta \subset \mathbb{A}^n_k \times \mathbb{A}^n_k$ is cut out by $x_i - y_i$ for $1 \le i \le n$ and has codimension n.

Example 31.22 (A codimension-two curve in three-space)**.** The z-axis $V(x, y) \subset \mathbb{A}^3_k$ has codimension 2.

Example 31.23 (A union with codimension one)**.** In $\mathbb{A}^2_k$, $V(xy) = V(x) \cup V(y)$ has two irreducible components, each codimension 1, so the closed subset has codimension 1.

Example 31.24 (A mixed closed subset)**.** In $\mathbb{A}^2_k$, $V(x) \cup \{(1, 0)\}$ has codimension 1: the line component has codimension 1, while the isolated point has codimension 2, and we take the infimum.

Example 31.25 (Localization at a height-one prime)**.** For $A = k[x, y]$ and $\mathfrak{p} = (x)$, $A_{\mathfrak{p}}$ has dimension 1, exactly the codimension of $V(x)$.

Example 31.26 (Localization at the origin)**.** $k[x, y]_{(x,y)}$ has dimension 2, equal to the codimension of the origin in $\mathbb{A}^2_k$.

Example 31.27 (Nested coordinate spaces)**.** Inside $\mathbb{A}^3_k$, the chain $V(x, y) \subset V(x) \subset \mathbb{A}^3_k$ has codimensions $1 + 1 = 2$.

Example 31.28 (The AG8 disconnected counterexample)**.** For $X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k$ and $Y = \operatorname{Spec} k$, one has $\operatorname{codim}(Y, X) = 0$, but $\dim Y + 0 = 0 < 1 = \dim X$.

Example 31.29 (Finite field extension)**.** $\operatorname{Spec} L$ inside itself has codimension 0 for every field extension L/k.

Example 31.30 (Reduction preserves codimension)**.** A nilpotent thickening does not change the underlying irreducible closed subsets, so corresponding closed subsets have the same codimension before and after reduction.

Example 31.31 (Affine three-space chain)**.** The chain $(0) \subsetneq (x) \subsetneq (x, y) \subsetneq (x, y, z)$ shows successive geometric codimensions $1, 1, 1$ and total codimension 3 for the origin.

Example 31.32 (Advanced 1: Height in a reducible affine scheme)**.** For $A = k[x, y]/(xy)$, the prime (x, y) has height 1. The closed origin therefore has codimension 1 even though two irreducible components pass through it.

Example 31.33 (Advanced 2: Closed subset with components of different codimension)**.** In $\mathbb{A}^3_k$, $W = V(x) \cup V(y, z)$ has components of codimensions 1 and 2. By convention $\operatorname{codim}(W, \mathbb{A}^3_k) = 1$ and $\dim W = 2$.

Example 31.34 (Advanced 3: Principal ideal theorem)**.** In a Noetherian domain, a minimal prime over a nonzero principal ideal has height exactly one. The nonzero condition excludes the generic component and the domain hypothesis makes height zero impossible.

Example 31.35 (Advanced 4: Two equations need not give codimension two)**.** $V(x, xy) \subset \mathbb{A}^2_k$ equals $V(x)$, so two displayed equations can define a codimension-one locus. Codimension counts prime chains, not equations.

Example 31.36 (Advanced 5: Three equations with codimension two)**.** $V(x, y, xy) \subset \mathbb{A}^3_k$ equals $V(x, y)$ and has codimension 2, again showing redundancy among equations.

Example 31.37 (Advanced 6: Product codimension)**.** For coordinate subspaces $L_r \subset \mathbb{A}^m$ and $L_s \subset \mathbb{A}^n$ of codimensions r, s, their product has codimension $r + s$ in $\mathbb{A}^{m+n}$.

Example 31.38 (Advanced 7: Graph of a morphism)**.** For a morphism $f : X \to \mathbb{A}^m_k$ from a variety, its graph is isomorphic to X inside $X \times \mathbb{A}^m_k$; when the graph is closed, its codimension is m.

Example 31.39 (Advanced 8: Diagonal of a smooth-looking ambient product)**.** For an n-dimensional separated variety X, the diagonal is closed in $X \times X$ and has codimension n whenever the product has dimension $2n$.

Example 31.40 (Advanced 9: Codimension and normalization)**.** Let $A \subseteq B$ be an integral extension, as in normalization. Contraction of a strict chain of primes in B remains strict by incomparability, while going-up lifts chains from A once a starting prime upstairs is chosen. Thus integral extensions preserve Krull dimension globally, but equality $\mathrm{ht}(\mathfrak{q}) = \mathrm{ht}(\mathfrak{q} \cap A)$ for every specified prime $\mathfrak{q}$ is not a consequence of integrality alone; extra hypotheses such as going-down are needed for that stronger prime-by-prime statement.

Example 31.41 (Advanced 10: Failure of global additivity from components)**.** The scheme $\operatorname{Spec} k \sqcup \mathbb{A}^2_k$ has dimension 2, but its point component has dimension and codimension both zero, so the additivity defect is 2.

Example 31.42 (Advanced 11: Nested additivity for varieties)**.** If $Z \subset Y \subset X$ are subvarieties with dimensions $r < s < n$, then their codimensions are $s - r$, $n - s$, and $n - r$, so the relative codimensions add.

Example 31.43 (Advanced 12: Codimension is not the number of generators)**.** The ideal $(x^2, xy, xz) \subset k[x, y, z]$ has radical (x), so its closed set has codimension 1 although the ideal is displayed with three generators.

Example 31.44 (Advanced 13: Generic-point minimum of local dimensions)**.** For irreducible closed Y, the function $y \mapsto \dim \mathcal{O}_{X,y}$ need not be constant, but its infimum on Y is attained at the generic point and equals $\mathrm{codim}(Y, X)$.

Example 31.45 (Advanced 14: Why catenary issues are deferred)**.** Concatenating prime chains always gives a lower bound for total codimension, but equality for arbitrary Noetherian rings is not formal. Finite-type algebras over fields enjoy the dimension formulas needed in the variety case.

31.13 Problem dossiers

Problem 31.46 (VI.31.P1: AG8 Problem 2: codimension statements)**.** Let X be a variety over k and let $Y \subseteq X$ be a closed irreducible subvariety. Prove

$$\mathrm{codim}(Y, X) = \inf_{y \in Y} \dim \mathcal{O}_{X,y} \quad \text{and} \quad \dim Y + \mathrm{codim}(Y, X) = \dim X.$$

Provenance. Legacy: AG8-P2b, the codimension portion of Problem 2. Global/open dimension is VI/30; the deeper local-dimension portion remains VI/32.

Problem 31.47 (VI.31.P2: AG8 Problem 3(c): failure of dimension–codimension additivity)**.** Let $X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k$ and let $Y = \operatorname{Spec} k$ be the zero-dimensional component. Show that

$$\dim Y + \operatorname{codim}(Y, X) \neq \dim X.$$

Provenance. Legacy: AG8-P3c, the exact part-(c) counterexample split from AG8 Problem 3. Part (d) is VI.30.P2; part (a) remains reserved for VI/32.

Problem 31.48 (VI.31.P3: Codimension equals height on affine schemes)**.** Let A be a ring and $\mathfrak{p} \in \operatorname{Spec} A$. Prove

$$\operatorname{codim}(V(\mathfrak{p}), \operatorname{Spec} A) = \operatorname{ht}(\mathfrak{p}).$$

Problem 31.49 (VI.31.P4: Codimension from the generic stalk)**.** Let $Y \subseteq X$ be irreducible closed with generic point η. Prove

$$\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,\eta}.$$

Problem 31.50 (VI.31.P5: Coordinate subspaces)**.** For

$$L = V(x_1, \dots, x_r) \subseteq \mathbb{A}^n_k,$$

compute $\dim L$ and $\operatorname{codim}(L, \mathbb{A}^n_k)$.

Problem 31.51 (VI.31.P6: Closed points of affine space)**.** Show that every k-rational point of $\mathbb{A}^n_k$ has codimension n.

Problem 31.52 (VI.31.P7: Irreducible hypersurface)**.** Let $0 \neq f \in k[x_1, \dots, x_n]$ be irreducible and nonconstant. Compute the codimension of $V(f)$.

Problem 31.53 (VI.31.P8: Redundant equations)**.** Compute the codimension of $V(x, xy) \subseteq \mathbb{A}^2_k$ and explain why counting equations fails.

Problem 31.54 (VI.31.P9: The diagonal)**.** Compute the codimension of the diagonal

$$\Delta \subseteq \mathbb{A}^n_k \times_k \mathbb{A}^n_k.$$

Problem 31.55 (VI.31.P10: Nested coordinate chain)**.** In $\mathbb{A}^4_k$, let

$$Z = V(x_1, x_2, x_3), \qquad Y = V(x_1).$$

Verify codimension additivity through $Z \subseteq Y \subseteq X$.

Problem 31.56 (VI.31.P11: A reducible closed subset)**.** Let

$$W = V(x) \cup V(y, z) \subseteq \mathbb{A}^3_k.$$

Compute $\dim W$ and $\operatorname{codim}(W, \mathbb{A}^3_k)$.

Problem 31.57 (VI.31.P12: Codimension zero)**.** Prove that an irreducible closed subset $Y \subseteq X$ has codimension zero exactly when it is an irreducible component of X.

Problem 31.58 (VI.31.P13: Reduction invariance)**.** Let $X_{\mathrm{red}} \to X$ be reduction and let $Y \subseteq X$ be irreducible closed. Compare codimensions of the corresponding subsets.

Problem 31.59 (VI.31.P14: Product of coordinate subspaces)**.** Let $L_r \subset \mathbb{A}^m$ and $M_s \subset \mathbb{A}^n$ be coordinate subspaces of codimensions r and s. Compute the codimension of $L_r \times M_s$.

Problem 31.60 (VI.31.P15: Infimum over a closed set)**.** Let W be a closed subset with finitely many irreducible components W_i. Show

$$\inf_{w\in W} \dim \mathcal{O}_{X,w} = \min_i \operatorname{codim}(W_i, X).$$

Problem 31.61 (VI.31.P16: Concatenation inequality)**.** For irreducible closed $Z \subseteq Y \subseteq X$ of finite codimension, prove the concatenation inequality for codimensions.

Problem 31.62 (VI.31.P17: Additivity for nested subvarieties)**.** Let $Z \subseteq Y \subseteq X$ be irreducible closed subvarieties of a variety. Prove exact codimension additivity.

Problem 31.63 (VI.31.P18: Codimension-one center)**.** Let A be a Noetherian domain and $f \neq 0$ a nonunit. If $\mathfrak{p}$ is minimal over (f), show $V(\mathfrak{p})$ has codimension one in $\operatorname{Spec} A$.

Problem 31.64 (VI.31.P19: Why equations do not equal codimension)**.** Give examples showing that the number of generators of an ideal can exceed the codimension of its zero set.

Problem 31.65 (VI.31.P20: VI/31 routing audit)**.** Explain why AG8 Problem 3 contributes only part (c) to VI/31 and why part (a) must remain for VI/32.

31.14 Exercises

Exercise 31.66 (VI.31.E1)**.** Compute $\operatorname{codim}(\mathbb{A}^2_k, \mathbb{A}^2_k)$.

Exercise 31.67 (VI.31.E2)**.** Compute the codimension of $V(x) \subset \mathbb{A}^3_k$.

Exercise 31.68 (VI.31.E3)**.** Compute the codimension of $V(x, y) \subset \mathbb{A}^3_k$.

Exercise 31.69 (VI.31.E4)**.** Compute the codimension of the origin in $\mathbb{A}^3_k$.

Exercise 31.70 (VI.31.E5)**.** If Y is an irreducible component of X, compute $\operatorname{codim}(Y, X)$.

Exercise 31.71 (VI.31.E6)**.** Let $A = k[x, y]$ and $\mathfrak{p} = (x)$. Compute $\dim A_{\mathfrak{p}}$.

Exercise 31.72 (VI.31.E7)**.** Let $A = k[x, y]$ and $\mathfrak{m} = (x, y)$. Compute $\dim A_{\mathfrak{m}}$.

Exercise 31.73 (VI.31.E8)**.** For an irreducible closed Y with generic point η, identify $\operatorname{codim}(Y, X)$ using η.

Exercise 31.74 (VI.31.E9)**.** Why is the infimum of $\dim \mathcal{O}_{X,y}$ on irreducible Y attained at η_Y?

Exercise 31.75 (VI.31.E10)**.** A variety X has dimension 5 and a subvariety Y has dimension 2. Compute its codimension.

Exercise 31.76 (VI.31.E11)**.** A codimension-one subvariety lies in a 7-dimensional variety. Compute its dimension.

Exercise 31.77 (VI.31.E12)**.** A closed point lies on a curve variety. Compute its codimension.

Exercise 31.78 (VI.31.E13)**.** A closed point lies on a 4-dimensional variety. Compute its codimension.

Exercise 31.79 (VI.31.E14)**.** Compute the codimension of the diagonal in $\mathbb{A}^2 \times \mathbb{A}^2$.

Exercise 31.80 (VI.31.E15)**.** Compute the codimension of $V(x_1, x_2, x_3) \subset \mathbb{A}^6$.

Exercise 31.81 (VI.31.E16)**.** Can two equations define a codimension-one subset? Give an example.

Exercise 31.82 (VI.31.E17)**.** Can a nonzero principal hypersurface in affine space have codimension two?

Exercise 31.83 (VI.31.E18)**.** In $X = \operatorname{Spec} k \sqcup \mathbb{A}^1$, compute the codimension of the point component.

Exercise 31.84 (VI.31.E19)**.** For the preceding X and point component Y, evaluate $\dim Y + \operatorname{codim}(Y, X)$.

Exercise 31.85 (VI.31.E20)**.** If $Z \subset Y \subset X$ are subvarieties with codimensions 2 and 3 successively, compute $\operatorname{codim}(Z, X)$.

Exercise 31.86 (VI.31.E21)**.** Does reduction change codimension?

Exercise 31.87 (VI.31.E22)**.** For $W = V(x) \cup V(y, z) \subset \mathbb{A}^3$, compute $\operatorname{codim}(W, \mathbb{A}^3)$.

Exercise 31.88 (VI.31.E23)**.** State a criterion for codimension zero of an irreducible closed subset.

Exercise 31.89 (VI.31.E24)**.** What later structure is attached to codimension-one subvarieties?

Exercise 31.90 (VI.31.E25)**.** Which AG8 Problem 3 part remains after VI/31?

Exercise 31.91 (VI.31.E26)**.** State the difference between $\dim X$ and $\operatorname{codim}(Y, X)$.

31.15 Challenges

Problem 31.92 (Challenge VI.31.C1: Chain geometry versus equations)**.** Construct three closed subsets of $\mathbb{A}^4_k$ each presented by three equations but having codimensions 1, 2, and 3. For each, compute the radical ideal and exhibit a prime chain realizing the codimension.

Problem 31.93 (Challenge VI.31.C2: Codimension through generic points)**.** Let X be a scheme and $Y \subseteq X$ irreducible closed. Starting only from the specialization order on points, reconstruct the proof that $\operatorname{codim}(Y, X) = \dim \mathcal{O}_{X,\eta_Y}$. Identify exactly where the scheme-theoretic sobriety/generic-point property enters.

Problem 31.94 (Challenge VI.31.C3: Failure modes of additivity)**.** Compare three settings: varieties over a field, a non-equidimensional disjoint union, and a general Noetherian spectrum. Determine which steps in the proof of $\dim Y + \operatorname{codim}(Y, X) = \dim X$ survive and which require extra hypotheses.

Problem 31.95 (Challenge VI.31.C4: Product and diagonal codimension)**.** For varieties X and Y under hypotheses ensuring the expected product dimensions, prove the product formula for codimension. Then, assuming X is separated over k, specialize to the closed diagonal

$$\Delta_X \subset X \times_k X$$

and compute its codimension.

Problem 31.96 (Challenge VI.31.C5: Codimension-one bridge to divisors)**.** For a normal Noetherian integral scheme, investigate why codimension-one points are the natural carriers of valuations. State only the codimension/height facts needed now and explicitly list the additional divisor-theoretic statements that must wait for VI/39.

31.16 Chapter synthesis

Codimension is the relative form of Krull dimension. An irreducible closed subset Y is measured by the height of its generic prime, equivalently by $\dim \mathcal{O}_{X,\eta_Y}$. In varieties, finite-type dimension theory forces the clean identity $\dim Y + \operatorname{codim}(Y, X) = \dim X$ and makes codimension additive along nested subvarieties. Outside that setting, component structure can break the identity immediately. The distinction is essential preparation for local dimension in VI/32 and for codimension-one divisor theory later in the volume.

$$\boxed{\text{height}} \longleftrightarrow \boxed{\text{codimension}} \longleftrightarrow \boxed{\text{generic local dimension}}$$

affine algebra $\longleftrightarrow$ global geometry $\longleftrightarrow$ local bridge

Figure 31.10: The three equivalent views that organize VI/31.

31.17 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 31.97 (Line plane)**.** Compute the codimension of a line in $\mathbb{A}^2$.

Hint. Use codimension and chains inside irreducible spaces. $\square$

Solution. 1. $\square$

Exercise 31.98 (Point plane)**.** Compute the codimension of a closed point in $\mathbb{A}^2$.

Hint. Use codimension and chains inside irreducible spaces. $\square$

Solution. 2. $\square$

Exercise 31.99 (Hyperplane)**.** Compute the codimension of a hyperplane in $\mathbb{A}^n$.

Hint. Use codimension and chains inside irreducible spaces. $\square$

Solution. 1. $\square$

Exercise 31.100 (Whole space)**.** Compute the codimension of X in itself.

Hint. Use codimension and chains inside irreducible spaces. $\square$

Solution. 0. $\square$

Exercise 31.101 (Prime height)**.** What algebraic invariant matches codimension of $V(\mathfrak{p})$ in an affine integral setting?

Hint. Use codimension and chains inside irreducible spaces. $\square$

Solution. The height of $\mathfrak{p}$ under standard hypotheses. $\square$

Proofs

Exercise 31.102 (Height chain)**.** Prove: Explain why height of a prime measures codimension through chains of primes below it.

Hint. Work directly from codimension and chains inside irreducible spaces. ☐

Solution. Translate prime chains into reverse chains of irreducible closed subsets from the ambient component down to the closure of the point. ☐

Exercise 31.103 (Hyperplane codimension)**.** Prove: Prove a coordinate hyperplane in affine n-space has codimension one.

Hint. Work directly from codimension and chains inside irreducible spaces. ☐

Solution. Its coordinate ring is a polynomial ring in $n-1$ variables. ☐

Exercise 31.104 (Point codimension)**.** Prove: Prove the origin in $\mathbb{A}^n$ has codimension n.

Hint. Work directly from codimension and chains inside irreducible spaces. ☐

Solution. Use the chain $(0) \subset (x_1) \subset (x_1, x_2) \subset \cdots \subset (x_1, \ldots, x_n)$. ☐

Exercise 31.105 (Isomorphism invariance)**.** Prove: Prove codimension is preserved by scheme isomorphisms.

Hint. Work directly from codimension and chains inside irreducible spaces. ☐

Solution. Isomorphisms preserve irreducible closed subsets, dimensions, and inclusion chains. ☐

Counterexamples and hypothesis tests

Exercise 31.106 (Always difference)**.** Test the claim: codimension always equals ambient dimension minus subset dimension without hypotheses.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in general noncatenary or reducible settings; care is required. ☐

Exercise 31.107 (Point codim one)**.** Test the claim: every closed point has codimension one.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False in dimensions greater than one. ☐

Exercise 31.108 (Component positive)**.** Test the claim: an irreducible component has positive codimension in the whole space.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. Typically it has codimension zero within itself and as a maximal irreducible component. ☐

Applications and investigations

Exercise 31.109 (Divisors)**.** Why is codimension one especially important?

Hint. Translate the question into codimension and chains inside irreducible spaces. □

Solution. Codimension-one subvarieties support divisor theory. □

Exercise 31.110 (Intersection theory)**.** How does codimension enter expected-intersection calculations?

Hint. Translate the question into codimension and chains inside irreducible spaces. □

Solution. Codimensions roughly add for transverse or properly intersecting conditions. □

Challenge problems

Exercise 31.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Codimension in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Codimension, translates it through codimension and chains inside irreducible spaces, and checks the correspondence in both algebraic and geometric language. □

Exercise 31.112 (Hypothesis challenge)**.** Choose one structural statement from Codimension, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Chapter 32

Local Dimension, Regularity, and Singularities

VI/29–VI/31 measured prime chains globally and relatively. VI/32 now fixes a point $x \in X$ and asks what the scheme looks like infinitesimally near that point. The local ring $\mathcal{O}_{X,x}$ supplies two basic numerical invariants:

$$\dim \mathcal{O}_{X,x} \qquad \text{and} \qquad \dim_{\kappa(x)} \mathfrak{m}_x/\mathfrak{m}_x^2.$$

The first counts prime chains inside the local ring; the second counts independent first-order directions. Their comparison detects regularity. For a Noetherian local ring, the second number is always at least the first, and equality is exactly the definition of a regular local ring. This turns singularity detection into a concrete comparison between Krull dimension and linear algebra.

Learning goals

By the end of the chapter, the reader should be able to:

- compute the local-ring dimension $\dim \mathcal{O}_{X,x}$ from an affine chart and identify it with prime height;
- prove the closed-point local-dimension theorem for varieties and recognize its failure for general schemes;
- compute the cotangent space $\mathfrak{m}_x/\mathfrak{m}_x^2$ and the embedding dimension;
- define the Zariski tangent space as the dual of the cotangent space and compute it from linearized equations;
- define regular local rings, regular points, and singular points;
- apply the Jacobian matrix to curves and hypersurfaces at rational points;
- distinguish a smooth-looking tangent direction from genuine regularity, especially for cusps, nodes, crossings, and nilpotent thickenings.

Conceptual roadmap

At one point x, the chapter follows the chain

$$x \in X \longrightarrow (\mathcal{O}_{X,x}, \mathfrak{m}_x, \kappa(x)) \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow T_xX \longrightarrow \text{regular or singular.}$$

The local ring retains all algebra visible in arbitrarily small neighborhoods. Passing to $\mathfrak{m}_x/\mathfrak{m}_x^2$ throws away quadratic and higher-order terms and keeps only first-order behavior. Regularity says that this first-order model has exactly as many independent directions as the local Krull dimension predicts.

32.1 Local-ring dimension at a point

Definition 32.1 (Local-ring dimension)**.** For a scheme X and a point $x \in X$, define

$$\dim_x^{\mathrm{loc}} X := \dim \mathcal{O}_{X,x}.$$

In this volume, "local dimension at x" means this local-ring dimension unless another convention is stated explicitly.

If x corresponds to $\mathfrak{p} \in \operatorname{Spec} A$ in an affine chart, then

$$\mathcal{O}_{X,x} \cong A_{\mathfrak{p}}, \qquad \dim \mathcal{O}_{X,x} = \operatorname{ht}(\mathfrak{p}).$$

Thus local dimension counts chains of generalizations ending at the chosen point. At a generic point of an irreducible component it is 0; at a closed point of an n-dimensional variety it is n.

$$\boxed{x \in X} \longrightarrow \boxed{\mathcal{O}_{X,x}} \longrightarrow \boxed{\dim \mathcal{O}_{X,x}}$$

Figure 32.1: Local dimension is the Krull dimension of the stalk at the chosen point.

32.2 Closed points of varieties

Let X be a variety over k and let $x \in X$ be closed. Choose an affine neighborhood $U = \operatorname{Spec} A$ of x, with maximal ideal $\mathfrak{m}$. Since A is a finitely generated k-domain, the dimension formula gives

$$\operatorname{ht}(\mathfrak{m}) + \operatorname{trdeg}_k \kappa(x) = \dim A.$$

Zariski's lemma makes $\kappa(x)/k$ algebraic, so the transcendence degree term is 0. Hence

Theorem 32.2 (Closed-point local dimension for varieties)**.** *If X is a variety over k and $x \in X$ is closed, then*

$$\boxed{\dim \mathcal{O}_{X,x} = \dim X.}$$

This is the remaining local-dimension statement from the AG8 Problem 2 stream.

$$\boxed{X \text{ variety}, \dim X = d} \longrightarrow \boxed{x \text{ closed}} \longrightarrow \boxed{\dim \mathcal{O}_{X,x} = d}$$

Figure 32.2: Every closed point of a variety has full local-ring dimension.

Example 32.3 (Tangent space of affine space)**.** At any point of $\mathbb{A}_k^n$, the maximal ideal modulo its square has dimension n, so the Zariski tangent space is k^n.

32.3 Why the closed-point formula can fail

The preceding theorem is not a formal fact about arbitrary schemes. Let

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k$$

and let x be the unique point of the $\operatorname{Spec} k$ component. Then x is closed and

$$\mathcal{O}_{X,x} \cong k, \qquad \dim \mathcal{O}_{X,x} = 0, \qquad \dim X = 1.$$

Thus the equality at closed points fails once components of different dimensions are allowed. This realizes the remaining part-(a) counterexample requirement of AG8 Problem 3 and completes the Problem 3 routing begun in VI/30 and VI/31.

$$\boxed{\operatorname{Spec} k} \qquad \boxed{\quad \mathbb{A}^1_k \quad}$$

$$\dim \mathcal{O}_{X,x} = 0 < 1 = \dim X$$

Figure 32.3: A closed point on a low-dimensional component need not see the global dimension.

32.4 The maximal ideal and the cotangent space

The local ring $(\mathcal{O}_{X,x}, \mathfrak{m}_x)$ distinguishes functions vanishing at x from units. Modulo $\mathfrak{m}_x^2$, products of two vanishing functions disappear, leaving their linear parts.

Definition 32.4 (Cotangent space and embedding dimension)**.** The cotangent space at x is the $\kappa(x)$-vector space

$$\mathfrak{m}_x/\mathfrak{m}_x^2.$$

Its dimension

$$\operatorname{edim}_x X := \dim_{\kappa(x)} \mathfrak{m}_x/\mathfrak{m}_x^2$$

is the embedding dimension at x.

For a Noetherian local ring, Nakayama's lemma identifies this number with the minimal number of generators of the maximal ideal.

$$\boxed{\mathfrak{m}_x} \xrightarrow{\text{linearize}} \boxed{\mathfrak{m}_x/\mathfrak{m}_x^2} \xrightarrow{\dim_{\kappa(x)}} \boxed{\text{embedding dimension}}$$

Figure 32.4: Quotienting by $\mathfrak{m}^2$ keeps first-order vanishing data.

32.5 Zariski tangent space

Definition 32.5 (Zariski tangent space)**.** The Zariski tangent space at x is

$$T_x X := \operatorname{Hom}_{\kappa(x)}(\mathfrak{m}_x/\mathfrak{m}_x^2, \kappa(x)).$$

Hence $\dim_{\kappa(x)} T_x X = \operatorname{edim}_x X$.

At a k-rational point of a k-scheme, these tangent vectors can be viewed as first-order maps from the dual numbers $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ whose closed point maps to x. The dual-number interpretation explains why the tangent space sees first-order infinitesimal directions rather than only ordinary nearby points.

$$\boxed{\mathfrak{m}/\mathfrak{m}^2} \xleftrightarrow{\text{dual over } \kappa(x)} \boxed{T_x X}$$

$$\operatorname{Spec} \kappa(x)[\varepsilon]/(\varepsilon^2)$$

Figure 32.5: The Zariski tangent space is dual to the cotangent space and records first-order directions.

32.6 Regular local rings and regular points

A basic theorem of Noetherian local algebra gives

$$\dim R \le \dim_\kappa \mathfrak{m}/\mathfrak{m}^2$$

for every Noetherian local ring $(R, \mathfrak{m}, \kappa)$.

Definition 32.6 (Regular local ring)**.** A Noetherian local ring $(R, \mathfrak{m}, \kappa)$ is regular if

$$\dim R = \dim_\kappa \mathfrak{m}/\mathfrak{m}^2.$$

Definition 32.7 (Regular and singular points)**.** A locally Noetherian scheme X is regular at x if $\mathcal{O}_{X,x}$ is a regular local ring. Otherwise x is singular. The scheme is regular if it is regular at every point.

Thus the numerical test is

$$\boxed{x \text{ regular} \iff \operatorname{edim}_x X = \dim \mathcal{O}_{X,x}.}$$

$$\boxed{\dim \mathcal{O}_{X,x}} \qquad \le \qquad \boxed{\operatorname{edim}_x X}$$

$$\text{equality} \iff \text{regular}$$

Figure 32.6: Regularity is the equality of local Krull dimension and embedding dimension.

Example 32.8 (A smooth parabola)**.** For $C = V(y - x^2)$, the gradient of $f = y - x^2$ at (a, a^2) is $(-2a, 1)$, which is never zero. The tangent space is the one-dimensional kernel of this linear equation.

32.7 Affine equations and the Jacobian matrix

Let

$$X = V(f_1, \ldots, f_r) \subseteq \mathbb{A}^n_k$$

and let $a \in X(k)$. Write $J(a)$ for the $r \times n$ matrix

$$J(a) = \left(\frac{\partial f_i}{\partial x_j}(a) \right).$$

Linearizing the equations at a gives

Proposition 32.9 (Tangent space from the Jacobian)**.** *For a k-rational point a,*

$$T_aX \cong \ker\big(J(a) : k^n \to k^r\big), \qquad \dim_k T_aX = n - \operatorname{rank} J(a).$$

This formula computes the tangent space directly. Regularity still requires comparison with the local Krull dimension; Jacobian rank alone is not a substitute for knowing the dimension of the local scheme.

$$\boxed{f_1, \dots, f_r} \xrightarrow{\text{differentiate}} \boxed{J(a)} \xrightarrow{\text{linear equations}} \boxed{\ker J(a) = T_aX}$$

Figure 32.7: The Jacobian matrix extracts the linearized equations defining the tangent space.

32.8 Hypersurfaces

For a nonzero nonconstant polynomial $f \in k[x_1, \dots, x_n]$, every component of the hypersurface $V(f)$ has dimension $n - 1$. At a k-rational point $a \in V(f)$,

$$T_aV(f) = \left\{ v \in k^n : \sum_{j=1}^{n} \frac{\partial f}{\partial x_j}(a) v_j = 0 \right\}.$$

Consequently, in this hypersurface setting,

Corollary 32.10 (Hypersurface regularity test)**.** *The point a is regular if at least one partial derivative of f is nonzero at a. If all partial derivatives vanish, then the tangent space has dimension $n > n - 1$ and a is singular.*

$$\boxed{\nabla f(a) \neq 0} \quad \Longrightarrow \quad \boxed{\dim T_a = n - 1} \quad = \quad \boxed{\dim \mathcal{O}_{X,a}}$$

Figure 32.8: For a hypersurface, a nonzero gradient forces regularity at a rational point.

32.9 Cusp, node, and crossing

Assume the relevant characteristics are excluded when factorization or derivatives require it. For the cusp

$$C : y^2 = x^3,$$

the origin has local dimension 1, while both first derivatives vanish there, so $\dim T_0C = 2$. The lowest-degree part is y^2, giving one tangent direction with multiplicity two.

For the node

$$N : y^2 = x^2(x + 1),$$

the origin also has local dimension 1 and two-dimensional Zariski tangent space, but the lowest-degree part $y^2 - x^2 = (y - x)(y + x)$ exhibits two distinct tangent directions. Both points are singular, but their first nonzero homogeneous terms distinguish the geometry beyond the mere dimension of the tangent space.

For the crossing $xy = 0$, the origin is again singular with local dimension 1 and tangent dimension 2, now because two irreducible components meet.

Example 32.11 (A singular node)**.** For $X = V(xy)$ at the origin, the gradient (y, x) vanishes. The linearized equation imposes no condition, so the Zariski tangent space has dimension 2, larger than the local dimension 1; this detects singularity.

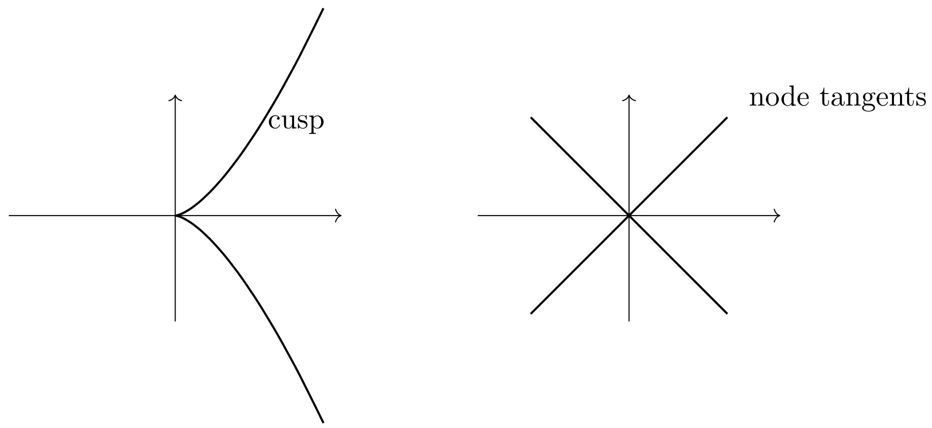


Figure 32.9: The cusp has one doubled tangent direction; the node has two distinct tangent directions.

32.10 A recurring multiple-lens example: the cusp

The cusp now supports several viewpoints developed across the volume:

lens	cusp data
equation	$y^2 - x^3 = 0$
coordinate ring	$k[x,y]/(y^2 - x^3) \cong k[t^2,t^3]$
function field	$k(t)$
normalization	$\operatorname{Spec} k[t] \to \operatorname{Spec} k[t^2,t^3]$
dimension	1
local ring at cusp	$k[t^2,t^3]_{(t^2,t^3)}$
embedding dimension	2
regularity	singular, since $2 > 1$

This is the first full “same geometry, multiple lenses” panel in the dimension arc.

32.11 Common pitfalls

- **Local-ring dimension is not the dimension of the closure of the point.** For a prime $\mathfrak{p}$ in an affine scheme, $\dim \mathcal{O}_{X,x} = \operatorname{ht}(\mathfrak{p})$, whereas $\dim \overline{\{x\}} = \dim A/\mathfrak{p}$ in the finite-type variety setting.
- **Terminology varies.** Some references use “dimension at x” for an infimum of neighborhood dimensions. This chapter explicitly uses the local-ring convention $\dim \mathcal{O}_{X,x}$.
- **Tangent dimension is not local dimension.** Equality characterizes regularity; a larger tangent space is a signal of singularity.
- **A tangent line is not enough to classify a singularity.** The cusp and node both have two-dimensional Zariski tangent spaces as plane curves, but their tangent cones differ.
- **Regular, normal, reduced, and smooth are different adjectives.** This chapter treats regularity. Their systematic comparison belongs to later property and morphism chapters.

32.12 Boundaries of VI/32

VI/32 closes the dimension/local-geometry arc by introducing local-ring dimension, cotangent and tangent spaces, embedding dimension, regularity, singularity, and elementary Jacobian calculations. It does not yet develop Kähler differentials, smooth morphisms, étale morphisms, completions, tangent cones in full generality, Serre's conditions, or divisor theory. Those require later structural chapters.

32.13 Examples

Example 32.12 (A field point)**.** For $X = \operatorname{Spec} k$, the unique local ring is k, so local dimension and embedding dimension are both 0. The point is regular.

Example 32.13 (Affine line at a closed point)**.** At $a \in \mathbb{A}^1_k(k)$, $\mathcal{O}_{X,a} = k[t]_{(t-a)}$ has dimension 1 and maximal ideal generated by one element. Hence the point is regular.

Example 32.14 (Affine plane at the origin)**.** $k[x,y]_{(x,y)}$ has dimension 2, while $(x,y)/(x,y)^2$ has basis given by the classes of x, y. Thus the embedding dimension is 2.

Example 32.15 (Generic point of an integral scheme)**.** At the generic point η of an integral scheme, $\mathcal{O}_{X,\eta} = K(X)$ is a field. Hence the local-ring dimension is 0.

Example 32.16 (A height-one point in the plane)**.** For $\mathfrak{p} = (x) \subset k[x,y]$, the local ring $k[x,y]_{(x)}$ has dimension 1.

Example 32.17 (Parabola at the origin)**.** For $y - x^2 = 0$, the Jacobian row at the origin is $(0,1)$, so the tangent space is one-dimensional. The curve has local dimension 1, hence the origin is regular.

Example 32.18 (Cusp at the origin)**.** For $y^2 - x^3 = 0$, both partial derivatives vanish at $(0,0)$. The tangent space is k^2 while the local dimension is 1, so the cusp point is singular.

Example 32.19 (Node at the origin)**.** For $y^2 - x^2(x+1) = 0$ in characteristic not 2, the origin has tangent dimension 2 and local dimension 1, hence is singular.

Example 32.20 (Crossing axes)**.** For $xy = 0$, the origin has tangent space all of k^2 but local dimension 1. The singularity comes from two components meeting.

Example 32.21 (Dual numbers)**.** $X = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ has one point and local dimension 0, but $\mathfrak{m}/\mathfrak{m}^2$ is one-dimensional. Thus the point is singular despite the one-point topology.

Example 32.22 (A smooth hypersurface point)**.** On $V(x_1) \subset \mathbb{A}^n_k$, the gradient is $(1,0,\ldots,0)$ everywhere, so the tangent space has dimension $n-1$, equal to the local dimension.

Example 32.23 (A singular quadric cone point)**.** For $xy - z^2 = 0$ in $\mathbb{A}^3$, the gradient vanishes at the origin. The local dimension is 2 while the tangent space has dimension 3.

Example 32.24 (Non-rational closed point)**.** For $X = \operatorname{Spec} \mathbb{R}[t]$ and $x = (t^2+1)$, the residue field is $\mathbb{C}$. The local ring has dimension 1 and $\mathfrak{m}/\mathfrak{m}^2$ is one-dimensional over $\mathbb{C}$.

Example 32.25 (Closed point on a variety)**.** If X is a d-dimensional variety and x is closed, then $\dim \mathcal{O}_{X,x} = d$, even when $\kappa(x) \neq k$.

Example 32.26 (Disconnected failure)**.** For $X = \operatorname{Spec} k \sqcup \mathbb{A}^2_k$, the point component has local dimension 0 while the global scheme has dimension 2.

Example 32.27 (Product affine space)**.** At the origin of $\mathbb{A}^m_k \times \mathbb{A}^n_k \cong \mathbb{A}^{m+n}_k$, both local and embedding dimensions equal $m + n$.

Example 32.28 (Coordinate subspace)**.** At a closed point of $V(x_1, \ldots, x_r) \subset \mathbb{A}^n$, the local and embedding dimensions are $n - r$.

Example 32.29 (A redundant equation)**.** $V(x, xy) \subset \mathbb{A}^2$ equals the line $V(x)$. At a closed point its local dimension and tangent dimension are both 1 despite the two displayed equations.

Example 32.30 (Nilpotent thickened line)**.** For $A = k[x, \varepsilon]/(\varepsilon^2)$ at (x, ε), the local dimension is 1 but the cotangent space has two independent classes. The thickening is singular.

Example 32.31 (Regularity after localization)**.** If a Noetherian ring A has $A_\mathfrak{p}$ regular, then the scheme $\operatorname{Spec} A$ is regular at the point $\mathfrak{p}$ by definition.

Example 32.32 (Advanced 1: Embedding dimension via Nakayama)**.** For a Noetherian local ring $(R, \mathfrak{m}, \kappa)$, a set generates $\mathfrak{m}$ minimally exactly when its classes form a basis of $\mathfrak{m}/\mathfrak{m}^2$. Thus embedding dimension is the minimal number of local parameters needed at first order.

Example 32.33 (Advanced 2: Regular parameters)**.** In a d-dimensional regular local ring, any minimal generating set of $\mathfrak{m}$ has exactly d elements. Such a set is called a regular system of parameters.

Example 32.34 (Advanced 3: Tangent space under redundant presentations)**.** Adding a redundant defining equation does not change $\mathfrak{m}/\mathfrak{m}^2$ or the intrinsic tangent space, although it adds a row to a chosen Jacobian matrix.

Example 32.35 (Advanced 4: Tangent cone of the cusp)**.** For $y^2 - x^3$, the first nonzero homogeneous part is y^2. The tangent cone is the doubled line $y = 0$, refining the information that the Zariski tangent space itself is two-dimensional.

Example 32.36 (Advanced 5: Tangent cone of the node)**.** For $y^2 - x^2 - x^3$, the lowest homogeneous part is $y^2 - x^2$. In characteristic not 2 it factors as $(y - x)(y + x)$ and displays two branches.

Example 32.37 (Advanced 6: Regular implies reduced locally)**.** A regular local ring is a domain, hence reduced. Therefore a nonreduced local scheme cannot be regular at a point where nilpotents survive.

Example 32.38 (Advanced 7: A normal singularity warning)**.** Normality and regularity are not equivalent in dimension at least two. Later chapters may use the quadric cone as a standard normal-but-singular example after the needed normality criterion has been established.

Example 32.39 (Advanced 8: Regular curves and normality)**.** For Noetherian integral curves, a regular local ring at a closed point is a DVR. Conversely, a one-dimensional Noetherian normal local domain is a DVR, so normal integral curves are regular.

Example 32.40 (Advanced 9: Base field and tangent spaces)**.** At a non-rational point, the intrinsic cotangent space is naturally a vector space over $\kappa(x)$, not necessarily over the ground field k. Relative tangent spaces over k require the corresponding differential formalism.

Example 32.41 (Advanced 10: Closed-point equality uses finite type)**.** The proof $\dim \mathcal{O}_{X,x} = \dim X$ at closed points uses that maximal ideals of finitely generated k-domains have residue fields algebraic over k and full height. It is not a theorem for arbitrary integral Noetherian schemes.

Example 32.42 (Advanced 11: Singular locus versus one singular point)**.** Computing one Jacobian rank identifies the behavior at one point only. Determining the whole singular locus means solving the defining equations together with the appropriate rank-drop conditions.

Example 32.43 (Advanced 12: Products of regular affine spaces)**.** At a rational point of $\mathbb{A}^m \times \mathbb{A}^n$, cotangent spaces split as a direct sum, so embedding dimensions add. This is the infinitesimal counterpart of product dimension.

Example 32.44 (Advanced 13: Completion preview)**.** Passing from a Noetherian local ring to its adic completion preserves dimension and regularity. This powerful local replacement is stated only as a preview here; completions are not developed in VI/32.

Example 32.45 (Advanced 14: Smoothness boundary)**.** For finite-type schemes over a perfect field, regularity and geometric smoothness agree at rational points under the standard hypotheses. Over imperfect fields, regular need not imply smooth over the base. Smooth morphisms are therefore deferred.

32.14 Problem dossiers

Problem 32.46 (VI.32.P1: AG8 local dimension at closed points)**.** Let X be a variety over k and let $x \in X$ be closed. Prove

$$\dim \mathcal{O}_{X,x} = \dim X.$$

Provenance. Legacy AG8 Problem 2 local-dimension slice; canonical destination VI/32.

Problem 32.47 (VI.32.P2: AG8 Problem 3(a): failure for a general scheme)**.** Let

$$X = \operatorname{Spec} k \sqcup \mathbb{A}^1_k$$

and let x be the point of the first component. Show that x is closed but

$$\dim \mathcal{O}_{X,x} \neq \dim X.$$

Provenance. Legacy AG8 Problem 3(a) counterexample requirement; canonical destination VI/32.

Problem 32.48 (VI.32.P3: Affine local dimension equals height)**.** Let $x = \mathfrak{p} \in \operatorname{Spec} A$. Prove

$$\dim \mathcal{O}_{X,x} = \operatorname{ht}(\mathfrak{p}).$$

Provenance. Canonical VI/32 synthesis.

Problem 32.49 (VI.32.P4: Cotangent space of affine space)**.** Compute $\mathfrak{m}/\mathfrak{m}^2$ at the origin of $\mathbb{A}^n_k$.

Provenance. Canonical VI/32 synthesis.

Problem 32.50 (VI.32.P5: Nakayama and embedding dimension)**.** For a Noetherian local ring $(R, \mathfrak{m}, \kappa)$, prove that

$$\dim_\kappa \mathfrak{m}/\mathfrak{m}^2$$

is the minimal number of generators of $\mathfrak{m}$.

Provenance. Canonical VI/32 synthesis.

Problem 32.51 (VI.32.P6: Parabola is regular)**.** At the origin of

$$V(y - x^2) \subset \mathbb{A}^2_k,$$

compute local dimension and tangent dimension and prove regularity.

Provenance. Canonical VI/32 synthesis.

Problem 32.52 (VI.32.P7: Cusp singularity)**.** Assume $\operatorname{char} k \neq 2, 3$. Prove the origin of

$$y^2 = x^3$$

is singular.

Provenance. Canonical VI/32 synthesis.

Problem 32.53 (VI.32.P8: Node singularity)**.** Assume $\operatorname{char} k \neq 2$. Prove the origin of

$$y^2 = x^2(x + 1)$$

is singular and identify the two tangent-cone directions.

Provenance. Canonical VI/32 synthesis.

Problem 32.54 (VI.32.P9: Crossing axes)**.** Analyze regularity at the origin of

$$V(xy) \subset \mathbb{A}^2_k.$$

Provenance. Canonical VI/32 synthesis.

Problem 32.55 (VI.32.P10: Dual numbers are singular)**.** Prove the unique point of

$$\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$$

is not regular.

Provenance. Canonical VI/32 synthesis.

Problem 32.56 (VI.32.P11: Nilpotent thickened line)**.** At the origin of

$$\operatorname{Spec} k[x, \varepsilon]/(\varepsilon^2),$$

compute local and embedding dimensions.

Provenance. Canonical VI/32 synthesis.

Problem 32.57 (VI.32.P12: Jacobian tangent formula)**.** Derive the tangent equations

$$J(a)v = 0$$

for

$$X = V(f_1, \dots, f_r) \subset \mathbb{A}^n$$

at a rational point a.

Provenance. Canonical VI/32 synthesis.

Problem 32.58 (VI.32.P13: Hypersurface criterion)**.** Let $f \neq 0$ be nonconstant and $a \in V(f)(k)$. Prove that a nonzero gradient at a implies regularity.

Provenance. Canonical VI/32 synthesis.

Problem 32.59 (VI.32.P14: Non-rational point over $\mathbb{R}$)**.** For

$$x = (t^2 + 1) \in \operatorname{Spec} \mathbb{R}[t],$$

compute $\kappa(x)$, local dimension, and embedding dimension.

Provenance. Canonical VI/32 synthesis.

Problem 32.60 (VI.32.P15: Local dimension and codimension bridge)**.** Let $Y \subseteq X$ be irreducible closed with generic point η_Y. Reconcile VI/31 with the present definition by proving

$$\operatorname{codim}(Y, X) = \dim^{\mathrm{loc}}_{\eta_Y} X.$$

Provenance. Canonical VI/32 synthesis.

Problem 32.61 (VI.32.P16: Cusp multiple lenses)**.** For

$$A = k[t^2, t^3],$$

compute the local dimension and embedding dimension at (t^2, t^3) and explain the normalization comparison.

Provenance. Canonical VI/32 synthesis.

Problem 32.62 (VI.32.P17: Regularity is intrinsic)**.** Explain why regularity does not depend on a chosen affine embedding of X.

Provenance. Canonical VI/32 synthesis.

Problem 32.63 (VI.32.P18: Regularity and reduction)**.** Give an example showing that reduction can change regularity even though it does not change Krull dimension.

Provenance. Canonical VI/32 synthesis.

Problem 32.64 (VI.32.P19: Tangent dimensions in a product)**.** Compute the tangent space at the origin of

$$\mathbb{A}^m_k \times \mathbb{A}^n_k$$

and relate its dimension to the factors.

Provenance. Canonical VI/32 synthesis.

Problem 32.65 (VI.32.P20: Legacy DG4 Problem 13 — unions of three lines)**.** Assume k is algebraically closed and $\operatorname{char} k \neq 2$. Compare

$$X = V(xy, xz, yz) \subset \mathbb{A}^3_k \qquad \text{and} \qquad Y = V(uv(u - v)) \subset \mathbb{A}^2_k.$$

Both are unions of three affine lines meeting at one point. Prove that X and Y are not isomorphic by comparing their Zariski tangent spaces at the common intersection point.

Provenance. Canonical recovery of legacy DG4 Problem 13, whose tangent-space invariant was reserved for VI/32. The two line-union models are the running examples already established in VI/05.

32.15 Exercises

Exercise 32.66 (VI.32.E1)**.** Compute $\dim \mathcal{O}_{\mathbb{A}^1,0}$.

Exercise 32.67 (VI.32.E2)**.** Compute $\dim \mathcal{O}_{\mathbb{A}^3,0}$.

Exercise 32.68 (VI.32.E3)**.** At $\mathfrak{p} = (x) \subset k[x, y, z]$, compute the local-ring dimension.

Exercise 32.69 (VI.32.E4)**.** At the generic point of an integral variety, compute the local-ring dimension.

Exercise 32.70 (VI.32.E5)**.** For a 5-dimensional variety and a closed point x, compute $\dim \mathcal{O}_{X,x}$.

Exercise 32.71 (VI.32.E6)**.** Compute the embedding dimension of $\mathbb{A}^4$ at the origin.

Exercise 32.72 (VI.32.E7)**.** For $k[t]_{(t)}$, compute $\dim_k \mathfrak{m}/\mathfrak{m}^2$.

Exercise 32.73 (VI.32.E8)**.** Is $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ regular?

Exercise 32.74 (VI.32.E9)**.** Compute the tangent dimension of $V(y-x^2)$ at the origin.

Exercise 32.75 (VI.32.E10)**.** Compute the tangent dimension of $V(y^2-x^3)$ at the origin in characteristic not $2,3$.

Exercise 32.76 (VI.32.E11)**.** Compute the tangent dimension of $V(xy)$ at the origin.

Exercise 32.77 (VI.32.E12)**.** For $V(x) \subset \mathbb{A}^3$, compute local and tangent dimensions at the origin.

Exercise 32.78 (VI.32.E13)**.** State the regular-local numerical criterion.

Exercise 32.79 (VI.32.E14)**.** What does Nakayama identify with embedding dimension?

Exercise 32.80 (VI.32.E15)**.** For a hypersurface point with nonzero gradient, regular or singular?

Exercise 32.81 (VI.32.E16)**.** Can all first derivatives vanish at a regular hypersurface point?

Exercise 32.82 (VI.32.E17)**.** Does local dimension equal tangent dimension at every point?

Exercise 32.83 (VI.32.E18)**.** Does reduction always preserve regularity?

Exercise 32.84 (VI.32.E19)**.** At $x=(t^2+1)$ in $\operatorname{Spec} \mathbb{R}[t]$, what is $\kappa(x)$?

Exercise 32.85 (VI.32.E20)**.** At that non-rational point, over which field is $\mathfrak{m}/\mathfrak{m}^2$ naturally a vector space?

Exercise 32.86 (VI.32.E21)**.** What is the local dimension at the point component of $\operatorname{Spec} k \sqcup \mathbb{A}^1$?

Exercise 32.87 (VI.32.E22)**.** What is the global dimension of that disjoint union?

Exercise 32.88 (VI.32.E23)**.** What does the cusp normalization change at the singular point?

Exercise 32.89 (VI.32.E24)**.** Do a cusp and a node have the same tangent cone?

Exercise 32.90 (VI.32.E25)**.** Which earlier chapter identifies $\dim \mathcal{O}_{X,\eta_Y}$ with codimension?

Exercise 32.91 (VI.32.E26)**.** Name two subjects explicitly deferred beyond VI/32.

32.16 Challenges

Problem 32.92 (Challenge VI.32.C1: Jacobian singular locus)**.** For a plane curve $f(x,y)=0$, compute the singular locus by solving $f=f_x=f_y=0$. Work this out for the cusp, the node, and one curve of your choice.

Problem 32.93 (Challenge VI.32.C2: Tangent vectors as dual-number points)**.** Prove directly that tangent vectors at a k-rational point correspond to lifts $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to X$ reducing to that point. Translate the lifting condition into derivations.

Problem 32.94 (Challenge VI.32.C3: Regularity under localization)**.** Let A be Noetherian. Investigate how regularity behaves under localization $A_{\mathfrak{p}} \to A_{\mathfrak{q}}$ for $\mathfrak{q} \subseteq \mathfrak{p}$ and formulate the regular-locus consequence.

Problem 32.95 (Challenge VI.32.C4: Normal versus regular in dimension two)**.** Using a normality criterion from earlier chapters or an allowed external theorem, analyze the quadric cone $xy - z^2 = 0$ and explain why it is the standard warning that normal need not mean regular in dimension at least two.

Problem 32.96 (Challenge VI.32.C5: From local geometry to smoothness)**.** Build a comparison table for regular point, nonsingular Jacobian point, smooth point over k, geometrically regular point, and normal point. State precisely which implications require perfectness or finite-type hypotheses.

32.17 Chapter synthesis

Local geometry compresses the neighborhood of a point into the local ring and then linearizes its maximal ideal. The Krull dimension $\dim \mathcal{O}_{X,x}$ measures specialization depth, while $\mathfrak{m}_x/\mathfrak{m}_x^2$ measures independent first-order directions. A Noetherian point is regular exactly when these numbers agree. The Jacobian matrix makes the comparison computable for affine equations, while the cusp, node, crossing, and dual numbers show how singularity can arise from ramification, branching, reducibility, or nilpotent structure. With VI/32, Part VI has moved from global prime chains to a pointwise diagnostic of geometric regularity.

$$\boxed{\text{local ring}} \longrightarrow \boxed{\mathfrak{m}/\mathfrak{m}^2} \longrightarrow \boxed{\text{tangent space}} \longrightarrow \boxed{\text{regularity}}$$

Figure 32.10: VI/32 turns local algebra into a first-order geometric regularity test.

32.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 32.97 (Affine space)**.** Compute $T_p\mathbb{A}^n_k$.

Hint. Use Zariski tangent spaces and cotangent quotients. $\square$

Solution. k^n. $\square$

Exercise 32.98 (Parabola tangent)**.** Find the tangent equation to $y - x^2 = 0$ at (a, a^2).

Hint. Use Zariski tangent spaces and cotangent quotients. $\square$

Solution. $-2a\,u + v = 0$. $\square$

Exercise 32.99 (Node origin)**.** Compute the tangent-space dimension of $xy = 0$ at the origin.

Hint. Use Zariski tangent spaces and cotangent quotients. $\square$

Solution. 2. $\square$

Exercise 32.100 (Cotangent)**.** What vector space dualizes to the tangent space at a k-rational point?

Hint. Use Zariski tangent spaces and cotangent quotients. ☐

Solution. $\mathfrak{m}/\mathfrak{m}^2$. ☐

Exercise 32.101 (Jacobian)**.** How are tangent equations obtained from defining equations?

Hint. Use Zariski tangent spaces and cotangent quotients. ☐

Solution. By evaluating their first derivatives at the point. ☐

Proofs

Exercise 32.102 (Cotangent definition)**.** Prove: Prove derivations at a k-rational point correspond to dual vectors of $\mathfrak{m}/\mathfrak{m}^2$.

Hint. Work directly from Zariski tangent spaces and cotangent quotients. ☐

Solution. A derivation kills constants and products of two maximal-ideal elements, so it factors through the cotangent quotient; conversely a linear functional defines a derivation on first-order classes. ☐

Exercise 32.103 (Jacobian kernel)**.** Prove: Prove the tangent space of an affine algebraic set is the kernel of the Jacobian matrix at the point.

Hint. Work directly from Zariski tangent spaces and cotangent quotients. ☐

Solution. Linearize each defining polynomial modulo second-order terms. ☐

Exercise 32.104 (Smooth parabola)**.** Prove: Prove the parabola has one-dimensional tangent space everywhere.

Hint. Work directly from Zariski tangent spaces and cotangent quotients. ☐

Solution. The gradient $(-2a, 1)$ has rank one at every point. ☐

Exercise 32.105 (Node singular)**.** Prove: Explain why the origin of $xy = 0$ is singular using tangent dimension.

Hint. Work directly from Zariski tangent spaces and cotangent quotients. ☐

Solution. The tangent space has dimension two while every irreducible component has local dimension one. ☐

Counterexamples and hypothesis tests

Exercise 32.106 (Tangent dimension)**.** Test the claim: tangent-space dimension always equals local dimension.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False at singular points. ☐

Exercise 32.107 (Gradient nonzero)**.** Test the claim: if all first derivatives vanish at a hypersurface point, the point must be smooth.

Hint. Test the claim on a concrete affine or local model before generalizing. ☐

Solution. False; vanishing gradient is the standard singularity warning. ☐

Exercise 32.108 (Embedding independent)**.** Test the claim: the Jacobian matrix itself is intrinsic.

Hint. Test the claim on a concrete affine or local model before generalizing. □

Solution. False; the intrinsic tangent space is independent of presentation, while a Jacobian depends on chosen equations and coordinates. □

Applications and investigations

Exercise 32.109 (Singularity detection)**.** How can tangent spaces detect singularities?

Hint. Translate the question into Zariski tangent spaces and cotangent quotients. □

Solution. An unexpectedly large tangent dimension indicates failure of smooth local geometry. □

Exercise 32.110 (First-order motion)**.** What does a tangent vector represent?

Hint. Translate the question into Zariski tangent spaces and cotangent quotients. □

Solution. A first-order infinitesimal direction compatible with the defining equations. □

Challenge problems

Exercise 32.111 (Synthesis challenge)**.** Combine the main algebraic and geometric viewpoints of Tangent Spaces and Local Geometry in one explicit argument, using at least one localization, quotient, stalk, fiber, or prime-ideal calculation when appropriate.

Hint. Start from one of the worked examples, then reformulate it using the chapter's structural correspondence. □

Solution. A complete solution identifies the concrete model from Tangent Spaces and Local Geometry, translates it through Zariski tangent spaces and cotangent quotients, and checks the correspondence in both algebraic and geometric language. □

Exercise 32.112 (Hypothesis challenge)**.** Choose one structural statement from Tangent Spaces and Local Geometry, remove one essential hypothesis, and construct or explain a counterexample.

Hint. Use one of the three hypothesis tests above as a starting point and sharpen it to the theorem under discussion. □

Solution. The solution must name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, sheaf, or morphism where the conclusion no longer holds. □

Part VII

Modules and Projective Geometry

Chapter 33

$\mathcal{O}_X$-Modules and Quasi-Coherent Sheaves

The structure sheaf turns a scheme into a space on which algebra can vary from open set to open set. The next step is to let modules vary in the same way. An $\mathcal{O}_X$-module is the scheme-theoretic analogue of a module over a ring; quasi-coherent sheaves are the modules that, on affine charts, really do come from ordinary modules. This is the correct language for ideal sheaves, vector bundles, closed subschemes, and eventually twisting sheaves on projective space.

This chapter deliberately comes before `Proj`. Projective geometry is built from graded rings, but its geometry is expressed through sheaves. Developing the module-sheaf dictionary first makes the later construction of $\widetilde{M}$ on $\operatorname{Proj} S$, the sheaves $\mathcal{O}_{\mathbf{P}^n}(d)$, and coherent ideal sheaves conceptually natural rather than abrupt.

Learning goals

By the end of the chapter, the reader should be able to:

- define sheaves of $\mathcal{O}_X$-modules and their morphisms;
- compute stalks and use the fact that exactness is stalkwise;
- construct the affine sheaf $\widetilde{M}$ attached to an A-module M;
- prove $\widetilde{M}(D(f)) \cong M_f$ and $(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}$;
- explain why $M \mapsto \widetilde{M}$ is exact;
- define quasi-coherent sheaves and state the affine equivalence

$$A\text{-Mod} \simeq \operatorname{QCoh}(\operatorname{Spec} A);$$

- identify ideal sheaves and quotient structure sheaves on affine closed subschemes;
- recognize finite-type, coherent, and locally free sheaves in the locally Noetherian setting;
- compute supports of finite modules and understand their geometric meaning;
- compute pullback of a quasi-coherent sheaf on affine schemes by tensor product.

Conceptual roadmap

The central dictionary is

$$\boxed{\begin{array}{c} A\text{-module } M \\ \downarrow\ M \mapsto \widetilde{M} \\ \mathcal{O}_{\operatorname{Spec} A}\text{-module} \\ \downarrow\ \text{affine-local model} \\ \text{quasi-coherent sheaf on a scheme.} \end{array}}$$

Localization is the mechanism underneath every arrow: the section module on $D(f)$ is M_f, and the stalk at $\mathfrak{p}$ is $M_{\mathfrak{p}}$.

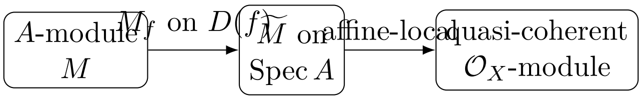


Figure 33.1: The module–sheaf bridge: localization turns algebra into geometry.

33.1 Modules over the structure sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space.

Definition 33.1 ($\mathcal{O}_X$-module)**.** A sheaf $\mathcal{F}$ of abelian groups on X is an $\mathcal{O}_X$-module if, for every open set $U \subseteq X$, the group $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module and every restriction map

$$\mathcal{F}(U) \longrightarrow \mathcal{F}(V), \qquad V \subseteq U,$$

is linear with respect to the restriction homomorphism $\mathcal{O}_X(U) \to \mathcal{O}_X(V)$.

The structure sheaf itself is an $\mathcal{O}_X$-module. So are finite free sheaves $\mathcal{O}_X^{\oplus r}$, ideal sheaves $\mathcal{I} \subseteq \mathcal{O}_X$, quotients $\mathcal{O}_X/\mathcal{I}$, kernels, cokernels, and tensor products of module sheaves.

A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of $\mathcal{O}_X$-modules is a morphism of sheaves whose maps on sections are $\mathcal{O}_X(U)$-linear for every open U.

33.2 Stalks are modules over local rings

For every $x \in X$, the stalk $\mathcal{F}_x$ is naturally a module over the local ring $\mathcal{O}_{X,x}$. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ induces an $\mathcal{O}_{X,x}$-linear map

$$\varphi_x : \mathcal{F}_x \longrightarrow \mathcal{G}_x.$$

Thus module-sheaf questions can be tested pointwise.

Theorem 33.2 (Stalkwise criterion)**.** *A morphism of $\mathcal{O}_X$-modules is an isomorphism if and only if it is an isomorphism on every stalk. A sequence*

$$\mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}''$$

is exact if and only if the induced sequence on every stalk is exact.

This is the module-theoretic form of the stalkwise exactness developed earlier for sheaves of abelian groups.

Example 33.3 (The sheaf associated to a module)**.** For $X = \operatorname{Spec} A$ and an A-module M, the associated sheaf $\widetilde{M}$ satisfies $\widetilde{M}(D(f)) \cong M_f$. Thus localization of modules is exactly restriction of quasi-coherent data to distinguished affine opens.

$$
\begin{tikzcd}
\mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \arrow[d] \\
\mathcal{F}_x \arrow[r, "\varphi_x"'] & \mathcal{G}_x
\end{tikzcd}
$$

Figure 33.2: Morphisms of module sheaves induce linear maps on every stalk.

33.3 Kernels, cokernels, and exact sequences

The kernel and cokernel of a morphism of $\mathcal{O}_X$-modules are formed in the category of sheaves, with the inherited module structures. The image must be sheafified when necessary; the pointwise presheaf image is not automatically a sheaf.

A short exact sequence

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

therefore means exactness on all stalks. This language will later control ideal sheaves, divisor sequences, and cohomology.

33.4 Restriction and locally free sheaves

If $U \subseteq X$ is open, restriction gives an $\mathcal{O}_U$-module $\mathcal{F}|_U$.

Definition 33.4 (Locally free sheaf)**.** An $\mathcal{O}_X$-module $\mathcal{E}$ is *locally free of rank r* if every point has an open neighborhood U such that

$$\mathcal{E}|_U \cong \mathcal{O}_U^{\oplus r}.$$

Finite locally free sheaves are the sheaf-theoretic form of algebraic vector bundles. Rank is locally constant, so on a connected scheme a finite locally free sheaf has constant rank.

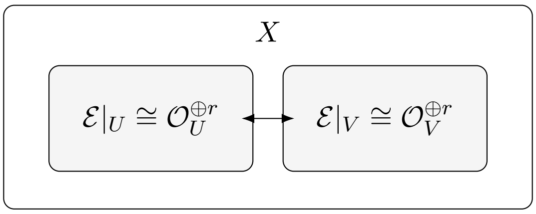


Figure 33.3: A locally free sheaf is free on local charts, with transition data on overlaps.

33.5 From an affine module to a sheaf

Let A be a ring, $X = \operatorname{Spec} A$, and let M be an A-module. On the distinguished-open basis define

$$D(f) \longmapsto M_f.$$

If $D(g) \subseteq D(f)$, the restriction is the canonical localization map $M_f \to M_g$.

Theorem 33.5 (The affine module sheaf)**.** *There is a unique sheaf of $\mathcal{O}_X$-modules, denoted $\widetilde{M}$, whose sections on distinguished opens satisfy*

$$\boxed{\widetilde{M}(D(f)) \cong M_f.}$$

For every prime $\mathfrak{p} \subseteq A$,

$$\boxed{(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}.}$$

Moreover

$$\Gamma(X, \widetilde{M}) \cong M.$$

The construction is the module analogue of the structure sheaf itself: $\widetilde{A} \cong \mathcal{O}_X$.

$$\begin{tikzcd}
M \arrow[r, "\text{localize}"] \arrow[d, "\text{sheafify}"'] & M_f \arrow[d, leftrightarrow, "\cong"] \\
\widetilde{M} \arrow[r, "\text{restrict}"'] & \widetilde{M}(D(f))
\end{tikzcd}$$

Figure 33.4: Localization commutes with the affine module-sheaf construction.

33.6 Localization makes the construction exact

Localization of modules is exact. Therefore an exact sequence

$$0 \to M' \to M \to M'' \to 0$$

of A-modules yields an exact sequence

$$0 \to \widetilde{M'} \to \widetilde{M} \to \widetilde{M''} \to 0$$

of $\mathcal{O}_X$-modules.

Proposition 33.6 (Compatibility with standard operations)**.** *For A-modules M, N,*

$$\widetilde{M \oplus N} \cong \widetilde{M} \oplus \widetilde{N}, \qquad \widetilde{M \otimes_A N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}.$$

Arbitrary direct sums are also preserved.

The tensor identity can be checked on $D(f)$, where both sides identify with $M_f \otimes_{A_f} N_f$.

Example 33.7 (An ideal sheaf on affine space)**.** For $A = k[x, y]$ and $I = (x, y)$, the quasi-coherent ideal sheaf $\widetilde{I}$ restricts on $D(x)$ to $I_x = A_x$ because x becomes a unit. On $D(y)$ the same phenomenon occurs, while at the origin the stalk remains the maximal ideal $(x, y)A_{(x,y)}$.

33.7 Quasi-coherent sheaves

The affine construction is local, and this leads to the central definition.

Definition 33.8 (Quasi-coherent sheaf)**.** An $\mathcal{O}_X$-module $\mathcal{F}$ on a scheme X is *quasi-coherent* if every point has an affine open neighborhood $U = \operatorname{Spec} A$ for which

$$\mathcal{F}|_U \cong \widetilde{M}$$

for some A-module M.

Equivalently, one may require this on an affine open cover. The definition says that quasi-coherent sheaves are exactly those module sheaves whose local behavior is governed by ordinary commutative algebra.

Theorem 33.9 (Affine module–sheaf equivalence)**.** *For $X = \operatorname{Spec} A$, the functors*

$$M \longmapsto \widetilde{M}, \qquad \mathcal{F} \longmapsto \Gamma(X, \mathcal{F})$$

give quasi-inverse equivalences

$$\boxed{A\text{-Mod} \simeq \mathrm{QCoh}(X).}$$

In particular, a quasi-coherent sheaf on an affine scheme is completely determined by its global sections.

Corollary 33.10 (Exactness of global sections on affine quasi-coherent sheaves)**.** *If*

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

is an exact sequence of quasi-coherent sheaves on an affine scheme, then

$$0 \to \Gamma(X, \mathcal{F}') \to \Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}'') \to 0$$

is exact.

On a general scheme, global sections are only left exact. The affine result is stronger, and its higher-cohomological form will later become affine vanishing.

$$\begin{tikzcd}
\boxed{A\text{-Mod}} \arrow[rr, bend left, "M \mapsto \widetilde{M}"] & & \boxed{\mathrm{QCoh}(\operatorname{Spec} A)} \arrow[ll, bend left, "\Gamma(X, -)"]
\end{tikzcd}$$

Figure 33.5: On an affine scheme, modules and quasi-coherent sheaves contain the same information.

33.8 Ideal sheaves and closed subschemes

Let $X = \operatorname{Spec} A$ and $I \subseteq A$ be an ideal. Write $i : Z = \operatorname{Spec}(A/I) \hookrightarrow X$ for the corresponding closed immersion. The inclusion $I \hookrightarrow A$ sheafifies to a quasi-coherent ideal sheaf

$$\widetilde{I} \hookrightarrow \mathcal{O}_X.$$

Exactness gives

$$0 \to \widetilde{I} \to \mathcal{O}_X \to \widetilde{A/I} \to 0.$$

The quotient is canonically $i_*\mathcal{O}_Z$, the structure sheaf of the affine closed subscheme viewed as an $\mathcal{O}_X$-module. Thus the ring-theoretic sequence

$$0 \to I \to A \to A/I \to 0$$

becomes the geometric sequence

$$\boxed{0 \to \mathcal{I} \to \mathcal{O}_X \to i_*\mathcal{O}_Z \to 0.}$$

More generally, quasi-coherent ideal sheaves are the correct global objects for closed subschemes of an arbitrary scheme.

$$\begin{tikzcd} 0 \arrow[r] & \widetilde{I} \arrow[r] & \mathcal{O}_X \arrow[r] & i_*\mathcal{O}_Z \arrow[r] & 0 \end{tikzcd}$$

$$Z = V(I) = \operatorname{Spec}(A/I) \subseteq X = \operatorname{Spec} A$$

Figure 33.6: An ideal module becomes the ideal sheaf cutting out a closed subscheme.

33.9 Finite type and coherent sheaves

A quasi-coherent sheaf $\mathcal{F}$ is of finite type if locally on affine opens it is represented by finitely generated modules. On a locally Noetherian scheme, this is the standard setting in which quasi-coherent sheaves of finite type are coherent.

Definition 33.11 (Coherent sheaf in the locally Noetherian setting)**.** Let X be locally Noetherian. A quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ is *coherent* if every point has an affine neighborhood $U = \operatorname{Spec} A$ such that

$$\mathcal{F}|_U \cong \widetilde{M}$$

with M a finitely generated A-module.

Because A is Noetherian, finitely generated modules are finitely presented. This is exactly the finiteness class needed for most projective and divisor constructions later in the volume.

Example 33.12 (Free modules give trivial bundles)**.** The sheaf $\widetilde{A^r}$ is isomorphic to $\mathcal{O}_X^{\oplus r}$. This is the affine model for a trivial rank-r vector bundle and shows how module rank becomes local geometric rank.

33.10 Support and annihilators

For any $\mathcal{O}_X$-module $\mathcal{F}$, define

$$\operatorname{Supp}(\mathcal{F}) = \{x \in X : \mathcal{F}_x \neq 0\}.$$

For $X = \operatorname{Spec} A$ and a finitely generated A-module M,

$$\boxed{\operatorname{Supp}(\widetilde{M}) = V(\operatorname{Ann}_A M).}$$

Finite generation is important for this clean closed-set formula. For example, $M = A/I$ has support exactly $V(I)$.

33.11 Pullback and tensor product

Let $f : X \to Y$ be a morphism of schemes. The pullback of an $\mathcal{O}_Y$-module is

$$f^*\mathcal{F} = \mathcal{O}_X \otimes_{f^{-1}\mathcal{O}_Y} f^{-1}\mathcal{F}.$$

Pullback preserves quasi-coherence.

On affine schemes this is just extension of scalars. If $A \to B$ corresponds to

$$f : \operatorname{Spec} B \to \operatorname{Spec} A$$

and M is an A-module, then

$$\boxed{f^*(\widetilde{M}) \cong \widetilde{B \otimes_A M}.}$$

This formula is the module-theoretic companion of base change and fiber products.

$$\begin{tikzcd}
\boxed{M \text{ over } A} \arrow[r, "\text{extend scalars}"] \arrow[d] & \boxed{B \otimes_A M \text{ over } B} \arrow[d] \\
\boxed{\widetilde{M} \text{ on } \operatorname{Spec} A} \arrow[r, "\text{pull back}"'] & \boxed{f^*\widetilde{M} \text{ on } \operatorname{Spec} B}
\end{tikzcd}$$

Figure 33.7: Affine pullback of quasi-coherent sheaves is extension of scalars.

33.12 Why this chapter precedes projective geometry

Projective geometry will replace an ordinary ring by a graded ring and an ordinary module by a graded module. The notation $\widetilde{M}$ will reappear on $\operatorname{Proj} S$, but with degree-zero localizations on the basic opens $D_+(f)$. Twisting sheaves $\mathcal{O}(d)$, homogeneous ideal sheaves, and projective embeddings all depend on the module-sheaf language developed here.

The conceptual order is therefore

$$\boxed{\text{modules} \to \mathcal{O}_X\text{-modules} \to \text{quasi-coherent sheaves} \to \text{graded modules} \to \operatorname{Proj}.}$$

33.13 Common pitfalls

- **A sheaf of modules is more than modules on opens.** The restriction maps must be compatible with scalar restriction.
- **The presheaf image need not be a sheaf.** Kernels behave sectionwise, but images and cokernels belong to the sheaf category.
- **Quasi-coherent does not mean locally constant.** It means affine-locally represented by modules.
- **Coherent is a finiteness condition.** In this chapter the clean finite-generation criterion is used only on locally Noetherian schemes.
- **Global sections do not usually recover an arbitrary sheaf.** The equivalence with modules is special to quasi-coherent sheaves on affine schemes.

- **Support need not be closed for an arbitrary module.** The formula $\operatorname{Supp}(\widetilde{M}) = V(\operatorname{Ann} M)$ is stated here for finitely generated M.
- **Pullback is tensorial; pushforward is subtler.** Preservation of quasi-coherence under pushforward requires hypotheses and is deferred.

33.14 Worked examples

Example 33.13 (The structure sheaf)**.** For $X = \operatorname{Spec} A$, one has $\widetilde{A} \cong \mathcal{O}_X$.

Example 33.14 (A free module)**.** For $M = A^{\oplus r}$, the associated sheaf is $\widetilde{M} \cong \mathcal{O}_X^{\oplus r}$.

Example 33.15 (A principal ideal)**.** For $I = (f) \subseteq A$, the sheaf $\widetilde{I}$ is the image of multiplication by f on $\mathcal{O}_X$.

Example 33.16 (A quotient module)**.** For $M = A/I$, the associated sheaf is supported on $V(I)$ and is the structure sheaf of the affine closed subscheme $\operatorname{Spec}(A/I)$.

Example 33.17 (A closed point of the affine line)**.** Let $A = k[t]$ and $M = A/(t-a)$. Then $\widetilde{M}$ is supported at the closed point $(t-a)$.

Example 33.18 (Localization)**.** For $M = A/(x)$ on $X = \operatorname{Spec} k[x,y]$, restriction to $D(x)$ is zero because $M_x = 0$.

Example 33.19 (Restriction to $D(y)$)**.** For the same module $M = A/(x)$, one has $M_y \cong k[y, y^{-1}]$ as a module over $k[x,y]_y/(x)$.

Example 33.20 (A torsion module)**.** For $A = k[t]$ and $M = A/(t^n)$, the associated coherent sheaf is supported at the origin but remembers the order-n nilpotent thickness.

Example 33.21 (Multiplication by a function)**.** The module map $A \xrightarrow{\cdot f} A$ gives a sheaf morphism $\mathcal{O}_X \xrightarrow{\cdot f} \mathcal{O}_X$ whose cokernel is $\widetilde{A/(f)}$.

Example 33.22 (Tensoring quotient modules)**.** For ideals $I, J \subseteq A$,

$$(A/I) \otimes_A (A/J) \cong A/(I+J),$$

so the tensor product of the corresponding affine quasi-coherent sheaves is supported on $V(I+J) = V(I) \cap V(J)$.

Example 33.23 (Support of a free sheaf)**.** If $M = A^{\oplus r}$ with $r > 0$, then $\operatorname{Supp}(\widetilde{M}) = \operatorname{Spec} A$.

Example 33.24 (Finite locally free from projective modules)**.** If P is a finitely generated projective A-module, then $\widetilde{P}$ is finite locally free on $\operatorname{Spec} A$.

Example 33.25 (Pullback along a quotient)**.** For $A \to A/I$ and an A-module M, the pullback to $\operatorname{Spec}(A/I)$ corresponds to

$$(A/I) \otimes_A M \cong M/IM.$$

Example 33.26 (Pullback along localization)**.** For $A \to A_f$, pullback of $\widetilde{M}$ to $D(f)$ is $\widetilde{M_f}$.

Example 33.27 (Coherent sheaf on a Noetherian affine scheme)**.** If A is Noetherian and M is finitely generated, then $\widetilde{M}$ is coherent.

Example 33.28 (Global sections on affine space)**.** For $X = \mathbb{A}^n_k$ and $M = k[x_1, \ldots, x_n]/I$, global sections of $\widetilde{M}$ recover the module M itself.

33.15 Exercises

Exercise 33.29. Show that $\mathcal{O}_X^{\oplus r}$ is an $\mathcal{O}_X$-module and compute its stalk at x.

Exercise 33.30. For $X = \operatorname{Spec} A$, prove $\widetilde{A}(D(f)) = A_f$.

Exercise 33.31. Show that $(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}$.

Exercise 33.32. Compute $\widetilde{A/I}(D(f))$.

Exercise 33.33. Show that $\widetilde{A/I}|_{D(f)} = 0$ exactly when $(A/I)_f = 0$.

Exercise 33.34. Prove that $M \mapsto \widetilde{M}$ preserves direct sums.

Exercise 33.35. Prove the tensor identity $\widetilde{M \otimes_A N} \cong \widetilde{M} \otimes \widetilde{N}$.

Exercise 33.36. Let $I \subseteq A$. Identify the stalk of $\widetilde{I}$ at $\mathfrak{p}$.

Exercise 33.37. Show that the sequence $0 \to \widetilde{I} \to \mathcal{O}_X \to \widetilde{A/I} \to 0$ is exact.

Exercise 33.38. For $A = k[t]$ and $M = A/(t^3)$, determine the support of $\widetilde{M}$.

Exercise 33.39. For $A = k[x, y]$ and $M = A/(x, y)$, compute the restrictions of $\widetilde{M}$ to $D(x)$ and $D(y)$.

Exercise 33.40. Show that a finite free sheaf is coherent on a locally Noetherian scheme.

Exercise 33.41. If M is finitely generated, prove $\operatorname{Supp}(\widetilde{M}) = V(\operatorname{Ann} M)$.

Exercise 33.42. Compute the support of $\widetilde{A/(xy)}$ on $\operatorname{Spec} k[x, y]$.

Exercise 33.43. Let $A \to B$ be a ring map. Verify on distinguished opens the formula $f^*\widetilde{M} \cong \widetilde{B \otimes_A M}$.

Exercise 33.44. For $A \to A/I$, show that pullback sends $\widetilde{M}$ to $\widetilde{M/IM}$.

Exercise 33.45. Show that a quasi-coherent sheaf is determined by its restrictions to an affine open cover together with compatible gluing maps.

Exercise 33.46. On an affine scheme, show that a morphism $\widetilde{M} \to \widetilde{N}$ is determined by the induced map on global sections.

Exercise 33.47. Deduce that $\operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \widetilde{N}) \cong \operatorname{Hom}_A(M, N)$ for $X = \operatorname{Spec} A$.

Exercise 33.48. Show that if P is finitely generated projective, then every prime has a distinguished neighborhood on which $\widetilde{P}$ is free.

Exercise 33.49. For $M = A/(f)$ and $N = A/(g)$, compute $\widetilde{M} \otimes \widetilde{N}$.

Exercise 33.50. Explain why the global-section functor is exact on affine quasi-coherent sheaves but need not be exact on arbitrary sheaves.

Exercise 33.51. Let X be locally Noetherian. Show that a quotient of a coherent sheaf by a coherent subsheaf is coherent.

Exercise 33.52. Explain how the construction $M \mapsto \widetilde{M}$ foreshadows the graded-module construction on $\operatorname{Proj} S$.

Exercise 33.53. Let $0 \to M' \to M \to M'' \to 0$ be exact. Prove directly on every distinguished open $D(f)$ that the corresponding sequence of tilde sheaves is exact, and explain why this does not assert right exactness of sections on arbitrary opens for arbitrary sheaves.

Exercise 33.54. For an A-linear map $u : M \to N$, prove

$$\widetilde{\ker u} \cong \ker(\widetilde{u}), \qquad \widetilde{\operatorname{coker} u} \cong \operatorname{coker}(\widetilde{u}).$$

33.16 Problem dossiers

Problem 33.55 (VI.33.P1: Affine reconstruction for quasi-coherent sheaves)**.** Let $X = \operatorname{Spec} A$ and let $\mathcal{F}$ be quasi-coherent. Prove that the canonical map

$$\widetilde{\Gamma(X, \mathcal{F})} \longrightarrow \mathcal{F}$$

is an isomorphism.

Problem 33.56 (VI.33.P2: Exactness of the tilde functor)**.** Prove that $M \mapsto \widetilde{M}$ is exact.

Problem 33.57 (VI.33.P3: Closed subscheme sequence)**.** Let $I \subseteq A$, let

$$i : Z = \operatorname{Spec}(A/I) \hookrightarrow X = \operatorname{Spec} A,$$

and derive the exact sequence

$$0 \to \widetilde{I} \to \mathcal{O}_X \to i_*\mathcal{O}_Z \to 0.$$

Problem 33.58 (VI.33.P4: Support of a finite module)**.** Let M be a finitely generated A-module. Prove

$$\operatorname{Supp}(\widetilde{M}) = V(\operatorname{Ann} M).$$

Problem 33.59 (VI.33.P5: Pullback is extension of scalars)**.** Let $A \to B$ and

$$f : \operatorname{Spec} B \to \operatorname{Spec} A.$$

Prove

$$f^*\widetilde{M} \cong \widetilde{B \otimes_A M}.$$

Problem 33.60 (VI.33.P6: Finite locally free versus projective)**.** Let P be a finitely generated A-module. Show that P is projective if and only if $\widetilde{P}$ is finite locally free on $\operatorname{Spec} A$.

Problem 33.61 (VI.33.P7: Tensor product and scheme-theoretic intersection)**.** For ideals $I, J \subseteq A$, prove

$$\widetilde{A/I} \otimes \widetilde{A/J} \cong \widetilde{A/(I + J)}$$

and interpret the support.

Problem 33.62 (VI.33.P8: Affine exactness of global sections)**.** Let X be affine and

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

an exact sequence of quasi-coherent sheaves. Prove that global sections remain exact on the right.

Problem 33.63 (VI.33.P9: A nilpotent thickening as a coherent module)**.** Let

$$A = k[t, \varepsilon]/(\varepsilon^2) \quad \text{and} \quad M = A/(t).$$

Describe $\operatorname{Supp}(\widetilde{M})$ and explain what module data remain invisible in the support.

Problem 33.64 (VI.33.P10: Local criterion for zero quasi-coherent sheaf)**.** Let $X = \operatorname{Spec} A$ and $\mathcal{F} = \widetilde{M}$. Prove that

$$\mathcal{F} = 0 \iff M = 0,$$

and that it is enough to check vanishing on a distinguished cover.

Problem 33.65 (VI.33.P11: Morphisms are module maps on affine schemes)**.** For $X = \operatorname{Spec} A$, prove
$$\operatorname{Hom}_{\mathcal{O}_X}(\widetilde{M}, \widetilde{N}) \cong \operatorname{Hom}_A(M, N).$$

Problem 33.66 (VI.33.P12: Bridge to graded modules)**.** Let S be a graded ring and M a graded S-module. Explain which part of the affine construction must change on $\operatorname{Proj} S$.

Problem 33.67 (VI.33.P13: Restriction to a principal affine open)**.** Let $X = \operatorname{Spec} A$, M an A-module, and $f \in A$. Prove
$$\widetilde{M}|_{D(f)} \cong \widetilde{M_f}$$
where the right side is formed as an A_f-module sheaf on $\operatorname{Spec} A_f = D(f)$.

Provenance. Canonical VI/33 extension.

Problem 33.68 (VI.33.P14: Kernels and cokernels under tilde)**.** Let $u : M \to N$ be A-linear. Prove
$$\ker(\widetilde{u}) \cong \widetilde{\ker u}, \qquad \operatorname{coker}(\widetilde{u}) \cong \widetilde{\operatorname{coker} u}.$$

Provenance. Canonical VI/33 extension.

Problem 33.69 (VI.33.P15: Quasi-coherent kernels and cokernels)**.** Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of quasi-coherent sheaves on a scheme X. Prove that $\ker \varphi$ and $\operatorname{coker} \varphi$ are quasi-coherent.

Provenance. Canonical VI/33 extension.

Problem 33.70 (VI.33.P16: Isomorphisms detected by localization)**.** Let $u : M \to N$. Prove that the following are equivalent:
$$\begin{aligned} &u \text{ is an isomorphism} \\ \iff\ & u_{\mathfrak{p}} \text{ is an isomorphism for every } \mathfrak{p} \\ \iff\ & \widetilde{u} \text{ is an isomorphism.} \end{aligned}$$

Provenance. Canonical VI/33 extension.

Problem 33.71 (VI.33.P17: Modules living on a closed subscheme)**.** Let $I \subseteq A$ and M an A-module with $IM = 0$. Show that M is naturally an A/I-module and that $\widetilde{M}$ on $\operatorname{Spec} A$ is the pushforward of the corresponding quasi-coherent sheaf on $Z = \operatorname{Spec}(A/I)$.

Provenance. Canonical VI/33 extension.

Problem 33.72 (VI.33.P18: Coherent kernels on a Noetherian affine scheme)**.** Let A be Noetherian and $u : M \to N$ a morphism of finitely generated A-modules. Prove that the kernel, image, and cokernel sheaves of $\widetilde{u}$ are coherent.

Provenance. Canonical VI/33 extension.

Problem 33.73 (VI.33.P19: Stalk of affine pullback)**.** Let $A \to B$, let $f : \operatorname{Spec} B \to \operatorname{Spec} A$, and let $\mathfrak{q} \in \operatorname{Spec} B$ with $\mathfrak{p} = \mathfrak{q} \cap A$. Prove
$$(f^* \widetilde{M})_{\mathfrak{q}} \cong B_{\mathfrak{q}} \otimes_{A_{\mathfrak{p}}} M_{\mathfrak{p}}.$$

Provenance. Canonical VI/33 extension.

Problem 33.74 (VI.33.P20: Support of a tensor product of finite modules)**.** Let M, N be finitely generated A-modules. Prove
$$\operatorname{Supp}(\widetilde{M \otimes_A N}) = \operatorname{Supp}(\widetilde{M}) \cap \operatorname{Supp}(\widetilde{N}).$$

Provenance. Canonical VI/33 extension.

33.17 Challenges

Problem 33.75 (Challenge 1: Quasi-coherent ideals and closed subschemes)**.** Formulate and prove the affine-local gluing statement that quasi-coherent ideal sheaves on a scheme determine closed subschemes.

Problem 33.76 (Challenge 2: Projective modules from geometry)**.** Develop the equivalence between finite locally free sheaves on an affine scheme and finitely generated projective modules, including rank functions on connected components.

Problem 33.77 (Challenge 3: Support beyond finite generation)**.** Find an A-module M for which $\operatorname{Supp}(\widetilde{M}) \neq V(\operatorname{Ann} M)$, and identify exactly where finite generation entered the proof of the closed-support formula.

Problem 33.78 (Challenge 4: Base change of closed subschemes)**.** For $A \to B$ and $I \subseteq A$, prove that the pullback of $\operatorname{Spec}(A/I)$ is $\operatorname{Spec}(B/IB)$ and recover the same statement from pullback of the ideal sheaf.

Problem 33.79 (Challenge 5: Toward Proj)**.** Starting only from the affine $M \mapsto \widetilde{M}$ construction, design the degree-zero localization recipe that should define a sheaf from a graded module on the standard opens of $\operatorname{Proj} S$.

33.18 Chapter synthesis

An A-module becomes geometry on $\operatorname{Spec} A$ by localization:

$$\widetilde{M}(D(f)) = M_f, \qquad (\widetilde{M})_{\mathfrak{p}} = M_{\mathfrak{p}}.$$

Quasi-coherent sheaves are precisely the module sheaves that look like this on affine charts, and on an affine scheme the correspondence is an equivalence of categories. Ideal sheaves, quotient structure sheaves, finite locally free sheaves, coherent sheaves, supports, and pullbacks all fit into this dictionary. The book is now ready to introduce graded modules and Proj without postponing the sheaf language on which projective geometry depends.

33.19 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 33.80 (Basic restriction)**.** Compute $\widetilde{M}(D(f))$.

Hint. Use quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. It is M_f. ☐

Exercise 33.81 (Free sheaf)**.** Identify $\widetilde{A^3}$.

Hint. Use quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. It is $\mathcal{O}_X^{\oplus 3}$. ☐

Exercise 33.82 (Ideal on $D(x)$)**.** For $I = (x, y) \subset k[x, y]$, compute I_x.

Hint. Use quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. It is the whole ring $k[x, y]_x$ because x is a unit. ☐

Exercise 33.83 (Stalk)**.** Compute $(\widetilde{M})_{\mathfrak{p}}$.

Hint. Use quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. It is $M_{\mathfrak{p}}$. ☐

Exercise 33.84 (Zero module)**.** What sheaf is associated to the zero module?

Hint. Use quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. The zero sheaf. ☐

Proofs

Exercise 33.85 (Distinguished-open formula)**.** Prove: Prove $\widetilde{M}(D(f)) \cong M_f$.

Hint. Work from quasi-coherent sheaves, localization, and module-sheaf correspondence and state the hypotheses explicitly. ☐

Solution. Use the affine sheaf construction and the localization compatibility on the distinguished-open basis. ☐

Exercise 33.86 (Stalk formula)**.** Prove: Prove $(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}$.

Hint. Work from quasi-coherent sheaves, localization, and module-sheaf correspondence and state the hypotheses explicitly. ☐

Solution. Take the direct limit of M_f over distinguished neighborhoods with $f \notin \mathfrak{p}$. ☐

Exercise 33.87 (Exactness)**.** Prove: Prove $M \mapsto \widetilde{M}$ is exact on affine schemes.

Hint. Work from quasi-coherent sheaves, localization, and module-sheaf correspondence and state the hypotheses explicitly. ☐

Solution. Localization is exact, so exactness can be checked on all stalks or distinguished opens. ☐

Exercise 33.88 (Affine recovery)**.** Prove: For a quasi-coherent sheaf $\mathcal{F}$ on affine X, prove $\mathcal{F} \cong \widetilde{\Gamma(X, \mathcal{F})}$.

Hint. Work from quasi-coherent sheaves, localization, and module-sheaf correspondence and state the hypotheses explicitly. ☐

Solution. Use a presentation by free modules and compare on distinguished opens. ☐

Counterexamples and hypothesis tests

Exercise 33.89 (All sheaves quasi-coherent)**.** Test the claim: every sheaf of $\mathcal{O}_X$-modules is quasi-coherent.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; quasi-coherence imposes affine-local module presentations. ☐

Exercise 33.90 (Global sections exact)**.** Test the claim: global sections is exact on arbitrary schemes for quasi-coherent sheaves.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it is exact on affine schemes but not generally. ☐

Exercise 33.91 (Same module everywhere)**.** Test the claim: a quasi-coherent sheaf must be globally free.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; quasi-coherent only means affine-locally associated to modules. ☐

Applications and investigations

Exercise 33.92 (Vector bundles)**.** How do finite projective modules appear geometrically on an affine scheme?

Hint. Translate the question into quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. They correspond to finite locally free quasi-coherent sheaves. ☐

Exercise 33.93 (Ideal geometry)**.** What does a quasi-coherent ideal sheaf encode?

Hint. Translate the question into quasi-coherent sheaves, localization, and module-sheaf correspondence. ☐

Solution. It encodes a closed subscheme through local quotient structure sheaves. ☐

Challenge problems

Exercise 33.94 (Synthesis challenge)**.** Connect two central viewpoints of Quasi-Coherent Modules in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Quasi-Coherent Modules and then translates it through quasi-coherent sheaves, localization, and module-sheaf correspondence, checking both descriptions agree. ☐

Exercise 33.95 (Hypothesis challenge)**.** Choose one theorem-level statement from Quasi-Coherent Modules, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. □

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Chapter 34

Graded Rings

Projective geometry begins with an algebraic asymmetry: scalar multiples should describe the same projective point, so polynomial equations must respect degree. Graded rings are the algebraic device that records this degree information. The legacy AG4 source places graded rings, homogeneous primes, the irrelevant ideal, and degree-zero homogeneous localization at the entrance to the construction of Proj. This chapter extracts that algebra into a self-contained bridge and extends it with graded modules, shifts, exact sequences, Veronese subrings, weighted gradings, and affine-cone geometry. The actual space $\operatorname{Proj} S$ is reserved for VI/35.

The preceding chapter developed the affine dictionary $M \mapsto \widetilde{M}$. The projective replacement will use a graded module M, localize at a homogeneous element f, and then retain only the degree-zero part:

$$\boxed{(A, M, M_f) \quad \rightsquigarrow \quad (S, M, (S_f)_0, (M_f)_0).}$$

Learning goals

By the end of the chapter, the reader should be able to:

- define $\mathbb{Z}$-graded and nonnegatively graded rings and identify homogeneous elements;
- recognize homogeneous ideals, quotients, radicals, and homogeneous prime ideals;
- use the irrelevant ideal $S_+ = \bigoplus_{n>0} S_n$;
- define graded modules, homogeneous maps, and shifts $M(d)$;
- localize graded rings and modules at homogeneous elements;
- compute $S_{(f)} = (S_f)_0$ and $M_{(f)} = (M_f)_0$;
- calculate $k[x_0, \dots, x_n]_{(x_i)}$;
- construct Veronese subrings and compare standard with weighted gradings;
- interpret $\operatorname{Spec} S$ as an affine cone in standard homogeneous situations;
- state exactly which graded-algebra data VI/35 needs to construct $\operatorname{Proj} S$.

Conceptual roadmap

$$\boxed{\begin{array}{c}\text{graded ring} \to \text{homogeneous algebra} \to \text{graded modules} \\ \to \text{homogeneous localization} \to \text{degree zero} \to \text{future Proj charts}\end{array}}$$

Degree is not decoration: degree-zero fractions are precisely the fractions unchanged by simultaneous scaling.

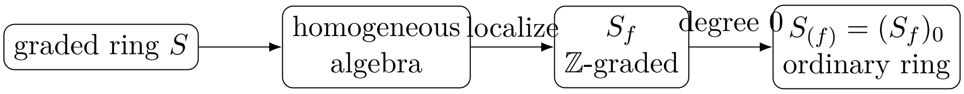


Figure 34.1: From graded algebra to the ordinary rings used on future projective charts.

34.1 Graded rings and homogeneous elements

Definition 34.1 (Graded ring)**.** A $\mathbb{Z}$*-graded ring* is a ring S with additive subgroups S_n, $n \in \mathbb{Z}$, such that

$$S = \bigoplus_{n\in\mathbb{Z}} S_n, \qquad S_m S_n \subseteq S_{m+n}.$$

If $S_n = 0$ for $n < 0$, the grading is *nonnegative*. A nonzero element of S_n is homogeneous of degree n.

Every $s \in S$ has a unique finite decomposition $s = \sum_n s_n$, $s_n \in S_n$. If a, b are homogeneous and $ab \neq 0$, then $\deg(ab) = \deg a + \deg b$.

The standard example is

$$S = k[x_0, \ldots, x_n], \qquad \deg x_i = 1,$$

where S_d consists of homogeneous polynomials of total degree d. Positive weights $\deg x_i = w_i$ give weighted gradings.

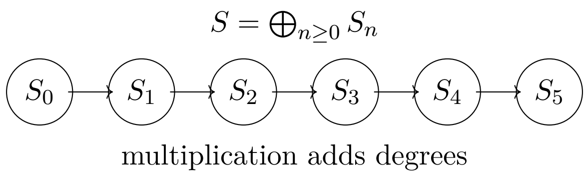


Figure 34.2: The grading decomposes the ring into degree pieces.

34.2 Graded maps and standard graded algebras

A ring homomorphism $\varphi : S \to T$ is graded of degree zero if $\varphi(S_n) \subseteq T_n$ for all n.

A nonnegatively graded ring S is *standard graded over* S_0 if it is generated as an S_0-algebra by S_1. Thus $k[x_0, \ldots, x_n]$ with all variables of degree one is standard graded, while $k[x, y]$ with $\deg x = 1, \deg y = 2$ is not.

34.3 Homogeneous ideals and quotients

Definition 34.2 (Homogeneous ideal)**.** An ideal $I \subseteq S$ is homogeneous if it is generated by homogeneous elements.

Proposition 34.3 (Component criterion)**.** *For an ideal $I \subseteq S$, the following are equivalent:*

1. *I is homogeneous;*

2. *if $f \in I$, every homogeneous component of f lies in I;*
3. $I = \bigoplus_n (I \cap S_n)$.

If I is homogeneous, then S/I inherits a grading. Conversely, the natural quotient map is graded for the quotient grading only when the kernel is homogeneous.

Proposition 34.4 (Homogeneous radicals). *If I is homogeneous, then $\sqrt{I}$ is homogeneous.*

$$\begin{tikzcd}
\boxed{\text{homogeneous } I \subseteq S} \arrow[r] \arrow[d] & \boxed{\text{graded } S/I} \arrow[d] \\
\boxed{\sqrt{I} \text{ homogeneous}} \arrow[r] & \boxed{\text{homogeneous primes}}
\end{tikzcd}$$

Figure 34.3: Homogeneous ideals remain compatible with quotient and radical operations.

Example 34.5 (Standard polynomial grading). The ring $k[x_0, \ldots, x_n]$ is graded by total degree. A polynomial such as $x_0^2 + x_1 x_2$ is homogeneous of degree 2, while $x_0 + x_1^2$ is not homogeneous.

34.4 Homogeneous prime ideals

A prime ideal need not be homogeneous. The primes compatible with projective grading are the homogeneous ones.

Proposition 34.6 (Homogeneous primality test). *Let $\mathfrak{p} \subsetneq S$ be homogeneous. Then $\mathfrak{p}$ is prime if and only if*

$$ab \in \mathfrak{p} \Longrightarrow a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}$$

for all homogeneous $a, b \in S$.

Proof. Pass to $S/\mathfrak{p}$. Suppose products of nonzero homogeneous elements are nonzero. If nonzero f, g satisfy $fg = 0$, choose the least-degree nonzero homogeneous components f_r, g_s. Then $f_r g_s$ is the least-degree component of fg and cannot cancel with any other term, contradiction. □

34.5 The irrelevant ideal

Assume $S = \bigoplus_{n \geq 0} S_n$.

Definition 34.7 (Irrelevant ideal). The irrelevant ideal is

$$S_+ := \bigoplus_{n>0} S_n.$$

In $S = k[x_0, \ldots, x_n]$ with the standard grading,

$$S_+ = (x_0, \ldots, x_n).$$

The terminology anticipates VI/35: homogeneous primes containing all of S_+ correspond to the cone-vertex direction and are excluded from the projective spectrum.

$$\begin{tikzcd} \boxed{S_+ = S_1 \oplus S_2 \oplus S_3 \oplus \cdots} \arrow[d, "\text{quotient by } S_+"] \\ \boxed{S_0} \end{tikzcd}$$

standard polynomial case:
$S_+ = (x_0, \dots, x_n)$

Figure 34.4: The irrelevant ideal contains all positive-degree pieces.

34.6 Graded modules

Definition 34.8 (Graded module)**.** A graded S-module is an S-module M with

$$M = \bigoplus_{n \in \mathbb{Z}} M_n, \qquad S_m M_n \subseteq M_{m+n}.$$

A map $\varphi : M \to N$ is homogeneous of degree r if $\varphi(M_n) \subseteq N_{n+r}$. Degree-zero maps are the usual morphisms of graded modules. Kernels, images, and cokernels of degree-zero maps are graded, so exactness is degreewise.

34.7 Degree shifts

Definition 34.9 (Shift)**.** For $d \in \mathbb{Z}$, define

$$M(d)_n := M_{n+d}.$$

An element originally of degree r has degree $r - d$ in $M(d)$.

If $f \in S_d$ is homogeneous, multiplication by f is a degree-zero map

$$S(-d) \xrightarrow{\cdot f} S,$$

because $S(-d)_n = S_{n-d}$ maps to S_n.

$$\begin{tikzcd} \boxed{\text{graded module } M} \arrow[r, "\text{reindex}"] & \boxed{\text{shift } M(d)} \arrow[r, "\text{homogeneous maps}"] & \boxed{S(-d) \xrightarrow{\cdot f} S} \end{tikzcd}$$

Figure 34.5: Shifts turn nonzero-degree multiplication into degree-zero graded maps.

Example 34.10 (A homogeneous ideal)**.** The ideal (x_0^2, x_0x_1, x_1^2) in $k[x_0, x_1]$ is homogeneous because it is generated by homogeneous elements. Its quotient inherits a grading.

34.8 Homogeneous localization

Let $f \in S_d$ be homogeneous of positive degree. The ordinary localization S_f becomes $\mathbb{Z}$-graded by

$$\deg\left(\frac{a}{f^r}\right) = \deg a - rd$$

for homogeneous a. Negative degrees appear because f becomes invertible.

For a graded module M, define the grading on M_f similarly:

$$\deg\left(\frac{m}{f^r}\right) = \deg m - rd.$$

Localization is exact and preserves homogeneous degree-zero maps.

34.9 Degree-zero localization

Definition 34.11 (Degree-zero homogeneous localization)**.** For homogeneous f of positive degree, write

$$S_{(f)} := (S_f)_0, \qquad M_{(f)} := (M_f)_0.$$

Thus $S_{(f)}$ consists of fractions a/f^r satisfying $\deg a = r \deg f$. It is an ordinary ring, and $M_{(f)}$ is an ordinary $S_{(f)}$-module.

For the standard polynomial ring and $f = x_i$,

$$S_{(x_i)} = k\left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i}\right] \cong k[t_1, \dots, t_n].$$

This is the algebraic reason projective n-space will have affine n-space charts.

$$\begin{tikzcd} \boxed{k[x_0, \dots, x_n]} \arrow[r, "\text{invert } x_i"] & \boxed{S_{x_i}} \arrow[r, "\text{degree zero}"] & \boxed{\begin{array}{c} S_{(x_i)} \\ = k[x_j/x_i : j \neq i] \end{array}} \end{tikzcd}$$

Figure 34.6: Standard degree-zero localization already has the algebra of affine n-space.

Proposition 34.12 (Exactness of degree-zero localization)**.** *For an exact sequence of graded modules with degree-zero maps,*

$$0 \to M' \to M \to M'' \to 0,$$

and homogeneous f, the sequence

$$0 \to M'_{(f)} \to M_{(f)} \to M''_{(f)} \to 0$$

is exact.

Proof. Ordinary localization is exact and yields an exact sequence of graded modules. Exactness of a sequence of graded modules with degree-zero maps holds in each degree, hence in degree zero. □

34.10 Localization in stages and overlap algebra

When homogeneous f, g have the same positive degree, g/f is degree zero in S_f, and

$$(S_{(f)})_{g/f} \cong S_{(fg)}.$$

More generally one uses suitable equal-degree powers of f and g. This compatibility is exactly what the affine chart rings will need on overlaps in VI/35.

34.11 Localized shifted modules

For a graded module M,

$$M(d)_{(f)} = (M(d)_f)_0$$

is an $S_{(f)}$-module. In particular, the modules $S(d)_{(f)}$ are the local algebra from which twisting sheaves will later be built. This chapter constructs the modules but does not yet glue them into sheaves on $\operatorname{Proj} S$.

Example 34.13 (Veronese subring)**.** The second Veronese subring of $k[x, y]$ is $\bigoplus_{d \geq 0} A_{2d} = k[x^2, xy, y^2]$. It packages only even-degree pieces and later gives the same projective scheme through a different homogeneous coordinate ring.

34.12 Veronese subrings

Definition 34.14 (Veronese subring)**.** For $d \geq 1$, the d-th Veronese subring is

$$S^{(d)} := \bigoplus_{n \geq 0} S_{nd},$$

regraded by $(S^{(d)})_n = S_{nd}$.

For $S = k[x, y]$,

$$S^{(2)} = k[x^2, xy, y^2] \cong k[u, v, w]/(uw - v^2).$$

Thus a regrading can turn a polynomial ring into the coordinate ring of a quadratic cone. VI/35 will explain the projective invariance behind Veronese regrading.

For a graded module M, one likewise defines

$$M^{(d)} := \bigoplus_{n \in \mathbb{Z}} M_{nd}.$$

$$\begin{tikzcd} \boxed{k[x,y]} \arrow[r, "\text{even degrees}"] & \boxed{k[x^2, xy, y^2]} \arrow[r, "\text{presentation}"] & \boxed{k[u,v,w]/(uw - v^2)} \end{tikzcd}$$

Figure 34.7: The second Veronese is the coordinate ring of a quadratic cone.

34.13 Weighted gradings

Let

$$S = k[x_0, \ldots, x_n], \qquad \deg x_i = w_i > 0.$$

A monomial has weighted degree $\sum_i a_i w_i$. Weighted gradings obey the same formal definitions, but S_1 can vanish and the chart rings $S_{(x_i)}$ need not look like ordinary polynomial rings. This is why hypotheses such as "standard graded" must be stated rather than assumed.

34.14 Affine cones and scaling

Let

$$S = k[x_0, \ldots, x_n]/I$$

with I homogeneous. Then $C = \operatorname{Spec} S$ is naturally an affine cone: for every homogeneous F of degree d,

$$F(\lambda a) = \lambda^d F(a).$$

Hence homogeneous zero loci are stable under scalar multiplication.

In the standard case over a field, S_+ cuts out the cone vertex. Projectivization will discard that vertex and encode the remaining scaling directions without choosing representatives.

34.15 The bridge to Proj

VI/35 will define $\operatorname{Proj} S$ from homogeneous prime ideals not containing S_+. For homogeneous $f \in S_+$, the basic projective open will have coordinate ring

$$\boxed{S_{(f)} = (S_f)_0.}$$

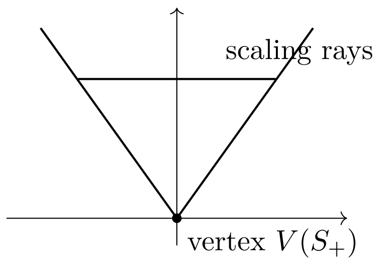


Figure 34.8: Homogeneous equations define cones stable along scaling rays.

For a graded module M, the corresponding local module will be

$$\boxed{M_{(f)} = (M_f)_0.}$$

The purpose of VI/34 is to make these formulas routine before topology and sheaf gluing are introduced.

34.16 Common pitfalls

- A general element of a graded ring need not have a degree; only nonzero homogeneous elements do.
- Not every ideal is homogeneous.
- A prime ideal need not be homogeneous.
- In a standard polynomial ring, S_+ is $(x_0, \dots, x_n)$, not the zero ideal.
- S_f and $S_{(f)}$ are different: the latter keeps only degree zero.
- Homogeneous localization naturally creates negative degrees.
- Shift conventions vary. This chapter uses $M(d)_n = M_{n+d}$.
- Standard and weighted gradings are not interchangeable.
- VI/34 does not define Proj; it prepares the algebra that VI/35 will assemble geometrically.

34.17 Worked examples

Example 34.15 (Standard polynomial grading)**.** For $S = k[x, y, z]$ with $\deg x = \deg y = \deg z = 1$, $x^2 + yz$ is homogeneous of degree two, while $x + y^2$ is not.

Example 34.16 (A homogeneous ideal)**.** The ideal $I = (x^2, xy, y^3) \subset k[x, y]$ is homogeneous, so the quotient inherits the standard grading.

Example 34.17 (A nonhomogeneous ideal)**.** The ideal $(x + 1) \subset k[x]$ is not homogeneous because it contains $x + 1$ but not its components x and 1.

Example 34.18 (Homogeneous prime)**.** In $k[x, y, z]$, (x, y) is homogeneous and prime since the quotient is $k[z]$.

Example 34.19 (Irrelevant ideal)**.** For $k[x_0, x_1, x_2]$ standard graded, $S_+ = (x_0, x_1, x_2)$ and $S/S_+ \cong k$.

Example 34.20 (A graded cone quotient)**.** The quotient $k[x,y,z]/(xy-z^2)$ is graded because the relation is homogeneous of degree two.

Example 34.21 (Shifted free module)**.** With $S(-2)_n = S_{n-2}$, the element $1 \in S_0$ has degree 2 in $S(-2)$. Multiplication by a quadratic gives a degree-zero map $S(-2) \to S$.

Example 34.22 (Graded hypersurface sequence)**.** If homogeneous $f \in S_d$ is a non-zero-divisor, then

$$0 \to S(-d) \xrightarrow{\cdot f} S \to S/(f) \to 0$$

is exact in graded modules.

Example 34.23 (Homogeneous localization)**.** For $S = k[x,y]$, $y^3/x^2 \in S_x$ has degree 1, while y^2/x^2 has degree 0.

Example 34.24 (A standard chart ring)**.** For $S = k[x_0, x_1, x_2]$,

$$S_{(x_0)} = k[x_1/x_0, x_2/x_0] \cong k[u,v].$$

Example 34.25 (A quadratic cone chart)**.** For $S = k[x,y,z]/(xy - z^2)$, localizing at x and setting $u = z/x$ gives $y/x = u^2$, hence $S_{(x)} \cong k[u]$.

Example 34.26 (Localized shifted module)**.** For $S = k[x,y]$ and $M = S(-1)$, $M_{(x)}$ is free of rank one over $k[y/x]$.

Example 34.27 (Second Veronese)**.**

$$k[x,y]^{(2)} = k[x^2, xy, y^2] \cong k[u,v,w]/(uw - v^2).$$

Example 34.28 (Weighted homogeneous polynomial)**.** If $\deg x = 2$ and $\deg y = 3$, then $x^3 + y^2$ is weighted homogeneous of degree six.

Example 34.29 (Cone scaling)**.** If F is homogeneous of degree d, then $F(\lambda a) = \lambda^d F(a)$, so $V(F)$ is stable under scalar multiplication.

Example 34.30 (Ordinary versus degree-zero localization)**.** $k[x,y]_x$ contains x^{-1} of degree -1, whereas $k[x,y]_{(x)}$ retains only degree-zero expressions such as y/x.

34.18 Exercises

Exercise 34.31 (Homogeneous decomposition)**.** Prove uniqueness of the homogeneous decomposition of an element of a graded ring.

Exercise 34.32 (Degrees of products)**.** For nonzero homogeneous a, b with $ab \neq 0$, prove $\deg(ab) = \deg a + \deg b$.

Exercise 34.33 (Standard grading dimensions)**.** For $S = k[x,y,z]$, compute $\dim_k S_d$.

Exercise 34.34 (Component criterion)**.** Prove that an ideal generated by homogeneous elements contains every homogeneous component of each of its elements.

Exercise 34.35 (Nonhomogeneous ideal)**.** Show $(x+1) \subset k[x]$ is not homogeneous.

Exercise 34.36 (Graded quotient)**.** Verify that a quotient by a homogeneous ideal inherits a grading.

Exercise 34.37 (Homogeneous radical)**.** Prove $\sqrt{I}$ is homogeneous when I is homogeneous.

Exercise 34.38 (Homogeneous prime test)**.** Prove the homogeneous primality test from the chapter.

Exercise 34.39 (Irrelevant ideal)**.** Compute S_+ for $k[x_0, \dots, x_n]$ with standard grading.

Exercise 34.40 (Weighted irrelevant ideal)**.** For $k[x, y]$ with $\deg x = 2, \deg y = 3$, compute S_+ and S_1.

Exercise 34.41 (Graded kernels)**.** Show the kernel of a degree-zero map of graded modules is homogeneous.

Exercise 34.42 (Shift convention)**.** Under $M(d)_n = M_{n+d}$, find the new degree of an element originally in M_r.

Exercise 34.43 (Multiplication map)**.** For homogeneous $f \in S_d$, verify $S(-d) \xrightarrow{\cdot f} S$ has degree zero.

Exercise 34.44 (Localization degree)**.** Show the grading on S_f is independent of the homogeneous fraction representation.

Exercise 34.45 (Compute $S_{(x)}$)**.** Prove $k[x, y]_{(x)} = k[y/x]$.

Exercise 34.46 (Three-variable chart)**.** Compute $k[x_0, x_1, x_2]_{(x_2)}$.

Exercise 34.47 (Quadric cone chart)**.** For $S = k[x, y, z]/(xy - z^2)$, compute $S_{(x)}$.

Exercise 34.48 (Module localization)**.** Show $M_{(f)}$ is an $S_{(f)}$-module.

Exercise 34.49 (Exactness after localization)**.** Show homogeneous localization is exact in graded modules.

Exercise 34.50 (Degree-zero functor)**.** Show taking the degree-zero part of an exact sequence of graded modules with degree-zero maps is exact.

Exercise 34.51 (Second Veronese)**.** Show $k[x, y]^{(2)} \cong k[u, v, w]/(uw - v^2)$.

Exercise 34.52 (Weighted degree 12)**.** With $\deg x = 2, \deg y = 3$, list monomials of weighted degree 12.

Exercise 34.53 (Cone invariance)**.** Prove the zero locus of a homogeneous ideal is stable under scalar multiplication.

Exercise 34.54 (Why degree zero?)**.** Explain why S_f is too large for a projective chart and $(S_f)_0$ is natural.

Exercise 34.55 (Localization in stages)**.** Let f, g be homogeneous of the same positive degree. Prove

$$(S_{(f)})_{g/f} \cong S_{(fg)},$$

and formulate the equal-power variant when $\deg f \neq \deg g$.

Exercise 34.56 (Shifted degree-zero localization)**.** Let $f \in S_e$ be homogeneous of positive degree. Prove

$$S(d)_{(f)} = (S_f)_d.$$

If $d = qe$, show multiplication by f^{-q} gives

$$S(d)_{(f)} \cong S_{(f)}.$$

34.19 Problem dossiers

Problem 34.57 (VI.34.P1: Homogeneous ideals from components)**.** Let $S = \bigoplus S_n$ and $I \subseteq S$. Prove the equivalence of the three conditions in the component criterion.

Problem 34.58 (VI.34.P2: Homogeneous radicals)**.** Prove that $\sqrt{I}$ is homogeneous whenever I is homogeneous.

Problem 34.59 (VI.34.P3: Homogeneous primality)**.** Let $\mathfrak{p}$ be homogeneous. Prove that it is prime if and only if it passes the prime test on homogeneous elements.

Problem 34.60 (VI.34.P4: Standard chart algebra)**.** For $S = k[x_0, \dots, x_n]$, prove $S_{(x_i)} \cong k[t_1, \dots, t_n]$ by explicit inverse maps.

Problem 34.61 (VI.34.P5: A conic chart)**.** Let $S = k[x, y, z]/(xz - y^2)$. Compute $S_{(x)}$, $S_{(z)}$, and their common overlap.

Problem 34.62 (VI.34.P6: Graded hypersurface sequence)**.** Let $f \in S_d$ be homogeneous and a non-zero-divisor. Prove

$$0 \to S(-d) \xrightarrow{\cdot f} S \to S/(f) \to 0$$

is exact in graded modules.

Problem 34.63 (VI.34.P7: Exact degree-zero localization)**.** Prove

$$0 \to M'_{(f)} \to M_{(f)} \to M''_{(f)} \to 0$$

for a short exact sequence of graded modules.

Problem 34.64 (VI.34.P8: Veronese quadratic cone)**.** Prove

$$k[x, y]^{(2)} \cong k[u, v, w]/(uw - v^2)$$

and interpret the affine spectrum.

Problem 34.65 (VI.34.P9: Weighted localization)**.** Let $S = k[x, y]$ with $\deg x = 2, \deg y = 3$. Describe $S_{(x)}$ and $S_{(y)}$.

Problem 34.66 (VI.34.P10: Cone vertex and irrelevant ideal)**.** For standard graded $S = k[x_0, \dots, x_n]/I$ over a field with I homogeneous and proper, prove S_+ cuts out the cone vertex.

Problem 34.67 (VI.34.P11: Degree-zero scaling invariance)**.** Let F/G be a homogeneous rational expression. Explain algebraically why equal numerator and denominator degree is the scaling-invariant condition.

Problem 34.68 (VI.34.P12: Design of the future standard open)**.** Using VI/33 and this chapter, formulate the ring and module that should live on a future basic projective open and state the required overlap compatibility.

Problem 34.69 (VI.34.P13: Homogeneous localization in stages)**.** Let f, g be homogeneous of positive degrees. Prove the equal-power overlap formula that compares the degree-zero rings obtained by localizing first at f and then along the g-direction.

Provenance. Canonical VI/34 extension.

Problem 34.70 (VI.34.P14: Module overlap algebra)**.** Under the hypotheses of P13, prove the corresponding formula

$$(M_{(f)})_{g^b/f^a} \cong M_{(f^a g^b)}$$

for a graded S-module M.

Provenance. Canonical VI/34 extension.

Problem 34.71 (VI.34.P15: Shift localization and local triviality)**.** Let $f \in S_e$ be homogeneous of positive degree. Show

$$S(d)_{(f)} = (S_f)_d.$$

If $e \mid d$, prove $S(d)_{(f)} \cong S_{(f)}$.

Provenance. Canonical VI/34 extension.

Problem 34.72 (VI.34.P16: Veronese chart equality)**.** Let $S^{(d)} = \bigoplus_{n \geq 0} S_{nd}$. If $f \in S_{md}$ is homogeneous, compare the chart rings obtained from S and $S^{(d)}$.

Provenance. Canonical VI/34 extension.

Problem 34.73 (VI.34.P17: A weighted chart that is not affine space)**.** Let

$$S = k[x, y, z], \qquad \deg x = \deg y = 1, \quad \deg z = 2.$$

Compute $S_{(z)}$ and identify its affine spectrum.

Provenance. Canonical VI/34 extension.

Problem 34.74 (VI.34.P18: Saturation does not change projective localizations)**.** Let $I \subseteq S$ be homogeneous and define

$$I^{\text{sat}} = I : S_+^{\infty}.$$

Prove that I^{sat} is homogeneous and that for every homogeneous $f \in S_+$,

$$(I^{\text{sat}})_f = I_f.$$

Provenance. Canonical VI/34 extension.

Problem 34.75 (VI.34.P19: Finite graded presentations)**.** Let M be generated by homogeneous elements m_i of degrees d_i. Construct a graded free presentation by shifts, assuming the relation module is finitely generated by homogeneous elements.

Provenance. Canonical VI/34 extension.

Problem 34.76 (VI.34.P20: Degree zero as the intrinsic invariant piece)**.** Explain intrinsically why the degree-zero subring is the part of a graded localization invariant under the scaling action, without relying on the size of the ground field.

Provenance. Canonical VI/34 extension.

34.20 Challenges

Problem 34.77 (Challenge 1: Saturation preview)**.** For a homogeneous ideal I in a standard graded ring, study $I : S_+^{\infty} = \bigcup_m (I : S_+^m)$. Prove it is homogeneous and investigate why removing the cone vertex could make saturation relevant.

Problem 34.78 (Challenge 2: Veronese chart comparison)**.** Compare degree-zero localizations of S and $S^{(d)}$ at corresponding homogeneous elements and identify hypotheses under which the chart rings agree.

Problem 34.79 (Challenge 3: Weighted projective preview)**.** For a weighted polynomial ring, compute several $S_{(x_i)}$ and identify phenomena absent from standard affine-space charts.

Problem 34.80 (Challenge 4: Graded finite presentations)**.** Develop graded finite presentations and show that finitely generated graded modules admit presentations by finite direct sums of shifts when the relations are finitely generated.

Problem 34.81 (Challenge 5: Reconstruct the Proj recipe)**.** Using only homogeneous primes, S_+, $S_{(f)}$, $M_{(f)}$, and VI/33 sheaf gluing, propose the definitions and theorems needed in VI/35. Separate definitions from statements requiring proof.

Source and scope audit

The inspected project audit describes legacy 'theory-of-algebraic-geometry-4.tex' as containing graded rings, homogeneous primes, the irrelevant ideal, degree-zero localization, Proj S, its structure sheaf, standard opens, projective space, and closed projective subschemes. The canonical architecture isolates the 'graded ring|homogeneous' thematic block for VI/34. This chapter therefore consumes that graded-algebra bridge and does not claim a separately numbered legacy problem. Graded modules, shifts, Veronese subrings, weighted gradings, and the affine-cone synthesis are canonical extensions needed to make the next chapter self-contained.

Chapter synthesis

A graded ring remembers how algebra transforms under scaling. Homogeneous ideals preserve that structure, homogeneous primes are the primes compatible with it, and the irrelevant ideal isolates the positive-degree direction removed by projective geometry. After localizing at homogeneous f, taking degree zero gives

$$S_{(f)} = (S_f)_0, \qquad M_{(f)} = (M_f)_0.$$

These ordinary rings and modules are the affine algebra from which VI/35 will build Proj S and its sheaves.

34.21 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 34.82 (Degree)**.** Find the degree of x^2y^3 in the standard grading.

Hint. Use graded rings, homogeneous ideals, and degree bookkeeping. □

Solution. It has degree 5. □

Exercise 34.83 (Homogeneous test)**.** Is $x^2 + xy + y^2$ homogeneous?

Hint. Use graded rings, homogeneous ideals, and degree bookkeeping. □

Solution. Yes, of degree 2. □

Exercise 34.84 (Nonhomogeneous test)**.** Is $x + y^2$ homogeneous?

Hint. Use graded rings, homogeneous ideals, and degree bookkeeping. ☐

Solution. No. ☐

Exercise 34.85 (Graded quotient)**.** When does A/I inherit a grading?

Hint. Use graded rings, homogeneous ideals, and degree bookkeeping. ☐

Solution. When I is a homogeneous ideal. ☐

Exercise 34.86 (Veronese generators)**.** Give generators of the second Veronese of $k[x, y]$.

Hint. Use graded rings, homogeneous ideals, and degree bookkeeping. ☐

Solution. x^2, xy, y^2. ☐

Proofs

Exercise 34.87 (Homogeneous decomposition)**.** Prove: Prove every element of a graded ring has a unique finite sum of homogeneous components.

Hint. Work from graded rings, homogeneous ideals, and degree bookkeeping and state the hypotheses explicitly. ☐

Solution. Use the direct-sum definition of the grading. ☐

Exercise 34.88 (Homogeneous ideal quotient)**.** Prove: Prove A/I is graded when I is homogeneous.

Hint. Work from graded rings, homogeneous ideals, and degree bookkeeping and state the hypotheses explicitly. ☐

Solution. Define the degree-d piece as $(A_d + I)/I$ and check products respect degree. ☐

Exercise 34.89 (Homogeneous membership)**.** Prove: Prove an ideal is homogeneous iff it contains every homogeneous component of each of its elements.

Hint. Work from graded rings, homogeneous ideals, and degree bookkeeping and state the hypotheses explicitly. ☐

Solution. One direction uses homogeneous generators; the other reconstructs each element from components. ☐

Exercise 34.90 (Veronese grading)**.** Prove: Prove the Veronese subring is naturally graded.

Hint. Work from graded rings, homogeneous ideals, and degree bookkeeping and state the hypotheses explicitly. ☐

Solution. Set the new degree-d piece equal to the old degree-nd piece and check multiplication. ☐

Counterexamples and hypothesis tests

Exercise 34.91 (Arbitrary ideal)**.** Test the claim: every ideal in a graded ring is homogeneous.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; $(x+1)$ in $k[x]$ is not homogeneous. ☐

Exercise 34.92 (Degree cancellation)**.** Test the claim: a sum of homogeneous elements of different degrees is homogeneous.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False unless all but one summand vanish. ☐

Exercise 34.93 (Units positive degree)**.** Test the claim: a standard positively graded domain has nonzero units of positive degree.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False in the usual connected grading; units lie in degree zero. ☐

Applications and investigations

Exercise 34.94 (Projective equations)**.** Why must projective defining ideals be homogeneous?

Hint. Translate the question into graded rings, homogeneous ideals, and degree bookkeeping. ☐

Solution. Scaling homogeneous coordinates preserves vanishing only for homogeneous equations. ☐

Exercise 34.95 (Coordinate changes)**.** What does the Veronese construction enable?

Hint. Translate the question into graded rings, homogeneous ideals, and degree bookkeeping. ☐

Solution. It changes the embedding degree while preserving the underlying projective geometry in standard settings. ☐

Challenge problems

Exercise 34.96 (Synthesis challenge)**.** Connect two central viewpoints of Graded Rings in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Graded Rings and then translates it through graded rings, homogeneous ideals, and degree bookkeeping, checking both descriptions agree. ☐

Exercise 34.97 (Hypothesis challenge)**.** Choose one theorem-level statement from Graded Rings, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. □

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Chapter 35

Proj

VI/34 prepared the graded algebra needed to pass from an affine cone to projective geometry. The decisive operation is not merely to remove the cone vertex from $\operatorname{Spec} S$, but to retain only homogeneous prime ideals and then replace homogeneous localizations by their degree-zero parts. The legacy AG4 source identifies exactly this package: $\operatorname{Proj} S$, homogeneous primes, the structure sheaf, degree-zero localization, and the standard opens $D_+(f)$. This chapter reconstructs that package as a self-contained scheme construction. Projective space and projective subschemes are reserved for VI/36 and VI/37.

Throughout the core construction, let

$$S = \bigoplus_{n \geq 0} S_n$$

be a nonnegatively graded ring and write $S_+ = \bigoplus_{n>0} S_n$. When an assertion needs a stronger hypothesis, such as standard grading or finite generation, it is stated explicitly.

Learning goals

By the end of the chapter, the reader should be able to:

- define $\operatorname{Proj} S$ as homogeneous primes not containing S_+;
- define the closed sets $V_+(I)$ and standard opens $D_+(f)$;
- prove $D_+(f) \cap D_+(g) = D_+(fg)$ and recognize the standard opens as a basis;
- prove the affine chart theorem $D_+(f) \cong \operatorname{Spec} S_{(f)}$;
- construct the structure sheaf by gluing the affine chart sheaves;
- compute stalks as degree-zero homogeneous localizations $S_{(\mathfrak{p})}$;
- associate a sheaf $\widetilde{M}$ to a graded module using $M_{(f)}$;
- explain why exact sequences of graded modules give exact sequences of associated sheaves;
- compute concrete charts for small graded rings and detect empty, reducible, and nonreduced Proj examples;
- distinguish the general Proj construction from the later special theory of projective space and projective embeddings.

Conceptual roadmap

$$\boxed{\begin{array}{c} S \text{ graded} \longrightarrow \{\text{homogeneous primes } \mathfrak{p} \not\supseteq S_+\} \\ \longrightarrow D_+(f) \longleftrightarrow \operatorname{Spec} S_{(f)} \longrightarrow \mathcal{O}_{\operatorname{Proj} S} \longrightarrow \widetilde{M} \end{array}}$$

VI/34 supplied the rings $S_{(f)} = (S_f)_0$ and modules $M_{(f)} = (M_f)_0$. VI/35 now glues those affine pieces into geometry.

$$\begin{tikzcd} \boxed{\text{graded ring } S} \arrow[r, "\text{exclude } S_+"] & \boxed{\text{homogeneous primes}} \arrow[r, "\text{localize}"] & \boxed{D_+(f)} \arrow[r, leftrightarrow, "\text{affine chart}"] & \boxed{\operatorname{Spec} S_{(f)}} \end{tikzcd}$$

Figure 35.1: The Proj construction turns homogeneous-prime data into affine degree-zero charts.

35.1 The projective spectrum

Definition 35.1 (Projective spectrum)**.** The *projective spectrum* of S is the set

$$\operatorname{Proj} S := \{\mathfrak{p} \subset S : \mathfrak{p} \text{ is homogeneous prime and } S_+ \not\subseteq \mathfrak{p}\}.$$

Thus the irrelevant ideal is excluded: a point of $\operatorname{Proj} S$ must miss at least one positive-degree homogeneous element.

If $S = k[x_0, \dots, x_n]$ with the standard grading, the excluded homogeneous maximal ideal $(x_0, \dots, x_n)$ is the cone vertex. But $\operatorname{Proj} S$ is not defined by first forming a quotient of $\operatorname{Spec} S$ by scalar multiplication; it is defined algebraically from homogeneous primes.

Remark 35.2 (Underlying topology versus structure sheaf). As a set, $\operatorname{Proj} S \subseteq \operatorname{Spec} S$, and the topology below agrees with the subspace topology. The structure sheaf, however, is *not* the restriction of $\mathcal{O}_{\operatorname{Spec} S}$: projective charts use $S_{(f)} = (S_f)_0$, not the whole localization S_f.

35.2 Closed sets and the Zariski topology

For a homogeneous ideal $I \subseteq S$, define

$$V_+(I) := \{\mathfrak{p} \in \operatorname{Proj} S : I \subseteq \mathfrak{p}\}.$$

These sets satisfy

$$V_+(0) = \operatorname{Proj} S, \qquad V_+(S) = \varnothing,$$

$$V_+(I) \cup V_+(J) = V_+(IJ), \qquad \bigcap_\alpha V_+(I_\alpha) = V_+\left(\sum_\alpha I_\alpha\right).$$

Hence they are the closed sets of a topology on $\operatorname{Proj} S$.

Proposition 35.3 (Radicals do not change the closed set)**.** *For homogeneous I,*

$$V_+(I) = V_+(\sqrt{I}).$$

Proof. A prime contains I exactly when it contains $\sqrt{I}$. VI/34 showed that $\sqrt{I}$ is homogeneous. □

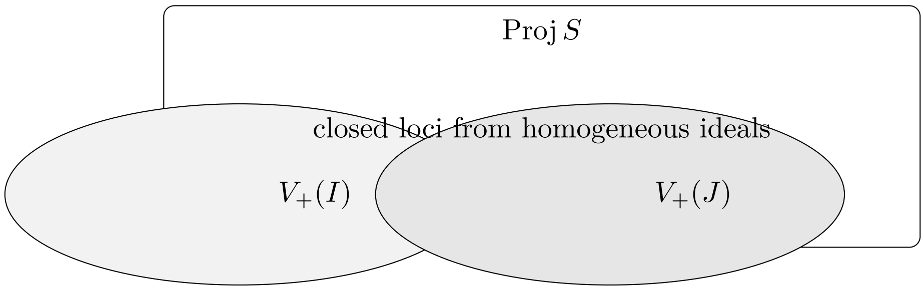


Figure 35.2: Projective closed sets are cut out by homogeneous ideals.

35.3 Standard opens

Let $f \in S$ be homogeneous of positive degree.

Definition 35.4 (Standard open).

$$D_+(f) := \operatorname{Proj} S \setminus V_+((f)) = \{\mathfrak{p} \in \operatorname{Proj} S : f \notin \mathfrak{p}\}.$$

The standard opens satisfy the projective analogue of the affine basic-open identities:

Proposition 35.5 (Standard-open calculus). *For homogeneous f, g of positive degree,*

$$D_+(f) \cap D_+(g) = D_+(fg).$$

If homogeneous elements f_i generate an ideal whose radical contains S_+, then

$$\operatorname{Proj} S = \bigcup_i D_+(f_i).$$

Proof. A prime misses fg exactly when it misses both f and g. For the cover statement, a point outside every $D_+(f_i)$ contains all f_i, hence their radical and therefore S_+, contradicting the definition of $\operatorname{Proj} S$. $\square$

Corollary 35.6 (Standard opens form a basis). *The sets $D_+(f)$, for positive-degree homogeneous f, form a basis for the topology of* Proj S.

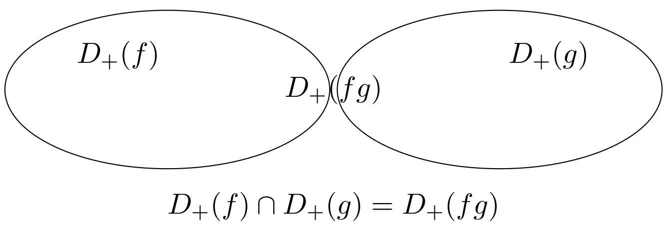


Figure 35.3: The standard-open basis is closed under finite intersections.

Example 35.7 (Proj of a polynomial ring). For $S = k[x_0, x_1]$ with irrelevant ideal $S_+ = (x_0, x_1)$, Proj S is the projective line. Its standard opens $D_+(x_0)$ and $D_+(x_1)$ are affine lines.

35.4 The affine chart theorem

VI/34 defined

$$S_{(f)} = (S_f)_0.$$

The central theorem of the construction is that this ordinary ring is exactly the coordinate ring of $D_+(f)$.

Theorem 35.8 (Affine chart theorem)**.** *For every homogeneous $f \in S$ of positive degree there is a canonical homeomorphism*

$$\boxed{D_+(f) \cong \operatorname{Spec} S_{(f)}.}$$

Under this correspondence,

$$\mathfrak{p} \longmapsto (\mathfrak{p} S_f)_0.$$

Proof. A homogeneous prime $\mathfrak{p}$ with $f \notin \mathfrak{p}$ localizes to a homogeneous prime $\mathfrak{p}S_f \subset S_f$. Since a power of the homogeneous element f is invertible in S_f, a homogeneous prime of S_f is determined by its degree-zero part; that degree-zero part is a prime of $(S_f)_0 = S_{(f)}$. Conversely, a prime $\mathfrak{q} \subset S_{(f)}$ determines the homogeneous prime of S_f generated degreewise by $\mathfrak{q}$, whose contraction to S is homogeneous and misses f. These operations are inverse.

For a homogeneous g, the condition $g \notin \mathfrak{p}$ becomes the nonvanishing of a suitable degree-zero fraction obtained by balancing powers of g and f. Thus standard opens inside $D_+(f)$ correspond to distinguished opens in $\operatorname{Spec} S_{(f)}$, proving continuity in both directions. □

When $S = k[x_0, \dots, x_n]$ is standard graded and $f = x_i$, VI/34 computed

$$S_{(x_i)} \cong k\left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i}\right].$$

Thus the standard chart is an ordinary affine scheme. VI/36 will organize these charts into projective space.

$$\begin{tikzcd}
D_+(f) \arrow[rr, "\sim"] \arrow[d, hook] & & \operatorname{Spec} S_{(f)} \\
 & & \deg a = r \deg f \\
\operatorname{Proj} S & & \text{degree-zero fractions } a/f^r
\end{tikzcd}$$

Figure 35.4: The affine chart theorem: projective localization means localize and then take degree zero.

35.5 Overlaps and localization in stages

The affine chart theorem must be compatible on overlaps. Suppose for simplicity that f and g have the same positive degree. Then $g/f \in S_{(f)}$, and VI/34 gives

$$(S_{(f)})_{g/f} \cong S_{(fg)}.$$

Therefore

$$D_+(f) \cap D_+(g) = D_+(fg)$$

is identified from either side with the same affine scheme

$$\operatorname{Spec} S_{(fg)}.$$

If the degrees differ, choose positive powers f^a, g^b with equal degree and use the corresponding degree-zero ratios. This is the algebraic cycle that makes the global construction possible.

35.6 The structure sheaf on Proj

On every standard open define

$$\mathcal{O}_{\operatorname{Proj} S}(D_+(f)) := S_{(f)}.$$

On overlaps, use the localization maps supplied by the previous section. Since the standard opens form a basis and the restrictions satisfy the cocycle identities, these affine structure sheaves glue.

Theorem 35.9 (Proj is a scheme)**.** *There is a unique structure sheaf $\mathcal{O}_{\operatorname{Proj} S}$ whose restriction to each standard open agrees under theorem 35.8 with the affine structure sheaf of* $\operatorname{Spec} S_{(f)}$. *Consequently*

$$\boxed{(\operatorname{Proj} S, \mathcal{O}_{\operatorname{Proj} S}) \text{ is a scheme.}}$$

Proof. The standard opens cover $\operatorname{Proj} S$. Their affine scheme structures agree on pairwise intersections because both restrictions identify with the structure sheaf of $\operatorname{Spec} S_{(fg)}$, after replacing by equal-degree powers when necessary. The scheme-gluing theorem of VI/20 therefore produces a scheme. Uniqueness follows because the standard opens are a basis. ☐

$$\begin{tikzcd}
\operatorname{Spec} S_{(fg)} \arrow[r, hook] \arrow[d, hook] & \operatorname{Spec} S_{(f)} \arrow[d, hook] \\
\operatorname{Spec} S_{(g)} \arrow[r, hook] & \operatorname{Proj} S
\end{tikzcd}$$

Figure 35.5: Affine chart sheaves agree on the degree-zero localization describing the overlap.

35.7 Stalks and homogeneous local rings

Let $\mathfrak{p} \in \operatorname{Proj} S$. Localize S at the multiplicative system of homogeneous elements outside $\mathfrak{p}$, preserving the grading, and take degree zero. Denote the resulting local ring by

$$S_{(\mathfrak{p})}.$$

Equivalently, its elements are represented by fractions a/b with a, b homogeneous of the same degree and $b \notin \mathfrak{p}$.

Proposition 35.10 (Stalk formula)**.**

$$\boxed{\mathcal{O}_{\operatorname{Proj} S,\mathfrak{p}} \cong S_{(\mathfrak{p})}.}$$

Its maximal ideal consists of the degree-zero fractions a/b with $a \in \mathfrak{p}$.

Proof. Choose homogeneous $f \notin \mathfrak{p}$. Inside the affine chart $D_+(f) \cong \operatorname{Spec} S_{(f)}$, the stalk is the ordinary localization of $S_{(f)}$ at the corresponding prime. Unwinding the fractions gives exactly the displayed homogeneous degree-zero localization. ☐

Thus the residue field is

$$\kappa(\mathfrak{p}) = S_{(\mathfrak{p})}/\mathfrak{m}_{\mathfrak{p}}.$$

Example 35.11 (Affine chart from degree zero)**.** On $D_+(x_0) \subset \operatorname{Proj} k[x_0, x_1]$, the coordinate ring is the degree-zero part of S_{x_0}, namely $k[x_1/x_0]$.

$$\boxed{\mathfrak{p} \in \operatorname{Proj} S} \xrightarrow{\text{stalk}} \boxed{S_{(\mathfrak{p})}} \longrightarrow \boxed{\mathfrak{m}_{\mathfrak{p}}} \xrightarrow{\text{quotient}} \boxed{\kappa(\mathfrak{p})}$$

Figure 35.6: Pointwise local algebra on Proj is homogeneous localization followed by degree zero.

35.8 Sheaves from graded modules

Let $M = \bigoplus_{n \in \mathbb{Z}} M_n$ be a graded S-module. VI/34 defined

$$M_{(f)} = (M_f)_0.$$

On the affine chart $D_+(f) \cong \operatorname{Spec} S_{(f)}$, VI/33 turns this module into a quasi-coherent sheaf. The overlap identifications from homogeneous localization let these affine sheaves glue.

Definition 35.12 (Associated sheaf of a graded module)**.** The sheaf associated to M on $\operatorname{Proj} S$, denoted $\widetilde{M}$, is characterized on standard opens by

$$\boxed{\Gamma(D_+(f), \widetilde{M}) = M_{(f)}}$$

and by the corresponding affine quasi-coherent sheaves on those opens.

Proposition 35.13 (Stalks of an associated graded sheaf)**.** *For* $\mathfrak{p} \in \operatorname{Proj} S$*,*

$$\widetilde{M}_{\mathfrak{p}} \cong M_{(\mathfrak{p})}.$$

Theorem 35.14 (Quasi-coherence and exactness)**.** *The sheaf* $\widetilde{M}$ *is quasi-coherent. Moreover, an exact sequence of graded modules with degree-zero maps*

$$0 \to M' \to M \to M'' \to 0$$

induces an exact sequence

$$0 \to \widetilde{M'} \to \widetilde{M} \to \widetilde{M''} \to 0.$$

Proof. Quasi-coherence is affine-local and holds on every standard chart by VI/33. VI/34 proved exactness of degree-zero homogeneous localization, so the induced sequence is exact on every standard chart, hence on stalks. $\square$

$$\begin{tikzcd} M \arrow[r, "\text{localize at } f"] \arrow[d, mapsto] & M_f \arrow[r, "(\cdot)_0"] & M_{(f)} \arrow[d, mapsto] \\ \widetilde{M} \arrow[rr, "\text{restrict}"'] & & \widetilde{M}|_{D_+(f)} \end{tikzcd}$$

Figure 35.7: A graded module becomes a quasi-coherent sheaf chart by chart.

35.9 Shifts and the first twisting preview

For $d \in \mathbb{Z}$, VI/34 defined the shifted graded module $S(d)$. Applying the present construction gives sheaves

$$\widetilde{S(d)}.$$

It is customary to write

$$\mathcal{O}_{\operatorname{Proj} S}(d) := \widetilde{S(d)}.$$

At this stage this notation records a family of quasi-coherent sheaves. Their invertibility, tensor identities, sections, and role in projective embeddings belong to later projective and divisor chapters.

A key warning is that the graded shift matters globally even when some affine charts make the localized module look free of rank one. Projective information is encoded in how those local trivializations transform on overlaps.

35.10 Homogeneous quotients and closed loci: topological form

If $I \subseteq S$ is homogeneous, then homogeneous primes of S/I correspond to homogeneous primes of S containing I. Therefore there is a canonical homeomorphism

$$\operatorname{Proj}(S/I) \cong V_+(I) \subseteq \operatorname{Proj} S.$$

This chapter uses this only as a topological and chart-level statement. The systematic closed-subscheme interpretation, homogeneous coordinate ideals, and projective subschemes are reserved for VI/37.

The projective closed set depends only on the saturation of I with respect to S_+ in standard Noetherian situations; this explains why equations supported only at the cone vertex can disappear after passing to Proj. Full saturation theory is deferred.

35.11 Veronese invariance

VI/34 introduced the d-th Veronese subring

$$S^{(d)} = \bigoplus_{n \geq 0} S_{nd}.$$

Projective geometry is insensitive to this regrading in the expected sense.

Theorem 35.15 (Veronese invariance)**.** *Assume S is standard graded over S_0. For every $d \geq 1$, there is a canonical isomorphism of schemes*

$$\boxed{\operatorname{Proj} S \cong \operatorname{Proj} S^{(d)}}.$$

Proof idea. A homogeneous prime of S contracts to a homogeneous prime of $S^{(d)}$, and the projectively relevant primes correspond. On a standard open indexed by a homogeneous element f of degree divisible by d, both sides have the same degree-zero chart ring: passing to the Veronese changes the bookkeeping of degrees but not the balanced fractions. These chart identifications agree on overlaps and glue. □

Example 35.16 (Irrelevant prime excluded)**.** The homogeneous ideal (x_0, x_1) is prime in $k[x_0, x_1]$, but it is excluded from Proj because it contains the irrelevant ideal.

35.12 Prototype computations

Several small examples already show the range of the construction.

One variable. For $S = k[x]$ with $\deg x = 1$, the only homogeneous prime missing $S_+ = (x)$ is (0). Hence $\operatorname{Proj} S$ is one point, and

$$D_+(x) = \operatorname{Proj} S \cong \operatorname{Spec} S_{(x)} = \operatorname{Spec} k.$$

Two variables. For $S = k[x, y]$,

$$D_+(x) \cong \operatorname{Spec} k[y/x], \qquad D_+(y) \cong \operatorname{Spec} k[x/y].$$

Their overlap is obtained by inverting y/x on the first chart or x/y on the second. VI/36 will identify the resulting scheme as projective one-space.

A reducible example. Let $S = k[x, y]/(xy)$. The projectively relevant homogeneous primes are (x) and (y). The two standard charts are each $\operatorname{Spec} k$, so $\operatorname{Proj} S$ is a disjoint union of two reduced points.

A nonreduced example. Let $S = k[x, y]/(y^2)$. Topologically $\operatorname{Proj} S$ has the homogeneous prime (y). On $D_+(x)$,

$$S_{(x)} \cong k[t]/(t^2), \qquad t = y/x,$$

so the projective scheme is a one-point nonreduced scheme. Proj retains nilpotent structure even when the underlying point set is tiny.

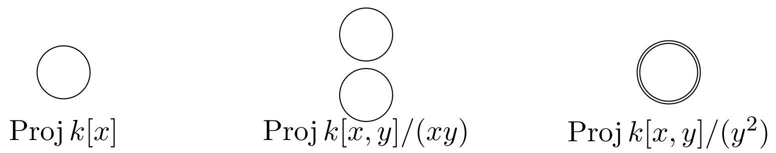


Figure 35.8: Proj can be one reduced point, two reduced points, or a one-point nonreduced scheme.

35.13 Common pitfalls

- **Proj is not all homogeneous primes.** Primes containing S_+ are excluded.
- **The projective structure sheaf is not $\mathcal{O}_{\operatorname{Spec} S}|_{\operatorname{Proj} S}$.** Its chart rings are $S_{(f)}$, not S_f.
- **Only homogeneous equations define $V_+(I)$ directly.** Homogeneity is what makes vanishing compatible with scaling.
- **A graded ring map does not automatically give an everywhere-defined morphism on Proj.** Additional hypotheses are needed to control the irrelevant ideals; projective morphisms are deferred to VI/38.
- **Global sections need not simply recover the original graded ring.** The affine reconstruction identity $A = \Gamma(\operatorname{Spec} A, \mathcal{O})$ has no naive graded analogue for $\operatorname{Proj} S$.
- **The notation $\mathcal{O}(d)$ does not yet prove invertibility.** The associated shifted sheaf is defined here; its line-bundle theory comes later.
- **Projective space is an example of Proj, not the definition of Proj.** VI/36 specializes the general construction to polynomial rings.

35.14 Boundaries of VI/35

VI/35 owns the general construction of $\operatorname{Proj} S$: its homogeneous-prime topology, standard-open basis, affine chart theorem, structure sheaf, stalks, and sheaves associated to graded modules. It does not yet develop projective n-space systematically, homogeneous coordinates and projective varieties, closed projective subschemes, projective morphisms, very ample sheaves, or projective embeddings. Those are the jobs of VI/36–VI/38 and the later line-bundle chapters.

35.15 Examples

Example 35.17 (Proj of a field in degree zero)**.** If $S = k$ is concentrated in degree zero, then $S_+ = 0$. Every prime contains S_+, so $\operatorname{Proj} S = \varnothing$.

Example 35.18 (One standard generator)**.** For $S = k[x]$, $\operatorname{Proj} S = \{(0)\}$ and $D_+(x) = \operatorname{Proj} S \cong \operatorname{Spec} k$.

Example 35.19 (Two-variable chart)**.** For $S = k[x, y]$, $S_{(x)} = k[y/x]$, so $D_+(x) \cong \mathbb{A}^1_k$. Similarly $D_+(y) \cong \mathbb{A}^1_k$.

Example 35.20 (The overlap)**.** On $D_+(x) \cap D_+(y)$, the coordinate $t = y/x$ is invertible. Hence the overlap is $\operatorname{Spec} k[t, t^{-1}]$.

Example 35.21 (A projectively empty nilpotent positive part)**.** Let $S = k[\varepsilon]$ with $\deg \varepsilon = 1$ and $\varepsilon^2 = 0$. Every prime contains ε, hence contains S_+, so $\operatorname{Proj} S = \varnothing$.

Example 35.22 (Reducible Proj)**.** For $S = k[x, y]/(xy)$, the projectively relevant primes are (x) and (y). Each standard chart is one reduced point.

Example 35.23 (Nonreduced Proj)**.** For $S = k[x, y]/(y^2)$, the chart $D_+(x)$ has ring

$$k[t]/(t^2), \qquad t = y/x.$$

The topology is one point but the local ring is nonreduced.

Example 35.24 (A homogeneous quotient)**.** For homogeneous $I \subset S$, points of $\operatorname{Proj}(S/I)$ correspond exactly to points of $V_+(I) \subseteq \operatorname{Proj} S$.

Example 35.25 (A nonhomogeneous equation warning)**.** In $k[x, y]$, the polynomial $x + y^2$ mixes degrees. Its vanishing is not invariant under simultaneous scalar rescaling, so it is not a projective equation for the standard grading.

Example 35.26 (Stalk on a chart)**.** If $\mathfrak{p} \in D_+(f)$, then $\mathcal{O}_{\operatorname{Proj} S, \mathfrak{p}}$ is the localization of $S_{(f)}$ at the prime corresponding to $\mathfrak{p}$, hence equals $S_{(\mathfrak{p})}$.

Example 35.27 (Structure sheaf from the module S)**.** For the graded module $M = S$, one has $M_{(f)} = S_{(f)}$. Thus $\widetilde{S} \cong \mathcal{O}_{\operatorname{Proj} S}$.

Example 35.28 (A quotient module)**.** For homogeneous $I \subseteq S$, $\widetilde{S/I}$ restricts on $D_+(f)$ to the affine sheaf associated to $(S/I)_{(f)}$.

Example 35.29 (Shifted module)**.** The shift $S(d)$ gives $\mathcal{O}_{\operatorname{Proj} S}(d) = \widetilde{S(d)}$. On $D_+(f)$ its sections are $S(d)_{(f)}$.

Example 35.30 (Exactness chart by chart)**.** If $0 \to M' \to M \to M'' \to 0$ is exact in graded modules, then on every $D_+(f)$ the degree-zero localized sequence is exact, so the associated sheaf sequence is exact.

Example 35.31 (Veronese chart)**.** For $S = k[x, y]$ and $S^{(2)} = k[x^2, xy, y^2]$, the chart indexed by x^2 has degree-zero ring $k[y/x]$, the same affine chart as $D_+(x)$ for S.

Example 35.32 (Global-section warning)**.** The recipe $S \mapsto \operatorname{Proj} S$ discards and reorganizes graded information. One should not expect to recover S simply as $\Gamma(\operatorname{Proj} S, \mathcal{O})$; later projective theory instead studies all twists and their sections.

35.16 Exercises

Exercise 35.33 (Point-set definition)**.** For $S = k[x, y, z]$, explain why (x, y, z) is not a point of $\operatorname{Proj} S$.

Exercise 35.34 (Homogeneous prime)**.** Show $(x) \subset k[x, y]$ defines a point of $\operatorname{Proj} k[x, y]$.

Exercise 35.35 (Closed-set union)**.** Prove $V_+(I) \cup V_+(J) = V_+(IJ)$ for homogeneous ideals.

Exercise 35.36 (Closed-set intersection)**.** Prove $\bigcap_\alpha V_+(I_\alpha) = V_+(\sum I_\alpha)$.

Exercise 35.37 (Standard-open intersection)**.** Prove $D_+(f) \cap D_+(g) = D_+(fg)$.

Exercise 35.38 (Standard cover)**.** If $S_+ = (f_0, \ldots, f_r)$ with homogeneous f_i, prove $\operatorname{Proj} S = \bigcup_i D_+(f_i)$.

Exercise 35.39 (One-variable Proj)**.** Compute $\operatorname{Proj} k[x]$ and its structure sheaf.

Exercise 35.40 (Empty Proj)**.** Compute $\operatorname{Proj} k[\varepsilon]/(\varepsilon^3)$ when $\deg \varepsilon = 1$.

Exercise 35.41 (Affine chart)**.** Compute $k[x, y, z]_{(x)}$ in the standard grading.

Exercise 35.42 (Overlap ring)**.** For $S = k[x, y]$, compute the coordinate ring of $D_+(x) \cap D_+(y)$.

Exercise 35.43 (Homogeneous quotient)**.** Show set-theoretically $\operatorname{Proj}(S/I) \cong V_+(I)$ for homogeneous I.

Exercise 35.44 (Stalk fractions)**.** Describe elements of $S_{(\mathfrak{p})}$ as fractions.

Exercise 35.45 (Locality)**.** Prove the fractions with numerator in $\mathfrak{p}$ form the maximal ideal of $S_{(\mathfrak{p})}$.

Exercise 35.46 (Associated sheaf)**.** For a graded module M, state the sections of $\widetilde{M}$ on $D_+(f)$.

Exercise 35.47 (Stalk of a graded sheaf)**.** Show $\widetilde{M}_\mathfrak{p} \cong M_{(\mathfrak{p})}$.

Exercise 35.48 (Structure sheaf)**.** Prove $\widetilde{S} \cong \mathcal{O}_{\operatorname{Proj} S}$.

Exercise 35.49 (Exactness)**.** Explain why the functor $M \mapsto \widetilde{M}$ is exact on degree-zero exact sequences of graded modules.

Exercise 35.50 (Shift notation)**.** Write the chart sections of $\mathcal{O}(d) = \widetilde{S(d)}$.

Exercise 35.51 (Reducible example)**.** Compute $\operatorname{Proj} k[x, y]/(xy)$.

Exercise 35.52 (Nonreduced example)**.** Compute the x-chart of $\operatorname{Proj} k[x, y]/(y^2)$.

Exercise 35.53 (Veronese prototype)**.** For $S = k[x, y]$, compare the x-chart of $\operatorname{Proj} S$ with the x^2-chart of $\operatorname{Proj} S^{(2)}$.

Exercise 35.54 (Nonhomogeneous warning)**.** Explain why $x + y^2$ is not a valid homogeneous projective equation in standard $k[x, y]$.

Exercise 35.55 (Subspace topology)**.** Show the V_+-topology agrees with the subspace topology from $\operatorname{Spec} S$.

Exercise 35.56 (Boundary to VI/36)**.** For $S = k[x_0, \dots, x_n]$, state without developing projective-space theory what the standard charts are.

Exercise 35.57 (Irrelevant torsion disappears)**.** Let M be a graded S-module and suppose $S_+^N M = 0$ for some N. Prove

$$\widetilde{M} = 0 \quad \text{on } \operatorname{Proj} S.$$

Exercise 35.58 (Domain of a graded map)**.** Let $\varphi : S \to T$ be a degree-preserving graded ring map. Show that contraction defines a point of $\operatorname{Proj} S$ precisely on

$$U = \operatorname{Proj} T \setminus V_+(\varphi(S_+)T) = \bigcup_{f \in S_+ \text{ homogeneous}} D_+(\varphi(f)).$$

35.17 Problem dossiers

Problem 35.59 (VI.35.P1: The standard-open basis)**.** Prove from the definitions that the sets $D_+(f)$ form a basis and give an explicit basis refinement inside an arbitrary $V_+(I)^c$.

Problem 35.60 (VI.35.P2: The affine chart correspondence)**.** Construct explicitly the prime correspondence in $D_+(f) \cong \operatorname{Spec} S_{(f)}$ and verify that it respects basic opens.

Problem 35.61 (VI.35.P3: Structure-sheaf gluing)**.** Using VI/20, prove that the affine schemes $\operatorname{Spec} S_{(f)}$ glue along the standard overlaps to a scheme structure on $\operatorname{Proj} S$.

Problem 35.62 (VI.35.P4: Stalk computation)**.** Prove $\mathcal{O}_{\operatorname{Proj} S, \mathfrak{p}} \cong S_{(\mathfrak{p})}$ and identify its maximal ideal and residue field.

Problem 35.63 (VI.35.P5: Graded modules become quasi-coherent sheaves)**.** Construct $\widetilde{M}$ from the modules $M_{(f)}$ and prove quasi-coherence.

Problem 35.64 (VI.35.P6: Exactness of the tilde construction)**.** Prove that a short exact sequence of graded modules with degree-zero maps remains exact after applying $M \mapsto \widetilde{M}$.

Problem 35.65 (VI.35.P7: A two-chart prototype)**.** For $S = k[x, y]$, compute both standard charts and their overlap, including the transition $t = y/x \leftrightarrow t^{-1} = x/y$.

Problem 35.66 (VI.35.P8: Reducible and nonreduced Proj)**.** Compare $\operatorname{Proj} k[x, y]/(xy)$ and $\operatorname{Proj} k[x, y]/(y^2)$ as topological spaces and schemes.

Problem 35.67 (VI.35.P9: Homogeneous quotient topology)**.** For homogeneous I, prove that the natural prime correspondence identifies

$$\operatorname{Proj}(S/I) \longrightarrow \operatorname{Proj} S$$

homeomorphically with $V_+(I)$.

Problem 35.68 (VI.35.P10: Veronese invariance on charts)**.** Give a chartwise proof that $\operatorname{Proj} S \cong \operatorname{Proj} S^{(d)}$ in the standard setting.

Problem 35.69 (VI.35.P11: Why the Spec sheaf restriction is wrong)**.** Give a concrete example showing that $\mathcal{O}_{\operatorname{Spec} S}(D(f)) = S_f$ is too large for projective geometry, while $S_{(f)}$ has the correct scaling-invariant functions.

Problem 35.70 (VI.35.P12: Boundary to projective morphisms)**.** Explain why a graded map $S \to T$ can fail to define a morphism $\operatorname{Proj} T \to \operatorname{Proj} S$ everywhere, and formulate the obstruction in terms of irrelevant ideals.

Problem 35.71 (VI.35.P13: Quotient charts)**.** Let $I \subseteq S$ be homogeneous and let $f \in S_+$ be homogeneous. Prove

$$(S/I)_{(f)} \cong S_{(f)}/(I_f)_0.$$

Interpret this as the chart-level compatibility between $\operatorname{Proj}(S/I)$ and $V_+(I)$.

Provenance. Canonical VI/35 extension.

Problem 35.72 (VI.35.P14: Kernels and cokernels after projective sheafification)**.** Let $u : M \to N$ be a degree-zero homomorphism of graded S-modules. Prove

$$\widetilde{\ker u} \cong \ker(\widetilde{u}), \qquad \widetilde{\operatorname{coker} u} \cong \operatorname{coker}(\widetilde{u}).$$

Provenance. Canonical VI/35 extension.

Problem 35.73 (VI.35.P15: Irrelevant torsion sheafifies to zero)**.** Let M be graded and suppose $S_+^N M = 0$ for some N. Prove $\widetilde{M} = 0$. Give a concrete example.

Provenance. Canonical VI/35 extension.

Problem 35.74 (VI.35.P16: Veronese invariance of the structure sheaf)**.** Under the standard hypotheses of the Veronese theorem, prove that the chartwise homeomorphism

$$\operatorname{Proj} S \cong \operatorname{Proj} S^{(d)}$$

identifies the structure sheaves.

Provenance. Canonical VI/35 extension.

Problem 35.75 (VI.35.P17: Weighted chart rings for weights $1, 2, 3$)**.** Let

$$S = k[x, y, z], \qquad \deg x = 1, \quad \deg y = 2, \quad \deg z = 3.$$

Compute $S_{(x)}, S_{(y)}, S_{(z)}$.

Provenance. Canonical VI/35 extension.

Problem 35.76 (VI.35.P18: Saturation is invisible on every standard chart)**.** Let $I \subseteq S$ be homogeneous and set

$$I^{\mathrm{sat}} = I : S_+^{\infty}.$$

Prove that for every homogeneous $f \in S_+$,

$$(I^{\mathrm{sat}})_f = I_f,$$

and deduce equality of the corresponding V_+-closed sets.

Provenance. Canonical VI/35 extension.

Problem 35.77 (VI.35.P19: Local dimension of the polynomial Proj charts)**.** Let

$$S = k[x_0, \ldots, x_n]$$

with standard grading. Show every standard chart $D_+(x_i)$ is an integral n-dimensional affine scheme.

Provenance. Canonical VI/35 extension.

Problem 35.78 (VI.35.P20: The maximal domain of a graded-map morphism)**.** Let $\varphi : S \to T$ be degree preserving. Define

$$U = \operatorname{Proj} T \setminus V_+(\varphi(S_+)T).$$

Prove that contraction and the degree-zero chart maps glue to a morphism

$$U \longrightarrow \operatorname{Proj} S,$$

and that U is maximal for this contraction construction.

Provenance. Canonical VI/35 extension.

35.18 Challenges

Problem 35.79 (Challenge 1: Saturation and invisible cone-vertex data)**.** For a standard Noetherian graded ring and homogeneous ideal I, investigate the saturation $I : S_+^\infty$. Prove that replacing I by its saturation does not change $V_+(I)$, and determine additional hypotheses needed for equality of the corresponding closed projective subschemes.

Problem 35.80 (Challenge 2: Global sections versus graded pieces)**.** Study when the natural map

$$S_n \longrightarrow \Gamma(\operatorname{Proj} S, \mathcal{O}(n))$$

is an isomorphism. Produce examples showing why no unconditional statement of this form should be built into the definition of Proj.

Problem 35.81 (Challenge 3: Weighted Proj charts)**.** Take $S = k[x, y, z]$ with weights $1, 2, 3$. Compute representative standard chart rings $S_{(x)}, S_{(y)}, S_{(z)}$ and identify how they differ from ordinary affine-space charts.

Problem 35.82 (Challenge 4: Functoriality domain)**.** Given a graded map $S \to T$, characterize the largest open subset of $\operatorname{Proj} T$ on which contraction defines a morphism to $\operatorname{Proj} S$, and compare your construction with the standard-open charts.

Problem 35.83 (Challenge 5: Recover projective space from the machine)**.** Using only the results of VI/35, reconstruct the chart atlas for $\operatorname{Proj} k[x_0, \ldots, x_n]$, its overlaps, and its k-valued points. Stop before developing homogeneous coordinates globally; that is the starting point of VI/36.

Source and scope audit

The repository migration ledger assigns the legacy AG4 thematic selector `Proj` and its inherited theorem/problem descendants to VI/35. The broader audit describes the same source as containing $\operatorname{Proj} S$, its structure sheaf, degree-zero localization, $D_+(f)$, projective space, and closed projective subschemes. This chapter consumes the general Proj construction and its local sheaf theory. The source's projective-space and projective-subscheme material remains reserved for VI/36 and VI/37. The additional treatments of graded-module sheaves, exactness, shifts, Veronese invariance, reducible/nonreduced prototypes, and saturation warnings are canonical extensions justified by VI/33–VI/34; they are not presented as separately numbered legacy problems.

Chapter synthesis

Projective geometry is obtained from graded algebra by a precise two-step reduction:

$$\boxed{\text{homogeneous primes } \mathfrak{p} \not\supseteq S_+ \quad + \quad \text{degree-zero localizations.}}$$

The topology is controlled by

$$D_+(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}, \qquad D_+(f) \cap D_+(g) = D_+(fg),$$

and every standard open is affine:

$$\boxed{D_+(f) \cong \operatorname{Spec} S_{(f)}.}$$

Gluing these charts gives $\mathcal{O}_{\operatorname{Proj} S}$, with stalks $S_{(\mathfrak{p})}$. A graded module becomes a quasi-coherent sheaf by the parallel formula

$$\Gamma(D_+(f), \widetilde{M}) = M_{(f)}, \qquad \widetilde{M}_{\mathfrak{p}} = M_{(\mathfrak{p})}.$$

The general Proj machine is now complete. VI/36 can specialize it to polynomial rings and projective space without hiding any of the underlying algebra or sheaf theory.

35.19 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 35.84 (Standard chart)**.** Compute the coordinate ring of $D_+(x_0)$ in $\operatorname{Proj} k[x_0, x_1, x_2]$.

Hint. Use homogeneous prime ideals, irrelevant ideals, and standard opens. □

Solution. $k[x_1/x_0, x_2/x_0]$. □

Exercise 35.85 (Irrelevant ideal)**.** What is the irrelevant ideal of $k[x_0, \dots, x_n]$?

Hint. Use homogeneous prime ideals, irrelevant ideals, and standard opens. □

Solution. $(x_0, \dots, x_n)$. □

Exercise 35.86 (Point condition)**.** Which homogeneous primes define points of $\operatorname{Proj} S$?

Hint. Use homogeneous prime ideals, irrelevant ideals, and standard opens. □

Solution. Those not containing S_+. □

Exercise 35.87 (Intersection)**.** Compute $D_+(f) \cap D_+(g)$.

Hint. Use homogeneous prime ideals, irrelevant ideals, and standard opens. □

Solution. $D_+(fg)$. □

Exercise 35.88 (Projective line charts)**.** How many standard coordinate charts cover $\mathbb{P}^1$?

Hint. Use homogeneous prime ideals, irrelevant ideals, and standard opens. □

Solution. Two. □

Proofs

Exercise 35.89 (Standard-open affineness)**.** Prove: Prove $D_+(f) \cong \operatorname{Spec}(S_f)_0$ for homogeneous f of positive degree.

Hint. Work from homogeneous prime ideals, irrelevant ideals, and standard opens and state the hypotheses explicitly. □

Solution. Use homogeneous prime correspondence under localization and restrict to degree-zero fractions. □

Exercise 35.90 (Basis)**.** Prove: Prove standard opens form a basis for $\operatorname{Proj} S$.

Hint. Work from homogeneous prime ideals, irrelevant ideals, and standard opens and state the hypotheses explicitly. □

Solution. A point not containing the irrelevant ideal avoids some positive-degree homogeneous element. □

Exercise 35.91 (Intersection)**.** Prove: Prove $D_+(f) \cap D_+(g) = D_+(fg)$.

Hint. Work from homogeneous prime ideals, irrelevant ideals, and standard opens and state the hypotheses explicitly. □

Solution. A homogeneous prime avoids the product iff it avoids both factors. □

Exercise 35.92 (Polynomial Proj)**.** Prove: Prove $\operatorname{Proj} k[x_0, \ldots, x_n]$ is covered by $n+1$ affine n-spaces.

Hint. Work from homogeneous prime ideals, irrelevant ideals, and standard opens and state the hypotheses explicitly. □

Solution. Compute each $D_+(x_i)$ as the spectrum of a polynomial ring in ratios. □

Counterexamples and hypothesis tests

Exercise 35.93 (Spec equality)**.** Test the claim: $\operatorname{Proj} S$ contains all prime ideals of S.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. □

Solution. False; only homogeneous primes avoiding the irrelevant ideal occur. □

Exercise 35.94 (Same localization)**.** Test the claim: $D_+(f)$ has coordinate ring S_f itself.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. □

Solution. False; use the degree-zero part $(S_f)_0$. □

Exercise 35.95 (Degree-zero primes)**.** Test the claim: every prime of S_0 automatically gives a point of $\operatorname{Proj} S$.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. □

Solution. False without compatible homogeneous data. □

Applications and investigations

Exercise 35.96 (Projective construction)**.** Why is Proj the projective analogue of Spec?

Hint. Translate the question into homogeneous prime ideals, irrelevant ideals, and standard opens. ☐

Solution. It glues affine charts from homogeneous localizations while discarding the irrelevant origin. ☐

Exercise 35.97 (Embeddings)**.** How do graded quotients produce projective subschemes?

Hint. Translate the question into homogeneous prime ideals, irrelevant ideals, and standard opens. ☐

Solution. A homogeneous quotient S/I gives $\mathrm{Proj}(S/I)$ as a closed projective subscheme. ☐

Challenge problems

Exercise 35.98 (Synthesis challenge)**.** Connect two central viewpoints of Proj in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Proj and then translates it through homogeneous prime ideals, irrelevant ideals, and standard opens, checking both descriptions agree. ☐

Exercise 35.99 (Hypothesis challenge)**.** Choose one theorem-level statement from Proj, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 36

Projective Space

VI/35 constructed $\operatorname{Proj} S$ for a general graded ring. Projective space is the most important special case:

$$\mathbb{P}^n_k := \operatorname{Proj} k[x_0, \dots, x_n],$$

where every variable has degree one. The general machinery now becomes concrete. The standard open $D_+(x_i)$ is an affine n-space, ratios x_j/x_i become affine coordinates, and the overlap maps record exactly how homogeneous coordinates change from one normalization to another.

The legacy migration ledger assigns the projective-space section of the AG4 source, together with its theorem-like and exercise/problem descendants, to VI/36. The raw legacy source is not present in the accessible repository archive, so this chapter uses that ledger as provenance while developing a new canonical treatment compatible with VI/34–VI/35. Projective subschemes and homogeneous coordinate rings are reserved for VI/37; projective morphisms and closed embeddings are reserved for VI/38.

Learning goals

By the end of the chapter, the reader should be able to:

- define $\mathbb{P}^n_k$ as $\operatorname{Proj} k[x_0, \dots, x_n]$;
- identify K-valued points with one-dimensional subspaces of K^{n+1} and use homogeneous coordinates;
- compute the standard affine charts $U_i = D_+(x_i) \cong \mathbb{A}^n_k$;
- write the transition maps on $U_i \cap U_j$;
- reconstruct $\mathbb{P}^1$ from two affine lines and $\mathbb{P}^2$ from three affine planes;
- identify $\mathbb{A}^n_k$ as the complement of a hyperplane at infinity;
- describe projective linear subspaces by homogeneous linear equations;
- prove $\dim \mathbb{P}^n_k = n$;
- distinguish closed points from k-rational points and compute a basic non-rational example;
- understand base extension $\mathbb{P}^n_k \times_k K \cong \mathbb{P}^n_K$;
- trivialize $\mathcal{O}_{\mathbb{P}^n}(d)$ on standard charts and compute its transition factors;

- compute global sections of $\mathcal{O}(d)$ for $d \geq 0$;
- use the Veronese and Segre maps as first concrete projective-coordinate constructions without yet invoking the closed-embedding theory of VI/38.

Conceptual roadmap

$$\boxed{\begin{array}{c} S = k[x_0, \dots, x_n] \quad \longrightarrow \quad \mathbb{P}^n_k = \operatorname{Proj} S \\ U_i = D_+(x_i) \cong \operatorname{Spec} S_{(x_i)} \cong \mathbb{A}^n_k \\ \text{homogeneous coordinates} \longleftrightarrow \text{affine ratios} \longleftrightarrow \text{transition maps} \end{array}}$$

Projective space is therefore not an additional primitive object: it is the standard graded polynomial ring interpreted through the Proj construction.

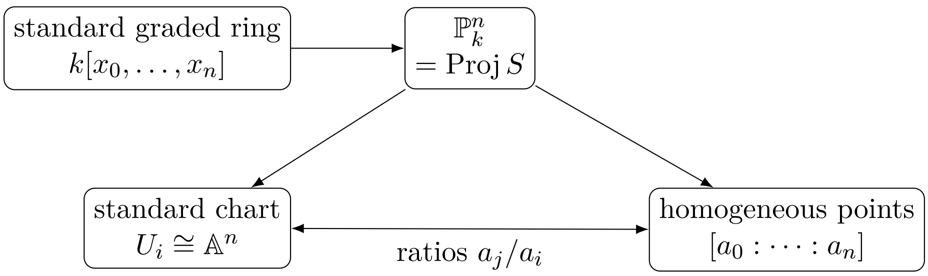


Figure 36.1: Projective space turns standard graded polynomial algebra into an atlas of affine charts linked by homogeneous coordinate ratios.

36.1 Definition and homogeneous coordinates

Let

$$S = k[x_0, \dots, x_n]$$

with the standard grading. By definition,

$$\mathbb{P}^n_k = \operatorname{Proj} S.$$

For a field extension K/k, the K-valued points admit the familiar coordinate description.

Theorem 36.1 (Homogeneous coordinates over a field)**.** *For every field extension K/k, there is a natural bijection*

$$\mathbb{P}^n_k(K) \cong \{\textit{one-dimensional } K\textit{-subspaces of } K^{n+1}\}.$$

Equivalently, every K-point can be written

$$[a_0 : \cdots : a_n], \qquad (a_0, \dots, a_n) \neq 0,$$

with

$$[a_0 : \cdots : a_n] = [\lambda a_0 : \cdots : \lambda a_n]$$

for every $\lambda \in K^\times$.

Proof. A nonzero vector $a = (a_0, \dots, a_n)$ determines a homogeneous prime kernel in the usual coordinate description after choosing a chart containing it; multiplying a by $\lambda \neq 0$ does not change any ratio a_j/a_i. Conversely, a K-point lies in some standard chart U_i, where it is determined by the affine ratios a_j/a_i. Normalizing $a_i = 1$ produces a nonzero vector unique up to nonzero scalar. The chart descriptions agree on overlaps by the transition formulas proved below. □

Remark 36.2 (What homogeneous coordinates do not mean). The symbols $[a_0 : \cdots : a_n]$ are coordinates for field-valued points, not a replacement for the scheme. The scheme also contains points with larger residue fields and generic points of irreducible closed subsets.

36.2 The standard affine charts

For $0 \leq i \leq n$, set

$$U_i := D_+(x_i) \subseteq \mathbb{P}^n_k.$$

VI/35 gives

$$U_i \cong \operatorname{Spec} S_{(x_i)}.$$

Since

$$S_{(x_i)} = k\left[\frac{x_0}{x_i}, \dots, \widehat{\frac{x_i}{x_i}}, \dots, \frac{x_n}{x_i}\right],$$

we obtain the fundamental atlas.

Theorem 36.3 (Standard affine atlas)**.** *Each standard open is canonically an affine n-space:*

$$\boxed{U_i \cong \mathbb{A}^n_k.}$$

The $n+1$ opens $U_0, \dots, U_n$ cover $\mathbb{P}^n_k$.

On U_i, write

$$t^{(i)}_j := \frac{x_j}{x_i} \qquad (j \neq i).$$

For a field-valued point with $a_i \neq 0$, the chart map is

$$[a_0 : \cdots : a_n] \longmapsto \left(\frac{a_0}{a_i}, \dots, \widehat{\frac{a_i}{a_i}}, \dots, \frac{a_n}{a_i}\right).$$

Its inverse inserts 1 in the i-th position.

$$\begin{tikzcd}
\boxed{[a_0 : \cdots : a_i : \cdots : a_n]} \arrow[d, shift right, "{a_i \neq 0,\ \text{normalize } a_i = 1}"'] \\
\boxed{(a_0/a_i, \dots, \widehat{1}, \dots, a_n/a_i)} \arrow[u, shift right, "\text{insert } 1"']
\end{tikzcd}$$

Figure 36.2: A standard projective chart is obtained by choosing one nonzero coordinate and dividing all coordinates by it.

36.3 Transition maps on overlaps

Suppose $i \neq j$. On $U_i \cap U_j$, the coordinate $t_j^{(i)} = x_j/x_i$ is invertible. Passing from the i-chart to the j-chart gives

$$t_i^{(j)} = \frac{x_i}{x_j} = \frac{1}{t_j^{(i)}},$$

and for $\ell \neq i, j$,

$$t_\ell^{(j)} = \frac{x_\ell}{x_j} = \frac{x_\ell/x_i}{x_j/x_i} = \frac{t_\ell^{(i)}}{t_j^{(i)}}.$$

Thus the change of coordinates is regular precisely on the principal open where $t_j^{(i)} \neq 0$.

Proposition 36.4 (Overlap algebra)**.** *Under $U_i \cong \mathbb{A}_k^n$, the overlap $U_i \cap U_j$ is the distinguished open*

$$D(t_j^{(i)}) \subseteq \mathbb{A}_k^n,$$

and its coordinate ring is

$$k[t_\ell^{(i)} : \ell \neq i]_{t_j^{(i)}}.$$

The transition formulas above induce the canonical isomorphism with the corresponding localization in the j-chart.

$$\begin{tikzcd}
\boxed{\begin{array}{c} U_i \cong \mathbb{A}^n \\ \text{coordinates } t_\ell^{(i)} \end{array}} \arrow[rr, leftrightarrow, "\text{change normalization}"] \arrow[dr] & & \boxed{\begin{array}{c} U_j \cong \mathbb{A}^n \\ \text{coordinates } t_\ell^{(j)} \end{array}} \arrow[dl] \\
& \boxed{\begin{array}{c} \text{overlap: } t_j^{(i)} \neq 0 \\ t_i^{(j)} = 1/t_j^{(i)} \\ t_\ell^{(j)} = t_\ell^{(i)}/t_j^{(i)} \end{array}} &
\end{tikzcd}$$

Figure 36.3: Transition maps are regular on the principal open where the new normalizing coordinate is nonzero.

Example 36.5 (The affine chart x_0e0)**.** A point $[x_0 : \cdots : x_n]$ with $x_0 \neq 0$ has unique coordinates $[1 : u_1 : \cdots : u_n]$. Thus $D_+(x_0) \cong \mathbb{A}^n$ with $u_i = x_i/x_0$.

36.4 The projective line from two affine lines

For $\mathbb{P}_k^1 = \operatorname{Proj} k[x_0, x_1]$, set

$$U_0 = D_+(x_0), \qquad U_1 = D_+(x_1).$$

Then

$$U_0 \cong \operatorname{Spec} k[t], \qquad t = \frac{x_1}{x_0},$$

and

$$U_1 \cong \operatorname{Spec} k[u], \qquad u = \frac{x_0}{x_1}.$$

Their overlap is

$$\operatorname{Spec} k[t, t^{-1}] \cong \operatorname{Spec} k[u, u^{-1}],$$

with transition

$$u = t^{-1}.$$

Therefore $\mathbb{P}^1_k$ is obtained by gluing two affine lines along their punctured lines by inversion. The point outside U_0 is

$$[0 : 1],$$

which is the standard point at infinity for the affine coordinate t.

Corollary 36.6 (Affine line plus infinity)**.**

$$\mathbb{P}^1_k = \mathbb{A}^1_k \cup \{\infty\}$$

as a mnemonic for the open embedding $U_0 \cong \mathbb{A}^1_k$ and its one-point complement. The scheme structure near ∞ is still an affine-line chart, not a topological one-point compactification.

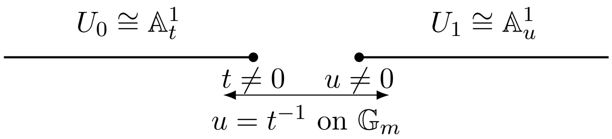


Figure 36.4: The projective line is two affine lines glued along their punctured lines by inversion.

36.5 The projective plane from three affine planes

For $\mathbb{P}^2_k$, the charts

$$U_0, U_1, U_2 \cong \mathbb{A}^2_k$$

have coordinates

$$U_0 : (x_1/x_0, x_2/x_0), \quad U_1 : (x_0/x_1, x_2/x_1), \quad U_2 : (x_0/x_2, x_1/x_2).$$

A point $[a : b : c]$ belongs to whichever charts correspond to its nonzero coordinates. On a double overlap, the transition divides all remaining coordinates by the new normalizing coordinate. Triple overlap means $abc \neq 0$.

This chart description is the practical computational model for projective plane geometry: equations are homogenized globally and dehomogenized on a chosen affine chart.

36.6 Affine space and the hyperplane at infinity

The chart $U_0 = D_+(x_0)$ identifies with $\mathbb{A}^n_k$ by

$$[a_0 : a_1 : \cdots : a_n] \longmapsto \left(\frac{a_1}{a_0}, \ldots, \frac{a_n}{a_0}\right).$$

The complementary closed set is

$$H_\infty := V_+(x_0).$$

Its underlying projective locus is naturally identified with

$$\mathbb{P}^{n-1}_k$$

using the remaining coordinates $[a_1 : \cdots : a_n]$. Thus projective space decomposes geometrically into an affine chart and a lower-dimensional hyperplane at infinity:

$$\boxed{\mathbb{P}^n_k = \mathbb{A}^n_k \sqcup H_\infty, \qquad H_\infty \simeq \mathbb{P}^{n-1}_k.}$$

The full closed-subscheme interpretation of $V_+(x_0)$ belongs to VI/37; here only the standard projective-space identification is used.

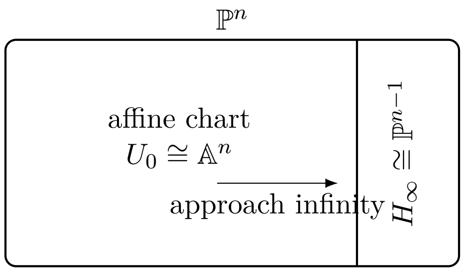


Figure 36.5: Choosing a standard affine chart singles out a complementary hyperplane at infinity.

36.7 Dimension

Every chart U_i is an affine n-space, hence has dimension n. Since the charts cover $\mathbb{P}^n_k$, dimension is determined locally.

Theorem 36.7 (Dimension of projective space)**.**

$$\boxed{\dim \mathbb{P}^n_k = n.}$$

Proof. The nonempty open subset $U_0 \cong \mathbb{A}^n_k$ has dimension n, so $\dim \mathbb{P}^n_k \geq n$. Conversely every point has a neighborhood isomorphic to an open subset of $\mathbb{A}^n_k$, so no chain of irreducible closed subsets can have length exceeding n locally. VI/30 then gives the equality. □

In particular, the hyperplane at infinity has codimension one, in agreement with VI/31.

Example 36.8 (A projective conic)**.** The equation $X_0X_2 - X_1^2 = 0$ is homogeneous of degree 2, so it defines a closed conic in $\mathbb{P}^2$. On $X_0 \neq 0$ it becomes $u_2 - u_1^2 = 0$, the affine parabola.

36.8 Projective linear subspaces

A system of homogeneous linear equations

$$L_1 = \cdots = L_r = 0$$

defines a projective linear locus. If the independent linear forms have rank r, a linear change of homogeneous coordinates reduces the system to

$$x_0 = \cdots = x_{r-1} = 0.$$

The remaining coordinates identify the locus with

$$\mathbb{P}^{n-r}_k.$$

Thus projective points, lines, planes, and hyperplanes form one uniform hierarchy.

Proposition 36.9 (Projective span of two rational points)**.** *If $p = [a_0 : \cdots : a_n]$ and $q = [b_0 : \cdots : b_n]$ are distinct k-rational points, their projective line consists of*

$$[\lambda a_0 + \mu b_0 : \cdots : \lambda a_n + \mu b_n], \qquad [\lambda : \mu] \in \mathbb{P}^1_k,$$

provided the displayed vector is nonzero. Equivalently it is the projectivization of the two-dimensional vector subspace spanned by representatives of p and q.

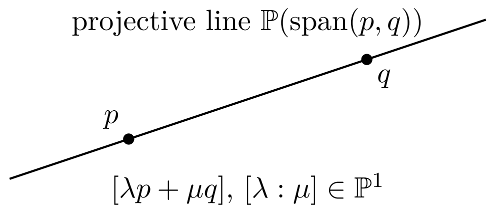


Figure 36.6: Two distinct rational points determine the projectivization of their two-dimensional vector span.

36.9 Closed points and rational points

A k-rational point has residue field k, but closed points can have larger finite extensions as residue fields. This is already visible on $\mathbb{P}^1_{\mathbb{R}}$. On the chart $U_0 \cong \operatorname{Spec} \mathbb{R}[t]$, the prime

$$(t^2 + 1)$$

is maximal and has residue field $\mathbb{C}$. In homogeneous form it corresponds to the irreducible quadratic

$$x_1^2 + x_0^2.$$

Hence it defines a closed point of $\mathbb{P}^1_{\mathbb{R}}$ that is not $\mathbb{R}$-rational.

Over an algebraically closed field, every closed point of $\mathbb{P}^n_k$ is k-rational by the affine-chart form of the Nullstellensatz.

36.10 Base extension

Let K/k be a field extension. Base change of the standard affine charts gives

$$U_i \times_k K \cong \mathbb{A}^n_K.$$

The overlap maps are obtained by the same rational formulas after extending scalars. Therefore the charts glue to the expected scheme.

Theorem 36.10 (Projective space commutes with field extension)**.**

$$\boxed{\mathbb{P}^n_k \times_{\operatorname{Spec} k} \operatorname{Spec} K \cong \mathbb{P}^n_K.}$$

This explains why $\mathbb{P}^n_k(K)$ is naturally the set of one-dimensional K-subspaces of K^{n+1}: the field-valued points are ordinary rational points after base extension to K.

36.11 Twisting sheaves on projective space

VI/35 defined

$$\mathcal{O}_{\mathbb{P}^n}(d) := \widetilde{S(d)}, \qquad S = k[x_0, \dots, x_n].$$

On U_i, the degree-zero localized module $S(d)_{(x_i)}$ is free of rank one over $S_{(x_i)}$. A convenient generator is the localized homogeneous element x_i^d, interpreted inside the graded localization for arbitrary $d \in \mathbb{Z}$.

On $U_i \cap U_j$, the two local generators satisfy

$$x_i^d = \left(\frac{x_i}{x_j}\right)^d x_j^d.$$

Thus $\mathcal{O}(d)$ is locally free of rank one, with transition factors

$$\left(\frac{x_i}{x_j}\right)^d.$$

This is the first concrete line-bundle calculation in projective geometry. The general theory of invertible sheaves and Picard groups is deferred to the later line-bundle chapter.

$$\begin{tikzcd}
\boxed{\text{frame } e_i = x_i^d \text{ on } U_i} \arrow[rr, leftrightarrow, "\text{transition}"] \arrow[dr] & & \boxed{\text{frame } e_j = x_j^d \text{ on } U_j} \arrow[dl] \\
& \begin{array}{c} \text{on } U_i \cap U_j\text{:} \\ e_i = (x_i/x_j)^d e_j \end{array} &
\end{tikzcd}$$

Figure 36.7: The twisting sheaf $\mathcal{O}(d)$ is locally trivial, but its local frames differ by projective coordinate ratios.

Example 36.11 (Point at infinity)**.** The projective closure of the affine parabola $y = x^2$ has equation $X_0X_2 - X_1^2 = 0$. Setting $X_0 = 0$ gives the single point $[0 : 0 : 1]$, the point at infinity.

36.12 Global sections of the basic twists

For $d \geq 0$, every homogeneous polynomial $F \in S_d$ determines a global section of $\mathcal{O}(d)$: on U_i, divide by x_i^d to obtain the degree-zero function

$$\frac{F}{x_i^d} \in S_{(x_i)}.$$

These local functions agree on overlaps after accounting for the transition factors.

Theorem 36.12 (Global sections in nonnegative degree)**.** *For $d \geq 0$,*

$$\boxed{\Gamma(\mathbb{P}_k^n, \mathcal{O}(d)) \cong k[x_0, \dots, x_n]_d.}$$

In particular,

$$\Gamma(\mathbb{P}_k^n, \mathcal{O}) = k.$$

Proof. The construction above gives a map $S_d \to \Gamma(\mathbb{P}^n, \mathcal{O}(d))$. Conversely, a global section is represented on each U_i by a degree-zero element of $S(d)_{x_i}$, hence by a homogeneous rational expression of total degree d. Agreement on overlaps forces these expressions to define one homogeneous polynomial of degree d; any possible denominator supported on a coordinate hyperplane is eliminated by comparing with a chart on which that coordinate vanishes. Thus the map is bijective. □

The notation $H^0(\mathbb{P}^n, \mathcal{O}(d))$ is often used for the same vector space, but no higher cohomology is needed here.

36.13 The Veronese map

Let $d \geq 1$, and let $N = \binom{n+d}{d} - 1$. List all monomials $m_0, \ldots, m_N$ of degree d in $x_0, \ldots, x_n$. Since each monomial scales by λ^d, the formula

$$\nu_d : \mathbb{P}^n_k \longrightarrow \mathbb{P}^N_k, \qquad [a_0 : \cdots : a_n] \longmapsto [m_0(a) : \cdots : m_N(a)]$$

is well defined on field-valued points and, chart by chart, defines a scheme morphism.

For $\mathbb{P}^1$ and $d = 2$,

$$\nu_2([s : t]) = [s^2 : st : t^2] \in \mathbb{P}^2.$$

Its image satisfies

$$XZ - Y^2 = 0.$$

VI/38 will prove that the Veronese map is a closed embedding; VI/37 will give the scheme-theoretic meaning of the displayed homogeneous equation.

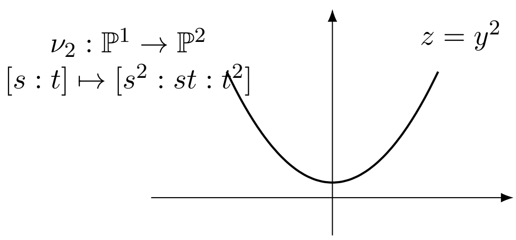


Figure 36.8: On the affine chart $s \neq 0$, the quadratic Veronese map is the parabola $u \mapsto (u, u^2)$; globally it closes into a projective conic.

36.14 The Segre map

For projective spaces over k, define

$$\sigma : \mathbb{P}^m_k \times_k \mathbb{P}^n_k \longrightarrow \mathbb{P}^{(m+1)(n+1)-1}_k$$

on field-valued points by

$$([x_0 : \cdots : x_m], [y_0 : \cdots : y_n]) \longmapsto [x_i y_j]_{0 \leq i \leq m,\ 0 \leq j \leq n}.$$

Independent rescaling of the two factors changes all coordinates by one common nonzero scalar, so the formula is projectively well defined. On affine charts it is polynomial, hence defines a morphism. The image satisfies the rank-one relations

$$z_{ij} z_{\ell r} - z_{ir} z_{\ell j} = 0.$$

Again, the closed-embedding theorem and scheme-theoretic image belong to VI/38 and VI/37 respectively.

36.15 Common pitfalls

- Homogeneous coordinates are defined only up to a common nonzero scalar.
- One may normalize $a_i = 1$ only on the chart where $a_i \neq 0$.
- $\mathbb{P}^n_k$ is not one affine scheme for $n > 0$; it requires the standard affine cover.
- The phrase "hyperplane at infinity" depends on the chosen affine chart. Another choice of distinguished coordinate produces another hyperplane.
- Closed points need not be k-rational when k is not algebraically closed.
- The coordinate formulas for Veronese and Segre maps are easy; proving that they are closed embeddings is a separate theorem reserved for VI/38.
- Writing a homogeneous equation inside projective space suggests a projective closed locus, but its full scheme structure belongs to VI/37.
- $\Gamma(\mathbb{P}^n, \mathcal{O}) = k$ does not imply $\mathbb{P}^n \cong \operatorname{Spec} k$; global sections do not recover a non-affine scheme.

36.16 Boundaries of VI/36

VI/36 owns projective space itself: homogeneous coordinates, standard affine charts, chart transitions, the hyperplane at infinity, projective linear loci, dimension, residue-field behavior, base extension, basic twists, and the first coordinate morphisms such as Veronese and Segre. VI/37 will develop projective schemes from homogeneous ideals and quotients, including scheme-theoretic intersections and homogeneous coordinate rings. VI/38 will develop projective morphisms and closed embeddings and will prove the embedding statements previewed here. General line bundles, divisors, and sheaf cohomology remain later topics.

36.17 Worked examples

Example 36.13 (Projective zero-space)**.** $\mathbb{P}^0_k = \operatorname{Proj} k[x_0]$ has one point and is isomorphic to $\operatorname{Spec} k$.

Example 36.14 (A point in two charts)**.** The point $[1:2:3] \in \mathbb{P}^2_k$ lies in all three standard charts when $2, 3 \neq 0$. In U_0 its coordinates are $(2, 3)$; in U_1 they are $(1/2, 3/2)$.

Example 36.15 (A point at infinity)**.** Relative to the affine chart $U_0 \subset \mathbb{P}^2$, the point $[0:1:2]$ lies on the line at infinity $x_0 = 0$.

Example 36.16 (Projective line transition)**.** On $\mathbb{P}^1$, $t = x_1/x_0$ and $u = x_0/x_1$ satisfy $u = t^{-1}$ on $\mathbb{G}_m$.

Example 36.17 (Dehomogenizing a line)**.** The homogeneous linear equation $x_0 + x_1 + x_2 = 0$ becomes $1 + t_1 + t_2 = 0$ on U_0.

Example 36.18 (Dehomogenizing a conic)**.** The equation $x_0x_2 - x_1^2 = 0$ becomes $t_2 - t_1^2 = 0$ on U_0, displaying the affine parabola chart of the projective conic.

Example 36.19 (The line at infinity of a plane curve)**.** If an affine polynomial $f(x, y)$ is homogenized to $F(x_0, x_1, x_2)$, its points at infinity are detected set-theoretically by $F(0, x_1, x_2) = 0$.

Example 36.20 (A projective coordinate change)**.** An invertible matrix $A \in \mathrm{GL}_{n+1}(k)$ sends a nonzero vector to a nonzero vector and therefore induces a bijection $[v] \mapsto [Av]$ on $\mathbb{P}^n(k)$. The corresponding scheme automorphism will be interpreted systematically in later projective-morphism language.

Example 36.21 (A non-rational real point)**.** The closed point of $\mathbb{P}^1_{\mathbb{R}}$ defined on U_0 by t^2+1 has residue field $\mathbb{C}$.

Example 36.22 (Base extension to the complex numbers)**.** After base changing the preceding point from $\mathbb{R}$ to $\mathbb{C}$, the polynomial t^2+1 factors and the geometric fiber separates into the two $\mathbb{C}$-points $[1 : i]$ and $[1 : -i]$.

Example 36.23 (A hyperplane)**.** The locus $x_n = 0$ in $\mathbb{P}^n$ has coordinates $[x_0 : \cdots : x_{n-1}]$ and is naturally a $\mathbb{P}^{n-1}$.

Example 36.24 (A projective line in $\mathbb{P}^3$)**.** The simultaneous linear equations $x_0 = x_1 = 0$ leave coordinates $[0 : 0 : x_2 : x_3]$, giving a copy of $\mathbb{P}^1$.

Example 36.25 (Sections of $\mathcal{O}(1)$)**.** $\Gamma(\mathbb{P}^n, \mathcal{O}(1))$ has basis $x_0, \ldots, x_n$ and dimension $n+1$.

Example 36.26 (Sections of $\mathcal{O}(2)$ on $\mathbb{P}^2$)**.** The six quadratic monomials $x_0^2, x_0x_1, x_0x_2, x_1^2, x_1x_2, x_2^2$ form a basis of $\Gamma(\mathbb{P}^2, \mathcal{O}(2))$.

Example 36.27 (Quadratic Veronese of $\mathbb{P}^1$)**.** The map $[s : t] \mapsto [s^2 : st : t^2]$ lands on the conic $XZ - Y^2 = 0$ in $\mathbb{P}^2$.

Example 36.28 (Segre coordinates for $\mathbb{P}^1 \times \mathbb{P}^1$)**.** The Segre map sends $([s : t], [u : v])$ to $[su : sv : tu : tv] \in \mathbb{P}^3$, whose coordinates satisfy $z_{00}z_{11} - z_{01}z_{10} = 0$.

36.18 Exercises

Exercise 36.29 (Homogeneous scaling)**.** Prove that the relation $(a_0, \ldots, a_n) \sim (\lambda a_0, \ldots, \lambda a_n)$ for $\lambda \in K^\times$ is an equivalence relation on $K^{n+1} \setminus \{0\}$.

Exercise 36.30 (Chart membership)**.** Determine exactly which standard charts contain $[2 : 0 : 3 : 0] \in \mathbb{P}^3_k$, assuming $2, 3 \neq 0$ in k.

Exercise 36.31 (The U_0 chart)**.** Write the explicit isomorphism $U_0 \to \mathbb{A}^n_k$ and its inverse on K-valued points.

Exercise 36.32 (Transition in $\mathbb{P}^2$)**.** Write the change of coordinates from U_0 coordinates $(u, v) = (x_1/x_0, x_2/x_0)$ to U_1 coordinates on $u \neq 0$.

Exercise 36.33 (The overlap of the two $\mathbb{P}^1$ charts)**.** Show that $U_0 \cap U_1 \cong \mathbb{G}_m$ and that the gluing map is inversion.

Exercise 36.34 (Projective zero-space)**.** Prove directly from Proj that $\mathbb{P}^0_k \cong \operatorname{Spec} k$.

Exercise 36.35 (Hyperplane at infinity)**.** Show that $\mathbb{P}^n \setminus U_0$ has underlying projective locus $\mathbb{P}^{n-1}$.

Exercise 36.36 (Dimension)**.** Use the standard affine cover to prove $\dim \mathbb{P}^n_k = n$.

Exercise 36.37 (A linear subspace)**.** Identify the projective locus $x_0 = x_1 = x_2 = 0$ in $\mathbb{P}^n$ when $n \geq 3$.

Exercise 36.38 (Line through two points)**.** Find a parametrization of the projective line through $[1:0:0]$ and $[0:1:0]$ in $\mathbb{P}^2$.

Exercise 36.39 (Closed non-rational point)**.** Compute the residue field of the closed point of $\mathbb{P}^1_{\mathbb{R}}$ given on U_0 by t^2+1.

Exercise 36.40 (Base extension)**.** Verify the base-change theorem for $\mathbb{P}^1$ by gluing the two base-changed affine charts.

Exercise 36.41 (Trivializing $\mathcal{O}(1)$)**.** Show that x_i gives a local generator of $\mathcal{O}(1)$ on U_i.

Exercise 36.42 (Transition for $\mathcal{O}(d)$)**.** Derive the transition factor between the generators x_i^d and x_j^d on $U_i \cap U_j$.

Exercise 36.43 (Global linear sections)**.** Compute $\dim_k \Gamma(\mathbb{P}^n, \mathcal{O}(1))$.

Exercise 36.44 (Global quadratic sections)**.** Compute $\dim_k \Gamma(\mathbb{P}^n, \mathcal{O}(2))$.

Exercise 36.45 (Dehomogenization)**.** Dehomogenize $x_0x_2 - x_1^2$ on each of the three standard charts of $\mathbb{P}^2$.

Exercise 36.46 (Homogenization)**.** Homogenize $y - x^2 \in k[x,y]$ to a degree-two polynomial in $k[x_0,x_1,x_2]$, and determine its set-theoretic points at infinity.

Exercise 36.47 (Veronese well-definedness)**.** Prove that the degree-d Veronese coordinate formula is independent of the representative of homogeneous coordinates.

Exercise 36.48 (Quadratic Veronese equation)**.** Verify directly that $[s^2 : st : t^2]$ satisfies $XZ - Y^2 = 0$.

Exercise 36.49 (Segre well-definedness)**.** Show that independent rescaling in the two factors changes all Segre coordinates by one common scalar.

Exercise 36.50 (Segre relation)**.** Verify the 2×2 minor relation for the Segre map $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$.

Exercise 36.51 (Global functions)**.** Use the theorem on $\mathcal{O}(d)$ to recover $\Gamma(\mathbb{P}^n, \mathcal{O}) = k$.

Exercise 36.52 (Affine versus projective)**.** Explain why $\Gamma(\mathbb{P}^n, \mathcal{O}) = k$ does not imply $\mathbb{P}^n \cong \operatorname{Spec} k$ for $n > 0$.

Exercise 36.53 (Projective linear action)**.** Let $A \in \mathrm{GL}_{n+1}(k)$. Prove that

$$[v] \longmapsto [Av]$$

defines an automorphism of $\mathbb{P}^n_k(k)$, and show that scalar matrices act trivially.

Exercise 36.54 (Finite-field points)**.** For $k = \mathbb{F}_q$, prove

$$\#\mathbb{P}^n(\mathbb{F}_q) = 1 + q + \cdots + q^n = \frac{q^{n+1}-1}{q-1}.$$

36.19 Problem dossiers

Problem 36.55 (VI.36.P1: Reconstruct $\mathbb{P}^1$ by gluing)**.** Starting from two copies $X_0 = \operatorname{Spec} k[t]$ and $X_1 = \operatorname{Spec} k[u]$, glue $D(t)$ to $D(u)$ by $u = t^{-1}$. Prove that the resulting scheme is isomorphic to $\mathbb{P}^1_k$.

Problem 36.56 (VI.36.P2: Three-chart atlas for $\mathbb{P}^2$)**.** Write all three transition maps among the standard affine charts of $\mathbb{P}^2$ and verify the cocycle identity on the triple overlap.

Problem 36.57 (VI.36.P3: Affine chart and hyperplane at infinity)**.** Show that choosing any nonzero linear form L on k^{n+1} produces an affine open $D_+(L) \cong \mathbb{A}^n_k$ whose complement is a projective hyperplane.

Problem 36.58 (VI.36.P4: Dimension and codimension of linear spaces)**.** Let $L \subset \mathbb{P}^n$ be the projective linear locus cut out by r independent linear forms. Compute its dimension and codimension.

Problem 36.59 (VI.36.P5: A closed point over a nonclosed field)**.** For an irreducible polynomial $f(t) \in k[t]$ of degree $d > 1$, construct the corresponding closed point of $\mathbb{P}^1_k$ in the chart U_0 and compute its residue field.

Problem 36.60 (VI.36.P6: Base change of a non-rational point)**.** Let x be the closed point of $\mathbb{P}^1_k$ associated with a separable irreducible polynomial f of degree d. Describe what happens after base change to a splitting field K.

Problem 36.61 (VI.36.P7: Global sections of $\mathcal{O}(d)$)**.** Give a chartwise proof that a homogeneous polynomial $F \in S_d$ defines a global section of $\mathcal{O}(d)$, and compute the transition compatibility explicitly.

Problem 36.62 (VI.36.P8: Quadratic Veronese chart computation)**.** Compute the quadratic Veronese map $\nu_2 : \mathbb{P}^1 \to \mathbb{P}^2$ on the affine chart $s \neq 0$, and show that its image there is the parabola $z = x^2$.

Problem 36.63 (VI.36.P9: Segre chart computation)**.** Compute the Segre map $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ on the chart $s \neq 0, u \neq 0$ and identify its affine image equation.

Problem 36.64 (VI.36.P10: Projective completion of a parabola)**.** Homogenize $y - x^2 = 0$ and determine the additional point at infinity of its projective closure set-theoretically.

Problem 36.65 (VI.36.P11: Projective completion of a hyperbola)**.** Homogenize $xy - 1 = 0$ in $\mathbb{A}^2$ and determine its set-theoretic points at infinity.

Problem 36.66 (VI.36.P12: Why projective space is not affine)**.** For $n > 0$, use global sections and dimension to prove that $\mathbb{P}^n_k$ is not affine.

Problem 36.67 (VI.36.P13: The PGL-action)**.** Show that $\mathrm{GL}_{n+1}(k)$ acts on $\mathbb{P}^n_k(k)$, identify the kernel, and deduce an action of $\mathrm{PGL}_{n+1}(k)$.

Provenance. Canonical VI/36 extension.

Problem 36.68 (VI.36.P14: Counting finite-field points)**.** For $k = \mathbb{F}_q$, compute $\#\mathbb{P}^n(k)$.

Provenance. Canonical VI/36 extension.

Problem 36.69 (VI.36.P15: Spans and intersections)**.** Let $A, B \subseteq k^{n+1}$ be nonzero vector subspaces. Write

$$L = \mathbb{P}(A), \qquad M = \mathbb{P}(B) \subseteq \mathbb{P}^n_k.$$

Express their projective span and intersection in terms of $A + B$ and $A \cap B$. Derive the dimension formula when $A \cap B \neq 0$.

Provenance. Canonical VI/36 extension.

Problem 36.70 (VI.36.P16: Homogenization and dehomogenization)**.** Let $f \in k[t_1, \dots, t_n]$ have total degree at most d. Define its degree-d homogenization

$$F(x_0, \dots, x_n) = x_0^d f(x_1/x_0, \dots, x_n/x_0).$$

Prove that F is homogeneous of degree d, and that dehomogenizing at $x_0 = 1$ recovers f.

Provenance. Canonical VI/36 extension.

Problem 36.71 (VI.36.P17: Transition cocycle for $\mathcal{O}(d)$)**.** Using the standard frames $e_i = x_i^d$ of $\mathcal{O}(d)$ on U_i, prove the transition factors satisfy the cocycle identity on triple overlaps.

Provenance. Canonical VI/36 extension.

Problem 36.72 (VI.36.P18: Tensoring the basic twists)**.** Prove directly from transition factors that

$$\mathcal{O}(d) \otimes \mathcal{O}(e) \cong \mathcal{O}(d+e)$$

on $\mathbb{P}_k^n$.

Provenance. Canonical VI/36 extension.

Problem 36.73 (VI.36.P19: Rational normal curve on a chart)**.** For the degree-d Veronese map

$$\nu_d : \mathbb{P}^1 \to \mathbb{P}^d, \qquad [s:t] \mapsto [s^d : s^{d-1}t : \cdots : t^d],$$

compute the map on $s \neq 0$ and prove it is injective on field-valued points.

Provenance. Canonical VI/36 extension.

Problem 36.74 (VI.36.P20: The main Segre affine chart)**.** For the Segre map

$$\sigma : \mathbb{P}^m \times \mathbb{P}^n \to \mathbb{P}^{(m+1)(n+1)-1},$$

compute the source chart $x_0 \neq 0,\ y_0 \neq 0$ and show its image inside $z_{00} \neq 0$ is isomorphic to $\mathbb{A}^{m+n}$.

Provenance. Canonical VI/36 extension.

36.20 Challenges

Problem 36.75 (Challenge 1: Projective frames)**.** A projective frame in $\mathbb{P}^n(k)$ consists of $n+2$ points in general linear position. Show that any two projective frames are related by a unique projective linear transformation, after making the appropriate scalar normalization.

Problem 36.76 (Challenge 2: Cross-ratio preview)**.** For four distinct points of $\mathbb{P}^1(k)$, derive the classical cross-ratio in an affine coordinate and investigate how it behaves under fractional linear transformations.

Problem 36.77 (Challenge 3: Veronese equations)**.** For $\nu_2 : \mathbb{P}^2 \to \mathbb{P}^5$, write the six quadratic coordinates and find a natural family of quadratic relations among them. Do not assume the closed-embedding theorem.

Problem 36.78 (Challenge 4: Segre rank-one locus)**.** For general m, n, prove set-theoretically over a field that every nonzero rank-one matrix (z_{ij}) is represented by Segre coordinates $z_{ij} = x_i y_j$, unique up to reciprocal rescaling of the two vectors.

Problem 36.79 (Challenge 5: Counting sections)**.** Show combinatorially that

$$\dim_k \Gamma(\mathbb{P}^n, \mathcal{O}(d)) = \binom{n+d}{n}$$

for $d \geq 0$, and interpret this number as the count of degree-d monomials in $n+1$ variables.

Chapter synthesis

Projective space is the first place where the Proj construction becomes geometric in coordinates. The ring $k[x_0, \ldots, x_n]$ yields $n+1$ affine charts $U_i \cong \mathbb{A}^n$, and their transition functions are exactly the ratios forced by homogeneous rescaling. This gives a single object containing affine n-space together with a hyperplane at infinity, while preserving enough symmetry that no affine chart is preferred intrinsically. The twists $\mathcal{O}(d)$, their homogeneous global sections, and the Veronese and Segre coordinate systems now prepare the passage from the ambient space $\mathbb{P}^n$ to projective schemes and projective embeddings in VI/37–VI/38.

36.21 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 36.80 (Affine chart)**.** Write coordinates on $D_+(X_0) \subset \mathbb{P}^n$.

Hint. Use projective coordinates, affine charts, and homogeneous equations. ☐

Solution. $u_i = X_i/X_0$. ☐

Exercise 36.81 (Homogenize)**.** Homogenize $y - x^2$ to degree 2.

Hint. Use projective coordinates, affine charts, and homogeneous equations. ☐

Solution. $X_0X_2 - X_1^2$ after corresponding coordinate naming. ☐

Exercise 36.82 (Hyperplane)**.** What equation defines a projective hyperplane?

Hint. Use projective coordinates, affine charts, and homogeneous equations. ☐

Solution. A nonzero homogeneous linear form. ☐

Exercise 36.83 (Dimension)**.** What is $\dim \mathbb{P}^n_k$?

Hint. Use projective coordinates, affine charts, and homogeneous equations. ☐

Solution. n. ☐

Exercise 36.84 (Standard cover)**.** Which opens cover $\mathbb{P}^n$?

Hint. Use projective coordinates, affine charts, and homogeneous equations. ☐

Solution. $D_+(X_i)$ for $0 \leq i \leq n$. ☐

Proofs

Exercise 36.85 (Chart isomorphism)**.** Prove: Prove $D_+(X_i) \cong \mathbb{A}^n_k$.

Hint. Work from projective coordinates, affine charts, and homogeneous equations and state the hypotheses explicitly. ☐

Solution. Compute the degree-zero localization ring generated by ratios X_j/X_i. ☐

Exercise 36.86 (Homogeneous well-definedness)**.** Prove: Prove a homogeneous equation gives a well-defined vanishing condition on projective points.

Hint. Work from projective coordinates, affine charts, and homogeneous equations and state the hypotheses explicitly. ☐

Solution. Scaling coordinates multiplies a degree-d polynomial by the scalar to the dth power. ☐

Exercise 36.87 (Hyperplane projective space)**.** Prove: Prove a coordinate hyperplane in $\mathbb{P}^n$ is isomorphic to $\mathbb{P}^{n-1}$.

Hint. Work from projective coordinates, affine charts, and homogeneous equations and state the hypotheses explicitly. ☐

Solution. Set one homogeneous coordinate equal to zero and retain the remaining coordinates. ☐

Exercise 36.88 (Projective dimension)**.** Prove: Prove $\dim \mathbb{P}^n = n$.

Hint. Work from projective coordinates, affine charts, and homogeneous equations and state the hypotheses explicitly. ☐

Solution. Use the affine standard cover and dimension of $\mathbb{A}^n$. ☐

Counterexamples and hypothesis tests

Exercise 36.89 (Ordinary equations)**.** Test the claim: any nonhomogeneous polynomial defines a projective subset by naive substitution.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the condition is not invariant under coordinate scaling. ☐

Exercise 36.90 (Unique coordinates)**.** Test the claim: a projective point has a unique coordinate tuple.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; tuples are defined up to nonzero scalar. ☐

Exercise 36.91 (One affine chart)**.** Test the claim: one standard affine chart covers all of $\mathbb{P}^n$.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False. ☐

Applications and investigations

Exercise 36.92 (Compactification)**.** How does projective space compactify affine space algebraically?

Hint. Translate the question into projective coordinates, affine charts, and homogeneous equations. ☐

Solution. It adds loci at infinity while preserving affine space as a standard open chart. ☐

Exercise 36.93 (Homogenization)**.** Why homogenize affine equations?

Hint. Translate the question into projective coordinates, affine charts, and homogeneous equations. ☐

Solution. Homogenization produces their projective closures and reveals behavior at infinity. ☐

Challenge problems

Exercise 36.94 (Synthesis challenge)**.** Connect two central viewpoints of Projective Space in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Projective Space and then translates it through projective coordinates, affine charts, and homogeneous equations, checking both descriptions agree. ☐

Exercise 36.95 (Hypothesis challenge)**.** Choose one theorem-level statement from Projective Space, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 37

Projective Schemes

VI/34 developed graded rings, VI/35 constructed $\operatorname{Proj} S$, and VI/36 specialized that construction to projective space. We now pass from the ambient space $\mathbb{P}^n_k$ to the schemes cut out by homogeneous equations. If

$$S = k[x_0, \dots, x_n]$$

and $I \subseteq S$ is homogeneous, then the quotient S/I is again graded and

$$X = \operatorname{Proj}(S/I)$$

is the projective scheme defined by I. On every standard chart this is ordinary affine scheme theory:

$$X \cap D_+(f) \cong \operatorname{Spec}\big((S/I)_{(f)}\big) \cong \operatorname{Spec}\big(S_{(f)}/I_{(f)}\big).$$

Thus projective equations become affine equations after dehomogenization, while the homogeneous grading guarantees that the chart descriptions glue.

The migration ledger assigns the projective-scheme section of the AG4 source, together with its theorem-like and exercise/problem descendants, to VI/37. The raw legacy source is not present in the accessible repository archive, so this chapter uses the ledger as provenance and gives a new canonical treatment compatible with VI/34–VI/36. General projective morphisms and the general theory of closed embeddings remain VI/38.

Learning goals

By the end of the chapter, the reader should be able to:

- construct $\operatorname{Proj}(S/I)$ from a homogeneous ideal $I \subseteq S$;
- identify its underlying closed set with $V_+(I) \subseteq \operatorname{Proj} S$;
- compute its affine charts by degree-zero localization and dehomogenization;
- define projective schemes over a field using finitely generated standard graded algebras;
- work with homogeneous coordinate rings while recognizing their dependence on an embedding;
- compute projective hypersurfaces and their affine charts;
- form scheme-theoretic intersections by adding homogeneous ideals;
- distinguish set-theoretic union from scheme-theoretic union and retain nilpotent structure;

- compute saturation with respect to the irrelevant ideal and explain why saturation does not change Proj;
- recognize empty projective schemes from powers of the irrelevant ideal;
- construct projective closures of affine schemes by homogenization and saturation;
- relate a projective scheme to its affine cone;
- detect reducedness, irreducibility, and components from homogeneous coordinate data, with the necessary caveats about irrelevant torsion.

Conceptual roadmap

$$\boxed{\begin{array}{ccc} & S = k[x_0,\dots,x_n],\ I \text{ homogeneous} & \\ & \downarrow & \\ S/I \text{ graded} & \longrightarrow & X = \operatorname{Proj}(S/I) \\ \downarrow & & \downarrow \\ (S/I)_{(f)} \cong S_{(f)}/I_{(f)} & \longleftrightarrow & X \cap D_+(f) \end{array}}$$

Projective scheme theory is therefore affine scheme theory performed coherently on homogeneous localizations.

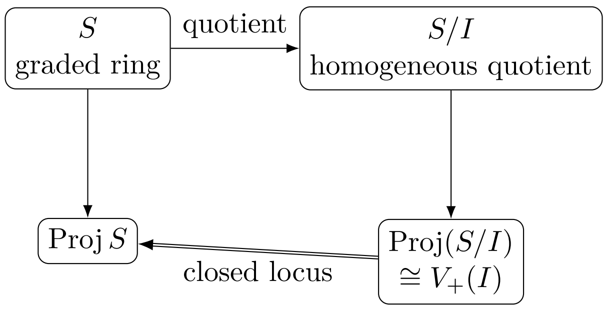


Figure 37.1: A homogeneous quotient produces a projective closed subscheme; the construction is checked on affine standard charts.

37.1 Homogeneous quotients and projective closed loci

Let $S = \bigoplus_{d\geq 0} S_d$ be a graded ring and $I \subseteq S$ a homogeneous ideal. Then S/I inherits a grading and the quotient map preserves degree.

Theorem 37.1 (Proj of a homogeneous quotient)**.** *There is a canonical identification of underlying topological spaces*

$$\operatorname{Proj}(S/I) \cong V_+(I) \subseteq \operatorname{Proj} S.$$

For every homogeneous $f \in S_+$, the corresponding chart is

$$D_+(\bar{f}) \subseteq \operatorname{Proj}(S/I) \cong \operatorname{Spec}(S_{(f)}/I_{(f)}),$$

where $I_{(f)}$ is the degree-zero part of the localized ideal IS_f. These affine closed subschemes glue to a canonical closed subscheme of $\operatorname{Proj} S$*.*

Proof. Homogeneous primes of S/I correspond to homogeneous primes $\mathfrak{p} \subseteq S$ containing I. The condition that the irrelevant ideal of S/I is not contained in the prime is exactly the condition $S_+ \not\subseteq \mathfrak{p}$. Hence the point set is $V_+(I)$. On $D_+(f)$, localization commutes with quotient:

$$(S/I)_f \cong S_f/IS_f.$$

Taking degree-zero parts gives

$$(S/I)_{(f)} \cong S_{(f)}/I_{(f)}.$$

These quotient descriptions are compatible on overlaps by localization in stages, so the affine closed subschemes glue. □

The theorem is the projective analogue of $V(I) = \mathrm{Spec}(A/I)$ in affine geometry. It is the special quotient construction needed in this chapter; VI/38 develops closed embeddings in general.

37.2 Projective schemes over a field

Let k be a field.

Definition 37.2 (Projective scheme in this volume)**.** A k-scheme X is called projective if it is isomorphic to

$$\mathrm{Proj}\, A$$

for some finitely generated standard graded k-algebra

$$A = \bigoplus_{d\geq 0} A_d, \qquad A_0 = k,$$

generated by A_1 over k.

Choosing generators of A_1 gives a surjection

$$k[x_0, \ldots, x_n] \twoheadrightarrow A$$

with homogeneous kernel I, so

$$X \cong \mathrm{Proj}(k[x_0, \ldots, x_n]/I).$$

Thus every projective scheme in this sense is presented by homogeneous equations inside some projective space. VI/38 will repackage this using the general language of closed embeddings and projective morphisms.

Example 37.3 (A projective hypersurface)**.** A homogeneous polynomial $F \in k[X_0, \ldots, X_n]$ defines the closed projective scheme $\mathrm{Proj}\, k[X_0, \ldots, X_n]/(F)$. Its affine charts are obtained by dehomogenizing F.

37.3 Homogeneous coordinate rings

If

$$X = \mathrm{Proj}(S/I) \subseteq \mathbb{P}^n_k, \qquad S = k[x_0, \ldots, x_n],$$

then

$$S(X) := S/I$$

is called the homogeneous coordinate ring of the displayed projective presentation.

Remark 37.4 (Coordinate ring depends on the embedding). Unlike an affine scheme, a projective scheme does not have one intrinsic global coordinate ring recovering the scheme by Spec. Different projective embeddings can produce different homogeneous coordinate rings. What is intrinsic is the scheme; the graded ring records the chosen embedding data.

The standard chart $U_i = D_+(x_i)$ has coordinate ring

$$\Gamma(X \cap U_i, \mathcal{O}_X) \cong (S/I)_{(x_i)}.$$

Computationally this is obtained by setting $x_i = 1$ in homogeneous equations after dividing to degree zero.

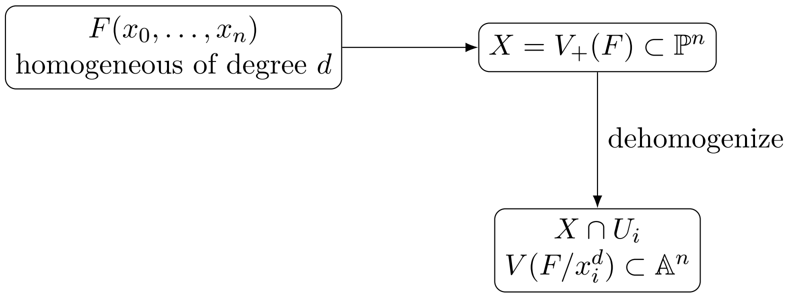


Figure 37.2: A projective hypersurface is computed chart by chart by dividing a homogeneous equation by the appropriate power of the normalized coordinate.

37.4 Projective hypersurfaces

Let $F \in k[x_0, \ldots, x_n]$ be a nonzero homogeneous polynomial of degree $d > 0$. The projective hypersurface defined by F is

$$X = \operatorname{Proj}\big(k[x_0, \ldots, x_n]/(F)\big).$$

On U_i, divide by x_i^d and write

$$f_i = \frac{F}{x_i^d}$$

in the ratio coordinates. Then

$$X \cap U_i \cong V(f_i) \subseteq \mathbb{A}^n_k.$$

Theorem 37.5 (Dimension of a projective hypersurface)**.** *Assume $n \geq 1$. If $F \neq 0$ is homogeneous of positive degree, then every irreducible component of*

$$V_+(F) \subseteq \mathbb{P}^n_k$$

has dimension $n - 1$. Consequently the projective hypersurface has dimension $n - 1$.

Proof. The polynomial ring $S = k[x_0, \ldots, x_n]$ has dimension $n + 1$. Every minimal prime over the principal ideal (F) has height one by the principal ideal theorem, since S is a domain and $F \neq 0$. Hence every corresponding graded quotient component has Krull dimension n. Passing from a nontrivial standard graded affine cone to Proj lowers the dimension by one, yielding dimension $n - 1$ on every projective component. $\square$

37.5 Scheme-theoretic intersections

Suppose two projective closed subschemes of $\mathbb{P}^n_k$ are presented by homogeneous ideals $I, J \subseteq S$. Their scheme-theoretic intersection is obtained chart by chart by taking the sum of the defining ideals.

Proposition 37.6 (Intersection by sum of ideals)**.** *If*

$$X = \operatorname{Proj}(S/I), \qquad Y = \operatorname{Proj}(S/J),$$

then their scheme-theoretic intersection inside $\mathbb{P}^n_k$ is

$$X \cap Y = \operatorname{Proj}\big(S/(I+J)\big).$$

On every standard chart this reduces to the affine formula

$$V(I_{(x_i)}) \cap V(J_{(x_i)}) = V(I_{(x_i)} + J_{(x_i)}).$$

This formula retains multiplicity and nilpotent structure. Replacing an ideal by its radical would generally change the scheme even when the underlying point set remains unchanged.

$$\boxed{I,\ J} \xrightarrow{\text{sum}} \boxed{I+J} \xrightarrow{\text{defines}} \boxed{X \cap Y}$$

$$\boxed{I,\ J} \xrightarrow{\text{intersection}} \boxed{I \cap J} \xrightarrow{\text{defines}} \boxed{X \cup Y}$$

Figure 37.3: Ideal operations reverse geometric behavior: sums define scheme-theoretic intersections, while intersections of ideals define unions.

37.6 Unions and nonreduced structure

For homogeneous ideals $I, J \subseteq S$, the quotient by $I \cap J$ gives the scheme-theoretic union of the two corresponding closed subschemes:

$$\boxed{X \cup Y = \operatorname{Proj}\big(S/(I \cap J)\big).}$$

Replacing $I \cap J$ by its saturation gives the same projective scheme. Set-theoretically,

$$V_+(I \cap J) = V_+(IJ) = V_+(I) \cup V_+(J).$$

Scheme-theoretically, however, $I \cap J$, IJ, and $\sqrt{IJ}$ need not define the same structure.

A basic example is the double hyperplane

$$\operatorname{Proj} S/(x_0^2).$$

Its underlying closed set is the hyperplane $x_0 = 0$, but its structure sheaf contains a nilpotent class coming from x_0. Projective schemes therefore remember infinitesimal thickness just as affine schemes do.

Example 37.7 (A doubled hyperplane)**.** The homogeneous ideal (X_0^2) defines a nonreduced projective subscheme supported on the hyperplane $X_0 = 0$. Its reduction is the ordinary hyperplane defined by (X_0).

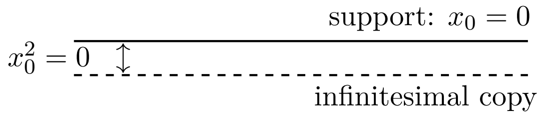


Figure 37.4: A double hyperplane has the same support as the reduced hyperplane but carries an additional nilpotent normal direction.

37.7 Saturation and irrelevant torsion

Let $S = k[x_0, \ldots, x_n]$ with irrelevant ideal

$$\mathfrak{m} = S_+ = (x_0, \ldots, x_n).$$

For a homogeneous ideal $I \subseteq S$, define its saturation by

$$I^{\mathrm{sat}} := I : \mathfrak{m}^\infty = \bigcup_{r \geq 0} (I : \mathfrak{m}^r).$$

Equivalently, a homogeneous element g lies in I^{sat} if some power of the irrelevant ideal sends g into I.

Theorem 37.8 (Saturation does not change Proj)**.** *For every homogeneous ideal $I \subseteq S$,*

$$\mathrm{Proj}(S/I) \cong \mathrm{Proj}(S/I^{\mathrm{sat}})$$

as closed subschemes of $\mathbb{P}^n_k$.

Proof. On the chart $D_+(x_i)$, the element x_i is invertible. If $g \in I^{\mathrm{sat}}$, then for some r one has $\mathfrak{m}^r g \subseteq I$, hence in particular $x_i^r g \in I$. After localizing at x_i, this gives $g \in IS_{x_i}$. Thus I and I^{sat} have the same localized ideal on every standard chart, hence the same degree-zero chart ideal. The resulting closed subschemes agree on the standard cover. $\square$

Saturation removes algebra supported only at the irrelevant cone vertex, which is absent from Proj.

Example 37.9 (Vertex-supported embedded structure)**.** Let

$$I = (x_0^2, x_0x_1, \ldots, x_0x_n) = x_0\mathfrak{m}.$$

Then $x_0 \in I^{\mathrm{sat}}$, and in fact

$$I^{\mathrm{sat}} = (x_0).$$

Thus I and (x_0) define the same projective hyperplane even though their affine cones differ at the origin.

37.8 When Proj is empty

The saturation viewpoint gives a useful criterion.

Proposition 37.10 (Empty projective scheme criterion)**.** *For a homogeneous ideal $I \subseteq S = k[x_0, \ldots, x_n]$, the following are equivalent:*

(i) $\mathrm{Proj}(S/I) = \varnothing$*;*

(ii) $I^{\mathrm{sat}} = S$*;*

$$\begin{tikzcd}
\boxed{\begin{array}{c} I \\ \text{may contain vertex torsion} \end{array}} \arrow[r, "\text{saturate}"] \arrow[d] & \boxed{\begin{array}{c} I^{\mathrm{sat}} \\ \text{projectively visible part} \end{array}} \arrow[d] \\
\boxed{\mathrm{Proj}(S/I)} \arrow[r, leftrightarrow, "\cong"'] & \boxed{\mathrm{Proj}(S/I^{\mathrm{sat}})}
\end{tikzcd}$$

Figure 37.5: Saturation removes homogeneous algebra supported only at the irrelevant cone vertex and leaves the projective scheme unchanged.

(iii) $\mathfrak{m}^r \subseteq I$ *for some* $r \geq 1$.

For example,

$$\mathrm{Proj}\, k[x_0, x_1]/(x_0, x_1)^2 = \varnothing,$$

even though the graded quotient ring is nonzero. All of its geometry is concentrated at the irrelevant vertex.

37.9 Projective closure of affine schemes

Let

$$Y = V(J) \subseteq \mathbb{A}^n_k$$

with coordinates $t_1, \ldots, t_n$. Introduce homogeneous coordinates $x_0, \ldots, x_n$ and identify the affine chart U_0 by $t_i = x_i/x_0$.

For a polynomial $f \in k[t_1, \ldots, t_n]$ of degree d, its homogenization is

$$f^h(x_0, \ldots, x_n) = x_0^d f(x_1/x_0, \ldots, x_n/x_0).$$

The projective closure of Y is defined by the homogenized ideal

$$J^h = (f^h : f \in J) \subseteq k[x_0, \ldots, x_n].$$

If one starts from a chosen generating set of J, merely homogenizing those generators can give an ideal smaller than J^h. The missing equations are recovered by saturation with respect to the homogenizing variable x_0: for a finite generating set $f_1, \ldots, f_r$, one computes

$$J^h = (f_1^h, \ldots, f_r^h) : x_0^\infty$$

after choosing a generating set for which this standard homogenization procedure is applied. A final saturation with respect to the irrelevant ideal does not change Proj and removes purely projectively invisible vertex torsion.

Remark 37.11 (Two different saturations). Saturation by x_0 repairs a naive homogenization of affine generators; it can remove spurious components contained entirely in the hyperplane at infinity. Saturation by the irrelevant ideal S_+ is different: it leaves the projective scheme unchanged and removes only algebra invisible on every standard projective chart. These operations should not be conflated.

The intersection of the closure with $x_0 = 0$ records the projective points at infinity.

Example 37.12 (Veronese image preview)**.** The degree-two monomials in X_0, X_1 define a morphism $\mathbb{P}^1 \to \mathbb{P}^2$, $[s : t] \mapsto [s^2 : st : t^2]$, whose image is the conic $Y_0 Y_2 - Y_1^2 = 0$.

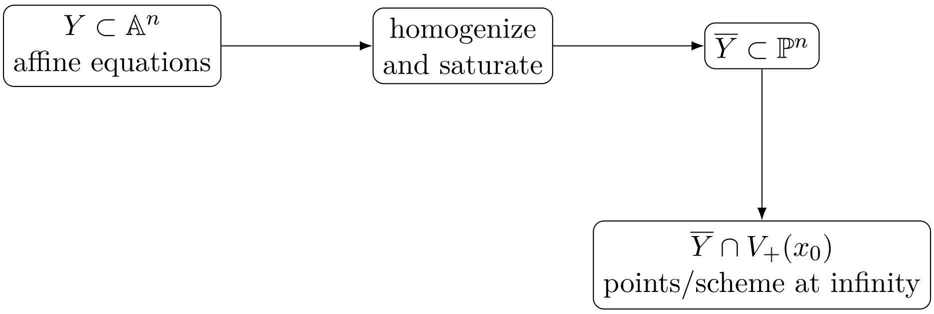


Figure 37.6: Projective closure converts affine equations into homogeneous ones; saturation is the safeguard against spurious projective structure.

37.10 Affine cones and projectivization

Let $A = S/I$ be a standard graded homogeneous coordinate ring. Its affine cone is

$$C(X) := \operatorname{Spec} A.$$

The ideal

$$A_+ = \bigoplus_{d>0} A_d$$

defines the cone vertex when $A_0 = k$. The projective scheme

$$X = \operatorname{Proj} A$$

forgets the vertex and records homogeneous directions away from it.

The cone and the projective scheme carry related but distinct local geometry. A smooth projective scheme can have a singular affine cone at the vertex; conversely, nilpotent or embedded structure supported only at the vertex can disappear after passing to Proj. Saturation is the algebraic mechanism that removes precisely this projectively invisible torsion.

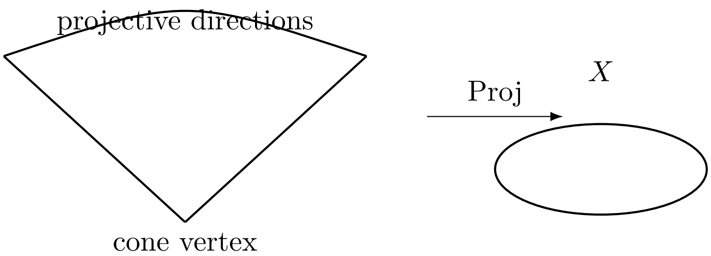


Figure 37.7: The affine cone remembers the vertex; passing to Proj keeps homogeneous directions away from that irrelevant point.

37.11 Reducedness, integrality, and components

If A is a reduced graded ring, then every chart ring $A_{(f)}$ is reduced, so $\operatorname{Proj} A$ is reduced. If A is a domain and $\operatorname{Proj} A \neq \varnothing$, then $\operatorname{Proj} A$ is integral.

The converses need not hold because irrelevant torsion can disappear on every projective chart. For example, let

$$A = k[x_0, x_1, z]/(z^2, x_0 z, x_1 z)$$

with all variables of degree one. The nilpotent class z is annihilated by the irrelevant ideal after passing to the quotient, so it vanishes on both standard charts $D_+(x_0)$ and $D_+(x_1)$. Consequently

$$\operatorname{Proj} A \cong \mathbb{P}^1_k$$

is reduced although A is not.

For a Noetherian standard graded ring, irreducible components of $\operatorname{Proj} A$ correspond to homogeneous minimal primes that do not contain the irrelevant ideal. Minimal primes supported only at the cone vertex contribute no projective component.

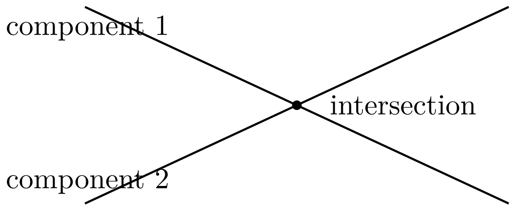


Figure 37.8: Homogeneous minimal primes not containing the irrelevant ideal correspond to projective irreducible components.

37.12 Common pitfalls

- A homogeneous ideal and its saturation can be different rings but define the same projective scheme.
- The homogeneous coordinate ring is attached to a chosen projective presentation; it is not an intrinsic replacement for $\Gamma(X, \mathcal{O}_X)$.
- Homogeneous equations must be dehomogenized separately on each affine chart.
- Taking radicals preserves the underlying closed set but can destroy nilpotent and multiplicity information.
- Scheme-theoretic intersection uses $I + J$, not $I \cap J$.
- Scheme-theoretic union is controlled by an intersection of ideals, not by their sum.
- Homogenizing only a convenient generating set can introduce the wrong projective closure unless saturation is checked.
- The affine cone contains the irrelevant vertex; Proj does not.
- Reducedness or integrality of a homogeneous coordinate ring implies the corresponding property for Proj, but the converse can fail because projectively invisible torsion may live at the irrelevant vertex.
- General closed-embedding and projective-morphism theorems are not proved here; they belong to VI/38.

37.13 Boundaries of VI/37

VI/37 owns projective schemes defined by homogeneous quotients, their affine-chart equations, homogeneous coordinate rings, projective hypersurfaces, scheme-theoretic intersections and unions, saturation, projective closure, affine cones, and basic component/reducedness diagnostics. VI/38 develops projective morphisms, general closed embeddings, and the embedding theorems for Veronese and Segre maps. Divisors, line bundles in general, Picard groups, and sheaf cohomology remain later topics.

37.14 Worked examples

Example 37.13 (A projective point)**.** In $\mathbb{P}^2_k$, the homogeneous ideal (x_1, x_2) gives $\operatorname{Proj} k[x_0] \cong \operatorname{Spec} k$, the point $[1 : 0 : 0]$.

Example 37.14 (A projective line)**.** The ideal $(x_2) \subset k[x_0, x_1, x_2]$ defines $\operatorname{Proj} k[x_0, x_1] \cong \mathbb{P}^1_k$.

Example 37.15 (A conic)**.** The quotient $k[X, Y, Z]/(XZ - Y^2)$ defines a projective conic in $\mathbb{P}^2$. On $X \neq 0$ the equation becomes $z - y^2 = 0$.

Example 37.16 (A cubic curve)**.** The homogeneous cubic $Y^2Z - X^3 - XZ^2$ defines a projective plane cubic. On $Z \neq 0$ it becomes $y^2 = x^3 + x$.

Example 37.17 (Two coordinate lines)**.** The ideal $(x_0x_1) \subset k[x_0, x_1, x_2]$ defines the union of the two projective lines $x_0 = 0$ and $x_1 = 0$.

Example 37.18 (Double line)**.** The ideals (x_0) and (x_0^2) have the same underlying projective line, but $S/(x_0^2)$ retains a nilpotent normal direction.

Example 37.19 (Intersection of two lines)**.** In $\mathbb{P}^2$, $V_+(x_0) \cap V_+(x_1) = V_+(x_0, x_1)$, the point $[0 : 0 : 1]$.

Example 37.20 (A saturated hyperplane ideal)**.** For $I = x_0(x_0, x_1, x_2)$, one has $I^{\mathrm{sat}} = (x_0)$, so both ideals define the same projective line.

Example 37.21 (An empty Proj)**.** The quotient $k[x_0, x_1]/(x_0, x_1)^3$ is nonzero, but its Proj is empty because a power of the irrelevant ideal vanishes.

Example 37.22 (Closure of an affine parabola)**.** The affine parabola $y = x^2$ homogenizes to $x_0x_2 - x_1^2 = 0$ in $\mathbb{P}^2$, producing the projective conic closure.

Example 37.23 (Points at infinity)**.** The projective closure of $xy = 1$ is $x_1x_2 - x_0^2 = 0$. On $x_0 = 0$ it meets infinity at $[0 : 1 : 0]$ and $[0 : 0 : 1]$.

Example 37.24 (Cone over a conic)**.** The affine cone over $XZ - Y^2 = 0 \subset \mathbb{P}^2$ is the quadric cone $V(XZ - Y^2) \subset \mathbb{A}^3$, singular at the origin when the characteristic does not create exceptional behavior.

Example 37.25 (A reducible projective scheme)**.** $\operatorname{Proj} k[x_0, x_1, x_2]/(x_0x_1)$ has two irreducible components, corresponding to the homogeneous minimal primes (x_0) and (x_1).

Example 37.26 (Nonreduced ring, reduced Proj)**.** For $A = k[x_0, x_1, z]/(z^2, x_0z, x_1z)$, the class of z vanishes after localizing at either x_0 or x_1, so $\operatorname{Proj} A \cong \mathbb{P}^1_k$.

Example 37.27 (Homogeneous coordinate ring changes with embedding)**.** The projective line has coordinate ring $k[s, t]$ in its standard embedding, while the quadratic Veronese presentation has coordinate ring $k[X, Y, Z]/(XZ - Y^2)$. The schemes are isomorphic although the graded rings are not.

Example 37.28 (A zero-dimensional projective scheme)**.** In $\mathbb{P}^1$, the homogeneous polynomial x_0x_1 defines two reduced points when the factors are distinct; the projective scheme has dimension zero.

37.15 Exercises

Exercise 37.29 (Quotient chart)**.** Prove directly that $(S/I)_{(f)} \cong S_{(f)}/I_{(f)}$ for homogeneous I and f.

Exercise 37.30 (Underlying closed set)**.** Show that homogeneous primes of S/I defining Proj correspond exactly to points of $V_+(I) \subseteq \operatorname{Proj} S$.

Exercise 37.31 (Coordinate line)**.** Identify $\operatorname{Proj} k[x_0, x_1, x_2]/(x_2)$.

Exercise 37.32 (Conic chart)**.** Dehomogenize $XZ - Y^2$ on each standard chart of $\mathbb{P}^2$.

Exercise 37.33 (Hypersurface dimension)**.** Use the principal ideal theorem to justify the dimension formula for a nonzero projective hypersurface in $\mathbb{P}^n$.

Exercise 37.34 (Intersection by sum)**.** Compute the scheme-theoretic intersection of the lines $x_0 = 0$ and $x_1 = 0$ in $\mathbb{P}^2$.

Exercise 37.35 (Union by intersection)**.** Show that $(x_0) \cap (x_1) = (x_0 x_1)$ in a polynomial ring and interpret the corresponding projective union.

Exercise 37.36 (Double hyperplane)**.** Show that $\operatorname{Proj} S/(x_0^2)$ and $\operatorname{Proj} S/(x_0)$ have the same underlying topological space but different structure sheaves.

Exercise 37.37 (Saturation calculation)**.** For $I = x_0(x_0, x_1, x_2) \subset k[x_0, x_1, x_2]$, prove $I^{\text{sat}} = (x_0)$.

Exercise 37.38 (Empty Proj)**.** Prove that $\operatorname{Proj} S/I = \varnothing$ when $S_+^r \subseteq I$ for some r.

Exercise 37.39 (Converse empty criterion)**.** For $S = k[x_0, \ldots, x_n]$, prove that empty Proj implies some power of S_+ lies in I.

Exercise 37.40 (Homogenize a parabola)**.** Homogenize $y - x^2$ and identify the point(s) of its projective closure at infinity.

Exercise 37.41 (Homogenize a hyperbola)**.** Homogenize $xy - 1$ and determine its points at infinity.

Exercise 37.42 (Affine cone)**.** Write the affine cone of the projective conic $XZ - Y^2 = 0$ and identify its vertex.

Exercise 37.43 (Reducedness descends)**.** Prove that if a graded ring A is reduced, then $\operatorname{Proj} A$ is reduced.

Exercise 37.44 (Domain descends)**.** Prove that if A is a domain and $\operatorname{Proj} A \neq \varnothing$, then $\operatorname{Proj} A$ is integral.

Exercise 37.45 (Projectively invisible nilpotent)**.** Verify that z vanishes on the standard charts of $\operatorname{Proj} k[x_0, x_1, z]/(z^2, x_0 z, x_1 z)$.

Exercise 37.46 (Components)**.** Determine the projective components of $\operatorname{Proj} k[x_0, x_1, x_2]/(x_0 x_1)$.

Exercise 37.47 (Two coordinate points)**.** Find a homogeneous ideal defining the two points $[1 : 0 : 0]$ and $[0 : 1 : 0]$ in $\mathbb{P}^2$.

Exercise 37.48 (A fat point)**.** Describe the projective scheme defined in $\mathbb{P}^2$ by $(x_1, x_2)^2$ near $[1 : 0 : 0]$.

Exercise 37.49 (Coordinate ring dependence)**.** Explain why $k[s,t]$ and $k[X,Y,Z]/(XZ-Y^2)$ can both serve as homogeneous coordinate rings of schemes isomorphic to $\mathbb{P}^1$.

Exercise 37.50 (Chartwise equality after saturation)**.** Prove directly that I and I^{sat} have the same localized ideal after inverting any variable x_i.

Exercise 37.51 (Intersection with infinity)**.** Let F be the homogenization of an affine polynomial. Explain how setting $x_0 = 0$ computes the scheme-theoretic intersection of its projective closure with the hyperplane at infinity, subject to the saturation caveat.

Exercise 37.52 (Why radicals lose information)**.** Compare $S/(x_0^2)$ and $S/(x_0)$ on a standard chart and identify the nilpotent killed by passing to the radical.

Exercise 37.53 (Saturated ideals from chart data)**.** Let $I, J \subseteq k[x_0, \dots, x_n]$ be homogeneous. Assume $IS_{x_i} = JS_{x_i}$ for every i. Prove

$$I^{\mathrm{sat}} = J^{\mathrm{sat}}.$$

Exercise 37.54 (Naive homogenization needs saturation)**.** In $k[x,y]$, let

$$J = (x^2 - y,\ x^3 - xy - x).$$

Show $J = (x,y)$. Homogenize the displayed generators using z, and prove saturation by z recovers the homogeneous ideal (X,Y) of the projective closure of the origin.

37.16 Problem dossiers

Problem 37.55 (VI.37.P1: Quotient chart theorem)**.** Let $S = k[x_0, \dots, x_n]$, let I be homogeneous, and let f be homogeneous of positive degree. Prove the chart identity

$$D_+(\bar{f}) \subseteq \mathrm{Proj}(S/I) \cong \mathrm{Spec}(S_{(f)}/I_{(f)}).$$

Problem 37.56 (VI.37.P2: Projective conic charts)**.** For $X = V_+(XZ - Y^2) \subseteq \mathbb{P}^2_k$, compute all three standard affine charts.

Problem 37.57 (VI.37.P3: Intersection multiplicity preview)**.** Inside $\mathbb{P}^2$, intersect the line $Y = 0$ with the conic $Y^2 - XZ = 0$. Determine the scheme-theoretic intersection.

Problem 37.58 (VI.37.P4: A tangent line creates a fat intersection)**.** Intersect the conic $XZ - Y^2 = 0$ with its tangent line $Z = 0$ at $[1:0:0]$.

Problem 37.59 (VI.37.P5: Saturation removes vertex torsion)**.** Let $I = (x^2, xy, xz) \subset k[x,y,z]$. Compute I^{sat} with respect to (x,y,z) and compare the associated projective schemes.

Problem 37.60 (VI.37.P6: Projective closure of a hyperbola)**.** Find the projective closure of $xy = 1 \subseteq \mathbb{A}^2$ and its intersection with the line at infinity.

Problem 37.61 (VI.37.P7: Two lines and their union)**.** Let $L_0 = V_+(x_0)$ and $L_1 = V_+(x_1)$ in $\mathbb{P}^2$. Compute their union and intersection scheme-theoretically.

Problem 37.62 (VI.37.P8: Empty projective scheme)**.** Show that $\mathrm{Proj}\, k[x_0,x_1]/(x_0,x_1)^2$ is empty although the quotient ring is nonzero.

Problem 37.63 (VI.37.P9: A nonreduced projective line)**.** Describe $X = \mathrm{Proj}\, k[x_0,x_1,x_2]/(x_2^2)$ and compare it with $V_+(x_2)$.

Problem 37.64 (VI.37.P10: Components from minimal primes)**.** Let

$$X = \operatorname{Proj}(k[x_0, x_1, x_2]/(x_0x_1x_2)).$$

Determine the irreducible components of X.

Problem 37.65 (VI.37.P11: Same Proj, different graded rings)**.** Let $A = k[x_0, x_1, z]/(z^2, x_0z, x_1z)$. Prove $\operatorname{Proj} A \cong \mathbb{P}^1_k$.

Problem 37.66 (VI.37.P12: A saturated zero-dimensional scheme)**.** In $\mathbb{P}^2$, let $I = (x_2, x_0x_1)$. Describe $\operatorname{Proj} S/I$, and show I is saturated.

Problem 37.67 (VI.37.P13: Union exact sequence)**.** Let $I, J \subseteq S = k[x_0, \ldots, x_n]$ be homogeneous. Prove the exact sequence

$$0 \to S/(I \cap J) \longrightarrow S/I \oplus S/J \longrightarrow S/(I + J) \to 0,$$

and explain its projective scheme meaning.

Provenance. Canonical VI/37 extension.

Problem 37.68 (VI.37.P14: Saturated ideals are determined by projective chart ideals)**.** Let $I, J \subseteq S = k[x_0, \ldots, x_n]$ be homogeneous. Assume

$$IS_{x_i} = JS_{x_i}$$

for every standard variable x_i. Prove

$$I^{\mathrm{sat}} = J^{\mathrm{sat}}.$$

Deduce that two saturated homogeneous ideals with the same projective ideal sheaf are equal.

Provenance. Canonical VI/37 extension.

Problem 37.69 (VI.37.P15: A length-three projective fat point)**.** In $\mathbb{P}^2$, let

$$I = (x_1, x_2)^2.$$

Show that $\operatorname{Proj} S/I$ is supported at $[1 : 0 : 0]$, compute its affine local ring there, and determine its length.

Provenance. Canonical VI/37 extension.

Problem 37.70 (VI.37.P16: Two homogeneous coordinate rings for $\mathbb{P}^1$)**.** Prove

$$\operatorname{Proj} k[s, t] \cong \operatorname{Proj} k[X, Y, Z]/(XZ - Y^2)$$

using the second Veronese ring.

Provenance. Canonical VI/37 extension.

Problem 37.71 (VI.37.P17: Projective closure of the cusp)**.** Let

$$Y = V(y^2 - x^3) \subseteq \mathbb{A}^2_k.$$

Compute its projective closure in $\mathbb{P}^2_k$, determine its point at infinity, and write the affine equation on the chart $y \neq 0$.

Provenance. Canonical VI/37 extension.

Problem 37.72 (VI.37.P18: Equivalent criteria for empty Proj)**.** For homogeneous $I \subseteq S = k[x_0, \ldots, x_n]$, prove the equivalence

$$\operatorname{Proj}(S/I) = \varnothing \iff I^{\mathrm{sat}} = S \iff S_+^N \subseteq I \text{ for some } N.$$

Provenance. Canonical VI/37 extension.

Problem 37.73 (VI.37.P19: Reduced Proj from a nonreduced coordinate ring)**.** Let

$$A = k[x_0, x_1, z]/(z^2, x_0 z, x_1 z).$$

Show that A is nonreduced but $\operatorname{Proj} A$ is reduced, and identify the projectively invisible nilpotent module.

Provenance. Canonical VI/37 extension.

Problem 37.74 (VI.37.P20: An embedded irrelevant associated prime)**.** Let

$$A = k[x, y]/(x^2, xy)$$

with the standard grading. Determine a minimal prime and an embedded associated prime, and explain which one contributes to $\operatorname{Proj} A$.

Provenance. Canonical VI/37 extension.

37.17 Challenges

Problem 37.75 (Challenge 1: Saturation and sheaf ideals)**.** Let $I \subseteq S = k[x_0, \ldots, x_n]$ be homogeneous. Prove directly from standard charts that the quasi-coherent ideal sheaves associated to I and I^{sat} on $\mathbb{P}^n$ are equal. Explain why this is stronger than equality of underlying closed sets.

Problem 37.76 (Challenge 2: A projective closure requiring saturation)**.** Find an explicit affine ideal with a chosen generating set whose naive homogenized generators do not yet give the saturated homogeneous ideal of the projective closure. Compute the saturation and identify the spurious component or embedded structure removed.

Problem 37.77 (Challenge 3: Length-two tangency)**.** For a smooth conic in $\mathbb{P}^2$, choose a rational point and compute the scheme-theoretic intersection with its tangent line on an affine chart. Show that the local intersection is isomorphic to $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ under suitable characteristic assumptions.

Problem 37.78 (Challenge 4: Cone versus projective smoothness)**.** Investigate the affine cone over the quadratic Veronese conic. Compute the Jacobian at the cone vertex and compare the singularity of the cone with the regularity of the projective conic.

Problem 37.79 (Challenge 5: Projective components and irrelevant associated primes)**.** Construct a Noetherian standard graded k-algebra with a homogeneous minimal prime not containing the irrelevant ideal and an embedded associated prime equal to the irrelevant ideal. Determine which associated prime contributes to Proj and explain the result chart by chart.

Correction note. The previous wording asked for one minimal prime containing the irrelevant ideal and another minimal prime not containing it. For a standard graded k-algebra with $A_0 = k$, that configuration is impossible: every proper homogeneous prime is contained in A_+, so if A_+ itself were minimal there could be no second homogeneous minimal prime below it.

Chapter synthesis

A projective scheme is affine scheme theory organized by homogeneous localization. Homogeneous equations define quotient graded rings, standard opens turn those quotients into ordinary affine coordinate rings, and the resulting pieces glue inside projective space. Saturation separates genuine projective geometry from algebra supported only at the irrelevant cone vertex. With this distinction in place, homogeneous coordinate rings can be used safely to compute hypersurfaces, intersections, closures, components, and nilpotent thickenings. The next chapter turns these constructions into a general theory of projective morphisms and closed embeddings.

37.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 37.80 (Hypersurface ring)**.** Write the homogeneous coordinate ring of $V_+(F) \subset \mathbb{P}^n$.

Hint. Use homogeneous ideals, projective subschemes, and Proj quotients. □

Solution. $k[X_0, \ldots, X_n]/(F)$. □

Exercise 37.81 (Reduction)**.** Reduce the projective subscheme defined by (X_0^2).

Hint. Use homogeneous ideals, projective subschemes, and Proj quotients. □

Solution. It becomes the hyperplane defined by (X_0). □

Exercise 37.82 (Conic ideal)**.** What equation cuts out the quadratic Veronese image of $\mathbb{P}^1$ in $\mathbb{P}^2$?

Hint. Use homogeneous ideals, projective subschemes, and Proj quotients. □

Solution. $Y_0Y_2 - Y_1^2$. □

Exercise 37.83 (Affine chart)**.** How do you obtain an affine equation from a homogeneous one on $X_i \neq 0$?

Hint. Use homogeneous ideals, projective subschemes, and Proj quotients. □

Solution. Set $X_i = 1$ after dividing by the appropriate power. □

Exercise 37.84 (Closed subscheme)**.** What kind of ideal defines a projective closed subscheme?

Hint. Use homogeneous ideals, projective subschemes, and Proj quotients. □

Solution. A homogeneous ideal. □

Proofs

Exercise 37.85 (Proj quotient)**.** Prove: Prove $\mathrm{Proj}(S/I)$ identifies with the closed subscheme $V_+(I) \subset \mathrm{Proj}\, S$ for homogeneous I.

Hint. Work from homogeneous ideals, projective subschemes, and Proj quotients and state the hypotheses explicitly. ☐

Solution. Use homogeneous prime correspondence and compare standard affine charts. ☐

Exercise 37.86 (Reduction)**.** Prove: Prove replacing I by $\sqrt{I}$ preserves the underlying projective closed set.

Hint. Work from homogeneous ideals, projective subschemes, and Proj quotients and state the hypotheses explicitly. ☐

Solution. Homogeneous primes contain I iff they contain its radical. ☐

Exercise 37.87 (Chart compatibility)**.** Prove: Prove dehomogenized affine charts glue to the projective subscheme.

Hint. Work from homogeneous ideals, projective subschemes, and Proj quotients and state the hypotheses explicitly. ☐

Solution. On overlaps, localizations by coordinate ratios agree through degree-zero localization. ☐

Exercise 37.88 (Veronese conic)**.** Prove: Prove the quadratic Veronese map lands on the conic.

Hint. Work from homogeneous ideals, projective subschemes, and Proj quotients and state the hypotheses explicitly. ☐

Solution. Substitute $Y_0 = s^2$, $Y_1 = st$, $Y_2 = t^2$ into the defining equation. ☐

Counterexamples and hypothesis tests

Exercise 37.89 (Reduced automatic)**.** Test the claim: every projective scheme is reduced.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; homogeneous ideals with powers give nilpotent thickenings. ☐

Exercise 37.90 (Variety only)**.** Test the claim: projective schemes must be irreducible varieties.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; they may be reducible or nonreduced. ☐

Exercise 37.91 (Affine equality)**.** Test the claim: every projective scheme is affine.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; positive-dimensional projective space is the standard counterexample. ☐

Applications and investigations

Exercise 37.92 (Projective closure)**.** How is the projective closure of an affine scheme computed?

Hint. Translate the question into homogeneous ideals, projective subschemes, and Proj quotients. ☐

Solution. Homogenize the defining ideal, with saturation when needed to remove spurious components. ☐

Exercise 37.93 (Embedded geometry)**.** What does a homogeneous coordinate ring encode?

Hint. Translate the question into homogeneous ideals, projective subschemes, and Proj quotients. ☐

Solution. It records a chosen projective embedding together with its scheme structure. ☐

Challenge problems

Exercise 37.94 (Synthesis challenge)**.** Connect two central viewpoints of Projective Schemes in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Projective Schemes and then translates it through homogeneous ideals, projective subschemes, and Proj quotients, checking both descriptions agree. ☐

Exercise 37.95 (Hypothesis challenge)**.** Choose one theorem-level statement from Projective Schemes, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 38

Projective Morphisms and Closed Embeddings

VI/34 developed graded algebra, VI/35 constructed Proj, VI/36 made projective space concrete, and VI/37 developed projective schemes cut out by homogeneous ideals. This chapter completes the projective arc by answering the functorial questions that were deliberately postponed: when does a graded ring map induce a morphism on Proj, what exactly is a projective morphism, how are closed embeddings recognized, and why are the Veronese and Segre maps genuine closed embeddings rather than merely injective formulas on points?

The central warning is that Proj is not contravariantly functorial for arbitrary graded maps on all points. If $\varphi : S \to T$ preserves degree and $\mathfrak{q} \in \operatorname{Proj} T$, the contracted prime $\varphi^{-1}(\mathfrak{q})$ belongs to $\operatorname{Proj} S$ only when it does not contain S_+. Thus one must first identify the open domain on which the irrelevant ideal survives. Once that issue is handled, the affine chart maps glue exactly as in VI/35.

The current migration ledger assigns six rules to VI/38: the AG4 projective-morphism/closed-embedding section with its inherited theorem/problem descendants, together with the projective-morphism overlap split from AG9 during the architecture synchronization. The accessible repository does not contain the raw legacy AG4/AG9 prose, so the ledger is used as provenance and scope evidence rather than as a basis for claiming verbatim recovery.

Learning goals

By the end of the chapter, the reader should be able to:

- determine the maximal open subset of $\operatorname{Proj} T$ on which a degree-preserving map $S \to T$ induces a morphism to $\operatorname{Proj} S$;
- state practical sufficient conditions ensuring that the induced Proj morphism is defined everywhere;
- handle homogeneous maps of positive degree by passing to a Veronese regrading;
- construct morphisms to projective space from homogeneous forms of one degree and identify their base locus;
- characterize closed immersions affine-locally by surjective ring maps;
- prove that a homogeneous quotient $S \twoheadrightarrow S/I$ induces the canonical closed immersion $\operatorname{Proj}(S/I) \hookrightarrow \operatorname{Proj} S$;
- define relative projective space $\mathbb{P}^n_Y$ and projective morphisms over a base scheme;

- prove stability of projective morphisms under arbitrary base change;
- prove stability under composition using the relative Segre embedding;
- prove the d-uple Veronese map is a closed embedding via the Veronese subring;
- prove the Segre map is a closed embedding and identify its ideal by the 2×2 minors;
- connect closed embeddings with very ample invertible sheaves and systems of homogeneous coordinates.

Conceptual roadmap

$$S \xrightarrow{\varphi} T \text{ graded}$$
$$\downarrow$$
$$U_\varphi = \{\mathfrak{q} \in \operatorname{Proj} T : \varphi(S_+) \not\subseteq \mathfrak{q}\} \xrightarrow{\operatorname{Proj}(\varphi)} \operatorname{Proj} S$$

surjection $\Rightarrow$ closed immersion   same-degree forms $\Rightarrow$ projective map

Veronese + Segre $\Rightarrow$ closed embeddings and composition of projective morphisms

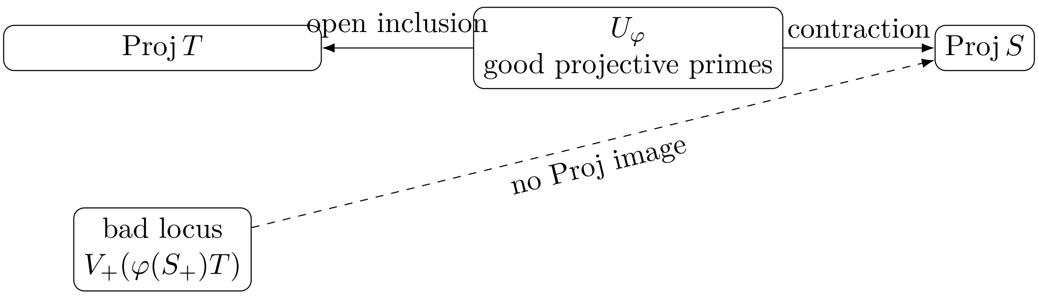


Figure 38.1: A graded map induces a Proj morphism only on the open locus where the image of the irrelevant ideal does not vanish simultaneously.

38.1 Graded maps and the domain of definition

Let $S = \bigoplus_{d \geq 0} S_d$ and $T = \bigoplus_{d \geq 0} T_d$ be graded rings, and let

$$\varphi : S \longrightarrow T$$

be degree preserving. For a homogeneous prime $\mathfrak{q} \subset T$, the contraction $\varphi^{-1}(\mathfrak{q})$ is homogeneous and prime. However it may contain S_+, in which case it is not a point of $\operatorname{Proj} S$.

Definition 38.1 (Proj domain of a graded map)**.** Define

$$U_\varphi := \{\mathfrak{q} \in \operatorname{Proj} T : \varphi(S_+) \not\subseteq \mathfrak{q}\} = \bigcup_{f \in S_+ \text{ homogeneous}} D_+(\varphi(f)).$$

This is the maximal open subset of $\operatorname{Proj} T$ on which contraction lands in $\operatorname{Proj} S$.

Theorem 38.2 (Functoriality on the correct domain)**.** *A degree-preserving map $\varphi : S \to T$ induces a canonical morphism*

$$\operatorname{Proj}(\varphi) : U_\varphi \longrightarrow \operatorname{Proj} S, \qquad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}).$$

For homogeneous $f \in S_+$, its restriction to $D_+(\varphi(f))$ is the affine morphism induced by

$$S_{(f)} \longrightarrow T_{(\varphi(f))}.$$

Proof. On $D_+(\varphi(f))$, contraction lands in $D_+(f)$. Localization gives a graded map $S_f \to T_{\varphi(f)}$, and taking degree-zero parts gives $S_{(f)} \to T_{(\varphi(f))}$. Hence VI/35 produces an affine morphism of the corresponding standard opens. If both f and g are nonvanishing, localization in stages shows that the two chart maps agree on $D_+(\varphi(fg))$. They therefore glue. Maximality follows because any point outside U_φ contracts to a prime containing S_+. $\square$

Corollary 38.3 (A useful global criterion)**.** *If*

$$T_+ \subseteq \sqrt{\varphi(S_+)T},$$

then $U_\varphi = \operatorname{Proj} T$, so φ induces a morphism on all of $\operatorname{Proj} T$. For finitely generated standard graded algebras over a field, this is equivalent to the absence of projective points containing $\varphi(S_+)T$.

Remark 38.4 (Why arbitrary graded maps fail). There is no contradiction with the contravariance of Spec. Spec uses every prime, while Proj deliberately discards primes containing the irrelevant ideal. Contraction can move a surviving projective prime into the discarded locus.

38.2 Maps defined by homogeneous forms and base loci

Let $S = k[y_0, \dots, y_m]$ and $T = k[x_0, \dots, x_n]$. Suppose $F_0, \dots, F_m \in T_d$ are homogeneous of one positive degree d. The assignment

$$y_i \longmapsto F_i$$

is degree preserving as a map

$$S \longrightarrow T^{(d)} := \bigoplus_{r \geq 0} T_{rd},$$

where $T^{(d)}$ is regraded so that T_{rd} has degree r. By Veronese invariance from VI/35,

$$\operatorname{Proj} T^{(d)} \cong \operatorname{Proj} T = \mathbb{P}^n_k.$$

Definition 38.5 (Base locus)**.** The base locus of $F_0, \dots, F_m$ is

$$B = V_+(F_0, \dots, F_m) \subseteq \mathbb{P}^n_k.$$

On $\mathbb{P}^n_k \setminus B$, the forms define a morphism

$$[x] \longmapsto [F_0(x) : \cdots : F_m(x)].$$

If $B = \varnothing$, the morphism is everywhere defined.

The empty-base-locus condition is exactly the Proj-domain condition for the induced map $S \to T^{(d)}$. Thus the familiar coordinate formula is not an extra construction: it is the graded-ring theorem above after the correct regrading.

Example 38.6 (Linear embedding)**.** The map $\mathbb{P}^m \hookrightarrow \mathbb{P}^n$ given by $[x_0 : \cdots : x_m] \mapsto [x_0 : \cdots : x_m : 0 : \cdots : 0]$ is a closed embedding cut out by the final coordinate functions.

$$\begin{tikzcd}
\boxed{\begin{array}{c} F_0,\dots,F_m \\ \text{same degree } d \end{array}} \arrow[r] \arrow[d] & \boxed{\begin{array}{c} B = V_+(F_0,\dots,F_m) \\ \text{base locus} \end{array}} \\
\boxed{\begin{array}{c} \mathbb{P}^n \setminus B \\ \longrightarrow \mathbb{P}^m \end{array}} & \boxed{k[y_0,\dots,y_m] \to k[x]^{(d)}} \arrow[l]
\end{tikzcd}$$

Figure 38.2: Same-degree homogeneous forms define a projective morphism precisely away from their common projective zero locus.

38.3 Closed immersions

Definition 38.7 (Closed immersion)**.** A morphism $i : X \to Y$ is a closed immersion if it identifies $|X|$ homeomorphically with a closed subset of $|Y|$ and the map of sheaves

$$\mathcal{O}_Y \longrightarrow i_*\mathcal{O}_X$$

is surjective.

Theorem 38.8 (Affine-local criterion)**.** *A morphism $i : X \to Y$ is a closed immersion if and only if every point of Y has an affine open neighborhood $V = \operatorname{Spec} A$ such that*

$$i^{-1}(V) = \operatorname{Spec} B$$

for a surjective ring map $A \twoheadrightarrow B$.

Proof. If i is a closed immersion with ideal sheaf $\mathcal{I}$, then on an affine open $V = \operatorname{Spec} A$ the restriction is cut out by the ideal $I = \Gamma(V, \mathcal{I})$, hence is $\operatorname{Spec}(A/I)$. Conversely, affine closed immersions glue because the defining quotient maps agree under localization; the resulting map is a homeomorphism onto a closed subspace and the structure-sheaf map is locally surjective. ☐

Closed immersions are stable under composition and arbitrary base change. They also preserve scheme-theoretic nilpotents: a double hyperplane embeds just as genuinely as its reduced support.

38.4 Homogeneous quotients give closed immersions

Let $\varphi : S \twoheadrightarrow T = S/I$ be a surjective degree-preserving map with homogeneous kernel I. Then $T_+ = \varphi(S_+)$, so the induced Proj morphism is defined everywhere.

Theorem 38.9 (Quotient closed immersion)**.** *The morphism*

$$\operatorname{Proj}(S/I) \longrightarrow \operatorname{Proj} S$$

induced by the quotient map is a closed immersion. On $D_+(f)$ it is induced by the surjection

$$S_{(f)} \twoheadrightarrow (S/I)_{(\bar{f})} \cong S_{(f)}/I_{(f)}.$$

Its image is the projective closed locus $V_+(I)$.

Proof. VI/37 identifies the chart of $\operatorname{Proj}(S/I)$ with $\operatorname{Spec}(S_{(f)}/I_{(f)})$. The chart map is therefore an affine closed immersion. These local closed immersions glue by theorem 38.8, and the underlying image is $V_+(I)$. ☐

$$\begin{tikzcd}
\operatorname{Proj}(S/I) \arrow[r, hook, "i"] \arrow[d, "\cong"'] & \operatorname{Proj} S \arrow[d] \\
\operatorname{Spec}(S_{(f)}/I_{(f)}) \arrow[r, hook] & \operatorname{Spec} S_{(f)}
\end{tikzcd}$$

Figure 38.3: The quotient closed immersion is affine-locally the spectrum of a surjective degree-zero localization map.

38.5 Relative projective space and projective morphisms

For any scheme Y, define

$$\mathbb{P}^n_Y := \mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec}\mathbb{Z}} Y.$$

Equivalently, it is obtained by gluing $n+1$ copies of $\mathbb{A}^n_Y$ with the same transition maps as ordinary projective space. The projection is denoted

$$\pi : \mathbb{P}^n_Y \longrightarrow Y.$$

Definition 38.10 (Projective morphism)**.** A morphism $f : X \to Y$ is projective if there exist an integer $n \geq 0$ and a factorization

$$X \xrightarrow{i} \mathbb{P}^n_Y \xrightarrow{\pi} Y$$

with i a closed immersion. All applications in this volume are in finite-type/Noetherian settings, so this agrees with the usual finite-type formulation used in algebraic geometry.

$$\begin{tikzcd}
\boxed{X} \arrow[r, "\text{closed immersion}"] \arrow[rr, bend right, "f"'] & \boxed{\mathbb{P}^n_Y} \arrow[r, "\pi"] & \boxed{Y}
\end{tikzcd}$$

Figure 38.4: A projective morphism is one that factors through a closed subscheme of relative projective space.

A closed immersion is projective: take $n = 0$, since $\mathbb{P}^0_Y \cong Y$. The structural morphism $\mathbb{P}^n_Y \to Y$ is projective by definition.

Theorem 38.11 (Base change stability)**.** *If $f : X \to Y$ is projective and $Y' \to Y$ is any morphism, then the base change*

$$f' : X \times_Y Y' \longrightarrow Y'$$

is projective.

Proof. Choose a closed immersion $X \hookrightarrow \mathbb{P}^n_Y$. Base change preserves closed immersions, and

$$\mathbb{P}^n_Y \times_Y Y' \cong \mathbb{P}^n_{Y'}.$$

Thus $X \times_Y Y'$ is a closed subscheme of $\mathbb{P}^n_{Y'}$. $\square$

38.6 Very ample invertible sheaves

Let $f : X \to Y$. In the projective setting considered here, we use the closed-embedding form of relative very ampleness: an invertible sheaf $\mathcal{L}$ on X is *very ample relative to Y* if there exists a closed immersion

$$i : X \hookrightarrow \mathbb{P}^n_Y$$

with

$$\mathcal{L} \cong i^*\mathcal{O}_{\mathbb{P}^n_Y}(1).$$

Thus projectivity can be encoded by an embedding together with the pullback of the universal twisting sheaf. Over a field, a choice of generating sections of a very ample line bundle produces homogeneous coordinates. The divisor and Picard chapters will develop the intrinsic line-bundle side further.

Example 38.12 (Quadratic Veronese embedding)**.** The map $[s : t] \mapsto [s^2 : st : t^2]$ embeds $\mathbb{P}^1$ as a smooth conic in $\mathbb{P}^2$. The homogeneous coordinate functions all have the same degree.

38.7 The Veronese embedding

Fix $d \geq 1$ and let

$$S = k[x_0, \dots, x_n], \qquad S^{(d)} = \bigoplus_{r \geq 0} S_{rd}.$$

Let $\mathcal{M}_d$ be the set of monomials $x^\alpha = x_0^{\alpha_0} \cdots x_n^{\alpha_n}$ of degree $|\alpha| = d$. Its cardinality is

$$N + 1 = \binom{n+d}{d}.$$

Introduce a polynomial ring

$$R = k[y_\alpha : |\alpha| = d].$$

Theorem 38.13 (Veronese closed embedding)**.** *The map*

$$\nu_d : \mathbb{P}^n_k \longrightarrow \mathbb{P}^N_k, \qquad [x_0 : \cdots : x_n] \longmapsto [x^\alpha]_{|\alpha|=d},$$

is a closed immersion.

Proof. Define a degree-preserving map

$$\psi : R \longrightarrow S^{(d)}, \qquad y_\alpha \longmapsto x^\alpha.$$

It is surjective: every monomial of degree rd can be written as a product of r monomials of degree d, so $S^{(d)}$ is generated in degree one by S_d. Therefore theorem 38.9 gives a closed immersion

$$\operatorname{Proj} S^{(d)} \hookrightarrow \operatorname{Proj} R = \mathbb{P}^N_k.$$

By Veronese invariance from VI/35, the canonical map $\operatorname{Proj} S \cong \operatorname{Proj} S^{(d)}$ is an isomorphism. Their composite is a closed immersion. On K-valued points it is exactly the displayed coordinate formula because the degree-one generators of $S^{(d)}$ are the degree-d monomials in the x_i. $\square$

For $n = 1, d = 2$, this gives the conic

$$[s : t] \longmapsto [s^2 : st : t^2]$$

with homogeneous ideal $(Y_0Y_2 - Y_1^2)$. For $n = 1$ and general d, the image is the rational normal curve of degree d in $\mathbb{P}^d$.

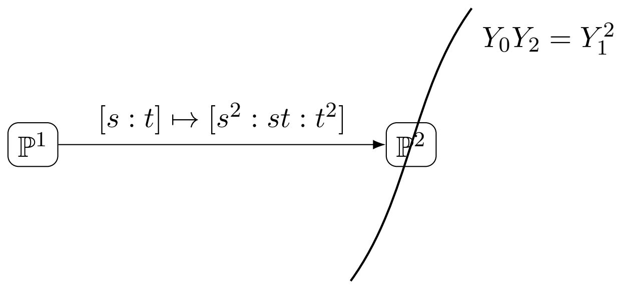


Figure 38.5: The quadratic Veronese map is a closed immersion because its coordinate ring is a quotient onto the second Veronese subring.

38.8 The Segre ring and product charts

Set

$$A = k[x_0, \ldots, x_m], \qquad B = k[y_0, \ldots, y_n].$$

Their Segre product is the standard graded ring

$$A\#B := \bigoplus_{d \geq 0} A_d \otimes_k B_d.$$

It is generated in degree one by the elements $x_i \otimes y_j$.

Proposition 38.14 (Proj of the Segre product)**.** *There is a canonical isomorphism*

$$\operatorname{Proj}(A\#B) \cong \mathbb{P}^m_k \times_k \mathbb{P}^n_k.$$

On the standard open defined by $x_i \otimes y_j$, the degree-zero ring is

$$(A\#B)_{(x_i \otimes y_j)} \cong A_{(x_i)} \otimes_k B_{(y_j)},$$

so the chart is $U_i \times V_j \cong \mathbb{A}^m \times \mathbb{A}^n$.

Proof. In the localization, the ratios

$$\frac{x_a \otimes y_j}{x_i \otimes y_j} = \frac{x_a}{x_i}, \qquad \frac{x_i \otimes y_b}{x_i \otimes y_j} = \frac{y_b}{y_j}$$

generate the degree-zero part. Hence the displayed chart ring is the tensor product of the two affine chart rings. These chart identifications agree on overlaps, because all transition maps are obtained by further localization. The charts cover both sides. □

38.9 The Segre embedding

Let

$$R = k[z_{ij} : 0 \leq i \leq m,\ 0 \leq j \leq n].$$

Define

$$\theta : R \longrightarrow A\#B, \qquad z_{ij} \longmapsto x_i \otimes y_j.$$

This map is surjective because the Segre product is generated by its degree-one elements.

Theorem 38.15 (Segre closed embedding and equations)**.** *The Segre map*

$$\sigma : \mathbb{P}^m_k \times \mathbb{P}^n_k \longrightarrow \mathbb{P}^{(m+1)(n+1)-1}_k, \qquad ([x_i],[y_j]) \longmapsto [x_i y_j]_{i,j},$$

is a closed immersion. Its homogeneous ideal is generated by the 2×2 *minors*

$$z_{ij} z_{\ell r} - z_{ir} z_{\ell j}.$$

Equivalently, the image consists scheme-theoretically of rank-one matrices (z_{ij})*.*

Proof. The surjection $\theta : R \twoheadrightarrow A\#B$ gives a closed immersion

$$\operatorname{Proj}(A\#B) \hookrightarrow \operatorname{Proj} R.$$

By theorem 38.14, the source is $\mathbb{P}^m \times \mathbb{P}^n$, and on points the map has coordinates $z_{ij} = x_i y_j$. It remains to identify the kernel.

Every 2×2 minor maps to zero, so the ideal J generated by the minors lies in $\ker\theta$. Conversely, a monomial $z_{i_1 j_1} \cdots z_{i_d j_d}$ maps to

$$(x_{i_1} \cdots x_{i_d}) \otimes (y_{j_1} \cdots y_{j_d}),$$

which depends only on the multiset of row indices and the multiset of column indices. The minor relations permit the swap

$$z_{ij} z_{\ell r} \equiv z_{ir} z_{\ell j} \pmod J,$$

so any two monomials with the same row and column multisets are congruent modulo J. Group a polynomial in $\ker\theta$ by its image monomial in $A\#B$; in each group the coefficients sum to zero, and the group is therefore a linear combination of differences of monomials with the same image. Each such difference lies in J. Hence $\ker\theta = J$, proving the equation statement and the closed-embedding theorem. ☐

$$\begin{tikzcd} \boxed{\begin{matrix} \mathbb{P}^m \times \mathbb{P}^n \\ ([x_i],[y_j]) \end{matrix}} \arrow[r, "\text{Segre}"] & \boxed{\begin{matrix} \text{matrix } (z_{ij}) \\ z_{ij} = x_i y_j \end{matrix}} \arrow[r] & \boxed{\begin{matrix} \text{rank one} \\ \text{all } 2 \times 2 \text{ minors vanish} \end{matrix}} \end{tikzcd}$$

Figure 38.6: The Segre embedding realizes a product of projective spaces as the determinantal rank-one locus.

The smallest nontrivial case is

$$\mathbb{P}^1 \times \mathbb{P}^1 \hookrightarrow \mathbb{P}^3, \qquad ([s:t],[u:v]) \longmapsto [su : sv : tu : tv],$$

whose image is the smooth quadric $Z_{00}Z_{11} - Z_{01}Z_{10} = 0$.

Example 38.16 (Graph of a projective morphism)**.** A morphism to projective space can often be encoded by a line bundle together with globally generated sections. The ratios of those sections on trivializing opens recover the usual coordinate formula.

38.10 Composition of projective morphisms

The Segre embedding is exactly what is needed to prove that projective morphisms compose.

Theorem 38.17 (Projective morphisms compose)**.** *If* $f : X \to Y$ *and* $g : Y \to Z$ *are projective, then* $g \circ f : X \to Z$ *is projective.*

Proof. Choose closed immersions over the bases

$$X \hookrightarrow \mathbb{P}^m_Y, \qquad Y \hookrightarrow \mathbb{P}^n_Z.$$

After base change along $Y \to Z$, the first ambient space is

$$\mathbb{P}^m_Y \cong \mathbb{P}^m_Z \times_Z Y,$$

which is a closed subscheme of

$$\mathbb{P}^m_Z \times_Z \mathbb{P}^n_Z.$$

The relative Segre embedding, obtained from theorem 38.15 by base change, is a closed immersion of this product into $\mathbb{P}^N_Z$, where $N = (m+1)(n+1) - 1$. Composing the closed immersions embeds X into $\mathbb{P}^N_Z$, proving that $X \to Z$ is projective. □

$$\begin{tikzcd}
X \arrow[r] & \mathbb{P}^m_Y \arrow[r] & \mathbb{P}^m_Z \times_Z \mathbb{P}^n_Z \arrow[d, "\text{Segre}"] \\
 & & \mathbb{P}^N_Z
\end{tikzcd}$$

Figure 38.7: Composition of projective morphisms is reduced to one closed immersion by the relative Segre embedding.

38.11 Consequences and geometric diagnostics

Projective morphisms are stable under arbitrary base change and composition. In the finite-type situations used throughout this volume they are finite type and separated. The separatedness can be seen from the fact that $\mathbb{P}^n_Y \to Y$ is separated and closed immersions are separated; the property is inherited by composites.

For maps to projective space, a practical diagnostic is therefore:

1. choose homogeneous forms of one degree;
2. compute the base locus;
3. if the base locus is empty, obtain a morphism;
4. if the induced graded map is surjective after the appropriate Veronese presentation, obtain a closed immersion;
5. determine equations of the image from the kernel of the coordinate-ring map.

38.12 Common pitfalls

- A graded map $S \to T$ does not automatically give a morphism $\operatorname{Proj} T \to \operatorname{Proj} S$ everywhere; first check the irrelevant-ideal condition.
- Homogeneous forms defining a coordinate formula must have one common degree. Different degrees require regrading or a different target construction.

$$
\begin{tikzcd}
\text{same-degree forms} \arrow[r] & \text{base locus empty?} \arrow[r] & \text{morphism} \arrow[d] \\
\text{closed embedding} & \text{surjective?} \arrow[l] & \text{coordinate-ring map} \arrow[l]
\end{tikzcd}
$$

Figure 38.8: A practical workflow for turning homogeneous coordinate formulas into scheme-theoretic projective morphisms and embeddings.

- A common projective zero of all coordinate forms is a base point; the formula is not a morphism there.
- Injectivity on k-points is much weaker than being a closed immersion. A closed immersion is a scheme-theoretic statement about topology and structure sheaves.
- A surjective homogeneous ring map gives a closed immersion in the opposite direction on Proj.
- The Veronese proof depends on the surjection onto the Veronese subring, not merely on the coordinate formula being injective.
- The Segre image is defined scheme-theoretically by all 2×2 minors, not merely by the set-theoretic rank-one condition.
- Projectivity is relative: a projective morphism $X \to Y$ means an embedding into some $\mathbb{P}^n_Y$, not merely that X is abstractly a projective scheme over a field.
- Composition of projective morphisms is not immediate from the definition; the Segre embedding supplies the required single projective ambient space.

38.13 Boundaries of VI/38

VI/38 completes the projective-geometry arc: functoriality of Proj, projective morphisms, closed immersions, very ampleness, and the Veronese/Segre embedding theorems. The next arc turns to divisors and line bundles in intrinsic form. Weil divisors, Cartier divisors, class groups, Picard groups, and cohomological ampleness criteria remain later chapters.

38.14 Worked examples

Example 38.18 (A graded map defined only on one chart)**.** Let $S = k[x, y]$, $T = k[u, v]$, and $x \mapsto u,\ y \mapsto 0$. The point $(u) \in \operatorname{Proj} T$ contracts to $(x, y) = S_+$, so no image point exists there. The induced map is defined on $D_+(u)$, not on all of $\mathbb{P}^1$.

Example 38.19 (A homogeneous quotient)**.** The map $k[X, Y, Z] \to k[s, t]$, $X \mapsto s^2$, $Y \mapsto st$, $Z \mapsto t^2$, has image exactly the second Veronese subring $k[s, t]^{(2)}$. Viewed as $k[X, Y, Z] \twoheadrightarrow k[s, t]^{(2)}$, it is a surjective degree-preserving map and gives the conic embedding.

Example 38.20 (Linear change of coordinates)**.** A matrix in $\mathrm{GL}_{n+1}(k)$ acts on the variables and extends to a graded automorphism of the polynomial ring, hence induces an automorphism of $\mathbb{P}^n$.

Example 38.21 (A morphism from two quadratic forms)**.** The forms s^2, t^2 have no common zero on $\mathbb{P}^1$, so $[s : t] \mapsto [s^2 : t^2]$ defines a morphism $\mathbb{P}^1 \to \mathbb{P}^1$.

Example 38.22 (A base point)**.** The forms x_0x_1, x_1^2 vanish simultaneously at $[1:0]$, so the formula $[x_0 : x_1] \mapsto [x_0x_1 : x_1^2]$ is undefined there. On $D_+(x_1)$ it reduces to $[x_0 : x_1] \mapsto [x_0 : x_1]$.

Example 38.23 (A hyperplane closed immersion)**.** The quotient $k[x_0, \dots, x_n] \twoheadrightarrow k[x_0, \dots, x_n]/(x_n)$ gives $\mathbb{P}^{n-1} \hookrightarrow \mathbb{P}^n$ as the hyperplane $x_n = 0$.

Example 38.24 (A nonreduced closed immersion)**.** The quotient by (x_n^2) gives a double hyperplane embedded as a closed subscheme of $\mathbb{P}^n$. Closed immersion does not imply reduced image.

Example 38.25 (Projective space over a base)**.** For any scheme Y, the projection $\mathbb{P}^n_Y \to Y$ is projective. Its fiber over $y \in Y$ is $\mathbb{P}^n_{\kappa(y)}$.

Example 38.26 (Base change of a projective family)**.** If $X \hookrightarrow \mathbb{P}^n_Y$ is cut out by homogeneous equations with coefficients on Y, then after $Y' \to Y$ the same equations with base-changed coefficients cut out $X \times_Y Y' \subseteq \mathbb{P}^n_{Y'}$.

Example 38.27 (The quadratic Veronese conic)**.** The map $[s : t] \mapsto [s^2 : st : t^2]$ is a closed immersion of $\mathbb{P}^1$ into $\mathbb{P}^2$, with image $Y_0Y_2 - Y_1^2 = 0$.

Example 38.28 (The cubic rational normal curve)**.** The third Veronese embedding sends $[s : t]$ to $[s^3 : s^2t : st^2 : t^3] \in \mathbb{P}^3$. Its ideal contains the 2×2 minors of $\begin{pmatrix} Y_0 & Y_1 & Y_2 \\ Y_1 & Y_2 & Y_3 \end{pmatrix}$.

Example 38.29 (The quadric surface)**.** The Segre map $\mathbb{P}^1 \times \mathbb{P}^1 \to \mathbb{P}^3$ has image $Z_{00}Z_{11} - Z_{01}Z_{10} = 0$, identifying the product with a smooth projective quadric surface.

Example 38.30 ($\mathbb{P}^1 \times \mathbb{P}^2$)**.** The Segre image in $\mathbb{P}^5$ is defined by the 2×2 minors of a 2×3 matrix (Z_{ij}).

Example 38.31 (A closed point)**.** A k-rational point $[1 : 0 : \cdots : 0]$ is the closed subscheme cut out by $(x_1, \dots, x_n)$, hence its inclusion into $\mathbb{P}^n$ is a closed immersion.

Example 38.32 (Composition via Segre)**.** If $X \subseteq \mathbb{P}^m_Y$ and $Y \subseteq \mathbb{P}^n_Z$, then X embeds into $\mathbb{P}^m_Z \times_Z \mathbb{P}^n_Z$ and hence, by relative Segre, into one projective space over Z.

Example 38.33 (A very ample twist)**.** On $\mathbb{P}^n_k$, $\mathcal{O}(1)$ is very ample by the identity embedding. The line bundle $\mathcal{O}(d)$ for $d \geq 1$ is very ample because the d-uple Veronese embedding pulls back $\mathcal{O}_{\mathbb{P}^N}(1)$ to $\mathcal{O}_{\mathbb{P}^n}(d)$.

38.15 Exercises

Exercise 38.34 (Domain of definition)**.** For a degree-preserving map $\varphi : S \to T$, prove that $U_\varphi = \bigcup_{f \in S_+} D_+(\varphi(f))$ is open in $\operatorname{Proj} T$.

Exercise 38.35 (Bad contraction)**.** For $k[x, y] \to k[u, v]$ with $x \mapsto u, y \mapsto 0$, identify explicitly the points where the Proj map is undefined.

Exercise 38.36 (Global radical criterion)**.** Prove that $T_+ \subseteq \sqrt{\varphi(S_+)T}$ implies the induced Proj morphism is defined everywhere.

Exercise 38.37 (Positive-degree regrading)**.** Show that a map satisfying $\varphi(S_r) \subseteq T_{dr}$ becomes degree preserving as $S \to T^{(d)}$.

Exercise 38.38 (Base locus of forms)**.** Show that same-degree forms $F_0, \dots, F_m$ define a morphism precisely on the complement of $V_+(F_0, \dots, F_m)$.

Exercise 38.39 (Affine criterion for closed immersion)**.** Verify directly that a surjection $A \twoheadrightarrow B$ induces a closed immersion $\operatorname{Spec} B \hookrightarrow \operatorname{Spec} A$.

Exercise 38.40 (Quotient chart)**.** For homogeneous $I \subset S$, show that $S_{(f)} \twoheadrightarrow (S/I)_{(\bar{f})}$ is surjective with kernel $I_{(f)}$.

Exercise 38.41 (Hyperplane inclusion)**.** Recover the standard inclusion $\mathbb{P}^{n-1} \hookrightarrow \mathbb{P}^n$ from a homogeneous quotient.

Exercise 38.42 (Closed immersions base change)**.** Prove affine-locally that closed immersions are stable under arbitrary base change.

Exercise 38.43 (Projective base change)**.** Derive base-change stability of projective morphisms from the previous exercise.

Exercise 38.44 (Quadratic Veronese)**.** Compute the kernel of $k[Y_0, Y_1, Y_2] \to k[s,t]^{(2)}$ and recover the conic equation.

Exercise 38.45 (Veronese generation)**.** Prove that every monomial of degree rd is a product of r monomials of degree d.

Exercise 38.46 (Veronese dimension)**.** For $n = 2, d = 2$, compute the target dimension N and list all degree-two coordinates.

Exercise 38.47 (Segre chart)**.** Compute the degree-zero ring on $D_+(z_{ij})$ of the Segre product and identify it with an affine product chart.

Exercise 38.48 (Segre minors vanish)**.** Check directly that every 2×2 minor vanishes under $z_{ij} \mapsto x_i y_j$.

Exercise 38.49 (Segre kernel)**.** Complete the monomial-swap proof that the 2×2 minors generate the kernel of the Segre coordinate map.

Exercise 38.50 (Quadric surface)**.** Show that the Segre image of $\mathbb{P}^1 \times \mathbb{P}^1$ is exactly the quadric $Z_{00}Z_{11} - Z_{01}Z_{10} = 0$.

Exercise 38.51 (Relative Segre)**.** Explain why the Segre equations commute with base change from $\mathbb{Z}$ to an arbitrary scheme Y.

Exercise 38.52 (Composition theorem)**.** Write out the chain of closed immersions used to prove the composition of two projective morphisms is projective.

Exercise 38.53 (Projectivity of a closed immersion)**.** Show that every closed immersion $X \hookrightarrow Y$ is projective using $\mathbb{P}^0_Y \cong Y$.

Exercise 38.54 (Very ample pullback)**.** For the Veronese embedding ν_d, prove $\nu_d^* \mathcal{O}(1) \cong \mathcal{O}(d)$ from transition functions.

Exercise 38.55 (Linear coordinate change)**.** Show that $\mathrm{PGL}_{n+1}(k)$ acts on $\mathbb{P}^n_k$ through graded automorphisms of the polynomial ring.

Exercise 38.56 (A rational but not global formula)**.** Give homogeneous forms with a nonempty base locus and describe the resulting rational map on its maximal domain.

Exercise 38.57 (Injective points versus closed immersion)**.** Explain why checking injectivity on k-points alone cannot prove that a morphism is a closed immersion.

38.16 Problem dossiers

Problem 38.58 (VI.38.P1: Domain theorem for Proj)**.** Let $\varphi : S \to T$ be degree preserving. Prove carefully that the standard-open maps $\operatorname{Spec} T_{(\varphi(f))} \to \operatorname{Spec} S_{(f)}$ glue on U_φ.

Problem 38.59 (VI.38.P2: A map with a base point)**.** Analyze the map of graded rings $k[a, b] \to k[x, y]^{(2)}$ given by $a \mapsto x^2$, $b \mapsto xy$. Determine the induced projective domain and the pointwise formula.

Problem 38.60 (VI.38.P3: Quotient maps and closed immersion)**.** Let $S \twoheadrightarrow T$ be a surjective map of standard graded k-algebras. Prove the induced morphism $\operatorname{Proj} T \to \operatorname{Proj} S$ is a closed immersion without appealing to the global quotient theorem of VI/37.

Problem 38.61 (VI.38.P4: Morphism from same-degree forms)**.** Let $F_0, \dots, F_m \in k[x_0, \dots, x_n]_d$. Prove that if their projective common zero locus is empty, they define a morphism $\mathbb{P}^n \to \mathbb{P}^m$.

Problem 38.62 (VI.38.P5: Relative projectivity survives base change)**.** Let $X \to Y$ be projective and $Y' \to Y$ arbitrary. Prove directly from a chosen projective embedding that $X_{Y'} \to Y'$ is projective.

Problem 38.63 (VI.38.P6: Quadratic Veronese embedding)**.** Prove from graded rings that $\nu_2 : \mathbb{P}^1 \to \mathbb{P}^2$, $[s : t] \mapsto [s^2 : st : t^2]$, is a closed immersion and compute its ideal.

Problem 38.64 (VI.38.P7: General Veronese embedding)**.** Give the complete ring-theoretic proof that the d-uple map $\mathbb{P}^n \to \mathbb{P}^N$ is a closed immersion.

Problem 38.65 (VI.38.P8: Proj of a Segre product)**.** Prove $\operatorname{Proj}(A\#B) \cong \operatorname{Proj} A \times \operatorname{Proj} B$ for polynomial rings $A = k[x_0, \dots, x_m]$, $B = k[y_0, \dots, y_n]$.

Problem 38.66 (VI.38.P9: Segre ideal)**.** Prove that the kernel of $k[z_{ij}] \to A\#B$, $z_{ij} \mapsto x_i \otimes y_j$, is generated by the 2×2 minors.

Problem 38.67 (VI.38.P10: Segre quadric)**.** Identify $\mathbb{P}^1 \times \mathbb{P}^1$ with a closed quadric in $\mathbb{P}^3$ and compute its four standard product charts.

Problem 38.68 (VI.38.P11: Composition of projective morphisms)**.** Prove projective morphisms are closed under composition, explicitly identifying the relative projective ambient dimension.

Problem 38.69 (VI.38.P12: Very ample twists on projective space)**.** Show that $\mathcal{O}_{\mathbb{P}^n}(d)$ is very ample for every $d \geq 1$.

38.17 Challenges

Problem 38.70 (Challenge 1: Universal base-locus criterion)**.** Let S be a finitely generated standard graded ring and $F_0, \dots, F_m$ homogeneous of one degree. Formulate and prove a criterion for the associated map to $\mathbb{P}^m$ to be everywhere defined using radicals of homogeneous ideals rather than pointwise coordinates.

Problem 38.71 (Challenge 2: Equations of the rational normal curve)**.** For the degree-d Veronese embedding of $\mathbb{P}^1$ into $\mathbb{P}^d$, prove that the homogeneous ideal is generated by the 2×2 minors of a catalecticant matrix.

Problem 38.72 (Challenge 3: Segre-Veronese embedding)**.** Combine the Segre and Veronese constructions to embed $\mathbb{P}^m \times \mathbb{P}^n$ using bihomogeneous forms of bidegree (a, b). Identify the corresponding very ample line bundle.

Problem 38.73 (Challenge 4: Projective graphs)**.** Let $f : X \to \mathbb{P}^n_Y$ be a Y-morphism. Prove that its graph $X \to X \times_Y \mathbb{P}^n_Y$ is a closed immersion because $\mathbb{P}^n_Y \to Y$ is separated, and investigate how graph embeddings can be combined with projective embeddings of $X \to Y$.

Problem 38.74 (Challenge 5: Closed subschemes and saturated ideals)**.** For $X \subseteq \mathbb{P}^n_k$, relate the ideal sheaf of the closed immersion to the saturated homogeneous ideal $\bigoplus_d H^0(\mathbb{P}^n, \mathcal{I}_X(d))$. Identify precisely which parts require the later coherent-sheaf or cohomological theory.

Chapter synthesis

The functorial behavior of Proj is controlled by the irrelevant ideal. Once the correct domain is identified, graded ring maps become ordinary affine morphisms on standard charts and glue globally. Surjective graded maps give closed immersions, same-degree homogeneous forms give projective coordinate maps away from their base locus, and relative projective space packages these constructions over arbitrary bases. The Veronese embedding is the quotient map onto a Veronese subring viewed contravariantly, while the Segre embedding is the quotient map onto the Segre product; its determinantal equations make the product geometry explicit. Together, base change and Segre show that projective morphisms form a robust geometric class closed under composition.

38.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 38.75 (Linear image)**.** What equations cut out the standard $\mathbb{P}^m \subset \mathbb{P}^n$?

Hint. Use projective morphisms, homogeneous coordinates, and closed embeddings. □

Solution. $X_{m+1} = \cdots = X_n = 0$. □

Exercise 38.76 (Veronese degree)**.** What common degree do the quadratic Veronese coordinates have?

Hint. Use projective morphisms, homogeneous coordinates, and closed embeddings. □

Solution. 2. □

Exercise 38.77 (Conic image)**.** What equation cuts out the quadratic Veronese image?

Hint. Use projective morphisms, homogeneous coordinates, and closed embeddings. □

Solution. $Y_0Y_2 - Y_1^2 = 0$. □

Exercise 38.78 (Embedding criterion)**.** What must a closed embedding do topologically?

Hint. Use projective morphisms, homogeneous coordinates, and closed embeddings. □

Solution. Identify the source homeomorphically with a closed subset. □

Exercise 38.79 (Sheaf condition)**.** What must happen to structure sheaves for a closed immersion?

Hint. Use projective morphisms, homogeneous coordinates, and closed embeddings. ☐

Solution. The target structure sheaf must surject onto the pushforward source sheaf locally. ☐

Proofs

Exercise 38.80 (Linear closed embedding)**.** Prove: Prove the standard coordinate inclusion $\mathbb{P}^m \to \mathbb{P}^n$ is a closed immersion.

Hint. Work from projective morphisms, homogeneous coordinates, and closed embeddings and state the hypotheses explicitly. ☐

Solution. Check it on standard affine charts or use the homogeneous quotient by the final variables. ☐

Exercise 38.81 (Veronese injectivity)**.** Prove: Prove the quadratic Veronese map on $\mathbb{P}^1$ is injective on points.

Hint. Work from projective morphisms, homogeneous coordinates, and closed embeddings and state the hypotheses explicitly. ☐

Solution. Recover $[s : t]$ from any nonzero pair of quadratic coordinates or use ratios on standard charts. ☐

Exercise 38.82 (Veronese immersion)**.** Prove: Prove the quadratic Veronese identifies $\mathbb{P}^1$ with the conic.

Hint. Work from projective morphisms, homogeneous coordinates, and closed embeddings and state the hypotheses explicitly. ☐

Solution. Construct inverse chart maps on the conic and verify compatibility. ☐

Exercise 38.83 (Composition)**.** Prove: Prove a composition of closed immersions is a closed immersion.

Hint. Work from projective morphisms, homogeneous coordinates, and closed embeddings and state the hypotheses explicitly. ☐

Solution. Compose the closed topological embeddings and the surjective maps of sheaves. ☐

Counterexamples and hypothesis tests

Exercise 38.84 (Same degree)**.** Test the claim: homogeneous coordinate formulas of different degrees define a projective morphism without adjustment.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; scaling would affect coordinates incompatibly. ☐

Exercise 38.85 (Injective enough)**.** Test the claim: every injective morphism is a closed embedding.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; topology and sheaf structure must also match. □

Exercise 38.86 (Projective affine)**.** Test the claim: projective morphism means the source is projective over the ground field only.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. □

Solution. False; projectivity is relative to a base. □

Applications and investigations

Exercise 38.87 (Linear systems)**.** How do sections of a line bundle produce projective maps?

Hint. Translate the question into projective morphisms, homogeneous coordinates, and closed embeddings. □

Solution. A basepoint-free collection of sections gives homogeneous coordinates locally and hence a morphism to projective space. □

Exercise 38.88 (Embedding design)**.** Why are Veronese embeddings useful?

Hint. Translate the question into projective morphisms, homogeneous coordinates, and closed embeddings. □

Solution. They convert higher-degree forms into linear hyperplane sections in a larger projective space. □

Challenge problems

Exercise 38.89 (Synthesis challenge)**.** Connect two central viewpoints of Projective Morphisms and Closed Embeddings in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. □

Solution. A complete solution makes one concrete computation in Projective Morphisms and Closed Embeddings and then translates it through projective morphisms, homogeneous coordinates, and closed embeddings, checking both descriptions agree. □

Exercise 38.90 (Hypothesis challenge)**.** Choose one theorem-level statement from Projective Morphisms and Closed Embeddings, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. □

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Part VIII

Divisors and Birational Geometry

Chapter 39

Weil Divisors

The projective chapters supplied a global language for schemes built from graded rings. The next layer of geometry asks a different question: how can codimension-one subvarieties record zeros and poles of rational functions? Weil divisors answer this by turning codimension-one geometry into an abelian group.

The cleanest valuation-theoretic theory is obtained on a normal Noetherian integral scheme. The definition of the free abelian group of codimension-one cycles makes sense more generally, but normality is the hypothesis that turns every codimension-one local ring into a discrete valuation ring. That distinction will be kept explicit throughout the chapter.

The migration ledger assigns eight rules to VI/39. The primary source is the AG9 block selected by “Weil divisor|prime divisor” together with its theorem and exercise descendants. A later AG10 file contains a broad divisor/Picard/line-bundle overlap and is marked compare-and-migrate rather than canonical. Accordingly, this chapter uses the ledger as provenance and scope evidence, retains Weil-divisor material, and leaves Cartier divisors, divisor class groups, and Picard theory to VI/40–VI/42.

Learning goals

By the end of the chapter, the reader should be able to:

- identify prime divisors as codimension-one integral closed subschemes;
- form the Weil divisor group $\mathrm{Div}(X)$ and manipulate effective, positive, and negative parts;
- explain why normality makes codimension-one local rings discrete valuation rings;
- compute $\mathrm{ord}_Z(f)$ for rational functions along prime divisors;
- prove that principal divisors have finite support and that $f \mapsto \mathrm{div}(f)$ is a homomorphism;
- compute principal divisors on affine space, projective space, and normal curves;
- restrict Weil divisors to open subsets and understand the loss of boundary components;
- recognize the failure of naive valuation language on nonnormal schemes;
- distinguish what is functorial for principal divisors from the non-functoriality of arbitrary Weil divisors;
- state the first bridge from Cartier divisors to Weil divisors without using the later class-group theory.

Conceptual roadmap

$$\begin{gathered}
\text{codimension-one points } \eta_Z \\
\Downarrow \\
\text{prime divisors } Z = \overline{\{\eta_Z\}} \\
\Downarrow \\
\operatorname{Div}(X) = \bigoplus_{Z \in X^{(1)}} \mathbb{Z}[Z] \\
X \text{ normal} \Rightarrow \mathcal{O}_{X,\eta_Z} \text{ DVR} \Rightarrow \operatorname{ord}_Z : K(X)^\times \to \mathbb{Z} \\
\operatorname{div}(f) = \sum_Z \operatorname{ord}_Z(f)[Z]
\end{gathered}$$

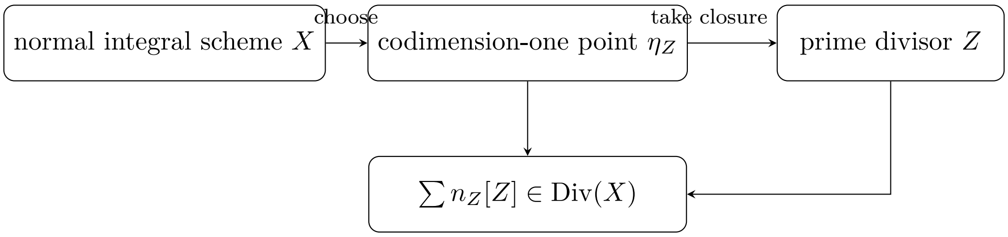


Figure 39.1: Weil divisors are finite integer combinations of codimension-one irreducible closed subsets.

39.1 Prime divisors and codimension-one points

Let X be a Noetherian integral scheme. Write $X^{(1)}$ for the set of points of codimension one.

Definition 39.1 (Prime divisor)**.** A *prime divisor* on X is an integral closed subscheme $Z \subseteq X$ of codimension one, endowed with its reduced induced structure. Equivalently, a prime divisor is the closure

$$Z = \overline{\{\eta_Z\}}$$

of a point $\eta_Z \in X^{(1)}$.

The word *prime* refers to irreducibility. A reducible hypersurface is not one prime divisor; it is a union of prime divisors and later contributes a sum of them.

Example 39.2 (Coordinate axes in the affine plane)**.** On $X = \mathbb{A}^2_k = \operatorname{Spec} k[x, y]$, the closed subsets $V(x)$ and $V(y)$ are prime divisors. Their generic points correspond to the height-one prime ideals (x) and (y). The union $V(xy)$ is codimension one but reducible, so it is not a prime divisor.

Example 39.3 (Height-one primes in a UFD)**.** If A is a UFD, every height-one prime is generated by an irreducible element. Thus the prime divisors on $\operatorname{Spec} A$ are exactly the irreducible hypersurfaces $V(p)$, with p irreducible up to associates.

39.2 The Weil divisor group

Definition 39.4 (Weil divisor)**.** A *Weil divisor* on X is a finite formal integer combination

$$D = \sum_{Z \in X^{(1)}} n_Z[Z], \qquad n_Z \in \mathbb{Z},$$

where all but finitely many n_Z vanish. The group of Weil divisors is

$$\mathrm{Div}(X) := \bigoplus_{Z \in X^{(1)}} \mathbb{Z}[Z].$$

Definition 39.5 (Support and effectivity)**.** For $D = \sum n_Z[Z]$, define

$$\mathrm{Supp}(D) = \bigcup_{n_Z \neq 0} Z.$$

The divisor is *effective*, written $D \geq 0$, if every $n_Z \geq 0$. Its positive and negative parts are

$$D^+ = \sum_Z \max(n_Z, 0)[Z], \qquad D^- = \sum_Z \max(-n_Z, 0)[Z],$$

so that $D = D^+ - D^-$ and the supports of D^+ and D^- have no common prime component.

$$\begin{tikzcd} & \boxed{D = 2Z_1 - 3Z_2 + Z_3} \arrow[dl] \arrow[dr] & \\ \boxed{D^+ = 2Z_1 + Z_3} & & \boxed{D^- = 3Z_2} \end{tikzcd}$$

$D = D^+ - D^-$, disjoint prime supports

Figure 39.2: Every Weil divisor decomposes uniquely into positive and negative effective parts.

Example 39.6 (Divisors on the affine line)**.** For $X = \mathbb{A}^1_k$, prime divisors are closed points. Over an algebraically closed field they are the points $[a] = V(t - a)$, so a Weil divisor is a finite sum $\sum_a n_a[a]$. Over a non-algebraically closed field, closed points correspond to irreducible polynomials, and their residue fields may have degree greater than one over k.

Example 39.7 (Divisor of a rational function on $\mathbb{A}^1$)**.** On $X = \mathbb{A}^1_k$ with coordinate t, the rational function $f = t^2(t-1)^{-1}$ has divisor $2[0] - [1]$ when viewed on the affine curve away from contributions at infinity.

39.3 Why normality matters in codimension one

For a normal Noetherian integral scheme, every codimension-one local ring is a DVR. This is the valuation engine behind Weil divisors.

Theorem 39.8 (Codimension one on a normal scheme)**.** *Let X be a normal Noetherian integral scheme and let Z be a prime divisor with generic point η_Z. Then*

$$\mathcal{O}_{X,\eta_Z}$$

is a discrete valuation ring with fraction field $K(X)$.

Proof. The local ring is a one-dimensional Noetherian local domain because η_Z has codimension one. Normality means it is integrally closed in its fraction field. A one-dimensional Noetherian local integrally closed domain is a DVR. $\square$

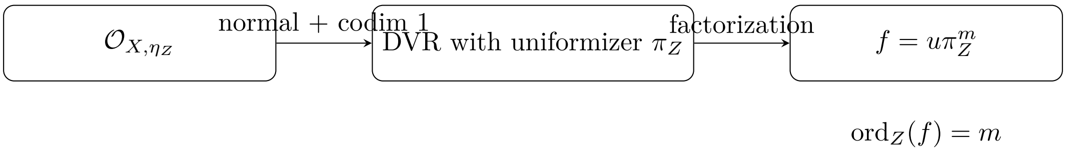


Figure 39.3: Normality turns codimension-one local algebra into a discrete valuation.

Definition 39.9 (Order along a prime divisor)**.** Assume X is normal Noetherian integral. For a prime divisor Z, let

$$\mathrm{ord}_Z : K(X)^\times \longrightarrow \mathbb{Z}$$

be the normalized discrete valuation of the DVR $\mathcal{O}_{X,\eta_Z}$. If π_Z is a uniformizer and

$$f = u\pi_Z^m$$

with u a unit, then $\mathrm{ord}_Z(f) = m$.

Thus positive order means a zero along Z, negative order means a pole, and order zero means that f is a unit at the generic point of Z.

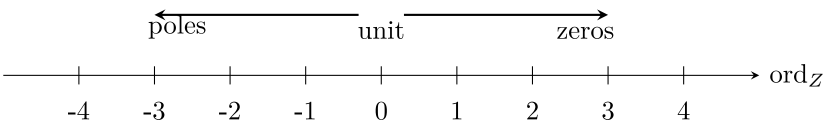


Figure 39.4: The sign of the codimension-one valuation distinguishes poles, units, and zeros.

Proposition 39.10 (Valuation rules)**.** *For $f, g \in K(X)^\times$,*

$$\mathrm{ord}_Z(fg) = \mathrm{ord}_Z(f) + \mathrm{ord}_Z(g), \qquad \mathrm{ord}_Z(f^{-1}) = -\,\mathrm{ord}_Z(f).$$

If $f + g \neq 0$, then

$$\mathrm{ord}_Z(f+g) \geq \min\{\mathrm{ord}_Z(f), \mathrm{ord}_Z(g)\}.$$

Example 39.11 (A basic order computation)**.** On $\mathbb{A}^1_k$ with coordinate t, let $Z = V(t)$. The local ring at Z is $k[t]_{(t)}$, with uniformizer t. Therefore

$$\mathrm{ord}_Z\left(\frac{t^3}{(1+t)^5}\right) = 3,$$

because $1 + t$ is a unit at (t).

39.4 Principal divisors

Definition 39.12 (Principal divisor)**.** Let X be normal Noetherian integral and $0 \neq f \in K(X)$. Its *principal Weil divisor* is

$$\mathrm{div}_X(f) := \sum_{Z \in X^{(1)}} \mathrm{ord}_Z(f)[Z].$$

The notation makes sense because the sum is finite.

Theorem 39.13 (Finite support)**.** *For every $f \in K(X)^\times$, only finitely many prime divisors Z satisfy* $\mathrm{ord}_Z(f) \neq 0$.

Proof. Choose a nonempty affine open $U = \operatorname{Spec} A \subseteq X$ on which $f = a/b$ with $a, b \in A \setminus \{0\}$. Any codimension-one point of U at which f has nonzero order lies in $V(a) \cup V(b)$. A Noetherian closed subset has only finitely many irreducible components, hence only finitely many codimension-one generic points arising this way. The complement $X \setminus U$ has finitely many irreducible components, and only its codimension-one components can contribute. Thus the support is finite. ☐

Proposition 39.14 (Principal divisors form a homomorphic image)**.** *The map*

$$K(X)^\times \longrightarrow \operatorname{Div}(X), \qquad f \longmapsto \operatorname{div}(f),$$

is a group homomorphism from multiplication to addition:

$$\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g).$$

In particular, $\operatorname{div}(f^{-1}) = -\operatorname{div}(f)$*.*

Example 39.15 (A rational function on the affine line)**.** On $\mathbb{A}^1_k$, assuming $0, 1 \in k$,

$$f(t) = \frac{t^3}{(t-1)^2}$$

has

$$\operatorname{div}_{\mathbb{A}^1}(f) = 3[0] - 2[1].$$

There is no contribution from infinity because infinity is not a point of $\mathbb{A}^1$.

39.5 The projective line: zeros must balance poles

Let $t = x_1/x_0$ be the usual rational coordinate on $\mathbb{P}^1_k$. The point at infinity is $\infty = [0 : 1]$.

Example 39.16 (The divisor of the coordinate on $\mathbb{P}^1$)**.** The rational function t has a simple zero at $0 = [1 : 0]$ and a simple pole at ∞. Hence

$$\operatorname{div}_{\mathbb{P}^1}(t) = [0] - [\infty].$$

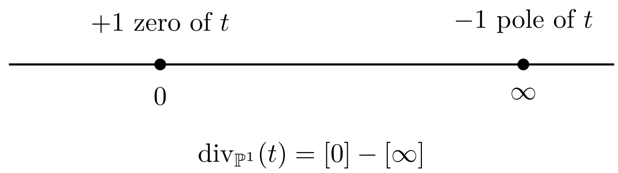


Figure 39.5: On the projective line a nonconstant rational coordinate acquires a pole at infinity.

Example 39.17 (Moving one point to another)**.** For distinct $a, b \in k$,

$$\operatorname{div}\left(\frac{t-a}{t-b}\right) = [a] - [b].$$

The orders at infinity cancel because numerator and denominator have the same degree.

Example 39.18 (Polynomial viewed projectively)**.** For a nonzero polynomial

$$p(t) = c \prod_i q_i(t)^{e_i}$$

with distinct monic irreducible $q_i \in k[t]$, let P_i be the corresponding closed points of $\mathbb{A}^1_k \subset \mathbb{P}^1_k$. If $d = \deg p$, then

$$\operatorname{div}_{\mathbb{P}^1}(p) = \sum_i e_i[P_i] - d[\infty].$$

Over a non-algebraically closed field, the degree of $[P_i]$ is $[\kappa(P_i) : k] = \deg q_i$, so the weighted degree of the divisor is zero.

Example 39.19 (Divisor on $\mathbb{P}^1$)**.** On $\mathbb{P}^1$, the rational function t has a simple zero at 0 and a simple pole at ∞, so $\operatorname{div}(t) = [0] - [\infty]$.

39.6 Factorization and divisors on affine space

On a factorial normal affine scheme, principal divisors can be read directly from irreducible factorization.

Proposition 39.20 (Principal divisors in a UFD)**.** *Let A be a Noetherian UFD and $X =$ $\operatorname{Spec} A$. If*

$$f = u \prod_{i=1}^r p_i^{e_i}$$

with $u \in A^\times$ and pairwise nonassociate irreducibles p_i, then

$$\operatorname{div}_X(f) = \sum_{i=1}^r e_i[V(p_i)].$$

For $f = a/b \in \operatorname{Frac}(A)^\times$, subtract the factorization divisor of b from that of a.

Example 39.21 (A divisor in $\mathbb{A}^2$)**.** In $k[x, y]$,

$$f = x^2 y(x - y)^3$$

has

$$\operatorname{div}_{\mathbb{A}^2}(f) = 2[V(x)] + [V(y)] + 3[V(x - y)].$$

The union of the three lines is the support; the multiplicities are encoded in the coefficients.

Example 39.22 (A rational function on $\mathbb{P}^2$)**.** Let $F, G \in k[x_0, x_1, x_2]$ be nonzero homogeneous forms of the same degree. Their ratio F/G is a rational function on $\mathbb{P}^2$. Factoring F and G into irreducible homogeneous factors yields

$$\operatorname{div}_{\mathbb{P}^2}(F/G) = \operatorname{div}(F) - \operatorname{div}(G),$$

where each irreducible factor contributes its projective hypersurface with the corresponding exponent.

39.7 Units and the dependence on the ambient open set

A principal divisor records codimension-one zeros and poles *inside the chosen scheme*. Boundary components disappear when one passes to an open subset.

Example 39.23 (A nonconstant function with zero divisor)**.** On $\mathbb{G}_m = \operatorname{Spec} k[t, t^{-1}]$, the function t is a global unit. Therefore

$$\operatorname{div}_{\mathbb{G}_m}(t) = 0,$$

even though t is nonconstant. Thus the kernel of $K(X)^\times \to \operatorname{Div}(X)$ need not consist only of constants on a nonproper scheme.

Proposition 39.24 (Restriction to an open subset)**.** *Let $j : U \hookrightarrow X$ be a nonempty open subset of a normal Noetherian integral scheme. For*

$$D = \sum_Z n_Z[Z],$$

define

$$D|_U := \sum_{Z \cap U \neq \varnothing} n_Z[\overline{Z \cap U}^U],$$

omitting the prime divisors contained entirely in $X \setminus U$. Then

$$\operatorname{div}_U(f) = \operatorname{div}_X(f)|_U$$

for every $f \in K(X)^\times = K(U)^\times$.

Example 39.25 (Removing infinity)**.** Take $X = \mathbb{P}^1_k$ and $U = \mathbb{A}^1_k = X \setminus \{\infty\}$. Since

$$\operatorname{div}_X(t) = [0] - [\infty],$$

restriction gives

$$\operatorname{div}_U(t) = [0].$$

The pole has not vanished analytically; its divisor component has been removed from the ambient scheme.

$$\boxed{X = \mathbb{P}^1\colon\ [0] - [\infty]} \xrightarrow{\text{restrict to } X \setminus \{\infty\}} \boxed{U = \mathbb{A}^1\colon\ [0]}$$

boundary components are omitted

Figure 39.6: Restriction of a Weil divisor to an open subset forgets prime components contained in the boundary.

39.8 A normal singular surface: multiplicity without smoothness

Normality does not mean smoothness. A codimension-one DVR calculation can remain valid even at a singular surface.

Example 39.26 (The quadric cone $xy = z^2$)**.** Let

$$A = k[x, y, z]/(xy - z^2), \qquad X = \operatorname{Spec} A.$$

This is a normal two-dimensional domain. Consider the height-one primes

$$P = (x, z), \qquad Q = (y, z).$$

At P, the element y is a unit and the relation gives $x = z^2/y$. Thus z is a uniformizer in A_P, so

$$\operatorname{ord}_P(z) = 1, \qquad \operatorname{ord}_P(x) = 2.$$

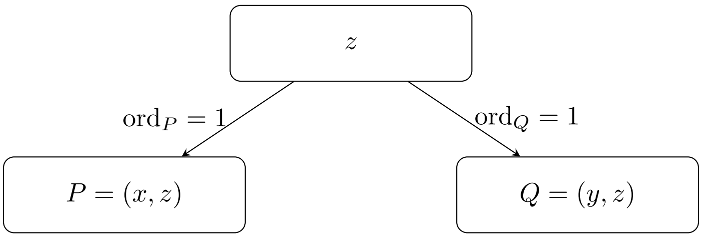
Similarly,

$$\operatorname{ord}_Q(z) = 1, \qquad \operatorname{ord}_Q(y) = 2.$$

Hence

$$\operatorname{div}(z) = [P] + [Q], \qquad \operatorname{div}(x) = 2[P], \qquad \operatorname{div}(y) = 2[Q].$$

The fact that $[P]$ occurs with coefficient two in $\operatorname{div}(x)$ is a genuine codimension-one multiplicity phenomenon.



$$xy = z^2 : \quad \operatorname{div}(z) = P + Q,\ \operatorname{div}(x) = 2P,\ \operatorname{div}(y) = 2Q$$

Figure 39.7: A normal singular surface still has DVR valuations in codimension one.

Example 39.27 (Prime divisor on a surface)**.** In $\mathbb{A}^2_k$, the irreducible curve $V(x)$ is codimension one and therefore defines a prime Weil divisor. A rational function contributes its order of vanishing along this curve.

39.9 What fails without normality

Example 39.28 (The cusp)**.** Let

$$A = k[t^2, t^3] \subset k[t]$$

and $X = \operatorname{Spec} A$. The singular closed point $P = (t^2, t^3)$ has codimension one because X is a curve, but A_P is not a DVR. In particular, there is no element of valuation one lying in A_P: the orders coming from the normalization $k[t]_{(t)}$ of elements of the maximal ideal begin at two and three. Thus codimension one alone does not produce the intrinsic DVR valuation used in the normal theory.

Remark 39.29 (Normalization restores codimension-one valuations). If $\nu : \widetilde{X} \to X$ is the normalization of an integral Noetherian scheme, the codimension-one points of $\widetilde{X}$ carry DVR valuations. Several points of $\widetilde{X}$ may lie over one codimension-one point of X, so passing to the normalization can split branches. This is one reason Weil divisors are especially natural on normal schemes.

39.10 Divisors on normal projective curves

Let C be a normal projective integral curve over a field k. Since every closed point has codimension one,

$$\operatorname{Div}(C) = \bigoplus_{P \in C_{\mathrm{cl}}} \mathbb{Z}[P].$$

Definition 39.30 (Degree on a proper curve)**.** For

$$D = \sum_P n_P[P],$$

define

$$\deg D := \sum_P n_P[\kappa(P) : k].$$

Theorem 39.31 (Principal divisors have degree zero)**.** *For every $f \in K(C)^\times$,*

$$\deg \operatorname{div}(f) = 0.$$

Proof. If $f \in k^\times$, the divisor is zero. Otherwise f defines a finite morphism $C \to \mathbb{P}^1_k$. The zero divisor and pole divisor are the scheme-theoretic fibers over 0 and ∞, with multiplicities given by the corresponding DVR ramification indices. Every fiber of a finite morphism of proper integral curves has the same weighted degree, namely $[K(C) : k(f)]$. Hence the total weighted order of zeros equals the total weighted order of poles. ☐

Example 39.32 (Degree on $\mathbb{P}^1$)**.** The divisor $[0] - [\infty]$ has degree zero. More generally, for an irreducible polynomial $q(t)$ of degree d, the corresponding closed point P_q satisfies

$$\deg[P_q] = d,$$

and

$$\operatorname{div}(q(t)) = [P_q] - d[\infty].$$

39.11 Principal pullback and the warning about general pullback

Let $g : Y \to X$ be a dominant morphism of normal integral schemes. It induces an inclusion of function fields

$$K(X) \hookrightarrow K(Y).$$

Therefore every rational function on X can be pulled back as a rational function on Y.

Proposition 39.33 (Pullback of principal divisors)**.** *For $f \in K(X)^\times$, the expression*

$$\operatorname{div}_Y(g^* f)$$

is always defined. It is the only pullback operation on Weil divisors needed in this chapter.

Warning 39.34 (Arbitrary Weil divisors do not pull back automatically). There is no canonical pullback $g^* : \operatorname{Div}(X) \to \operatorname{Div}(Y)$ for an arbitrary morphism of normal schemes. A prime divisor may map into codimension greater than one, a morphism may factor through its support, and multiplicities require extra hypotheses. Cartier divisors have a substantially better pullback theory; this is one motivation for VI/40.

39.12 Finite dominant morphisms: controlled pullback and pushforward

The warning above concerns arbitrary morphisms. For a finite dominant morphism between normal Noetherian integral schemes, codimension-one behavior is controlled by finite extensions of DVRs, and both pullback and pushforward admit clean formulas.

Proposition 39.35 (Zero divisor versus global unit)**.** *Let X be normal Noetherian integral and $u \in K(X)^\times$. Then*

$$\operatorname{div}_X(u) = 0 \quad \Longleftrightarrow \quad u \in \Gamma(X, \mathcal{O}_X)^\times.$$

Proof. A global unit has order zero at every codimension-one point. Conversely, suppose every order of u is zero. On a normal affine open $V = \operatorname{Spec} A$, the Noetherian normal domain A is a Krull domain and

$$A = \bigcap_{\operatorname{ht}\mathfrak{p}=1} A_{\mathfrak{p}} \subset \operatorname{Frac}(A).$$

Both u and u^{-1} have nonnegative order at every height-one prime, hence both lie in every $A_{\mathfrak{p}}$, and therefore in A. Thus u is a unit on every affine open and hence globally. □

Let $g : Y \to X$ now be finite dominant, with function-field extension $L = K(Y) \supset K = K(X)$. If $D \subset X$ is a prime divisor and $E \subset Y$ is a prime divisor lying over D, then the inclusion of DVRs

$$\mathcal{O}_{X,\eta_D} \subset \mathcal{O}_{Y,\eta_E}$$

defines the *ramification index*

$$e(E/D) := \operatorname{ord}_E(\pi_D),$$

where π_D is any uniformizer at D.

Definition 39.36 (Finite pullback and pushforward)**.** For a prime divisor D on X, define

$$g^*[D] := \sum_{E|D} e(E/D)[E].$$

For a prime divisor E on Y, let $D = g(E)$; finiteness implies that D is again codimension one. Define

$$g_*[E] := [\kappa(E) : \kappa(D)]\,[D].$$

Extend both maps linearly to Weil divisors.

Proposition 39.37 (Principal divisors and finite pullback)**.** *For every $f \in K(X)^\times$,*

$$\operatorname{div}_Y(g^* f) = g^* \operatorname{div}_X(f).$$

Proof. If E lies over D, extension of normalized DVR valuations gives

$$\operatorname{ord}_E(f) = e(E/D) \operatorname{ord}_D(f).$$

This is exactly the coefficient of $[E]$ on the right-hand side. □

Proposition 39.38 (Principal divisors and the field norm)**.** *For $h \in L^\times$,*

$$g_* \operatorname{div}_Y(h) = \operatorname{div}_X(N_{L/K}(h)).$$

Proof. For every prime divisor D of X, the valuation formula for the norm in a finite extension of discretely valued fields is

$$\operatorname{ord}_D(N_{L/K}(h)) = \sum_{E|D} [\kappa(E) : \kappa(D)] \operatorname{ord}_E(h).$$

The right-hand side is precisely the coefficient of $[D]$ in $g_* \operatorname{div}_Y(h)$. ☐

Example 39.39 (The power map on the affine line)**.** Let

$$g : \mathbb{A}^1_u \longrightarrow \mathbb{A}^1_x, \qquad x = u^m,$$

with $m \geq 1$. The divisor $[0]_x$ has one prime divisor above it, namely $[0]_u$, and

$$e(0/0) = \operatorname{ord}_{u=0}(u^m) = m.$$

Hence

$$g^*[0]_x = m[0]_u,$$

which agrees with

$$\operatorname{div}(g^*x) = \operatorname{div}(u^m) = m[0]_u.$$

Also $g_*[0]_u = [0]_x$ when the residue fields are both k.

Example 39.40 (The power map on $\mathbb{P}^1$)**.** The morphism

$$[s : t] \longmapsto [s^m : t^m]$$

is finite. It is totally ramified at 0 and ∞, so

$$g^*[0] = m[0], \qquad g^*[\infty] = m[\infty].$$

Since $\operatorname{div}(t/s) = [0] - [\infty]$ in the chosen coordinate convention, pulling back gives

$$\operatorname{div}((t/s)^m) = m[0] - m[\infty] = g^* \operatorname{div}(t/s).$$

This is the finite-morphism version of principal-divisor functoriality.

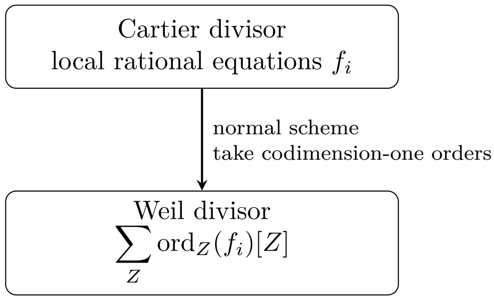
Warning 39.41 (Scope of the finite formulas). The formulas in this section use finiteness in an essential way. They should not be read as defining a pullback of arbitrary Weil divisors for every morphism. Flat pullback, proper pushforward of cycles, and the full intersection-theoretic formalism belong to later theory.

39.13 First bridge to Cartier divisors

A Cartier divisor is locally represented by nonzero rational functions modulo regular units. On a normal integral scheme, those local equations have well-defined orders along every prime divisor.

Proposition 39.42 (Cartier-to-Weil map: preview)**.** *Let X be normal Noetherian integral. If a Cartier divisor is represented near the generic point of a prime divisor Z by $f_Z \in K(X)^\times$, define its coefficient along Z by $\operatorname{ord}_Z(f_Z)$. These coefficients are independent of the chosen local equation and yield a Weil divisor.*

Proof. Two local equations differ by a regular unit. A unit in the DVR $\mathcal{O}_{X,\eta_Z}$ has order zero, so the order is unchanged. ☐



VI/40 develops when this passage is reversible

Figure 39.8: Preview of the Cartier-to-Weil construction on a normal integral scheme.

Example 39.43 (A hyperplane is both Cartier and Weil)**.** On $\mathbb{P}^n_k$, the hyperplane $H = V_+(x_0)$ is a prime Weil divisor. It is also locally cut out by one nonzerodivisor, so VI/40 will identify it with an effective Cartier divisor. The associated Weil divisor has coefficient one along H.

Example 39.44 (A Weil divisor that signals the Cartier distinction)**.** On the normal surface $xy = z^2$, the prime divisor $P = (x, z)$ is a perfectly good Weil divisor. The relation $\mathrm{div}(x) = 2[P]$ hints that $[P]$ need not itself be principal or locally cut out by one equation at the singular point. VI/40 will make the Cartier distinction precise, and VI/41 will measure the remaining obstruction in the divisor class group.

39.14 Worked examples

The following examples consolidate the chapter's main computational patterns.

Example 39.45 (Zero and pole orders from a local parameter)**.** Let C be a smooth curve, $P \in C$, and u a uniformizer at P. If

$$f = u^{-4}(1 + u + u^3),$$

then the parenthesized factor is a unit, so $\mathrm{ord}_P(f) = -4$.

Example 39.46 (Cancellation must be tested before taking orders)**.** If $\mathrm{ord}_P(f) = \mathrm{ord}_P(g) = 2$, the order of $f + g$ can exceed two. For instance, in $k[[u]]$, take $f = u^2$ and $g = -u^2 + u^5$. Then $f + g = u^5$, so the order jumps to five.

39.15 Exercises

Exercise 39.47. Show that the support of a Weil divisor is closed.

Exercise 39.48. Prove that $D = D^+ - D^-$ with $D^+, D^- \geq 0$ and no common prime component, and that this decomposition is unique.

Exercise 39.49. On $\mathbb{A}^1_k$, compute $\mathrm{div}(t^m)$ for $m \in \mathbb{Z}$.

Exercise 39.50. On $\mathbb{P}^1_k$, compute $\mathrm{div}(t^m)$.

Exercise 39.51. Compute the divisor of $(t - a)^r/(t - b)^s$ on $\mathbb{A}^1_k$ for distinct $a, b \in k$.

Exercise 39.52. Compute the same divisor on $\mathbb{P}^1_k$.

Exercise 39.53. Let $A = k[x, y]$. Compute $\operatorname{div}(x^4y^2/(x-y)^3)$ on $\operatorname{Spec} A$.

Exercise 39.54. Show that a unit in $\mathcal{O}_{X,\eta_Z}$ has order zero.

Exercise 39.55. Show directly that $\operatorname{div}(f/g) = \operatorname{div}(f) - \operatorname{div}(g)$.

Exercise 39.56. Let $U \subseteq X$ be nonempty open. Prove that every prime divisor of X meeting U restricts to a prime divisor of U.

Exercise 39.57. Explain why a codimension-two closed subset cannot itself occur as a prime component of a Weil divisor.

Exercise 39.58. Let X be normal and Z prime. Show that f is regular and nonvanishing at η_Z iff $\operatorname{ord}_Z(f) = 0$.

Exercise 39.59. Show that $f \in \mathcal{O}_{X,\eta_Z}$ iff $\operatorname{ord}_Z(f) \geq 0$.

Exercise 39.60. Let $D = 2[Z_1] - 3[Z_2] + [Z_3]$. Compute D^+, D^-, and $\operatorname{Supp}(D)$.

Exercise 39.61. For the quadric cone $xy = z^2$, verify $\operatorname{div}(z^2) = 2[P] + 2[Q]$ and compare it with $\operatorname{div}(xy)$.

Exercise 39.62. Show that the divisor of a global unit is zero.

Exercise 39.63. Give a nonconstant global unit on an integral scheme and verify that its principal divisor is zero.

Exercise 39.64. Let $q(t) \in k[t]$ be irreducible of degree d. Compute $\deg \operatorname{div}_{\mathbb{P}^1}(q(t))$.

Exercise 39.65. Prove that the degree map $\operatorname{Div}(C) \to \mathbb{Z}$ is a group homomorphism for a proper curve C.

Exercise 39.66. Why does $\operatorname{div}(f) = 0$ imply that f has no codimension-one zeros or poles, but not necessarily that f is constant?

Exercise 39.67. Let F, G be homogeneous forms of equal degree on $\mathbb{P}^n$. Explain why F/G is invariant under homogeneous rescaling of coordinates.

Exercise 39.68. Show that if a Cartier local equation is multiplied by a regular unit, all associated Weil coefficients remain unchanged.

Exercise 39.69. For a dominant morphism $g : Y \to X$, verify $\operatorname{div}_Y(g^*(fh)) = \operatorname{div}_Y(g^*f) + \operatorname{div}_Y(g^*h)$.

Exercise 39.70. Explain why the singular point of a normal surface does not itself define a prime Weil divisor.

39.16 Problem dossiers

Problem 39.71 (VI.39.P1: Factorization on an affine factorial scheme)**.** Let $X = \operatorname{Spec} k[x, y, z]$ and

$$f = \frac{x^3(y-z)^2}{y^4 z}.$$

Compute $\operatorname{div}(f)$.

Problem 39.72 (VI.39.P2: Restriction changes the visible divisor)**.** Let $X = \mathbb{P}^1_k$, $U = X \setminus \{0, \infty\}$, and $f = t$. Compute $\operatorname{div}_X(f)$ and $\operatorname{div}_U(f)$.

Problem 39.73 (VI.39.P3: A projective ratio)**.** On $\mathbb{P}^2_k$, compute the divisor of

$$\frac{x_0 x_1}{x_2^2}.$$

Problem 39.74 (VI.39.P4: Orders on the quadric cone)**.** On $X = V(xy - z^2)$, with $P = (x, z)$, compute the order along P of

$$f = \frac{x^3}{z^5}.$$

Problem 39.75 (VI.39.P5: Degree check on $\mathbb{P}^1$)**.** Let $p, q \in k[t]$ be nonzero polynomials of degrees r, s. Show directly that the principal divisor of p/q on $\mathbb{P}^1$ has degree zero.

Problem 39.76 (VI.39.P6: Effective principal divisors on a proper curve)**.** Let C be a normal projective integral curve. Show that if $\operatorname{div}(f) \geq 0$, then $\operatorname{div}(f) = 0$.

Problem 39.77 (VI.39.P7: Detecting a unit at a generic point)**.** Let Z be prime on normal X, and suppose $f, g \in K(X)^\times$ satisfy $\operatorname{ord}_Z(f) = 3$ and $\operatorname{ord}_Z(g) = -3$. What can be said about fg at η_Z?

Problem 39.78 (VI.39.P8: Why the cusp is not a DVR computation)**.** At the cusp $A = k[t^2, t^3]$, explain why t^2 cannot serve as a uniformizer of $A_{(t^2,t^3)}$.

Problem 39.79 (VI.39.P9: Same support, different multiplicity)**.** In $\mathbb{A}^2$, compare the principal divisors of xy and x^2y^3.

Problem 39.80 (VI.39.P10: A rational function with no divisor on an open torus)**.** Let $U = D(xy) \subset \mathbb{A}^2_k$. Compute the divisor on U of x^5/y^2.

Problem 39.81 (VI.39.P11: Principal pullback only)**.** Let $g : Y \to X$ be dominant between normal integral schemes and $f, h \in K(X)^\times$ with $\operatorname{div}_X(f) = \operatorname{div}_X(h)$. What can be concluded about their pulled-back principal divisors on Y?

Problem 39.82 (VI.39.P12: Cartier coefficients are well defined)**.** Suppose two Cartier local equations near η_Z are f and g, with $f = ug$ for $u \in \mathcal{O}^\times_{X,\eta_Z}$. Prove that they induce the same Weil coefficient along Z.

39.17 Challenges

Problem 39.83 (Challenge 1: Product formula on a projective curve)**.** Give a detailed proof of theorem 39.31 using the finite morphism $C \to \mathbb{P}^1$ defined by a nonconstant rational function and the equality of degrees of finite fibers.

Problem 39.84 (Challenge 2: Height-one primes on a normal affine domain)**.** Let A be a Noetherian normal domain. Starting from the theorem that a one-dimensional Noetherian integrally closed local domain is a DVR, derive the valuation $\operatorname{ord}_{\mathfrak{p}}$ for every height-one prime $\mathfrak{p} \subset A$ and reconstruct $\operatorname{div}(f)$ purely algebraically.

Problem 39.85 (Challenge 3: The quadric cone and non-principality)**.** On $A = k[x, y, z]/(xy - z^2)$, investigate whether the prime divisor $P = (x, z)$ can be principal. Use $\operatorname{div}(x) = 2[P]$ and the symmetry of the cone to anticipate the class-group computation of VI/41.

Problem 39.86 (Challenge 4: Boundary dependence)**.** Let X be normal projective integral and $U = X \setminus B$ where B is a union of prime divisors. Describe precisely how the kernel of $\operatorname{Div}(X) \to \operatorname{Div}(U)$ is generated by the components of B, and compare principal divisors before and after restriction.

Problem 39.87 (Challenge 5: Cartier versus Weil)**.** Using only the local-equation definition of a Cartier divisor and the DVR theory of this chapter, prove that the Cartier-to-Weil construction is additive. Then identify the extra local property needed for a prime Weil divisor itself to come from a Cartier divisor.

Chapter summary

For a normal Noetherian integral scheme X, codimension-one points are controlled by discrete valuation rings. They produce the free abelian group

$$\mathrm{Div}(X) = \bigoplus_{Z \in X^{(1)}} \mathbb{Z}[Z],$$

and every nonzero rational function determines the finite principal divisor

$$\mathrm{div}(f) = \sum_{Z} \mathrm{ord}_Z(f)[Z].$$

This turns zeros and poles into codimension-one geometry with multiplicity. Restriction to open subsets discards boundary components, normality is what makes the valuation theory intrinsic, and arbitrary Weil divisors do not admit unrestricted pullback. VI/40 now replaces formal codimension-one cycles by local rational equations and develops Cartier divisors, whose pullback behavior is substantially better.

39.18 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 39.88 (Projective-line divisor)**.** Compute $\mathrm{div}(t)$ on $\mathbb{P}^1$.

Hint. Use codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. $[0] - [\infty]$. ☐

Exercise 39.89 (Square)**.** Compute $\mathrm{div}(t^2)$ on $\mathbb{P}^1$.

Hint. Use codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. $2[0] - 2[\infty]$. ☐

Exercise 39.90 (Ratio)**.** Compute $\mathrm{div}((t-a)/(t-b))$ for distinct finite points.

Hint. Use codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. $[a] - [b]$. ☐

Exercise 39.91 (Prime divisor)**.** What geometric object defines a prime Weil divisor?

Hint. Use codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. An irreducible codimension-one closed subvariety. ☐

Exercise 39.92 (Principal)**.** What is a principal Weil divisor?

Hint. Use codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. The divisor of a nonzero rational function. ☐

Proofs

Exercise 39.93 (Additivity)**.** Prove: Prove $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$.

Hint. Work from codimension-one subvarieties, valuations, and principal divisors and state the hypotheses explicitly. ☐

Solution. Orders of vanishing are valuations, hence additive on products. ☐

Exercise 39.94 (Ratio)**.** Prove: Prove $\operatorname{div}(f/g) = \operatorname{div}(f) - \operatorname{div}(g)$.

Hint. Work from codimension-one subvarieties, valuations, and principal divisors and state the hypotheses explicitly. ☐

Solution. Apply additivity and the valuation of inverses. ☐

Exercise 39.95 (Finite support)**.** Prove: Explain why the divisor of a rational function on a Noetherian normal integral scheme has finite support.

Hint. Work from codimension-one subvarieties, valuations, and principal divisors and state the hypotheses explicitly. ☐

Solution. A rational function is regular and invertible on a dense open; only finitely many codimension-one components of the complement contribute locally. ☐

Exercise 39.96 (Degree zero on $\mathbb{P}^1$)**.** Prove: Prove every principal divisor on $\mathbb{P}^1$ has degree zero.

Hint. Work from codimension-one subvarieties, valuations, and principal divisors and state the hypotheses explicitly. ☐

Solution. Count zeros and poles of a rational function with multiplicity, including infinity. ☐

Counterexamples and hypothesis tests

Exercise 39.97 (All closed subsets)**.** Test the claim: every closed subset defines a Weil divisor.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; only codimension-one irreducible pieces generate Weil divisors. ☐

Exercise 39.98 (Effective principal)**.** Test the claim: every principal divisor is effective.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; poles contribute negative coefficients. ☐

Exercise 39.99 (Unique function)**.** Test the claim: a principal divisor determines its rational function uniquely.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; multiplying by a nonzero constant leaves the divisor unchanged. ☐

Applications and investigations

Exercise 39.100 (Zeros and poles)**.** What information does a Weil divisor package?

Hint. Translate the question into codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. Codimension-one zeros and poles with multiplicities. ☐

Exercise 39.101 (Birational geometry)**.** Why are Weil divisors natural on normal varieties?

Hint. Translate the question into codimension-one subvarieties, valuations, and principal divisors. ☐

Solution. Codimension-one local rings are valuation-like enough to measure rational functions. ☐

Challenge problems

Exercise 39.102 (Synthesis challenge)**.** Connect two central viewpoints of Weil Divisors in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Weil Divisors and then translates it through codimension-one subvarieties, valuations, and principal divisors, checking both descriptions agree. ☐

Exercise 39.103 (Hypothesis challenge)**.** Choose one theorem-level statement from Weil Divisors, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 40

Cartier Divisors

VI/39 encoded codimension-one geometry by Weil divisors. That construction is global and cycle-theoretic: one records the order of a rational function along each prime divisor. Cartier divisors start from the opposite direction. They are given locally by one rational equation, with two equations considered equivalent when they differ by a nowhere-vanishing regular function.

This local formulation is both more flexible and more functorial. It makes sense on schemes that need not be normal, it remembers when a codimension-one closed subscheme is locally principal, and it produces an invertible sheaf almost automatically. The price is an important technical point: pullback is not defined for every morphism unless the pulled-back local equations remain invertible in the total quotient sheaf.

The migration ledger assigns three canonical rules to VI/40: the AG9 Cartier-divisor section and its theorem-like and exercise/problem descendants. The later AG10 divisor/Picard/line-bundle block is mapped to VI/39 only as compare-and-migrate overlap. Accordingly, this chapter uses the AG9 mapping as provenance and develops the canonical Cartier theory afresh, while leaving divisor class groups to VI/41 and the full Picard-group equivalence to VI/42.

Learning goals

By the end of the chapter, the reader should be able to:

- construct the sheaf of total quotient rings $\mathcal{K}_X$ and interpret $\mathcal{K}_X^\times/\mathcal{O}_X^\times$;
- describe a Cartier divisor by compatible local rational equations (U_i, f_i);
- add, negate, restrict, and recognize principal Cartier divisors;
- characterize effective Cartier divisors as locally principal closed subschemes cut out by non-zero-divisors;
- identify the ideal sheaf of an effective Cartier divisor as an invertible sheaf;
- state exactly when a morphism admits pullback of a Cartier divisor and compute ramification multiplicities in examples;
- construct the Cartier-to-Weil map on normal Noetherian integral schemes and prove compatibility with principal divisors;
- recognize locally factorial and regular schemes as settings in which every Weil divisor is Cartier;

- exhibit a normal singular scheme carrying a Weil divisor that is not Cartier;
- construct $\mathcal{O}_X(D)$ from local equations and identify $\mathcal{O}_X(-D)$ with the ideal sheaf of an effective Cartier divisor;
- explain the first correspondence between Cartier divisors and invertible sheaves equipped with a rational trivialization.

Conceptual roadmap

$$\boxed{\begin{array}{ccc} \mathcal{K}_X^\times/\mathcal{O}_X^\times & \Longleftrightarrow & \text{local rational equations } (U_i, f_i),\ f_i/f_j \in \mathcal{O}_X^\times \\ D \text{ effective} & \Longleftrightarrow & f_i \in \mathcal{O}_X(U_i) \text{ non-zero-divisors} \\ & \Downarrow & \\ \multicolumn{3}{c}{\mathcal{O}_X(-D) = \mathcal{I}_D \text{ invertible}, \qquad \mathcal{O}_X(D) = \mathcal{I}_D^{-1}} \\ X \text{ normal integral} & \Longrightarrow & D \mapsto \displaystyle\sum_{Z \in X^{(1)}} \mathrm{ord}_Z(f_i)[Z]. \end{array}}$$

$$\begin{tikzcd}[column sep=small] & \boxed{\begin{array}{c}\mathcal{K}_X^\times/\mathcal{O}_X^\times \\ \text{Cartier divisor}\end{array}} \arrow[dl] \arrow[dr] & \\ \boxed{\begin{array}{c}\text{local equations} \\ f_i/f_j \in \mathcal{O}_X^\times\end{array}} \arrow[dr] & & \boxed{\begin{array}{c}\text{effective case} \\ f_i \in \mathcal{O}_X\text{, non-zero-divisor}\end{array}} \arrow[dl] \\ & \boxed{\begin{array}{c}\mathcal{O}_X(D) \\ \text{invertible sheaf}\end{array}} & \end{tikzcd}$$

Figure 40.1: Cartier divisors connect local rational equations, effective locally principal subschemes, and invertible sheaves.

40.1 The total quotient sheaf

Let X be a locally Noetherian scheme. On an open set $U \subseteq X$, call a section $s \in \mathcal{O}_X(U)$ *regular* in the ring-theoretic sense if multiplication by s is injective, equivalently if s is a non-zero-divisor on $\mathcal{O}_X(U)$. Localizing at all such sections gives the total quotient ring.

Definition 40.1 (Sheaf of total quotient rings)**.** The *sheaf of total quotient rings* $\mathcal{K}_X$ is the sheaf associated to the presheaf

$$U \longmapsto Q(\mathcal{O}_X(U)),$$

where $Q(A)$ denotes the localization of a ring A at its multiplicative set of non-zero-divisors. We write $\mathcal{K}_X^\times$ for its sheaf of units.

On an integral scheme X, every nonzero regular function is a non-zero-divisor and $\mathcal{K}_X$ is the constant sheaf of rational functions with value $K(X)$ on every nonempty sufficiently small open set. Thus the general language specializes to the familiar function field.

Example 40.2 (Integral affine scheme)**.** If $X = \operatorname{Spec} A$ with A a domain, then

$$\Gamma(X, \mathcal{K}_X) = \operatorname{Frac}(A).$$

For $A = k[x, y]$, the rational function $(x^2 + y)/(xy - 1)$ is a global section of $\mathcal{K}_X$ even though it is not regular everywhere on X.

Example 40.3 (A reducible ring)**.** For $A = k[x, y]/(xy)$, both x and y are zero-divisors and therefore are not inverted in $Q(A)$. This already signals why a principal ideal generated by x need not define an effective Cartier divisor.

40.2 Cartier divisors as a quotient sheaf

Definition 40.4 (Cartier divisor)**.** A *Cartier divisor* on X is a global section

$$D \in \Gamma\left(X, \mathcal{K}_X^\times / \mathcal{O}_X^\times\right).$$

The abelian group of Cartier divisors is denoted

$$\operatorname{CaDiv}(X) := \Gamma\left(X, \mathcal{K}_X^\times / \mathcal{O}_X^\times\right),$$

where multiplication of rational functions is written additively on divisors.

The quotient by $\mathcal{O}_X^\times$ expresses the geometric principle that multiplying a local equation by a unit does not change its zero or pole locus.

Proposition 40.5 (Local-equation description)**.** *A Cartier divisor can equivalently be described by an open cover $X = \bigcup_i U_i$ and sections*

$$f_i \in \Gamma(U_i, \mathcal{K}_X^\times)$$

such that on every overlap

$$\frac{f_i}{f_j} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times).$$

Two such collections define the same divisor exactly when, after passing to a common refinement, their corresponding equations differ by units.

$$\begin{tikzcd} \boxed{U_i,\ \text{equation } f_i} \arrow[r, "f_i"] & \boxed{U_i \cap U_j} & \boxed{U_j,\ \text{equation } f_j} \arrow[l, "f_j"'] \end{tikzcd}$$

$$f_i/f_j \in \mathcal{O}_X^\times(U_i \cap U_j)$$

Figure 40.2: Local Cartier equations agree only up to a regular unit on overlaps.

Example 40.6 (A point on the projective line)**.** Let $D = [\infty]$ on $\mathbb{P}^1_k$. On $U_0 = \mathbb{P}^1 \setminus \{\infty\}$ take the local equation $f_0 = 1$. On the chart U_∞ with coordinate $u = 1/t$, take $f_\infty = u$. On the overlap, u is a unit, so these equations define a Cartier divisor supported at infinity.

Example 40.7 (A principal Cartier divisor)**.** A nonzero rational function f on an integral scheme defines the Cartier divisor represented locally by the single equation f. Changing f by a unit leaves the local Cartier class unchanged.

40.3 Group operations and restriction

Suppose D has local equations f_i and E has local equations g_i on a common refinement. Define

$$D+E \quad \text{by} \quad f_i g_i, \qquad -D \quad \text{by} \quad f_i^{-1}.$$

Because ratios of corresponding equations are units, these operations are independent of all choices.

Proposition 40.8 (Restriction). *For every open subset $j : U \hookrightarrow X$, restriction of local equations defines a homomorphism*

$$j^* : \mathrm{CaDiv}(X) \longrightarrow \mathrm{CaDiv}(U).$$

If a divisor is supported entirely in $X \setminus U$, its restriction may become zero.

Example 40.9 (Forgetting infinity). The Cartier divisor $[\infty]$ on $\mathbb{P}^1$ restricts to zero on $\mathring{A}^1 = \mathbb{P}^1 \setminus \{\infty\}$, because its local equation there is the unit 1.

40.4 Principal Cartier divisors

Definition 40.10 (Principal Cartier divisor). A global rational unit $f \in \Gamma(X, \mathcal{K}_X^\times)$ determines the *principal Cartier divisor*

$$(f)_{\mathrm{Cart}} \in \mathrm{CaDiv}(X)$$

represented by the single local equation f. Two global rational units determine the same principal Cartier divisor exactly when their quotient is a global regular unit.

Thus there is an exact sequence at the left end

$$\Gamma(X, \mathcal{O}_X^\times) \longrightarrow \Gamma(X, \mathcal{K}_X^\times) \longrightarrow \mathrm{CaDiv}(X),$$

whose second arrow is $f \mapsto (f)_{\mathrm{Cart}}$. We postpone the quotient of Cartier divisors by principal Cartier divisors to VI/41.

Example 40.11 (The coordinate on $\mathbb{P}^1$). For the rational coordinate t on $\mathbb{P}^1$, the principal Cartier divisor $(t)_{\mathrm{Cart}}$ is locally represented by t. Under the Cartier-to-Weil map developed below, it becomes

$$[0] - [\infty].$$

Example 40.12 (A unit gives zero). On $\operatorname{Spec} k[x, x^{-1}]$, the function x is a global unit. Hence $(x)_{\mathrm{Cart}} = 0$, matching the VI/39 observation that $\mathrm{div}_{\mathbb{G}_m}(x) = 0$.

40.5 Effective Cartier divisors

The local equations of a general Cartier divisor may have poles. Effectivity means that they can be chosen to be genuine regular non-zero-divisors.

Definition 40.13 (Effective Cartier divisor). A Cartier divisor D is *effective* if it admits local equations $f_i \in \mathcal{O}_X(U_i)$ such that each f_i is a non-zero-divisor. The corresponding closed subschemes

$$V(f_i) \subseteq U_i$$

glue to a closed subscheme, also denoted $D \subseteq X$.

Theorem 40.14 (Ideal-sheaf characterization)**.** *A closed subscheme $D \hookrightarrow X$ is an effective Cartier divisor if and only if its ideal sheaf $\mathcal{I}_D$ is locally generated by one non-zero-divisor. In this case $\mathcal{I}_D$ is an invertible $\mathcal{O}_X$-module of rank one.*

Proof. If $D \cap U_i = V(f_i)$ with f_i a non-zero-divisor, then $\mathcal{I}_D|_{U_i} = f_i\mathcal{O}_{U_i}$. Multiplication by f_i gives an isomorphism $\mathcal{O}_{U_i} \xrightarrow{\sim} f_i\mathcal{O}_{U_i}$, so the ideal is locally free of rank one. Conversely, if the ideal is locally generated by one non-zero-divisor, those generators define compatible effective Cartier local equations because two generators of the same invertible ideal differ by a unit. $\square$

$$\boxed{f \in \mathcal{O}_X(U)} \xrightarrow{\text{non-zero-divisor}} \boxed{\text{ideal } (f)} \xrightarrow{\text{closed subscheme}} \boxed{D = V(f)}$$

$(f) \cong \mathcal{O}_U$ is invertible

Figure 40.3: An effective Cartier divisor is locally principal in the strong sense: its local generator is a non-zero-divisor.

Example 40.15 (Coordinate hyperplane)**.** On $\text{Å}^n_k = \operatorname{Spec} k[x_1, \dots, x_n]$, the equation $x_1 = 0$ cuts out an effective Cartier divisor. Since the coordinate ring is a domain, x_1 is a non-zero-divisor.

Example 40.16 (A principal closed subscheme that is not Cartier)**.** Let

$$X = \operatorname{Spec} k[x, y]/(xy).$$

The closed subscheme $V(x)$ is principal as an ideal, but x is a zero-divisor because $xy = 0$. Hence $V(x)$ is *not* an effective Cartier divisor. Local principality alone is insufficient; the generator must be a non-zero-divisor.

Example 40.17 (A doubled Cartier divisor)**.** If D is effective with local equation f, then $2D$ is effective with local equation f^2. Scheme-theoretically it is the thickening cut out by (f^2), not merely the same reduced support counted twice informally.

Example 40.18 (Hyperplane divisor)**.** On $\mathbb{P}^n$, the hyperplane $H = V(X_0)$ is Cartier. On each standard chart, H is cut out by one nonzerodivisor, and the local equations differ by units on overlaps.

40.6 Support and the exact sequence of an effective divisor

For an effective Cartier divisor $i : D \hookrightarrow X$, the defining ideal gives the exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_D \longrightarrow 0,$$

where $\mathcal{O}_X(-D)$ is initially notation for $\mathcal{I}_D$. The invertible-sheaf construction below will show that this notation is compatible with arbitrary Cartier divisors.

Proposition 40.19 (Support)**.** *For an effective Cartier divisor with local equations f_i, its underlying support is*

$$|D| = \{x \in X : (f_i)_x \notin \mathcal{O}_{X,x}^{\times}\}.$$

It is nowhere dense in every irreducible component on which the equation is defined by a non-zero-divisor.

40.7 Pullback: the correct domain of definition

Cartier divisors are more pullback-friendly than Weil divisors, but not completely automatic.

Definition 40.20 (Defined pullback)**.** Let $g : Y \to X$ be a morphism and let D have local equations (U_i, f_i). We say that g^*D is *defined* if each pulled-back equation

$$g^{\#}(f_i)|_{g^{-1}(U_i)}$$

represents a unit of $\mathcal{K}_Y$. Then these equations define a Cartier divisor g^*D on Y.

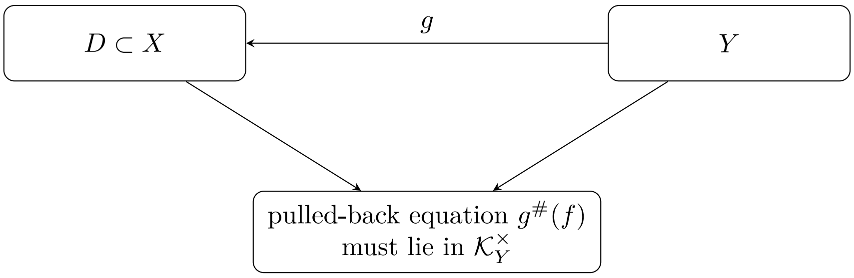
This condition is independent of the chosen local equations because multiplication by a regular unit remains multiplication by a regular unit after pullback.

Proposition 40.21 (Two useful sufficient criteria)**.** *The pullback g^*D is defined in each of the following situations.*

(a) X and Y are integral, g is dominant, and D is any Cartier divisor.

(b) Y is locally Noetherian, D is effective, and no associated point of Y maps into $|D|$.

Proof. For (a), a dominant morphism of integral schemes induces an inclusion of function fields on the generic point, so a nonzero rational equation remains a nonzero rational function. For (b), an effective local equation is a non-zero-divisor exactly when it avoids the associated primes in the locally Noetherian setting. If no associated point maps into the support, its pullback avoids all associated primes of Y, hence remains a non-zero-divisor. □



if the equation becomes 0 or a zero-divisor, Cartier pullback fails

Figure 40.4: Cartier pullback is defined precisely when local rational equations remain valid total-quotient units on the source.

Example 40.22 (Ramification multiplicity)**.** Let $g : \text{Å}^1_u \to \text{Å}^1_t$ be defined by $t = u^e$, and let $D = V(t)$. Then

$$g^*D = V(u^e) = e\,V(u)$$

as Cartier divisors. Pullback records multiplicity automatically.

Example 40.23 (Why pullback can fail)**.** Let $D = V(t)$ on Å^1_t, and let $g : \operatorname{Spec} k \to \text{Å}^1$ be the point $t = 0$. The local equation t pulls back to $0 \in k$, which is not a unit of the total quotient ring k. Thus g^*D is not a Cartier divisor on the point. Geometrically, the entire source lies inside the divisor.

Proposition 40.24 (Functoriality where defined)**.** *Whenever all relevant pullbacks are defined,*

$$g^*(D + E) = g^*D + g^*E, \qquad g^*(f)_{\mathrm{Cart}} = (g^*f)_{\mathrm{Cart}},$$

and for $Z \xrightarrow{h} Y \xrightarrow{g} X$,

$$(g \circ h)^*D = h^*(g^*D).$$

40.8 The Cartier-to-Weil map

Return now to the setting of VI/39: let X be a normal Noetherian integral scheme. Every codimension-one local ring is then a DVR.

Construction 40.25 (From Cartier to Weil)**.** Let D be represented near the generic point η_Z of a prime divisor Z by a local equation f. Define

$$n_Z(D) := \mathrm{ord}_Z(f).$$

If f' is another local equation, then f'/f is a unit, hence has order zero. Therefore $n_Z(D)$ is well defined. Set

$$[D]_{\mathrm{W}} := \sum_{Z \in X^{(1)}} n_Z(D)[Z].$$

Theorem 40.26 (Cartier-to-Weil comparison)**.** *For a normal Noetherian integral scheme X, the preceding construction defines an injective group homomorphism*

$$\mathrm{CaDiv}(X) \hookrightarrow \mathrm{Div}(X).$$

It sends principal Cartier divisors to principal Weil divisors:

$$[(f)_{\mathrm{Cart}}]_{\mathrm{W}} = \mathrm{div}_X(f).$$

If D is effective Cartier, then $[D]_{\mathrm{W}}$ is effective Weil.

Proof. Additivity follows from additivity of each DVR valuation, and compatibility with principal divisors is immediate from the definitions. Effectivity gives regular non-zero-divisor local equations, so their codimension-one orders are nonnegative.

For injectivity, work affine-locally on $U = \operatorname{Spec} A$ with A normal Noetherian domain. If every codimension-one order of a local equation $f \in \mathrm{Frac}(A)^\times$ is zero, then both f and f^{-1} lie in every height-one localization $A_{\mathfrak{p}}$. Normality gives

$$A = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_{\mathfrak{p}} \subseteq \mathrm{Frac}(A),$$

so $f, f^{-1} \in A$, hence $f \in A^\times$. Thus the Cartier divisor is locally zero and therefore globally zero. □

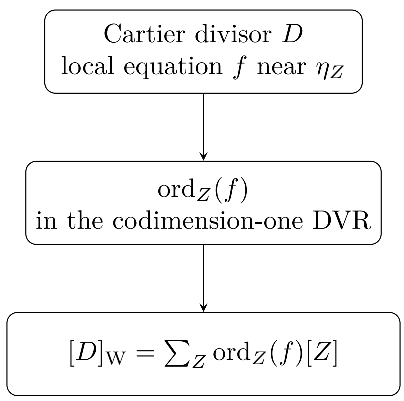


Figure 40.5: On a normal Noetherian integral scheme, codimension-one valuations send Cartier divisors injectively to Weil divisors.

Example 40.27 (The hyperplane in projective space)**.** Let $H = V_+(x_0) \subset \mathbb{P}^n_k$. On $U_i = D_+(x_i)$, a Cartier equation is

$$f_i = \frac{x_0}{x_i}.$$

On overlaps,

$$\frac{f_i}{f_j} = \frac{x_j}{x_i}$$

is a unit. The associated Weil divisor has coefficient one along the prime hyperplane H.

Example 40.28 (Cartier versus Weil on a smooth curve)**.** On a nonsingular curve every closed point is locally principal, so Weil and Cartier divisors agree. A point p is represented by a local parameter vanishing to order one.

40.9 When Weil and Cartier differ

The comparison map need not be surjective. A prime Weil divisor may fail to be locally principal at a singular point.

Definition 40.29 (Locally factorial)**.** A locally Noetherian integral scheme X is *locally factorial* if every local ring $\mathcal{O}_{X,x}$ is a UFD.

Theorem 40.30 (Locally factorial criterion)**.** *If X is locally factorial and Noetherian integral, every Weil divisor on X is Cartier. In particular, every regular Noetherian integral scheme is locally factorial, hence every Weil divisor is Cartier.*

Proof. A prime divisor is locally represented by a height-one prime ideal. In a UFD every height-one prime is principal, generated by an irreducible non-zero-divisor. These local equations make the prime divisor Cartier, and finite integer sums preserve the property. A regular local ring is a UFD, giving the final assertion. ☐

Example 40.31 (A normal singular surface with a non-Cartier Weil divisor)**.** Let

$$X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$$

and let $P = V(x, z)$. VI/39 computed

$$\operatorname{div}(x) = 2P.$$

Thus $2P$ is Cartier, indeed principal. But P itself is not Cartier at the vertex (x, y, z): its ideal (x, z) is not principal in the local ring there. Hence the Cartier-to-Weil map is not surjective. This example foreshadows a nontrivial divisor class in VI/41.

$$\boxed{P = V(x,z)} \xrightarrow{\text{double}} \boxed{2P = \operatorname{div}(x)} \to \boxed{\text{principal Cartier}}$$

$xy = z^2$: P is Weil but not Cartier at the vertex

Figure 40.6: On the quadric cone, a non-Cartier prime Weil divisor can become Cartier after multiplication.

40.10 The invertible sheaf associated to a Cartier divisor

A Cartier divisor automatically produces a line bundle. This is the bridge to VI/42.

Construction 40.32 (The sheaf associated to a Cartier divisor)**.** Let D have local equations $f_i \in \Gamma(U_i, \mathcal{K}_X^\times)$. Define a subsheaf of $\mathcal{K}_X$ by

$$\mathcal{O}_X(D)|_{U_i} := f_i^{-1}\mathcal{O}_{U_i}.$$

On overlaps,

$$f_i^{-1}\mathcal{O} = f_j^{-1}\mathcal{O}$$

because f_i/f_j is a unit. Thus the local modules glue to a globally defined invertible sheaf $\mathcal{O}_X(D)$.

Each f_i^{-1} is a local generator, so $\mathcal{O}_X(D)$ is locally free of rank one.

Proposition 40.33 (Basic identities)**.** *For Cartier divisors D, E, there are canonical isomorphisms*

$$\mathcal{O}_X(D+E) \cong \mathcal{O}_X(D) \otimes \mathcal{O}_X(E), \qquad \mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee.$$

If $D = (f)_{\mathrm{Cart}}$ is principal, multiplication by f identifies

$$\mathcal{O}_X(D) \cong \mathcal{O}_X.$$

Example 40.34 (Effective divisor and its ideal)**.** If D is effective with local equation f_i, then

$$\mathcal{O}_X(-D)|_{U_i} = f_i\mathcal{O}_{U_i}.$$

Therefore

$$\boxed{\mathcal{O}_X(-D) = \mathcal{I}_D}$$

and $\mathcal{O}_X(D) = \mathcal{I}_D^{-1}$. The familiar ideal sheaf of a hypersurface is therefore the negative of its divisor line bundle.

$$\boxed{\text{local equation } f_i} \to \boxed{\mathcal{O}_X(D)|_{U_i} = f_i^{-1}\mathcal{O}} \to \boxed{1 = f_i \cdot f_i^{-1}}$$

the canonical rational section has divisor D

Figure 40.7: A Cartier equation becomes the local coefficient of the canonical rational section of $\mathcal{O}_X(D)$.

40.11 The canonical rational section

Inside $\mathcal{O}_X(D) \subseteq \mathcal{K}_X$, the rational function 1 defines a canonical rational section s_D. Relative to the local generator $e_i = f_i^{-1}$, one has

$$1 = f_i e_i.$$

Thus the local coefficient of s_D is exactly the Cartier equation f_i.

Proposition 40.35 (Divisor of the canonical rational section)**.** *The canonical rational section $s_D = 1$ of $\mathcal{O}_X(D)$ has Cartier divisor D. If D is effective, then s_D is a regular global section and its zero subscheme is precisely D.*

Example 40.36 (The hyperplane bundle)**.** For the hyperplane $H \subset \mathbb{P}^n_k$, the construction gives

$$\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1).$$

The homogeneous coordinate x_0 becomes the canonical section whose zero divisor is H. VI/42 will generalize this relationship between divisors, line bundles, and sections.

40.12 Cartier divisors and rationally trivialized line bundles

The previous construction can be reversed once a line bundle is equipped with a rational trivialization.

Proposition 40.37 (First line-bundle correspondence)**.** *Let $\mathcal{L}$ be an invertible sheaf on X and suppose s is a rational section that generates $\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{K}_X$. Choose local generators e_i of $\mathcal{L}$ and write*

$$s = f_i e_i, \qquad f_i \in \Gamma(U_i, \mathcal{K}_X^\times).$$

Then the f_i define a Cartier divisor $\operatorname{div}(s)$*. Moreover*

$$\mathcal{L} \cong \mathcal{O}_X(\operatorname{div}(s)).$$

Changing s by a global rational unit changes the divisor by a principal Cartier divisor.

This is the first appearance of the bridge

$$\boxed{\text{Cartier divisors modulo principal divisors} \longleftrightarrow \text{invertible sheaves with isomorphism classes}},$$

but the quotient and its relation to $\operatorname{Pic}(X)$ belong to VI/41–VI/42.

$$\boxed{\begin{array}{c}\text{invertible sheaf } \mathcal{L}\\ + \text{ rational generator } s\end{array}} \xleftrightarrow{\text{local coefficients } f_i} \boxed{\begin{array}{c}\text{Cartier divisor}\\ \operatorname{div}(s)\end{array}}$$

$s \mapsto fs$ changes the divisor by the principal Cartier divisor (f)

Figure 40.8: A rationally trivialized line bundle carries exactly the local data of a Cartier divisor.

40.13 Worked examples

Example 40.38 (A divisor on Å^2)**.** On $X =$ Å^2_k, the polynomial $f = x^2y$ defines the principal effective Cartier divisor (f). Under Cartier-to-Weil,

$$[(f)]_{\mathrm{W}} = 2V(x) + V(y).$$

Its ideal sheaf is (x^2y), and $\mathcal{O}_X((f)) \cong \mathcal{O}_X$ because the divisor is principal.

Example 40.39 (Difference of two effective Cartier divisors)**.** On $\mathbb{P}^1$, let $D = [0]$ and $E = [\infty]$. Then $D - E$ is Cartier and principal:

$$D - E = (t)_{\mathrm{Cart}}.$$

Consequently

$$\mathcal{O}_{\mathbb{P}^1}(D - E) \cong \mathcal{O}_{\mathbb{P}^1}.$$

This is the line-bundle shadow of the relation $[0] \sim [\infty]$.

Example 40.40 (Pulling back a hyperplane)**.** Let $\phi : \mathbb{P}^n \to \mathbb{P}^m$ be a morphism defined by homogeneous forms $F_0, \dots, F_m$ of common degree d with no base points. If $H_j = V_+(y_j) \subset \mathbb{P}^m$ and $F_j \not\equiv 0$, then

$$\phi^* H_j = V_+(F_j)$$

as an effective Cartier divisor, with multiplicities determined by the factorization of F_j. Correspondingly,

$$\phi^* \mathcal{O}_{\mathbb{P}^m}(1) \cong \mathcal{O}_{\mathbb{P}^n}(d).$$

Example 40.41 (A smooth curve)**.** On a smooth integral curve, every closed point is an effective Cartier divisor because its local ring is a DVR. Hence every Weil divisor is Cartier. The distinction between the two theories appears only when singularities or higher-dimensional local factoriality enter.

40.14 Exercises

Exercise 40.42. Show directly from local equations that $\mathrm{CaDiv}(X)$ is an abelian group.

Exercise 40.43. Prove that a Cartier divisor represented everywhere by regular units is zero.

Exercise 40.44. On Å^1_k, compute the Cartier-to-Weil image of $(t^3(t-1)^{-2})_{\mathrm{Cart}}$.

Exercise 40.45. Show that $V(x^m) \subset \text{Å}^n$ is an effective Cartier divisor for every $m \geq 1$.

Exercise 40.46. In $k[x, y]/(xy)$, show that neither x nor y defines an effective Cartier divisor.

Exercise 40.47. Let D be effective Cartier. Prove that nD is effective for every $n \geq 1$.

Exercise 40.48. Prove $\mathcal{O}_X(-D) = \mathcal{I}_D$ for an effective Cartier divisor.

Exercise 40.49. Show that $\mathcal{O}_X(D + E) \cong \mathcal{O}_X(D) \otimes \mathcal{O}_X(E)$.

Exercise 40.50. For $g : \text{Å}^1_u \to \text{Å}^1_t$, $t = u^4$, compute $g^* V(t)$.

Exercise 40.51. Explain why the inclusion $V(t) \hookrightarrow \text{Å}^1_t$ does not admit pullback of the Cartier divisor $V(t)$.

Exercise 40.52. Show that the pullback of a principal Cartier divisor, when defined, is principal.

Exercise 40.53. On a normal integral scheme, prove that equivalent Cartier local equations have equal order along every prime divisor.

Exercise 40.54. Show that effective Cartier divisors map to effective Weil divisors on normal schemes.

Exercise 40.55. Prove that a principal Cartier divisor maps to the principal Weil divisor of the same rational function.

Exercise 40.56. Show that every prime divisor on $\mathbb{A}^n_k$ is Cartier.

Exercise 40.57. Let X be regular Noetherian integral. Explain why every Weil divisor is Cartier.

Exercise 40.58. On $\mathbb{P}^n$, verify explicitly that x_0/x_i and x_0/x_j differ by a unit on $U_i \cap U_j$.

Exercise 40.59. Prove that $\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1)$ for a hyperplane H.

Exercise 40.60. Show that if D is principal then $\mathcal{O}_X(D)$ is trivial.

Exercise 40.61. If D and E are linearly equivalent via $D - E = (f)$, show $\mathcal{O}_X(D) \cong \mathcal{O}_X(E)$.

Exercise 40.62. Let D be effective Cartier. Show that the canonical rational section of $\mathcal{O}_X(D)$ is regular.

Exercise 40.63. Show that its zero subscheme is exactly D.

Exercise 40.64. Let s be a rational trivializing section of a line bundle. Show that replacing s by fs changes its divisor by (f).

Exercise 40.65. Show that on a smooth integral curve every closed point is an effective Cartier divisor.

40.15 Problem dossiers

Problem 40.66 (Local equations on $\mathbb{P}^1$)**.** Construct explicit Cartier local equations for $[a] - [\infty]$ on $\mathbb{P}^1$.

Problem 40.67 (A nonreduced effective divisor)**.** Describe the effective Cartier divisor $2V(x) \subset \mathbb{A}^2$ scheme-theoretically.

Problem 40.68 (Pullback and multiplicity)**.** Let $g : \mathbb{A}^1_u \to \mathbb{A}^1_t$ be $t = u^3(u-1)^2$. Pull back $V(t)$.

Problem 40.69 (A failed pullback)**.** Let $Y = V(t) \subset \mathbb{A}^2_{t,u}$ and let $D = V(t)$ on $\mathbb{A}^2$. Explain why $D|_Y$ is not a Cartier pullback.

Problem 40.70 (Cartier-to-Weil on a UFD)**.** Let A be a UFD and $f = p_1^{a_1} \cdots p_r^{a_r} / q_1^{b_1} \cdots q_s^{b_s} \in \operatorname{Frac}(A)^\times$. Compute the associated Weil divisor.

Problem 40.71 (Why P on the quadric cone is not Cartier)**.** For $X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$ and $P = V(x, z)$, explain the obstruction at the vertex.

Problem 40.72 (Tensor law for divisor sheaves)**.** Prove the tensor identity for $\mathcal{O}(D + E)$ directly from transition functions.

Problem 40.73 (Hyperplane pullback under the Veronese map)**.** Let $\nu_d : \mathbb{P}^n \to \mathbb{P}^N$ be the d-uple Veronese map. Show that the pullback of a coordinate hyperplane is a degree-d hypersurface and identify the line bundle pullback.

Problem 40.74 (Cartier divisor from a line-bundle section)**.** Let s be a regular section of a line bundle $\mathcal{L}$ whose local coefficients are non-zero-divisors. Construct its zero divisor.

Problem 40.75 (Restriction and line bundles)**.** Let $j : U \hookrightarrow X$ be open. Show $j^*\mathcal{O}_X(D) \cong \mathcal{O}_U(D|_U)$.

Problem 40.76 (Effective Cartier divisors are pure codimension one)**.** Let X be locally Noetherian and $D \subset X$ effective Cartier. Explain why every irreducible component of D meeting an irreducible component of X has codimension one there.

Problem 40.77 (Recovering a divisor from $\mathcal{O}(D)$ plus rational section)**.** Suppose $\mathcal{L} \cong \mathcal{O}_X(D)$ but no rational section is specified. Explain why the line bundle alone does not recover D uniquely.

40.16 Challenges

Problem 40.78 (Challenge: codimension-one injectivity)**.** Give a complete affine proof that the Cartier-to-Weil map is injective on a normal Noetherian integral scheme, including the identity $A = \bigcap_{\operatorname{ht}\mathfrak{p}=1} A_{\mathfrak{p}}$ inside $\operatorname{Frac}(A)$.

Problem 40.79 (Challenge: associated points and pullback)**.** Prove carefully that if an effective Cartier divisor $D \subset X$ contains no image of an associated point of a locally Noetherian scheme Y, then its pullback along $Y \to X$ is effective Cartier.

Problem 40.80 (Challenge: the quadric cone)**.** For $X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$, prove directly that (x, z) is not principal at the vertex and that $2P = \operatorname{div}(x)$. Interpret the result as the first evidence for torsion in a divisor class group.

Problem 40.81 (Challenge: projective hypersurfaces)**.** Let F be a nonzero homogeneous polynomial of degree d on $\mathbb{P}^n_k$. Construct the effective Cartier divisor $V_+(F)$ from local equations, prove $\mathcal{O}(V_+(F)) \cong \mathcal{O}(d)$, and track factor multiplicities when F is reducible.

Problem 40.82 (Challenge: rational sections)**.** Formulate and prove the equivalence between Cartier divisors on an integral scheme and isomorphism classes of pairs $(\mathcal{L}, s)$, where $\mathcal{L}$ is invertible and s is a nonzero rational section generating $\mathcal{L} \otimes K(X)$. Identify how multiplying s by $f \in K(X)^\times$ changes the divisor.

Bridge to VI/41

Cartier divisors now have three equivalent-looking faces that will be separated and compared in the next chapters:

$$\boxed{\text{local rational equations} \longleftrightarrow \text{Cartier divisor} \longrightarrow \mathcal{O}_X(D)}$$

and, on normal schemes,

$$\boxed{\text{Cartier divisor} \hookrightarrow \text{Weil divisor.}}$$

VI/41 will quotient Weil divisors by principal divisors to construct the divisor class group and measure the failure of factoriality. VI/42 will then return to the invertible-sheaf side and identify the corresponding Picard group.

40.17 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 40.83 (Principal data)**.** What Cartier divisor is defined by a rational function f?

Hint. Use local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. The principal Cartier divisor (f). ☐

Exercise 40.84 (Unit change)**.** Do f and uf with u a unit define the same local Cartier divisor?

Hint. Use local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. Yes. ☐

Exercise 40.85 (Hyperplane)**.** What line bundle is associated to a hyperplane divisor on $\mathbb{P}^n$?

Hint. Use local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. $\mathcal{O}_{\mathbb{P}^n}(1)$. ☐

Exercise 40.86 (Local equation)**.** What property must define an effective Cartier divisor locally?

Hint. Use local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. A single nonzerodivisor equation. ☐

Exercise 40.87 (Curve point)**.** How is a smooth curve point represented Cartier-locally?

Hint. Use local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. By a local parameter. ☐

Proofs

Exercise 40.88 (Overlap condition)**.** Prove: Prove local equations differing by units glue to Cartier divisor data.

Hint. Work from local equations, invertible meromorphic functions, and Cartier data and state the hypotheses explicitly. ☐

Solution. The ratios are invertible regular functions and satisfy the cocycle relation on triple overlaps. ☐

Exercise 40.89 (Principal triviality)**.** Prove: Prove a principal Cartier divisor is zero in the Cartier divisor class group.

Hint. Work from local equations, invertible meromorphic functions, and Cartier data and state the hypotheses explicitly. ☐

Solution. It is globally represented by one rational function. ☐

Exercise 40.90 (Effective support)**.** Prove: Prove an effective Cartier divisor has codimension one on a reasonable Noetherian scheme when nonempty.

Hint. Work from local equations, invertible meromorphic functions, and Cartier data and state the hypotheses explicitly. ☐

Solution. Use the principal ideal theorem locally for a nonzerodivisor. ☐

Exercise 40.91 (Weil comparison)**.** Prove: On a normal integral scheme, explain how a Cartier divisor determines a Weil divisor.

Hint. Work from local equations, invertible meromorphic functions, and Cartier data and state the hypotheses explicitly. ☐

Solution. Take the valuation/order of each local defining rational function along codimension-one points; unit changes contribute zero. ☐

Counterexamples and hypothesis tests

Exercise 40.92 (Every Weil Cartier)**.** Test the claim: every Weil divisor is Cartier on every normal scheme.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; normality does not imply local factoriality. ☐

Exercise 40.93 (Zero divisor)**.** Test the claim: any single local equation defines an effective Cartier divisor.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the equation must be a nonzerodivisor. ☐

Exercise 40.94 (Global equation)**.** Test the claim: every Cartier divisor has one global defining equation.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; Cartier divisors are generally only locally principal. ☐

Applications and investigations

Exercise 40.95 (Line bundles)**.** How does a Cartier divisor produce a line bundle?

Hint. Translate the question into local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. Local equations define transition functions whose associated invertible sheaf is $\mathcal{O}(D)$. ☐

Exercise 40.96 (Intersection theory)**.** Why are Cartier divisors convenient for intersections?

Hint. Translate the question into local equations, invertible meromorphic functions, and Cartier data. ☐

Solution. Local principal equations allow pullback and intersection constructions with good functorial behavior. ☐

Challenge problems

Exercise 40.97 (Synthesis challenge)**.** Connect two central viewpoints of Cartier Divisors in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Cartier Divisors and then translates it through local equations, invertible meromorphic functions, and Cartier data, checking both descriptions agree. ☐

Exercise 40.98 (Hypothesis challenge)**.** Choose one theorem-level statement from Cartier Divisors, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 41

Divisor Class Groups

VI/39 attached an integer multiplicity to every codimension-one prime and constructed Weil divisors. VI/40 developed Cartier divisors from local rational equations and proved that, on a normal Noetherian integral scheme, every Cartier divisor determines a Weil divisor. The present chapter takes the decisive quotient: divisors differing by the divisor of one rational function represent the same class.

The resulting group
$$\mathrm{Cl}(X) = \mathrm{Div}(X)/\,\mathrm{Prin}(X)$$
measures a global obstruction. On an affine normal Noetherian domain it detects the failure of unique factorization; on an open subset it records exactly which codimension-one classes survive localization; and on projective space it recovers the integer degree generated by a hyperplane. The quadric cone

$$X = \operatorname{Spec} k[x, y, z]/(xy - z^2)$$

is the running example. Its prime divisor $L = V(x, z)$ satisfies $2L = \mathrm{div}(x)$ but is not principal, and the full computation gives

$$\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$$

This single example distinguishes normality from factoriality, shows how torsion enters the class group, and explains why the singular vertex matters even though the punctured cone is smooth.

The migration ledger assigns three canonical rules to VI/41: the AG9 class-group/divisor-class section and its inherited theorem-like and exercise/problem descendants. The accessible repository does not expose the raw AG9 prose as a canonical chapter source, so the ledger is used as provenance and scope evidence rather than as a basis for claiming verbatim recovery. The full Picard-group equivalence is deliberately deferred to VI/42.

Learning goals

By the end of the chapter, the reader should be able to:

- define linear equivalence of Weil divisors and construct $\mathrm{Cl}(X)$;
- interpret $\mathrm{Cl}(A)$ in terms of height-one primes of a normal Noetherian domain;
- prove that a normal Noetherian domain is a UFD exactly when its class group vanishes;
- distinguish factoriality from local factoriality and explain the role of Cartier divisors;

- prove the localization exact sequence for divisor class groups;
- compute $\mathrm{Cl}(A_f)$ by killing the classes of height-one primes containing f;
- compute the divisor class group of the quadric cone as $\mathbb{Z}/2\mathbb{Z}$;
- prove rigorously that the cone divisor $L = V(x, z)$ is not principal;
- explain why deleting a codimension-at-least-two subset does not change the Weil divisor class group of a normal scheme;
- compute $\mathrm{Cl}(\mathbb{A}^n) = 0$ and $\mathrm{Cl}(\mathbb{P}^n) \cong \mathbb{Z}$;
- recognize torsion divisor classes and the first appearance of $\mathbb{Q}$-Cartier phenomena;
- identify the precise bridge from divisor classes to the Picard group developed in VI/42.

Conceptual roadmap

$$\boxed{\begin{gathered} K(X)^\times \xrightarrow{\text{div}} \mathrm{Div}(X) \xrightarrow{\text{quotient}} \mathrm{Cl}(X) \longrightarrow 0 \\ X = \operatorname{Spec} A,\ A \text{ normal Noetherian} \\ \mathrm{Cl}(A) = 0 \quad \iff \quad A \text{ UFD} \\ S^{-1}A: \quad \mathrm{Cl}(A) \twoheadrightarrow \mathrm{Cl}(S^{-1}A), \quad \text{classes meeting } S \text{ die} \\ k[x,y,z]/(xy - z^2): \quad \mathrm{div}(x) = 2L \quad \Longrightarrow \quad \mathrm{Cl} \cong \mathbb{Z}/2. \end{gathered}}$$

$$\boxed{K(X)^\times} \xrightarrow{\text{div}} \boxed{\mathrm{Div}(X)} \xrightarrow{\text{quotient}} \boxed{\mathrm{Cl}(X)}$$

principal divisors become zero; surviving classes measure
global codimension-one obstructions

Figure 41.1: The divisor class group is the quotient of Weil divisors by the relations supplied by rational functions.

41.1 Linear equivalence and principal divisors

Let X be a normal Noetherian integral scheme with function field $K(X)$. By VI/39,

$$\mathrm{Div}(X) = \bigoplus_{Z \in X^{(1)}} \mathbb{Z}[Z],$$

where $X^{(1)}$ denotes the codimension-one points, equivalently the prime divisors.

Every $f \in K(X)^\times$ determines the principal Weil divisor

$$\mathrm{div}(f) = \sum_{Z \in X^{(1)}} \mathrm{ord}_Z(f)[Z].$$

The valuation identity

$$\mathrm{div}(fg) = \mathrm{div}(f) + \mathrm{div}(g)$$

shows that principal divisors form a subgroup.

Definition 41.1 (Principal divisor subgroup)**.** Define

$$\mathrm{Prin}(X) := \{\mathrm{div}(f) : f \in K(X)^\times\} \subseteq \mathrm{Div}(X).$$

Definition 41.2 (Linear equivalence)**.** Two Weil divisors $D, E \in \mathrm{Div}(X)$ are *linearly equivalent*, written

$$D \sim E,$$

if

$$D - E = \mathrm{div}(f)$$

for some $f \in K(X)^\times$.

Thus $D \sim 0$ exactly when D is principal. Linear equivalence is an equivalence relation compatible with addition.

Example 41.3 (The affine line)**.** On $\text{Å}^1_k = \operatorname{Spec} k[t]$, if $a, b \in k$, then

$$\mathrm{div}\left(\frac{t-a}{t-b}\right) = [a] - [b].$$

Hence every two k-rational closed points are linearly equivalent. In fact every Weil divisor on Å^1_k is principal because $k[t]$ is a PID.

41.2 The divisor class group

Definition 41.4 (Divisor class group)**.** The *divisor class group* of X is

$$\boxed{\mathrm{Cl}(X) := \mathrm{Div}(X)/\,\mathrm{Prin}(X).}$$

The class of a divisor D is denoted $[D]$.

Equivalently,

$$[D] = [E] \quad \Longleftrightarrow \quad D \sim E,$$

and

$$[D] = 0 \quad \Longleftrightarrow \quad D \text{ is principal.}$$

The definition packages the exact sequence

$$K(X)^\times \xrightarrow{\mathrm{div}} \mathrm{Div}(X) \longrightarrow \mathrm{Cl}(X) \longrightarrow 0.$$

The kernel of the divisor map consists of rational functions having no zeros or poles in codimension one. On many familiar complete varieties this kernel is just the constant functions; on affine schemes it may be larger because nonconstant units can occur.

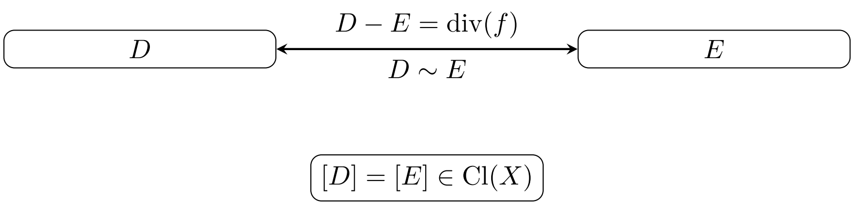


Figure 41.2: Linear equivalence identifies exactly those divisors whose difference is principal.

Remark 41.5 (What the class group forgets). The group $\mathrm{Div}(X)$ remembers every codimension-one prime separately. Passing to $\mathrm{Cl}(X)$ imposes exactly the relations produced by rational functions. It does not identify divisors merely because they have the same support, the same degree, or the same local multiplicities at selected points.

41.3 The affine description by height-one primes

Let A be a normal Noetherian domain and $X = \operatorname{Spec} A$. Prime divisors correspond to height-one prime ideals, so

$$\mathrm{Div}(A) = \bigoplus_{\mathrm{ht}\,\mathfrak{p}=1} \mathbb{Z}[\mathfrak{p}].$$

For $f \in \mathrm{Frac}(A)^\times$,

$$\mathrm{div}_A(f) = \sum_{\mathrm{ht}\,\mathfrak{p}=1} v_{\mathfrak{p}}(f)[\mathfrak{p}],$$

where $A_{\mathfrak{p}}$ is a DVR.

We write

$$\mathrm{Cl}(A) := \mathrm{Cl}(\operatorname{Spec} A).$$

Example 41.6 (A PID)**.** If A is a PID, every height-one prime is generated by one prime element. Hence every prime divisor is principal and

$$\mathrm{Cl}(A) = 0.$$

Example 41.7 (Polynomial rings)**.** The ring $k[x_1, \dots, x_n]$ is a UFD. Therefore

$$\mathrm{Cl}(\text{Å}^n_k) = 0.$$

The connection between UFDs and the class group will now be proved in general.

Example 41.8 (Class group of a UFD)**.** If A is a UFD, every height-one prime is principal. For a normal affine UFD this forces every Weil divisor to be linearly equivalent to zero, so $\mathrm{Cl}(A) = 0$.

41.4 Factoriality and the class group

Theorem 41.9 (Class-group criterion for unique factorization)**.** *Let A be a normal Noetherian domain. Then*

$$\boxed{A \text{ is a UFD} \iff \mathrm{Cl}(A) = 0.}$$

Proof. Suppose first that A is a UFD. Every height-one prime $\mathfrak{p}$ is generated by a prime element p. The principal divisor $\mathrm{div}(p)$ has coefficient one at $\mathfrak{p}$ and zero at every other height-one prime, so

$$\mathrm{div}(p) = [\mathfrak{p}].$$

Since the height-one prime divisors freely generate $\mathrm{Div}(A)$, every Weil divisor is principal and $\mathrm{Cl}(A) = 0$.

Conversely, assume $\mathrm{Cl}(A) = 0$. Then every height-one prime divisor $[\mathfrak{p}]$ is principal as a Weil divisor. Thus there exists $f \in \mathrm{Frac}(A)^\times$ with

$$\mathrm{div}(f) = [\mathfrak{p}].$$

The Krull-domain identity

$$A = \bigcap_{\mathrm{ht}\,\mathfrak{q}=1} A_{\mathfrak{q}} \subseteq \mathrm{Frac}(A)$$

shows that $f \in A$: all height-one valuations of f are nonnegative. If $a \in \mathfrak{p}$, then $v_{\mathfrak{p}}(a) \geq 1$, so a/f has nonnegative valuation at every height-one prime and hence lies in A. Therefore $\mathfrak{p} = (f)$.

A Noetherian domain is a UFD exactly when every height-one prime ideal is principal. Hence A is a UFD. $\square$

Corollary 41.10 (Affine factoriality)**.** *For a normal Noetherian affine scheme $X = \operatorname{Spec} A$,*

$$X \text{ factorial} \quad \Longleftrightarrow \quad \mathrm{Cl}(X) = 0.$$

Warning 41.11 (Normality is not optional)*.* The class group used here is the Weil divisor class group, whose clean valuation theory depends on normality. The slogan

$$\mathrm{Cl}(A) = 0 \Longleftrightarrow A \text{ is a UFD}$$

should not be applied blindly to arbitrary nonnormal domains.

$$\begin{tikzcd}
\boxed{A \text{ normal Noetherian UFD}} \arrow[d, leftrightarrow] \\
\boxed{\mathrm{Cl}(A) = 0} \arrow[d, leftrightarrow] \\
\boxed{\text{every height-one prime principal}}
\end{tikzcd}$$

Figure 41.3: For a normal Noetherian domain, vanishing of the divisor class group is exactly unique factorization.

41.5 Factorial versus locally factorial

Definition 41.12 (Locally factorial)**.** A scheme X is *locally factorial* if every local ring $\mathcal{O}_{X,x}$ is a UFD.

If A is a UFD, every localization $A_{\mathfrak{p}}$ is a UFD, so an affine factorial scheme is locally factorial. The converse fails: local principalness does not imply that a divisor is globally principal.

VI/40 proved that on a normal Noetherian integral scheme,

$$X \text{ locally factorial} \quad \Longleftrightarrow \quad \text{every Weil divisor is Cartier.}$$

Thus there are two different obstructions:

$$\begin{aligned}
\text{Weil divisor not Cartier} \quad &\Longleftrightarrow \quad \text{failure of local factoriality,} \\
\text{Cartier divisor not principal} \quad &\Longleftrightarrow \quad \text{global line-bundle obstruction.}
\end{aligned}$$

The second obstruction will become the Picard group in VI/42.

Example 41.13 (Smooth does not force global factoriality)**.** A smooth variety is locally factorial because regular local rings are UFDs. Nevertheless its global divisor class group can be nonzero. Projective space is the basic example:

$$\mathrm{Cl}(\mathbb{P}^n_k) \cong \mathbb{Z}.$$

Thus local factoriality and vanishing of the global class group are fundamentally different statements.

41.6 Localization of divisor classes

Let A be a normal Noetherian domain and $S \subseteq A$ a multiplicative subset. The localization $S^{-1}A$ is again normal. A height-one prime $\mathfrak{p}$ survives as a height-one prime of $S^{-1}A$ exactly when

$$\mathfrak{p} \cap S = \varnothing.$$

If $\mathfrak{p} \cap S \neq \varnothing$, then its divisor disappears after localization.

Theorem 41.14 (Localization exact sequence)**.** *There is a natural exact sequence*

$$\bigoplus_{\substack{\operatorname{ht}\mathfrak{p}=1 \\ \mathfrak{p}\cap S\neq\varnothing}} \mathbb{Z}[\mathfrak{p}] \longrightarrow \mathrm{Cl}(A) \longrightarrow \mathrm{Cl}(S^{-1}A) \longrightarrow 0.$$

Equivalently,

$$\boxed{\mathrm{Cl}(S^{-1}A) \cong \mathrm{Cl}(A)\Big/ \langle [\mathfrak{p}] : \operatorname{ht}\mathfrak{p} = 1,\ \mathfrak{p} \cap S \neq \varnothing \rangle\,.}$$

Proof. Every height-one prime of $S^{-1}A$ is uniquely of the form $S^{-1}\mathfrak{p}$ for a height-one prime $\mathfrak{p}$ disjoint from S. Thus restriction of divisors gives a surjection

$$\mathrm{Div}(A) \twoheadrightarrow \mathrm{Div}(S^{-1}A)$$

whose kernel is the free subgroup generated by height-one primes meeting S.

For $f \in \mathrm{Frac}(A)^\times = \mathrm{Frac}(S^{-1}A)^\times$, the principal divisor in the localization is obtained by deleting from $\mathrm{div}_A(f)$ the components meeting S. Hence restriction sends principal divisors to principal divisors and induces a surjection on class groups. Its kernel is precisely the subgroup generated by the deleted prime-divisor classes. □

$$\begin{tikzcd} \boxed{\mathrm{Cl}(A)} \arrow[r, "\text{surjective}"] & \boxed{\mathrm{Cl}(S^{-1}A)} \end{tikzcd}$$

kill $[\mathfrak{p}]$ whenever
$\mathfrak{p} \cap S \neq \varnothing$

Figure 41.4: Localization forgets precisely the codimension-one prime classes meeting the multiplicative set.

Corollary 41.15 (Principal localization)**.** *For $f \in A$,*

$$\mathrm{Cl}(A_f) \cong \mathrm{Cl}(A)\Big/ \langle [\mathfrak{p}] : \operatorname{ht}\mathfrak{p} = 1,\ f \in \mathfrak{p} \rangle\,.$$

Example 41.16 (Laurent polynomial rings)**.** Since $k[x]$ is a UFD,

$$\mathrm{Cl}(k[x, x^{-1}]) = 0.$$

More generally, localization of a UFD is a UFD, exactly as the class-group localization theorem predicts.

41.7 Open subsets and deleted divisors

The geometric form of localization is equally important.

Theorem 41.17 (Class group of an open subset)**.** *Let X be a normal Noetherian integral scheme and $j : U \hookrightarrow X$ a nonempty open subset. If $D_1, \dots, D_r$ are the codimension-one irreducible components of $X \setminus U$, then*

$$\bigoplus_{i=1}^{r} \mathbb{Z}[D_i] \longrightarrow \mathrm{Cl}(X) \xrightarrow{j^*} \mathrm{Cl}(U) \longrightarrow 0$$

is exact.

Corollary 41.18 (Removing codimension at least two)**.** *If*

$$\mathrm{codim}_X(X \setminus U) \geq 2,$$

then restriction induces an isomorphism

$$\boxed{\mathrm{Cl}(X) \xrightarrow{\sim} \mathrm{Cl}(U).}$$

Proof. There are no codimension-one components in the complement, so the leftmost direct sum in theorem 41.17 is zero. □

This codimension-two invariance is one of the characteristic features of Weil divisor theory: Weil divisors only see codimension one.

41.8 The quadric cone: setup and normality

Let

$$A = k[x, y, z]/(xy - z^2), \qquad X = \mathrm{Spec}\, A.$$

The ring has dimension two. Its singular point is the vertex

$$o = V(x, y, z).$$

Proposition 41.19 (Normality of the quadric cone)**.** *The ring A is a normal domain.*

Proof. The polynomial $xy - z^2$ is irreducible, so A is a domain. As a hypersurface, A is Cohen–Macaulay, hence satisfies Serre's condition S_2. Away from the vertex one of x, y, z is nonzero, and the corresponding local chart is regular; thus the singular locus has codimension two. Hence A satisfies R_1. Serre's criterion $R_1 + S_2$ gives normality. □

Define two height-one primes

$$\mathfrak{p}_L = (x, z), \qquad \mathfrak{p}_M = (y, z),$$

with corresponding prime divisors

$$L = V(x, z), \qquad M = V(y, z).$$

Example 41.20 (A nontrivial class on a quadric cone)**.** For the normal quadric cone $A = k[x, y, z]/(xy - z^2)$, the height-one prime (x, z) defines a divisor class that detects failure of unique factorization. It is the standard model where divisor class groups distinguish normality from factoriality.

$$\begin{tikzcd}
 & X = V(xy - z^2) \subset \text{Å}^3 \arrow[dl] \arrow[dr] & \\
L = V(x, z) \arrow[dr] & & M = V(y, z) \arrow[dl] \\
 & o &
\end{tikzcd}$$

the two prime divisors meet at the singular vertex

Figure 41.5: The two distinguished ruling divisors on the quadric cone meet at the vertex.

41.9 The three key divisor computations

Proposition 41.21 (Coordinate divisors on the cone)**.** *On X,*

$$\boxed{\operatorname{div}(x) = 2L, \qquad \operatorname{div}(y) = 2M, \qquad \operatorname{div}(z) = L + M.}$$

Proof. At the generic point of L, the element y is a unit and

$$x = \frac{z^2}{y}.$$

The DVR $A_{\mathfrak{p}_L}$ has uniformizer z, so

$$v_L(z) = 1, \qquad v_L(x) = 2.$$

The zero locus of x has only the codimension-one component L, hence

$$\operatorname{div}(x) = 2L.$$

By symmetry,

$$\operatorname{div}(y) = 2M.$$

Finally, $z = 0$ together with $xy = z^2$ gives $xy = 0$, so the codimension-one zero locus of z is $L \cup M$. At the generic point of either component, z is a uniformizer, giving multiplicity one. Thus

$$\operatorname{div}(z) = L + M.$$

□

Consequently,

$$2[L] = 0, \qquad 2[M] = 0, \qquad [L] + [M] = 0.$$

Since $2[L] = 0$, we have $-[L] = [L]$, and therefore

$$[M] = [L].$$

The computation already shows that the class of L, if nonzero, has order exactly two.

41.10 Why the cone class group is generated by L

Localize at x. Since $xy = z^2$,

$$y = \frac{z^2}{x}$$

in A_x, and therefore

$$A_x \cong k[x, x^{-1}, z].$$

This is a Laurent polynomial ring, hence a UFD, so

$$\mathrm{Cl}(A_x) = 0.$$

Which height-one primes disappear when x is inverted? The radical of (x) is

$$\sqrt{(x)} = (x, z) = \mathfrak{p}_L,$$

so L is the unique height-one prime containing x.

Applying theorem 41.15 gives

$$\mathrm{Cl}(A_x) \cong \mathrm{Cl}(A)/\langle [L] \rangle.$$

Since the left side is zero,

$$\boxed{\mathrm{Cl}(A) = \langle [L] \rangle.}$$

Thus it remains only to prove that $[L] \neq 0$.

41.11 Why L is not principal

The relation

$$\mathrm{div}(x) = 2L$$

does not by itself exclude L from being principal. We now prove nontriviality.

Theorem 41.22 (Nonprincipality of the ruling divisor)**.** *The prime divisor $L = V(x, z)$ is not principal as a Weil divisor. Hence*

$$[L] \neq 0 \in \mathrm{Cl}(A).$$

Proof. Suppose

$$L = \mathrm{div}(f)$$

for some $f \in K(X)^\times$. Then

$$\mathrm{div}\left(\frac{f^2}{x}\right) = 2L - 2L = 0.$$

For a normal affine domain, an element of the fraction field whose valuations at every height-one prime are all zero is a unit: both it and its inverse lie in

$$A = \bigcap_{\mathrm{ht}\,\mathfrak{p}=1} A_{\mathfrak{p}}.$$

The ring A is a connected positively graded k-domain, so

$$A^\times = k^\times.$$

Therefore

$$f^2 = cx$$

for some $c \in k^\times$.

But

$$K(X) = \mathrm{Frac}(A) = k(x, z),$$

because $y = z^2/x$. In the rational function field $k(x, z)$, the discrete valuation associated with the irreducible polynomial x satisfies

$$v_x(cx) = 1.$$

Every square has even v_x-valuation. Hence cx cannot be a square in $k(x, z)$, a contradiction.

Therefore L is not principal. ☐

This proof is stronger than the informal observation that the parametrizing square root of x is missing from the coordinate ring: it rules out every rational function, not merely every regular function, as a possible equation for L.

41.12 The class group of the quadric cone

Combining generation, the relation $2[L] = 0$, and nontriviality gives the central computation.

Theorem 41.23 (Class group of the quadric cone)**.** *For*

$$A = k[x, y, z]/(xy - z^2),$$

$$\boxed{\mathrm{Cl}(A) \cong \mathbb{Z}/2\mathbb{Z},}$$

with generator $[L]$, where $L = V(x, z)$.

Proof. Localization at x shows that $[L]$ generates $\mathrm{Cl}(A)$. The principal divisor

$$\mathrm{div}(x) = 2L$$

gives $2[L] = 0$. By theorem 41.22, $[L] \neq 0$. Therefore $[L]$ has exact order two. ☐

Corollary 41.24 (Normal but not factorial)**.** *The quadric cone is normal but not factorial.*

Proof. Normality is theorem 41.19; nonfactoriality follows from

$$\mathrm{Cl}(A) \cong \mathbb{Z}/2\mathbb{Z} \neq 0$$

and theorem 41.9. ☐

$$\begin{tikzcd}
\boxed{A_x \text{ is a UFD}} \arrow[r] & \boxed{\mathrm{Cl}(A) = \langle [L] \rangle} \arrow[r] \arrow[d] & \boxed{2[L] = 0} \arrow[d] \\
 & \boxed{[L] \neq 0} \arrow[r] & \boxed{\mathrm{Cl}(A) \cong \mathbb{Z}/2}
\end{tikzcd}$$

Figure 41.6: The quadric-cone computation: localization gives the generator, a principal divisor gives the relation, and nonprincipality proves the class survives.

41.13 The singular vertex and the punctured cone

Let

$$U = X \setminus \{o\}.$$

Since X has dimension two, the removed vertex has codimension two. Hence

$$\mathrm{Cl}(X) \xrightarrow{\sim} \mathrm{Cl}(U)$$

by theorem 41.18. Therefore

$$\boxed{\mathrm{Cl}(U) \cong \mathbb{Z}/2\mathbb{Z}.}$$

At the same time, U is smooth, so it is locally factorial. Thus every Weil divisor on U is Cartier, yet the class group is still nonzero. This is the cleanest warning against confusing local factoriality with global factoriality.

The two facts

$$\mathrm{Cl}(D(x)) = 0, \qquad \mathrm{Cl}(U) = \mathbb{Z}/2$$

are not contradictory. The principal open $D(x)$ deletes the codimension-one divisor L, so its class is killed. The punctured cone deletes only the codimension-two vertex, so no Weil prime divisor is removed.

Remark 41.25 (Preview of VI/42). Because U is locally factorial, every Weil divisor class on U is represented by a Cartier divisor. VI/42 will identify this class with a nontrivial line bundle and obtain

$$\mathrm{Pic}(U) \cong \mathrm{Cl}(U) \cong \mathbb{Z}/2\mathbb{Z}.$$

The present chapter needs only the divisor-class side of that statement.

Example 41.26 (Localization and class groups)**.** Localizing a normal domain can kill divisor classes supported on the removed codimension-one primes. Thus passing to an open subset can simplify the class group by discarding those generators.

41.14 The local class group at the vertex

Let

$$\mathfrak{m} = (x, y, z)$$

be the vertex. Localizing at $\mathfrak{m}$ gives the normal local domain $A_{\mathfrak{m}}$.

Proposition 41.27 (Local obstruction at the vertex)**.** *The class of $\mathfrak{p}_L A_{\mathfrak{m}}$ is nonzero of order two in*

$$\mathrm{Cl}(A_{\mathfrak{m}}).$$

In fact

$$\mathrm{Cl}(A_{\mathfrak{m}}) \cong \mathbb{Z}/2\mathbb{Z}.$$

Proof. The localization map

$$\mathrm{Cl}(A) \twoheadrightarrow \mathrm{Cl}(A_{\mathfrak{m}})$$

is surjective, so the local class group is a quotient of $\mathbb{Z}/2\mathbb{Z}$. It is therefore enough to show that the image of $[L]$ is nonzero.

Set $I = (x, z)A_{\mathfrak{m}}$. Modulo $\mathfrak{m}I$, the classes of x and z are linearly independent over the residue field k: every element of $\mathfrak{m}I$ has order at least two in the standard grading, while a nontrivial k-linear combination $ax + bz$ has degree one. Hence

$$\dim_k I/\mathfrak{m}I = 2.$$

By Nakayama's lemma, I needs two generators and is not principal. Thus L is not principal in the local ring, so its local class is nonzero. The only nonzero quotient of $\mathbb{Z}/2\mathbb{Z}$ is itself, giving

$$\mathrm{Cl}(A_{\mathfrak{m}}) \cong \mathbb{Z}/2\mathbb{Z}.$$

□

Thus the failure of local factoriality is concentrated at the singular point.

41.15 Projective space

The affine space calculation

$$\mathrm{Cl}(\text{Å}^n) = 0$$

should be contrasted with projective space.

Theorem 41.28 (Class group of projective space)**.** *Let k be a field and $n \geq 1$. Then*

$$\boxed{\mathrm{Cl}(\mathbb{P}^n_k) \cong \mathbb{Z},}$$

generated by the class H of a hyperplane.

Proof. Let

$$U = D_+(x_0) \cong \text{Å}^n_k.$$

Its class group is zero. The complement is the single hyperplane

$$H = V_+(x_0).$$

The open-subset exact sequence gives a surjection

$$\mathbb{Z}[H] \twoheadrightarrow \mathrm{Cl}(\mathbb{P}^n_k).$$

It remains to show that no nonzero multiple of H is principal.

For a nonzero rational function

$$f = \frac{F}{G}$$

with F, G homogeneous of the same degree after clearing common factors, the divisor of f has total degree zero: the zero divisor and pole divisor have the same homogeneous degree. Hence a principal divisor cannot equal mH unless $m = 0$. Therefore $[H]$ has infinite order and generates the class group. □

$$\boxed{\mathbb{P}^n} \xrightarrow{\text{remove } H} \boxed{\text{Å}^n = D_+(x_0)}$$

$\mathrm{Cl}(\text{Å}^n) = 0$ forces $[H]$ to generate $\mathrm{Cl}(\mathbb{P}^n)$;
degree shows $[H]$ has infinite order

Figure 41.7: Deleting one hyperplane reduces projective space to affine space and exposes the generator of its class group.

Corollary 41.29 (Hypersurface classes)**.** *If F is a nonzero homogeneous polynomial of degree d, then*

$$[V_+(F)] = d[H]$$

in $\mathrm{Cl}(\mathbb{P}^n_k)$*, with factor multiplicities included.*

Proof. Choose a linear form ℓ not dividing F. The rational function

$$\frac{F}{\ell^d}$$

has principal divisor

$$\mathrm{div}\left(\frac{F}{\ell^d}\right) = V_+(F) - dV_+(\ell).$$

Since $V_+(\ell) \sim H$, the claim follows. □

Thus degree is the class-group coordinate on projective space:

$$\deg : \mathrm{Cl}(\mathbb{P}^n_k) \xrightarrow{\sim} \mathbb{Z}.$$

41.16 Cartier classes inside the Weil class group

On a normal Noetherian integral scheme, VI/40 constructed an injective homomorphism

$$\mathrm{CaDiv}(X) \hookrightarrow \mathrm{Div}(X)$$

compatible with principal divisors. Passing to equivalence classes therefore gives a natural map from Cartier divisor classes into $\mathrm{Cl}(X)$.

If X is locally factorial, every Weil divisor is Cartier, so every class in $\mathrm{Cl}(X)$ has a Cartier representative. The remaining question is what quotient of Cartier divisors by principal Cartier divisors means intrinsically. VI/42 answers:

$$\{\text{Cartier divisors}\}/\{\text{principal Cartier divisors}\} \cong \mathrm{Pic}(X).$$

Consequently, on a normal locally factorial integral scheme, VI/42 will give

$$\mathrm{Pic}(X) \cong \mathrm{Cl}(X).$$

We record this here only as a bridge, not as a substitute for the line-bundle theory.

41.17 Torsion classes and $\mathbb{Q}$-Cartier divisors

The cone generator satisfies

$$2[L] = 0.$$

Equivalently,

$$2L$$

is principal and therefore Cartier even though L itself is not Cartier at the vertex.

Definition 41.30 ($\mathbb{Q}$-Cartier divisor)**.** A Weil divisor D on a normal integral scheme is $\mathbb{Q}$-*Cartier* if there exists an integer $m > 0$ such that mD is Cartier.

Thus the cone divisor L is $\mathbb{Q}$-Cartier of index two. This is the first appearance of a notion that becomes central in birational geometry: a singularity may fail to be locally factorial while still allowing rational multiples of divisors to behave like Cartier divisors.

Warning 41.31 (Torsion in $\mathrm{Cl}(X)$ versus $\mathbb{Q}$-factoriality)*.* A torsion class in the global class group means some multiple is *principal*, which is stronger than merely being Cartier. In general, $\mathbb{Q}$-factoriality is a local Cartier condition and should not be identified with the statement that the entire global class group is torsion.

41.18 A computation strategy

For an affine normal domain A, explicit class-group computations often follow the same pattern:

1. choose a convenient element $f \in A$;
2. prove A_f is a UFD, so $\mathrm{Cl}(A_f) = 0$;
3. list the height-one primes containing f;
4. conclude from localization that their classes generate $\mathrm{Cl}(A)$;
5. compute principal divisors of selected functions to obtain relations;
6. prove the remaining candidate generators are nonprincipal.

For the quadric cone this becomes

$$\boxed{A_x \text{ UFD} \Longrightarrow \mathrm{Cl}(A) = \langle [L] \rangle, \qquad \mathrm{div}(x) = 2L \Longrightarrow 2[L] = 0, \qquad L \text{ nonprincipal} \Longrightarrow \mathrm{Cl}(A) = \mathbb{Z}/2.}$$

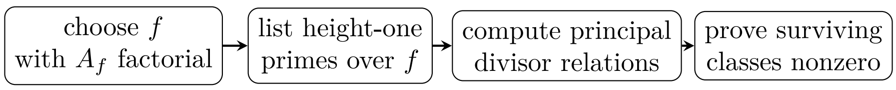


Figure 41.8: A reusable workflow for explicit affine divisor-class-group computations.

41.19 Exercises

Exercise 41.32. Prove that linear equivalence is an equivalence relation on $\mathrm{Div}(X)$.

Exercise 41.33. Show that $[D] = 0$ in $\mathrm{Cl}(X)$ exactly when D is principal.

Exercise 41.34. Compute $\mathrm{Cl}(\mathrm{Spec}\, k[t])$.

Exercise 41.35. Prove $\mathrm{Cl}(\text{Å}^n_k) = 0$.

Exercise 41.36. Let A be a normal Noetherian domain with $\mathrm{Cl}(A) = 0$. Prove directly that every height-one prime is principal.

Exercise 41.37. Show that localization of a UFD is again a UFD using divisor class groups.

Exercise 41.38. For the quadric cone, verify that (x, z) and (y, z) have height one.

Exercise 41.39. Compute $\mathrm{div}(x)$ on the quadric cone.

Exercise 41.40. Compute $\mathrm{div}(z)$ on the quadric cone.

Exercise 41.41. Show that $L \sim M$ on the cone.

Exercise 41.42. Prove $A_x \cong k[x, x^{-1}, z]$ for the cone ring A.

Exercise 41.43. Use localization at x to prove that $[L]$ generates $\mathrm{Cl}(A)$.

Exercise 41.44. Show that the only units of $A = k[x, y, z]/(xy - z^2)$ are $k^\times$.

Exercise 41.45. Show directly that cx is not a square in $k(x, z)$ for $c \in k^\times$.

Exercise 41.46. Deduce $\mathrm{Cl}(k[x, y, z]/(xy - z^2)) \cong \mathbb{Z}/2\mathbb{Z}$.

Exercise 41.47. Let $U = X \setminus \{o\}$ be the punctured cone. Compute $\mathrm{Cl}(U)$.

Exercise 41.48. Explain why U is locally factorial but has nonzero class group.

Exercise 41.49. Let $U = D_+(x_0) \subset \mathbb{P}^n$. Use the open-subset exact sequence to show that the hyperplane class generates $\mathrm{Cl}(\mathbb{P}^n)$.

Exercise 41.50. Prove that a principal divisor on $\mathbb{P}^n$ has degree zero.

Exercise 41.51. Conclude that $\mathrm{Cl}(\mathbb{P}^n) \cong \mathbb{Z}$.

Exercise 41.52. Let F be homogeneous of degree d. Prove $[V_+(F)] = d[H]$ in $\mathrm{Cl}(\mathbb{P}^n)$.

Exercise 41.53. Let $U \subset X$ have complement of codimension at least two. Prove directly, without invoking the theorem, that $\mathrm{Div}(X) \to \mathrm{Div}(U)$ is an isomorphism.

Exercise 41.54. Show that principal divisors are preserved under the identification in Exercise VI.41.22.

Exercise 41.55. Show that the cone divisor L is $\mathbb{Q}$-Cartier of index two.

41.20 Solved problem dossiers

Problem 41.56 (Class group of a localization)**.** Let A be a normal Noetherian domain with

$$\mathrm{Cl}(A) \cong \mathbb{Z}[D_1] \oplus \mathbb{Z}/3[D_2].$$

Suppose f lies in the height-one prime D_1 and in no height-one prime representing D_2. Describe what localization at f can do to the displayed generators.

Problem 41.57 (The cone relation from x/z)**.** Compute the principal divisor of x/z on the quadric cone and interpret it in the class group.

Problem 41.58 (Why $D(x)$ forgets the cone class)**.** Explain simultaneously algebraically and geometrically why

$$\mathrm{Cl}(D(x)) = 0$$

for the quadric cone.

Problem 41.59 (Why deleting the vertex does not forget the cone class)**.** Explain why removing only the vertex leaves the class group unchanged.

Problem 41.60 (A factoriality test)**.** Let A be a normal Noetherian domain. Suppose there are height-one primes $\mathfrak{p}_1, \dots, \mathfrak{p}_r$ and an element f such that A_f is a UFD and the $\mathfrak{p}_i$ are exactly the height-one primes containing f. Give a finite presentation of $\mathrm{Cl}(A)$ from principal divisors supported on these primes.

Problem 41.61 (Hyperplanes and rational functions)**.** Let $H_0 = V_+(x_0)$ and $H_1 = V_+(x_1)$ in $\mathbb{P}^n$. Exhibit a rational function proving $H_0 \sim H_1$.

Problem 41.62 (A degree-d hypersurface)**.** Let $F = F_1^{e_1} \cdots F_r^{e_r}$ be a factorization into distinct irreducible homogeneous polynomials. Compute its divisor class in $\mathbb{P}^n$.

Problem 41.63 (Local versus global factoriality)**.** Give a conceptual explanation for how a smooth scheme can have nonzero class group without contradicting the fact that regular local rings are UFDs.

Problem 41.64 (Torsion and a Cartier multiple)**.** Suppose D is a Weil divisor with $m[D] = 0 \in \mathrm{Cl}(X)$. What can be concluded about mD?

Problem 41.65 (Codimension-one deletion)**.** Let $X = \mathbb{P}^n$ and $U = \mathbb{A}^n = D_+(x_0)$. Use the open-subset exact sequence to recover $\mathrm{Cl}(U) = 0$ from $\mathrm{Cl}(X) \cong \mathbb{Z}$.

Problem 41.66 (The local cone obstruction)**.** Why does the nonprincipality of $(x, z)A_\mathfrak{m}$ show that the vertex is not locally factorial?

Problem 41.67 (A divisor-class diagnostic)**.** Suppose A is a normal Noetherian domain and one finds a height-one prime $\mathfrak{p}$ with no rational function having divisor exactly $[\mathfrak{p}]$. What does this prove?

41.21 Challenges

Problem 41.68 (Challenge: the localization sequence)**.** Give a complete proof of theorem 41.14 directly from the correspondence of height-one primes under localization, including the compatibility of all valuation multiplicities with principal divisors.

Problem 41.69 (Challenge: the quadric cone without parametrization)**.** Compute

$$\mathrm{Cl}(k[x, y, z]/(xy - z^2))$$

using only localization, DVR valuations, the Krull intersection formula, and the rational function field $k(x, z)$. Do not use the parametrization by s^2, st, t^2.

Problem 41.70 (Challenge: punctured cone and gluing)**.** Cover the punctured quadric cone by principal affine opens and track explicit local equations for L. Show that $L|_U$ is Cartier but not globally principal, thereby anticipating the nontrivial order-two line bundle of VI/42.

Problem 41.71 (Challenge: projective-space class group)**.** Prove $\mathrm{Cl}(\mathbb{P}^n_k) \cong \mathbb{Z}$ in two independent ways: first from the open-subset exact sequence with $\mathbb{A}^n$, and second from homogeneous factorization and the degree map. Compare exactly where unique factorization in $k[x_0, \dots, x_n]$ enters.

Problem 41.72 (Challenge: local class groups)**.** For a normal Noetherian domain A, study the maps

$$\mathrm{Cl}(A) \longrightarrow \mathrm{Cl}(A_\mathfrak{p})$$

as $\mathfrak{p}$ varies. Prove that a Weil divisor is Cartier precisely when its image is zero in the appropriate local divisor-class obstruction at every point, and explain why this criterion points naturally toward the Picard group.

Bridge to VI/42

The divisor sequence is now complete:

$$\boxed{K(X)^\times \xrightarrow{\mathrm{div}} \mathrm{Div}(X) \longrightarrow \mathrm{Cl}(X) \longrightarrow 0.}$$

For affine normal Noetherian domains,

$$\boxed{\mathrm{Cl}(A) = 0 \quad \Longleftrightarrow \quad A \text{ is a UFD.}}$$

For the quadric cone,

$$\boxed{\operatorname{div}(x) = 2L, \qquad [L] \neq 0, \qquad \mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.}$$

VI/42 moves from divisors to invertible sheaves. It will identify Cartier divisors modulo principal Cartier divisors with the Picard group, prove the locally factorial comparison

$$\mathrm{Pic}(X) \cong \mathrm{Cl}(X),$$

and reinterpret the order-two class on the punctured cone as an order-two line bundle.

41.22 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 41.73 (UFD class group)**.** What is $\mathrm{Cl}(A)$ for a normal UFD?

Hint. Use Weil divisors modulo principal divisors and factoriality. □

Solution. It is zero. □

Exercise 41.74 (Principal class)**.** What class does a principal divisor represent?

Hint. Use Weil divisors modulo principal divisors and factoriality. □

Solution. The zero class. □

Exercise 41.75 (Generator type)**.** What objects generate the Weil divisor group?

Hint. Use Weil divisors modulo principal divisors and factoriality. □

Solution. Prime codimension-one subvarieties. □

Exercise 41.76 (Factoriality)**.** What does $\mathrm{Cl}(A) = 0$ imply for a normal Noetherian domain under the standard criterion?

Hint. Use Weil divisors modulo principal divisors and factoriality. □

Solution. It is factorial. □

Exercise 41.77 (Localization)**.** What may happen to classes supported on inverted height-one primes?

Hint. Use Weil divisors modulo principal divisors and factoriality. □

Solution. They disappear after localization. □

Proofs

Exercise 41.78 (Principal subgroup)**.** Prove: Prove principal divisors form a subgroup of the Weil divisor group.

Hint. Work from Weil divisors modulo principal divisors and factoriality and state the hypotheses explicitly. ☐

Solution. Use $\operatorname{div}(fg) = \operatorname{div}(f) + \operatorname{div}(g)$ and inverses. ☐

Exercise 41.79 (UFD vanishing)**.** Prove: Prove the class group of a normal UFD is zero.

Hint. Work from Weil divisors modulo principal divisors and factoriality and state the hypotheses explicitly. ☐

Solution. Every height-one prime is generated by an irreducible element, so each prime divisor is principal. ☐

Exercise 41.80 (Factorial criterion)**.** Prove: Prove vanishing class group implies every height-one prime is principal.

Hint. Work from Weil divisors modulo principal divisors and factoriality and state the hypotheses explicitly. ☐

Solution. Each prime divisor has zero class, hence differs from zero by a principal divisor; compare coefficients to obtain a principal generator. ☐

Exercise 41.81 (Localization effect)**.** Prove: Explain the quotient description of the class group after localizing at a multiplicative set in the standard normal Noetherian setting.

Hint. Work from Weil divisors modulo principal divisors and factoriality and state the hypotheses explicitly. ☐

Solution. Prime divisors meeting the multiplicative set disappear, and their classes generate the kernel of the localization map. ☐

Counterexamples and hypothesis tests

Exercise 41.82 (Normal UFD)**.** Test the claim: every normal domain is a UFD.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the quadric cone supplies a standard counterexample. ☐

Exercise 41.83 (Class group Picard)**.** Test the claim: the Weil class group and Picard group always coincide.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; equality needs extra regularity or local factoriality hypotheses. ☐

Exercise 41.84 (Topological invariant)**.** Test the claim: divisor class group depends only on the underlying topology.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it depends on codimension-one local algebra. ☐

Applications and investigations

Exercise 41.85 (Factoriality diagnosis)**.** Why compute $\mathrm{Cl}(X)$?

Hint. Translate the question into Weil divisors modulo principal divisors and factoriality. □

Solution. It measures obstruction to representing Weil divisors by global principal divisors and hence to factoriality. □

Exercise 41.86 (Birational models)**.** How can class groups change under removing subsets?

Hint. Translate the question into Weil divisors modulo principal divisors and factoriality. □

Solution. Removing codimension-one components can quotient out their divisor classes. □

Challenge problems

Exercise 41.87 (Synthesis challenge)**.** Connect two central viewpoints of Divisor Class Groups in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. □

Solution. A complete solution makes one concrete computation in Divisor Class Groups and then translates it through Weil divisors modulo principal divisors and factoriality, checking both descriptions agree. □

Exercise 41.88 (Hypothesis challenge)**.** Choose one theorem-level statement from Divisor Class Groups, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. □

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Chapter 42

Line Bundles and Picard Groups

VI/40 attached an invertible sheaf $\mathcal{O}_X(D)$ to every Cartier divisor, and VI/41 passed from Weil divisors to the divisor class group

$$\mathrm{Cl}(X) = \mathrm{Div}(X)/\,\mathrm{Prin}(X).$$

The present chapter develops the intrinsic object behind the Cartier construction: a line bundle, equivalently an invertible $\mathcal{O}_X$-module. Tensor product turns isomorphism classes of line bundles into the Picard group

$$\mathrm{Pic}(X).$$

On an integral scheme, Cartier divisors modulo principal Cartier divisors recover this group:

$$\mathrm{CaCl}(X) \cong \mathrm{Pic}(X).$$

On a normal Noetherian integral scheme this embeds naturally into $\mathrm{Cl}(X)$, and on a locally factorial scheme the two groups agree.

The distinction is visible immediately on the quadric cone

$$X = \operatorname{Spec} k[x, y, z]/(xy - z^2).$$

VI/41 computed

$$\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$$

and showed that the nonzero class is represented by the non-Cartier ruling $L = V(x, z)$. Consequently

$$\mathrm{Pic}(X) = 0.$$

After deleting the singular vertex, however, the punctured cone U is regular and locally factorial, so

$$\mathrm{Pic}(U) \cong \mathrm{Cl}(U) \cong \mathbb{Z}/2\mathbb{Z}.$$

Thus a codimension-two deletion can leave the Weil class group unchanged while changing the Picard group.

The migration ledger records VI/42 as the canonical destination for line-bundle, invertible-sheaf, and Picard material. The directly inspectable AG9 selector consists of the thematic section together with its inherited theorem-like and exercise/problem descendants. The chapter-status ledger records six mapped rules in total; this reconstruction treats that number as provenance metadata and does not invent unavailable legacy prose. The mathematical content is rebuilt canonically and is not claimed to be a verbatim recovery.

Learning goals

By the end of the chapter, the reader should be able to:

- define line bundles and invertible $\mathcal{O}_X$-modules and pass freely between the two viewpoints;
- describe line bundles by transition functions $g_{ij} \in \mathcal{O}_X^\times(U_i \cap U_j)$;
- recognize the cocycle and coboundary relations arising from changes of local frame;
- construct tensor products and duals of line bundles;
- define the Picard group and identify its identity and inverse;
- pull back line bundles functorially along scheme morphisms;
- construct the homomorphism $D \mapsto \mathcal{O}_X(D)$ from Cartier divisors to line bundles;
- prove $\mathrm{CaCl}(X) \cong \mathrm{Pic}(X)$ for integral schemes;
- construct the natural injection $\mathrm{Pic}(X) \hookrightarrow \mathrm{Cl}(X)$ on normal Noetherian integral schemes;
- prove $\mathrm{Pic}(X) \cong \mathrm{Cl}(X)$ on locally factorial normal integral schemes;
- compute $\mathrm{Pic}(\mathring{A}^n_k) = 0$ and $\mathrm{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z}$, generated by $\mathcal{O}(1)$;
- distinguish $\mathrm{Pic}(X)$ from $\mathrm{Cl}(X)$ on the quadric cone and compute the Picard group of the punctured cone;
- relate regular sections of line bundles to effective Cartier divisors;
- construct morphisms to projective space from globally generating sections;
- define degree and Pic^0 on smooth projective curves as preparation for plane cubics.

Conceptual roadmap

$$\boxed{\mathrm{CaDiv}(X) \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Cl}(X), \qquad D \longmapsto [\mathcal{O}_X(D)]}$$

and, after quotienting Cartier divisors by principal divisors,

$$\boxed{\mathrm{CaCl}(X) \xrightarrow{\sim} \mathrm{Pic}(X).}$$

for integral X, with the right-hand map available when X is normal Noetherian. If X is locally factorial, every Weil divisor is Cartier and

$$\boxed{\mathrm{Pic}(X) \cong \mathrm{Cl}(X).}$$

42.1 Line bundles and invertible sheaves

Definition 42.1 (Invertible sheaf)**.** An $\mathcal{O}_X$-module $\mathcal{L}$ is *invertible* if every point $x \in X$ has an open neighborhood U such that

$$\mathcal{L}|_U \cong \mathcal{O}_U.$$

Equivalently, $\mathcal{L}$ is locally free of rank one.

$$\begin{tikzcd} \mathrm{CaCl}(X) \arrow[r, "D \mapsto \mathcal{O}(D)"] & \mathrm{Pic}(X) \arrow[r, "\text{Cartier class}"] & \mathrm{Cl}(X) \end{tikzcd}$$

left arrow is an isomorphism for integral schemes;
right arrow is injective for normal Noetherian integral schemes

Figure 42.1: The Picard group is the intrinsic line-bundle form of Cartier divisor classes and sits inside the Weil divisor class group under normality.

A geometric line bundle is a morphism $p : L \to X$ whose fibers are one-dimensional vector spaces and which is locally isomorphic to $U \times \mathbb{A}^1 \to U$. Its sheaf of regular sections is invertible. Conversely an invertible sheaf determines a line bundle by relative spectrum:

$$\mathbf{V}(\mathcal{L}^\vee) := \underline{\mathrm{Spec}}_X(\mathrm{Sym}_{\mathcal{O}_X} \mathcal{L}^\vee).$$

For the purposes of divisor theory, the invertible-sheaf language is usually more efficient.

Definition 42.2 (Local frame)**.** A *local frame* for an invertible sheaf $\mathcal{L}$ on $U \subseteq X$ is a section

$$e \in \Gamma(U, \mathcal{L})$$

such that multiplication by e defines an isomorphism

$$\mathcal{O}_U \xrightarrow{\sim} \mathcal{L}|_U.$$

If e_i and e_j are frames on U_i and U_j, then on the overlap there is a unique unit

$$g_{ij} \in \mathcal{O}_X^\times(U_i \cap U_j)$$

such that

$$e_i = g_{ij} e_j.$$

42.2 Transition functions and cocycles

The transition functions satisfy

$$g_{ii} = 1, \qquad g_{ij} = g_{ji}^{-1},$$

and on triple overlaps

$$g_{ij} g_{jk} = g_{ik}.$$

The last identity is the cocycle condition.

Theorem 42.3 (Gluing line bundles from units)**.** *Let $X = \bigcup_i U_i$ be an open cover and suppose units*

$$g_{ij} \in \mathcal{O}_X^\times(U_i \cap U_j)$$

satisfy

$$g_{ii} = 1, \qquad g_{ij} g_{jk} = g_{ik}.$$

Then the trivial bundles $\mathcal{O}_{U_i}$ glue to an invertible sheaf $\mathcal{L}$ with transition functions g_{ij}.

Proof. On $U_i \cap U_j$, identify a section a_j in the j-th trivialization with

$$a_i = g_{ij}^{-1} a_j$$

in the i-th trivialization, equivalently choose local frames satisfying $e_i = g_{ij} e_j$. The cocycle identity guarantees that the identifications agree on triple overlaps. The sheaf-gluing theorem then produces $\mathcal{L}$, and each U_i remains a trivializing open. □

Changing frames does not change the isomorphism class. If

$$e_i' = u_i e_i, \qquad u_i \in \mathcal{O}_X^\times(U_i),$$

then

$$g_{ij}' = u_i g_{ij} u_j^{-1}.$$

Thus transition functions differing by such a family of local units define isomorphic line bundles.

$$g_{ij} g_{jk} = g_{ik} \text{ on } U_i \cap U_j \cap U_k$$

$$\begin{tikzcd}
\boxed{U_i,\ e_i} \arrow[rr, leftrightarrow, "g_{ij}"] \arrow[dr, leftrightarrow, "g_{ik}"'] & & \boxed{U_j,\ e_j} \arrow[dl, leftrightarrow, "g_{jk}"] \\
& \boxed{U_k,\ e_k} &
\end{tikzcd}$$

Figure 42.2: Local frames glue to a line bundle exactly when the transition units satisfy the multiplicative cocycle condition.

Remark 42.4 (Čech viewpoint). For a fixed cover, line bundles are represented by multiplicative 1-cocycles with values in $\mathcal{O}_X^\times$, modulo multiplicative coboundaries. Later cohomology chapters package this as

$$\mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^\times),$$

but the present chapter needs only the explicit gluing calculation.

42.3 Tensor products, duals, and the Picard group

If $\mathcal{L}$ and $\mathcal{M}$ are invertible, then so is

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{M}.$$

If e_i and f_i are local frames with transition functions g_{ij} and h_{ij}, then $e_i \otimes f_i$ has transition function

$$g_{ij} h_{ij}.$$

The dual

$$\mathcal{L}^\vee := \mathcal{H}om_{\mathcal{O}_X}(\mathcal{L}, \mathcal{O}_X)$$

is invertible, with transition functions g_{ij}^{-1}, and evaluation gives

$$\mathcal{L} \otimes \mathcal{L}^\vee \cong \mathcal{O}_X.$$

Definition 42.5 (Picard group)**.** The *Picard group* of X is

$$\mathrm{Pic}(X) := \{\text{isomorphism classes of invertible } \mathcal{O}_X\text{-modules}\},$$

with group law

$$[\mathcal{L}] + [\mathcal{M}] := [\mathcal{L} \otimes \mathcal{M}].$$

The identity is $[\mathcal{O}_X]$, and

$$-[\mathcal{L}] = [\mathcal{L}^\vee].$$

Associativity and commutativity follow from the corresponding canonical tensor-product isomorphisms.

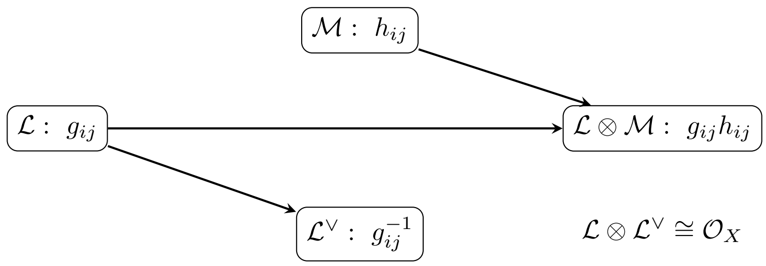


Figure 42.3: Tensor product multiplies transition functions, while duality inverts them; these operations give the Picard group law.

Example 42.6 (The twisting sheaf on projective space)**.** The line bundle $\mathcal{O}_{\mathbb{P}^n}(1)$ is obtained by gluing trivial line bundles on standard charts with transition functions X_j/X_i. Its tensor powers give $\mathcal{O}(d)$.

42.4 Pullback of line bundles

Let $f : Y \to X$ be a morphism. For an invertible sheaf $\mathcal{L}$ on X, define

$$f^*\mathcal{L} := f^{-1}\mathcal{L} \otimes_{f^{-1}\mathcal{O}_X} \mathcal{O}_Y.$$

Since local freeness of rank one is preserved by pullback, $f^*\mathcal{L}$ is invertible.

Proposition 42.7 (Functoriality of the Picard group)**.** *Every morphism $f : Y \to X$ induces a homomorphism*

$$f^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(Y).$$

Moreover,

$$\mathrm{id}_X^* = \mathrm{id}_{\mathrm{Pic}(X)}, \qquad (f \circ g)^* \cong g^* f^*.$$

Proof. Pullback commutes canonically with tensor products:

$$f^*(\mathcal{L} \otimes \mathcal{M}) \cong f^*\mathcal{L} \otimes f^*\mathcal{M}.$$

It also preserves the trivial bundle. The functorial identities are the standard identities for pullback of modules. □

If $\mathcal{L}$ is represented by transition functions g_{ij}, then $f^*\mathcal{L}$ is represented on $f^{-1}(U_i)$ by the pulled-back units $f^\#(g_{ij})$.

42.5 Cartier divisors produce line bundles

Let X be integral with function field $K(X)$. Suppose a Cartier divisor D is represented on U_i by local rational equations

$$f_i \in K(X)^\times$$

such that

$$\frac{f_i}{f_j} \in \mathcal{O}_X^\times(U_i \cap U_j).$$

VI/40 defined

$$\mathcal{O}_X(D)|_{U_i} = f_i^{-1}\mathcal{O}_{U_i} \subset K(X).$$

A local frame is $e_i = f_i^{-1}$, and

$$e_i = \frac{f_j}{f_i} e_j.$$

Thus the transition functions of $\mathcal{O}_X(D)$ are f_j/f_i.

Proposition 42.8 (Tensor law for divisor line bundles)**.** *For Cartier divisors D, E,*

$$\mathcal{O}_X(D+E) \cong \mathcal{O}_X(D) \otimes \mathcal{O}_X(E),$$

and

$$\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee.$$

Proof. If D and E have local equations f_i and g_i, then $D+E$ has equation $f_i g_i$. The local generator

$$(f_i g_i)^{-1}$$

is exactly $f_i^{-1} \otimes g_i^{-1}$, and the transition functions multiply. □

A principal Cartier divisor $D = \operatorname{div}(f)$ gives

$$\mathcal{O}_X(D) = f^{-1}\mathcal{O}_X \cong \mathcal{O}_X$$

by multiplication by f. Hence $D \mapsto \mathcal{O}_X(D)$ factors through Cartier divisor classes.

$$\begin{tikzcd}
\boxed{D=(f_i)} \arrow[r, "f_i^{-1}\mathcal{O}"] \arrow[dr] & \boxed{\mathcal{O}_X(D)} \arrow[r, "\text{frames}"] & \boxed{g_{ij}=f_j/f_i} \\
& \boxed{D=\operatorname{div}(f) \Rightarrow \mathcal{O}_X(D)\cong\mathcal{O}_X} &
\end{tikzcd}$$

Figure 42.4: Cartier local equations become transition functions of an invertible sheaf, and principal divisors become the trivial bundle.

42.6 Cartier divisor classes equal the Picard group

Write

$$\operatorname{CaCl}(X) := \operatorname{CaDiv}(X)/\{\text{principal Cartier divisors}\}.$$

Theorem 42.9 (Cartier–Picard equivalence)**.** *Let X be an integral scheme. Then*

$$\boxed{\operatorname{CaCl}(X) \xrightarrow{\sim} \operatorname{Pic}(X), \qquad [D] \longmapsto [\mathcal{O}_X(D)].}$$

Proof. The tensor law makes the map a homomorphism, and principal divisors map to the identity.

For surjectivity, let $\mathcal{L}$ be invertible. At the generic point η, the stalk

$$\mathcal{L}_\eta$$

is a one-dimensional $K(X)$-vector space. Choose a nonzero rational section s. On a trivializing open U_i, choose a frame e_i and write

$$s = f_i e_i, \qquad f_i \in K(X)^\times.$$

If $e_i = g_{ij} e_j$, then

$$f_j = f_i g_{ij},$$

so

$$\frac{f_i}{f_j} = g_{ij}^{-1} \in \mathcal{O}_X^\times(U_i \cap U_j).$$

Hence the f_i define a Cartier divisor $D = \operatorname{div}(s)$. The frames f_i^{-1} of $\mathcal{O}_X(D)$ have transition functions

$$\frac{f_j}{f_i} = g_{ij},$$

so

$$\mathcal{O}_X(D) \cong \mathcal{L}.$$

For injectivity, suppose $\mathcal{O}_X(D) \cong \mathcal{O}_X$. Under a chosen trivialization, the canonical rational section 1 of $\mathcal{O}_X(D)$ becomes a rational function $f \in K(X)^\times$. Since the divisor of the canonical rational section is D, we obtain

$$D = \operatorname{div}(f).$$

Thus D is principal. □

Remark 42.10 (What the theorem remembers). A line bundle does not remember a specific Cartier divisor, only its class modulo principal divisors. Choosing a nonzero rational section of $\mathcal{L}$ chooses one divisor representative; multiplying the section by $f \in K(X)^\times$ changes that representative by $\operatorname{div}(f)$.

42.7 From the Picard group to the Weil class group

Assume now that X is normal, Noetherian, and integral. VI/40 constructed an injective map

$$\operatorname{CaDiv}(X) \hookrightarrow \operatorname{Div}(X)$$

compatible with principal divisors. Passing to quotient groups and using theorem 42.9 gives:

Theorem 42.11 (Picard group inside the class group)**.** *There is a natural injective homomorphism*

$$\boxed{\operatorname{Pic}(X) \hookrightarrow \operatorname{Cl}(X).}$$

Its image consists exactly of Weil divisor classes admitting a Cartier representative.

Proof. A Cartier divisor determines a Weil divisor. If two Cartier divisors differ by a principal Cartier divisor, their Weil divisors differ by the same principal Weil divisor, so the map descends to classes.

If a Cartier class maps to zero in $\operatorname{Cl}(X)$, its associated Weil divisor is principal. By injectivity of the Cartier-to-Weil map on a normal Noetherian integral scheme, the Cartier divisor itself is principal. Hence the induced map on classes is injective.

The image statement is immediate from the construction. □

Corollary 42.12 (Locally factorial comparison)**.** *If X is normal, Noetherian, integral, and locally factorial, then*

$$\boxed{\mathrm{Pic}(X) \cong \mathrm{Cl}(X).}$$

In particular this holds for regular Noetherian integral schemes.

Proof. On a locally factorial scheme every Weil divisor is Cartier. Therefore every class in $\mathrm{Cl}(X)$ lies in the image of theorem 42.11. Regular local rings are UFDs, so regular Noetherian schemes are locally factorial. □

Example 42.13 (A Cartier divisor line bundle)**.** A Cartier divisor D with local equations f_i defines an invertible sheaf $\mathcal{O}(D)$ whose transition functions are f_i/f_j. Principal divisors yield trivial line bundles.

42.8 Affine schemes and invertible modules

For $X = \operatorname{Spec} A$, quasi-coherent sheaves correspond to A-modules. Under this correspondence, invertible sheaves correspond to finitely generated projective A-modules of rank one.

Proposition 42.14 (Affine description)**.** *There is a natural identification*

$$\mathrm{Pic}(\operatorname{Spec} A) \cong \{\textit{rank-one finitely generated projective } A\textit{-modules}\}/\cong.$$

An important warning is that affine does not mean Picard-trivial. Polynomial affine space is trivial because its coordinate ring is factorial, but general affine schemes may have nonzero Picard group.

If A is a local ring, every finitely generated projective A-module is free. Hence

$$\mathrm{Pic}(\operatorname{Spec} A) = 0.$$

For a Dedekind domain, every nonzero fractional ideal is invertible and

$$\mathrm{Pic}(\operatorname{Spec} A)$$

is the usual ideal class group.

42.9 Affine space

Theorem 42.15 (Picard group of affine space)**.** *For every field k and $n \geq 1$,*

$$\boxed{\mathrm{Pic}(\mathring{A}^n_k) = 0.}$$

Proof. The polynomial ring

$$k[x_1, \ldots, x_n]$$

is a UFD, so VI/41 gives

$$\mathrm{Cl}(\mathring{A}^n_k) = 0.$$

Affine space is regular, hence locally factorial, and theorem 42.12 gives

$$\mathrm{Pic}(\mathring{A}^n_k) \cong \mathrm{Cl}(\mathring{A}^n_k) = 0.$$

□

42.10 The quadric cone: Pic versus Cl

Let

$$X = \operatorname{Spec} A, \qquad A = k[x,y,z]/(xy - z^2),$$

and let

$$L = V(x,z).$$

VI/41 proved

$$\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$$

with nonzero generator $[L]$, and VI/40 proved that L is not Cartier at the vertex.

Theorem 42.16 (Picard group of the affine quadric cone)**.** *For the affine quadric cone,*

$$\boxed{\operatorname{Pic}(X) = 0}$$

while

$$\boxed{\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.}$$

Proof. By theorem 42.11, $\operatorname{Pic}(X)$ injects into the two-element group $\mathrm{Cl}(X)$. If the image were nonzero, it would contain $[L]$. Then $[L]$ would admit a Cartier representative D. Since

$$[D] = [L],$$

the difference $L - D$ would be principal, hence Cartier. Therefore L itself would be Cartier, contradicting the failure of local principality at the vertex. Thus the image is zero and injectivity forces $\operatorname{Pic}(X) = 0$. ☐

Now remove the vertex $o = (x,y,z)$:

$$U = X \setminus \{o\}.$$

VI/41 showed that deleting this codimension-two point does not change the Weil class group:

$$\mathrm{Cl}(U) \cong \mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$$

The scheme U is regular, hence locally factorial.

Corollary 42.17 (Picard group of the punctured cone)**.** *For the punctured quadric cone,*

$$\boxed{\operatorname{Pic}(U) \cong \mathrm{Cl}(U) \cong \mathbb{Z}/2\mathbb{Z}.}$$

The nonzero line bundle on U is represented by the restriction

$$\mathcal{O}_U(L|_U),$$

because the ruling becomes Cartier after the singular vertex is removed.

Warning 42.18 (Codimension two behaves differently for Pic). For normal schemes, removing a codimension-at-least-two subset leaves Cl unchanged. The example above shows that Pic need not be unchanged:

$$\operatorname{Pic}(X) = 0, \qquad \operatorname{Pic}(U) = \mathbb{Z}/2.$$

The reason is that Cartierness is a local condition sensitive to the removed singular point.

$$\begin{tikzcd} \boxed{\begin{array}{c} X = \operatorname{Spec} k[x,y,z]/(xy-z^2) \\ \operatorname{Pic}(X) = 0 \\ \operatorname{Cl}(X) = \mathbb{Z}/2 \end{array}} \arrow[r, "\text{delete codim-2 vertex}"] & \boxed{\begin{array}{c} U = X \setminus \{o\} \\ \operatorname{Pic}(U) = \mathbb{Z}/2 \\ \operatorname{Cl}(U) = \mathbb{Z}/2 \end{array}} \end{tikzcd}$$

L is non-Cartier at o, but $L|_U$ is Cartier;
Cl survives while Pic changes

Figure 42.5: The quadric cone separates Weil divisor classes from line bundles and shows the sensitivity of Cartierness to a codimension-two singular point.

42.11 Projective space and the twisting sheaves

VI/36 introduced the twisting sheaves $\mathcal{O}_{\mathbb{P}^n}(d)$. Their tensor law is

$$\mathcal{O}(a) \otimes \mathcal{O}(b) \cong \mathcal{O}(a+b), \qquad \mathcal{O}(d)^\vee \cong \mathcal{O}(-d).$$

Let $H \subset \mathbb{P}^n$ be a hyperplane. VI/40 identified

$$\mathcal{O}_{\mathbb{P}^n}(H) \cong \mathcal{O}_{\mathbb{P}^n}(1),$$

and VI/41 computed

$$\operatorname{Cl}(\mathbb{P}^n_k) \cong \mathbb{Z}[H].$$

Since projective space is regular, theorem 42.12 immediately gives:

Theorem 42.19 (Picard group of projective space)**.** *For $n \geq 1$,*

$$\boxed{\operatorname{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z},}$$

generated by $[\mathcal{O}_{\mathbb{P}^n}(1)]$. Under this isomorphism,

$$d \longleftrightarrow \mathcal{O}_{\mathbb{P}^n}(d).$$

If F is a nonzero homogeneous polynomial of degree d, then its zero hypersurface satisfies

$$\mathcal{O}_{\mathbb{P}^n}(V_+(F)) \cong \mathcal{O}_{\mathbb{P}^n}(d).$$

$$\begin{tikzcd} \boxed{\mathcal{O}(-2)} \arrow[r] & \boxed{\mathcal{O}(-1)} \arrow[r] & \boxed{\mathcal{O}} \arrow[r] & \boxed{\mathcal{O}(1)} \arrow[r] & \boxed{\mathcal{O}(2)} \end{tikzcd}$$

$$\operatorname{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z}, \qquad \mathcal{O}(a) \otimes \mathcal{O}(b) \cong \mathcal{O}(a+b)$$

Figure 42.6: The twisting sheaves form the integer lattice of the Picard group of projective space, generated by $\mathcal{O}(1)$.

Example 42.20 (Picard group of affine space)**.** For affine space over a field, every line bundle is trivial, so $\operatorname{Pic}(\mathbb{A}^n_k) = 0$. This contrasts with $\operatorname{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z}$ generated by $\mathcal{O}(1)$.

42.12 Regular sections and effective Cartier divisors

Let $\mathcal{L}$ be invertible on an integral scheme X, and let

$$s \in \Gamma(X, \mathcal{L})$$

be nonzero at the generic point. Choose a local frame e_i and write

$$s = f_i e_i, \qquad f_i \in \mathcal{O}_X(U_i).$$

Because X is integral and s is generically nonzero, each nonzero local coefficient f_i is a non-zero-divisor wherever the frame is defined. The ideals (f_i) glue to an effective Cartier divisor.

Theorem 42.21 (Zero divisor of a section)**.** *Let X be integral, $\mathcal{L}$ invertible, and s a nonzero global section. Then s determines an effective Cartier divisor*

$$D = (s)_0$$

and a canonical isomorphism

$$\boxed{\mathcal{L} \cong \mathcal{O}_X(D)}$$

carrying s to the canonical section of $\mathcal{O}_X(D)$.

Proof. Write $s = f_i e_i$. On overlaps, if $e_i = g_{ij} e_j$, then

$$f_j = f_i g_{ij},$$

so the local equations f_i differ by units and define an effective Cartier divisor D. The transition function of $\mathcal{O}_X(D)$ between frames f_i^{-1} and f_j^{-1} is

$$\frac{f_j}{f_i} = g_{ij},$$

the same as for $\mathcal{L}$. Under the induced isomorphism, the canonical section $1 = f_i(f_i^{-1})$ corresponds to $f_i e_i = s$. $\square$

Conversely every effective Cartier divisor produces $\mathcal{O}_X(D)$ with its canonical global section. Thus effective Cartier divisors are line bundles together with a distinguished section satisfying the non-zero-divisor condition.

$$\boxed{\text{nonzero section } s \in \Gamma(X, \mathcal{L})} \xrightarrow{\text{zeros}} \boxed{D = (s)_0} \xrightarrow{D \mapsto \mathcal{O}(D)} \boxed{\mathcal{L} \cong \mathcal{O}_X(D)}$$

effective Cartier divisors are line bundles
equipped with their canonical vanishing section

Figure 42.7: A generically nonzero regular section converts line-bundle data into an effective Cartier divisor and back.

42.13 Globally generated line bundles and projective maps

An invertible sheaf $\mathcal{L}$ is *globally generated* if the evaluation map

$$\Gamma(X, \mathcal{L}) \otimes \mathcal{O}_X \longrightarrow \mathcal{L}$$

is surjective.

Suppose

$$s_0, \ldots, s_n \in \Gamma(X, \mathcal{L})$$

have no common zero. On a trivializing open where e is a frame, write

$$s_i = f_i e.$$

The ratios define

$$x \longmapsto [f_0(x) : \cdots : f_n(x)].$$

Changing the frame multiplies every f_i by the same unit, so the projective point is unchanged.

Theorem 42.22 (Line bundle and sections define a projective morphism)**.** *If $s_0, \ldots, s_n$ generate $\mathcal{L}$ at every point, they define a morphism*

$$\varphi : X \longrightarrow \mathbb{P}^n$$

such that

$$\varphi^* \mathcal{O}_{\mathbb{P}^n}(1) \cong \mathcal{L}.$$

Conversely, a morphism $\varphi : X \to \mathbb{P}^n$ pulls back the coordinate sections of $\mathcal{O}(1)$ to $n+1$ global sections generating $\varphi^*\mathcal{O}(1)$.

This is the line-bundle formulation of the homogeneous-coordinate constructions from VI/38. Very ample line bundles are those for which a suitable finite-dimensional space of sections gives a closed immersion.

42.14 Restriction to open subsets

If $j : U \hookrightarrow X$ is open, restriction gives

$$j^* : \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(U).$$

Line bundles can become trivial after deleting the support of a divisor. In the locally factorial normal setting, the class-group localization sequence from VI/41 gives an efficient description.

Proposition 42.23 (Restriction across divisors)**.** *Let X be normal, Noetherian, integral, and locally factorial, and let*

$$U = X \setminus \bigcup_{\alpha=1}^{r} Z_\alpha$$

where the Z_α are the codimension-one irreducible components of the complement. Then

$$\bigoplus_{\alpha=1}^{r} \mathbb{Z}[Z_\alpha] \longrightarrow \mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(U) \longrightarrow 0$$

is exact.

Proof. Use Pic $\cong$ Cl on X and U, then apply the open-subset exact sequence for class groups from VI/41. $\square$

For example,

$$\mathring{\mathbb{A}}^1 = \mathbb{P}^1 \setminus \{\infty\}.$$

The class of $\mathcal{O}_{\mathbb{P}^1}(1)$ is generated by the removed point and therefore becomes trivial on $\mathring{\mathbb{A}}^1$.

42.15 Degree on smooth projective curves

Let C be a smooth projective integral curve over a field. Every Weil divisor is Cartier, so

$$\mathrm{Pic}(C) \cong \mathrm{Cl}(C).$$

For a divisor

$$D = \sum_P n_P[P]$$

over an algebraically closed field, define

$$\deg D = \sum_P n_P.$$

More generally one weights closed points by residue-field degrees.

Principal divisors have degree zero, so degree descends to

$$\deg : \mathrm{Pic}(C) \longrightarrow \mathbb{Z}.$$

Definition 42.24 (Degree-zero Picard group)**.** The subgroup

$$\mathrm{Pic}^0(C) := \ker\bigl(\deg : \mathrm{Pic}(C) \to \mathbb{Z}\bigr)$$

is the degree-zero Picard group.

For $\mathbb{P}^1$,

$$\mathrm{Pic}(\mathbb{P}^1) \cong \mathbb{Z}, \qquad \mathrm{Pic}^0(\mathbb{P}^1) = 0.$$

The next chapter studies the first dramatically different case: a smooth plane cubic E. After choosing a rational point O, one obtains

$$P \longmapsto [\mathcal{O}_E(P - O)] \in \mathrm{Pic}^0(E),$$

and VI/43 will connect this map with the geometric group law on the cubic.

$$\begin{tikzcd}
\boxed{P \in C} \arrow[r, "P \mapsto P - O"] & \boxed{\mathcal{O}_C(P - O)} \arrow[r, "\text{class}"] & \boxed{\mathrm{Pic}^0(C)}
\end{tikzcd}$$

$\deg(P - O) = 0$
for a smooth plane cubic, VI/43 turns this map
into the chord-and-tangent group law

Figure 42.8: Degree-zero line bundles provide the bridge from divisor theory to the geometry and group law of plane cubics.

42.16 A practical dictionary

The constructions of VI/40–VI/42 can be summarized as follows:

object	local data	equivalence
Cartier divisor	$f_i \in K(X)^\times,\ f_i/f_j \in \mathcal{O}^\times$	multiply globally by f
line bundle	$g_{ij} \in \mathcal{O}^\times,\ g_{ij}g_{jk} = g_{ik}$	$g'_{ij} = u_i g_{ij} u_j^{-1}$
divisor line bundle	$g_{ij} = f_j/f_i$	$[D] \mapsto [\mathcal{O}(D)]$
Weil class	$\sum n_Z[Z]$	mod principal divisors

On an integral scheme,

$$\mathrm{CaCl}(X) \cong \mathrm{Pic}(X).$$

On a normal Noetherian integral scheme,

$$\mathrm{Pic}(X) \hookrightarrow \mathrm{Cl}(X).$$

On a locally factorial one,

$$\mathrm{Pic}(X) \cong \mathrm{Cl}(X).$$

42.17 Exercises

Exercise 42.25. Show that an invertible sheaf is locally free of rank one and that a locally free sheaf of rank one is invertible.

Exercise 42.26. If $e_i = g_{ij}e_j$, prove $g_{ij}g_{jk} = g_{ik}$ on triple overlaps.

Exercise 42.27. Show that replacing e_i by $u_i e_i$ changes g_{ij} to $u_i g_{ij} u_j^{-1}$.

Exercise 42.28. Prove that $\mathcal{L} \otimes \mathcal{M}$ is invertible when $\mathcal{L}, \mathcal{M}$ are.

Exercise 42.29. Prove $\mathcal{L} \otimes \mathcal{L}^\vee \cong \mathcal{O}_X$.

Exercise 42.30. Verify directly from transition functions that the inverse of $[\mathcal{L}]$ in $\mathrm{Pic}(X)$ is $[\mathcal{L}^\vee]$.

Exercise 42.31. Show that pullback defines a group homomorphism $f^* : \mathrm{Pic}(X) \to \mathrm{Pic}(Y)$.

Exercise 42.32. Let D have local equations f_i. Compute the transition functions of $\mathcal{O}_X(D)$.

Exercise 42.33. Show that if $D = \mathrm{div}(f)$, then $\mathcal{O}_X(D) \cong \mathcal{O}_X$.

Exercise 42.34. Prove $\mathcal{O}_X(D + E) \cong \mathcal{O}_X(D) \otimes \mathcal{O}_X(E)$ from local equations.

Exercise 42.35. Let $\mathcal{L}$ be invertible on an integral scheme. Explain why it has a nonzero rational section.

Exercise 42.36. Show that changing a nonzero rational section s to fs changes its Cartier divisor by $\mathrm{div}(f)$.

Exercise 42.37. Prove $\mathrm{Pic}(\operatorname{Spec} A) = 0$ when A is local.

Exercise 42.38. Prove $\mathrm{Pic}(\text{Å}^1_k) = 0$ directly from $k[t]$.

Exercise 42.39. Show that the restriction of $\mathcal{O}_{\mathbb{P}^1}(1)$ to $\text{Å}^1 = \mathbb{P}^1 \setminus \{\infty\}$ is trivial.

Exercise 42.40. Prove $\mathcal{O}_{\mathbb{P}^n}(a) \otimes \mathcal{O}_{\mathbb{P}^n}(b) \cong \mathcal{O}_{\mathbb{P}^n}(a + b)$.

Exercise 42.41. Compute $\mathrm{Pic}(\mathbb{P}^1_k)$.

Exercise 42.42. Let F be homogeneous of degree d. Show $\mathcal{O}_{\mathbb{P}^n}(V_+(F)) \cong \mathcal{O}(d)$.

Exercise 42.43. Why does the nonzero class in $\mathrm{Cl}(X)$ for the quadric cone not lie in the image of $\mathrm{Pic}(X)$?

Exercise 42.44. Compute $\mathrm{Pic}(U)$ for the punctured quadric cone.

Exercise 42.45. Let s be a nonzero global section of a line bundle on an integral scheme. Prove that its local coefficients define an effective Cartier divisor.

Exercise 42.46. Suppose $s_0, \ldots, s_n$ generate $\mathcal{L}$. Explain why $[s_0 : \cdots : s_n]$ is independent of the local frame.

Exercise 42.47. Show that principal divisors on a smooth projective curve have degree zero.

Exercise 42.48. Compute $\mathrm{Pic}^0(\mathbb{P}^1)$.

42.18 Solved problem dossiers

Problem 42.49 (Two-chart gluing)**.** Let $X = U \cup V$ and $g \in \mathcal{O}_X^\times(U \cap V)$. Construct the corresponding line bundle and determine when it is trivial.

Problem 42.50 (The standard bundle on $\mathbb{P}^1$)**.** Using $U_0 = D_+(x_0)$ and $U_1 = D_+(x_1)$, write a transition function for $\mathcal{O}_{\mathbb{P}^1}(1)$.

Problem 42.51 (Tensor powers)**.** Prove that $\mathcal{L}^{\otimes m}$ has transition functions g_{ij}^m, including negative m.

Problem 42.52 (Recovering a Cartier divisor from a line bundle)**.** Let X be integral and $\mathcal{L}$ invertible. Construct explicitly a Cartier divisor D with $\mathcal{L} \cong \mathcal{O}_X(D)$.

Problem 42.53 (Why the rational section matters)**.** Explain why $\mathcal{L}$ alone determines only a Cartier divisor class, not a unique Cartier divisor.

Problem 42.54 (The cone loses its obstruction after puncturing)**.** For the quadric cone, reconcile

$$\mathrm{Pic}(X) = 0, \qquad \mathrm{Pic}(U) = \mathbb{Z}/2.$$

Problem 42.55 (Restriction of $\mathcal{O}(d)$ to affine space)**.** Show that $\mathcal{O}_{\mathbb{P}^n}(d)$ restricts trivially to $D_+(x_0) \cong \mathring{\mathbb{A}}^n$.

Problem 42.56 (A morphism from two sections)**.** Let $\mathcal{L}$ be generated by two global sections s_0, s_1. Construct $X \to \mathbb{P}^1$ and identify the inverse images of the coordinate points.

Problem 42.57 (Dedekind domains)**.** Let A be a Dedekind domain. Relate $\mathrm{Pic}(\mathrm{Spec}\, A)$ to fractional ideals.

Problem 42.58 (Hyperplane pullback)**.** Let $f : X \to \mathbb{P}^n$ and set $\mathcal{L} = f^*\mathcal{O}(1)$. Interpret the pullback of a hyperplane $H = V_+(\ell)$.

Problem 42.59 (Degree of $\mathcal{O}_{\mathbb{P}^1}(d)$)**.** Compute the degree of $\mathcal{O}_{\mathbb{P}^1}(d)$.

Problem 42.60 (The first elliptic preview)**.** Let E be a smooth plane cubic with a chosen point O. Show that

$$\mathcal{L}_P := \mathcal{O}_E(P - O)$$

lies in $\mathrm{Pic}^0(E)$ for every closed point P of degree one.

42.19 Challenges

Problem 42.61 (Challenge: Cartier–Picard without shortcuts)**.** Give a complete proof of $\mathrm{CaCl}(X) \cong \mathrm{Pic}(X)$ for an integral scheme, including the construction of a rational section from the generic stalk and the verification that different rational sections change the divisor only by a principal divisor.

Problem 42.62 (Challenge: local-to-global classification)**.** Starting from an open cover $\{U_i\}$, prove directly that isomorphism classes of line bundles trivialized by the cover are multiplicative 1-cocycles $g_{ij} \in \mathcal{O}^\times(U_{ij})$ modulo coboundaries $u_i u_j^{-1}$, and show that tensor product corresponds to multiplication of cocycles.

Problem 42.63 (Challenge: Picard versus class group on the cone)**.** For $X = \mathrm{Spec}\, k[x, y, z]/(xy - z^2)$, reconstruct from first principles the chain

$$\mathrm{Pic}(X) = 0, \qquad \mathrm{Cl}(X) = \mathbb{Z}/2, \qquad \mathrm{Pic}(X \setminus \{o\}) = \mathbb{Z}/2.$$

Identify exactly which implication uses normality, which uses local factoriality, and which uses codimension-two invariance of Cl.

Problem 42.64 (Challenge: projective space from two methods)**.** Compute $\mathrm{Pic}(\mathbb{P}^n_k)$ twice: first from $\mathrm{Pic} \cong \mathrm{Cl}$ and the class-group computation of VI/41, and second directly from transition functions on the standard affine cover. Compare the two generators.

Problem 42.65 (Challenge: bridge to plane cubics)**.** Let $E \subset \mathbb{P}^2$ be a smooth plane cubic with $O \in E(k)$. Prove as much as possible, using only divisor and line-bundle methods developed so far, about the map

$$E(k) \longrightarrow \mathrm{Pic}^0(E), \qquad P \longmapsto [\mathcal{O}_E(P - O)].$$

Isolate precisely the additional geometric input needed in VI/43 to prove that this map realizes the chord-and-tangent group law.

Bridge to VI/43

The divisor and line-bundle arcs now meet:

$$\boxed{\text{Cartier divisors/principal} \cong \mathrm{Pic}(X) \hookrightarrow \mathrm{Cl}(X).}$$

For smooth varieties the last arrow is an isomorphism, so divisor classes and line bundles become interchangeable. On smooth projective curves, degree splits the discussion into

$$\mathrm{Pic}(C) \xrightarrow{\deg} \mathbb{Z}, \qquad \mathrm{Pic}^0(C) = \ker(\deg).$$

For $\mathbb{P}^1$, $\mathrm{Pic}^0 = 0$. VI/43 turns to a smooth plane cubic, where Pic^0 becomes a nontrivial geometric object and the divisor relation cut out by a line,

$$P + Q + R \sim 3O,$$

will be converted into the chord-and-tangent group law.

42.20 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 42.66 (Tensor)**.** Compute $\mathcal{O}(a) \otimes \mathcal{O}(b)$ on projective space.

Hint. Use invertible sheaves, transition functions, and tensor product. □

Solution. $\mathcal{O}(a + b)$. □

Exercise 42.67 (Dual)**.** What is the dual of $\mathcal{O}(d)$?

Hint. Use invertible sheaves, transition functions, and tensor product. □

Solution. $\mathcal{O}(-d)$. □

Exercise 42.68 (Identity)**.** What line bundle represents the identity in the Picard group?

Hint. Use invertible sheaves, transition functions, and tensor product. □

Solution. The trivial line bundle $\mathcal{O}_X$. □

Exercise 42.69 (Principal divisor)**.** What line bundle comes from a principal Cartier divisor?

Hint. Use invertible sheaves, transition functions, and tensor product. ☐

Solution. A trivial line bundle. ☐

Exercise 42.70 (Projective generator)**.** What generates $\mathrm{Pic}(\mathbb{P}^n_k)$?

Hint. Use invertible sheaves, transition functions, and tensor product. ☐

Solution. $\mathcal{O}(1)$ under the standard theorem. ☐

Proofs

Exercise 42.71 (Group law)**.** Prove: Prove isomorphism classes of line bundles form an abelian group under tensor product.

Hint. Work from invertible sheaves, transition functions, and tensor product and state the hypotheses explicitly. ☐

Solution. Tensor product is associative and commutative up to canonical isomorphism; duals give inverses and $\mathcal{O}_X$ is identity. ☐

Exercise 42.72 (Divisor map)**.** Prove: Prove linearly equivalent Cartier divisors define isomorphic line bundles.

Hint. Work from invertible sheaves, transition functions, and tensor product and state the hypotheses explicitly. ☐

Solution. Changing by a principal divisor multiplies local equations by a global rational function, producing a coboundary change of transition functions. ☐

Exercise 42.73 (Tensor twists)**.** Prove: Prove $\mathcal{O}(a) \otimes \mathcal{O}(b) \cong \mathcal{O}(a+b)$.

Hint. Work from invertible sheaves, transition functions, and tensor product and state the hypotheses explicitly. ☐

Solution. Multiply the standard transition functions on overlaps. ☐

Exercise 42.74 (Principal trivial)**.** Prove: Prove $\mathcal{O}(\operatorname{div} f)$ is trivial.

Hint. Work from invertible sheaves, transition functions, and tensor product and state the hypotheses explicitly. ☐

Solution. Use f itself as a global rational trivializing section. ☐

Counterexamples and hypothesis tests

Exercise 42.75 (All line bundles trivial)**.** Test the claim: every scheme has trivial Picard group.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; projective space has nontrivial twists. ☐

Exercise 42.76 (Sections imply trivial)**.** Test the claim: any line bundle with a nonzero global section is trivial.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the section may vanish. ☐

Exercise 42.77 (Picard topology)**.** Test the claim: Picard group is determined solely by the underlying topological space.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the structure sheaf and its units matter. ☐

Applications and investigations

Exercise 42.78 (Divisors)**.** Why are line bundles the natural global form of Cartier divisors?

Hint. Translate the question into invertible sheaves, transition functions, and tensor product. ☐

Solution. Local divisor equations become transition functions for an invertible sheaf. ☐

Exercise 42.79 (Projective embeddings)**.** How can a line bundle produce a map to projective space?

Hint. Translate the question into invertible sheaves, transition functions, and tensor product. ☐

Solution. A basepoint-free finite-dimensional space of global sections gives homogeneous coordinates. ☐

Challenge problems

Exercise 42.80 (Synthesis challenge)**.** Connect two central viewpoints of Line Bundles and Picard Groups in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Line Bundles and Picard Groups and then translates it through invertible sheaves, transition functions, and tensor product, checking both descriptions agree. ☐

Exercise 42.81 (Hypothesis challenge)**.** Choose one theorem-level statement from Line Bundles and Picard Groups, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 43

Plane Cubics

VI/42 ended with the degree-zero Picard group of a smooth projective curve,

$$\mathrm{Pic}^0(C) = \ker\bigl(\deg : \mathrm{Pic}(C) \to \mathbb{Z}\bigr),$$

and with the map

$$P \longmapsto [\mathcal{O}_C(P - O)].$$

For a smooth plane cubic this map is not merely a convenient source of degree-zero line bundles: it is the mechanism that turns the geometry of secant and tangent lines into an abelian group law on the curve itself.

The key divisor calculation is almost visible before any group law is defined. If a line $L \subset \mathbb{P}^2$ meets a smooth cubic E in the divisor

$$L \cdot E = P + Q + R$$

with intersection multiplicities understood, then every line section represents the same hyperplane class H, so

$$P + Q + R \sim H.$$

If a chosen origin $O \in E$ is a flex, its tangent line cuts the divisor $3O$, hence

$$H \sim 3O$$

and therefore

$$\boxed{P + Q + R \sim 3O.}$$

This is the divisor-theoretic heart of the familiar rule "the three collinear points sum to zero."

The present chapter develops this carefully. It first treats plane cubics and their singularities, then line sections, tangency and flexes, and finally the Abel–Picard map

$$\alpha_O : E \longrightarrow \mathrm{Pic}^0(E), \qquad P \longmapsto [\mathcal{O}_E(P - O)].$$

The chord-and-tangent law is constructed so that α_O carries it to tensor product in $\mathrm{Pic}^0(E)$. Commutativity and associativity are then inherited from the Picard group rather than proved by a long synthetic incidence argument.

Standing conventions

Unless explicitly stated otherwise, k is algebraically closed and $E \subset \mathbb{P}^2_k$ is a smooth plane cubic. Statements over an arbitrary field are collected later. Formulas in short Weierstrass form are stated under $\mathrm{char}\, k \neq 2, 3$. All line–curve intersections are counted with scheme-theoretic intersection multiplicity.

Learning goals

By the end of the chapter, the reader should be able to:

- define a plane cubic and test smoothness from the homogeneous partial derivatives;
- distinguish smooth, nodal and cuspidal cubic behavior in standard examples;
- use Bézout's theorem to count line–cubic intersections with multiplicity;
- interpret a tangent as intersection multiplicity at least two and a flex as multiplicity three;
- identify the hyperplane divisor class on a cubic and prove $P + Q + R \sim H$ for a line section;
- explain why a flex origin gives $H \sim 3O$;
- construct the Abel–Picard map $P \mapsto [\mathcal{O}_E(P - O)]$;
- use the plane-cubic degree-one rigidity lemma to prove injectivity of the Abel–Picard map;
- reduce degree-zero divisors geometrically and obtain surjectivity onto $\mathrm{Pic}^0(E)$;
- define the chord-and-tangent law for an arbitrary chosen origin O;
- specialize the construction to a flex origin and recover the usual "third point, then inverse" rule;
- deduce associativity and commutativity from $\mathrm{Pic}^0(E)$;
- put a cubic with a flex into Weierstrass form and compute the explicit addition formulas when $\mathrm{char}\, k \neq 2, 3$;
- identify geometric descriptions of 2-torsion and 3-torsion;
- understand how nodal and cuspidal cubics degenerate toward multiplicative and additive group laws;
- separate geometric statements over $\bar{k}$ from rational-point statements over a nonclosed field.

Conceptual roadmap

$$\boxed{\text{line sections} \Longrightarrow \text{divisor relations} \Longrightarrow \mathrm{Pic}^0(E) \Longrightarrow \text{group law on } E}$$

$$\begin{tikzcd}
\begin{array}{c}\text{line section}\\ P+Q+R\end{array} \arrow[r] \arrow[d] & \begin{array}{c}\text{divisor relation}\\ P+Q+R \sim H\end{array} \arrow[d] \\
\begin{array}{c}\mathrm{Pic}^0(E)\\ {[P-O]}\end{array} \arrow[r] & \begin{array}{c}\text{cubic group law}\\ P \oplus Q\end{array}
\end{tikzcd}$$

Figure 43.1: Incidence becomes divisor theory, then Picard addition, then the group law on the cubic.

43.1 Plane cubics and the smoothness test

Definition 43.1 (Plane cubic)**.** A *plane cubic* over k is a closed subscheme

$$C = V_+(F) \subset \mathbb{P}^2_k$$

defined by a nonzero homogeneous polynomial

$$F(x, y, z) \in k[x, y, z]$$

of degree 3.

The Jacobian criterion gives the basic smoothness test. A geometric point $P \in C(\bar{k})$ is singular precisely when

$$F(P) = F_x(P) = F_y(P) = F_z(P) = 0.$$

Because Euler's relation for a cubic is

$$3F = xF_x + yF_y + zF_z,$$

one should not use it to discard $F = 0$ in characteristic 3 without checking the hypotheses separately.

Proposition 43.2 (Smooth plane cubic criterion)**.** *The plane cubic $C = V_+(F)$ is smooth over k if and only if the homogeneous partial derivatives F_x, F_y, F_z have no common geometric zero on C.*

Proof. This is the projective Jacobian criterion for a hypersurface. At a point of a hypersurface, singularity means that the differential dF vanishes in the cotangent space, equivalently all homogeneous partial derivatives vanish there. □

A smooth plane cubic is geometrically integral. Indeed a reducible cubic over $\bar{k}$ would contain components of positive degree, and distinct components meet in $\mathbb{P}^2$; every such intersection is singular on the union. A nonreduced cubic is likewise singular along the repeated component.

Example 43.3 (Fermat cubic)**.** If $\operatorname{char} k \neq 3$, the cubic

$$x^3 + y^3 + z^3 = 0$$

is smooth because simultaneous vanishing of the partial derivatives forces $x = y = z = 0$, which is not a projective point. In characteristic 3, the polynomial is $(x + y + z)^3$, so the scheme is nonreduced and certainly not smooth.

Example 43.4 (A nodal and a cuspidal model)**.** In characteristic different from 2, the affine cubic

$$y^2 = x^2(x + 1)$$

has a node at the origin: the quadratic tangent cone has two distinct lines. The cubic

$$y^2 = x^3$$

has a cusp: the tangent cone is a repeated line. Their projective closures are singular plane cubics and will reappear in the degeneration section.

Example 43.5 (Fermat cubic smoothness)**.** For $F = X^3 + Y^3 + Z^3$ in characteristic not equal to 3, the partial derivatives are $3X^2, 3Y^2, 3Z^2$. They cannot vanish simultaneously at a projective point, so the Fermat cubic is smooth.

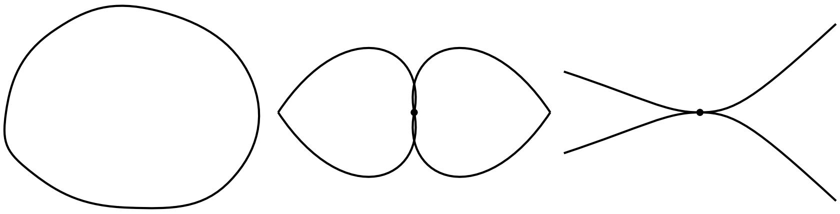


Figure 43.2: Smooth, nodal, and cuspidal plane-cubic behavior. The singular sketches are schematic local pictures.

43.2 Lines, Bézout, and the hyperplane divisor

Let $E \subset \mathbb{P}^2$ be smooth of degree 3. If a line $L = V_+(\ell)$ is not a component of E, Bézout's theorem gives

$$\sum_{P \in E \cap L} I_P(E, L) = 3.$$

Thus a line meets a smooth cubic in three points counted with multiplicity.

The restriction of ℓ to E is a global section of

$$\mathcal{O}_E(1) := \mathcal{O}_{\mathbb{P}^2}(1)|_E.$$

Its zero divisor is exactly the scheme-theoretic intersection divisor $L \cdot E$. Consequently every line section is linearly equivalent to the same degree-three divisor class.

Definition 43.6 (Hyperplane class on a cubic)**.** The divisor class

$$H := [L \cdot E] \in \operatorname{Pic}(E)$$

for any line $L \subset \mathbb{P}^2$ is the *hyperplane class*. Equivalently

$$\mathcal{O}_E(H) \cong \mathcal{O}_E(1), \qquad \deg H = 3.$$

Theorem 43.7 (Collinear divisor relation)**.** *If a line meets E in the degree-three divisor*

$$P + Q + R$$

with multiplicities, then

$$P + Q + R \sim H.$$

More generally, any two line sections of E are linearly equivalent.

Proof. If $L = V_+(\ell)$ and $M = V_+(m)$, then the rational function ℓ/m on E has divisor

$$\operatorname{div}_E(\ell/m) = (L \cdot E) - (M \cdot E).$$

Hence the two intersection divisors are linearly equivalent. □

This proof is the first place where the divisor theory of VI/39–VI/42 becomes computational geometry: a ratio of linear forms converts an incidence statement in $\mathbb{P}^2$ into a principal divisor on E.

43.3 Tangents, residual points, and flexes

At a smooth point $P = [a : b : c] \in E$, the tangent line is

$$T_P E : F_x(P)x + F_y(P)y + F_z(P)z = 0,$$

In small characteristic, interpret this formula through the differential dF; this avoids relying on Euler identities that may degenerate.

Definition 43.8 (Tangent and flex)**.** A line L is tangent to E at P if

$$I_P(E, L) \geq 2.$$

The point P is a *flex* (inflection point) if its tangent line satisfies

$$I_P(E, T_P E) = 3.$$

Thus the tangent section at a flex is the divisor $3P$.

For a nonflex point P, the tangent meets E in

$$2P + R$$

for a unique residual point R. If a secant passes through distinct P, Q, its intersection divisor is

$$P + Q + R.$$

These residual points remain meaningful in all limiting cases once multiplicity is used consistently.

Proposition 43.9 (Flex origin trivializes the hyperplane shift)**.** *If $O \in E$ is a flex, then*

$$H \sim 3O, \qquad \mathcal{O}_E(1) \cong \mathcal{O}_E(3O).$$

Hence every line section $P + Q + R$ satisfies

$$\boxed{P + Q + R \sim 3O.}$$

Proof. The tangent line at O cuts out $3O$, and every line section represents H. □

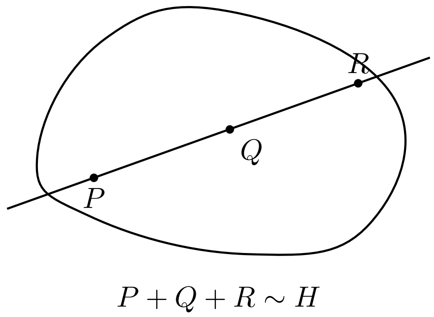


Figure 43.3: A line cuts a cubic in a divisor of degree three. Multiplicities replace distinct points in tangent cases.

43.4 The Abel–Picard map

Fix a point $O \in E(k)$, not necessarily a flex for the moment.

Definition 43.10 (Abel–Picard map for a cubic)**.** Define

$$\alpha_O : E(k) \longrightarrow \mathrm{Pic}^0(E), \qquad \alpha_O(P) = [\mathcal{O}_E(P - O)].$$

Over an algebraically closed field we also write this as a map on geometric points

$$\alpha_O : E \longrightarrow \mathrm{Pic}^0(E).$$

The degree is zero because $\deg(P - O) = 0$. The nontrivial statement is that on a smooth cubic this map captures every degree-zero divisor class exactly once.

Theorem 43.11 (Degree-one rigidity on a smooth cubic)**.** *If $P, Q \in E$ and*

$$P \sim Q,$$

then $P = Q$. Equivalently, no nonconstant rational function on a smooth plane cubic has a single simple pole.

Proof. If $P \neq Q$ and $P - Q = \mathrm{div}(f)$, then the rational function f defines a morphism

$$f : E \longrightarrow \mathbb{P}^1$$

of degree 1. A finite degree-one morphism between smooth projective curves is an isomorphism. But a smooth plane cubic has genus

$$g = \frac{(3-1)(3-2)}{2} = 1,$$

whereas $\mathbb{P}^1$ has genus 0. This is impossible. Thus $P = Q$. $\square$

The genus formula is the only curve-genus input needed for injectivity here. Later cohomological machinery packages the same fact as the degree-one case of Riemann–Roch.

Proposition 43.12 (Geometric reduction of two points)**.** *For any $A, B \in E$, there exists a unique point $S \in E$ such that*

$$A + B \sim O + S.$$

It is obtained as follows: let the secant or tangent through A, B have residual point R, then let the line through R, O have residual point S.

Proof. The first line gives

$$A + B + R \sim H,$$

and the second gives

$$R + O + S \sim H.$$

Subtracting the two relations yields $A + B \sim O + S$. Uniqueness follows from degree-one rigidity: if also $A + B \sim O + S'$, then $S \sim S'$, hence $S = S'$. $\square$

This reduction already contains the group law, but it is useful first to finish the Picard statement.

Theorem 43.13 (Abel–Picard theorem for a smooth plane cubic)**.** *Let k be algebraically closed and $E \subset \mathbb{P}^2_k$ a smooth plane cubic with $O \in E(k)$. Then*

$$\alpha_O : E(k) \xrightarrow{\sim} \mathrm{Pic}^0(E)$$

is a bijection.

Proof. Injectivity is immediate from degree-one rigidity: if $\alpha_O(P) = \alpha_O(Q)$, then $P - O \sim Q - O$, so $P \sim Q$ and hence $P = Q$.

For surjectivity, let $[D] \in \mathrm{Pic}^0(E)$. Because E is smooth, divisor classes and line bundles agree, and over an algebraically closed field we may write a degree-zero divisor as a difference of effective divisors of the same degree:

$$D = (P_1 + \cdots + P_n) - (Q_1 + \cdots + Q_n).$$

Repeatedly applying Proposition 43.12 reduces

$$P_1 + \cdots + P_n \sim P + (n-1)O, \qquad Q_1 + \cdots + Q_n \sim Q + (n-1)O$$

for some $P, Q \in E$. Thus $D \sim P - Q$.

It remains to replace $P - Q$ by a divisor of the form $S - O$. Let the tangent at O cut

$$2O + A \sim H.$$

Let the line through Q, A have residual point B, so

$$Q + A + B \sim H.$$

Subtracting gives

$$Q + B \sim 2O.$$

Now reduce $P + B$: there is S with

$$P + B \sim O + S.$$

Using $B \sim 2O - Q$, we obtain

$$P - Q \sim S - O.$$

Therefore $[D] = \alpha_O(S)$. □

$$\begin{tikzcd}
\boxed{P \in E} \arrow[r] \arrow[dr, "\alpha_O"'] & \boxed{P - O \in \mathrm{Div}^0(E)} \arrow[r] & \boxed{\mathcal{O}_E(P-O)} \arrow[dl] \\
& \boxed{[\mathcal{O}_E(P-O)] \in \mathrm{Pic}^0(E)} &
\end{tikzcd}$$

Figure 43.4: For a smooth plane cubic, the Abel–Picard map identifies points with degree-zero line-bundle classes.

43.5 The chord-and-tangent group law with arbitrary origin

We can now define addition intrinsically by transporting the group law of $\mathrm{Pic}^0(E)$ back to E.

Definition 43.14 (Cubic group law). For $P, Q \in E$, define $P \oplus Q$ to be the unique point S such that

$$\alpha_O(S) = \alpha_O(P) + \alpha_O(Q).$$

Equivalently,

$$P + Q \sim O + S.$$

Proposition 43.12 gives the geometric construction without first computing in Pic^0:

1. draw the secant through P, Q, or the tangent if $P = Q$, and call the residual intersection R;
2. draw the line through R, O, using the tangent if $R = O$, and call its residual intersection S;
3. then $P \oplus Q = S$.

Theorem 43.15 (Group law from Picard theory). *The operation $\oplus$ makes E into an abelian group with identity O. Under the Abel–Picard bijection,*

$$(E, \oplus, O) \xrightarrow{\sim} \mathrm{Pic}^0(E)$$

is an isomorphism of abelian groups.

Proof. The operation was defined by transport of structure along the bijection α_O. Hence associativity and commutativity follow from tensor product of line bundles, the identity is the unique point mapping to $[\mathcal{O}_E]$, namely O, and inverses are transported from dual line bundles. □

This proof is conceptually important. A direct coordinate verification of associativity is possible but unenlightening. Divisor theory explains why the apparently ad hoc chord construction is forced by an already existing abelian group.

Proposition 43.16 (Geometric inverse for arbitrary origin). *Let the tangent at O meet E in $2O + A$. For $P \in E$, let the line through P, A meet E residually at P^-. Then*

$$P \oplus P^- = O.$$

Equivalently,

$$P + P^- \sim 2O.$$

Proof. The tangent relation gives $2O + A \sim H$, while the second line gives $P + A + P^- \sim H$. Subtraction yields $P + P^- \sim 2O$, which is exactly $\alpha_O(P^-) = -\alpha_O(P)$. □

Example 43.17 (A nodal cubic). The affine cubic $y^2 = x^2(x + 1)$ has a singular point at the origin because both partial derivatives vanish there. Its normalization separates the two tangent directions of the node.

43.6 The flex-origin form of the law

Suppose now that O is a flex. Then the tangent at O has intersection divisor $3O$, so the auxiliary point A in the preceding construction is simply O.

Corollary 43.18 (Three collinear points sum to zero). *With a flex origin O, if a line cuts E in $P+Q+R$, then*

$$P \oplus Q \oplus R = O.$$

Equivalently,

$$\alpha_O(P) + \alpha_O(Q) + \alpha_O(R) = 0.$$

Proof. Since $P + Q + R \sim H \sim 3O$, subtract $3O$ and pass to $\mathrm{Pic}^0(E)$. □

Thus to add P and Q, find the third point R of the secant or tangent and then take the inverse of R. With a flex origin, the inverse of R is the third point on the line through R and O.

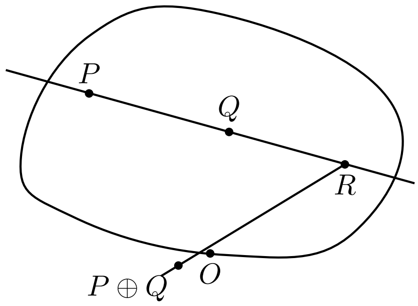


Figure 43.5: With a flex origin, add P and Q by taking the third secant point R and then the third point on the line through R and O.

43.7 Weierstrass form

A smooth cubic with a chosen flex can be put into a form adapted to the group law. After a projective change of coordinates placing

$$O = [0:1:0]$$

and its tangent at infinity, one obtains a generalized Weierstrass equation

$$y^2z + a_1xyz + a_3yz^2 = x^3 + a_2x^2z + a_4xz^2 + a_6z^3.$$

If $\mathrm{char}\, k \neq 2, 3$, completing the square and cube yields the short form

$$E: \quad y^2z = x^3 + Axz^2 + Bz^3,$$

or on the affine chart $z = 1$,

$$y^2 = x^3 + Ax + B.$$

Proposition 43.19 (Smoothness discriminant in short Weierstrass form). *Assume* $\mathrm{char}\, k \neq 2, 3$. *The cubic*

$$y^2z = x^3 + Axz^2 + Bz^3$$

is smooth if and only if

$$\Delta = -16(4A^3 + 27B^2) \neq 0.$$

Proof. On the affine chart, a singular point must satisfy

$$y = 0, \qquad 3x^2 + A = 0, \qquad x^3 + Ax + B = 0.$$

Such a common solution exists exactly when the cubic polynomial $x^3 + Ax + B$ has a repeated root, equivalently when its discriminant $-4A^3 - 27B^2$ vanishes. The conventional elliptic discriminant differs by the nonzero factor 16. □

The point $O = [0 : 1 : 0]$ is a flex and acts as the identity.

43.8 Explicit addition formulas

Assume again $\operatorname{char} k \neq 2, 3$ and

$$E : y^2 = x^3 + Ax + B.$$

Let $P = (x_1, y_1)$, $Q = (x_2, y_2)$.

If $P \neq Q$ and $x_1 \neq x_2$, the secant slope is

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

If $P = Q$ and $y_1 \neq 0$, the tangent slope is

$$m = \frac{3x_1^2 + A}{2y_1}.$$

Substitution of the line into the cubic shows that the x-coordinates of the three intersection points have sum m^2. Therefore the group sum $P \oplus Q = (x_3, y_3)$ is

$$\boxed{x_3 = m^2 - x_1 - x_2,}$$

$$\boxed{y_3 = m(x_1 - x_3) - y_1.}$$

If $x_1 = x_2$ and $y_2 = -y_1$, then the secant is vertical and

$$P \oplus Q = O.$$

In particular

$$-(x, y) = (x, -y).$$

If $y_1 = 0$, then $P = -P$, so $2P = O$.

Proposition 43.20 (Coordinate formulas agree with Picard addition)**.** *The above formulas compute the unique point S satisfying*

$$[\mathcal{O}_E(S - O)] = [\mathcal{O}_E(P - O)] \otimes [\mathcal{O}_E(Q - O)].$$

Proof. The secant or tangent meets E at a third point R. In short Weierstrass form the inverse map is reflection across the x-axis, so $S = -R$. The collinear divisor relation gives $P + Q + R \sim 3O$, hence $P + Q \sim O + S$, which is the Picard identity. □

Example 43.21 (Weierstrass form)**.** A smooth cubic in Weierstrass form $Y^2Z = X^3 + aXZ^2 + bZ^3$ has discriminant $\Delta = -16(4a^3 + 27b^2)$. Nonvanishing of Δ is the smoothness criterion in characteristic away from the usual exceptional cases.

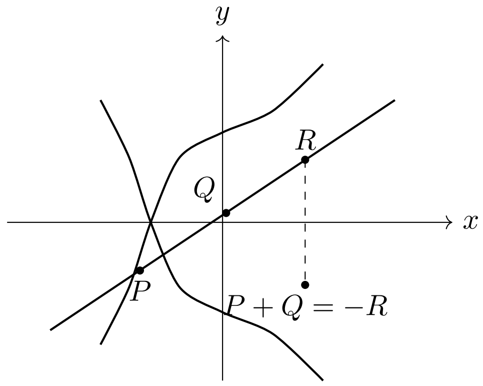


Figure 43.6: In short Weierstrass form, inversion is reflection across the x-axis.

43.9 Torsion seen in the plane

The geometry of low-order torsion is especially transparent.

Proposition 43.22 (Two-torsion)**.** *In short Weierstrass form, a finite point $P = (x, y)$ satisfies $2P = O$ if and only if $y = 0$. Thus nontrivial 2-torsion points correspond to roots of*

$$x^3 + Ax + B.$$

Proof. A point has order dividing two exactly when $P = -P$. In short Weierstrass form, inversion sends (x, y) to $(x, -y)$. Hence a finite point is fixed by inversion exactly when $y = 0$. □

Proposition 43.23 (Flexes and three-torsion)**.** *Let O be a flex origin. A point $P \in E$ is a flex if and only if*

$$3P = O.$$

Thus flexes are exactly the geometric 3-torsion points for the embedding determined by $\mathcal{O}_E(3O)$.

Proof. If P is a flex, its tangent cuts $3P$, so

$$3P \sim H \sim 3O,$$

and hence $3P = O$ in the group law. Conversely, if $3P = O$, then $3P \sim 3O \sim H$. The canonical section of $\mathcal{O}_E(3P) \cong \mathcal{O}_E(1)$ therefore gives a section of $\mathcal{O}_E(1)$ vanishing exactly to order three at P. Degree-one sections of a plane cubic are restrictions of linear forms, so this section is cut out by a line. That line has intersection divisor $3P$, hence it is the tangent line and P is a flex. □

When $\operatorname{char} k \neq 3$, a smooth cubic over an algebraically closed field has nine geometric flexes, matching

$$E[3] \cong (\mathbb{Z}/3\mathbb{Z})^2.$$

In characteristic 3, the group scheme $E[3]$ need not have nine distinct geometric points, so the naive count of flexes must be modified.

43.10 Singular cubics as degenerations

Smoothness is essential for the elliptic-curve picture. An irreducible singular plane cubic has normalization $\mathbb{P}^1$; in characteristic away from the small exceptional cases, the singularity is either a node or a cusp.

The smooth locus still carries a natural algebraic group structure after choosing an identity, but the group changes type:

$$\text{nodal cubic smooth locus} \sim \mathbb{G}_m, \qquad \text{cuspidal cubic smooth locus} \sim \mathbb{G}_a.$$

This is the simplest geometric explanation for why multiplicative and additive groups occur as boundary degenerations of elliptic curves.

Example 43.24 (Nodal parameter)**.** For a nodal cubic, normalization separates the two branches above the node. After choosing coordinates on $\mathbb{P}^1$ sending the two preimages to 0 and ∞, the smooth locus corresponds to

$$\mathbb{P}^1 \setminus \{0, \infty\} = \mathbb{G}_m.$$

The chord law becomes multiplication in a suitable parameter.

Example 43.25 (Cuspidal parameter)**.** For a cuspidal cubic, the normalization has one point lying above the cusp. Removing it gives

$$\mathbb{P}^1 \setminus \{\infty\} = \mathbb{A}^1 = \mathbb{G}_a,$$

and the chord law becomes addition in a suitable parameter.

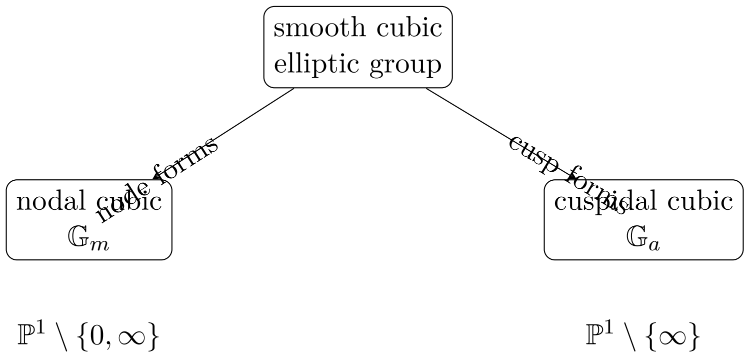


Figure 43.7: The two basic singular degenerations of an elliptic group law.

43.11 Over a nonclosed field

Let now k be arbitrary and suppose E/k is a smooth plane cubic with a rational point $O \in E(k)$. For rational points $P, Q \in E(k)$, the line through them is defined over k. Its intersection polynomial with E has two known k-rational roots counted with multiplicity, so the residual third point is again k-rational. The same applies to tangent lines. Consequently the chord-and-tangent construction closes on $E(k)$.

The correct scheme-level statement is that E with a chosen rational origin is an elliptic curve: a smooth proper geometrically connected genus-one curve together with a rational point. Its group law is defined over k, and the Abel–Picard identification is defined over k. In Picard-scheme notation,

$$E(k) \cong \mathrm{Pic}^0_{E/k}(k).$$

Because the chosen rational point rigidifies the degree-zero Picard functor, these points correspond to the expected degree-zero line-bundle classes. Without a rational base point, one must distinguish rational divisor classes, Galois-fixed geometric classes, and the Picard variety; descent can matter.

If $E(k)$ contains a rational flex, the plane embedding may be arranged in Weierstrass form over k. More generally a rational point can be chosen as the identity even if it is not a flex; the two-line construction from the arbitrary-origin section remains valid.

43.12 A practical divisor dictionary for cubics

For a smooth cubic E and a chosen flex origin O:

geometry	divisor/Picard statement
$L \cap E = P + Q + R$	$P + Q + R \sim H \sim 3O$
$T_P E = 2P + R$	$2P + R \sim 3O$
P flex	$3P \sim 3O$
$P \oplus Q = S$	$P + Q \sim O + S$
$-P = R$	$P + R \sim 2O$
$3P = O$	P is a flex

The entire group law is encoded by linear equivalence of degree-three line sections.

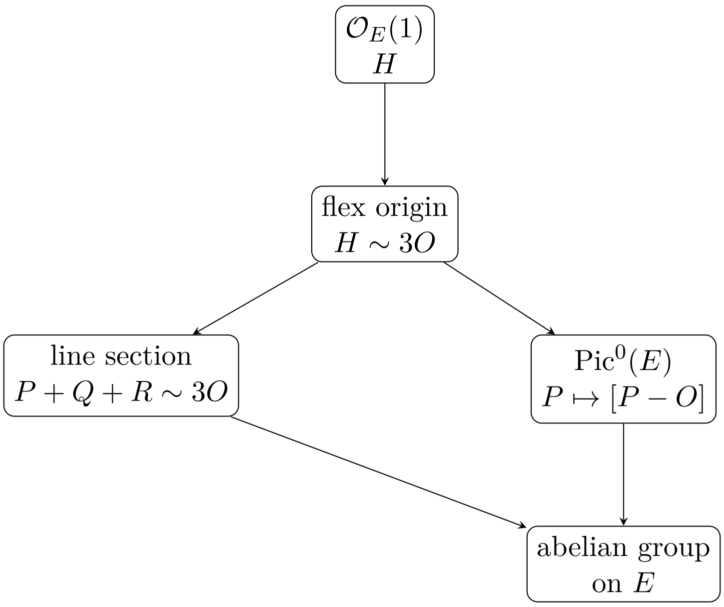


Figure 43.8: The flex-origin dictionary that turns degree-three plane geometry into degree-zero Picard addition.

43.13 Exercises

Exercise 43.26. Let F be a homogeneous cubic. State the projective Jacobian criterion for singular points of $V_+(F)$.

Exercise 43.27. Assume $\operatorname{char} k \neq 3$. Prove that $x^3 + y^3 + z^3 = 0$ is smooth.

Exercise 43.28. Why must a reducible plane cubic over an algebraically closed field be singular somewhere?

Exercise 43.29. Show that a line not contained in a cubic meets it in total intersection multiplicity 3.

Exercise 43.30. If two lines L, M cut divisors D_L, D_M on E, prove $D_L \sim D_M$.

Exercise 43.31. Prove that the hyperplane class on a plane cubic has degree 3.

Exercise 43.32. If O is a flex, prove $\mathcal{O}_E(1) \cong \mathcal{O}_E(3O)$.

Exercise 43.33. Let P, Q, R be collinear and let O be a flex origin. Show that $[P - O] + [Q - O] + [R - O] = 0$ in $\mathrm{Pic}^0(E)$.

Exercise 43.34. Explain why $P \mapsto [\mathcal{O}_E(P - O)]$ is injective once degree-one rigidity is known.

Exercise 43.35. Carry out the two-line construction and prove directly that its output S satisfies $P + Q \sim O + S$.

Exercise 43.36. Using the Picard description, prove that O is the identity for $\oplus$.

Exercise 43.37. Prove commutativity of the chord law without referring to a picture.

Exercise 43.38. Prove associativity of the chord law using $\mathrm{Pic}^0(E)$.

Exercise 43.39. For a flex origin, show that the inverse of P is the residual point on the line through P and O.

Exercise 43.40. In short Weierstrass form, show that $-(x, y) = (x, -y)$.

Exercise 43.41. Derive $x_3 = m^2 - x_1 - x_2$ for the sum of two affine points in short Weierstrass form.

Exercise 43.42. Derive the tangent slope $m = (3x_1^2 + A)/(2y_1)$.

Exercise 43.43. Show that a finite point in short Weierstrass form has order two exactly when $y = 0$.

Exercise 43.44. With flex origin O, prove that every flex is a 3-torsion point.

Exercise 43.45. Assume $\mathrm{char}\, k \neq 2, 3$. Verify that $y^2 = x^3 + Ax + B$ is singular exactly when $4A^3 + 27B^2 = 0$.

Exercise 43.46. Explain why the residual third point of a line through two k-rational points is again k-rational.

Exercise 43.47. Show that the normalization of a nodal cubic has two points above the node, while the normalization of a cuspidal cubic has one.

Exercise 43.48. Why does removing the two points above a node from $\mathbb{P}^1$ produce $\mathbb{G}_m$?

Exercise 43.49. Why does removing one point from $\mathbb{P}^1$ produce $\mathbb{A}^1$, the underlying variety of $\mathbb{G}_a$?

43.14 Solved problem dossiers

Problem 43.50 (A line divisor from a rational function)**.** Let $L = (\ell = 0)$ and $M = (m = 0)$ be lines meeting E properly. Compute $\mathrm{div}_E(\ell/m)$.

Problem 43.51 (Doubling as a tangent construction)**.** Let O be a flex and $P \in E$. Describe $2P$ using only tangent and secant lines.

Problem 43.52 (Inverse for a nonflex origin)**.** Let O be any chosen point. Give a two-line construction of $-P$.

Problem 43.53 (Why the two-line law is independent of choices)**.** Explain why the residual point construction gives a unique output even in tangent cases.

Problem 43.54 (Three collinear points and Picard addition)**.** Assume O is a flex. If P, Q, R are collinear, compute $\alpha_O(R)$ in terms of $\alpha_O(P), \alpha_O(Q)$.

Problem 43.55 (Concrete addition on $y^2 = x^3 - x$)**.** Over a field of characteristic different from 2, compute the nontrivial 2-torsion points on

$$y^2 = x^3 - x.$$

Problem 43.56 (A doubling formula)**.** On $E : y^2 = x^3 + Ax + B$, derive the coordinates of $2P$ for $P = (x, y)$ with $y \neq 0$.

Problem 43.57 (Flexes as three-torsion)**.** Assume $\mathrm{char}\, k \neq 3$. Explain conceptually why a smooth cubic has nine flexes over $\bar{k}$.

Problem 43.58 (Degree-zero divisor reduction)**.** Let

$$D = P_1 + P_2 + P_3 - Q_1 - Q_2 - Q_3.$$

Describe an algorithm using only secants and tangents that produces S with $D \sim S - O$.

Problem 43.59 (Nodal degeneration)**.** Why is $\mathbb{G}_m$, rather than an elliptic curve, the natural group on the smooth locus of a nodal cubic?

Problem 43.60 (Cuspidal degeneration)**.** Give the analogous explanation for $\mathbb{G}_a$ on a cuspidal cubic.

Problem 43.61 (Rational closure of the chord law)**.** Let E/k be a smooth cubic and $P, Q, O \in E(k)$. Prove that the chord-and-tangent construction produces a point of $E(k)$.

43.15 Challenges

Problem 43.62 (Challenge: associativity without coordinates)**.** Starting only from VI/42 divisor and Picard theory, the line-section relation, and degree-one rigidity, give a complete proof that the chord-and-tangent construction is associative. Your proof should identify exactly where injectivity of α_O is required.

Problem 43.63 (Challenge: arbitrary origin versus flex origin)**.** Develop the group law first with an arbitrary origin O, then choose a flex F and compare the two group structures explicitly. Determine the translation that identifies them and explain why the one-line relation $P + Q + R \sim 3F$ is unavailable for a general origin.

Problem 43.64 (Challenge: Weierstrass transformation)**.** Assume $\operatorname{char} k \neq 2, 3$. Starting from a smooth plane cubic with a chosen flex, carry out the projective coordinate changes leading to

$$y^2 z = x^3 + Axz^2 + Bz^3,$$

and track the flex and tangent line throughout the transformation.

Problem 43.65 (Challenge: singular boundary of elliptic addition)**.** For explicit one-parameter families of smooth cubics degenerating to a node and to a cusp, follow the discriminant, normalization, and chord law. Recover multiplicative and additive parameters in the limits and explain the appearance of $\mathbb{G}_m$ and $\mathbb{G}_a$.

Problem 43.66 (Challenge: bridge to Cremona transformations)**.** Study how a quadratic plane transformation acts on a cubic passing through its base points. Track degree, multiplicities, and divisor classes carefully enough to predict which parts of the cubic group-law geometry survive birational modification. Use this as preparation for VI/44.

Bridge to VI/44

For a smooth plane cubic, a line is no longer just a projective-geometric object. Its intersection divisor is a degree-three representative of $\mathcal{O}_E(1)$, and once a flex origin is chosen,

$$\mathcal{O}_E(1) \cong \mathcal{O}_E(3O).$$

The Picard group then turns the elementary incidence relation

$$P + Q + R \sim 3O$$

into an abelian group law on the curve. The striking point is that associativity comes from tensor product of line bundles, not from a hidden miracle of Euclidean-looking chord diagrams.

VI/44 changes the ambient geometry rather than the curve. It studies *Cremona transformations*: birational self-maps of the plane, generated in the basic case by quadratic coordinate functions. The needed bookkeeping has already appeared here: degree, base points, multiplicities, strict transforms, and divisor classes. VI/44 uses these data to follow plane curves through birational operations.

43.16 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 43.67 (Fermat gradient)**.** Compute the gradient of $X^3 + Y^3 + Z^3$.

Hint. Use smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. $(3X^2, 3Y^2, 3Z^2)$. ☐

Exercise 43.68 (Node test)**.** Check the origin of $y^2 - x^2(x+1) = 0$.

Hint. Use smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. Both partial derivatives vanish, so it is singular. ☐

Exercise 43.69 (Weierstrass discriminant)**.** What condition on $4a^3 + 27b^2$ gives smoothness?

Hint. Use smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. It must be nonzero in the standard characteristic assumptions. ☐

Exercise 43.70 (Degree)**.** What is the degree of a plane cubic?

Hint. Use smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. 3. ☐

Exercise 43.71 (Genus)**.** What is the genus of a smooth plane cubic?

Hint. Use smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. 1. ☐

Proofs

Exercise 43.72 (Jacobian criterion)**.** Prove: Prove the Fermat cubic is smooth in characteristic not 3.

Hint. Work from smooth plane cubics, singularity tests, and genus-one geometry and state the hypotheses explicitly. ☐

Solution. Simultaneous vanishing of the three partials would force all projective coordinates to vanish. ☐

Exercise 43.73 (Singular point)**.** Prove: Verify the nodal cubic has a singularity at the origin.

Hint. Work from smooth plane cubics, singularity tests, and genus-one geometry and state the hypotheses explicitly. ☐

Solution. Evaluate the defining polynomial and both first partial derivatives at the origin. ☐

Exercise 43.74 (Line intersection count)**.** Prove: Explain why a line meets a plane cubic in three points counted with multiplicity over an algebraically closed field.

Hint. Work from smooth plane cubics, singularity tests, and genus-one geometry and state the hypotheses explicitly. ☐

Solution. Restrict the cubic equation to the line and apply the degree-three polynomial count, including multiplicity. ☐

Exercise 43.75 (Genus formula)**.** Prove: Use the plane-curve genus formula to compute the genus of a smooth cubic.

Hint. Work from smooth plane cubics, singularity tests, and genus-one geometry and state the hypotheses explicitly. ☐

Solution. For degree d, $g = (d-1)(d-2)/2$; insert $d = 3$. ☐

Counterexamples and hypothesis tests

Exercise 43.76 (Every cubic smooth)**.** Test the claim: every irreducible plane cubic is smooth.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; nodal and cuspidal irreducible cubics are singular. ☐

Exercise 43.77 (Genus one singular)**.** Test the claim: a singular plane cubic has the same geometric genus as a smooth cubic.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; normalization lowers the geometric genus. ☐

Exercise 43.78 (One tangent)**.** Test the claim: a node has only one tangent direction.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; a node has two distinct tangent directions. ☐

Applications and investigations

Exercise 43.79 (Elliptic curves)**.** When does a plane cubic become an elliptic curve?

Hint. Translate the question into smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. When it is smooth and equipped with a chosen rational base point. ☐

Exercise 43.80 (Normalization)**.** What does normalization do to a singular cubic?

Hint. Translate the question into smooth plane cubics, singularity tests, and genus-one geometry. ☐

Solution. It replaces the singular curve by a smooth curve with a finite birational map to the cubic. ☐

Challenge problems

Exercise 43.81 (Synthesis challenge)**.** Connect two central viewpoints of Plane Cubics in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Plane Cubics and then translates it through smooth plane cubics, singularity tests, and genus-one geometry, checking both descriptions agree. ☐

Exercise 43.82 (Hypothesis challenge)**.** Choose one theorem-level statement from Plane Cubics, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Chapter 44

Cremona Transformations

VI/43 used projective lines to study a fixed plane cubic. The present chapter changes the viewpoint: now the ambient plane itself is allowed to move birationally. A *Cremona transformation* is a birational self-map of $\mathbb{P}^2$. Such a map is not required to be defined everywhere, and that single relaxation produces the central phenomena of the chapter: base points, contracted curves, inverse base points, degree change, and linear systems with assigned multiplicities.

The model example is the standard quadratic transformation

$$\sigma : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2, \qquad [x : y : z] \longmapsto [yz : xz : xy].$$

It is undefined at the three coordinate points

$$p_1 = [1 : 0 : 0], \qquad p_2 = [0 : 1 : 0], \qquad p_3 = [0 : 0 : 1],$$

contracts the three coordinate lines, and satisfies $\sigma^2 = \mathrm{id}$ wherever both sides are defined. Thus a map given by quadratic polynomials can nevertheless have a rational inverse of the same degree.

This example is the prototype for the entire subject. A general rational map is encoded by homogeneous forms of the same degree; their common zero set is the base locus. A birational map sends a general curve to a birational transform, but the degree of the image is corrected by the multiplicities of the curve at the base points. For the standard quadratic transformation the basic formula is

$$\boxed{\deg \sigma_*(C) = 2 \deg C - m_1 - m_2 - m_3,}$$

provided the coordinate lines are not components of C, where $m_i = \mathrm{mult}_{p_i}(C)$.

The formula already points toward VI/45. The expression

$$2d - m_1 - m_2 - m_3$$

looks like intersection against a divisor class $2H - E_1 - E_2 - E_3$. In this chapter that notation is only a preview. VI/45 will create the surface on which the exceptional classes E_i actually live and on which σ becomes an honest morphism.

Standing conventions

Unless stated otherwise, k is a field and all projective spaces are over k. When geometric base points are discussed, we pass to $\bar{k}$ if necessary. A rational map is identified with its restriction to any dense open set on which it is represented by regular functions. Curves are assumed reduced and without components contracted by the map when a degree formula requires that hypothesis.

Learning goals

By the end of the chapter, the reader should be able to:

- distinguish rational maps, morphisms, birational maps, and Cremona transformations;
- represent a rational map of projective space by homogeneous forms of one degree;
- compute its base locus and explain why a common factor must be removed;
- analyze the standard quadratic transformation $[x : y : z] \dashrightarrow [yz : xz : xy]$;
- identify its three base points and three contracted coordinate lines;
- verify directly that the standard quadratic transformation is an involution on a dense open set;
- compute the image of a general line and the inverse image of a general line;
- interpret the map through the net of conics passing through three noncollinear points;
- compute algebraic degree and recognize degree drop under composition caused by cancellation;
- define the birational transform of a curve away from contracted components;
- derive the degree formula $2d - m_1 - m_2 - m_3$ for the standard quadratic transformation;
- track smooth cubics, singular cubics, and curves passing through the base triangle;
- compare projective automorphisms with genuine Cremona transformations;
- understand how source and target base points are exchanged by inversion;
- recognize the divisor-class shadow $H \mapsto 2H - E_1 - E_2 - E_3$ without using blow-ups prematurely;
- explain why resolving indeterminacy is the natural next step.

Conceptual roadmap

$$\boxed{\begin{array}{c}\text{homogeneous forms} \Longrightarrow \text{base locus} \Longrightarrow \text{birational plane map}\\ \Downarrow \\ \text{degree/multiplicity bookkeeping} \Longrightarrow \text{resolution}\end{array}}$$

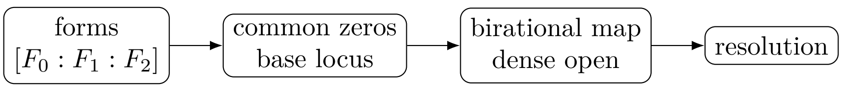


Figure 44.1: The production chain for a plane Cremona transformation.

44.1 Rational maps of the projective plane

Definition 44.1 (Rational map)**.** A *rational map*

$$\phi : X \dashrightarrow Y$$

between integral varieties is an equivalence class of morphisms $U \to Y$ defined on nonempty open subsets $U \subseteq X$, where two representatives are equivalent if they agree on a nonempty open subset of their common domain.

The maximal open subset on which the rational map is represented by a morphism is its *domain of definition*. Its complement is the *indeterminacy locus*. For maps from a smooth surface to projective space, indeterminacy is typically concentrated at finitely many points.

Definition 44.2 (Cremona transformation)**.** A *Cremona transformation of the plane* is a birational map

$$\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2.$$

The set of such transformations, under composition of rational maps, is the plane Cremona group

$$\mathrm{Cr}_2(k) = \mathrm{Bir}_k(\mathbb{P}^2).$$

Every projective automorphism belongs to $\mathrm{Cr}_2(k)$, but Cremona transformations are much more general: they may fail to be defined at points and may contract curves.

Proposition 44.3 (Homogeneous representation)**.** *A rational map*

$$\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$$

can be represented as

$$[x : y : z] \longmapsto [F_0(x, y, z) : F_1(x, y, z) : F_2(x, y, z)],$$

where the F_i are homogeneous forms of the same degree and have no nonconstant common factor. Two such triples represent the same rational map exactly when they differ by multiplication by a common nonzero rational function; after choosing primitive homogeneous triples, this reduces to a common scalar.

Proof. A rational map to projective space is determined by three rational functions, not all zero, up to common multiplication. Clearing denominators and homogenizing gives a triple of homogeneous forms of one degree. Removing their gcd does not change the rational map on the dense open set where the common factor is nonzero. Conversely, a primitive homogeneous triple defines a morphism wherever the three forms do not vanish simultaneously. □

Definition 44.4 (Base locus)**.** For a primitive representation $\phi = [F_0 : F_1 : F_2]$, the *base locus* is

$$\mathrm{Bs}(\phi) = V_+(F_0, F_1, F_2) \subset \mathbb{P}^2.$$

Its points are the base points of the rational map.

The primitive condition matters. The triple

$$[xz : yz : z^2]$$

has a common factor z and represents the identity map on $z \neq 0$; after cancellation it becomes $[x : y : z]$. Treating the whole line $z = 0$ as a genuine base locus would therefore be an artifact of a nonminimal representation.

Example 44.5 (Standard quadratic Cremona map)**.** The rational map $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ defined by $[x : y : z] \mapsto [yz : xz : xy]$ is undefined exactly at the three coordinate points. Applying the same formula twice recovers the original point away from the coordinate triangle, so the map is birational and involutive.

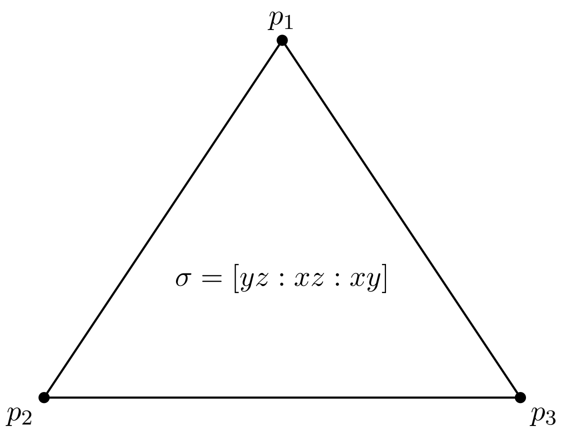


Figure 44.2: The base triangle of the standard quadratic transformation.

44.2 The standard quadratic transformation

Consider

$$\sigma[x : y : z] = [yz : xz : xy].$$

The three coordinate quadrics have no common factor, so this is a primitive quadratic representation.

Proposition 44.6 (Domain and base points of the standard map)**.** *The base locus of σ is precisely*

$$\{p_1, p_2, p_3\} = \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\}.$$

On the dense torus $xyz \neq 0$, the map is regular and can be written as

$$[x : y : z] \longmapsto [1/x : 1/y : 1/z]$$

up to a common projective scalar.

Proof. The equations $yz = xz = xy = 0$ force at least two coordinates to vanish, hence give exactly the three coordinate points. If $xyz \neq 0$, multiplying $[1/x : 1/y : 1/z]$ by xyz gives $[yz : xz : xy]$. ☐

Proposition 44.7 (Quadratic involution)**.** *The standard quadratic transformation is a birational involution:*

$$\sigma \circ \sigma = \mathrm{id}_{\mathbb{P}^2}$$

as rational maps.

Proof. On the dense open set $xyz \neq 0$,

$$\sigma^2[x : y : z] = \sigma[yz : xz : xy] = [x^2yz : xy^2z : xyz^2] = [x : y : z].$$

Equality on a dense open subset is equality of rational maps. Thus σ is its own rational inverse. ☐

This calculation also illustrates degree cancellation. Naively substituting one quadratic triple into another produces quartics, but the three quartics have the common cubic factor xyz. After cancellation, the composition has degree one.

$$
\begin{tikzcd}
\boxed{[x : y : z]} \arrow[rr, bend left, "\sigma"] & & \boxed{[yz : xz : xy]} \arrow[ll, bend left, "\sigma"]
\end{tikzcd}
$$

quartics after substitution
$xyz[x : y : z]$

Figure 44.3: The standard quadratic map is an involution after cancellation.

44.3 Exceptional curves and inverse base points

A birational map need not preserve dimension on every closed subset. For σ, each side of the coordinate triangle is contracted.

Proposition 44.8 (Contracted coordinate lines)**.** *Let*

$$L_x = V_+(x), \qquad L_y = V_+(y), \qquad L_z = V_+(z).$$

Then away from the two base points lying on each line,

$$\sigma(L_x) = p_1, \qquad \sigma(L_y) = p_2, \qquad \sigma(L_z) = p_3.$$

Thus the three coordinate lines are exceptional curves for σ, and the three target coordinate points are base points of the inverse map as well.

Proof. If $x = 0$ and $yz \neq 0$, then

$$\sigma[0 : y : z] = [yz : 0 : 0] = [1 : 0 : 0] = p_1.$$

The other two cases are cyclic permutations. Since $\sigma^{-1} = \sigma$, the same three points are the inverse base points. □

This is the first sign that a birational map does not simply "delete" information at a contracted line. The inverse map must remember that line through an indeterminacy point. VI/45 will replace each such base point by an exceptional projective line, making the exchange geometrically visible.

44.4 Lines become conics

Let $M \subset \mathbb{P}^2$ be the line

$$aX + bY + cZ = 0$$

in the target. Pulling its equation back through σ gives

$$ayz + bxz + cxy = 0.$$

This is a conic through p_1, p_2, p_3.

Proposition 44.9 (Line–conic correspondence)**.** *Under the standard quadratic transformation, the inverse image of a general target line is a conic through all three base points. Conversely, every conic in the three-dimensional vector space*

$$\langle yz, xz, xy \rangle$$

is the pullback of a line. If the conic is irreducible and avoids exceptional degeneracies, its birational transform is a line.

Proof. The pullback calculation above gives the first assertion. Every conic in the displayed span has the form $ayz + bxz + cxy$, so it arises from $aX + bY + cZ$. On the dense torus σ is an isomorphism, hence an irreducible member not contracted by σ maps birationally to the corresponding line. □

$$\boxed{\begin{array}{c}\text{target line}\\ aX+bY+cZ=0\end{array}} \xleftrightarrow{\quad\sigma\quad} \boxed{\begin{array}{c}\text{source conic}\\ ayz+bxz+cxy=0\end{array}}$$

passes through p_1, p_2, p_3

Figure 44.4: Lines correspond to conics in the defining net.

44.5 Linear systems and the map they define

The three quadrics yz, xz, xy do more than provide coordinates. Their span is a linear system of conics.

Proposition 44.10 (The net of conics through three points)**.** *The projective linear system of conics through the three noncollinear coordinate points is*

$$\mathbb{P}\langle yz, xz, xy\rangle \cong \mathbb{P}^2.$$

It defines the rational map σ. More generally, after a projective change of coordinates, the net of conics through any three noncollinear points defines a quadratic Cremona transformation.

Proof. A general conic has equation

$$ax^2 + by^2 + cz^2 + dxy + exz + fyz = 0.$$

Vanishing at p_1, p_2, p_3 forces $a = b = c = 0$, leaving the three-dimensional vector space spanned by xy, xz, yz. Choosing these three basis sections as target coordinates gives σ. Any ordered triple of noncollinear points can be carried to the coordinate points by an element of $\mathrm{PGL}_3(k)$. □

The word *net* means a projective linear system of dimension two. Thus the standard Cremona transformation is the map associated with the complete net of conics through its three base points.

Proposition 44.11 (General quadratic transformation with three proper base points)**.** *Let $q_1, q_2, q_3 \in \mathbb{P}^2$ be noncollinear points. The complete net of conics through them defines a quadratic Cremona transformation. Up to pre- and post-composition by projective automorphisms, it is the standard quadratic transformation.*

Proof. Choose $A \in \mathrm{PGL}_3(k)$ taking (q_1, q_2, q_3) to (p_1, p_2, p_3). In the new coordinates the net is $\langle yz, xz, xy\rangle$, hence defines σ. Changing the basis of the three-dimensional vector space of sections corresponds to a projective automorphism B of the target. Therefore the original map has the form $B^{-1}\sigma A$. □

Example 44.12 (Exceptional coordinate lines)**.** Under the standard Cremona map, the line $x = 0$ away from the base points is sent to $[1 : 0 : 0]$. Thus each coordinate line is contracted to the opposite base point.

44.6 Algebraic degree and cancellation

Definition 44.13 (Algebraic degree)**.** If a rational map $\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ has a primitive homogeneous representation $[F_0 : F_1 : F_2]$ with forms of common degree d, then

$$\deg(\phi) := d$$

is its *algebraic degree*.

Projective automorphisms have degree one. The standard Cremona involution has degree two. Under composition, degrees satisfy

$$\deg(\phi \circ \psi) \leq \deg(\phi) \deg(\psi),$$

and the inequality can be strict because a common factor may appear after substitution.

Proposition 44.14 (Degree drop records exceptional geometry)**.** *Let $\phi = [F_0 : F_1 : F_2]$ and $\psi = [G_0 : G_1 : G_2]$ be primitive rational maps of degrees d and e. The substituted triple $[F_0(G) : F_1(G) : F_2(G)]$ has degree de. If its gcd has degree r, then*

$$\deg(\phi \circ \psi) = de - r.$$

For σ^2, the common factor is xyz of degree three, so $4 - 3 = 1$.

Proof. All substituted forms have degree de. Dividing by the common homogeneous gcd of degree r produces a primitive representative of degree $de - r$. The calculation for σ^2 was given above. ☐

Degree cancellation is not merely formal simplification. Common factors often encode curves that one map contracts into base points of the next map.

$$\boxed{\text{curve } C, \text{ degree } d} \longrightarrow \boxed{\text{pullback conic}} \longrightarrow \boxed{2d - (m_1 + m_2 + m_3)} \longrightarrow \boxed{\deg C'}$$

Figure 44.5: Bézout produces the quadratic degree-transform formula.

44.7 Birational transforms of curves

Definition 44.15 (Birational transform of a curve)**.** Let $\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$ be birational and let $C \subset \mathbb{P}^2$ be an irreducible curve not contracted by ϕ. The *birational transform* of C is

$$\phi_*(C) := \overline{\phi(C \cap U)},$$

where U is any dense open subset on which ϕ is an isomorphism onto its image.

This definition avoids pretending that the image of points where ϕ is undefined is already meaningful. The closure is taken only after applying ϕ on a dense open set.

For the standard transformation, one can derive the degree of $\sigma_*(C)$ by intersecting it with a general target line. Its pullback is a conic through the three base points.

Theorem 44.16 (Degree transform under the standard quadratic map)**.** *Let $C \subset \mathbb{P}^2$ be an irreducible plane curve of degree d, not equal to a coordinate line, and let*

$$m_i = \operatorname{mult}_{p_i}(C)$$

for the three base points of σ. Assume the chosen general member of the conic net has no common component with C. Then the birational transform $C' = \sigma_*(C)$ has degree

$$\boxed{\deg C' = 2d - m_1 - m_2 - m_3.}$$

Proof. Let M be a general line in the target. Its inverse image is a general conic Q through p_1, p_2, p_3. By Bézout,

$$C \cdot Q = 2d.$$

At p_i, the general conic is smooth and has a general tangent direction, so the local intersection contributes exactly m_i. The remaining intersection points lie in the open set where σ is an isomorphism and correspond bijectively, with multiplicity, to intersections of C' with M. Hence

$$\deg C' = 2d - (m_1 + m_2 + m_3).$$

□

The formula is the elementary predecessor of a divisor calculation on a blown-up plane. It is also immediately useful:

- a general line $(d = 1, m_i = 0)$ becomes a conic;
- a line through one base point $(d = 1, m_i = 1)$ becomes a line;
- a conic through all three base points $(d = 2, m_1 = m_2 = m_3 = 1)$ becomes a line;
- a cubic through all three base points simply $(d = 3, m_i = 1)$ remains cubic.

44.8 What happens to the base triangle

The degree formula does not apply naively to a coordinate line because that line is contracted. For instance, L_x has degree one and contains p_2, p_3, so the formal expression gives

$$2 - 1 - 1 = 0,$$

which correctly signals that the image has dimension zero rather than being a degree-zero curve.

This is a useful diagnostic rule: when the transformed degree drops to zero, first ask whether the curve is exceptional. The formula is not a license to regard points as curves of degree zero; it is a numerical shadow of contraction.

44.9 Cubic curves under a quadratic transformation

VI/43 makes cubic curves a natural test family. The transformation can preserve, raise, or effectively lower plane degree depending on the cubic's incidence with the base points.

Proposition 44.17 (Cubic transform bookkeeping)**.** *Let C be an irreducible cubic not containing a coordinate line.*

(a) If C avoids all three base points, then $\deg \sigma_*(C) = 6$.

(b) If C passes simply through exactly one, two, or three base points, the transformed degrees are respectively $5, 4, 3$.

(c) If a singular cubic has multiplicity two at one base point and passes simply through the other two, then its transform has degree two.

In every noncontracted case the restriction of σ gives a birational equivalence between C and its transform.

Proof. Apply Theorem 44.16. Birationality of the restrictions follows because σ is an isomorphism between dense open subsets of the source and target planes; intersecting these opens with a noncontracted irreducible curve gives dense opens of the curves. □

A smooth cubic through all three base points therefore transforms to another plane cubic. Its chord-and-tangent drawing changes, but the function field of the curve does not. If a smooth cubic avoids the base points, its plane image is a sextic with singularities forced by the three contracted coordinate lines. Thus plane degree is not a birational invariant.

$$\boxed{\text{cubic, no base points}} \xrightarrow{\sigma} \boxed{\text{sextic}}$$

$$\boxed{\text{cubic through } p_1, p_2, p_3} \xrightarrow{\sigma} \boxed{\text{cubic}}$$

Figure 44.6: Plane degree depends on incidence with the base points.

Example 44.18 (Affine chart formula)**.** On the chart $xyz \neq 0$, dividing coordinates shows the Cremona map is equivalent to $[x : y : z] \mapsto [1/x : 1/y : 1/z]$ up to common scaling.

44.10 Function fields and birational invariance

Proposition 44.19 (Function-field criterion)**.** *For integral varieties X and Y, a dominant rational map $\phi : X \dashrightarrow Y$ induces an inclusion*

$$\phi^* : k(Y) \hookrightarrow k(X).$$

The map is birational if and only if this inclusion is an isomorphism. Hence every Cremona transformation of $\mathbb{P}^2$ acts by an automorphism of

$$k(\mathbb{P}^2) \cong k(u, v).$$

Proof. Pullback of rational functions along a dominant rational map is injective. A rational inverse induces the inverse field homomorphism. Conversely, an isomorphism of function fields identifies dense affine opens and therefore yields a birational correspondence. $\square$

On the affine chart $z = 1$, with coordinates $u = x/z$ and $v = y/z$, the standard quadratic transformation becomes

$$(u, v) \dashrightarrow \left(\frac{1}{u}, \frac{1}{v}\right)$$

after identifying the appropriate target chart. This makes birationality transparent at the function-field level.

44.11 Projective automorphisms versus quadratic transformations

A projective automorphism has no base points, contracts no curve, and preserves degree of every plane curve. A genuine quadratic Cremona transformation has base points and exceptional curves. These differences are not coordinate accidents; they survive conjugation by projective automorphisms.

For a quadratic map with three noncollinear proper base points, the picture is therefore rigid up to coordinates:

$$\text{three base points} \longleftrightarrow \text{three opposite contracted lines.}$$

The source and target configurations are interchanged by the inverse map.

44.12 The divisor-class shadow

Although blow-ups have not yet been constructed, the numerical patterns strongly suggest a more stable bookkeeping system. Let H denote the class of a line. A general target line pulls back to a conic through all three base points, so one expects a symbolic transformation

$$H \longmapsto 2H - E_1 - E_2 - E_3.$$

Similarly, the coordinate line through p_2, p_3 should have a transform recorded by

$$H - E_2 - E_3.$$

In the plane itself the symbols E_i have nowhere to live. They are placeholders for the exceptional divisors that will appear after blowing up the base points.

Theorem 44.20 (Classical generation theorem — preview only)**.** *Over an algebraically closed field, the classical Noether–Castelnuovo theorem says, under the standard hypotheses of plane birational geometry, that the plane Cremona group is generated by projective automorphisms together with a standard quadratic transformation.*

No proof is attempted here. A serious proof requires the multiplicity theory of base points, including infinitely near points, and repeated reduction of degree. The theorem is included only to explain why the quadratic model deserves such attention: it is not an isolated curiosity but the elementary generator of the classical theory.

$$\boxed{H} \xrightarrow{\text{quadratic pullback}} \boxed{2H - E_1 - E_2 - E_3}$$

E_i are only symbolic here;
they become divisors in VI/45

Figure 44.7: The divisor-class shadow of the standard transformation.

44.13 Why indeterminacy must be resolved

Proposition 44.21 (Resolution preview for the standard map)**.** *For the standard quadratic transformation, any geometric resolution of the three source base points must replace each p_i by data recording tangent directions, because the exceptional line opposite p_i is recovered by the inverse map from those directions. After resolving the three source base points and the three target base points, the birational correspondence becomes an isomorphism between the resolved surfaces.*

Explanation. Approaching a base point along different curves produces different limiting values under σ; a single target point cannot encode these limits. The missing parameter is a projective line of directions. The blow-up construction of VI/45 replaces the point precisely by such a $\mathbb{P}^1$. Doing this at the three base points separates the coordinate functions and turns the rational map into a morphism. Since $\sigma^{-1} = \sigma$, the analogous target resolution yields the inverse morphism. $\square$

This is the geometric reason blow-ups are next. They are not an auxiliary trick attached to Cremona theory; they are the operation that makes the hidden geometry of a rational map visible.

$$p \xrightarrow{\text{replace directions}} E \cong \mathbb{P}^1 \xrightarrow{\text{resolve}} \boxed{\text{regular map}}$$

Figure 44.8: VI/45 replaces a base point by its projective line of directions.

44.14 Exercises

Exercise 44.22. Show that $[xz : yz : z^2]$ represents the identity rational map on $\mathbb{P}^2$.

Exercise 44.23. Compute the base locus of $[xy : xz : yz]$.

Exercise 44.24. Verify $\sigma[1 : 2 : 3] = [6 : 3 : 2]$ and $\sigma[6 : 3 : 2] = [1 : 2 : 3]$.

Exercise 44.25. Determine the image of the line $x = 0$ away from its base points.

Exercise 44.26. Find the inverse image of the target line $X + Y + Z = 0$.

Exercise 44.27. Show that the vector space of conics through p_1, p_2, p_3 has dimension three.

Exercise 44.28. Show that a projective automorphism has empty base locus.

Exercise 44.29. Why is $\deg(\sigma^2) \neq 4$ even though $\deg \sigma = 2$?

Exercise 44.30. Let L be a general line avoiding the base points. Compute $\deg \sigma_*(L)$.

Exercise 44.31. Let L be a line through p_1 but not through p_2, p_3. Compute the transformed degree.

Exercise 44.32. Let Q be an irreducible conic through all three base points. Compute $\deg \sigma_*(Q)$.

Exercise 44.33. Explain why the coordinate line $x = 0$ is excluded from the curve degree formula as an ordinary noncontracted curve.

Exercise 44.34. Compute the transformed degree of a cubic passing simply through all three base points.

Exercise 44.35. Compute the transformed degree of a smooth cubic avoiding all three base points.

Exercise 44.36. On the affine chart $z = 1$, express σ in affine coordinates on the target chart where $XY \neq 0$.

Exercise 44.37. Show directly on function fields that $(u, v) \mapsto (1/u, 1/v)$ is birational.

Exercise 44.38. Let $A, B \in \mathrm{PGL}_3(k)$. Show that $B\sigma A$ is again a quadratic Cremona transformation.

Exercise 44.39. Why must the three proper base points of a standard quadratic transformation be noncollinear?

Exercise 44.40. Suppose an irreducible quartic has multiplicities $(2, 1, 1)$ at the three base points. Compute its transformed degree.

Exercise 44.41. Find the image degree of a quintic with multiplicities $(3, 2, 1)$ at the three base points.

Exercise 44.42. Explain why plane degree cannot be a birational invariant of curves.

Exercise 44.43. What geometric event can make $\deg(\phi \circ \psi) < \deg \phi \deg \psi$?

Exercise 44.44. Why does a birational map induce an isomorphism of function fields?

Exercise 44.45. Explain informally why one projective line of tangent directions should replace a base point in a resolution.

44.15 Solved problem dossiers

Problem 44.46 (Recovering the standard map from its conic net)**.** Starting from the complete linear system of conics through p_1, p_2, p_3, reconstruct the standard quadratic transformation.

Problem 44.47 (A full involution calculation)**.** Compute σ^2 as a homogeneous triple and explain every cancellation.

Problem 44.48 (Transform of a general line)**.** Let $L : ax + by + cz = 0$ avoid the base points. Derive an equation for its transformed conic in target coordinates.

Problem 44.49 (Transform of a conic through the base triangle)**.** Let

$$Q : ayz + bxz + cxy = 0$$

be irreducible. Determine its image.

Problem 44.50 (Degree formula by a general line)**.** Give a self-contained Bézout proof of the formula $2d - m_1 - m_2 - m_3$.

Problem 44.51 (A cubic preserved in degree)**.** Let E be a smooth cubic through p_1, p_2, p_3, with transverse passage through each. What can be said about $E' = \sigma_*(E)$?

Problem 44.52 (A smooth cubic becomes a sextic)**.** Let E be a smooth cubic avoiding the base points. Explain why its sextic image must be singular.

Problem 44.53 (Conjugating the base triangle)**.** Given three noncollinear points q_1, q_2, q_3, construct a quadratic Cremona transformation having them as base points.

Problem 44.54 (Function-field form of the standard map)**.** Describe the automorphism of $k(u, v)$ induced by σ and verify its order.

Problem 44.55 (Why composition degree can collapse)**.** Relate the degree collapse in σ^2 to the exceptional triangle.

Problem 44.56 (Numerical preview of exceptional divisors)**.** Explain why the degree formula suggests the class $2H - E_1 - E_2 - E_3$ for the pullback of a line.

Problem 44.57 (Why blow-up directions are necessary)**.** Near $p_1 = [1 : 0 : 0]$, use affine coordinates $u = y/x, v = z/x$ to see direction dependence of σ.

44.16 Challenges

Problem 44.58 (Challenge: classify quadratic base configurations)**.** Analyze primitive triples of quadratic forms defining birational plane maps. Determine how the standard three-proper-point configuration changes when base points collide or become infinitely near, and identify precisely which cases require VI/45 machinery to formulate correctly.

Problem 44.59 (Challenge: degree dynamics of compositions)**.** Study sequences of projective automorphisms and standard quadratic transformations. Develop an algorithm that predicts the algebraic degree after each composition from base-point/exceptional-curve incidence, and compare the prediction with direct gcd cancellation in coordinates.

Problem 44.60 (Challenge: Cremona action on cubic models)**.** Start with a smooth plane cubic and vary the position of the three quadratic base points: off the cubic, one point on it, two points on it, and all three on it. Track image degree, singularities, normalization, and the induced birational map of the genus-one curve.

Problem 44.61 (Challenge: reconstruct the divisor action)**.** Assuming only the geometric behavior established in this chapter, predict the full linear action of the standard quadratic involution on the symbols H, E_1, E_2, E_3. Check formally that the predicted action is an involution and preserves the quadratic intersection form that VI/45 will construct.

Problem 44.62 (Challenge: resolve the standard transformation)**.** Using local coordinates near all three base points, design the charts needed to replace each point by its projective line of directions. Show chart-by-chart how $[yz : xz : xy]$ becomes regular after this replacement. Treat the result as a blueprint for VI/45 rather than invoking the blow-up theorem in advance.

Bridge to VI/45

The standard quadratic transformation compresses the essential paradox of birational geometry into one formula:

$$[x : y : z] \dashrightarrow [yz : xz : xy].$$

It is invertible on a dense open set, yet it is undefined at three points and contracts three whole lines. The degree formula

$$d \longmapsto 2d - m_1 - m_2 - m_3$$

shows that the correct invariants must remember not only plane degree but also multiplicities at base points. The symbolic pullback

$$H \longmapsto 2H - E_1 - E_2 - E_3$$

then announces the missing geometry: the plane must be replaced by a surface carrying exceptional divisors E_i.

VI/45 performs exactly that replacement. A blow-up replaces a point by the projective line of tangent directions through it. On the blow-up, rational maps such as σ can be resolved into morphisms; strict transforms replace naive curve images; exceptional divisors become genuine curves with computable intersection numbers; and the degree/multiplicity formulas of the present chapter become linear algebra in the Picard group of the resolved surface.

44.17 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 44.63 (Base points)**.** List the base points of $[yz : xz : xy]$.

Hint. Use birational maps, base points, and the standard quadratic transformation. ☐

Solution. $[1 : 0 : 0]$, $[0 : 1 : 0]$, and $[0 : 0 : 1]$. ☐

Exercise 44.64 (Contracted line)**.** Where does the line $x = 0$ map generically?

Hint. Use birational maps, base points, and the standard quadratic transformation. ☐

Solution. To $[1 : 0 : 0]$. ☐

Exercise 44.65 (Involution)**.** What is the square of the standard Cremona transformation on its common domain?

Hint. Use birational maps, base points, and the standard quadratic transformation. ☐

Solution. The identity. ☐

Exercise 44.66 (Dense torus)**.** How does the map act where $xyz \neq 0$?

Hint. Use birational maps, base points, and the standard quadratic transformation. ☐

Solution. By coordinatewise inversion up to projective scaling. ☐

Exercise 44.67 (Degree)**.** What degree are the defining homogeneous coordinates?

Hint. Use birational maps, base points, and the standard quadratic transformation. ☐

Solution. 2. ☐

Proofs

Exercise 44.68 (Base locus)**.** Prove: Prove the standard Cremona map is undefined exactly at the three coordinate points.

Hint. Work from birational maps, base points, and the standard quadratic transformation and state the hypotheses explicitly. ☐

Solution. All three quadratic coordinates vanish simultaneously exactly when at least two of x, y, z vanish. ☐

Exercise 44.69 (Involution)**.** Prove: Prove the standard Cremona transformation is birational and involutive.

Hint. Work from birational maps, base points, and the standard quadratic transformation and state the hypotheses explicitly. ☐

Solution. Apply the formula twice and factor out the common scalar xyz. ☐

Exercise 44.70 (Line contraction)**.** Prove: Prove each coordinate line is contracted to the opposite coordinate point.

Hint. Work from birational maps, base points, and the standard quadratic transformation and state the hypotheses explicitly. ☐

Solution. Set the corresponding coordinate to zero and inspect the remaining output. ☐

Exercise 44.71 (Dense isomorphism)**.** Prove: Prove the map restricts to an automorphism of the torus $xyz \neq 0$.

Hint. Work from birational maps, base points, and the standard quadratic transformation and state the hypotheses explicitly. ☐

Solution. Coordinatewise inversion is regular on the torus and self-inverse. ☐

Counterexamples and hypothesis tests

Exercise 44.72 (Everywhere morphism)**.** Test the claim: the Cremona transformation is a morphism on all of $\mathbb{P}^2$.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it has three base points. ☐

Exercise 44.73 (Isomorphism)**.** Test the claim: every birational map is a global isomorphism.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the Cremona map contracts curves and is undefined at points. ☐

Exercise 44.74 (No exceptional curves)**.** Test the claim: a birational plane map cannot contract a curve.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the coordinate lines are contracted. ☐

Applications and investigations

Exercise 44.75 (Birational simplification)**.** Why study Cremona transformations?

Hint. Translate the question into birational maps, base points, and the standard quadratic transformation. ☐

Solution. They change projective models while preserving the function field and can simplify configurations of curves. ☐

Exercise 44.76 (Blow-up bridge)**.** How are the base points resolved?

Hint. Translate the question into birational maps, base points, and the standard quadratic transformation. ☐

Solution. Blowing up the three base points turns the rational map into a morphism. ☐

Challenge problems

Exercise 44.77 (Synthesis challenge)**.** Connect two central viewpoints of Cremona Transformations in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Cremona Transformations and then translates it through birational maps, base points, and the standard quadratic transformation, checking both descriptions agree. ☐

Exercise 44.78 (Hypothesis challenge)**.** Choose one theorem-level statement from Cremona Transformations, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Chapter 45

Blow-Ups

VI/44 ended with a symbolic calculation. For the standard quadratic Cremona transformation

$$\sigma : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2,$$
$$[x : y : z] \longmapsto [yz : xz : xy].$$

the pullback of a general target line behaves numerically like

$$2H - E_1 - E_2 - E_3.$$

The symbols E_i were deliberately left unexplained: the plane itself contains no such divisors. The present chapter constructs the surface on which they become genuine curves.

A *blow-up* replaces a subvariety by the projectivized family of normal directions along it. For a point on a smooth surface this is especially concrete. The point disappears and is replaced by a copy of $\mathbb{P}^1$, one point for each tangent direction through the original center. This exceptional curve remembers precisely the directional data that a rational map loses at a base point.

For the origin of $\mathbb{A}^2$, the blow-up is the incidence surface

$$\widetilde{\mathbb{A}^2} = \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 : xt = ys\}.$$

Away from $(0, 0)$, the direction $[s : t] = [x : y]$ is forced, so the projection to $\mathbb{A}^2$ is an isomorphism. Over the origin, the equation becomes $0 = 0$, leaving the whole $\mathbb{P}^1$. The blow-up therefore changes nothing away from the center but separates all directions through it.

The chapter develops this local model first, then gives the intrinsic Rees-algebra construction

$$\operatorname{Bl}_Z X = \operatorname{Proj}_X \left(\bigoplus_{n \geq 0} \mathcal{I}_Z^n \right).$$

After that, strict transforms and multiplicities turn local equations into divisor formulas. For the blow-up of $\mathbb{P}^2$ at a point,

$$\operatorname{Pic}(\operatorname{Bl}_p \mathbb{P}^2) \cong \mathbb{Z}H \oplus \mathbb{Z}E,$$

with

$$H^2 = 1, \qquad H \cdot E = 0, \qquad E^2 = -1.$$

Finally we blow up the three base points of σ. The rational map becomes regular, the symbolic classes E_i become actual exceptional divisors, and the degree formula from VI/44 becomes an intersection calculation on the resolved surface.

Standing conventions

Unless stated otherwise, k is an algebraically closed field. Surfaces are integral, separated, and of finite type over k. When local coordinates at a smooth point are used, the completed or local ring is regular of dimension two. The blow-up of an ideal sheaf $\mathcal{I}$ means the relative Proj of its Rees algebra. For a curve C through a blown-up point, $\widetilde{C}$ denotes its strict transform, while π^*C denotes its total transform as a Cartier divisor whenever this notation is defined.

Learning goals

By the end of the chapter, the reader should be able to:

- construct the blow-up of $\mathbb{A}^2$ at the origin as an incidence subvariety of $\mathbb{A}^2 \times \mathbb{P}^1$;
- compute the two standard affine charts of this blow-up and their transition map;
- identify the exceptional divisor with $\mathbb{P}^1$ and interpret its points as tangent directions;
- define $\mathrm{Bl}_Z X$ intrinsically using the Rees algebra of the ideal sheaf of Z;
- prove that the blow-up is an isomorphism away from its center;
- use the universal property of blow-ups when an ideal becomes invertible;
- distinguish total transform, strict transform, and exceptional components;
- factor a curve equation after blow-up using its multiplicity at the center;
- recover tangent-cone directions from the intersection $\widetilde{C} \cap E$;
- see explicitly how a blow-up separates the two branches of an ordinary node;
- compute divisor classes on $\mathrm{Bl}_p \mathbb{P}^2$;
- derive $H^2 = 1$, $H \cdot E = 0$, and $E^2 = -1$;
- prove that a degree-d plane curve of multiplicity m has strict-transform class $dH - mE$;
- understand the canonical-divisor correction $K_{\widetilde{X}} = \pi^* K_X + E$ for a point blow-up of a smooth surface;
- compute the Picard group and intersection form after blowing up several distinct points;
- resolve the three base points of the standard quadratic Cremona transformation;
- identify the strict transforms of the coordinate lines as the curves contracted by the resolved map;
- compute the lifted Cremona action on H, E_1, E_2, E_3;
- recover the degree and multiplicity formulas of VI/44 from the intersection lattice;
- explain how repeated blow-ups create and resolve infinitely near base points.

Conceptual roadmap

$$\boxed{\begin{array}{c} \text{point} \rightsquigarrow \mathbb{P}^1 \text{ of directions} \Longrightarrow \text{strict transforms} \\ \Longrightarrow \mathrm{Pic} \text{ and intersections} \Longrightarrow \text{resolved Cremona map} \end{array}}$$

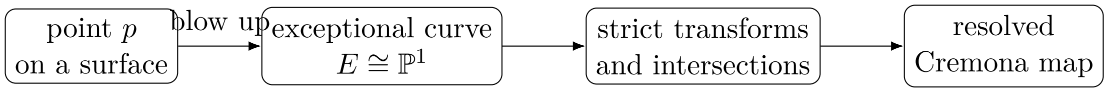


Figure 45.1: The chapter's progression from a point replacement to a resolved birational map.

45.1 Blowing up the affine plane at the origin

We begin with the local model that controls every blow-up of a smooth surface at a point.

Definition 45.1 (Incidence model of the point blow-up)**.** The blow-up of $\mathbb{A}^2 = \operatorname{Spec} k[x, y]$ at the origin is

$$\operatorname{Bl}_0 \mathbb{A}^2 := \{((x, y), [s : t]) \in \mathbb{A}^2 \times \mathbb{P}^1 : xt = ys\}.$$

The projection onto the first factor is denoted

$$\pi : \operatorname{Bl}_0 \mathbb{A}^2 \longrightarrow \mathbb{A}^2.$$

The equation says that $[s : t]$ records the direction of the vector (x, y). If $(x, y) \neq (0, 0)$, then $[s : t] = [x : y]$ is uniquely determined. Over the origin there is no constraint on $[s : t]$.

Proposition 45.2 (The two affine charts)**.** *The blow-up* $\operatorname{Bl}_0 \mathbb{A}^2$ *is covered by two affine planes.*

On $s \neq 0$, write $u = t/s$. Then

$$y = xu,$$

so this chart has coordinates (x, u) and projection

$$(x, u) \longmapsto (x, xu).$$

On $t \neq 0$, write $v = s/t$. Then

$$x = yv,$$

so this chart has coordinates (v, y) and projection

$$(v, y) \longmapsto (yv, y).$$

On the overlap, $u = v^{-1}$ and $y = xu$.

Proof. On $s \neq 0$, divide the incidence equation $xt = ys$ by s to obtain $xu = y$. Thus y is eliminated and the chart is $\operatorname{Spec} k[x, u]$. The second chart is symmetric. Where both $s, t \neq 0$, one has $u = t/s$ and $v = s/t$, hence $uv = 1$. □

Definition 45.3 (Exceptional divisor)**.** The inverse image of the center,

$$E := \pi^{-1}(0),$$

is the *exceptional divisor*. In the incidence model it is

$$E = \{(0, 0)\} \times \mathbb{P}^1 \cong \mathbb{P}^1.$$

In the (x, u)-chart, E is the divisor $x = 0$; in the (v, y)-chart, it is $y = 0$. The coordinate u records the slope y/x, while v records the reciprocal slope. The two charts on E are exactly the two standard affine charts of $\mathbb{P}^1$.

Example 45.4 (Blow-up of the affine plane at the origin)**.** The blow-up of $\mathbb{A}^2$ at $(0, 0)$ is covered by charts $U_x = \operatorname{Spec} k[x, u]$ with $y = xu$ and $U_y = \operatorname{Spec} k[v, y]$ with $x = vy$. The exceptional divisor is $x = 0$ in the first chart and $y = 0$ in the second, gluing to $\mathbb{P}^1$.

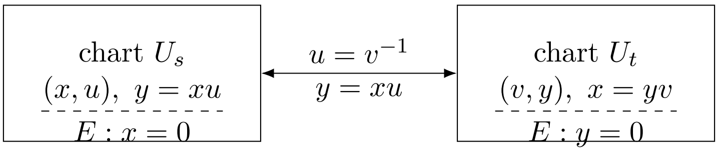


Figure 45.2: The two affine charts of $\mathrm{Bl}_0 \mathbb{A}^2$; their exceptional axes glue to $\mathbb{P}^1$.

45.2 Why the exceptional divisor is a space of directions

Fix a line through the origin,

$$L_a : \ y = ax.$$

In the (x, u)-chart its inverse image is defined by

$$x(u - a) = 0.$$

This equation has two components: the exceptional divisor $x = 0$ and the closure $u = a$ of the inverse image of $L_a \setminus \{0\}$. The latter is the strict transform of the line. It meets E at the single point $(x, u) = (0, a)$, so the point of E records the direction of L_a.

Proposition 45.5 (Blow-up is unchanged off the center)**.** *The restriction*

$$\pi : \mathrm{Bl}_0 \mathbb{A}^2 \setminus E \longrightarrow \mathbb{A}^2 \setminus \{0\}$$

is an isomorphism.

Proof. The inverse sends $(x, y) \neq (0, 0)$ to $((x, y), [x : y])$. This map is regular on the standard opens $x \neq 0$ and $y \neq 0$, and the two formulas agree on their overlap. □

Thus a blow-up is not a local surgery that destroys the surrounding surface. It is a birational modification supported exactly at the center.

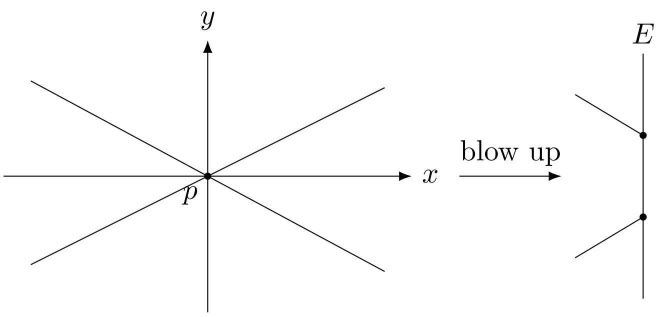


Figure 45.3: Two directions through one point become two distinct points on the exceptional divisor.

45.3 The intrinsic Rees-algebra construction

The incidence equation is not the definition in general. Let X be a scheme and $Z \subset X$ a closed subscheme with quasi-coherent ideal sheaf $\mathcal{I}$. The graded $\mathcal{O}_X$-algebra

$$\mathcal{R}(\mathcal{I}) := \mathcal{O}_X \oplus \mathcal{I} \oplus \mathcal{I}^2 \oplus \cdots$$

is the *Rees algebra* of $\mathcal{I}$. Here the summand $\mathcal{I}^n$ is placed in degree n.

Definition 45.6 (Blow-up along a closed subscheme)**.** The blow-up of X along Z is

$$\operatorname{Bl}_Z X := \operatorname{Proj}_X \mathcal{R}(\mathcal{I}),$$

with its canonical projective morphism

$$\pi : \operatorname{Bl}_Z X \longrightarrow X.$$

For an affine scheme $X = \operatorname{Spec} A$ and ideal $I \subset A$, this is the ordinary Proj of the graded Rees algebra

$$A[It] = A \oplus It \oplus I^2t^2 \oplus \cdots \subset A[t].$$

If $I = (f_0, \dots, f_r)$, the blow-up comes with a natural closed immersion into $X \times \mathbb{P}^r$, and the projective coordinates record the ratios of the generators of I. For $I = (x, y)$, this recovers $xt = ys$.

Theorem 45.7 (Universal property of the blow-up)**.** *Let $\pi : \operatorname{Bl}_Z X \to X$ be the blow-up of the ideal sheaf $\mathcal{I}$. If $f : Y \to X$ is a morphism such that*

$$\mathcal{I}\mathcal{O}_Y$$

is an invertible ideal sheaf on Y, then there is a unique morphism

$$\widetilde{f} : Y \longrightarrow \operatorname{Bl}_Z X$$

with $f = \pi \circ \widetilde{f}$.

Proof. An invertible quotient of the pullback of the degree-one piece of the Rees algebra determines, by the universal property of relative Proj, a morphism to $\operatorname{Proj}_X \mathcal{R}(\mathcal{I})$. The construction is forced by the invertible ideal $\mathcal{I}\mathcal{O}_Y$, giving uniqueness. □

This property explains why blow-ups resolve base ideals: after blowing up the ideal, the ideal becomes locally principal and therefore defines a line bundle rather than an indeterminate ratio.

45.4 Total transforms and strict transforms

Let $\pi : \widetilde{X} \to X$ be the blow-up of a smooth surface at a point p, with exceptional divisor E. If $C \subset X$ is a curve, two different transforms appear.

Definition 45.8 (Total and strict transforms)**.** The *total transform* of a Cartier divisor C is the pullback divisor π^*C. The *strict transform* $\widetilde{C}$ is the closure in $\widetilde{X}$ of

$$\pi^{-1}(C \setminus \{p\}).$$

The total transform usually contains the exceptional divisor; the strict transform removes that forced exceptional component.

Suppose local parameters at p are x, y, and C has local equation

$$f(x, y) = f_m(x, y) + f_{m+1}(x, y) + \cdots,$$

where f_j is homogeneous of degree j and $f_m \neq 0$. Then $m = \operatorname{mult}_p(C)$.

Proposition 45.9 (Multiplicity becomes exceptional multiplicity)**.** *With the preceding notation,*

$$\boxed{\pi^* C = \widetilde{C} + mE.}$$

In the chart $y = xu$, one has

$$f(x, xu) = x^m (f_m(1, u) + x f_{m+1}(1, u) + \cdots).$$

The factor x^m is the exceptional component of the total transform.

Proof. Every monomial of total degree at least m acquires at least a factor x^m after substituting $y = xu$, and the degree-m part contributes exactly $x^m f_m(1, u)$. Since $x = 0$ defines E in this chart, the order along E is m. Removing that factor gives the local equation of the strict transform. $\square$

Proposition 45.10 (Tangent cone on the exceptional divisor)**.** *In the chart $y = xu$, the intersection $\widetilde{C} \cap E$ is cut out by*

$$f_m(1, u) = 0.$$

Thus the points where $\widetilde{C}$ meets E, counted with multiplicity and completed by the second chart, are exactly the projective tangent directions of C at p.

Proof. The strict-transform equation is the factor in parentheses above. Restricting to E, where $x = 0$, leaves $f_m(1, u) = 0$. Homogenizing across the two charts gives the zero scheme of the tangent cone $f_m(s, t)$ in $\mathbb{P}^1$. $\square$

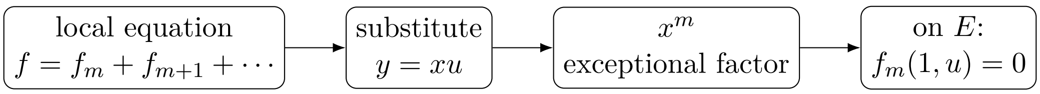


Figure 45.4: Multiplicity becomes exceptional multiplicity, while the tangent cone controls $\widetilde{C} \cap E$.

45.5 Blowing up singular plane curves

The tangent-cone proposition gives an immediate geometric use of blow-ups. If a curve has several tangent directions at a singular point, those directions become different points of the exceptional divisor.

For example, consider an ordinary node whose quadratic tangent cone factors into two distinct lines. The two branches of the strict transform meet E at two distinct points, so the blow-up separates them. For the cusp

$$C : y^2 = x^3,$$

the chart $y = xu$ gives

$$x^2(u^2 - x) = 0.$$

The strict transform $u^2 = x$ is smooth, but it meets $E : x = 0$ with multiplicity two. Thus one blow-up removes the singularity of the curve while still remembering its high-order tangency to the exceptional divisor.

Proposition 45.11 (Ordinary multiple points separate by direction)**.** *Suppose the tangent cone of a plane curve singularity of multiplicity m is a product of m distinct linear forms. Then after blowing up the singular point, the strict transform meets the exceptional divisor in m distinct points. In particular, the local branches are separated at the first blow-up.*

Proof. By Proposition 45.10, the intersection with E is the projectivized zero set of the tangent cone. Distinct linear factors give distinct points of $\mathbb{P}^1$. □

This is the first glimpse of embedded resolution: repeated blow-ups can replace a complicated singular curve by a configuration of smooth strict transforms and exceptional curves with controlled intersections.

Example 45.12 (Separating tangent directions)**.** For a plane curve through the origin, the strict transform meets the exceptional divisor at points corresponding to tangent directions. Distinct tangents of a node meet the exceptional $\mathbb{P}^1$ at distinct points.

45.6 The Picard group of the blown-up plane

Now let

$$\pi : S = \mathrm{Bl}_p \mathbb{P}^2 \longrightarrow \mathbb{P}^2$$

be the blow-up of a point. Write

$$H := \pi^*[\text{line}],$$

and let E be the exceptional divisor.

Theorem 45.13 (Picard group after one point blow-up)**.** *There is an isomorphism*

$$\mathrm{Pic}(S) \cong \mathbb{Z}H \oplus \mathbb{Z}E.$$

Every divisor class on S can be written uniquely as

$$dH - mE, \qquad d, m \in \mathbb{Z}.$$

Proof. A divisor on S not supported on E pushes forward to a divisor on $\mathbb{P}^2$, whose class is a multiple of a line. Subtracting the pullback of that class leaves a divisor supported on E, hence an integral multiple of E. Uniqueness follows from the intersection numbers computed below: if $dH + mE \sim 0$, intersecting first with H gives $d = 0$, and then with E gives $m = 0$. □

Theorem 45.14 (Intersection form)**.** *On $S = \mathrm{Bl}_p \mathbb{P}^2$,*

$$\boxed{H^2 = 1, \qquad H \cdot E = 0, \qquad E^2 = -1.}$$

Consequently

$$(dH - mE) \cdot (d'H - m'E) = dd' - mm'.$$

Proof. Choose a line in $\mathbb{P}^2$ avoiding p. Its inverse image is isomorphic to the line and does not meet E, so $H \cdot E = 0$. Two general such lines meet once, giving $H^2 = 1$.

Now choose two distinct lines through p. Their strict transforms are disjoint because their only intersection downstairs was p, and on the blow-up they meet E at two different direction points. Each strict transform has class $H - E$. Hence

$$0 = (H - E)^2 = H^2 - 2H \cdot E + E^2 = 1 + E^2,$$

so $E^2 = -1$. □

Proposition 45.15 (Class of a strict transform)**.** *Let $C \subset \mathbb{P}^2$ be a degree-d curve with multiplicity m at p, and assume C does not contain an exceptional artifact because it lives downstairs. Then*

$$[\widetilde{C}] = dH - mE.$$

Moreover,

$$\widetilde{C} \cdot E = m.$$

Proof. Downstairs, $[C] = d[\text{line}]$, so $\pi^* C \sim dH$. Proposition 45.9 gives $\pi^* C = \widetilde{C} + mE$, hence $[\widetilde{C}] = dH - mE$. Intersecting with E yields

$$(dH - mE) \cdot E = m.$$

□

The final equality is a global version of the tangent-cone calculation: the total number of tangent directions, counted with multiplicity, is the multiplicity of the original curve at the blown-up point.

45.7 Canonical divisors and the exceptional correction

Blow-ups also change canonical divisors in a controlled way. Let $\pi : \widetilde{X} \to X$ blow up a smooth point of a smooth surface.

Theorem 45.16 (Canonical divisor under a point blow-up)**.** *If E is the exceptional divisor, then*

$$\boxed{K_{\widetilde{X}} = \pi^* K_X + E.}$$

For $S = \mathrm{Bl}_p \mathbb{P}^2$, this becomes

$$K_S = -3H + E.$$

Proof. Take local coordinates (x, y) at the center. On the blow-up chart $y = xu$,

$$dx \wedge dy = dx \wedge (u\, dx + x\, du) = x\, dx \wedge du.$$

Thus the pullback of a local generator of the canonical bundle vanishes to order one along $E : x = 0$. This gives the discrepancy $+E$. Since $K_{\mathbb{P}^2} = -3[\text{line}]$, the displayed formula follows for the blown-up plane. □

The coefficient $+1$ is one of the simplest examples of a discrepancy in birational geometry. Here it is visible directly from the Jacobian of the chart substitution.

45.8 Blowing up three noncollinear points

Return to the three coordinate points

$$p_1 = [1 : 0 : 0], \qquad p_2 = [0 : 1 : 0], \qquad p_3 = [0 : 0 : 1].$$

Let

$$\pi : S = \mathrm{Bl}_{p_1, p_2, p_3} \mathbb{P}^2 \longrightarrow \mathbb{P}^2$$

be the blow-up of all three points. Since the centers are distinct, the order of these three blow-ups does not matter up to canonical isomorphism. Let E_i be the exceptional divisor over p_i.

Theorem 45.17 (Picard lattice after three point blow-ups)**.** *The Picard group is*

$$\operatorname{Pic}(S) \cong \mathbb{Z}H \oplus \mathbb{Z}E_1 \oplus \mathbb{Z}E_2 \oplus \mathbb{Z}E_3,$$

with intersection form

$$H^2 = 1, \qquad H \cdot E_i = 0, \qquad E_i \cdot E_j = 0 \ (i \neq j), \qquad E_i^2 = -1.$$

If C has degree d and multiplicities m_i at the three points, then

$$[\widetilde{C}] = dH - m_1E_1 - m_2E_2 - m_3E_3.$$

Proof. Apply the one-point statement successively. Because later blow-up centers do not lie on the earlier exceptional divisors, the exceptional curves remain pairwise disjoint and retain self-intersection -1. The strict-transform formula is likewise obtained by subtracting the multiplicity at each center. ☐

Let $L_x = V_+(x)$, $L_y = V_+(y)$, $L_z = V_+(z)$. Their strict transforms have classes

$$\widetilde{L}_x = H - E_2 - E_3, \qquad \widetilde{L}_y = H - E_1 - E_3, \qquad \widetilde{L}_z = H - E_1 - E_2.$$

Each has self-intersection -1, and the three are pairwise disjoint on S.

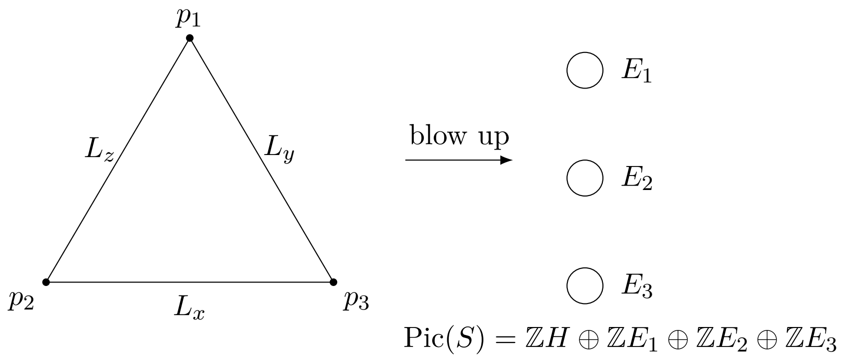


Figure 45.5: Blowing up the three vertices turns the base-point data into three divisor classes.

45.9 Resolving the standard quadratic transformation

The standard Cremona map is defined by the net

$$\langle yz, xz, xy \rangle.$$

Every member is a conic through p_1, p_2, p_3, so after blowing up those base points the corresponding divisor class is

$$D = 2H - E_1 - E_2 - E_3.$$

The key point is that the proper transform of this linear system has no base points.

Theorem 45.18 (Resolution of the standard Cremona map)**.** *Let $S = \operatorname{Bl}_{p_1,p_2,p_3} \mathbb{P}^2$. The rational map*

$$\sigma[x : y : z] = [yz : xz : xy]$$

lifts to a morphism

$$\widetilde{\sigma} : S \longrightarrow \mathbb{P}^2$$

defined by the base-point-free linear system $|2H - E_1 - E_2 - E_3|$.

For each i*, the exceptional divisor* E_i *maps isomorphically onto the coordinate line opposite* p_i*, while each strict transform* $\widetilde{L}_x, \widetilde{L}_y, \widetilde{L}_z$ *is contracted to the corresponding coordinate point.*

Proof. Near p_1, use affine coordinates $u = y/x$, $v = z/x$. Then

$$\sigma[1 : u : v] = [uv : v : u].$$

On the blow-up chart $v = ut$, this becomes

$$[u^2 t : ut : u] = [ut : t : 1],$$

which is regular even at $u = 0$. On E_1, where $u = 0$, the image is $[0 : t : 1]$, varying along the target line $X = 0$. The complementary blow-up chart covers the missing point of that line. Thus the indeterminacy at p_1 is removed and E_1 maps isomorphically to $X = 0$. The same calculation applies cyclically at p_2, p_3.

Away from the base points the blow-up is an isomorphism and the map agrees with σ, so these local extensions glue to a global morphism. Finally, on $L_x \setminus \{p_2, p_3\}$, one has $\sigma = [1 : 0 : 0] = p_1$, so $\widetilde{L}_x$ is contracted to p_1, and similarly for the other two coordinate lines. $\square$

There is a numerical check. Since

$$D \cdot E_i = 1,$$

the map has degree one on each E_i. Meanwhile

$$D \cdot (H - E_2 - E_3) = 2 - 1 - 1 = 0,$$

so $\widetilde{L}_x$ is contracted, exactly as the coordinate calculation shows.

$$\begin{tikzcd}
\begin{array}{c} E_i \\ \text{contracted by } \pi \end{array} & S = \mathrm{Bl}_{p_1,p_2,p_3}\mathbb{P}^2 \arrow[dl, "\pi"'] \arrow[dr, "\widetilde{\sigma}"] & \begin{array}{c} \widetilde{L}_i \\ \text{contracted by } \widetilde{\sigma} \end{array} \\
\mathbb{P}^2_{\text{source}} & & \mathbb{P}^2_{\text{target}}
\end{tikzcd}$$

Figure 45.6: The common resolution of the standard quadratic Cremona transformation.

Example 45.19 (Blow-up as Proj)**.** If $I \subset A$ is an ideal, the blow-up of $\operatorname{Spec} A$ along I is $\operatorname{Proj} \bigoplus_{n \geq 0} I^n$. The Rees algebra packages all powers of the center ideal into one graded ring.

45.10 The lifted Cremona involution on the blown-up surface

The morphism $\widetilde{\sigma} : S \to \mathbb{P}^2$ still contracts three curves. Blow up the three target coordinate points as well. Because the original Cremona map is an involution, the source and target blow-ups are naturally isomorphic, and the rational involution becomes a genuine automorphism

$$\widehat{\sigma} : S \xrightarrow{\sim} S.$$

It exchanges exceptional divisors with strict transforms of opposite coordinate lines.

Theorem 45.20 (Cremona action on the Picard lattice)**.** *For the lifted involution $\widehat{\sigma}$, the pullback action on* Pic(S) *is*

$$\begin{aligned} H &\longmapsto 2H - E_1 - E_2 - E_3, \\ E_1 &\longmapsto H - E_2 - E_3, \\ E_2 &\longmapsto H - E_1 - E_3, \\ E_3 &\longmapsto H - E_1 - E_2. \end{aligned}$$

This linear transformation is an involution and preserves the intersection form.

Proof. A general target line pulls back to a conic through all three source base points, giving the first formula. The target exceptional divisor over p_1 pulls back to the strict transform of the source line $x = 0$, because that line was contracted to p_1; this gives the second formula, and the remaining two are cyclic.

Applying the transformation twice to the basis returns each basis element. Preservation of the intersection form can be checked on the basis using

$$H^2 = 1, \qquad E_i^2 = -1,$$

and all mixed products zero. Geometrically, it must also preserve intersections because $\widehat{\sigma}$ is an automorphism. □

Proposition 45.21 (Degree and multiplicity transformation)**.** *Let a curve have strict-transform class*

$$C \sim dH - m_1E_1 - m_2E_2 - m_3E_3,$$

and suppose it contains none of the three coordinate lines as a component. Then its noncontracted Cremona transform has coefficients

$$\boxed{\begin{aligned} d' &= 2d - m_1 - m_2 - m_3, \\ m_1' &= d - m_2 - m_3, \\ m_2' &= d - m_1 - m_3, \\ m_3' &= d - m_1 - m_2. \end{aligned}}$$

In particular, the degree formula of VI/44 is the H-coefficient of the Picard-lattice action.

Proof. Apply Theorem 45.20 to

$$dH - m_1E_1 - m_2E_2 - m_3E_3$$

and collect coefficients of H, E_1, E_2, E_3. Since the transformation is involutive, the same formulas recover (d, m_1, m_2, m_3) from the primed data. □

45.11 Blowing up a base ideal resolves a projective rational map

The Cremona calculation is a special case of a general mechanism. Suppose X is integral and

$$\phi : X \dashrightarrow \mathbb{P}^r$$

is represented by sections $f_0, \ldots, f_r$ of a line bundle after choosing a common twist. Their common vanishing defines a base ideal $\mathcal{I}$.

$$\boxed{H} \longleftrightarrow \boxed{2H - E_1 - E_2 - E_3}$$

$$\boxed{E_1} \longleftrightarrow \boxed{H - E_2 - E_3}$$

cyclically for E_2, E_3

Figure 45.7: The lifted involution exchanges exceptional classes with strict transforms of opposite lines.

Theorem 45.22 (Resolution by the base ideal)**.** *Let*

$$\pi : \widetilde{X} = \mathrm{Bl}_{\mathcal{I}} X \longrightarrow X$$

be the blow-up of the base ideal of $\phi = [f_0 : \cdots : f_r]$. Then the pullback ideal $\mathcal{I}\mathcal{O}_{\widetilde{X}}$ is invertible, and the ratios of the pulled-back sections define a morphism

$$\widetilde{\phi} : \widetilde{X} \longrightarrow \mathbb{P}^r$$

that agrees with $\phi \circ \pi$ over the original domain of definition.

Proof. By construction of the blow-up, the pullback of the base ideal becomes the tautological invertible ideal. Locally one generator of the ideal is chosen as a denominator on each standard Proj chart, and all ratios f_i/f_j become regular there. These local projective-coordinate maps agree on overlaps and therefore glue to $\widetilde{\phi}$. □

One should distinguish this statement from a complete resolution theorem. Blowing up the base ideal resolves the given map to projective space, but when one wants a particularly smooth source or a factorization into point blow-ups, further normalization or repeated blow-ups may be required.

45.12 Infinitely near points and repeated blow-ups

After one blow-up, a transformed linear system can still have a base point on the exceptional divisor. Such a point represents a direction at the original center and is not a point of the original surface.

Definition 45.23 (Infinitely near point)**.** A point on an exceptional divisor created by a blow-up is called a point *infinitely near* to the original center. More generally, a point appearing on any surface obtained by a sequence of blow-ups is infinitely near to its image on the original surface.

If a strict transform still has a singularity or a linear system still has a base point there, one blows up again. The new exceptional divisor records directions through the infinitely near point. Iterating produces a tree of exceptional curves whose intersection pattern records the history of the resolution.

For plane curve singularities this process leads toward embedded resolution. For plane Cremona maps it leads to the complete cluster of proper and infinitely near base points. The elementary three-proper-point quadratic transformation is the first nontrivial case where the entire process terminates after one blow-up at each base point.

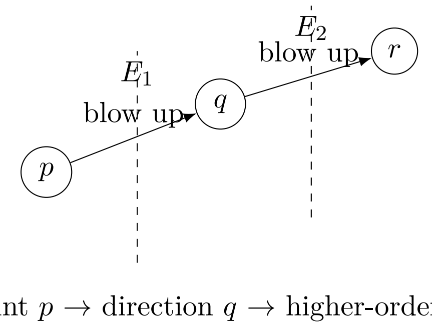


Figure 45.8: Repeated blow-ups create infinitely near points and an exceptional configuration recording resolution history.

45.13 Exceptional curves and blow-downs

A point blow-up introduces a smooth rational curve $E \cong \mathbb{P}^1$ with self-intersection -1. Conversely, under suitable hypotheses a smooth rational (-1)-curve on a smooth surface can be contracted to a smooth point. This is the geometric inverse of blowing up.

For the resolved Cremona transformation, this inverse viewpoint is essential. Starting from

$$S = \mathrm{Bl}_{p_1,p_2,p_3} \mathbb{P}^2,$$

one can obtain the target plane by contracting the three disjoint (-1)-curves

$$\widetilde{L}_x, \qquad \widetilde{L}_y, \qquad \widetilde{L}_z.$$

Thus the standard quadratic transformation factors schematically as

$$\mathbb{P}^2 \xleftarrow{\pi} S \xrightarrow{\widetilde{\sigma}} \mathbb{P}^2,$$

where the left arrow blows up three points and the right arrow blows down three different curves.

This factorization is the basic pattern of birational surface geometry: replace an indeterminate rational map by a diagram of honest morphisms, then study those morphisms through exceptional curves and the intersection lattice.

45.14 Exercises

Exercise 45.24. Verify directly that $\mathrm{Bl}_0 \mathbb{A}^2 \subset \mathbb{A}^2 \times \mathbb{P}^1$ is covered by two copies of $\mathbb{A}^2$.

Exercise 45.25. Show that the exceptional divisor is isomorphic to $\mathbb{P}^1$.

Exercise 45.26. Find the strict transform of the line $y = ax$ in the chart $y = xu$.

Exercise 45.27. Show that two distinct lines through the blown-up point have disjoint strict transforms.

Exercise 45.28. For $f(x, y) = y^2 - x^3$, compute the strict transform in the chart $y = xu$.

Exercise 45.29. Let C have multiplicity m at the origin. Explain why $\widetilde{C} \cdot E = m$.

Exercise 45.30. Prove that the blow-up morphism is projective.

Exercise 45.31. If the center ideal is already invertible, show that blowing it up changes nothing.

Exercise 45.32. Compute the class of the strict transform of a conic through p with multiplicity one.

Exercise 45.33. Compute the self-intersection of the strict transform of a line through p.

Exercise 45.34. Compute K_S^2 for $S = \mathrm{Bl}_p \mathbb{P}^2$.

Exercise 45.35. Let S be the blow-up of three distinct points of $\mathbb{P}^2$. Compute K_S^2.

Exercise 45.36. Show that $H - E_2 - E_3$ has self-intersection -1.

Exercise 45.37. Let $D = 2H - E_1 - E_2 - E_3$. Compute D^2.

Exercise 45.38. Compute $D \cdot E_1$ and explain its geometric meaning.

Exercise 45.39. Compute $D \cdot (H - E_2 - E_3)$.

Exercise 45.40. Apply the Cremona Picard action twice to H and verify that H is recovered.

Exercise 45.41. A cubic passes simply through all three Cremona base points. Compute the transformed degree and transformed multiplicities.

Exercise 45.42. A quartic has multiplicities $(2, 1, 1)$ at the three base points. Compute its transformed numerical type.

Exercise 45.43. Why does a point of E deserve to be called infinitely near to the original center?

Exercise 45.44. Show that the strict transforms of the three coordinate lines are pairwise disjoint after blowing up the three vertices.

Exercise 45.45. Let C be a degree-d curve with multiplicity m at p. Compute $\widetilde{C}^2$ on $\mathrm{Bl}_p \mathbb{P}^2$.

Exercise 45.46. Explain why blowing up a Cartier divisor is an isomorphism.

Exercise 45.47. In the chart near p_1 where $v = ut$, compute the image of the exceptional divisor under the resolved Cremona map.

45.15 Solved problem dossiers

Problem 45.48 (Deriving the incidence equation from the Rees algebra)**.** Starting with $I = (x, y) \subset k[x, y]$, recover the equation $xt = ys$ for $\mathrm{Bl}_0 \mathbb{A}^2$.

Problem 45.49 (The exceptional divisor as projectivized tangent space)**.** For a smooth surface X and point p, identify the exceptional divisor of $\mathrm{Bl}_p X$ intrinsically.

Problem 45.50 (Separating a node)**.** Let a plane curve have local equation

$$y^2 - x^2 - x^3 = 0$$

at the origin, in characteristic not two. Show that one blow-up separates its two tangent branches.

Problem 45.51 (Why the exceptional curve has square minus one)**.** Give a geometric derivation of $E^2 = -1$ without using the normal bundle of E.

Problem 45.52 (A curve class and its genus bookkeeping)**.** Let $C \subset \mathbb{P}^2$ be a degree-d curve with an ordinary point of multiplicity m at p. Compute the basic intersection data of its strict transform.

Problem 45.53 (The conic system that resolves Cremona)**.** Show numerically that $D = 2H - E_1 - E_2 - E_3$ has the behavior required of the pullback of a target line.

Problem 45.54 (Local regularity at a Cremona base point)**.** Carry out both blow-up charts near p_1 and show that no point of E_1 remains indeterminate.

Problem 45.55 (Recovering the degree formula by intersection)**.** Let C have class $dH - m_1E_1 - m_2E_2 - m_3E_3$. Derive its Cremona image degree without coordinates.

Problem 45.56 (The full Picard action as a matrix)**.** Write the lifted Cremona action as a matrix in the ordered basis (H, E_1, E_2, E_3) and verify that its square is the identity.

Problem 45.57 (Blow-up versus blow-down factorization)**.** Describe the standard Cremona transformation as a common resolution diagram and identify all six distinguished (-1)-curves.

Problem 45.58 (A first infinitely near base point)**.** Suppose a pencil of plane curves has multiplicity at least two at p and, after blowing up p, all strict transforms still pass through one fixed point $q \in E$. Explain the next resolution step and its geometric meaning.

Problem 45.59 (Universal resolution of a projective ratio)**.** Let $\phi = [f_0 : f_1 : f_2] : X \dashrightarrow \mathbb{P}^2$ with base ideal $I = (f_0, f_1, f_2)$. Explain locally why $\mathrm{Bl}_I X$ carries a regular map to $\mathbb{P}^2$.

45.16 Challenges

Problem 45.60 (Challenge: resolve a cusp to normal crossings)**.** Starting from $y^2 = x^3$, continue blowing up after the first blow-up until the total transform has only normal-crossing components. Record every exceptional divisor, its self-intersection after subsequent blow-ups, and the dual intersection graph.

Problem 45.61 (Challenge: infinitely near Cremona base points)**.** Construct a quadratic or higher-degree plane birational map whose base cluster contains an infinitely near point. Resolve it by an explicit sequence of point blow-ups and compare the resulting Picard-lattice action with the three-proper-point standard quadratic case.

Problem 45.62 (Challenge: characterize the six (-1)-curves)**.** On the blow-up of three noncollinear points of $\mathbb{P}^2$, prove that the three exceptional divisors and the three strict transforms of lines through pairs of points are exactly the six obvious (-1)-curves. Investigate whether any other effective divisor class can have self-intersection -1 and anticanonical degree one.

Problem 45.63 (Challenge: Weyl reflection viewpoint)**.** Let

$$\alpha = H - E_1 - E_2 - E_3.$$

Show that $\alpha^2 = -2$ and reinterpret the Cremona Picard action as the reflection

$$D \longmapsto D + (D \cdot \alpha)\alpha.$$

Determine which classes are fixed and explain why this reflection preserves both the intersection form and the canonical class.

Problem 45.64 (Challenge: design a resolution algorithm)**.** Given a primitive homogeneous triple defining a rational map $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$, design a symbolic procedure that finds proper base points, blows them up in charts, detects residual infinitely near base points, and outputs a divisor-class description of the resolved map. Test the procedure first on the standard quadratic involution and then on a nontrivial composition of quadratic maps.

Bridge to VI/46

The blow-up chapter completes the first birational-geometry arc of Volume VI. Weil divisors, Cartier divisors, class groups, line bundles, Picard groups, plane cubics, Cremona transformations, and blow-ups now fit into one chain:

$$\text{local equations} \Longrightarrow \text{divisor classes}$$
$$\Longrightarrow \text{birational modifications} \Longrightarrow \text{intersection bookkeeping.}$$

The symbolic exceptional classes of VI/44 have become actual curves, and the indeterminacy of the standard Cremona map has been replaced by a diagram of morphisms.

The next chapter changes direction. VI/46 begins the cohomological arc with *flabby sheaves.* The geometric constructions developed here will later return through line bundles, divisor sequences, and cohomology, but the immediate task is categorical: identify sheaves whose restriction maps are surjective and understand why they are acyclic tools for computing derived invariants.

45.17 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 45.65 (Plane chart)**.** Write the x-chart of the blow-up of $\mathbb{A}^2$ at the origin.

Hint. Use blow-ups, exceptional divisors, and affine charts. ☐

Solution. $k[x, u]$ with $y = xu$. ☐

Exercise 45.66 (Other chart)**.** Write the y-chart.

Hint. Use blow-ups, exceptional divisors, and affine charts. ☐

Solution. $k[v, y]$ with $x = vy$. ☐

Exercise 45.67 (Exceptional divisor)**.** What is the exceptional divisor over the origin?

Hint. Use blow-ups, exceptional divisors, and affine charts. ☐

Solution. $\mathbb{P}^1_k$. ☐

Exercise 45.68 (Rees algebra)**.** What graded algebra defines the blow-up of I?

Hint. Use blow-ups, exceptional divisors, and affine charts. ☐

Solution. $\bigoplus_{n\geq 0} I^n$. ☐

Exercise 45.69 (Away from center)**.** How does a blow-up behave away from the center?

Hint. Use blow-ups, exceptional divisors, and affine charts. ☐

Solution. It is an isomorphism. ☐

Proofs

Exercise 45.70 (Chart cover)**.** Prove: Derive the two affine charts of the blow-up of $\mathbb{A}^2$ at the origin.

Hint. Work from blow-ups, exceptional divisors, and affine charts and state the hypotheses explicitly. ☐

Solution. Take Proj of the Rees algebra and localize at each degree-one generator. ☐

Exercise 45.71 (Exceptional divisor)**.** Prove: Prove the fiber over the origin is $\mathbb{P}^1$.

Hint. Work from blow-ups, exceptional divisors, and affine charts and state the hypotheses explicitly. ☐

Solution. Set the base coordinates to zero while retaining the projective ratio of the two ideal generators. ☐

Exercise 45.72 (Isomorphism away)**.** Prove: Prove the blow-up is an isomorphism off the center.

Hint. Work from blow-ups, exceptional divisors, and affine charts and state the hypotheses explicitly. ☐

Solution. Where one generator of the center ideal is invertible, the Rees algebra chart collapses to the original localization. ☐

Exercise 45.73 (Universal property)**.** Prove: State and prove the factorization property for maps on which the pulled-back center ideal becomes invertible.

Hint. Work from blow-ups, exceptional divisors, and affine charts and state the hypotheses explicitly. ☐

Solution. Use the invertible quotient of the pulled-back ideal to construct the induced map to the Proj of the Rees algebra. ☐

Counterexamples and hypothesis tests

Exercise 45.74 (Fiber point)**.** Test the claim: blowing up a point replaces it by one point.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; in a smooth surface it is replaced by a projective line of directions. ☐

Exercise 45.75 (Global isomorphism)**.** Test the claim: a nontrivial blow-up is an isomorphism everywhere.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it changes the center. ☐

Exercise 45.76 (Dimension increase)**.** Test the claim: blow-up increases the dimension of the variety.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it is birational and preserves dimension. ☐

Applications and investigations

Exercise 45.77 (Resolve indeterminacy)**.** How do blow-ups help rational maps?

Hint. Translate the question into blow-ups, exceptional divisors, and affine charts. ☐

Solution. They can replace base points by exceptional divisors so the map extends regularly. ☐

Exercise 45.78 (Singularity analysis)**.** Why inspect strict transforms after blow-up?

Hint. Translate the question into blow-ups, exceptional divisors, and affine charts. ☐

Solution. Exceptional intersections reveal tangent directions and can simplify singularities. ☐

Challenge problems

Exercise 45.79 (Synthesis challenge)**.** Connect two central viewpoints of Blow-Ups in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Blow-Ups and then translates it through blow-ups, exceptional divisors, and affine charts, checking both descriptions agree. ☐

Exercise 45.80 (Hypothesis challenge)**.** Choose one theorem-level statement from Blow-Ups, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Part IX

Sheaf Cohomology

Chapter 46

Flabby Sheaves

VI/45 completed the birational geometry arc. VI/46 begins the cohomology arc by isolating sheaves whose sections extend across every open inclusion. This elementary condition is strong enough to force higher cohomology to vanish and makes the global section functor computationally useful.

The governing chain is

$$\boxed{\begin{array}{c}\text{surjective restrictions} \Longrightarrow \text{exact overlap complexes}\\ \Downarrow \\ \text{flabby resolutions} \Longrightarrow H^{>0}=0\end{array}}$$

Standing conventions

Unless stated otherwise, X is a topological space and all sheaves are sheaves of abelian groups. For a scheme, the underlying topology is the Zariski topology unless another site is explicitly named.

Learning goals

By the end of the chapter, the reader should be able to:

- define flabby sheaves and use the global-extension criterion;
- recognize skyscraper, all-functions, and rational-function examples;
- distinguish valid permanence properties from tempting false ones;
- prove that direct image preserves flabbiness;
- analyze short exact sequences with flabby kernel and middle term;
- write the two-open and general Čech complexes;
- construct a Godement-style flabby resolution;
- explain why flabby sheaves are acyclic and why injective sheaves are flabby;
- prepare for explicit Čech computations in VI/47.

Example 46.1 (Definition in action)**.** A sheaf $\mathcal{F}$ is flabby if every restriction $\mathcal{F}(U) \to \mathcal{F}(V)$ for $V \subseteq U$ is surjective. Any section on a smaller open extends to the larger open.

Conceptual roadmap

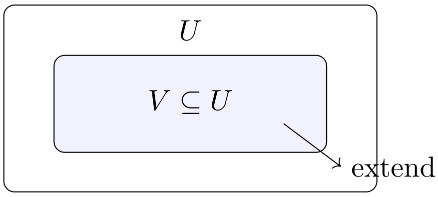


Figure 46.1: Flabbiness says every section on the smaller open extends to the larger open.

Example 46.2 (Skyscraper sheaf)**.** A skyscraper sheaf supported at one point is flabby: restriction maps are either identities, maps to zero, or projections that remain surjective.

46.1 Definition and first examples

Definition 46.3 (Flabby sheaf)**.** A sheaf $\mathcal{F}$ of abelian groups on a topological space X is *flabby* (or *flasque*) if for every inclusion of open sets $V \subseteq U \subseteq X$ the restriction map

$$\rho_V^U : \Gamma(U, \mathcal{F}) \longrightarrow \Gamma(V, \mathcal{F})$$

is surjective.

$$\Gamma(U, \mathcal{F}) \xrightarrow{\rho_V^U} \mathrel{\mkern-14mu}\twoheadrightarrow \Gamma(V, \mathcal{F})$$

Figure 46.2: The defining restriction map of a flabby sheaf is surjective.

Proposition 46.4 (Global-extension criterion)**.** *A sheaf $\mathcal{F}$ is flabby if and only if for every open $U \subseteq X$ the map*

$$\Gamma(X, \mathcal{F}) \longrightarrow \Gamma(U, \mathcal{F})$$

is surjective.

Proof. The forward implication is the defining property with the larger open equal to X. Conversely, if $V \subseteq U$ and $s \in \Gamma(V, \mathcal{F})$, extend s first to a global section and then restrict that global extension to U. □

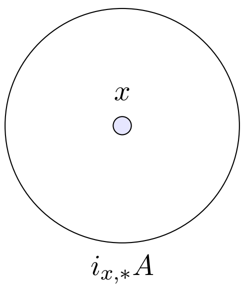


Figure 46.3: A skyscraper sheaf is a basic flabby example.

Example 46.5 (Skyscraper sheaf)**.** Let $i : \{x\} \hookrightarrow X$ and let A be an abelian group. The skyscraper sheaf i_*A is flabby. A section over an open set containing x is simply an element of A, while an open not containing x has only the zero section.

Example 46.6 (Sheaf of all functions)**.** Fix an abelian group A. The sheaf assigning to U the group of all functions $U \to A$ is flabby: extend any function by choosing arbitrary values, for instance 0, on the complement inside the larger open.

Example 46.7 (Rational functions on an integral scheme)**.** If X is an integral scheme and $K(X)$ its function field, the sheaf assigning $K(X)$ to every nonempty open set, with identity restrictions, is flabby.

Example 46.8 (A constant-sheaf nonexample)**.** Let $X = \mathbb{R}$ and let A contain two distinct elements a, b. On

$$V = (-2, -1) \cup (1, 2)$$

the constant sheaf $\underline{A}$ has the locally constant section equal to a on the first component and b on the second. It cannot extend to a locally constant section on the connected space $\mathbb{R}$. Hence $\underline{A}$ is not flabby.

Example 46.9 (Flabby resolution)**.** Embedding a sheaf into a flabby sheaf and iterating on cokernels produces a flabby resolution. Such resolutions compute sheaf cohomology because flabby sheaves are acyclic.

46.2 Permanence and exact sequences

Proposition 46.10 (Products and finite sums)**.** *Arbitrary products of flabby sheaves are flabby. Finite direct sums of flabby sheaves are flabby.*

Proof. Restrictions are computed componentwise, and products of surjective maps are surjective. ☐

Proposition 46.11 (Direct images preserve flabbiness)**.** *Let $f : X \to Y$ be continuous. If $\mathcal{F}$ is flabby on X, then $f_*\mathcal{F}$ is flabby on Y.*

Proof. For $V \subseteq U \subseteq Y$,

$$\Gamma(U, f_*\mathcal{F}) = \Gamma(f^{-1}U, \mathcal{F}) \longrightarrow \Gamma(f^{-1}V, \mathcal{F}) = \Gamma(V, f_*\mathcal{F})$$

is surjective because $f^{-1}V \subseteq f^{-1}U$. ☐

Warning 46.12 (Subobjects and quotients require care)*.* A subsheaf of a flabby sheaf need not be flabby. Moreover, from flabbiness of the middle term alone in a short exact sequence one cannot conclude that the quotient is flabby. A useful stable statement requires flabbiness of the kernel as well.

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \twoheadrightarrow \mathcal{F}'' \longrightarrow 0$$

Figure 46.4: If both $\mathcal{F}'$ and $\mathcal{F}$ are flabby, then $\mathcal{F}''$ is flabby and sections remain exact.

Theorem 46.13 (Flabby kernel and middle term)**.** *Suppose*

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

is exact and both $\mathcal{F}'$ and $\mathcal{F}$ are flabby. Then $\mathcal{F}''$ is flabby, and for every open $U \subseteq X$ the sequence

$$0 \to \Gamma(U, \mathcal{F}') \to \Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{F}'') \to 0$$

is exact.

Proof. The only issue is surjectivity on the right. Start with a section of $\mathcal{F}''$ over U and a maximal compatible local lift to $\mathcal{F}$. If the lift is not defined on all of U, choose a new local lift near a missing point. The difference of the two lifts on the overlap lies in $\mathcal{F}'$. Flabbiness of $\mathcal{F}'$ extends this difference, allowing the two lifts to be modified and glued, contradicting maximality. Thus a global lift over U exists. Flabbiness of $\mathcal{F}''$ follows from the global-extension criterion. □

Definition 46.14 (Flabby resolution)**.** A *flabby resolution* of $\mathcal{F}$ is an exact complex

$$0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \mathcal{I}^2 \to \cdots$$

with every $\mathcal{I}^q$ flabby.

46.3 Overlaps and Čech exactness

Proposition 46.15 (Global sections are left exact)**.** *For every short exact sequence*

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

the sequence

$$0 \to \Gamma(X, \mathcal{F}') \to \Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}'')$$

is exact at the first two terms. Surjectivity of the last map can fail; higher cohomology measures this failure.

Lemma 46.16 (Extension across one overlap)**.** *If $X = U \cup V$ and $\mathcal{F}$ is flabby, then the map*

$$\Gamma(U, \mathcal{F}) \oplus \Gamma(V, \mathcal{F}) \longrightarrow \Gamma(U \cap V, \mathcal{F}), \quad (s, t) \longmapsto s|_{U\cap V} - t|_{U\cap V}$$

is surjective.

Proof. Extend a prescribed overlap section from $U \cap V$ to U and take the second component to be zero. □

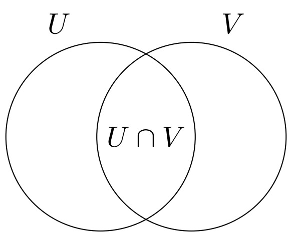


Figure 46.5: For a two-open cover, the first Čech obstruction lives on the overlap.

Proposition 46.17 (Two-open-set exact sequence)**.** *If $X = U \cup V$ and $\mathcal{F}$ is flabby, then*

$$0 \to \Gamma(X, \mathcal{F}) \to \Gamma(U, \mathcal{F}) \oplus \Gamma(V, \mathcal{F}) \to \Gamma(U \cap V, \mathcal{F}) \to 0$$

is exact.

Proof. The sheaf axiom gives exactness at the first two terms, and the previous lemma gives surjectivity at the last term. □

Definition 46.18 (The Čech cochain complex)**.** For an open cover $\mathfrak{U} = \{U_i\}_{i\in I}$, define

$$C^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0<\cdots<i_p} \Gamma(U_{i_0} \cap \cdots \cap U_{i_p}, \mathcal{F}),$$

with the usual alternating restriction differential $\delta : C^p \to C^{p+1}$.

$$C^0(\mathfrak{U}, \mathcal{F}) \xrightarrow{\delta^0} C^1(\mathfrak{U}, \mathcal{F}) \xrightarrow{\delta^1} C^2(\mathfrak{U}, \mathcal{F})$$

Figure 46.6: The Čech cochain complex associated with an open cover.

Theorem 46.19 (Flabby sheaves are Čech-acyclic)**.** *If $\mathcal{F}$ is flabby, then for every open cover $\mathfrak{U}$,*

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) = 0 \qquad (p > 0).$$

Equivalently, the augmented Čech complex is exact.

Proof. Flabbiness lets one extend cochain data from multiple intersections to larger unions of such intersections. Successive extension kills every positive-degree cocycle by a coboundary. The degree-zero exactness is the ordinary sheaf gluing axiom. □

46.4 Flabby resolutions and acyclicity

Definition 46.20 (The discontinuous-section sheaf)**.** For any sheaf $\mathcal{F}$, define

$$G^0(\mathcal{F})(U) = \prod_{x\in U} \mathcal{F}_x.$$

Restrictions are coordinate projections. The natural map $\mathcal{F} \to G^0(\mathcal{F})$ sends a section to its family of germs.

Proposition 46.21 (The first Godement term is flabby)**.** *The sheaf $G^0(\mathcal{F})$ is flabby, and the germ map $\mathcal{F} \to G^0(\mathcal{F})$ is injective.*

Proof. A family indexed by a smaller open extends to a larger open by assigning arbitrary elements, say zeros, at the new points. Injectivity of the germ map is exactly the locality axiom for a sheaf. □

$$0 \to \mathcal{F} \longrightarrow \mathcal{I}^0 \longrightarrow \mathcal{I}^1 \longrightarrow \mathcal{I}^2 \to \cdots$$

Figure 46.7: A flabby resolution replaces a sheaf by acyclic objects.

Proposition 46.22 (Existence of flabby resolutions)**.** *Every sheaf of abelian groups admits a flabby resolution.*

Proof. Embed $\mathcal{F}$ into $G^0(\mathcal{F})$, take the cokernel $\mathcal{Q}^1$, embed $\mathcal{Q}^1$ into $G^0(\mathcal{Q}^1)$, and continue. The successive G^0 terms are flabby. □

Definition 46.23 (Acyclic sheaf)**.** A sheaf $\mathcal{F}$ is Γ-*acyclic* if

$$H^p(X, \mathcal{F}) = 0 \qquad \text{for all } p > 0.$$

Theorem 46.24 (Flabby sheaves are acyclic)**.** *Every flabby sheaf is* Γ*-acyclic. The same is true after restriction to any open subset.*

Proof. The short exact theorem above verifies the standard acyclic resolution criterion for $\Gamma(X, -)$. Thus flabby resolutions compute the derived functors. For a flabby sheaf itself, the resolution may be concentrated in degree zero. ☐

$$\text{surjective restrictions} \longleftarrow \text{exact overlap complexes} \longleftarrow H^{>0} = 0$$

Figure 46.8: The conceptual route from flabbiness to acyclicity.

Proposition 46.25 (Injective sheaves are flabby)**.** *Every injective sheaf of abelian groups is flabby.*

Proof. A section on $V \subseteq U$ is represented by a morphism from the extension-by-zero free abelian sheaf supported on V. Injectivity extends this morphism across the corresponding monomorphism for $V \subseteq U$, producing the desired extension of the section. ☐

Corollary 46.26 (Flabby resolutions compute sheaf cohomology)**.** *If*

$$0 \to \mathcal{F} \to \mathcal{I}^0 \to \mathcal{I}^1 \to \cdots$$

is any flabby resolution, then

$$H^p(X, \mathcal{F}) \cong H^p(\Gamma(X, \mathcal{I}^\bullet)).$$

Proposition 46.27 (Higher direct-image preview)**.** *If* $f : X \to Y$ *is a morphism of ringed spaces and* $\mathcal{F}$ *is flabby, then*

$$R^p f_* \mathcal{F} = 0 \qquad (p > 0).$$

This is the relative analogue of flabby acyclicity and follows because the restriction of a flabby sheaf to $f^{-1}(V)$ *is acyclic for every open* $V \subseteq Y$*.*

Remark 46.28 (Soft and fine sheaves)*.* Other geometrically important acyclic classes include soft and fine sheaves. Their definitions use additional topology, while flabbiness is expressed purely through restriction maps.

Remark 46.29 (Bridge to VI/47)*.* VI/47 will turn the Čech complex into a computational tool. The present chapter supplies its easiest vanishing test: whenever the coefficient sheaf is flabby, every positive-degree Čech class dies.

46.5 Exercises

Exercise 46.30. Check that the zero sheaf is flabby.

Exercise 46.31. Prove the global-extension criterion directly from the definition.

Exercise 46.32. Verify that a skyscraper sheaf is flabby.

Exercise 46.33. Show that the sheaf of all functions $U \to A$ is flabby.

Exercise 46.34. On $X = \mathbb{R}$, show explicitly that the constant sheaf with at least two values is not flabby.

Exercise 46.35. Prove that arbitrary products of flabby sheaves are flabby.

Exercise 46.36. Prove that $f_*\mathcal{F}$ is flabby whenever $\mathcal{F}$ is flabby.

Exercise 46.37. Why is it unsafe to claim that every quotient of a flabby sheaf is flabby?

Exercise 46.38. In a short exact sequence with $\mathcal{F}'$ and $\mathcal{F}$ flabby, prove that sections over every open remain exact on the right.

Exercise 46.39. Deduce from the previous exercise that $\mathcal{F}''$ is flabby.

Exercise 46.40. Write out the map in the two-open exact sequence.

Exercise 46.41. Show directly that the two-open map onto $\Gamma(U \cap V, \mathcal{F})$ is surjective for flabby $\mathcal{F}$.

Exercise 46.42. Write the Čech differential $C^0 \to C^1$.

Exercise 46.43. Write the Čech differential $C^1 \to C^2$.

Exercise 46.44. Explain why a flabby sheaf has zero first Čech cohomology for a two-open cover.

Exercise 46.45. State the positive-degree Čech vanishing theorem for a flabby sheaf.

Exercise 46.46. Show that $G^0(\mathcal{F})$ is flabby.

Exercise 46.47. Show that the germ map $\mathcal{F} \to G^0(\mathcal{F})$ is injective.

Exercise 46.48. Construct the first two nontrivial terms of a flabby resolution.

Exercise 46.49. What does it mean for a sheaf to be Γ-acyclic?

Exercise 46.50. Show that the restriction of a flabby sheaf to an open subset is flabby.

Exercise 46.51. Explain the implication injective $\Rightarrow$ flabby.

Exercise 46.52. Why can a flabby resolution be used instead of an injective resolution to compute H^p?

Exercise 46.53. Show that if $\mathcal{F}$ is flabby, then $R^p f_*\mathcal{F} = 0$ for $p > 0$.

46.6 Solved problems

Problem 46.54. Let X be integral. Prove that the rational-function sheaf $\mathcal{K}_X$ is flabby and determine $H^p(X, \mathcal{K}_X)$ for $p > 0$.

Problem 46.55. Give a complete proof of the constant-sheaf nonexample on $\mathbb{R}$.

Problem 46.56. Prove the short-exact theorem for flabby kernel and middle term using Zorn's lemma.

Problem 46.57. Show that the direct image under an arbitrary continuous map preserves flabbiness.

Problem 46.58. For $X = U \cup V$, compute the augmented Čech complex of a flabby sheaf and prove it is exact.

Problem 46.59. For a three-open cover, write C^0, C^1, and C^2 explicitly.

Problem 46.60. Construct a flabby resolution functorially up to the first two cokernels.

Problem 46.61. Explain carefully why the statement "quotients of flabby sheaves are flabby" needs an extra hypothesis.

Problem 46.62. Prove that injective sheaves are flabby and deduce that enough injectives provide another existence proof for flabby resolutions.

Problem 46.63. Explain why flabby acyclicity on every open subset is stronger than merely knowing $H^{>0}(X, \mathcal{F}) = 0$.

Problem 46.64. Show how a flabby resolution produces the cochain complex computing $H^p(X, \mathcal{F})$.

Problem 46.65. Explain the precise bridge from VI/46 to VI/47.

46.7 Challenges

Problem 46.66 (Challenge)**.** Compare flabby, soft, and fine sheaves on paracompact spaces, making clear which implications use additional topology.

Problem 46.67 (Challenge)**.** Give a complete functorial construction of the Godement resolution and prove each term is flabby.

Problem 46.68 (Challenge)**.** Prove the arbitrary-cover Čech vanishing theorem for flabby sheaves without using derived categories.

Problem 46.69 (Challenge)**.** Develop the relative statement $R^p f_* \mathcal{F} = 0$ for flabby $\mathcal{F}$ directly from the definition of higher direct image.

Problem 46.70 (Challenge)**.** On an integral curve, compare $\mathcal{O}_X$ with the flabby rational-function sheaf $\mathcal{K}_X$ and interpret the quotient $\mathcal{K}_X / \mathcal{O}_X$.

46.8 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 46.71 (Restriction)**.** What property defines a flabby sheaf?

Hint. Use restriction surjectivity, acyclicity, and resolutions. ☐

Solution. All restriction maps are surjective. ☐

Exercise 46.72 (Extension)**.** What does flabbiness say about a section on $V \subseteq U$?

Hint. Use restriction surjectivity, acyclicity, and resolutions. ☐

Solution. It extends to a section on U. ☐

Exercise 46.73 (Skyscraper)**.** Is a skyscraper sheaf flabby?

Hint. Use restriction surjectivity, acyclicity, and resolutions. ☐

Solution. Yes. ☐

Exercise 46.74 (Acyclicity)**.** What are the positive cohomology groups of a flabby sheaf?

Hint. Use restriction surjectivity, acyclicity, and resolutions. ☐

Solution. They vanish. ☐

Exercise 46.75 (Resolution)**.** What kind of resolution can compute sheaf cohomology?

Hint. Use restriction surjectivity, acyclicity, and resolutions. ☐

Solution. A flabby resolution. ☐

Proofs

Exercise 46.76 (Skyscraper flabby)**.** Prove: Prove a skyscraper sheaf is flabby.

Hint. Work from restriction surjectivity, acyclicity, and resolutions and state the hypotheses explicitly. ☐

Solution. Inspect all possible inclusions depending on whether the support point lies in the opens. ☐

Exercise 46.77 (Quotient stability)**.** Prove: Show a quotient of a flabby sheaf is flabby under the standard sheaf-of-abelian-groups setting.

Hint. Work from restriction surjectivity, acyclicity, and resolutions and state the hypotheses explicitly. ☐

Solution. Lift a target quotient section locally/globally through the flabby source and descend. ☐

Exercise 46.78 (Acyclicity)**.** Prove: Prove flabby sheaves are acyclic for global sections.

Hint. Work from restriction surjectivity, acyclicity, and resolutions and state the hypotheses explicitly. ☐

Solution. Use the standard extension argument in the derived-functor construction or a flabby resolution of a flabby sheaf. ☐

Exercise 46.79 (Enough flabbies)**.** Prove: Explain how the canonical sheaf of discontinuous sections gives an embedding into a flabby sheaf.

Hint. Work from restriction surjectivity, acyclicity, and resolutions and state the hypotheses explicitly. ☐

Solution. Take products of stalks over opens; arbitrary germs extend by choosing values independently. ☐

Counterexamples and hypothesis tests

Exercise 46.80 (Every sheaf flabby)**.** Test the claim: every sheaf has surjective restrictions.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False. ☐

Exercise 46.81 (Flabby constant)**.** Test the claim: the constant sheaf is always flabby.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False on general spaces. ☐

Exercise 46.82 (Acyclic means zero)**.** Test the claim: a flabby sheaf must be the zero sheaf because higher cohomology vanishes.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; acyclicity concerns derived global sections, not the sheaf itself. ☐

Applications and investigations

Exercise 46.83 (Cohomology computation)**.** Why are flabby sheaves useful?

Hint. Translate the question into restriction surjectivity, acyclicity, and resolutions. ☐

Solution. They are acyclic objects that can resolve arbitrary sheaves for computing cohomology. ☐

Exercise 46.84 (Extension philosophy)**.** What geometric intuition does flabbiness encode?

Hint. Translate the question into restriction surjectivity, acyclicity, and resolutions. ☐

Solution. Local sections have no obstruction to extension across larger opens. ☐

Challenge problems

Exercise 46.85 (Synthesis challenge)**.** Connect two central viewpoints of Flabby Sheaves in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Flabby Sheaves and then translates it through restriction surjectivity, acyclicity, and resolutions, checking both descriptions agree. ☐

Exercise 46.86 (Hypothesis challenge)**.** Choose one theorem-level statement from Flabby Sheaves, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 47

Čech Cohomology

VI/46 introduced flabby sheaves and explained why their extension property kills higher cohomology. The present chapter turns that abstract vanishing mechanism into a computational machine. Given an open cover, one records sections on the opens, differences on pairwise overlaps, compatibility on triple overlaps, and so on. The resulting alternating complex is the Čech complex.

The basic progression is

$$\boxed{\begin{array}{c}\text{open cover} \Longrightarrow \text{cochains on overlaps}\\ \Downarrow \\ \text{cocycles modulo coboundaries} \Longrightarrow \check{H}^p\end{array}}$$

and the first major calculation is the standard two-chart cover of $\mathbb{P}^1$, where line-bundle cohomology becomes a quotient of Laurent polynomials.

Standing conventions

All sheaves are sheaves of abelian groups unless stated otherwise. Covers are indexed by totally ordered sets so that signs in alternating formulas are unambiguous. Empty intersections contribute the zero group. When comparing Čech and sheaf cohomology, the comparison hypotheses will always be stated explicitly.

Learning goals

By the end of the chapter, the reader should be able to:

- build the Čech cochain complex of an open cover;
- prove $\delta^2 = 0$ and compute cocycles and coboundaries;
- identify degree-zero Čech cohomology with global sections;
- compute two-open and three-open examples explicitly;
- use refinements and understand cover dependence;
- state the Leray comparison criterion;
- recover the flabby-sheaf vanishing result from VI/46;
- compute $\check{H}^0$ and $\check{H}^1$ of $\mathcal{O}_{\mathbb{P}^1}(n)$ on the standard cover;

- reinterpret line-bundle transition functions as $\check{H}^1(\mathcal{O}_X^\times)$;
- prepare for connecting homomorphisms and long exact sequences in VI/48.

Example 47.1 (Two-open cover)**.** For a cover $X = U \cup V$, a Cech 1-cochain is a section on $U \cap V$. It is a coboundary when it can be written as $s_V|_{U\cap V} - s_U|_{U\cap V}$ for sections on the two opens.

Conceptual roadmap

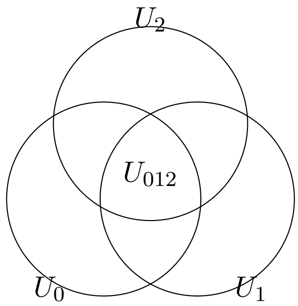


Figure 47.1: A cover is converted into data on opens, pairwise overlaps, triple overlaps, and higher intersections.

47.1 The Čech cochain complex

Definition 47.2 (Ordered open cover)**.** An *ordered open cover* of a space X is a family $\mathfrak{U} = \{U_i\}_{i\in I}$ whose union is X, together with a total order on I. For $i_0 < \cdots < i_p$ write

$$U_{i_0\cdots i_p} = U_{i_0} \cap \cdots \cap U_{i_p}.$$

Definition 47.3 (Čech cochains)**.** For a sheaf $\mathcal{F}$ of abelian groups and an ordered open cover $\mathfrak{U}$, define

$$C^p(\mathfrak{U}, \mathcal{F}) = \prod_{i_0<\cdots<i_p} \Gamma(U_{i_0\cdots i_p}, \mathcal{F}).$$

An element of C^p is called a *p-cochain.*

$$C^0(\mathfrak{U}, \mathcal{F}) \xrightarrow{\delta^0} C^1(\mathfrak{U}, \mathcal{F}) \xrightarrow{\delta^1} C^2(\mathfrak{U}, \mathcal{F})$$

$$\delta^{p+1}\delta^p = 0$$

Figure 47.2: The alternating restriction maps form a cochain complex.

Proposition 47.4 (The alternating differential satisfies $\delta^2 = 0$)**.** *Define*

$$(\delta c)_{i_0\cdots i_{p+1}} = \sum_{r=0}^{p+1} (-1)^r c_{i_0\cdots\widehat{i_r}\cdots i_{p+1}}|_{U_{i_0\cdots i_{p+1}}}.$$

Then $\delta^{p+1} \circ \delta^p = 0$.

Proof. Every term obtained by deleting indices i_r and i_s occurs twice. The two occurrences have opposite signs because deleting i_r first shifts the position of i_s by one exactly when $r < s$. The corresponding restrictions agree by functoriality of restriction maps, so the terms cancel pairwise. □

Definition 47.5 (Čech cocycles, coboundaries, and cohomology)**.** Set

$$Z^p(\mathfrak{U}, \mathcal{F}) = \ker(\delta : C^p \to C^{p+1}), \qquad B^p(\mathfrak{U}, \mathcal{F}) = \operatorname{im}(\delta : C^{p-1} \to C^p).$$

The pth Čech cohomology group of the cover is

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) = Z^p / B^p.$$

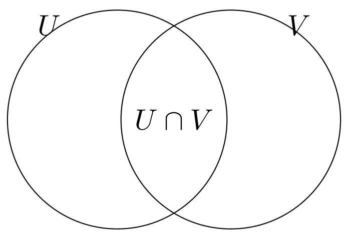
Proposition 47.6 (Degree zero is global sections)**.** *For every open cover* $\mathfrak{U}$ *there is a canonical isomorphism*

$$\check{H}^0(\mathfrak{U}, \mathcal{F}) \cong \Gamma(X, \mathcal{F}).$$

Proof. A zero-cocycle is a family $s_i \in \Gamma(U_i, \mathcal{F})$ whose restrictions agree on every pairwise overlap. The sheaf gluing axiom produces a unique global section, and every global section gives such a compatible family. □

Example 47.7 (Projective line standard cover)**.** Cover $\mathbb{P}^1$ by $U_0 \cong \mathbb{A}^1_t$ and $U_1 \cong \mathbb{A}^1_s$, with overlap $k[t, t^{-1}]$. Cech computations for $\mathcal{O}(d)$ reduce to Laurent polynomials and the transition factor t^d.

47.2 Two-open and three-open covers



$$C^0 = \mathcal{F}(U) \oplus \mathcal{F}(V), \quad C^1 = \mathcal{F}(U \cap V)$$

Figure 47.3: For a two-open cover, first cohomology is the obstruction to representing overlap data as a difference of restrictions.

Proposition 47.8 (The two-open complex)**.** *If* $X = U \cup V$*, then*

$$C^0 = \Gamma(U, \mathcal{F}) \oplus \Gamma(V, \mathcal{F}), \qquad C^1 = \Gamma(U \cap V, \mathcal{F}), \qquad C^p = 0 \ (p \geq 2),$$

and

$$\check{H}^1(\mathfrak{U}, \mathcal{F}) \cong \frac{\Gamma(U \cap V, \mathcal{F})}{\Gamma(U, \mathcal{F})|_{U \cap V} - \Gamma(V, \mathcal{F})|_{U \cap V}}.$$

Proof. There are only two indices, so no strictly increasing triple exists. The displayed quotient is therefore $C^1 / \operatorname{im} \delta^0$. □

Example 47.9 (An overlap obstruction)**.** A section $g \in \Gamma(U \cap V, \mathcal{F})$ determines a class in $\check{H}^1(\{U, V\}, \mathcal{F})$. Its class is zero exactly when

$$g = s_U|_{U\cap V} - s_V|_{U\cap V}$$

for sections s_U and s_V on the two opens. Thus first Čech cohomology records whether overlap data can be split into data coming from the members of the cover.

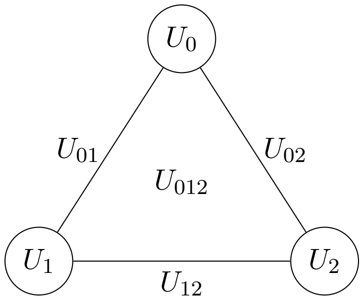


Figure 47.4: The nerve of a three-open cover: vertices, edges, and a 2-simplex encode the first three Čech degrees.

Proposition 47.10 (The three-open complex)**.** *For* $X = U_0 \cup U_1 \cup U_2$*,*

$$C^0 = \prod_i \Gamma(U_i, \mathcal{F}), \qquad C^1 = \prod_{i<j} \Gamma(U_{ij}, \mathcal{F}), \qquad C^2 = \Gamma(U_{012}, \mathcal{F}).$$

For $c = (c_{01}, c_{02}, c_{12})$*,*

$$(\delta c)_{012} = c_{12} - c_{02} + c_{01}$$

after restriction to the triple overlap.

Proof. This is the general alternating formula specialized to three indices. □

47.3 Refinements and intrinsic Čech cohomology

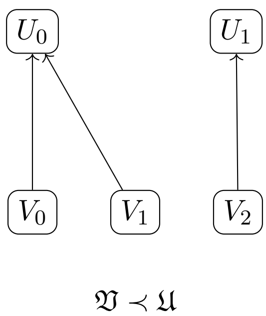


Figure 47.5: A refinement replaces each open by a smaller open contained in one member of the original cover.

Definition 47.11 (Refinement of a cover)**.** A cover $\mathfrak{V} = \{V_j\}_{j\in J}$ *refines* $\mathfrak{U} = \{U_i\}_{i\in I}$ if there is a function $\tau : J \to I$ such that

$$V_j \subseteq U_{\tau(j)}$$

for all j. We write $\mathfrak{V} \prec \mathfrak{U}$.

Theorem 47.12 (Refinements induce maps on cohomology)**.** *A refinement map τ induces a cochain map*

$$C^\bullet(\mathfrak{U}, \mathcal{F}) \longrightarrow C^\bullet(\mathfrak{V}, \mathcal{F})$$

by restriction, hence homomorphisms

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \longrightarrow \check{H}^p(\mathfrak{V}, \mathcal{F}).$$

Different choices of refinement map induce the same map on cohomology.

Proof. Restriction commutes with the alternating differential. Two refinement choices are joined by the standard prism cochain homotopy, so their induced maps differ by a coboundary and therefore agree on cohomology. ☐

Definition 47.13 (Global Čech cohomology)**.** The *global Čech cohomology* of $\mathcal{F}$ is the filtered colimit over open covers ordered by refinement:

$$\check{H}^p(X, \mathcal{F}) = \varinjlim_{\mathfrak{U}} \check{H}^p(\mathfrak{U}, \mathcal{F}).$$

Remark 47.14 (Cover dependence before refinement)*.* The group $\check{H}^p(\mathfrak{U}, \mathcal{F})$ can depend strongly on the chosen cover. Passing to refinements separates the intrinsic Čech theory of the space from artifacts of one coarse cover.

Example 47.15 (Cocycle condition on triple overlaps)**.** For three opens, a 1-cochain (g_{ij}) is a cocycle when $g_{jk} - g_{ik} + g_{ij} = 0$ on triple intersections. The alternating signs are exactly the Cech differential.

47.4 Flabby sheaves and Leray covers

$$\begin{tikzcd} \boxed{z \in Z^p} \arrow[r] & \boxed{\text{extend sections}} \arrow[r] & \boxed{z = \delta b} \end{tikzcd}$$

$\mathcal{F}$ flabby

Figure 47.6: Flabbiness kills positive-degree Čech cocycles by successive extension.

Theorem 47.16 (Flabby sheaves are Čech-acyclic)**.** *If $\mathcal{F}$ is flabby, then for every open cover $\mathfrak{U}$,*

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) = 0 \qquad (p > 0).$$

Proof. This is the exactness theorem developed in VI/46: successive extension of sections from intersections produces a contracting procedure for positive-degree cocycles. ☐

Definition 47.17 (Acyclic (Leray) cover for a sheaf)**.** An open cover $\mathfrak{U} = \{U_i\}$ is *$\mathcal{F}$-acyclic* if every nonempty finite intersection $U_{i_0 \cdots i_p}$ satisfies

$$H^q(U_{i_0 \cdots i_p}, \mathcal{F}) = 0 \qquad (q > 0).$$

Theorem 47.18 (Leray comparison criterion)**.** *If $\mathfrak{U}$ is $\mathcal{F}$-acyclic, then the natural comparison map is an isomorphism*

$$\check{H}^p(\mathfrak{U}, \mathcal{F}) \xrightarrow{\sim} H^p(X, \mathcal{F})$$

for all $p \geq 0$.

Proof. Resolve $\mathcal{F}$ by flabby sheaves and form the double complex obtained by taking Čech cochains degreewise. Exactness in one direction comes from flabbiness, while the acyclicity hypothesis collapses the other spectral filtration to the zeroth row. Both filtrations therefore compute the same total cohomology. □

Example 47.19 (Affine-cover preview for schemes)**.** If a scheme is covered by affine opens whose finite intersections are affine, then a quasi-coherent sheaf will be acyclic on those intersections once the affine vanishing theorem is available. Thus the corresponding affine cover becomes a Leray cover. The required affine vanishing result is proved systematically in VI/49.

47.5 The standard cover of $\mathbb{P}^1$

$$\begin{tikzcd}
\boxed{U_0 = D_+(x_0) \cong \mathbb{A}^1_t} & & \boxed{U_1 = D_+(x_1) \cong \mathbb{A}^1_u} \\
& \boxed{U_{01} \cong \mathbb{G}_m, \quad u = t^{-1}} \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

Figure 47.7: The standard two-affine cover of $\mathbb{P}^1$ reduces Čech calculations to Laurent polynomials.

Definition 47.20 (The standard cover of $\mathbb{P}^1$)**.** Let

$$\mathbb{P}^1_k = \operatorname{Proj} k[x_0, x_1], \qquad U_0 = D_+(x_0), \quad U_1 = D_+(x_1).$$

Writing $t = x_1/x_0$ on U_0 and $u = x_0/x_1 = t^{-1}$ on U_1, we have

$$U_0 \cong \mathbb{A}^1_t, \qquad U_1 \cong \mathbb{A}^1_u, \qquad U_{01} \cong \operatorname{Spec} k[t, t^{-1}].$$

Proposition 47.21 (Transition function for $\mathcal{O}(n)$)**.** *Choose the standard trivializing generators $e_0 = x_0^n$ on U_0 and $e_1 = x_1^n$ on U_1. On U_{01},*

$$e_1 = t^n e_0.$$

Consequently a section $b(u)e_1$ from U_1 is represented in the e_0 trivialization by

$$t^n b(t^{-1}) e_0.$$

Proof. The relation follows immediately from $x_1 = t x_0$ on $D_+(x_0 x_1)$. □

Proposition 47.22 (The two-chart complex for $\mathcal{O}(n)$)**.** *With the e_0 trivialization on the overlap, the Čech complex is*

$$k[t] \oplus k[t^{-1}] \longrightarrow k[t, t^{-1}], \qquad (a(t), b(t^{-1})) \longmapsto t^n b(t^{-1}) - a(t).$$

Hence

$$\check{H}^1(\mathfrak{U}, \mathcal{O}(n)) \cong \frac{k[t, t^{-1}]}{k[t] + t^n k[t^{-1}]}.$$

Proof. This is Proposition 47.8 with the transition function inserted. □

Theorem 47.23 (Degree-zero calculation for $\mathcal{O}(n)$)**.** *For the standard cover,*

$$\dim_k \check{H}^0(\mathfrak{U}, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

For $n \geq 0$, a basis corresponds to the Laurent exponents $0, 1, \ldots, n$, equivalently to the homogeneous monomials

$$x_0^n, x_0^{n-1}x_1, \ldots, x_1^n.$$

Proof. A zero-cocycle corresponds to an element of the intersection

$$k[t] \cap t^n k[t^{-1}] \subset k[t, t^{-1}].$$

For $n \geq 0$ this is spanned by $1, t, \ldots, t^n$; for $n < 0$ the intersection is zero. $\square$

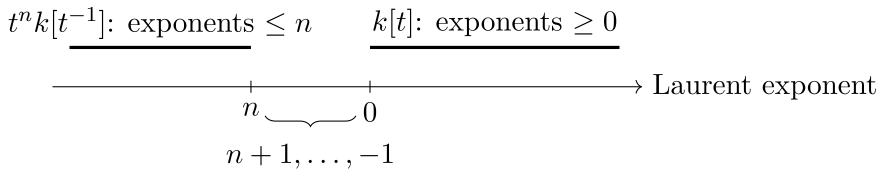


Figure 47.8: For $n \leq -2$, the missing Laurent exponents $n+1, \ldots, -1$ form a basis of $\check{H}^1(\mathbb{P}^1, \mathcal{O}(n))$.

Theorem 47.24 (First Čech cohomology of $\mathcal{O}(n)$)**.** *For the standard cover,*

$$\check{H}^1(\mathfrak{U}, \mathcal{O}(n)) = 0 \qquad (n \geq -1),$$

while for $n \leq -2$ it has basis

$$t^{n+1}, t^{n+2}, \ldots, t^{-1}.$$

Thus

$$\dim_k \check{H}^1(\mathfrak{U}, \mathcal{O}(n)) = -n-1 \qquad (n \leq -2).$$

Proof. The subspace $k[t]$ contains all Laurent monomials with exponent ≥ 0, while $t^n k[t^{-1}]$ contains all exponents $\leq n$. If $n \geq -1$, these two ranges cover every integer exponent. If $n \leq -2$, the missing exponents are exactly $n+1, \ldots, -1$. $\square$

Corollary 47.25 (Dimension pattern)**.** *For the standard cover of $\mathbb{P}^1$,*

$$\dim_k \check{H}^0(\mathcal{O}(n)) = \max\{n+1, 0\}, \qquad \dim_k \check{H}^1(\mathcal{O}(n)) = \max\{-n-1, 0\}.$$

Once affine acyclicity of quasi-coherent sheaves is established in VI/49, the Leray criterion identifies these groups with the sheaf cohomology groups H^0 and H^1.

Example 47.26 (The first negative case)**.** For $n = -2$,

$$\check{H}^1(\mathfrak{U}, \mathcal{O}(-2)) \cong k \cdot t^{-1}.$$

The single missing Laurent mode t^{-1} is the first nonzero line-bundle cohomology class visible in the standard two-chart calculation.

47.6 Line bundles and comparison issues

Proposition 47.27 (Line bundles as unit-valued 1-cocycles)**.** *For the sheaf of units $\mathcal{O}_X^\times$, a Čech 1-cocycle is a family of invertible transition functions g_{ij}. On triple overlaps they satisfy*

$$g_{ij}g_{jk} = g_{ik}.$$

Changing local trivializations multiplies the cocycle by a coboundary. Therefore line bundles trivialized by the cover are classified by

$$\check{H}^1(\mathfrak{U}, \mathcal{O}_X^\times).$$

Passing to refinements recovers the usual Čech description of $\operatorname{Pic}(X)$*.*

Proof. This is precisely the gluing construction for line bundles from VI/42, rewritten in cohomological language. ☐

Warning 47.28 (Čech and sheaf cohomology need not always coincide). For an arbitrary topological space, arbitrary sheaf, and arbitrary cover, the groups $\check{H}^p(\mathfrak{U}, \mathcal{F})$ need not equal the derived-functor groups $H^p(X, \mathcal{F})$. The Leray criterion gives a clean sufficient hypothesis. This distinction is essential whenever a computation depends on the choice of cover.

Remark 47.29 (Bridge to exact sequences). VI/48 studies what happens to cohomology under a short exact sequence

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0.$$

The connecting map $H^0(\mathcal{F}'') \to H^1(\mathcal{F}')$ will acquire a concrete interpretation: it is the Čech class of the failure of local lifts of a section of $\mathcal{F}''$ to glue globally.

47.7 Exercises

Exercise 47.30. For a two-open cover $X = U \cup V$, write the differential $C^0 \to C^1$ with a fixed sign convention.

Exercise 47.31. Verify directly that $\delta^1\delta^0 = 0$ for a three-open cover.

Exercise 47.32. Show that $\check{H}^0(\mathfrak{U}, \mathcal{F})$ is independent of the cover.

Exercise 47.33. Compute the Čech groups of the one-set cover $\{X\}$.

Exercise 47.34. For a three-open cover, write the 1-cocycle condition explicitly.

Exercise 47.35. Show that if $U \cap V = \varnothing$, then the two-open cover has $\check{H}^1 = 0$.

Exercise 47.36. Show that a refinement induces the identity map on $\check{H}^0$ under the identification with global sections.

Exercise 47.37. Explain why the direct limit over refinements removes dependence on a single coarse cover.

Exercise 47.38. Prove that a flabby sheaf has zero $\check{H}^1$ on a two-open cover.

Exercise 47.39. State the condition that makes a cover Leray for $\mathcal{F}$.

Exercise 47.40. On $\mathbb{P}^1$, verify $U_0 \cap U_1 \cong \operatorname{Spec} k[t, t^{-1}]$.

Exercise 47.41. For $\mathcal{O}(2)$, list a basis of $\check{H}^0$ on the standard cover.

Exercise 47.42. Compute $\check{H}^1(\mathbb{P}^1, \mathcal{O}(-1))$ on the standard cover.

Exercise 47.43. Compute $\check{H}^1(\mathbb{P}^1, \mathcal{O}(-3))$ and give a basis.

Exercise 47.44. Compute $\check{H}^0(\mathbb{P}^1, \mathcal{O}(-2))$.

Exercise 47.45. For $n = 0$, compute both Čech groups of $\mathcal{O}$ on the standard cover.

Exercise 47.46. Explain why $\check{H}^1(\mathfrak{U}, \mathcal{O}_X^\times)$ describes line bundles trivial on the cover.

Exercise 47.47. Show that the transition function for $\mathcal{O}(n + m)$ is the product of those for $\mathcal{O}(n)$ and $\mathcal{O}(m)$.

Exercise 47.48. Why is the Leray criterion only a sufficient condition in the form stated?

Exercise 47.49. Give a common refinement of two covers $\{U_i\}$ and $\{V_j\}$.

Exercise 47.50. For $n = -4$, identify the missing Laurent modes in

$$\check{H}^1(\mathcal{O}(n)).$$

Exercise 47.51. Check the dimension identity

$$\dim \check{H}^0(\mathcal{O}(n)) - \dim \check{H}^1(\mathcal{O}(n)) = n + 1.$$

Exercise 47.52. Explain why one should not automatically replace $\check{H}^p(\mathfrak{U}, \mathcal{F})$ by $H^p(X, \mathcal{F})$ in an arbitrary computation.

Exercise 47.53. Describe the future connecting class of local lifts in Čech language.

47.8 Solved problem dossiers

Problem 47.54. Derive the quotient formula for $\check{H}^1$ of a two-open cover and interpret its zero class geometrically.

Problem 47.55. For a three-open cover, prove directly that the coboundary of every zero-cochain satisfies the triangle cocycle condition.

Problem 47.56. Show that two refinement maps between the same covers induce the same map on cohomology.

Problem 47.57. Explain in detail how flabbiness kills a Čech 1-cocycle.

Problem 47.58. Compute $\check{H}^0$ and $\check{H}^1$ of $\mathcal{O}(3)$ on the standard cover of $\mathbb{P}^1$.

Problem 47.59. Compute $\check{H}^0$ and $\check{H}^1$ of $\mathcal{O}(-5)$ on the standard cover.

Problem 47.60. Recover the familiar $n + 1$ homogeneous polynomials as Čech zero-cocycles of $\mathcal{O}(n)$ for $n \geq 0$.

Problem 47.61. Use transition functions to show $\mathcal{O}(n) \otimes \mathcal{O}(m) \cong \mathcal{O}(n + m)$ from the Čech viewpoint.

Problem 47.62. Explain why the standard cover of $\mathbb{P}^1$ is expected to compute sheaf cohomology of $\mathcal{O}(n)$ once affine vanishing is known.

Problem 47.63. Construct the Čech 1-cocycle measuring failure of local lifts in a short exact sequence.

Problem 47.64. Compare the one-set cover and a nontrivial cover of the same space and explain why fixed-cover Čech cohomology can behave differently.

Problem 47.65. Show how the line-bundle discussion reconnects VI/47 to the Picard group of VI/42.

47.9 Challenges

Problem 47.66 (Challenge)**.** Prove the refinement-independence theorem by writing an explicit prism cochain homotopy.

Problem 47.67 (Challenge)**.** Generalize the $\mathbb{P}^1$ Laurent-polynomial calculation to the standard affine cover of projective space and identify what combinatorial object replaces the one-dimensional exponent interval.

Problem 47.68 (Challenge)**.** Work out $\check{H}^1$ of the constant sheaf $\mathbb{Z}$ on a good cover of the circle and compare it with a bad one-set cover.

Problem 47.69 (Challenge)**.** Prove the Leray comparison criterion using a double complex built from a flabby resolution.

Problem 47.70 (Challenge)**.** Use the calculation of $\mathcal{O}(n)$ to conjecture and then verify the Euler characteristic formula on $\mathbb{P}^1$.

47.10 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 47.71 (Zero cochains)**.** What is a Cech 0-cochain?

Hint. Use Cech cochains, cocycles, coboundaries, and affine covers. ☐

Solution. A tuple of sections, one on each open set. ☐

Exercise 47.72 (One cochains)**.** What is a Cech 1-cochain?

Hint. Use Cech cochains, cocycles, coboundaries, and affine covers. ☐

Solution. A tuple of sections on pairwise intersections. ☐

Exercise 47.73 (Two-cover differential)**.** What is the 0-to-1 differential for U, V?

Hint. Use Cech cochains, cocycles, coboundaries, and affine covers. ☐

Solution. $(s_U, s_V) \mapsto s_V|_{U\cap V} - s_U|_{U\cap V}$. ☐

Exercise 47.74 (Cocycle)**.** What equation defines a Cech 1-cocycle on triple overlaps?

Hint. Use Cech cochains, cocycles, coboundaries, and affine covers. ☐

Solution. The alternating sum vanishes. ☐

Exercise 47.75 (Cohomology)**.** How is Cech cohomology formed?

Hint. Use Cech cochains, cocycles, coboundaries, and affine covers. ☐

Solution. As cocycles modulo coboundaries. ☐

Proofs

Exercise 47.76 (Differential squares zero)**.** Prove: Prove the Cech differential satisfies $\delta^2 = 0$.

Hint. Work from Cech cochains, cocycles, coboundaries, and affine covers and state the hypotheses explicitly. ☐

Solution. Each restriction term appears twice with opposite signs in the alternating sum. ☐

Exercise 47.77 (H0)**.** Prove: Prove $\check{H}^0$ equals global sections for a sheaf.

Hint. Work from Cech cochains, cocycles, coboundaries, and affine covers and state the hypotheses explicitly. ☐

Solution. A zero cocycle is exactly a compatible family of local sections, which glues uniquely. ☐

Exercise 47.78 (Refinement map)**.** Prove: Explain how a refinement of covers induces a map of Cech complexes.

Hint. Work from Cech cochains, cocycles, coboundaries, and affine covers and state the hypotheses explicitly. ☐

Solution. Restrict each cochain component to the chosen refined intersections and check compatibility with δ. ☐

Exercise 47.79 (Two-cover H1)**.** Prove: Derive the quotient description of $\check{H}^1$ for a two-open cover.

Hint. Work from Cech cochains, cocycles, coboundaries, and affine covers and state the hypotheses explicitly. ☐

Solution. All 1-cochains are cocycles because there are no triple intersections; quotient by differences of restricted zero-cochains. ☐

Counterexamples and hypothesis tests

Exercise 47.80 (Cover independent automatically)**.** Test the claim: Cech cohomology for one arbitrary cover always equals sheaf cohomology.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False without acyclicity or suitable hypotheses on the cover. ☐

Exercise 47.81 (Differential no signs)**.** Test the claim: the Cech differential can omit alternating signs.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; signs are required for $\delta^2 = 0$. ☐

Exercise 47.82 (H0 quotient)**.** Test the claim: $\check{H}^0$ is unrelated to global sections.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it recovers compatible global sections. ☐

Applications and investigations

Exercise 47.83 (Line bundles)**.** How do Cech cocycles describe line bundles?

Hint. Translate the question into Cech cochains, cocycles, coboundaries, and affine covers. □

Solution. Transition functions form multiplicative 1-cocycles modulo coboundary changes of trivialization. □

Exercise 47.84 (Projective calculations)**.** Why is the standard affine cover useful?

Hint. Translate the question into Cech cochains, cocycles, coboundaries, and affine covers. □

Solution. Intersections are explicit Laurent-affine schemes where sections can be computed algebraically. □

Challenge problems

Exercise 47.85 (Synthesis challenge)**.** Connect two central viewpoints of Cech Cohomology in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. □

Solution. A complete solution makes one concrete computation in Cech Cohomology and then translates it through Cech cochains, cocycles, coboundaries, and affine covers, checking both descriptions agree. □

Exercise 47.86 (Hypothesis challenge)**.** Choose one theorem-level statement from Cech Cohomology, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. □

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. □

Chapter 48

Exact Sequences and Cohomology

VI/47 made cohomology computational by organizing local data on overlaps. The present chapter explains how cohomology behaves when sheaves themselves sit in an exact sequence. The essential phenomenon is that taking global sections is left exact but need not preserve surjectivity. The missing surjectivity is measured by a connecting class, and continuing this construction produces the long exact sequence in cohomology.

The first boundary map has a concrete geometric meaning. Given

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

and a global section s of $\mathcal{F}''$, choose local lifts t_i to $\mathcal{F}$. Their differences $t_j - t_i$ lie in $\mathcal{F}'$ on overlaps. Those differences form a Čech cocycle, and its class is precisely the obstruction to gluing the local lifts into one global lift.

$$\boxed{\text{local lifts} \Longrightarrow \text{overlap cocycle} \Longrightarrow \delta(s) \Longrightarrow \text{long exact sequence}}$$

Standing conventions

All sheaves are sheaves of abelian groups unless stated otherwise. The notation $H^q(X, -)$ denotes sheaf cohomology. When a Čech representative is used, the hypotheses needed for comparison with sheaf cohomology are stated explicitly. When X is fixed, we write $H^q(\mathcal{F})$ for $H^q(X, \mathcal{F})$.

Learning goals

By the end of the chapter, the reader should be able to:

- explain why global sections are left exact but not generally exact;
- construct the connecting homomorphism from local lifts;
- state and use the long exact sequence in sheaf cohomology;
- prove the lifting criterion $\delta(s) = 0$;
- use naturality and split exact sequences;
- distinguish a sheaf epimorphism from surjectivity on sections of a fixed open set;
- derive long exact sequences from effective Cartier divisor sequences;
- recover the cohomology of $\mathcal{O}_{\mathbb{P}^1}(n)$ recursively;

- perform dimension shifting with flabby sheaves;
- use vanishing hypotheses to turn obstruction problems into extension theorems.

Example 48.1 (From a short exact sequence to a long one)**.** A short exact sequence $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ yields $0 \to H^0(\mathcal{F}') \to H^0(\mathcal{F}) \to H^0(\mathcal{F}'') \to H^1(\mathcal{F}') \to \cdots$. The connecting map measures failure of a global section of $\mathcal{F}''$ to lift globally.

Conceptual roadmap

$$s \in H^0(X, \mathcal{F}'') \mapsto \text{local lifts } t_i \longrightarrow t_i - t_j \mapsto \delta(s) \in H^1(X, \mathcal{F}')$$

Figure 48.1: The connecting class records the failure of local lifts to glue globally.

48.1 Short exact sequences and the first obstruction

Definition 48.2 (Short exact sequence of sheaves)**.** A sequence

$$0 \longrightarrow \mathcal{F}' \xrightarrow{\alpha} \mathcal{F} \xrightarrow{\beta} \mathcal{F}'' \longrightarrow 0$$

is *short exact* if it is exact as a sequence of sheaves: α is a monomorphism, β is an epimorphism, and $\ker \beta = \operatorname{im} \alpha$. Surjectivity of β is a stalkwise or local statement; it does not imply that every section of $\mathcal{F}''$ on a fixed open set lifts on that whole open set.

$$\begin{tikzcd}
0 \arrow[r] & \mathcal{F}' \arrow[r] & \mathcal{F} \arrow[r, two heads] \arrow[d] & \mathcal{F}'' \arrow[r] & 0 \\
 & & \Gamma(X, -) \text{ is left exact, not usually exact} & &
\end{tikzcd}$$

Figure 48.2: A short exact sequence of sheaves need not remain surjective after taking global sections.

Proposition 48.3 (Global sections are left exact)**.** *For every short exact sequence as above and every open $U \subseteq X$, the sequence*

$$0 \to \Gamma(U, \mathcal{F}') \to \Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{F}'')$$

is exact at the first two terms.

Proof. Injectivity is immediate. A section of $\mathcal{F}$ maps to zero in $\mathcal{F}''$ exactly when all of its germs lie in the image of $\mathcal{F}'$. Those local preimages glue uniquely because $\mathcal{F}' \to \mathcal{F}$ is a monomorphism. □

Definition 48.4 (The connecting homomorphism in degree zero)**.** Given a short exact sequence, the first connecting homomorphism is the map

$$\delta : H^0(X, \mathcal{F}'') \longrightarrow H^1(X, \mathcal{F}')$$

that assigns to a quotient section the obstruction to lifting it to a global section of $\mathcal{F}$.

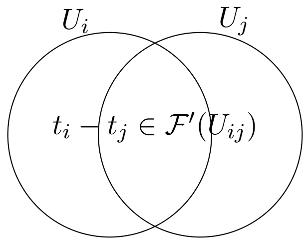


Figure 48.3: Two local lifts of the same quotient section differ by a section of the kernel on the overlap.

Proposition 48.5 (Local lifts produce a cocycle)**.** *Let $s \in \Gamma(X, \mathcal{F}'')$. Choose an open cover $\{U_i\}$ and local lifts $t_i \in \Gamma(U_i, \mathcal{F})$ of $s|_{U_i}$. On overlaps,*

$$a_{ij} = t_j|_{U_{ij}} - t_i|_{U_{ij}} \in \Gamma(U_{ij}, \mathcal{F}').$$

Then (a_{ij}) is a Čech 1-cocycle. Changing the local lifts changes this cocycle by a coboundary.

Proof. On a triple overlap,

$$a_{jk} - a_{ik} + a_{ij} = (t_k - t_j) - (t_k - t_i) + (t_j - t_i) = 0.$$

If t_i is replaced by $t_i + b_i$ with $b_i \in \mathcal{F}'(U_i)$, then a_{ij} changes by $b_j - b_i$, the Čech coboundary of (b_i). □

Corollary 48.6 (Vanishing of the boundary is the lifting criterion)**.** *For $s \in H^0(X, \mathcal{F}'')$,*

$$\delta(s) = 0 \iff s \text{ lifts to a section of } \mathcal{F} \text{ on } X.$$

Proof. If a global lift exists, take all local lifts to be its restrictions. Conversely, if the local-lift cocycle is a coboundary, modify the t_i by kernel-valued 0-cochains so that the corrected lifts agree on overlaps; they then glue to a global lift. □

Example 48.7 (Lifting obstruction)**.** If a section of $\mathcal{F}''$ lifts locally to $\mathcal{F}$, the differences of local lifts lie in $\mathcal{F}'$ on overlaps and form a Cech 1-cocycle. Its cohomology class is the connecting image.

48.2 The long exact sequence

$$H^0(\mathcal{F}') \to H^0(\mathcal{F}) \to H^0(\mathcal{F}'') \overset{\delta}{\to} H^1(\mathcal{F}') \to H^1(\mathcal{F})$$

Figure 48.4: The first part of the long exact cohomology sequence.

Theorem 48.8 (The long exact sequence in sheaf cohomology)**.** *Every short exact sequence*

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

induces a natural long exact sequence

$$\begin{aligned} 0 &\to H^0(X, \mathcal{F}') \to H^0(X, \mathcal{F}) \to H^0(X, \mathcal{F}'') \\ &\xrightarrow{\delta^0} H^1(X, \mathcal{F}') \to H^1(X, \mathcal{F}) \to H^1(X, \mathcal{F}'') \\ &\xrightarrow{\delta^1} H^2(X, \mathcal{F}') \to \cdots . \end{aligned}$$

Proof. Choose compatible injective resolutions of the three sheaves by the injective horseshoe construction. Injective sheaves are flabby, so global sections preserve the resulting short exact sequence of resolution complexes. The standard connecting-homomorphism construction for a short exact sequence of complexes then gives the displayed long exact sequence. Independence of the chosen resolutions follows from the usual comparison theorem. □

Proposition 48.9 (What exactness says in every degree). *At each degree q, exactness gives*

$$\ker\Big(H^q(\mathcal{F}'') \xrightarrow{\delta^q} H^{q+1}(\mathcal{F}')\Big) = \operatorname{im}\big(H^q(\mathcal{F}) \to H^q(\mathcal{F}'')\big)\,.$$

Thus a cohomology class in the quotient sheaf comes from the middle sheaf exactly when its connecting obstruction vanishes.

$$\begin{tikzcd}
H^q(\mathcal{F}'') \arrow[r, "\delta_{\mathcal{F}}"] \arrow[d] & H^{q+1}(\mathcal{F}') \arrow[d] \\
H^q(\mathcal{G}'') \arrow[r, "\delta_{\mathcal{G}}"'] & H^{q+1}(\mathcal{G}')
\end{tikzcd}$$

Figure 48.5: Naturality makes connecting homomorphisms commute with morphisms of short exact sequences.

Theorem 48.10 (Naturality of the long exact sequence). *A commutative diagram of short exact sequences of sheaves induces a commutative ladder of long exact cohomology sequences. In particular, for every q the square involving the connecting maps*

$$H^q(\mathcal{F}'') \xrightarrow{\delta^q_{\mathcal{F}}} H^{q+1}(\mathcal{F}')$$

and

$$H^q(\mathcal{G}'') \xrightarrow{\delta^q_{\mathcal{G}}} H^{q+1}(\mathcal{G}')$$

commutes.

Proof. Choose compatible maps between injective resolutions. The connecting maps of a short exact sequence of complexes are natural, hence so are the induced maps in sheaf cohomology. □

Corollary 48.11 (Split exact sequences split in cohomology). *If the short exact sequence splits as sheaves, so that $\mathcal{F} \cong \mathcal{F}' \oplus \mathcal{F}''$, then every connecting homomorphism is zero and*

$$H^q(X, \mathcal{F}) \cong H^q(X, \mathcal{F}') \oplus H^q(X, \mathcal{F}'')$$

for all $q \geq 0$.

Proposition 48.12 (Vanishing of H^1 forces global lifting). *If $H^1(X, \mathcal{F}') = 0$, then every global section of $\mathcal{F}''$ lifts to a global section of $\mathcal{F}$.*

Proof. The connecting map

$$H^0(X, \mathcal{F}'') \longrightarrow H^1(X, \mathcal{F}')$$

has zero target. Exactness therefore makes

$$H^0(X, \mathcal{F}) \longrightarrow H^0(X, \mathcal{F}'')$$

surjective. □

48.3 The Čech realization of the boundary

Proposition 48.13 (When a cover gives a short exact sequence of Čech complexes)**.** *Let $\mathfrak{U} = \{U_i\}$ be a cover such that for every nonempty finite intersection U_I the map*

$$\Gamma(U_I, \mathcal{F}) \longrightarrow \Gamma(U_I, \mathcal{F}'')$$

is surjective. Then

$$0 \to C^\bullet(\mathfrak{U}, \mathcal{F}') \to C^\bullet(\mathfrak{U}, \mathcal{F}) \to C^\bullet(\mathfrak{U}, \mathcal{F}'') \to 0$$

is a short exact sequence of cochain complexes.

Proof. Left exactness holds on every intersection. The stated lifting hypothesis supplies termwise surjectivity, and restriction maps commute with the Čech differential. □

Theorem 48.14 (The Čech boundary agrees with the cohomological boundary)**.** *Under the hypotheses of the previous proposition, the connecting map of the short exact sequence of Čech complexes sends a q-cocycle in $\mathcal{F}''$ to the class obtained by lifting it to an $\mathcal{F}$-cochain, applying the Čech differential, and viewing the result as an $\mathcal{F}'$-cocycle. If the cover is Leray for the three sheaves, this connecting map identifies with the connecting homomorphism in sheaf cohomology.*

Proof. This is precisely the standard boundary construction for a short exact sequence of cochain complexes. Leray comparison identifies the three Čech cohomology groups with the corresponding sheaf cohomology groups, and naturality identifies the boundary maps. □

Warning 48.15 (Do not assume termwise surjectivity on an arbitrary cover). An epimorphism of sheaves $\mathcal{F} \to \mathcal{F}''$ need not make

$$\Gamma(U, \mathcal{F}) \to \Gamma(U, \mathcal{F}'')$$

surjective for every open U. Therefore an arbitrary short exact sequence of sheaves does not automatically give a short exact sequence of Čech cochain complexes for a fixed cover. One must either refine to obtain local lifts or use resolutions.

Example 48.16 (Acyclic middle term)**.** When $\mathcal{F}$ is flabby, $H^i(\mathcal{F}) = 0$ for $i > 0$. The long exact sequence then converts cohomology of $\mathcal{F}''$ into shifted cohomology of $\mathcal{F}'$ in positive degrees.

48.4 Divisor sequences and geometric restriction

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow\!\!\!\!\!\rightarrow \mathcal{O}_D \longrightarrow 0$$

Figure 48.6: An effective Cartier divisor produces a basic geometric short exact sequence.

Definition 48.17 (The sequence attached to an effective Cartier divisor)**.** Let $D \subseteq X$ be an effective Cartier divisor with invertible ideal sheaf $\mathcal{I}_D \cong \mathcal{O}_X(-D)$. The quotient sheaf $\mathcal{O}_D = \mathcal{O}_X/\mathcal{I}_D$ records functions on the divisor.

Proposition 48.18 (The basic divisor exact sequence). *For an effective Cartier divisor D there is a canonical short exact sequence*

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0.$$

More generally, tensoring by a line bundle $\mathcal{L}$ gives

$$0 \longrightarrow \mathcal{L}(-D) \longrightarrow \mathcal{L} \longrightarrow \mathcal{L}|_D \longrightarrow 0.$$

Proof. The first sequence is the defining ideal sequence. Since $\mathcal{L}$ is locally free of rank one, tensoring by $\mathcal{L}$ preserves exactness. $\square$

Corollary 48.19 (The divisor sequence produces restriction obstructions). *The twisted divisor sequence yields*

$$0 \to H^0(X, \mathcal{L}(-D)) \to H^0(X, \mathcal{L}) \to H^0(D, \mathcal{L}|_D) \to H^1(X, \mathcal{L}(-D)) \to \cdots.$$

Hence a section prescribed on D extends to a global section of $\mathcal{L}$ whenever $H^1(X, \mathcal{L}(-D)) = 0$.

48.5 A recursive calculation on $\mathbb{P}^1$

$$\begin{tikzcd}
\mathcal{O}_{\mathbb{P}^1}(n-1) \arrow[r] & \mathcal{O}_{\mathbb{P}^1}(n) \arrow[r, two heads] \arrow[d] & k_p \\
& H^0 \text{ and } H^1 \text{ are linked recursively} &
\end{tikzcd}$$

Figure 48.7: A rational point on $\mathbb{P}^1$ gives the recurrence sequence for $\mathcal{O}(n)$.

Example 48.20 (A point on $\mathbb{P}^1$). Let $p \in \mathbb{P}^1_k$ be a k-rational point. Since $\mathcal{O}_{\mathbb{P}^1}(-p) \cong \mathcal{O}_{\mathbb{P}^1}(-1)$, tensoring the point sequence by $\mathcal{O}(n)$ gives

$$0 \to \mathcal{O}(n-1) \to \mathcal{O}(n) \to k_p \to 0,$$

where k_p is the skyscraper sheaf with fiber k at p.

Proposition 48.21 (The $\mathbb{P}^1$ recurrence in degrees zero and one). *For the point sequence on $\mathbb{P}^1$ the long exact sequence begins*

$$0 \to H^0(\mathcal{O}(n-1)) \to H^0(\mathcal{O}(n)) \to k \to H^1(\mathcal{O}(n-1)) \to H^1(\mathcal{O}(n)) \to 0.$$

The terminal zero follows from $H^1(\mathbb{P}^1, k_p) = 0$.

Proof. The displayed segment is the long exact sequence. A skyscraper sheaf is flabby, hence acyclic by VI/46. $\square$

Corollary 48.22 (Cohomology dimensions on $\mathbb{P}^1$). *Using $H^1(\mathcal{O}(-1)) = 0$ from the Čech calculation in VI/47, the recurrence gives*

$$H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0 \quad (n \geq -1),$$

and for $n \leq -2$,

$$\dim_k H^1(\mathbb{P}^1, \mathcal{O}(n)) = -n - 1.$$

Proof. For $n \geq 0$, evaluation $H^0(\mathcal{O}(n)) \to k$ is surjective, so $H^1(\mathcal{O}(n-1)) \cong H^1(\mathcal{O}(n))$. For $n < 0$, both adjacent H^0 groups vanish and the exact sequence becomes

$$0 \to k \to H^1(\mathcal{O}(n-1)) \to H^1(\mathcal{O}(n)) \to 0,$$

which increases the dimension by one as n decreases. □

Example 48.23 (Evaluation as an exact-sequence map)**.** For $n \geq 0$, the map

$$H^0(\mathbb{P}^1, \mathcal{O}(n)) \longrightarrow k_p$$

is evaluation at p after choosing a trivialization of the fiber. Its surjectivity is the elementary geometric fact that a degree-n homogeneous polynomial can be chosen with any prescribed value at a single point.

48.6 Dimension shifting and obstruction calculus

$$\begin{tikzcd} \mathcal{F} \arrow[r] & \text{flabby } \mathcal{I} \arrow[r, two heads] & \mathcal{Q} \arrow[d] \\ & & H^q(\mathcal{Q}) \cong H^{q+1}(\mathcal{F}) \end{tikzcd}$$

Figure 48.8: Dimension shifting moves cohomology one degree by embedding into a flabby sheaf.

Proposition 48.24 (Dimension shifting with a flabby sheaf)**.** *Let*

$$0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$$

be exact with $\mathcal{I}$ flabby. Then for every $q \geq 1$ the connecting map gives a canonical isomorphism

$$H^q(X, \mathcal{Q}) \xrightarrow{\sim} H^{q+1}(X, \mathcal{F}).$$

Proof. Since $H^q(X, \mathcal{I}) = H^{q+1}(X, \mathcal{I}) = 0$ for $q \geq 1$, the relevant part of the long exact sequence has zero terms on both sides of the connecting map. □

Corollary 48.25 (The first obstruction from a flabby embedding)**.** *In the same situation there is an exact sequence*

$$0 \to H^0(X, \mathcal{F}) \to H^0(X, \mathcal{I}) \to H^0(X, \mathcal{Q}) \to H^1(X, \mathcal{F}) \to 0.$$

Thus $H^1(X, \mathcal{F})$ is the cokernel of the global lifting map into the first flabby quotient.

Definition 48.26 (Cohomology as an obstruction space)**.** In this chapter, an *obstruction class* means a cohomology class produced by a connecting homomorphism. Its vanishing is equivalent to solving the corresponding lifting or extension problem one step earlier in the long exact sequence.

Proposition 48.27 (Euler-characteristic additivity)**.** *Assume the cohomology groups of $\mathcal{F}'$, $\mathcal{F}$, and $\mathcal{F}''$ are finite-dimensional over a field and vanish for $q \gg 0$. Define*

$$\chi(X, \mathcal{G}) = \sum_q (-1)^q \dim H^q(X, \mathcal{G}).$$

Then every short exact sequence gives

$$\chi(X, \mathcal{F}) = \chi(X, \mathcal{F}') + \chi(X, \mathcal{F}'').$$

Proof. The long exact cohomology sequence is a finite exact sequence of finite-dimensional vector spaces after truncation. The alternating sum of dimensions of an exact sequence is zero, which rearranges to the displayed identity. □

Example 48.28 (Principal parts as quotient data)**.** If D is an effective Cartier divisor, the sequence

$$0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to \mathcal{O}_X(D)|_D \to 0$$

interprets the quotient as allowed principal-part data along D. The connecting map sends prescribed polar data to the obstruction to realizing it by a global meromorphic section with poles bounded by D.

Remark 48.29 (Bridge to VI/49). VI/49 turns these exact-sequence mechanisms into vanishing tools. Once one knows that certain higher groups vanish, long exact sequences collapse into short exact sequences, restriction maps become surjective, and dimension shifting reduces unfamiliar cohomology groups to familiar ones.

48.7 Exercises

Exercise 48.30. Prove directly that $\Gamma(U, -)$ is left exact for every open U.

Exercise 48.31. Give an example of an epimorphism of sheaves that is not surjective on global sections.

Exercise 48.32. For local lifts t_i of a quotient section, verify explicitly that $a_{ij} = t_j - t_i$ is a 1-cocycle.

Exercise 48.33. Show that changing local lifts changes the local-lift cocycle by a coboundary.

Exercise 48.34. Prove that $\delta(s) = 0$ if a global lift of s exists.

Exercise 48.35. Prove the converse lifting criterion using a vanishing cocycle class.

Exercise 48.36. Write the first seven nonzero terms of the long exact sequence attached to a short exact sequence.

Exercise 48.37. If $H^1(X, \mathcal{F}') = 0$, prove that every quotient section lifts globally.

Exercise 48.38. Show that every connecting map vanishes for a split exact sequence.

Exercise 48.39. Explain why a fixed Čech cover may fail to give a short exact sequence of cochain groups.

Exercise 48.40. Under the cover-lifting hypothesis, construct the Čech connecting map in degree q.

Exercise 48.41. Prove naturality of the degree-zero connecting map using local lifts.

Exercise 48.42. Derive the short exact divisor sequence for an effective Cartier divisor.

Exercise 48.43. Tensor the divisor sequence by a line bundle and justify exactness.

Exercise 48.44. What obstruction controls extension of a section from D to X?

Exercise 48.45. For a k-rational point $p \in \mathbb{P}^1$, derive $0 \to \mathcal{O}(n-1) \to \mathcal{O}(n) \to k_p \to 0$.

Exercise 48.46. Why is $H^q(\mathbb{P}^1, k_p) = 0$ for $q > 0$?

Exercise 48.47. Use the point sequence to prove $H^1(\mathbb{P}^1, \mathcal{O}(n)) = 0$ for $n \geq -1$, assuming the base case $n = -1$.

Exercise 48.48. Use the same sequence to compute $\dim H^1(\mathcal{O}(-m))$ for $m \geq 2$.

Exercise 48.49. Prove the dimension-shifting isomorphism when the middle sheaf is flabby.

Exercise 48.50. Interpret $H^1(\mathcal{F})$ using a flabby embedding $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$.

Exercise 48.51. Prove additivity of Euler characteristic from a long exact sequence.

Exercise 48.52. If two of three sheaves in a short exact sequence are acyclic, what can you infer about the third?

Exercise 48.53. Explain why VI/49 can turn a long exact sequence into a practical calculation.

48.8 Solved problem dossiers

Problem 48.54. A quotient section is locally liftable but not globally liftable. Show how to construct its obstruction class and prove independence of all choices.

Problem 48.55. Given a morphism between two short exact sequences, prove that the degree-zero connecting homomorphisms commute without using resolutions.

Problem 48.56. Let D be an effective Cartier divisor and $\mathcal{L}$ a line bundle. Formulate and prove an extension criterion for a section of $\mathcal{L}|_D$.

Problem 48.57. Recover the formula $h^1(\mathbb{P}^1, \mathcal{O}(n)) = \max\{-n-1, 0\}$ using only the point sequence and the base case $H^1(\mathcal{O}(-1)) = 0$.

Problem 48.58. Show that a split short exact sequence has zero obstruction maps in every degree and identify its long exact sequence.

Problem 48.59. Let $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{Q} \to 0$ with $\mathcal{I}$ flabby. Express all higher cohomology of $\mathcal{F}$ in terms of $\mathcal{Q}$.

Problem 48.60. On a Leray cover for a short exact sequence, show that the connecting map computed in Čech cohomology agrees with the derived sheaf-cohomology boundary.

Problem 48.61. Assume $H^1(X, \mathcal{L}(-D)) = 0$ for an effective Cartier divisor D. Determine the kernel and cokernel of $H^0(X, \mathcal{L}) \to H^0(D, \mathcal{L}|_D)$.

Problem 48.62. Let $0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$ be exact and suppose $\mathcal{F}'$ and $\mathcal{F}$ are acyclic. Prove that $\mathcal{F}''$ is acyclic and that global sections form a short exact sequence.

Problem 48.63. Use Euler-characteristic additivity and the point sequence on $\mathbb{P}^1$ to show $\chi(\mathcal{O}(n)) = \chi(\mathcal{O}(n-1)) + 1$.

Problem 48.64. Explain why a sheaf epimorphism can fail to be onto on sections and show how the first connecting homomorphism measures exactly that failure.

Problem 48.65. For the principal-parts sequence $0 \to \mathcal{O}_X \to \mathcal{O}_X(D) \to \mathcal{O}_X(D)|_D \to 0$, interpret the connecting homomorphism geometrically.

48.9 Challenges

Problem 48.66 (Challenge)**.** Prove the long exact sequence directly from a short exact sequence of cochain complexes and then explain precisely where injective or flabby resolutions enter for sheaves.

Problem 48.67 (Challenge)**.** Develop the 3×3 lemma consequences for cohomology by arranging three compatible short exact sequences and comparing their connecting maps.

Problem 48.68 (Challenge)**.** On a smooth projective curve, use the point exact sequence

$$0 \to \mathcal{L}(-p) \to \mathcal{L} \to \mathcal{L}|_p \to 0$$

to analyze how h^0 and h^1 change when one point is added to a divisor.

Problem 48.69 (Challenge)**.** Construct an explicit example where a fixed open cover does not give termwise surjectivity for a sheaf epimorphism, then refine the cover until local lifts exist.

Problem 48.70 (Challenge)**.** Show how repeated dimension shifting along a flabby resolution identifies $H^n(X, \mathcal{F})$ with a first cohomology group of a successive cokernel, and explain why this viewpoint is useful for vanishing proofs.

48.10 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 48.71 (First terms)**.** Write the first four terms after $H^0(\mathcal{F}'')$ in the long exact sequence.

Hint. Use long exact cohomology sequences and connecting homomorphisms. □

Solution. $H^1(\mathcal{F}') \to H^1(\mathcal{F}) \to H^1(\mathcal{F}'') \to H^2(\mathcal{F}')$. □

Exercise 48.72 (Connecting target)**.** Where does the connecting map from $H^0(\mathcal{F}'')$ land?

Hint. Use long exact cohomology sequences and connecting homomorphisms. □

Solution. In $H^1(\mathcal{F}')$. □

Exercise 48.73 (Flabby vanishing)**.** What happens to positive cohomology of a flabby middle sheaf?

Hint. Use long exact cohomology sequences and connecting homomorphisms. □

Solution. It vanishes. □

Exercise 48.74 (Exact H0)**.** Is global sections left exact?

Hint. Use long exact cohomology sequences and connecting homomorphisms. □

Solution. Yes. □

Exercise 48.75 (Surjectivity criterion)**.** When is $H^0(\mathcal{F}) \to H^0(\mathcal{F}'')$ surjective from the long exact sequence?

Hint. Use long exact cohomology sequences and connecting homomorphisms. ☐

Solution. When the connecting map to $H^1(\mathcal{F}')$ vanishes on all sections, for example if $H^1(\mathcal{F}') = 0$. ☐

Proofs

Exercise 48.76 (Connecting map)**.** Prove: Construct the first connecting homomorphism using local lifts or a resolution.

Hint. Work from long exact cohomology sequences and connecting homomorphisms and state the hypotheses explicitly. ☐

Solution. Lift a cocycle/section to the middle complex, apply the differential, and identify the result in the subobject. ☐

Exercise 48.77 (Exactness at H0)**.** Prove: Prove the long sequence begins with the usual left-exact global-section sequence.

Hint. Work from long exact cohomology sequences and connecting homomorphisms and state the hypotheses explicitly. ☐

Solution. Global sections preserves kernels, and the derived construction extends this exactness. ☐

Exercise 48.78 (Naturality)**.** Prove: Prove connecting homomorphisms are natural with respect to maps of short exact sequences.

Hint. Work from long exact cohomology sequences and connecting homomorphisms and state the hypotheses explicitly. ☐

Solution. Choose compatible lifts or use functorial resolutions and chase the resulting diagram. ☐

Exercise 48.79 (Acyclic shift)**.** Prove: Derive the dimension-shifting isomorphism when the middle sheaf is acyclic.

Hint. Work from long exact cohomology sequences and connecting homomorphisms and state the hypotheses explicitly. ☐

Solution. Read exactness between zero neighboring cohomology groups in the long exact sequence. ☐

Counterexamples and hypothesis tests

Exercise 48.80 (H0 exact)**.** Test the claim: global sections sends every short exact sequence to a short exact sequence.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; right exactness can fail. ☐

Exercise 48.81 (Connecting arbitrary)**.** Test the claim: the connecting homomorphism is an arbitrary extra map.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; it is canonically determined and natural. ☐

Exercise 48.82 (Higher independent)**.** Test the claim: higher cohomology groups are unrelated across a short exact sequence.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False; the long exact sequence links them. ☐

Applications and investigations

Exercise 48.83 (Obstructions)**.** What does the connecting map measure geometrically?

Hint. Translate the question into long exact cohomology sequences and connecting homomorphisms. ☐

Solution. It measures the obstruction to globally lifting compatible local data. ☐

Exercise 48.84 (Computations)**.** How can a known acyclic term help compute unknown cohomology?

Hint. Translate the question into long exact cohomology sequences and connecting homomorphisms. ☐

Solution. Long exactness shifts the problem to neighboring sheaves. ☐

Challenge problems

Exercise 48.85 (Synthesis challenge)**.** Connect two central viewpoints of Exact Sequences and Cohomology in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Exact Sequences and Cohomology and then translates it through long exact cohomology sequences and connecting homomorphisms, checking both descriptions agree. ☐

Exercise 48.86 (Hypothesis challenge)**.** Choose one theorem-level statement from Exact Sequences and Cohomology, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐

Chapter 49

Basic Vanishing Results

VI/46 introduced flabby sheaves, VI/47 made cohomology computable by Čech complexes, and VI/48 showed how exact sequences transport cohomological information. This final chapter of Volume VI turns those tools into vanishing theorems. The central geometric fact is that quasi-coherent sheaves on affine schemes have no higher cohomology.

$$\boxed{X = \operatorname{Spec} A,\ \mathcal{F} \text{ quasi-coherent} \quad \Longrightarrow \quad H^q(X, \mathcal{F}) = 0 \ (q > 0)}$$

Once affine vanishing is known, affine covers become computational engines. On separated schemes their finite intersections remain affine, so the Čech groups of an affine cover compute actual sheaf cohomology. The line-bundle calculation on $\mathbb{P}^1$ from VI/47 therefore becomes a genuine sheaf-cohomology theorem rather than merely a fixed-cover calculation.

Standing conventions

Cohomology means sheaf cohomology on the Zariski topology. Unless stated otherwise, schemes are schemes over a fixed base and quasi-coherent sheaves are $\mathcal{O}_X$-modules. We use the results on flabby resolutions, Čech comparison, and long exact sequences established in VI/46–VI/48.

Learning goals

By the end of the chapter, the reader should be able to:

- recognize flabby and injective sheaves as acyclic objects;
- prove exactness of the localization Čech complex for a principal affine cover;
- state and use affine vanishing for quasi-coherent sheaves;
- identify affine covers of separated schemes as Leray covers;
- derive cohomological bounds from the number of affine opens in a cover;
- compute the complete cohomology of $\mathcal{O}_{\mathbb{P}^1}(n)$;
- propagate vanishing through long exact sequences;
- use divisor sequences and dimension shifting;
- distinguish basic affine vanishing from deeper projective vanishing theorems.

Example 49.1 (Affine quasi-coherent vanishing)**.** For an affine scheme $X = \operatorname{Spec} A$ and a quasi-coherent sheaf $\widetilde{M}$, one has $H^i(X, \widetilde{M}) = 0$ for every $i > 0$. This is the basic cohomological reason affine geometry is computationally simple.

Conceptual roadmap

$$\text{extension / localization} \longrightarrow \text{acyclicity} \longrightarrow H^{>0} = 0$$

Figure 49.1: Vanishing turns local extension or localization properties into global cohomological information.

49.1 Acyclic sheaves revisited

Definition 49.2 (Acyclic sheaf)**.** A sheaf $\mathcal{F}$ on X is called Γ-*acyclic* if

$$H^q(X, \mathcal{F}) = 0 \qquad (q > 0).$$

Acyclicity depends on the space on which the sheaf is considered; when necessary we say "acyclic on X."

$$\text{injective} \longrightarrow \text{flabby} \longrightarrow \Gamma\text{-acyclic}$$

Figure 49.2: The basic implication chain used throughout Part IX.

Theorem 49.3 (Flabby vanishing)**.** *Every flabby sheaf is* Γ*-acyclic. More generally, the restriction of a flabby sheaf to any open subset is flabby and hence acyclic.*

Proof. VI/46 constructed flabby resolutions and proved that global sections applied to a flabby sheaf already have no positive derived cohomology. Restriction preserves the surjectivity of restriction maps, so the same argument applies on every open subset. □

Corollary 49.4 (Injective vanishing)**.** *Every injective sheaf of abelian groups is acyclic.*

Proof. Injective sheaves are flabby by VI/46, so the previous theorem applies. □

Example 49.5 (Skyscraper sheaves)**.** A skyscraper sheaf $i_{x,*}A$ is flabby, hence

$$H^q(X, i_{x,*}A) = 0 \qquad (q > 0).$$

Finite direct sums of skyscraper sheaves are therefore acyclic as well.

Definition 49.6 (Affine quasi-coherent situation)**.** Let $X = \operatorname{Spec} A$ and let M be an A-module. We write $\widetilde{M}$ for the associated quasi-coherent sheaf. On a principal open $D(f)$,

$$\Gamma(D(f), \widetilde{M}) \cong M_f.$$

Example 49.7 (Flabby vanishing)**.** A flabby sheaf has vanishing higher cohomology on any space. Thus a flabby resolution computes the cohomology of an arbitrary sheaf from global sections of an acyclic complex.

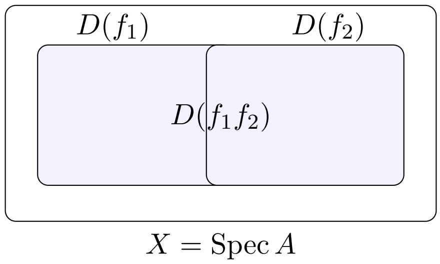


Figure 49.3: A principal affine cover and its affine intersections.

49.2 Affine vanishing

Proposition 49.8 (Exact localization Čech complex)**.** *Suppose $f_1, \ldots, f_r \in A$ generate the unit ideal. For every A-module M, the augmented localization complex*

$$0 \to M \to \prod_i M_{f_i} \to \prod_{i<j} M_{f_i f_j} \to \cdots \to M_{f_1 \cdots f_r} \to 0$$

with the alternating Čech differential is exact.

Proof. Exactness may be checked after localizing at each f_i, because the principal opens $D(f_i)$ cover $\operatorname{Spec} A$. After localizing at f_i, one member of the induced cover is the whole space, and the resulting augmented Čech complex admits the standard contracting homotopy obtained by inserting the distinguished index i. Hence every localized homology module vanishes. Since the $D(f_i)$ cover, the original homology modules vanish as well. ☐

$$M \longrightarrow \prod M_{f_i} \longrightarrow \prod M_{f_i f_j} \longrightarrow \cdots$$

Figure 49.4: The localization Čech complex underlying affine vanishing.

Theorem 49.9 (Affine vanishing theorem)**.** *Let $X = \operatorname{Spec} A$ be affine and let $\mathcal{F}$ be quasi-coherent. Then*

$$\boxed{H^q(X, \mathcal{F}) = 0 \qquad (q > 0).}$$

Proof. Write $\mathcal{F} \cong \widetilde{M}$. A finite principal cover of X gives the exact localization Čech complex of the preceding proposition. The standard Čech-to-derived comparison on the principal affine basis identifies this exact complex with the cohomology of $\widetilde{M}$. Thus all positive cohomology groups vanish. ☐

Corollary 49.10 (Principal opens are acyclic)**.** *If $X = \operatorname{Spec} A$, $f \in A$, and $\mathcal{F}$ is quasi-coherent, then*

$$H^q(D(f), \mathcal{F}|_{D(f)}) = 0 \qquad (q > 0).$$

Proof. The principal open $D(f)$ is affine and the restriction of a quasi-coherent sheaf is quasi-coherent. ☐

Proposition 49.11 (Čech exactness on a principal cover)**.** *Let $X = \operatorname{Spec} A$ and let $\mathfrak{U} = \{D(f_i)\}$ be a finite principal cover. For every quasi-coherent $\mathcal{F}$, the augmented Čech complex of $\mathfrak{U}$ is exact in positive degree.*

Proof. If $\mathcal{F} = \widetilde{M}$, its Čech terms are precisely the localizations $M_{f_{i_0}\cdots f_{i_p}}$, so this is the localization proposition. □

Definition 49.12 (Acyclic cover)**.** An open cover $\mathfrak{U} = \{U_i\}$ is $\mathcal{F}$-*acyclic* (or Leray for $\mathcal{F}$) if every nonempty finite intersection satisfies

$$H^q(U_{i_0\cdots i_p}, \mathcal{F}) = 0 \qquad (q > 0).$$

49.3 Affine covers and Leray comparison

Theorem 49.13 (Affine Leray covers)**.** *Let X be a separated scheme and let $\mathfrak{U} = \{U_i\}$ be an affine open cover. For every quasi-coherent sheaf $\mathcal{F}$, the cover is Leray, and hence*

$$\check{H}^q(\mathfrak{U}, \mathcal{F}) \cong H^q(X, \mathcal{F}).$$

Proof. On a separated scheme, every finite intersection of affine opens is affine. The restriction of $\mathcal{F}$ to such an intersection is quasi-coherent, so affine vanishing applies. VI/47's Leray comparison theorem now identifies Čech and sheaf cohomology. □

Proposition 49.14 (Why separatedness matters)**.** *If X is separated and $U, V \subseteq X$ are affine opens, then $U \cap V$ is affine.*

Proof. The intersection is the inverse image of $U \times V$ under the diagonal $\Delta_X : X \to X \times X$. For a separated scheme the diagonal is a closed immersion, hence affine; therefore its inverse image over the affine scheme $U \times V$ is affine. □

$$U_i \text{ affine} \xrightarrow{\text{separated}} U_{ij} \text{ affine} \xrightarrow{\text{iterate}} U_{ijk} \text{ affine}$$

Figure 49.5: On a separated scheme, finite intersections of affine opens remain affine.

Corollary 49.15 (Cover-length vanishing bound)**.** *Let X be separated and covered by r affine opens. For every quasi-coherent sheaf $\mathcal{F}$ one has*

$$H^q(X, \mathcal{F}) = 0 \qquad (q \geq r).$$

Proof. The affine cover is Leray. Its Čech complex has no terms in degrees $q \geq r$, because there are no strictly increasing $(q+1)$-tuples of indices. □

Corollary 49.16 (Two-affine vanishing)**.** *If a separated scheme is the union of two affine opens, then every quasi-coherent sheaf $\mathcal{F}$ satisfies*

$$H^q(X, \mathcal{F}) = 0 \qquad (q \geq 2).$$

Proposition 49.17 (Global sections of $\mathcal{O}_{\mathbb{P}^1}(n)$)**.** *For the projective line over a field k,*

$$\dim_k H^0(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} n+1, & n \geq 0, \\ 0, & n < 0. \end{cases}$$

For $n \geq 0$, the homogeneous monomials $x_0^{n-j} x_1^j$, $0 \leq j \leq n$, form a basis.

Proof. Use the standard affine cover $U_0 \cup U_1$ and write a section as compatible polynomials in t and $u = t^{-1}$. Compatibility forces a Laurent polynomial to have exponents between 0 and n. Such exponents exist exactly when $n \geq 0$. □

49.4 The complete cohomology of line bundles on $\mathbb{P}^1$

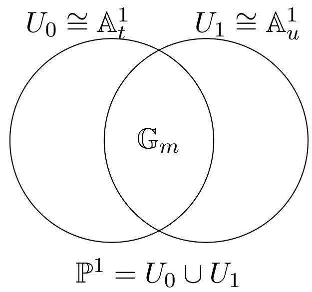


Figure 49.6: The standard two-affine cover of $\mathbb{P}^1$.

Theorem 49.18 (First cohomology of $\mathcal{O}_{\mathbb{P}^1}(n)$)**.** *For every integer n,*

$$H^1(\mathbb{P}^1, \mathcal{O}(n)) \cong \frac{k[t, t^{-1}]}{k[t] + t^n k[t^{-1}]}.$$

Consequently,

$$\dim_k H^1(\mathbb{P}^1, \mathcal{O}(n)) = \begin{cases} 0, & n \geq -1, \\ -n-1, & n \leq -2. \end{cases}$$

Proof. The standard cover consists of two affine opens, and their intersection is affine, so the cover is Leray by affine vanishing. The Čech computation from VI/47 therefore computes actual sheaf cohomology. The quotient leaves precisely the Laurent monomials $t^{n+1}, \ldots, t^{-1}$ when $n \leq -2$. □

Corollary 49.19 (Higher cohomology on $\mathbb{P}^1$)**.** *For every integer n,*

$$H^q(\mathbb{P}^1, \mathcal{O}(n)) = 0 \qquad (q \geq 2).$$

Proof. The projective line is separated and covered by two affine opens, so the two-affine vanishing result applies. □

Proposition 49.20 (Euler characteristic on $\mathbb{P}^1$)**.** *For every integer n,*

$$\chi(\mathbb{P}^1, \mathcal{O}(n)) = \dim H^0 - \dim H^1 = n + 1.$$

Proof. Combine the two preceding formulas in the cases $n \geq 0$, $n = -1$, and $n \leq -2$. □

Proposition 49.21 (Acyclic kernel and middle term)**.** *Suppose*

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0$$

is exact. If $\mathcal{F}'$ and $\mathcal{F}$ are acyclic, then $\mathcal{F}''$ is acyclic and

$$\Gamma(X, \mathcal{F}) \twoheadrightarrow \Gamma(X, \mathcal{F}'').$$

Proof. Read the statement directly from the long exact cohomology sequence of VI/48. □

Example 49.22 (Projective line basic twist)**.** On $\mathbb{P}^1$, $H^0(\mathcal{O}(d))$ has dimension $d + 1$ for $d \geq 0$, while the first cohomology vanishes for $d \geq -1$ and becomes nonzero for sufficiently negative twists. The standard two-chart Cech complex makes these ranges explicit.

$$H^q(\mathcal{F}') \longrightarrow H^q(\mathcal{F}) \longrightarrow H^q(\mathcal{F}'') \to H^{q+1}(\mathcal{F}')$$

Figure 49.7: Vanishing in a long exact sequence propagates information to neighboring terms.

49.5 Vanishing and exact sequences

Proposition 49.23 (Recovering acyclicity of a kernel)**.** *In the same short exact sequence, assume $\mathcal{F}$ and $\mathcal{F}''$ are acyclic and that*

$$\Gamma(X, \mathcal{F}) \to \Gamma(X, \mathcal{F}'')$$

is surjective. Then $\mathcal{F}'$ is acyclic.

Proof. Surjectivity forces $H^1(X, \mathcal{F}') = 0$. For $q \geq 2$, exactness places $H^q(X, \mathcal{F}')$ between two zero groups coming from $\mathcal{F}''$ and $\mathcal{F}$. □

Proposition 49.24 (Dimension shifting)**.** *If*

$$0 \to \mathcal{F}' \to \mathcal{I} \to \mathcal{F}'' \to 0$$

is exact and $\mathcal{I}$ is acyclic, then for every $q \geq 1$ there are natural isomorphisms

$$H^q(X, \mathcal{F}'') \cong H^{q+1}(X, \mathcal{F}').$$

Proof. The relevant portion of the long exact sequence has zero middle terms $H^q(X, \mathcal{I})$ and $H^{q+1}(X, \mathcal{I})$. □

Proposition 49.25 (Vanishing in the divisor sequence)**.** *Let D be an effective Cartier divisor on X and $\mathcal{L}$ a line bundle. Assume*

$$H^q(D, \mathcal{L}|_D) = 0 \qquad (q > 0),$$

for example when D is a finite scheme over a field. Then the exact sequence

$$0 \to \mathcal{L}(-D) \to \mathcal{L} \to i_*(\mathcal{L}|_D) \to 0$$

gives isomorphisms

$$H^q(X, \mathcal{L}(-D)) \xrightarrow{\sim} H^q(X, \mathcal{L}) \qquad (q \geq 2),$$

and the low-degree exact sequence controls the difference between H^0 and H^1.

Proof. Closed-immersion pushforward preserves cohomology, so the higher cohomology of the quotient term vanishes. Apply the long exact sequence. □

Corollary 49.26 (Point recurrence on $\mathbb{P}^1$)**.** *For a k-rational point $p \in \mathbb{P}^1$,*

$$0 \to \mathcal{O}(n-1) \to \mathcal{O}(n) \to k_p \to 0$$

relates the cohomology of adjacent twists. Since k_p is acyclic, this sequence recursively recovers the formulas for H^0 and H^1 of $\mathcal{O}(n)$.

Remark 49.27 (What is not proved here: Serre vanishing). A deeper projective theorem says that for a coherent sheaf $\mathcal{F}$ on a projective scheme and a sufficiently positive twist, $H^q(X, \mathcal{F}(n)) = 0$ for $q > 0$ and $n \gg 0$. That theorem requires substantially more projective machinery than the basic affine, Leray, and exact-sequence vanishing principles developed here.

$$
\begin{tikzcd}
\text{sheaves} \arrow[r] & \text{cohomology} \arrow[d] & \text{divisors / line bundles} \arrow[l] \\
& \text{vanishing} \arrow[ur] &
\end{tikzcd}
$$

Figure 49.8: Volume VI closes with cohomology reconnecting the sheaf and divisor viewpoints.

49.6 Synthesis and limits of the basic theory

Theorem 49.28 (Part IX synthesis)**.** *Let X be a separated scheme with a finite affine open cover $\mathfrak{U}$ of size r, and let $\mathcal{F}$ be quasi-coherent. Then*

$$H^q(X, \mathcal{F}) \cong \check{H}^q(\mathfrak{U}, \mathcal{F})$$

for all q, and

$$H^q(X, \mathcal{F}) = 0 \qquad (q \geq r).$$

In particular, affine schemes have no positive quasi-coherent cohomology, while two-affine schemes have no cohomology in degrees at least two.

Proof. Finite intersections of members of $\mathfrak{U}$ are affine by separatedness. Affine vanishing makes the cover Leray, so VI/47 identifies Čech and sheaf cohomology. The Čech complex has length at most $r - 1$. □

Remark 49.29 (Closing Volume VI)*.* The route from affine algebraic sets to schemes, divisors, line bundles, blow-ups, and sheaf cohomology is now complete. The final lesson is structural: local algebra supplies the sections, sheaves organize their compatibility, exact sequences measure obstructions, and vanishing theorems identify the situations in which those obstructions disappear.

49.7 Exercises

Exercise 49.30. Show directly from the definition that a flabby sheaf is acyclic on every open subset.

Exercise 49.31. Why is every skyscraper sheaf acyclic?

Exercise 49.32. Let $f_1, \ldots, f_r$ generate the unit ideal. Explain why a module N with $N_{f_i} = 0$ for all i must be zero.

Exercise 49.33. Write the localization Čech complex for a two-element principal cover $D(f) \cup D(g)$.

Exercise 49.34. Prove the two-element localization complex is exact when $(f, g) = A$.

Exercise 49.35. Let $X = \operatorname{Spec} A$ and $\mathcal{F} = \widetilde{M}$. Compute $H^q(X, \mathcal{F})$ for $q > 0$.

Exercise 49.36. Show that $D(f_1) \cap \cdots \cap D(f_p) = D(f_1 \cdots f_p)$.

Exercise 49.37. Why does a principal affine cover of an affine scheme automatically form a Leray cover for a quasi-coherent sheaf?

Exercise 49.38. Prove that the intersection of two affine opens in a separated scheme is affine.

Exercise 49.39. A separated scheme is covered by three affine opens. What cohomology groups of a quasi-coherent sheaf vanish for degree reasons?

Exercise 49.40. A separated scheme is the union of two affine opens. Show $H^2(X, \mathcal{F}) = 0$ for every quasi-coherent $\mathcal{F}$.

Exercise 49.41. Compute $H^0(\mathbb{P}^1, \mathcal{O}(2))$ and give a basis.

Exercise 49.42. Compute $H^0(\mathbb{P}^1, \mathcal{O}(-1))$.

Exercise 49.43. Compute $H^1(\mathbb{P}^1, \mathcal{O}(-1))$.

Exercise 49.44. Compute a basis of $H^1(\mathbb{P}^1, \mathcal{O}(-4))$.

Exercise 49.45. Verify $\chi(\mathbb{P}^1, \mathcal{O}(-4)) = -3$.

Exercise 49.46. Show that if $\mathcal{F}'$ and $\mathcal{F}$ are acyclic in a short exact sequence, then the quotient is acyclic.

Exercise 49.47. Give the extra hypothesis needed to conclude that a kernel is acyclic when the middle and quotient sheaves are acyclic.

Exercise 49.48. Derive the dimension-shifting isomorphism when the middle sheaf is acyclic.

Exercise 49.49. Why is a coherent sheaf supported on finitely many closed points of a variety over a field acyclic?

Exercise 49.50. Use $0 \to \mathcal{O}(n-1) \to \mathcal{O}(n) \to k_p \to 0$ to compare $H^1(\mathcal{O}(n-1))$ and $H^1(\mathcal{O}(n))$.

Exercise 49.51. Why does affine vanishing not imply that every sheaf on an affine scheme is acyclic?

Exercise 49.52. Explain why separatedness appears in the affine-cover theorem.

Exercise 49.53. Distinguish affine vanishing from Serre vanishing.

49.8 Solved problems

Problem 49.54. Prove the exactness of the augmented localization Čech complex by checking it after localization at each member of the cover.

Problem 49.55. Let X be separated and covered by affine opens U, V. Express $H^1(X, \mathcal{F})$ for a quasi-coherent sheaf in terms of sections.

Problem 49.56. Compute the cohomology of $\mathcal{O}_{\mathbb{P}^1}(-3)$ in all degrees.

Problem 49.57. Compute the cohomology of $\mathcal{O}_{\mathbb{P}^1}(3)$ in all degrees.

Problem 49.58. Use the point exact sequence on $\mathbb{P}^1$ to prove $\chi(\mathcal{O}(n)) = \chi(\mathcal{O}(n-1)) + 1$.

Problem 49.59. Let X be a separated scheme covered by four affine opens. Give the strongest degree-only vanishing conclusion supplied by this chapter.

Problem 49.60. Suppose $0 \to \mathcal{F}' \to \mathcal{I} \to \mathcal{F}'' \to 0$ with $\mathcal{I}$ flabby. Show $H^2(X, \mathcal{F}') \cong H^1(X, \mathcal{F}'')$.

Problem 49.61. Let D be a finite effective Cartier divisor on a projective curve and $\mathcal{L}$ a line bundle. Explain the effect of twisting by $-D$ on H^q for $q \geq 2$.

Problem 49.62. Let X be affine. Prove that global sections are exact on short exact sequences of quasi-coherent sheaves.

Problem 49.63. Let $X = \operatorname{Spec} A$ and M an A-module. Show that the sheaf-cohomological vanishing is compatible with localization to $D(f)$.

Problem 49.64. Explain how the separated affine-cover theorem converts VI/47's fixed-cover calculation on $\mathbb{P}^1$ into actual sheaf cohomology.

Problem 49.65. Summarize the three main vanishing mechanisms of Part IX and state when each applies.

49.9 Challenges

Problem 49.66 (Challenge)**.** Prove carefully that the standard contracting homotopy makes a Čech complex exact whenever one member of the cover is the entire space.

Problem 49.67 (Challenge)**.** Investigate what can fail for an affine cover on a nonseparated scheme and give a concrete nonseparated example where an intersection of affine opens is not affine.

Problem 49.68 (Challenge)**.** Generalize the $\mathbb{P}^1$ line-bundle calculation to predict the shape of $H^q(\mathbb{P}^n, \mathcal{O}(d))$.

Problem 49.69 (Challenge)**.** Use repeated dimension shifting to express $H^q(X, \mathcal{F})$ as an H^1 group of a successive cokernel in a flabby resolution.

Problem 49.70 (Challenge)**.** Formulate a precise boundary between the basic vanishing theorems of this chapter and Serre vanishing on projective schemes.

49.10 Graded supplementary exercises

These exercises extend the protected chapter with computations, proofs, hypothesis tests, applications, and challenges.

Standard computations

Exercise 49.71 (Affine positive cohomology)**.** Compute $H^i(\operatorname{Spec} A, \widetilde{M})$ for $i > 0$.

Hint. Use affine vanishing, acyclic sheaves, and projective-space basic cases. ☐

Solution. It is zero. ☐

Exercise 49.72 (Flabby positive cohomology)**.** Compute $H^i(X, \mathcal{F})$ for flabby $\mathcal{F}$ and $i > 0$.

Hint. Use affine vanishing, acyclic sheaves, and projective-space basic cases. ☐

Solution. It is zero. ☐

Exercise 49.73 (Projective line H0)**.** What is $\dim H^0(\mathbb{P}^1, \mathcal{O}(d))$ for $d \geq 0$?

Hint. Use affine vanishing, acyclic sheaves, and projective-space basic cases. ☐

Solution. $d + 1$. ☐

Exercise 49.74 (O global)**.** Compute $H^0(\mathbb{P}^1, \mathcal{O})$.

Hint. Use affine vanishing, acyclic sheaves, and projective-space basic cases. □

Solution. k. □

Exercise 49.75 (Affine H0)**.** Compute $H^0(\operatorname{Spec} A, \widetilde{M})$.

Hint. Use affine vanishing, acyclic sheaves, and projective-space basic cases. □

Solution. M. □

Proofs

Exercise 49.76 (Affine vanishing)**.** Prove: Prove higher cohomology of quasi-coherent sheaves vanishes on an affine scheme in the standard theorem.

Hint. Work from affine vanishing, acyclic sheaves, and projective-space basic cases and state the hypotheses explicitly. □

Solution. Use a Cech complex on a finite distinguished-open cover or the exactness of global sections for quasi-coherent sheaves on affine schemes. □

Exercise 49.77 (Flabby vanishing)**.** Prove: Prove flabby sheaves are acyclic.

Hint. Work from affine vanishing, acyclic sheaves, and projective-space basic cases and state the hypotheses explicitly. □

Solution. Use extension of sections to solve cocycle equations or the standard derived-functor acyclicity theorem. □

Exercise 49.78 (Projective-line H0)**.** Prove: Compute $H^0(\mathbb{P}^1, \mathcal{O}(d))$ for $d \geq 0$.

Hint. Work from affine vanishing, acyclic sheaves, and projective-space basic cases and state the hypotheses explicitly. □

Solution. Global sections correspond to homogeneous degree-d polynomials in two variables. □

Exercise 49.79 (Two-chart vanishing range)**.** Prove: Use the standard Cech cover to show $H^1(\mathbb{P}^1, \mathcal{O}(d)) = 0$ for $d \geq -1$.

Hint. Work from affine vanishing, acyclic sheaves, and projective-space basic cases and state the hypotheses explicitly. □

Solution. Every Laurent overlap term can be split into contributions extending to the two affine charts in that degree range. □

Counterexamples and hypothesis tests

Exercise 49.80 (All schemes affine-acyclic)**.** Test the claim: every quasi-coherent sheaf has zero higher cohomology on every scheme.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. □

Solution. False; projective schemes have nontrivial cohomology. □

Exercise 49.81 (Negative twist H0)**.** Test the claim: $\mathcal{O}(d)$ on $\mathbb{P}^1$ has nonzero global sections for every negative d.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False. ☐

Exercise 49.82 (Vanishing sheaf zero)**.** Test the claim: vanishing higher cohomology means the sheaf itself is zero.

Hint. Test the claim on a concrete projective, divisor, blow-up, or sheaf model before generalizing. ☐

Solution. False. ☐

Applications and investigations

Exercise 49.83 (Affine computations)**.** Why is affine vanishing powerful?

Hint. Translate the question into affine vanishing, acyclic sheaves, and projective-space basic cases. ☐

Solution. It reduces many exact-sequence and derived calculations to ordinary module algebra. ☐

Exercise 49.84 (Projective invariants)**.** Why study nonvanishing on projective schemes?

Hint. Translate the question into affine vanishing, acyclic sheaves, and projective-space basic cases. ☐

Solution. Higher cohomology carries global information invisible on affine charts and controls embeddings, dimensions of linear systems, and obstructions. ☐

Challenge problems

Exercise 49.85 (Synthesis challenge)**.** Connect two central viewpoints of Basic Vanishing Results in one explicit calculation or proof.

Hint. Start from one worked example and reformulate it using the chapter's structural correspondence. ☐

Solution. A complete solution makes one concrete computation in Basic Vanishing Results and then translates it through affine vanishing, acyclic sheaves, and projective-space basic cases, checking both descriptions agree. ☐

Exercise 49.86 (Hypothesis challenge)**.** Choose one theorem-level statement from Basic Vanishing Results, remove one essential hypothesis, and give a concrete failure.

Hint. Use one of the hypothesis tests above as the seed counterexample. ☐

Solution. Name the removed hypothesis, identify the proof step that fails, and exhibit a concrete ring, scheme, divisor, sheaf, or morphism where the conclusion is false. ☐